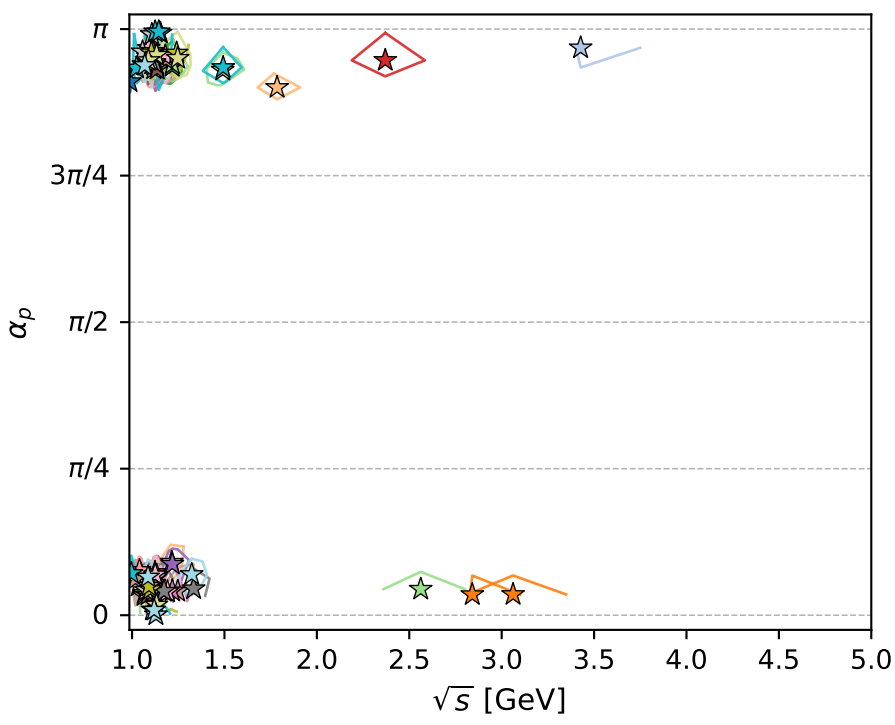
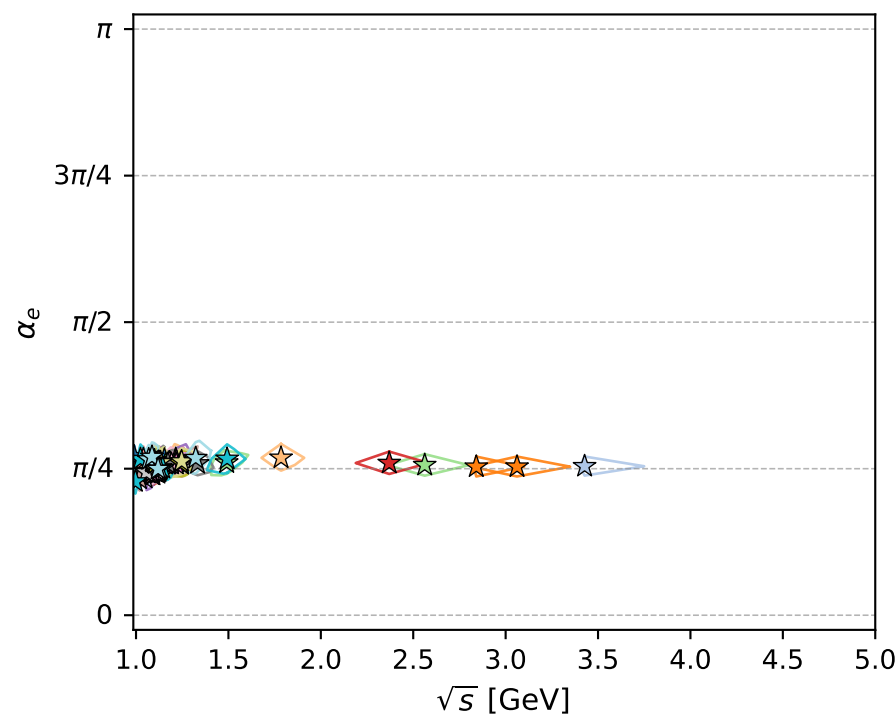
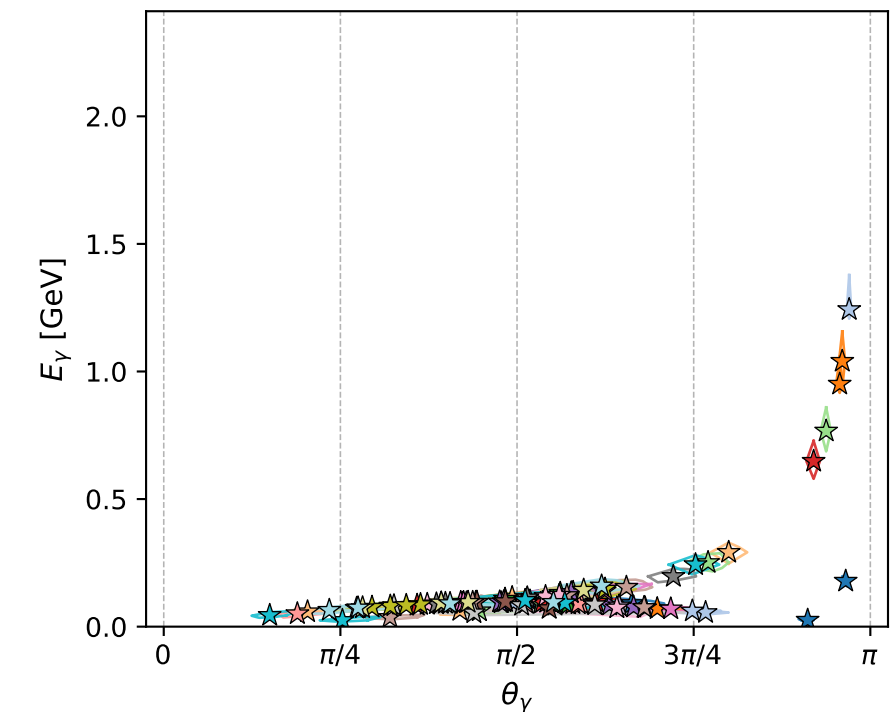
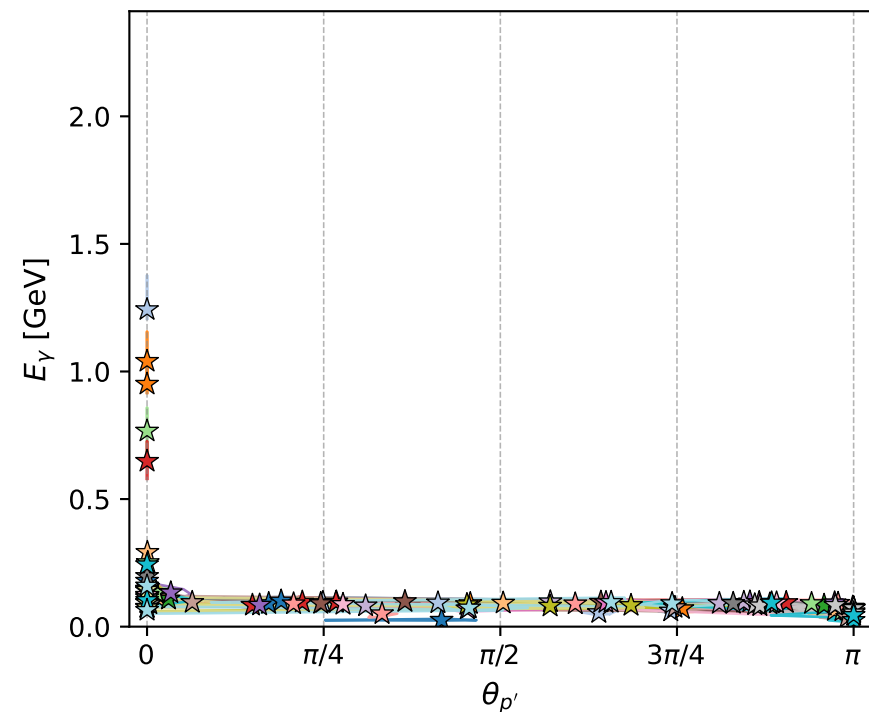
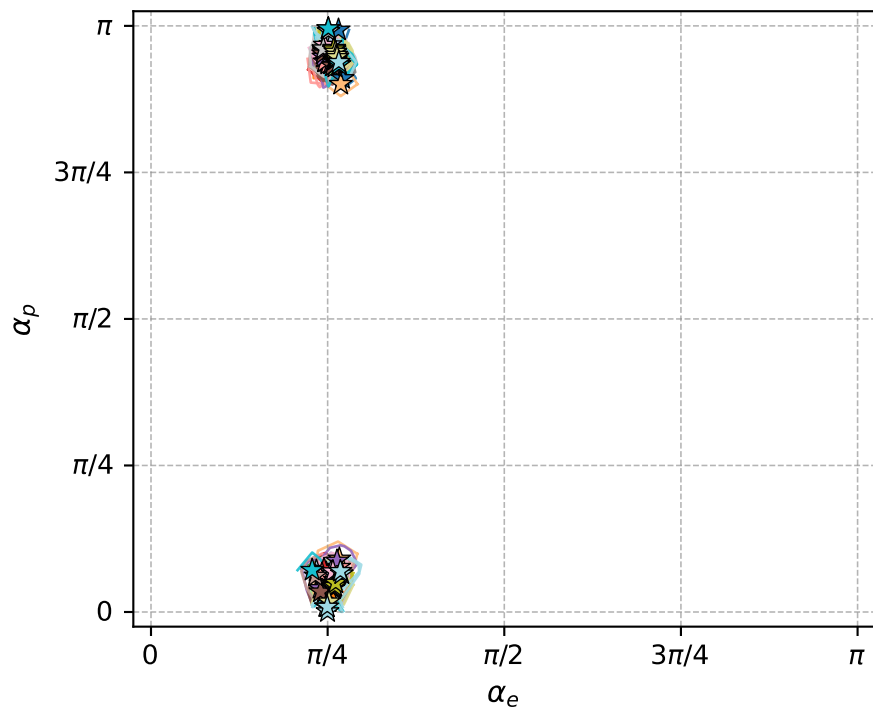
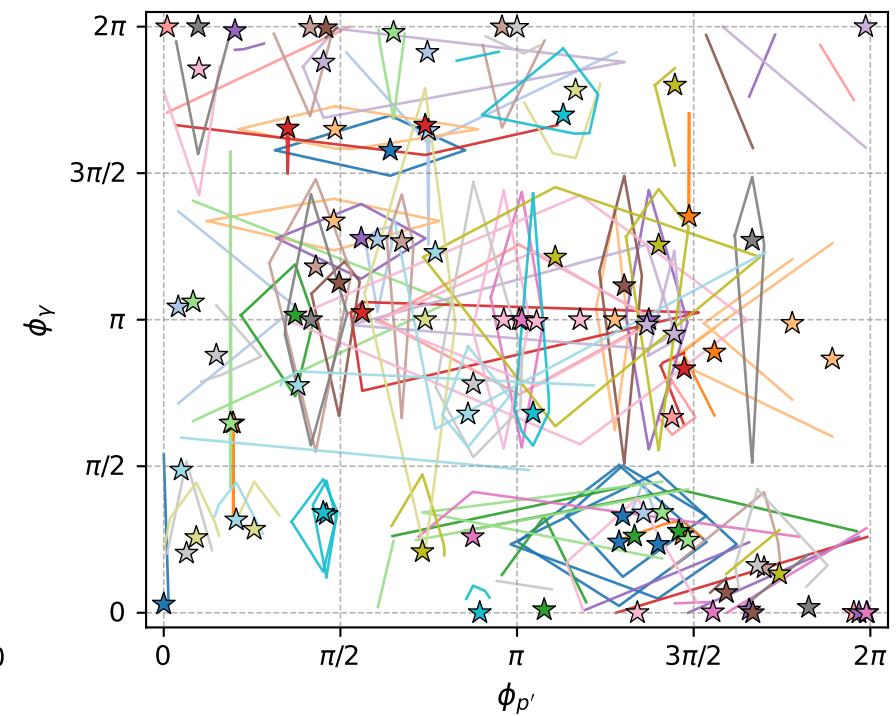
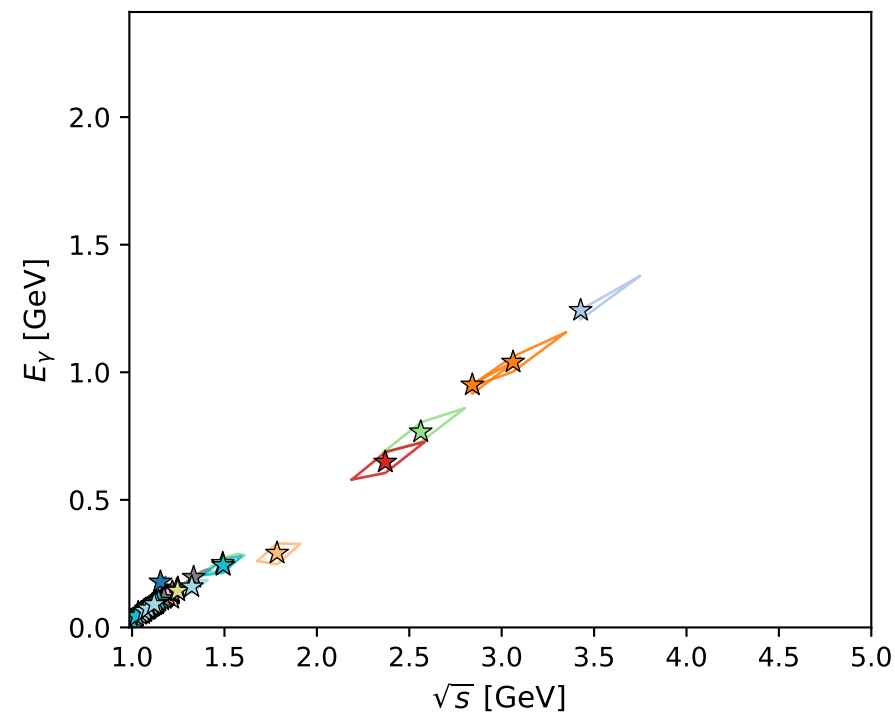
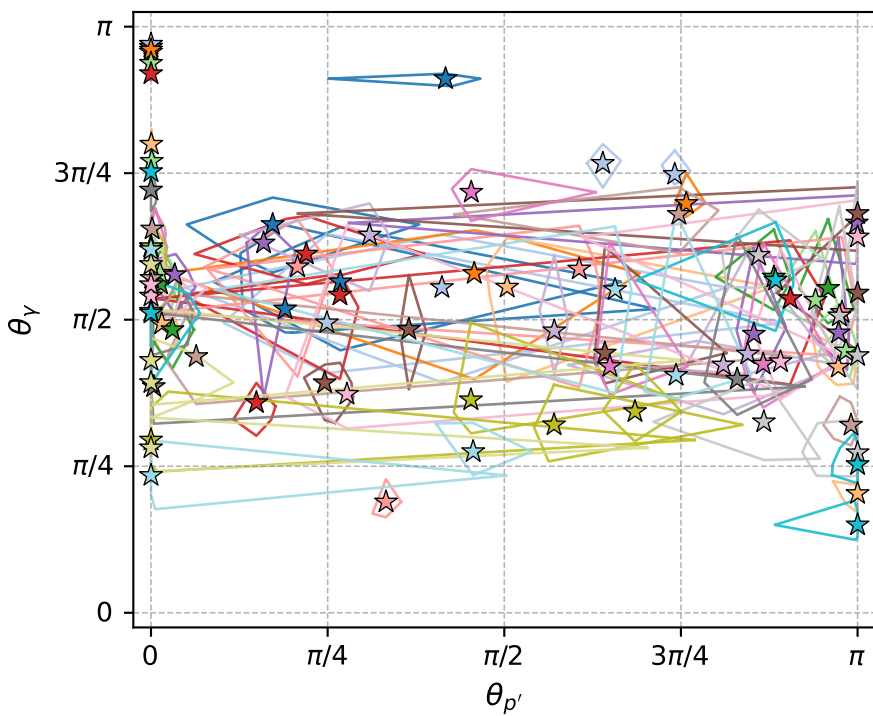


electron: polarization cluster P3; all local-minimum configurations and pairwise projections of their 8D contours



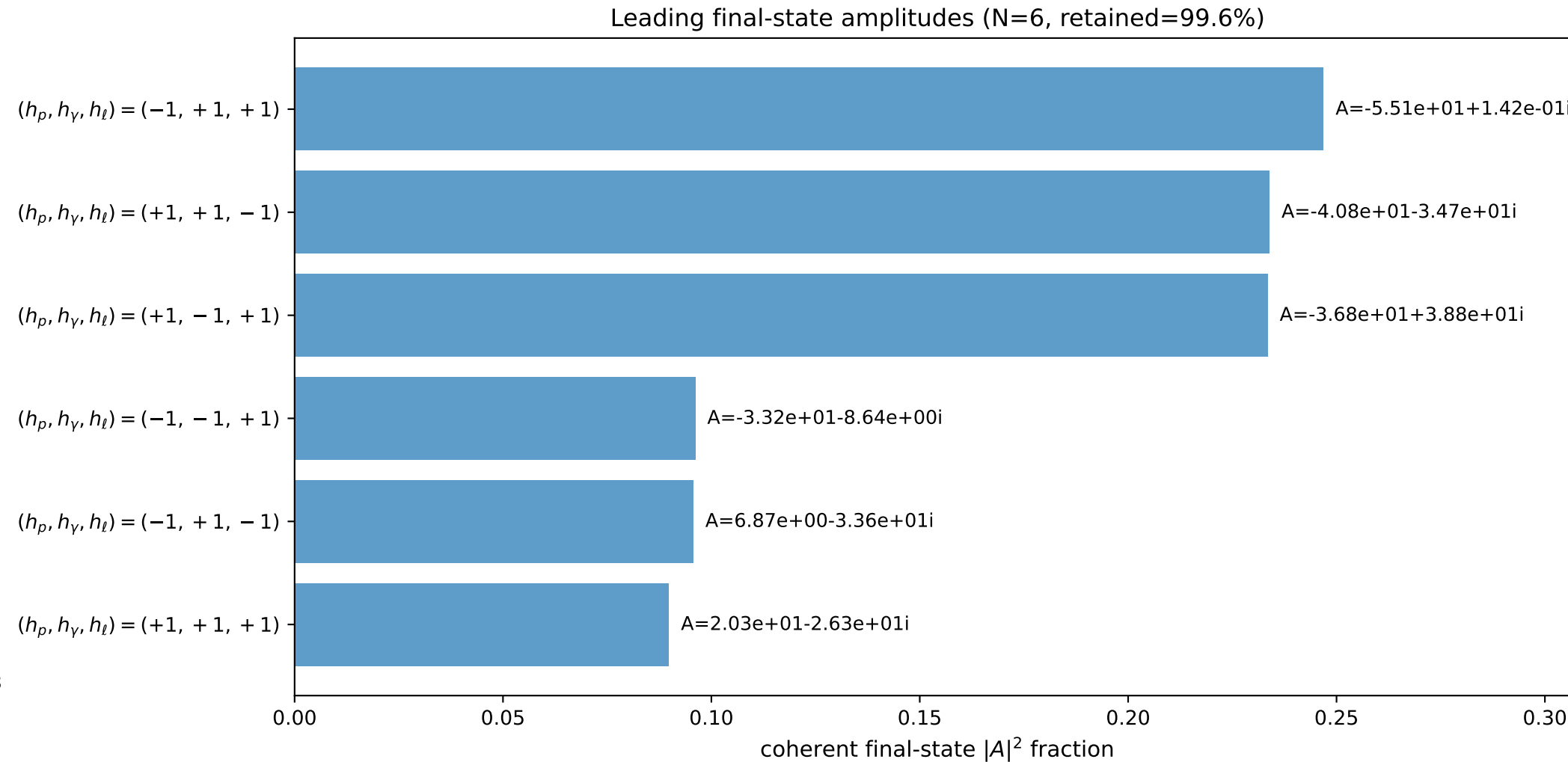
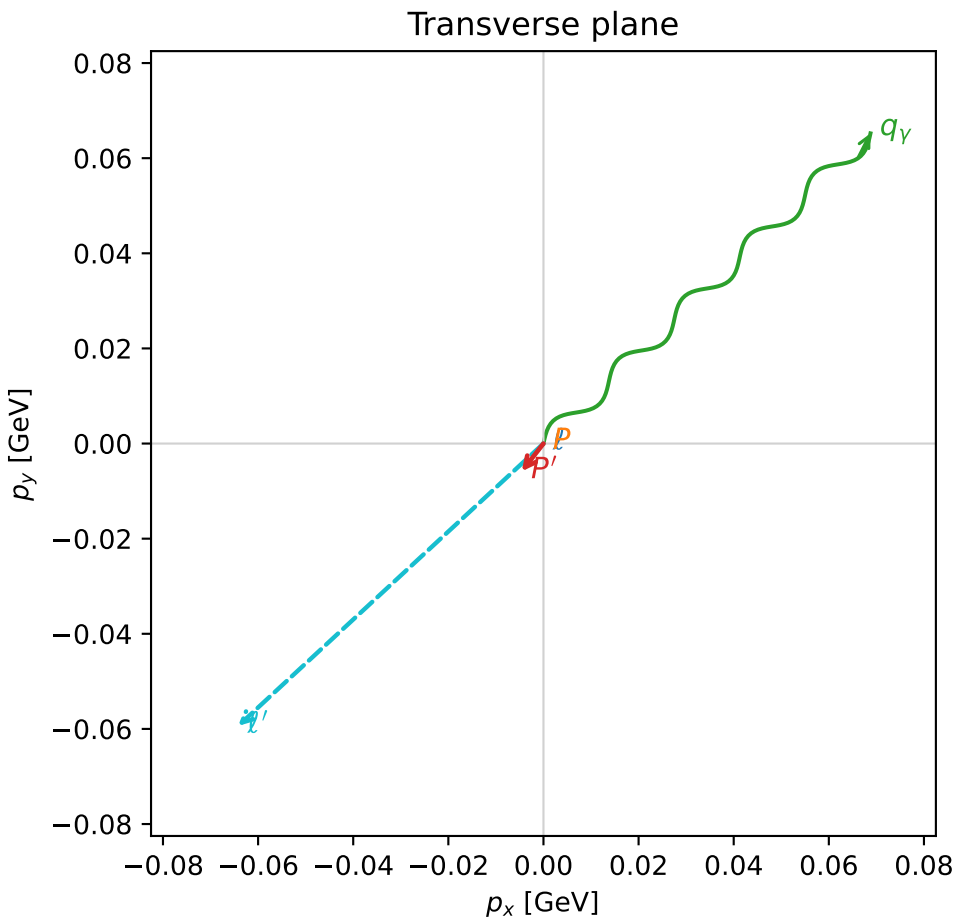
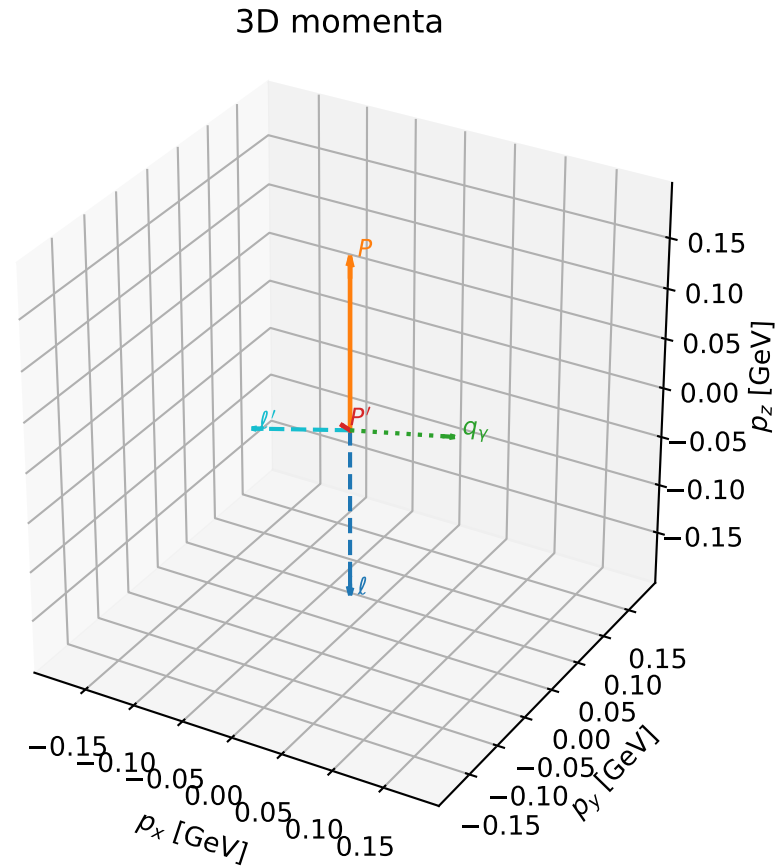
polarization cluster P3: $\alpha_e = \pi/4, \alpha_p = \pi$
 93 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=256
 D_W range=0.0004278916 to 0.03355944
 global-minimum offsets=0.0004114588 to 0.03354301

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

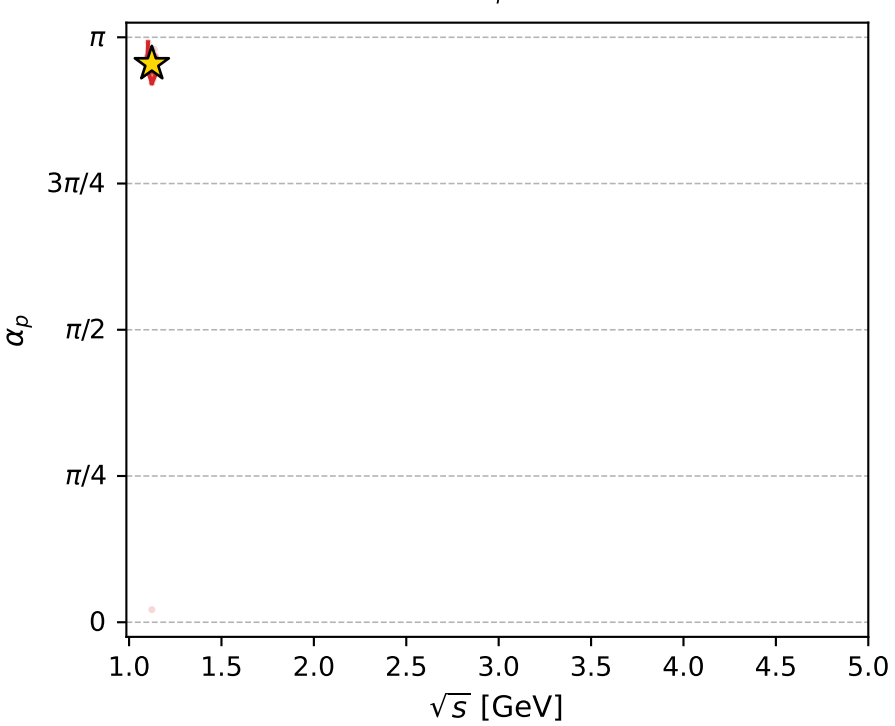
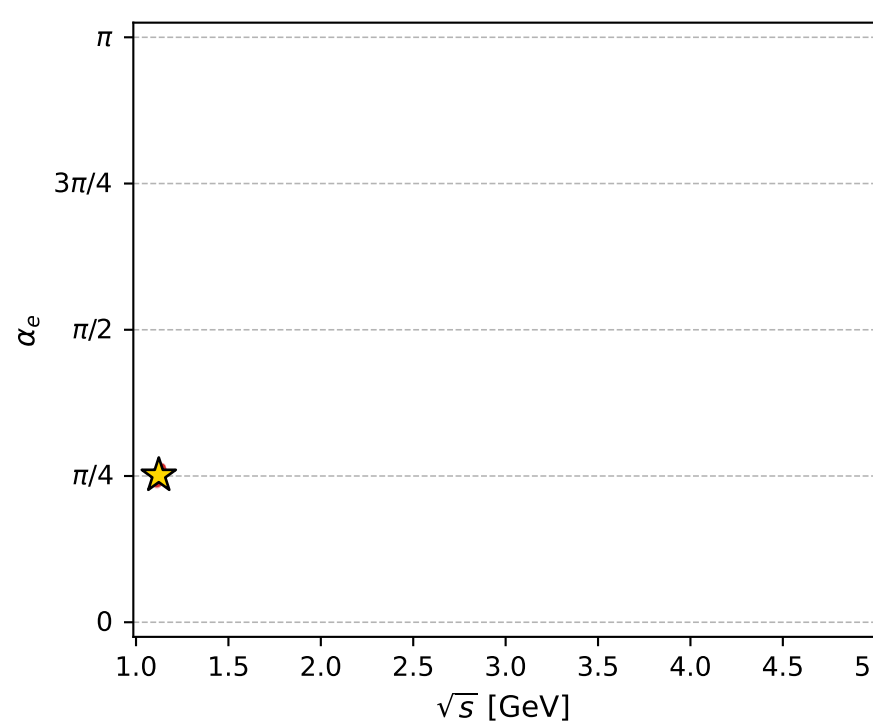
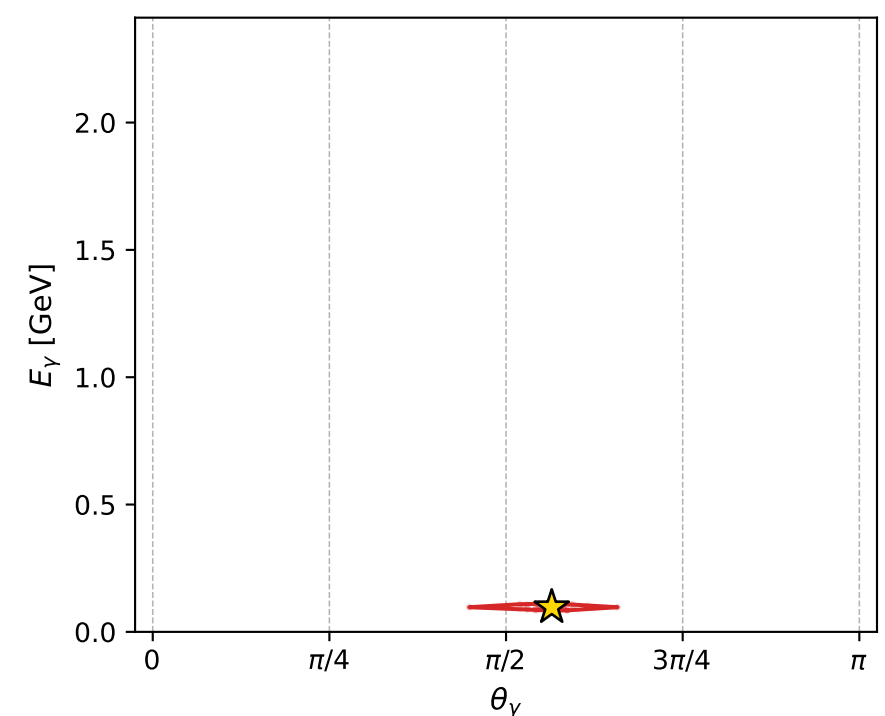
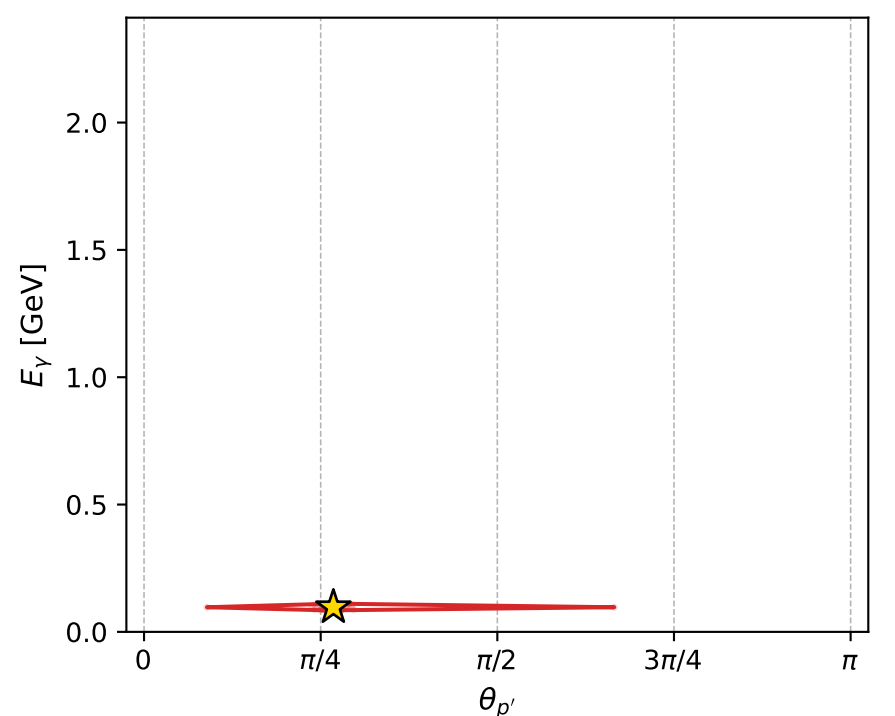
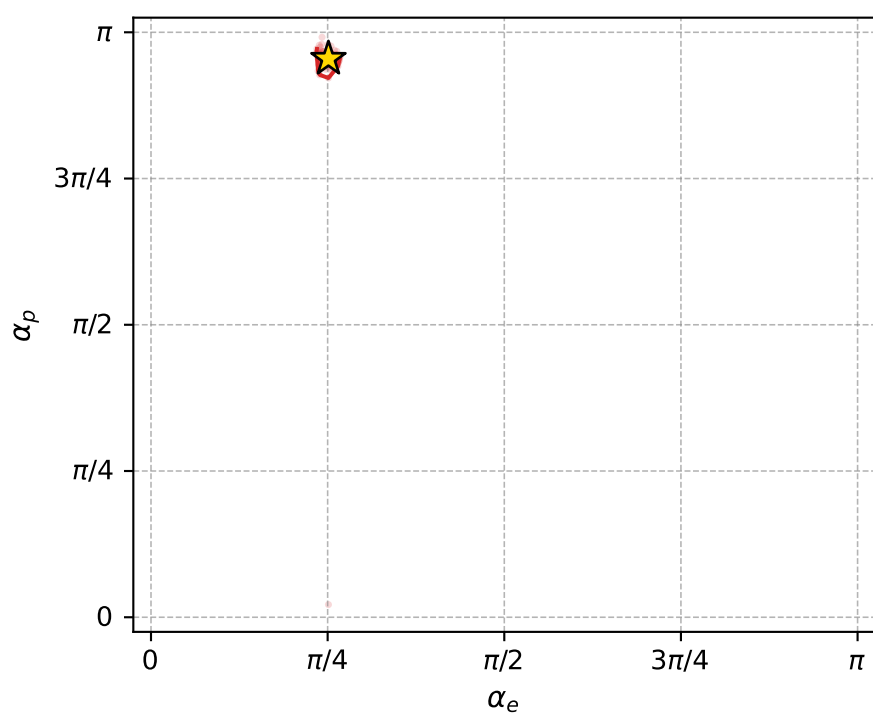
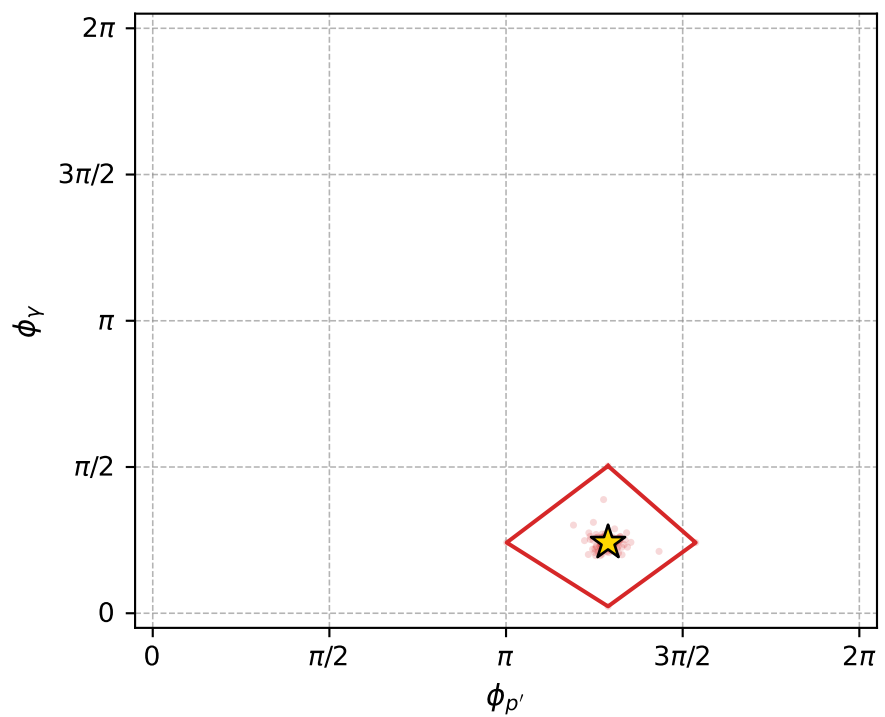
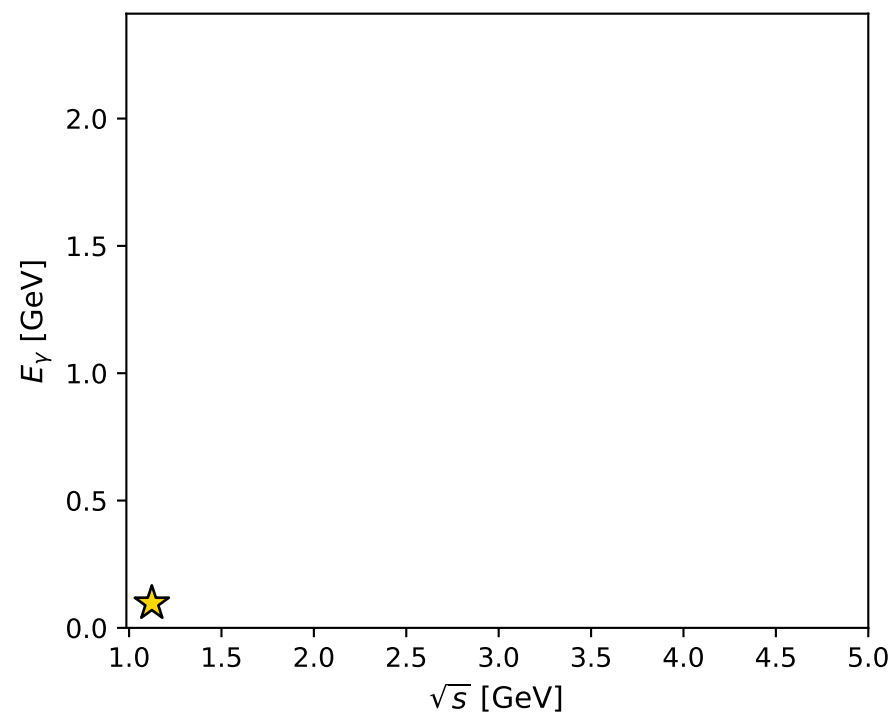
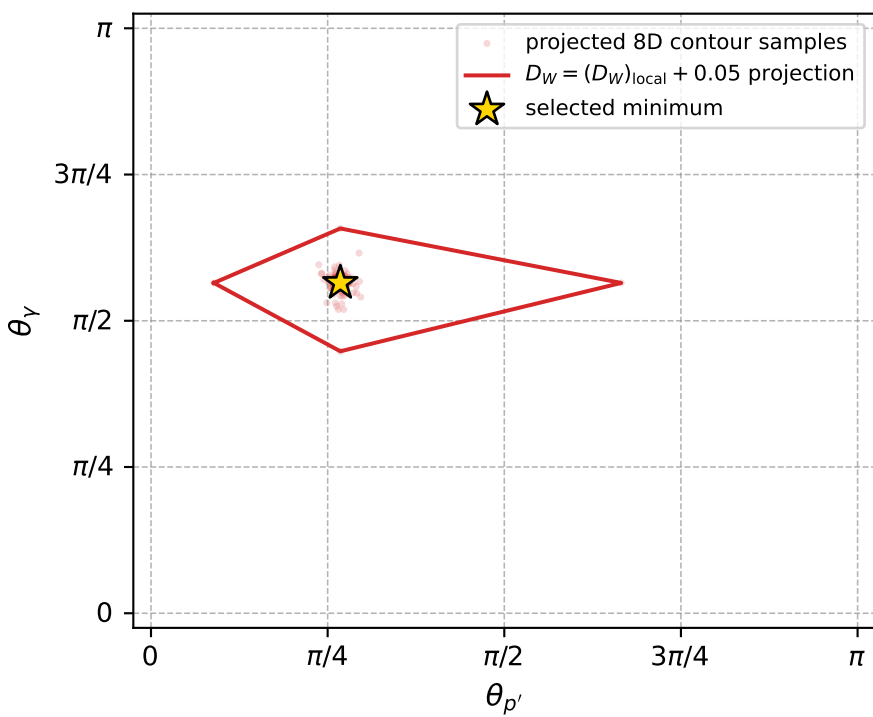
dw_mixing_angles_polarization_03_local_minimum_178 D_W=0.000427892
 region: polarization_cluster_03_local_minimum_178
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78894548$ rad
 incoming lepton: $0.704594|+\rangle + 0.709611|-\rangle$
 proton mixing angle: $\alpha_p=2.9993591$ rad
 incoming proton: $-0.989902|+\rangle + 0.141754|-\rangle$

$s=1.26154$, $\sqrt{s}=1.12318$, $|\vec{P}|=0.169918$, $|\vec{P}'|=0.00995884$
 $\theta_{P'}=0.841821$, $\theta_Y=1.77403$, $\phi_{P'}=4.04874$, $\phi_Y=0.758965$
 $E_Y=0.0967574$, $\theta_{\ell'}=1.42435$, $\phi_{\ell'}=3.88801$
 $Q^2=0.0344145$, $x_B=0.15816$, $t=-0.0264854$, $W^2=1.06302$, $y=0.570065$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1699$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1699$
 P $E= 0.9533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1699$
 ℓ' $E= 0.0884$ $p_x= -0.0642$ $p_y= -0.0594$ $p_z= 0.0129$
 P' $E= 0.9381$ $p_x= -0.0046$ $p_y= -0.0059$ $p_z= 0.0066$
 q_Y $E= 0.0968$ $p_x= 0.0688$ $p_y= 0.0652$ $p_z= -0.0195$



electron: polarization cluster P3, local minimum 178; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.00042789165$
 $D_W \text{ contour} = 0.050427892$
 above global minimum = 0.00041145881
 8D contour samples = 256

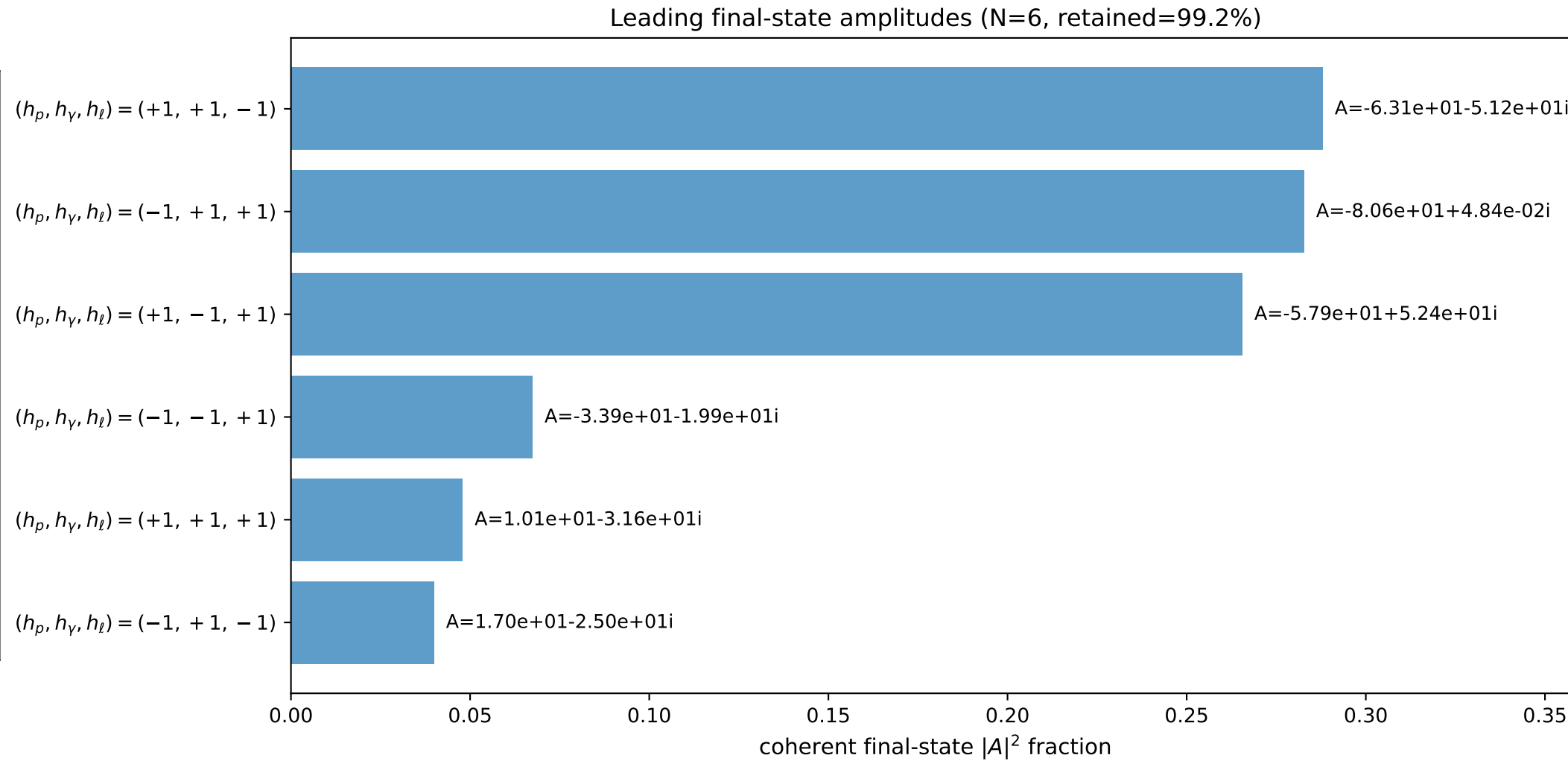
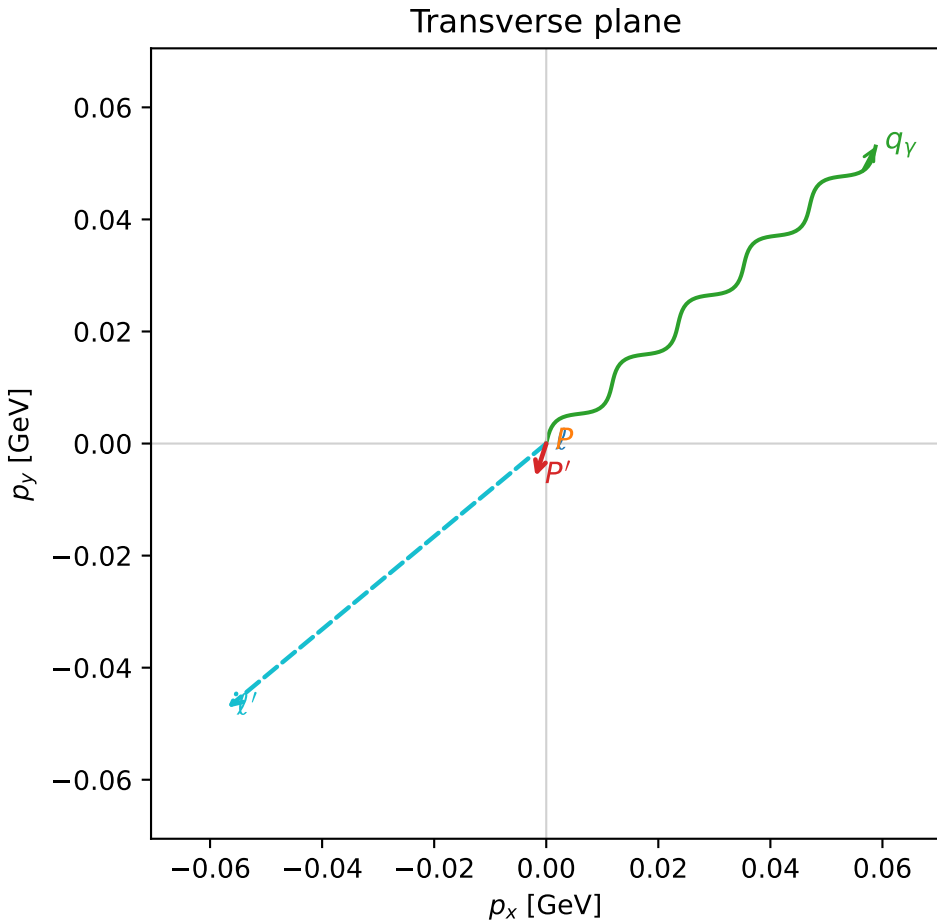
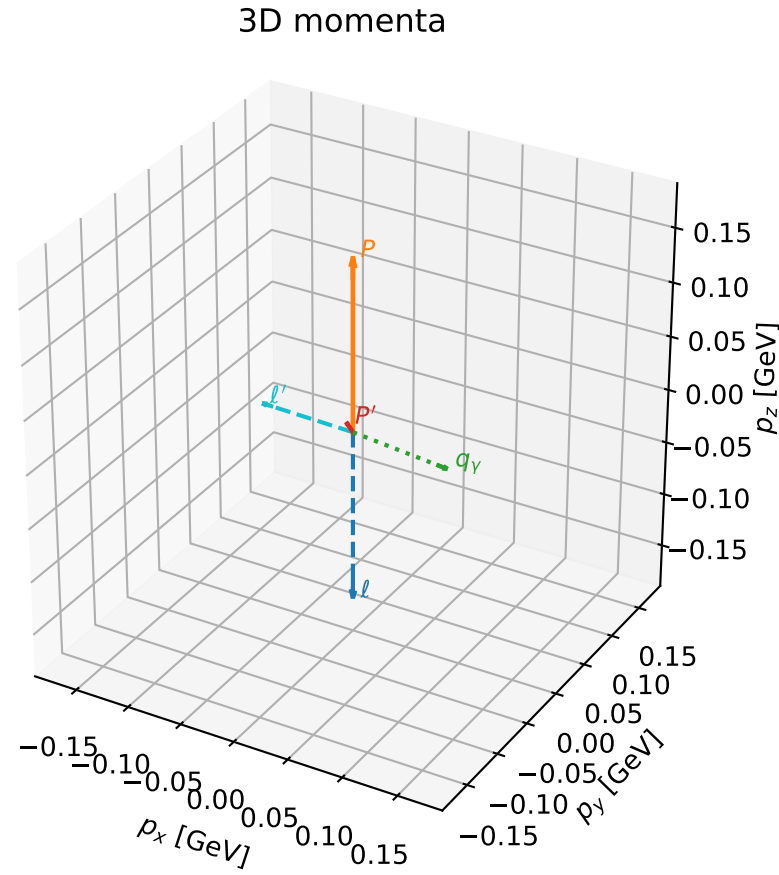
selected phase-space point:
 $\text{sqrt_s} = 1.123184$
 $\text{theta_p_out} = 0.8418209$
 $\text{theta_gamma_out} = 1.774027$
 $\text{q0out} = 0.09675744$
 $\text{phi_p_out} = 4.04874$
 $\text{phi_gamma_out} = 0.7589654$
 $\text{alpha_e} = 0.7889455$
 $\text{alpha_p} = 2.999359$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

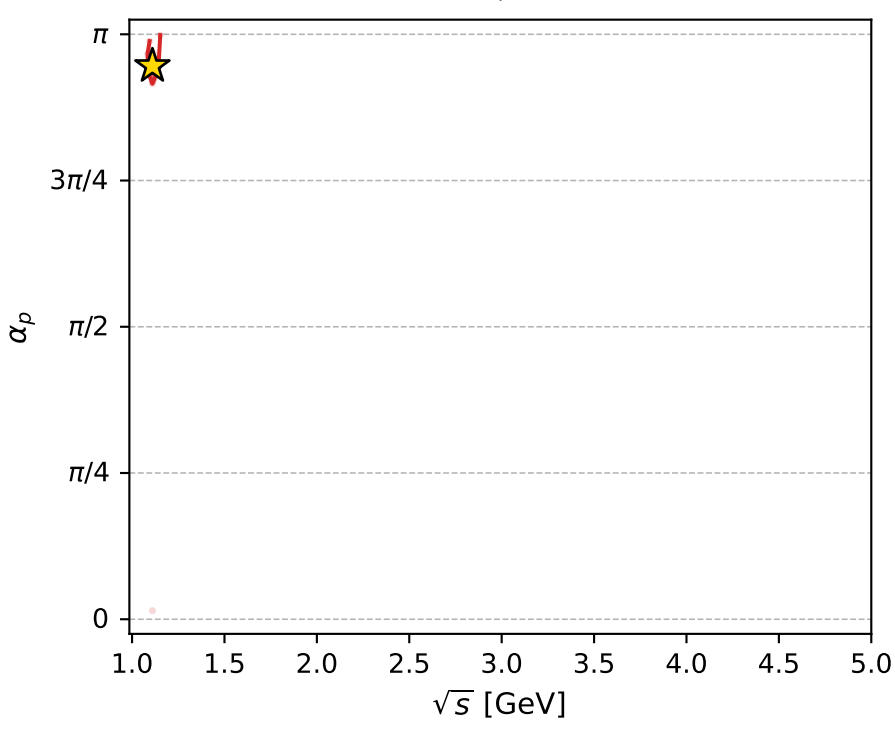
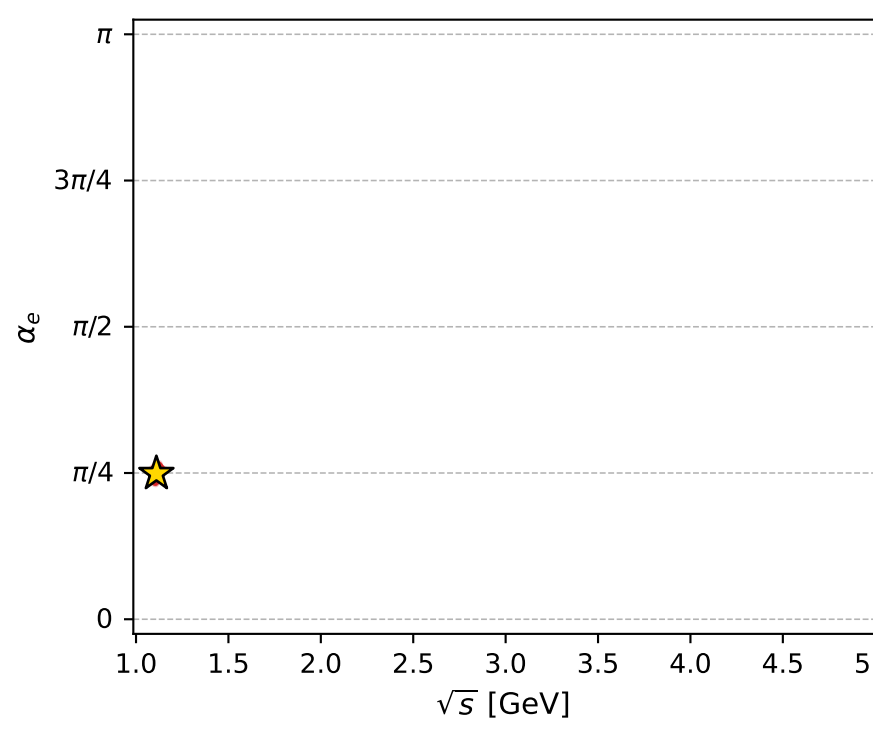
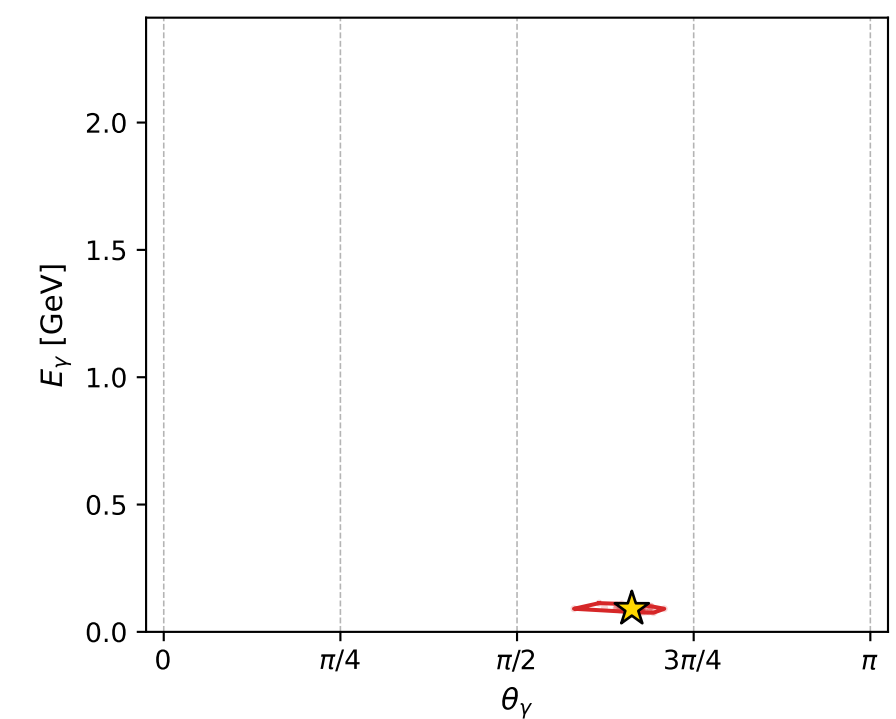
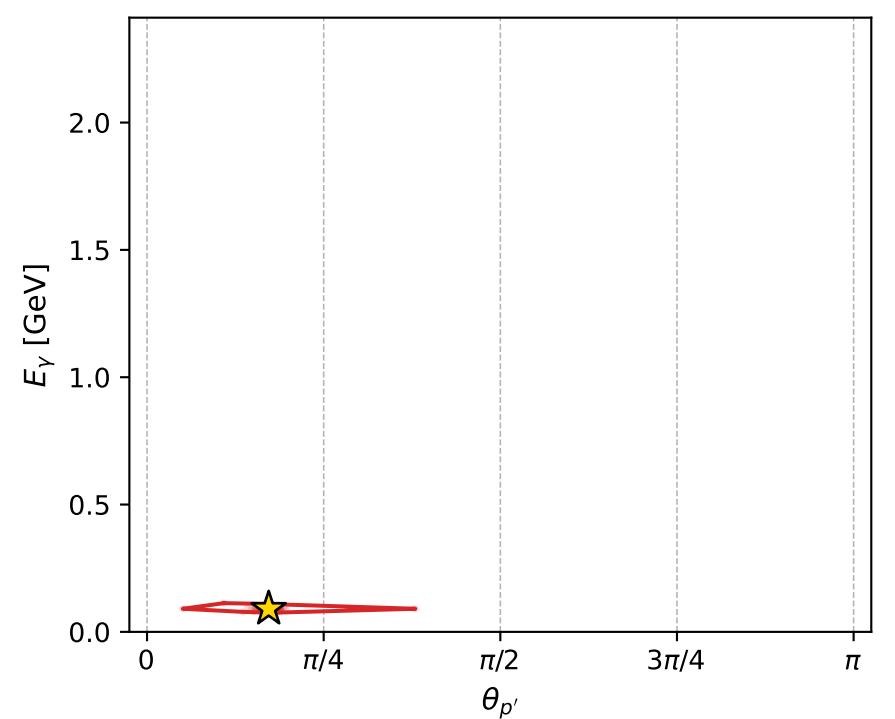
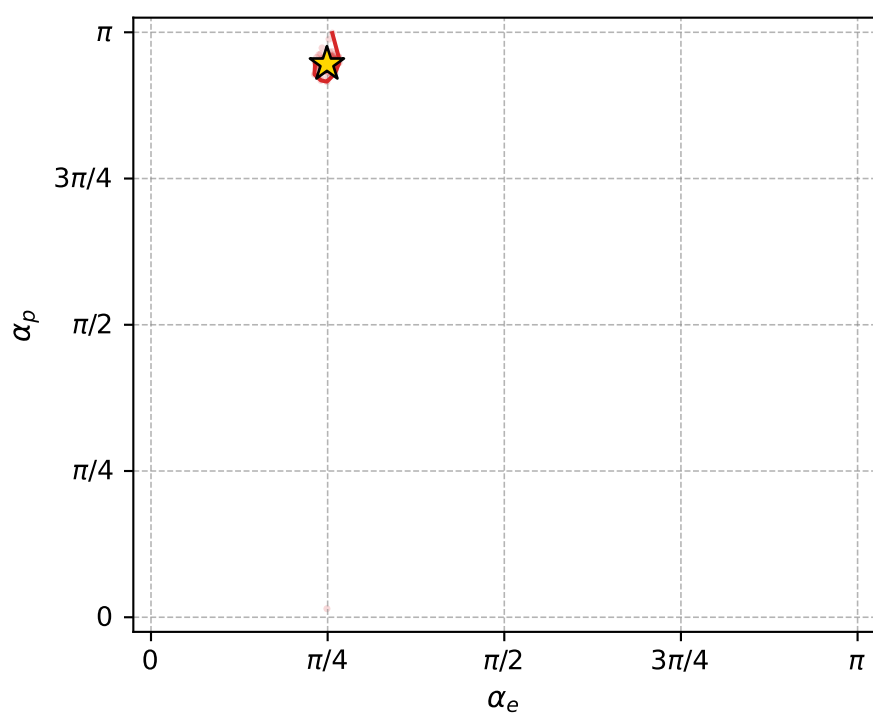
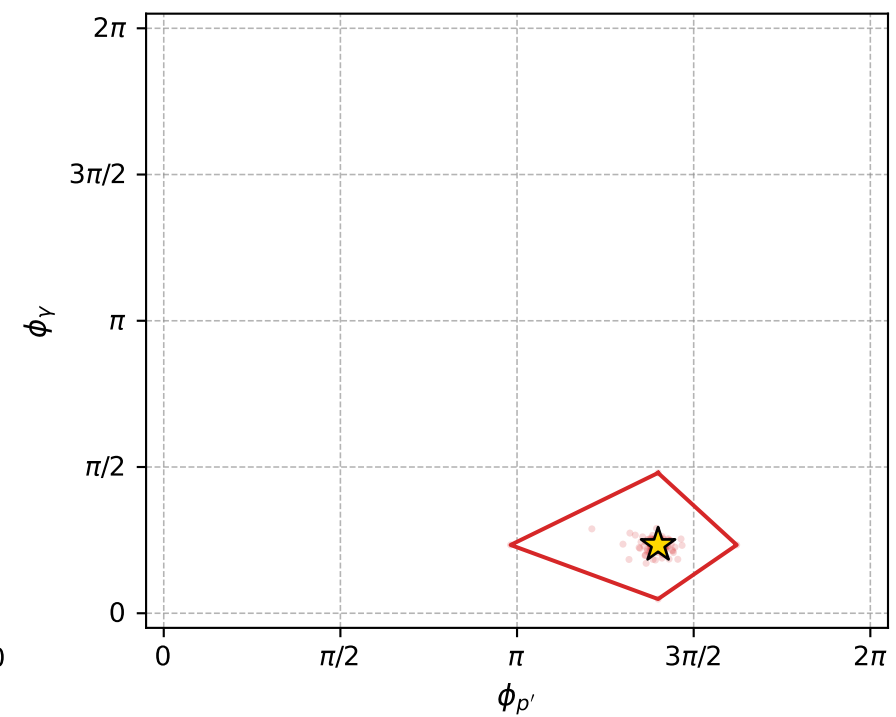
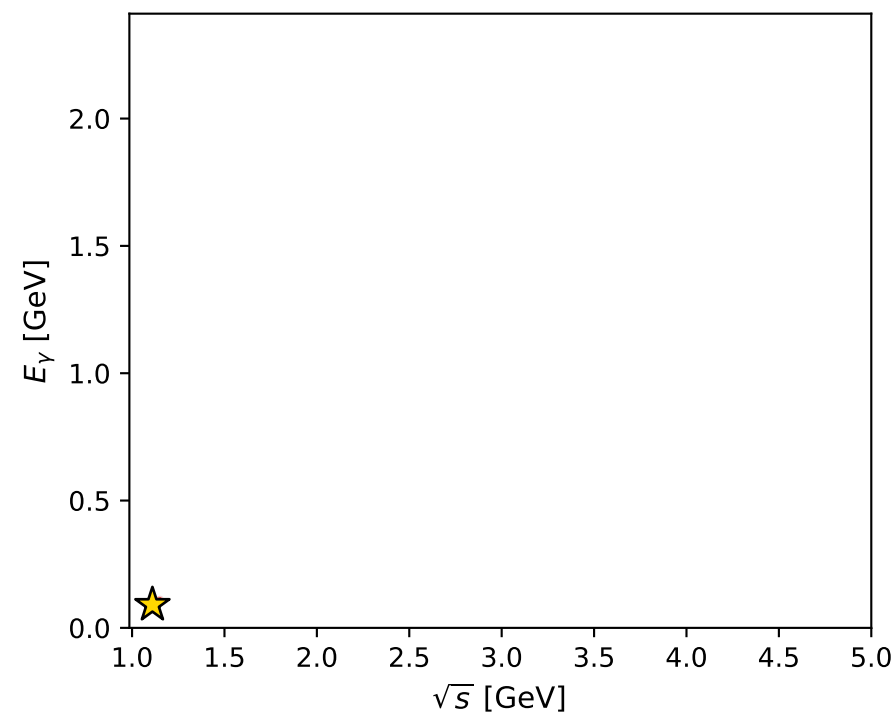
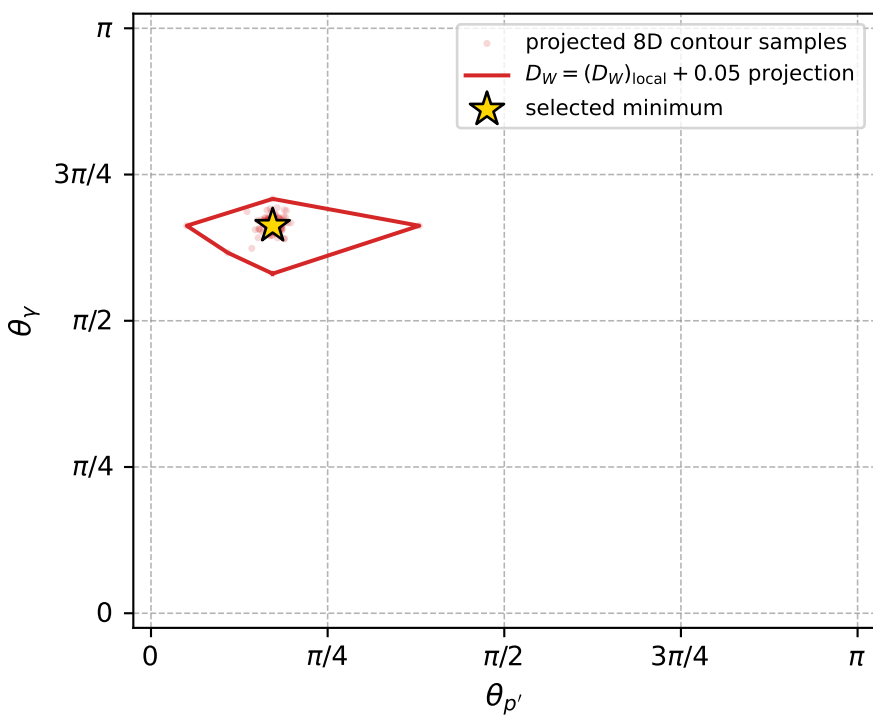
dw_mixing_angles_polarization_03_local_minimum_263 D_W=0.000665052
 region: polarization_cluster_03_local_minimum_263
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78220715$ rad
 incoming lepton: $0.70936|+\rangle +0.704847|-\rangle$
 proton mixing angle: $\alpha_p=2.9711339$ rad
 incoming proton: $-0.985507|+\rangle +0.169634|-\rangle$

$s=1.23237$, $\sqrt{s}=1.11012$, $|\vec{P}|=0.158778$, $|\vec{P}'|=0.0119428$
 $\theta_{P'}=0.540946$, $\theta_Y=2.08113$, $\phi_{P'}=4.39592$, $\phi_Y=0.733739$
 $E_Y=0.0906981$, $\theta_{\ell'}=1.13871$, $\phi_{\ell'}=3.83391$
 $Q^2=0.0366496$, $x_B=0.175723$, $t=-0.0219259$, $W^2=1.05176$, $y=0.591628$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1588$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1588$
 P $E= 0.9513$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1588$
 ℓ' $E= 0.0813$ $p_x= -0.0569$ $p_y= -0.0472$ $p_z= 0.0341$
 P' $E= 0.9381$ $p_x= -0.0019$ $p_y= -0.0058$ $p_z= 0.0102$
 q_Y $E= 0.0907$ $p_x= 0.0588$ $p_y= 0.0530$ $p_z= -0.0443$



electron: polarization cluster P3, local minimum 263; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.00066505249$
 $D_W \text{ contour} = 0.050665052$
 above global minimum = 0.00064861965
 8D contour samples = 256

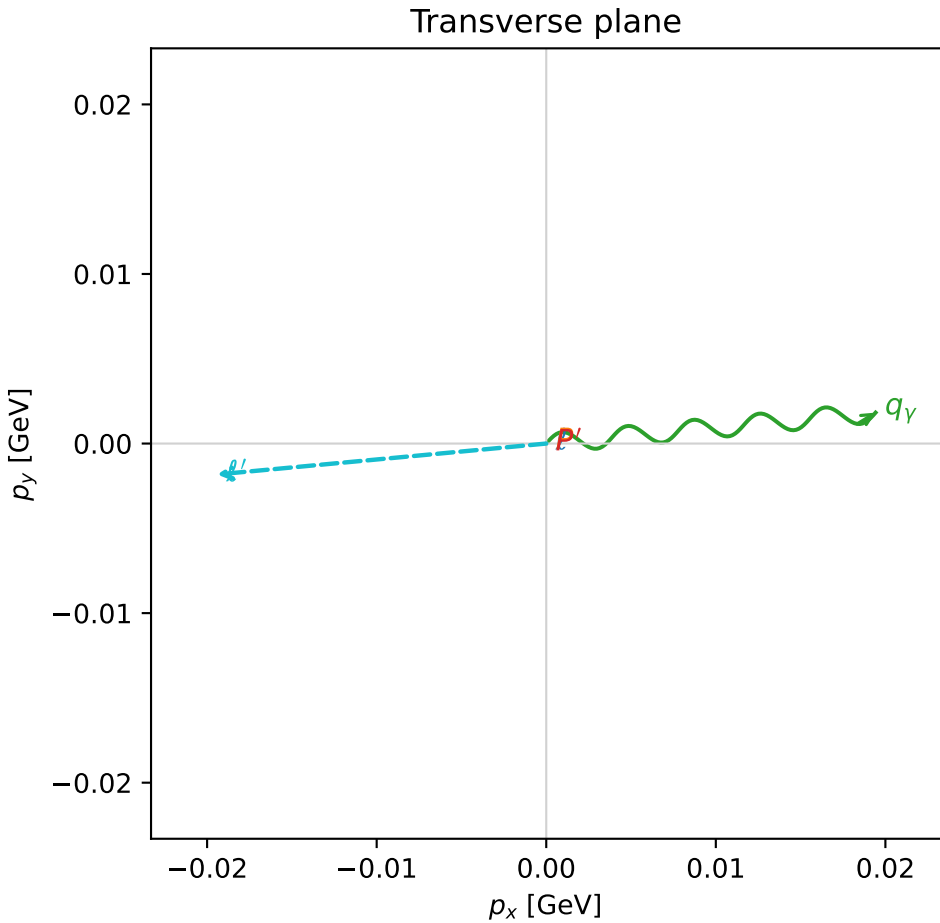
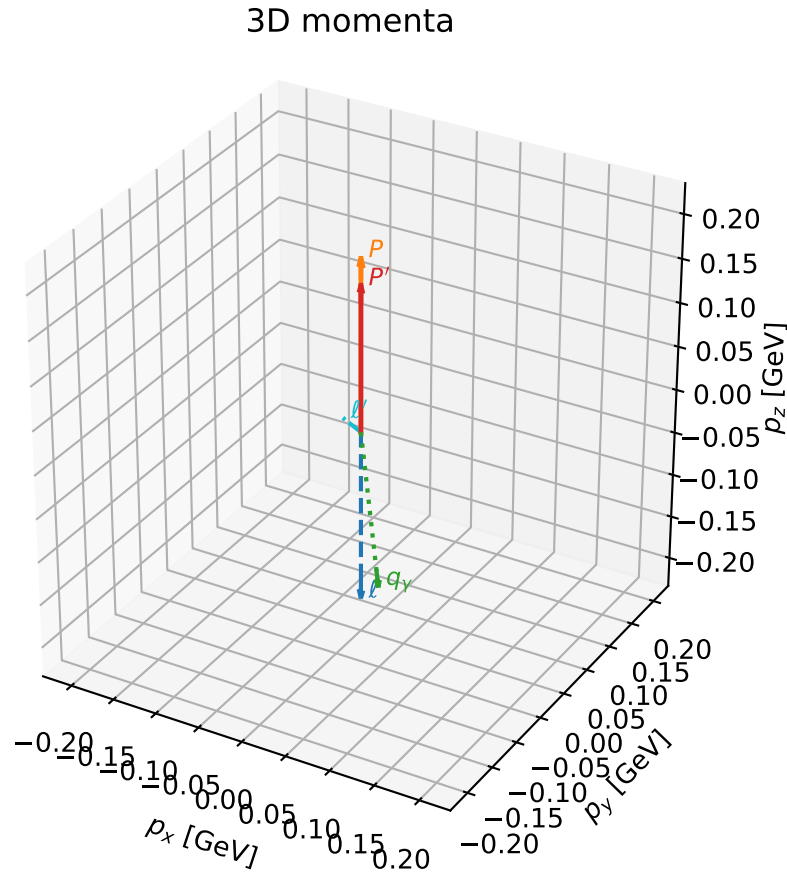
selected phase-space point:
 $\text{sqrt_s} = 1.110122$
 $\text{theta_p_out} = 0.5409461$
 $\text{theta_gamma_out} = 2.081125$
 $\text{qOut} = 0.09069808$
 $\text{phi_p_out} = 4.395924$
 $\text{phi_gamma_out} = 0.7337387$
 $\text{alpha_e} = 0.7822071$
 $\text{alpha_p} = 2.971134$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_378 D_W=0.0012061
 region: polarization_cluster_03_local_minimum_378
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.8336692$ rad
 incoming lepton: $0.672164|+\rangle + 0.740403|-\rangle$
 proton mixing angle: $\alpha_p=3.1197285$ rad
 incoming proton: $-0.999761|+\rangle + 0.0218624|-\rangle$

$s=1.33084$, $\sqrt{s}=1.15362$, $|\vec{P}|=0.195471$, $|\vec{P}'|=0.16574$
 $\theta_{P'}=0$, $\theta_V=3.03202$, $\phi_{P'}=0.000172031$, $\phi_V=0.0936696$
 $E_V=0.178397$, $\theta_{\ell'}=1.03482$, $\phi_{\ell'}=3.23526$
 $Q^2=0.0134016$, $x_B=0.0325252$, $t=-0.000852303$, $W^2=1.27848$, $y=0.913613$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1955$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1955$
 P $E= 0.9582$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1955$
 ℓ' $E= 0.0227$ $p_x= -0.0194$ $p_y= -0.0018$ $p_z= 0.0116$
 P' $E= 0.9525$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1657$
 q_Y $E= 0.1784$ $p_x= 0.0194$ $p_y= 0.0018$ $p_z= -0.1773$

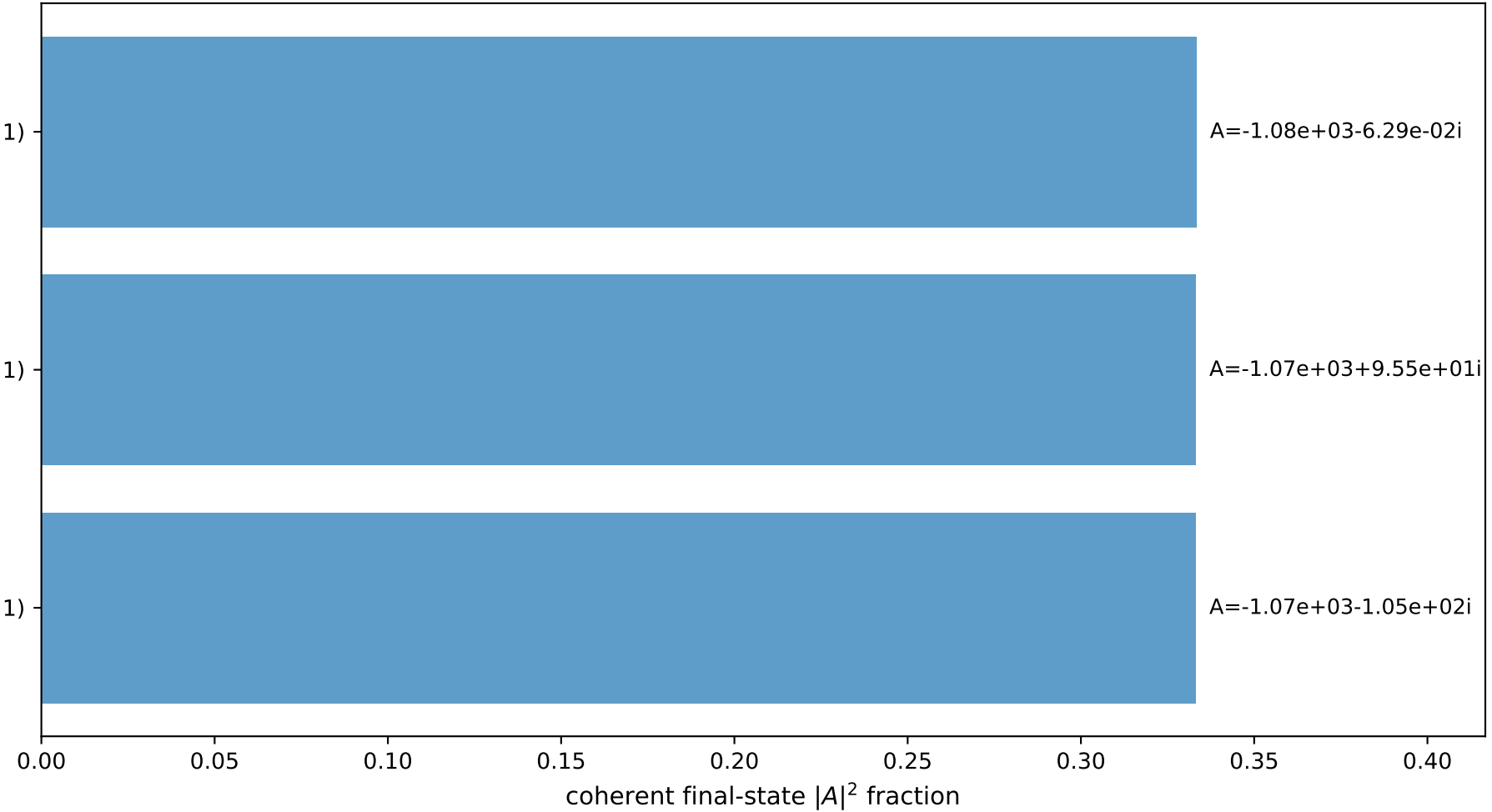


$(h_p, h_V, h_{\ell}) = (-1, +1, +1)$

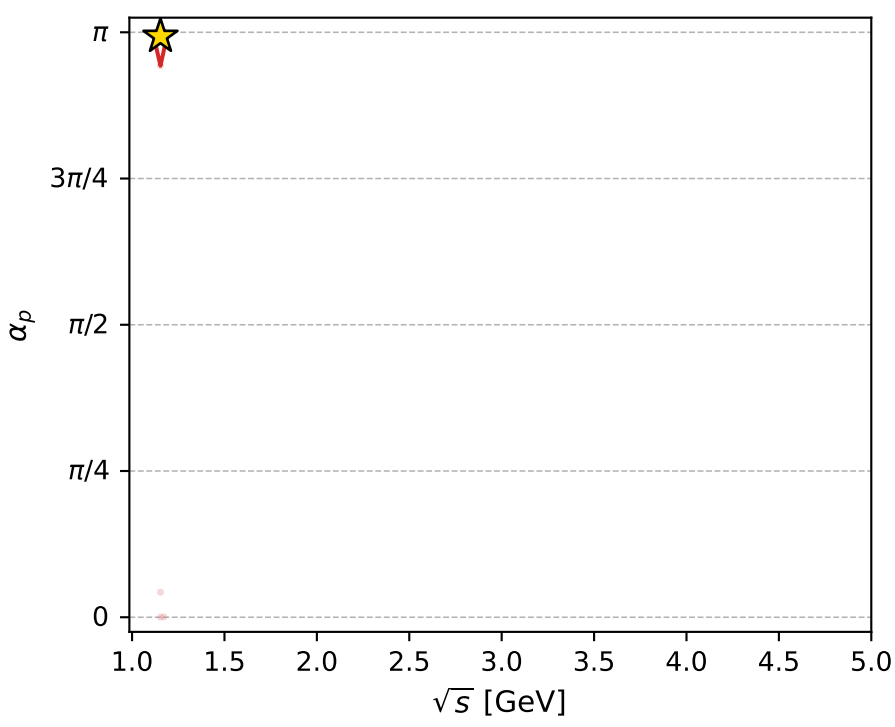
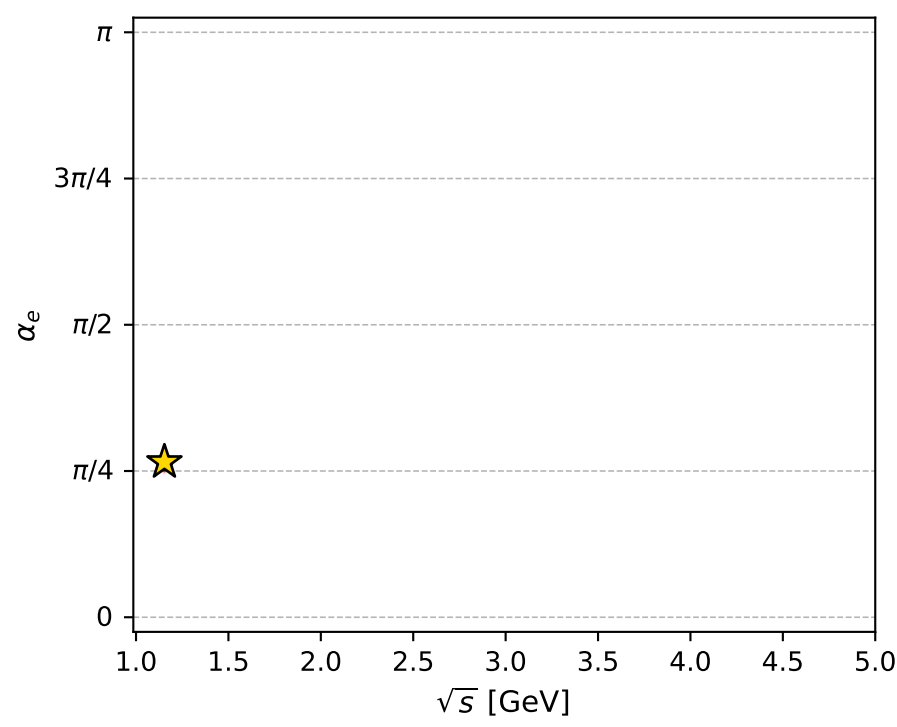
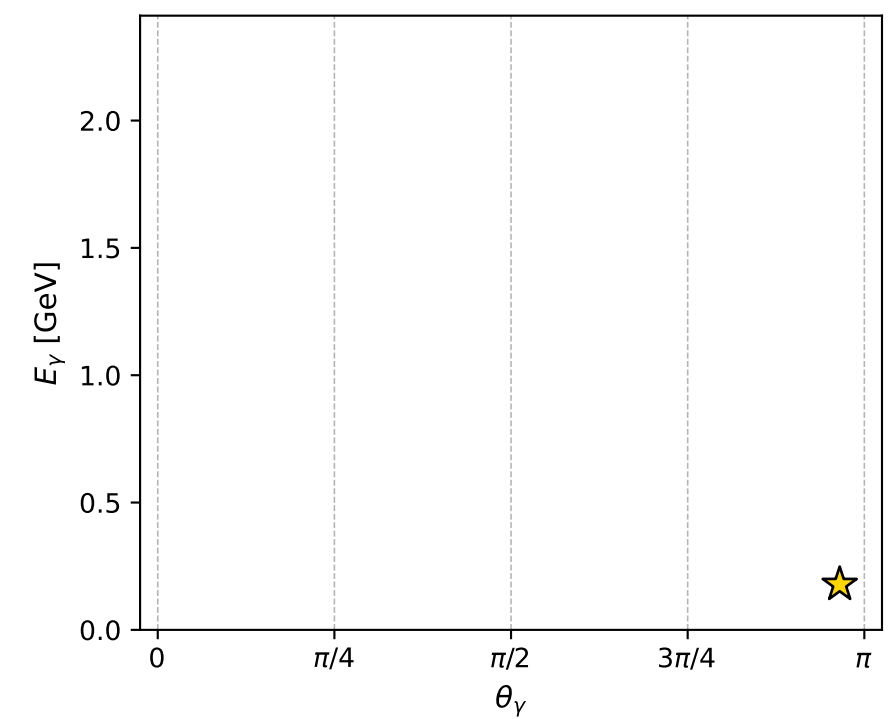
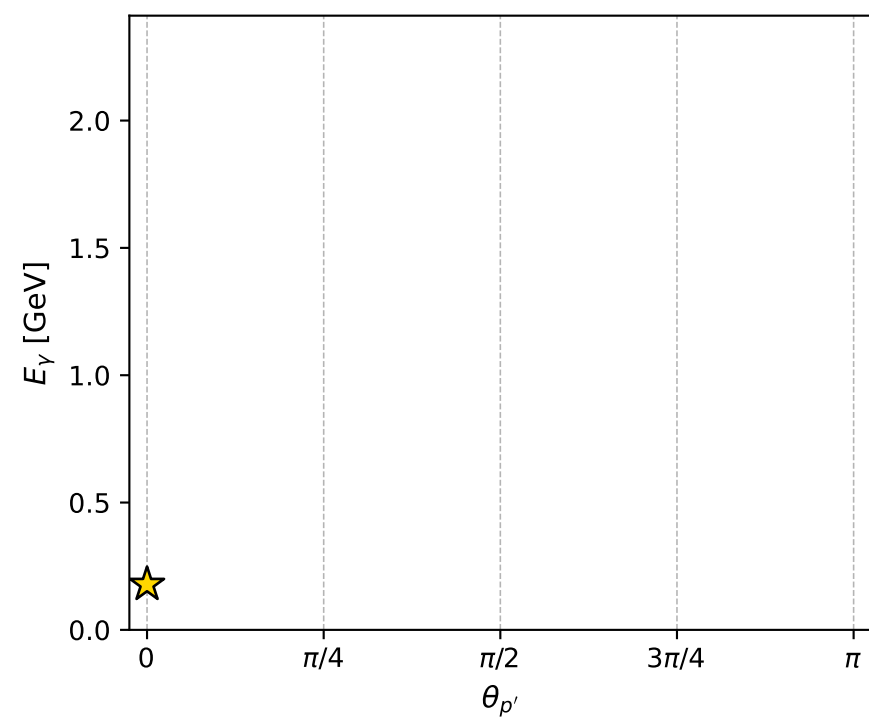
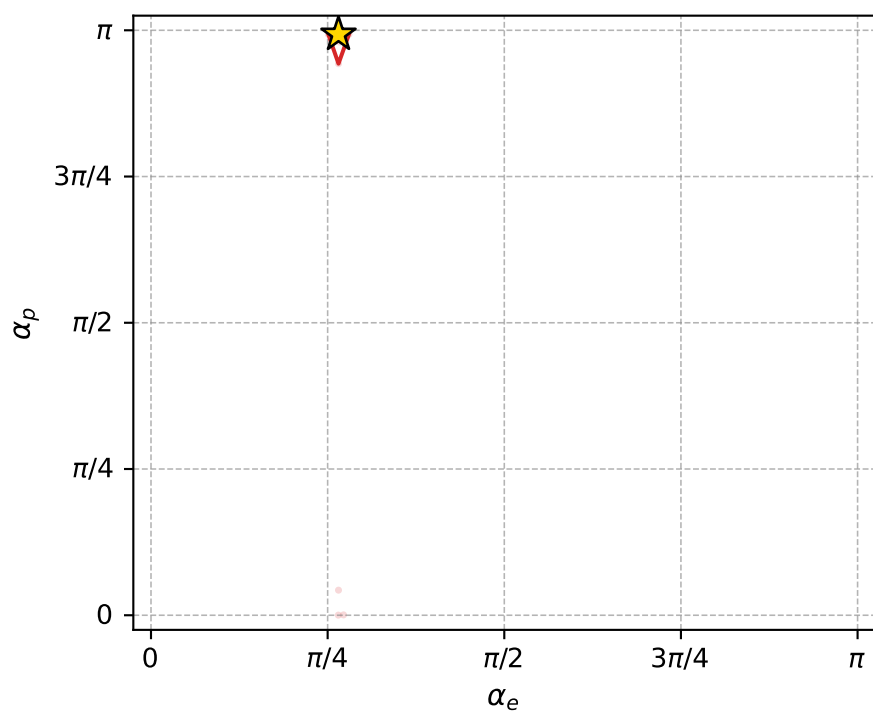
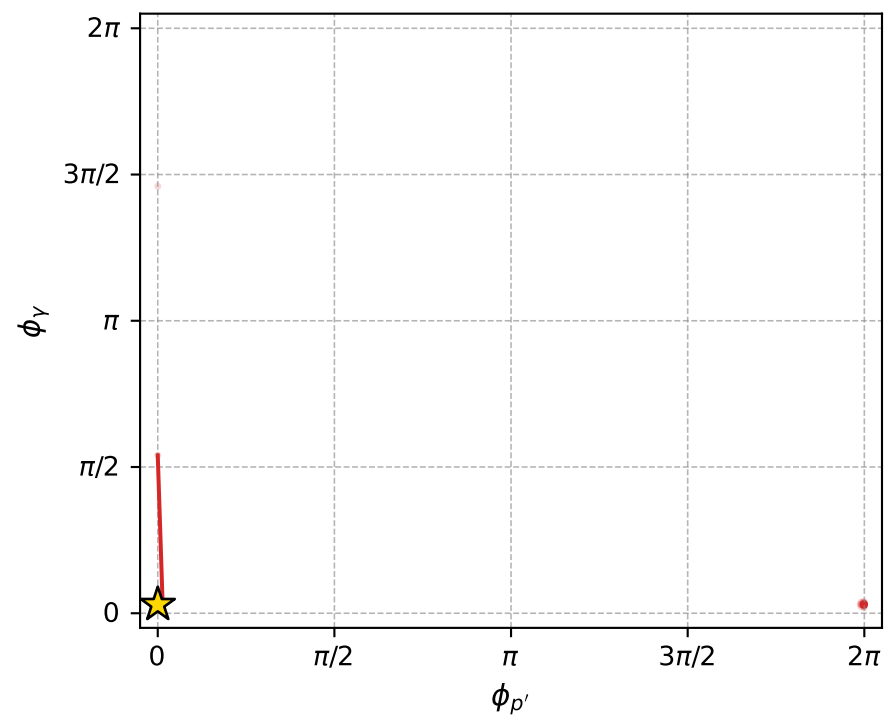
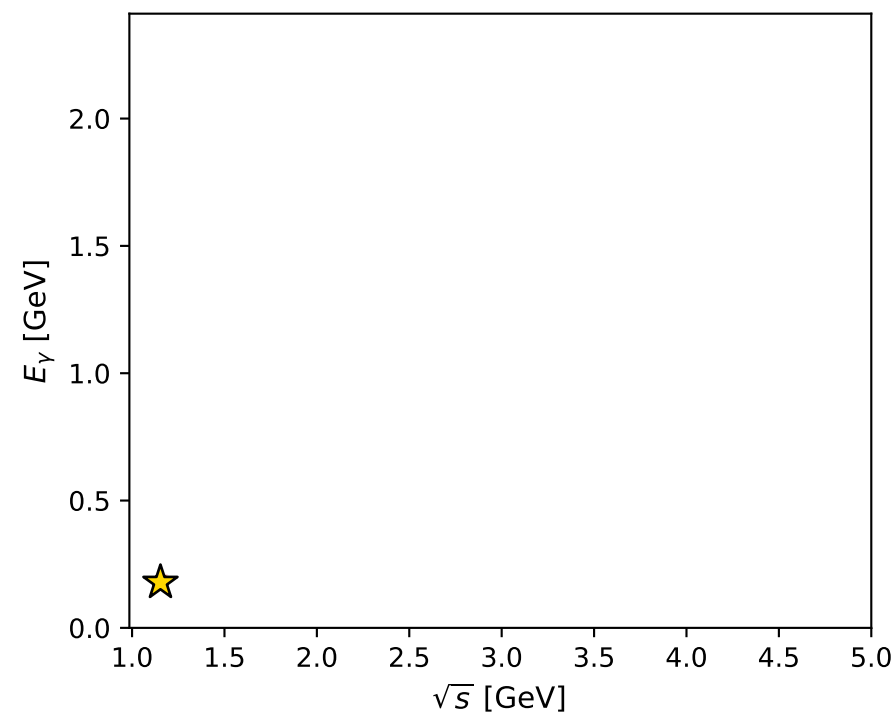
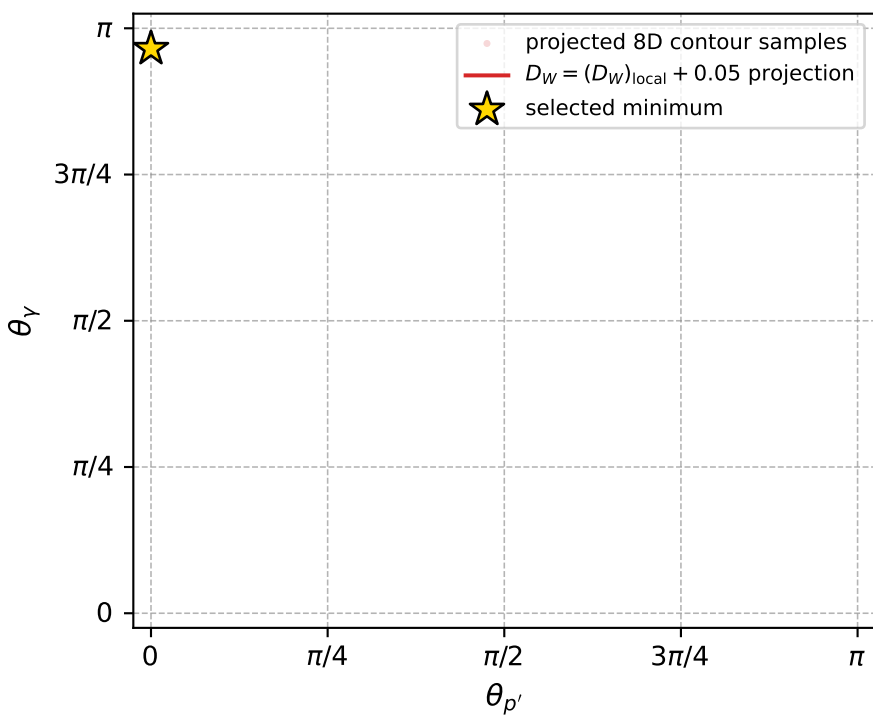
$(h_p, h_V, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_V, h_{\ell}) = (+1, +1, -1)$

Leading final-state amplitudes (N=3, retained=100.0%)



electron: polarization cluster P3, local minimum 378; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0012060997$

$D_W \text{ contour} = 0.0512061$

above global minimum = 0.0011896668

8D contour samples = 123

selected phase-space point:

$\text{sqrt}_s = 1.153622$

$\text{theta}_p \text{ out} = 0$

$\text{theta}_\gamma \text{ out} = 3.032025$

$q_{\text{out}} = 0.178397$

$\text{phi}_p \text{ out} = 0.0001720313$

$\text{phi}_\gamma \text{ out} = 0.09366959$

$\alpha_e = 0.8336692$

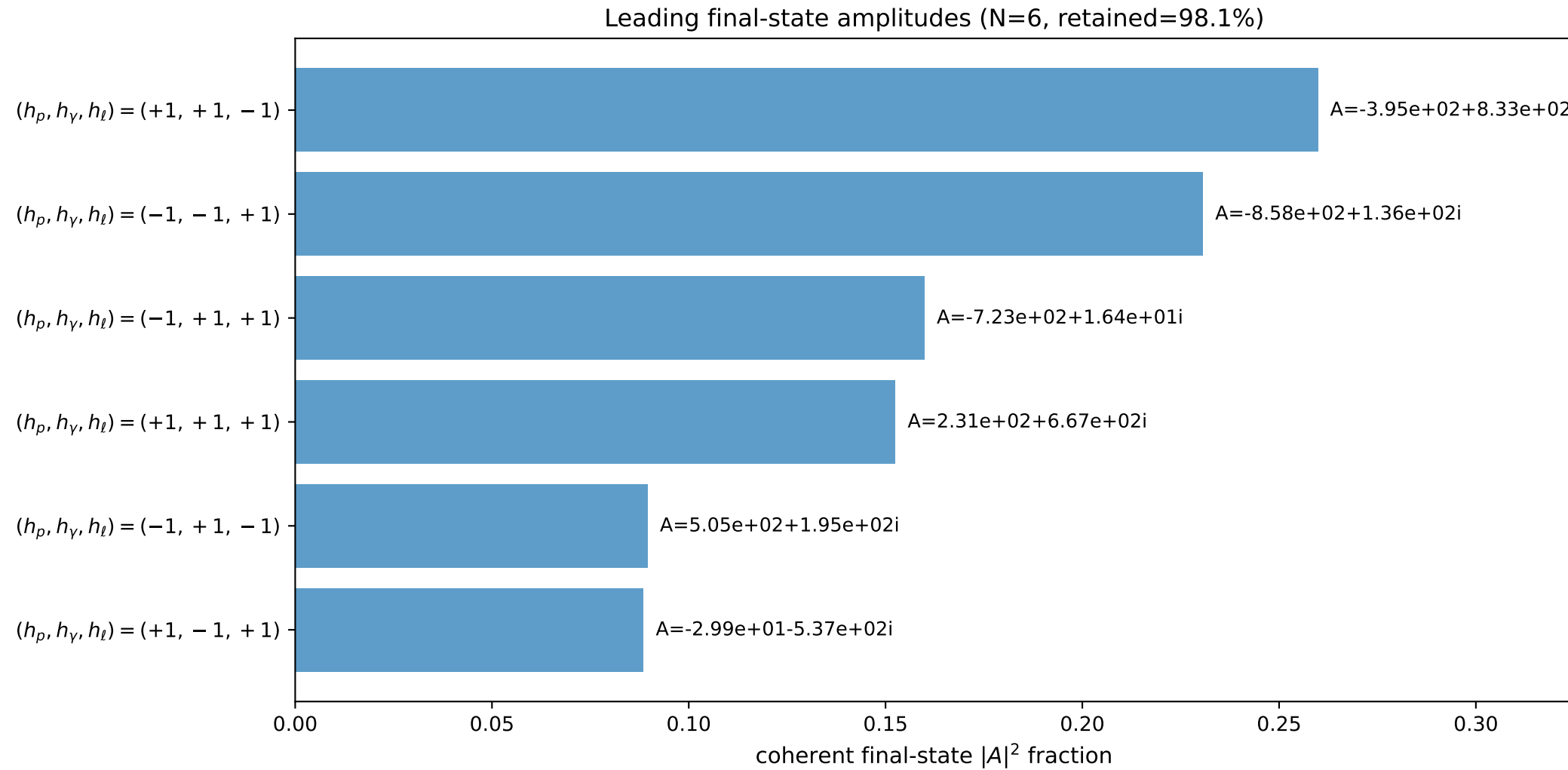
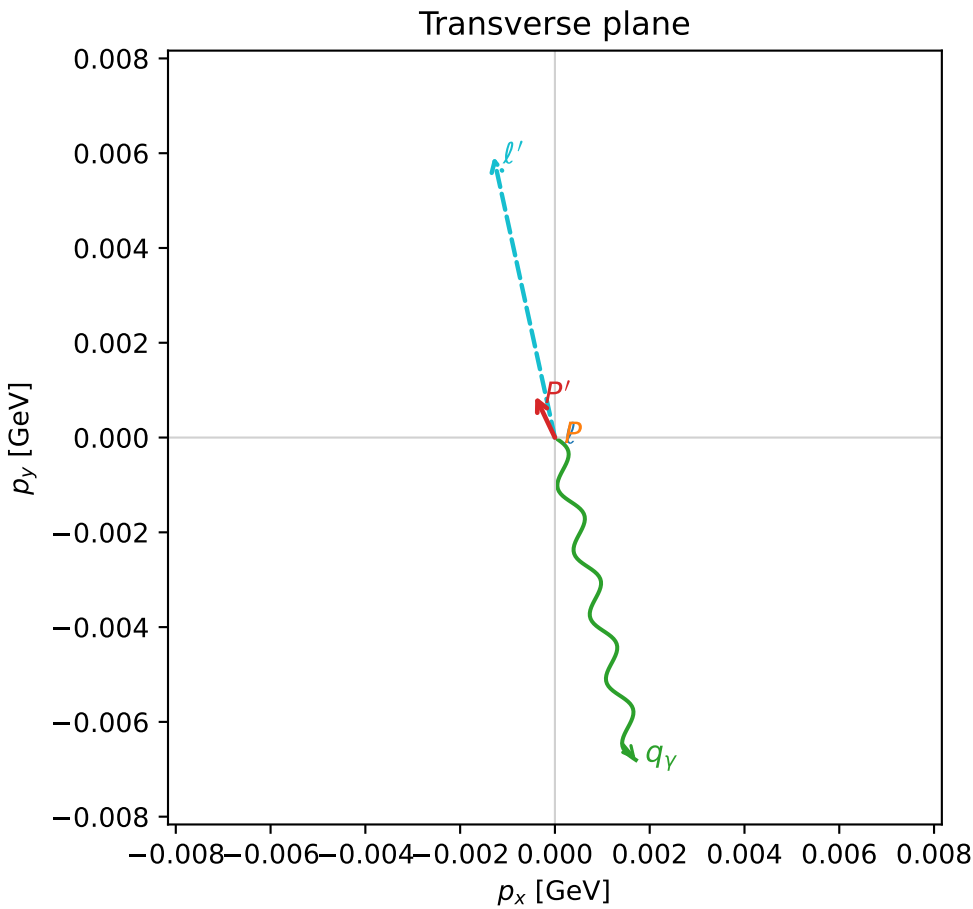
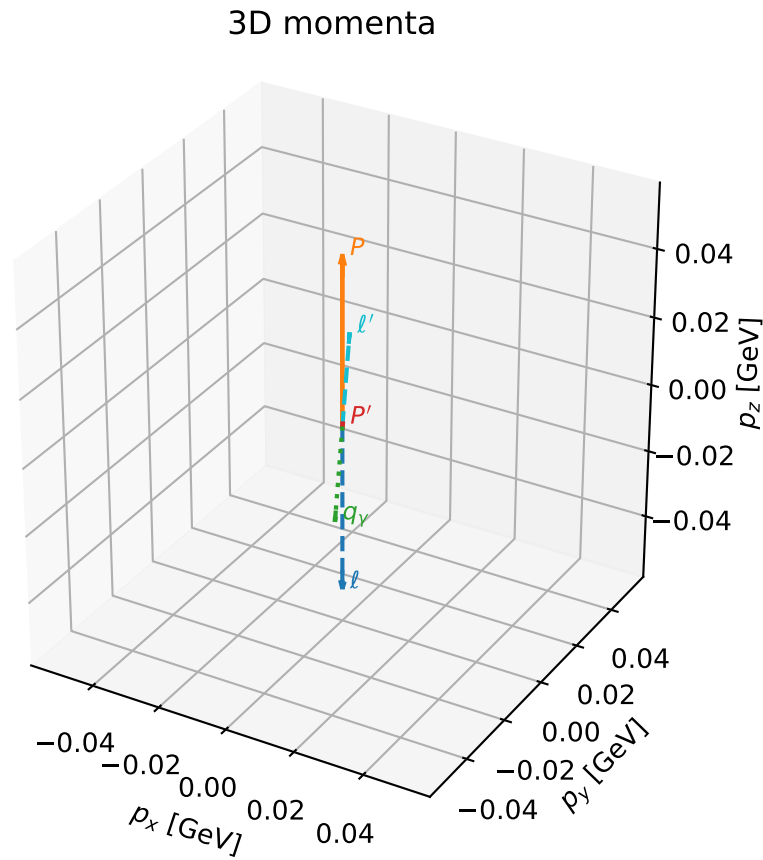
$\alpha_p = 3.119729$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

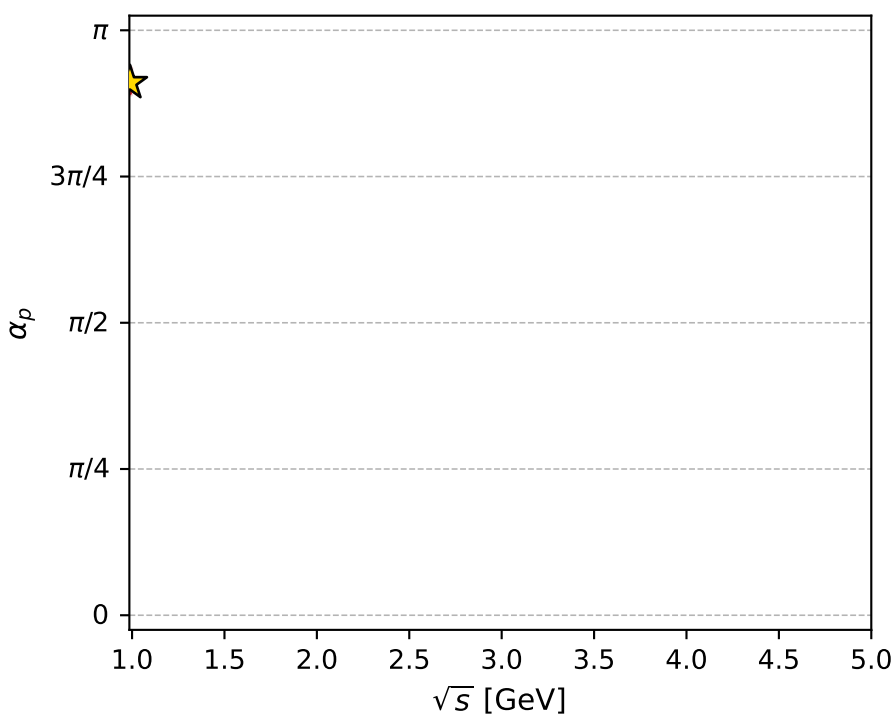
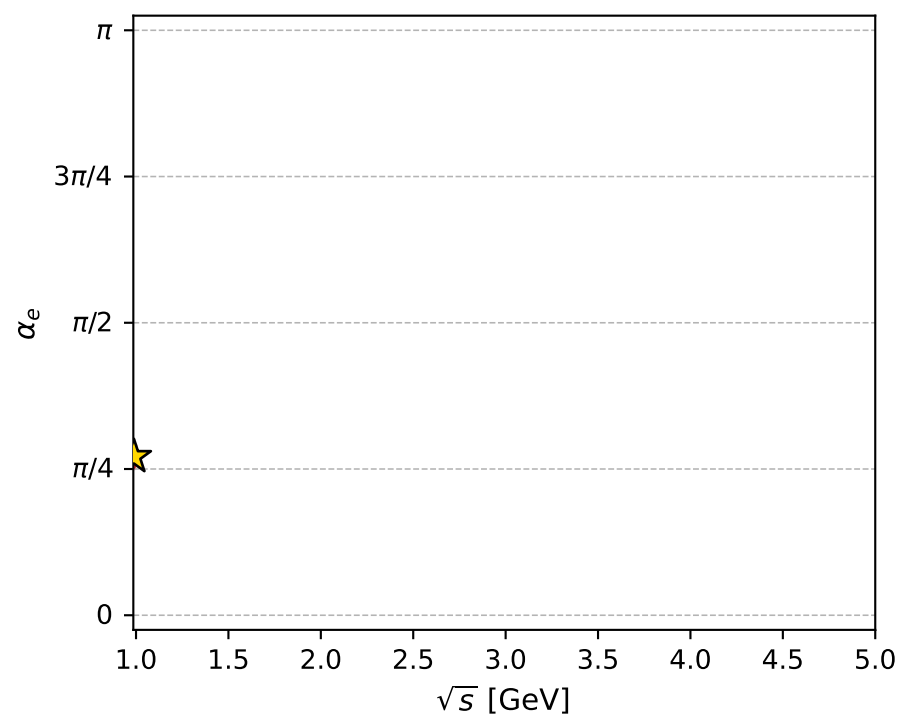
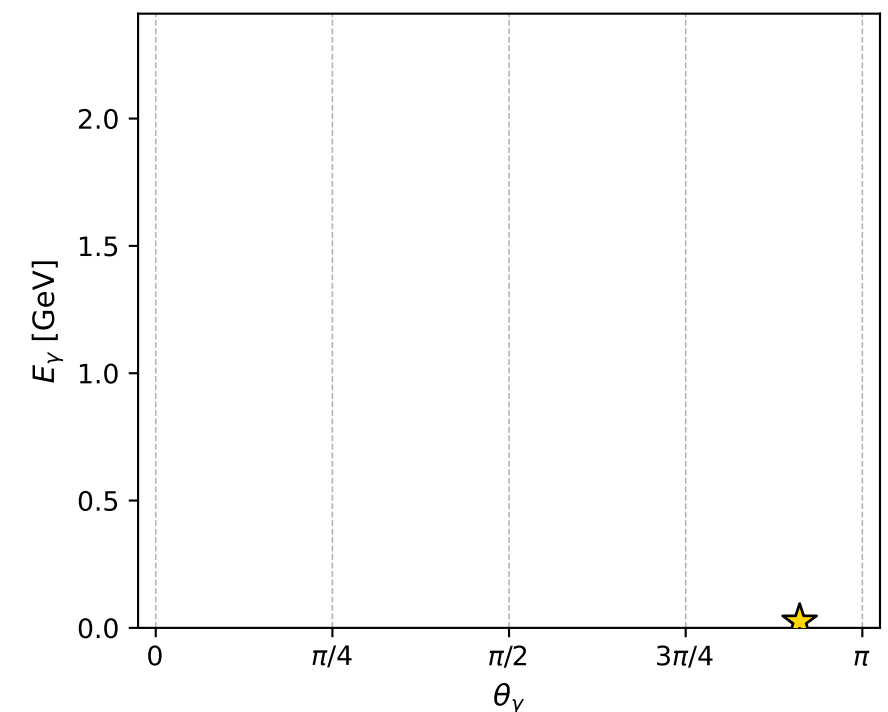
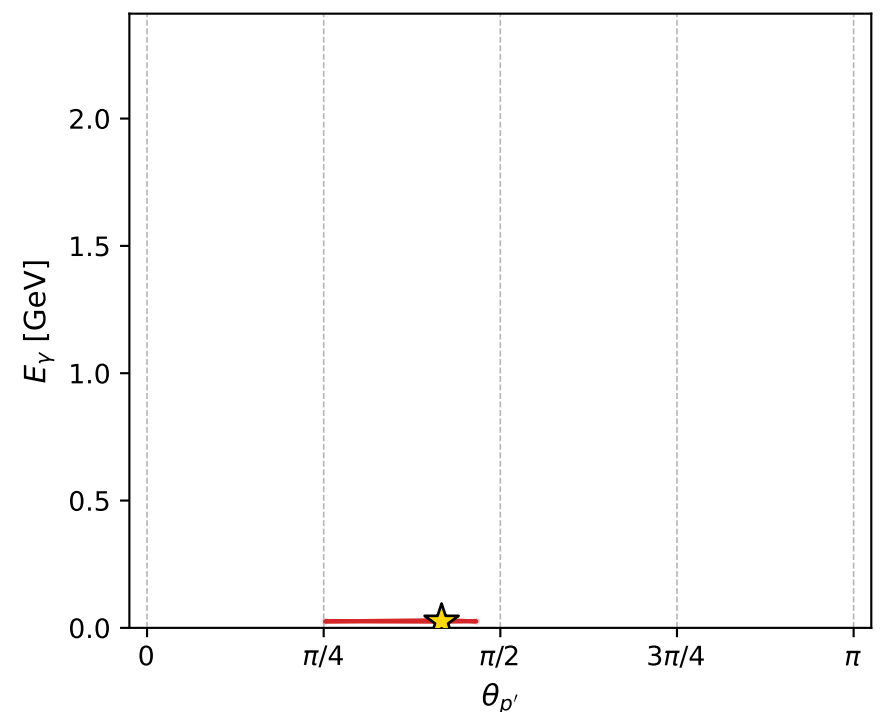
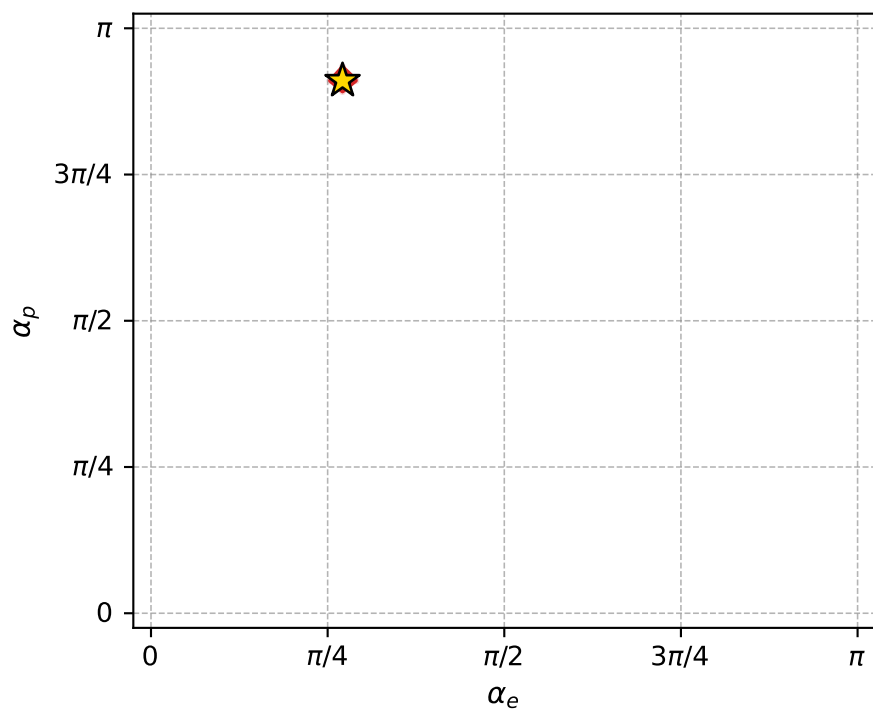
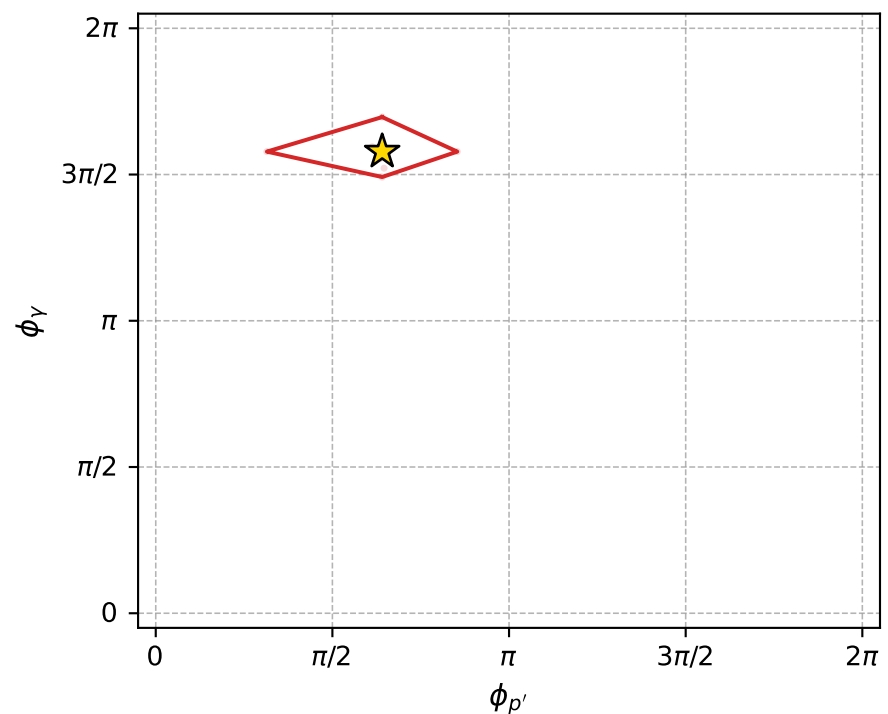
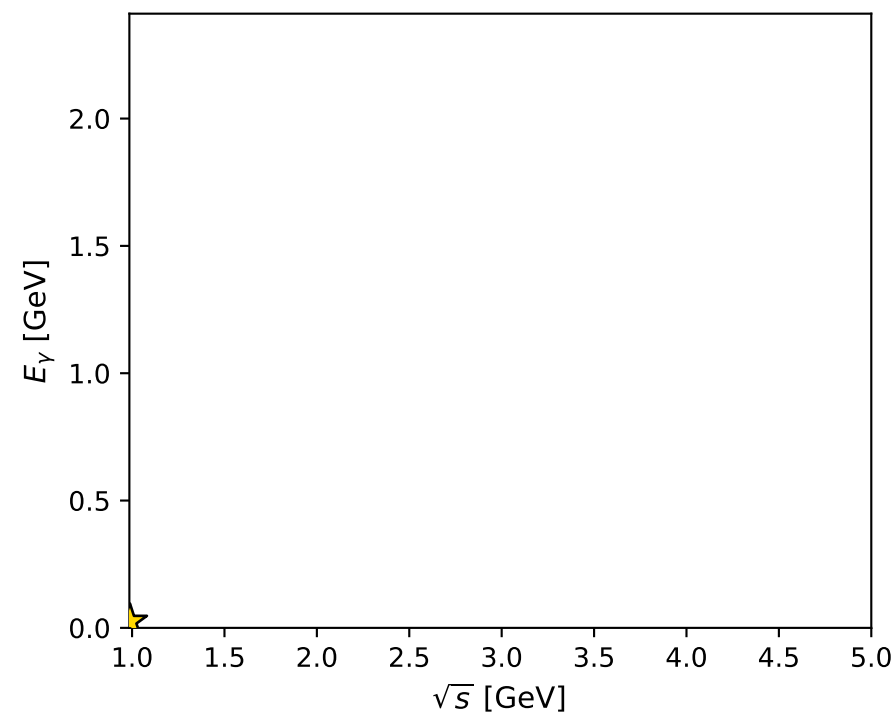
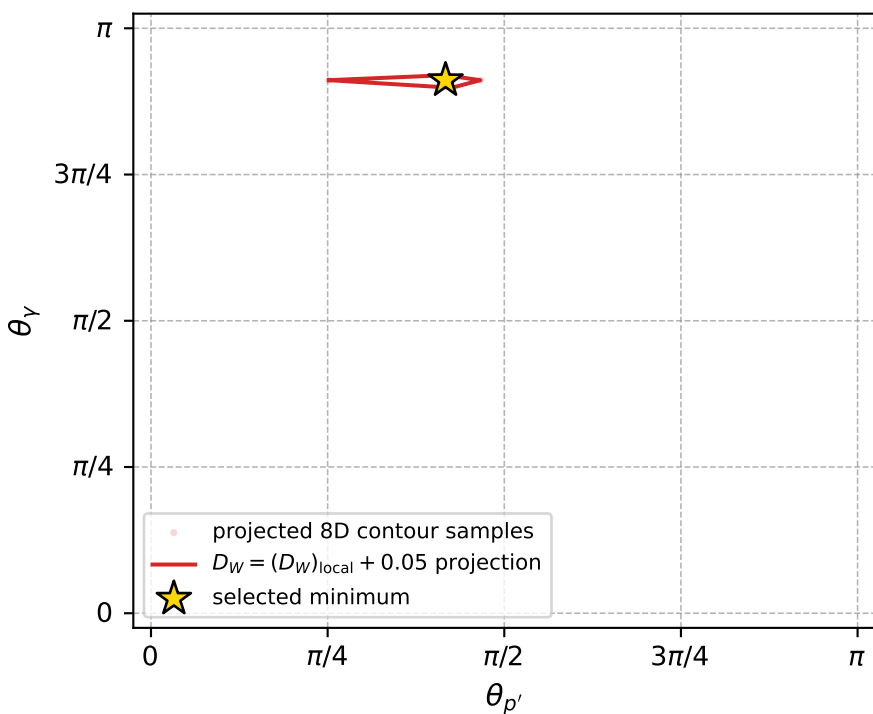
dw_mixing_angles_polarization_03_local_minimum_539 D_W=0.00307214
 region: polarization_cluster_03_local_minimum_539
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.85159301$ rad
 incoming lepton: $0.658786|+\rangle + 0.752331|-\rangle$
 proton mixing angle: $\alpha_p=2.8601868$ rad
 incoming proton: $-0.960666|+\rangle + 0.277706|-\rangle$

$s=0.977152$, $\sqrt{s}=0.98851$, $|\vec{P}|=0.0492172$, $|\vec{P}'|=0.00100159$
 $\theta_{P'}=1.30959$, $\theta_Y=2.8629$, $\phi_{P'}=2.01344$, $\phi_Y=4.95845$
 $E_Y=0.0254997$, $\theta_{\ell'}=0.245162$, $\phi_{\ell'}=1.78571$
 $Q^2=0.00484919$, $x_B=0.091993$, $t=-0.00239621$, $W^2=0.927707$, $y=0.541708$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0492$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0492$
 P $E= 0.9393$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0492$
 ℓ' $E= 0.0250$ $p_x= -0.0013$ $p_y= 0.0059$ $p_z= 0.0243$
 P' $E= 0.9380$ $p_x= -0.0004$ $p_y= 0.0009$ $p_z= 0.0003$
 q_Y $E= 0.0255$ $p_x= 0.0017$ $p_y= -0.0068$ $p_z= -0.0245$



electron: polarization cluster P3, local minimum 539; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0030721431$
 $D_W \text{ contour} = 0.053072143$
 above global minimum = 0.0030557103
 8D contour samples = 174

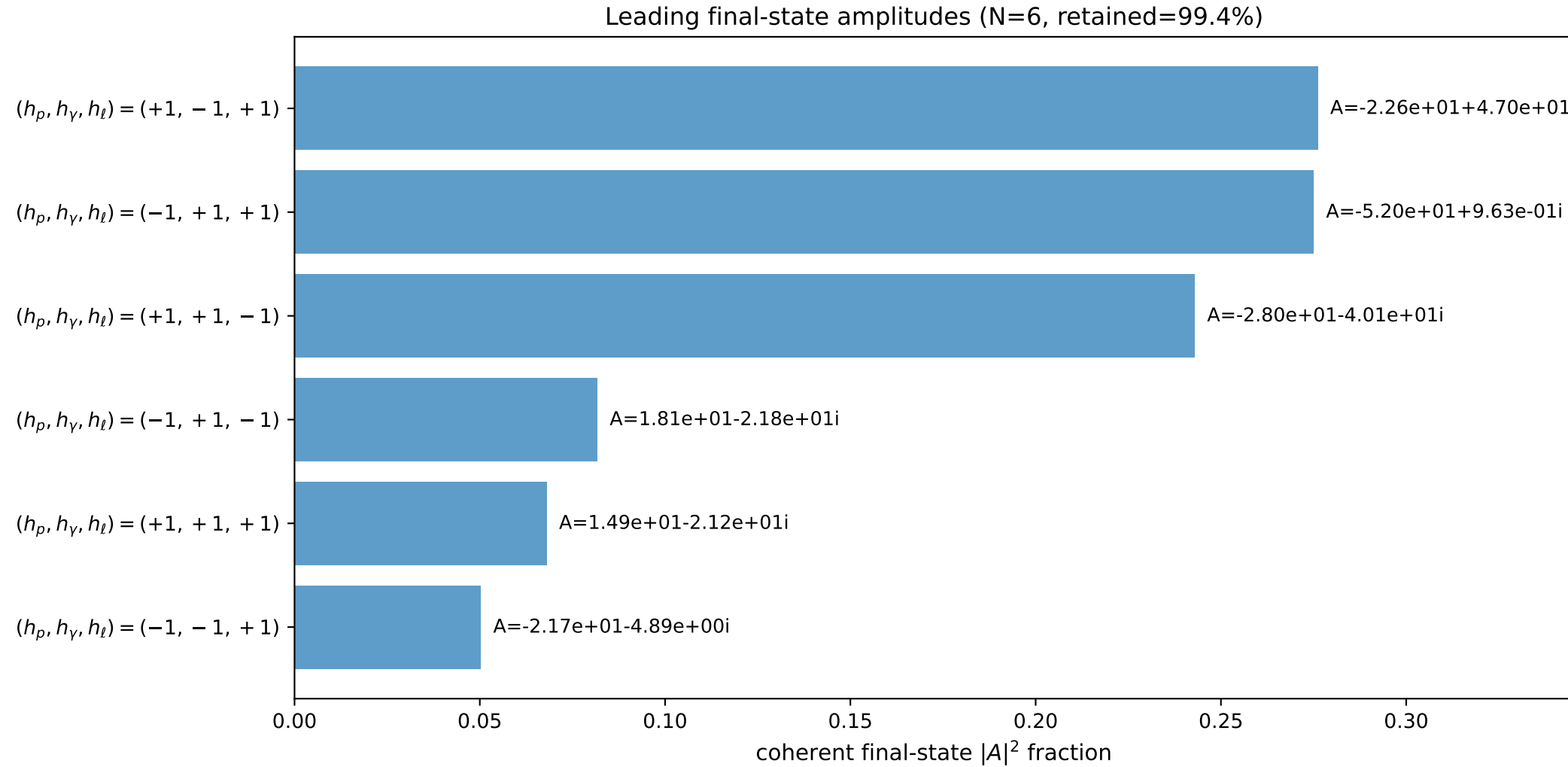
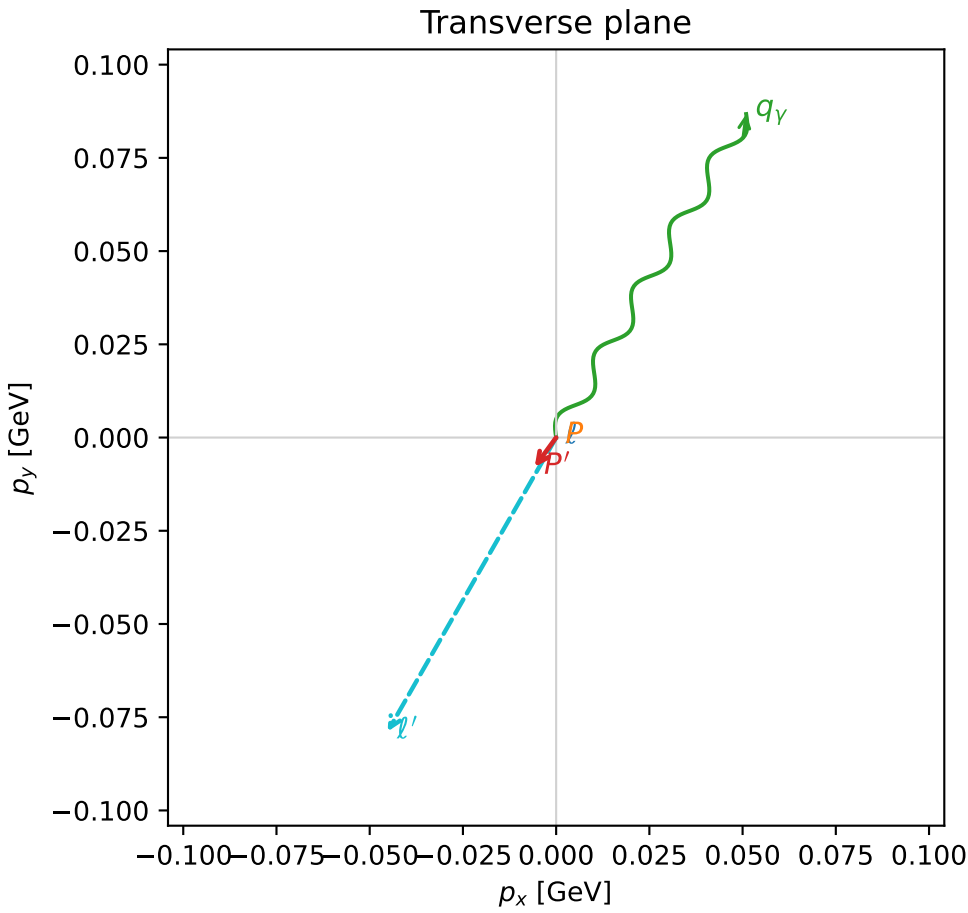
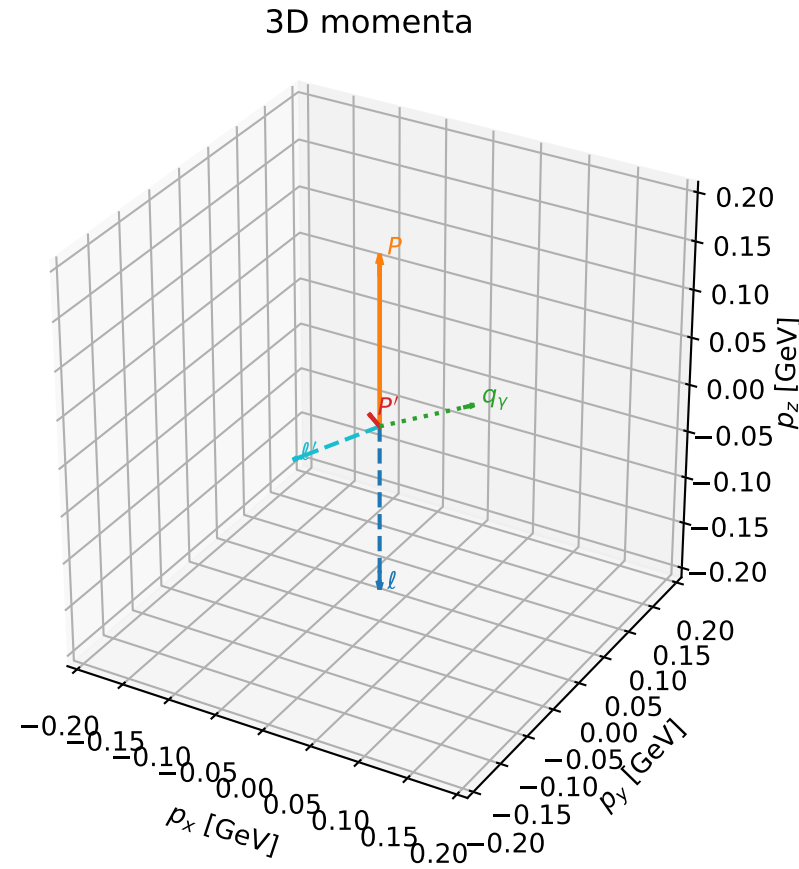
selected phase-space point:
 $\text{sqrt}_s = 0.9885102$
 $\text{theta}_p_{\text{out}} = 1.309588$
 $\text{theta}_\gamma_{\text{out}} = 2.862896$
 $q_{\text{out}} = 0.02549966$
 $\text{phi}_p_{\text{out}} = 2.013444$
 $\text{phi}_\gamma_{\text{out}} = 4.95845$
 $\alpha_e = 0.851593$
 $\alpha_p = 2.860187$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

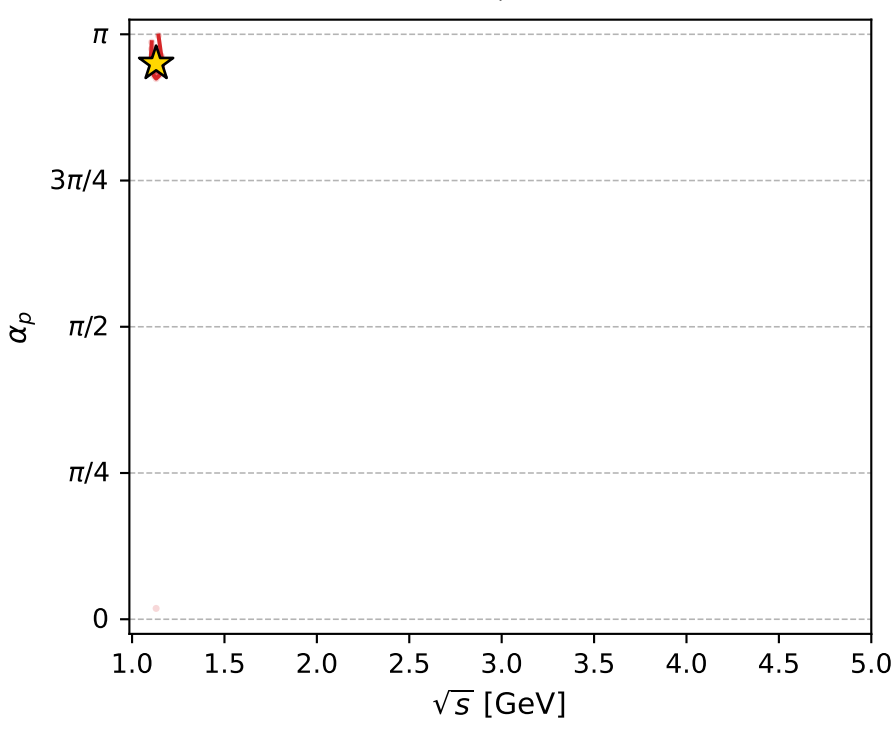
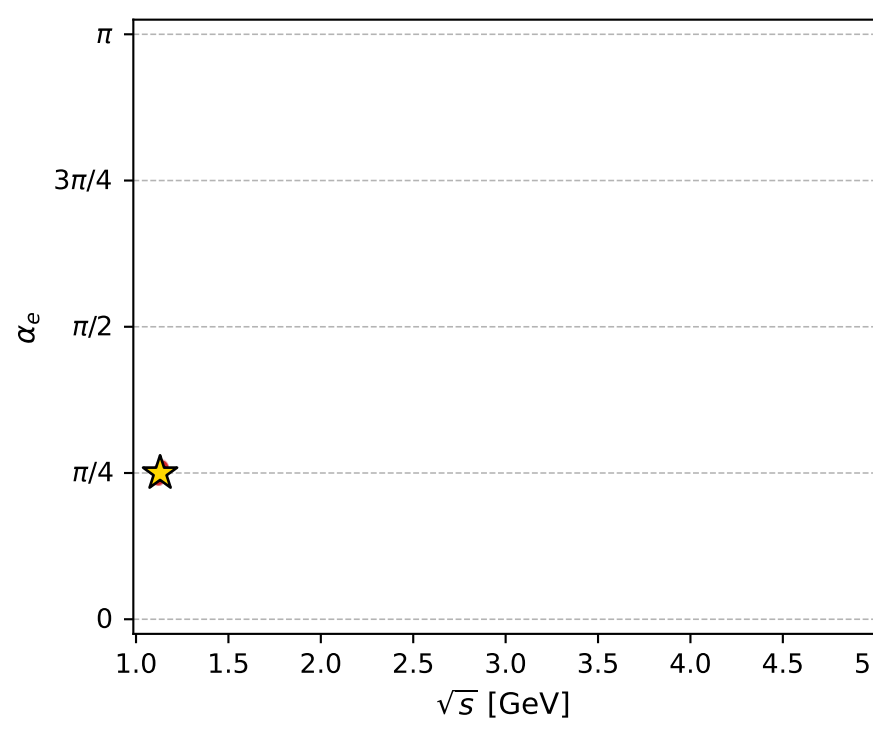
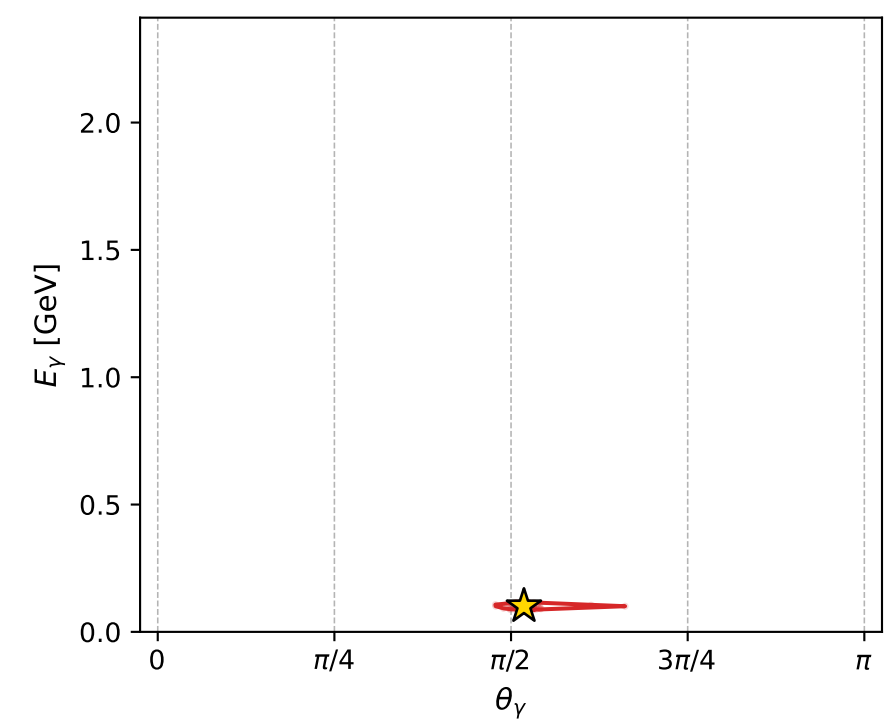
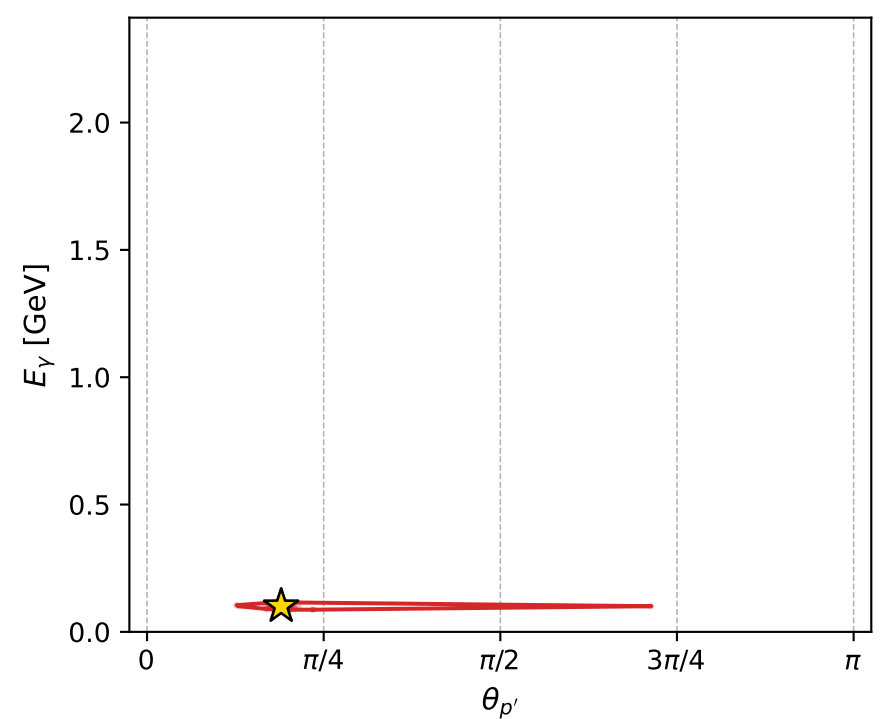
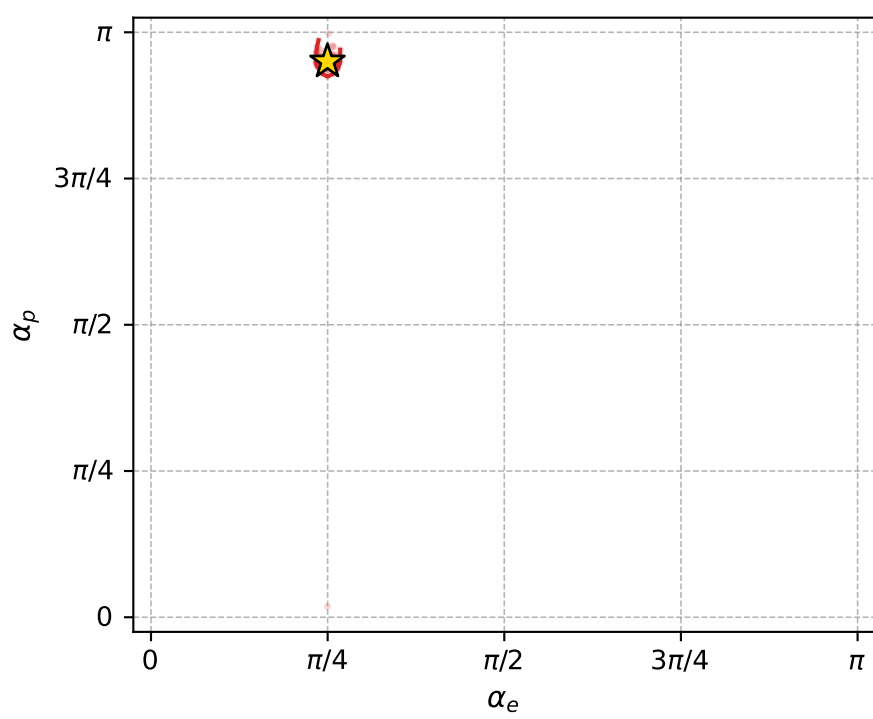
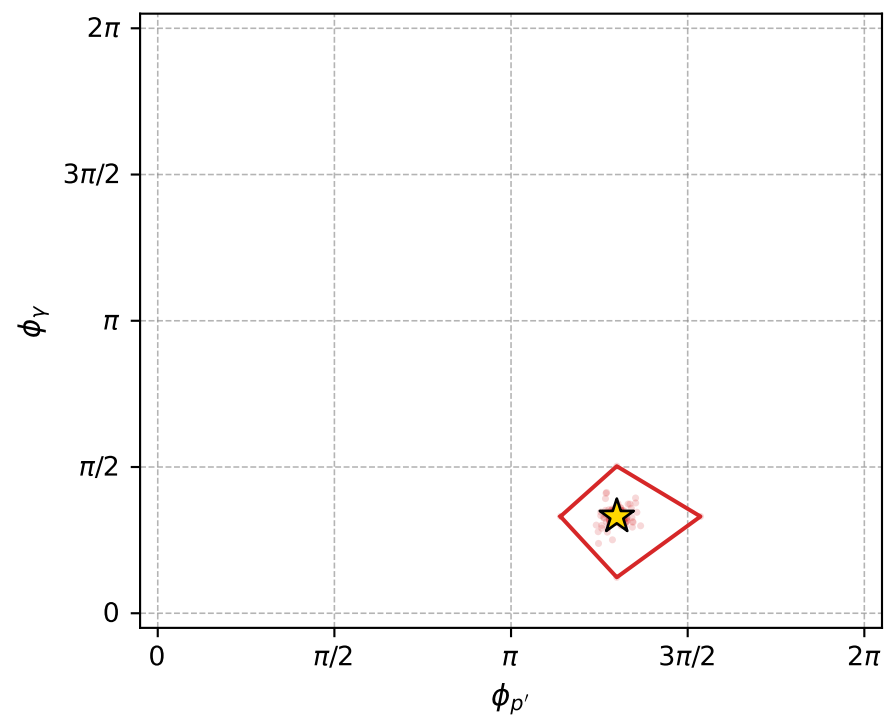
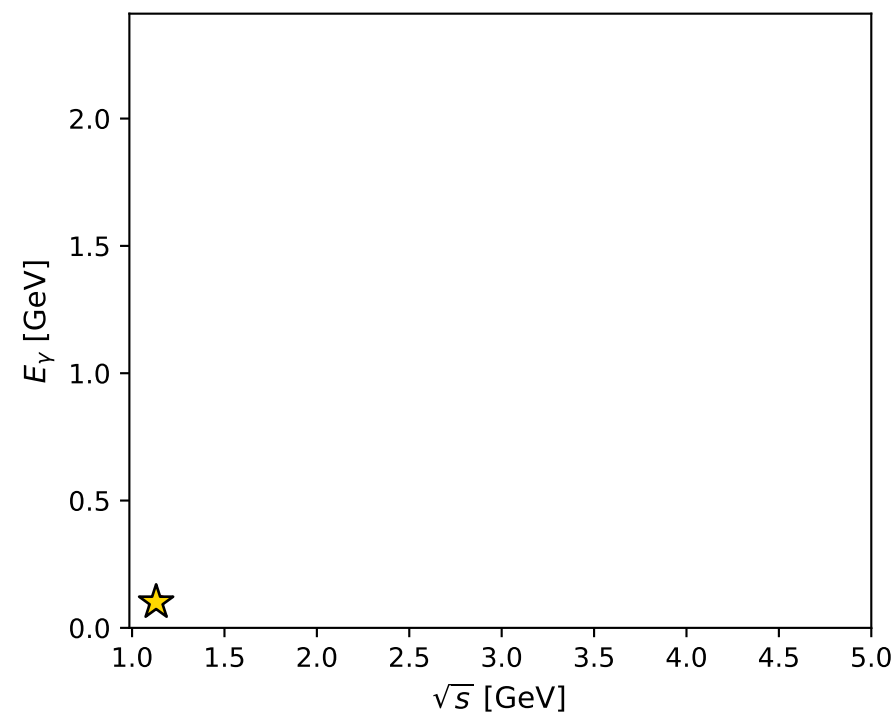
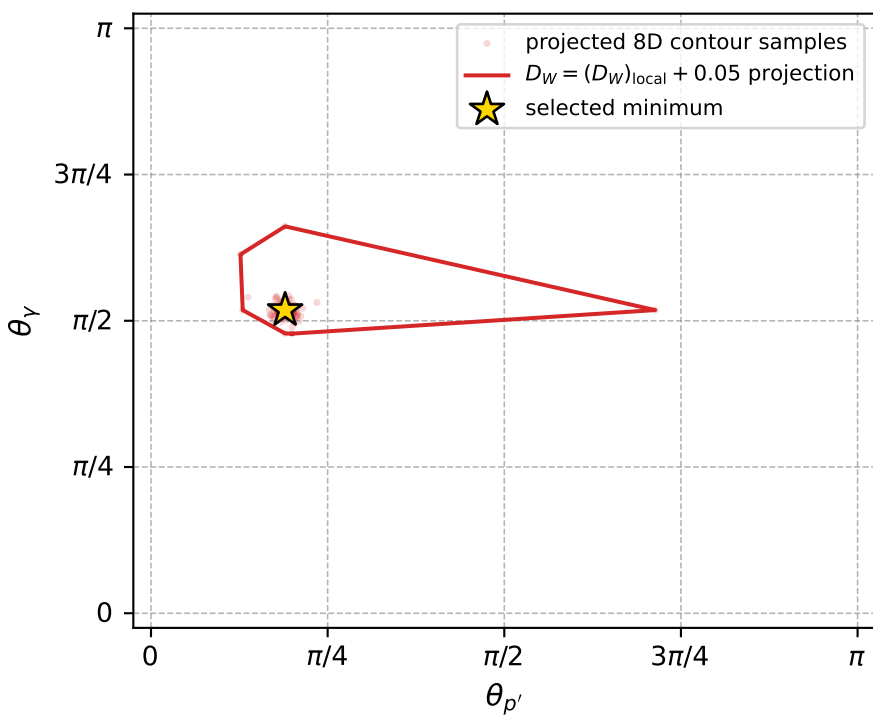
dw_mixing_angles_polarization_03_local_minimum_561 D_W=0.00333889
 region: polarization_cluster_03_local_minimum_561
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78469657$ rad
 incoming lepton: $0.707603|+\rangle +0.706611|-\rangle$
 proton mixing angle: $\alpha_p=2.984805$ rad
 incoming proton: $-0.987734|+\rangle +0.156146|-\rangle$

$s=1.2774$, $\sqrt{s}=1.13022$, $|\vec{P}|=0.175876$, $|\vec{P}'|=0.0174103$
 $\theta_{P'}=0.596276$, $\theta_V=1.62819$, $\phi_{P'}=4.08195$, $\phi_V=1.03963$
 $E_V=0.10077$, $\theta_{l'}=1.66542$, $\phi_{l'}=4.19188$
 $Q^2=0.0290772$, $x_B=0.132002$, $t=-0.0259062$, $W^2=1.07105$, $y=0.554079$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1759$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1759$
 P $E= 0.9543$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1759$
 l' $E= 0.0913$ $p_x= -0.0452$ $p_y= -0.0788$ $p_z= -0.0086$
 P' $E= 0.9382$ $p_x= -0.0058$ $p_y= -0.0079$ $p_z= 0.0144$
 q_V $E= 0.1008$ $p_x= 0.0510$ $p_y= 0.0867$ $p_z= -0.0058$



electron: polarization cluster P3, local minimum 561; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0033388942$
 $D_W \text{ contour} = 0.053338894$
 above global minimum = 0.0033224614
 8D contour samples = 256

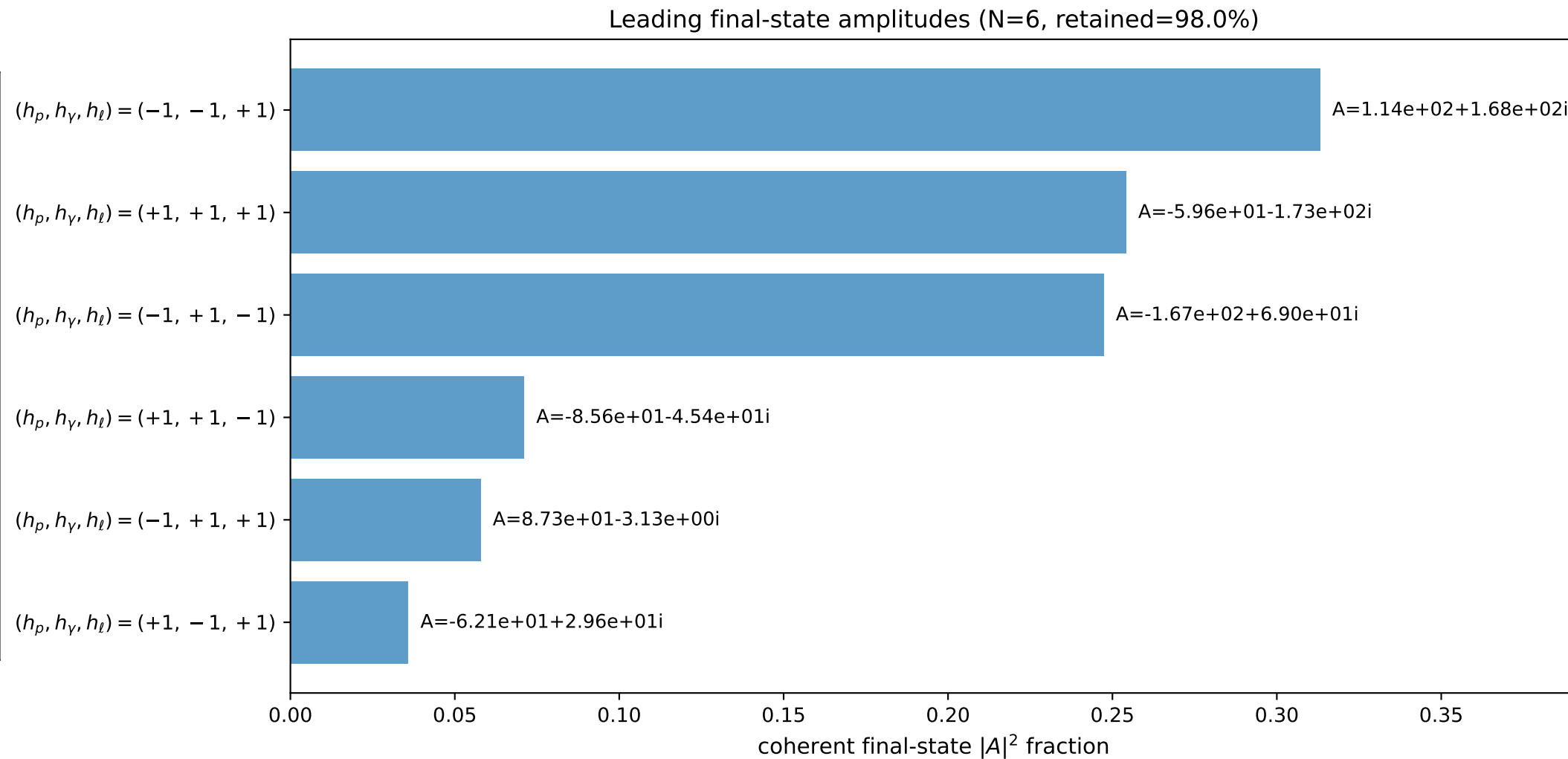
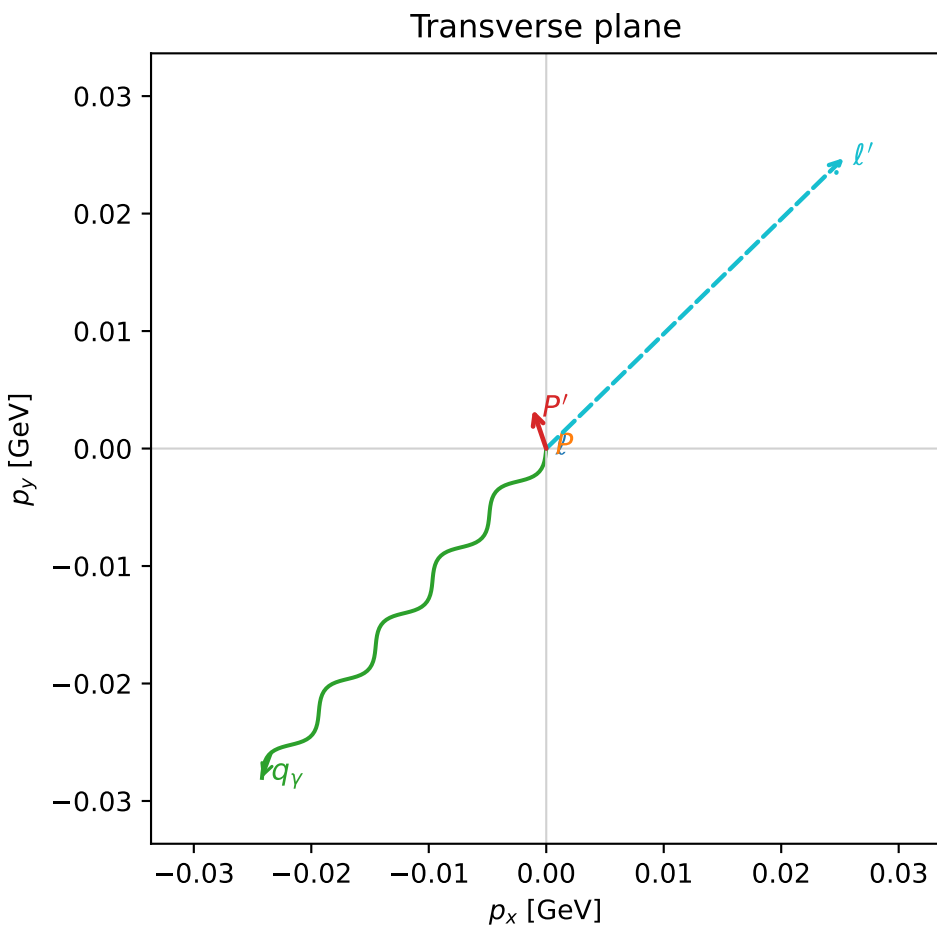
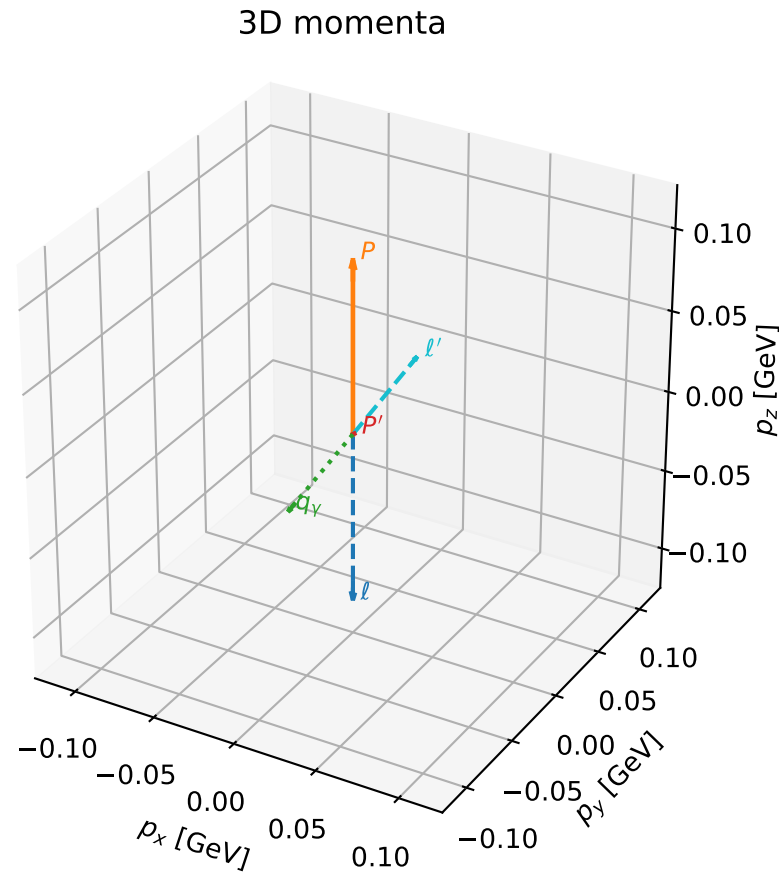
selected phase-space point:
 $\text{sqrt_s} = 1.130223$
 $\text{theta_p_out} = 0.596276$
 $\text{theta_gamma_out} = 1.628189$
 $\text{qOut} = 0.1007704$
 $\text{phi_p_out} = 4.081953$
 $\text{phi_gamma_out} = 1.039629$
 $\text{alpha_e} = 0.7846966$
 $\text{alpha_p} = 2.984805$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

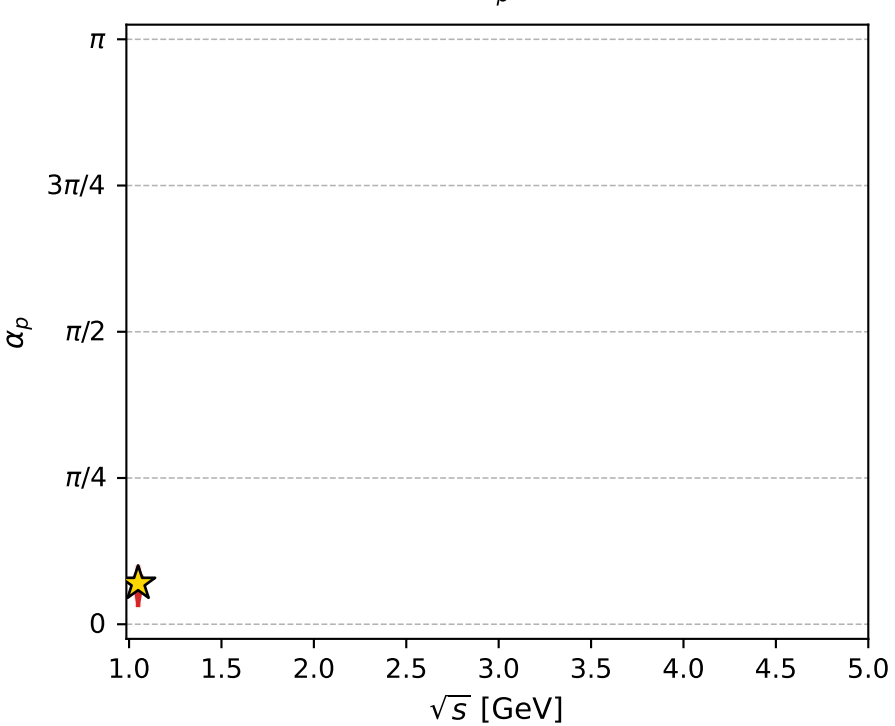
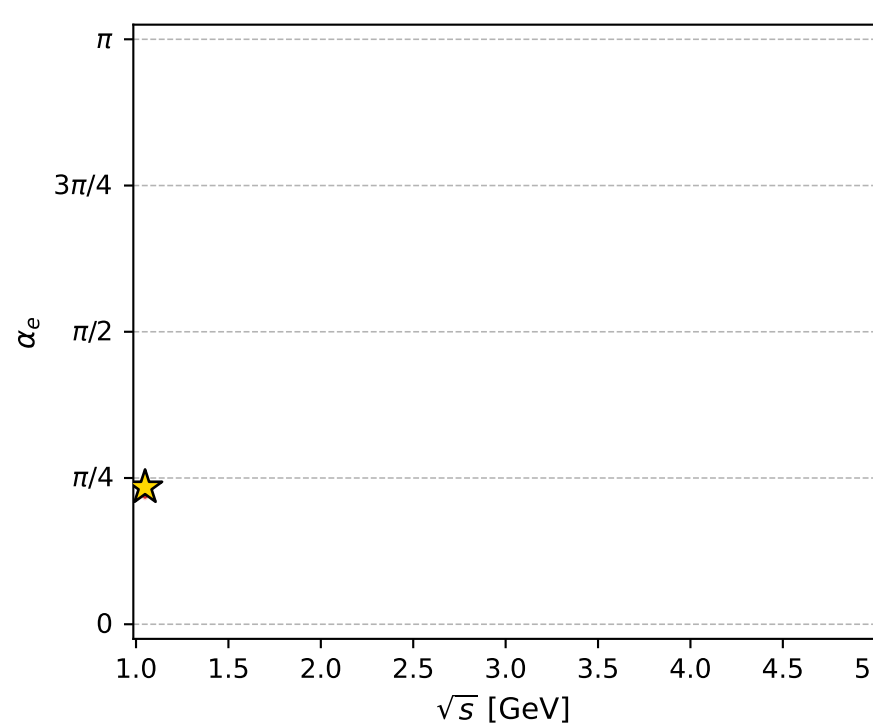
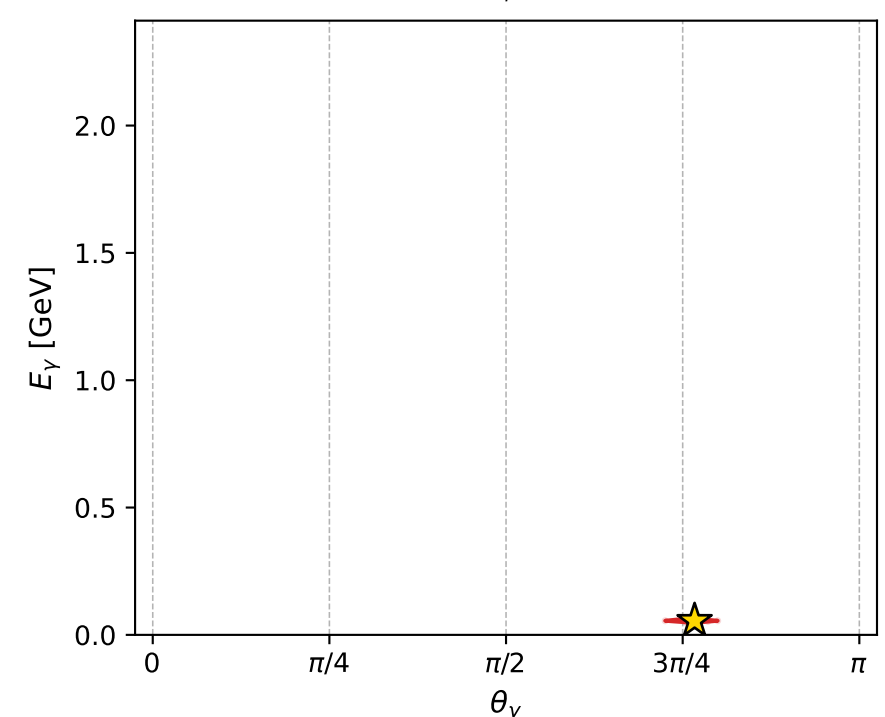
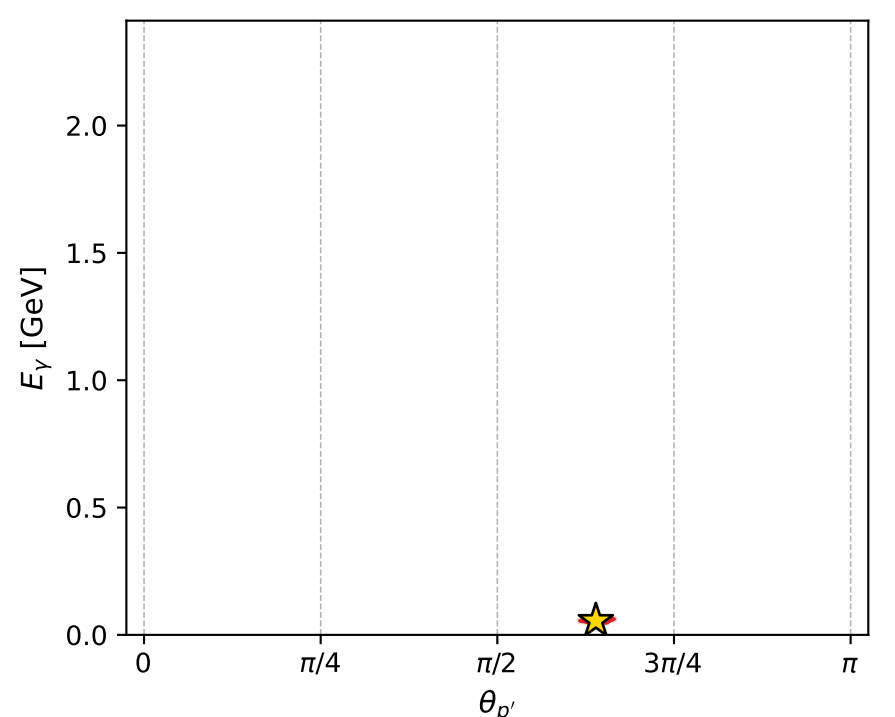
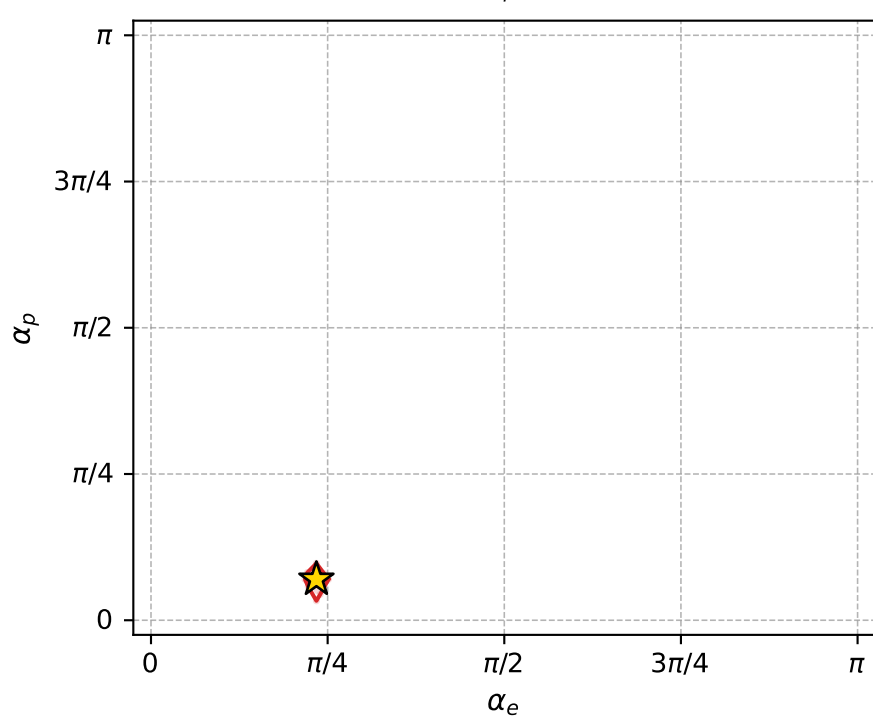
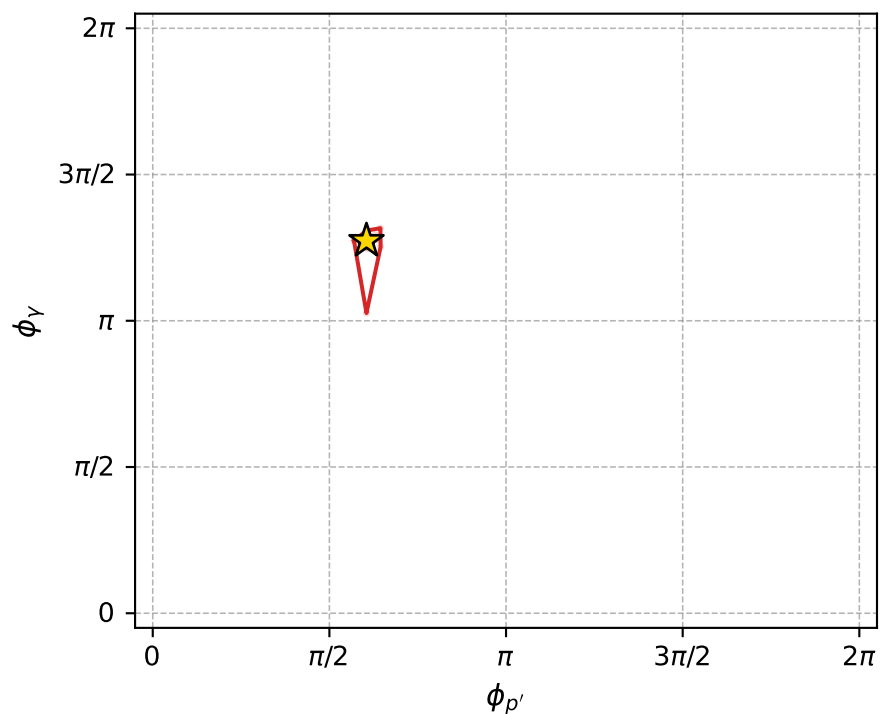
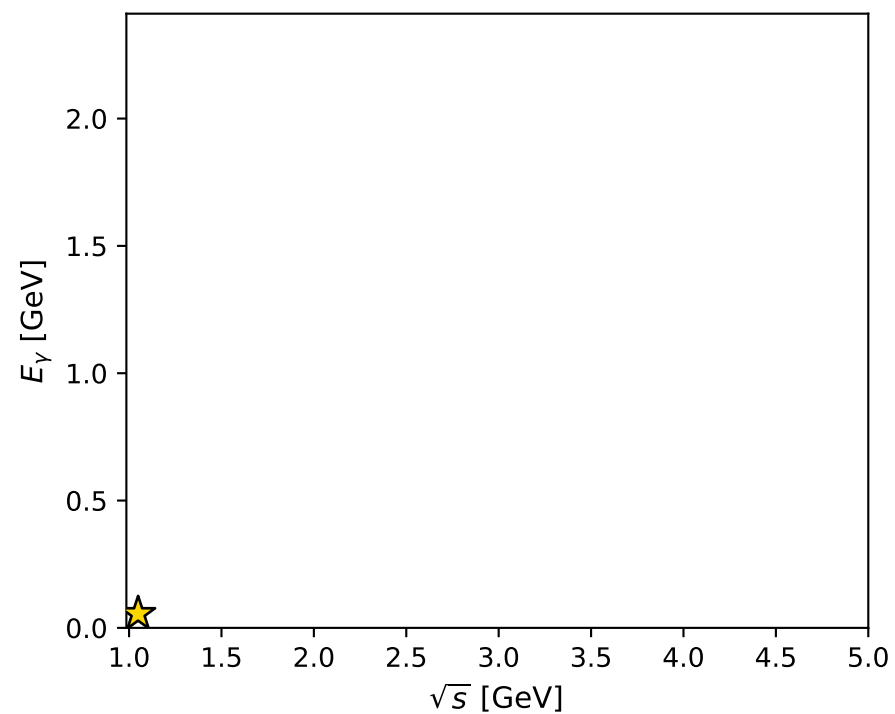
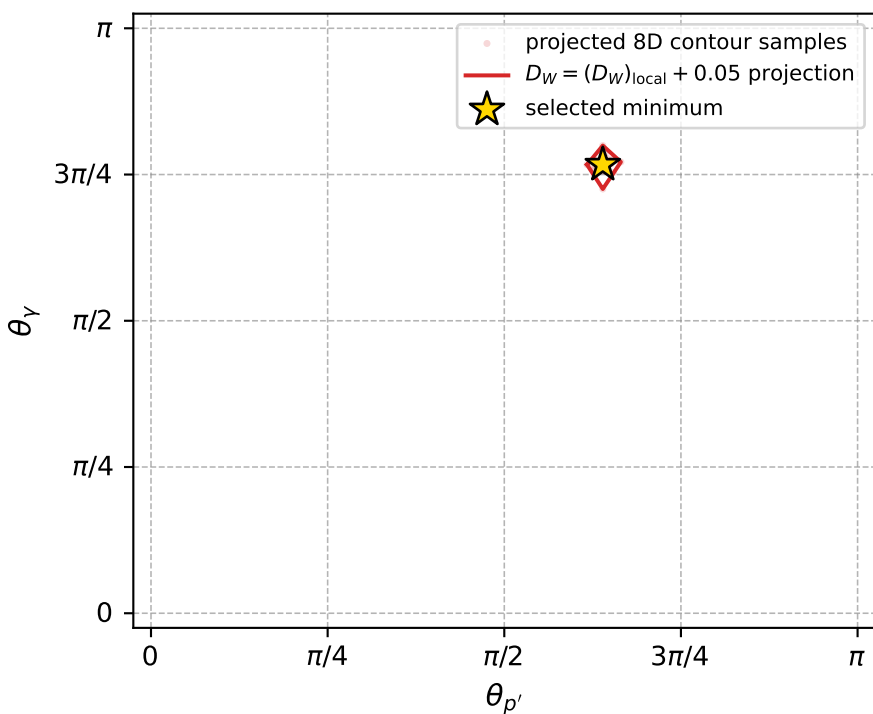
dw_mixing_angles_polarization_03_local_minimum_577 D_W=0.00365148
region: polarization_cluster_03_local_minimum_577
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.7354556$ rad
incoming lepton: $0.741525|+\rangle + 0.670925|-\rangle$
proton mixing angle: $\alpha_p=0.2197378$ rad
incoming proton: $0.975955|+\rangle + 0.217974|-\rangle$

$s=1.10027$, $\sqrt{s}=1.04894$, $|\vec{P}|=0.10507$, $|\vec{P}'|=0.00385496$
 $\theta_{P'}=2.00884$, $\theta_V=2.40911$, $\phi_{P'}=1.9005$, $\phi_V=4.00032$
 $E_V=0.055378$, $\theta_{\ell'}=0.690927$, $\phi_{\ell'}=0.773489$
 $Q^2=0.0206693$, $x_B=0.165945$, $t=-0.0113639$, $W^2=0.98373$, $y=0.565065$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1051$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1051$
 P $E= 0.9439$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1051$
 ℓ' $E= 0.0556$ $p_x= 0.0253$ $p_y= 0.0247$ $p_z= 0.0428$
 P' $E= 0.9380$ $p_x= -0.0011$ $p_y= 0.0033$ $p_z= -0.0016$
 q_V $E= 0.0554$ $p_x= -0.0242$ $p_y= -0.0280$ $p_z= -0.0412$



electron: polarization cluster P3, local minimum 577; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0036514775$
 $D_W \text{ contour} = 0.053651478$
 above global minimum = 0.0036350447
 8D contour samples = 254

selected phase-space point:

$\text{sqrt_s} = 1.048938$
 $\text{theta_p_out} = 2.008836$
 $\text{theta_gamma_out} = 2.409108$
 $\text{qOut} = 0.05537795$
 $\text{phi_p_out} = 1.900504$
 $\text{phi_gamma_out} = 4.000322$
 $\text{alpha_e} = 0.7354556$
 $\text{alpha_p} = 0.2197378$

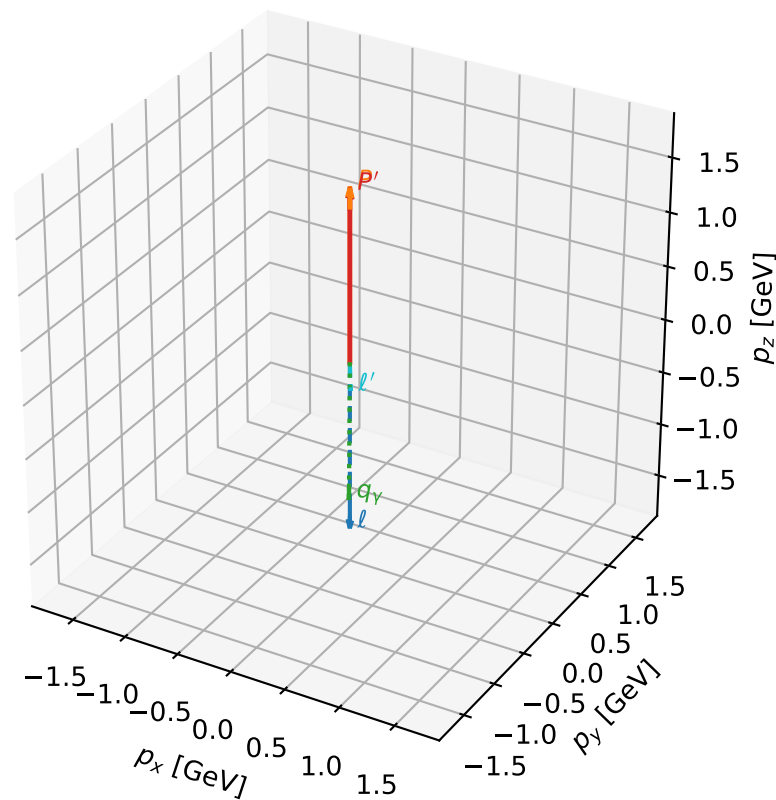
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_583 D_W=0.00377731
 region: polarization_cluster_03_local_minimum_583
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79847104$ rad
 incoming lepton: $0.697803|+\rangle + 0.71629|-\rangle$
 proton mixing angle: $\alpha_p=3.0412699$ rad
 incoming proton: $-0.994972|+\rangle + 0.100155|-\rangle$

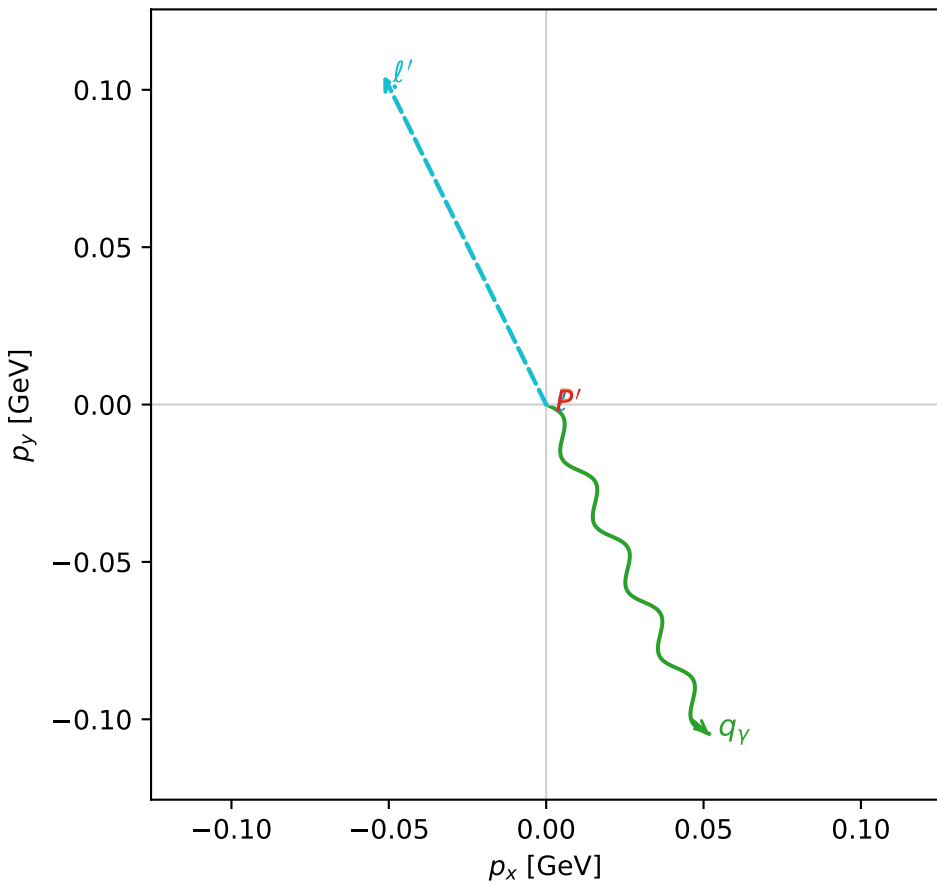
$s=11.746$, $\sqrt{s}=3.42724$, $|\vec{P}|=1.58526$, $|\vec{P}'|=1.57167$
 $\theta_{P'}=0$, $\theta_V=3.04754$, $\phi_{P'}=2.35408$, $\phi_V=5.17152$
 $E_V=1.24274$, $\theta_{\ell'}=2.80583$, $\phi_{\ell'}=2.02993$
 $Q^2=0.0627118$, $x_B=0.00737703$, $t=-4.81958e-05$, $W^2=9.31808$, $y=0.782334$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.5853$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.5853$
 P $E= 1.8420$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.5853$
 ℓ' $E= 0.3542$ $p_x= -0.0517$ $p_y= 0.1046$ $p_z= -0.3344$
 P' $E= 1.8303$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 1.5717$
 q_V $E= 1.2427$ $p_x= 0.0517$ $p_y= -0.1046$ $p_z= -1.2372$

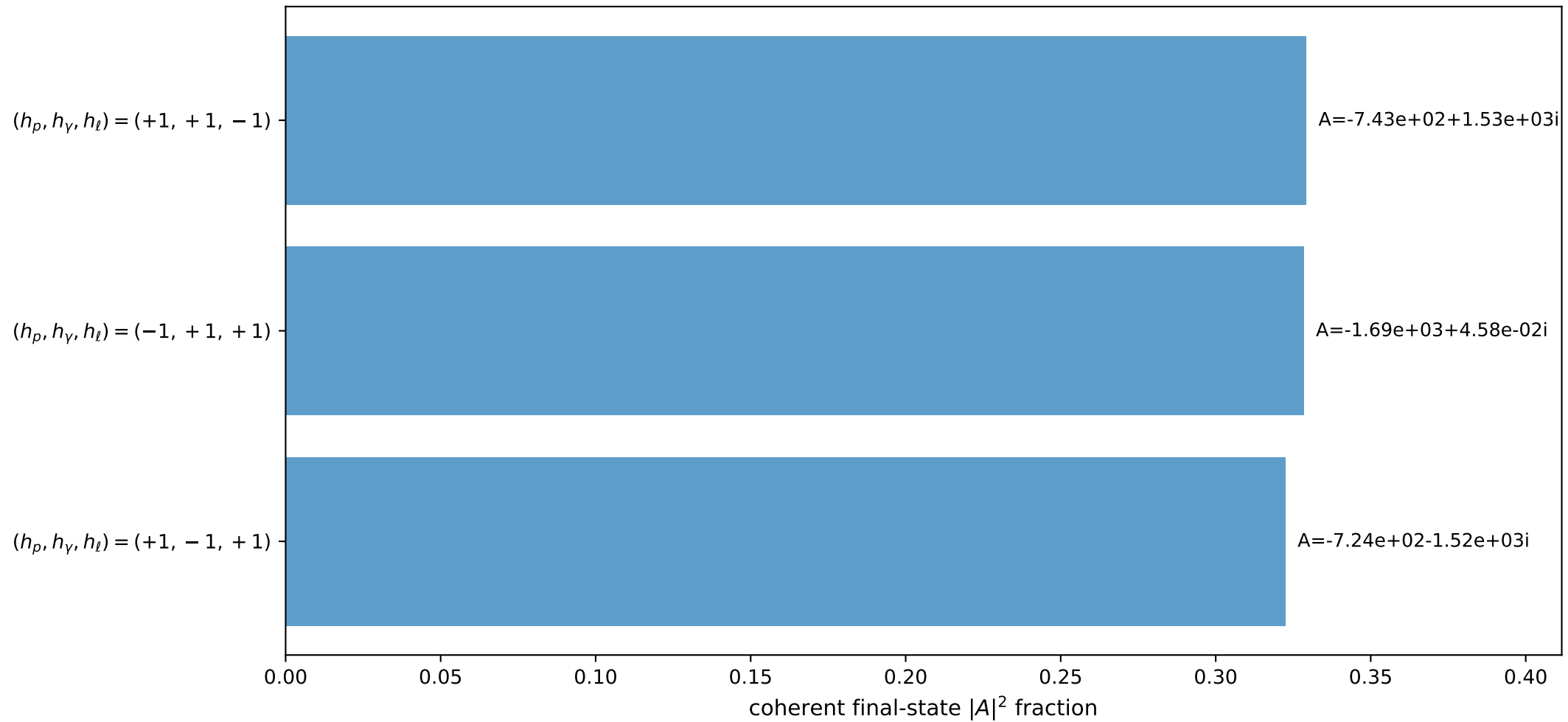
3D momenta



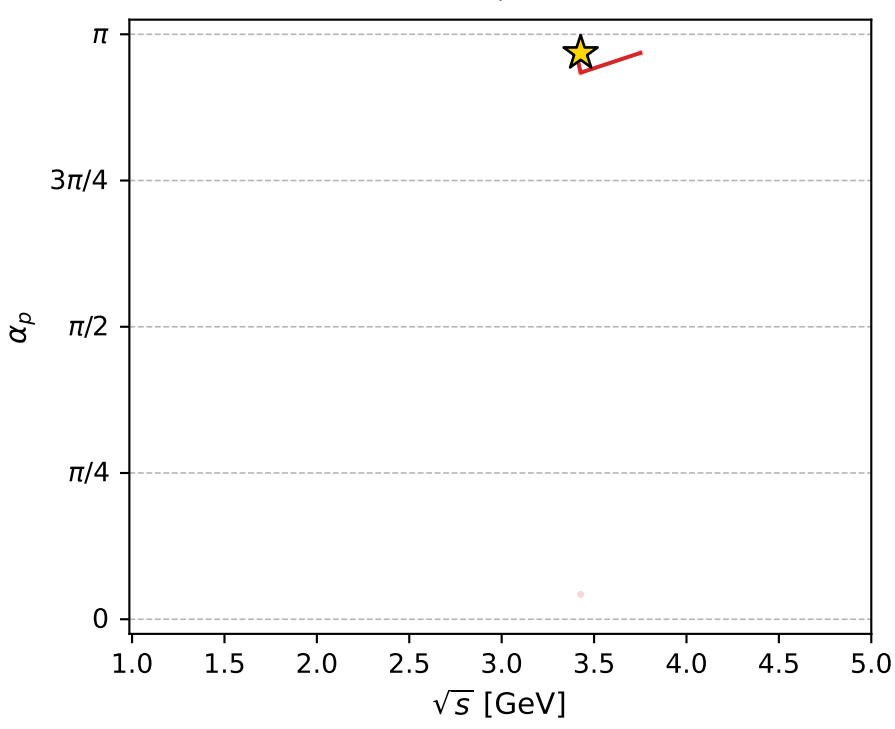
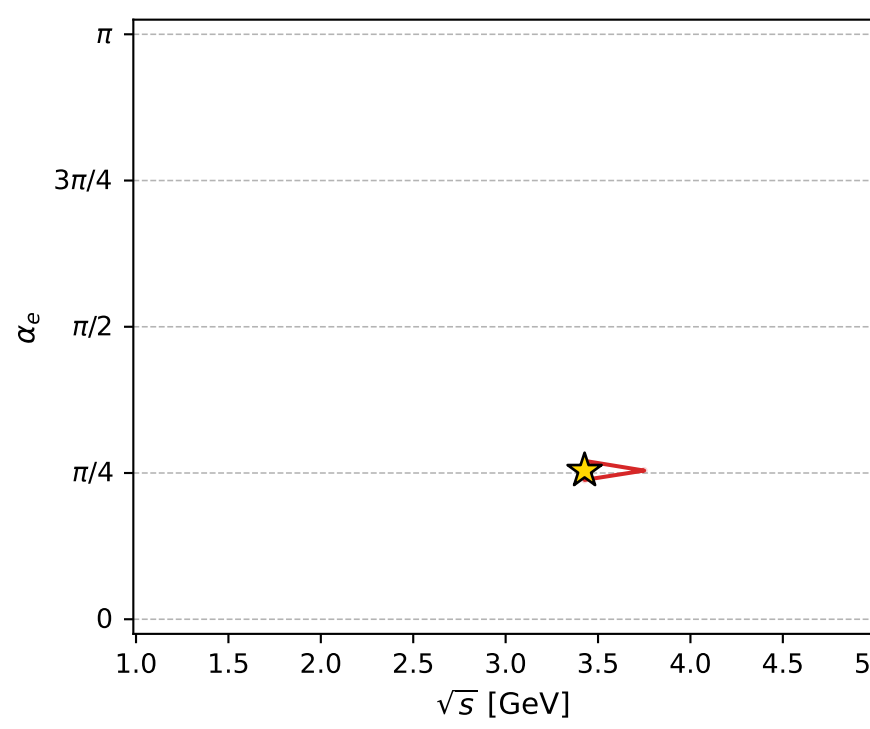
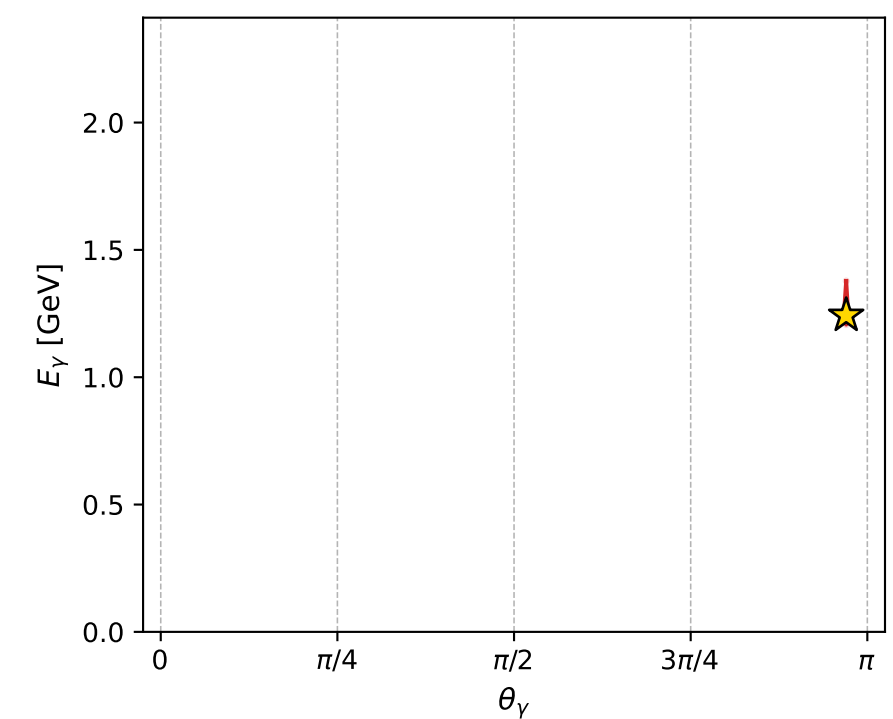
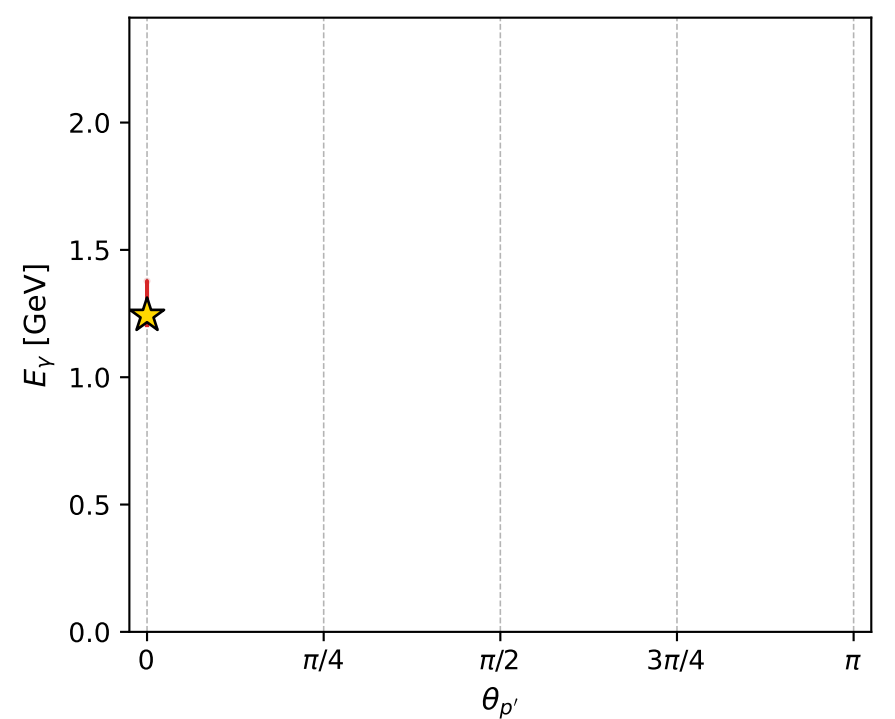
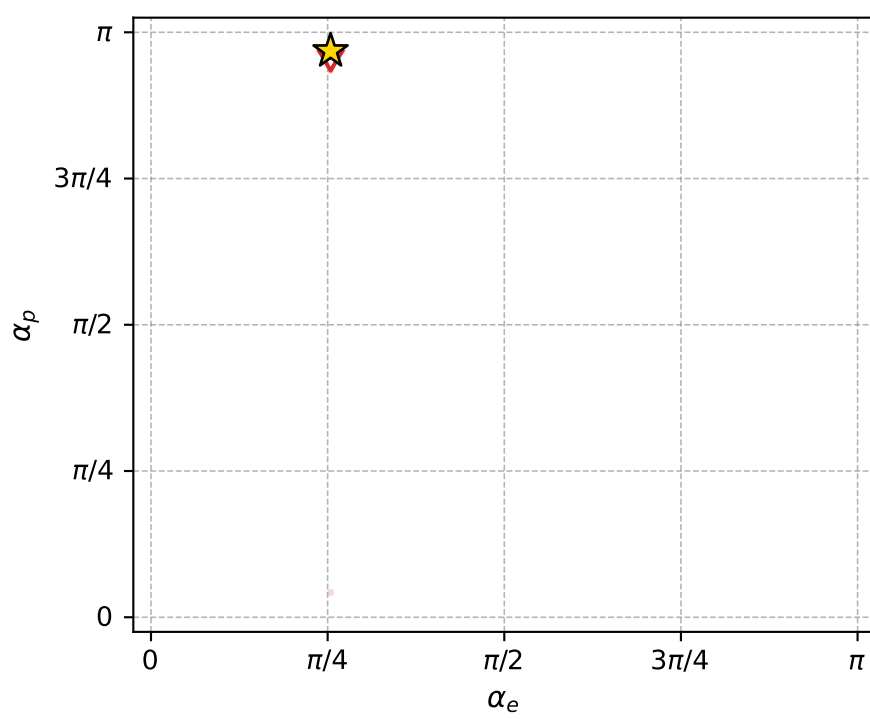
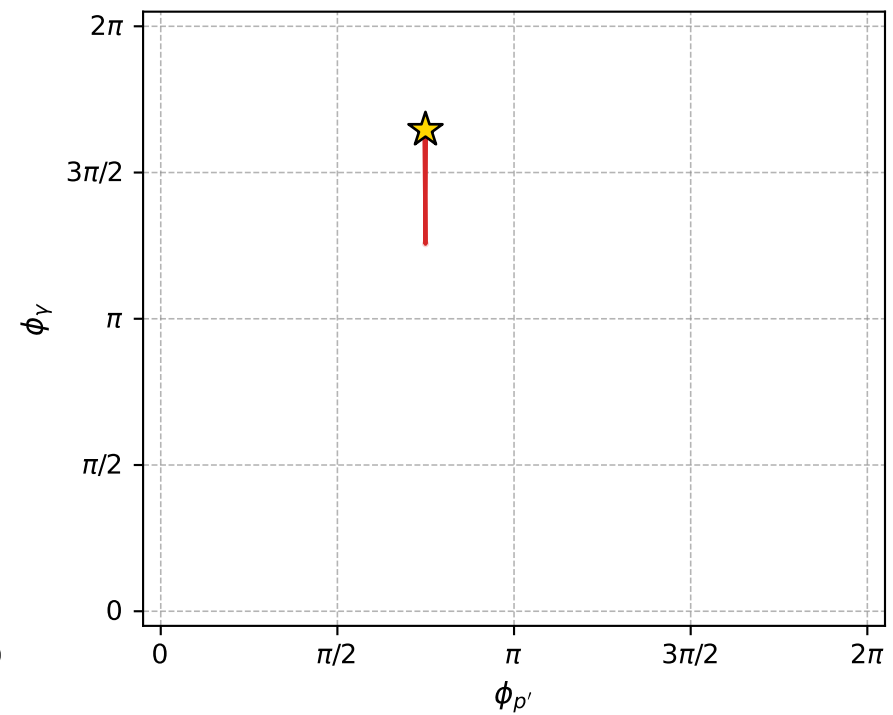
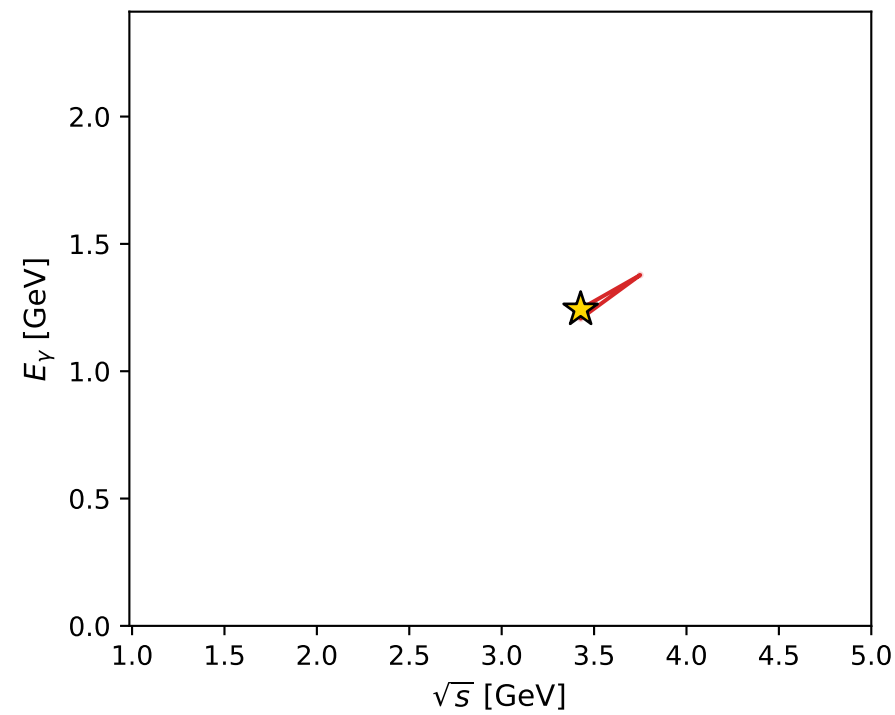
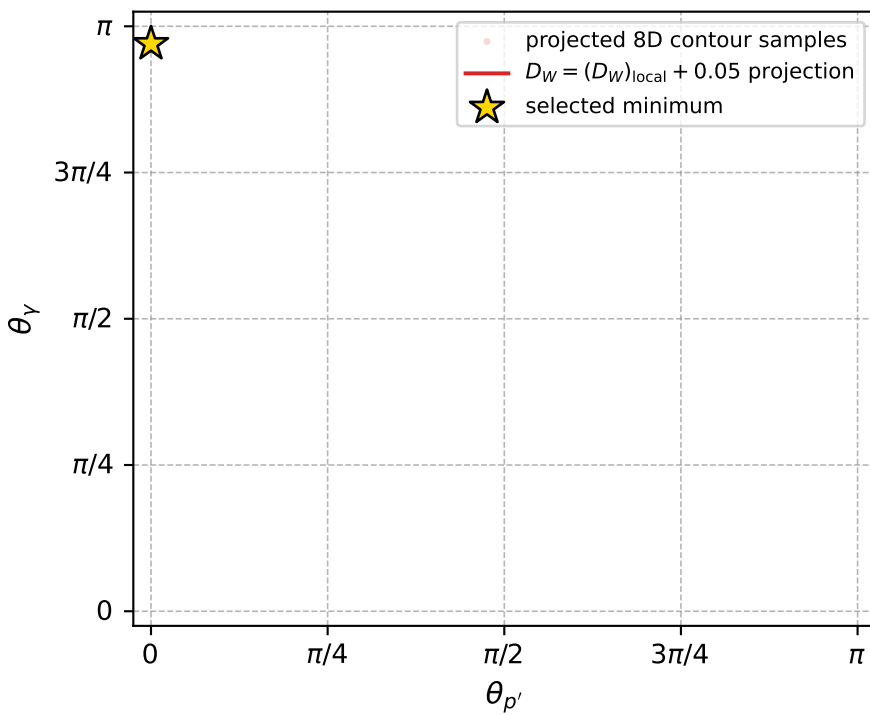
Transverse plane



Leading final-state amplitudes (N=3, retained=98.0%)



electron: polarization cluster P3, local minimum 583; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0037773078$
 $D_W \text{ contour} = 0.053777308$
 above global minimum = 0.0037608749
 8D contour samples = 139

selected phase-space point:
 $\text{sqrt_s} = 3.427242$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.047542$
 $\text{q0ut} = 1.242736$
 $\text{phi_p_out} = 2.354084$
 $\text{phi_gamma_out} = 5.171523$
 $\text{alpha_e} = 0.798471$
 $\text{alpha_p} = 3.04127$

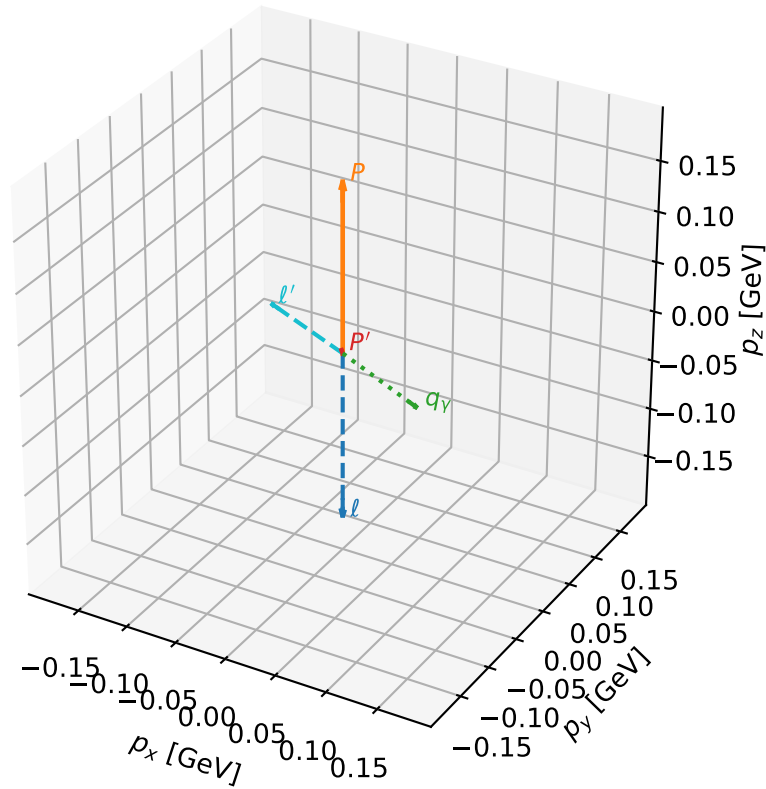
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_605 D_W=0.00421543
 region: polarization_cluster_03_local_minimum_605
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78670421$ rad
 incoming lepton: $0.706183|+\rangle + 0.70803|-\rangle$
 proton mixing angle: $\alpha_p=2.9846664$ rad
 incoming proton: $-0.987712|+\rangle + 0.156283|-\rangle$

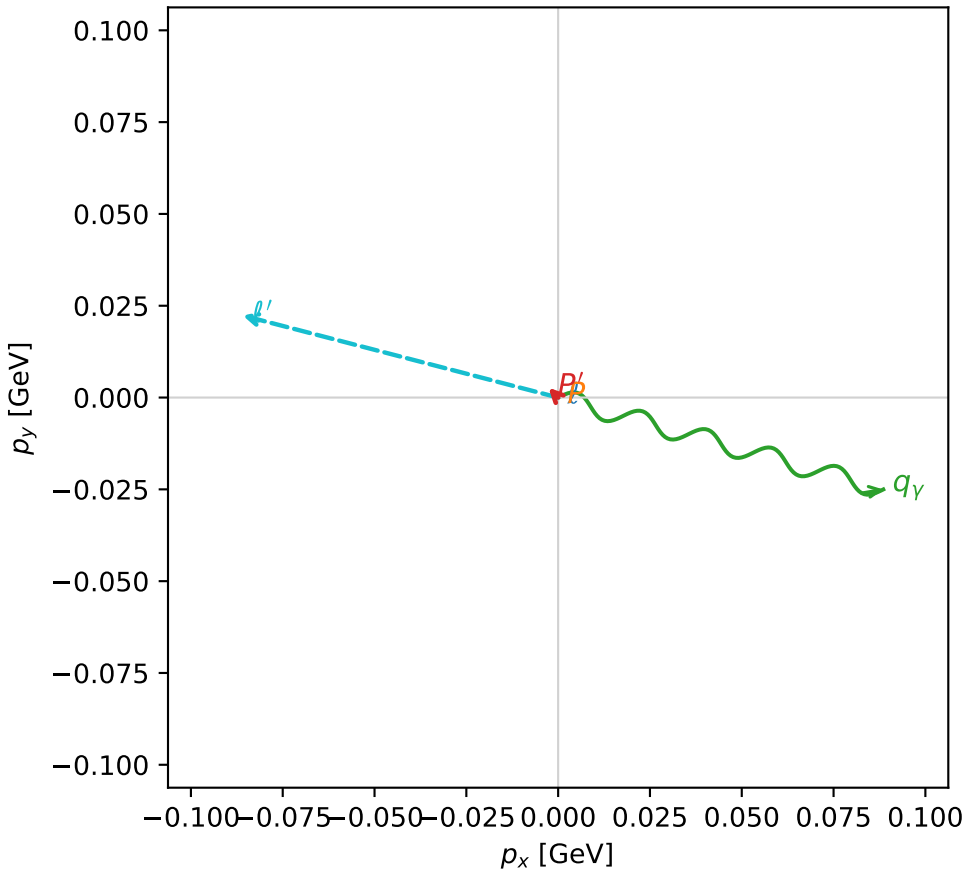
$s=1.25759$, $\sqrt{s}=1.12143$, $|\vec{P}|=0.168424$, $|\vec{P}'|=0.00390362$
 $\theta_{P'}=1.29341$, $\theta_V=1.7435$, $\phi_{P'}=2.3442$, $\phi_V=6.00771$
 $E_V=0.0933923$, $\theta_{l'}=1.40361$, $\phi_{l'}=2.88721$
 $Q^2=0.0353703$, $x_B=0.167466$, $t=-0.0277969$, $W^2=1.05568$, $y=0.559123$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1684$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1684$
 P $E= 0.9530$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1684$
 l' $E= 0.0900$ $p_x= -0.0859$ $p_y= 0.0223$ $p_z= 0.0150$
 P' $E= 0.9380$ $p_x= -0.0026$ $p_y= 0.0027$ $p_z= 0.0011$
 q_V $E= 0.0934$ $p_x= 0.0885$ $p_y= -0.0250$ $p_z= -0.0160$

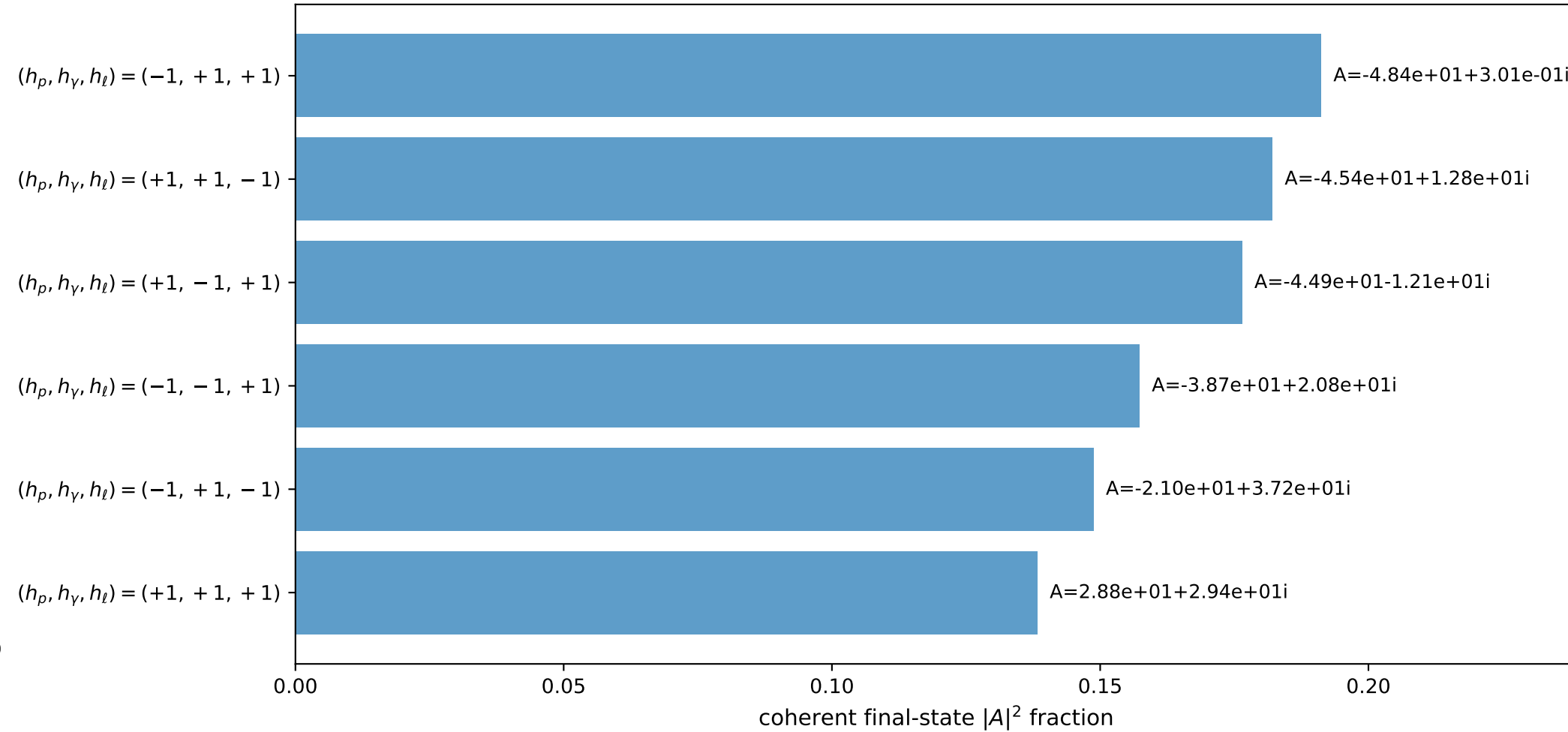
3D momenta



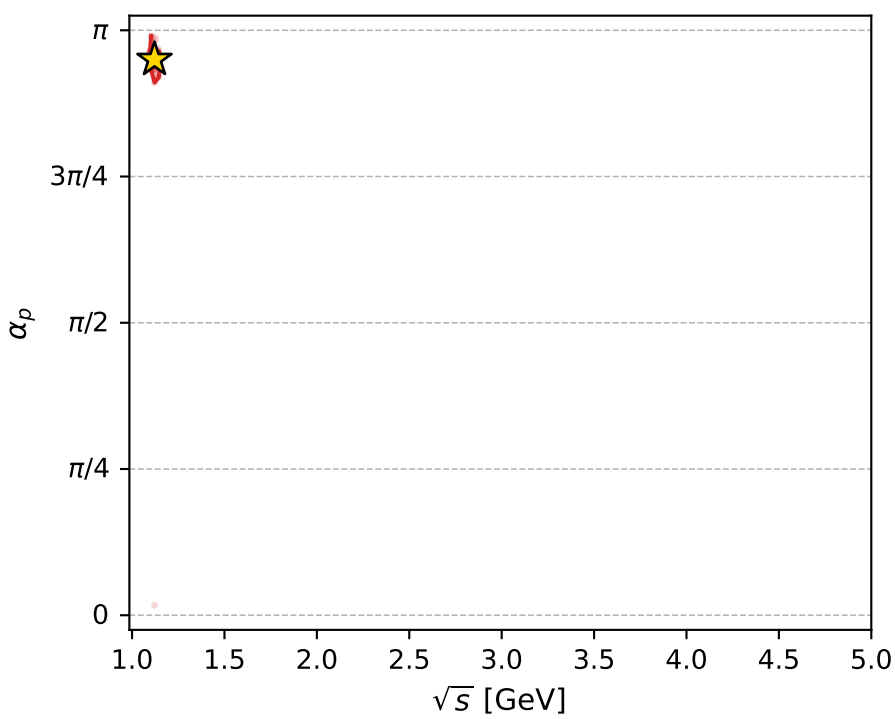
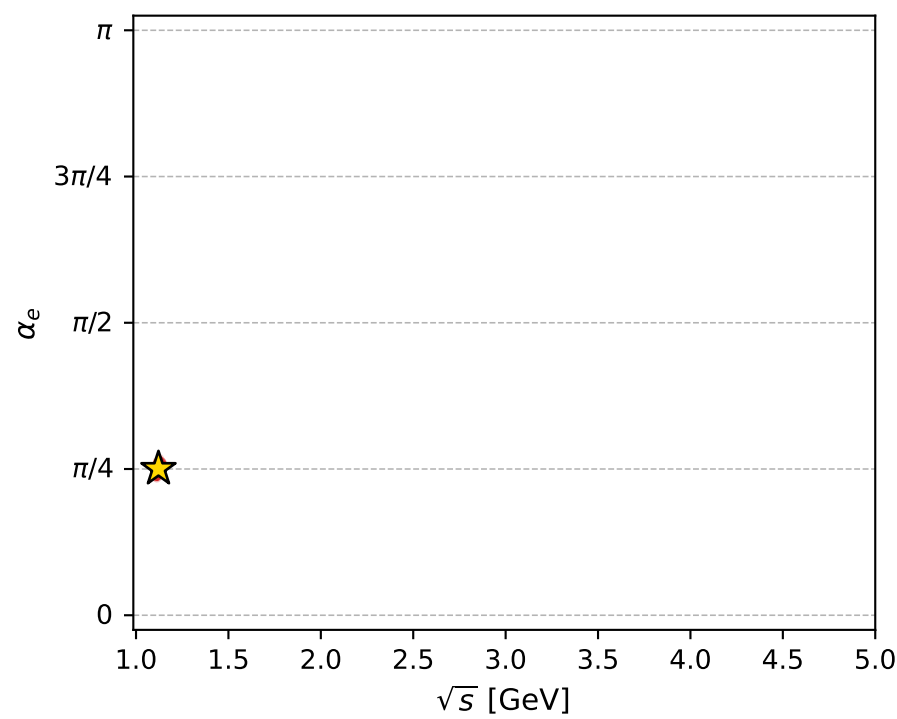
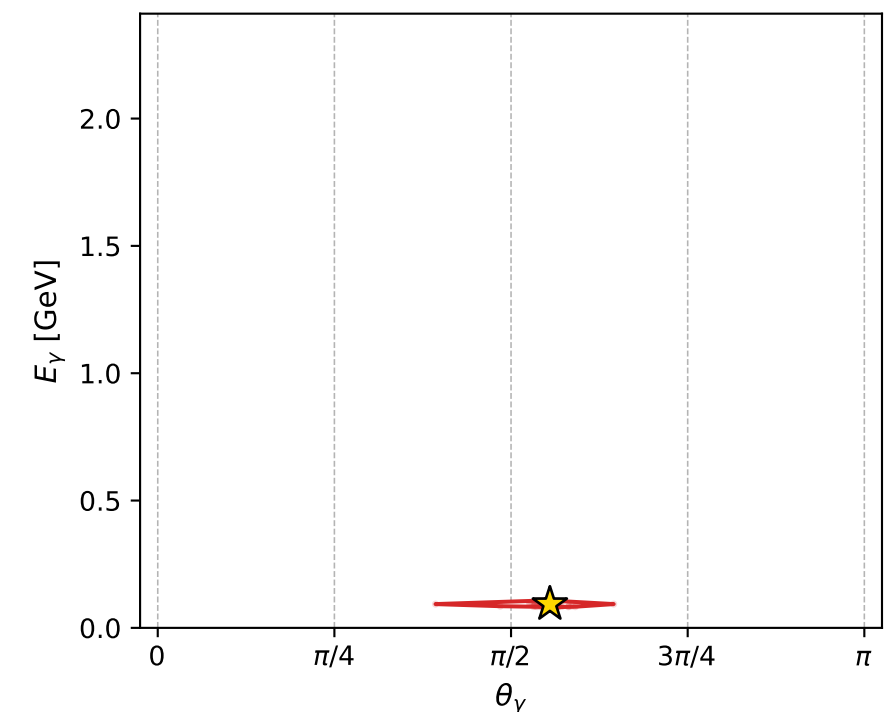
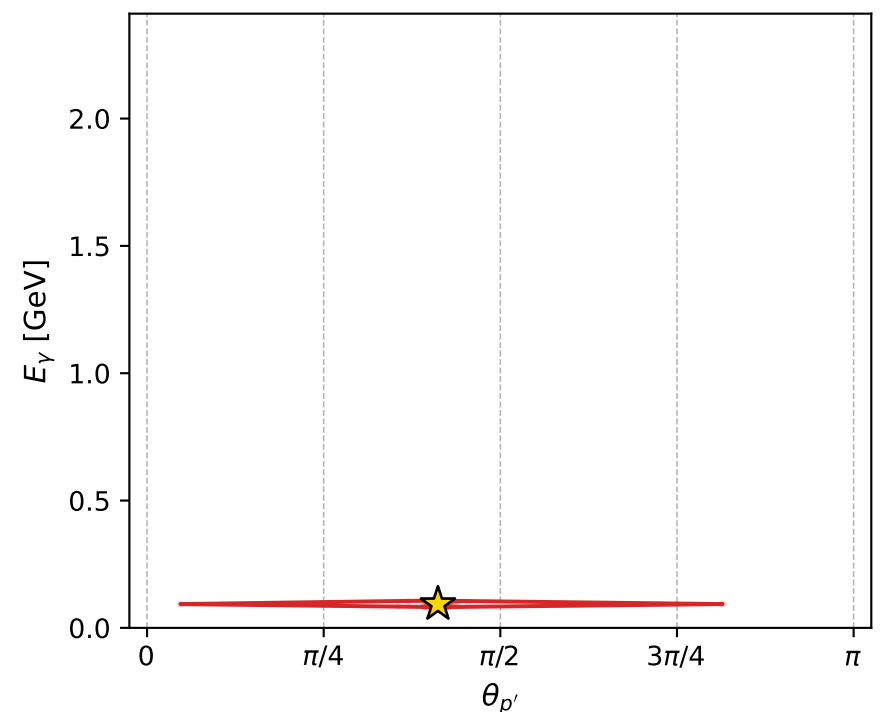
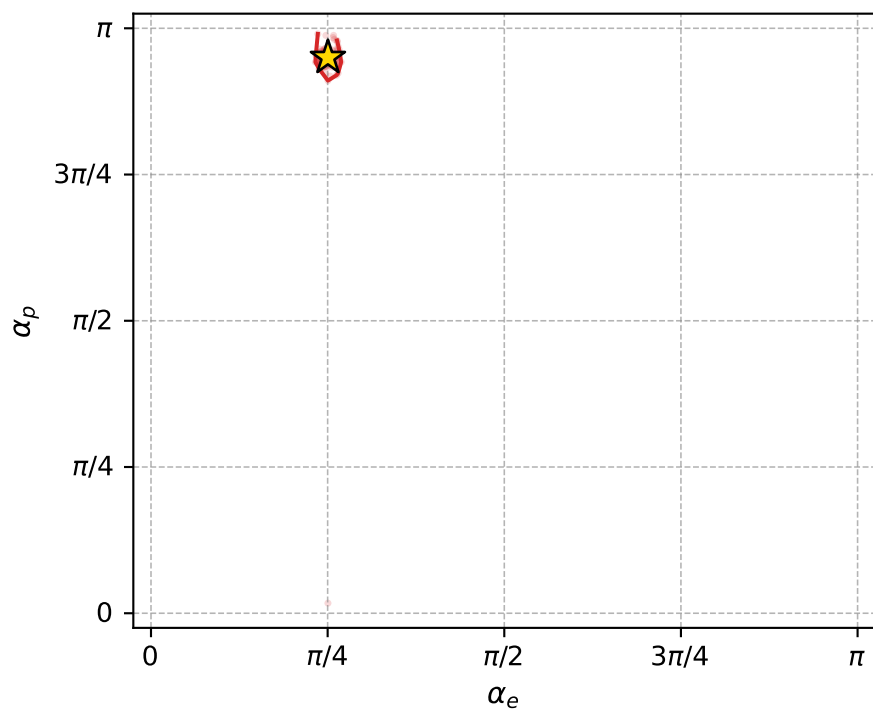
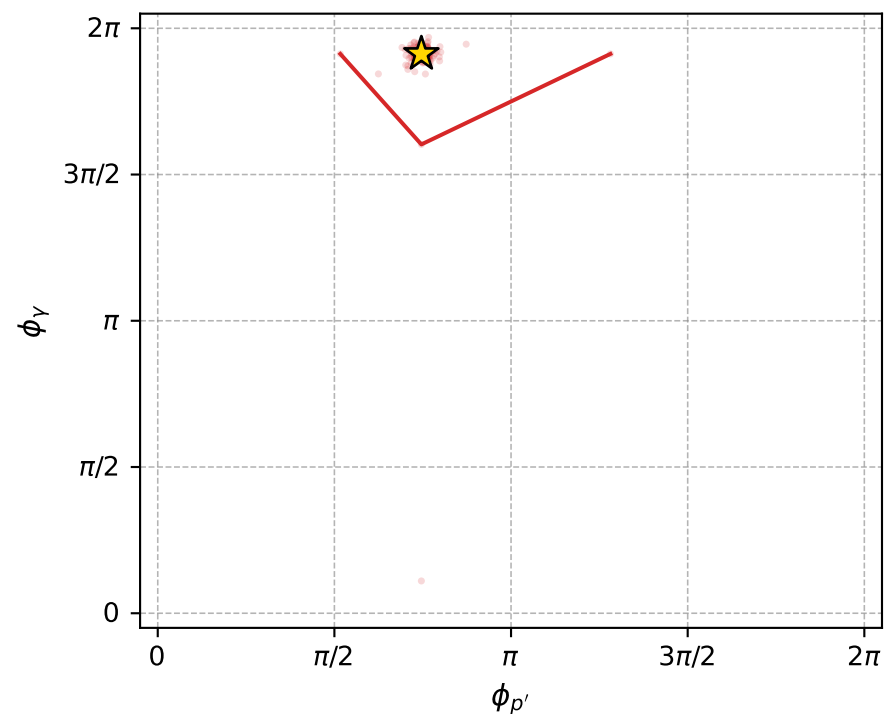
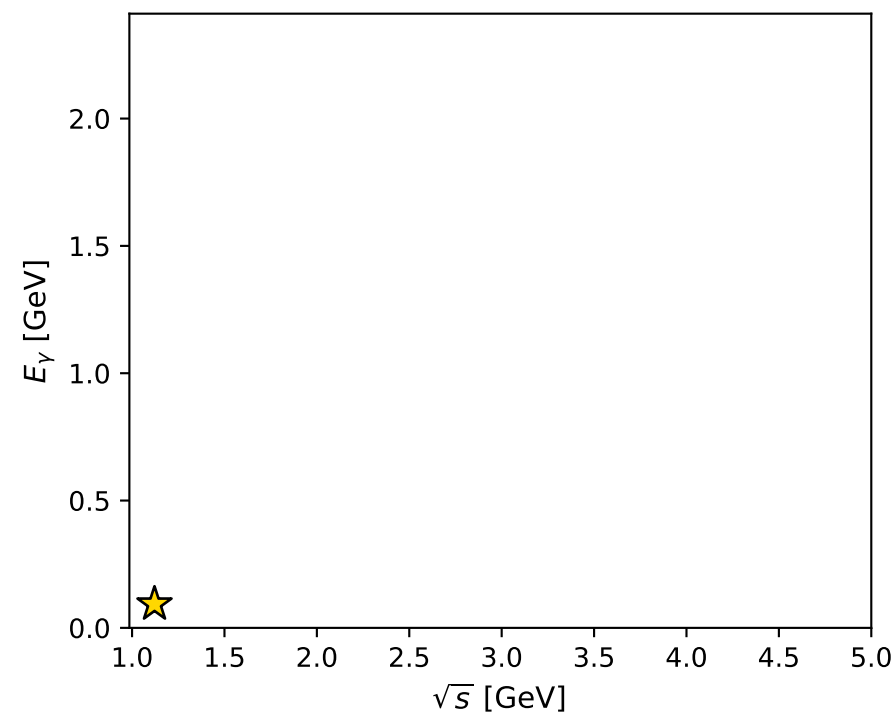
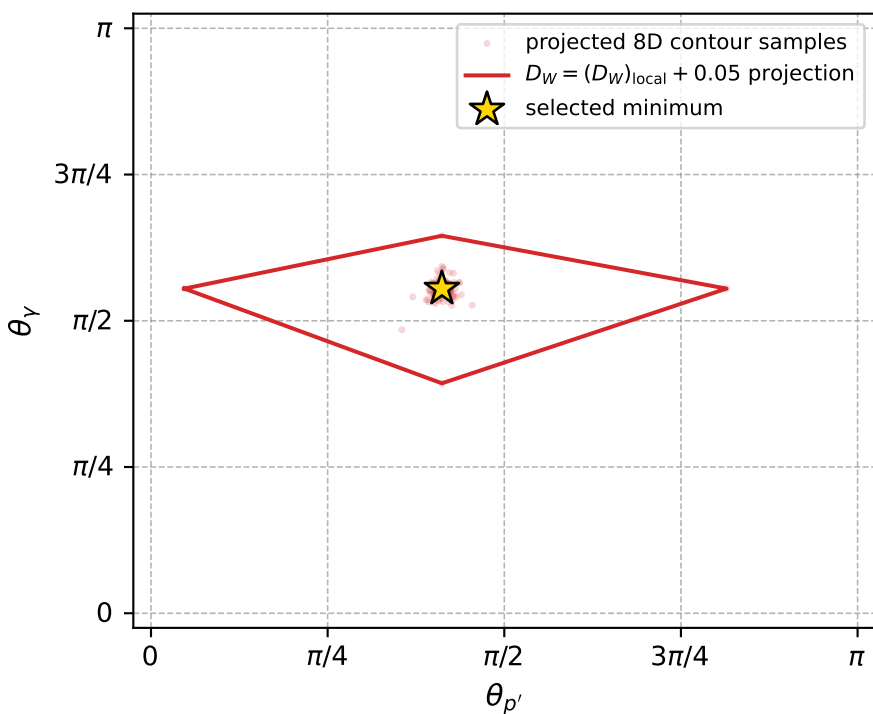
Transverse plane



Leading final-state amplitudes (N=6, retained=99.4%)



electron: polarization cluster P3, local minimum 605; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0042154292$
 $D_W \text{ contour} = 0.054215429$
 above global minimum = 0.0041989964
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.121425$
 $\text{theta_p_out} = 1.293414$
 $\text{theta_gamma_out} = 1.743502$
 $\text{qOut} = 0.09339226$
 $\text{phi_p_out} = 2.344204$
 $\text{phi_gamma_out} = 6.007712$
 $\text{alpha_e} = 0.7867042$
 $\text{alpha_p} = 2.984666$

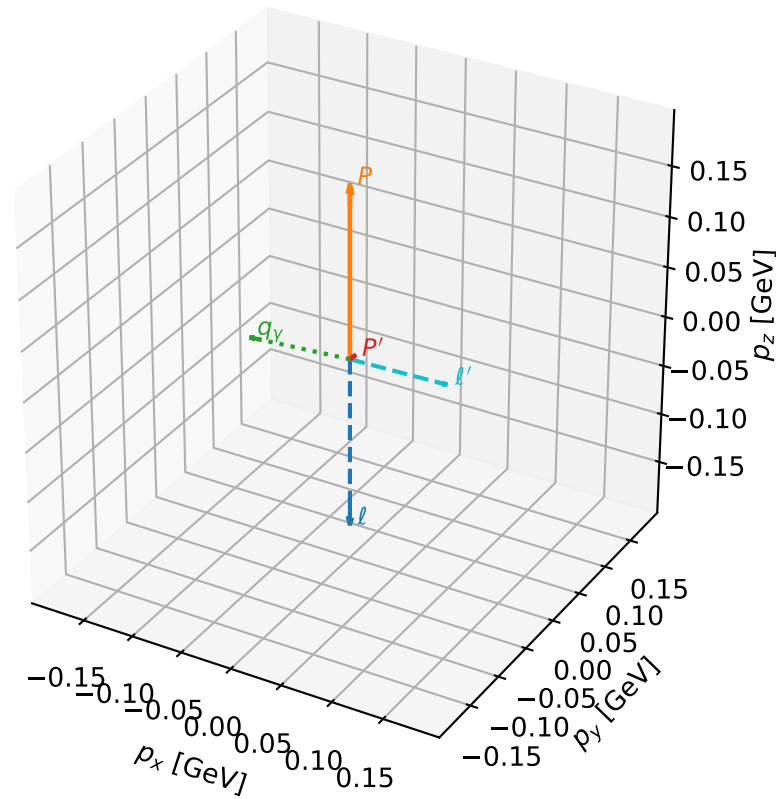
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_614 D_W=0.00451842
region: polarization_cluster_03_local_minimum_614
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.78573315$ rad
incoming lepton: $0.70687|+\rangle +0.707344|-\rangle$
proton mixing angle: $\alpha_p=0.09904092$ rad
incoming proton: $0.995099|+\rangle +0.0988791|-\rangle$

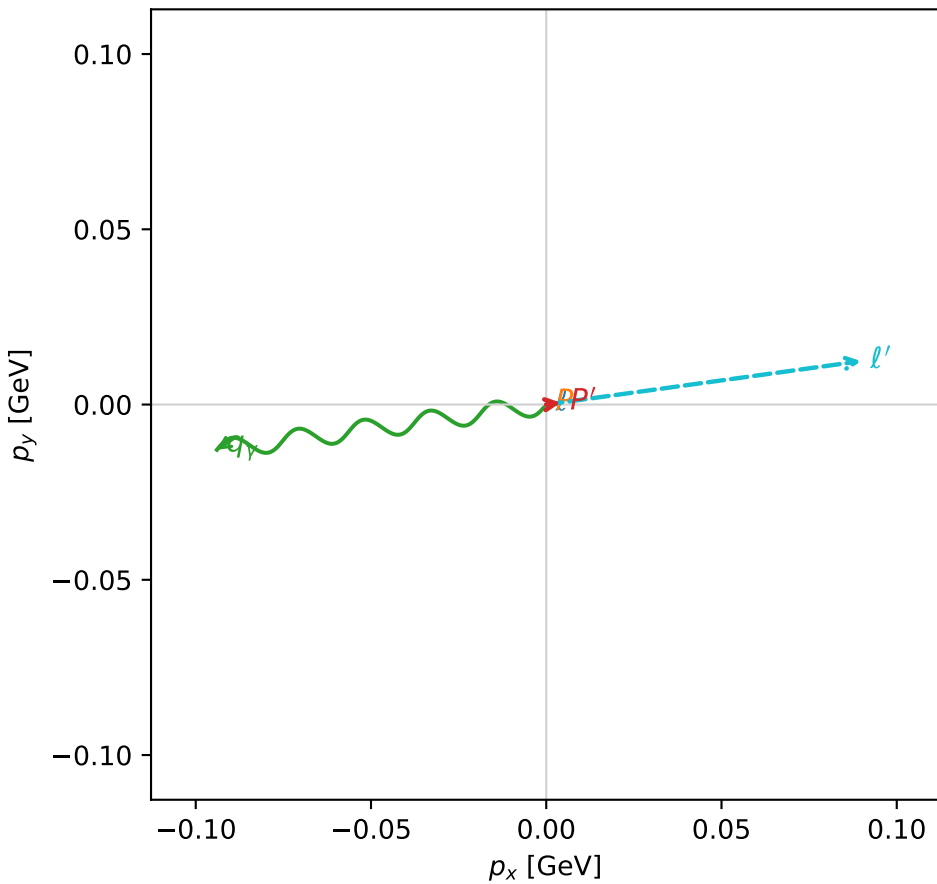
$s=1.26276$, $\sqrt{s}=1.12372$, $|\vec{P}|=0.170376$, $|\vec{P}'|=0.00593269$
 $\theta_{P'}=0.782019$, $\theta_V=1.55455$, $\phi_{P'}=0.129107$, $\phi_V=3.27819$
 $E_V=0.0948574$, $\theta_{\ell'}=1.63414$, $\phi_{\ell'}=0.136939$
 $Q^2=0.0289967$, $x_B=0.139586$, $t=-0.0273938$, $W^2=1.05858$, $y=0.542509$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1704$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1704$
 P $E= 0.9533$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1704$
 ℓ' $E= 0.0908$ $p_x= 0.0898$ $p_y= 0.0124$ $p_z= -0.0058$
 P' $E= 0.9380$ $p_x= 0.0041$ $p_y= 0.0005$ $p_z= 0.0042$
 q_V $E= 0.0949$ $p_x= -0.0940$ $p_y= -0.0129$ $p_z= 0.0015$

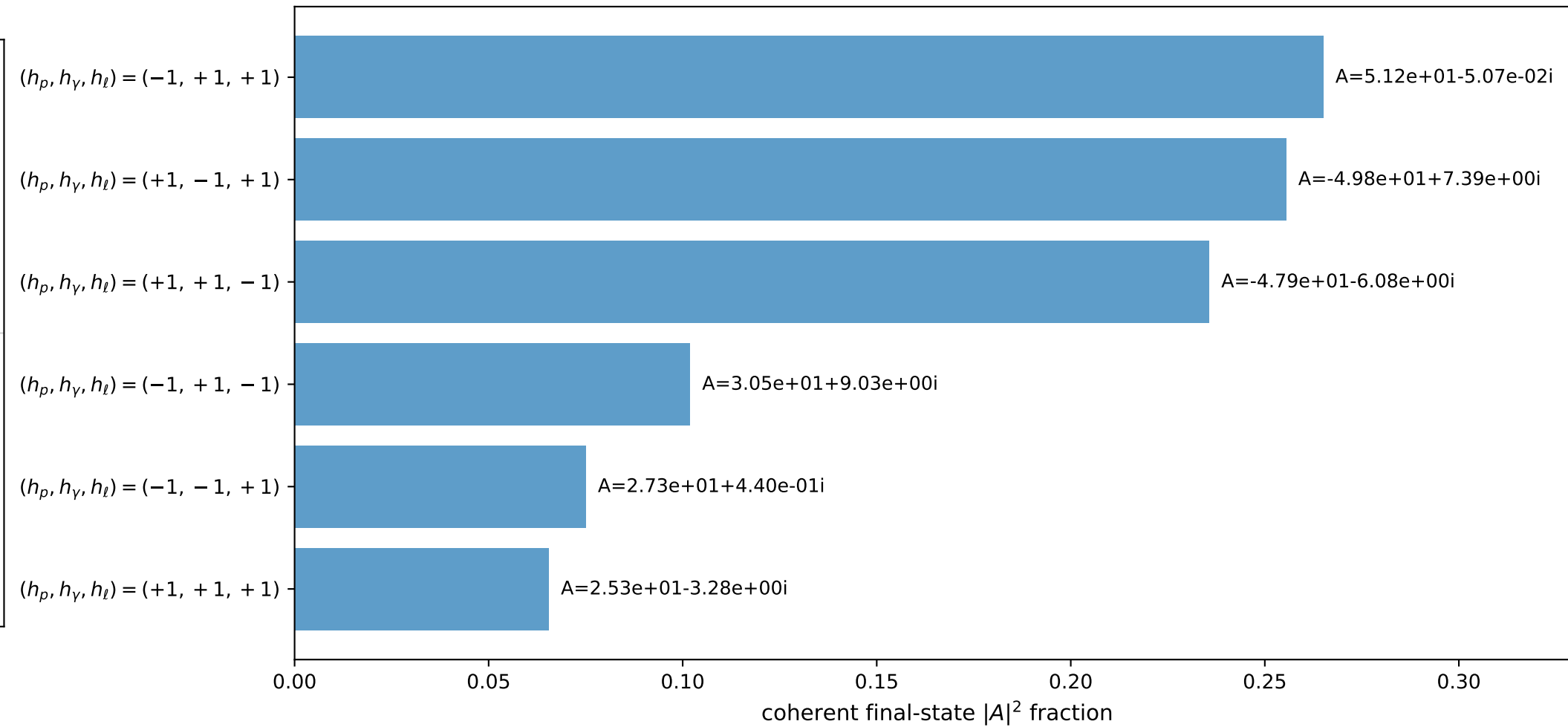
3D momenta



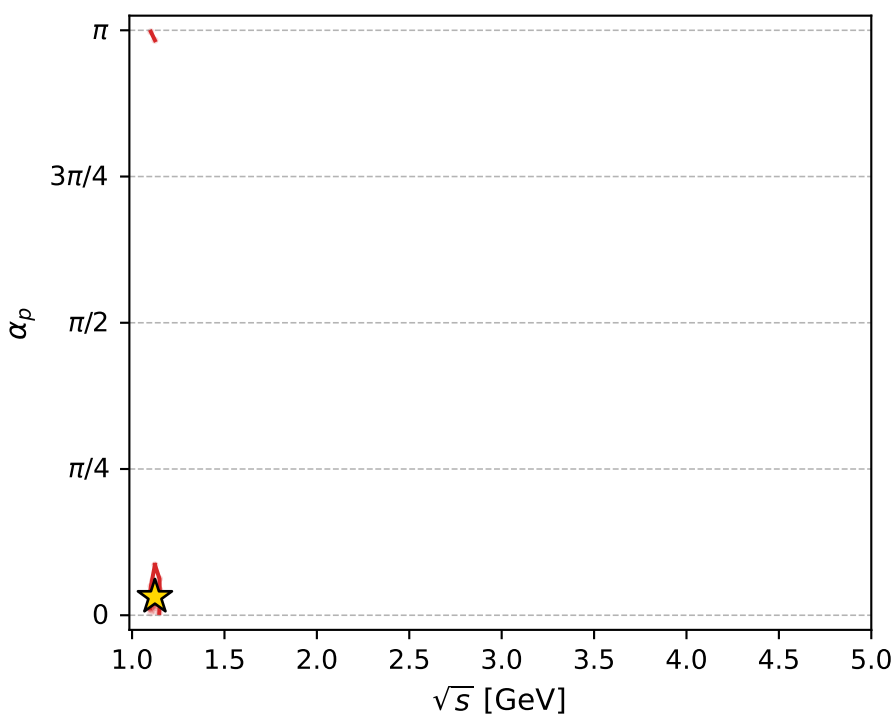
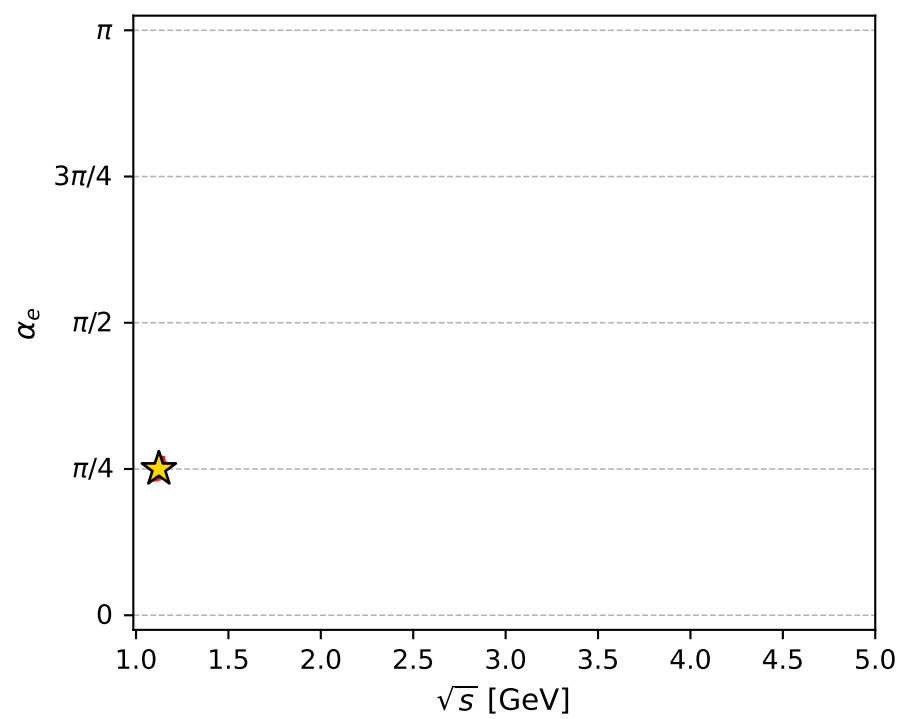
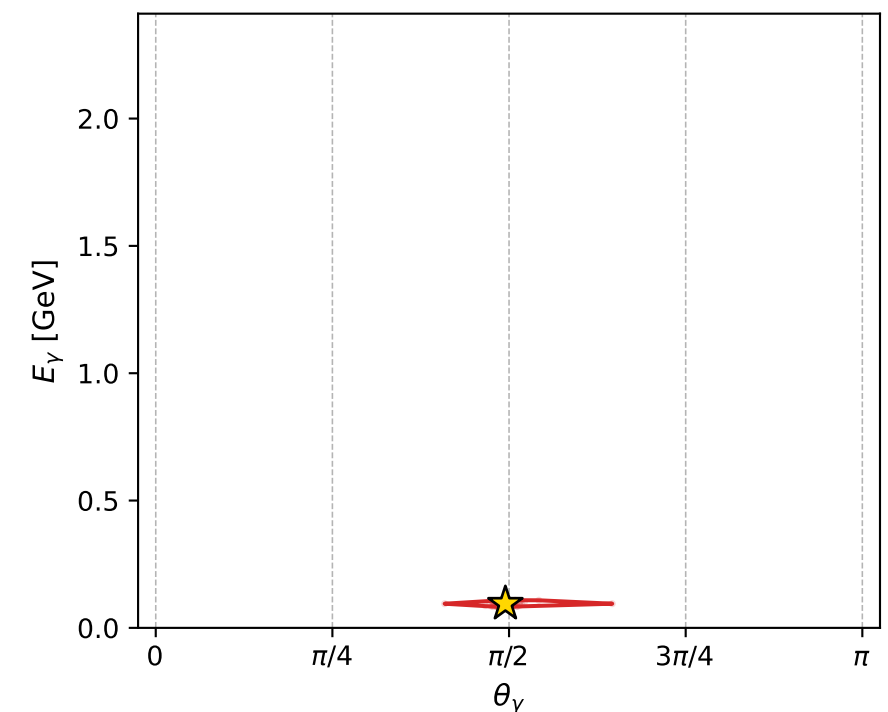
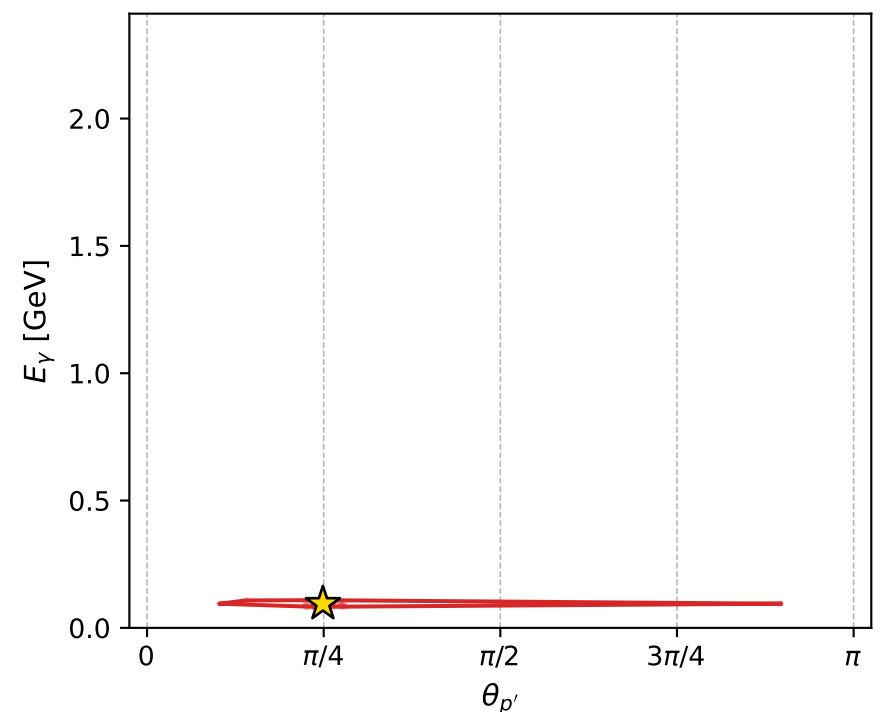
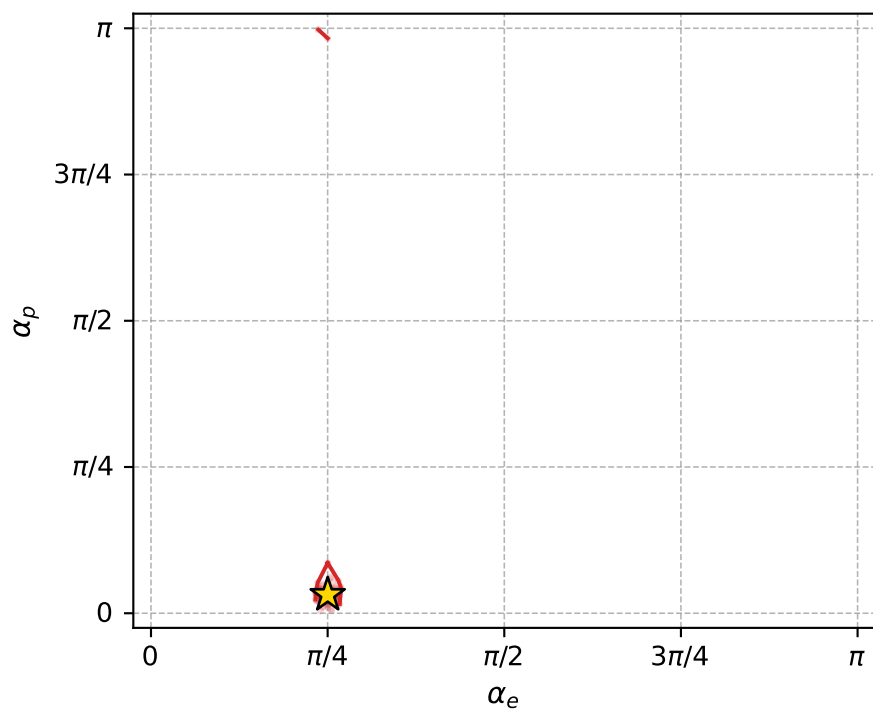
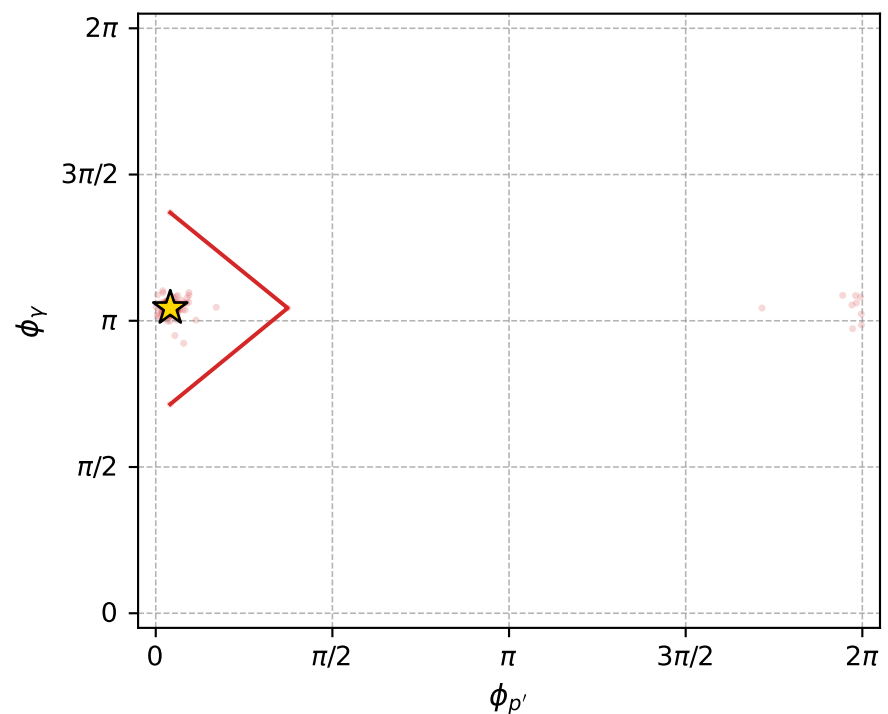
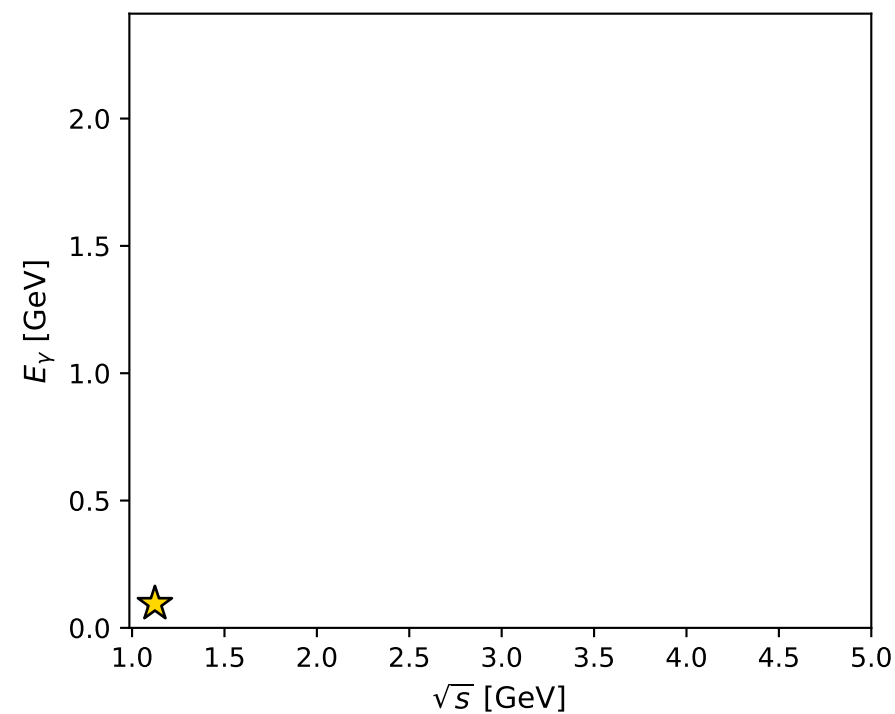
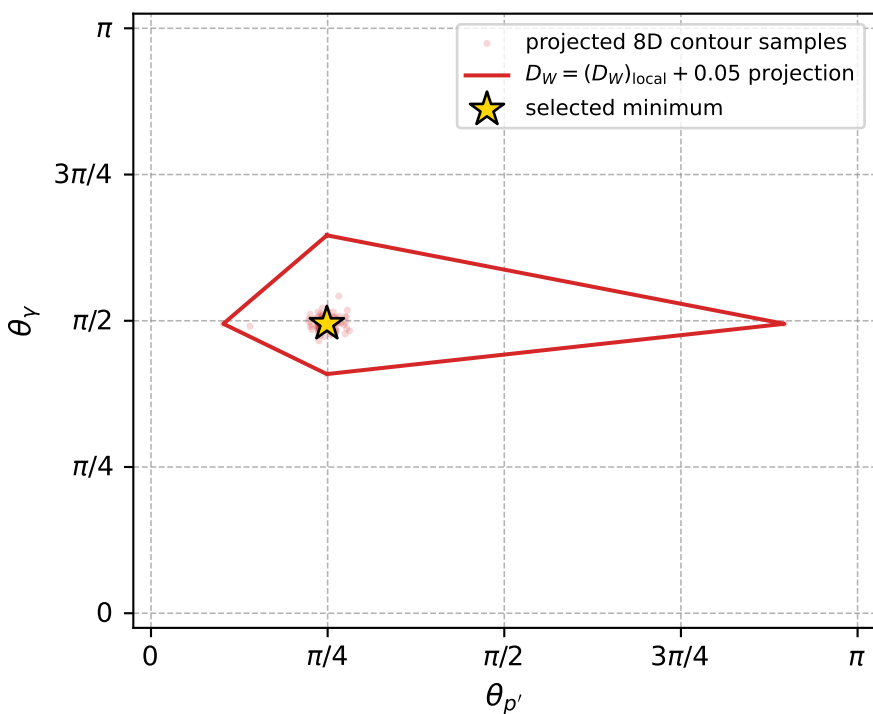
Transverse plane



Leading final-state amplitudes (N=6, retained=99.9%)



electron: polarization cluster P3, local minimum 614; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0045184216$
 $D_W \text{ contour} = 0.054518422$
 above global minimum = 0.0045019888
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.123724$
 $\text{theta_p_out} = 0.7820192$
 $\text{theta_gamma_out} = 1.554548$
 $\text{q0ut} = 0.09485743$
 $\text{phi_p_out} = 0.1291069$
 $\text{phi_gamma_out} = 3.278186$
 $\text{alpha_e} = 0.7857331$
 $\text{alpha_p} = 0.09904092$

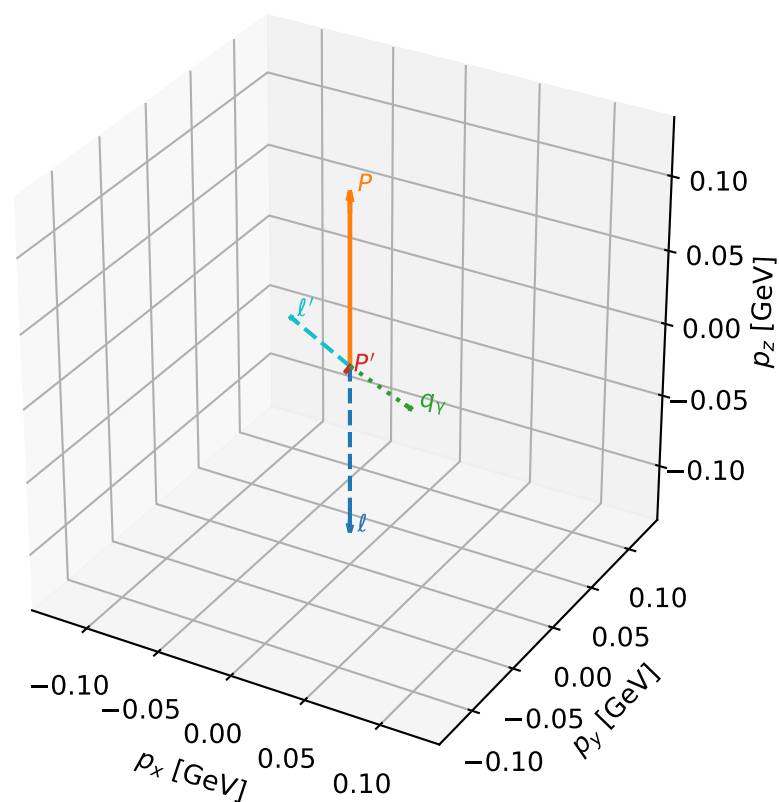
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_627 D_W=0.00481348
 region: polarization_cluster_03_local_minimum_627
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77941215$ rad
 incoming lepton: $0.711327|+\rangle + 0.702861|-\rangle$
 proton mixing angle: $\alpha_p=2.9513104$ rad
 incoming proton: $-0.981951|+\rangle + 0.189136|-\rangle$

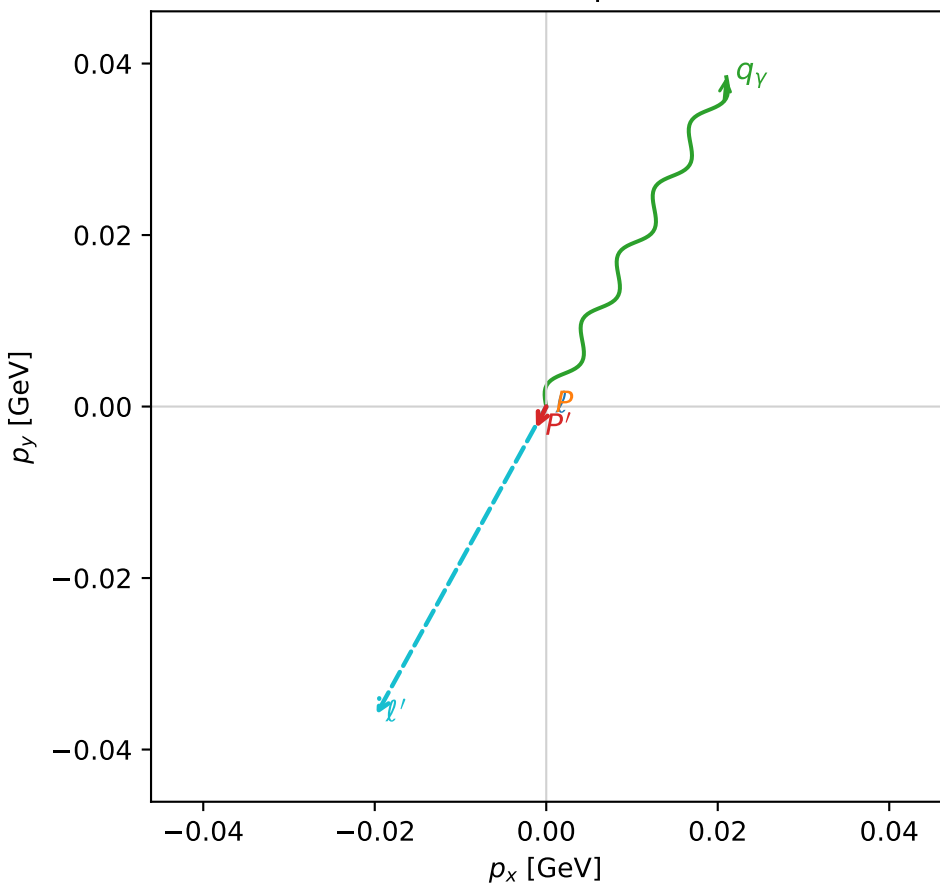
$s=1.12592$, $\sqrt{s}=1.06109$, $|\vec{P}|=0.115954$, $|\vec{P}'|=0.00381307$
 $\theta_{P'}=2.3286$, $\theta_Y=2.35026$, $\phi_{P'}=4.26555$, $\phi_Y=1.0703$
 $E_Y=0.0615458$, $\theta_l=0.729415$, $\phi_{l'}=4.20827$
 $Q^2=0.0249116$, $x_B=0.177449$, $t=-0.0140167$, $W^2=0.99532$, $y=0.570503$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1160$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1160$
 P $E= 0.9451$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1160$
 ℓ' $E= 0.0615$ $p_x= -0.0198$ $p_y= -0.0359$ $p_z= 0.0459$
 P' $E= 0.9380$ $p_x= -0.0012$ $p_y= -0.0025$ $p_z= -0.0026$
 q_Y $E= 0.0615$ $p_x= 0.0210$ $p_y= 0.0384$ $p_z= -0.0433$

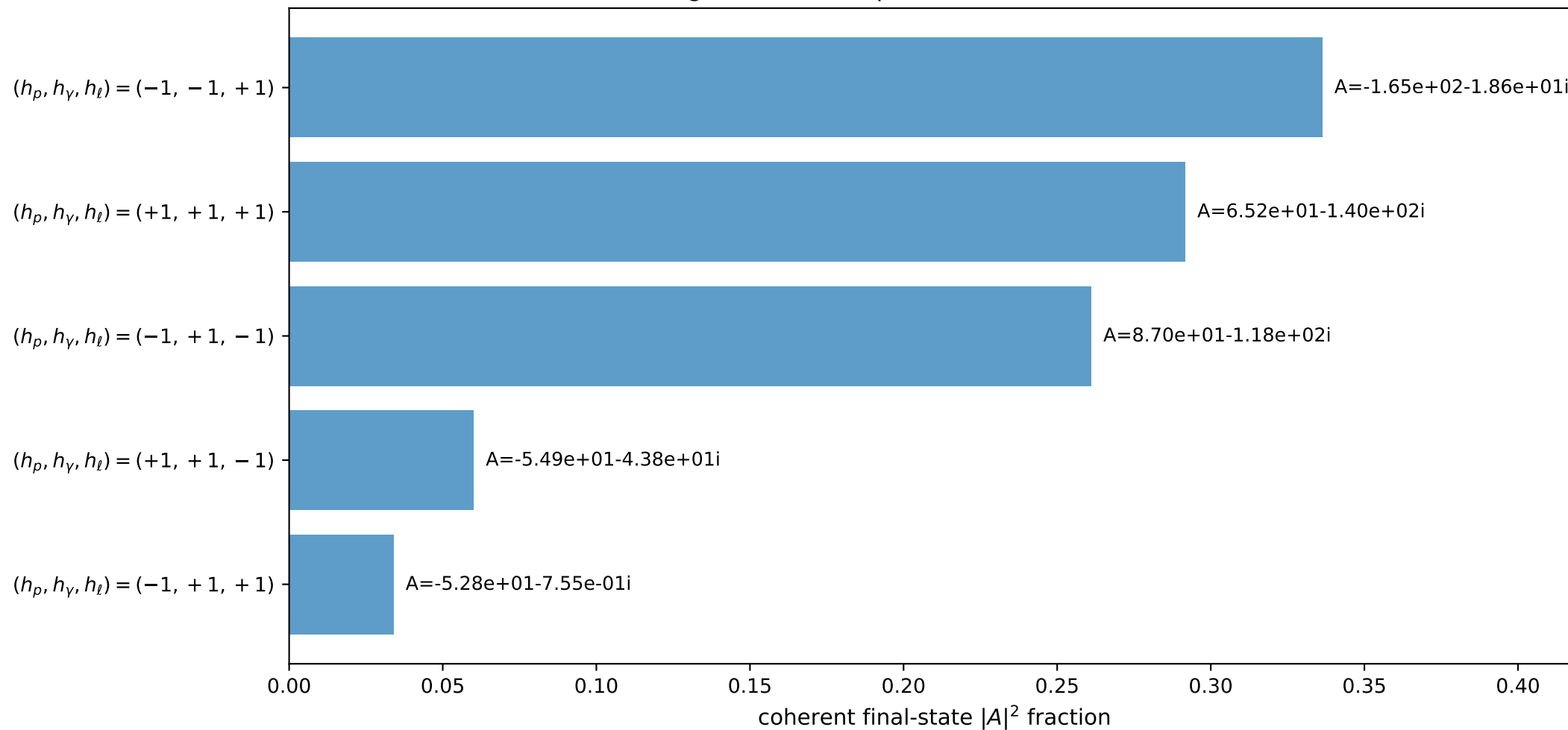
3D momenta



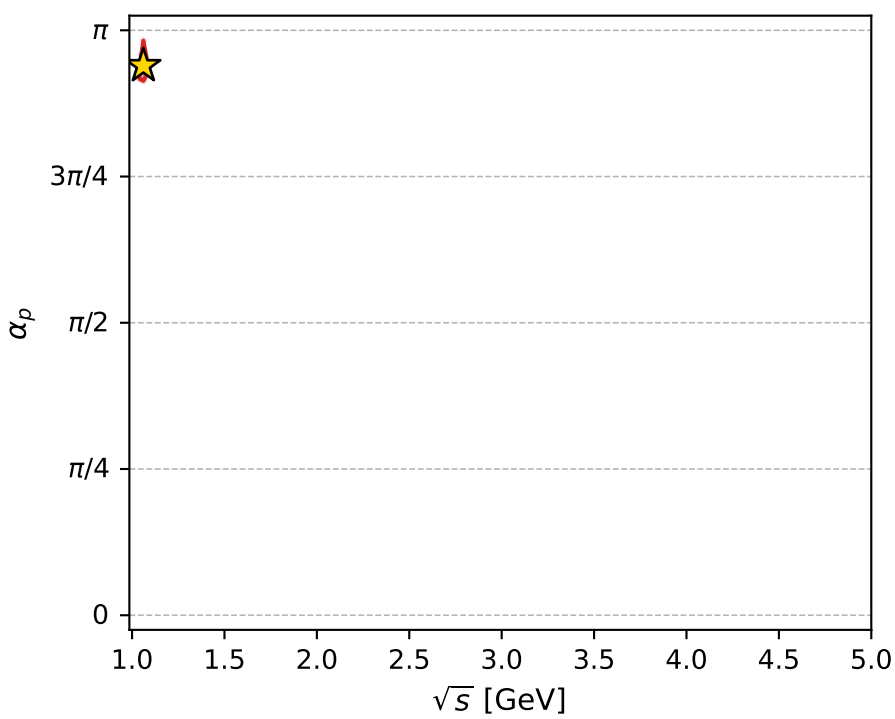
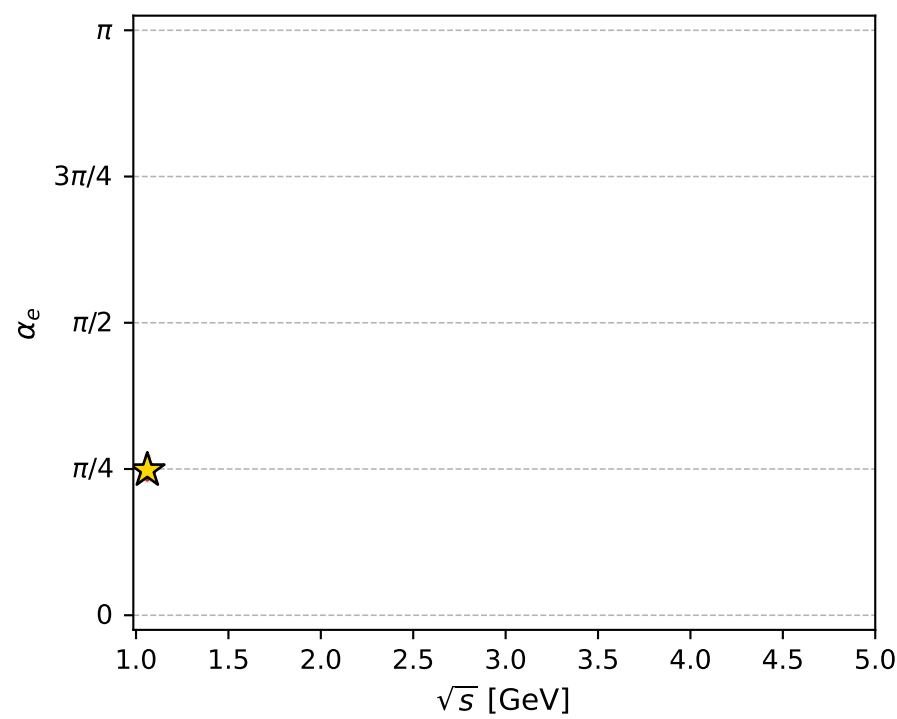
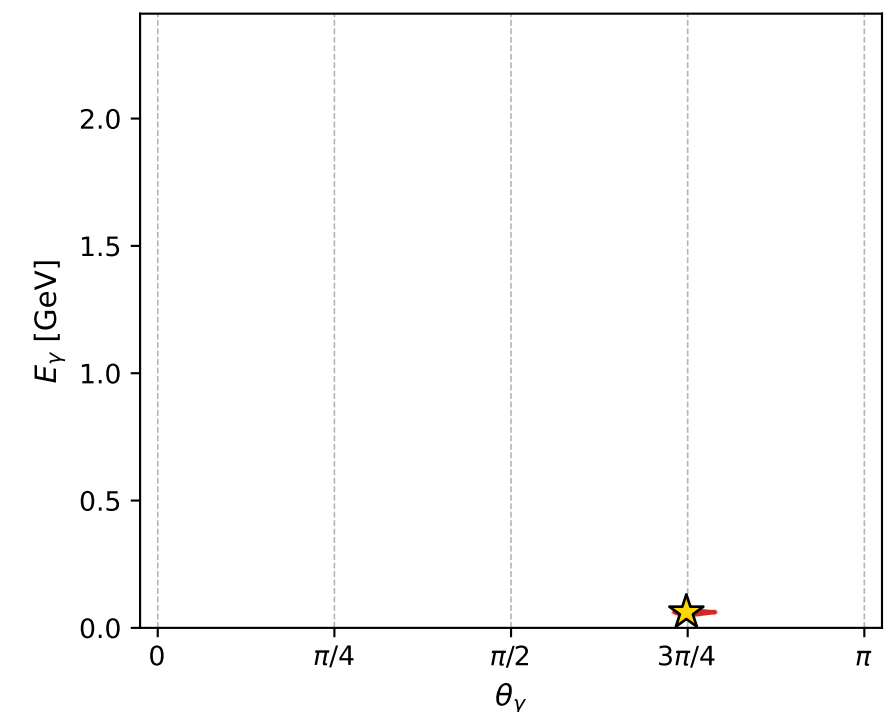
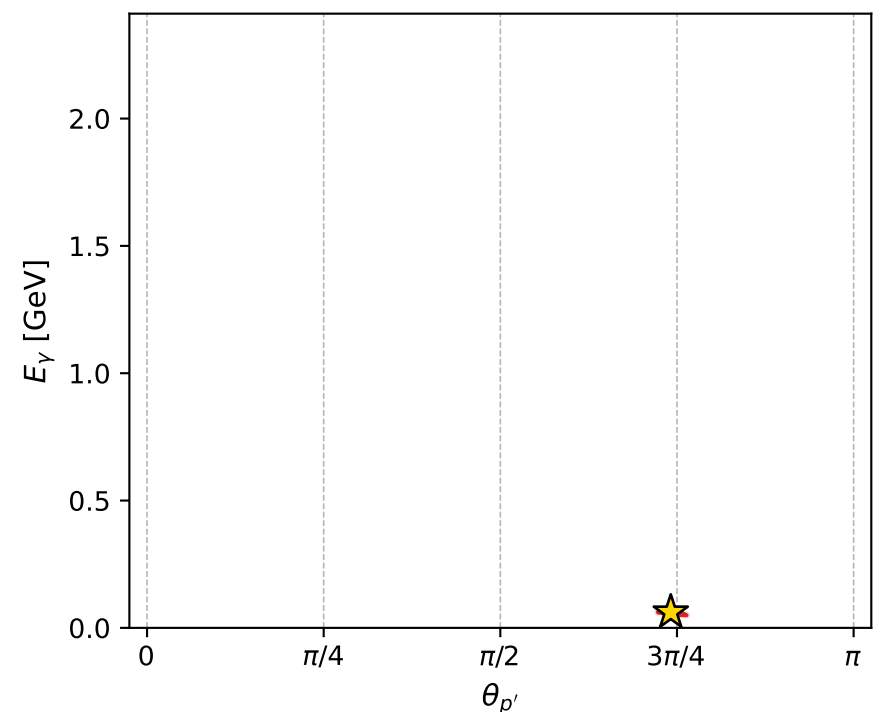
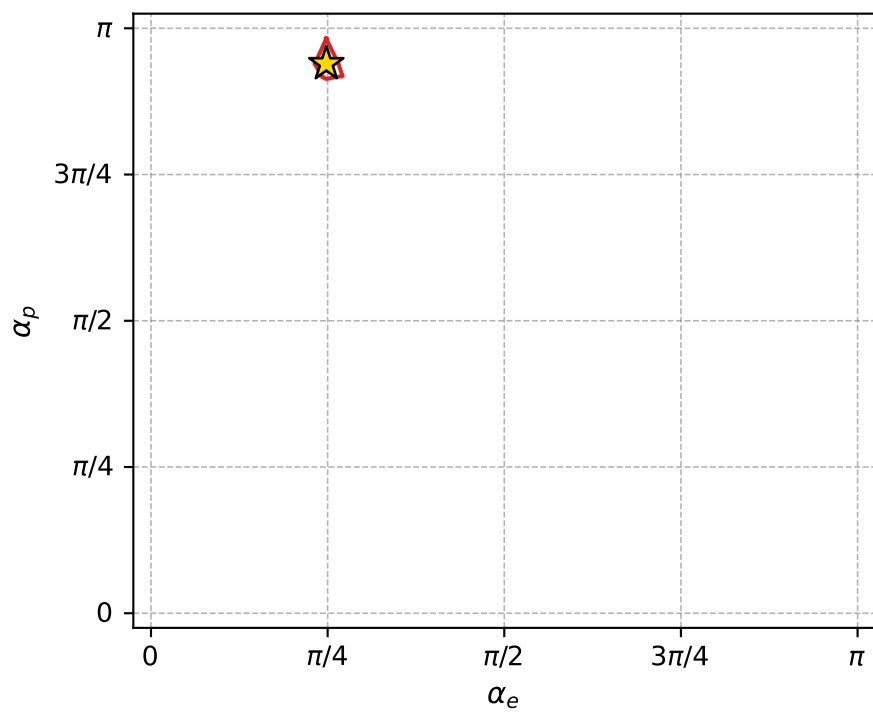
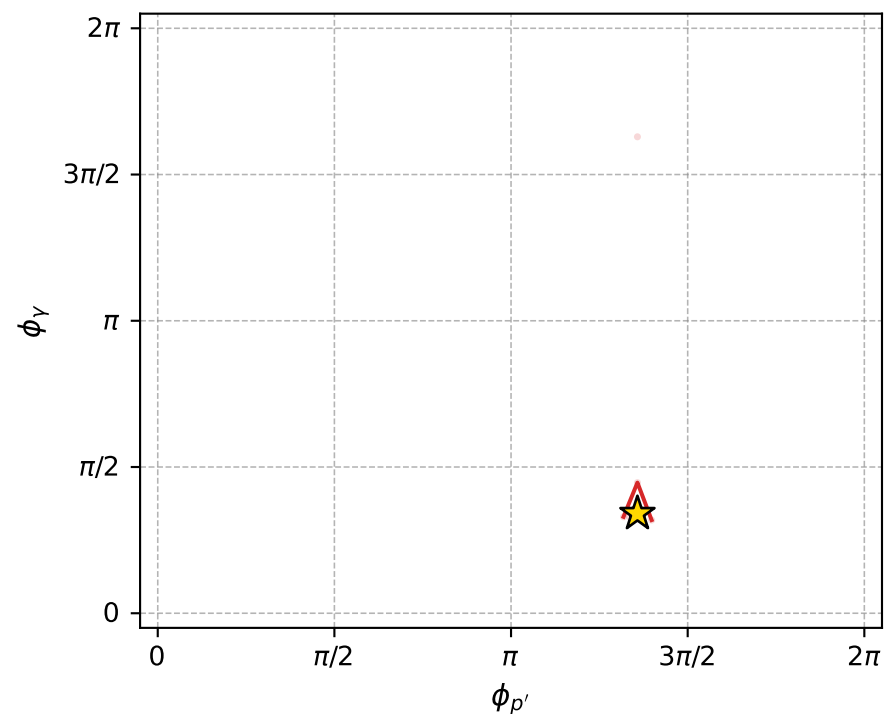
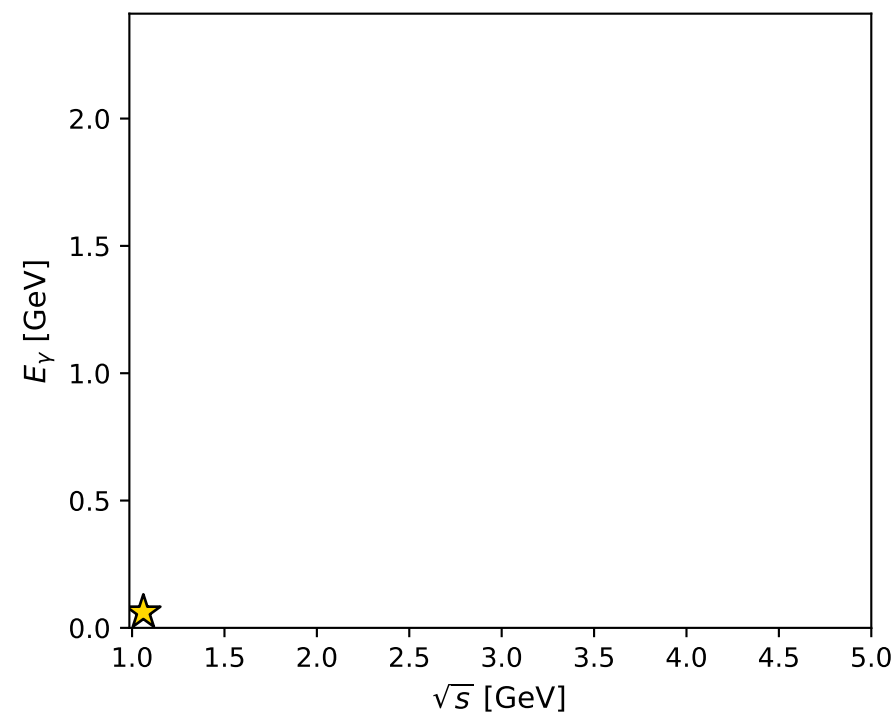
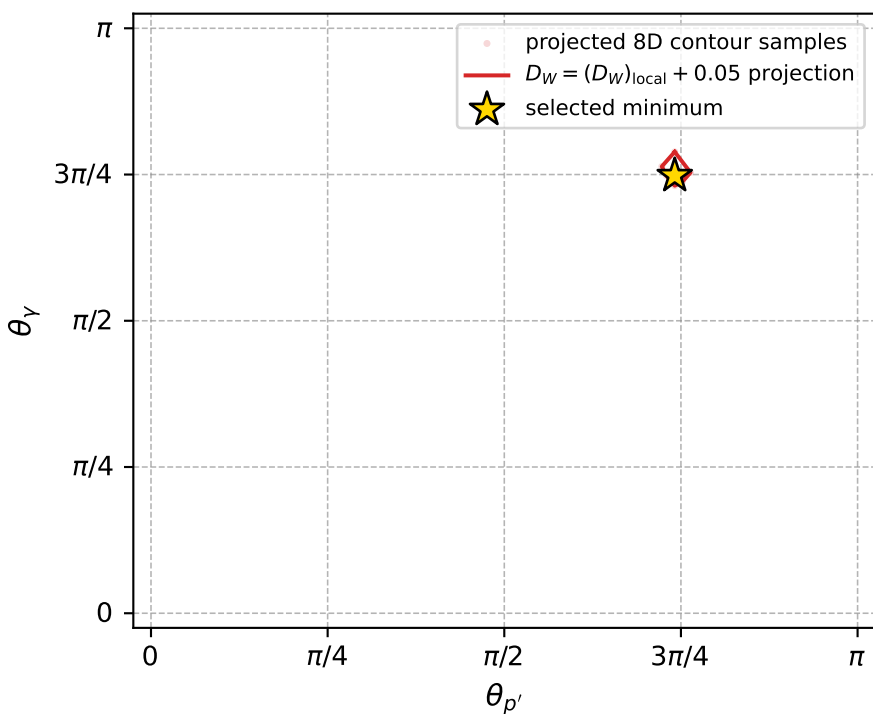
Transverse plane



Leading final-state amplitudes (N=5, retained=98.3%)



electron: polarization cluster P3, local minimum 627; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.004813483$
 $D_W \text{ contour} = 0.054813483$
 above global minimum = 0.0047970502
 8D contour samples = 253

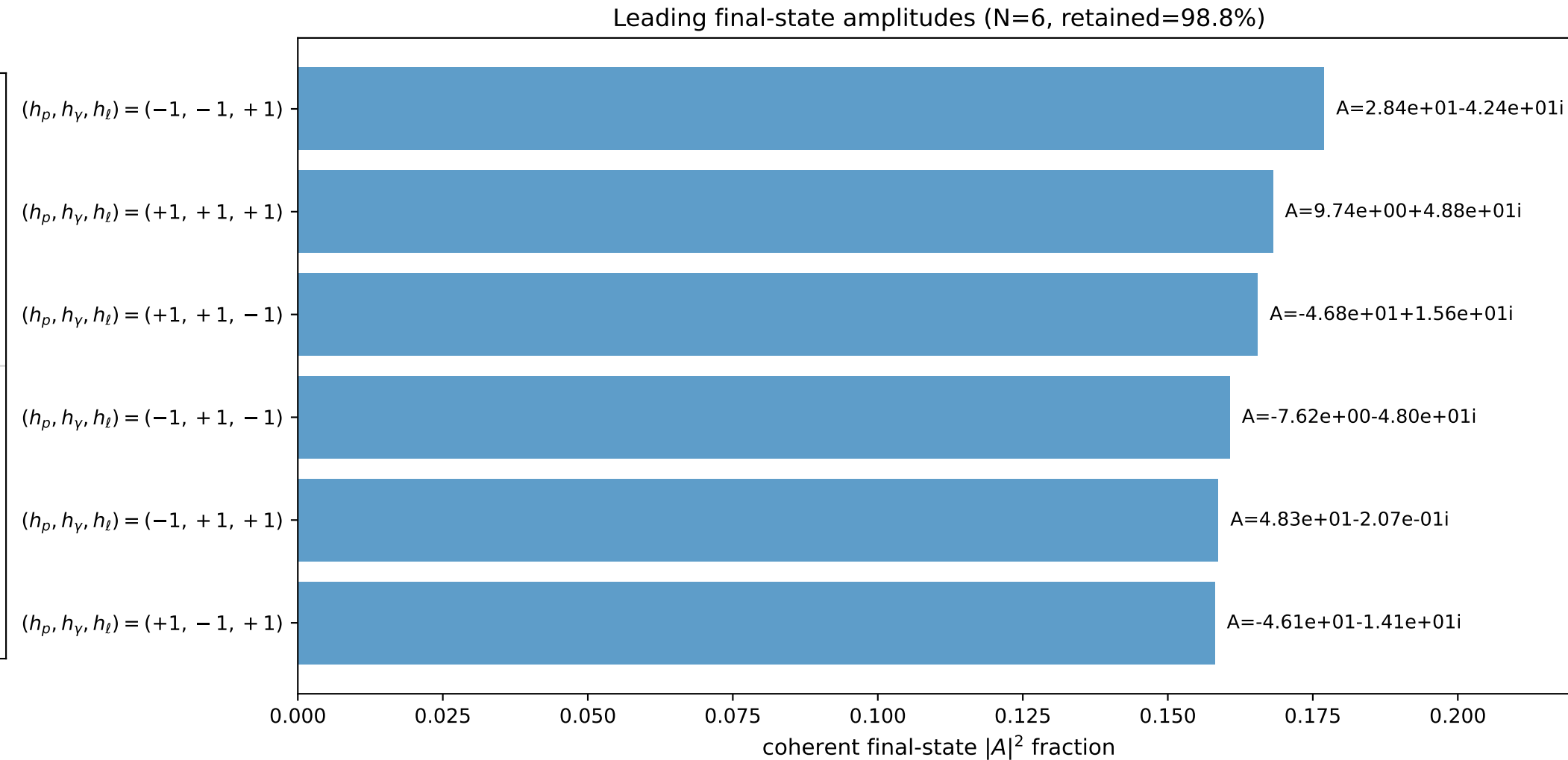
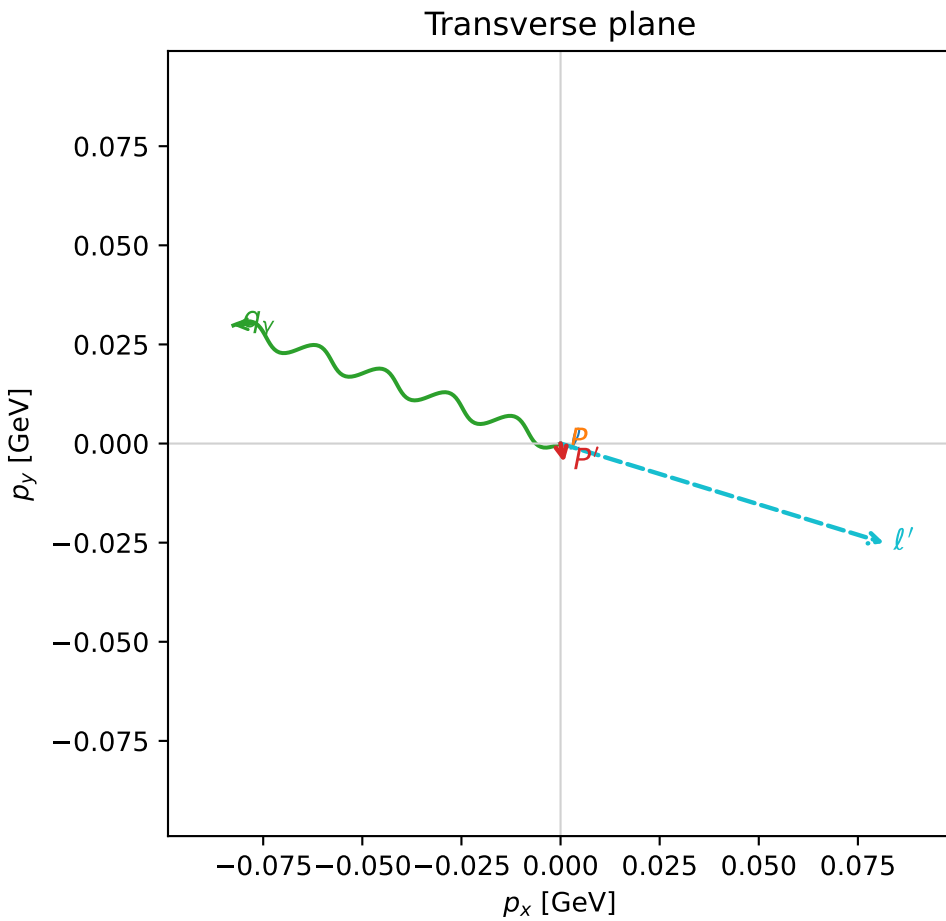
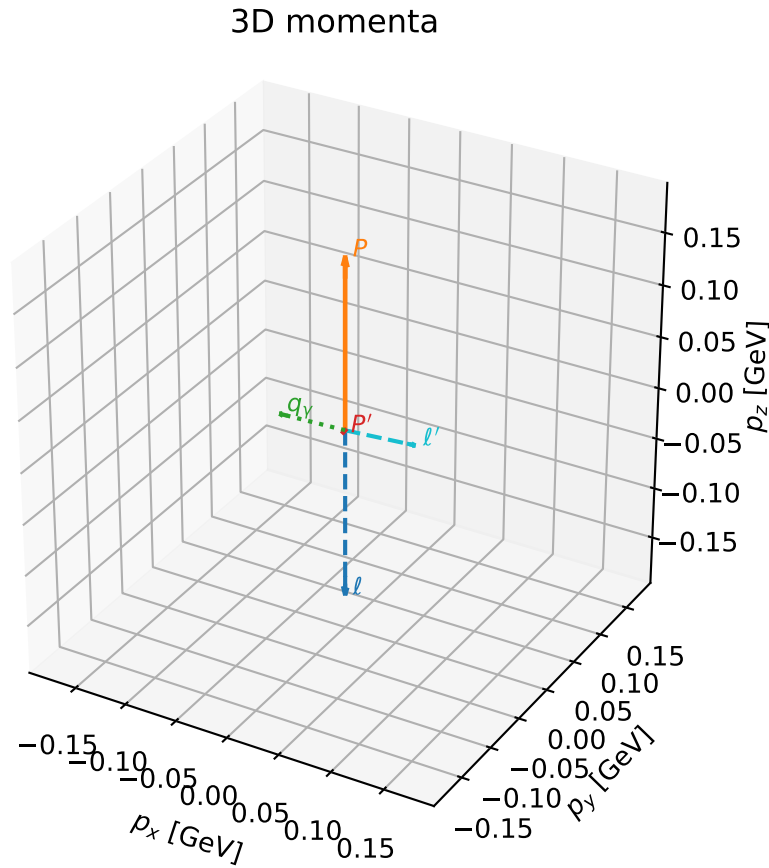
selected phase-space point:
 $\text{sqrt_s} = 1.061095$
 $\text{theta_p_out} = 2.328603$
 $\text{theta_gamma_out} = 2.35026$
 $\text{qOut} = 0.06154579$
 $\text{phi_p_out} = 4.265552$
 $\text{phi_gamma_out} = 1.070302$
 $\text{alpha_e} = 0.7794121$
 $\text{alpha_p} = 2.95131$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

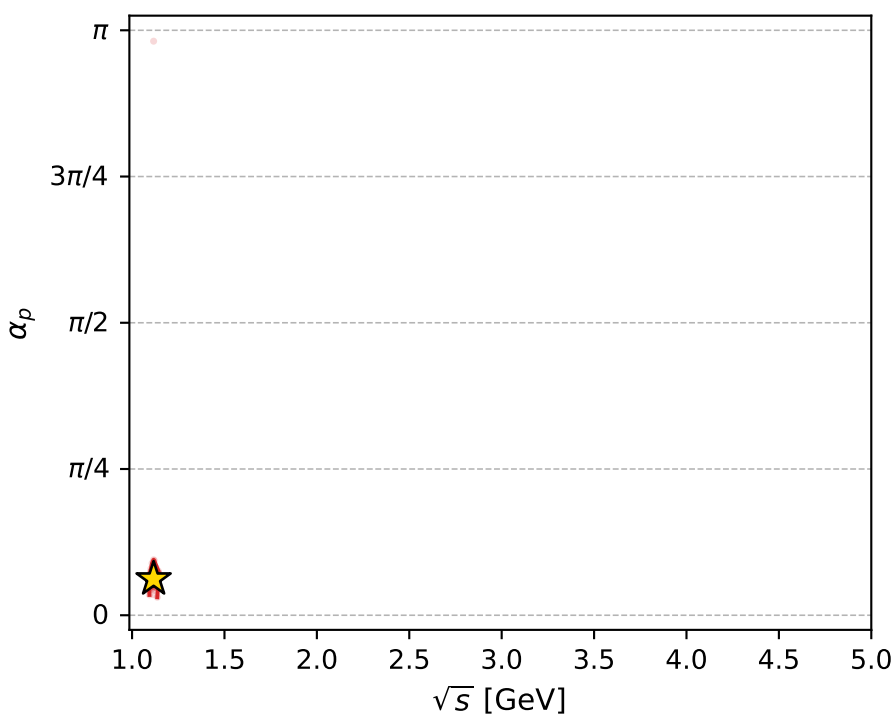
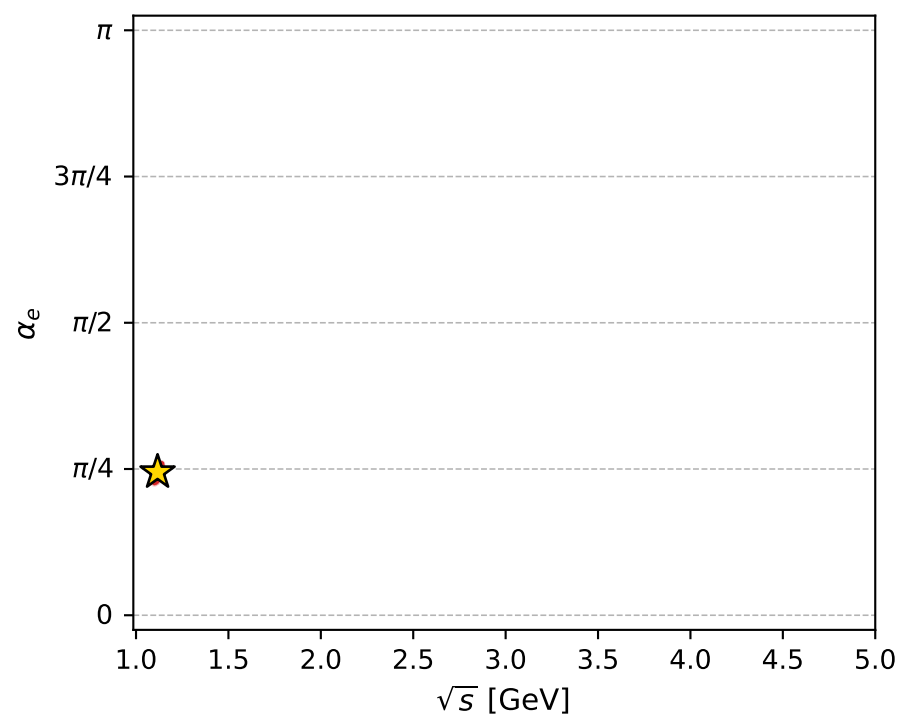
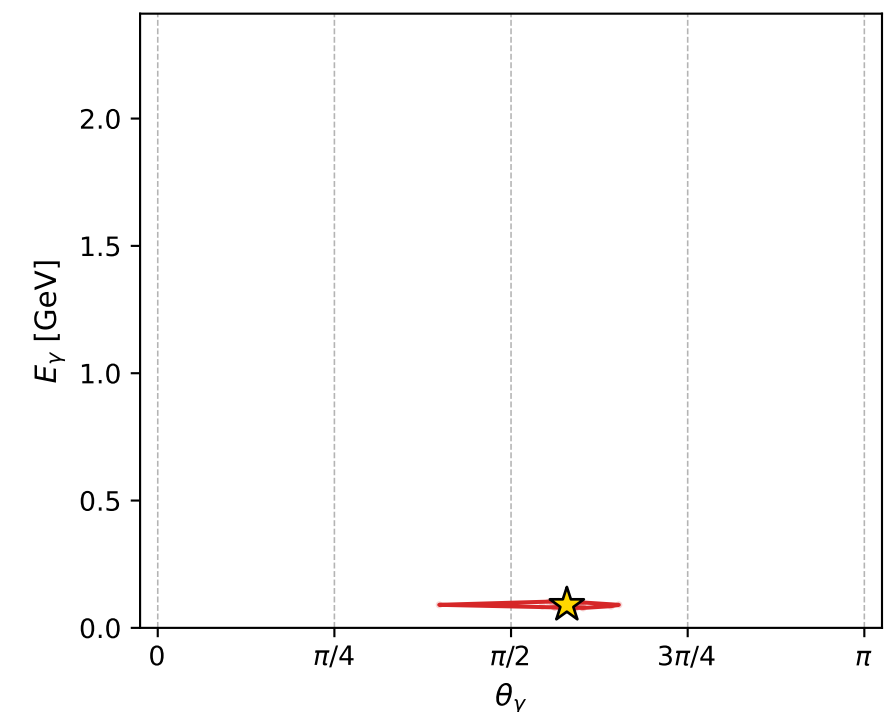
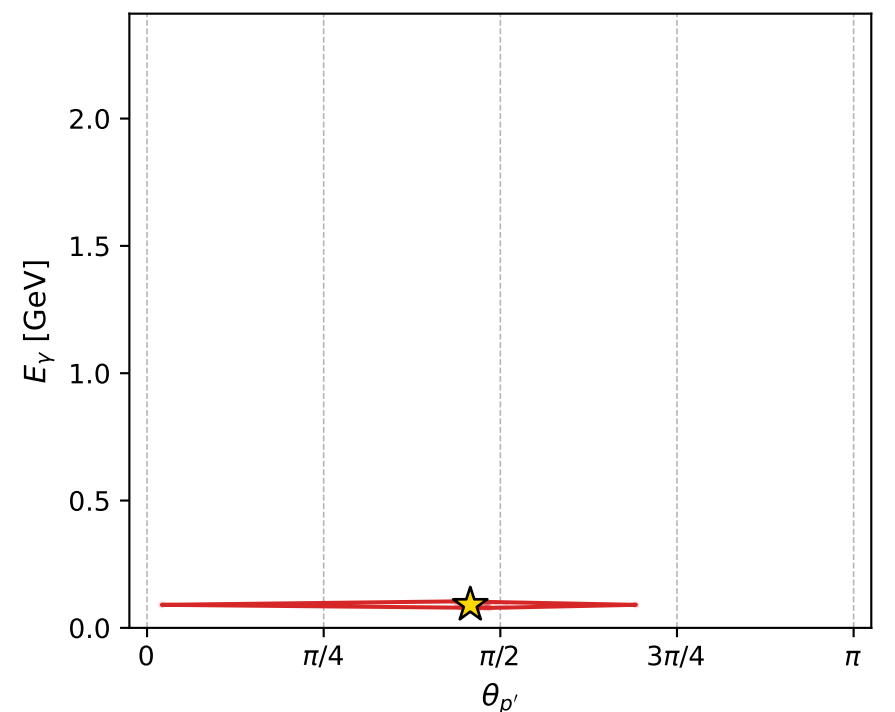
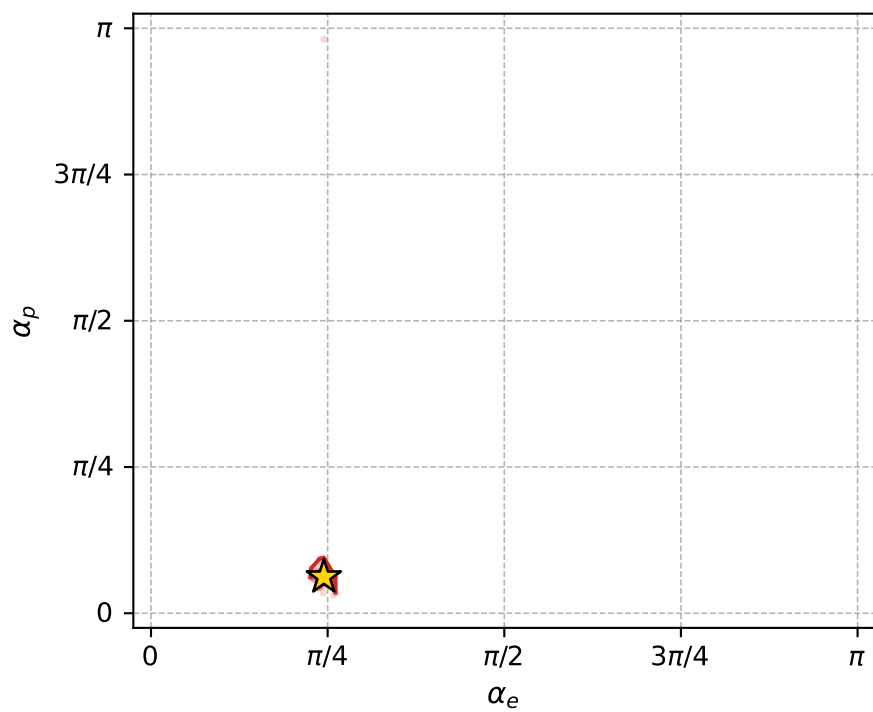
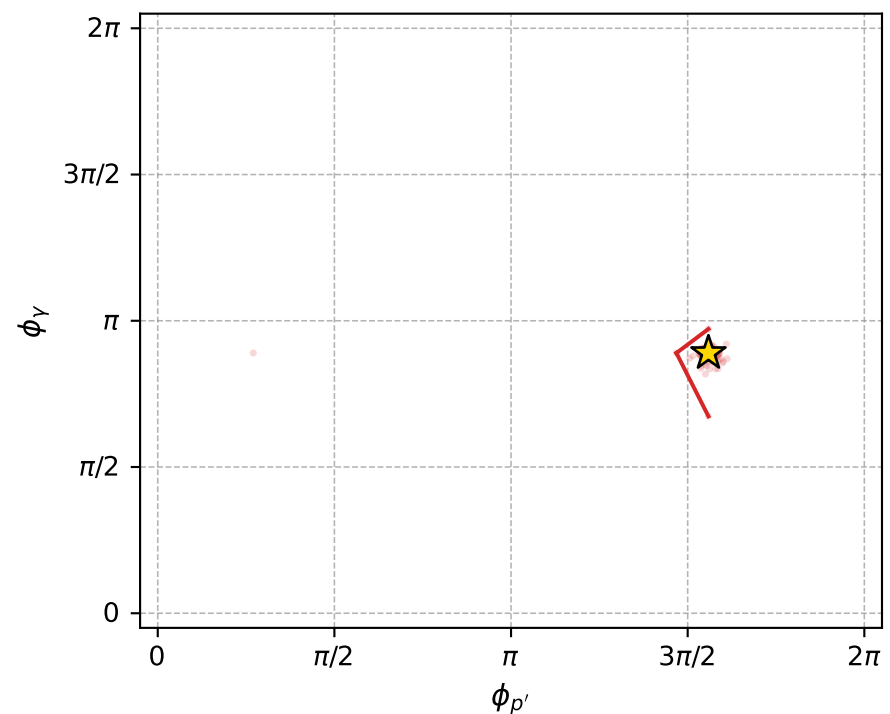
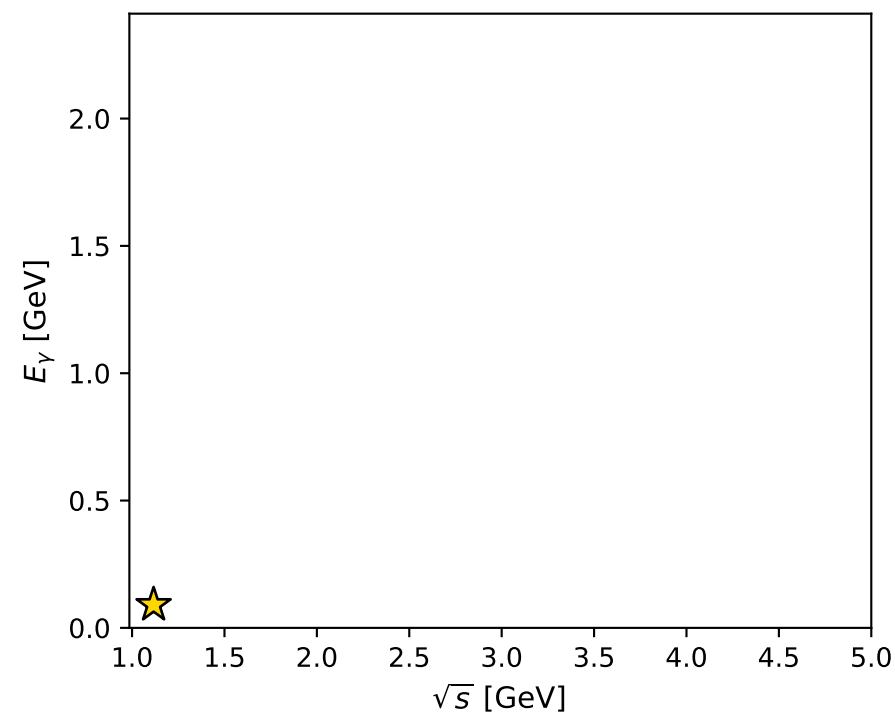
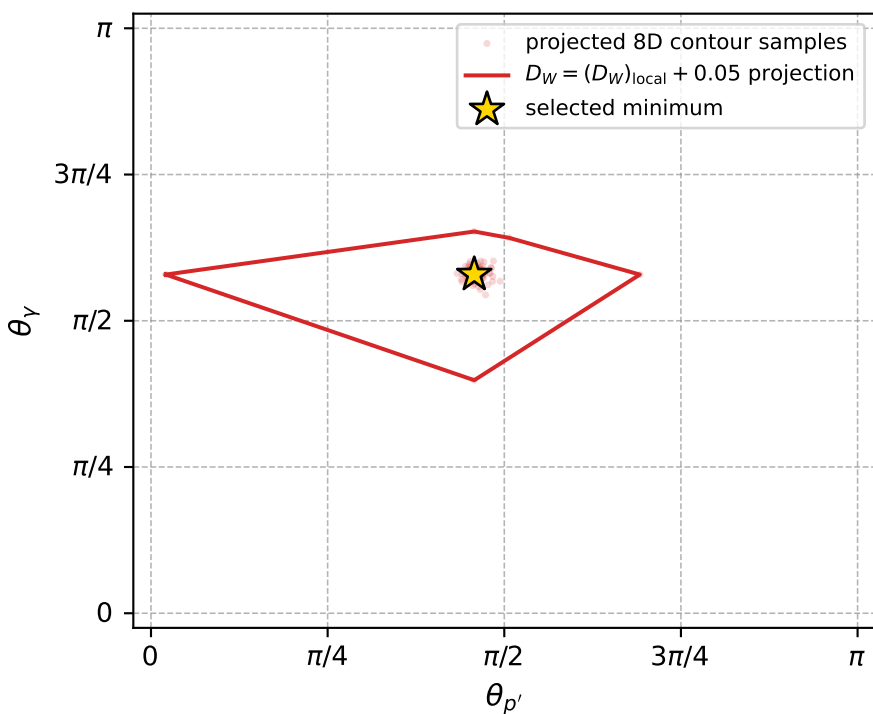
dw_mixing_angles_polarization_03_local_minimum_635 D_W=0.0050142
 region: polarization_cluster_03_local_minimum_635
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76858809$ rad
 incoming lepton: $0.718893|+\rangle +0.695121|-\rangle$
 proton mixing angle: $\alpha_p=0.19478437$ rad
 incoming proton: $0.981089|+\rangle +0.193555|-\rangle$

$s=1.24674$, $\sqrt{s}=1.11658$, $|\vec{P}|=0.164296$, $|\vec{P}'|=0.00487216$
 $\theta_{p'}=1.43722$, $\theta_Y=1.81898$, $\phi_{p'}=4.89688$, $\phi_Y=2.79505$
 $E_Y=0.090497$, $\theta_{\ell'}=1.32322$, $\phi_{\ell'}=5.98543$
 $Q^2=0.036029$, $x_B=0.174676$, $t=-0.0266002$, $W^2=1.05008$, $y=0.562175$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1643$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1643$
 P $E=0.9523$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1643$
 ℓ' $E=0.0881$ $p_x=0.0816$ $p_y=-0.0250$ $p_z=0.0216$
 P' $E=0.9380$ $p_x=0.0009$ $p_y=-0.0047$ $p_z=0.0006$
 q_Y $E=0.0905$ $p_x=-0.0825$ $p_y=0.0298$ $p_z=-0.0222$



electron: polarization cluster P3, local minimum 635; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0050141986$
 $D_W \text{ contour} = 0.055014199$
 above global minimum = 0.0049977657
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.116577$
 $\text{theta_p_out} = 1.437217$
 $\text{theta_gamma_out} = 1.818975$
 $\text{qOut} = 0.09049704$
 $\text{phi_p_out} = 4.89688$
 $\text{phi_gamma_out} = 2.79505$
 $\text{alpha_e} = 0.7685881$
 $\text{alpha_p} = 0.1947844$

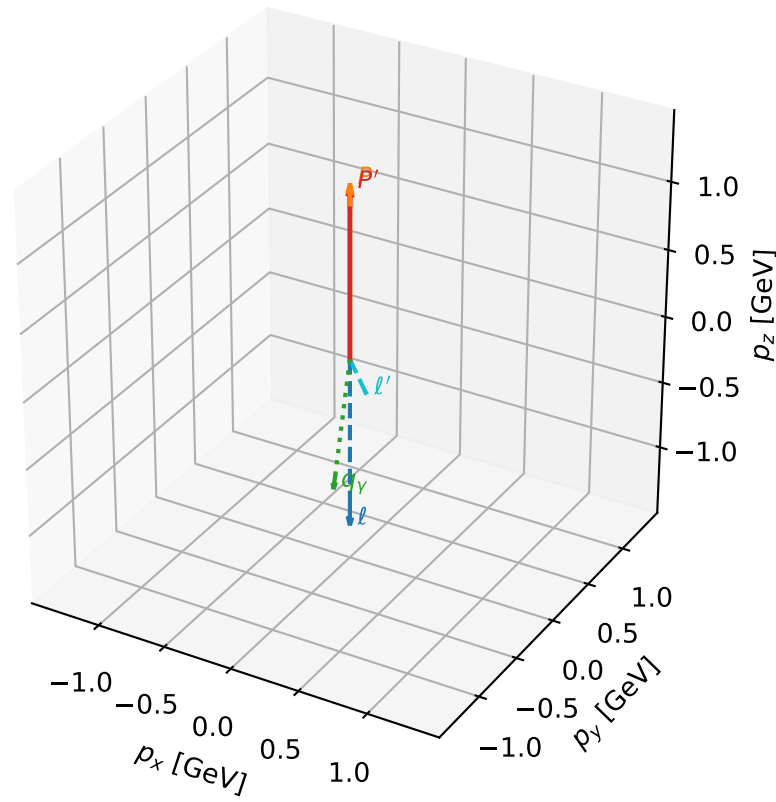
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_644 D_W=0.00536814
 region: polarization_cluster_03_local_minimum_644
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79589385$ rad
 incoming lepton: $0.699646|+\rangle +0.714489|-\rangle$
 proton mixing angle: $\alpha_p=0.11066879$ rad
 incoming proton: $0.993882|+\rangle +0.110443|-\rangle$

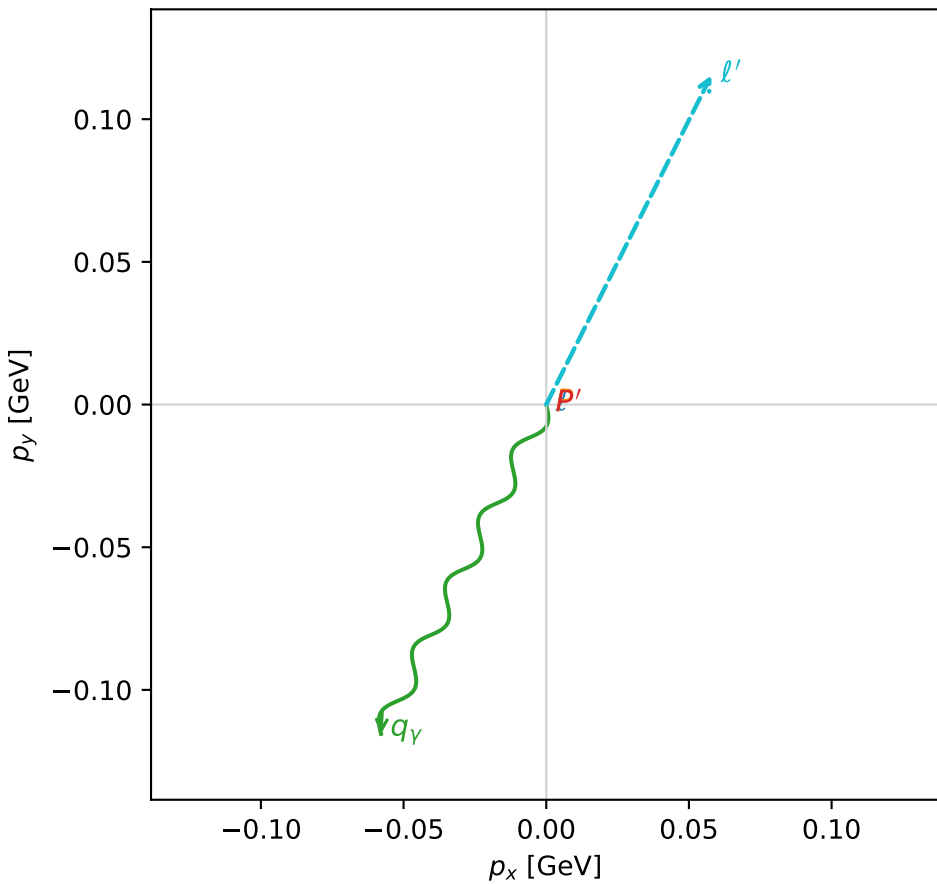
$s=8.07157$, $\sqrt{s}=2.84105$, $|\vec{P}|=1.26568$, $|\vec{P}'|=1.24622$
 $\theta_{P'}=0$, $\theta_V=3.00532$, $\phi_{P'}=4.67144$, $\phi_V=4.24742$
 $E_V=0.95051$, $\theta_{\ell'}=2.74054$, $\phi_{\ell'}=1.10583$
 $Q^2=0.066438$, $x_B=0.012352$, $t=-0.000135646$, $W^2=6.19213$, $y=0.747905$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.2657$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.2657$
 P $E= 1.5754$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.2657$
 ℓ' $E= 0.3308$ $p_x= 0.0579$ $p_y= 0.1154$ $p_z= -0.3045$
 P' $E= 1.5598$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 1.2462$
 q_V $E= 0.9505$ $p_x= -0.0579$ $p_y= -0.1154$ $p_z= -0.9417$

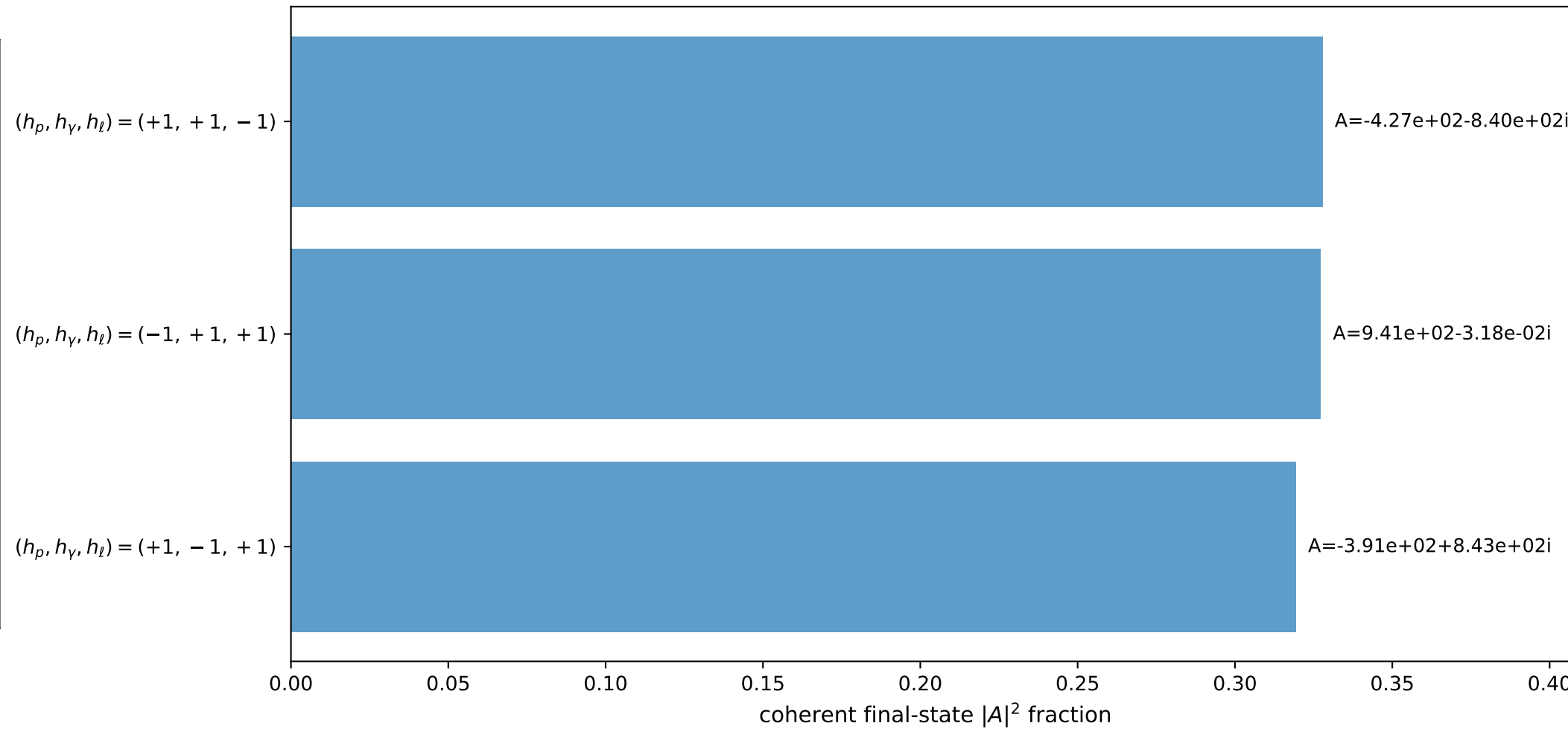
3D momenta



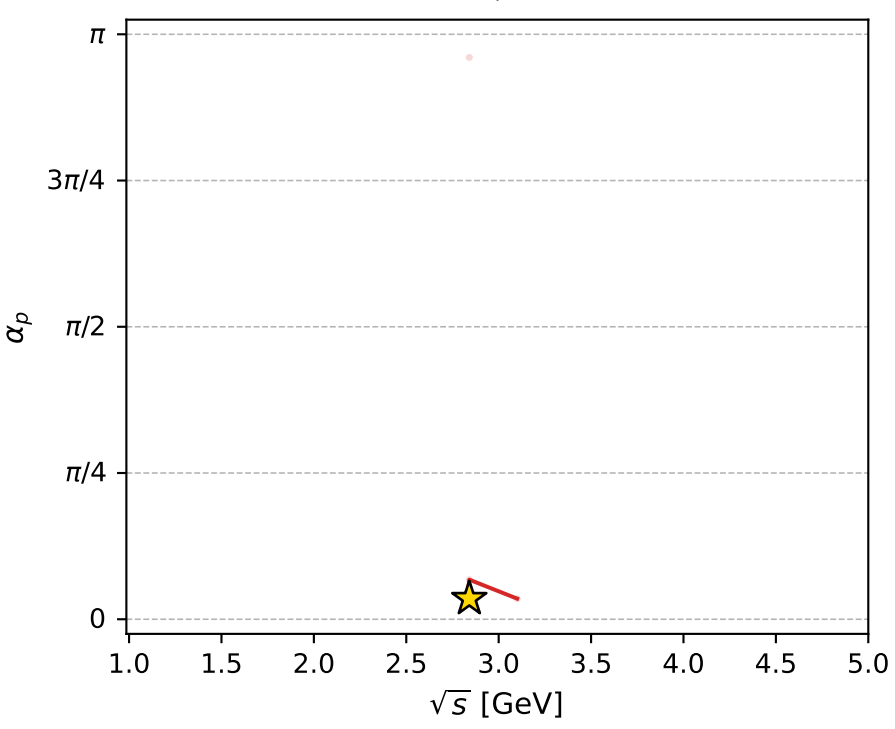
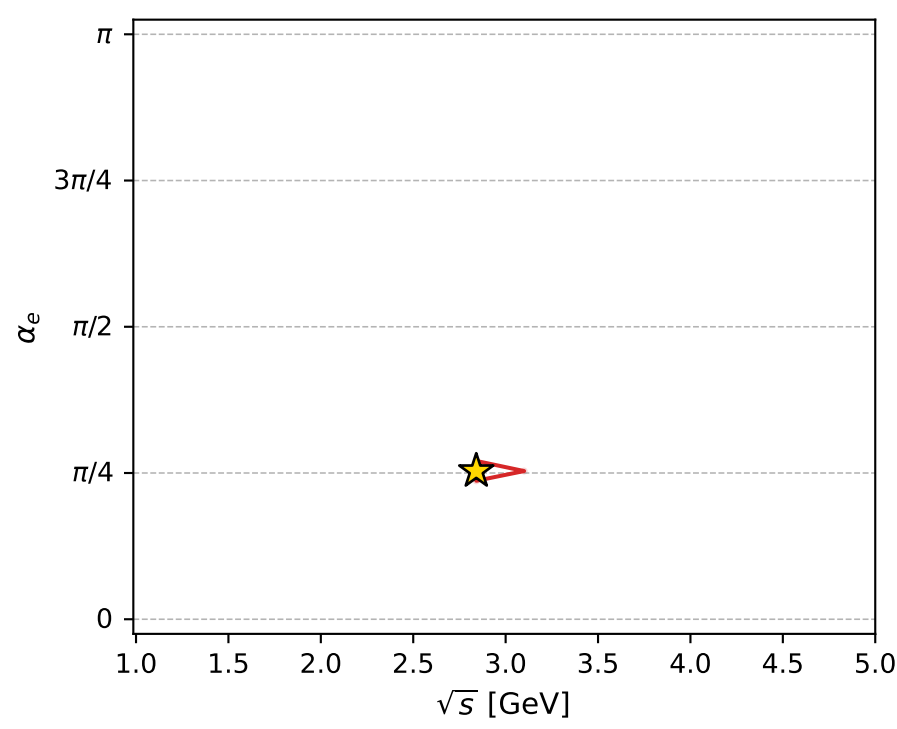
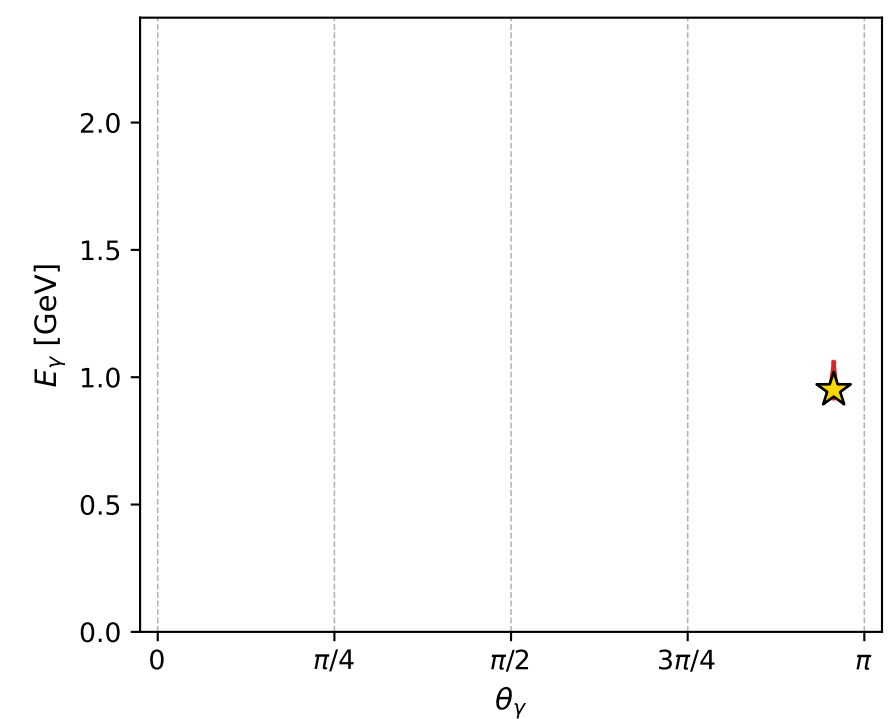
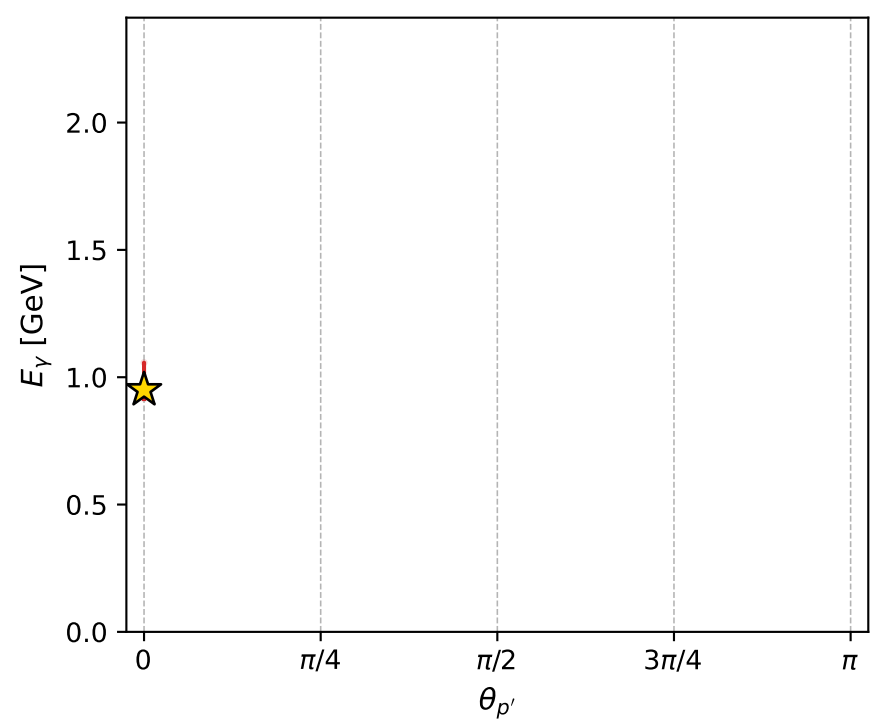
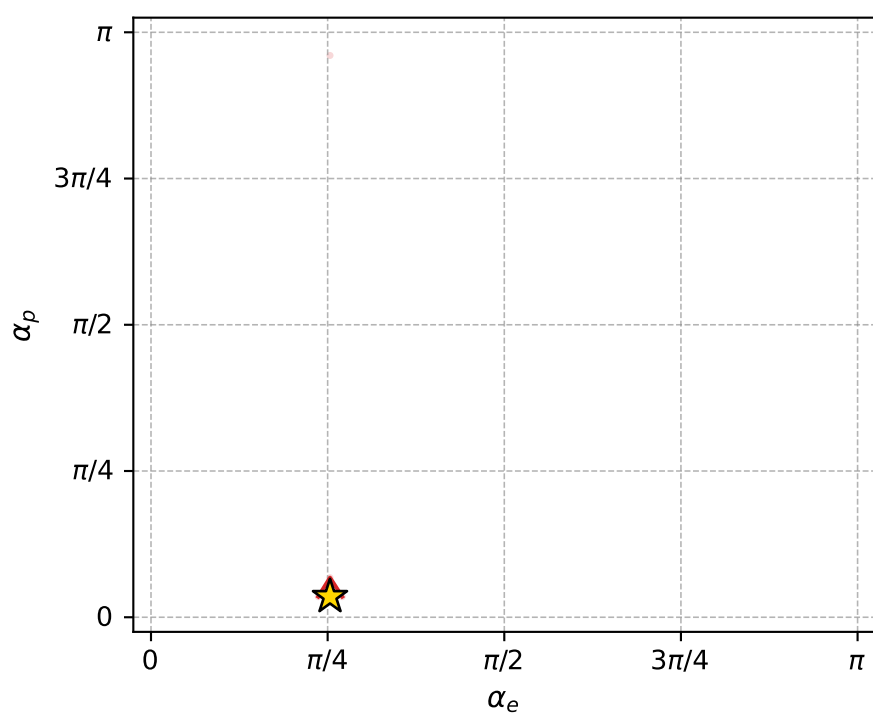
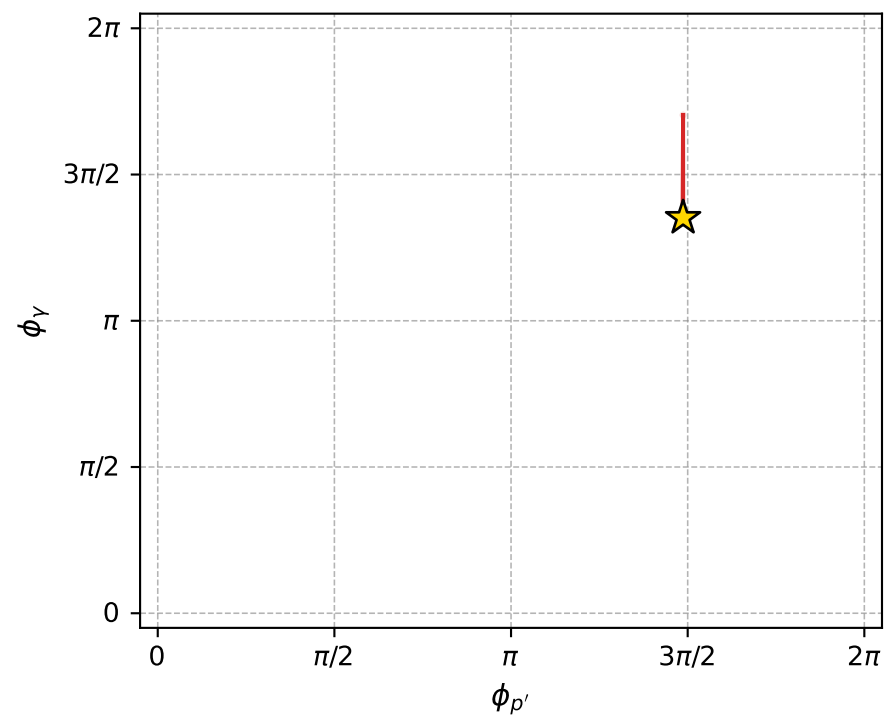
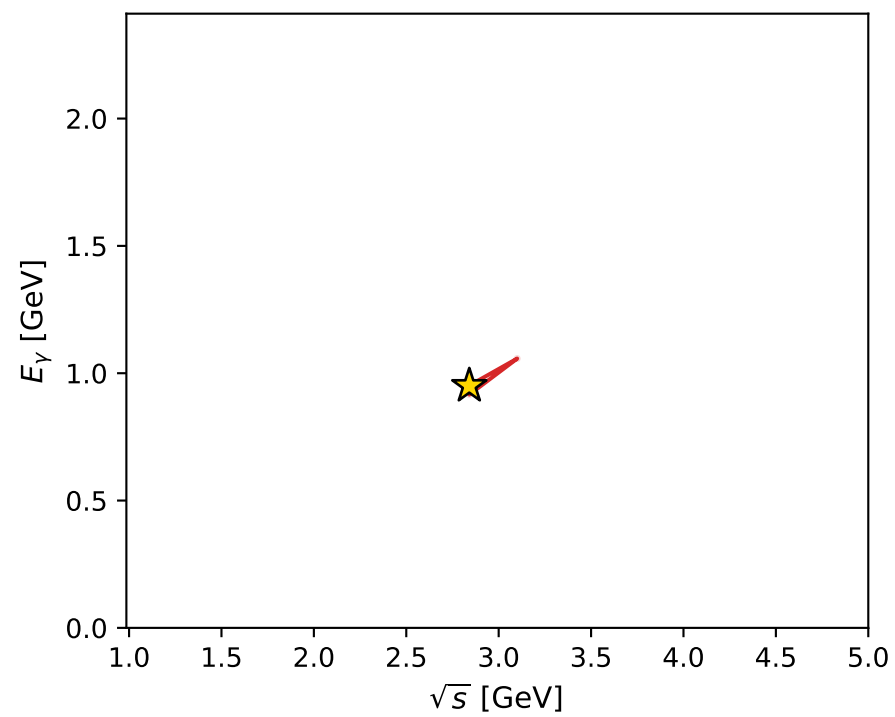
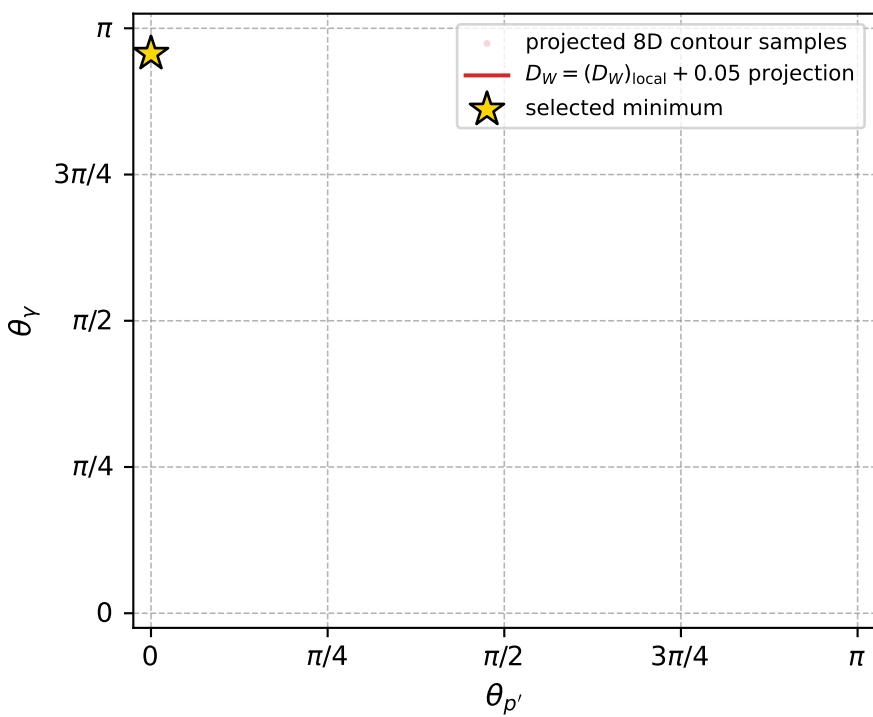
Transverse plane



Leading final-state amplitudes (N=3, retained=97.5%)



electron: polarization cluster P3, local minimum 644; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.005368136$
 $D_W \text{ contour} = 0.055368136$
 above global minimum = 0.0053517031
 8D contour samples = 125

selected phase-space point:
 $\text{sqrt}_s = 2.841051$
 $\text{theta}_p_{\text{out}} = 0$
 $\text{theta}_\gamma_{\text{out}} = 3.005322$
 $q_{\text{out}} = 0.9505103$
 $\text{phi}_p_{\text{out}} = 4.671441$
 $\text{phi}_\gamma_{\text{out}} = 4.247421$
 $\alpha_e = 0.7958939$
 $\alpha_p = 0.1106688$

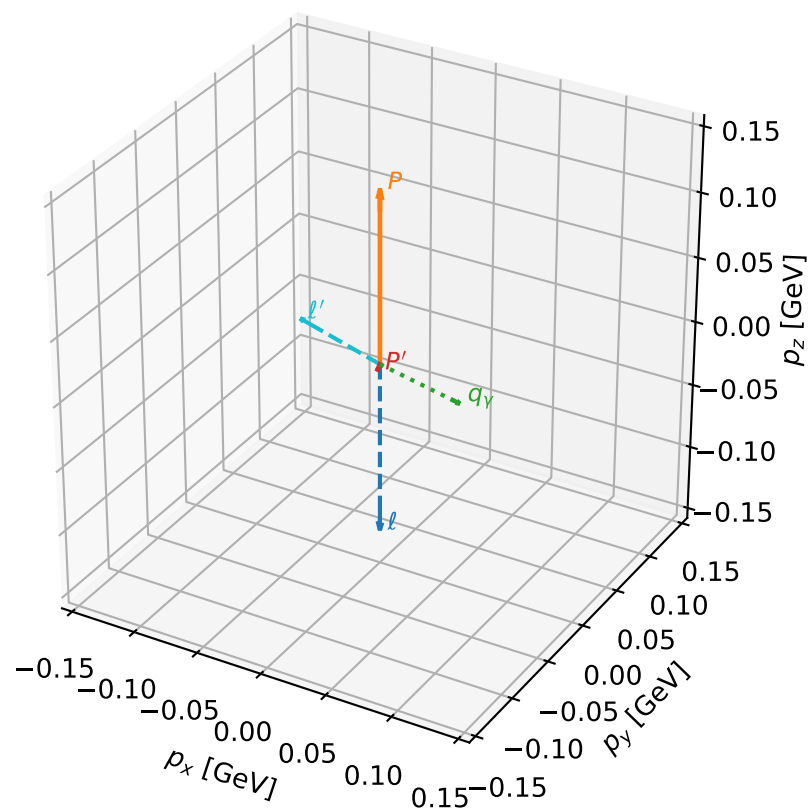
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_646 D_W=0.0053915
 region: polarization_cluster_03_local_minimum_646
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77145386$ rad
 incoming lepton: $0.716898|+\rangle +0.697178|-\rangle$
 proton mixing angle: $\alpha_p=2.9752731$ rad
 incoming proton: $-0.986201|+\rangle +0.165554|-\rangle$

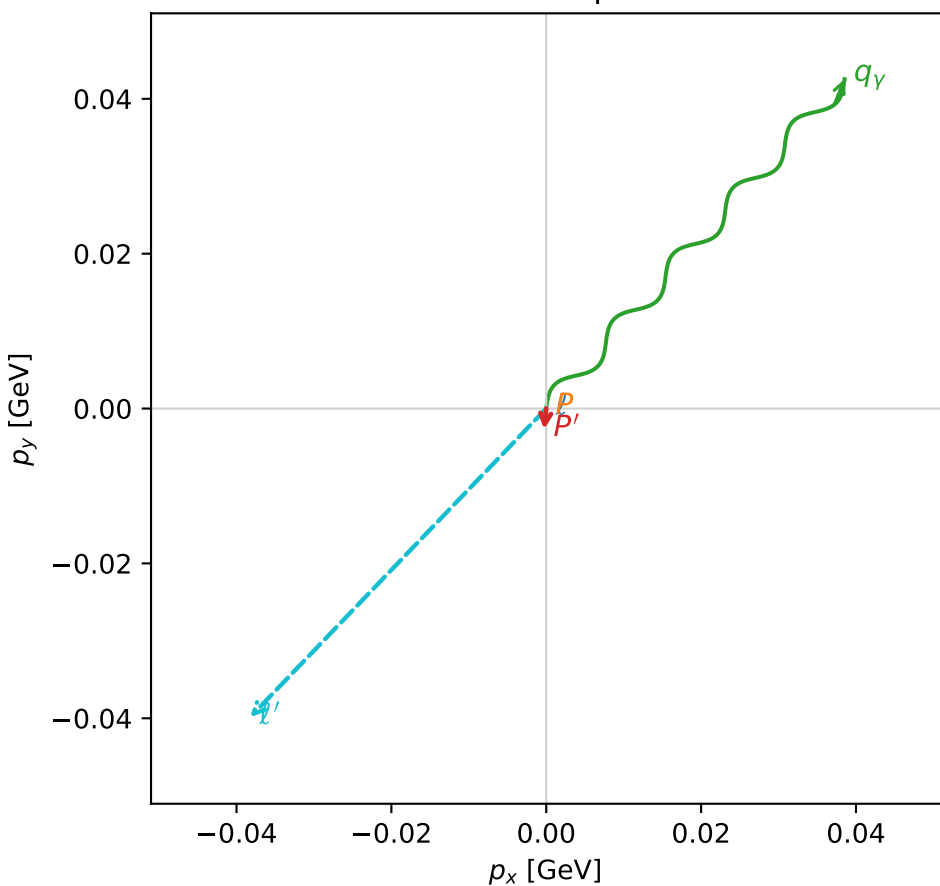
$s=1.16501$, $\sqrt{s}=1.07935$, $|\vec{P}|=0.132098$, $|\vec{P}'|=0.00389451$
 $\theta_{P'}=2.37993$, $\theta_\gamma=2.19414$, $\phi_{P'}=4.6097$, $\phi_\gamma=0.83485$
 $E_\gamma=0.0706756$, $\theta_{\ell'}=0.89734$, $\phi_{\ell'}=3.94765$
 $Q^2=0.0303152$, $x_B=0.186074$, $t=-0.0181241$, $W^2=1.01245$, $y=0.571323$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1321$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1321$
 P $E= 0.9473$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1321$
 ℓ' $E= 0.0707$ $p_x= -0.0382$ $p_y= -0.0399$ $p_z= 0.0441$
 P' $E= 0.9380$ $p_x= -0.0003$ $p_y= -0.0027$ $p_z= -0.0028$
 q_γ $E= 0.0707$ $p_x= 0.0385$ $p_y= 0.0425$ $p_z= -0.0413$

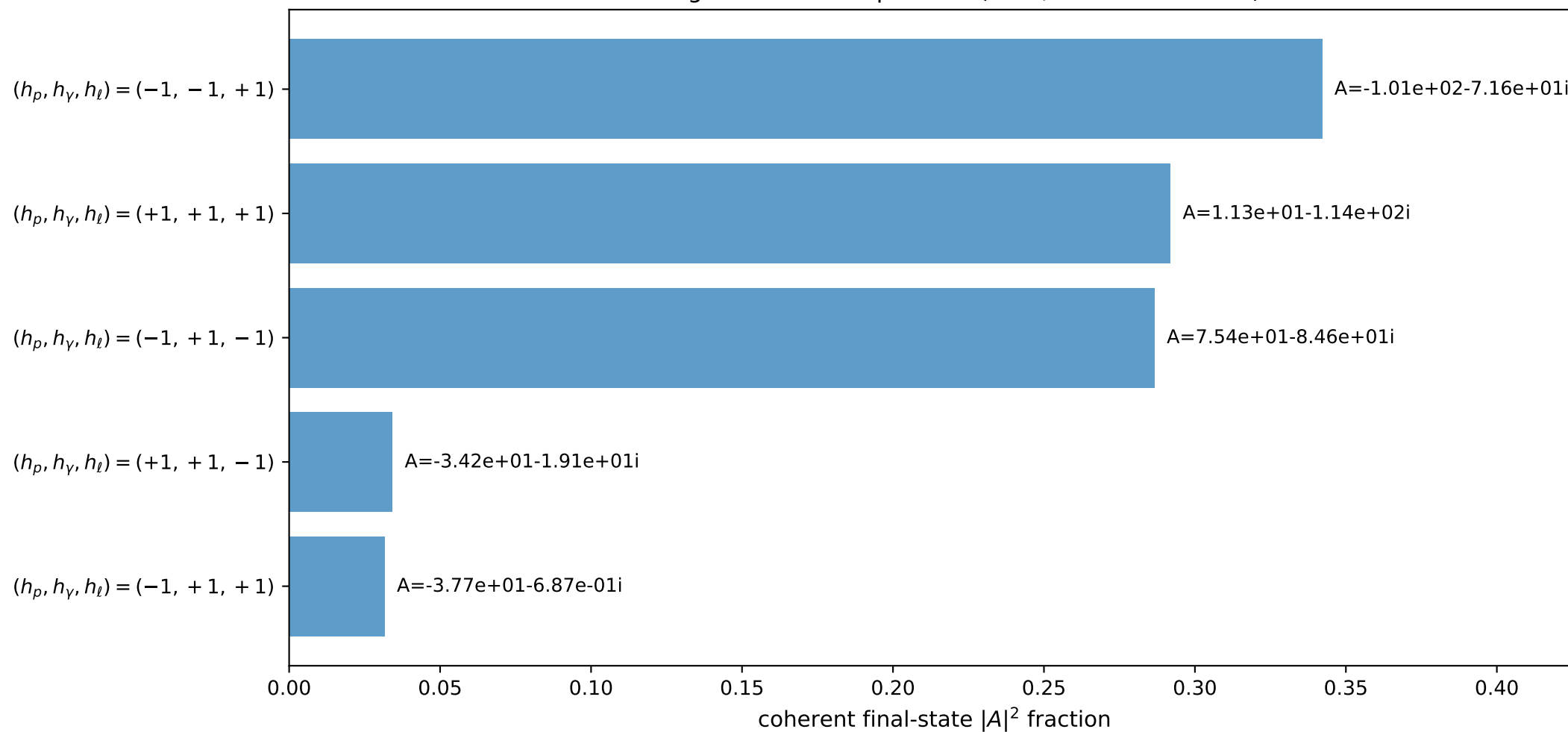
3D momenta



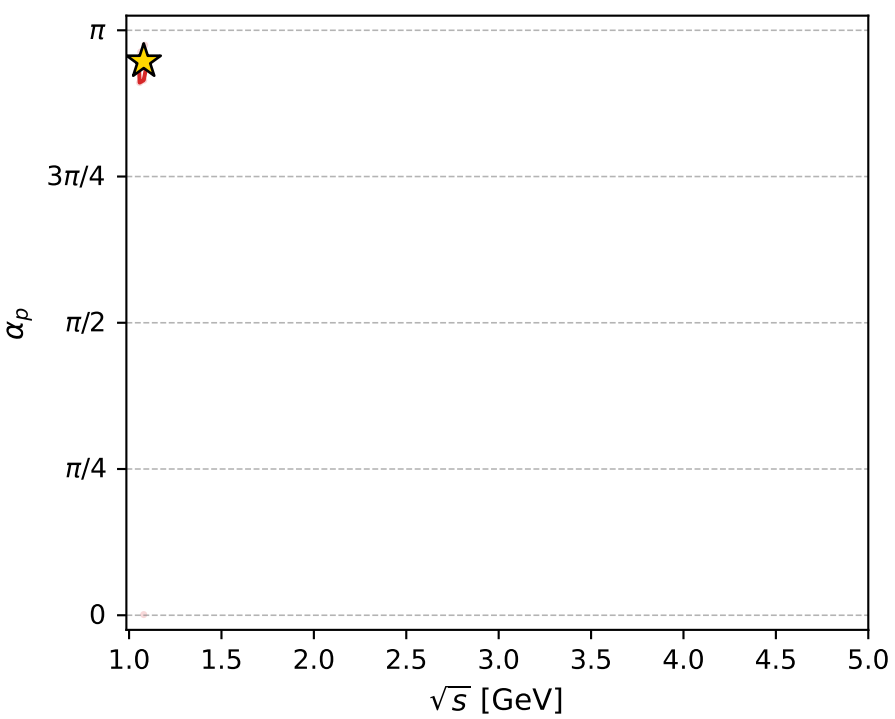
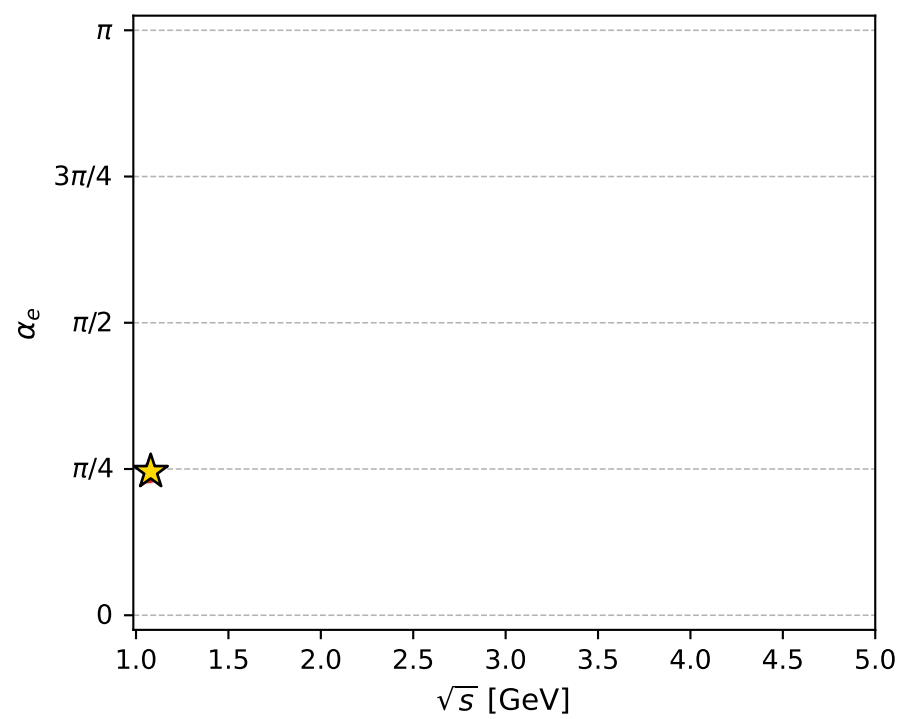
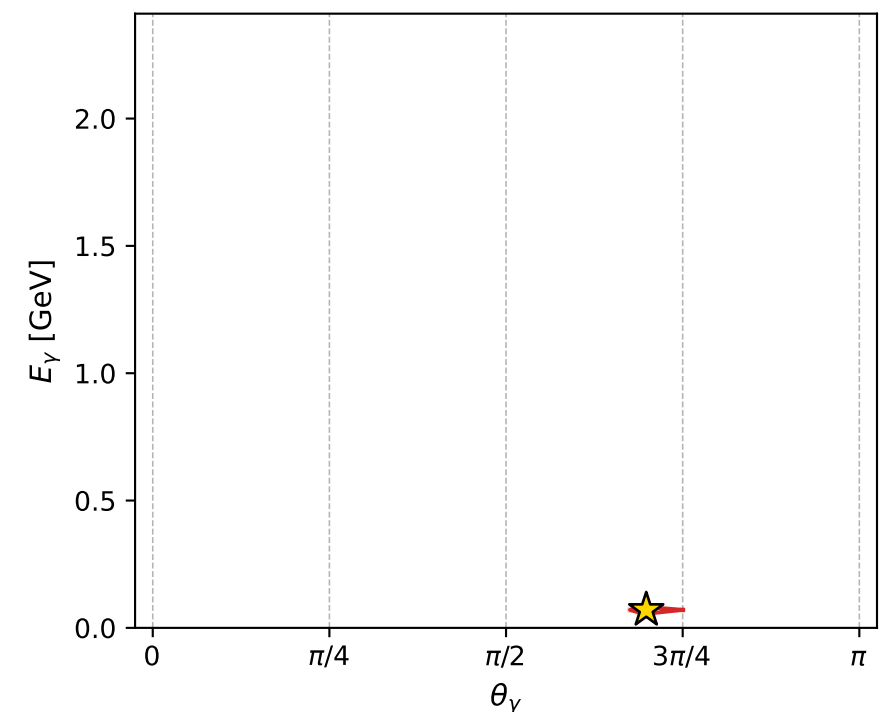
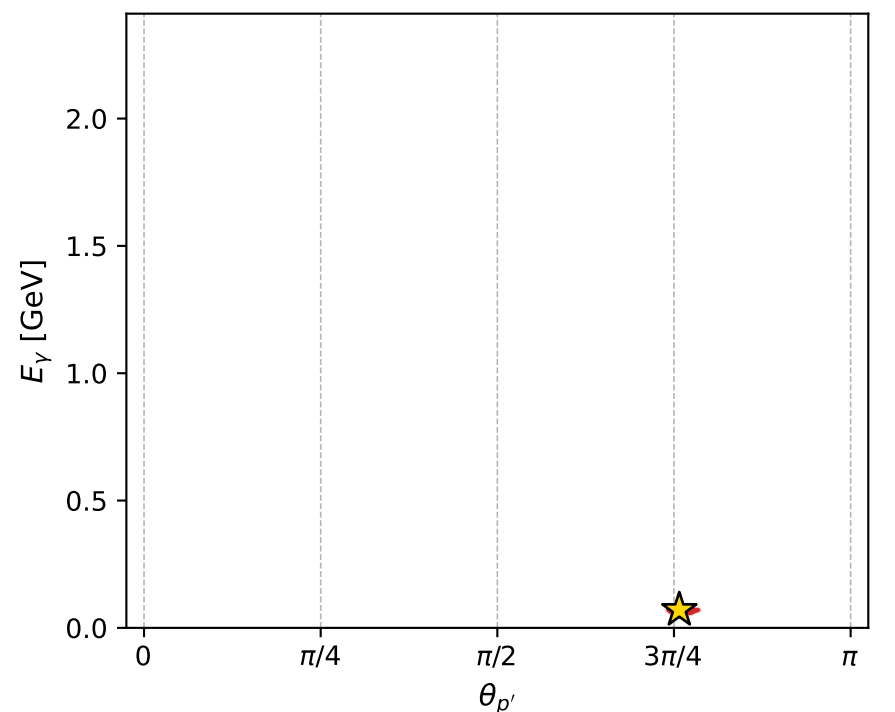
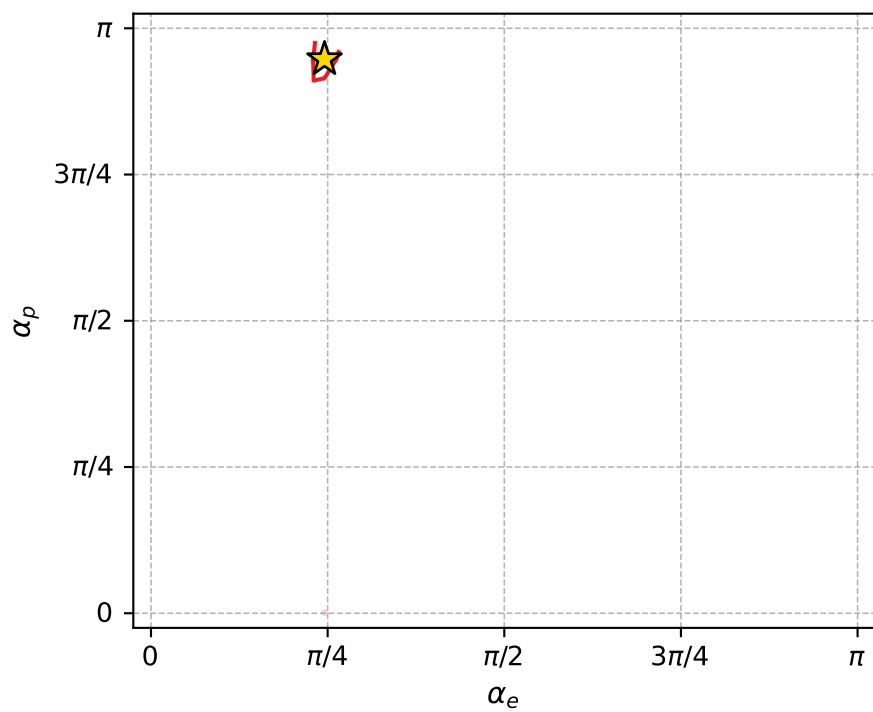
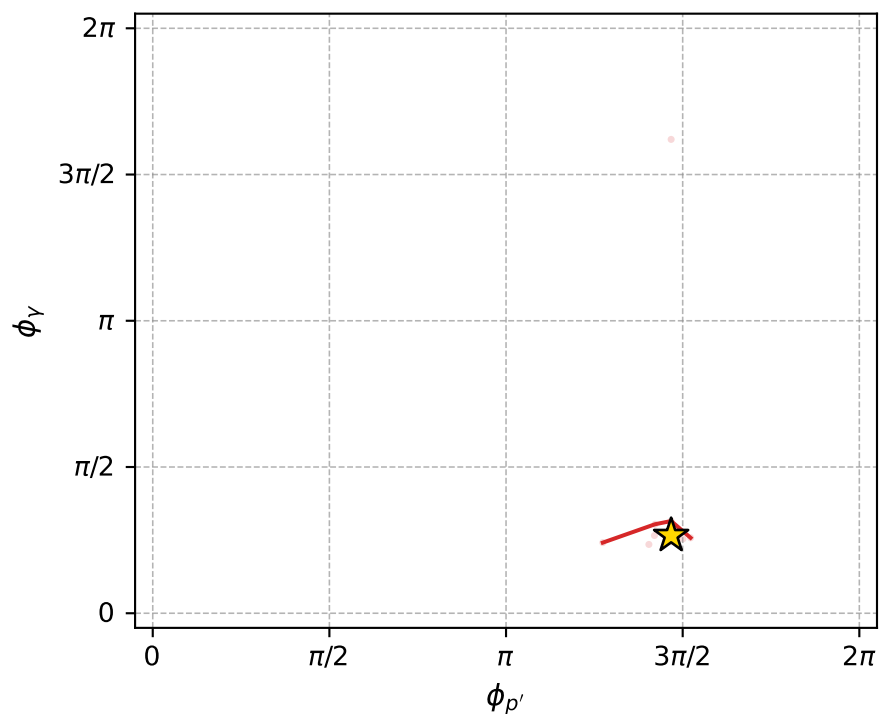
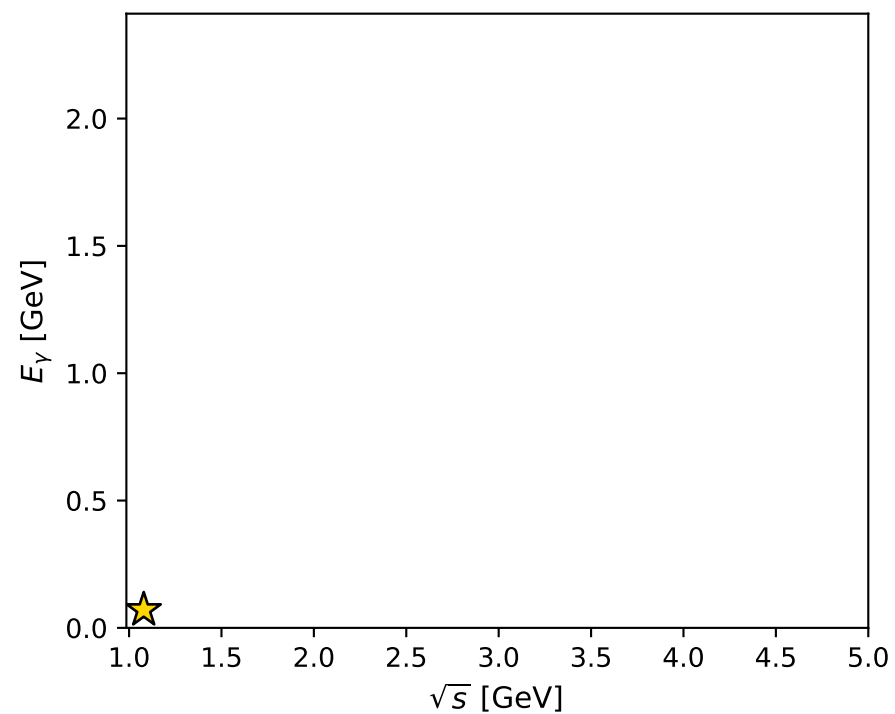
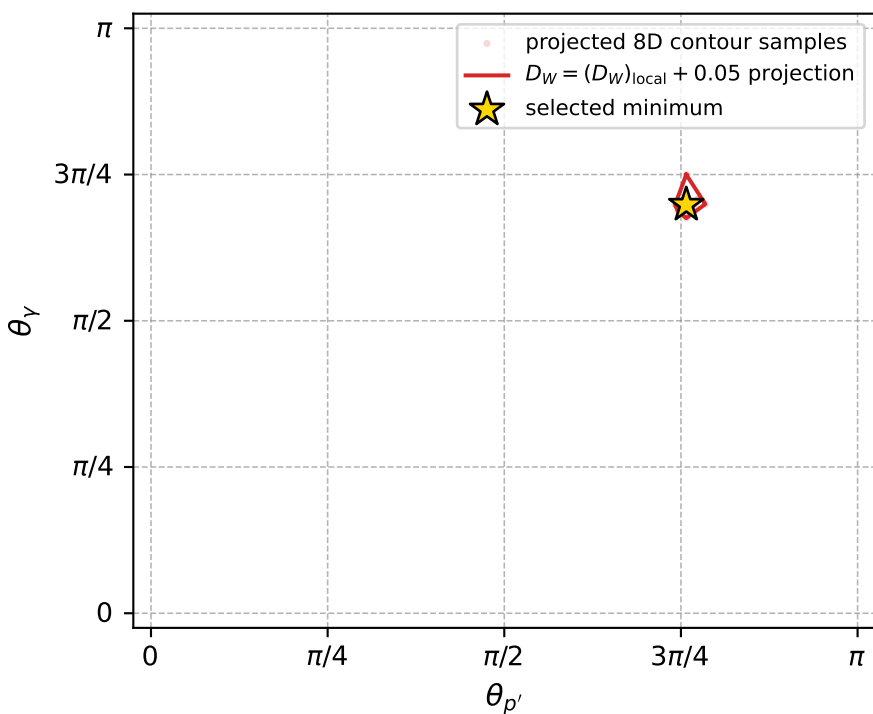
Transverse plane



Leading final-state amplitudes (N=5, retained=98.7%)



electron: polarization cluster P3, local minimum 646; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0053915039$
 $D_W \text{ contour} = 0.055391504$
 above global minimum = 0.005375071
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.079355$
 $\text{theta_p_out} = 2.379933$
 $\text{theta_gamma_out} = 2.194142$
 $\text{qOut} = 0.07067562$
 $\text{phi_p_out} = 4.609697$
 $\text{phi_gamma_out} = 0.8348504$
 $\text{alpha_e} = 0.7714539$
 $\text{alpha_p} = 2.975273$

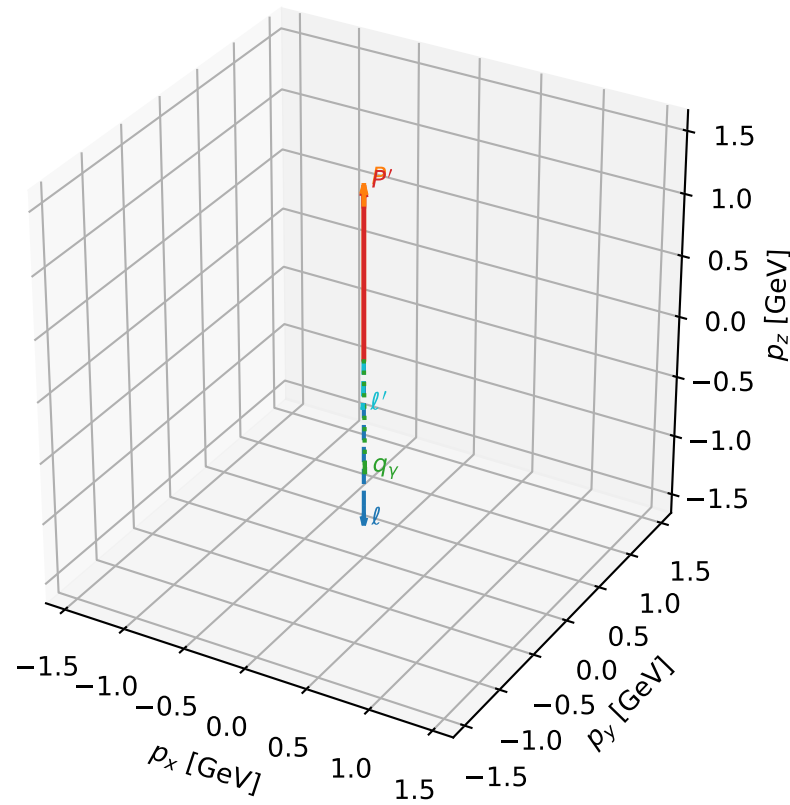
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_648 D_W=0.0054095
region: polarization_cluster_03_local_minimum_648
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79612311$ rad
incoming lepton: $0.699483|+\rangle + 0.71465|-\rangle$
proton mixing angle: $\alpha_p=0.11130992$ rad
incoming proton: $0.993811|+\rangle + 0.11108|-\rangle$

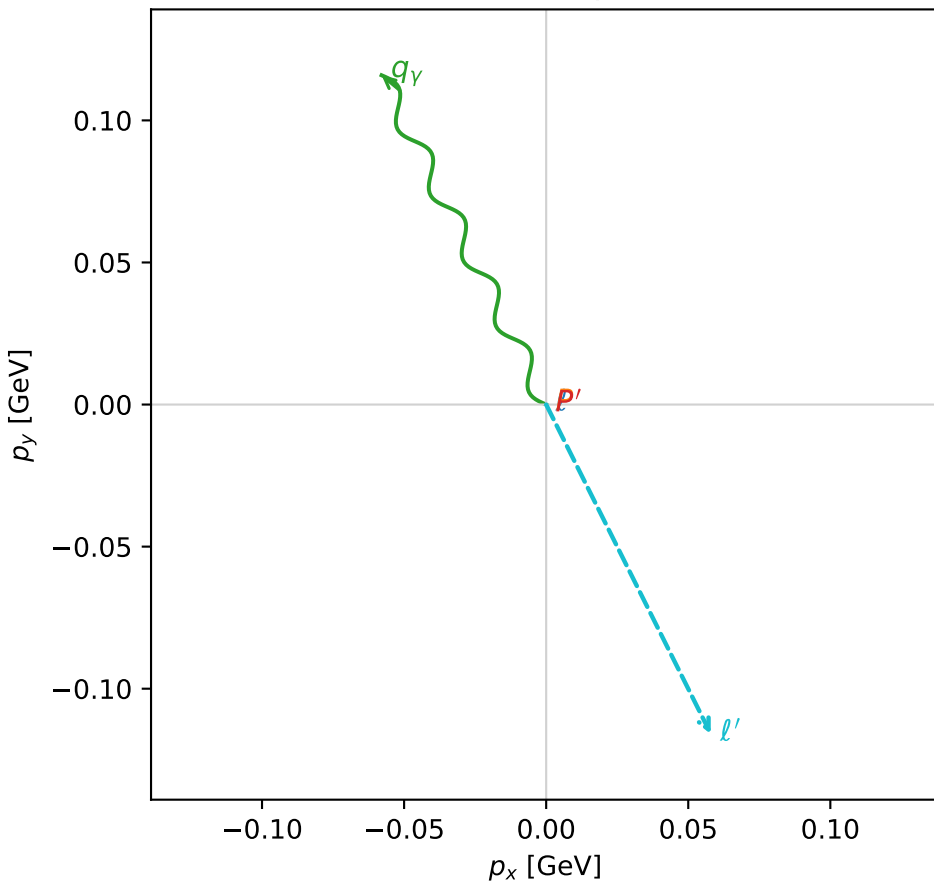
$s=9.37335$, $\sqrt{s}=3.06159$, $|\vec{P}|=1.38711$, $|\vec{P}'|=1.36953$
 $\theta_{P'}=0$, $\theta_V=3.01671$, $\phi_{P'}=0.61743$, $\phi_V=2.03429$
 $E_V=1.04008$, $\theta_{\ell'}=2.77514$, $\phi_{\ell'}=5.17588$
 $Q^2=0.0665991$, $x_B=0.0104943$, $t=-9.77838e-05$, $W^2=7.15947$, $y=0.747186$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 1.3871$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.3871$
 P $E= 1.6745$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3871$
 ℓ' $E= 0.3616$ $p_x= 0.0579$ $p_y= -0.1159$ $p_z= -0.3375$
 P' $E= 1.6600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.3695$
 q_V $E= 1.0401$ $p_x= -0.0579$ $p_y= 0.1159$ $p_z= -1.0320$

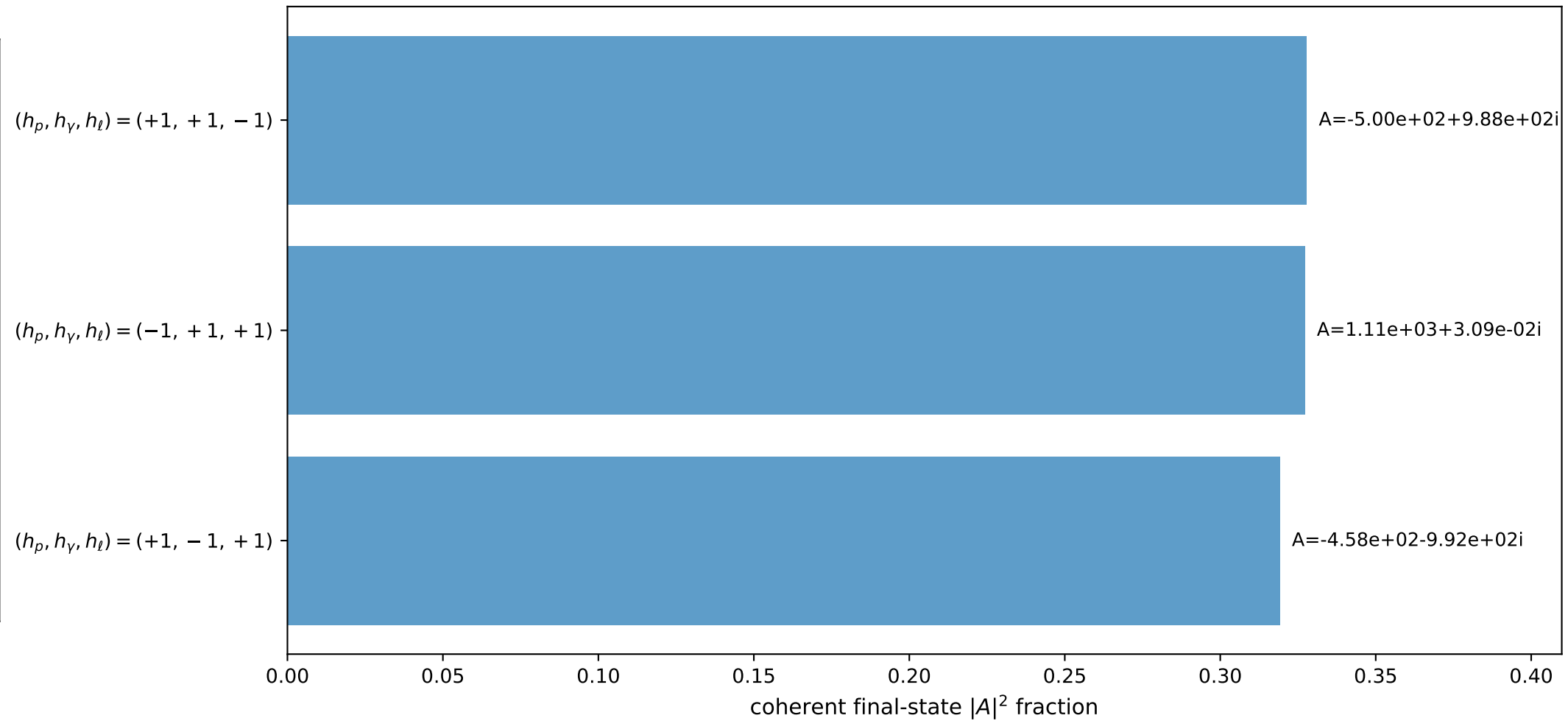
3D momenta



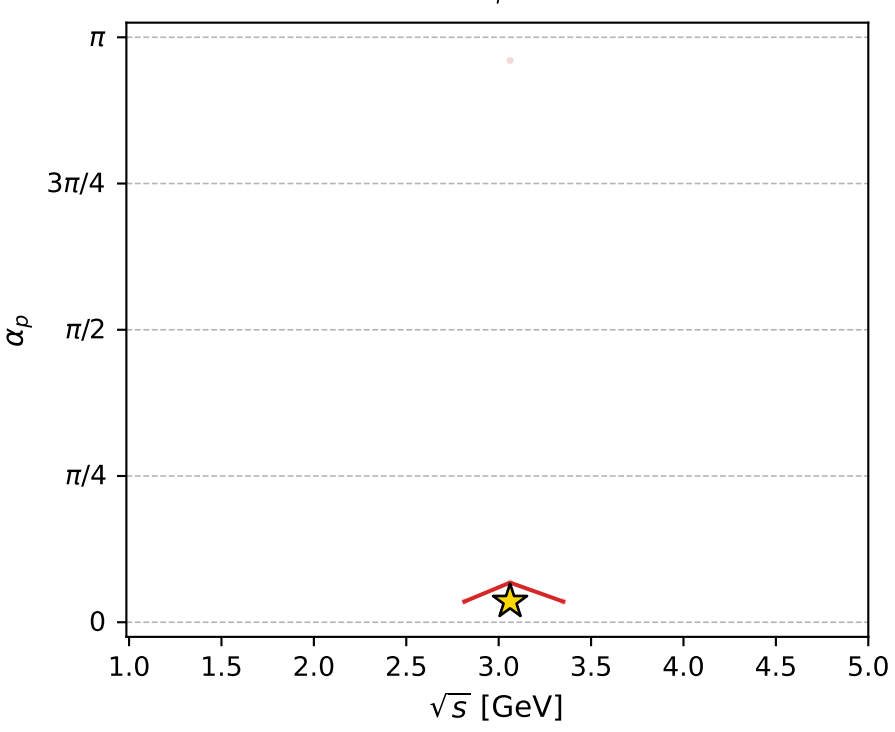
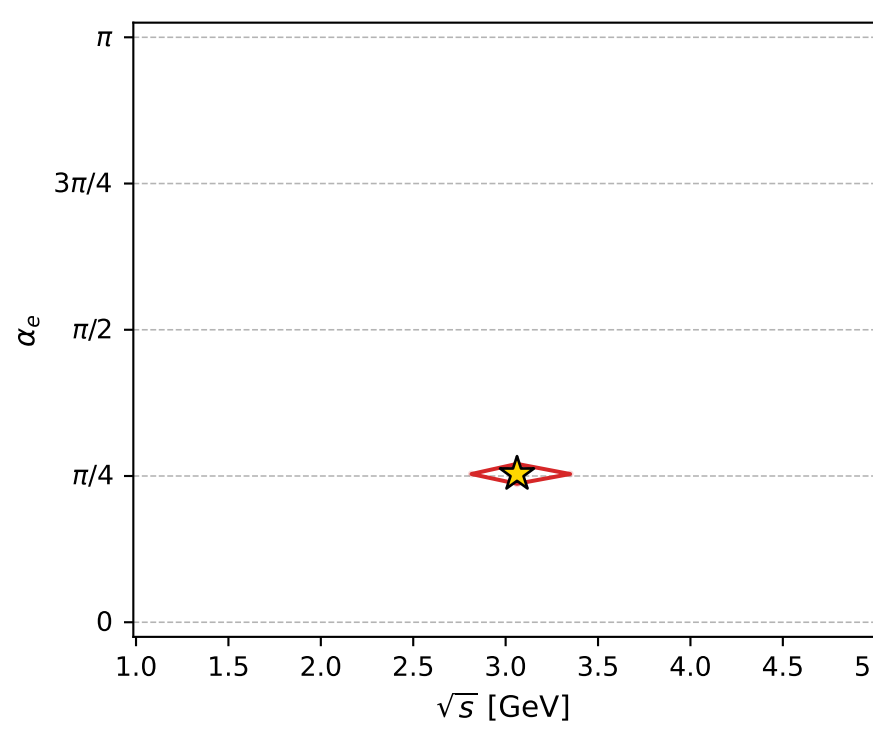
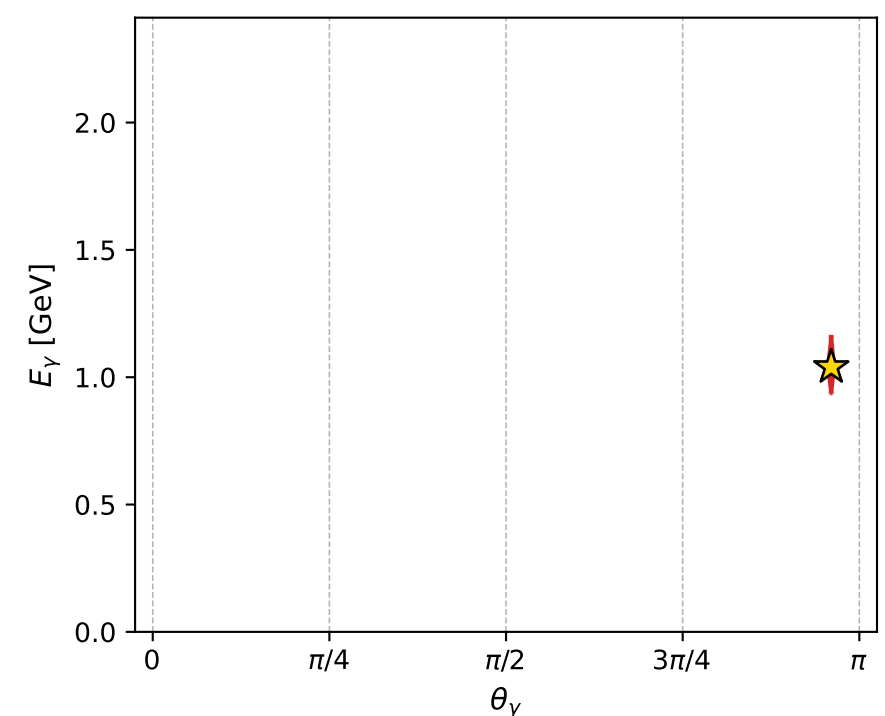
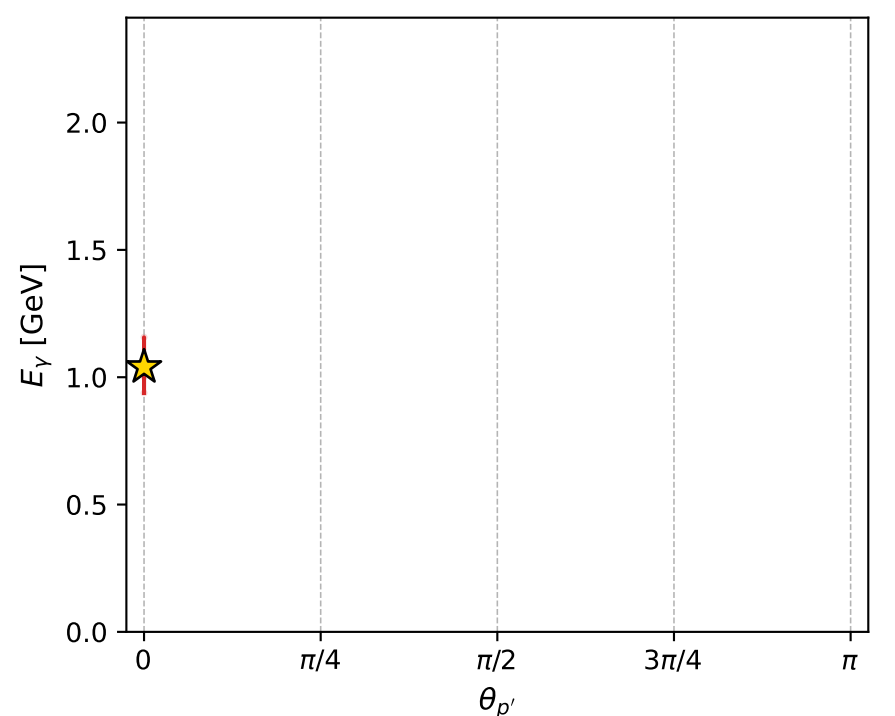
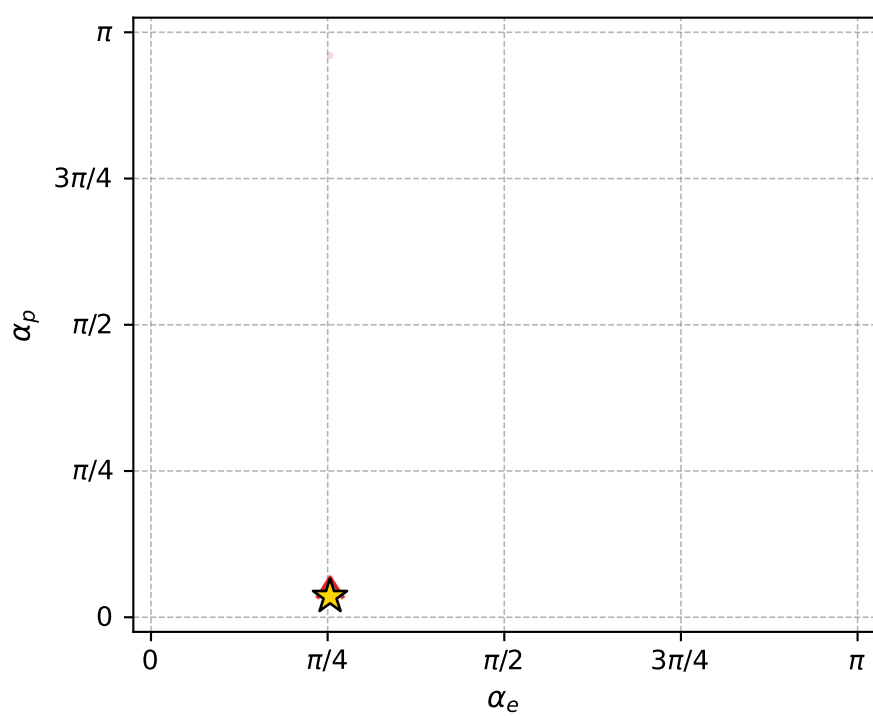
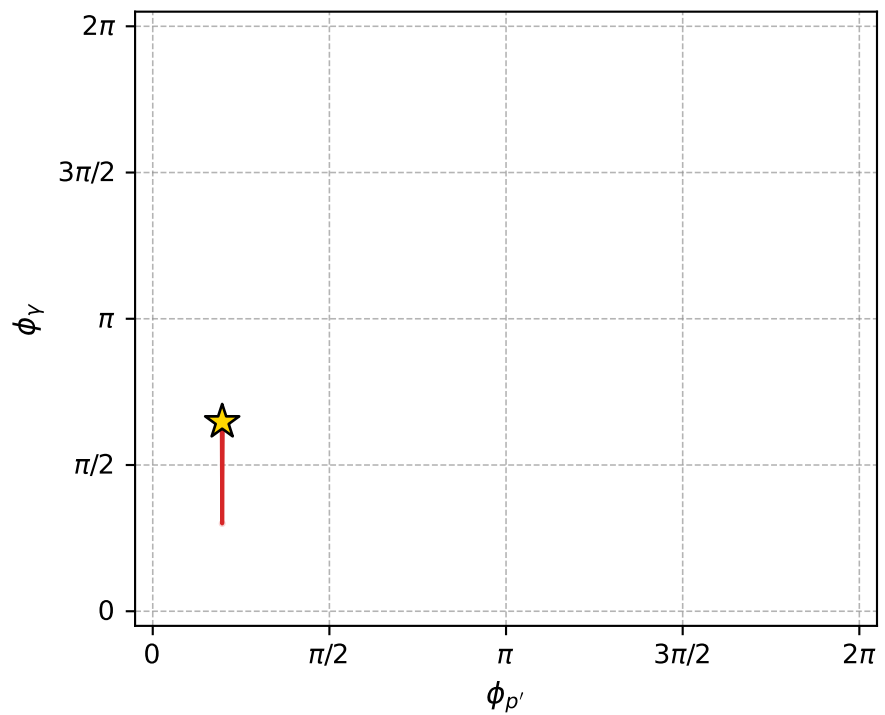
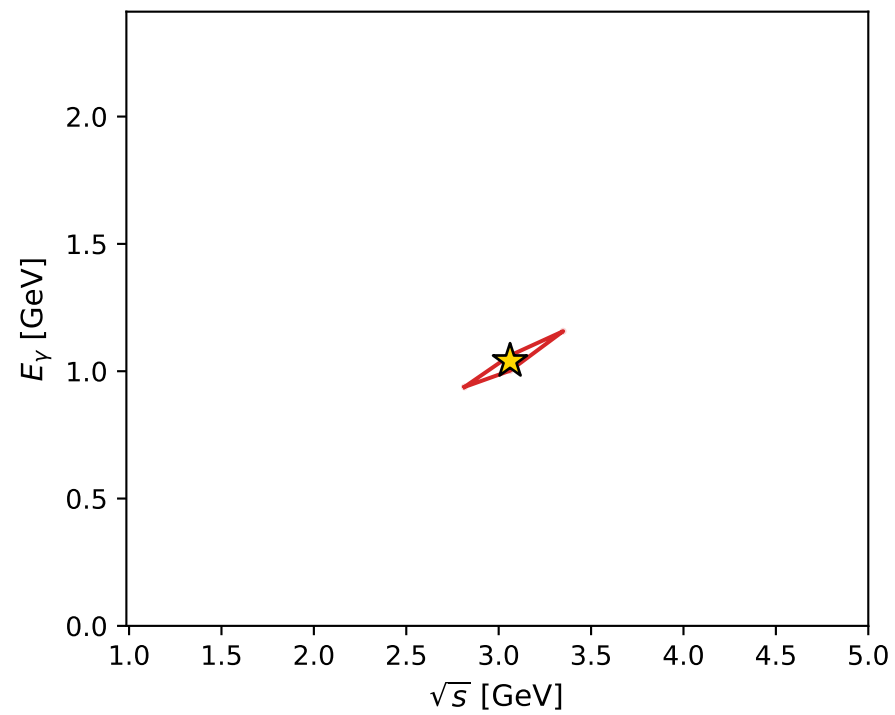
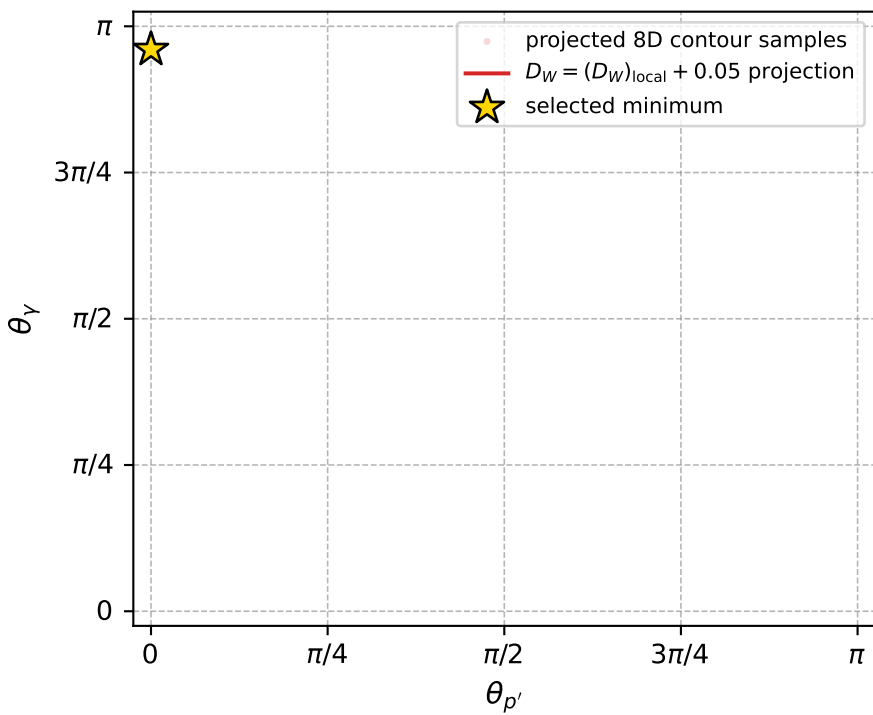
Transverse plane



Leading final-state amplitudes (N=3, retained=97.4%)



electron: polarization cluster P3, local minimum 648; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0054094995$
 $D_W \text{ contour} = 0.0554095$
 above global minimum = 0.0053930667
 8D contour samples = 140

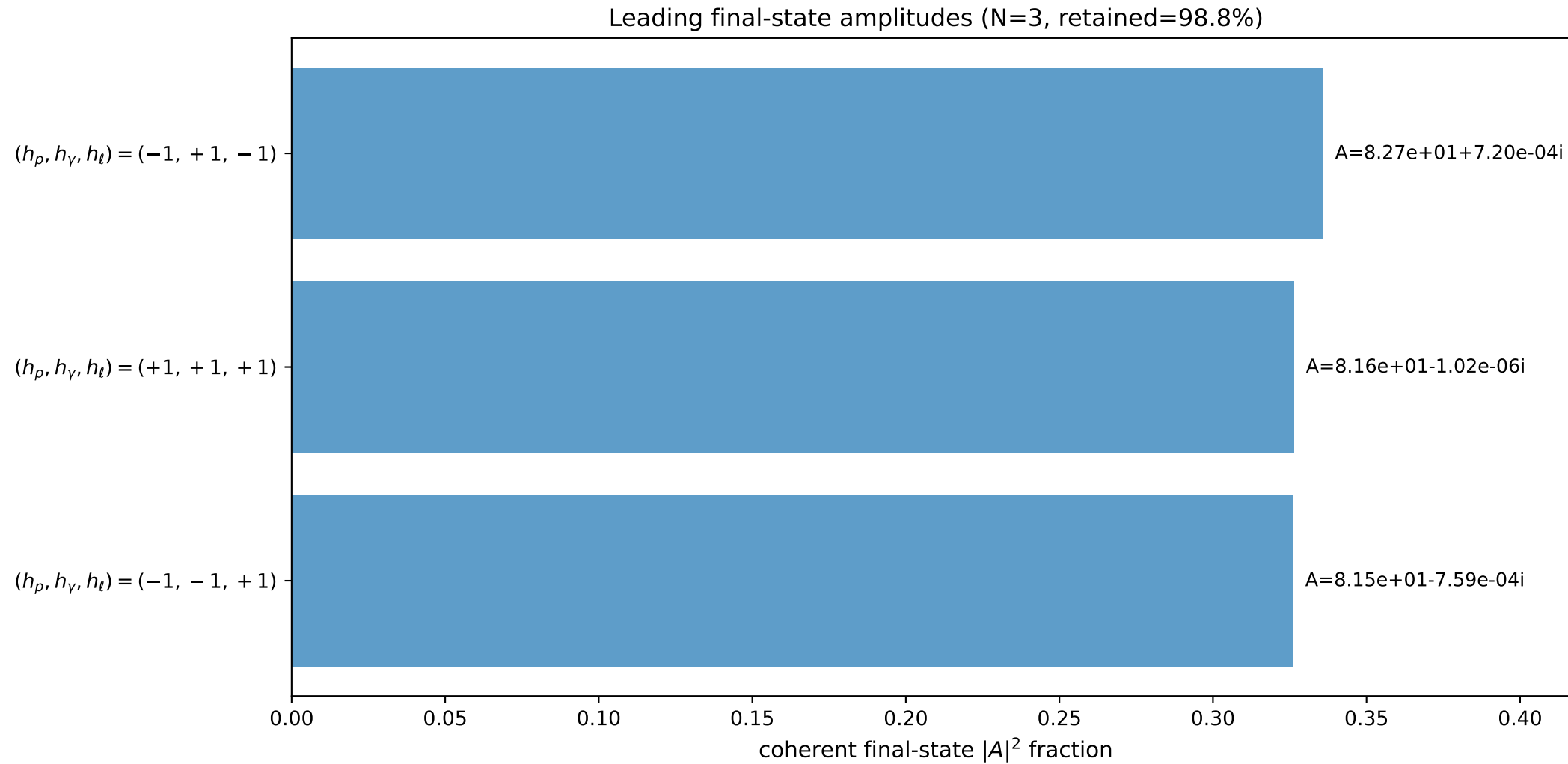
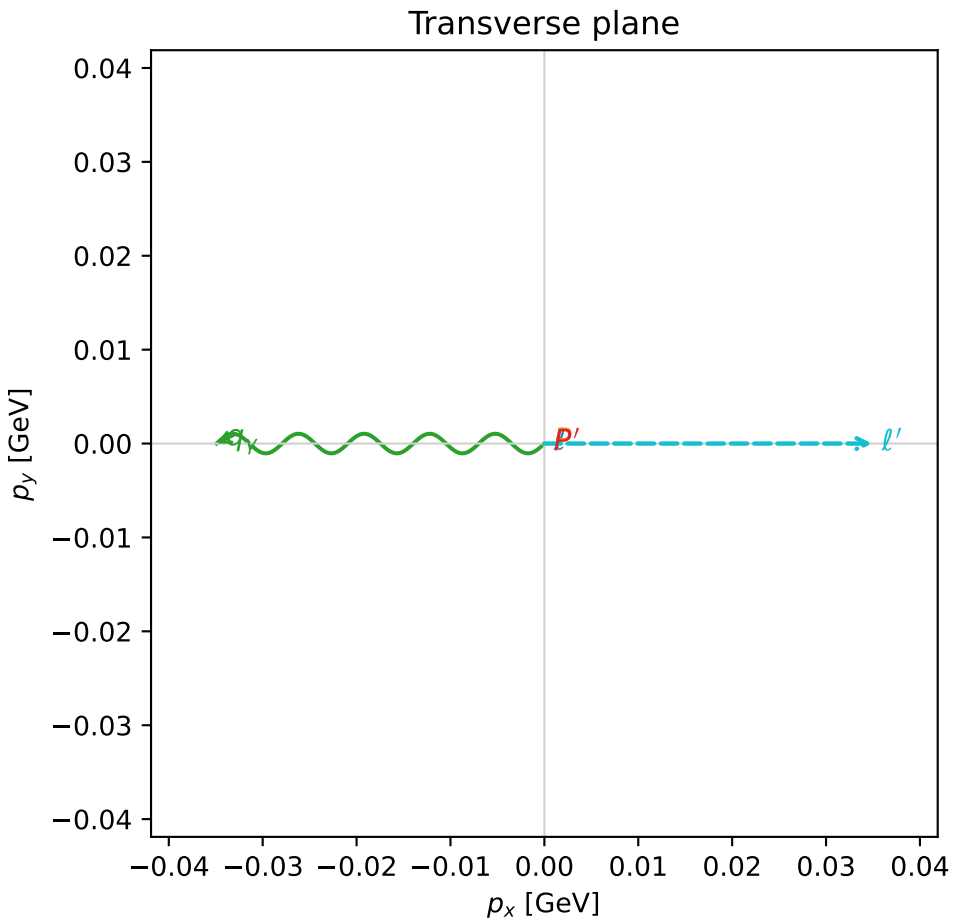
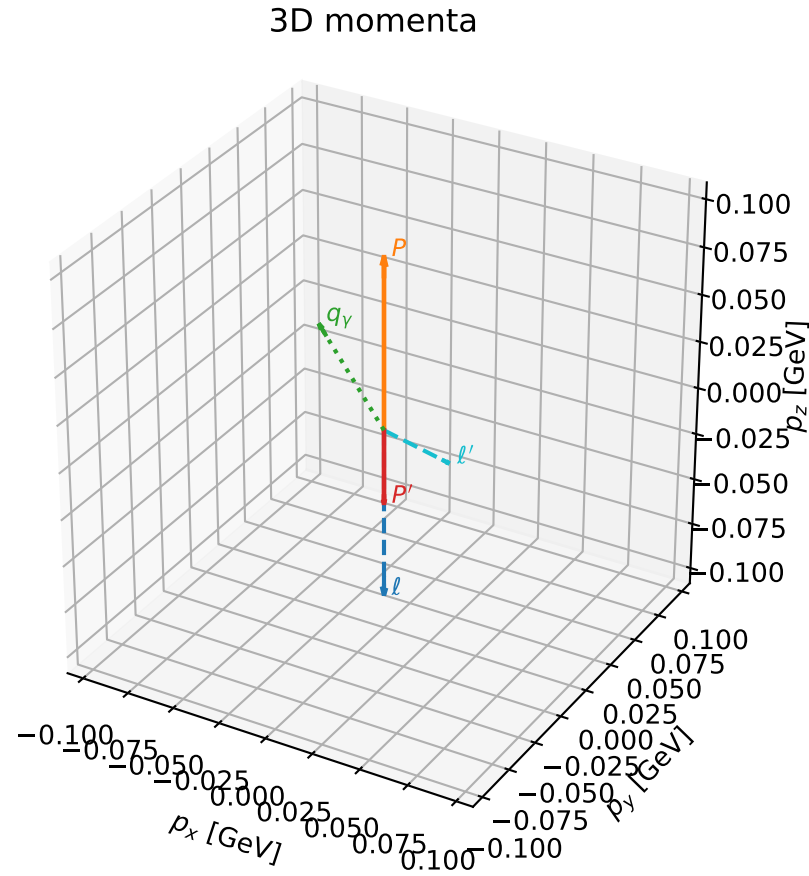
selected phase-space point:
 $\text{sqrt_s} = 3.061593$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 3.016712$
 $\text{qOut} = 1.04008$
 $\text{phi_p_out} = 0.61743$
 $\text{phi_gamma_out} = 2.034291$
 $\text{alpha_e} = 0.7961231$
 $\text{alpha_p} = 0.1113099$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

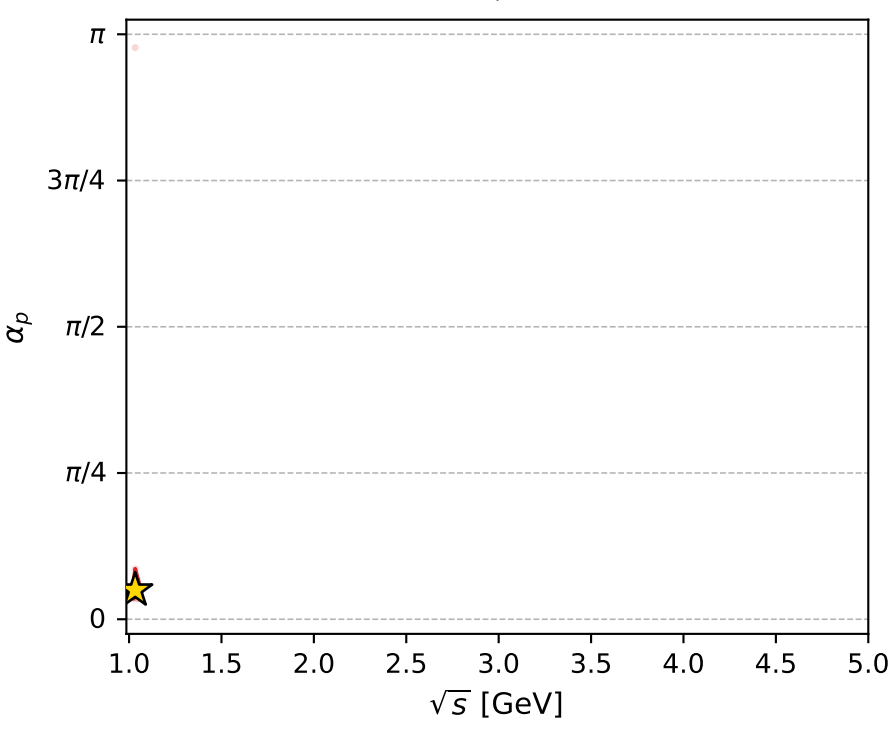
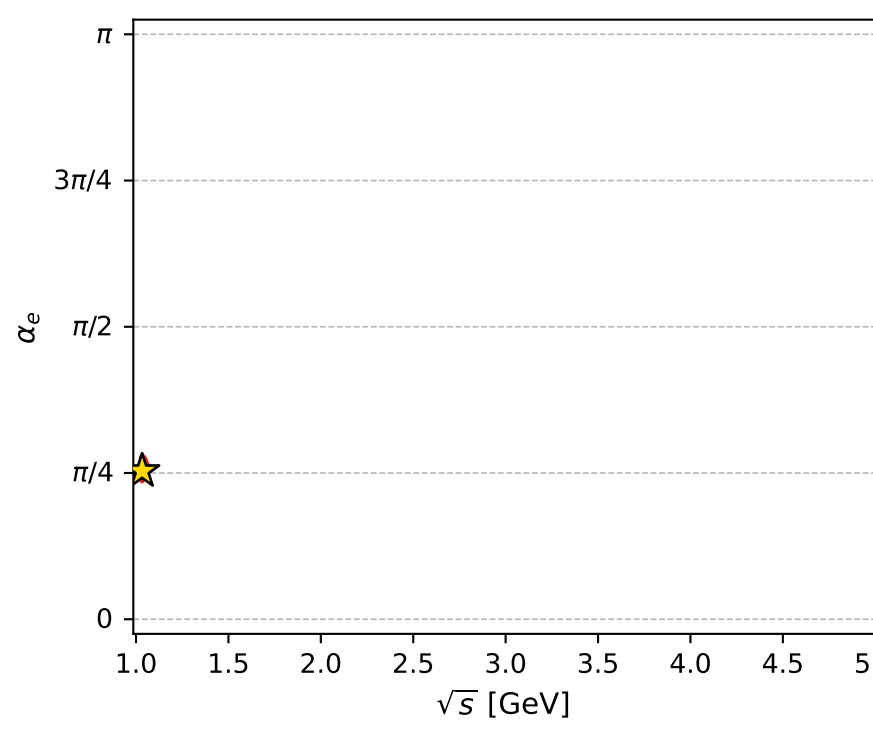
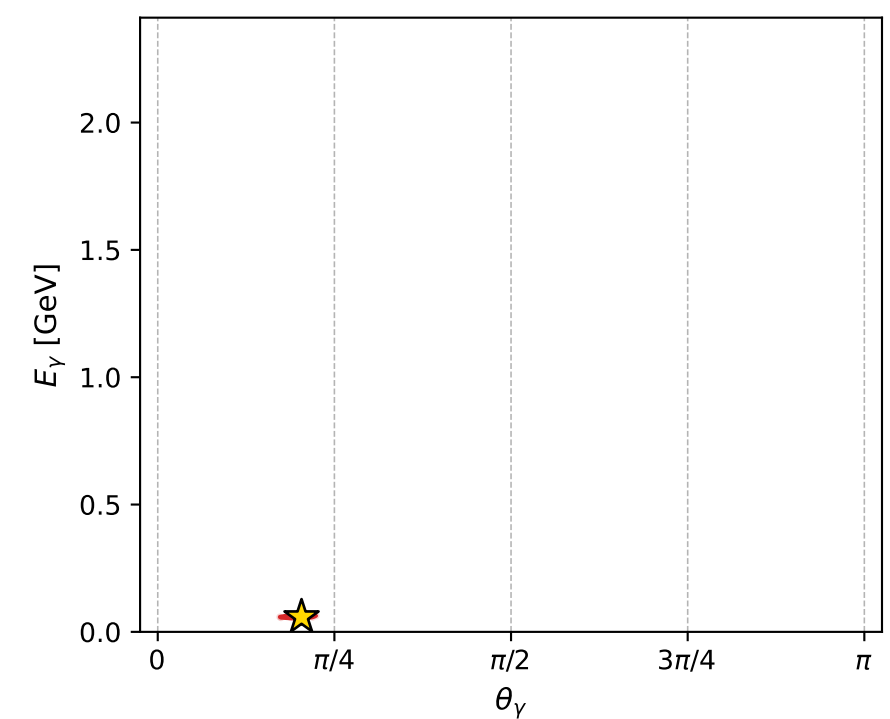
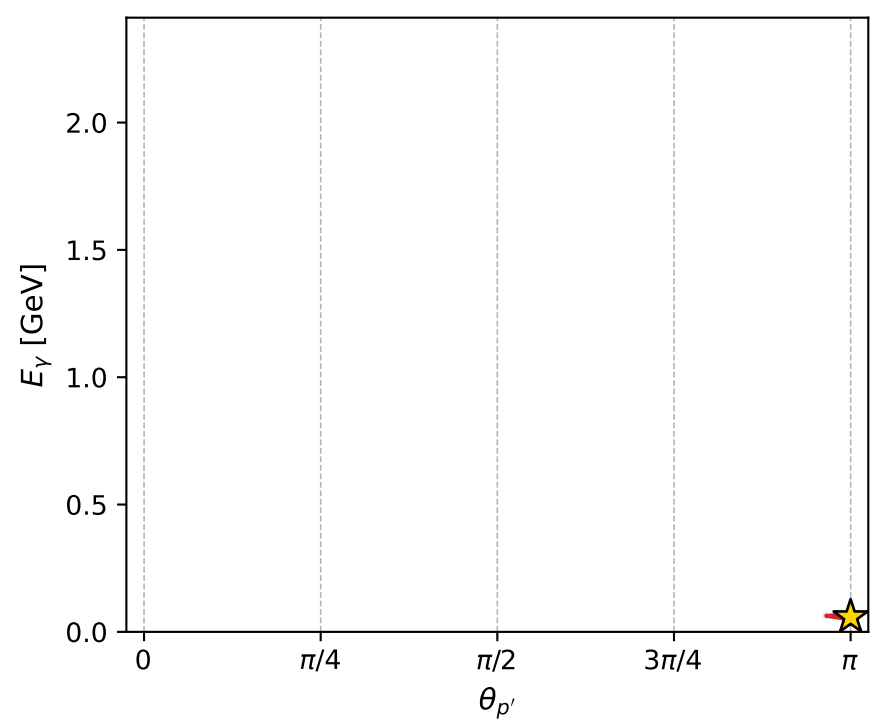
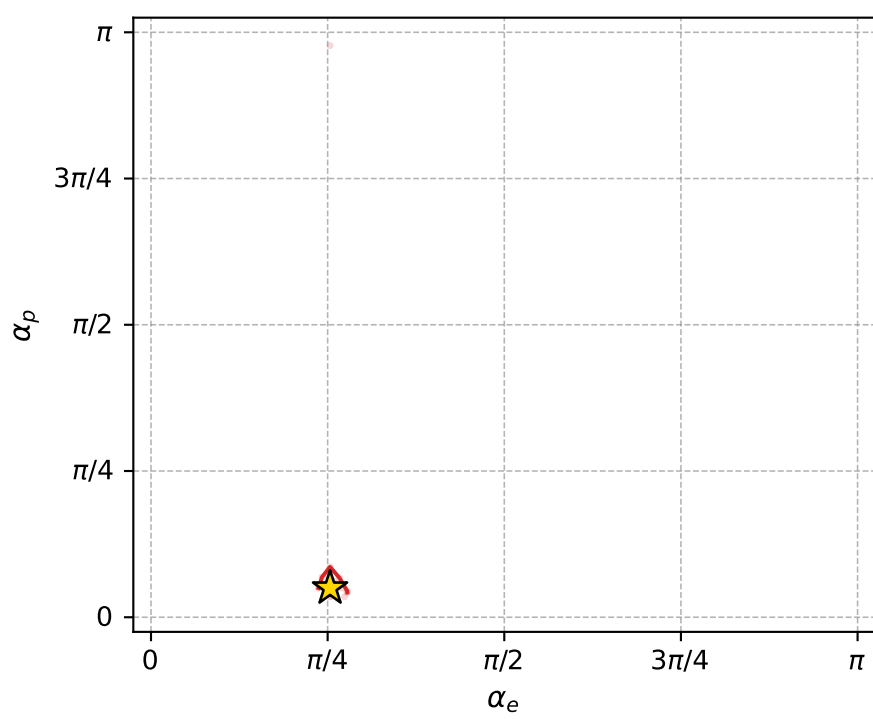
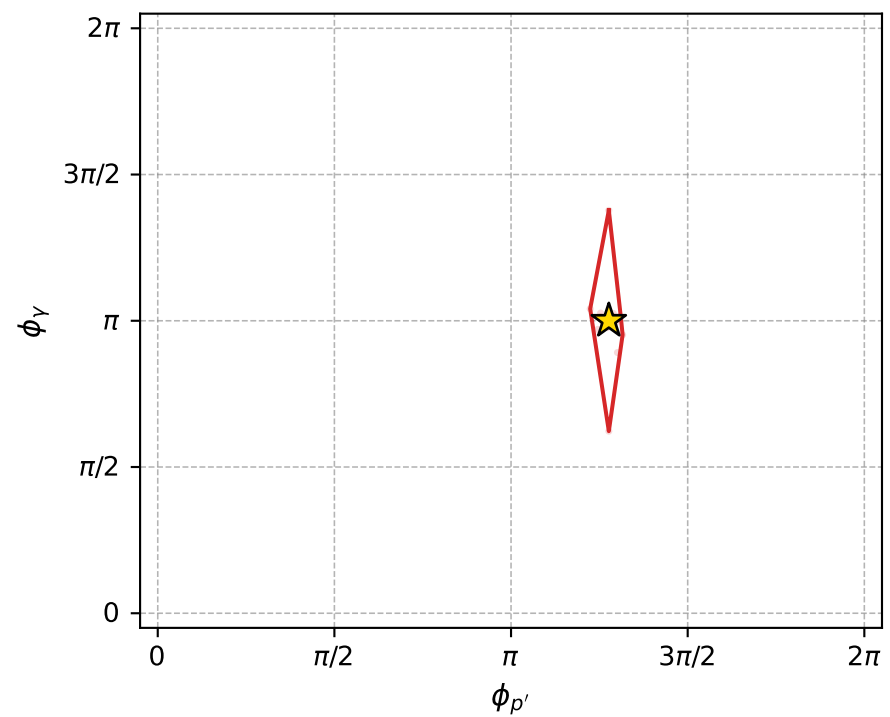
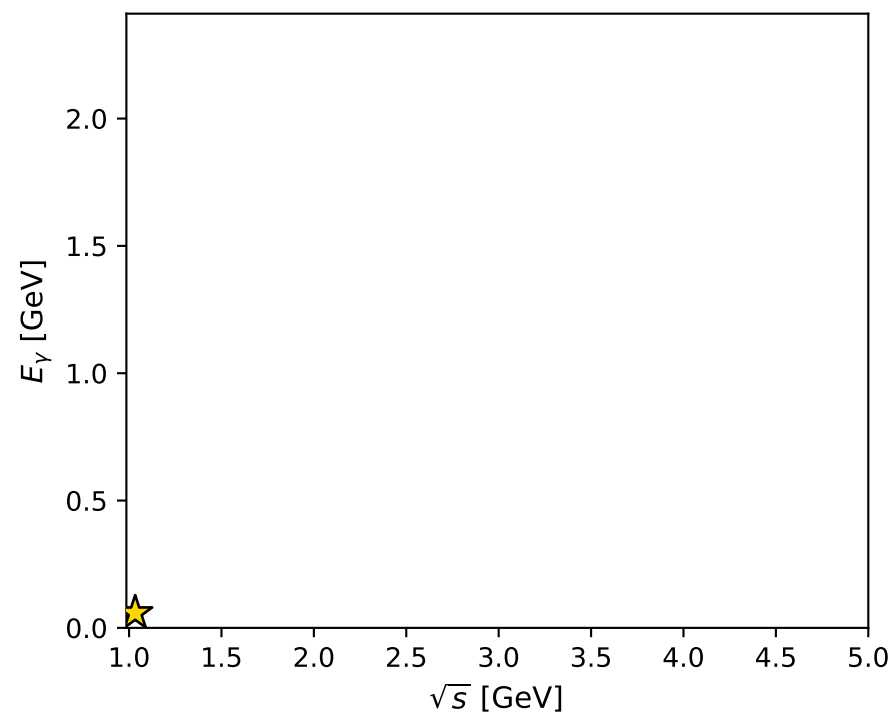
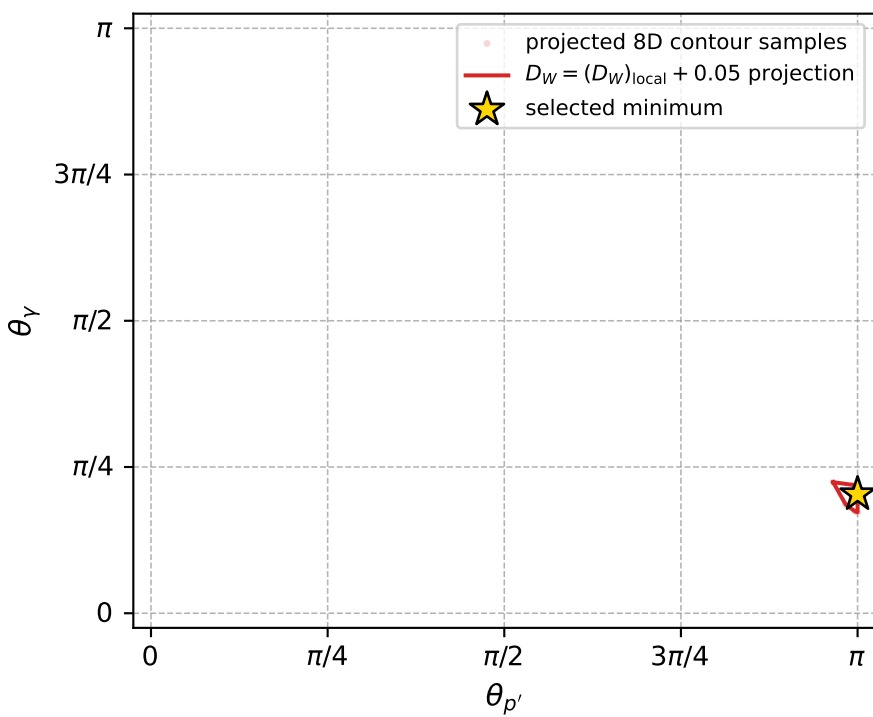
dw_mixing_angles_polarization_03_local_minimum_651 D_W=0.00554297
region: polarization_cluster_03_local_minimum_651
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.79607924$ rad
incoming lepton: $0.699514|+\rangle + 0.714619|-\rangle$
proton mixing angle: $\alpha_p=0.15671114$ rad
incoming proton: $0.987746|+\rangle + 0.156071|-\rangle$

$s=1.06724$, $\sqrt{s}=1.03307$, $|\vec{P}|=0.0906976$, $|\vec{P}'|=0.0393062$
 $\theta_{p'}=3.14159$, $\theta_\gamma=0.639302$, $\phi_{p'}=4.0112$, $\phi_\gamma=3.1416$
 $E_\gamma=0.0585098$, $\theta_{\ell'}=1.78649$, $\phi_{\ell'}=8.94111e-06$
 $Q^2=0.00509536$, $x_B=0.0429455$, $t=-0.0168884$, $W^2=0.993396$, $y=0.633133$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0907$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0907$
 P $E= 0.9424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0907$
 ℓ' $E= 0.0357$ $p_x= 0.0349$ $p_y= 0.0000$ $p_z= -0.0076$
 P' $E= 0.9388$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0393$
 q_γ $E= 0.0585$ $p_x= -0.0349$ $p_y= -0.0000$ $p_z= 0.0470$



electron: polarization cluster P3, local minimum 651; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0055429691$
 $D_W \text{ contour} = 0.055542969$
 above global minimum = 0.0055265362
 8D contour samples = 119

selected phase-space point:
 $\text{sqrt}_s = 1.033074$
 $\text{theta}_p_{\text{out}} = 3.141593$
 $\text{theta}_\gamma_{\text{out}} = 0.6393024$
 $q_{\text{out}} = 0.05850979$
 $\text{phi}_p_{\text{out}} = 4.011204$
 $\text{phi}_\gamma_{\text{out}} = 3.141602$
 $\alpha_e = 0.7960792$
 $\alpha_p = 0.1567111$

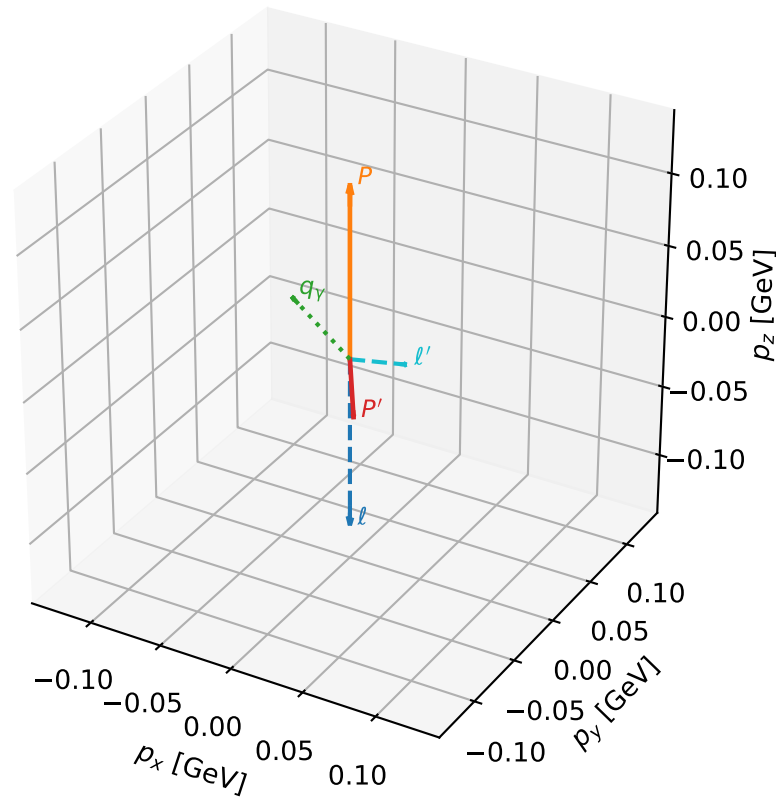
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_657 D_W=0.0057073
 region: polarization_cluster_03_local_minimum_657
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77071917$ rad
 incoming lepton: $0.71741|+\rangle +0.696651|-\rangle$
 proton mixing angle: $\alpha_p=0.10578762$ rad
 incoming proton: $0.99441|+\rangle +0.10559|-\rangle$

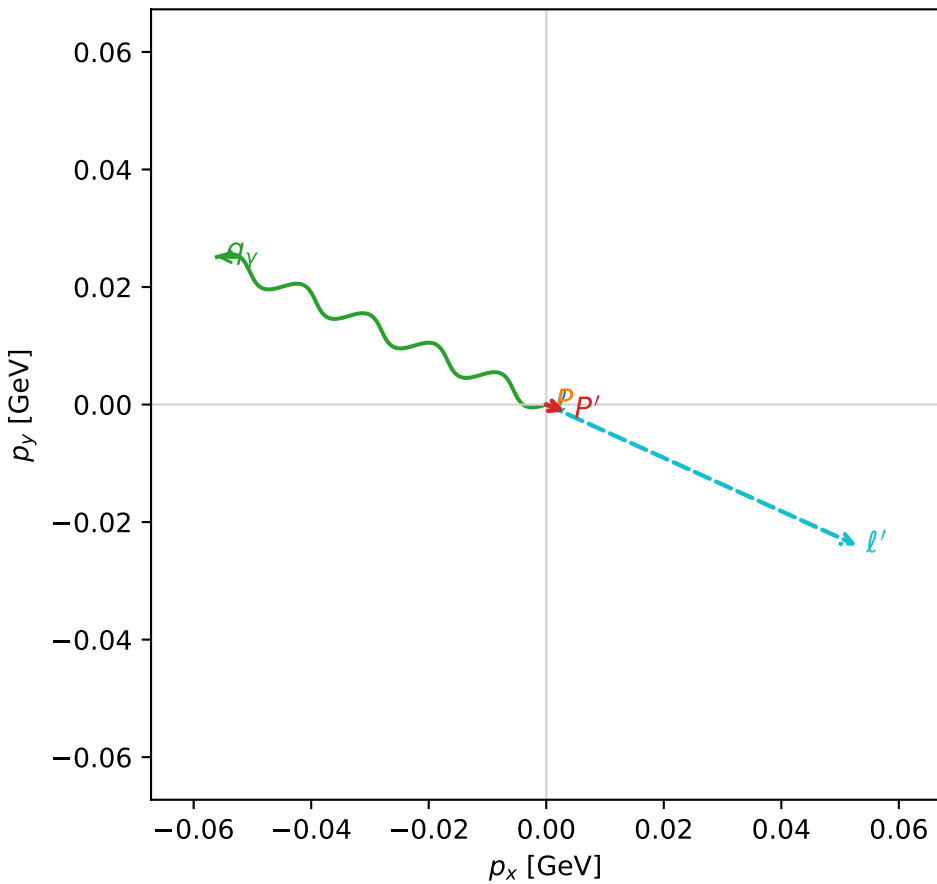
$s=1.13399$, $\sqrt{s}=1.06489$, $|\vec{P}|=0.11933$, $|\vec{P}'|=0.0395815$
 $\theta_{P'}=3.05612$, $\theta_V=1.31674$, $\phi_{P'}=5.94507$, $\phi_V=2.72054$
 $E_V=0.0634418$, $\theta_\ell=1.18619$, $\phi_{\ell'}=5.8573$
 $Q^2=0.0205499$, $x_B=0.14539$, $t=-0.0251732$, $W^2=1.00064$, $y=0.556145$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1193$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1193$
 P $E= 0.9456$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1193$
 ℓ' $E= 0.0626$ $p_x= 0.0529$ $p_y= -0.0240$ $p_z= 0.0235$
 P' $E= 0.9388$ $p_x= 0.0032$ $p_y= -0.0011$ $p_z= -0.0394$
 q_V $E= 0.0634$ $p_x= -0.0560$ $p_y= 0.0251$ $p_z= 0.0159$

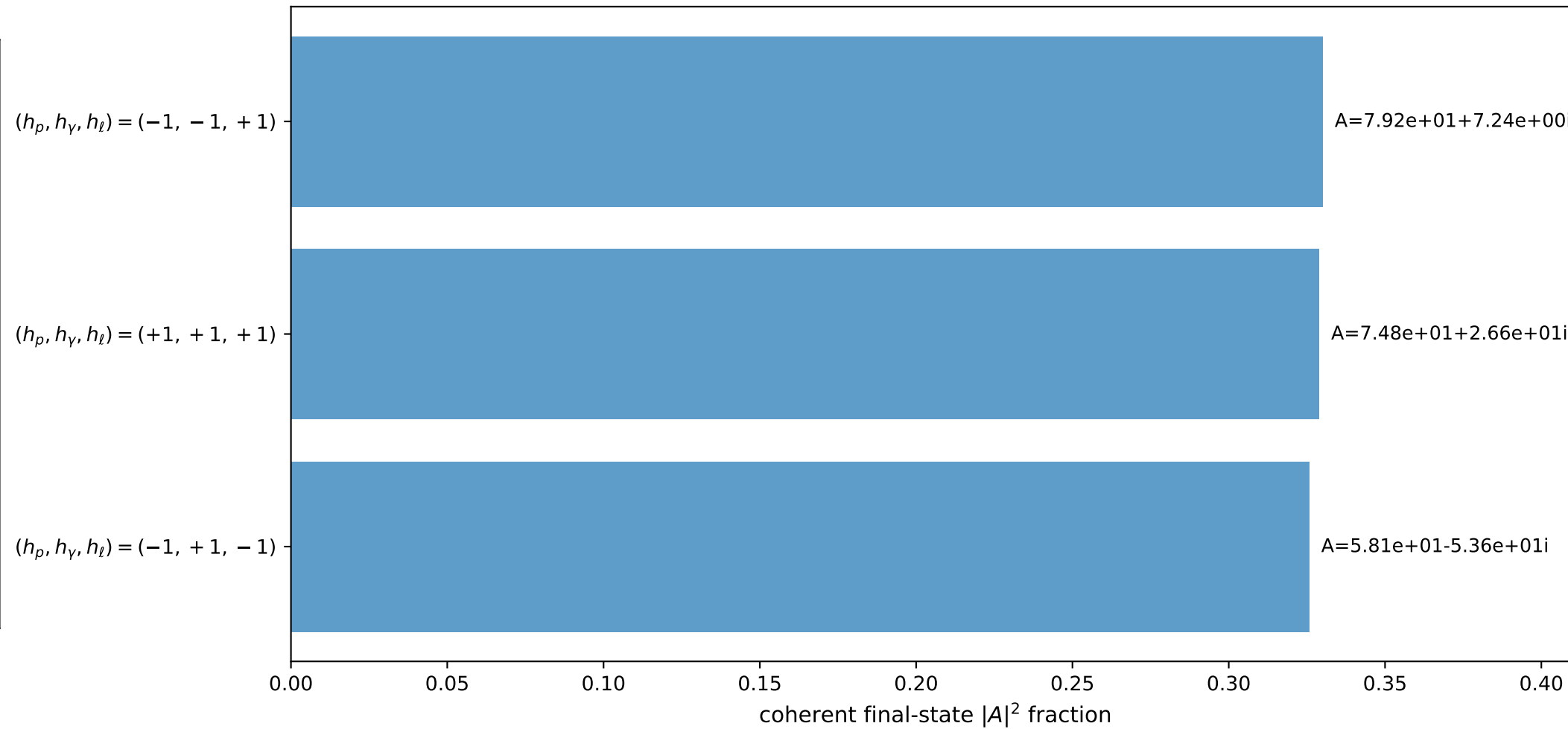
3D momenta



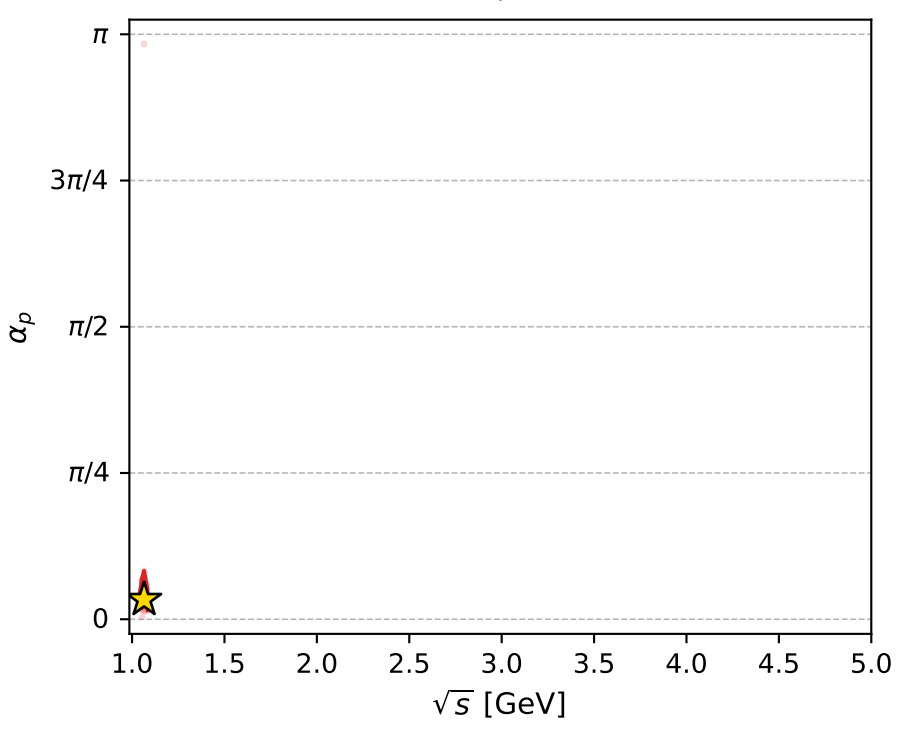
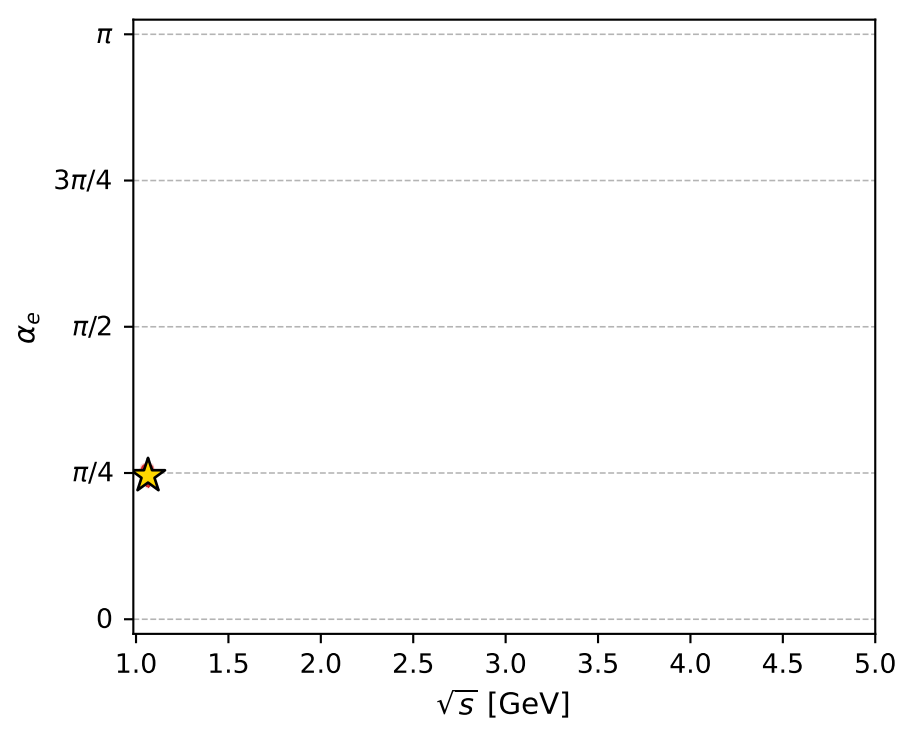
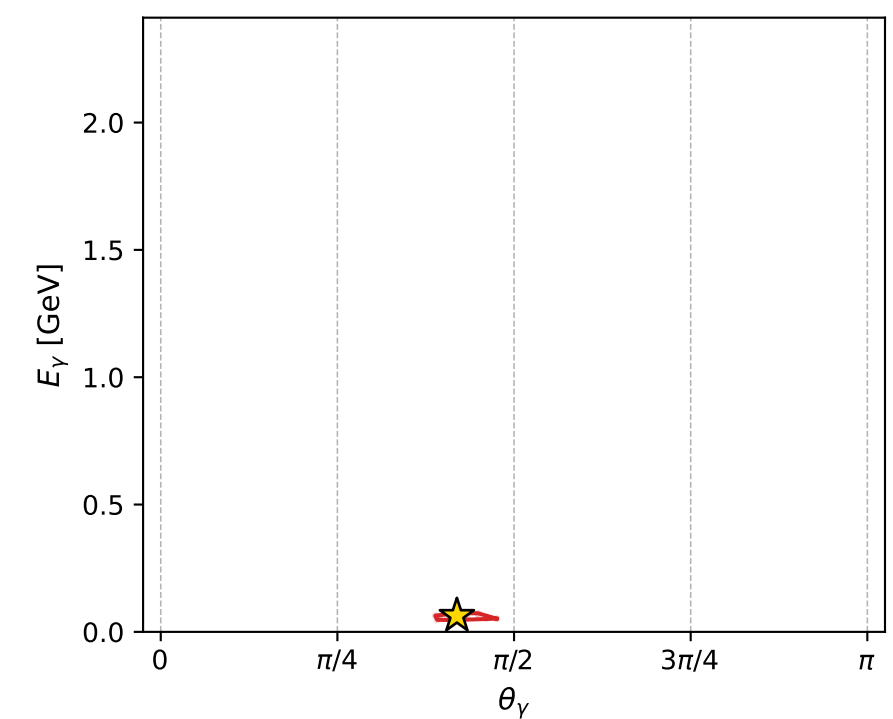
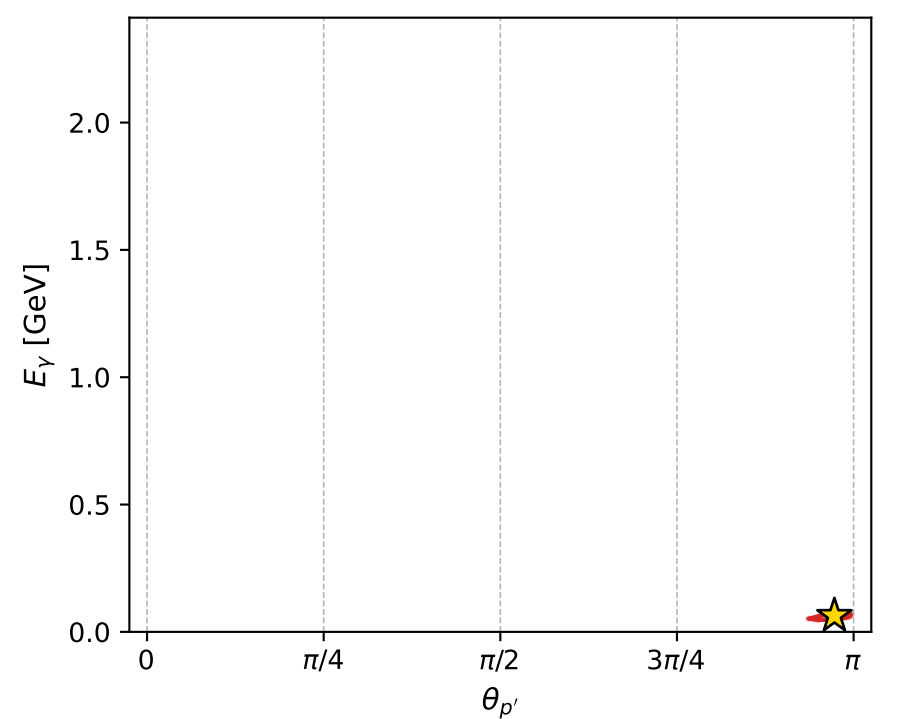
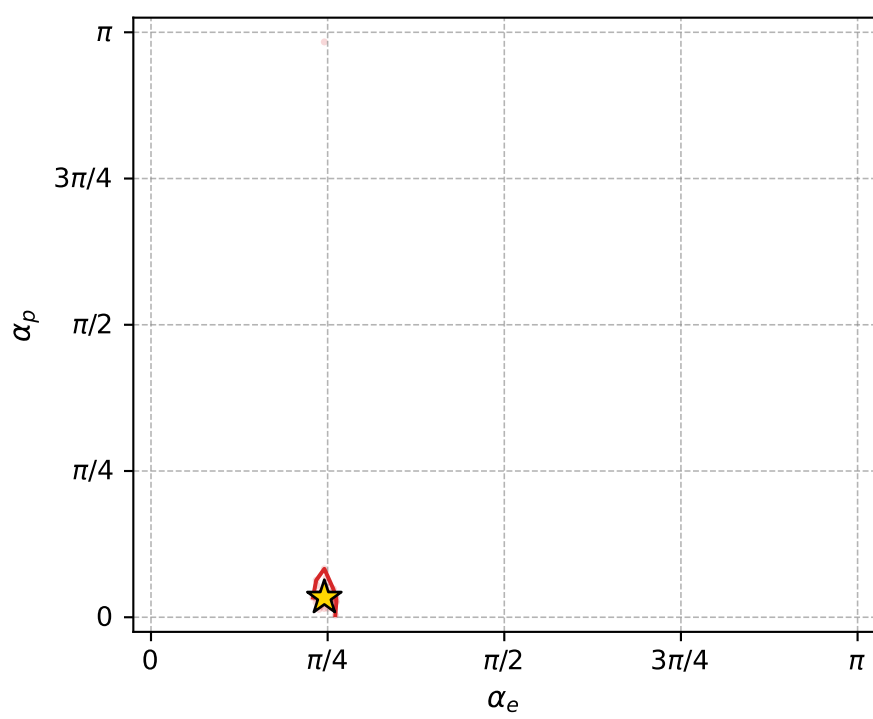
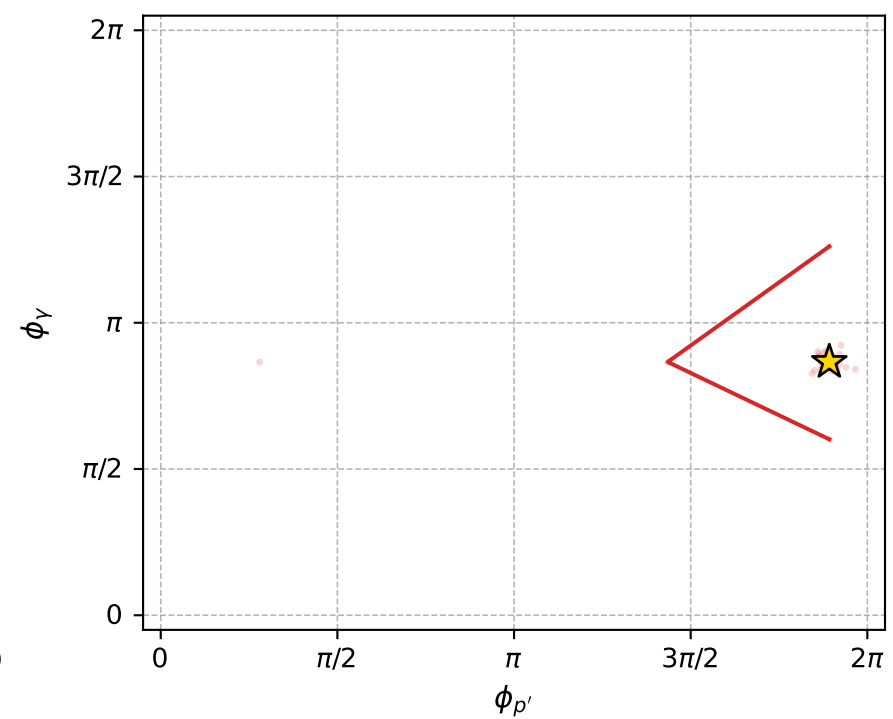
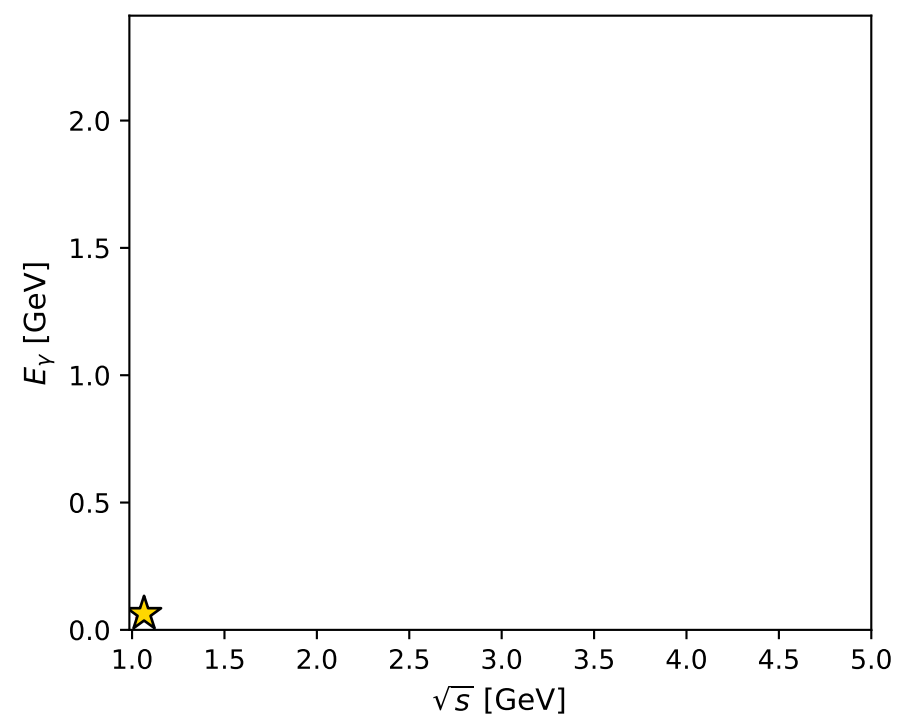
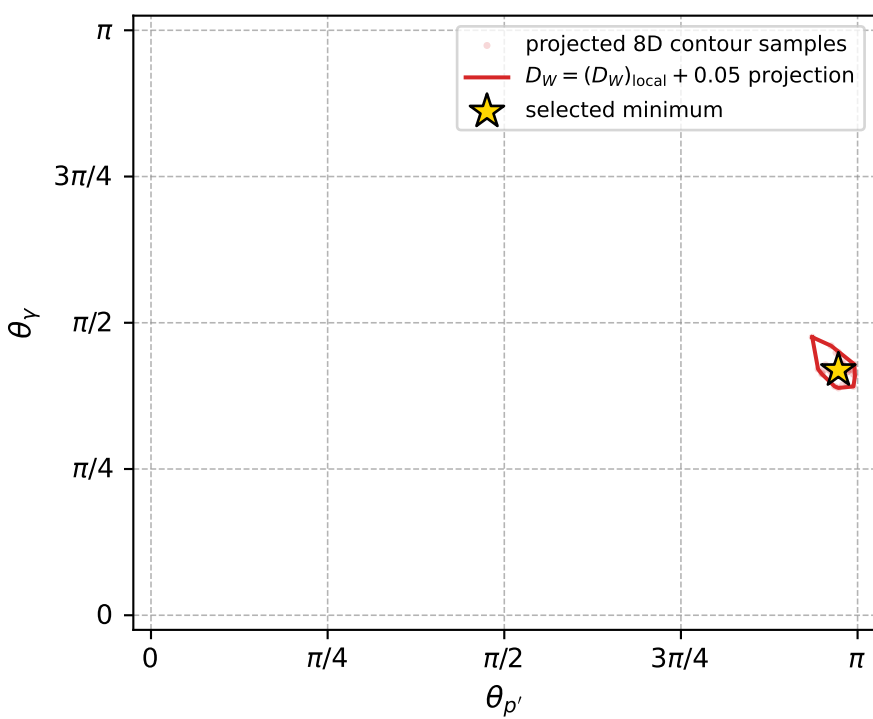
Transverse plane



Leading final-state amplitudes (N=3, retained=98.5%)



electron: polarization cluster P3, local minimum 657; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

```
D_W(local) = 0.0057073037
D_W contour = 0.055707304
above global minimum = 0.0056908709
8D contour samples = 255
```

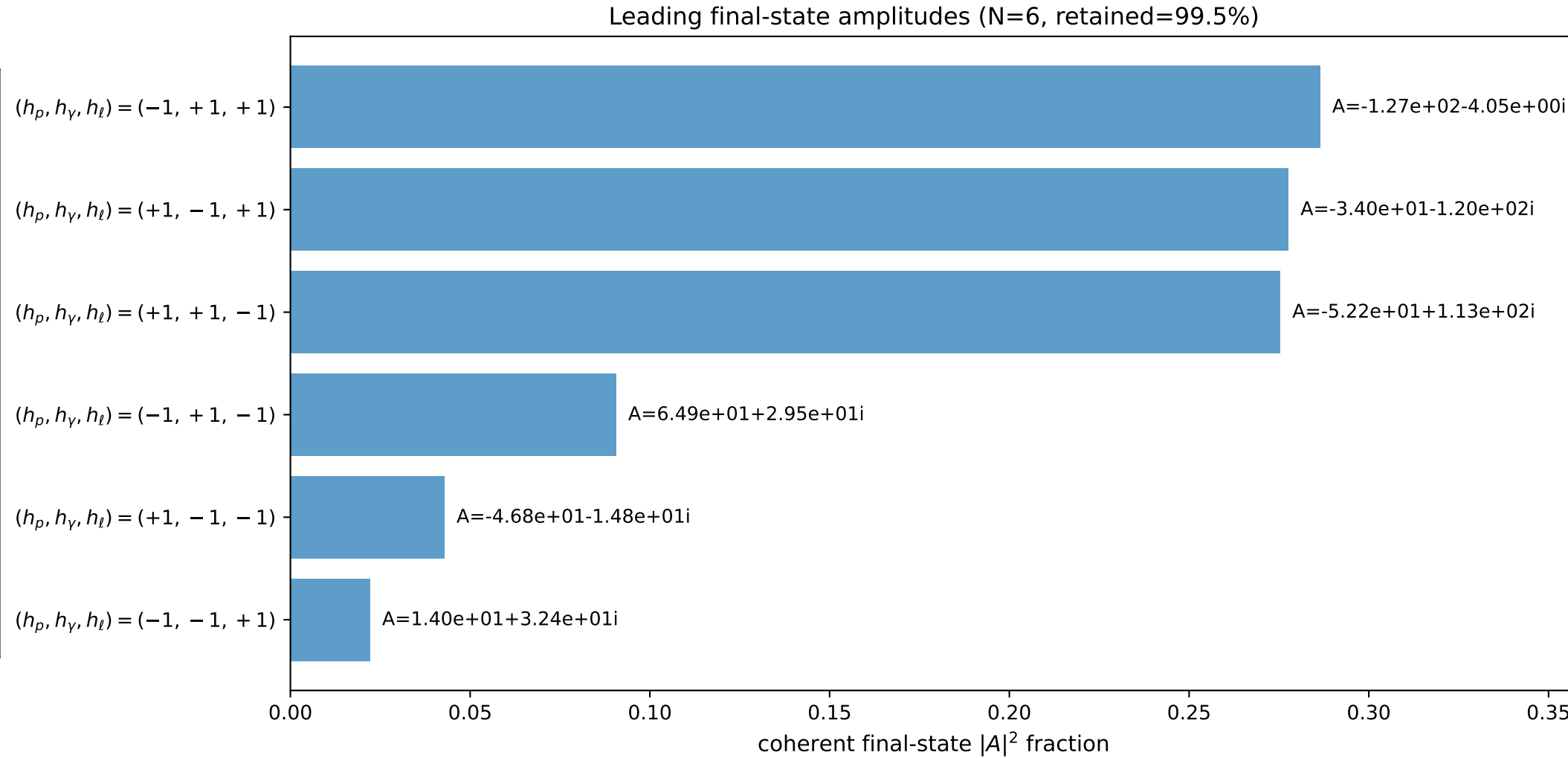
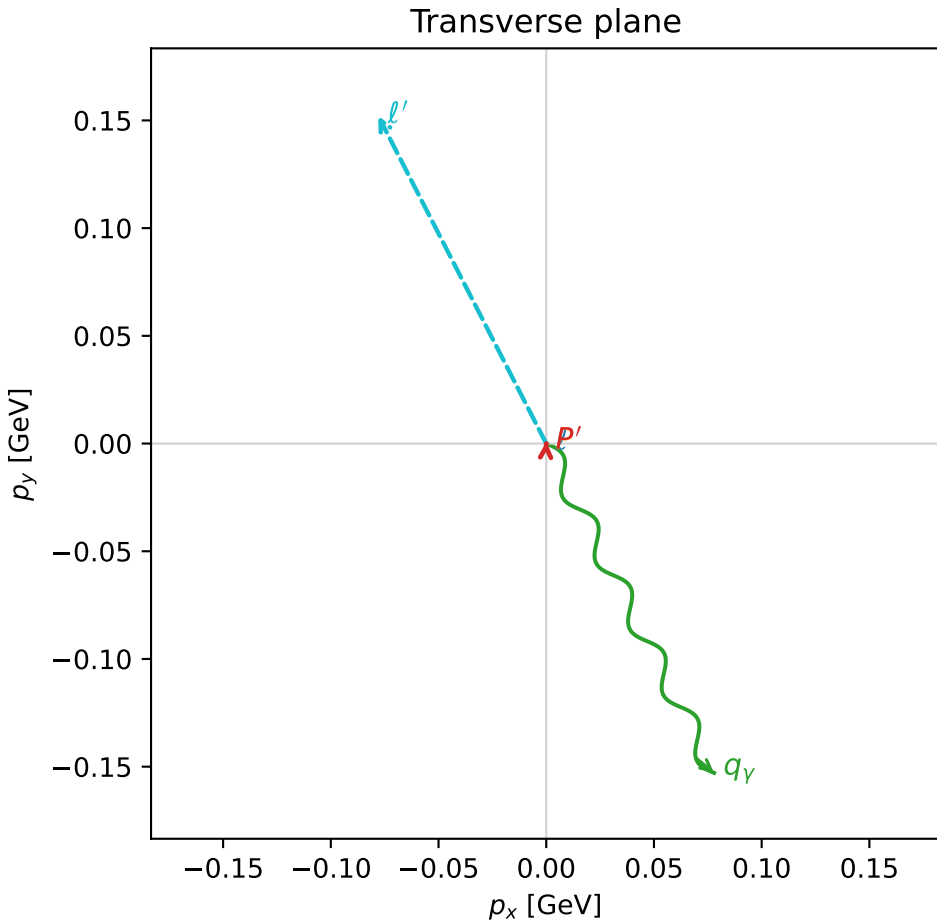
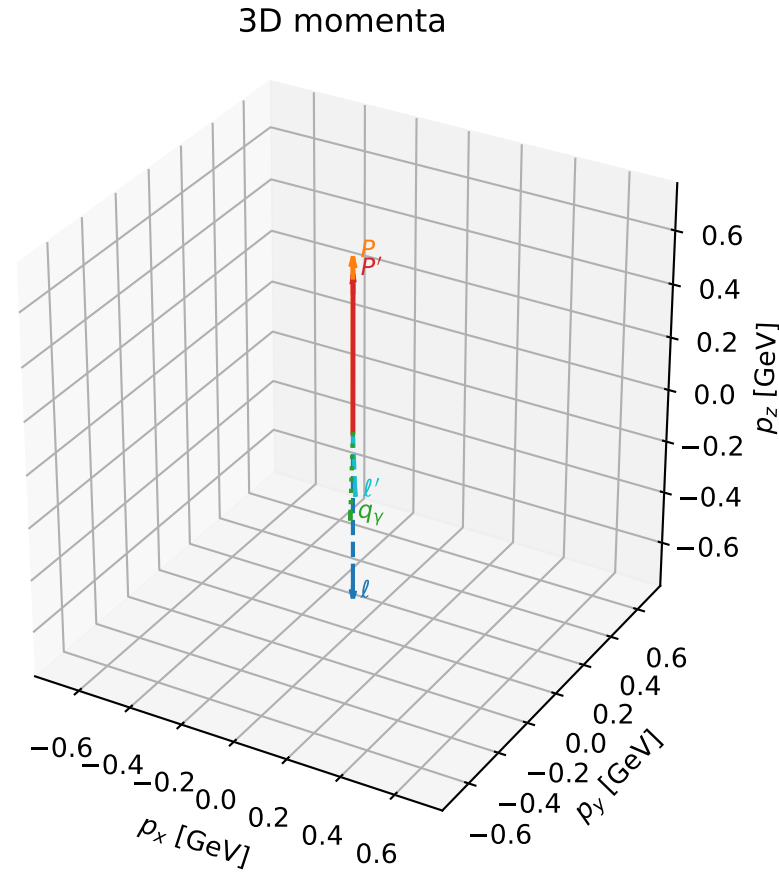
```
selected phase-space point:
    sqrt_s = 1.064891
    theta_p_out = 3.056115
    theta_gamma_out = 1.316742
    q0Out = 0.06344182
    phi_p_out = 5.945069
    phi_gamma_out = 2.720535
    alpha_e = 0.7707192
    alpha_p = 0.1057876
```

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

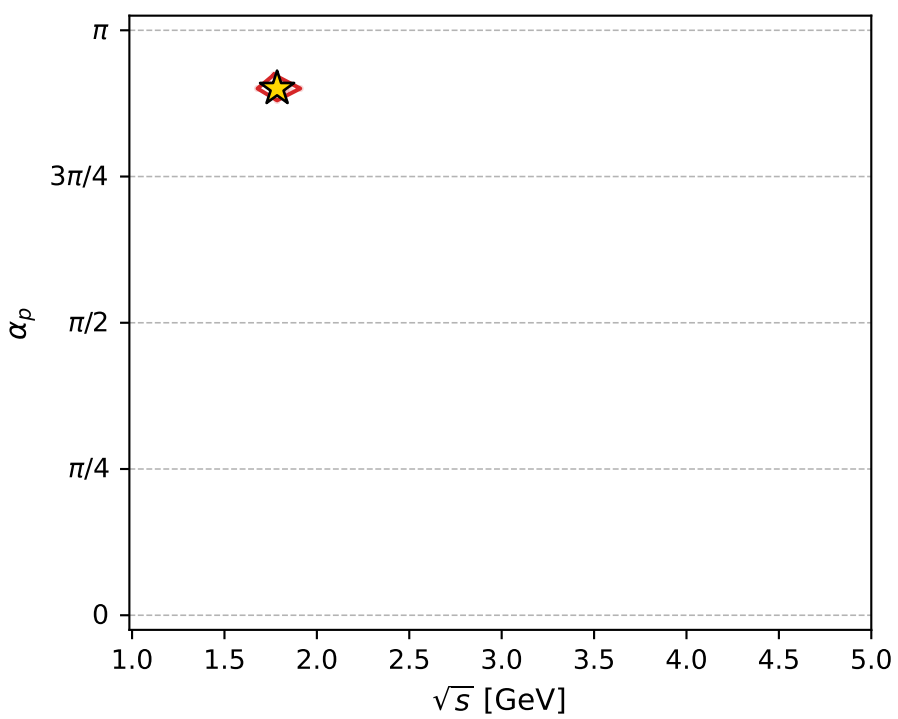
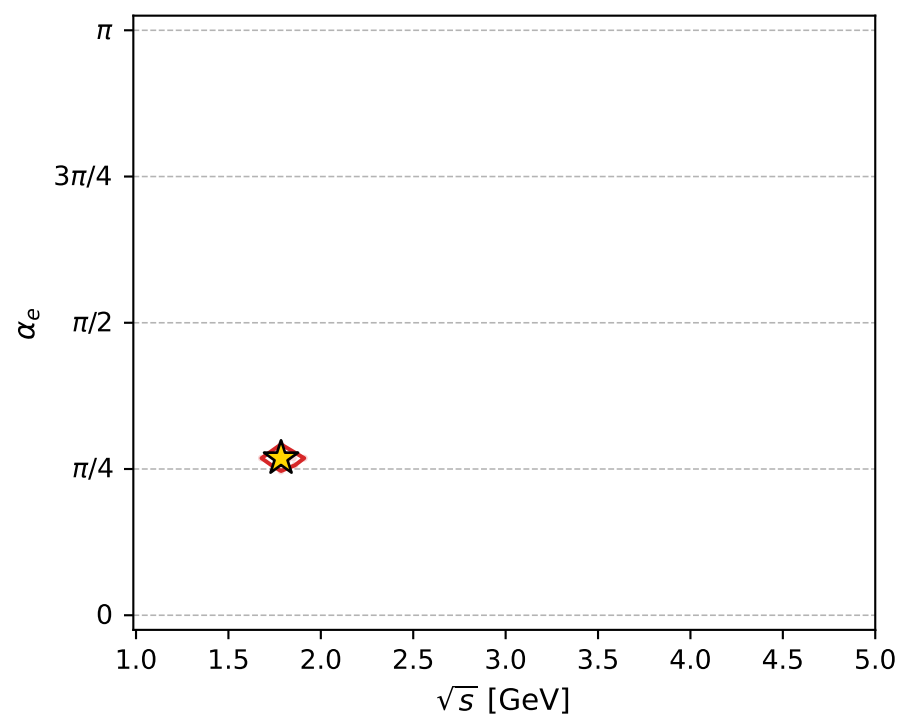
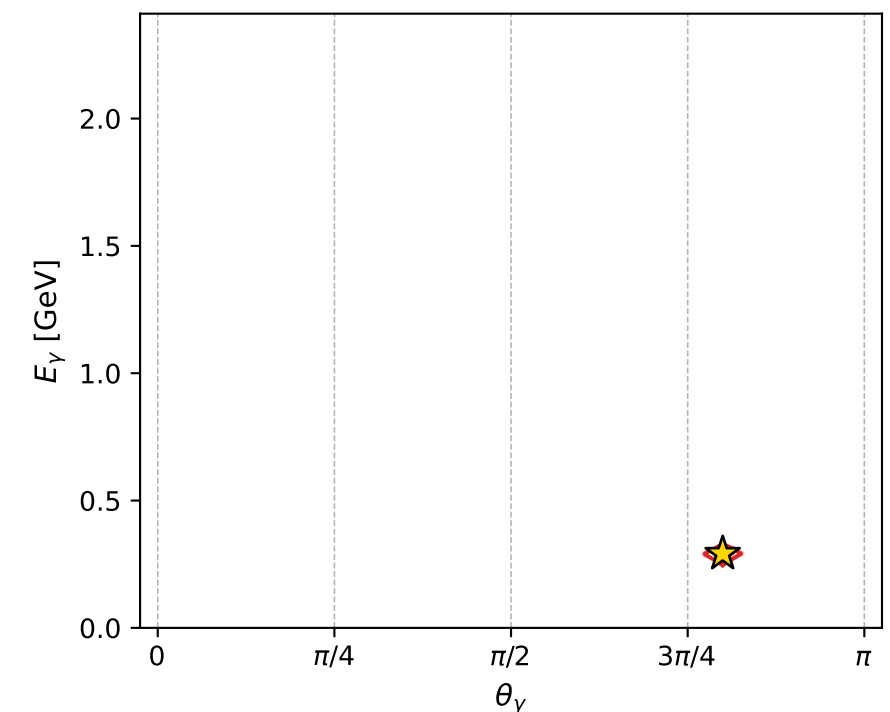
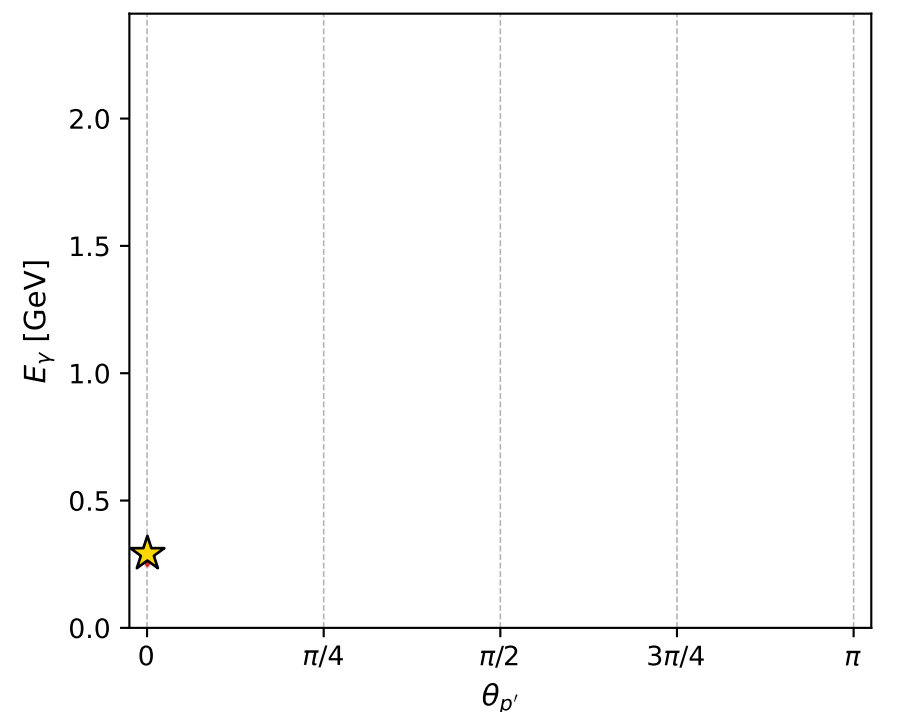
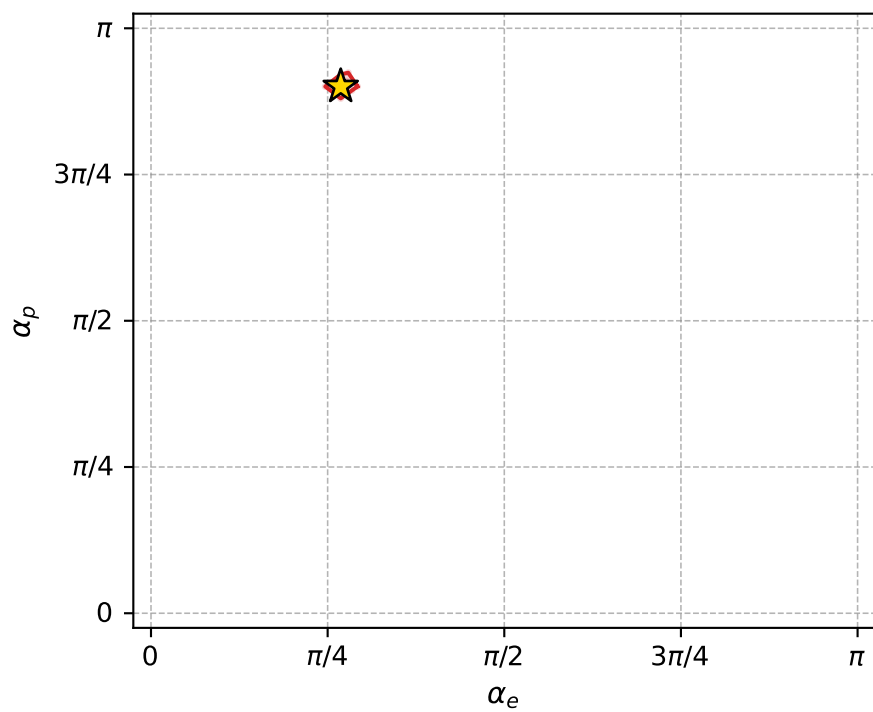
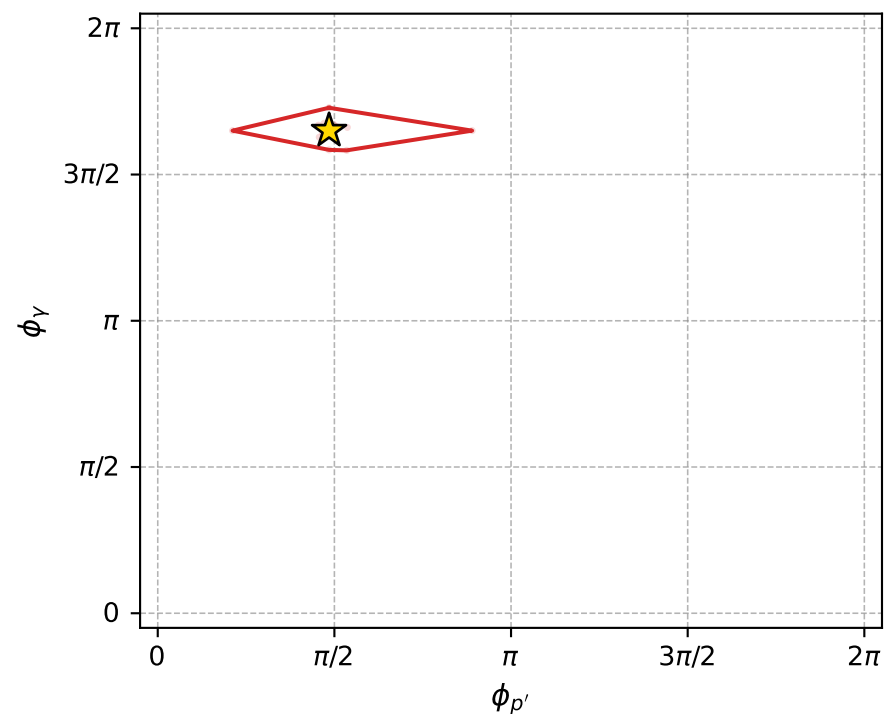
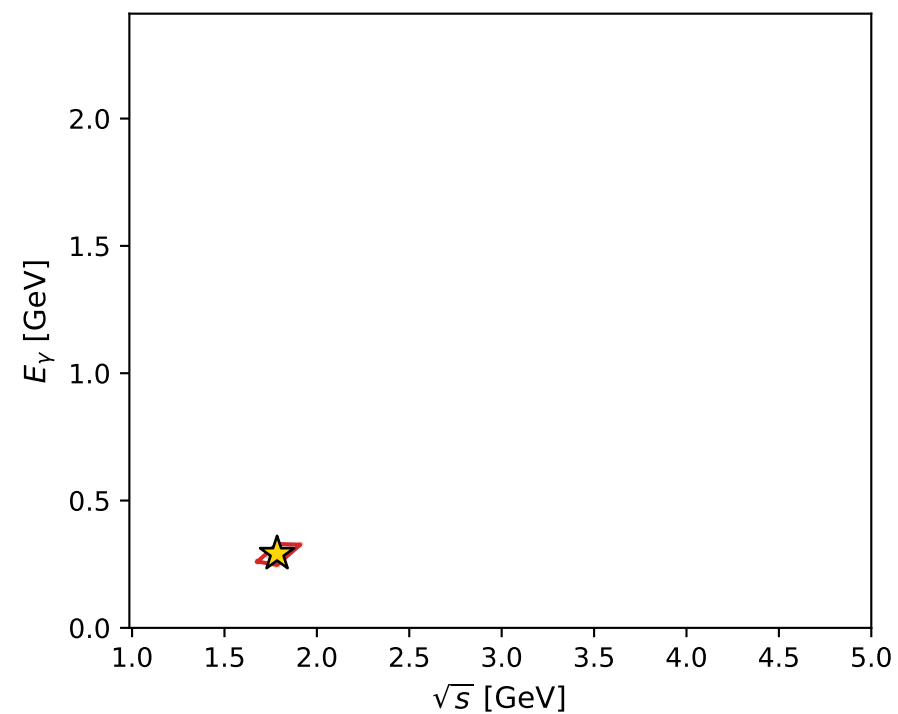
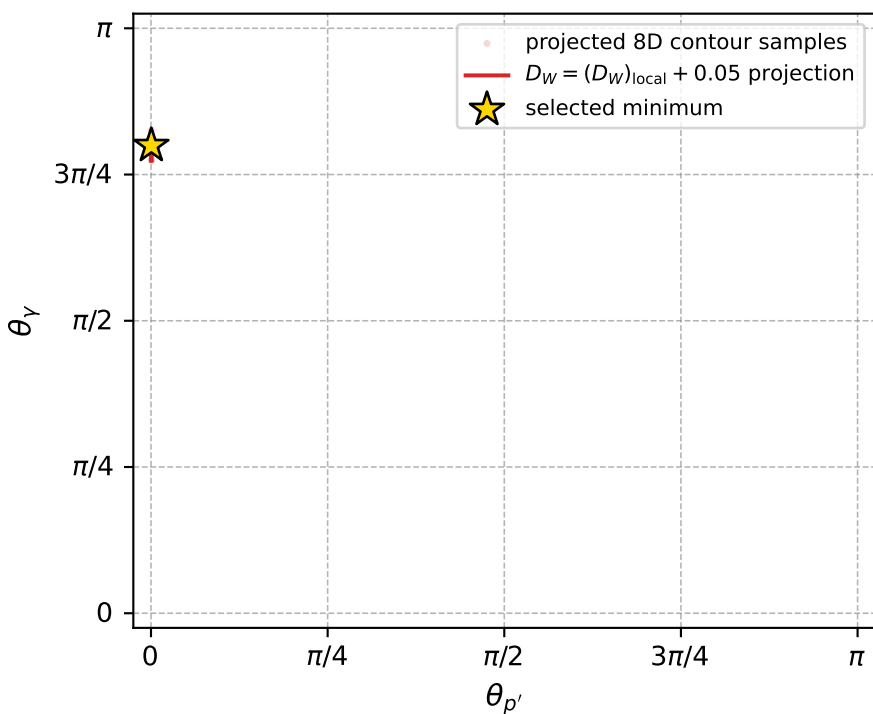
dw_mixing_angles_polarization_03_local_minimum_677 D_W=0.00627242
 region: polarization_cluster_03_local_minimum_677
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.84357583$ rad
 incoming lepton: $0.664796|+\rangle + 0.747025|-\rangle$
 proton mixing angle: $\alpha_p=2.8289662$ rad
 incoming proton: $-0.951529|+\rangle + 0.307559|-\rangle$

$s=3.18513$, $\sqrt{s}=1.78469$, $|\vec{P}|=0.645849$, $|\vec{P}'|=0.584171$
 $\theta_{P'}=0.00148014$, $\theta_Y=2.51177$, $\phi_{P'}=1.52339$, $\phi_Y=5.18367$
 $E_Y=0.291319$, $\theta_{\ell'}=2.68609$, $\phi_{\ell'}=2.04458$
 $Q^2=0.0511451$, $x_B=0.0527109$, $t=-0.00266195$, $W^2=1.79899$, $y=0.4209$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.6458$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.6458$
 P $E= 1.1388$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.6458$
 ℓ' $E= 0.3883$ $p_x= -0.0779$ $p_y= 0.1520$ $p_z= -0.3487$
 P' $E= 1.1050$ $p_x= 0.0000$ $p_y= 0.0009$ $p_z= 0.5842$
 q_Y $E= 0.2913$ $p_x= 0.0779$ $p_y= -0.1529$ $p_z= -0.2354$



electron: polarization cluster P3, local minimum 677; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0062724247$
 $D_W \text{ contour} = 0.056272425$
 above global minimum = 0.0062559918
 8D contour samples = 256

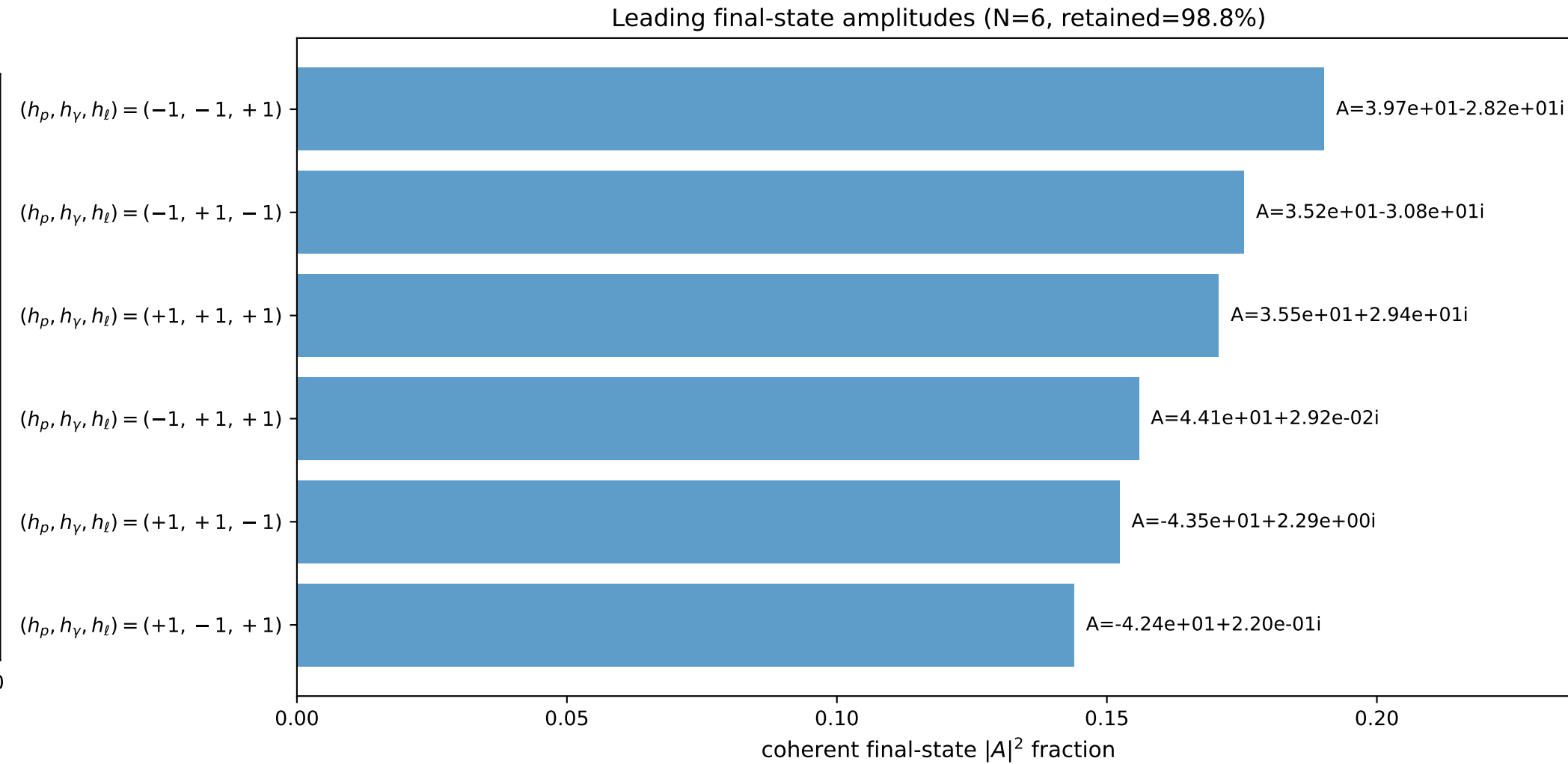
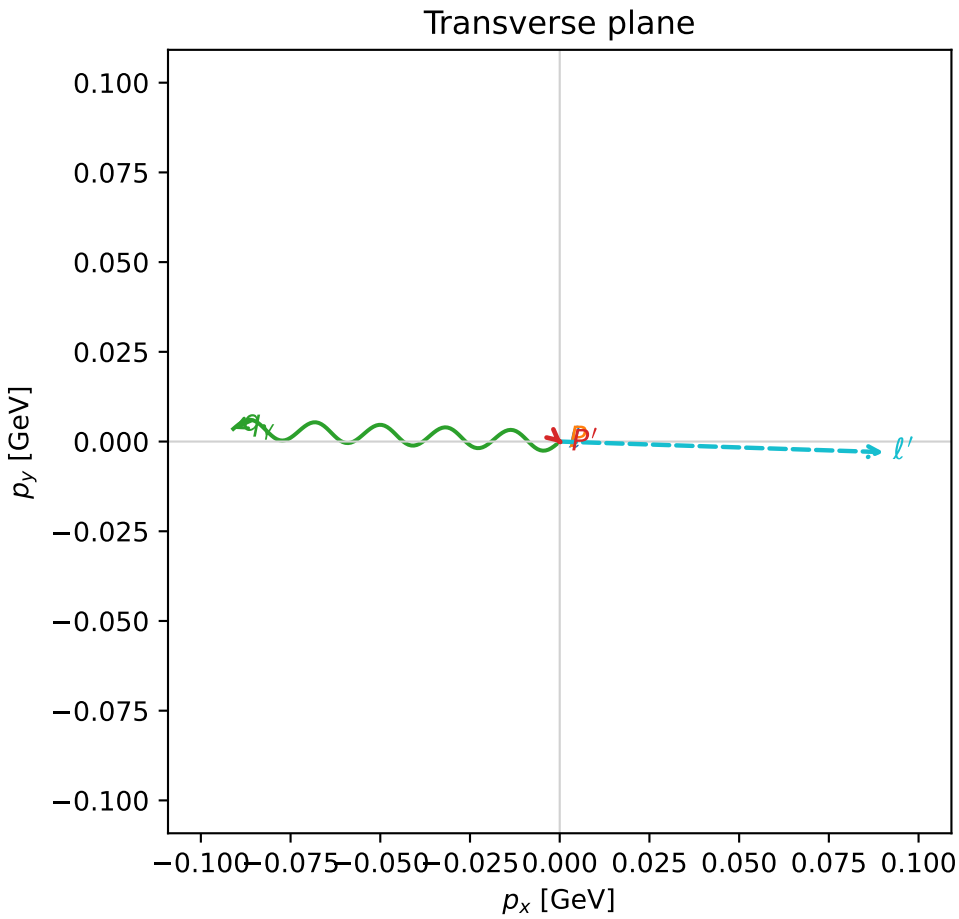
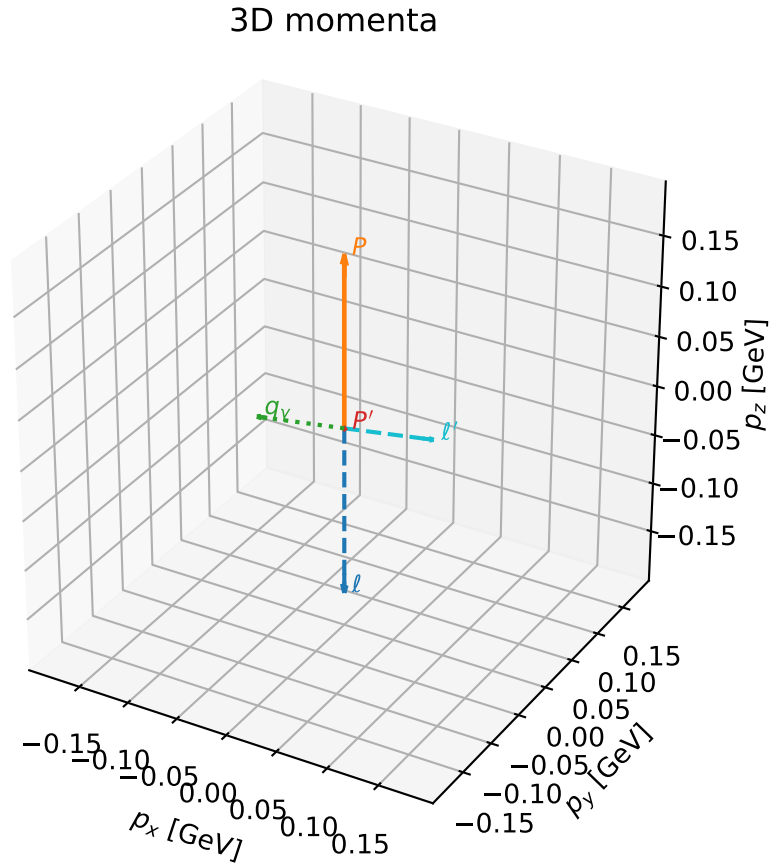
selected phase-space point:
 $\text{sqrt_s} = 1.784693$
 $\text{theta_p_out} = 0.001480142$
 $\text{theta_gamma_out} = 2.511774$
 $\text{qOut} = 0.2913193$
 $\text{phi_p_out} = 1.523387$
 $\text{phi_gamma_out} = 5.183667$
 $\text{alpha_e} = 0.8435758$
 $\text{alpha_p} = 2.828966$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

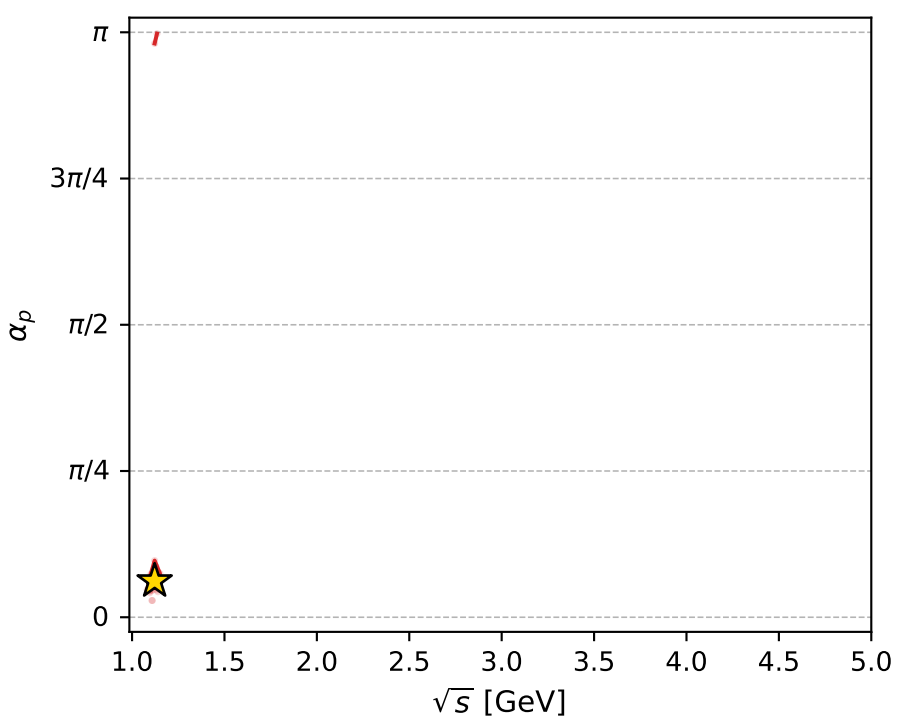
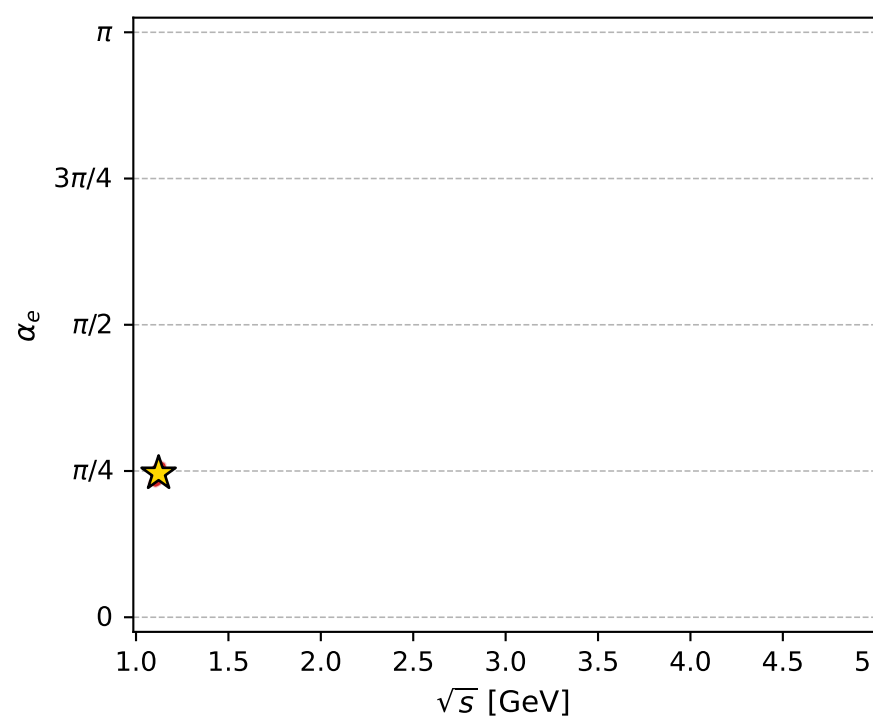
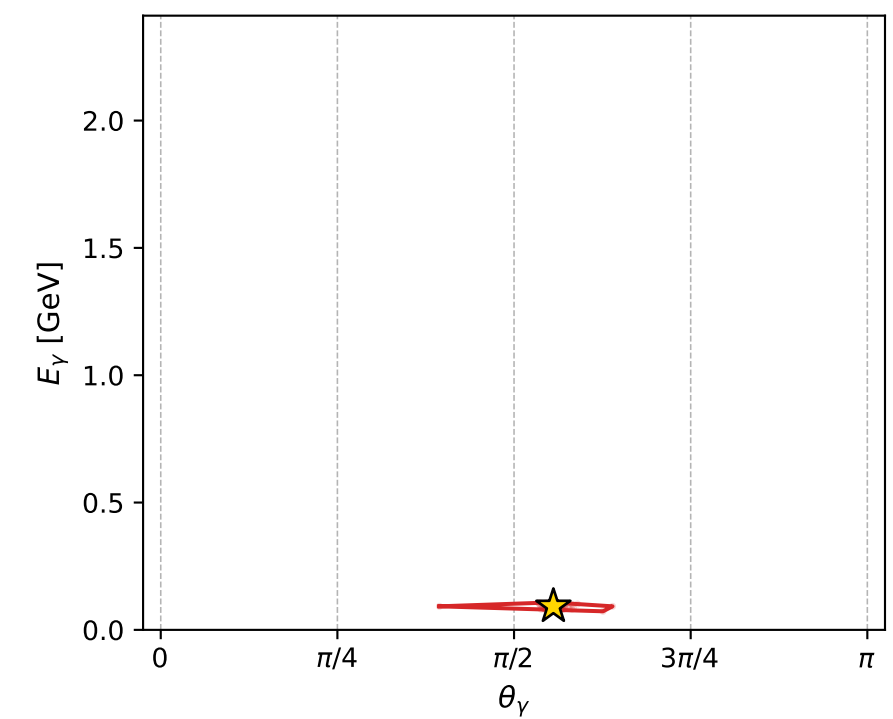
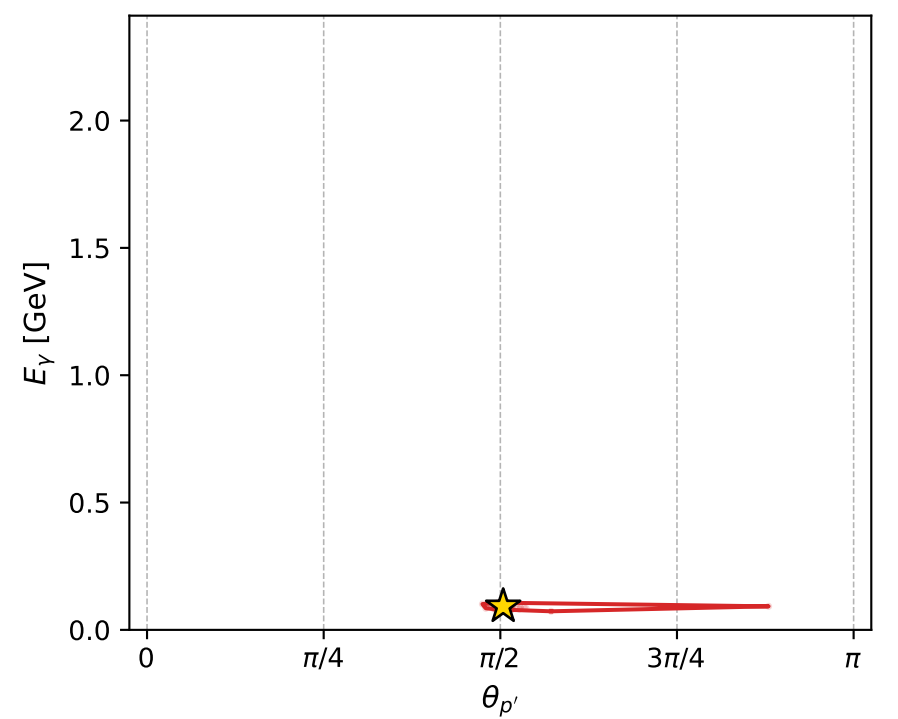
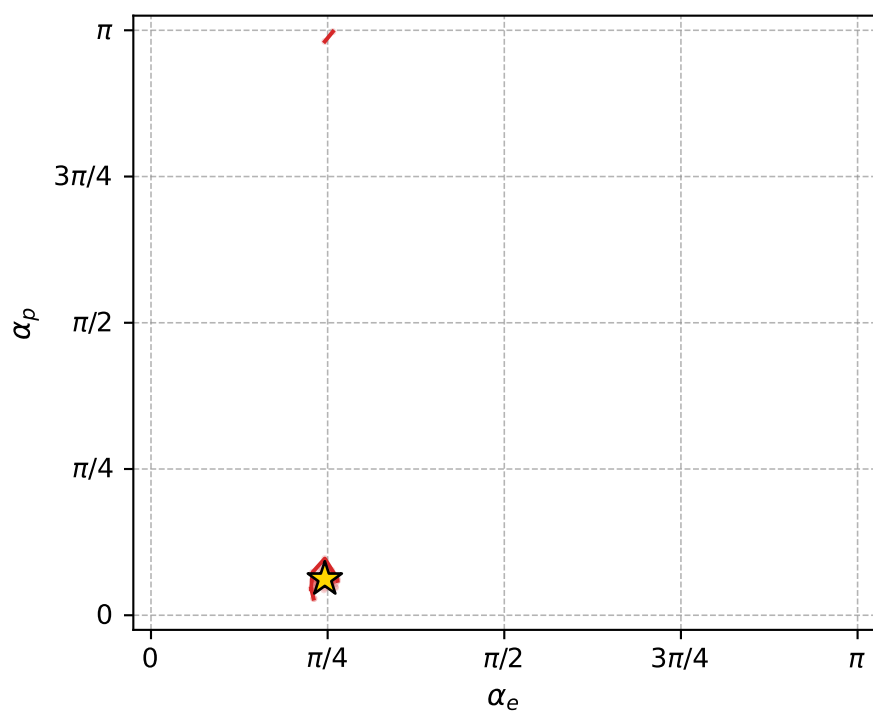
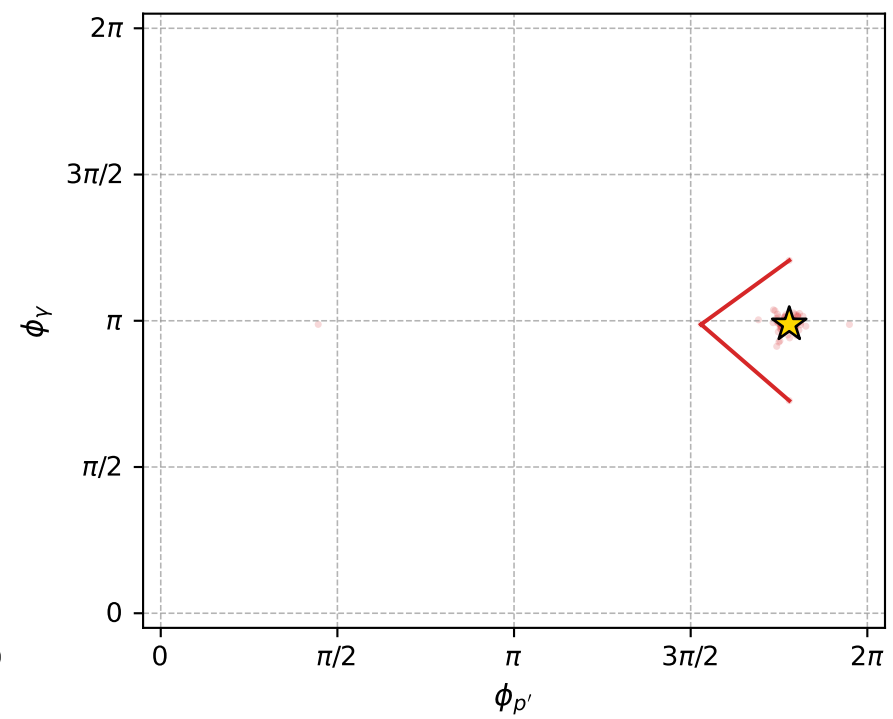
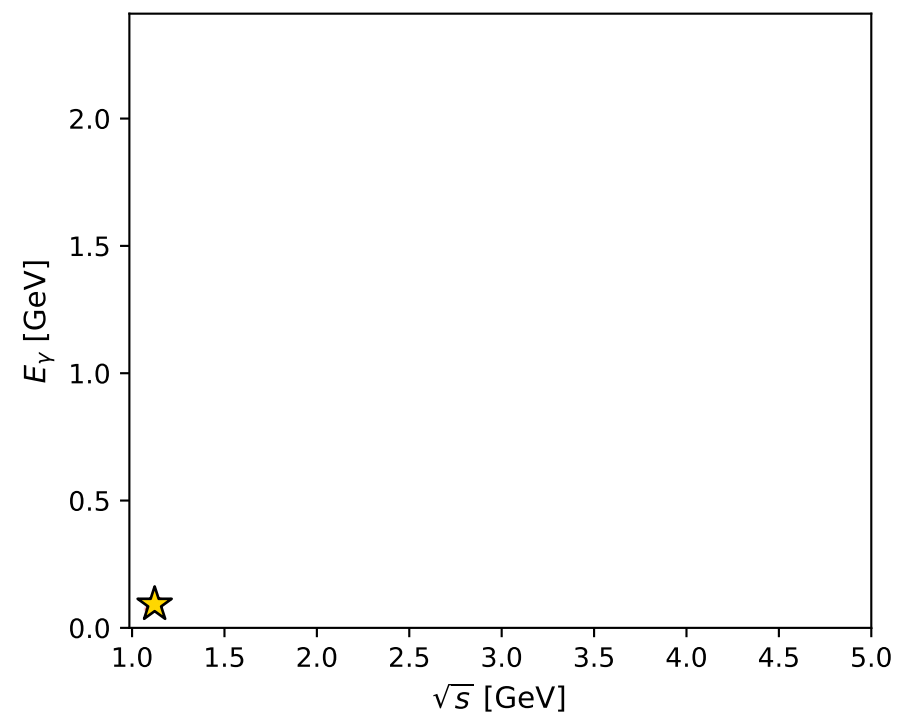
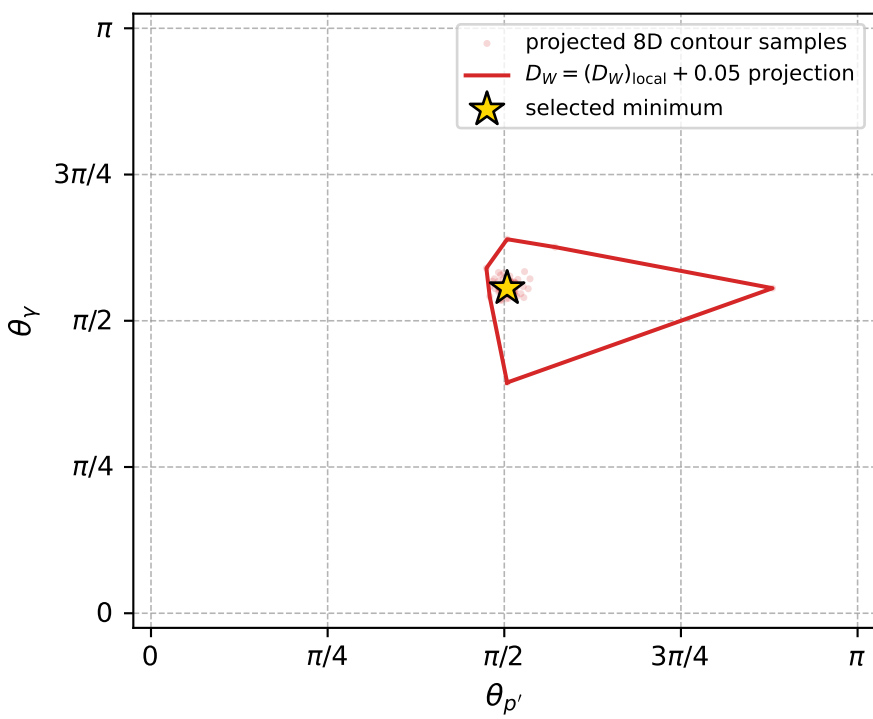
dw_mixing_angles_polarization_03_local_minimum_692 D_W=0.00686646
 region: polarization_cluster_03_local_minimum_692
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77290511$ rad
 incoming lepton: $0.715885|+\rangle +0.698218|-\rangle$
 proton mixing angle: $\alpha_p=0.19433853$ rad
 incoming proton: $0.981176|+\rangle +0.193118|-\rangle$

$s=1.25934$, $\sqrt{s}=1.1222$, $|\vec{P}|=0.169083$, $|\vec{P}'|=0.000887195$
 $\theta_{P'}=1.58343$, $\theta_Y=1.7457$, $\phi_{P'}=5.58881$, $\phi_Y=3.10279$
 $E_Y=0.0924443$, $\theta_{\ell'}=1.39445$, $\phi_{\ell'}=6.25037$
 $Q^2=0.0364723$, $x_B=0.173657$, $t=-0.0283652$, $W^2=1.0534$, $y=0.553439$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1691$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1691$
 P $E=0.9531$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1691$
 ℓ' $E=0.0918$ $p_x=0.0903$ $p_y=-0.0030$ $p_z=0.0161$
 P' $E=0.9380$ $p_x=0.0007$ $p_y=-0.0006$ $p_z=-0.0000$
 q_Y $E=0.0924$ $p_x=-0.0910$ $p_y=0.0035$ $p_z=-0.0161$



electron: polarization cluster P3, local minimum 692; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0068664611$
 $D_W \text{ contour} = 0.056866461$
 above global minimum = 0.0068500283
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.122202$
 $\text{theta_p_out} = 1.583433$
 $\text{theta_gamma_out} = 1.745696$
 $\text{qOut} = 0.09244428$
 $\text{phi_p_out} = 5.588808$
 $\text{phi_gamma_out} = 3.102795$
 $\text{alpha_e} = 0.7729051$
 $\text{alpha_p} = 0.1943385$

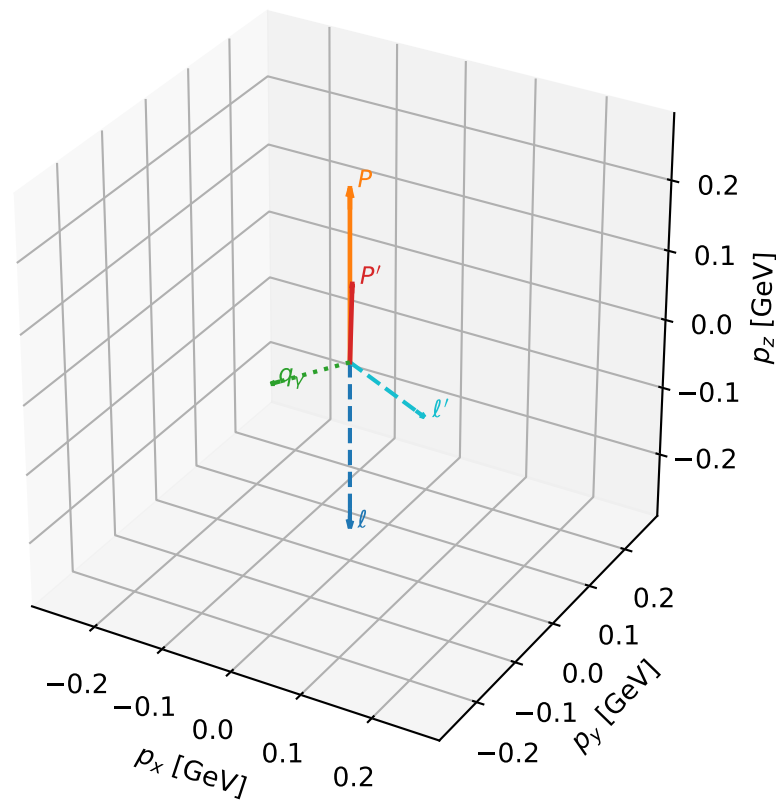
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_702 D_W=0.00731293
 region: polarization_cluster_03_local_minimum_702
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.83771976$ rad
 incoming lepton: $0.669159|+\rangle + 0.743119|-\rangle$
 proton mixing angle: $\alpha_p=0.28447884$ rad
 incoming proton: $0.959808|+\rangle + 0.280657|-\rangle$

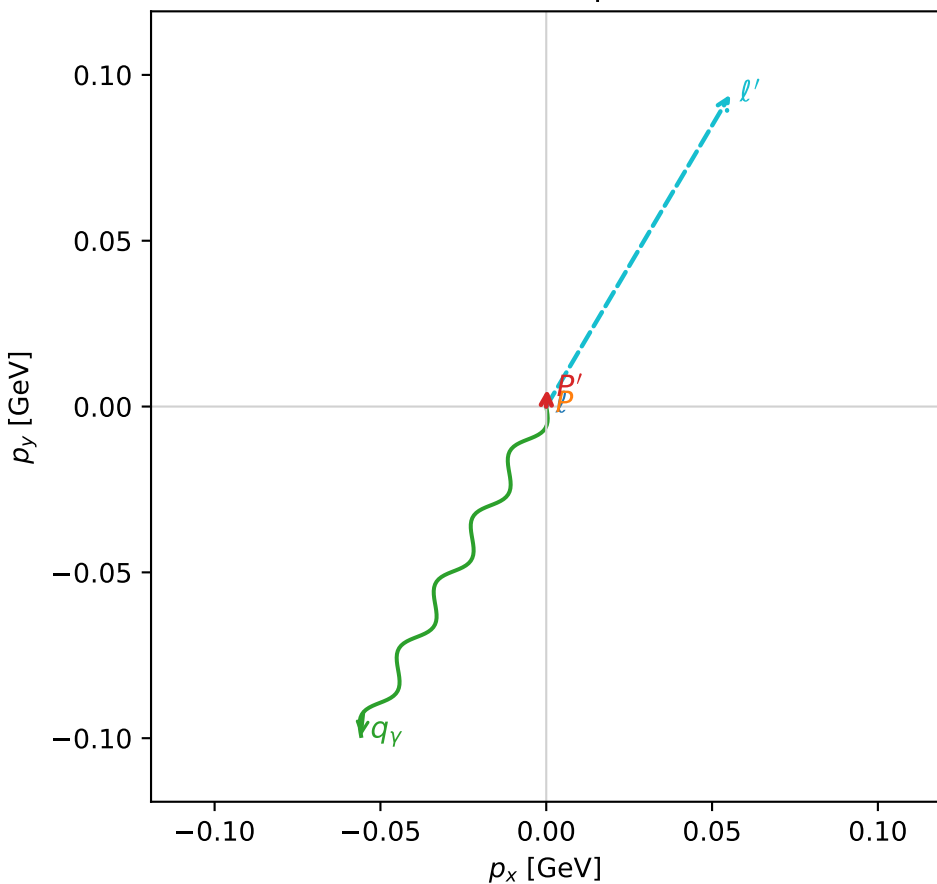
$s=1.47534$, $\sqrt{s}=1.21464$, $|\vec{P}|=0.245133$, $|\vec{P}'|=0.109465$
 $\theta_{P'}=0.0478135$, $\theta_Y=1.54913$, $\phi_{P'}=1.51206$, $\phi_Y=4.20024$
 $E_Y=0.11395$, $\theta_{\ell'}=2.3678$, $\phi_{\ell'}=1.03766$
 $Q^2=0.0218217$, $x_B=0.0918519$, $t=-0.0178352$, $W^2=1.0956$, $y=0.398954$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2451$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2451$
 P $E= 0.9695$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2451$
 ℓ' $E= 0.1563$ $p_x= 0.0555$ $p_y= 0.0941$ $p_z= -0.1118$
 P' $E= 0.9444$ $p_x= 0.0003$ $p_y= 0.0052$ $p_z= 0.1093$
 q_Y $E= 0.1140$ $p_x= -0.0558$ $p_y= -0.0993$ $p_z= 0.0025$

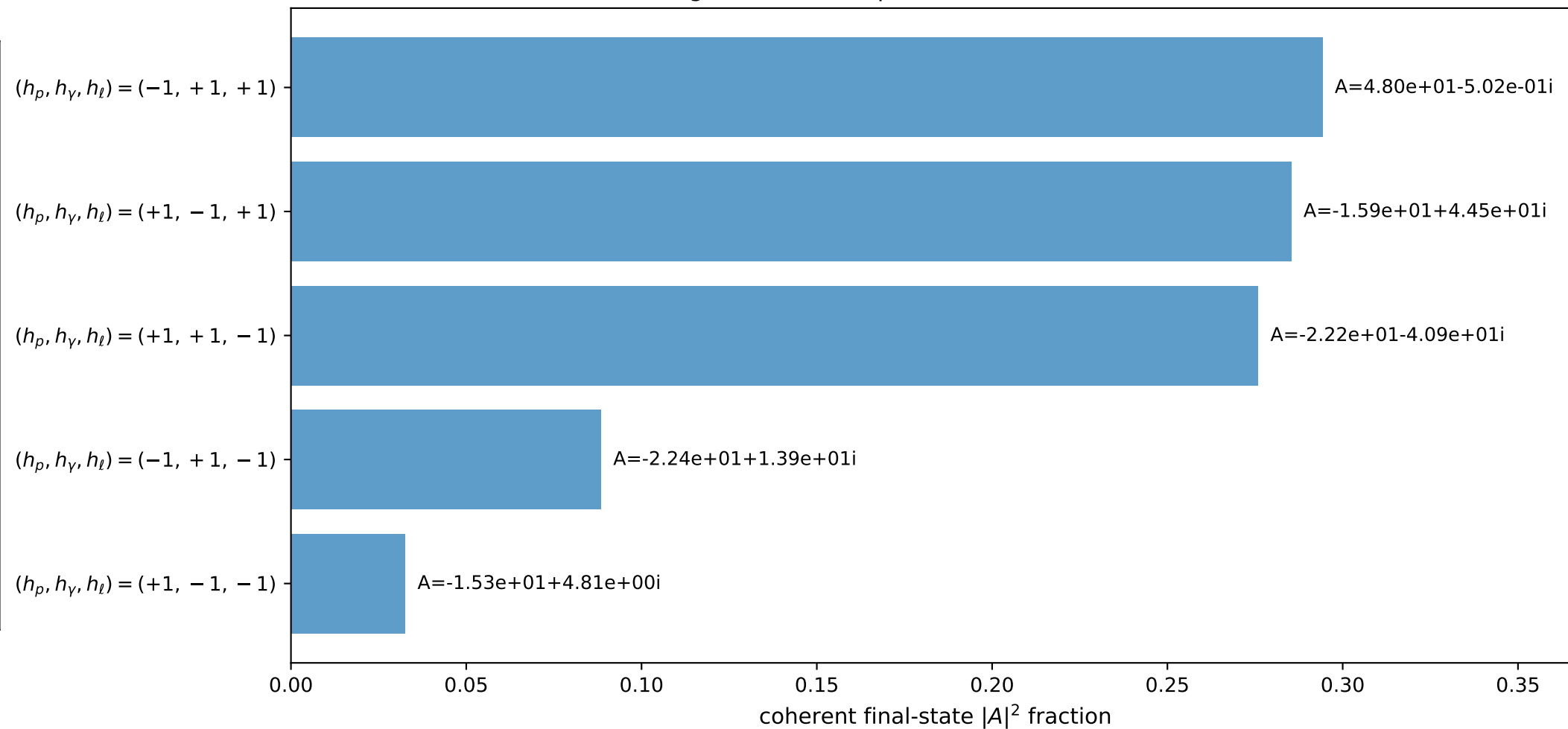
3D momenta



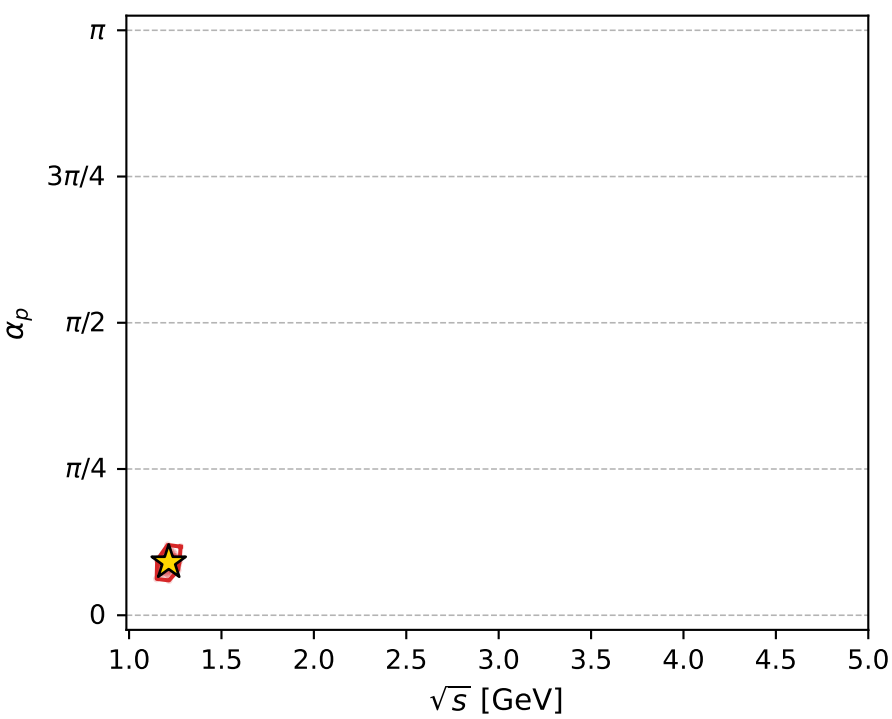
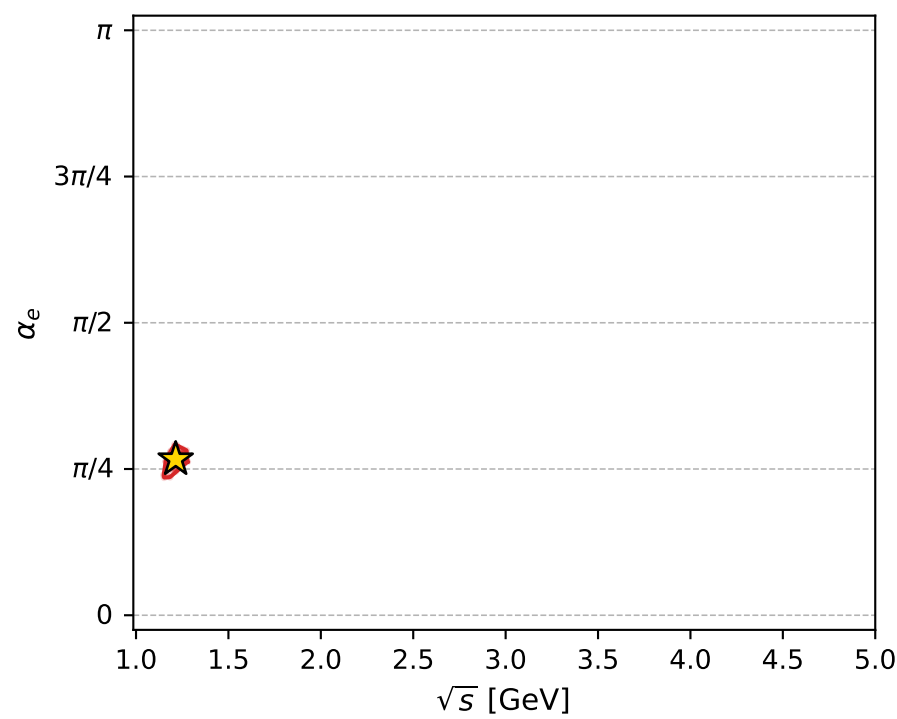
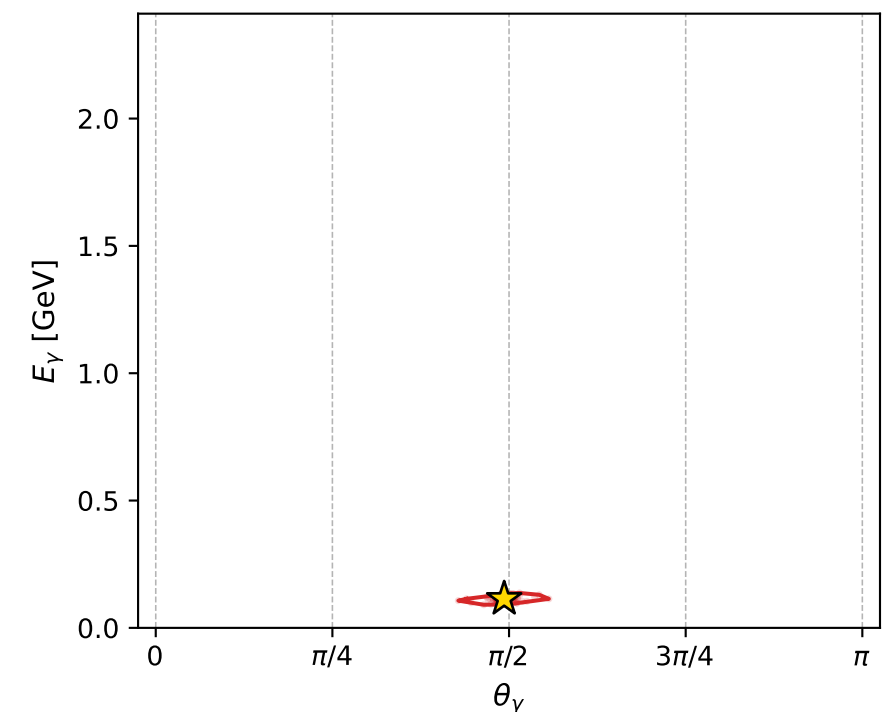
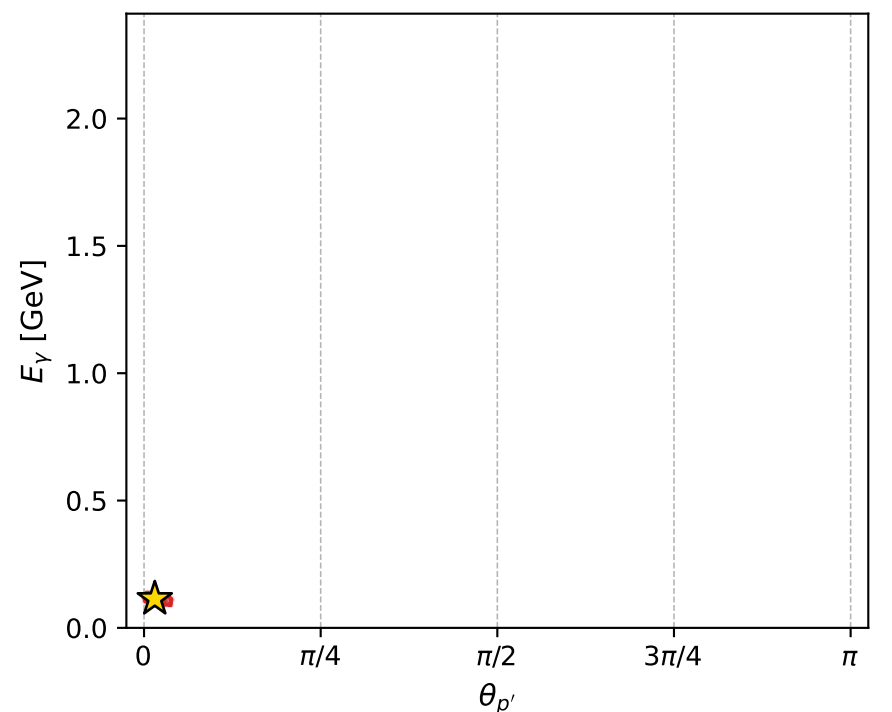
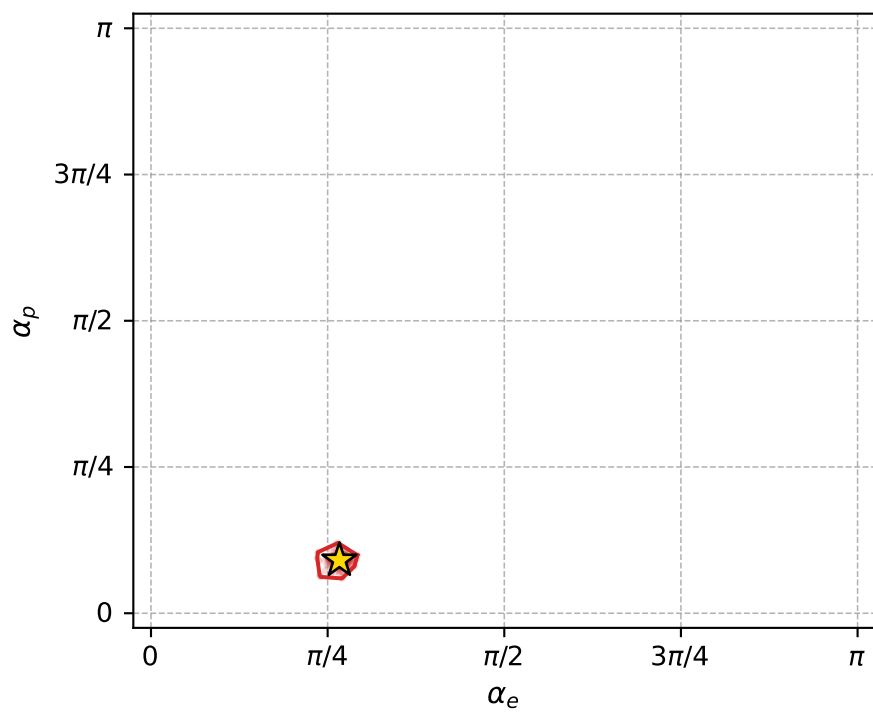
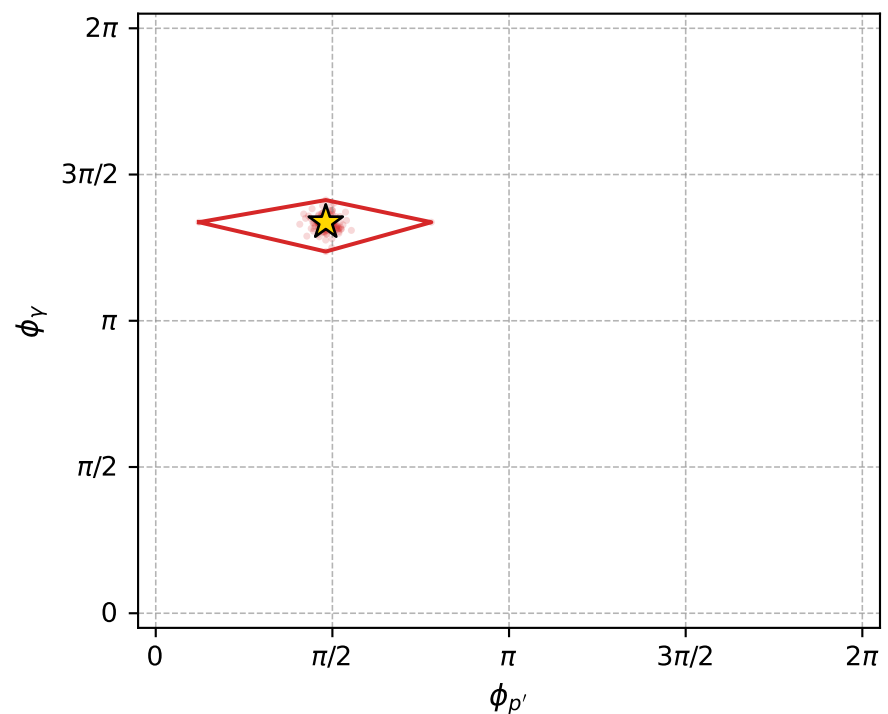
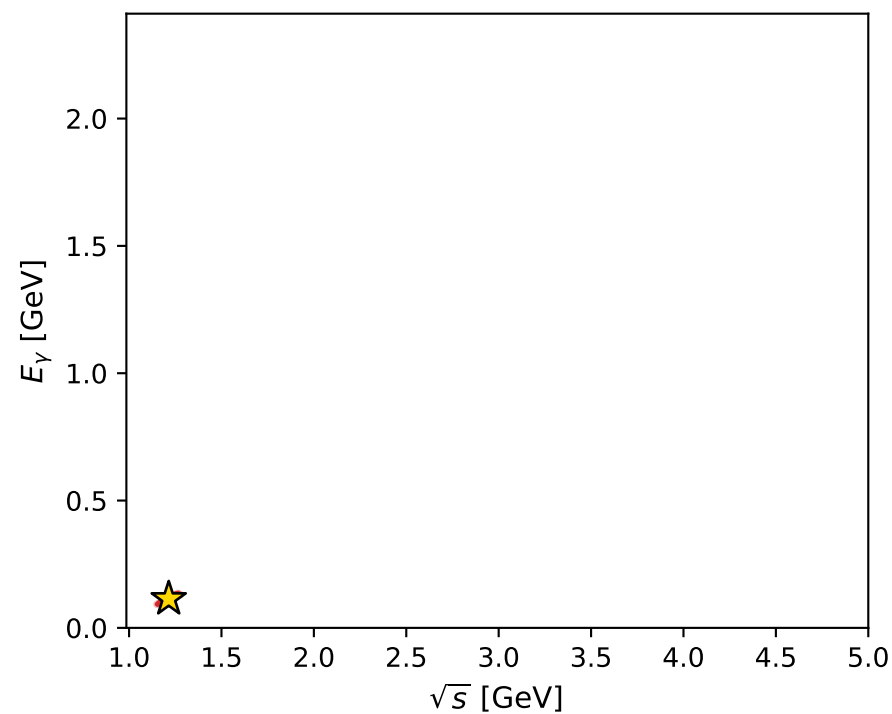
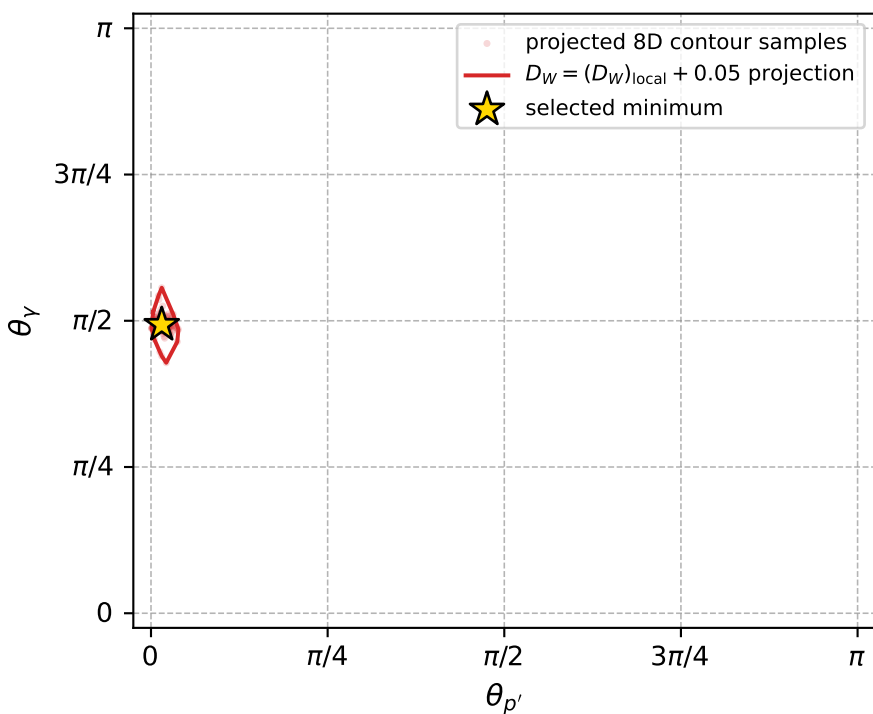
Transverse plane



Leading final-state amplitudes (N=5, retained=97.7%)



electron: polarization cluster P3, local minimum 702; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0073129331$
 $D_W \text{ contour} = 0.057312933$
 above global minimum = 0.0072965003
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.214635$
 $\text{theta_p_out} = 0.04781348$
 $\text{theta_gamma_out} = 1.54913$
 $\text{qOut} = 0.1139501$
 $\text{phi_p_out} = 1.512062$
 $\text{phi_gamma_out} = 4.200235$
 $\text{alpha_e} = 0.8377198$
 $\text{alpha_p} = 0.2844788$

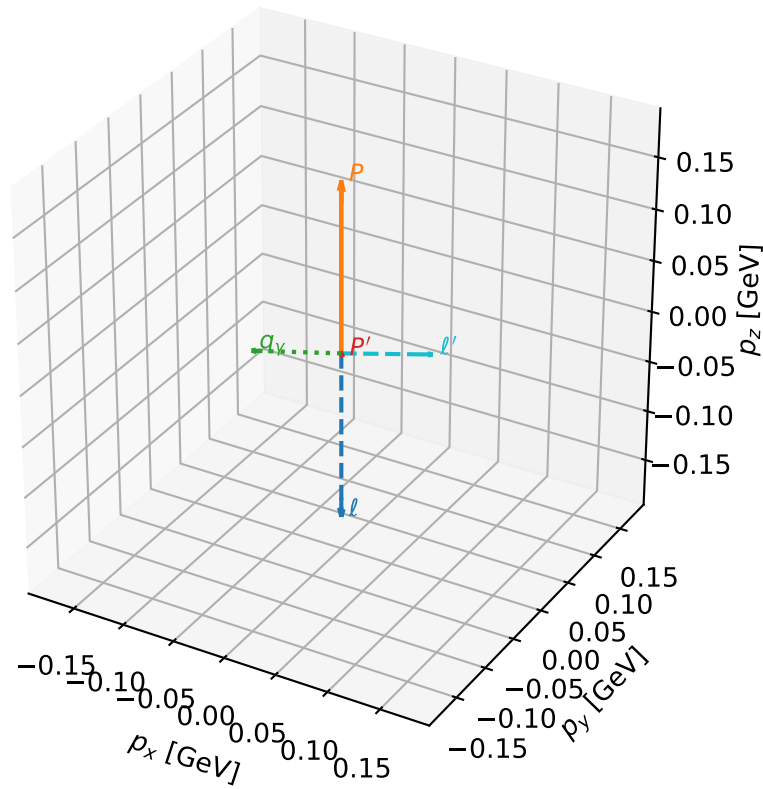
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_704 D_W=0.00743594
 region: polarization_cluster_03_local_minimum_704
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7651863$ rad
 incoming lepton: $0.721253|+\rangle + 0.692671|-\rangle$
 proton mixing angle: $\alpha_p=0.19293922$ rad
 incoming proton: $0.981445|+\rangle + 0.191744|-\rangle$

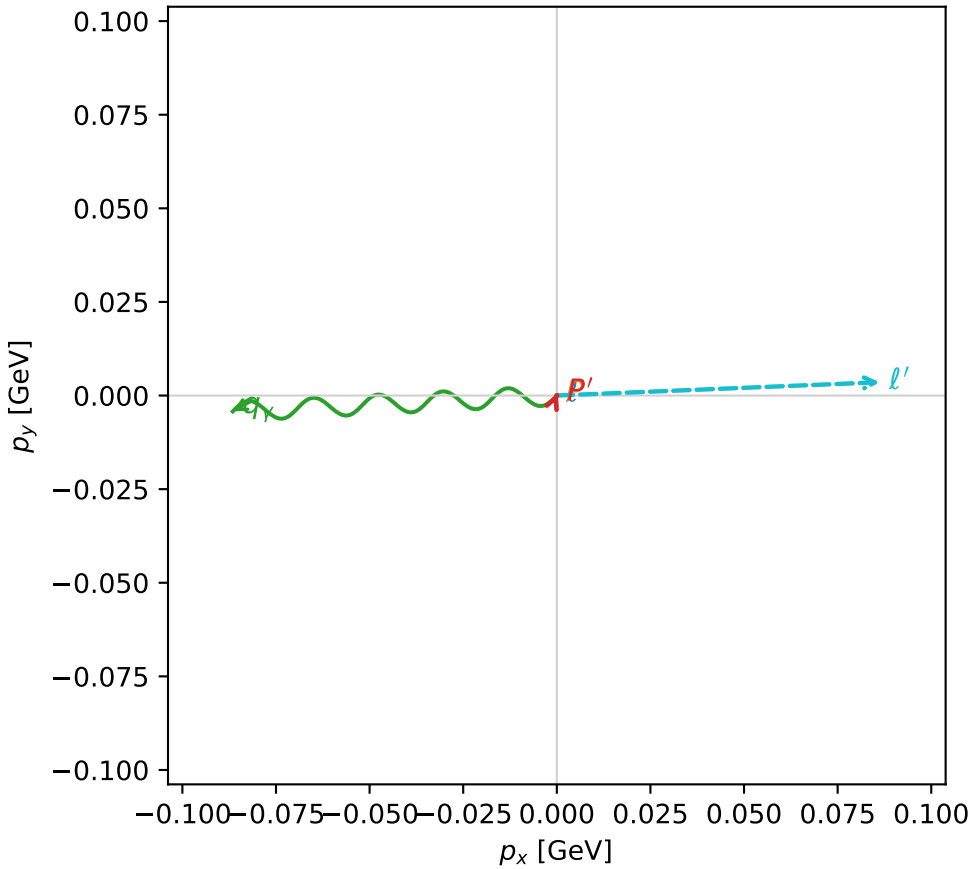
$s=1.24557$, $\sqrt{s}=1.11605$, $|\vec{P}|=0.163846$, $|\vec{P}'|=0.00200459$
 $\theta_{p'}=2.76927$, $\theta_Y=1.79959$, $\phi_{p'}=1.16918$, $\phi_Y=3.19046$
 $E_Y=0.0889545$, $\theta_{\ell'}=1.32079$, $\phi_{\ell'}=0.0412666$
 $Q^2=0.0364176$, $x_B=0.179153$, $t=-0.0272598$, $W^2=1.0467$, $y=0.555822$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1638$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1638$
 P $E= 0.9522$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1638$
 ℓ' $E= 0.0891$ $p_x= 0.0862$ $p_y= 0.0036$ $p_z= 0.0220$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= 0.0007$ $p_z= -0.0019$
 q_Y $E= 0.0890$ $p_x= -0.0865$ $p_y= -0.0042$ $p_z= -0.0202$

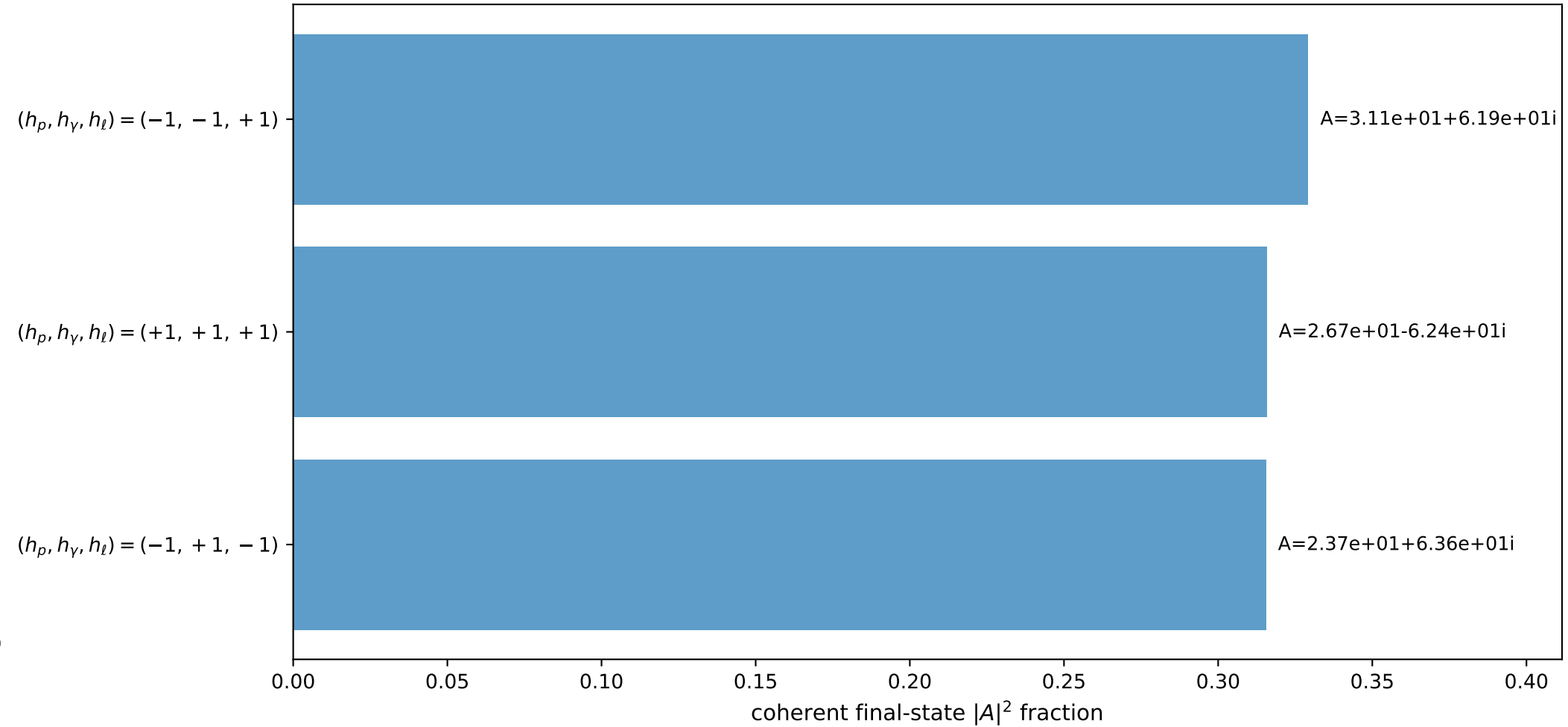
3D momenta



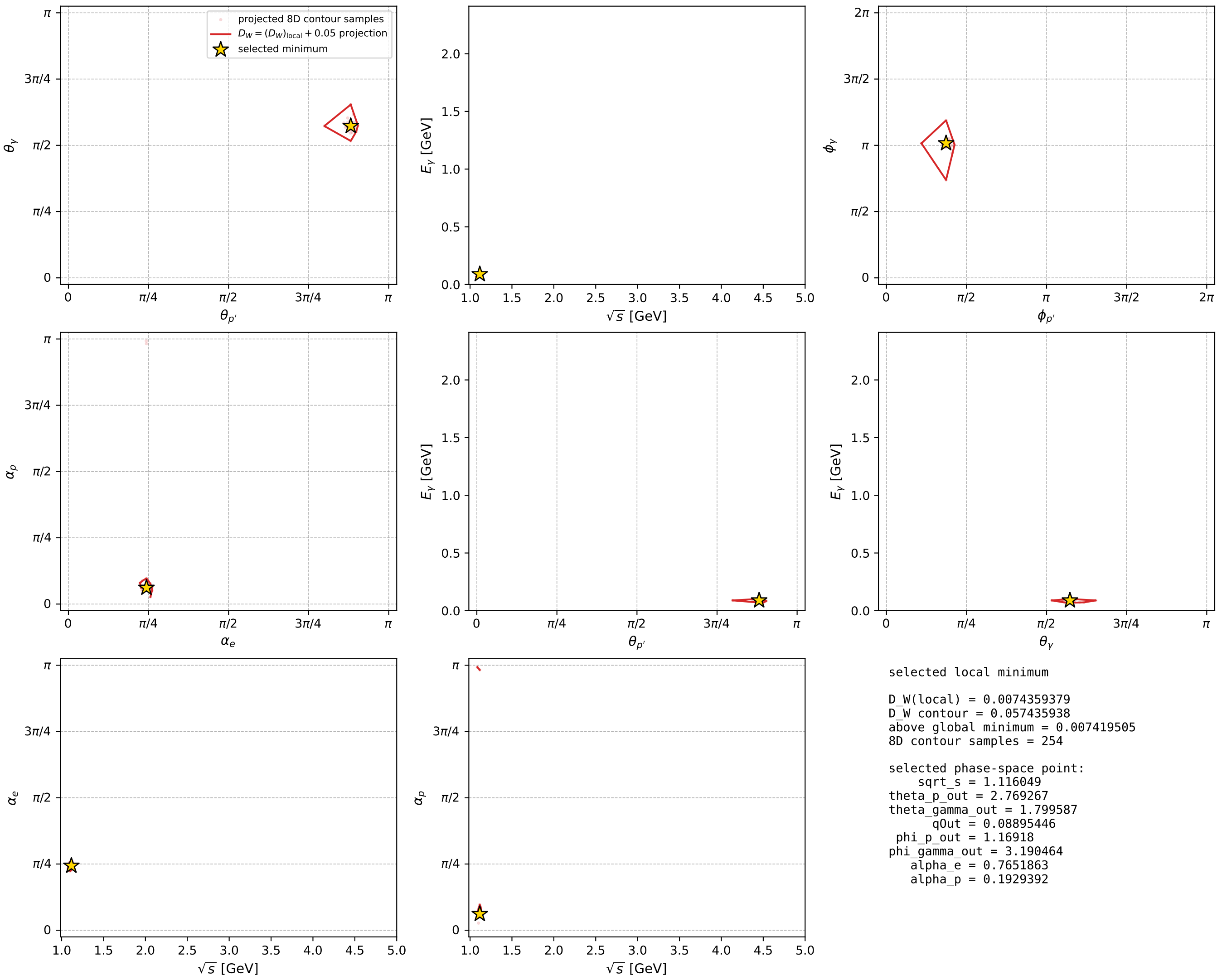
Transverse plane



Leading final-state amplitudes (N=3, retained=96.1%)



electron: polarization cluster P3, local minimum 704; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

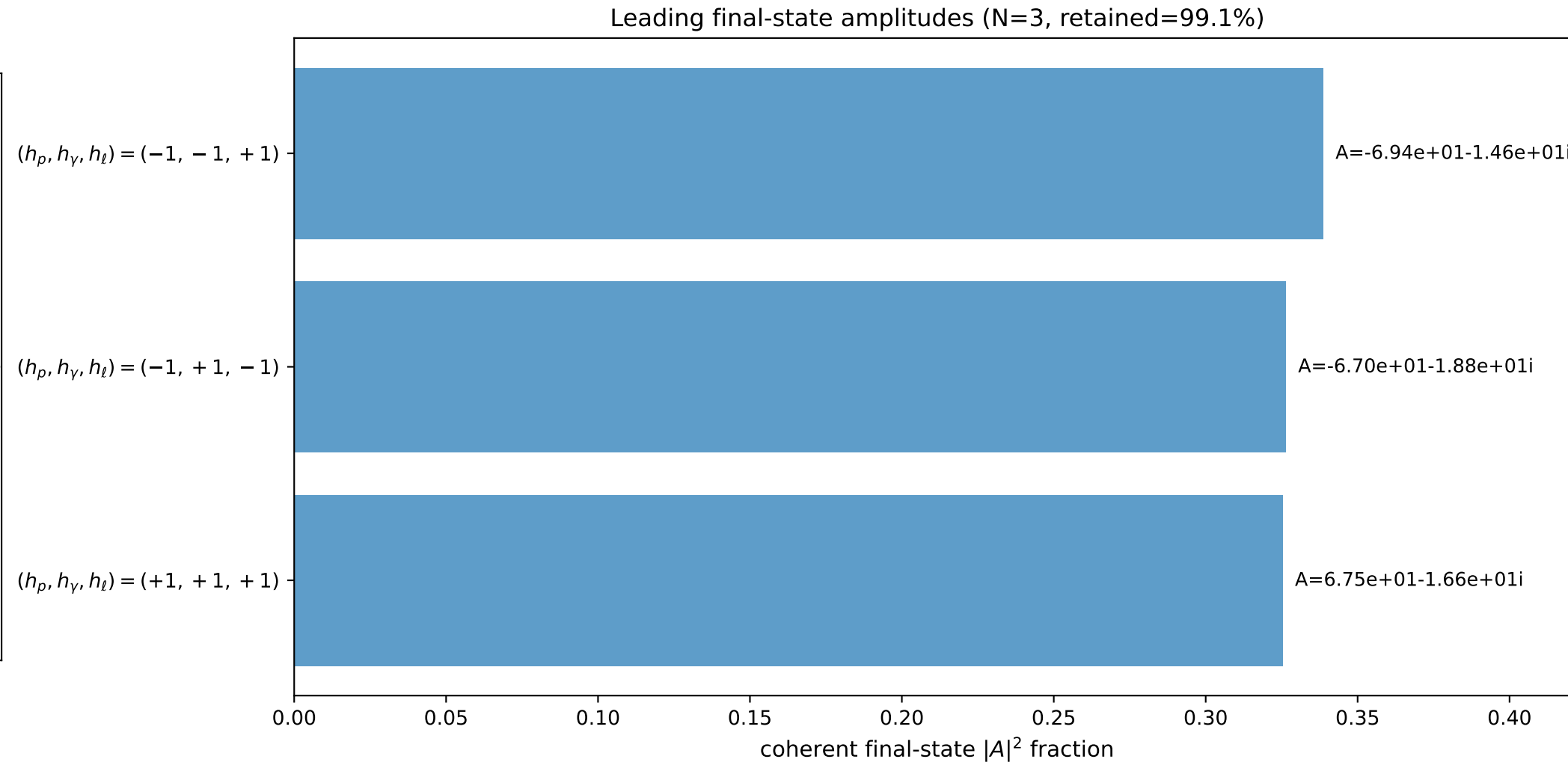
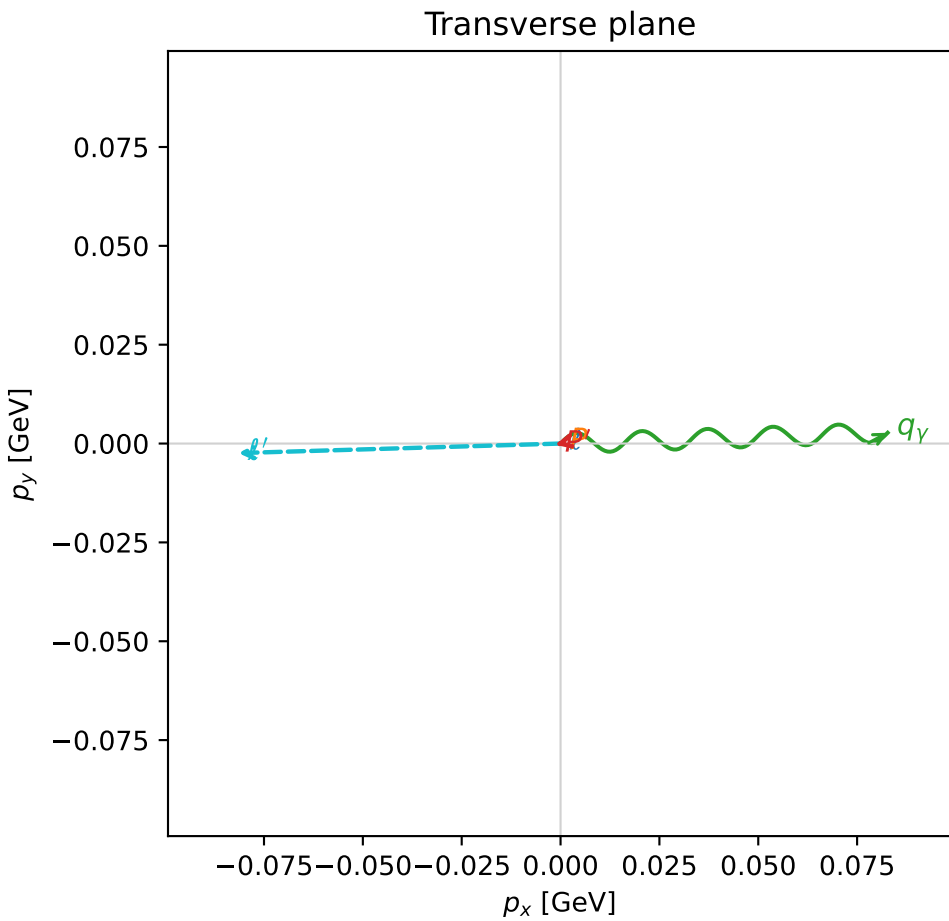
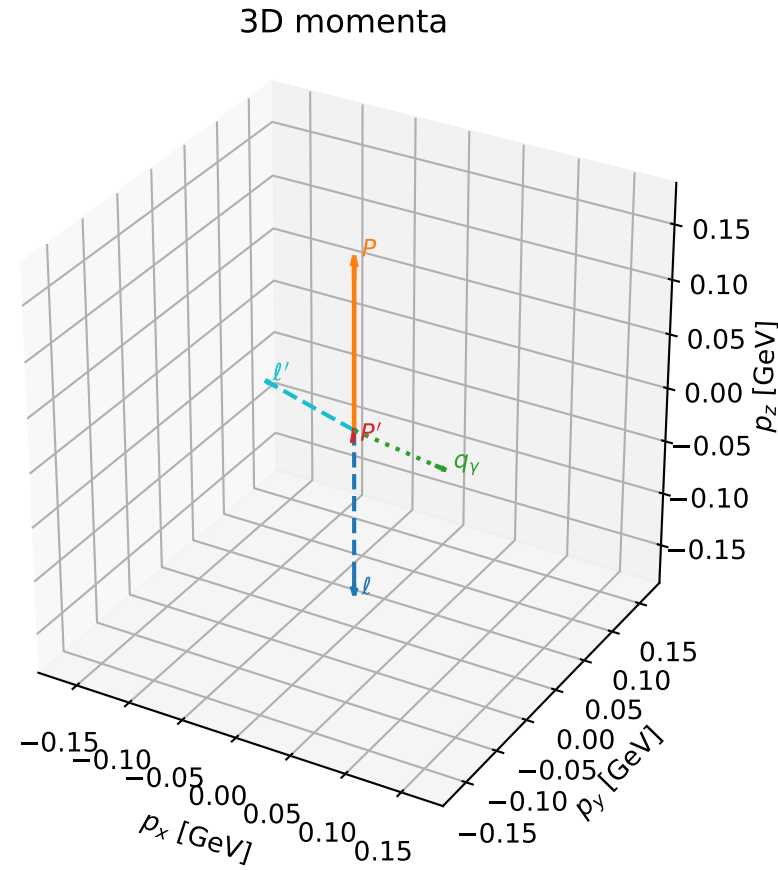


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

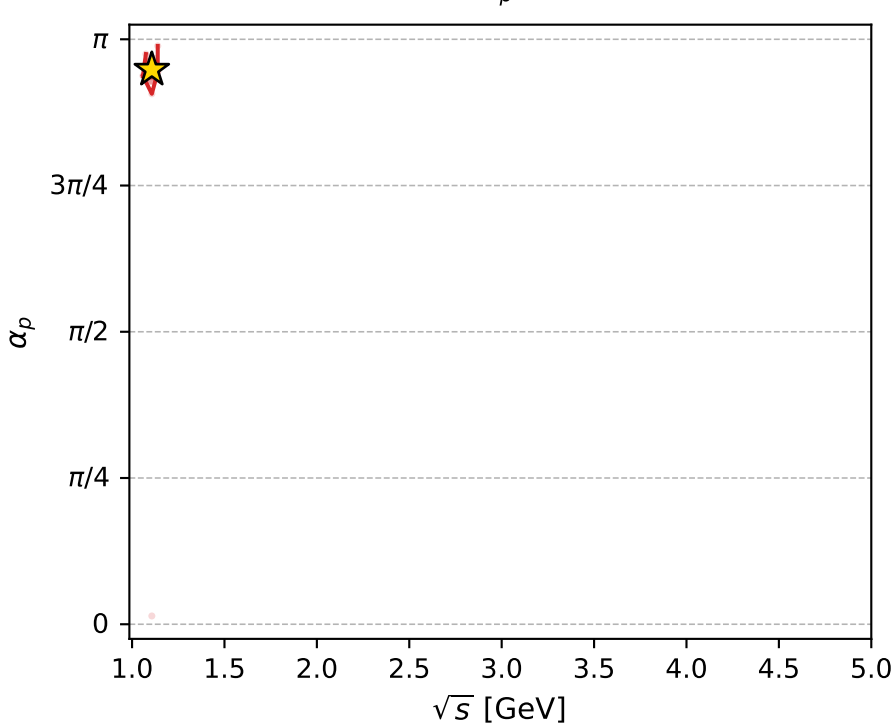
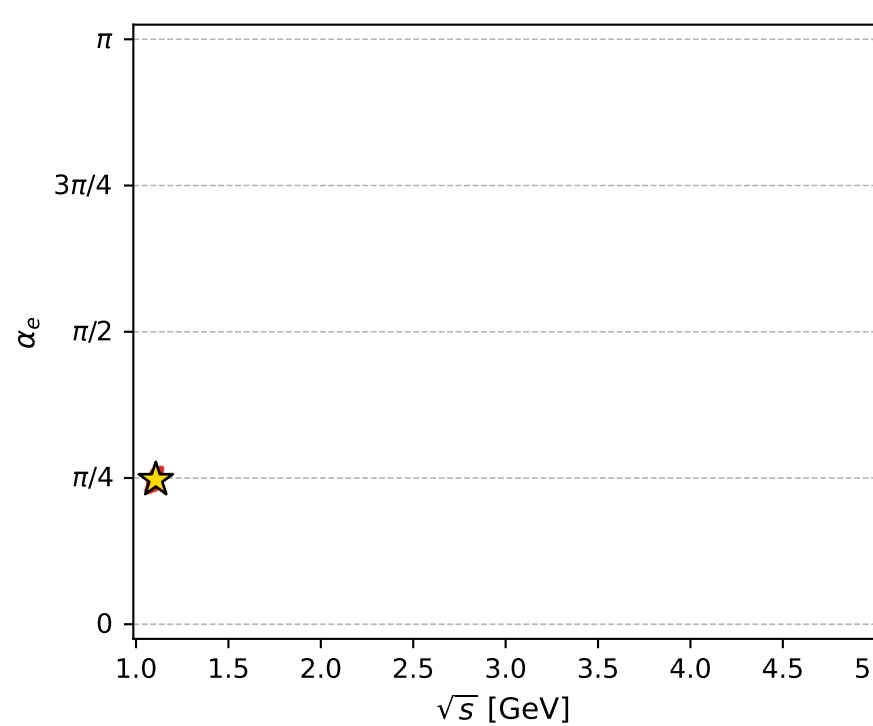
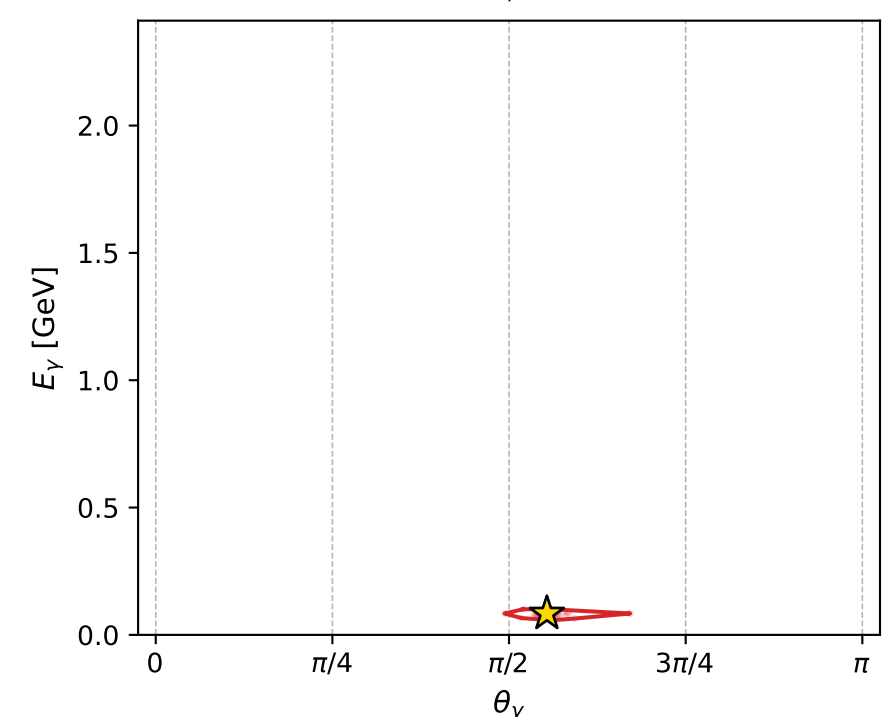
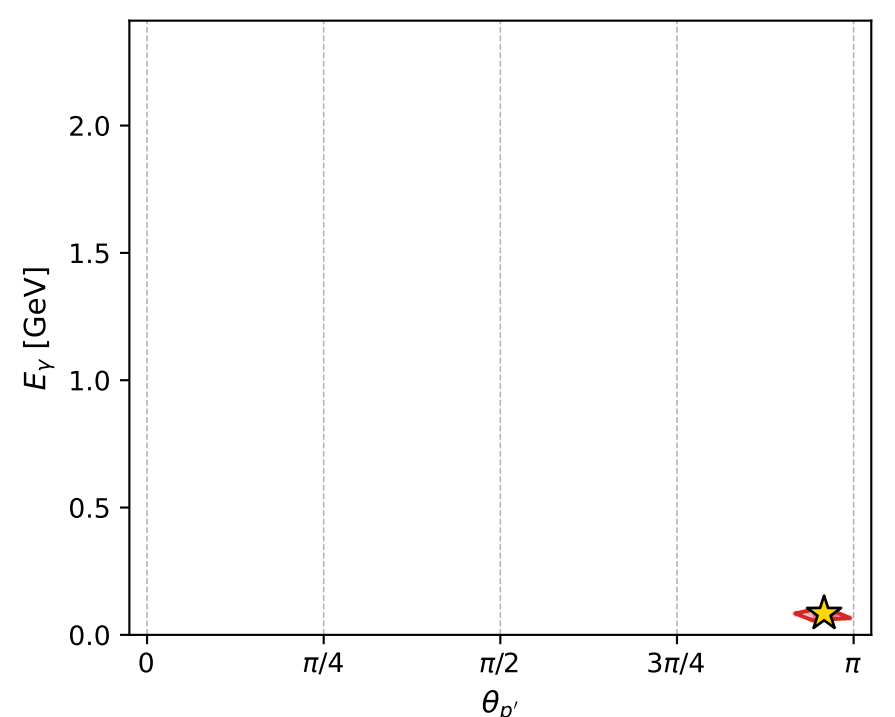
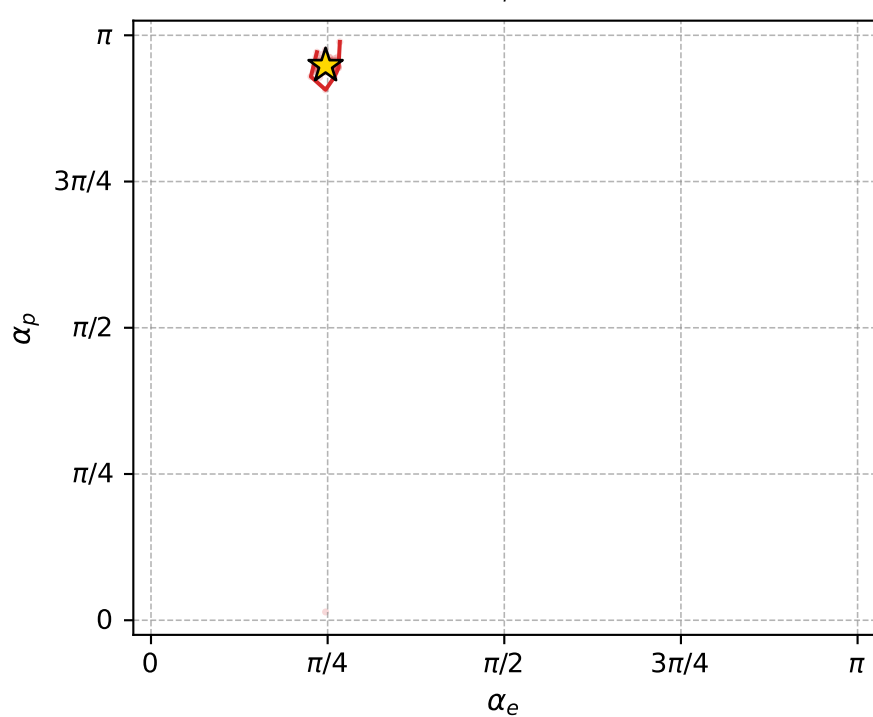
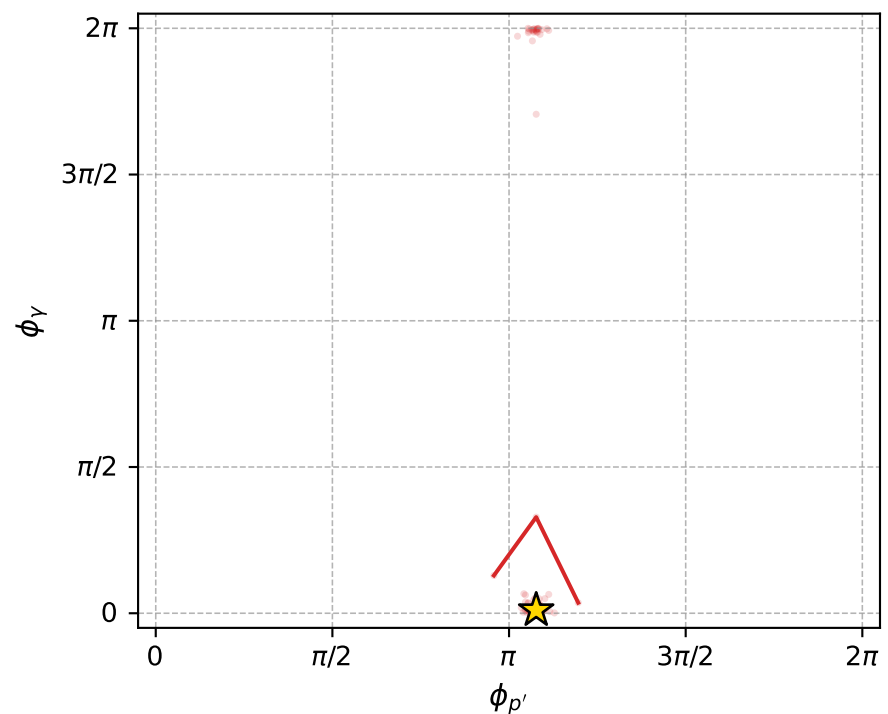
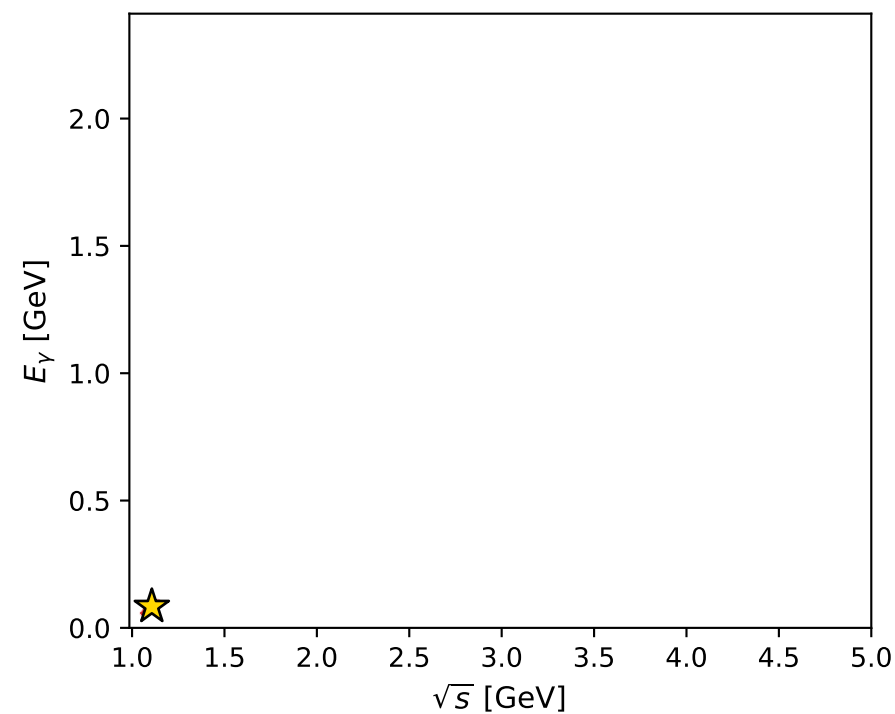
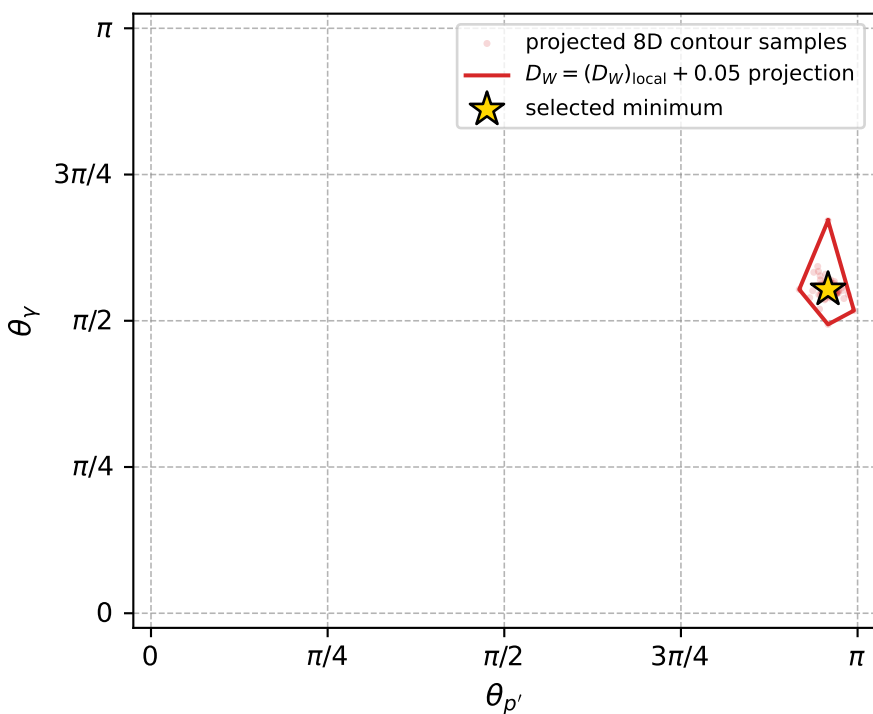
dw_mixing_angles_polarization_03_local_minimum_707 D_W=0.00749259
 region: polarization_cluster_03_local_minimum_707
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77624153$ rad
 incoming lepton: $0.713552|+\rangle +0.700603|-\rangle$
 proton mixing angle: $\alpha_p=2.9784057$ rad
 incoming proton: $-0.986715|+\rangle +0.162464|-\rangle$

$s=1.22541$, $\sqrt{s}=1.10698$, $|\vec{P}|=0.156084$, $|\vec{P}'|=0.0100748$
 $\theta_{p'}=3.01025$, $\theta_Y=1.73909$, $\phi_{p'}=3.38298$, $\phi_Y=0.0329562$
 $E_Y=0.0839642$, $\theta_{\ell'}=1.28379$, $\phi_{\ell'}=3.1712$
 $Q^2=0.0340311$, $x_B=0.177718$, $t=-0.0274168$, $W^2=1.0373$, $y=0.554133$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1561$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1561$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1561$
 ℓ' $E= 0.0850$ $p_x= -0.0815$ $p_y= -0.0024$ $p_z= 0.0241$
 P' $E= 0.9381$ $p_x= -0.0013$ $p_y= -0.0003$ $p_z= -0.0100$
 q_Y $E= 0.0840$ $p_x= 0.0827$ $p_y= 0.0027$ $p_z= -0.0141$



electron: polarization cluster P3, local minimum 707; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0074925878$
 $D_W \text{ contour} = 0.057492588$
 above global minimum = 0.0074761549
 8D contour samples = 252

selected phase-space point:
 $\text{sqrt_s} = 1.106983$
 $\text{theta_p_out} = 3.010249$
 $\text{theta_gamma_out} = 1.739086$
 $\text{qOut} = 0.08396415$
 $\text{phi_p_out} = 3.382983$
 $\text{phi_gamma_out} = 0.03295616$
 $\text{alpha_e} = 0.7762415$
 $\text{alpha_p} = 2.978406$

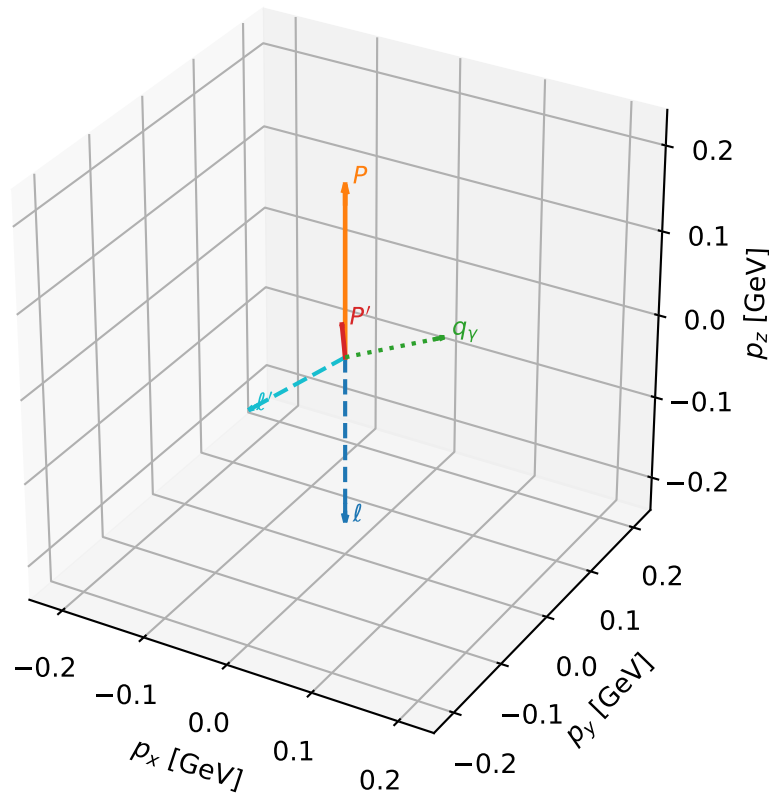
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_711 D_W=0.00781589
 region: polarization_cluster_03_local_minimum_711
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80950599$ rad
 incoming lepton: $0.689856|+\rangle +0.723946|-\rangle$
 proton mixing angle: $\alpha_p=2.9748593$ rad
 incoming proton: $-0.986132|+\rangle +0.165962|-\rangle$

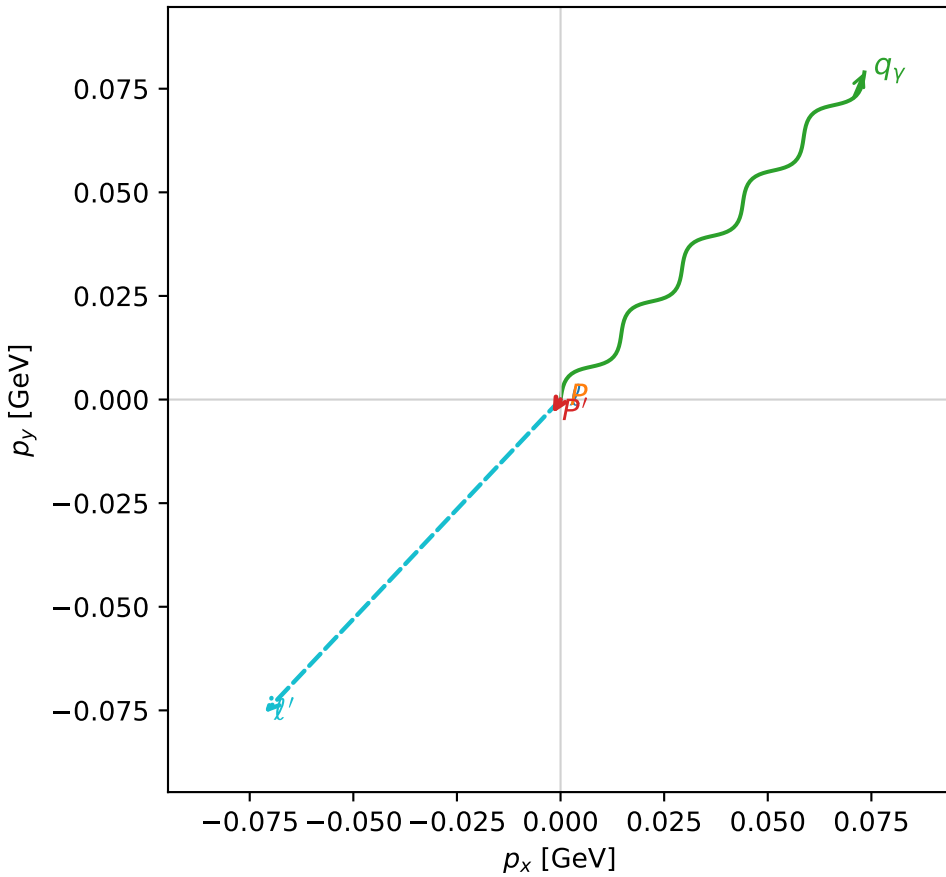
$s=1.3464$, $\sqrt{s}=1.16035$, $|\vec{P}|=0.201042$, $|\vec{P}'|=0.0401436$
 $\theta_{p'}=0.0955346$, $\theta_Y=1.51737$, $\phi_{p'}=4.18615$, $\phi_Y=0.822285$
 $E_Y=0.107882$, $\theta_{\ell'}=1.98501$, $\phi_{\ell'}=3.95576$
 $Q^2=0.0272942$, $x_B=0.118562$, $t=-0.0255438$, $W^2=1.08276$, $y=0.493424$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2010$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2010$
 P $E= 0.9593$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2010$
 ℓ' $E= 0.1136$ $p_x= -0.0714$ $p_y= -0.0756$ $p_z= -0.0457$
 P' $E= 0.9389$ $p_x= -0.0019$ $p_y= -0.0033$ $p_z= 0.0400$
 q_Y $E= 0.1079$ $p_x= 0.0733$ $p_y= 0.0789$ $p_z= 0.0058$

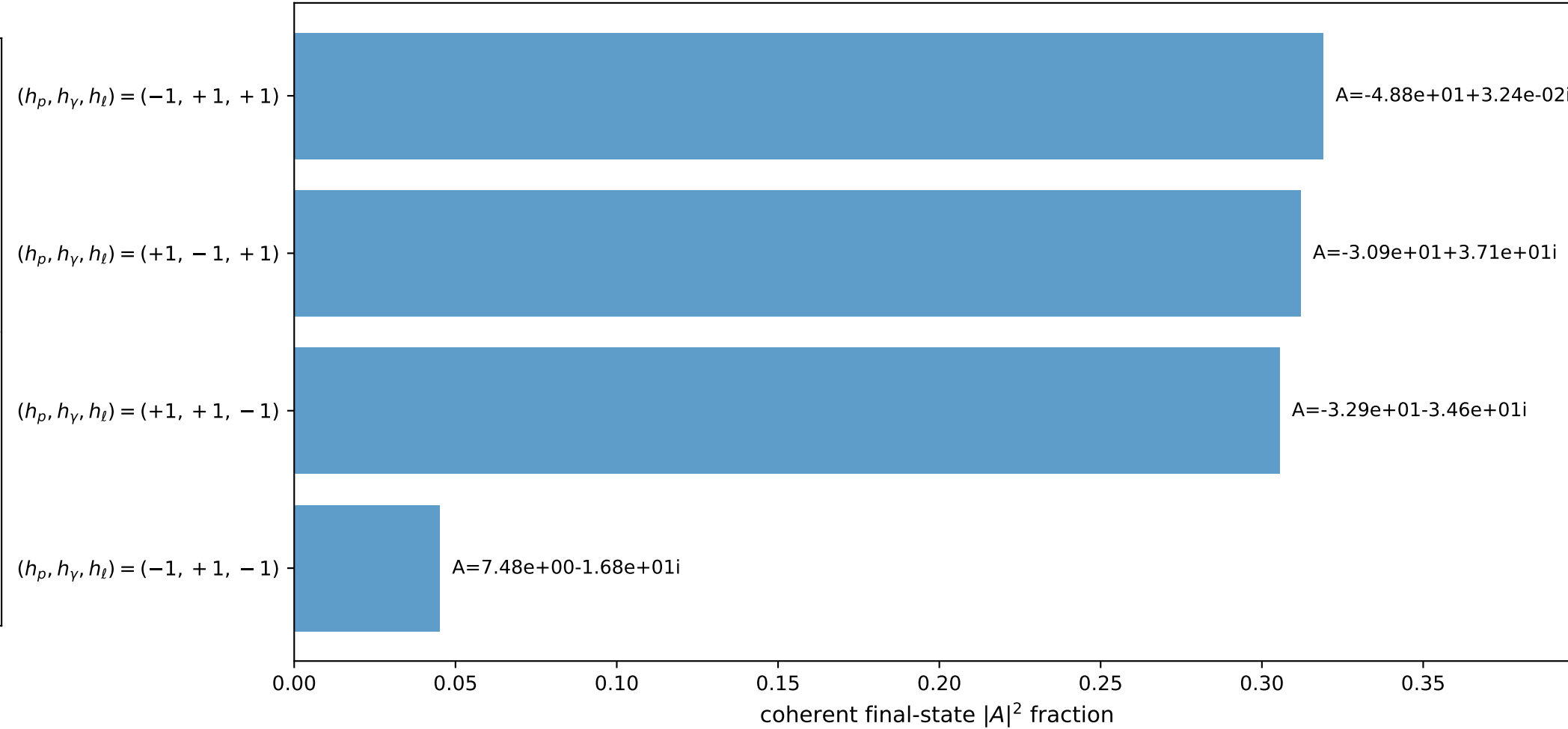
3D momenta



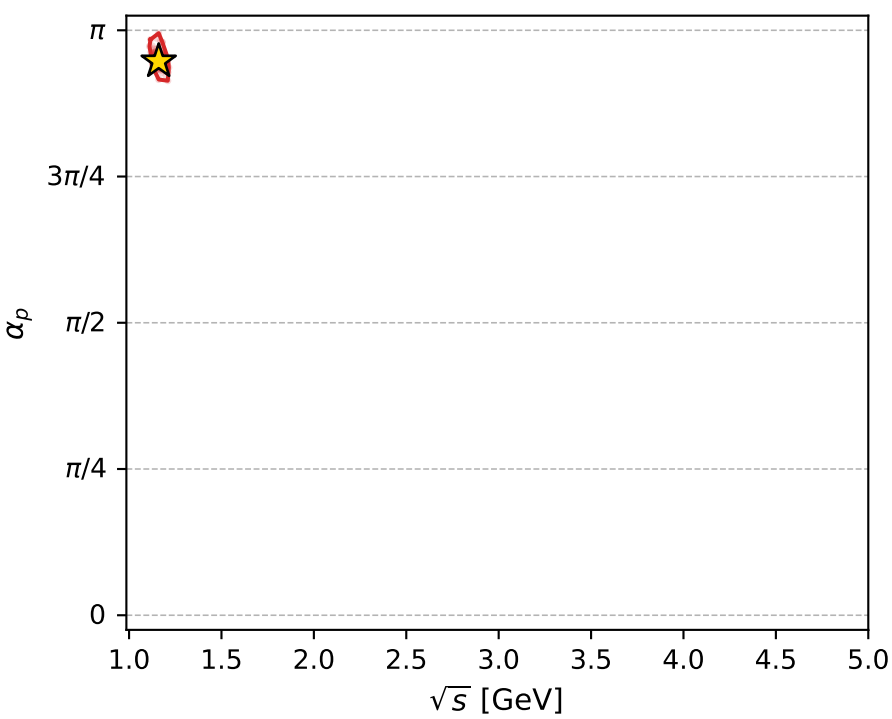
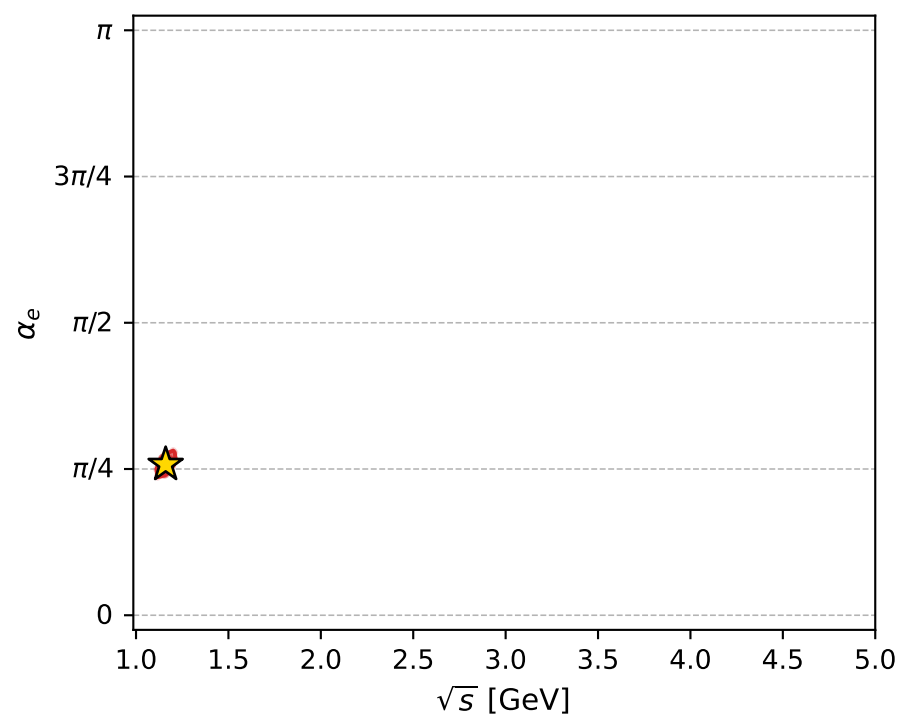
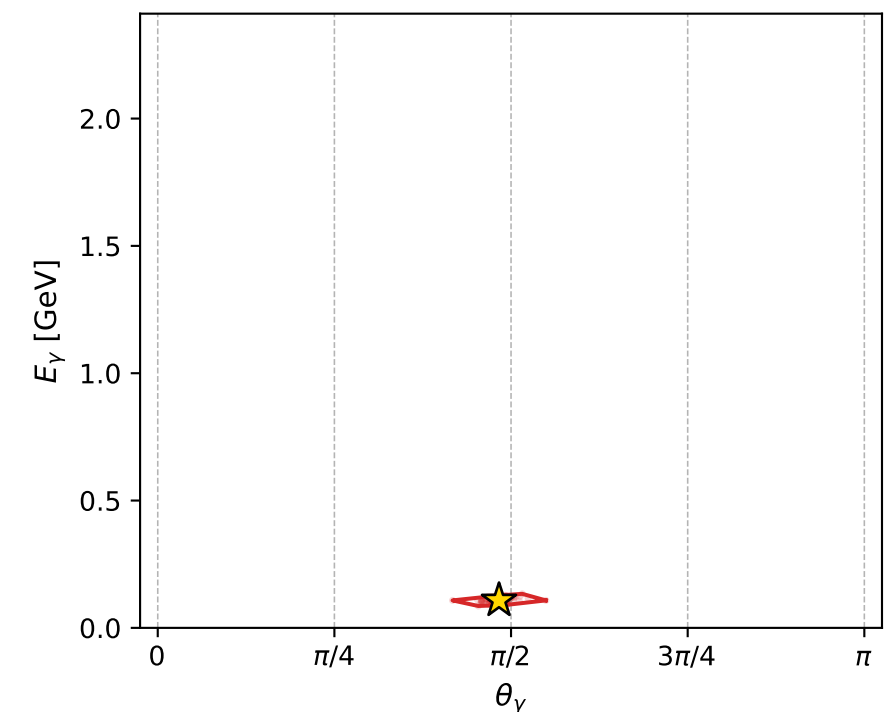
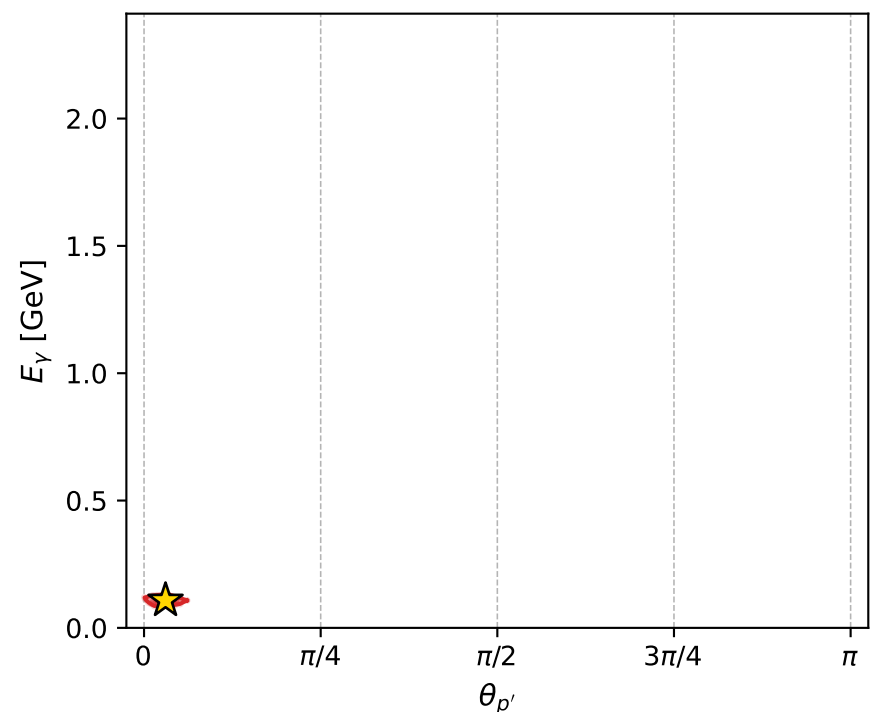
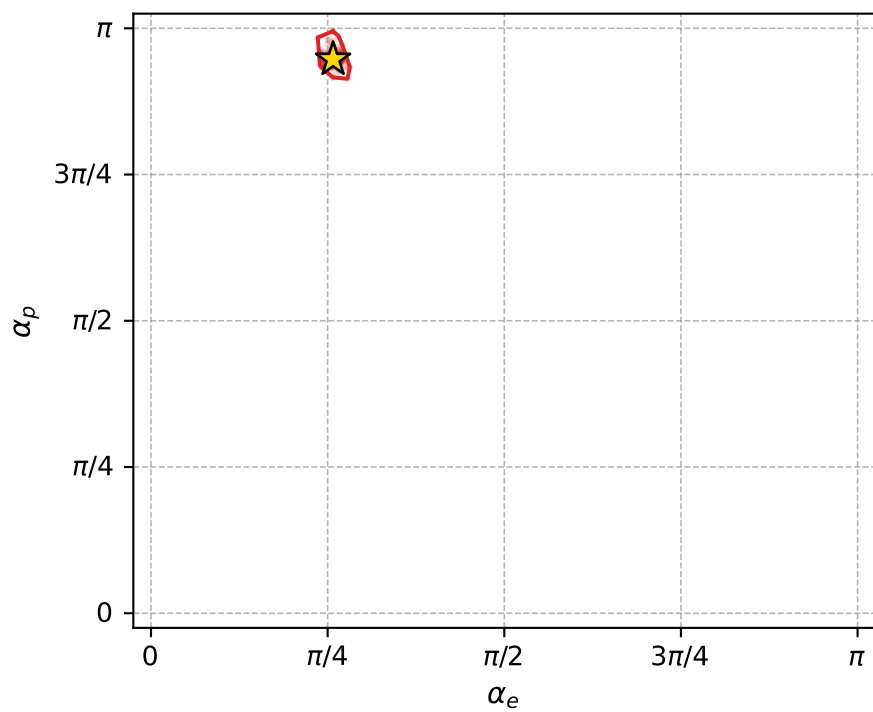
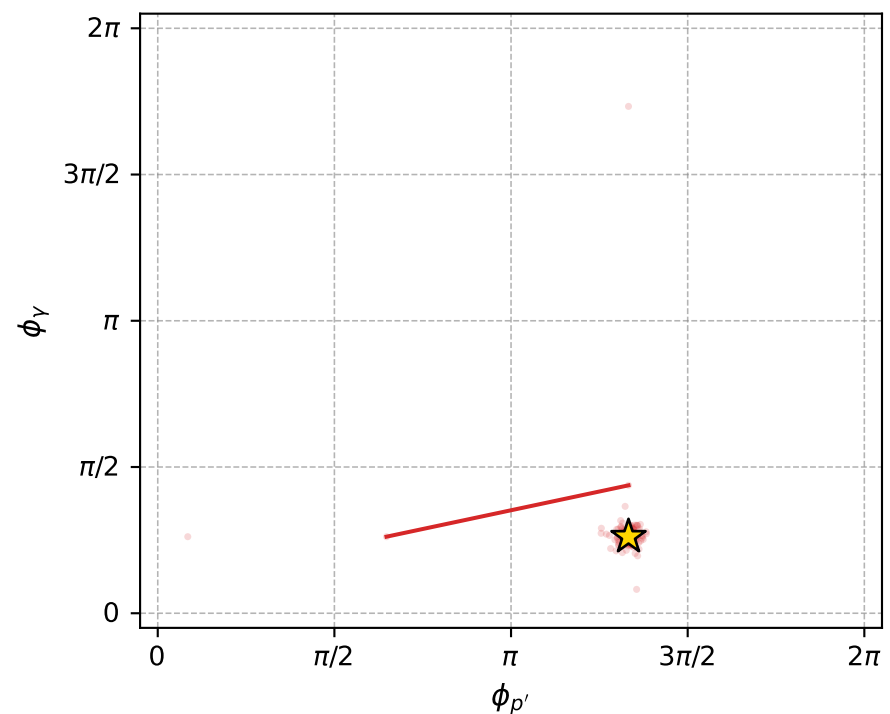
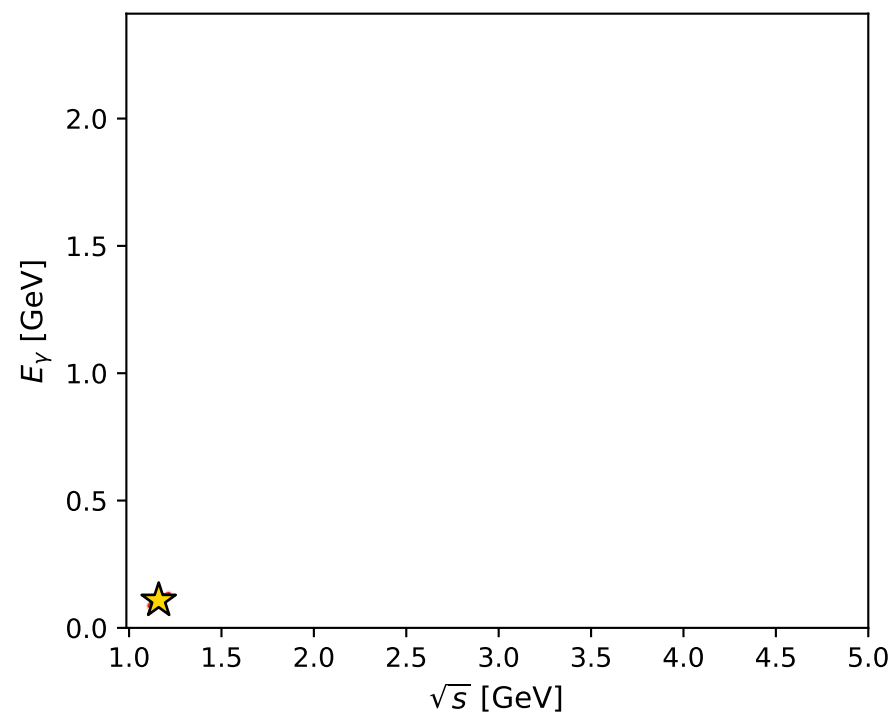
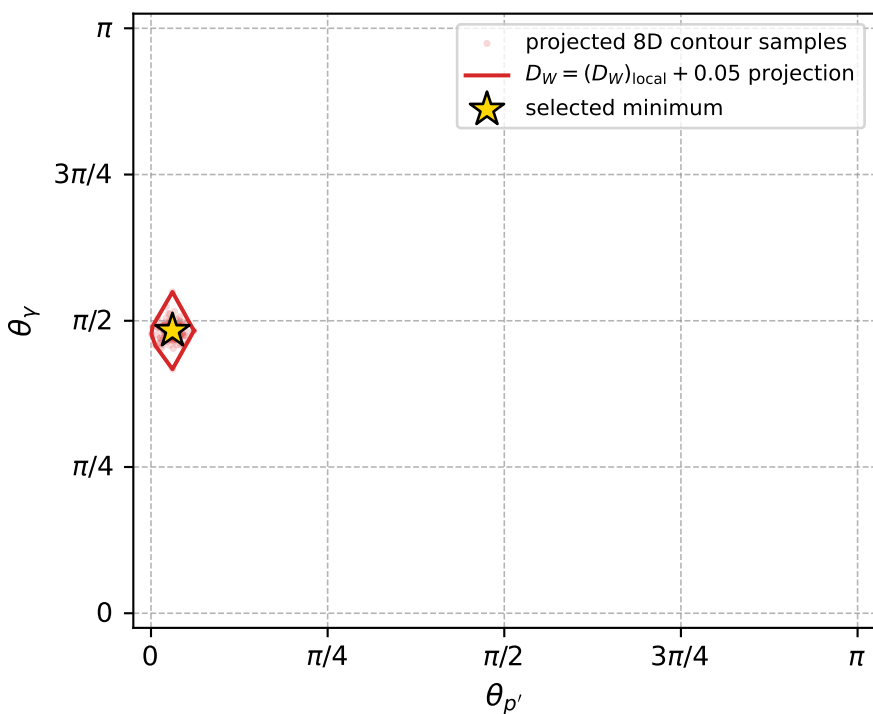
Transverse plane



Leading final-state amplitudes (N=4, retained=98.2%)



electron: polarization cluster P3, local minimum 711; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0078158924$
 $D_W \text{ contour} = 0.057815892$
 above global minimum = 0.0077994595
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.160345$
 $\text{theta_p_out} = 0.09553463$
 $\text{theta_gamma_out} = 1.517368$
 $\text{qOut} = 0.1078819$
 $\text{phi_p_out} = 4.186153$
 $\text{phi_gamma_out} = 0.8222845$
 $\text{alpha_e} = 0.809506$
 $\text{alpha_p} = 2.974859$

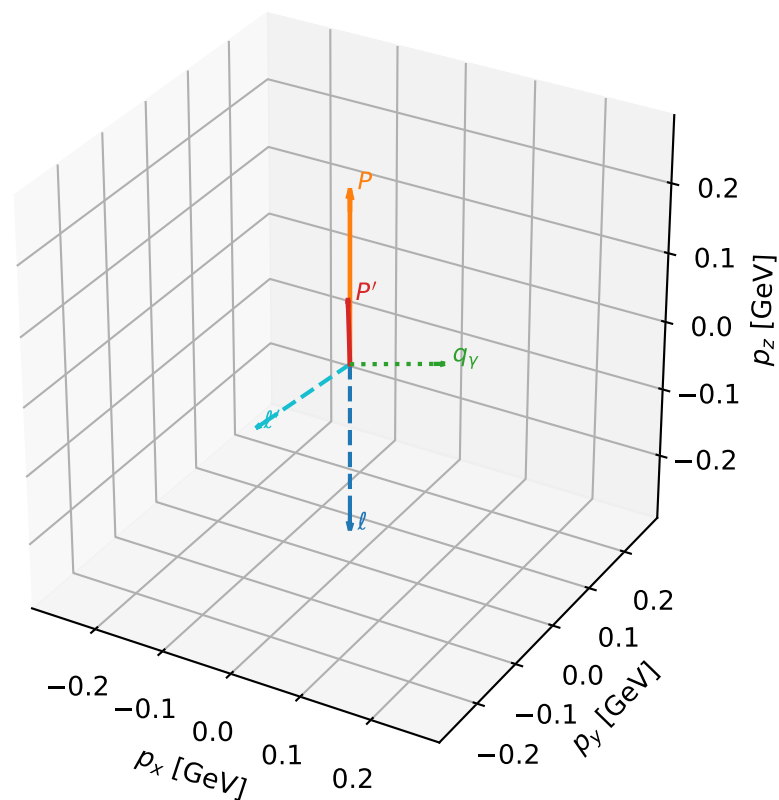
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_712 D_W=0.00783606
region: polarization_cluster_03_local_minimum_712
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.82127251$ rad
incoming lepton: $0.68129|+\rangle +0.732013|-\rangle$
proton mixing angle: $\alpha_p=2.9419744$ rad
incoming proton: $-0.980142|+\rangle +0.198295|-\rangle$

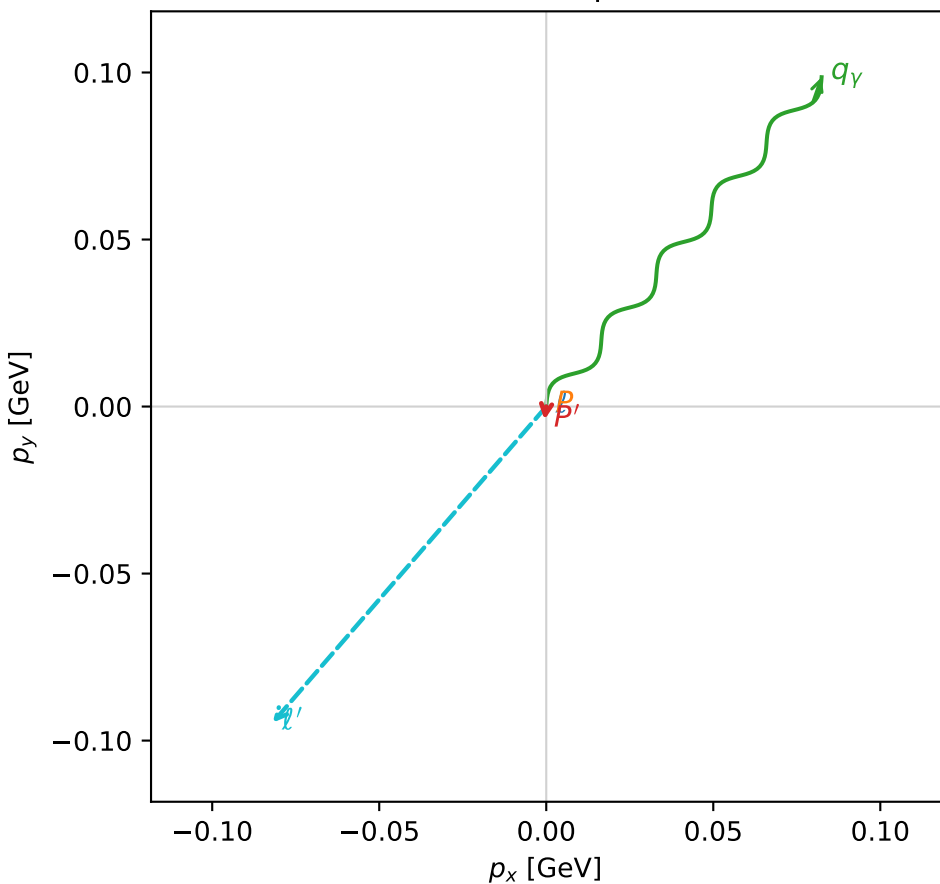
$s=1.4801$, $\sqrt{s}=1.21659$, $|\vec{P}|=0.246696$, $|\vec{P}'|=0.0941713$
 $\theta_{P'}=0.0424345$, $\theta_Y=1.76284$, $\phi_{P'}=4.5803$, $\phi_Y=0.874453$
 $E_Y=0.130917$, $\theta_{\ell'}=2.07527$, $\phi_{\ell'}=3.99897$
 $Q^2=0.0364429$, $x_B=0.126167$, $t=-0.0225667$, $W^2=1.13225$, $y=0.481205$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2467$
 P $E= 0.9699$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2467$
 ℓ' $E= 0.1430$ $p_x= -0.0819$ $p_y= -0.0946$ $p_z= -0.0691$
 P' $E= 0.9427$ $p_x= -0.0005$ $p_y= -0.0040$ $p_z= 0.0941$
 q_Y $E= 0.1309$ $p_x= 0.0824$ $p_y= 0.0986$ $p_z= -0.0250$

3D momenta



Transverse plane



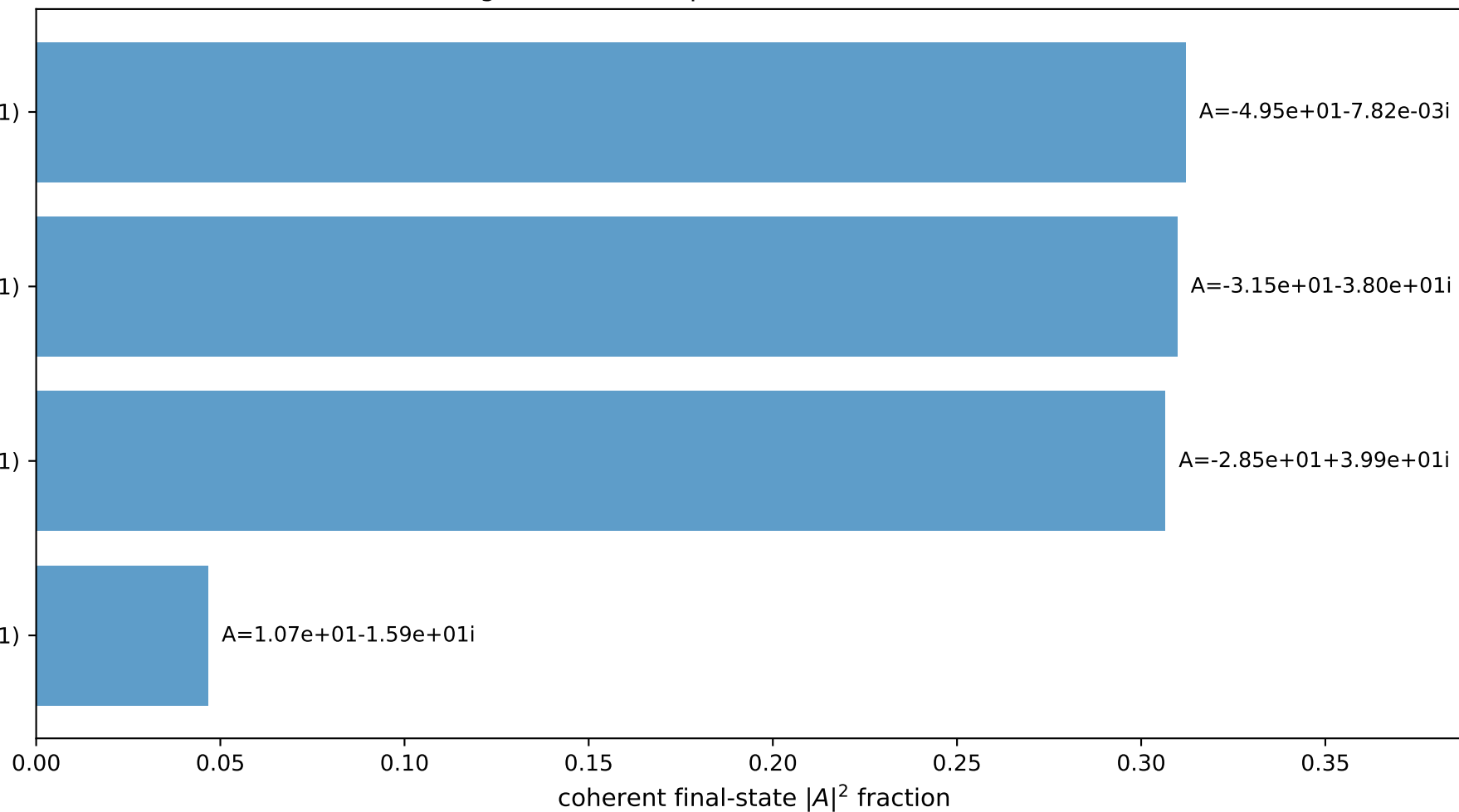
$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

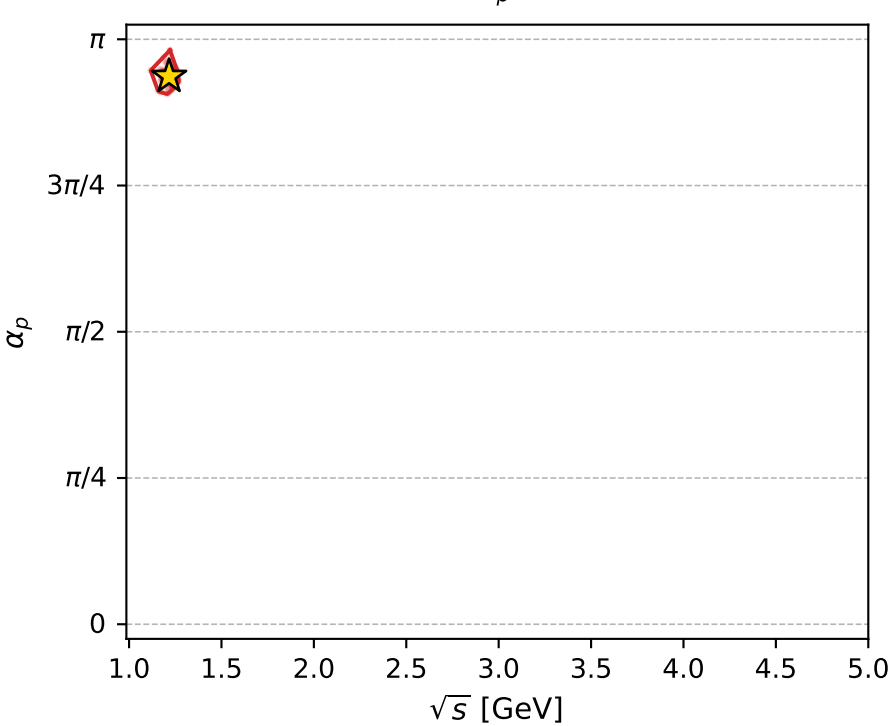
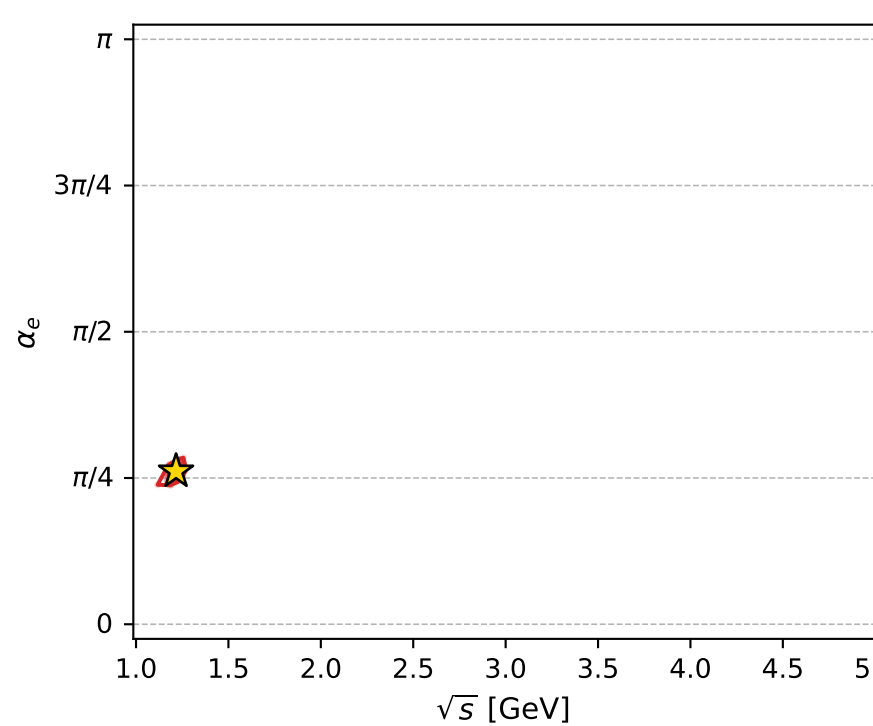
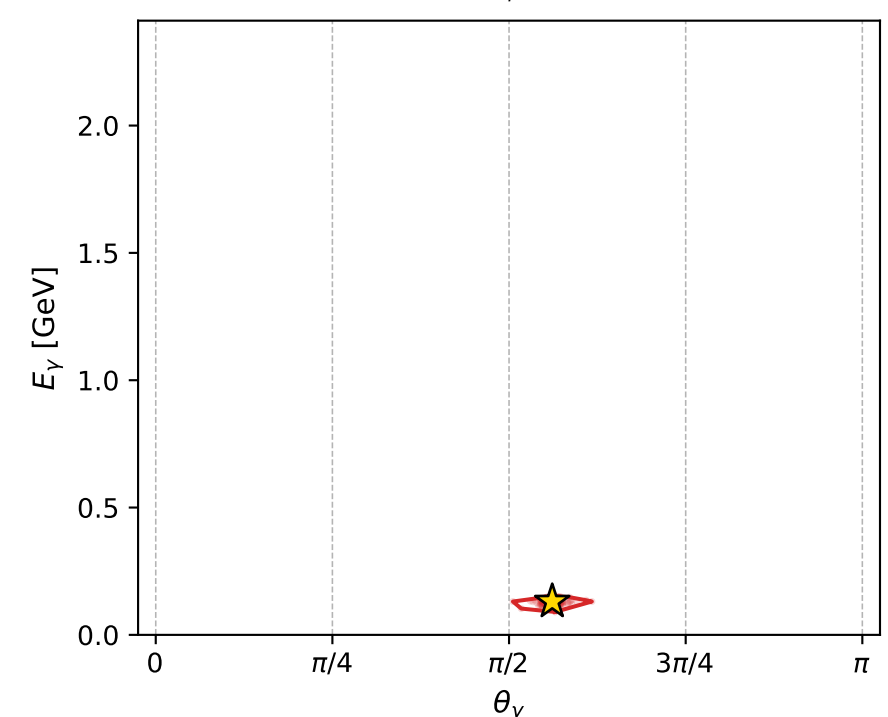
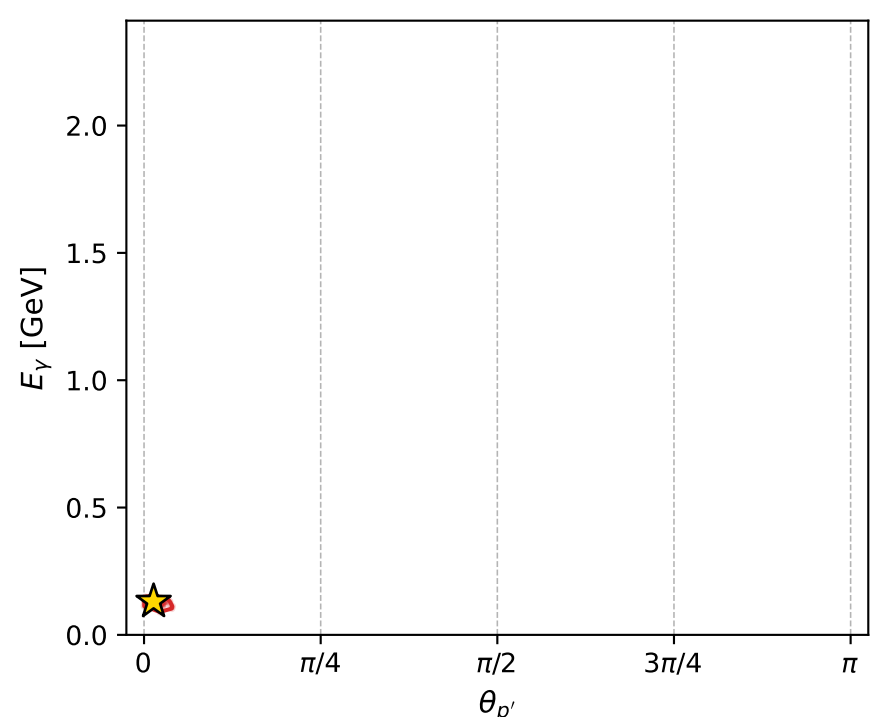
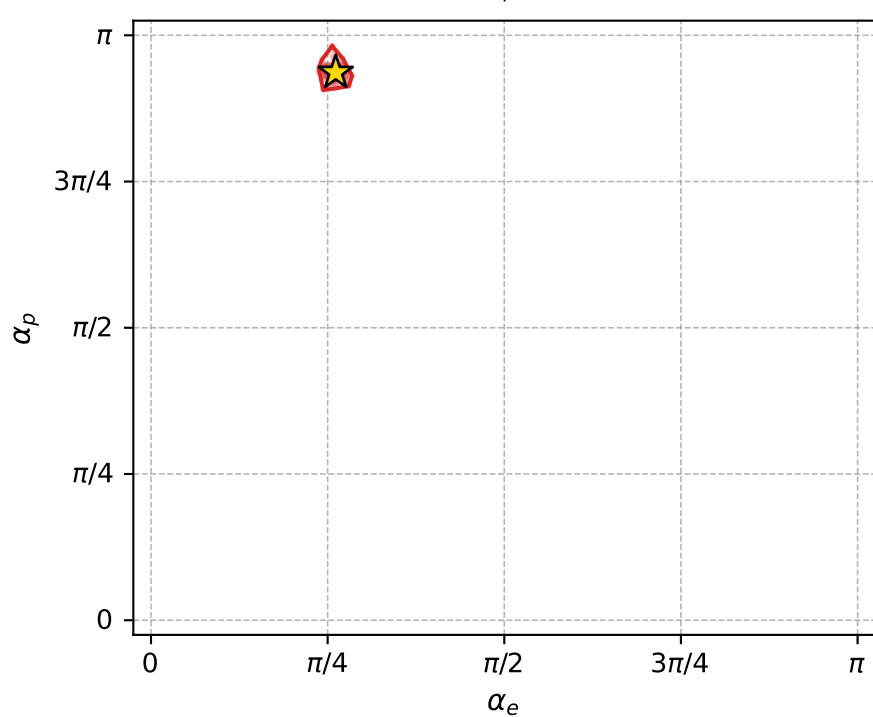
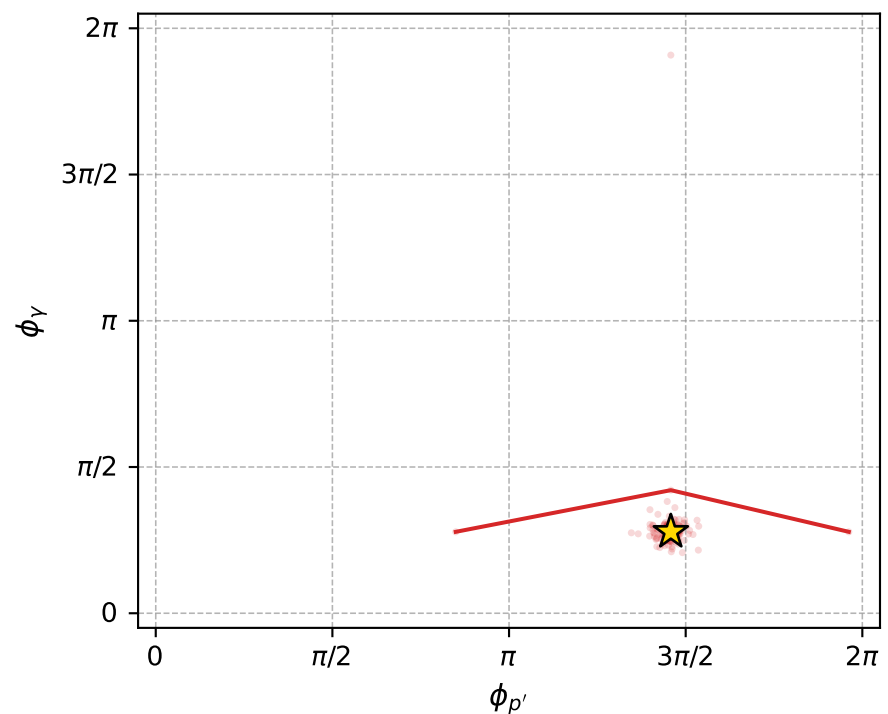
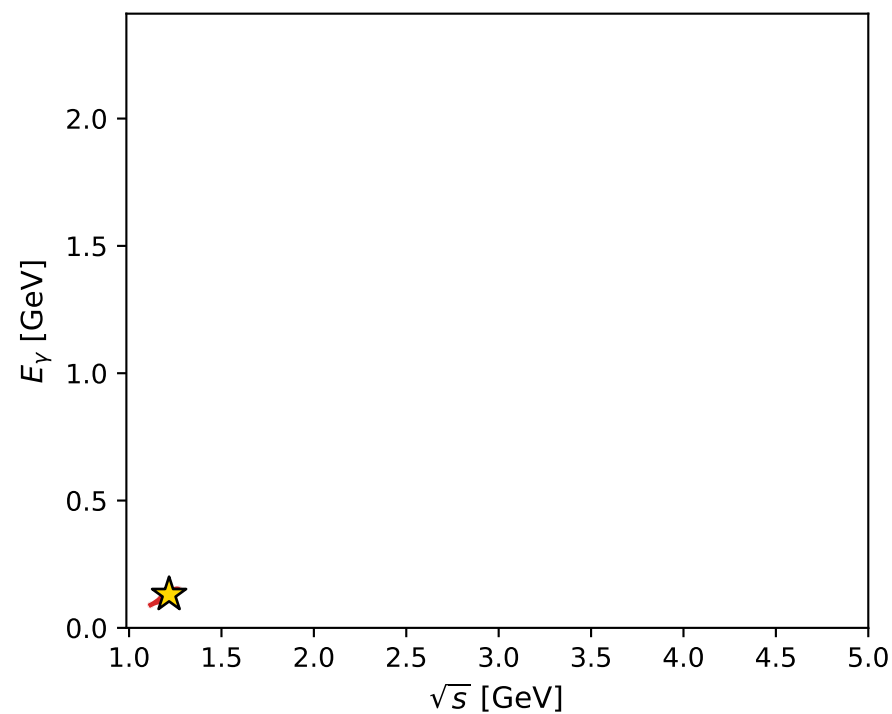
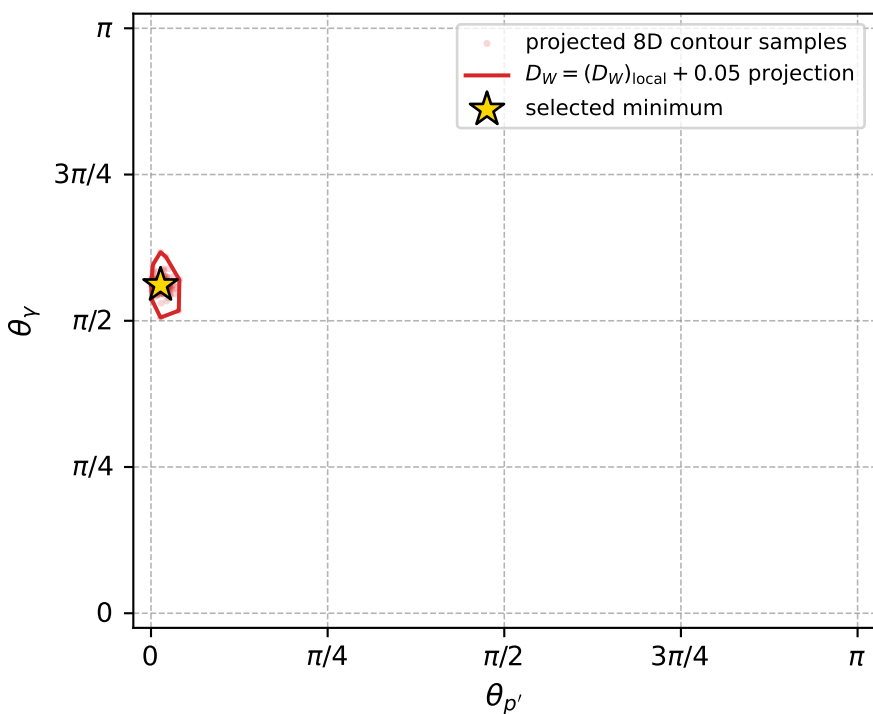
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=4, retained=97.5%)



electron: polarization cluster P3, local minimum 712; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0078360551$
 $D_W \text{ contour} = 0.057836055$
 above global minimum = 0.0078196222
 8D contour samples = 239

selected phase-space point:
 $\text{sqrt_s} = 1.216595$
 $\text{theta_p_out} = 0.04243449$
 $\text{theta_gamma_out} = 1.762835$
 $\text{qOut} = 0.1309169$
 $\text{phi_p_out} = 4.5803$
 $\text{phi_gamma_out} = 0.8744532$
 $\text{alpha_e} = 0.8212725$
 $\text{alpha_p} = 2.941974$

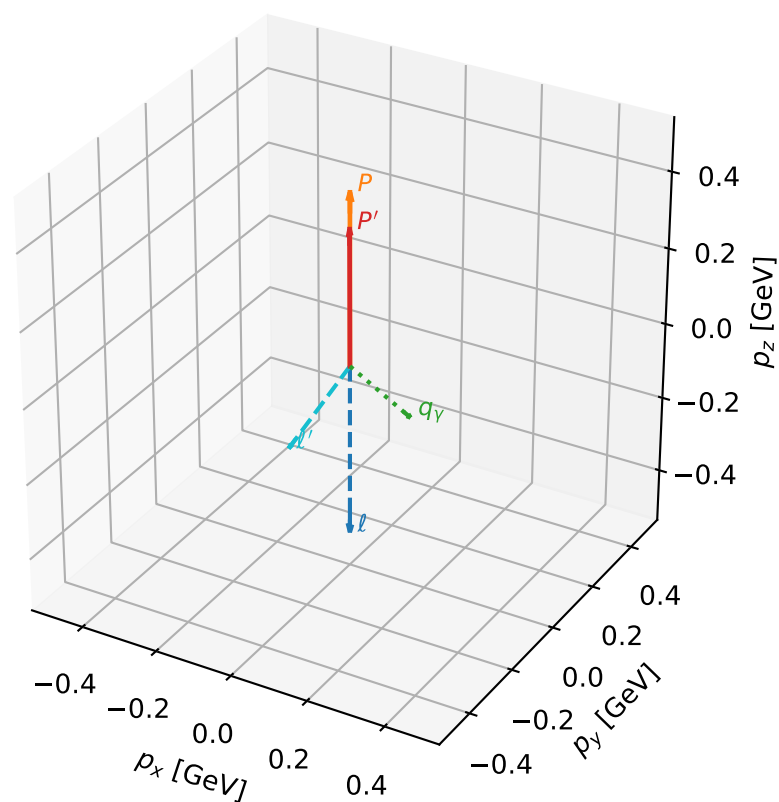
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_721 D_W=0.00804979
 region: polarization_cluster_03_local_minimum_721
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81789499$ rad
 incoming lepton: $0.683759|+\rangle + 0.729708|-\rangle$
 proton mixing angle: $\alpha_p=2.9213961$ rad
 incoming proton: $-0.975855|+\rangle + 0.218421|-\rangle$

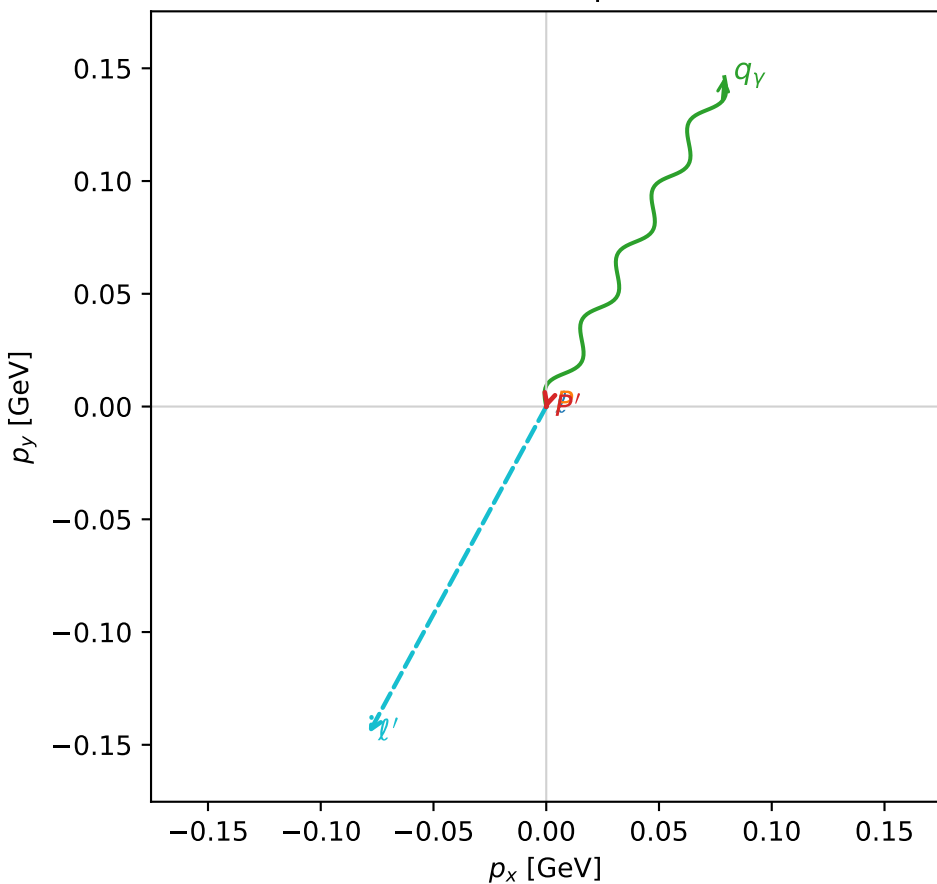
$s=2.22194$, $\sqrt{s}=1.49062$, $|\vec{P}|=0.450182$, $|\vec{P}'|=0.35698$
 $\theta_{P'}=0.00323455$, $\theta_Y=2.42007$, $\phi_{P'}=4.43178$, $\phi_Y=1.07485$
 $E_Y=0.251397$, $\theta_{\ell'}=2.3661$, $\phi_{\ell'}=4.21494$
 $Q^2=0.0606482$, $x_B=0.0865905$, $t=-0.00733381$, $W^2=1.5196$, $y=0.52187$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4502$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4502$
 P $E= 1.0404$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4502$
 ℓ' $E= 0.2356$ $p_x= -0.0787$ $p_y= -0.1449$ $p_z= -0.1682$
 P' $E= 1.0036$ $p_x= -0.0003$ $p_y= -0.0011$ $p_z= 0.3570$
 q_Y $E= 0.2514$ $p_x= 0.0790$ $p_y= 0.1460$ $p_z= -0.1887$

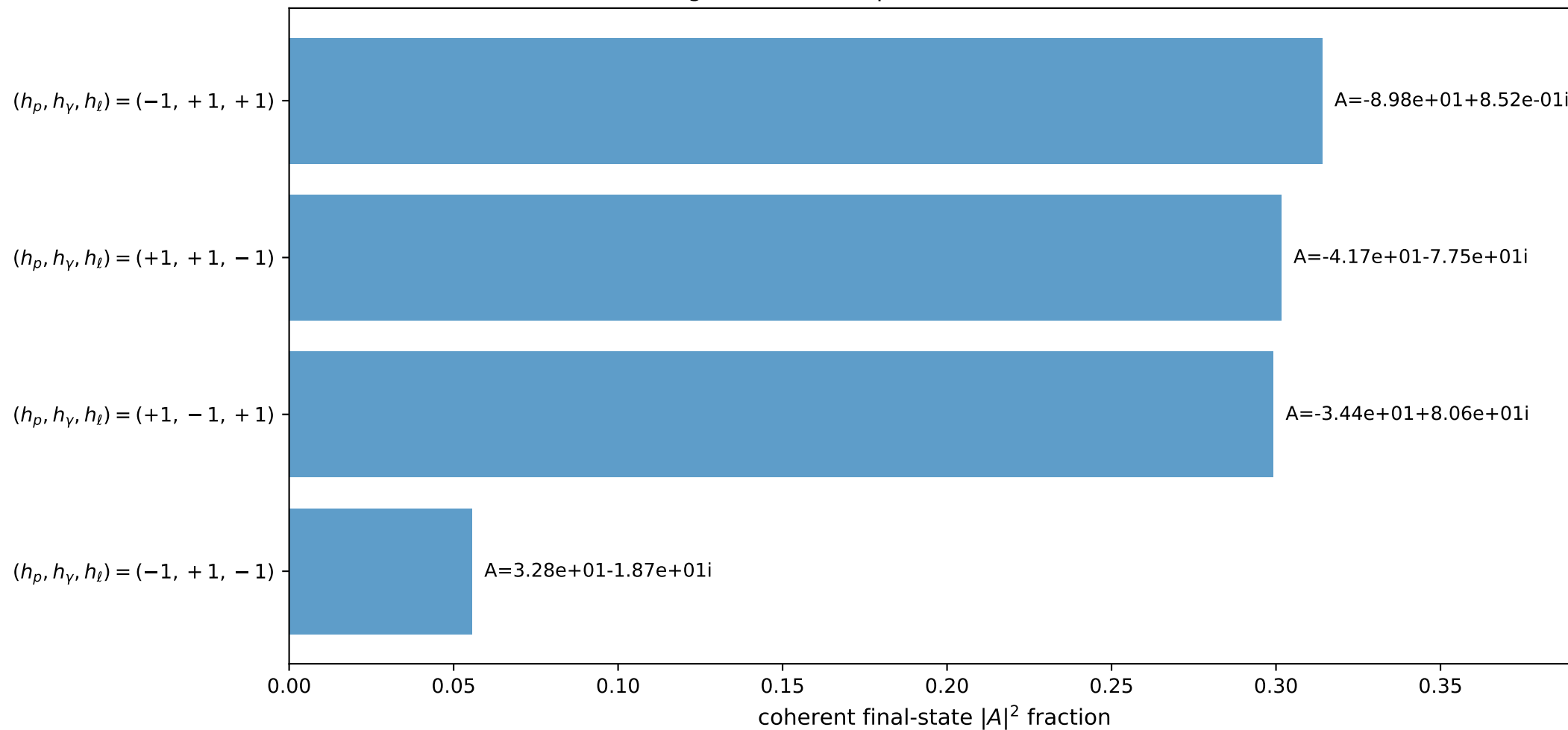
3D momenta



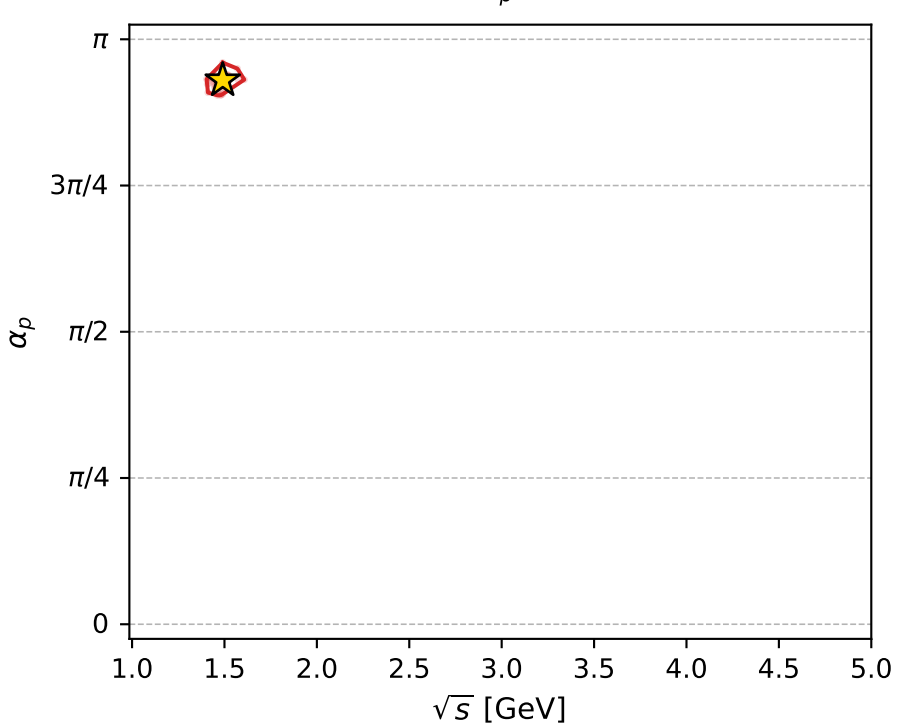
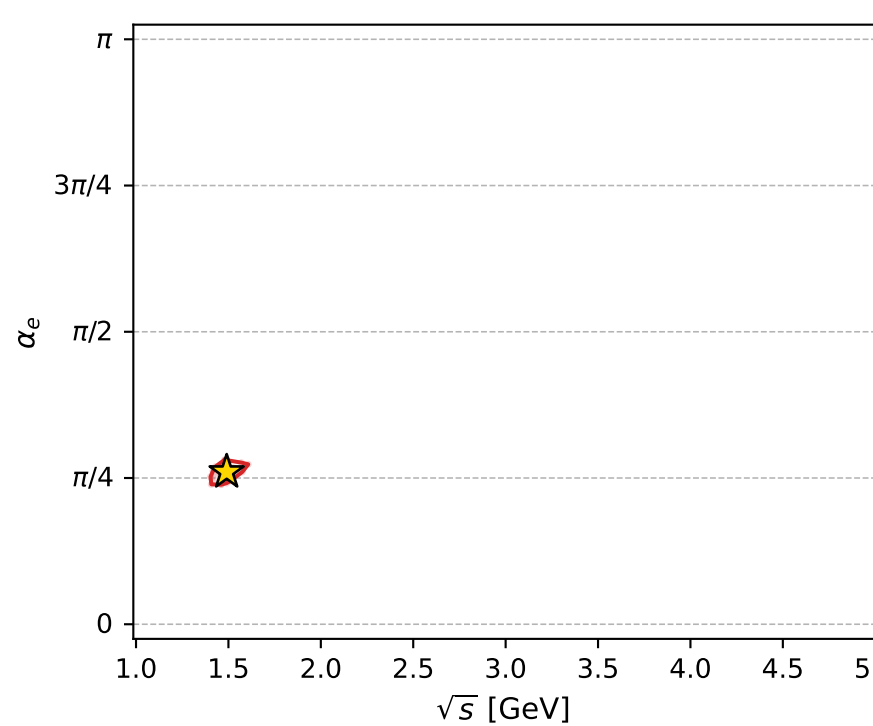
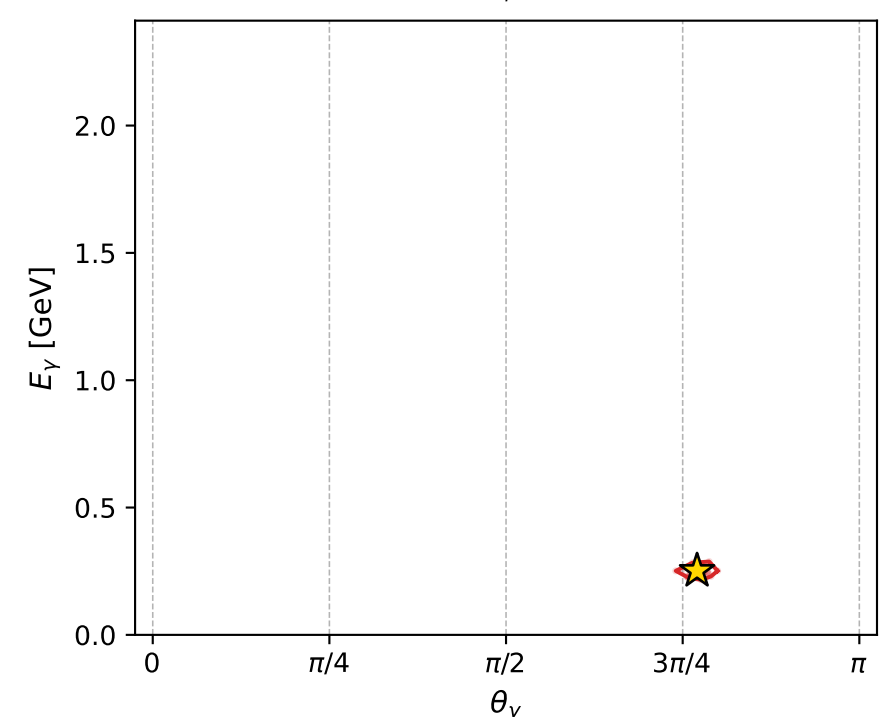
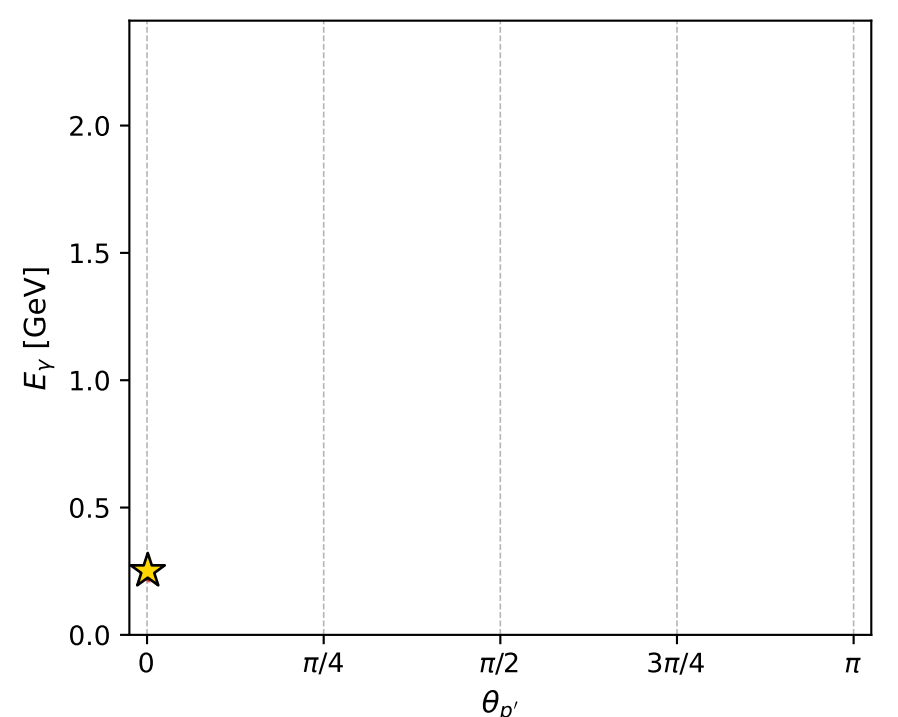
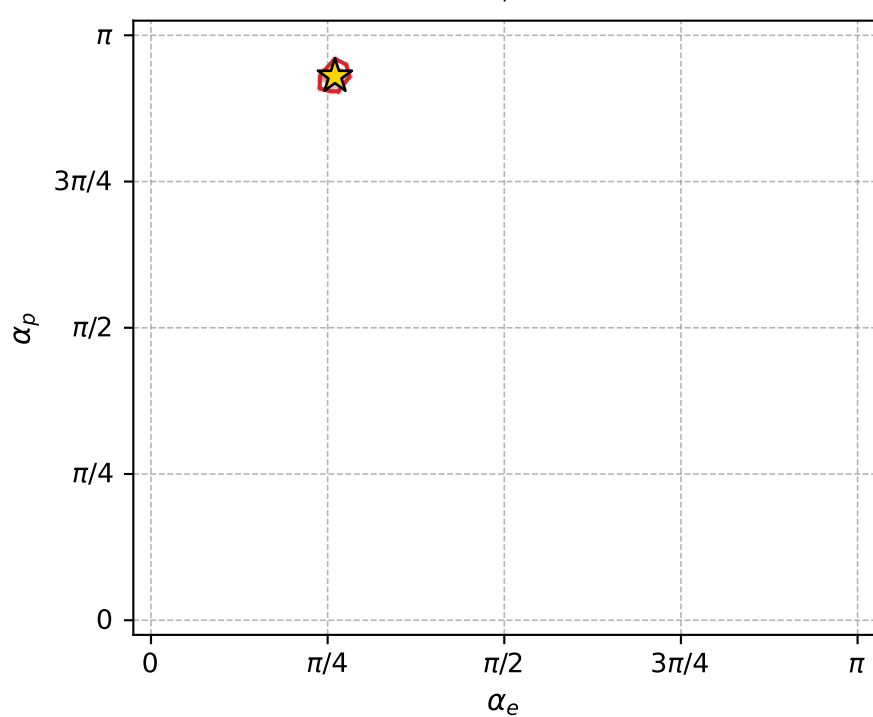
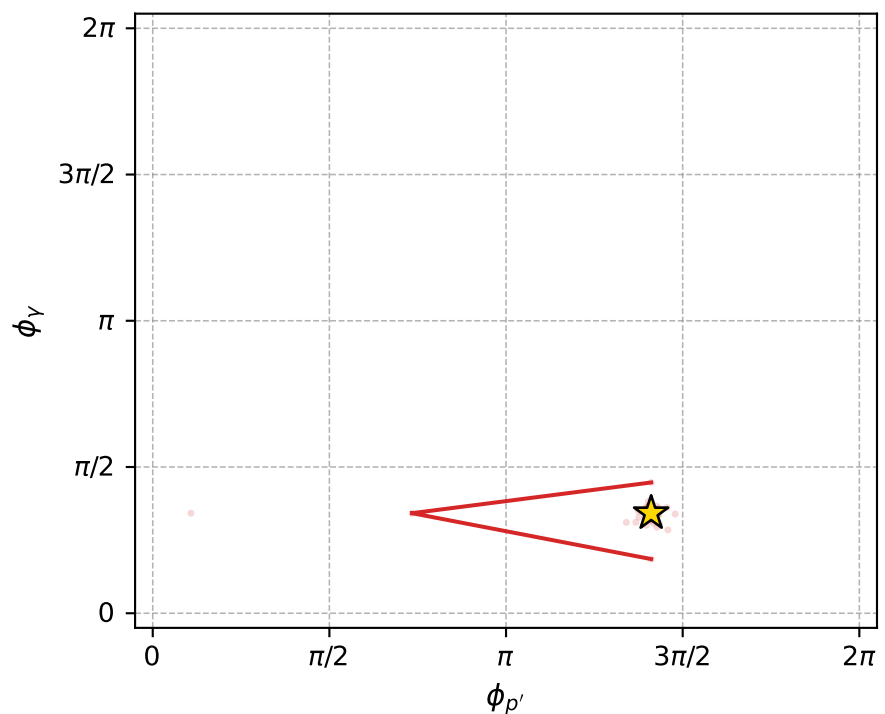
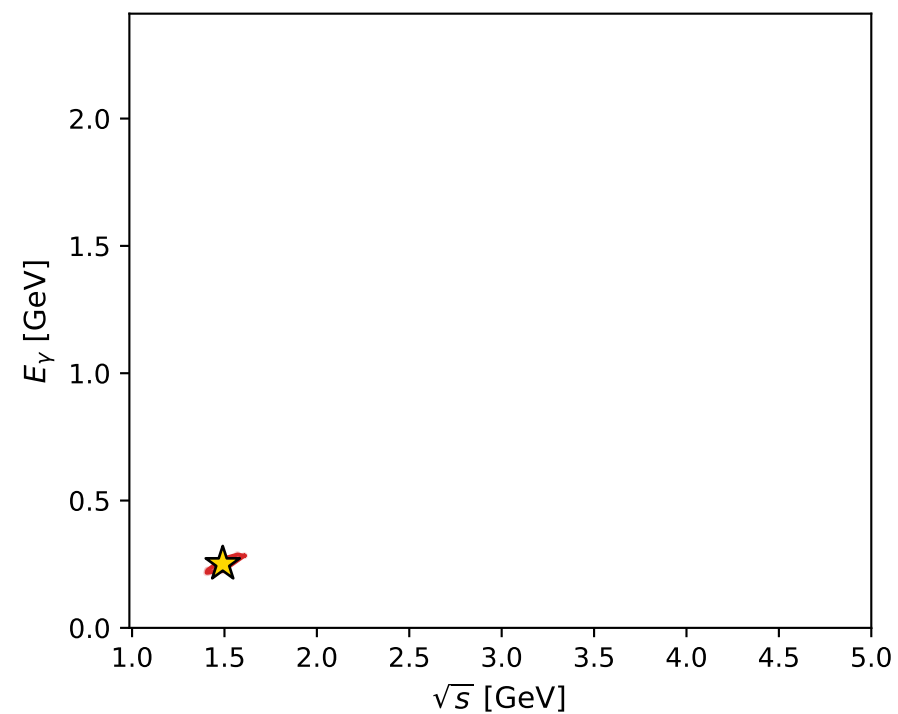
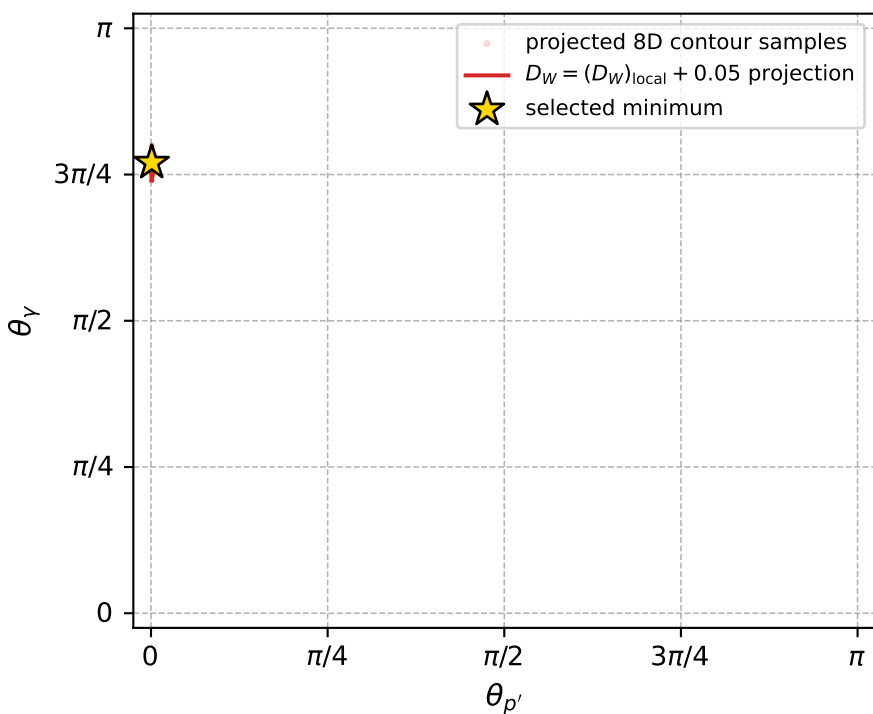
Transverse plane



Leading final-state amplitudes (N=4, retained=97.1%)



electron: polarization cluster P3, local minimum 721; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0080497877$
 $D_W \text{ contour} = 0.058049788$
 above global minimum = 0.0080333549
 8D contour samples = 147

selected phase-space point:
 $\text{sqrt_s} = 1.490618$
 $\text{theta_p_out} = 0.003234548$
 $\text{theta_gamma_out} = 2.420071$
 $\text{qOut} = 0.2513969$
 $\text{phi_p_out} = 4.431781$
 $\text{phi_gamma_out} = 1.074845$
 $\text{alpha_e} = 0.817895$
 $\text{alpha_p} = 2.921396$

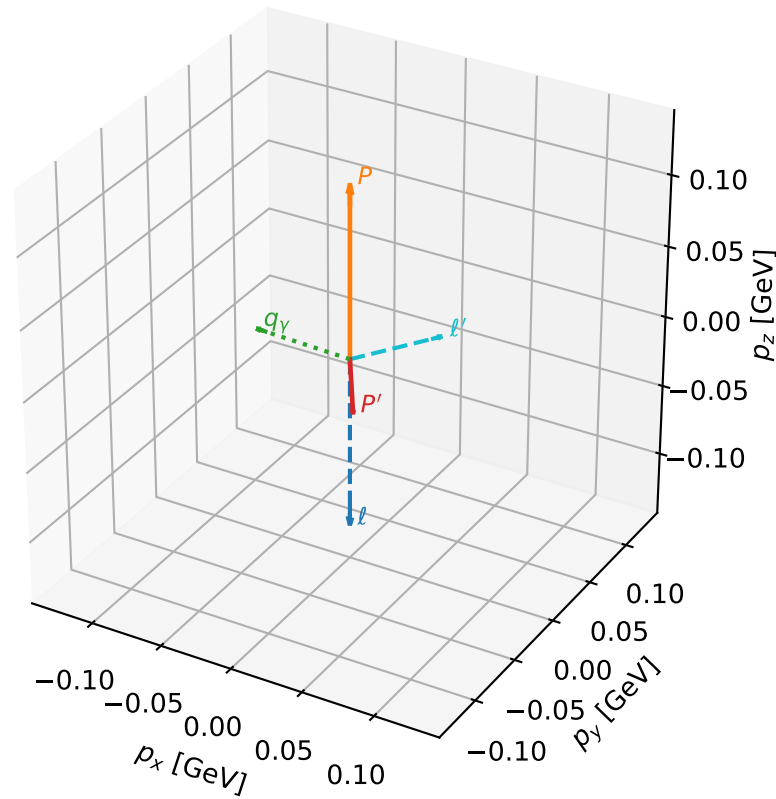
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_722 D_W=0.00805759
 region: polarization_cluster_03_local_minimum_722
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76206041$ rad
 incoming lepton: $0.723415|+\rangle + 0.690413|-\rangle$
 proton mixing angle: $\alpha_p=0.11757213$ rad
 incoming proton: $0.993096|+\rangle + 0.117301|-\rangle$

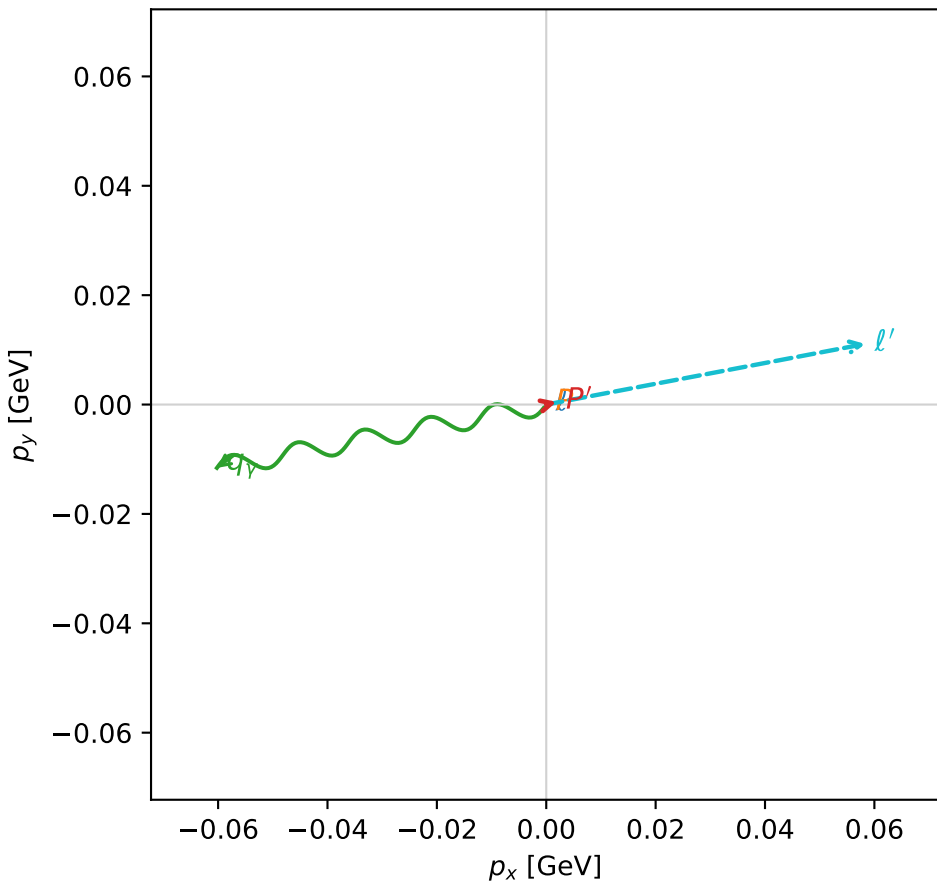
$s=1.13702$, $\sqrt{s}=1.06631$, $|\vec{P}|=0.120589$, $|\vec{P}'|=0.0377302$
 $\theta_{P'}=3.08974$, $\theta_V=1.40324$, $\phi_{P'}=0.260976$, $\phi_V=3.33166$
 $E_V=0.0621959$, $\theta_{\ell'}=1.13973$, $\phi_{\ell'}=0.187734$
 $Q^2=0.0223475$, $x_B=0.159463$, $t=-0.0250042$, $W^2=0.997639$, $y=0.544938$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1206$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1206$
 P $E= 0.9457$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1206$
 ℓ' $E= 0.0654$ $p_x= 0.0583$ $p_y= 0.0111$ $p_z= 0.0273$
 P' $E= 0.9388$ $p_x= 0.0019$ $p_y= 0.0005$ $p_z= -0.0377$
 q_Y $E= 0.0622$ $p_x= -0.0602$ $p_y= -0.0116$ $p_z= 0.0104$

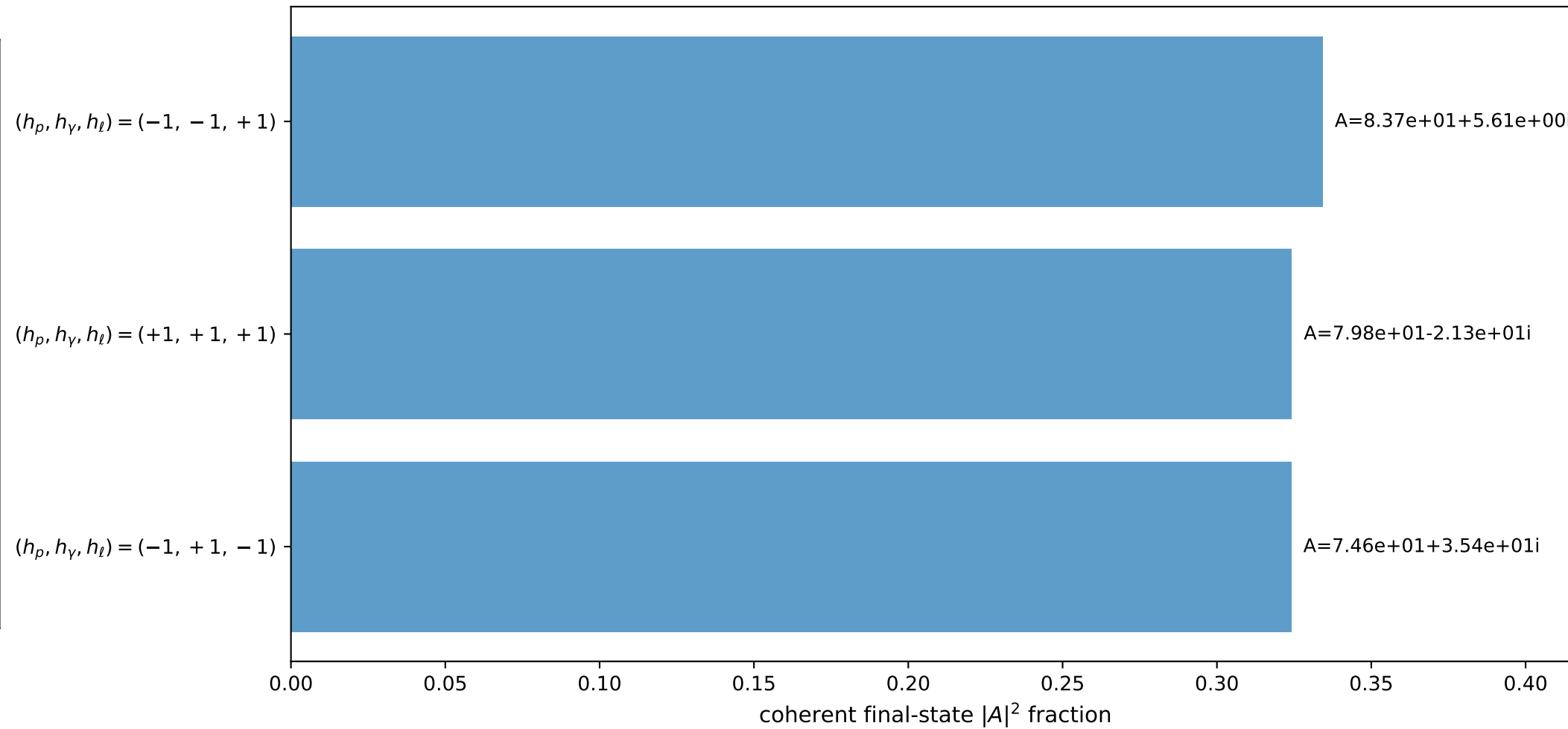
3D momenta



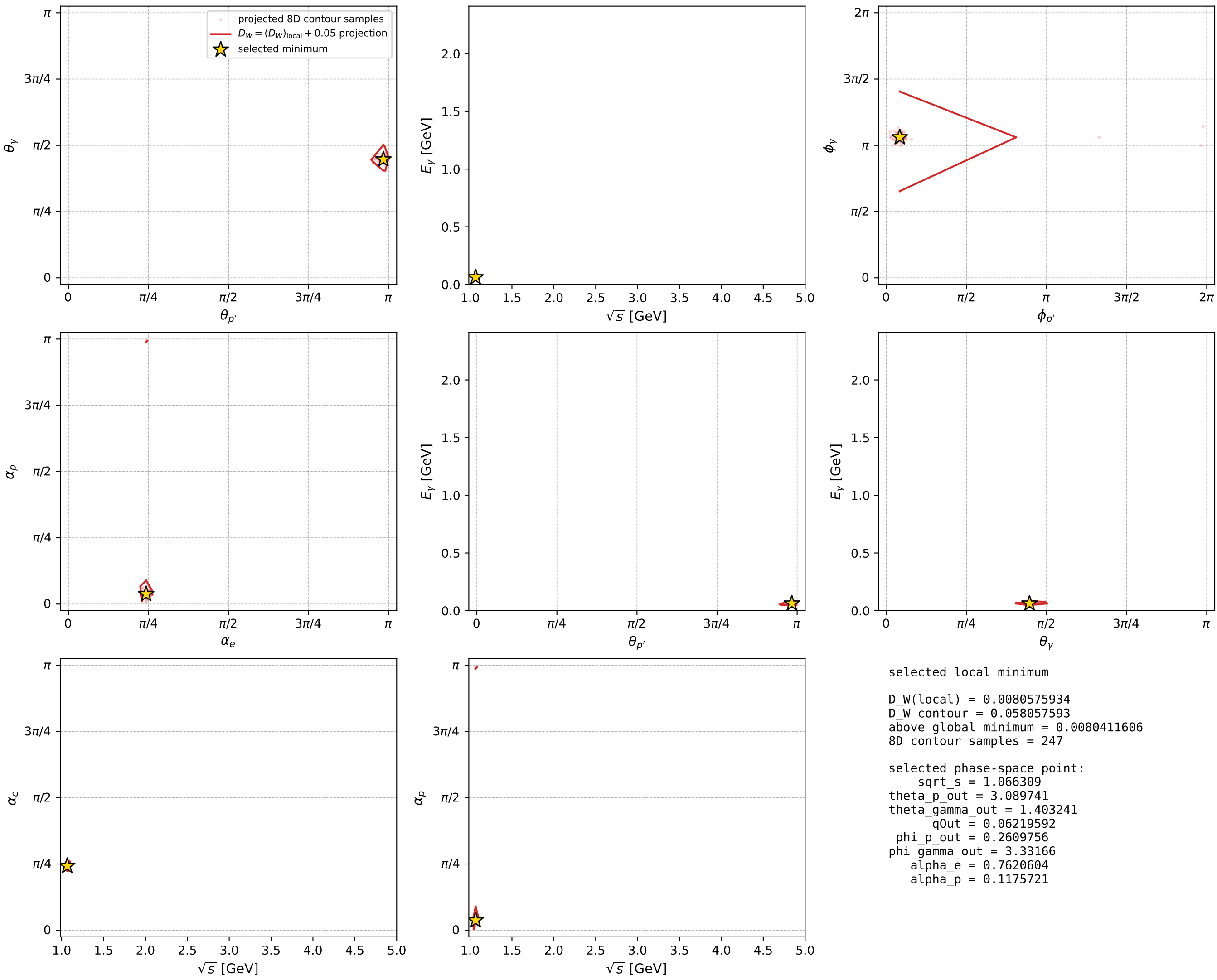
Transverse plane



Leading final-state amplitudes (N=3, retained=98.3%)



electron: polarization cluster P3, local minimum 722; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

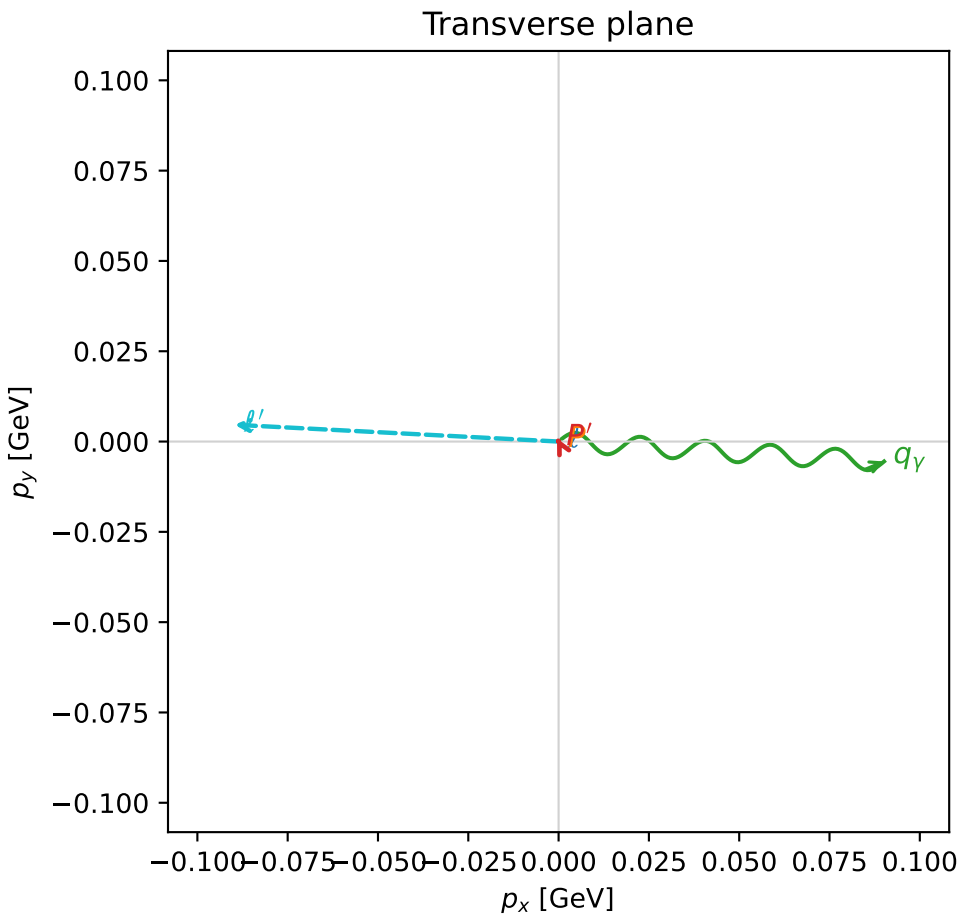
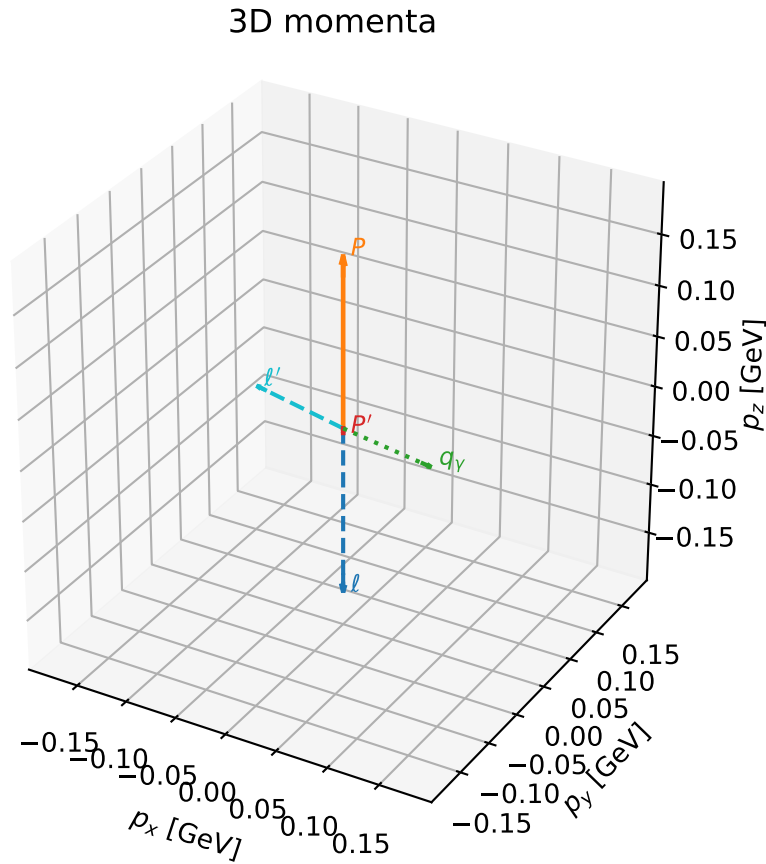


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_735 D_W=0.00848145
 region: polarization_cluster_03_local_minimum_735
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78322937$ rad
 incoming lepton: $0.708639|+\rangle + 0.705572|-\rangle$
 proton mixing angle: $\alpha_p=2.9506423$ rad
 incoming proton: $-0.981824|+\rangle + 0.189792|-\rangle$

$s=1.25386$, $\sqrt{s}=1.11976$, $|\vec{P}|=0.167008$, $|\vec{P}'|=0.0052075$
 $\theta_{P'}=2.95455$, $\theta_V=1.67547$, $\phi_{P'}=2.04328$, $\phi_V=6.22212$
 $E_V=0.0907752$, $\theta_{\ell'}=1.4096$, $\phi_{\ell'}=3.08981$
 $Q^2=0.0352621$, $x_B=0.171549$, $t=-0.0294106$, $W^2=1.05013$, $y=0.549574$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1670$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1670$
 P $E= 0.9528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1670$
 ℓ' $E= 0.0910$ $p_x= -0.0897$ $p_y= 0.0046$ $p_z= 0.0146$
 P' $E= 0.9380$ $p_x= -0.0004$ $p_y= 0.0009$ $p_z= -0.0051$
 q_V $E= 0.0908$ $p_x= 0.0901$ $p_y= -0.0055$ $p_z= -0.0095$

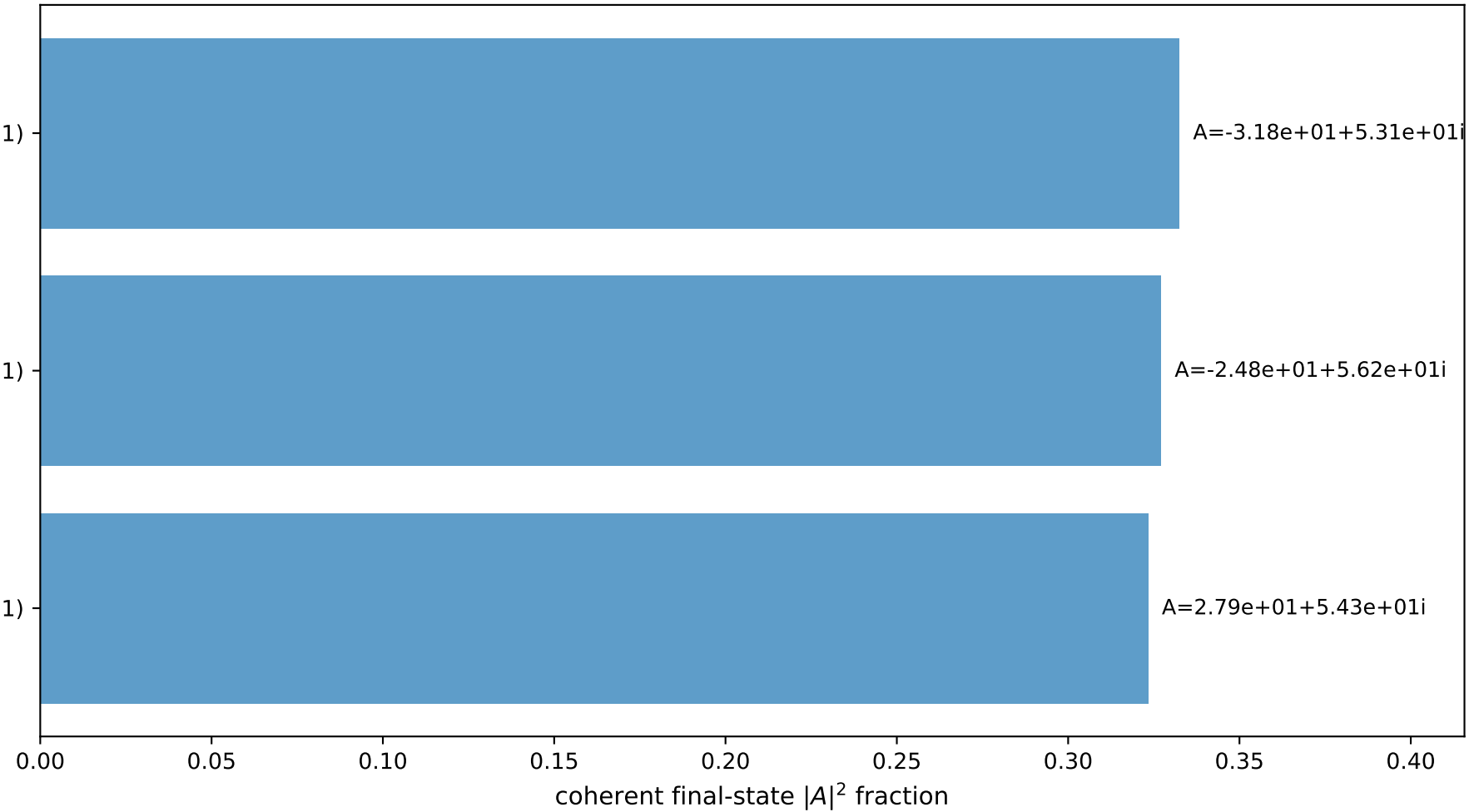


$(h_p, h_V, h_\ell) = (-1, -1, +1)$

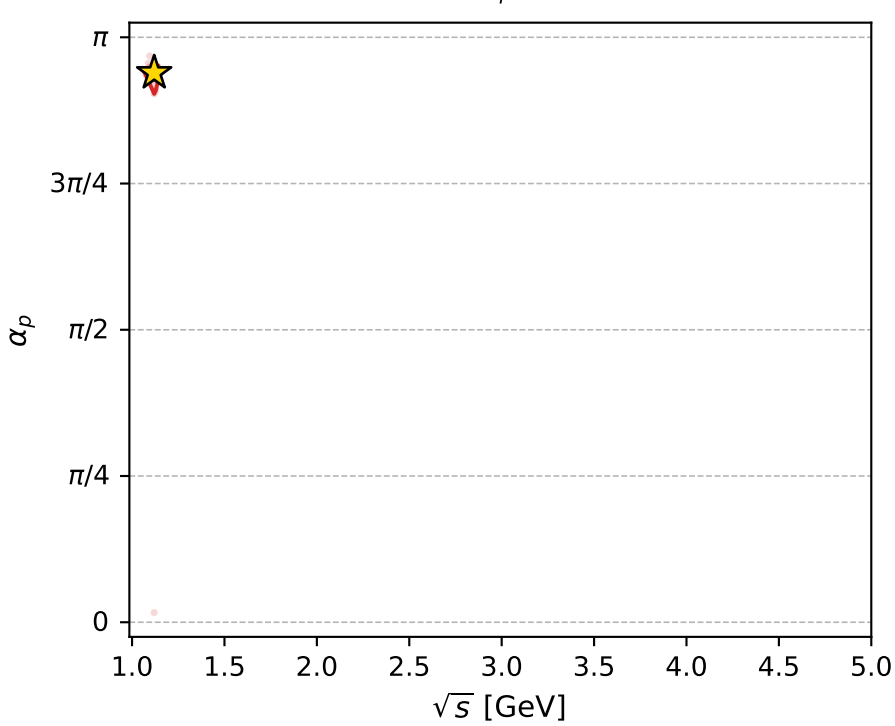
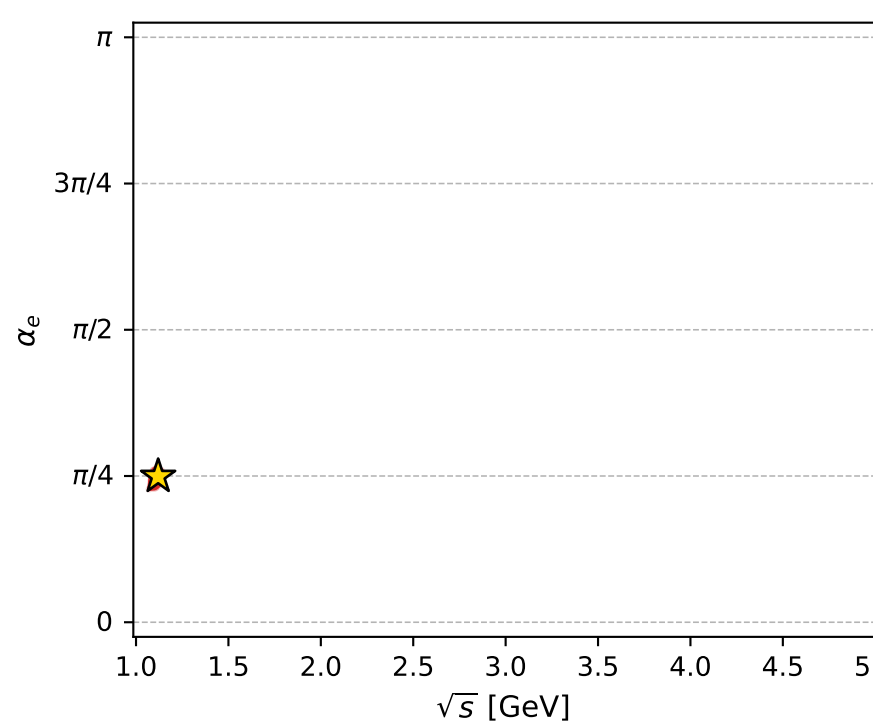
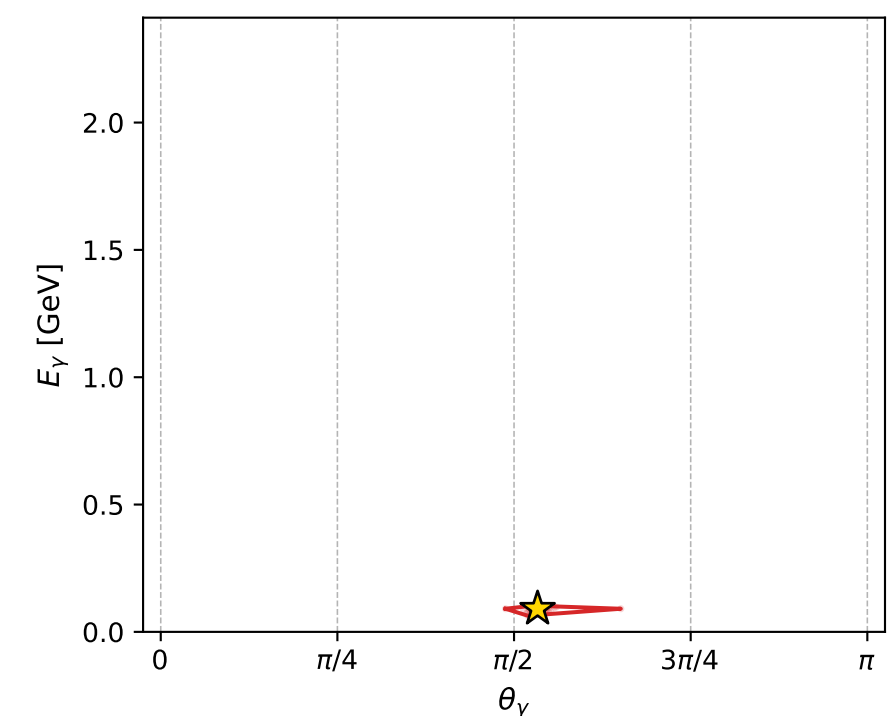
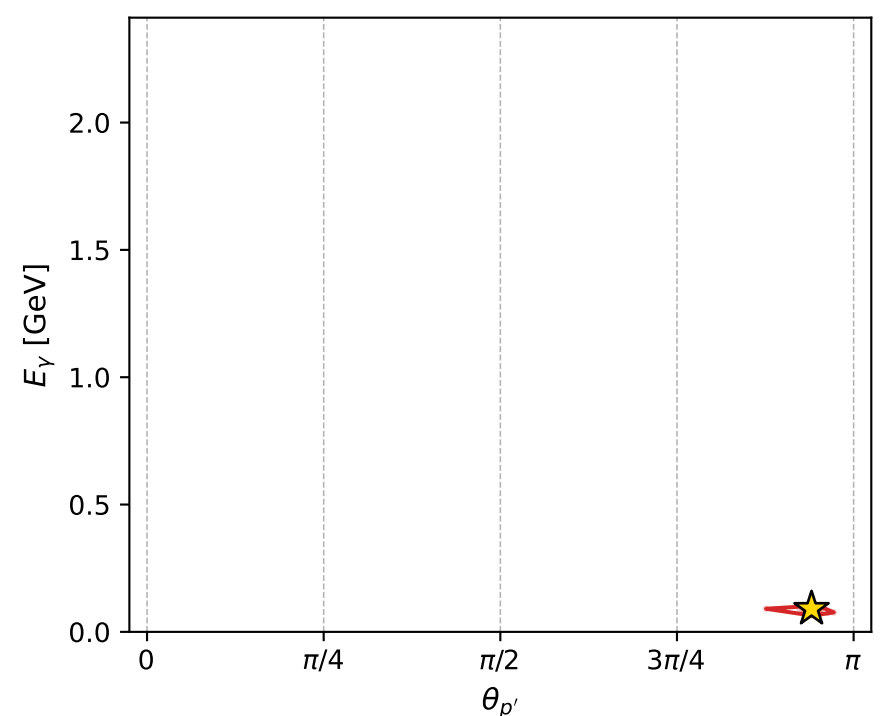
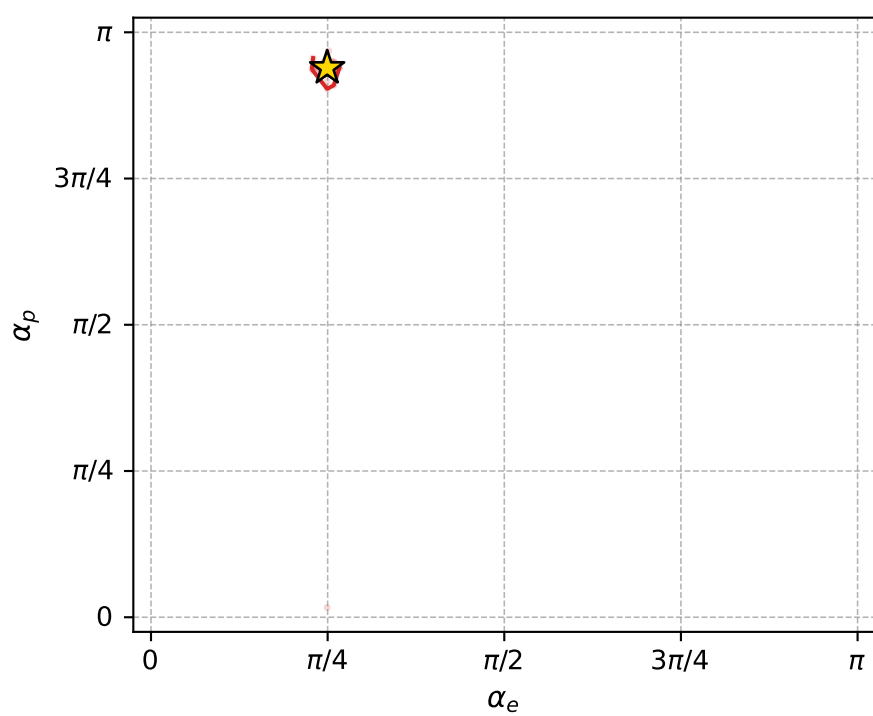
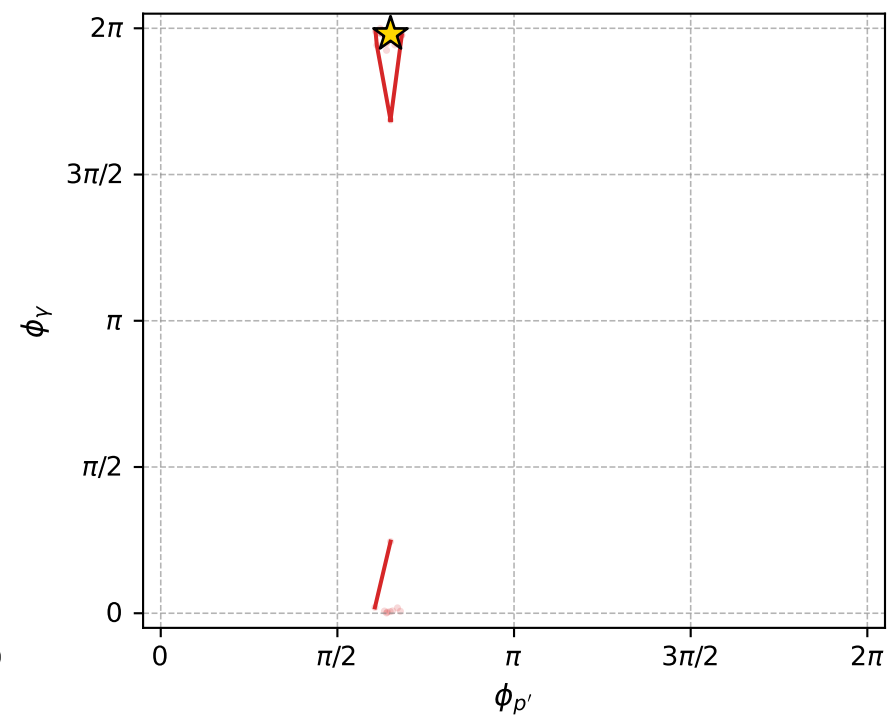
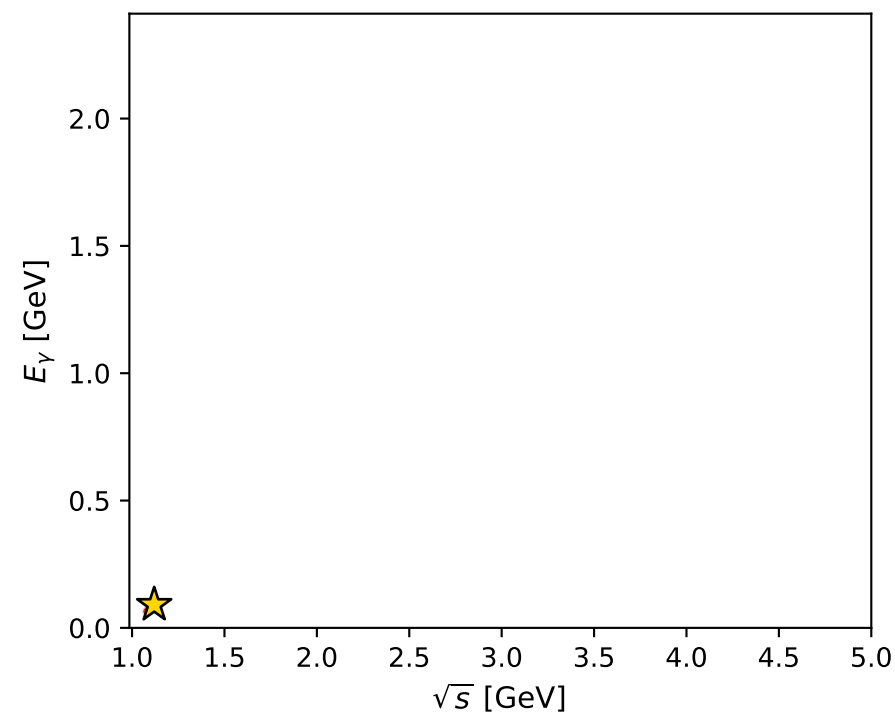
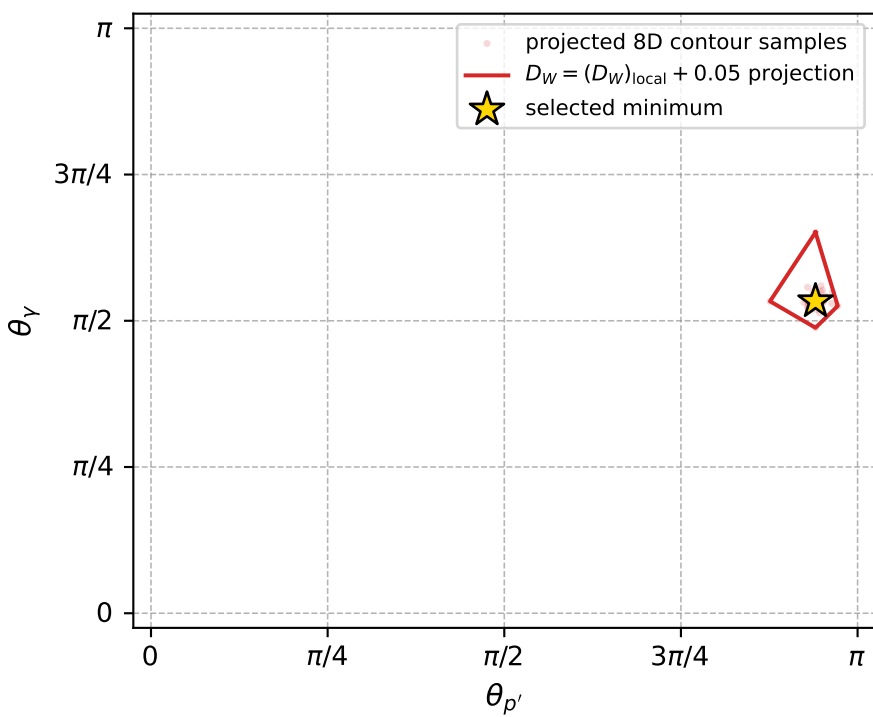
$(h_p, h_V, h_\ell) = (-1, +1, -1)$

$(h_p, h_V, h_\ell) = (+1, +1, +1)$

Leading final-state amplitudes (N=3, retained=98.3%)



electron: polarization cluster P3, local minimum 735; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0084814539$
 $D_W \text{ contour} = 0.058481454$
 above global minimum = 0.0084650211
 8D contour samples = 253

selected phase-space point:
 $\text{sqrt_s} = 1.11976$
 $\text{theta_p_out} = 2.954553$
 $\text{theta_gamma_out} = 1.675467$
 $\text{qOut} = 0.0907752$
 $\text{phi_p_out} = 2.043278$
 $\text{phi_gamma_out} = 6.222122$
 $\text{alpha_e} = 0.7832294$
 $\text{alpha_p} = 2.950642$

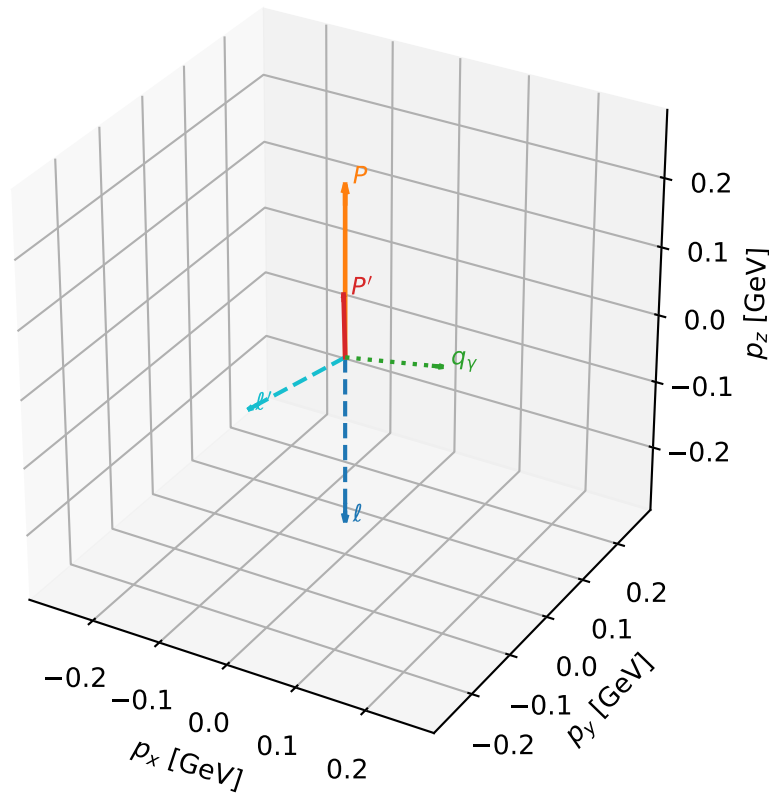
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_738 D_W=0.00854335
 region: polarization_cluster_03_local_minimum_738
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82149703$ rad
 incoming lepton: $0.681126|+\rangle + 0.732166|-\rangle$
 proton mixing angle: $\alpha_p=2.9553339$ rad
 incoming proton: $-0.982704|+\rangle + 0.185184|-\rangle$

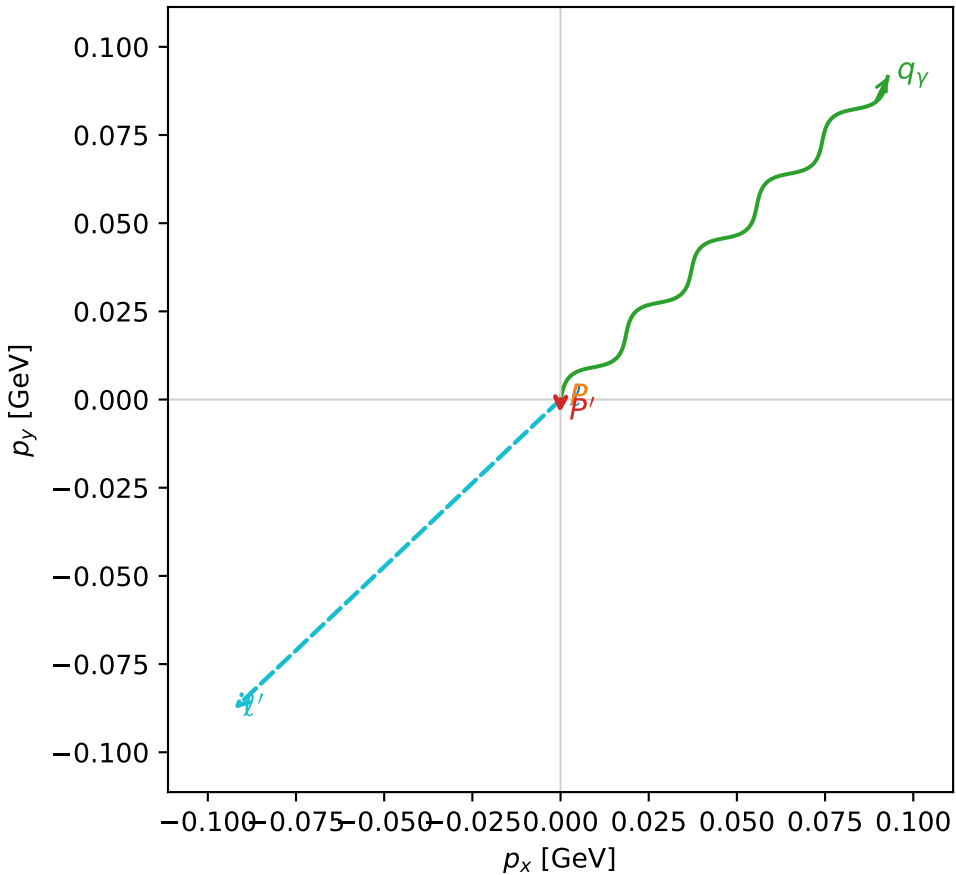
$s=1.4835$, $\sqrt{s}=1.21799$, $|\vec{P}|=0.247807$, $|\vec{P}'|=0.0932118$
 $\theta_{p'}=0.0400469$, $\theta_Y=1.82018$, $\phi_{p'}=4.66093$, $\phi_Y=0.778563$
 $E_Y=0.134433$, $\theta_{\ell'}=2.01023$, $\phi_{\ell'}=3.90041$
 $Q^2=0.0401335$, $x_B=0.133569$, $t=-0.0231771$, $W^2=1.14018$, $y=0.497752$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2478$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2478$
 P $E= 0.9702$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2478$
 ℓ' $E= 0.1409$ $p_x= -0.0926$ $p_y= -0.0878$ $p_z= -0.0600$
 P' $E= 0.9426$ $p_x= -0.0002$ $p_y= -0.0037$ $p_z= 0.0931$
 q_Y $E= 0.1344$ $p_x= 0.0927$ $p_y= 0.0915$ $p_z= -0.0332$

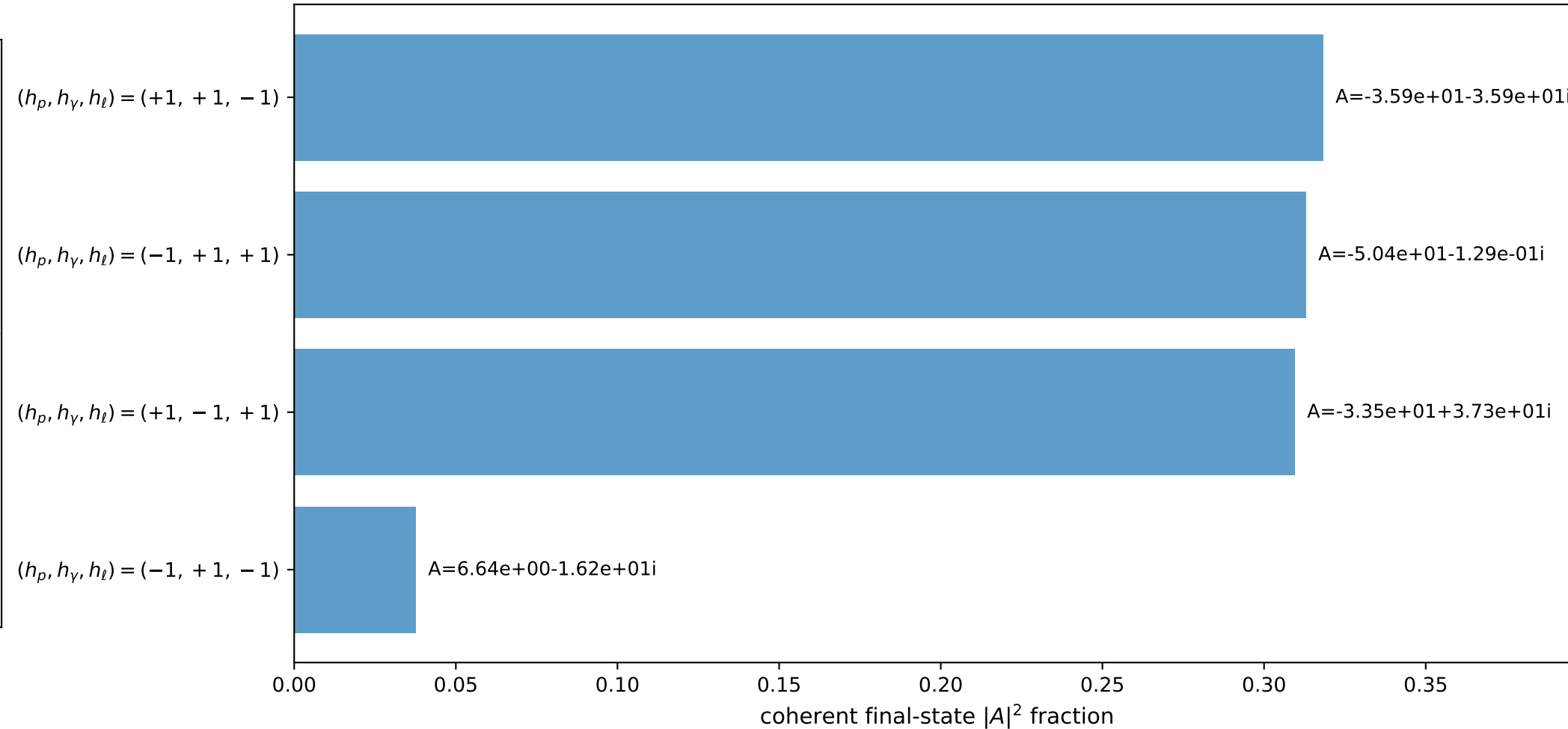
3D momenta



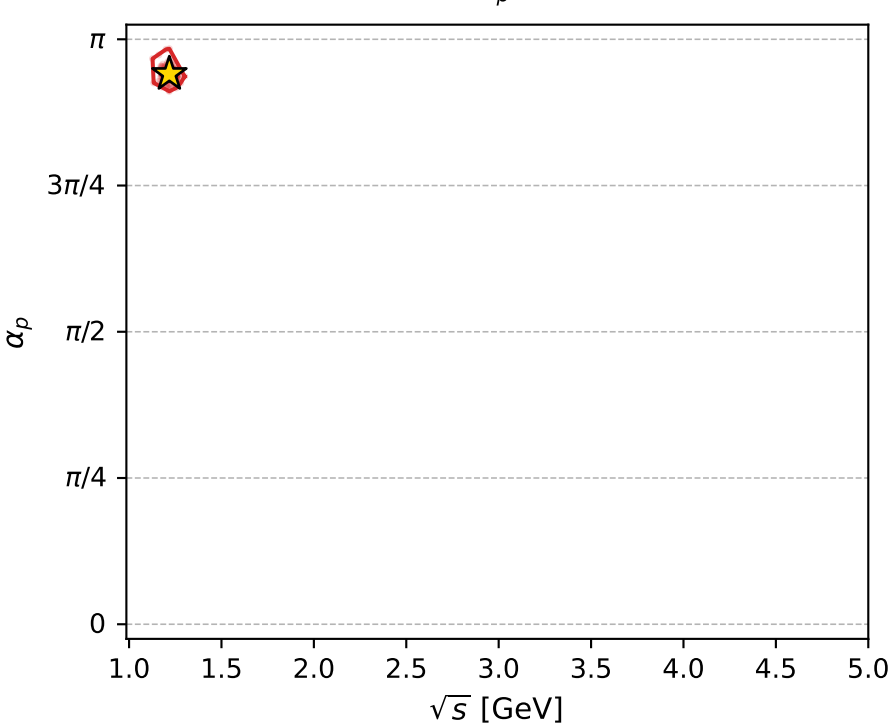
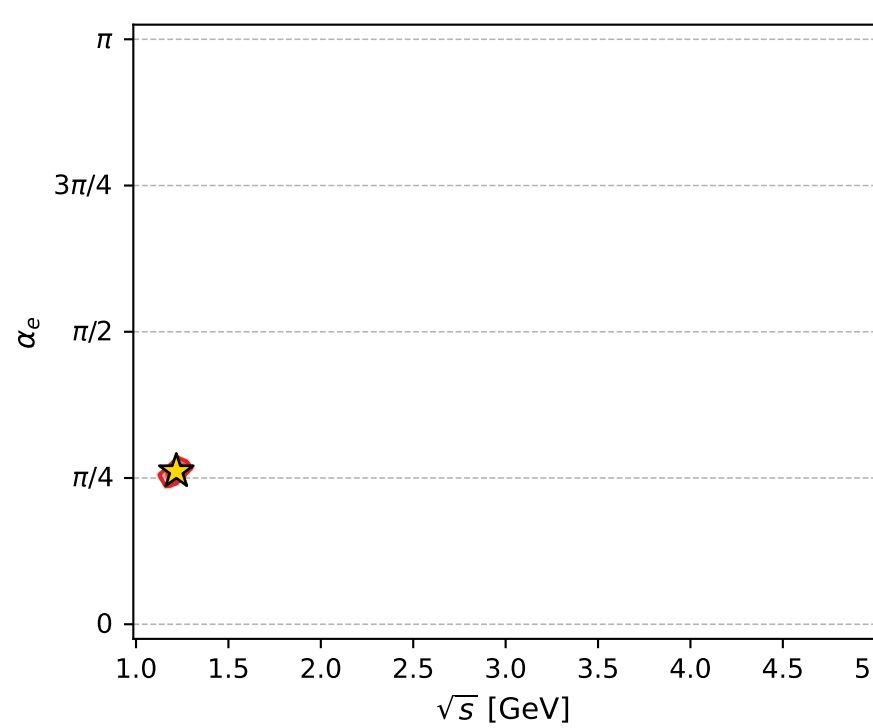
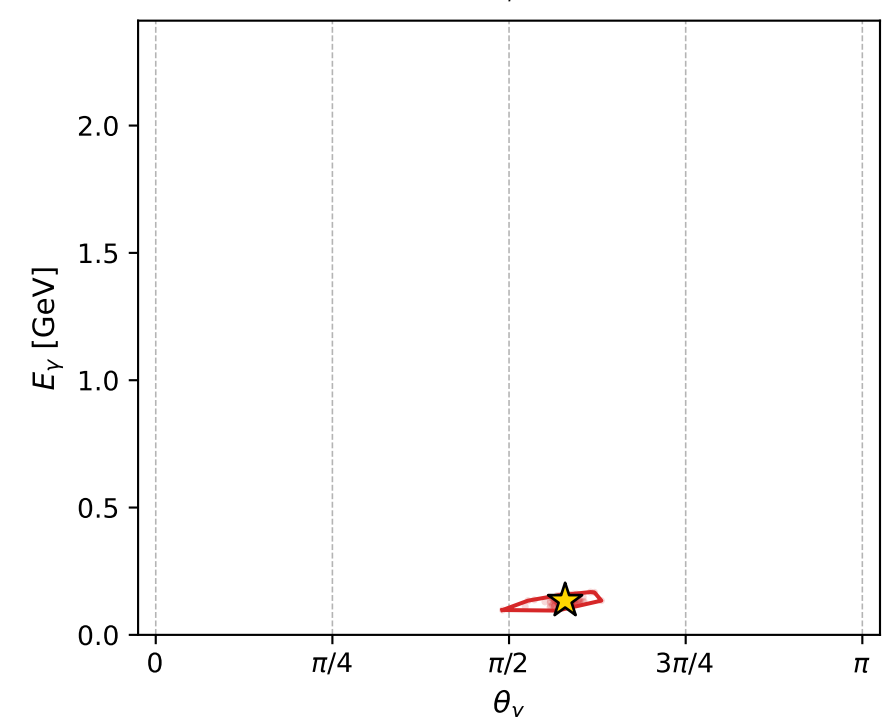
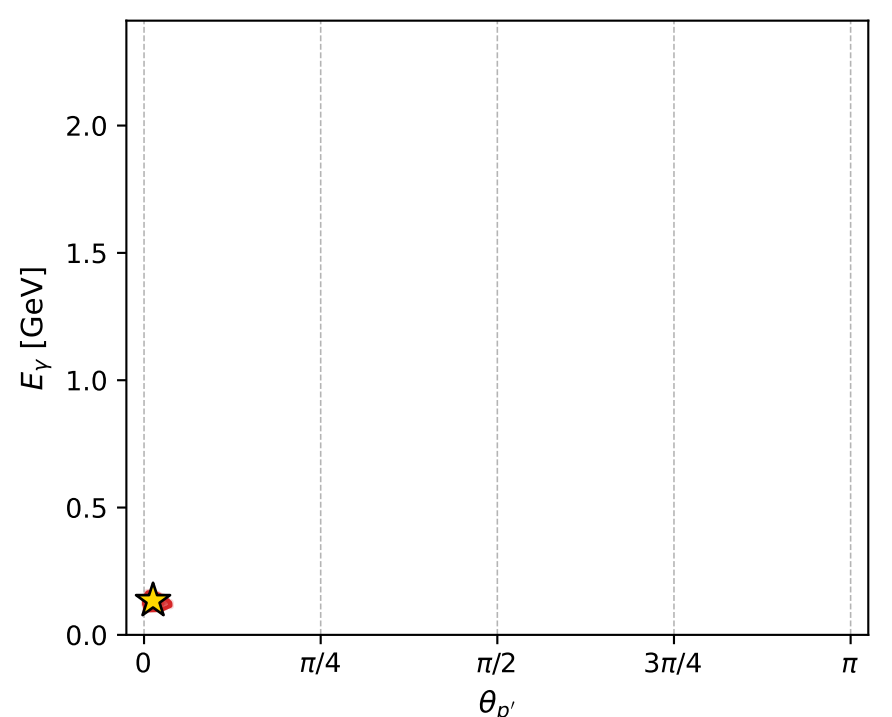
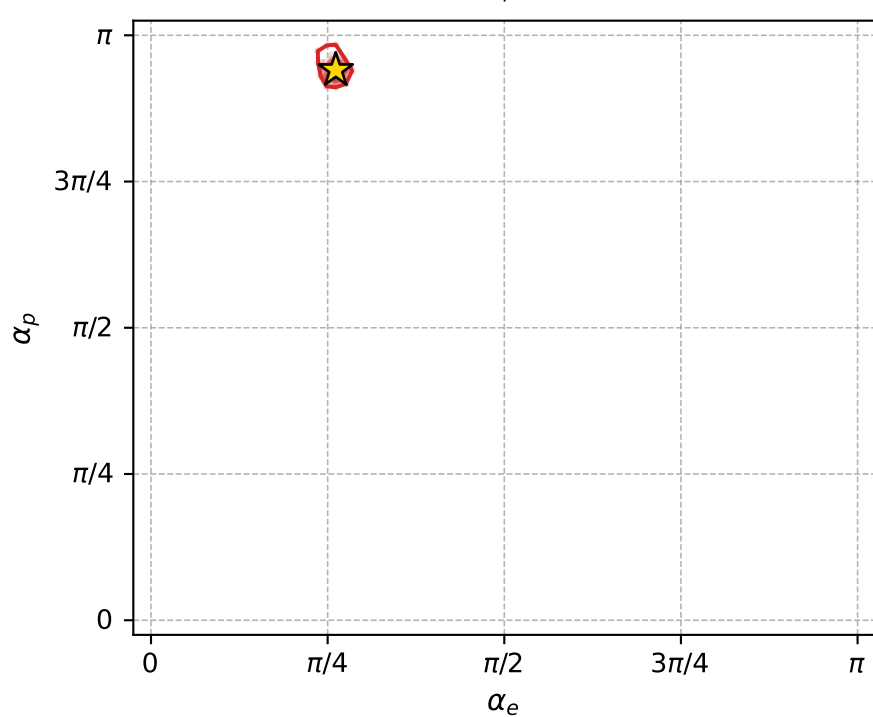
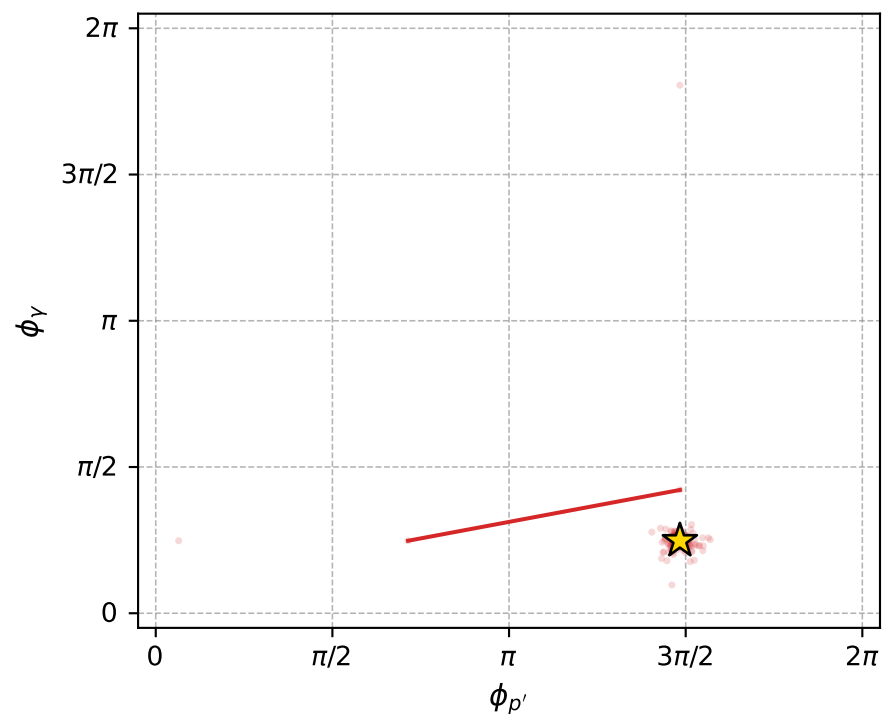
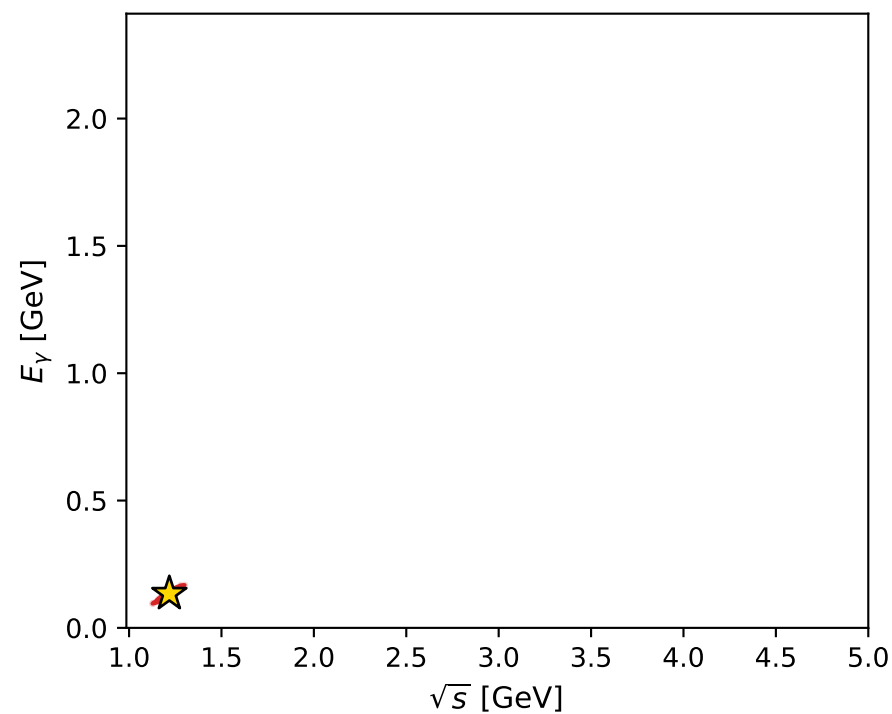
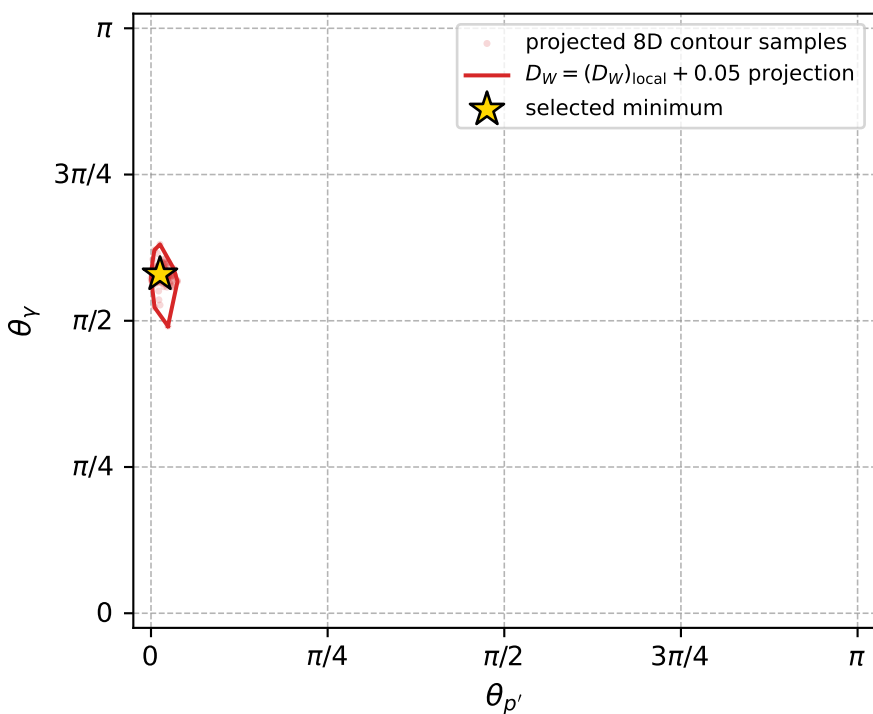
Transverse plane



Leading final-state amplitudes (N=4, retained=97.9%)



electron: polarization cluster P3, local minimum 738; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0085433537$
 $D_W \text{ contour} = 0.058543354$
 above global minimum = 0.0085269209
 8D contour samples = 229

selected phase-space point:
 $\text{sqrt_s} = 1.217989$
 $\text{theta_p_out} = 0.04004689$
 $\text{theta_gamma_out} = 1.820181$
 $\text{qOut} = 0.134433$
 $\text{phi_p_out} = 4.660927$
 $\text{phi_gamma_out} = 0.7785627$
 $\text{alpha_e} = 0.821497$
 $\text{alpha_p} = 2.955334$

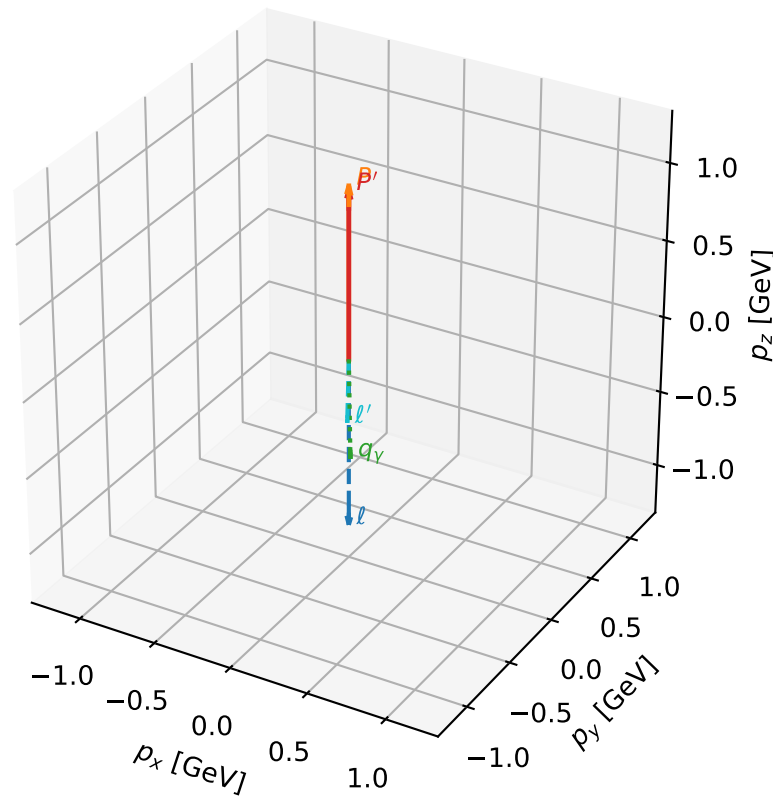
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_744 D_W=0.00869438
 region: polarization_cluster_03_local_minimum_744
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80437967$ rad
 incoming lepton: $0.693558|+\rangle +0.720401|-\rangle$
 proton mixing angle: $\alpha_p=0.13957319$ rad
 incoming proton: $0.990275|+\rangle +0.13912|-\rangle$

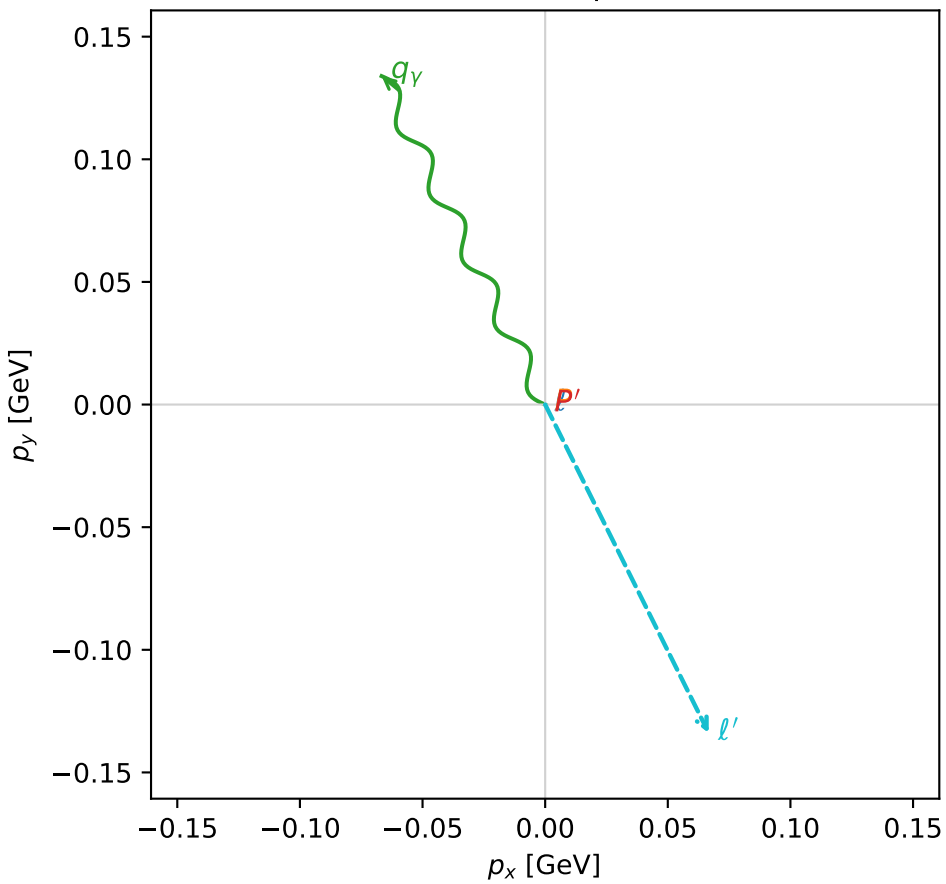
$s=6.56399$, $\sqrt{s}=2.56203$, $|\vec{P}|=1.10931$, $|\vec{P}'|=1.08259$
 $\theta_{p'}=0$, $\theta_Y=2.94521$, $\phi_{p'}=0.59395$, $\phi_Y=2.03282$
 $E_Y=0.766728$, $\theta_{l'}=2.71664$, $\phi_{l'}=5.17441$
 $Q^2=0.0716046$, $x_B=0.0183773$, $t=-0.00030178$, $W^2=4.7046$, $y=0.68548$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 1.1093$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -1.1093$
 P $E= 1.4527$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.1093$
 l' $E= 0.3629$ $p_x= 0.0667$ $p_y= -0.1339$ $p_z= -0.3306$
 P' $E= 1.4324$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 1.0826$
 q_Y $E= 0.7667$ $p_x= -0.0667$ $p_y= 0.1339$ $p_z= -0.7520$

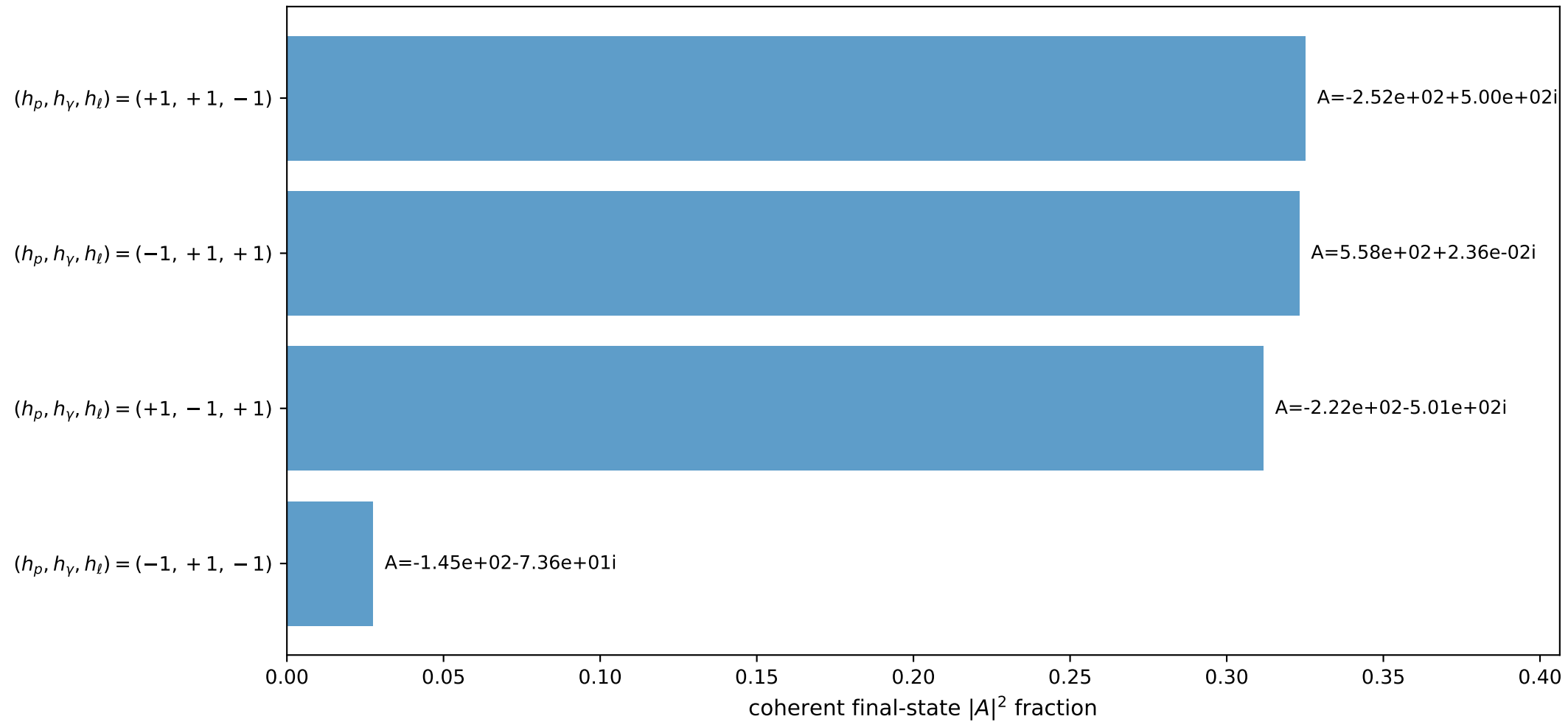
3D momenta



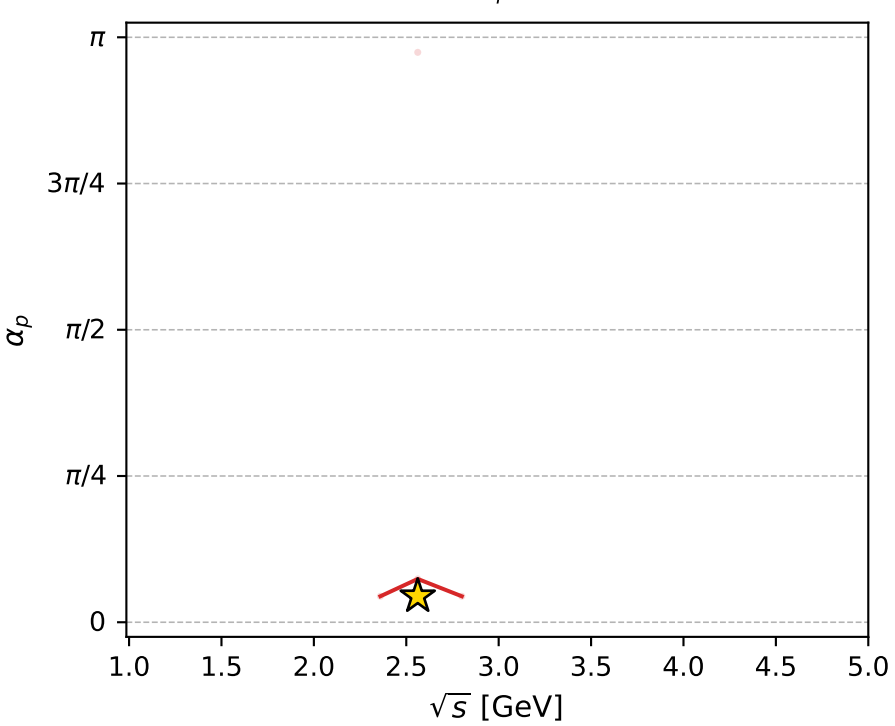
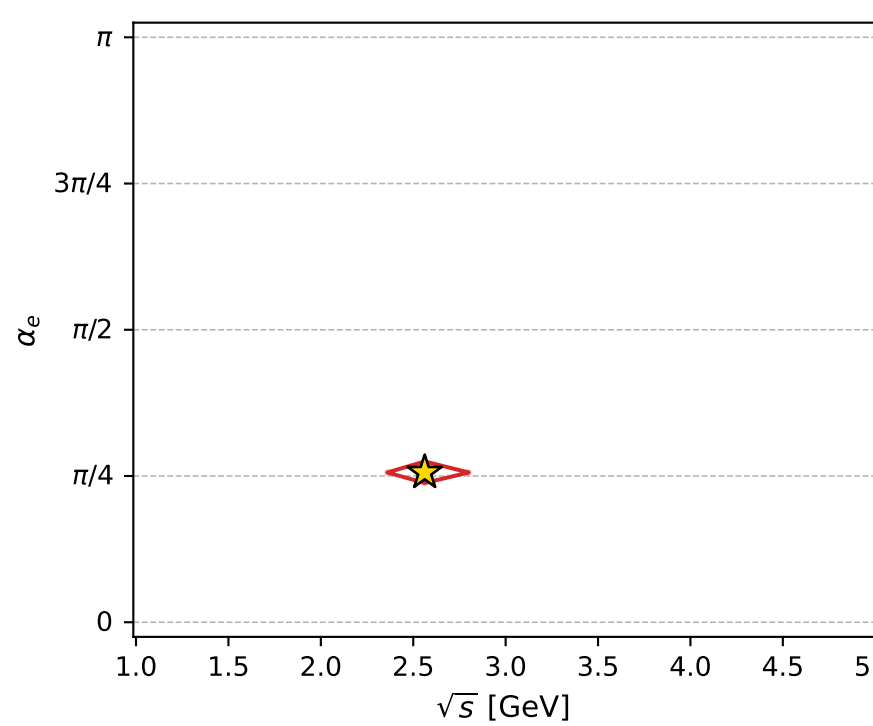
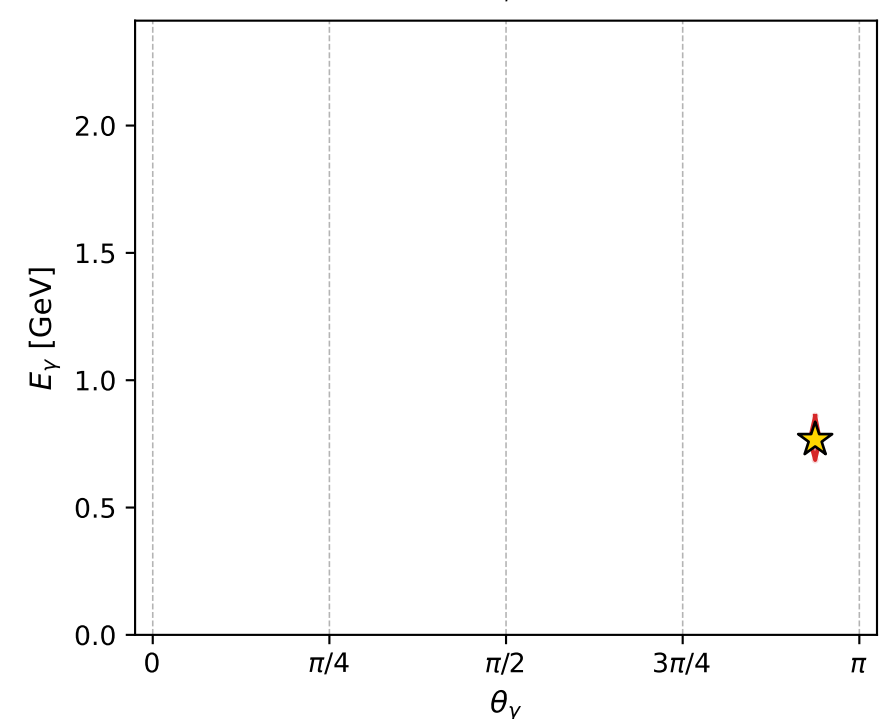
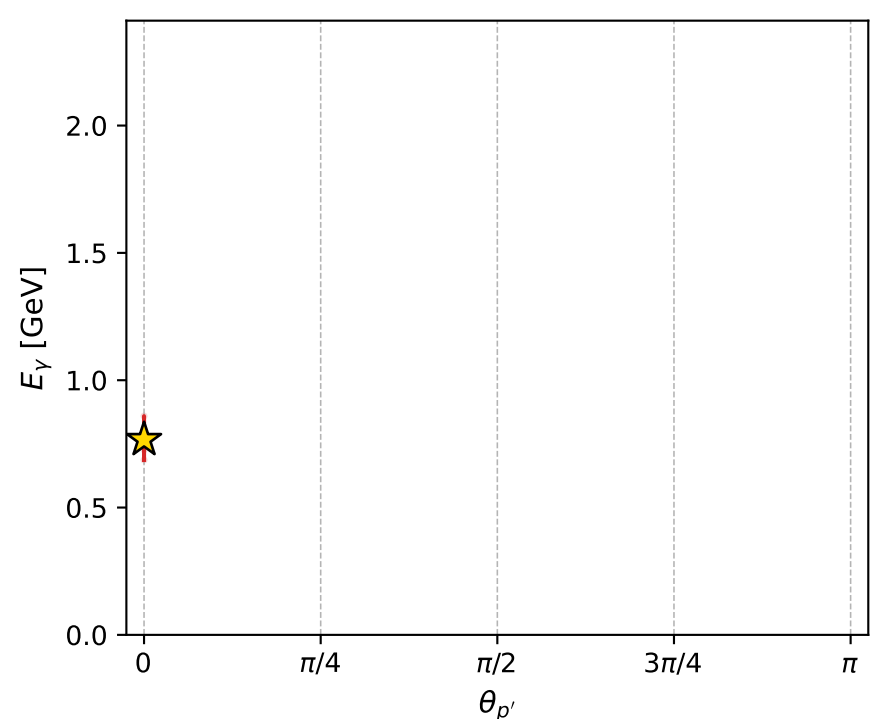
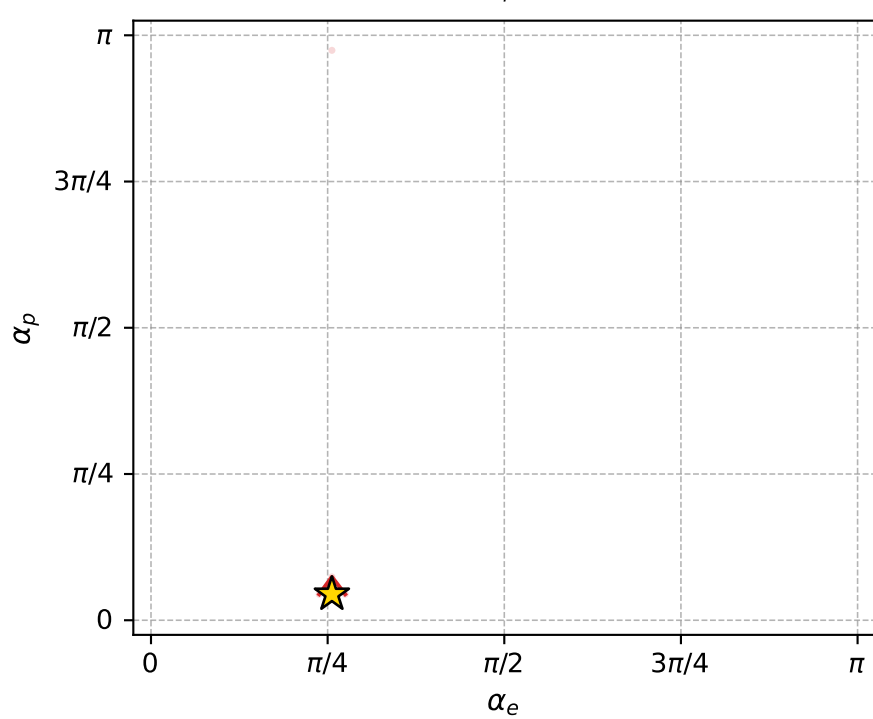
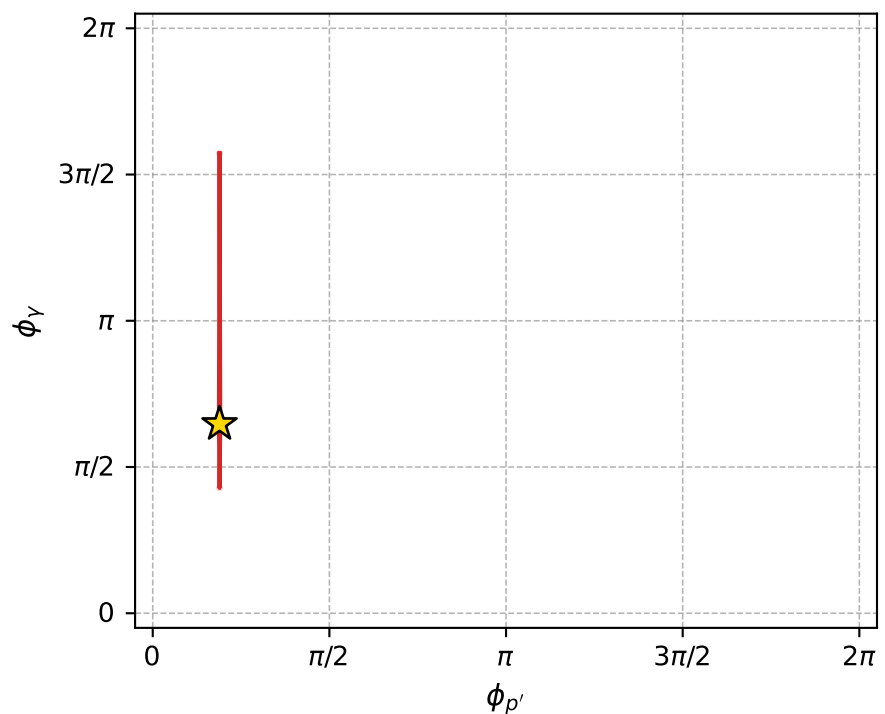
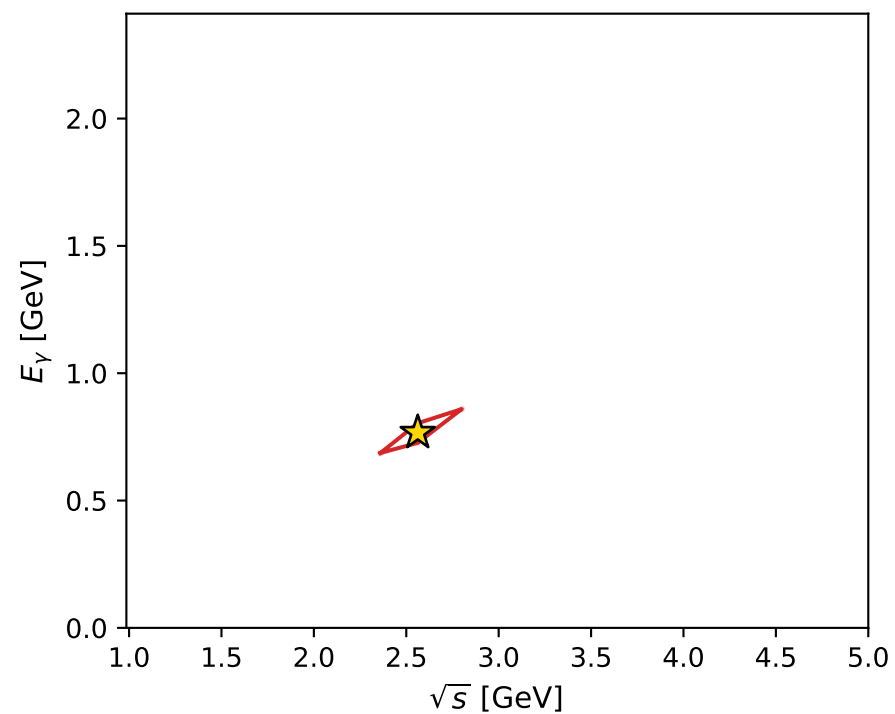
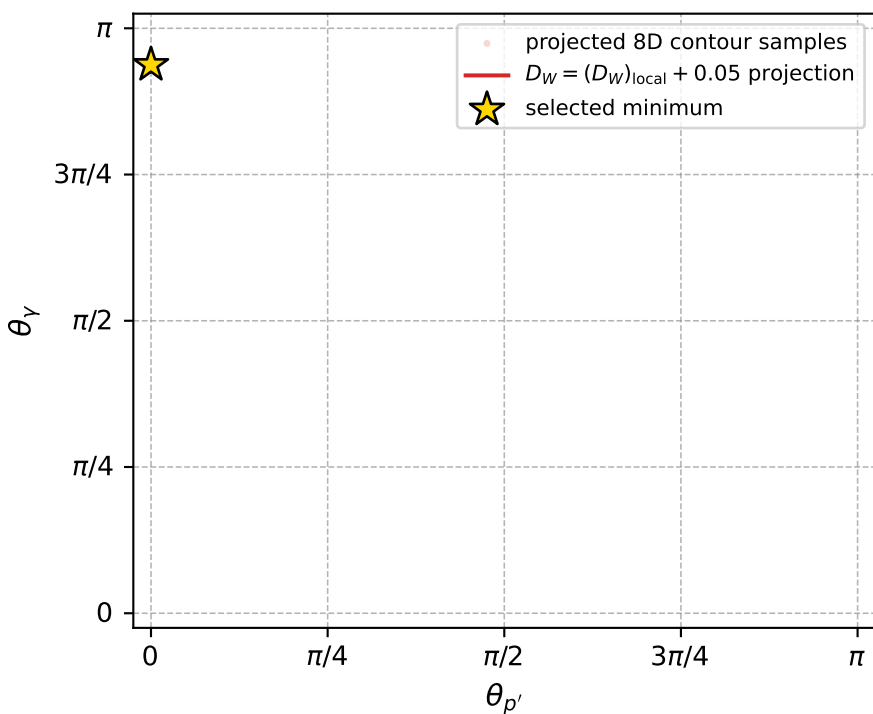
Transverse plane



Leading final-state amplitudes (N=4, retained=98.7%)



electron: polarization cluster P3, local minimum 744; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0086943828$
 $D_W \text{ contour} = 0.058694383$
 above global minimum = 0.00867795
 8D contour samples = 139

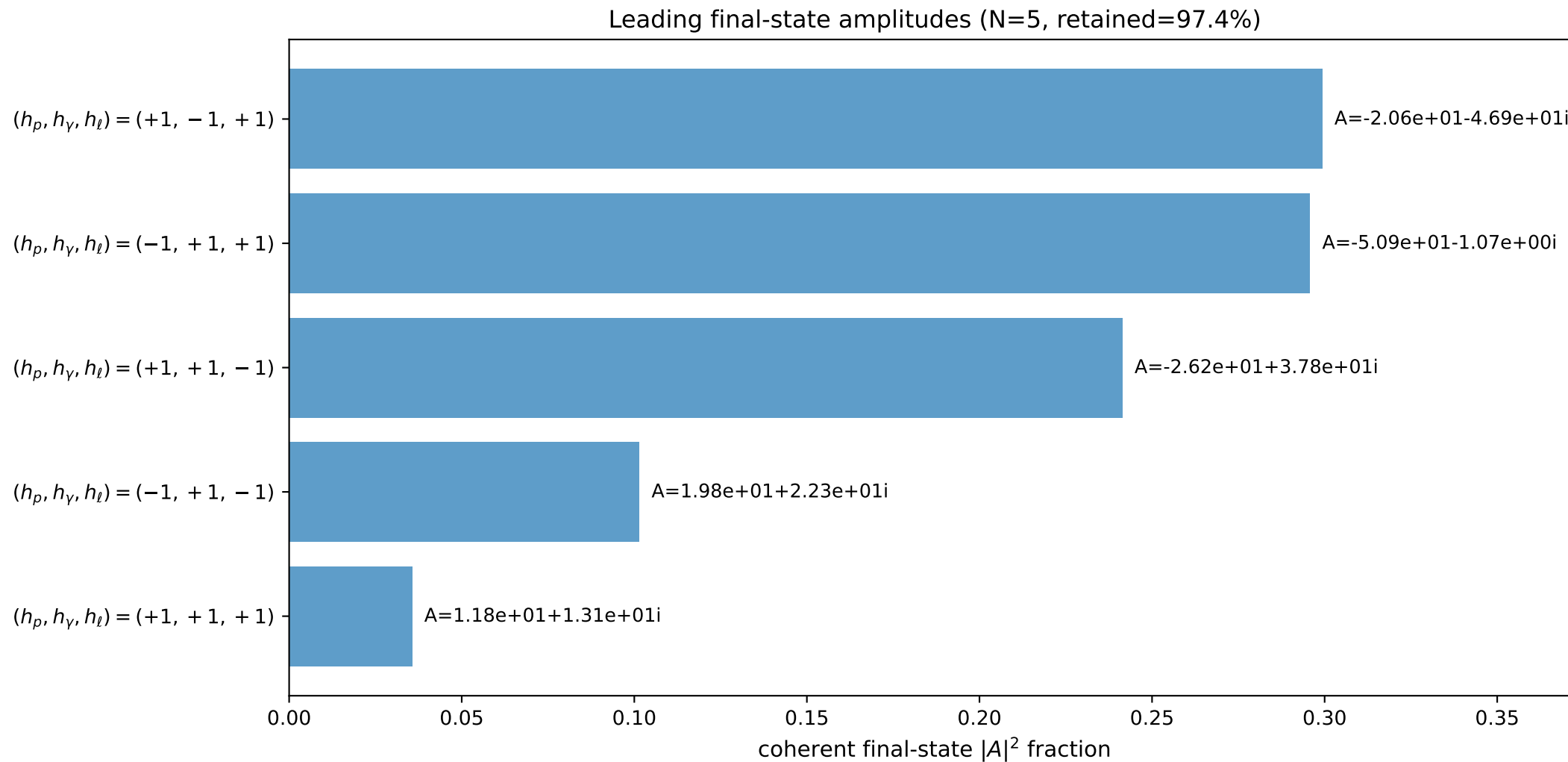
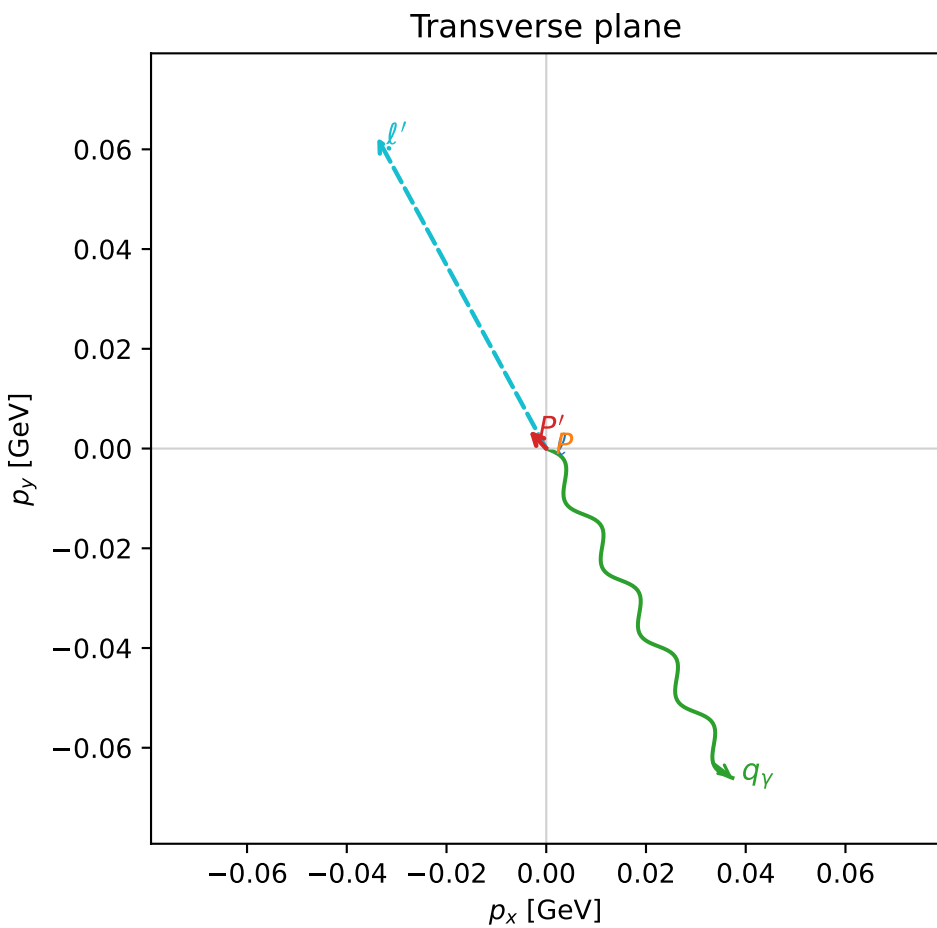
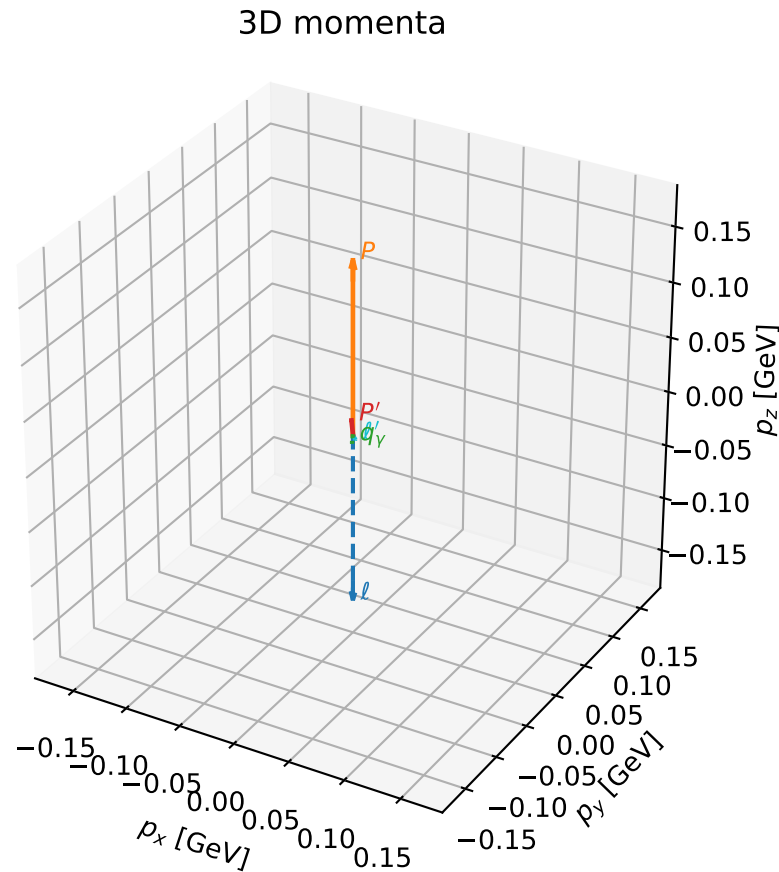
selected phase-space point:
 $\text{sqrt_s} = 2.562028$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 2.945213$
 $\text{q0ut} = 0.7667281$
 $\text{phi_p_out} = 0.5939497$
 $\text{phi_gamma_out} = 2.032819$
 $\text{alpha_e} = 0.8043797$
 $\text{alpha_p} = 0.1395732$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

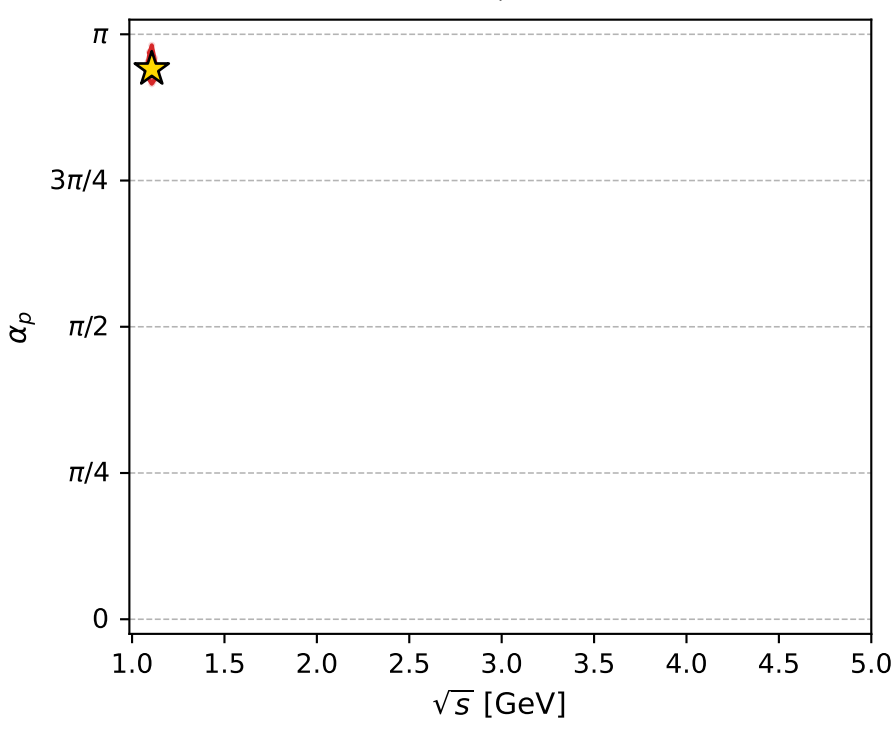
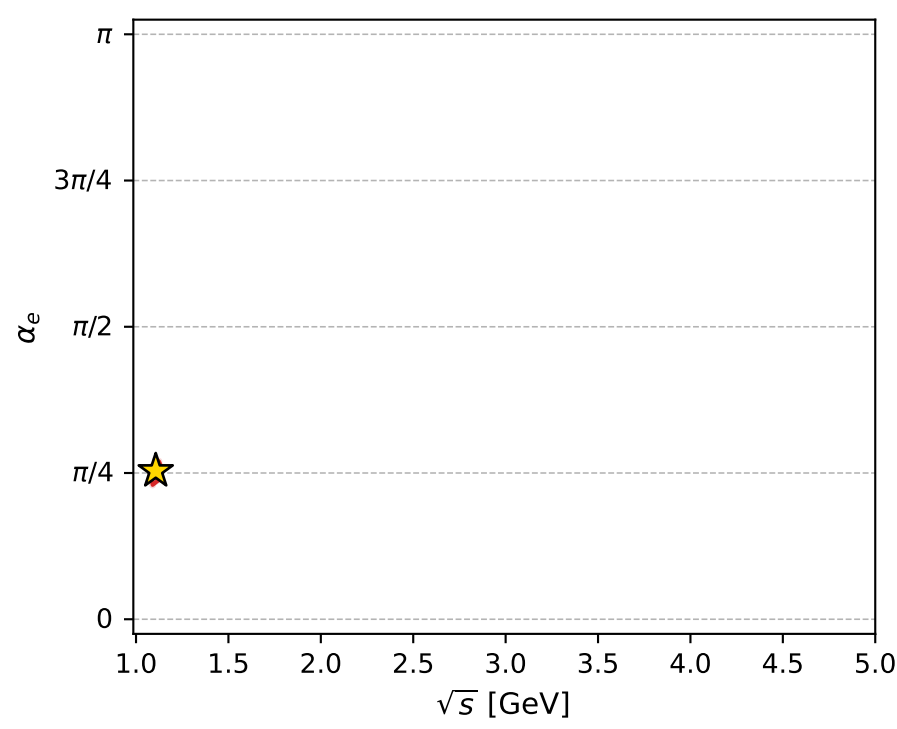
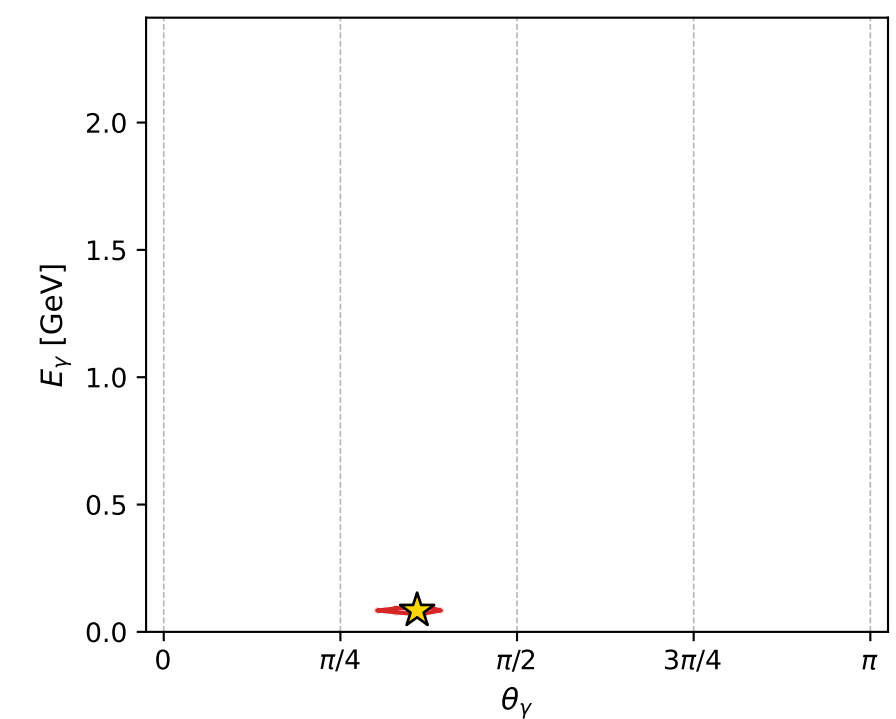
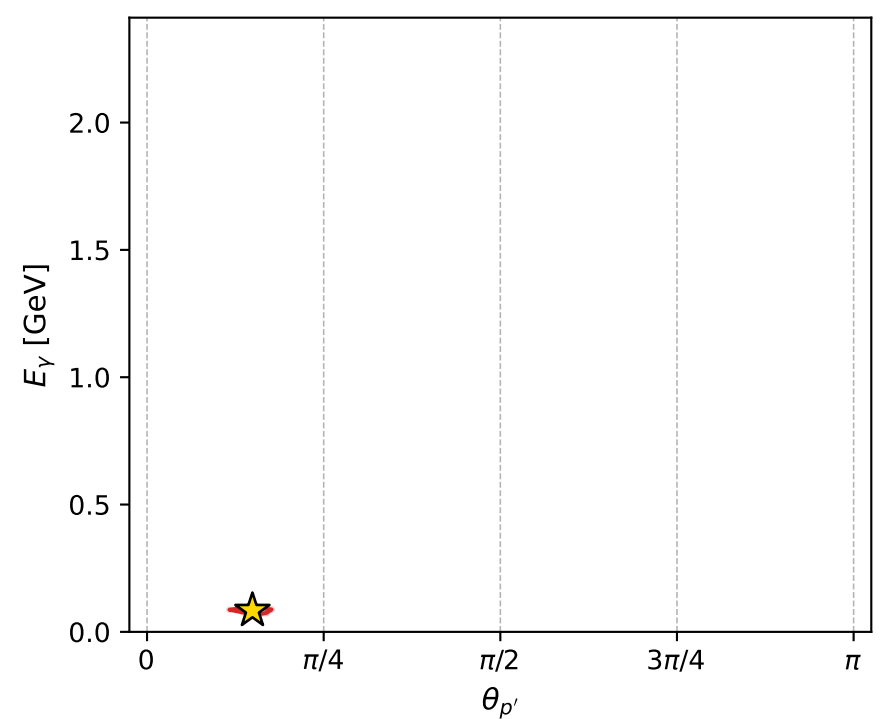
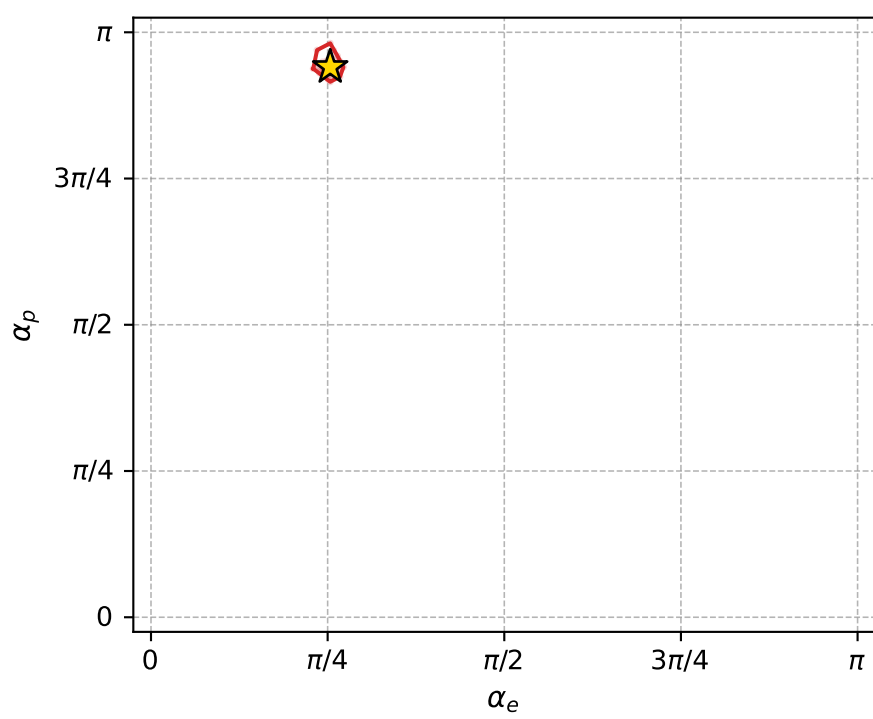
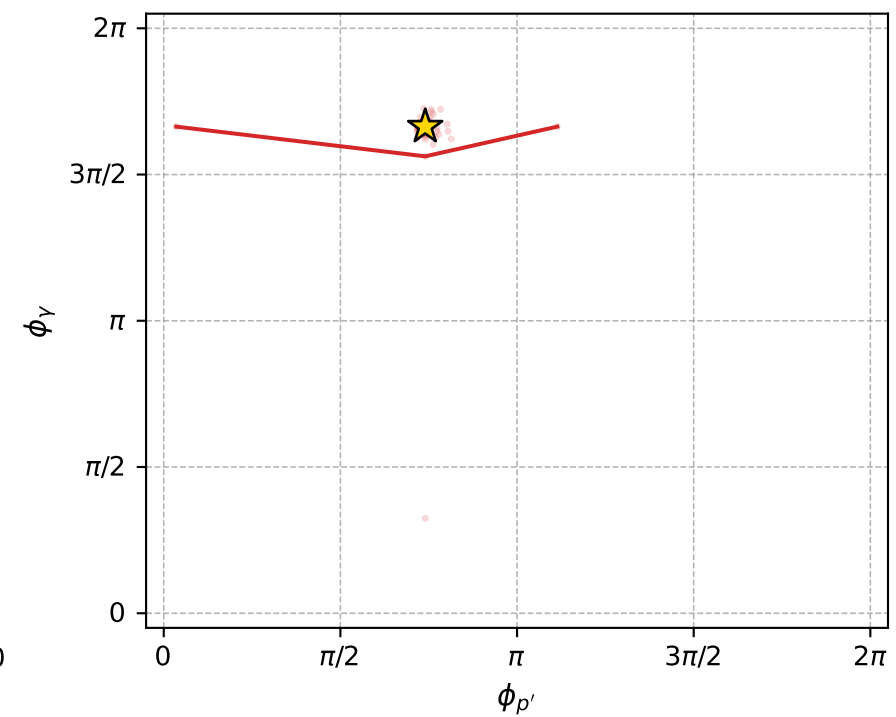
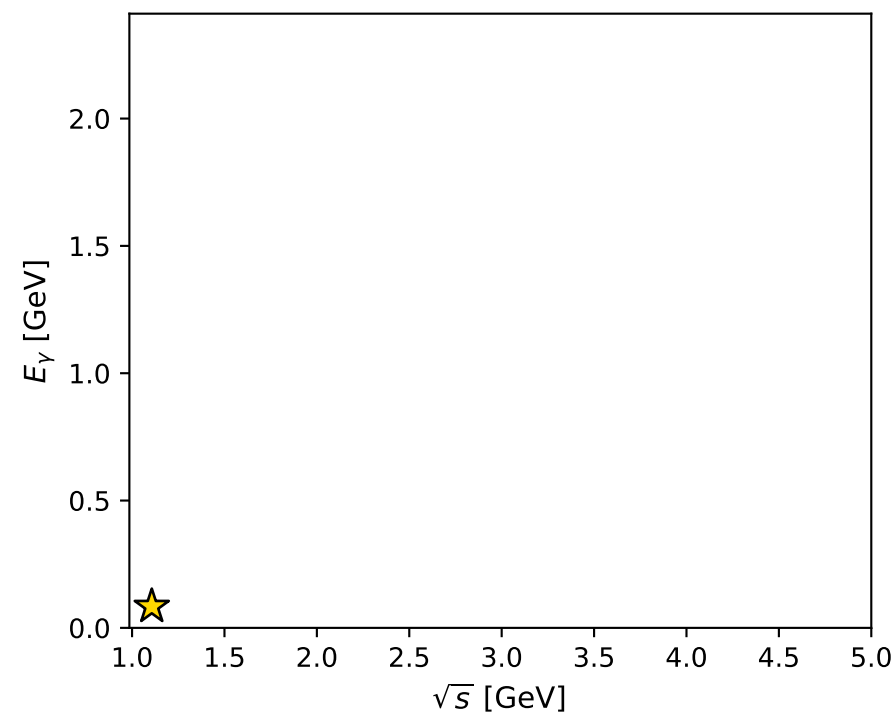
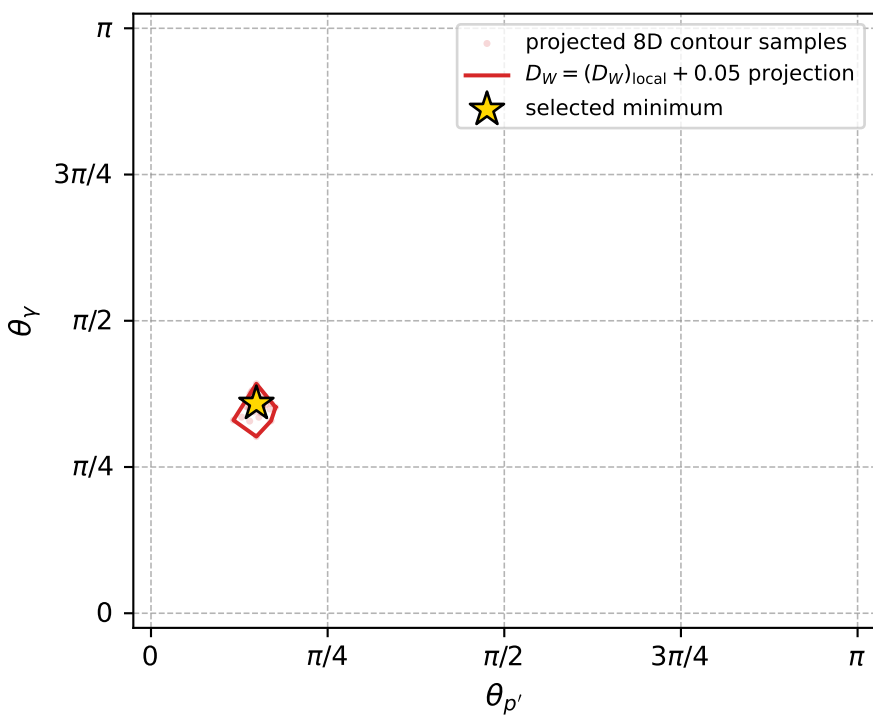
dw_mixing_angles_polarization_03_local_minimum_745 D_W=0.00871633
 region: polarization_cluster_03_local_minimum_745
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79674515$ rad
 incoming lepton: $0.699038|+\rangle + 0.715085|-\rangle$
 proton mixing angle: $\alpha_p=2.955029$ rad
 incoming proton: $-0.982647|+\rangle + 0.185483|-\rangle$

$s=1.22476$, $\sqrt{s}=1.10669$, $|\vec{P}|=0.15583$, $|\vec{P}'|=0.0110224$
 $\theta_{P'}=0.468579$, $\theta_Y=1.1265$, $\phi_{P'}=2.32518$, $\phi_Y=5.22687$
 $E_Y=0.0840194$, $\theta_{\ell'}=2.14493$, $\phi_{\ell'}=2.06862$
 $Q^2=0.0120469$, $x_B=0.0709888$, $t=-0.021176$, $W^2=1.0375$, $y=0.492013$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1558$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1558$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1558$
 ℓ' $E= 0.0846$ $p_x= -0.0339$ $p_y= 0.0624$ $p_z= -0.0459$
 P' $E= 0.9381$ $p_x= -0.0034$ $p_y= 0.0036$ $p_z= 0.0098$
 q_Y $E= 0.0840$ $p_x= 0.0373$ $p_y= -0.0660$ $p_z= 0.0361$



electron: polarization cluster P3, local minimum 745; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0087163285$
 $D_W \text{ contour} = 0.058716328$
 above global minimum = 0.0086998957
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.106687$
 $\text{theta_p_out} = 0.4685786$
 $\text{theta_gamma_out} = 1.126497$
 $\text{qOut} = 0.0840194$
 $\text{phi_p_out} = 2.325182$
 $\text{phi_gamma_out} = 5.226866$
 $\text{alpha_e} = 0.7967451$
 $\text{alpha_p} = 2.955029$

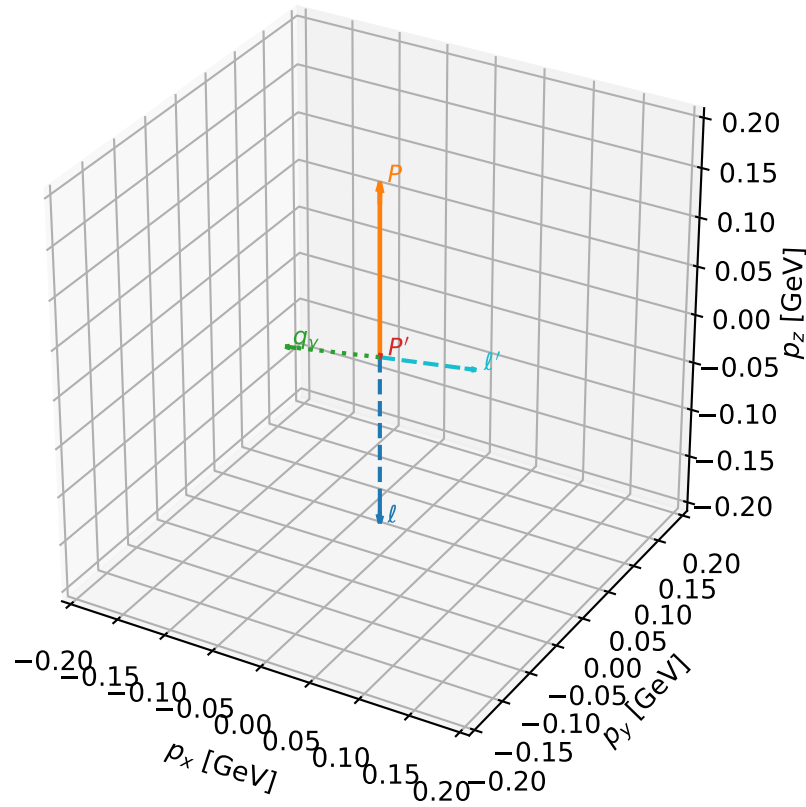
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_754 D_W=0.00921934
 region: polarization_cluster_03_local_minimum_754
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77896007$ rad
 incoming lepton: $0.711645|+\rangle + 0.70254|-\rangle$
 proton mixing angle: $\alpha_p=0.20936607$ rad
 incoming proton: $0.978163|+\rangle + 0.20784|-\rangle$

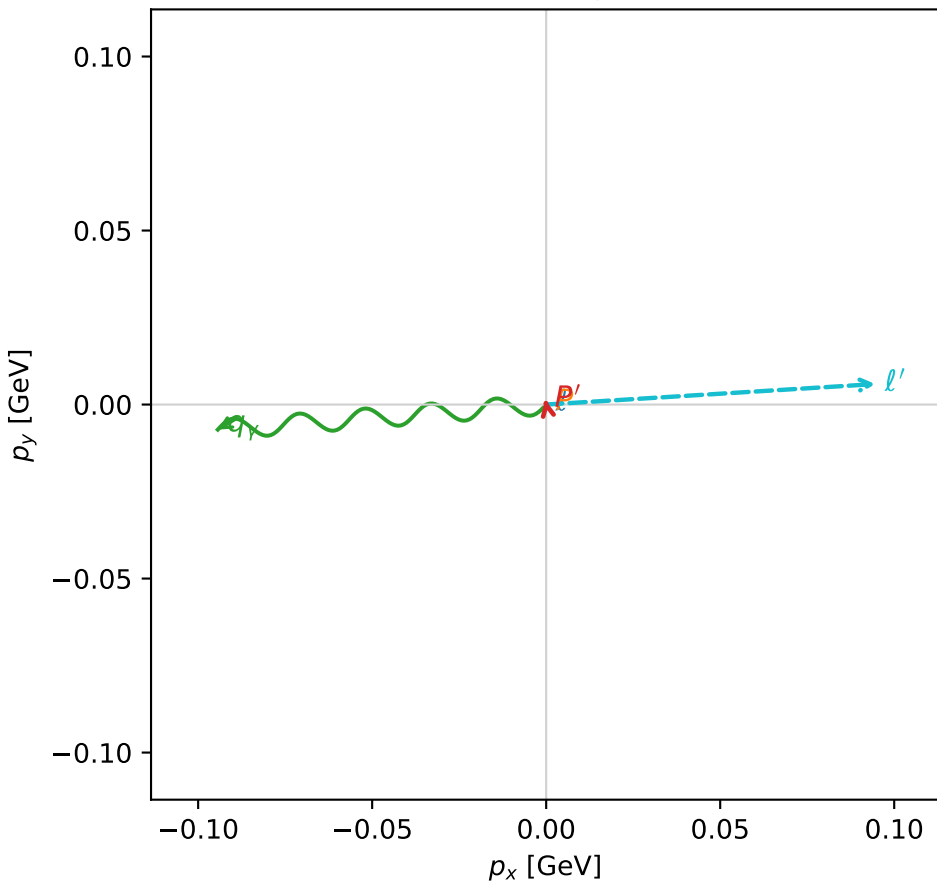
$s=1.27452$, $\sqrt{s}=1.12895$, $|\vec{P}|=0.174798$, $|\vec{P}'|=0.00180454$
 $\theta_{P'}=0.840143$, $\theta_V=1.70314$, $\phi_{P'}=1.76549$, $\phi_V=3.21788$
 $E_V=0.0954639$, $\theta_{\ell'}=1.45118$, $\phi_{\ell'}=0.0622055$
 $Q^2=0.0373623$, $x_B=0.172612$, $t=-0.0298758$, $W^2=1.05893$, $y=0.548433$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1748$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1748$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1748$
 ℓ' $E= 0.0955$ $p_x= 0.0946$ $p_y= 0.0059$ $p_z= 0.0114$
 P' $E= 0.9380$ $p_x= -0.0003$ $p_y= 0.0013$ $p_z= 0.0012$
 q_V $E= 0.0955$ $p_x= -0.0944$ $p_y= -0.0072$ $p_z= -0.0126$

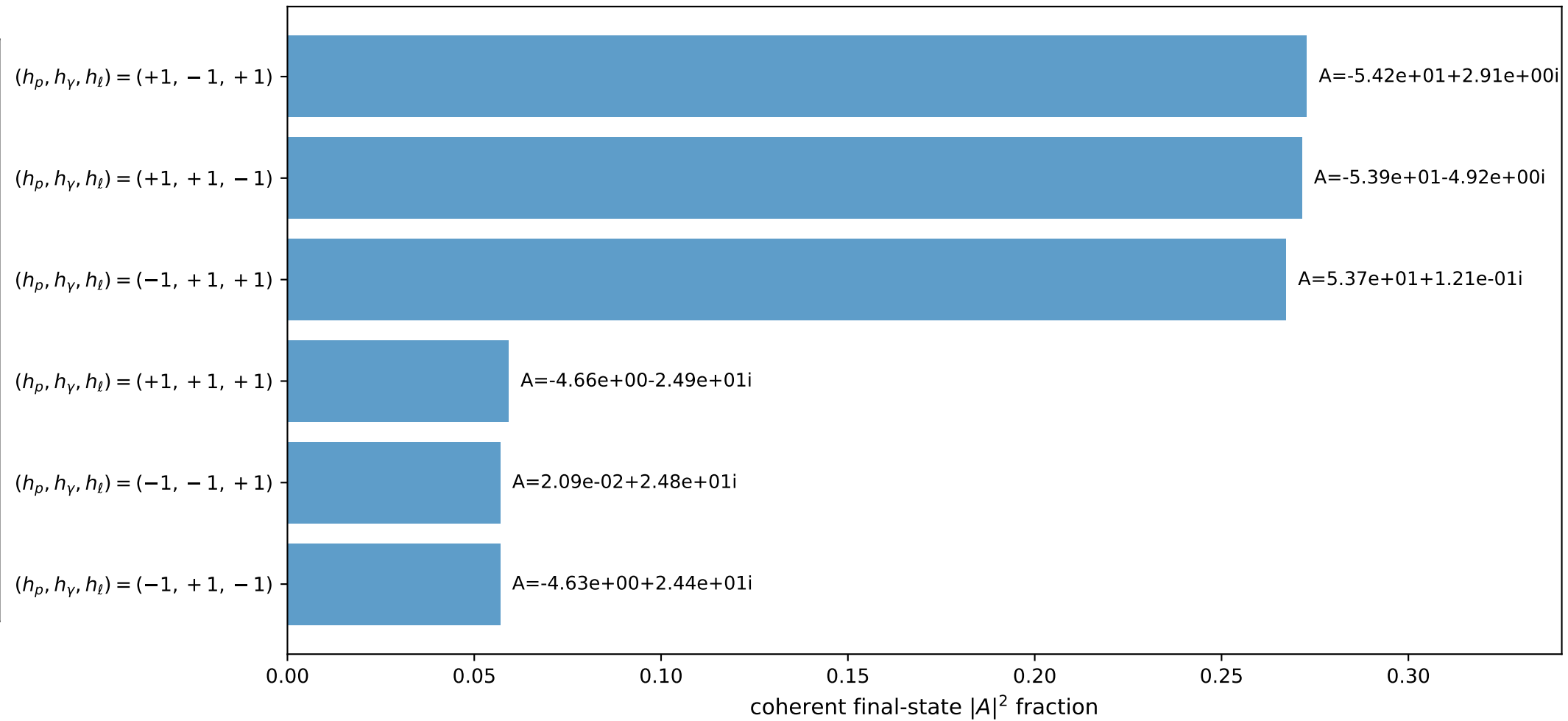
3D momenta



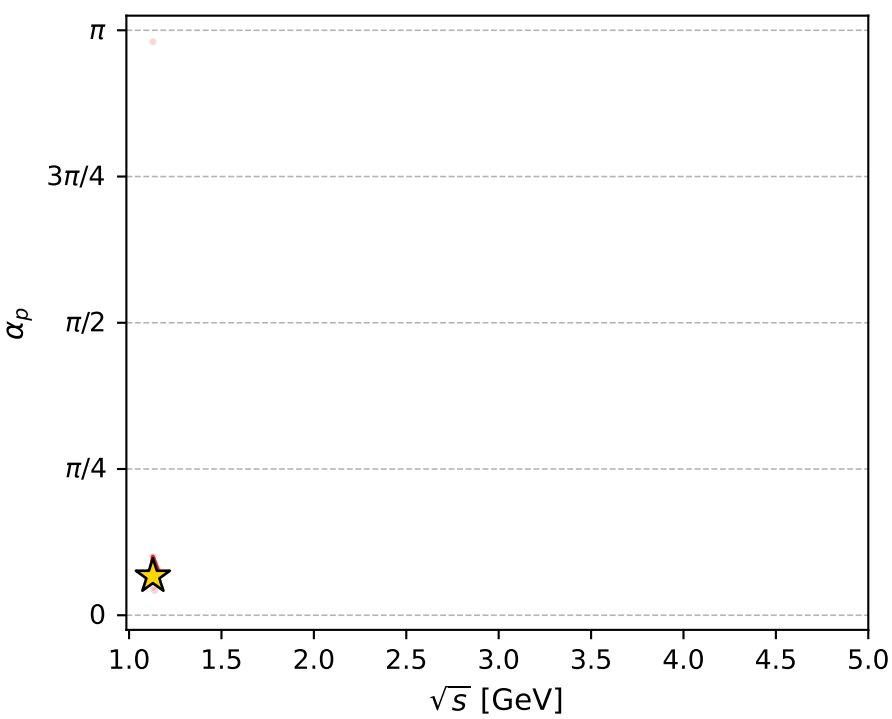
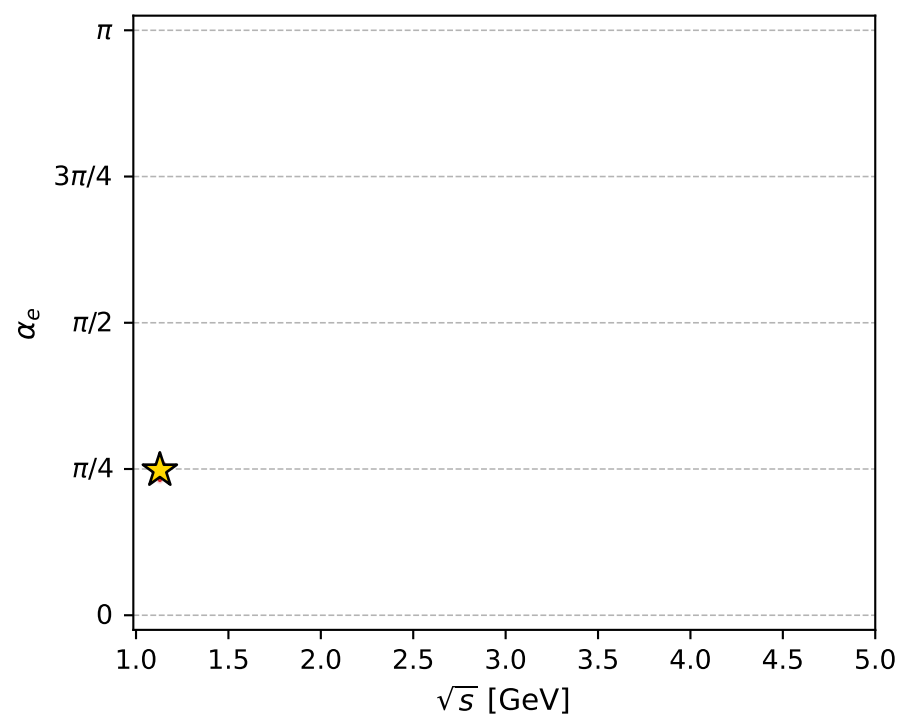
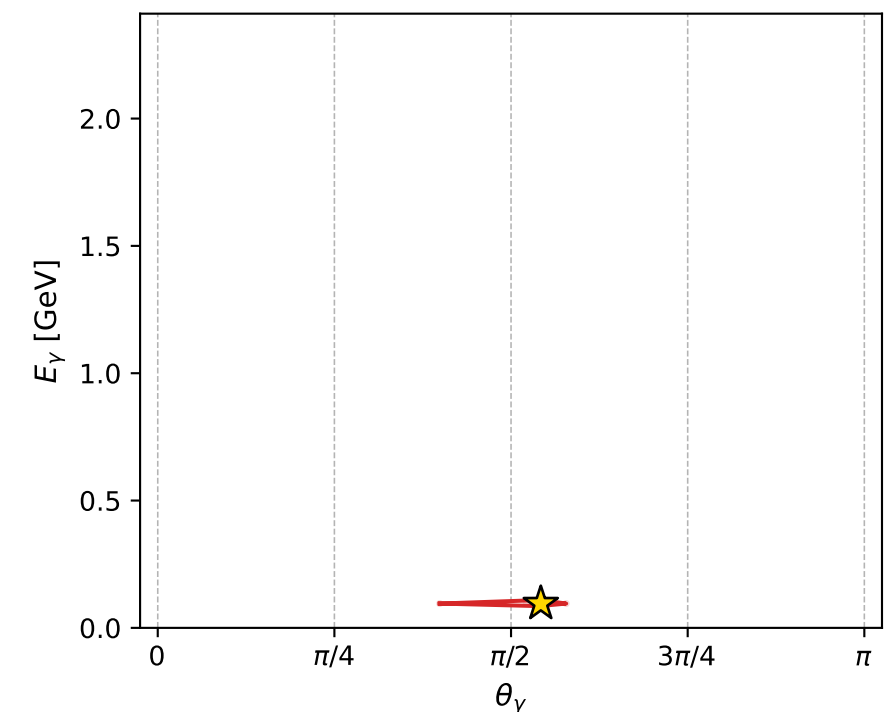
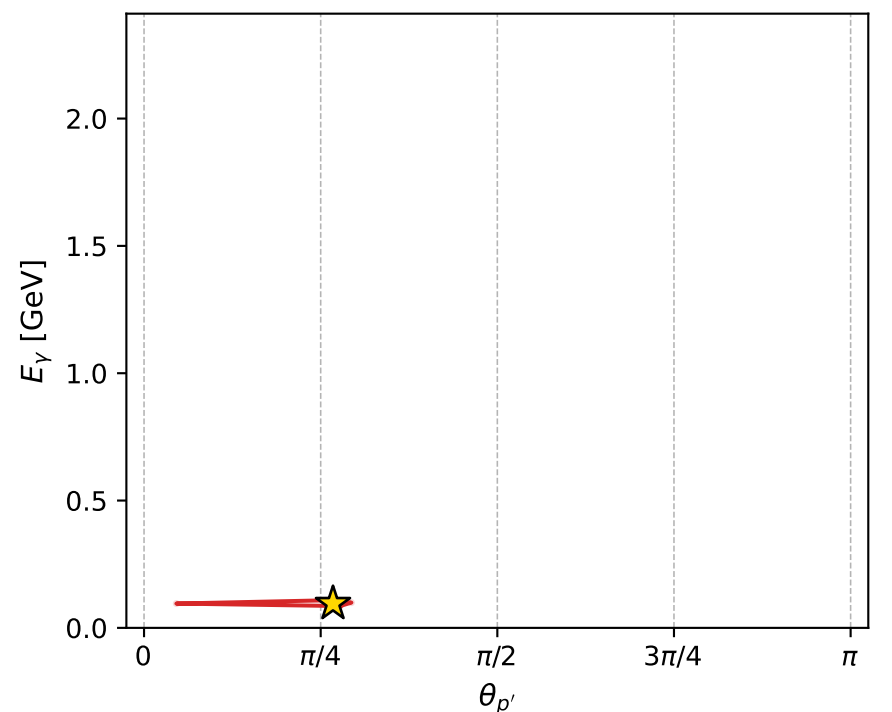
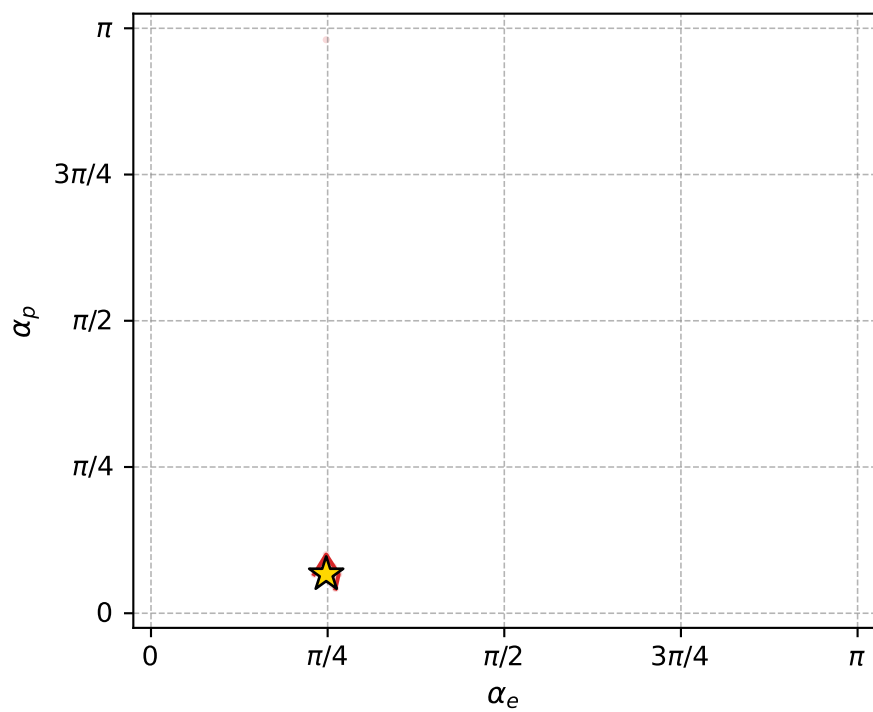
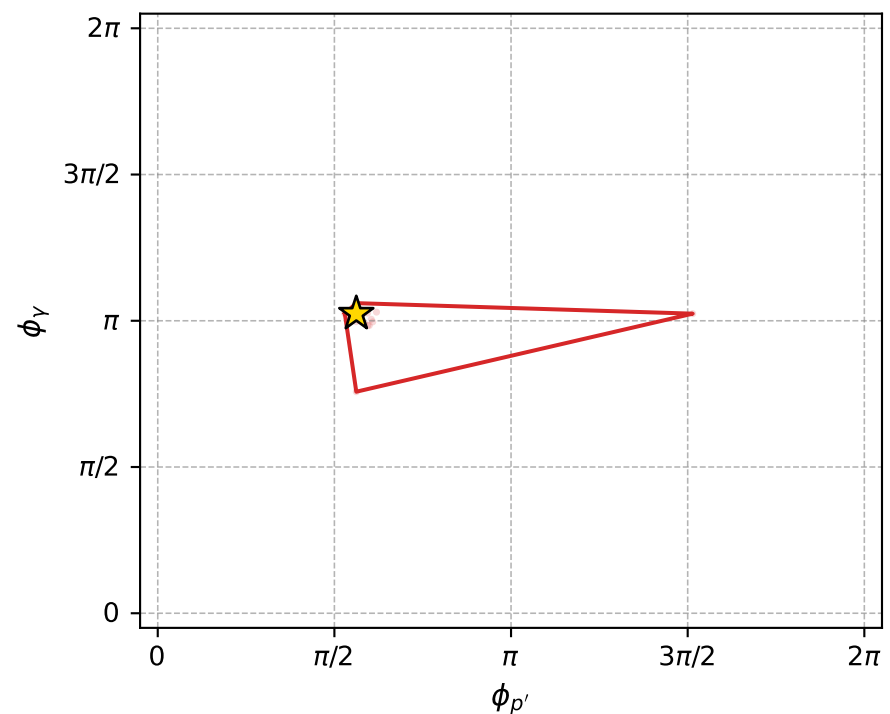
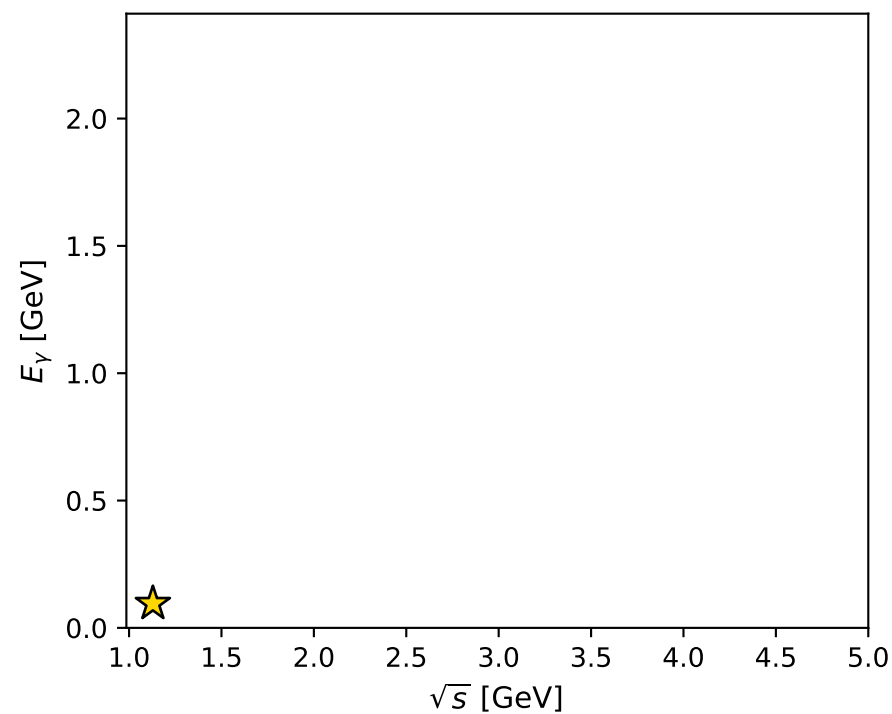
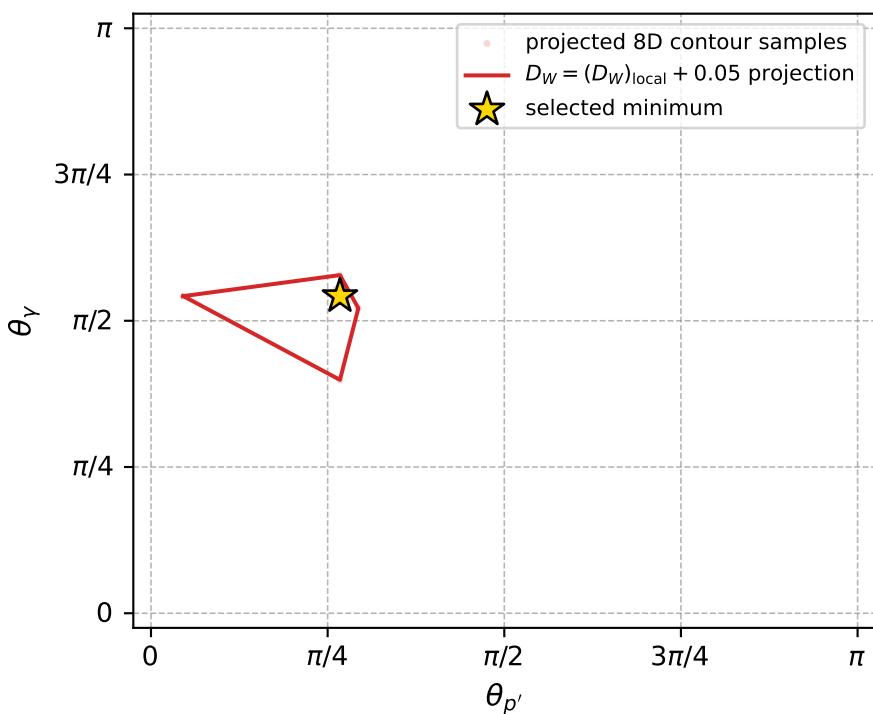
Transverse plane



Leading final-state amplitudes (N=6, retained=98.5%)



electron: polarization cluster P3, local minimum 754; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0092193371$
 $D_W \text{ contour} = 0.059219337$
 above global minimum = 0.0092029043
 8D contour samples = 255

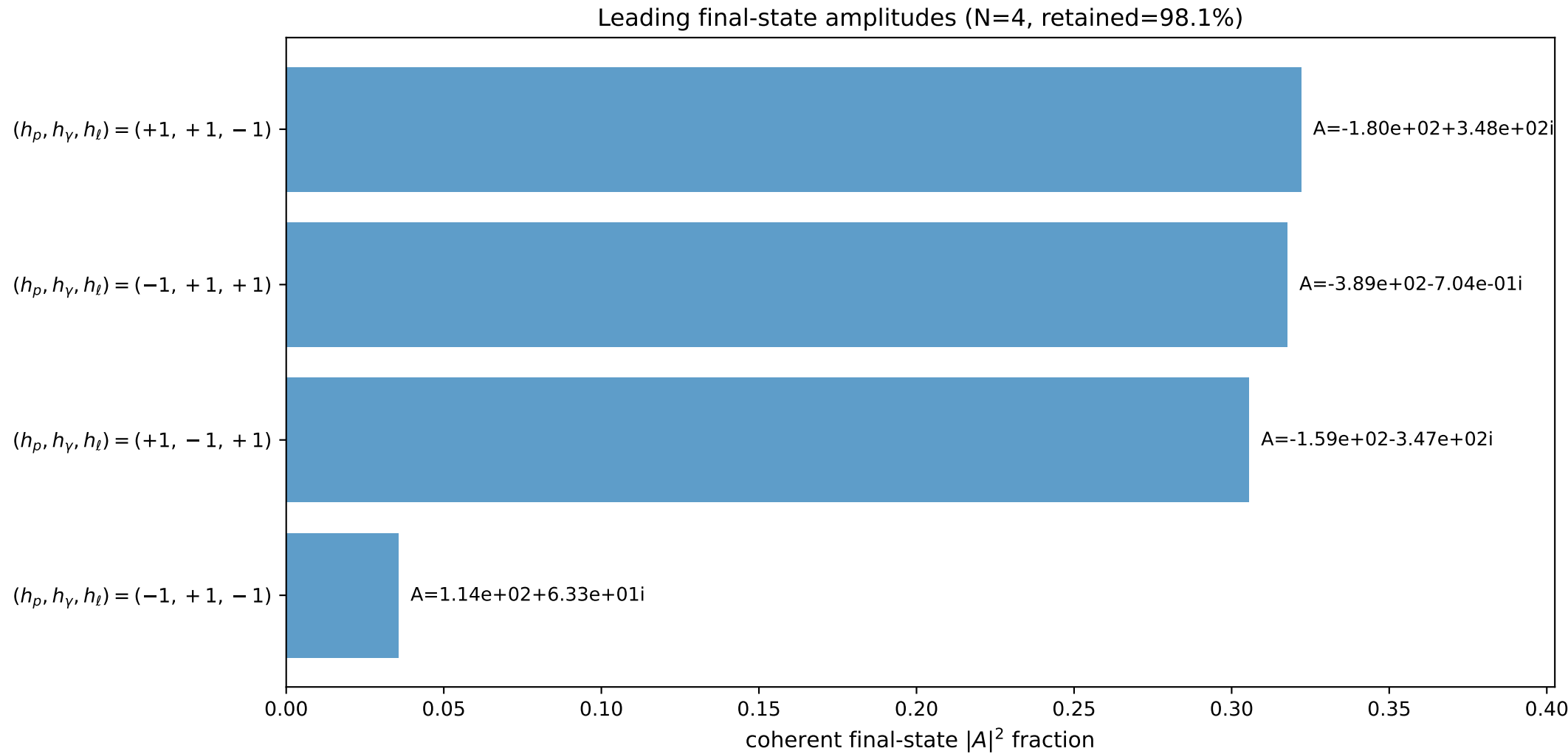
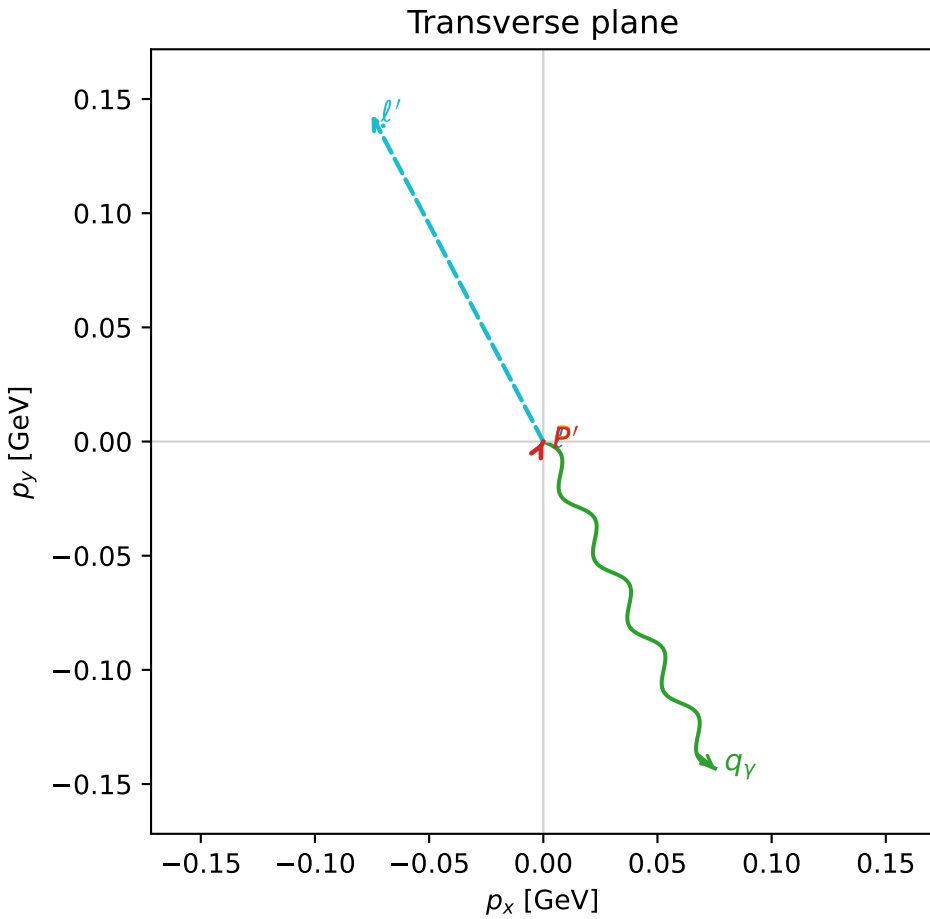
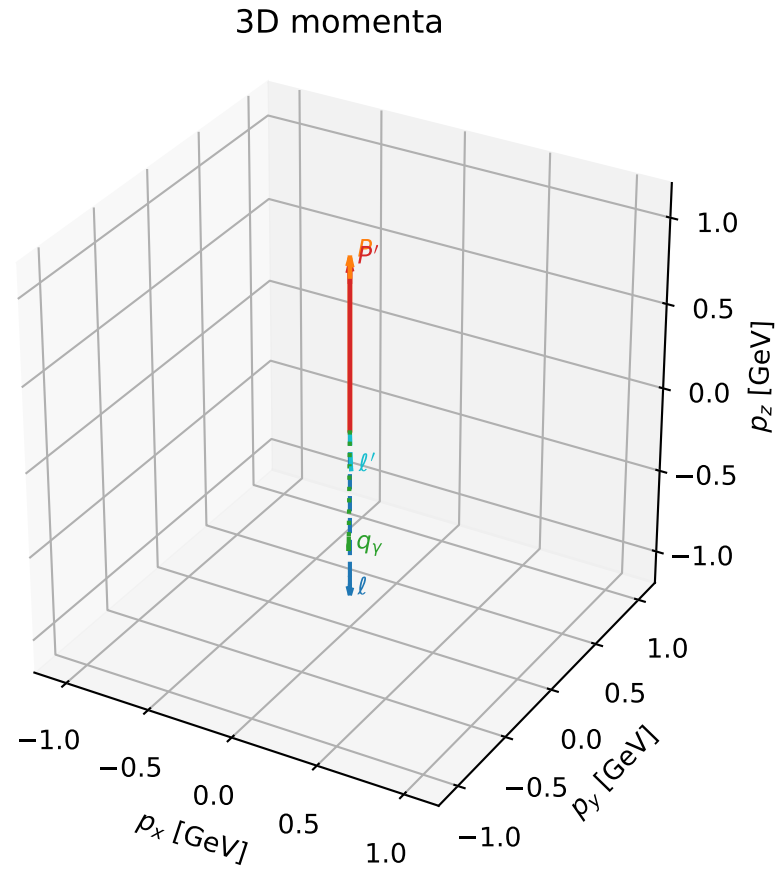
selected phase-space point:
 $\text{sqrt}_s = 1.12896$
 $\text{theta}_{p_out} = 0.8401427$
 $\text{theta}_{\gamma_out} = 1.703144$
 $q_{out} = 0.09546389$
 $\text{phi}_{p_out} = 1.765493$
 $\text{phi}_{\gamma_out} = 3.217876$
 $\alpha_e = 0.7789601$
 $\alpha_p = 0.2093661$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

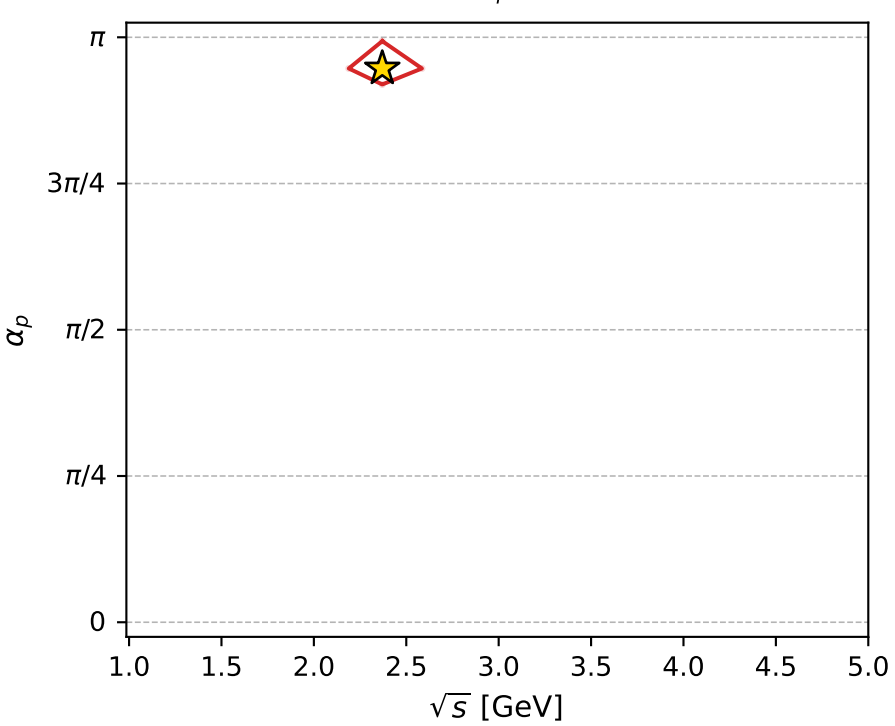
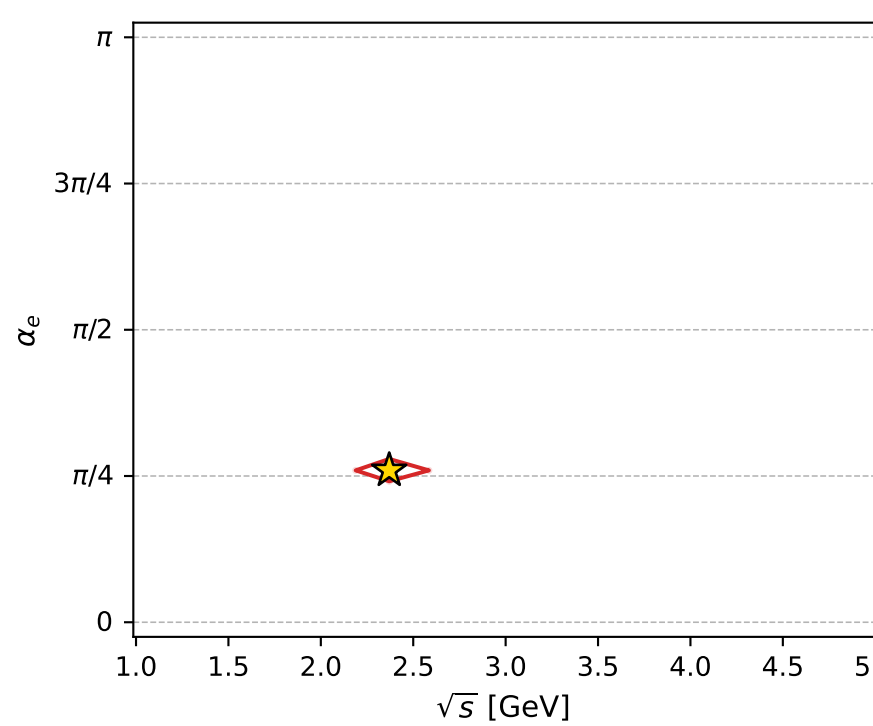
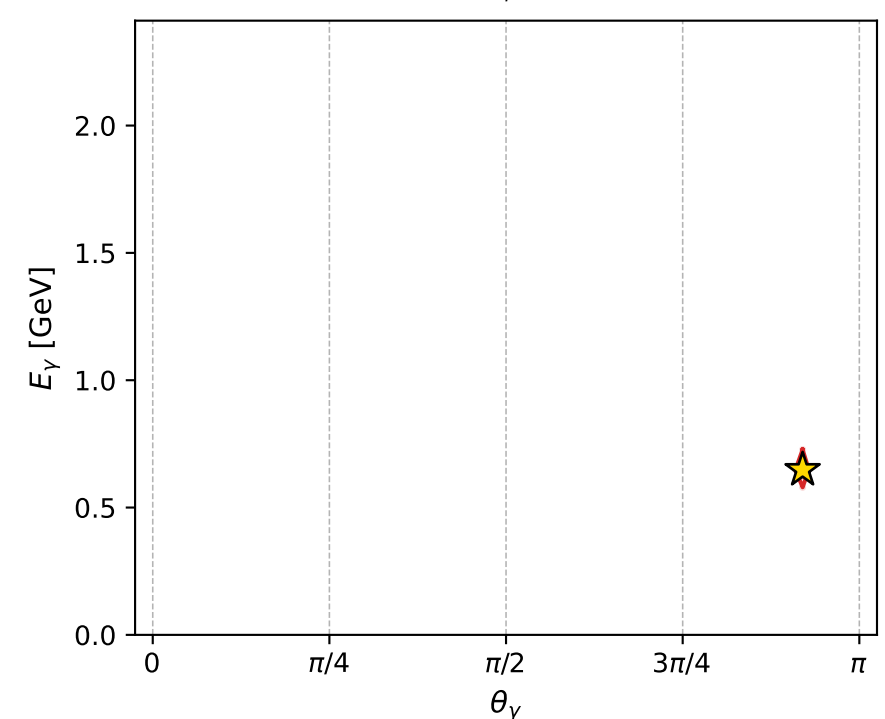
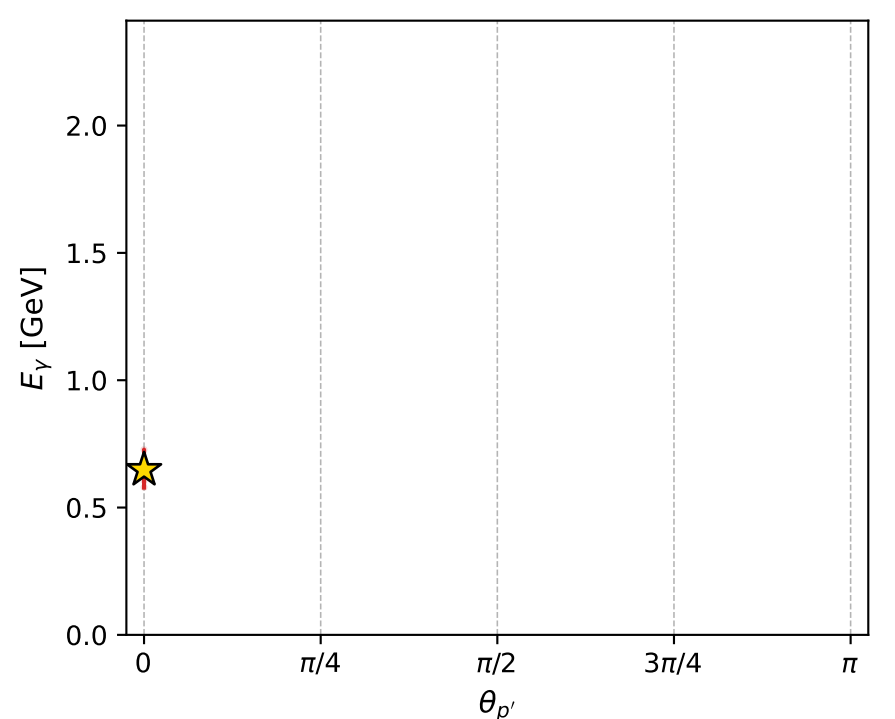
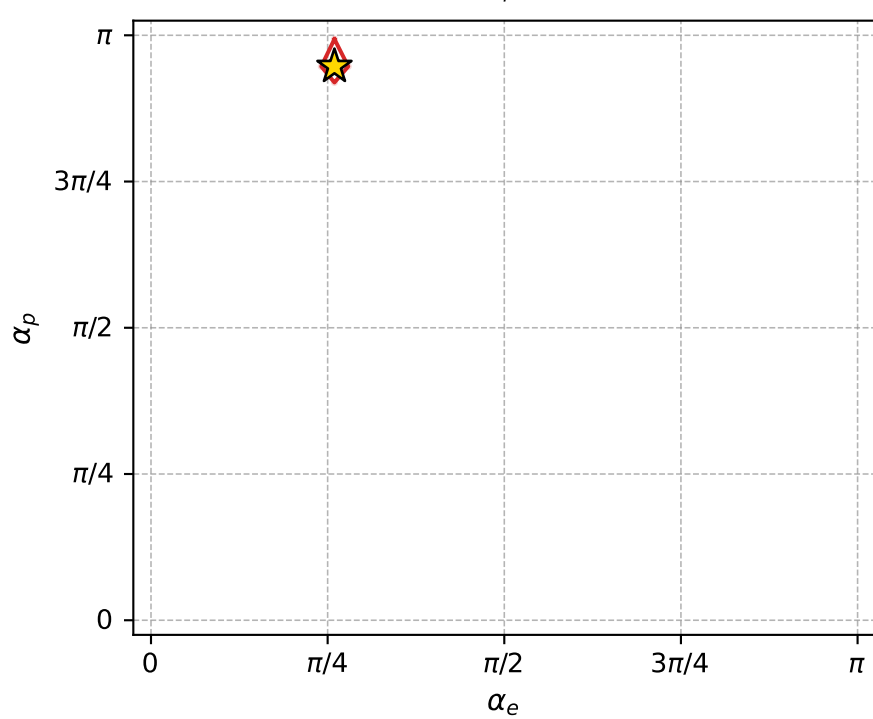
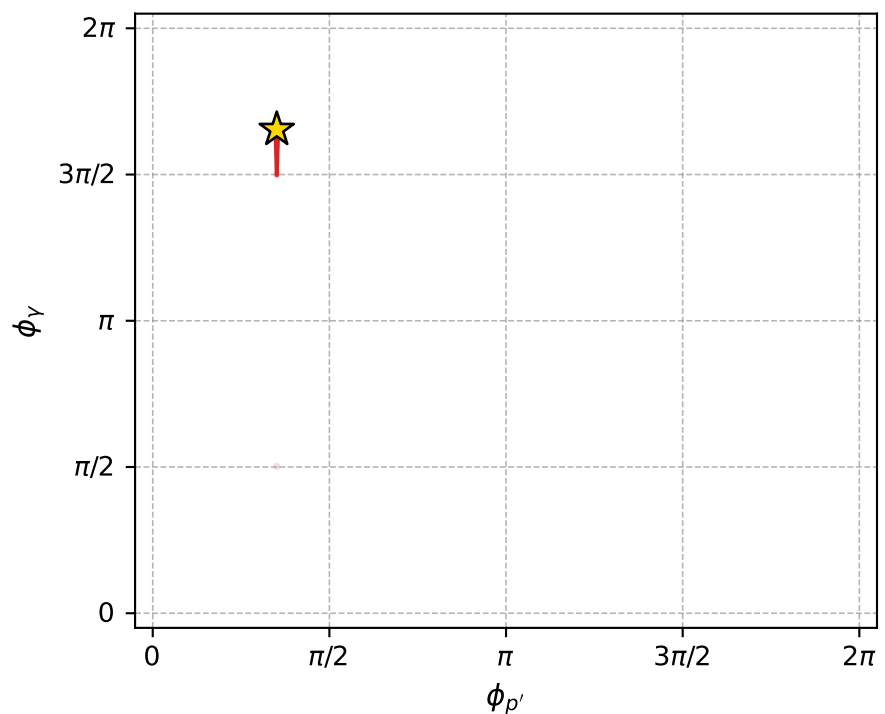
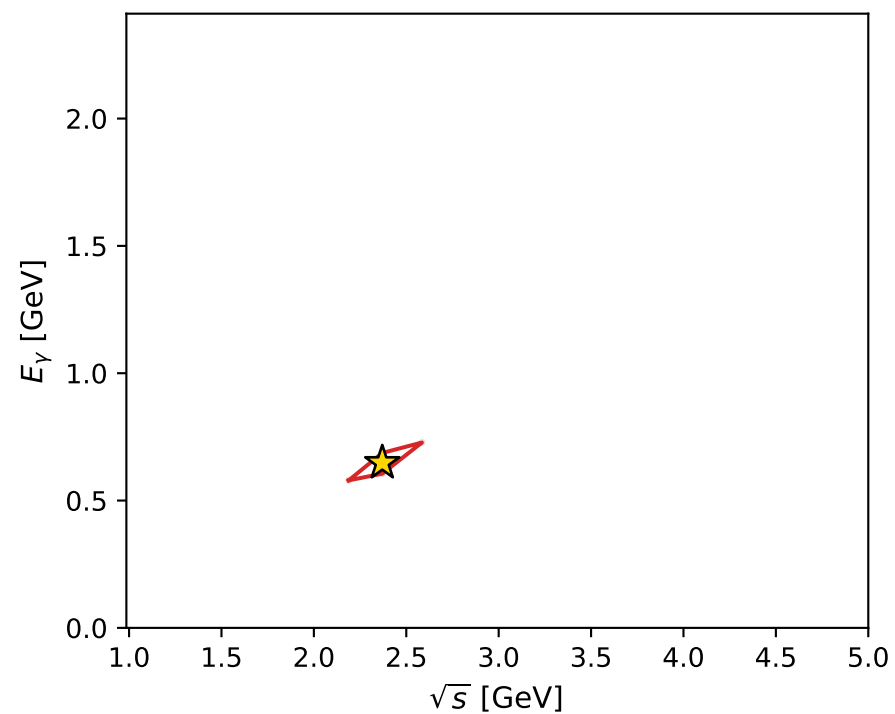
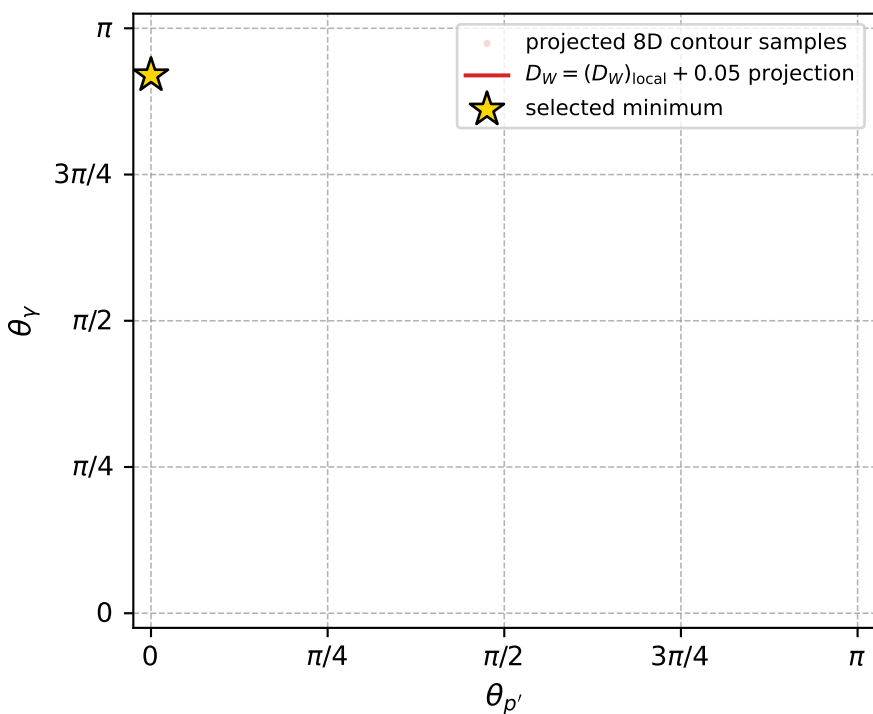
dw_mixing_angles_polarization_03_local_minimum_756 D_W=0.00926699
 region: polarization_cluster_03_local_minimum_756
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81564982$ rad
 incoming lepton: $0.685395|+\rangle +0.728171|-\rangle$
 proton mixing angle: $\alpha_p=2.9741722$ rad
 incoming proton: $-0.986018|+\rangle +0.166639|-\rangle$

$s=5.61751$, $\sqrt{s}=2.37013$, $|\vec{P}|=0.999453$, $|\vec{P}'|=0.966304$
 $\theta_{p'}=2.0761e-05$, $\theta_Y=2.88944$, $\phi_{p'}=1.10269$, $\phi_Y=5.19621$
 $E_Y=0.648198$, $\theta_{\ell'}=2.69605$, $\phi_{\ell'}=2.05472$
 $Q^2=0.0732236$, $x_B=0.0241489$, $t=-0.000523817$, $W^2=3.8388$, $y=0.640015$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.9995$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.9995$
 P $E= 1.3707$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9995$
 ℓ' $E= 0.3752$ $p_x= -0.0752$ $p_y= 0.1431$ $p_z= -0.3386$
 P' $E= 1.3467$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.9663$
 q_Y $E= 0.6482$ $p_x= 0.0752$ $p_y= -0.1432$ $p_z= -0.6277$



electron: polarization cluster P3, local minimum 756; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0092669914$
 $D_W \text{ contour} = 0.059266991$
 above global minimum = 0.0092505585
 8D contour samples = 135

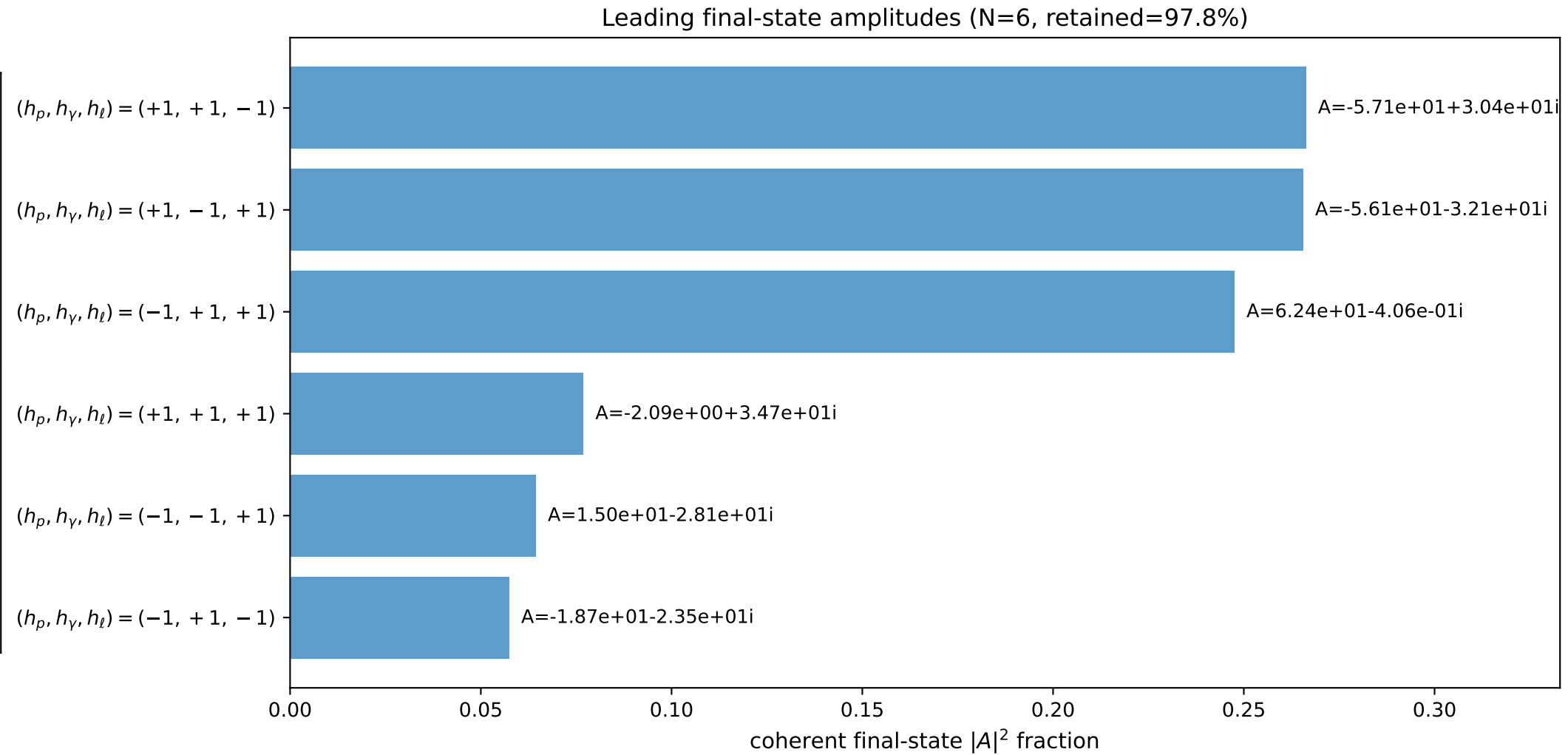
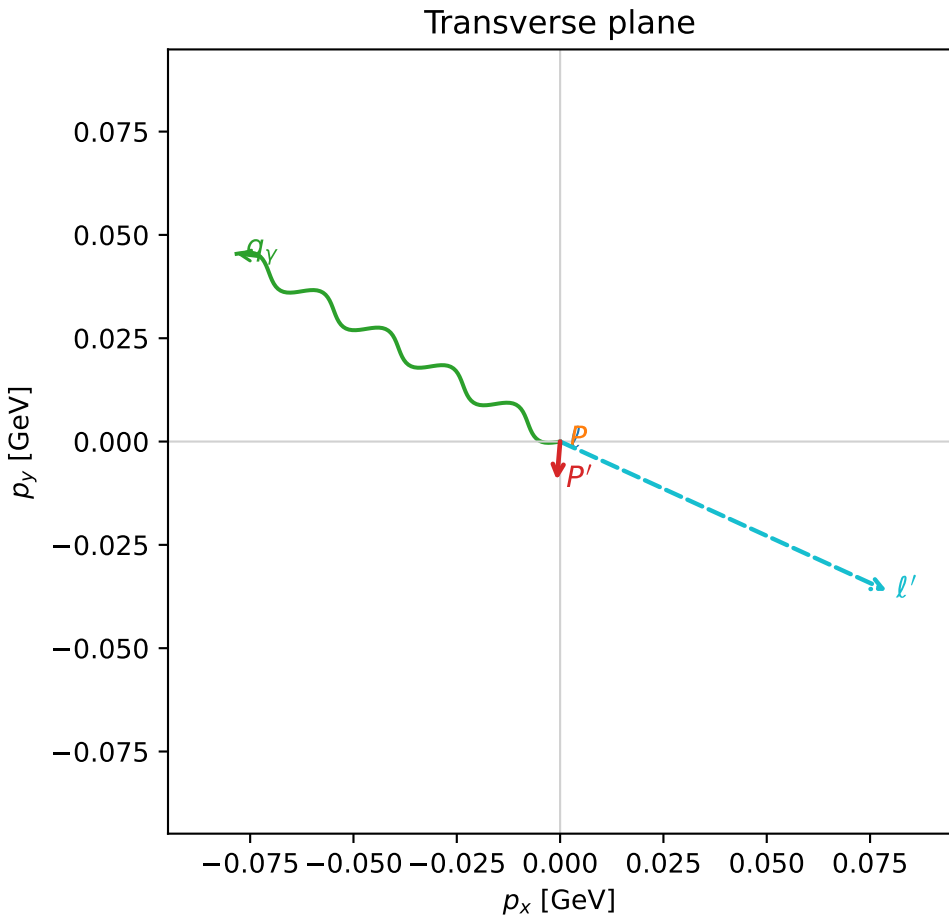
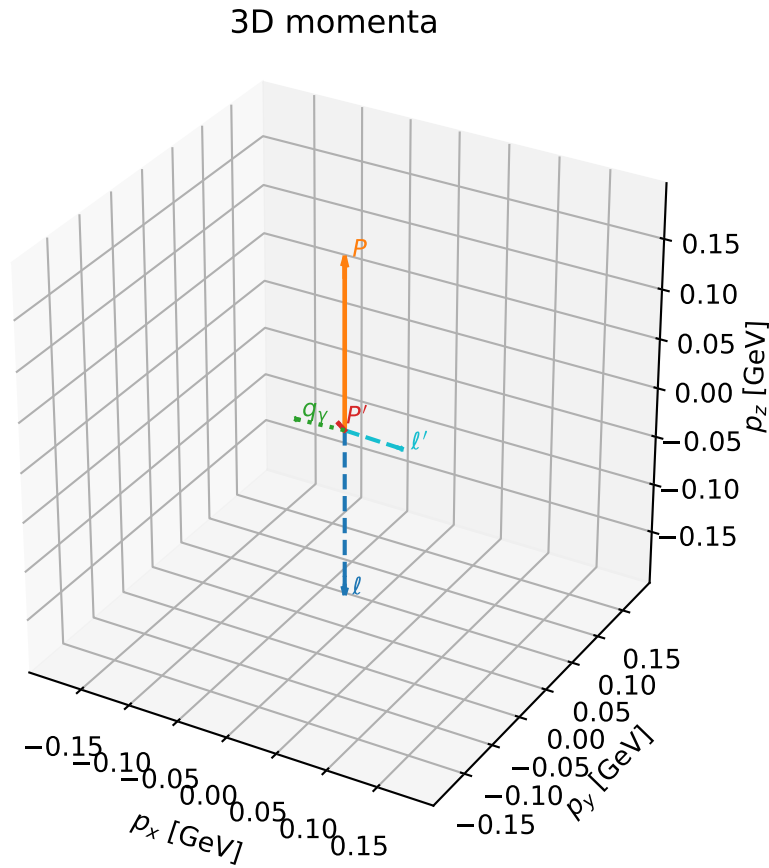
selected phase-space point:
 $\text{sqrt_s} = 2.370128$
 $\text{theta_p_out} = 2.076103\text{e-}05$
 $\text{theta_gamma_out} = 2.88944$
 $\text{qOut} = 0.6481984$
 $\text{phi_p_out} = 1.102692$
 $\text{phi_gamma_out} = 5.196207$
 $\text{alpha_e} = 0.8156498$
 $\text{alpha_p} = 2.974172$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

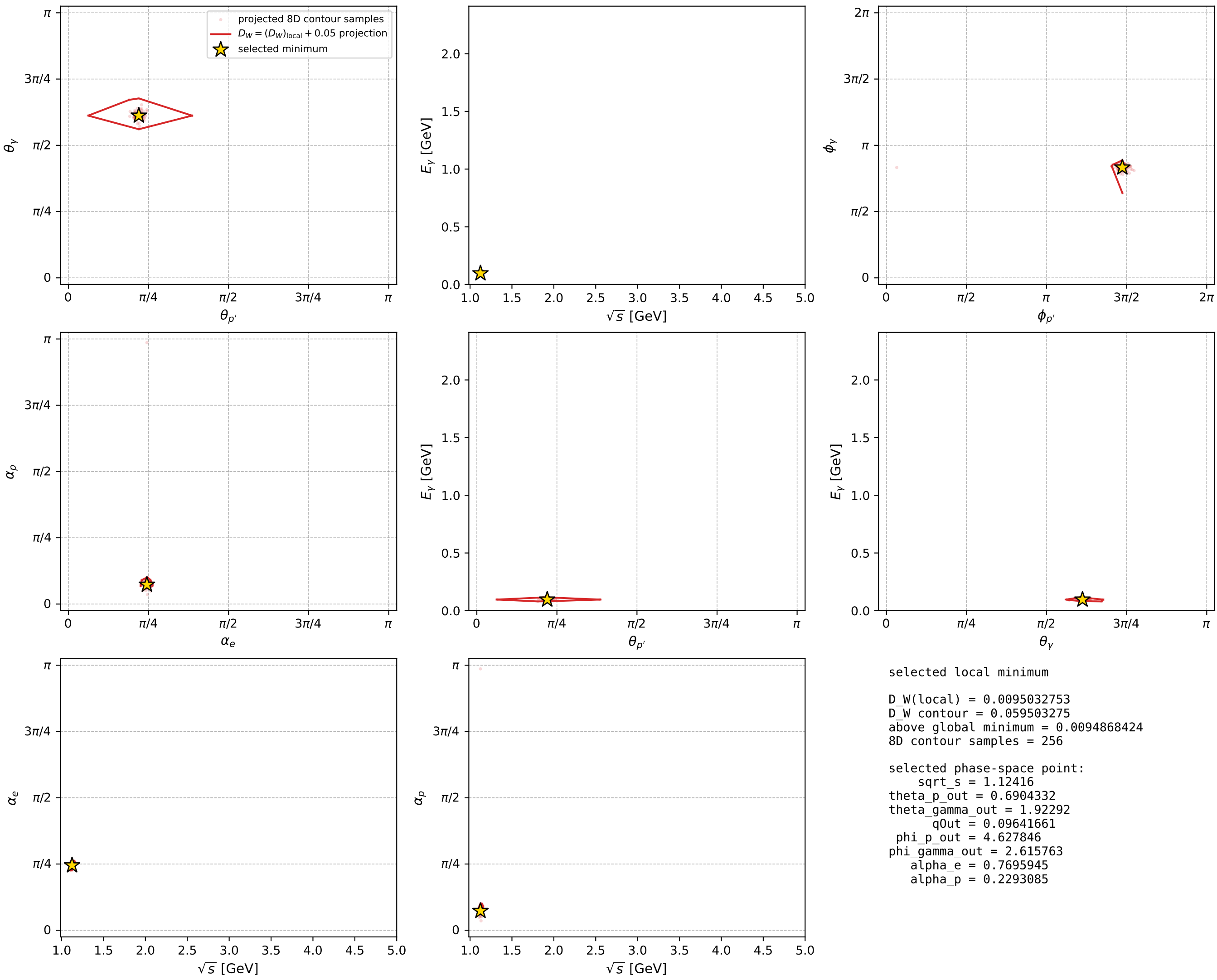
dw_mixing_angles_polarization_03_local_minimum_769 D_W=0.00950328
region: polarization_cluster_03_local_minimum_769
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.76959454$ rad
incoming lepton: $0.718193|+> +0.695844|->$
proton mixing angle: $\alpha_p=0.22930855$ rad
incoming proton: $0.973824|+> +0.227304|->$

$s=1.26374$, $\sqrt{s}=1.12416$, $|\vec{P}|=0.170745$, $|\vec{P}'|=0.0147265$
 $\theta_{p'}=0.690433$, $\theta_Y=1.92292$, $\phi_{p'}=4.62785$, $\phi_Y=2.61576$
 $E_Y=0.0964166$, $\theta_{\ell'}=1.32395$, $\phi_{\ell'}=5.85509$
 $Q^2=0.0380852$, $x_B=0.172749$, $t=-0.0252597$, $W^2=1.06222$, $y=0.57429$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1707$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1707$
 P $E= 0.9534$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1707$
 ℓ' $E= 0.0896$ $p_x= 0.0791$ $p_y= -0.0361$ $p_z= 0.0219$
 P' $E= 0.9381$ $p_x= -0.0008$ $p_y= -0.0093$ $p_z= 0.0114$
 q_Y $E= 0.0964$ $p_x= -0.0783$ $p_y= 0.0454$ $p_z= -0.0333$



electron: polarization cluster P3, local minimum 769; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



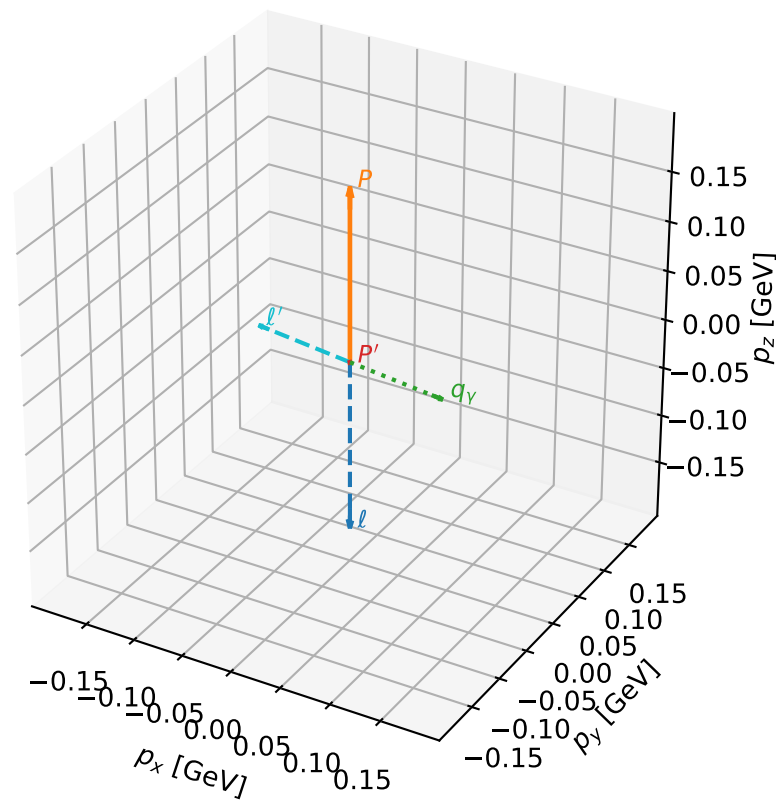
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_772 D_W=0.0095849
 region: polarization_cluster_03_local_minimum_772
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78701964$ rad
 incoming lepton: $0.705959|+\rangle + 0.708252|-\rangle$
 proton mixing angle: $\alpha_p=2.9424013$ rad
 incoming proton: $-0.980227|+\rangle + 0.197877|-\rangle$

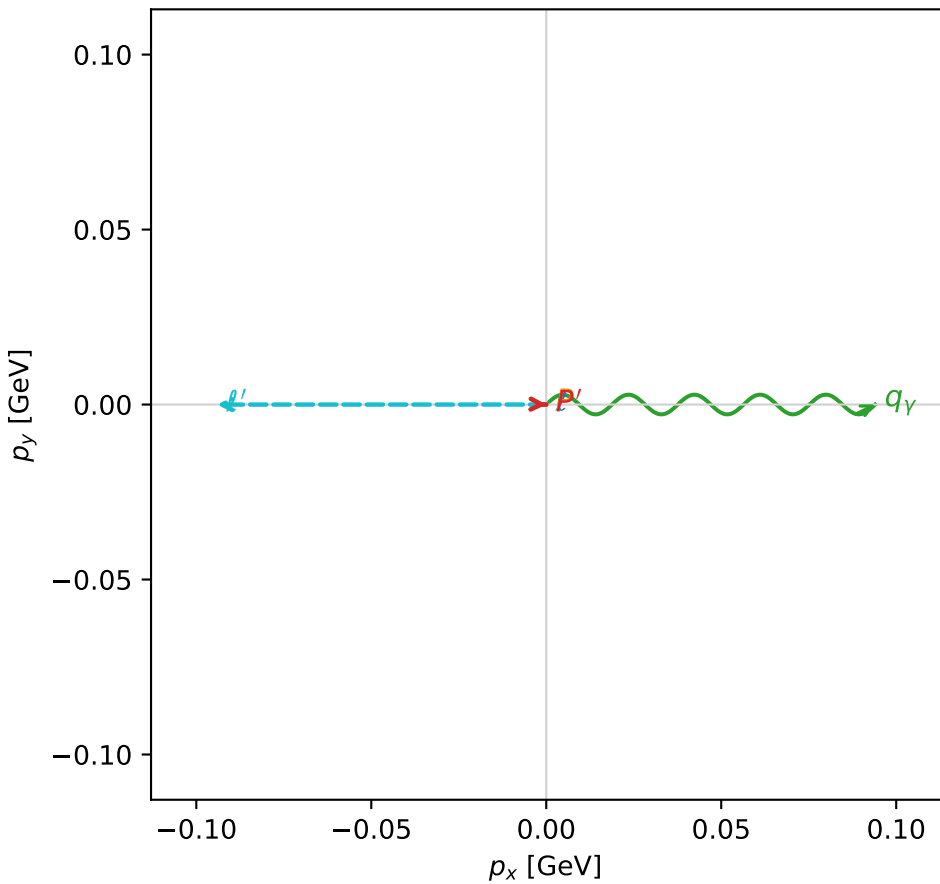
$s=1.27096$, $\sqrt{s}=1.12737$, $|\vec{P}|=0.173463$, $|\vec{P}'|=0.000335174$
 $\theta_{P'}=2.84197$, $\theta_Y=1.68397$, $\phi_{P'}=6.25528$, $\phi_Y=0.000170876$
 $E_Y=0.0946156$, $\theta_{\ell'}=1.45439$, $\phi_{\ell'}=3.14173$
 $Q^2=0.0366894$, $x_B=0.171315$, $t=-0.0299475$, $W^2=1.05732$, $y=0.547572$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1735$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1735$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1735$
 ℓ' $E= 0.0948$ $p_x= -0.0941$ $p_y= -0.0000$ $p_z= 0.0110$
 P' $E= 0.9380$ $p_x= 0.0001$ $p_y= -0.0000$ $p_z= -0.0003$
 q_Y $E= 0.0946$ $p_x= 0.0940$ $p_y= 0.0000$ $p_z= -0.0107$

3D momenta



Transverse plane

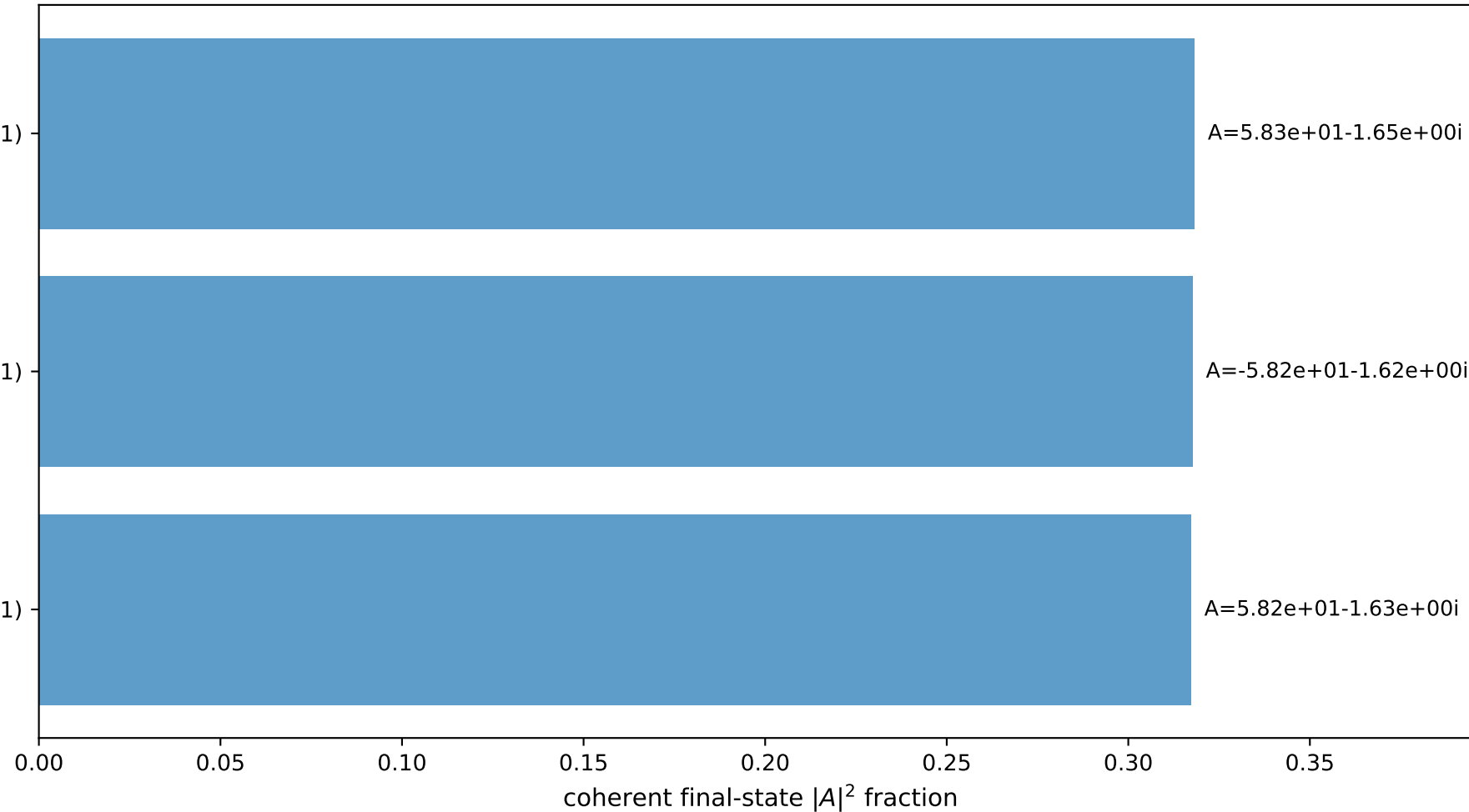


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

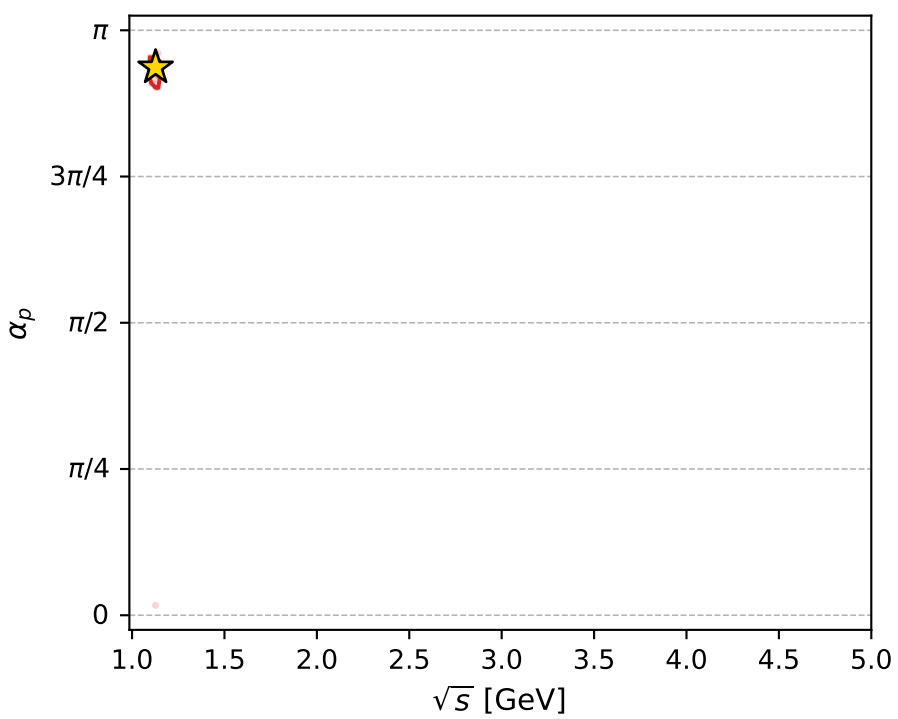
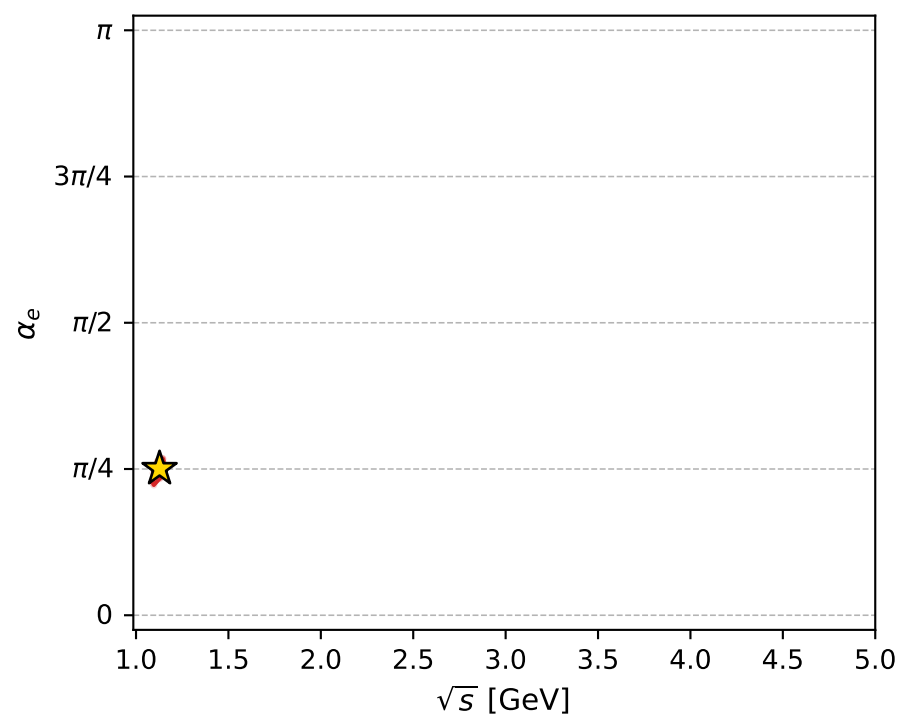
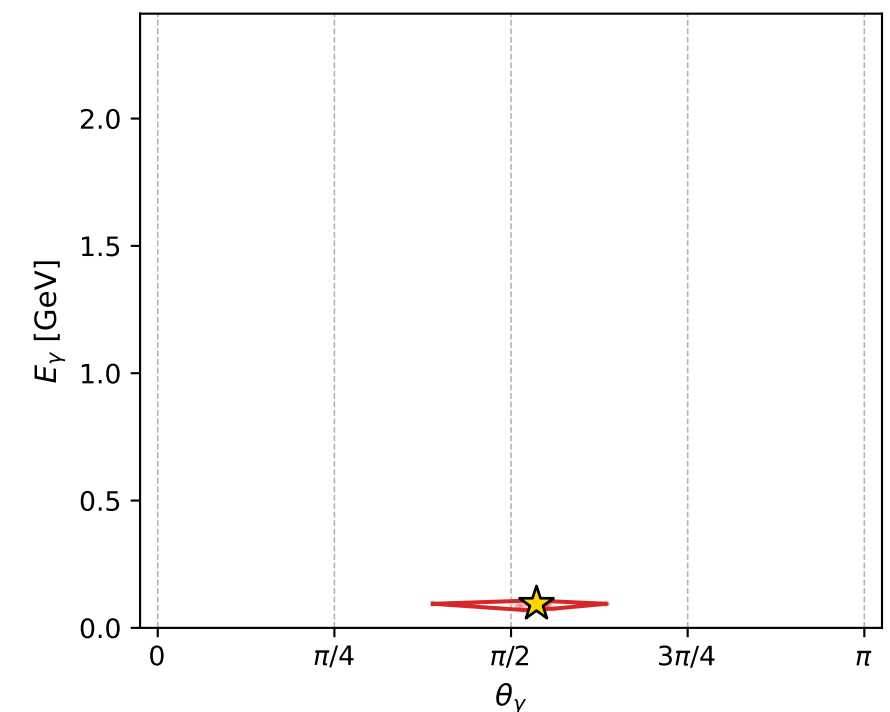
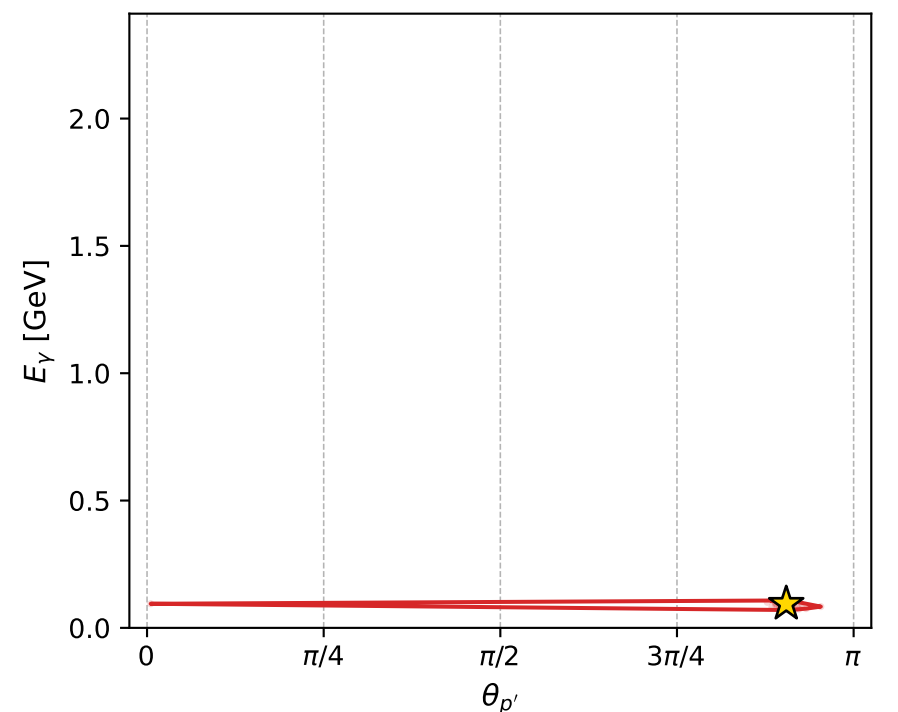
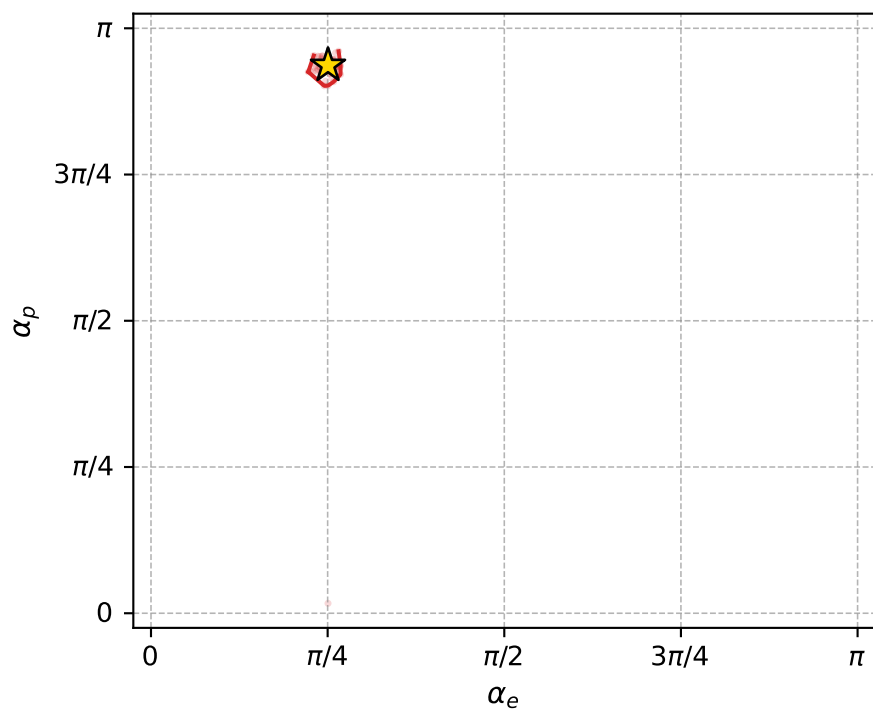
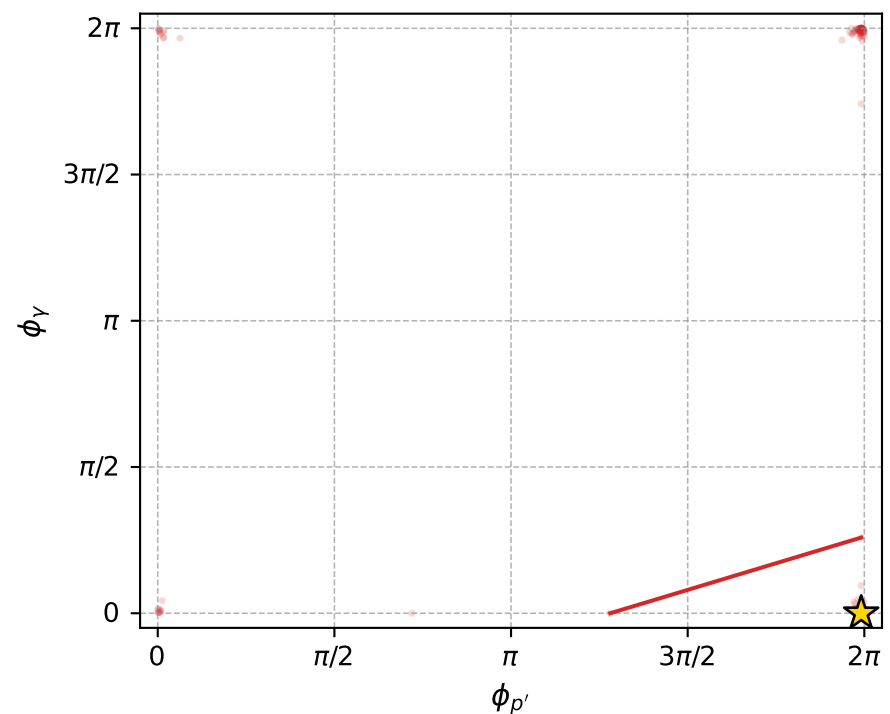
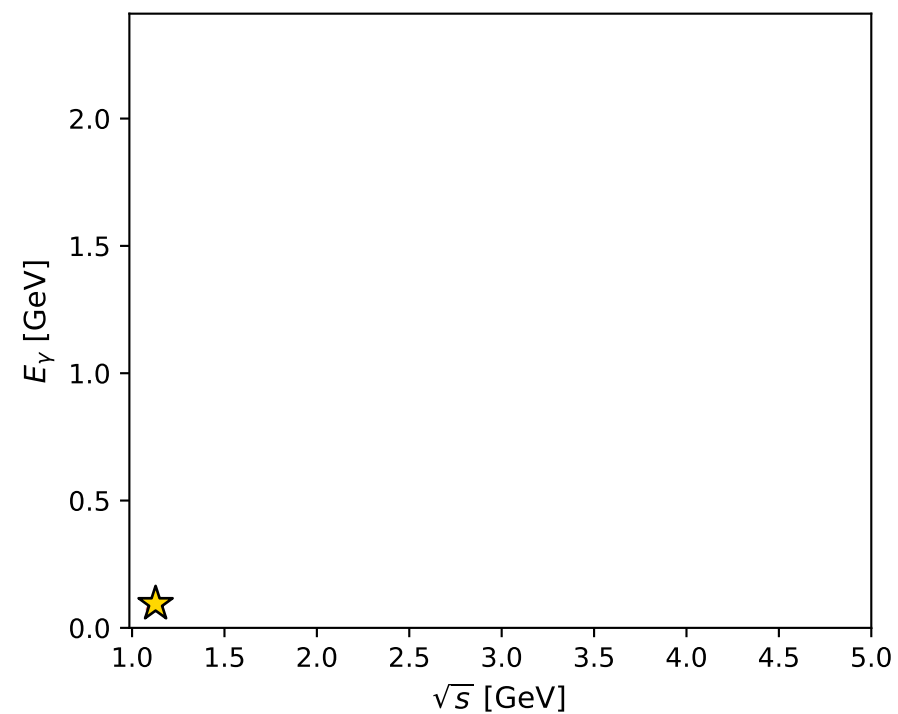
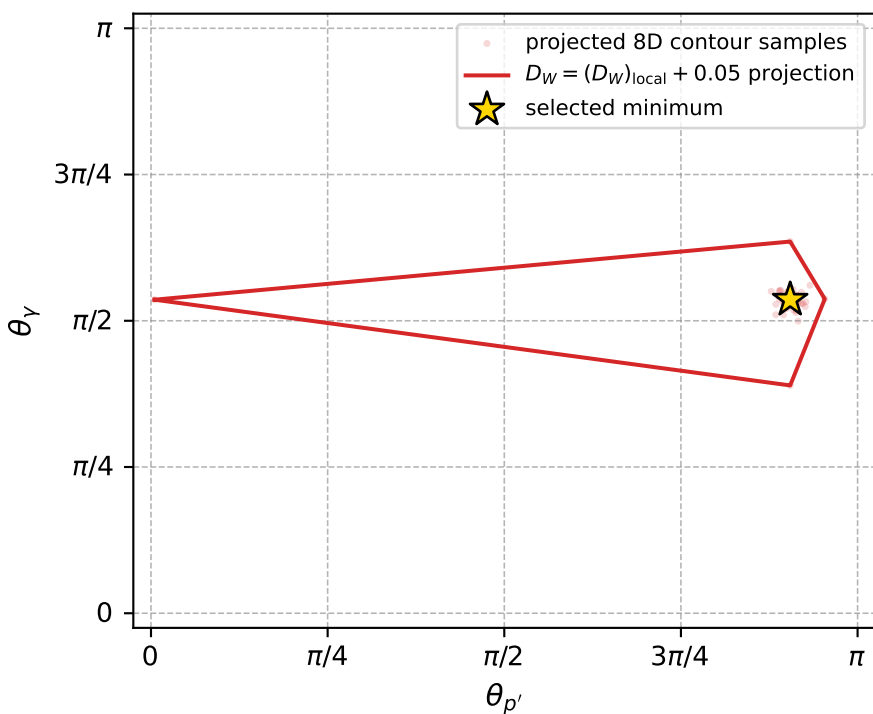
$(h_p, h_Y, h_{\ell}) = (+1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=3, retained=95.3%)



electron: polarization cluster P3, local minimum 772; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0095848951$
 $D_W \text{ contour} = 0.059584895$
 above global minimum = 0.0095684623
 8D contour samples = 255

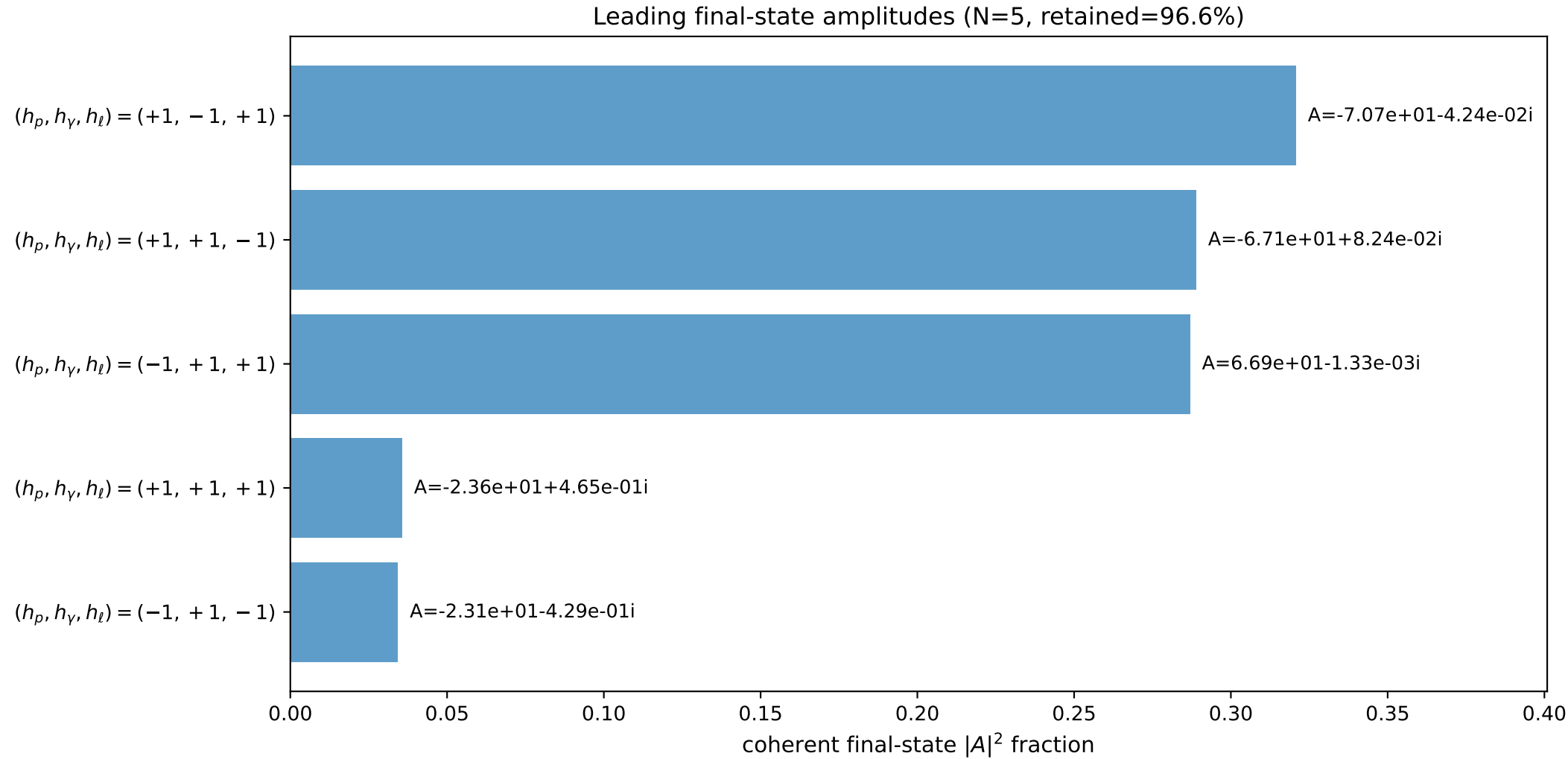
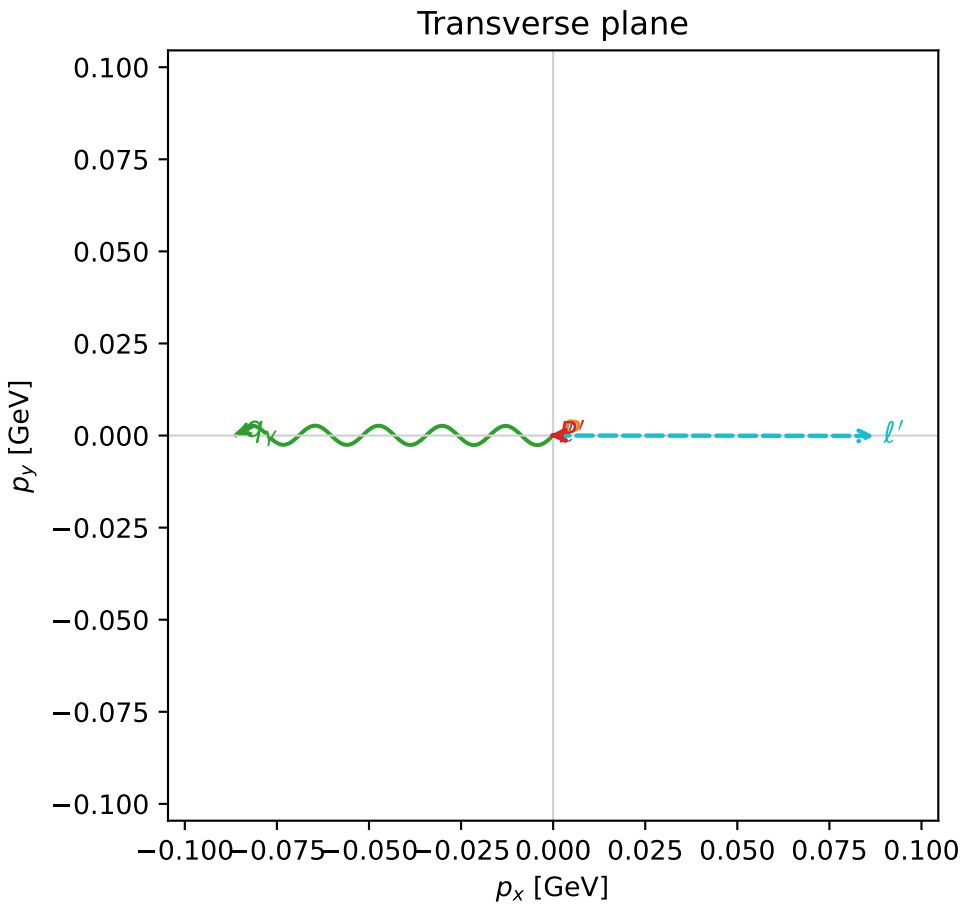
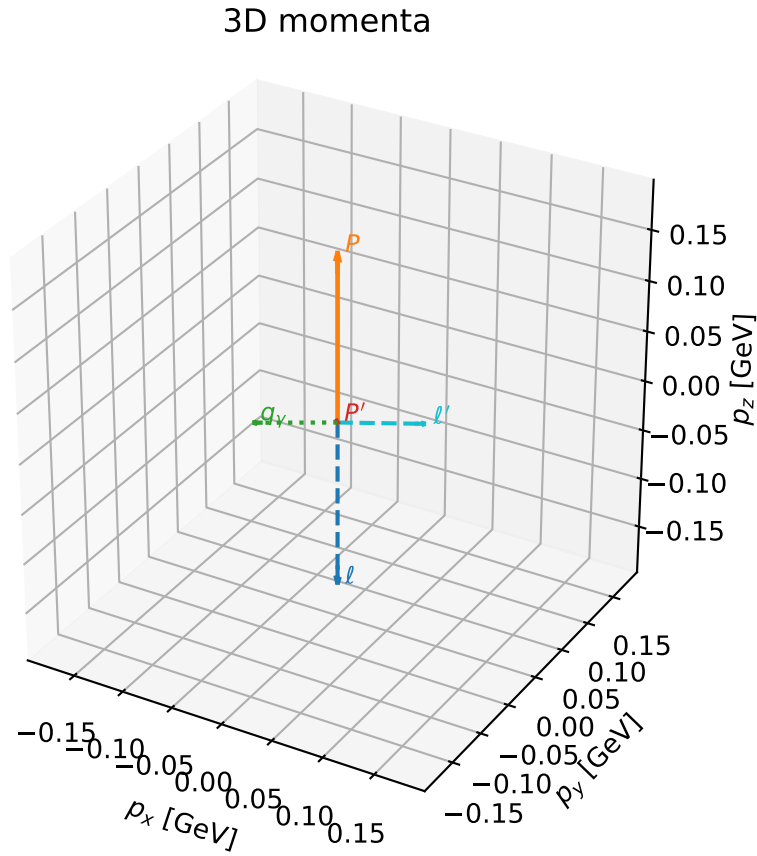
selected phase-space point:
 $\text{sqrt_s} = 1.127368$
 $\text{theta_p_out} = 2.84197$
 $\text{theta_gamma_out} = 1.683967$
 $\text{qOut} = 0.09461559$
 $\text{phi_p_out} = 6.255278$
 $\text{phi_gamma_out} = 0.0001708758$
 $\text{alpha_e} = 0.7870196$
 $\text{alpha_p} = 2.942401$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

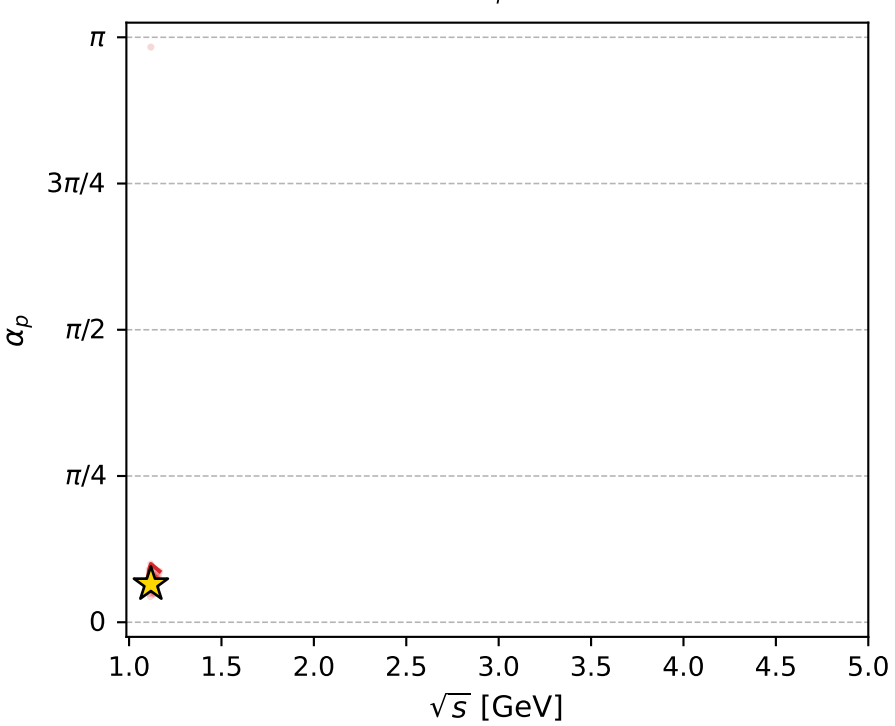
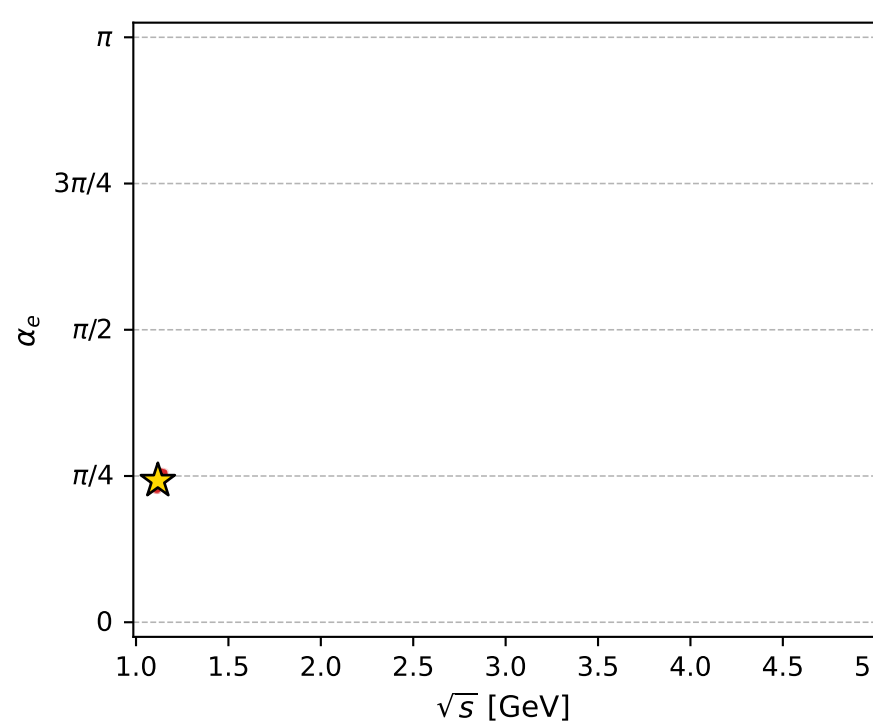
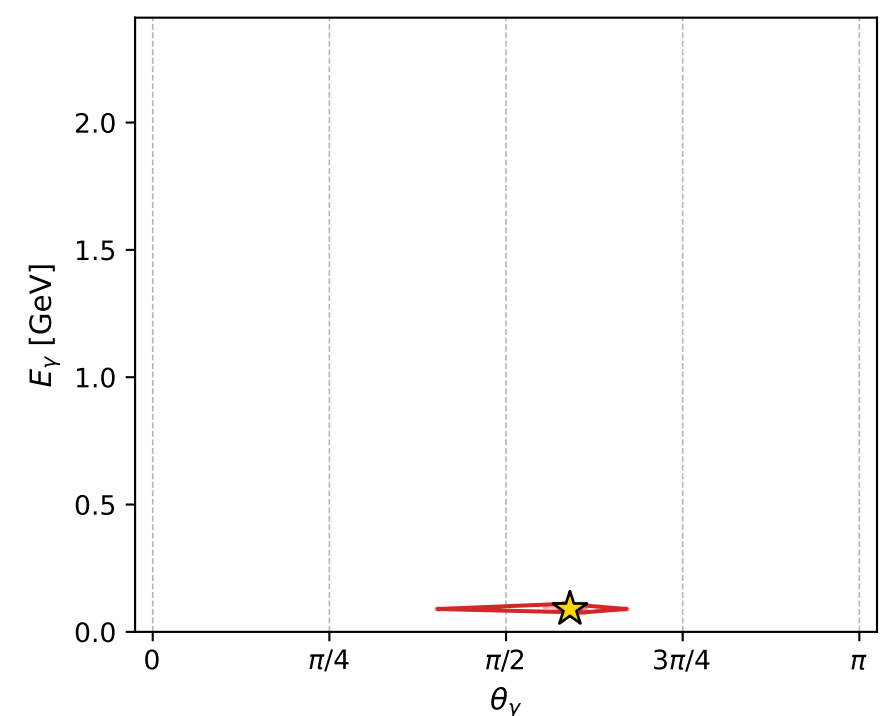
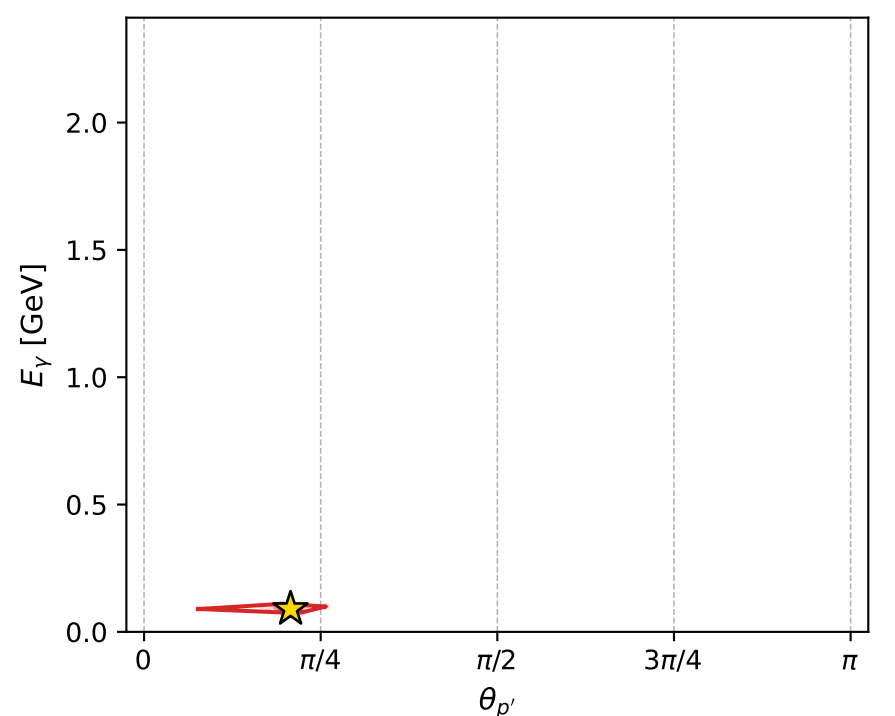
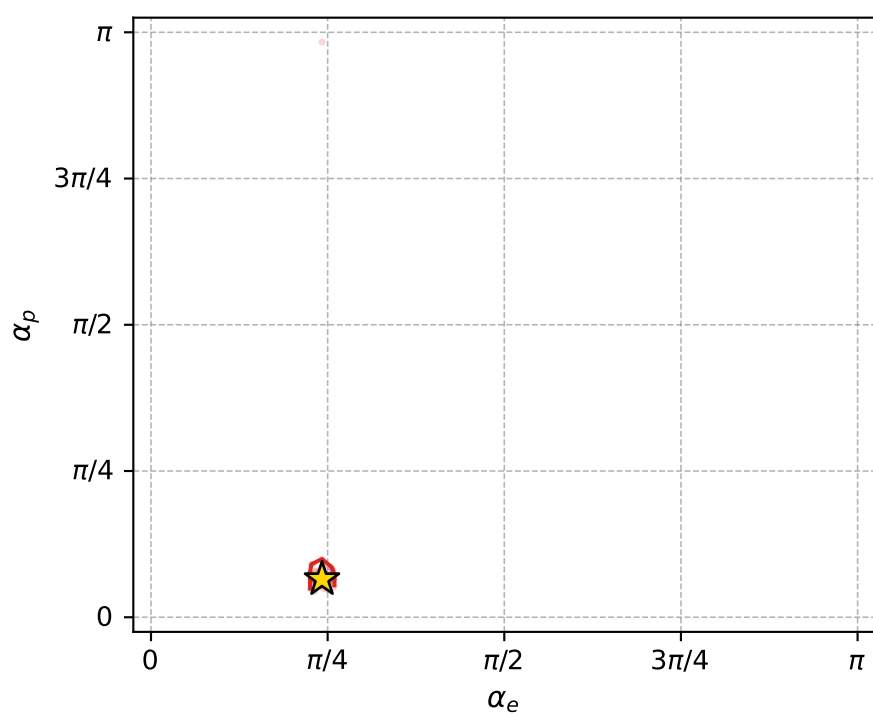
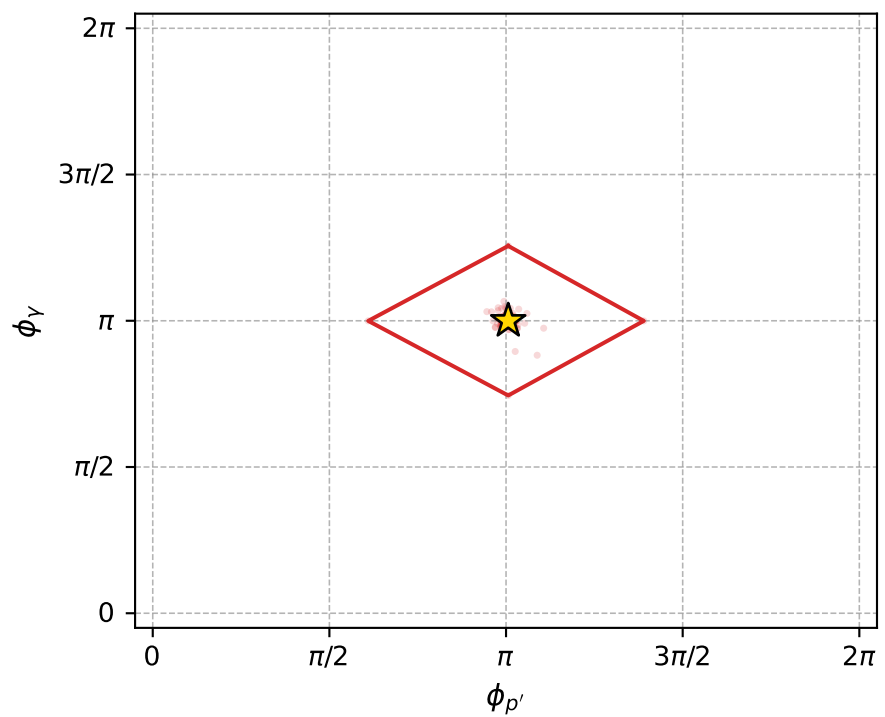
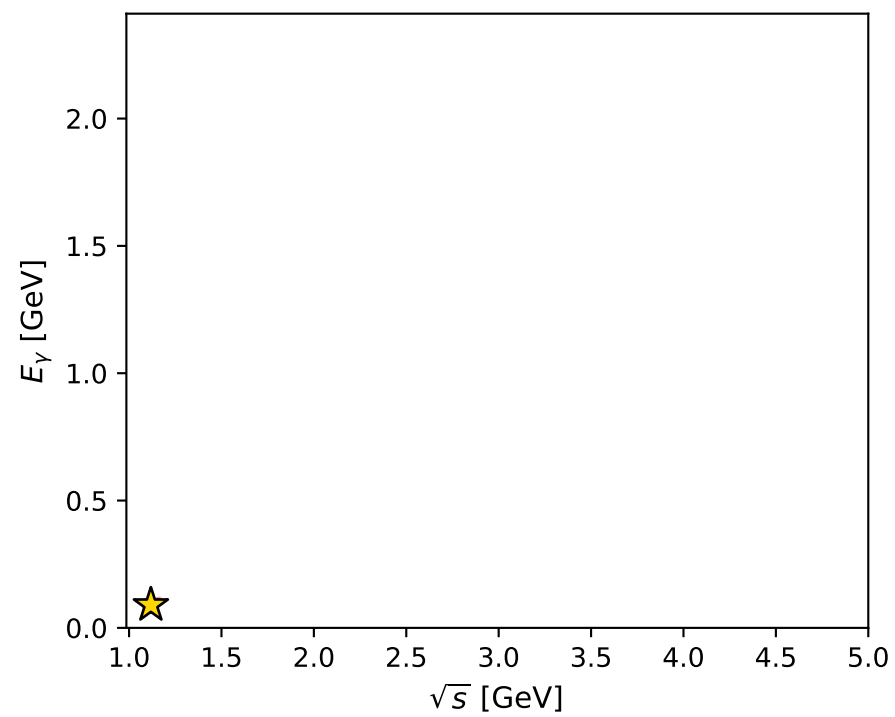
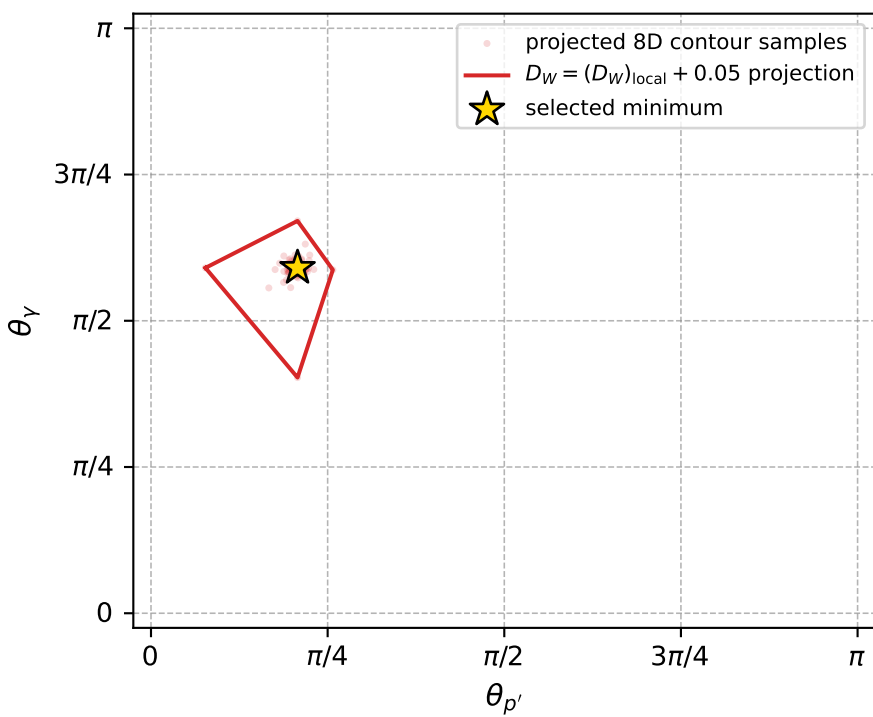
dw_mixing_angles_polarization_03_local_minimum_779 D_W=0.00976823
 region: polarization_cluster_03_local_minimum_779
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76042185$ rad
 incoming lepton: $0.724545|+\rangle +0.689227|-\rangle$
 proton mixing angle: $\alpha_p=0.2044224$ rad
 incoming proton: $0.979178|+\rangle +0.203002|-\rangle$

$s=1.25018$, $\sqrt{s}=1.11812$, $|\vec{P}|=0.165608$, $|\vec{P}'|=0.00161271$
 $\theta_{P'}=0.651558$, $\theta_V=1.85491$, $\phi_{P'}=3.1611$, $\phi_V=3.14033$
 $E_V=0.0897607$, $\theta_{l'}=1.30334$, $\phi_{l'}=6.28216$
 $Q^2=0.037835$, $x_B=0.183556$, $t=-0.0267935$, $W^2=1.04813$, $y=0.556578$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1656$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1656$
 P $E= 0.9525$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1656$
 l' $E= 0.0904$ $p_x= 0.0871$ $p_y= -0.0001$ $p_z= 0.0239$
 P' $E= 0.9380$ $p_x= -0.0010$ $p_y= -0.0000$ $p_z= 0.0013$
 q_V $E= 0.0898$ $p_x= -0.0862$ $p_y= 0.0001$ $p_z= -0.0252$



electron: polarization cluster P3, local minimum 779; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0097682315$
 $D_W \text{ contour} = 0.059768232$
 above global minimum = 0.0097517987
 8D contour samples = 255

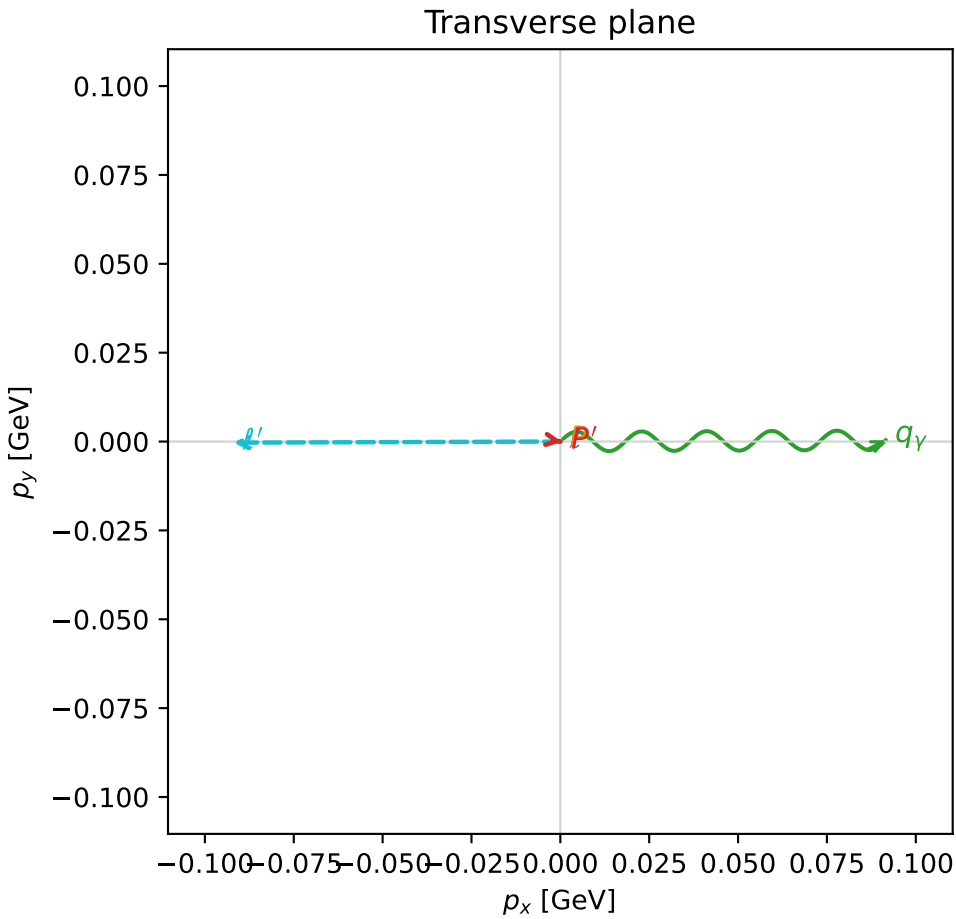
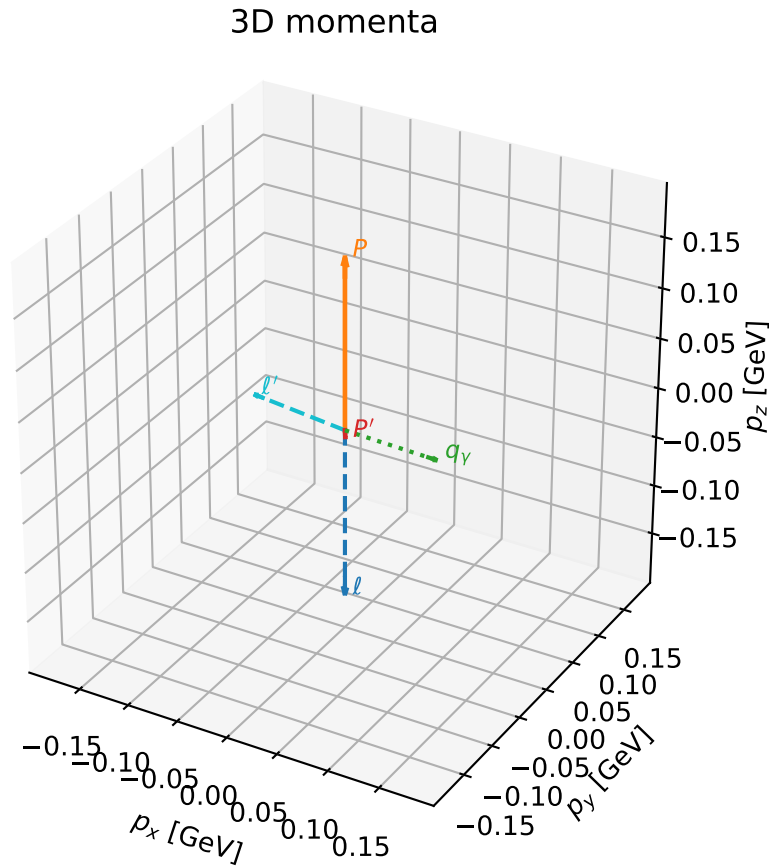
selected phase-space point:
 $\text{sqrt}_s = 1.118116$
 $\text{theta}_p_{\text{out}} = 0.6515581$
 $\text{theta}_\gamma_{\text{out}} = 1.854907$
 $q_{\text{out}} = 0.08976075$
 $\text{phi}_p_{\text{out}} = 3.161099$
 $\text{phi}_\gamma_{\text{out}} = 3.140332$
 $\alpha_e = 0.7604218$
 $\alpha_p = 0.2044224$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_800 D_W=0.0103956
 region: polarization_cluster_03_local_minimum_800
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78689655$ rad
 incoming lepton: $0.706046|+\rangle + 0.708166|-\rangle$
 proton mixing angle: $\alpha_p=2.9368835$ rad
 incoming proton: $-0.97912|+\rangle + 0.203282|-\rangle$

$s=1.25926$, $\sqrt{s}=1.12217$, $|\vec{P}|=0.169055$, $|\vec{P}'|=0.00653172$
 $\theta_{P'}=3.07246$, $\theta_Y=1.61004$, $\phi_{P'}=6.13637$, $\phi_Y=0.00404817$
 $E_Y=0.0916075$, $\theta_{\ell'}=1.46132$, $\phi_{\ell'}=3.1449$
 $Q^2=0.0347062$, $x_B=0.16812$, $t=-0.0305978$, $W^2=1.05158$, $y=0.54409$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1691$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1691$
 P $E=0.9531$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1691$
 ℓ' $E=0.0925$ $p_x=-0.0920$ $p_y=-0.0003$ $p_z=0.0101$
 P' $E=0.9380$ $p_x=0.0004$ $p_y=-0.0001$ $p_z=-0.0065$
 q_Y $E=0.0916$ $p_x=0.0915$ $p_y=0.0004$ $p_z=-0.0036$

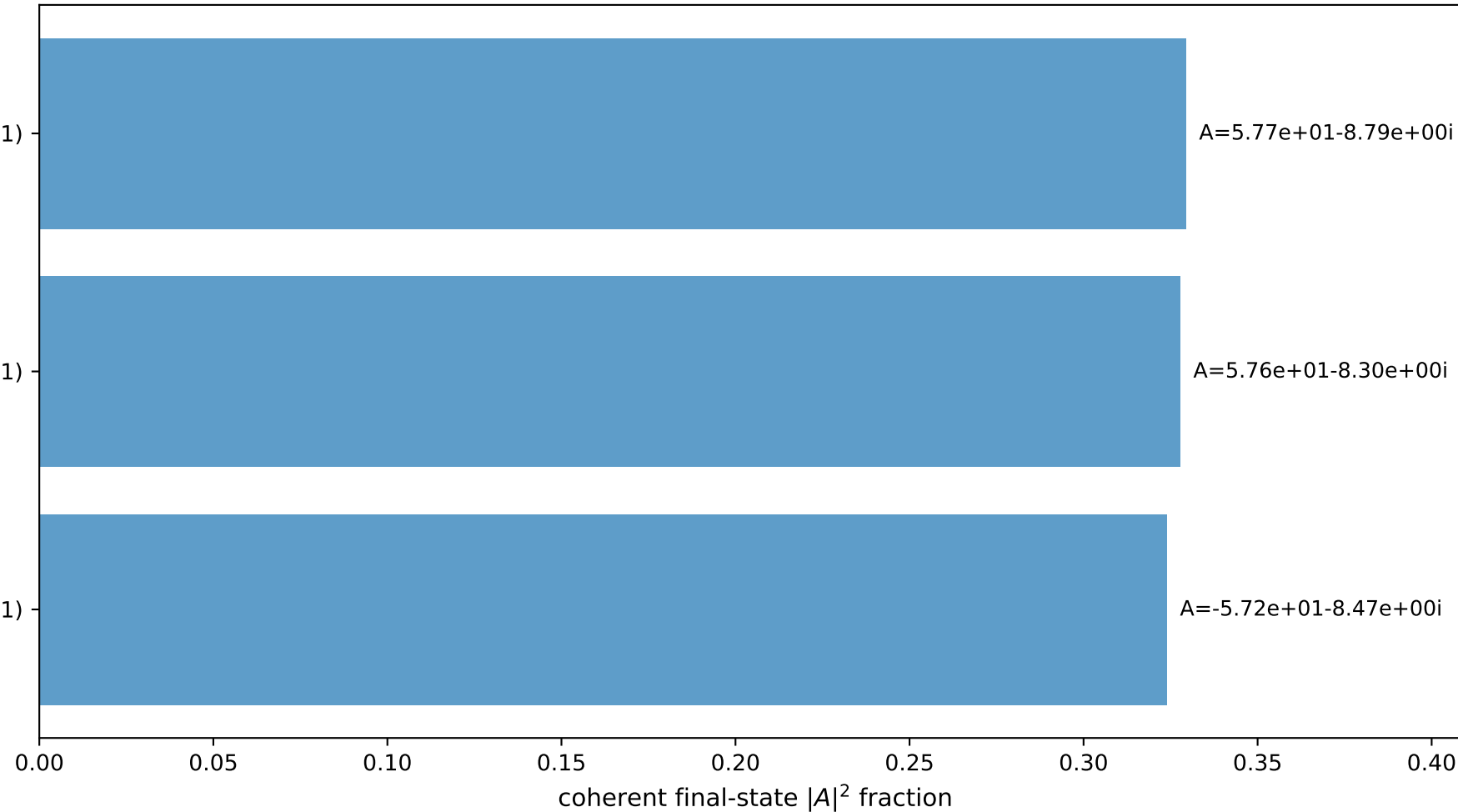


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

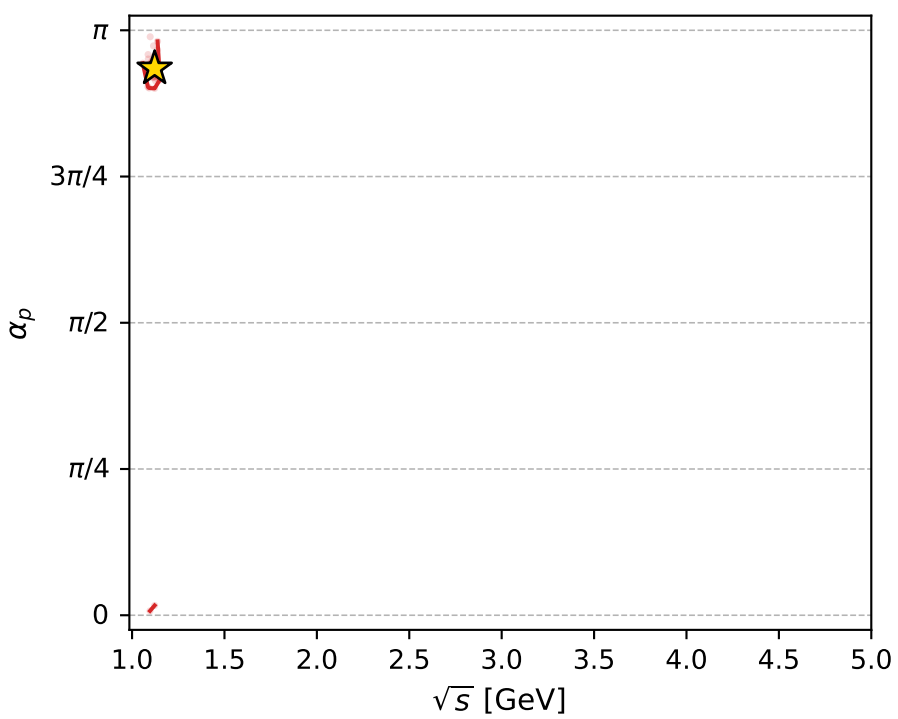
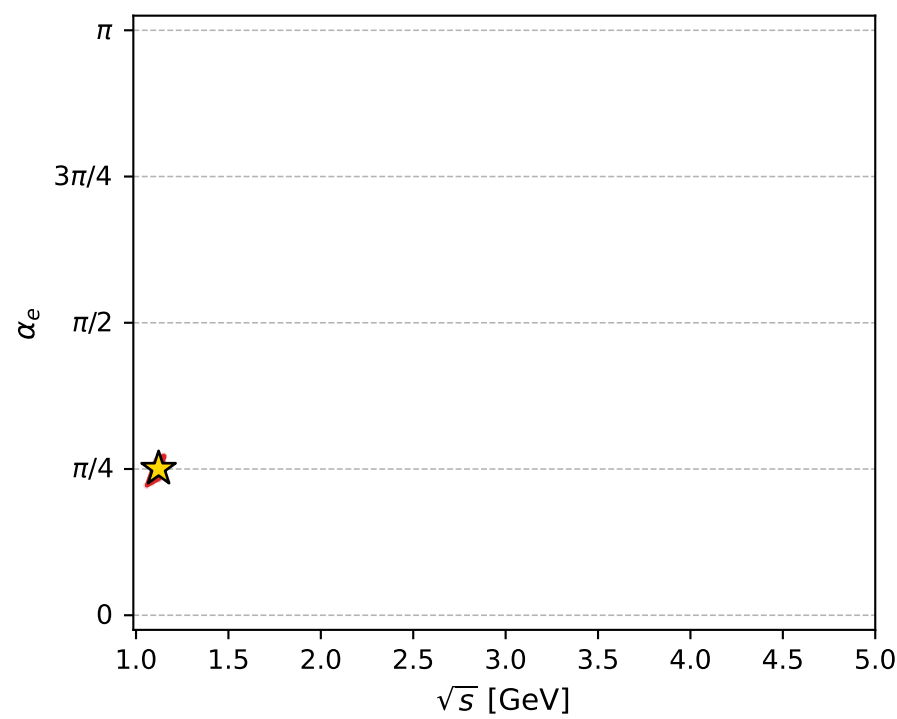
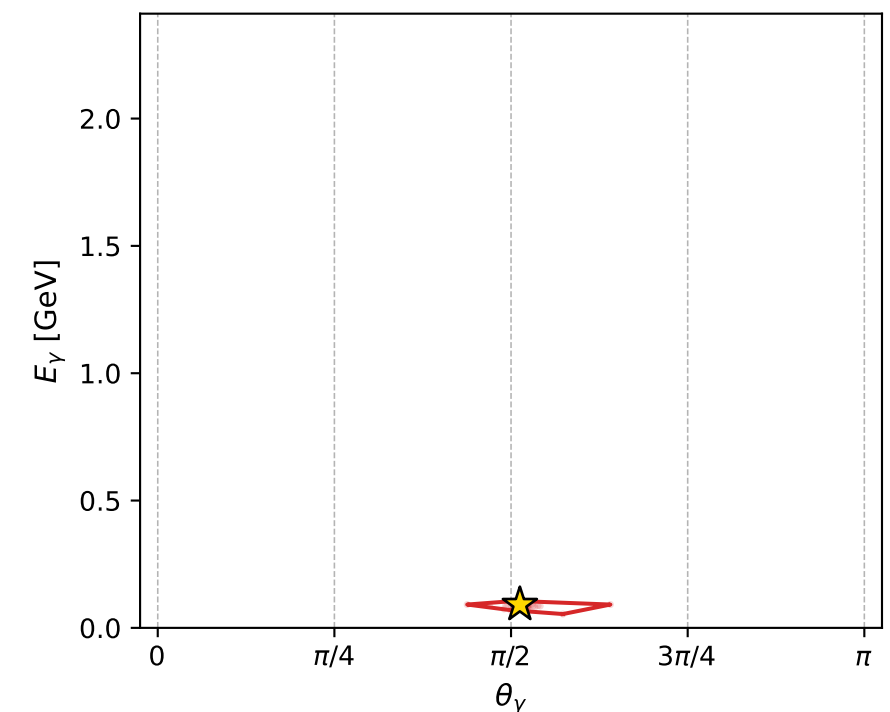
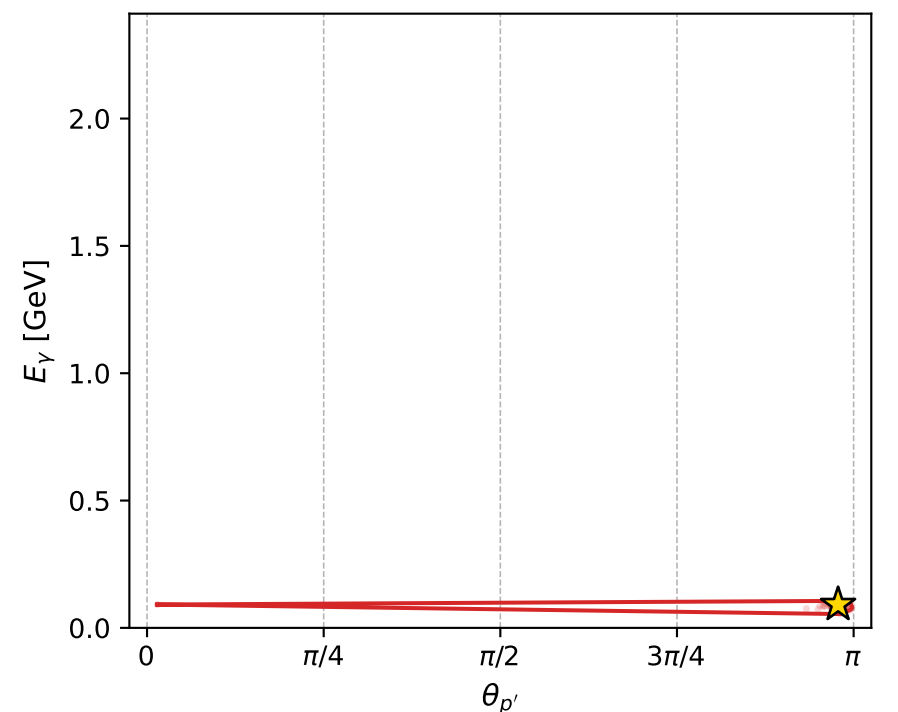
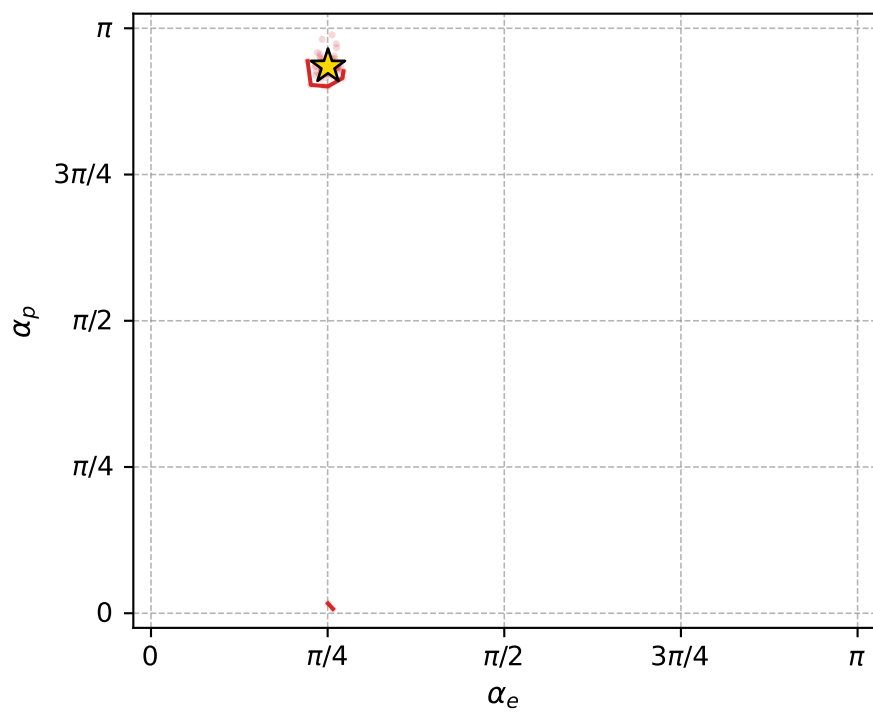
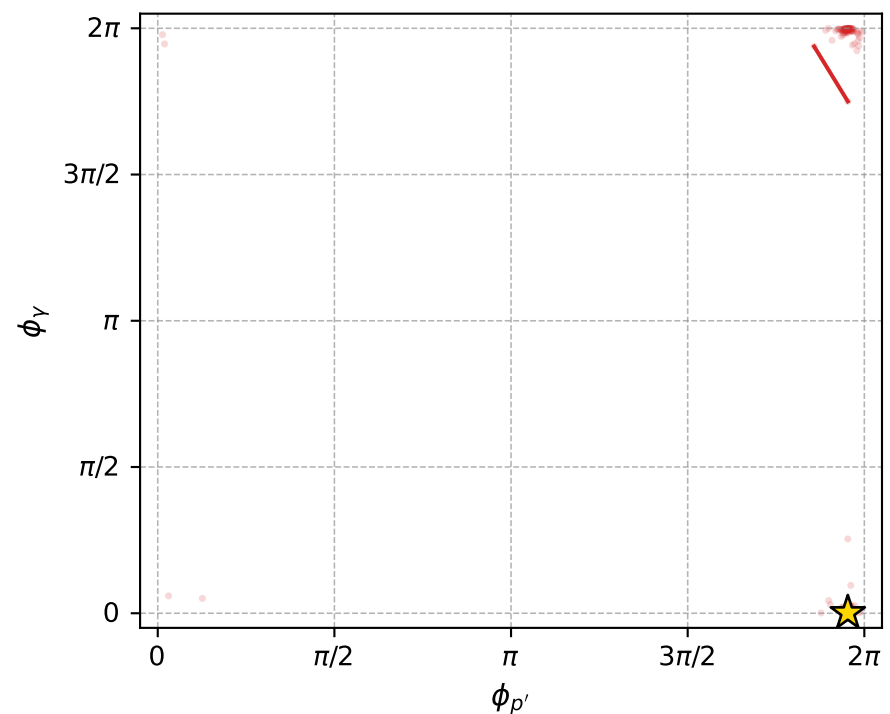
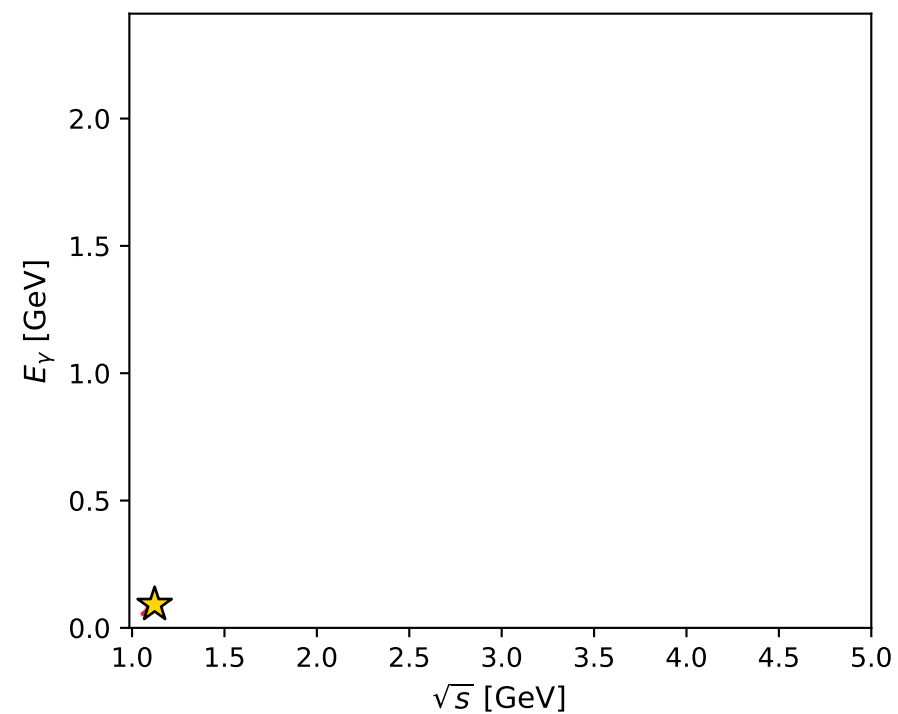
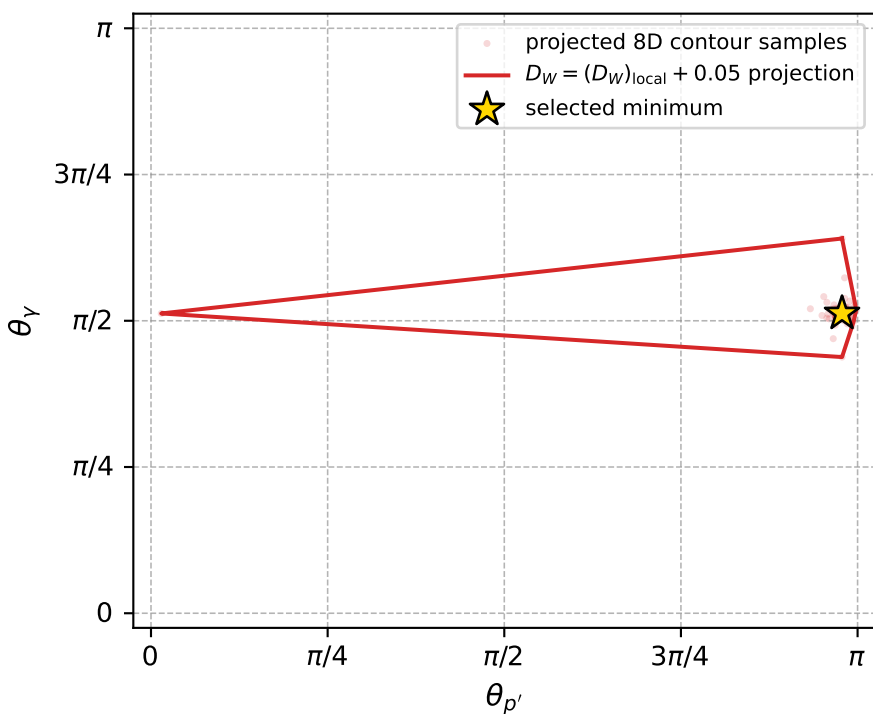
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, +1)$

Leading final-state amplitudes (N=3, retained=98.1%)



electron: polarization cluster P3, local minimum 800; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.010395558$
 $D_W \text{ contour} = 0.060395558$
 above global minimum = 0.010379125
 8D contour samples = 252

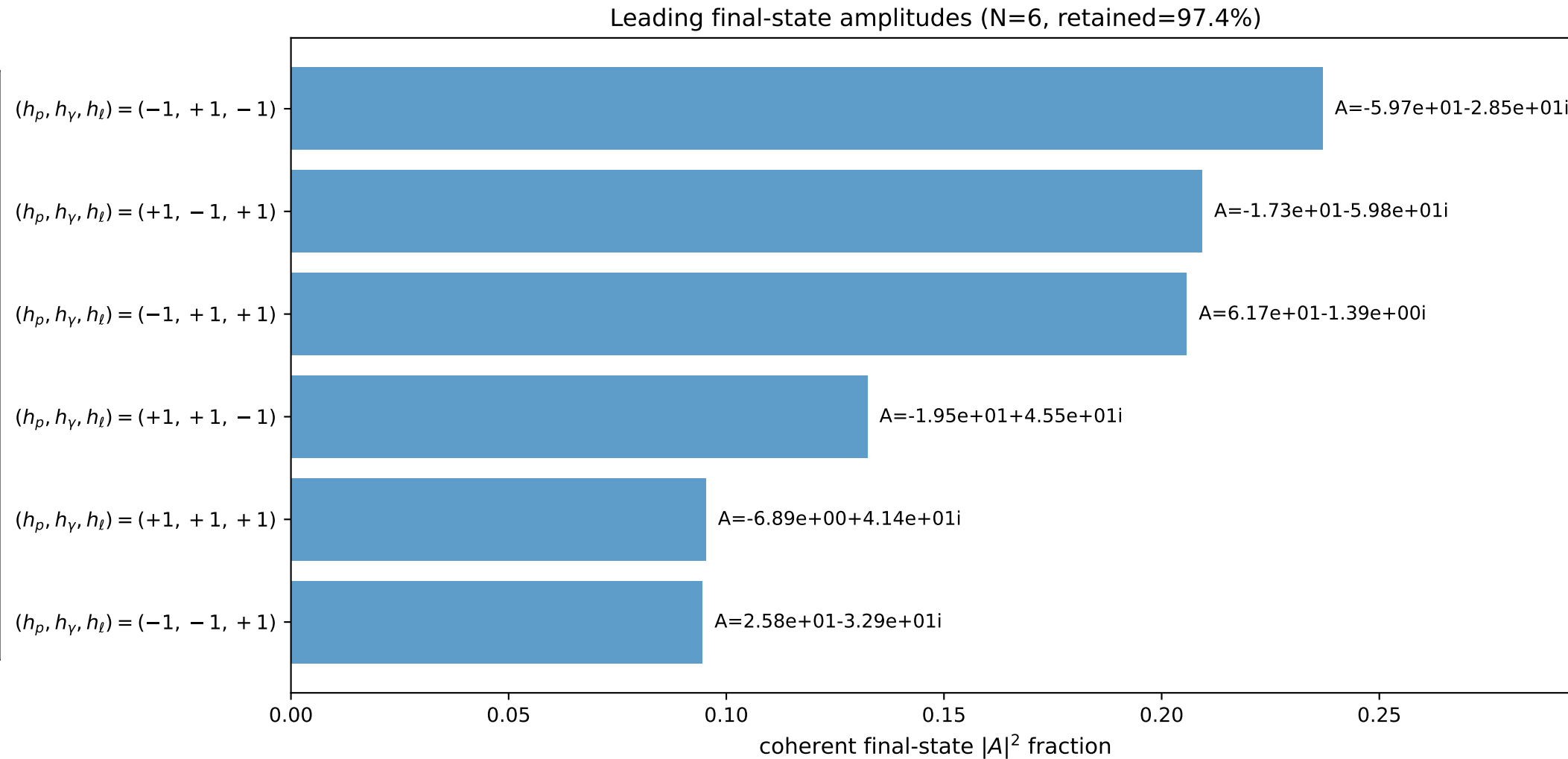
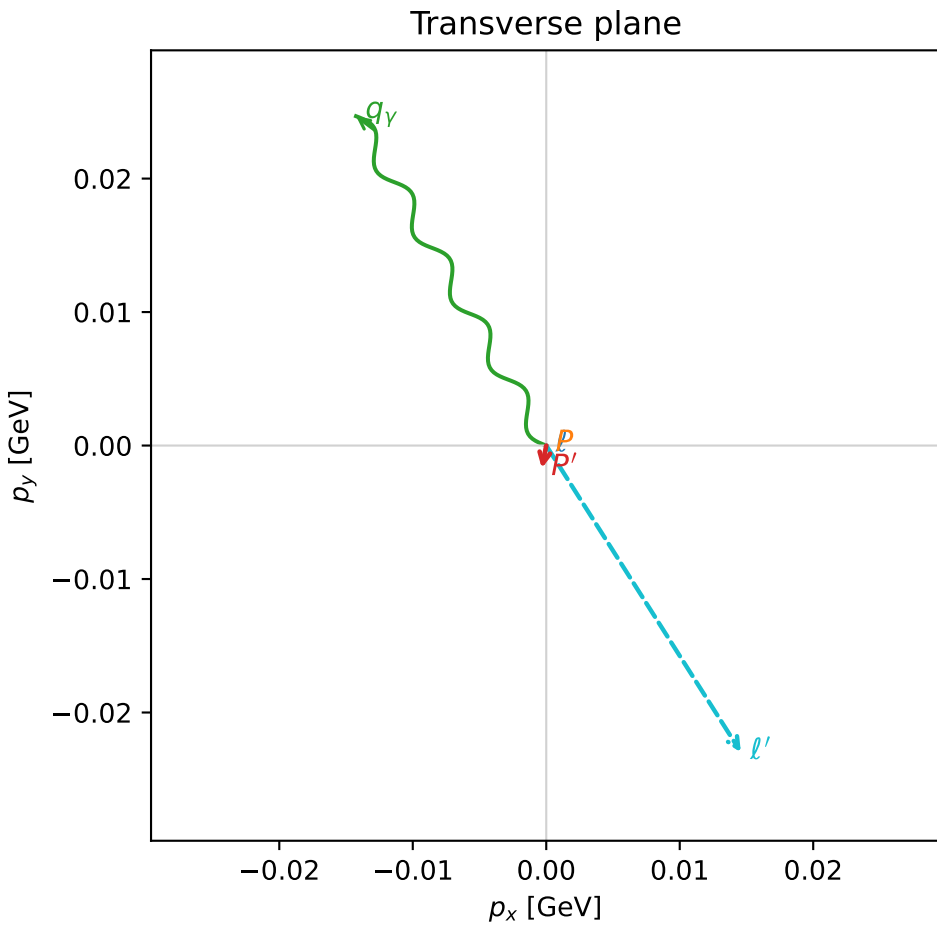
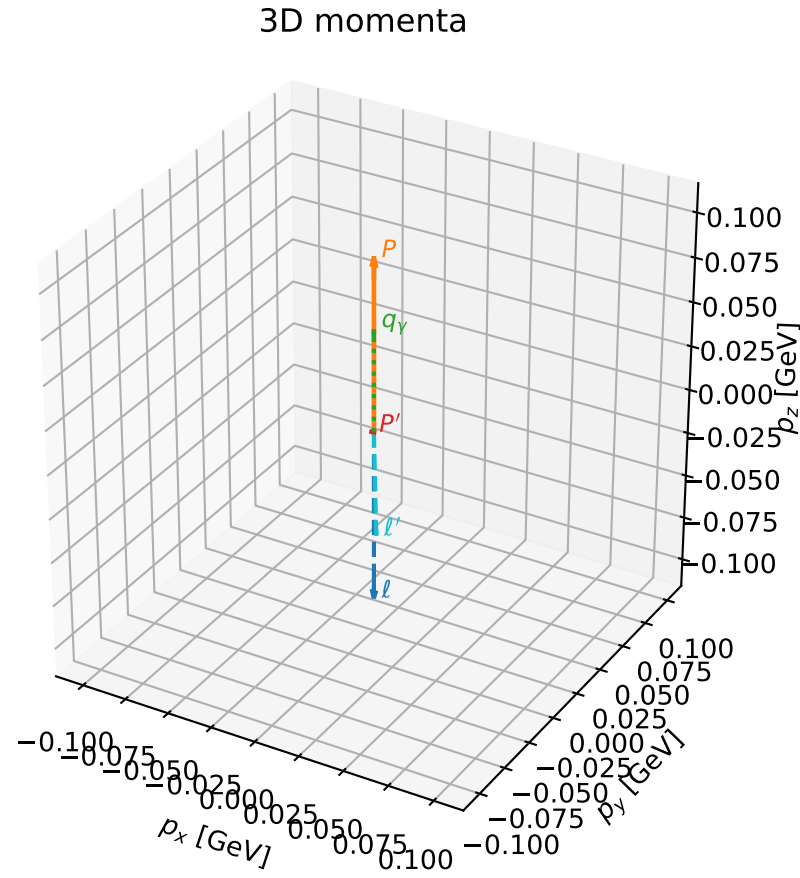
selected phase-space point:
 $\text{sqrt_s} = 1.122169$
 $\text{theta_p_out} = 3.072464$
 $\text{theta_gamma_out} = 1.61004$
 $\text{qOut} = 0.09160754$
 $\text{phi_p_out} = 6.136369$
 $\text{phi_gamma_out} = 0.004048169$
 $\text{alpha_e} = 0.7868965$
 $\text{alpha_p} = 2.936883$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

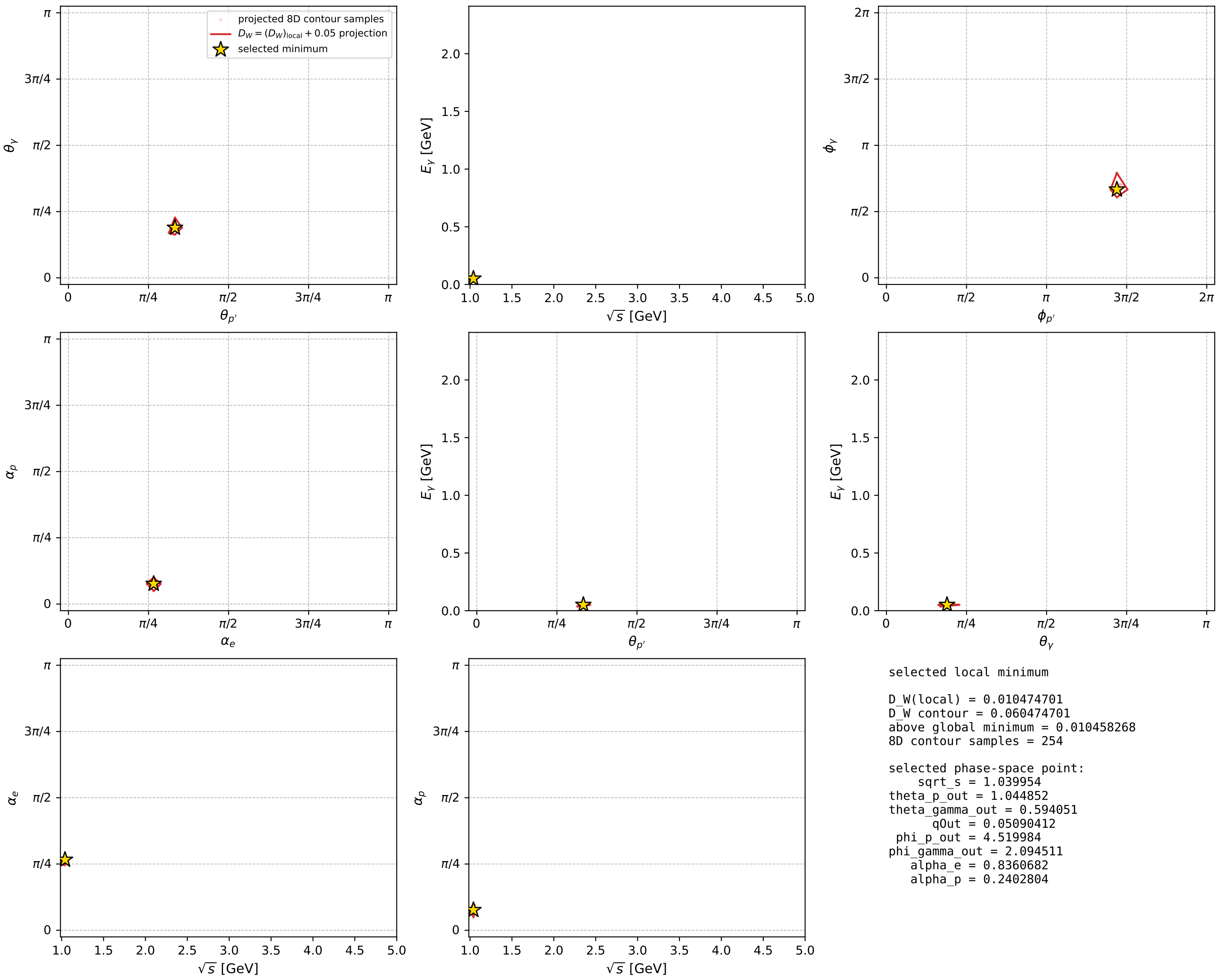
dw_mixing_angles_polarization_03_local_minimum_805 D_W=0.0104747
 region: polarization_cluster_03_local_minimum_805
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.8360682$ rad
 incoming lepton: $0.670385|+\rangle + 0.742013|-\rangle$
 proton mixing angle: $\alpha_p=0.24028043$ rad
 incoming proton: $0.971271|+\rangle + 0.237975|-\rangle$

$s=1.0815$, $\sqrt{s}=1.03995$, $|\vec{P}|=0.0969552$, $|\vec{P}'|=0.00200315$
 $\theta_{P'}=1.04485$, $\theta_V=0.594051$, $\phi_{P'}=4.51998$, $\phi_V=2.09451$
 $E_V=0.0509041$, $\theta_{\ell'}=2.5794$, $\phi_{\ell'}=5.27791$
 $Q^2=0.00152354$, $x_B=0.015705$, $t=-0.00918436$, $W^2=0.97533$, $y=0.481055$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0970$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0970$
 P $E= 0.9430$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0970$
 ℓ' $E= 0.0510$ $p_x= 0.0146$ $p_y= -0.0230$ $p_z= -0.0432$
 P' $E= 0.9380$ $p_x= -0.0003$ $p_y= -0.0017$ $p_z= 0.0010$
 q_V $E= 0.0509$ $p_x= -0.0142$ $p_y= 0.0247$ $p_z= 0.0422$



electron: polarization cluster P3, local minimum 805; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

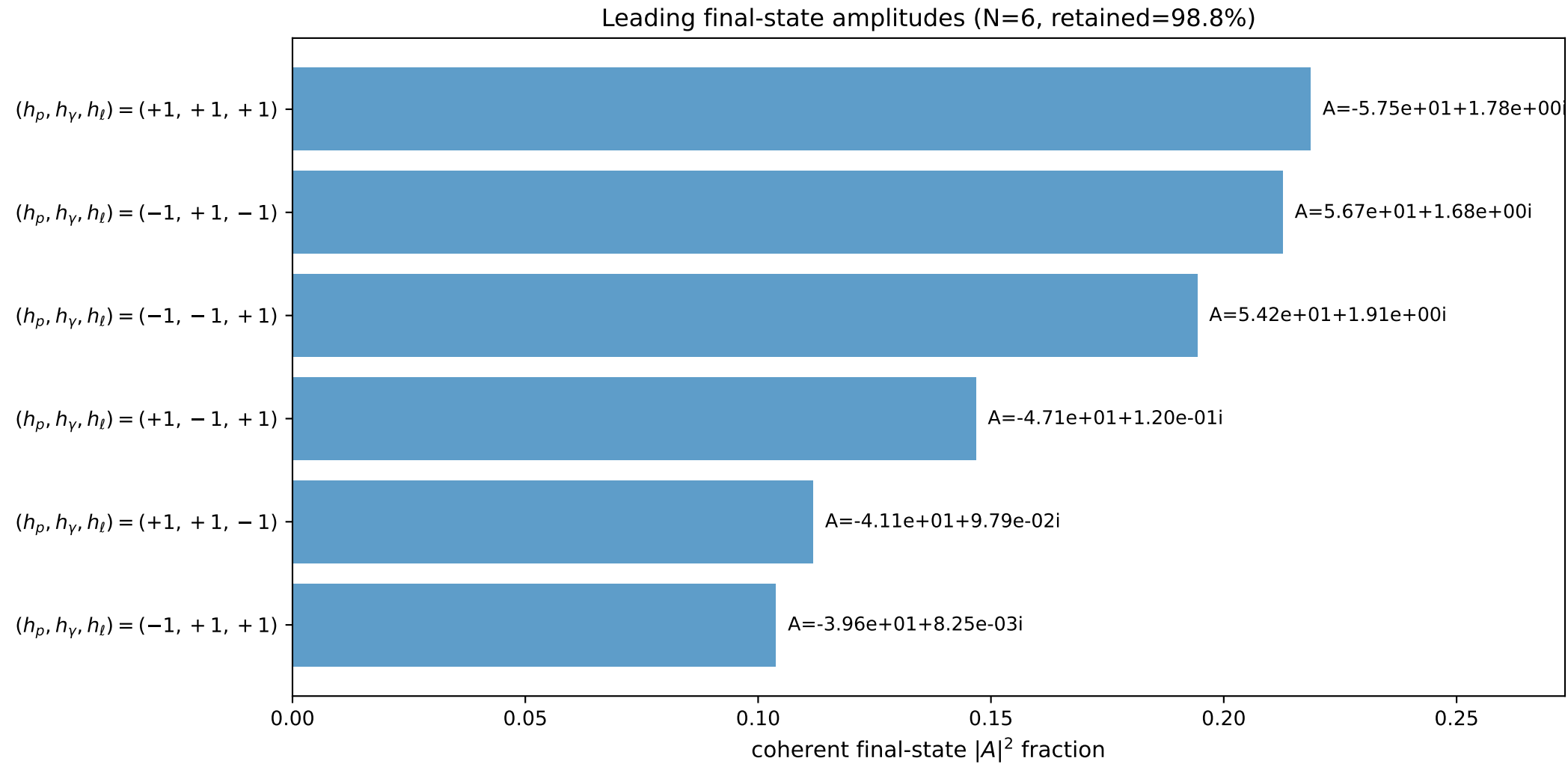
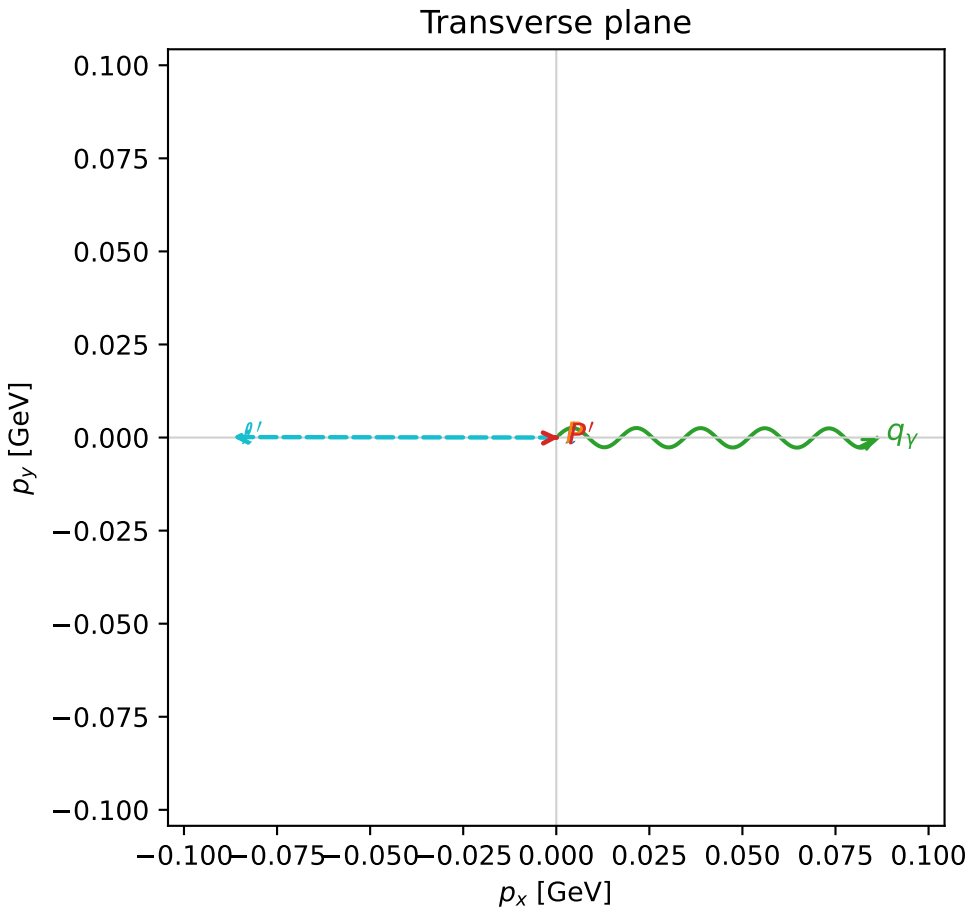
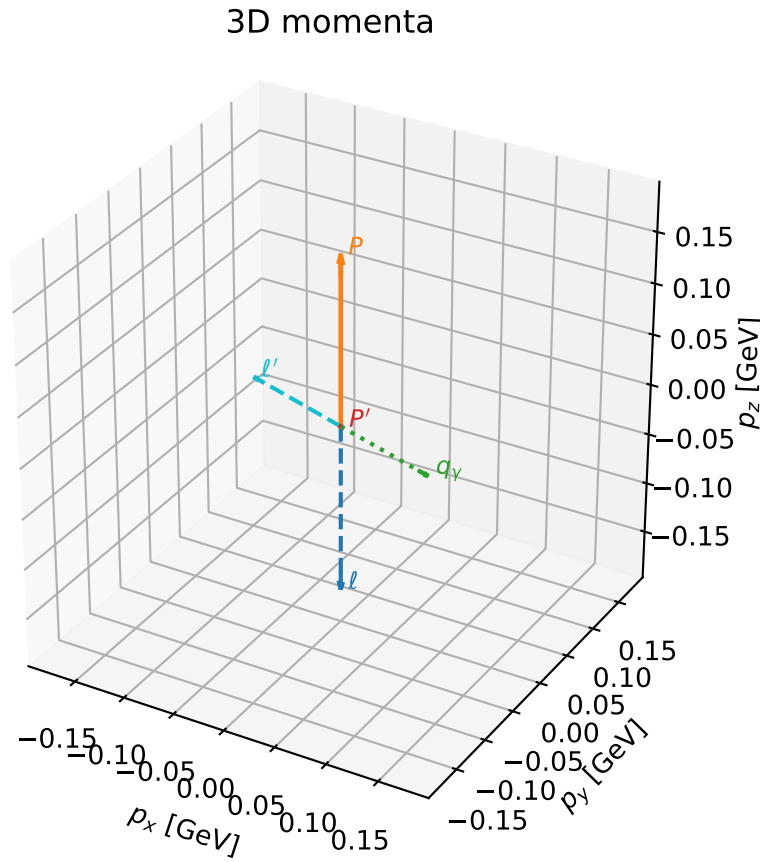


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

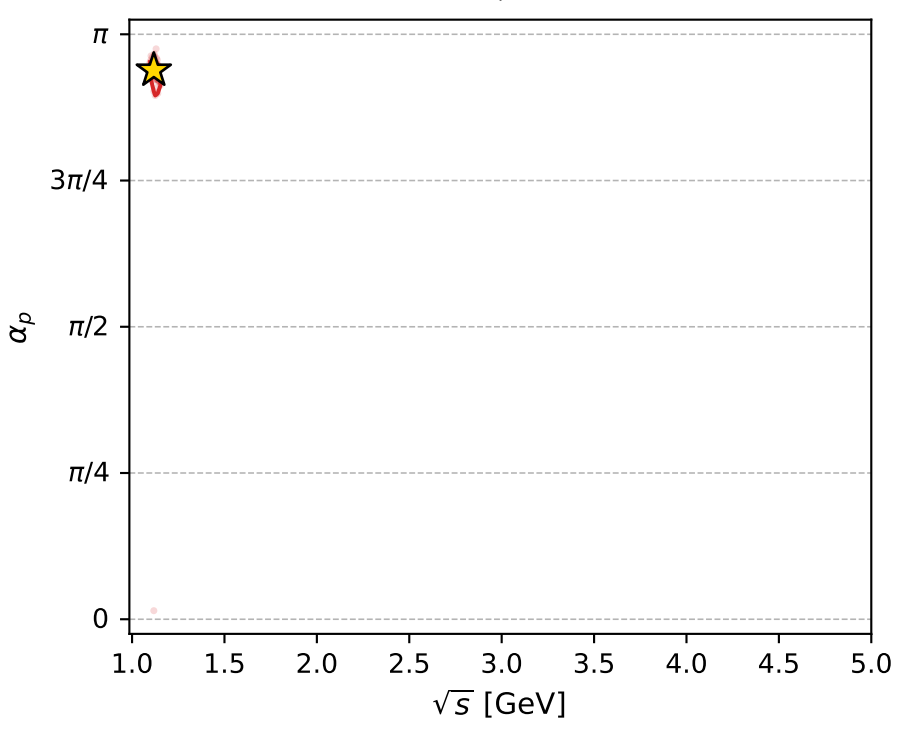
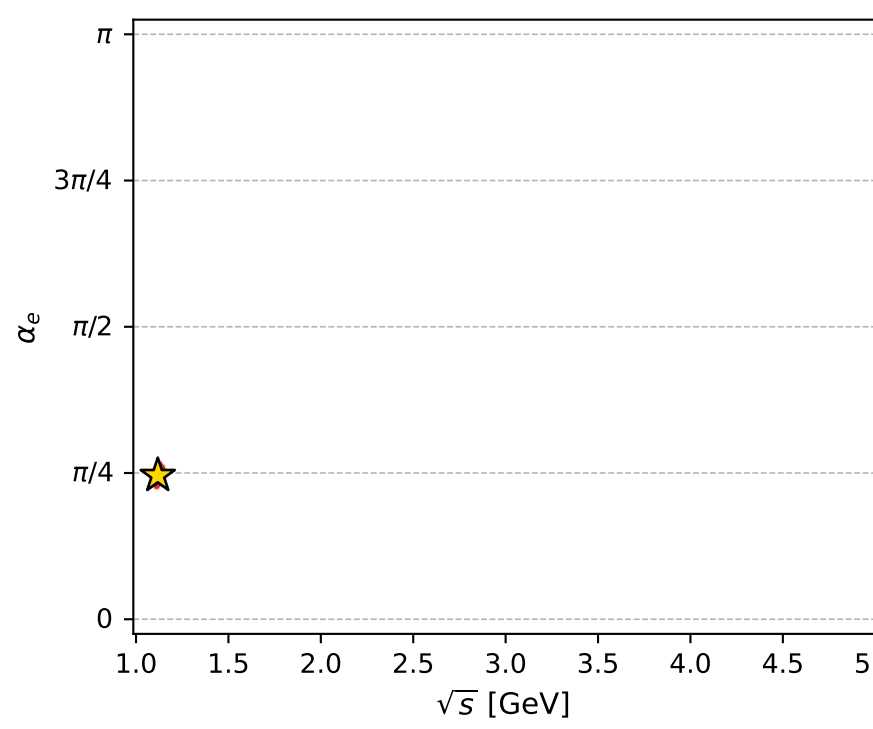
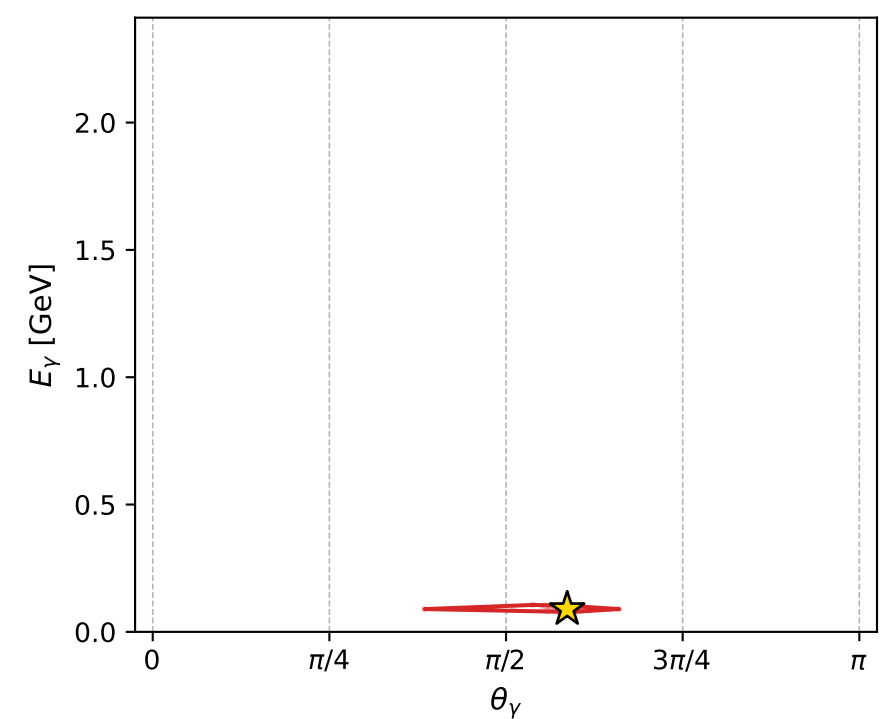
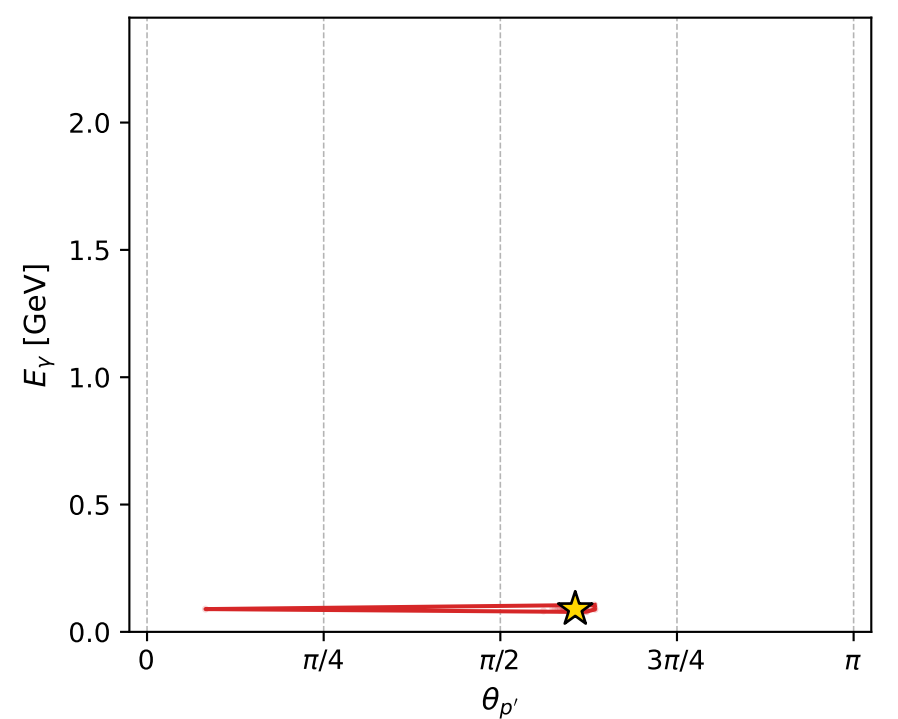
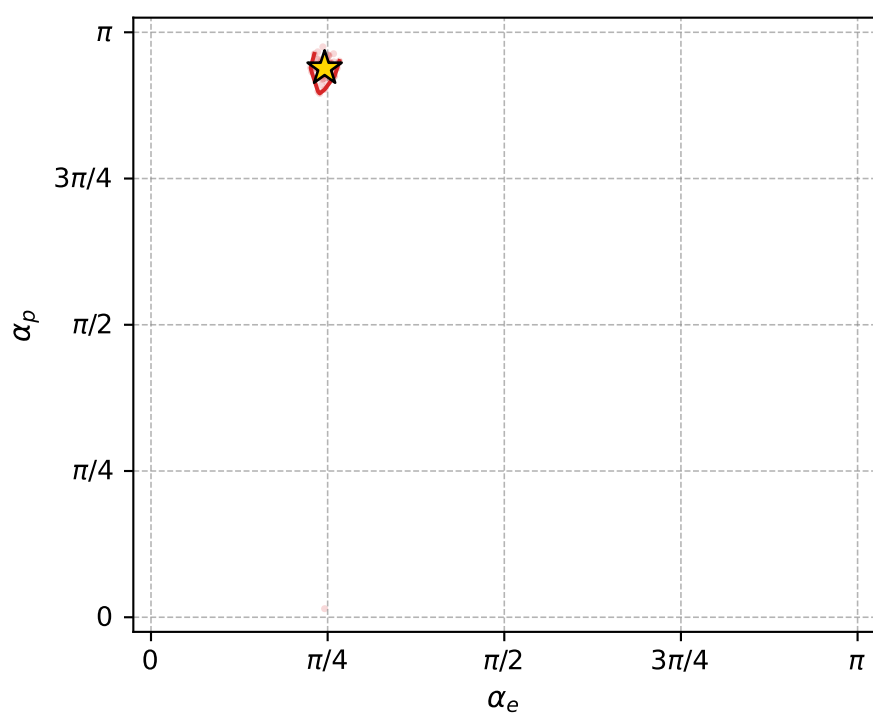
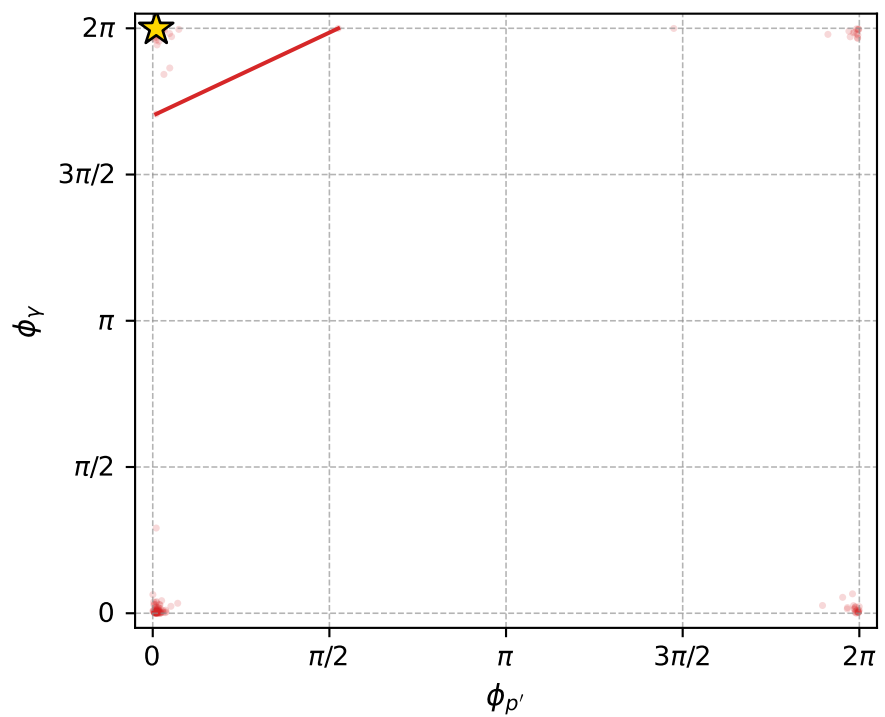
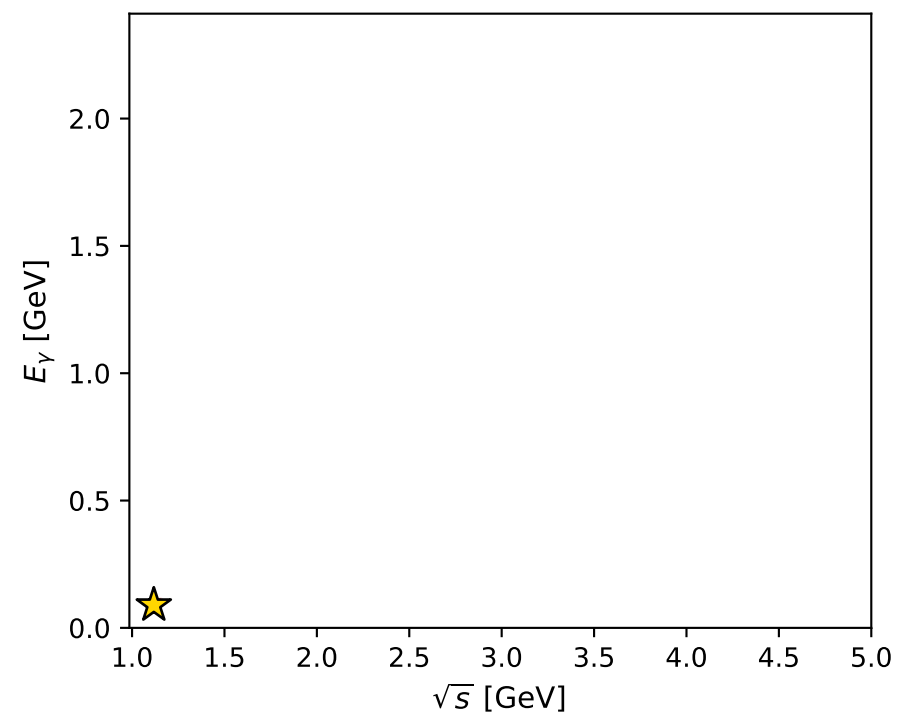
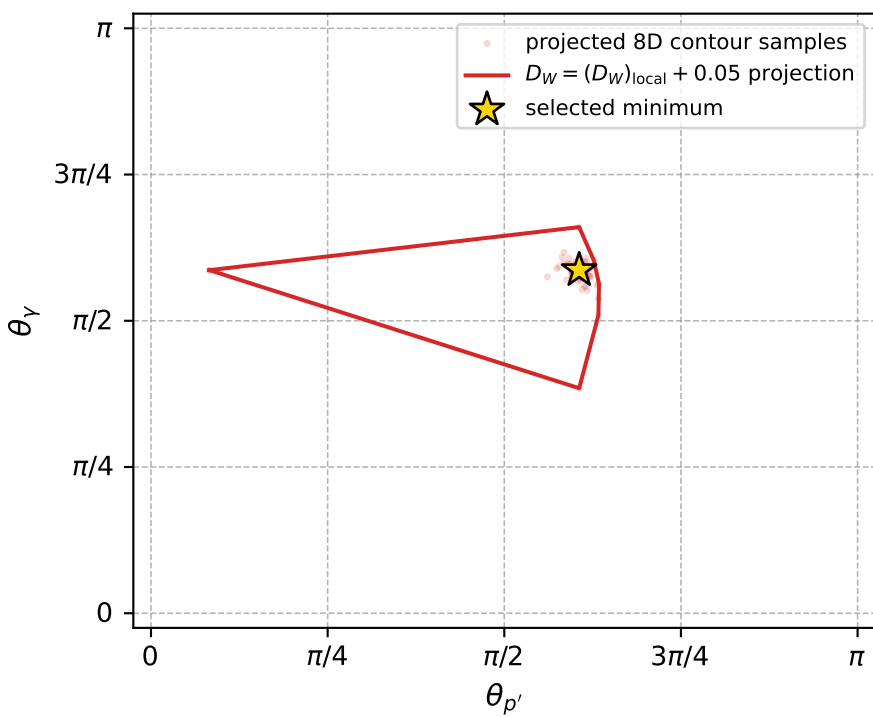
dw_mixing_angles_polarization_03_local_minimum_807 D_W=0.0106393
 region: polarization_cluster_03_local_minimum_807
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77197023$ rad
 incoming lepton: $0.716538|+\rangle +0.697548|-\rangle$
 proton mixing angle: $\alpha_p=2.9485193$ rad
 incoming proton: $-0.981419|+\rangle +0.191876|-\rangle$

$s=1.24926$, $\sqrt{s}=1.1177$, $|\vec{P}|=0.165254$, $|\vec{P}'|=0.000777436$
 $\theta_{p'}=1.90384$, $\theta_V=1.84262$, $\phi_{p'}=0.0309366$, $\phi_V=6.28149$
 $E_V=0.0894618$, $\theta_{l'}=1.29844$, $\phi_{l'}=3.14018$
 $Q^2=0.037847$, $x_B=0.184136$, $t=-0.0271849$, $W^2=1.04754$, $y=0.556395$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1653$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1653$
 P $E= 0.9524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1653$
 l' $E= 0.0902$ $p_x= -0.0869$ $p_y= 0.0001$ $p_z= 0.0243$
 P' $E= 0.9380$ $p_x= 0.0007$ $p_y= 0.0000$ $p_z= -0.0003$
 q_V $E= 0.0895$ $p_x= 0.0862$ $p_y= -0.0001$ $p_z= -0.0240$



electron: polarization cluster P3, local minimum 807; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.010639282$
 $D_W \text{ contour} = 0.060639282$
 above global minimum = 0.010622849
 8D contour samples = 255

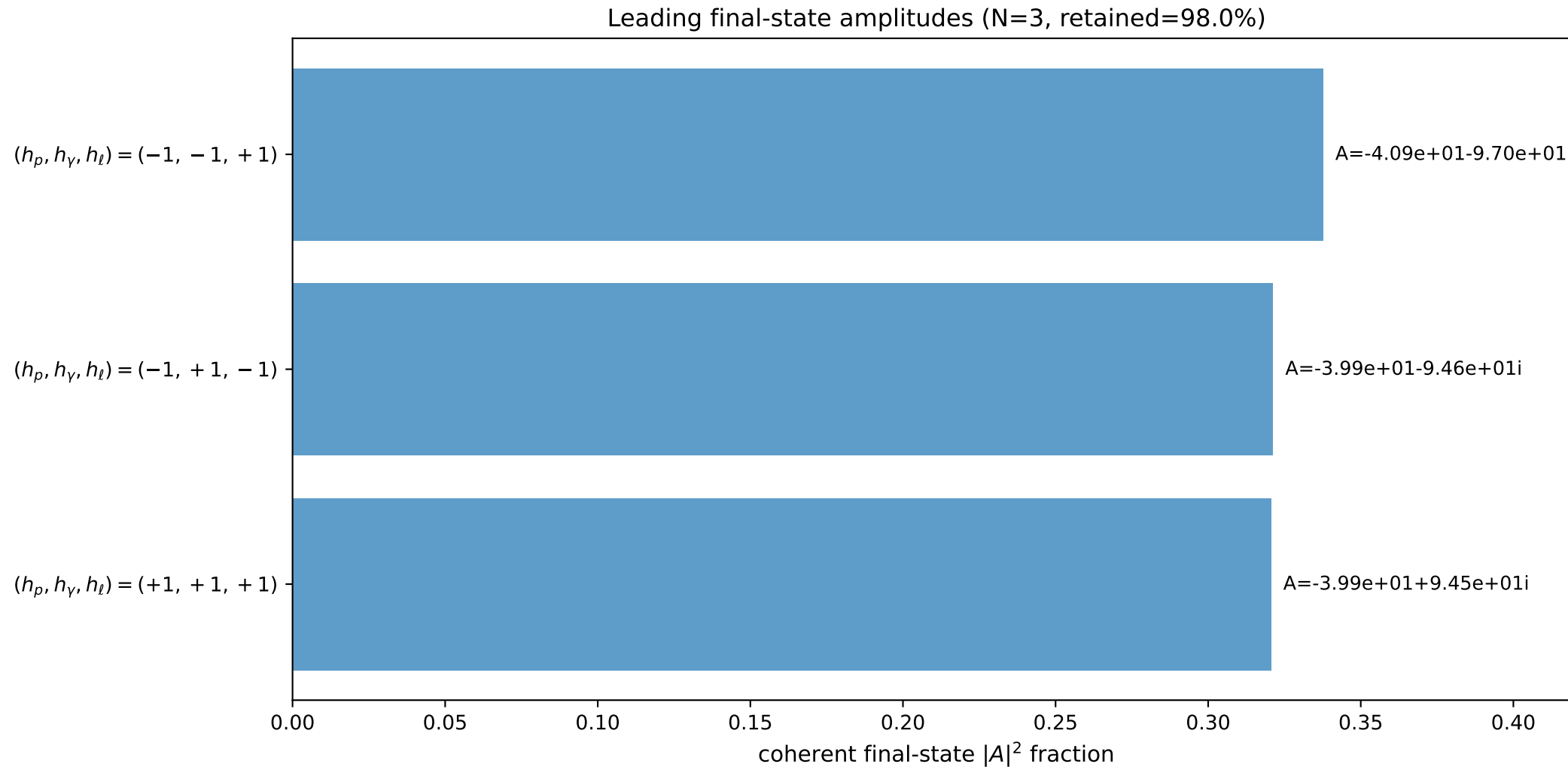
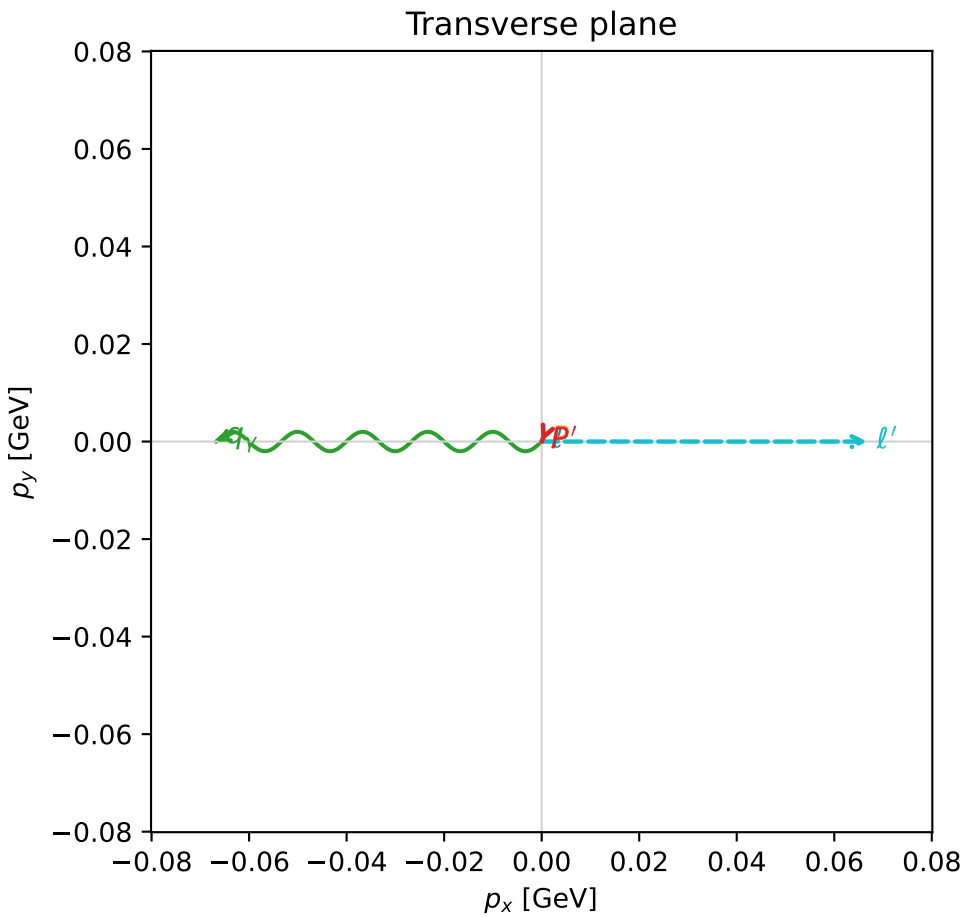
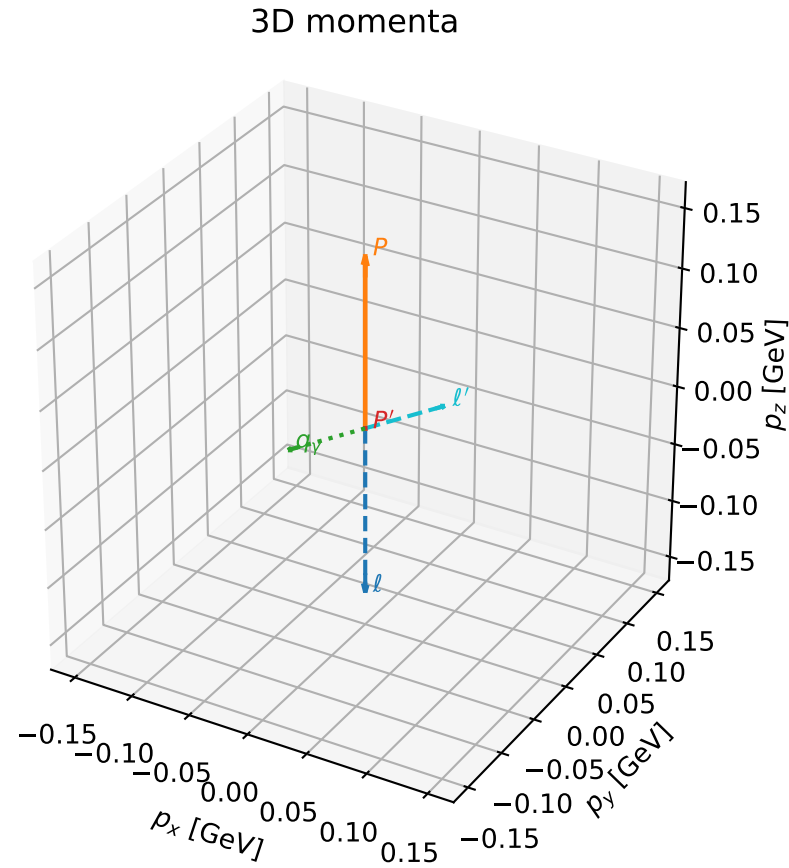
selected phase-space point:
 $\text{sqrt_s} = 1.117701$
 $\text{theta_p_out} = 1.903838$
 $\text{theta_gamma_out} = 1.842621$
 $\text{qOut} = 0.08946177$
 $\text{phi_p_out} = 0.03093664$
 $\text{phi_gamma_out} = 6.281492$
 $\text{alpha_e} = 0.7719702$
 $\text{alpha_p} = 2.948519$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

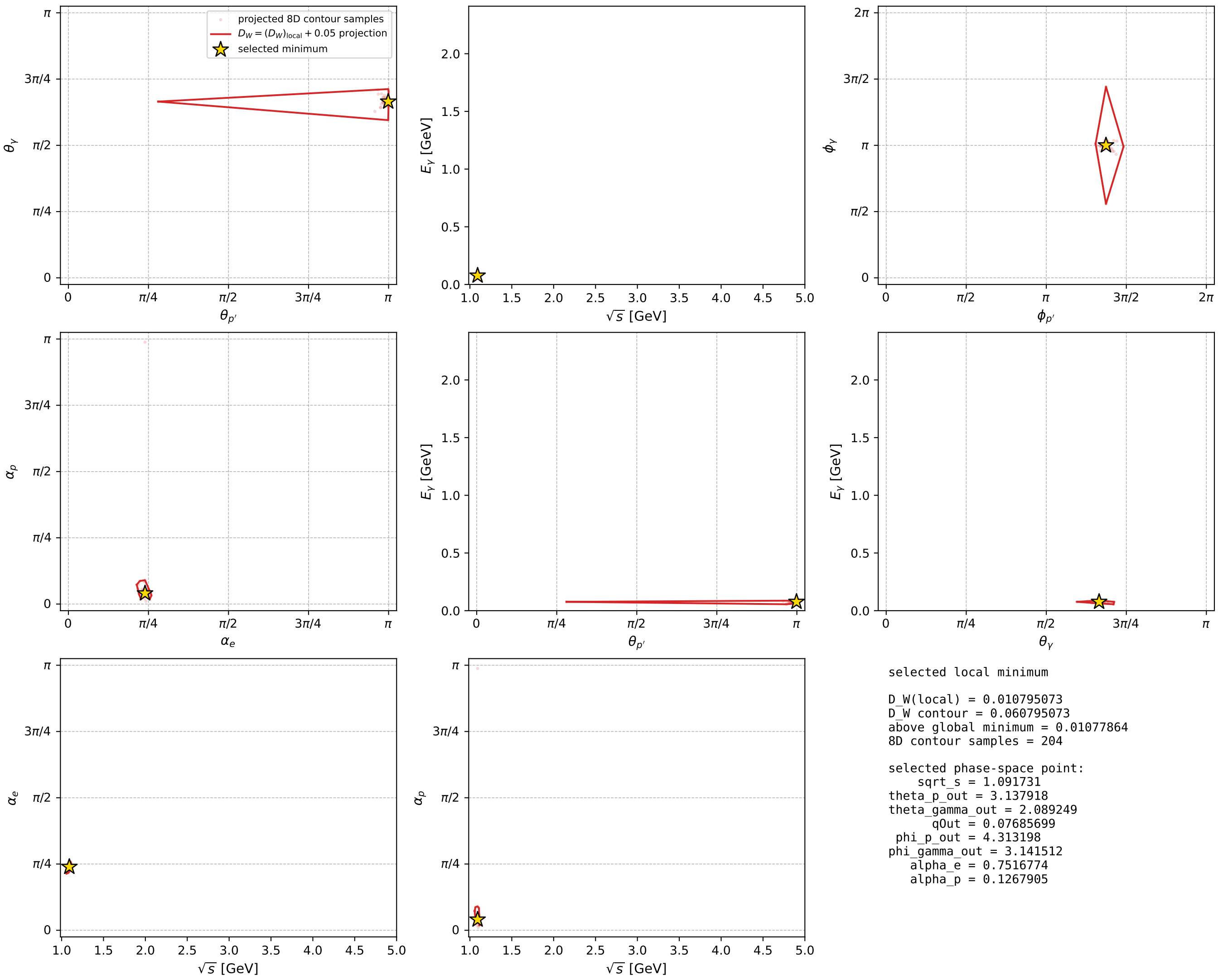
dw_mixing_angles_polarization_03_local_minimum_813 D_W=0.0107951
 region: polarization_cluster_03_local_minimum_813
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.75167736$ rad
 incoming lepton: $0.730544|+> +0.682865|->$
 proton mixing angle: $\alpha_p=0.12679047$ rad
 incoming proton: $0.991973|+> +0.126451|->$

$s=1.19188$, $\sqrt{s}=1.09173$, $|\vec{P}|=0.142907$, $|\vec{P}'|=3.11807\text{e-}05$
 $\theta_{P'}=3.13792$, $\theta_V=2.08925$, $\phi_{P'}=4.3132$, $\phi_V=3.14151$
 $E_V=0.076857$, $\theta_{\ell'}=1.05199$, $\phi_{\ell'}=6.28311$
 $Q^2=0.0328655$, $x_B=0.185632$, $t=-0.0203141$, $W^2=1.02403$, $y=0.567398$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1429$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1429$
 P $E= 0.9488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1429$
 ℓ' $E= 0.0769$ $p_x= 0.0668$ $p_y= -0.0000$ $p_z= 0.0381$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_V $E= 0.0769$ $p_x= -0.0668$ $p_y= 0.0000$ $p_z= -0.0381$



electron: polarization cluster P3, local minimum 813; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

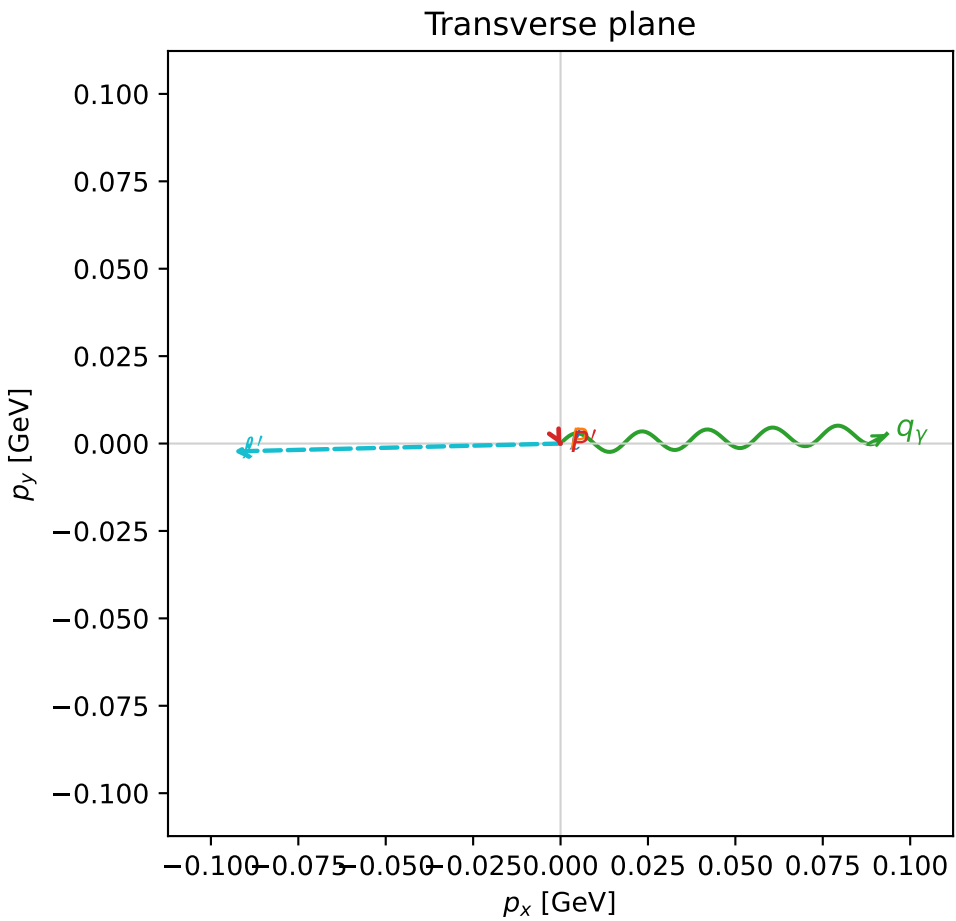
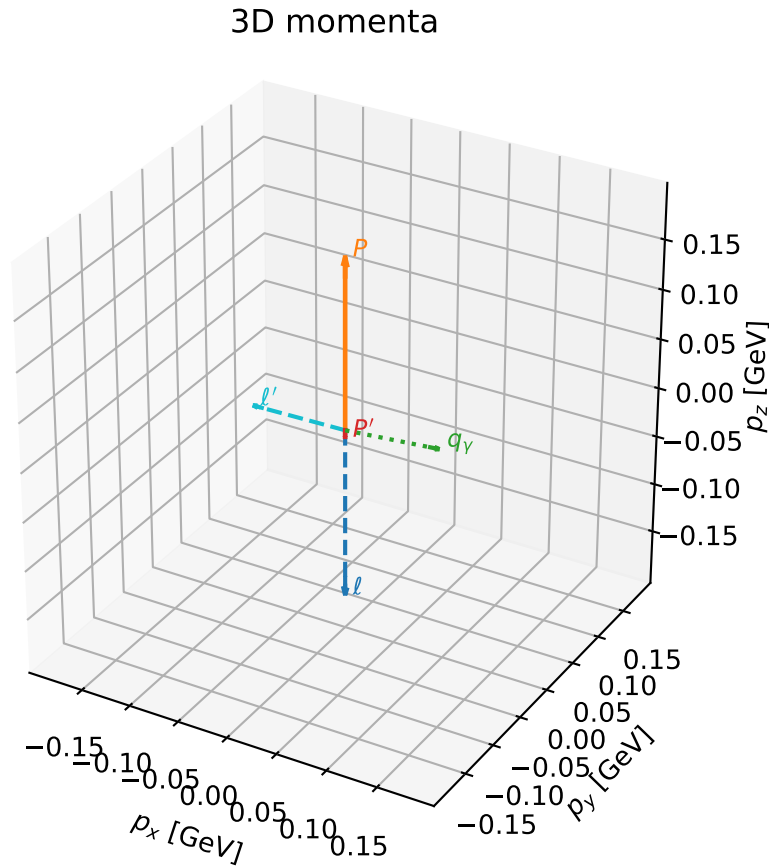


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_815 D_W=0.010864
 region: polarization_cluster_03_local_minimum_815
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79883948$ rad
 incoming lepton: $0.697539|+\rangle +0.716547|-\rangle$
 proton mixing angle: $\alpha_p=2.9379666$ rad
 incoming proton: $-0.97934|+\rangle +0.202222|-\rangle$

$s=1.26605$, $\sqrt{s}=1.12519$, $|\vec{P}|=0.171619$, $|\vec{P}'|=0.00636176$
 $\theta_{p'}=3.05711$, $\theta_Y=1.50047$, $\phi_{p'}=5.20611$, $\phi_Y=0.0292564$
 $E_Y=0.0935787$, $\theta_{\ell'}=1.57333$, $\phi_{\ell'}=3.16572$
 $Q^2=0.0320423$, $x_B=0.154318$, $t=-0.0314278$, $W^2=1.05544$, $y=0.53763$

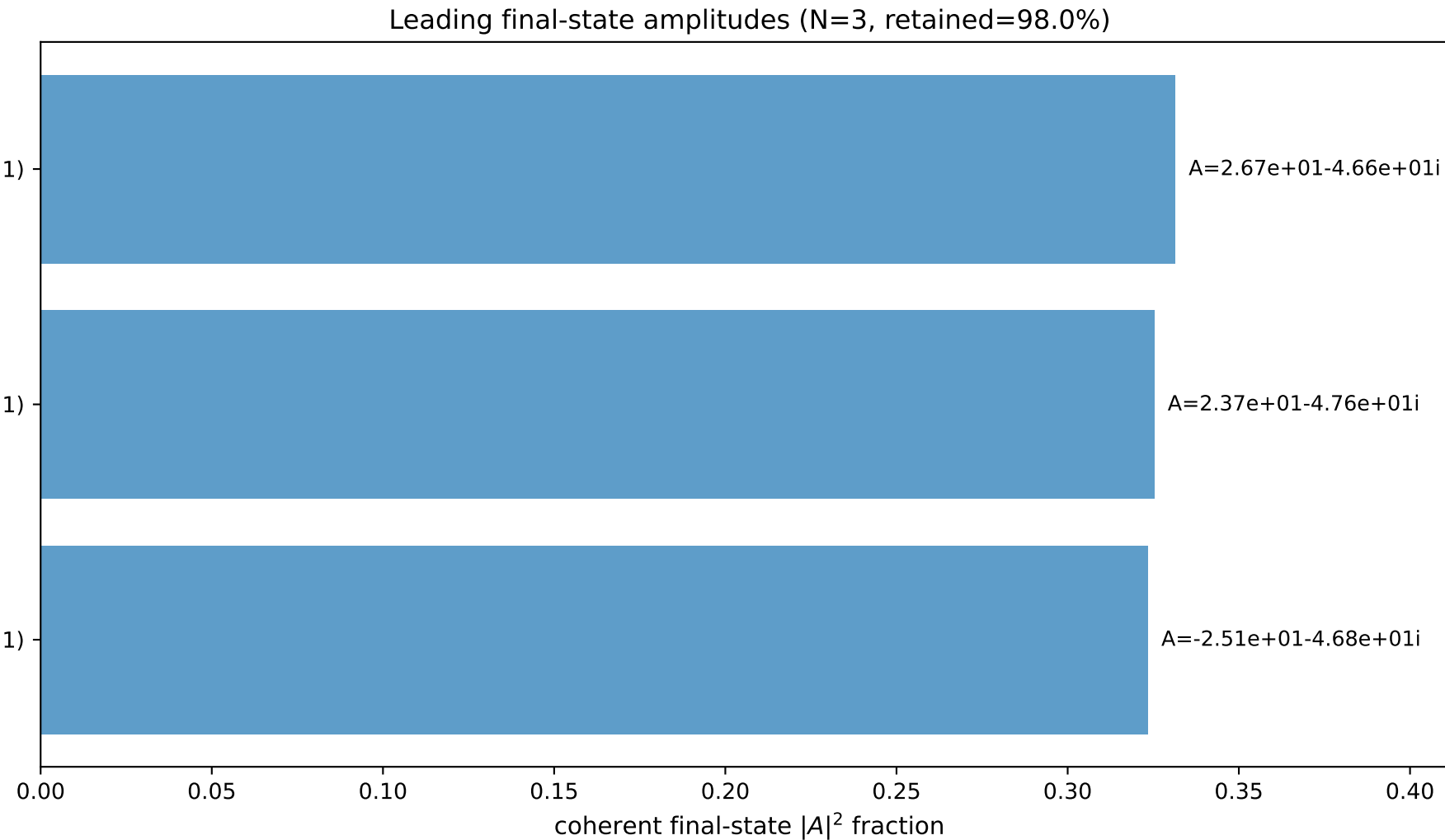
four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1716$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1716$
 P $E= 0.9536$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1716$
 ℓ' $E= 0.0936$ $p_x= -0.0936$ $p_y= -0.0023$ $p_z= -0.0002$
 P' $E= 0.9380$ $p_x= 0.0003$ $p_y= -0.0005$ $p_z= -0.0063$
 q_Y $E= 0.0936$ $p_x= 0.0933$ $p_y= 0.0027$ $p_z= 0.0066$



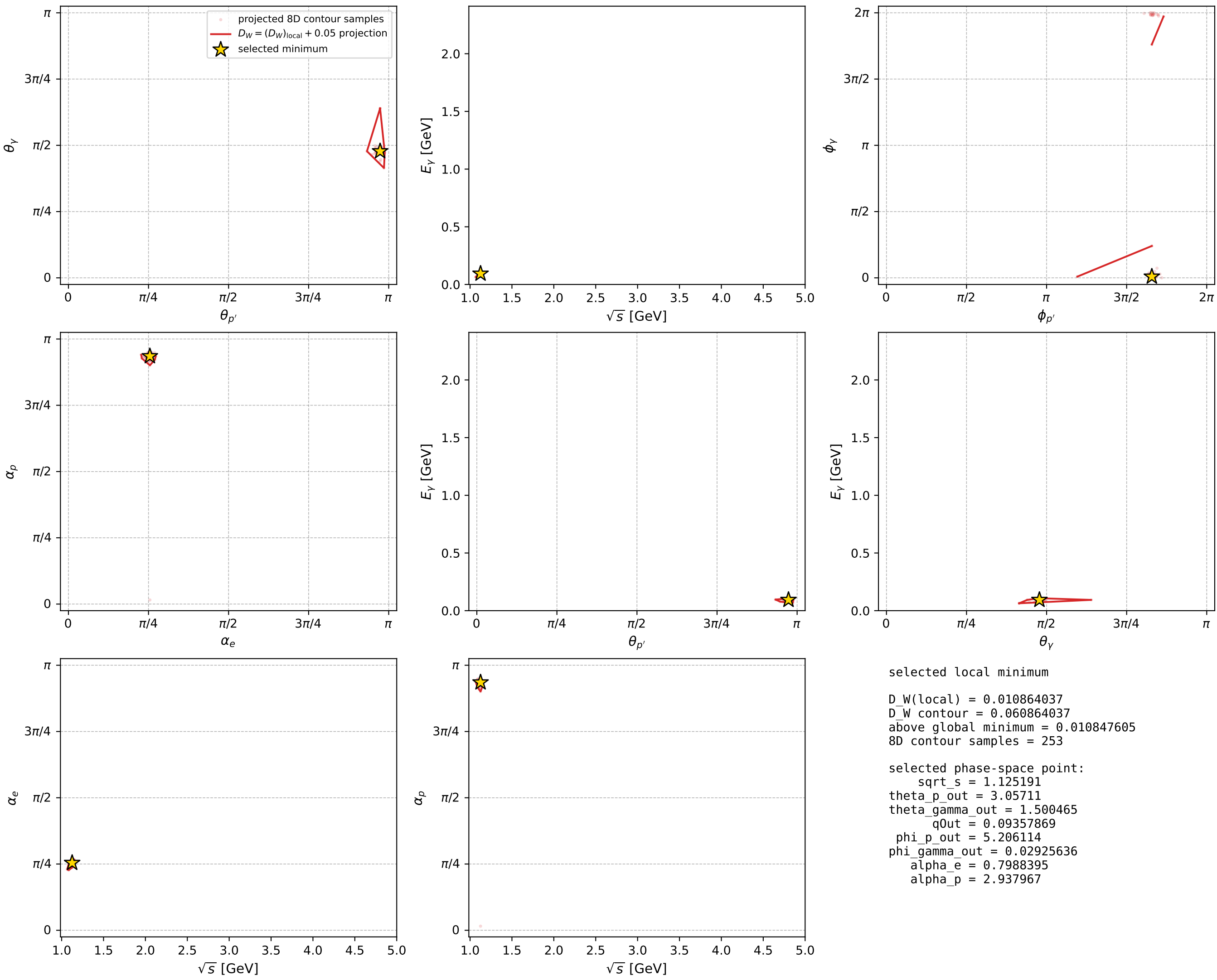
$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (+1, +1, +1)$



electron: polarization cluster P3, local minimum 815; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

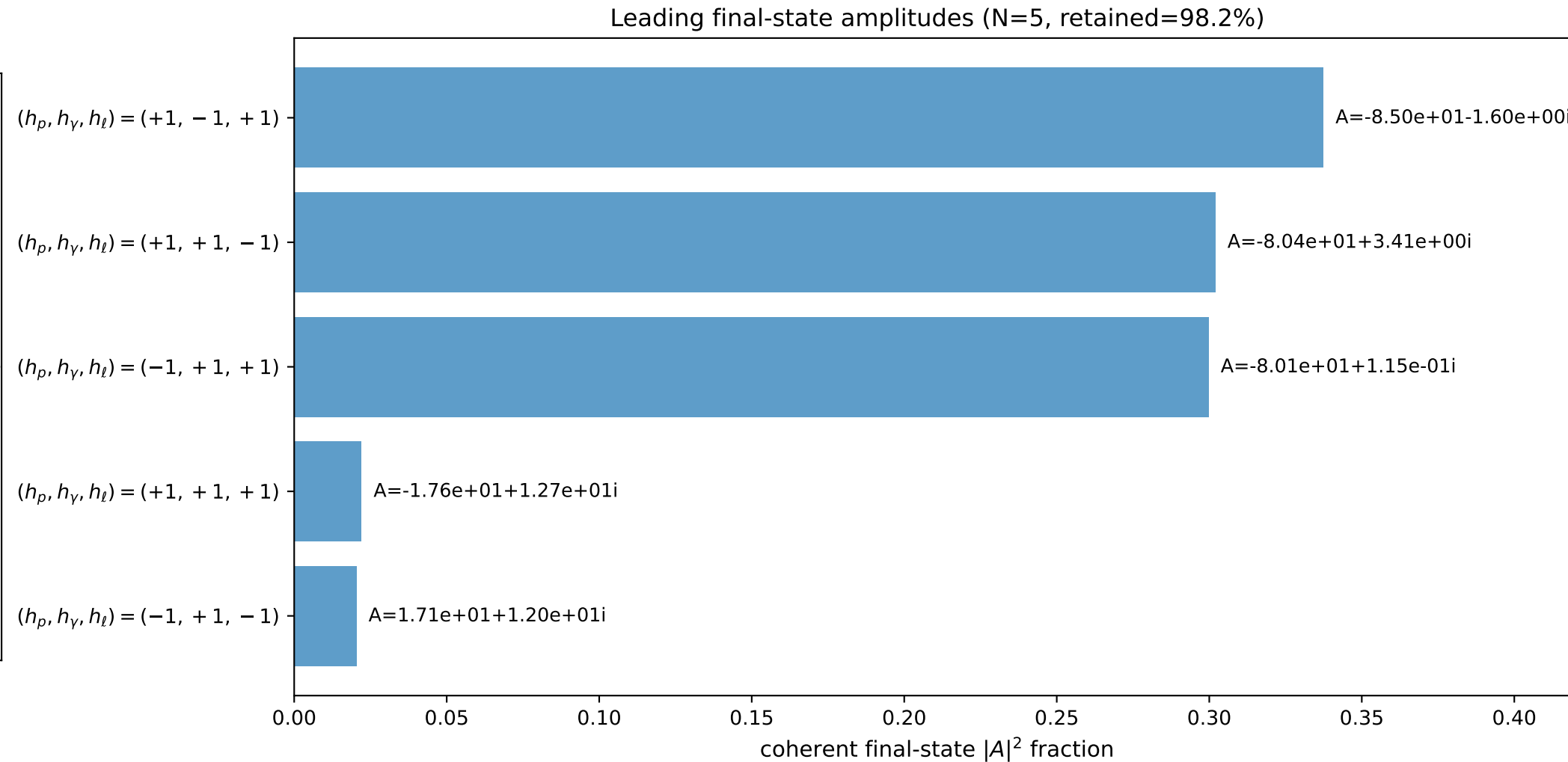
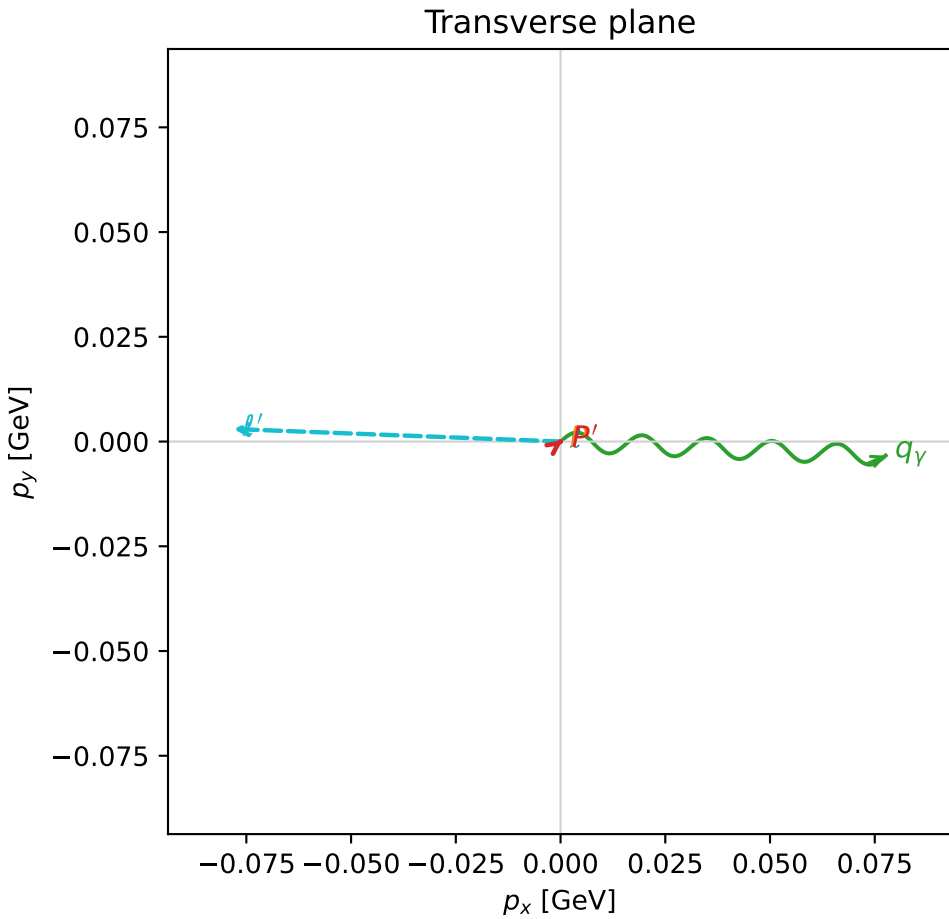
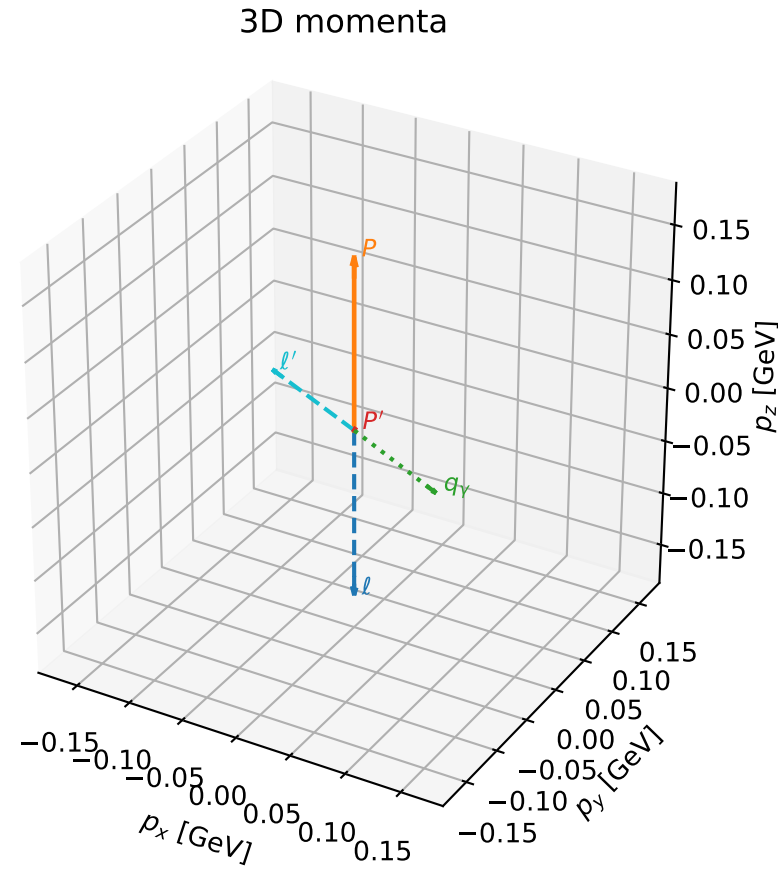


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

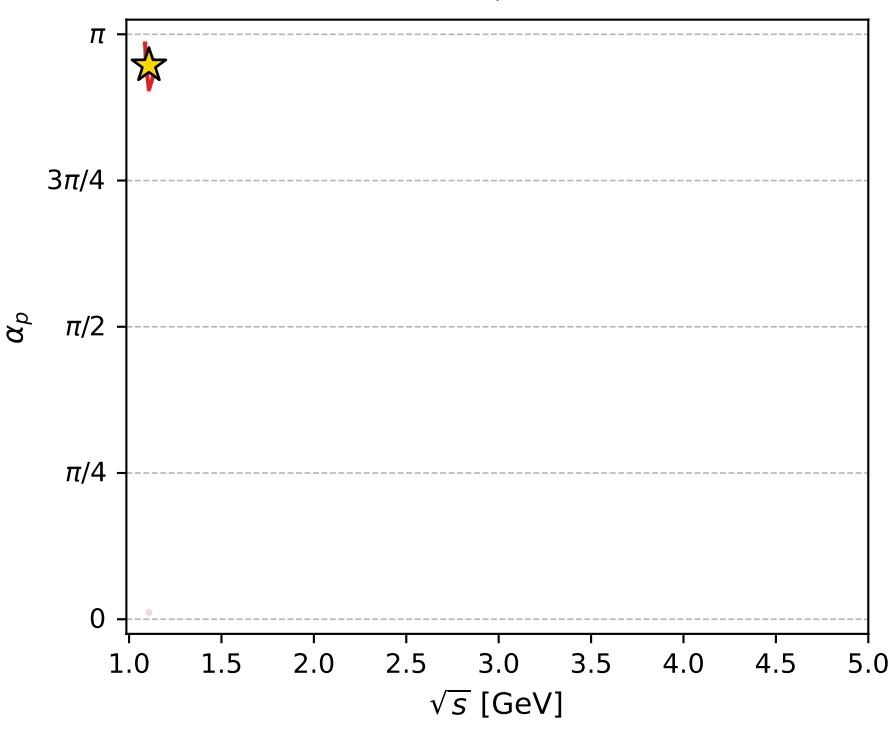
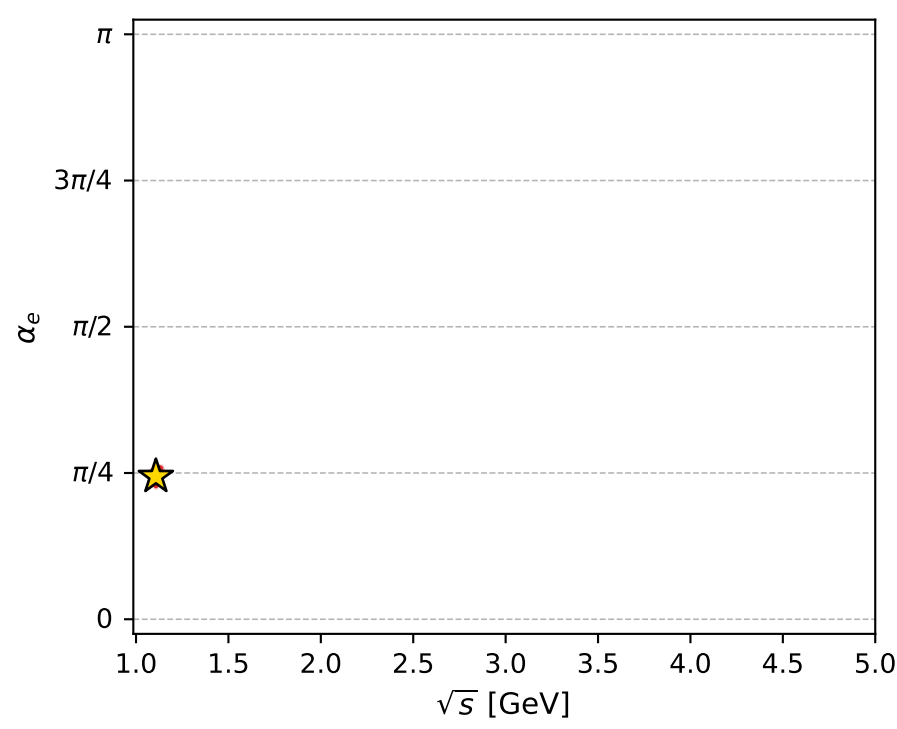
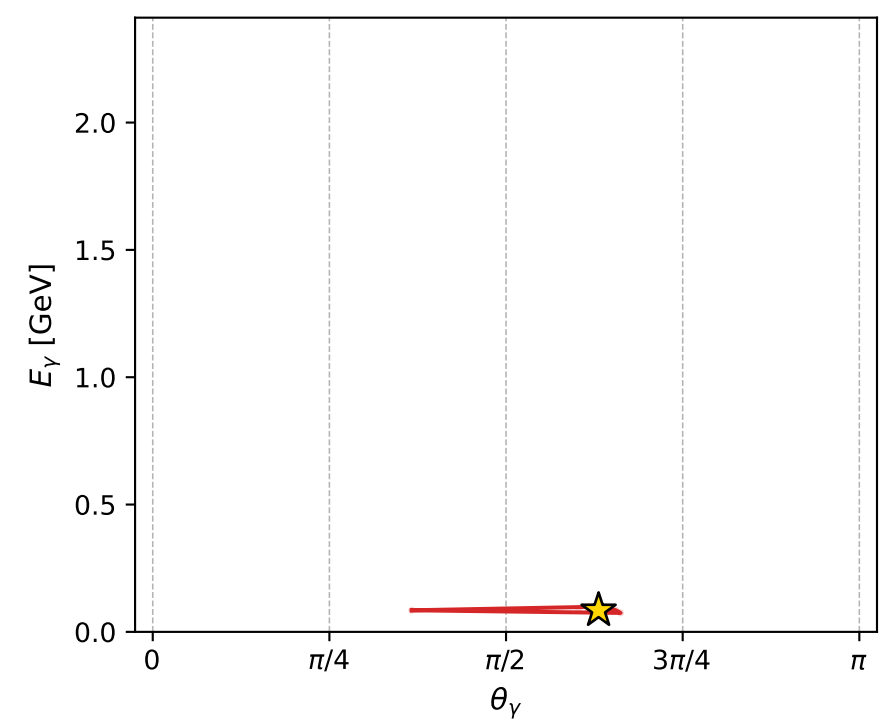
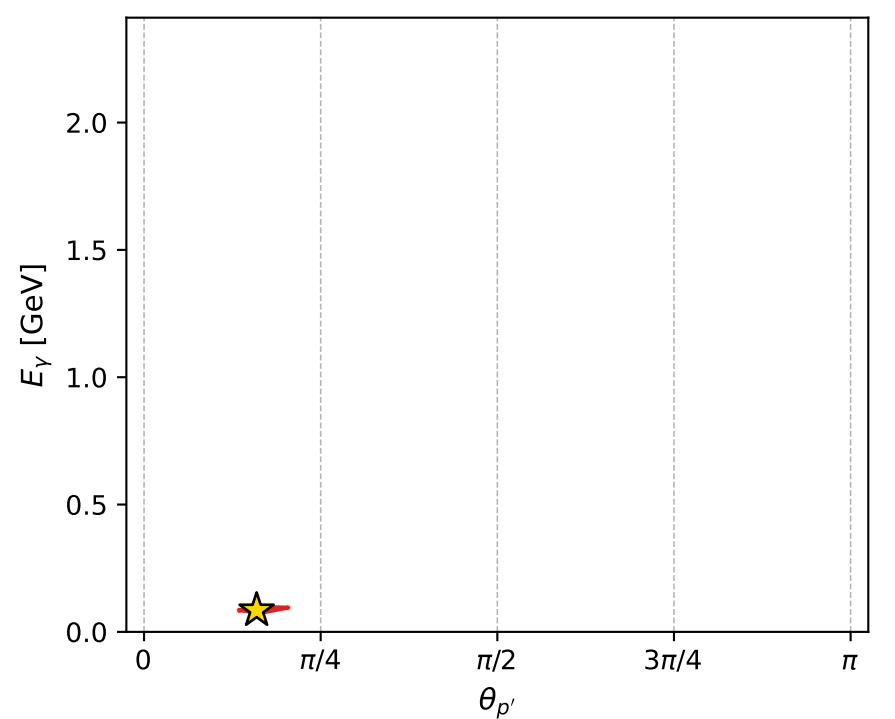
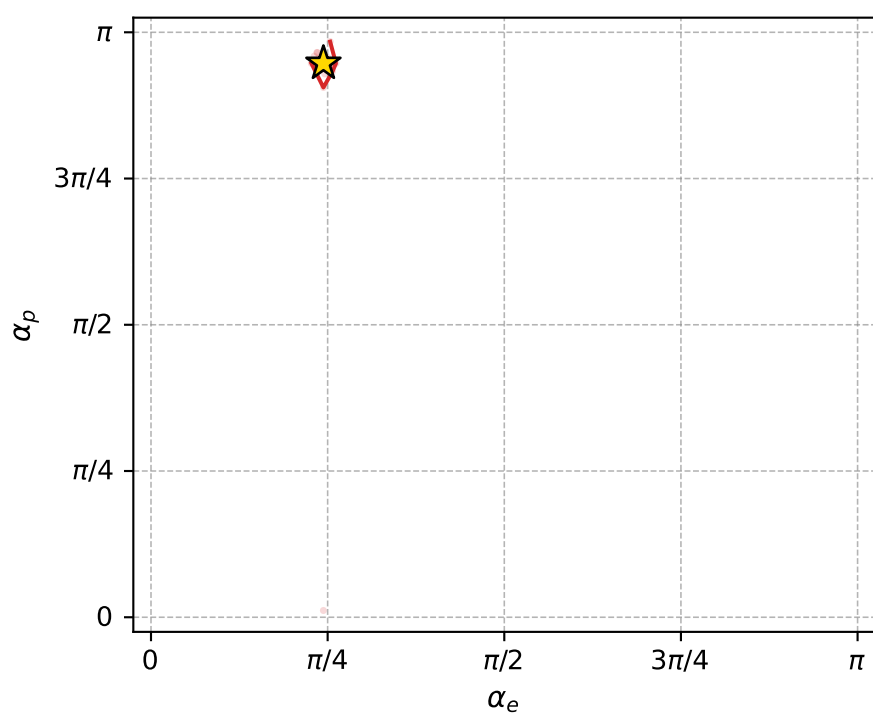
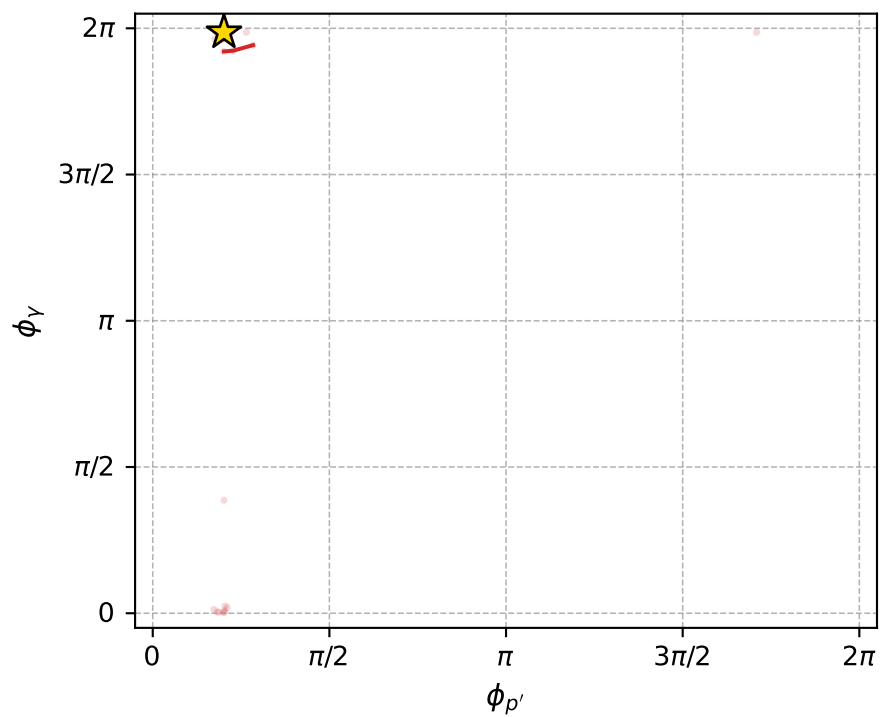
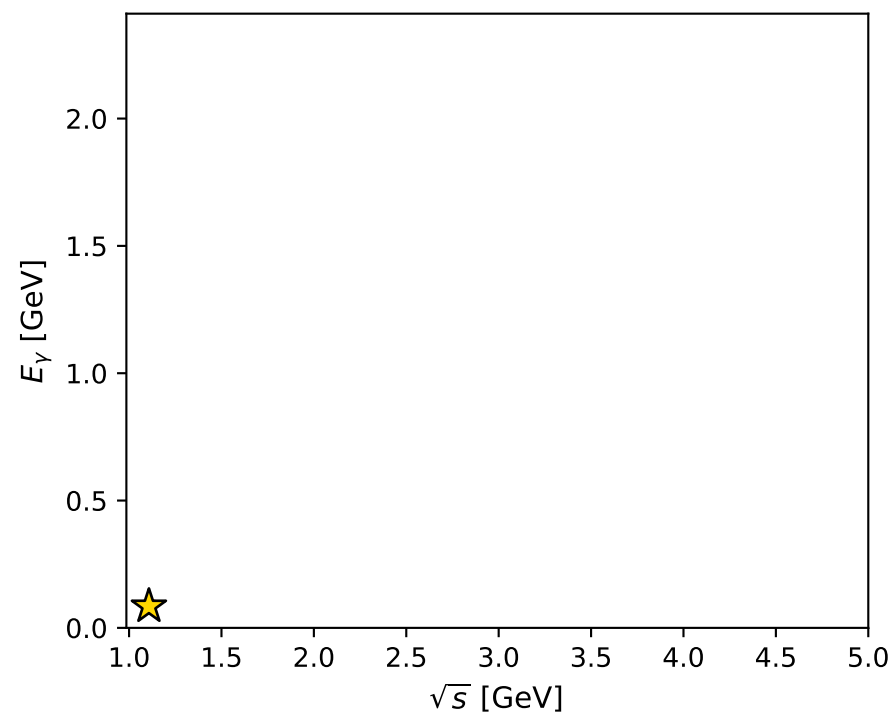
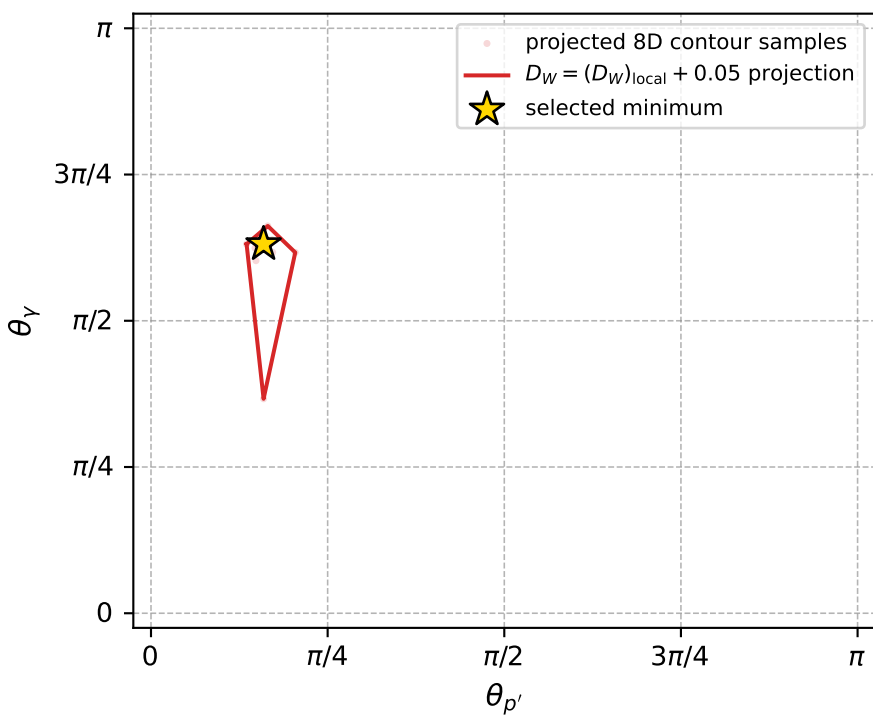
dw_mixing_angles_polarization_03_local_minimum_817 D_W=0.0109479
 region: polarization_cluster_03_local_minimum_817
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7666195$ rad
 incoming lepton: $0.72026|+\rangle + 0.693704|-\rangle$
 proton mixing angle: $\alpha_p=2.9747292$ rad
 incoming proton: $-0.986111|+\rangle + 0.16609|-\rangle$

$s=1.22651$, $\sqrt{s}=1.10748$, $|\vec{P}|=0.156512$, $|\vec{P}'|=0.00126651$
 $\theta_{p'}=0.500592$, $\theta_Y=1.9823$, $\phi_{p'}=0.633092$, $\phi_Y=6.23996$
 $E_Y=0.0847393$, $\theta_{\ell'}=1.17356$, $\phi_{\ell'}=3.10323$
 $Q^2=0.0367873$, $x_B=0.18792$, $t=-0.0239816$, $W^2=1.03882$, $y=0.564689$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1565$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1565$
 P $E= 0.9510$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1565$
 ℓ' $E= 0.0847$ $p_x= -0.0781$ $p_y= 0.0030$ $p_z= 0.0328$
 P' $E= 0.9380$ $p_x= 0.0005$ $p_y= 0.0004$ $p_z= 0.0011$
 q_Y $E= 0.0847$ $p_x= 0.0776$ $p_y= -0.0034$ $p_z= -0.0339$



electron: polarization cluster P3, local minimum 817; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.010947899$
 $D_W \text{ contour} = 0.060947899$
 above global minimum = 0.010931466
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt}_s = 1.107481$
 $\theta_{p_out} = 0.5005923$
 $\theta_{\gamma_out} = 1.982296$
 $q_{out} = 0.08473934$
 $\phi_{p_out} = 0.633092$
 $\phi_{\gamma_out} = 6.239956$
 $\alpha_e = 0.7666195$
 $\alpha_p = 2.974729$

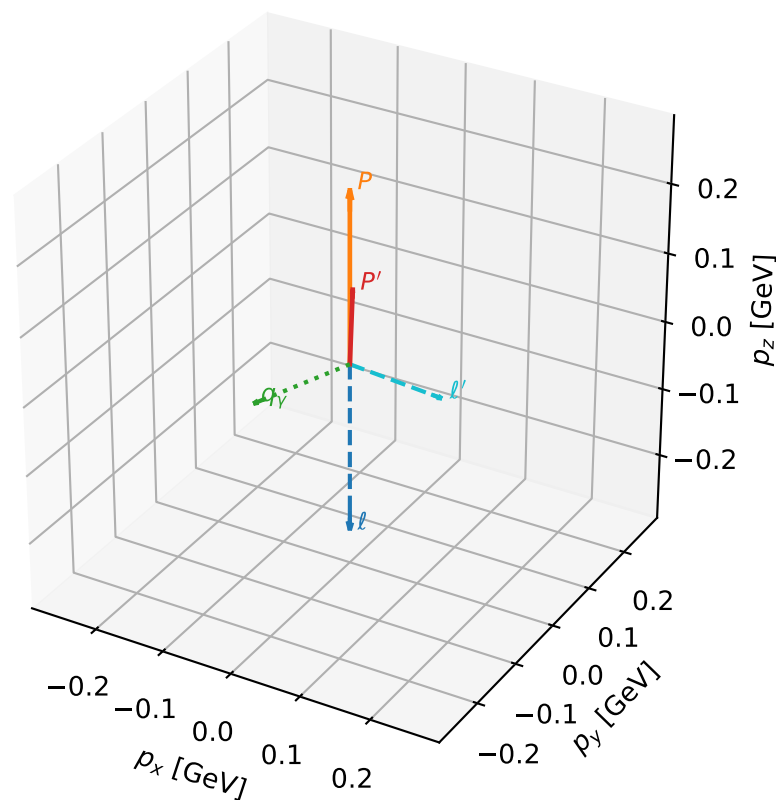
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_821 D_W=0.0111156
 region: polarization_cluster_03_local_minimum_821
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82700481$ rad
 incoming lepton: $0.677083|+\rangle + 0.735907|-\rangle$
 proton mixing angle: $\alpha_p=0.27388883$ rad
 incoming proton: $0.962726|+\rangle + 0.270477|-\rangle$

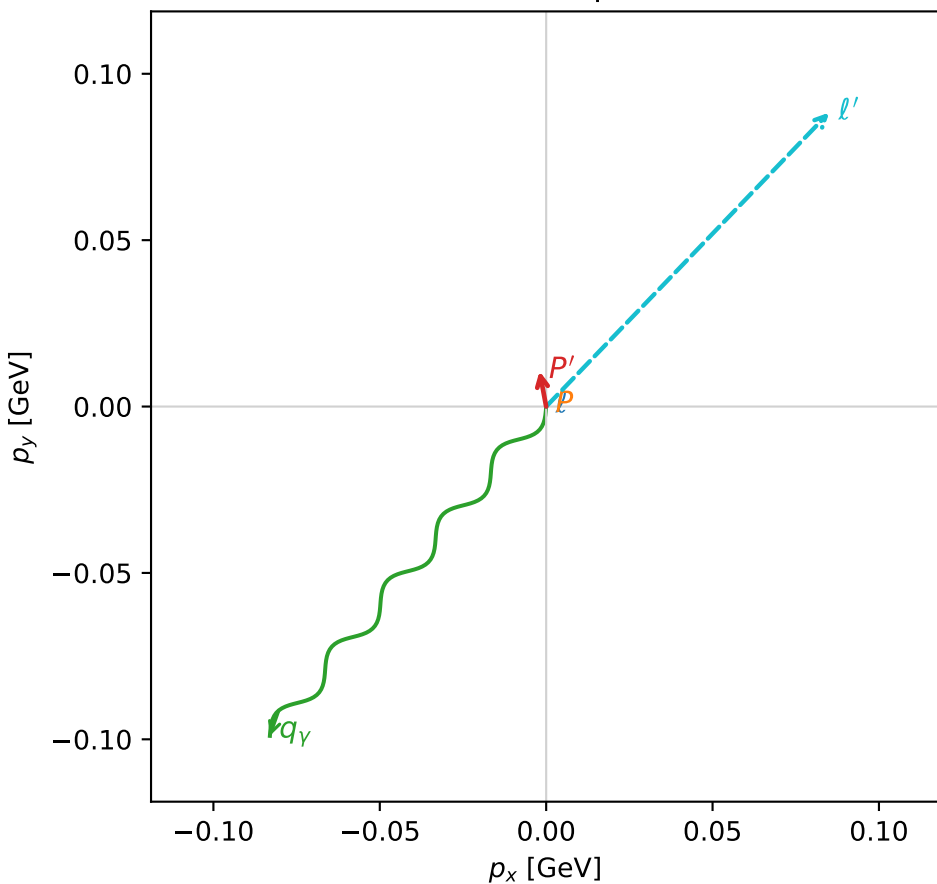
$s=1.48325$, $\sqrt{s}=1.21789$, $|\vec{P}|=0.247728$, $|\vec{P}'|=0.102547$
 $\theta_{P'}=0.105555$, $\theta_Y=1.81197$, $\phi_{P'}=1.75636$, $\phi_Y=4.01428$
 $E_Y=0.13302$, $\theta_{\ell'}=2.09085$, $\phi_{\ell'}=0.80451$
 $Q^2=0.0352141$, $x_B=0.119574$, $t=-0.020654$, $W^2=1.13913$, $y=0.488053$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2477$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2477$
 P $E= 0.9702$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2477$
 ℓ' $E= 0.1413$ $p_x= 0.0850$ $p_y= 0.0883$ $p_z= -0.0702$
 P' $E= 0.9436$ $p_x= -0.0020$ $p_y= 0.0106$ $p_z= 0.1020$
 q_Y $E= 0.1330$ $p_x= -0.0830$ $p_y= -0.0990$ $p_z= -0.0318$

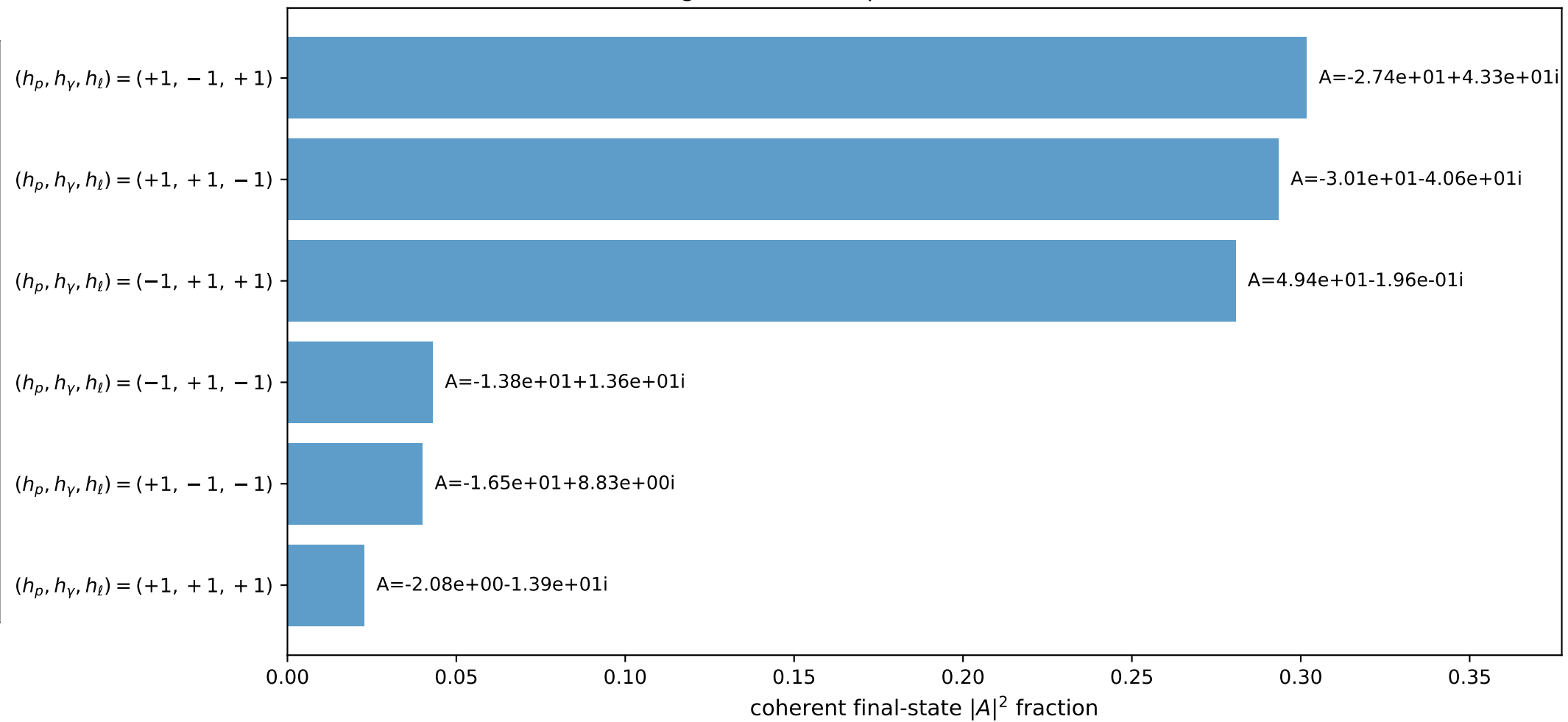
3D momenta



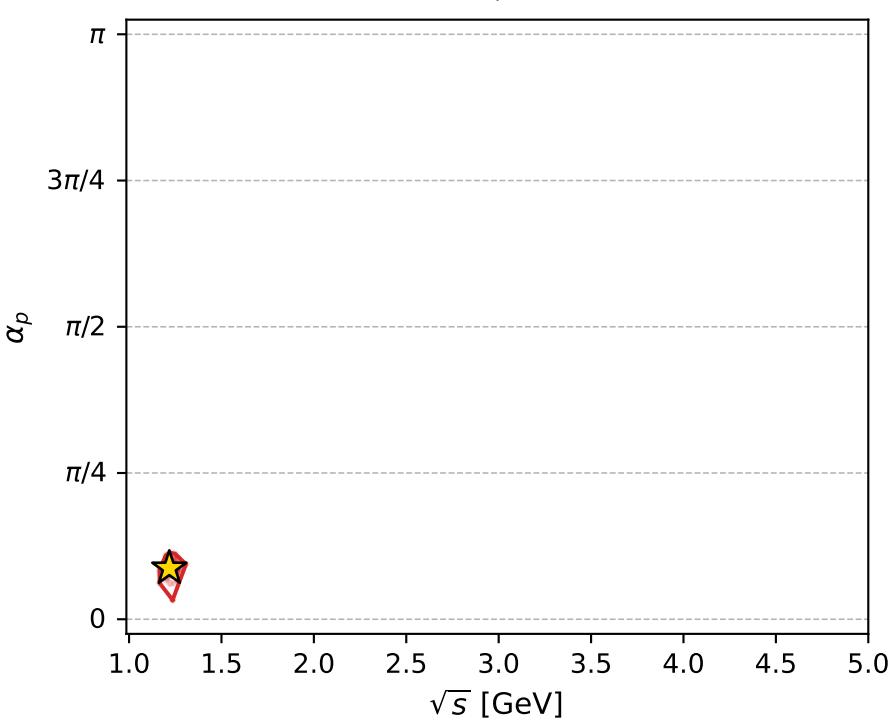
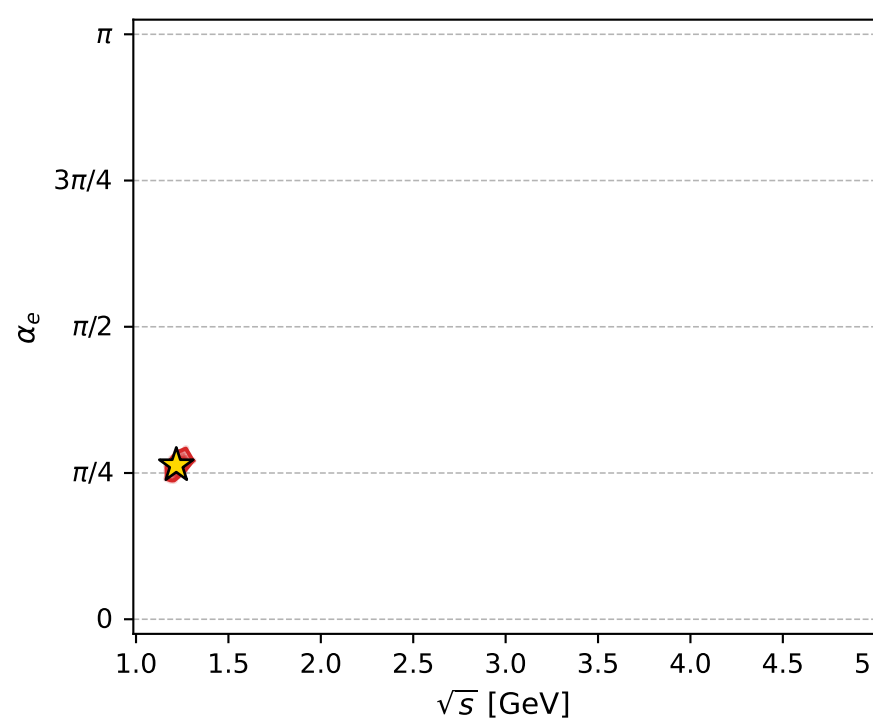
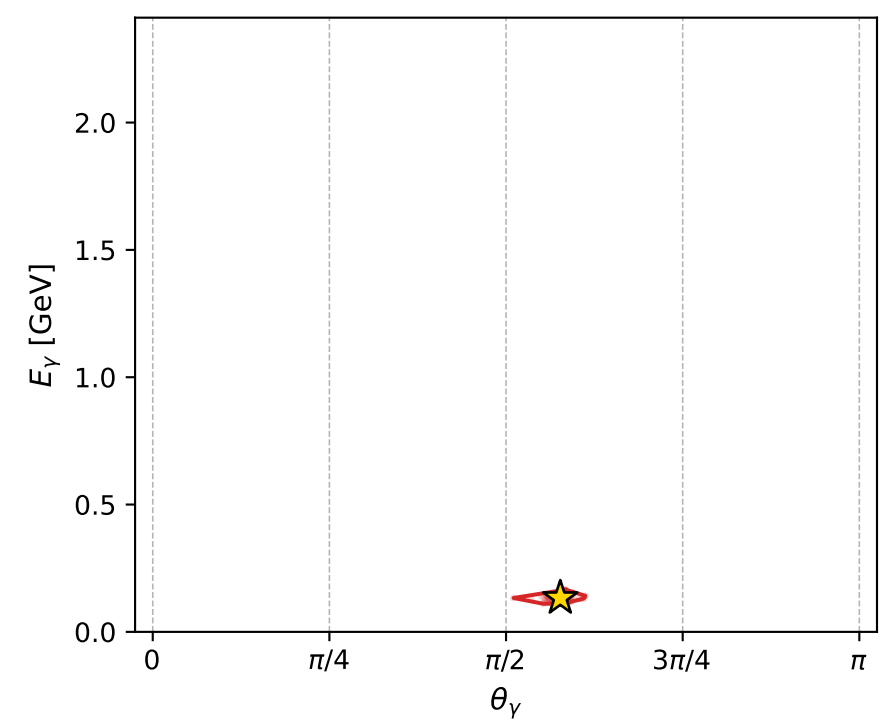
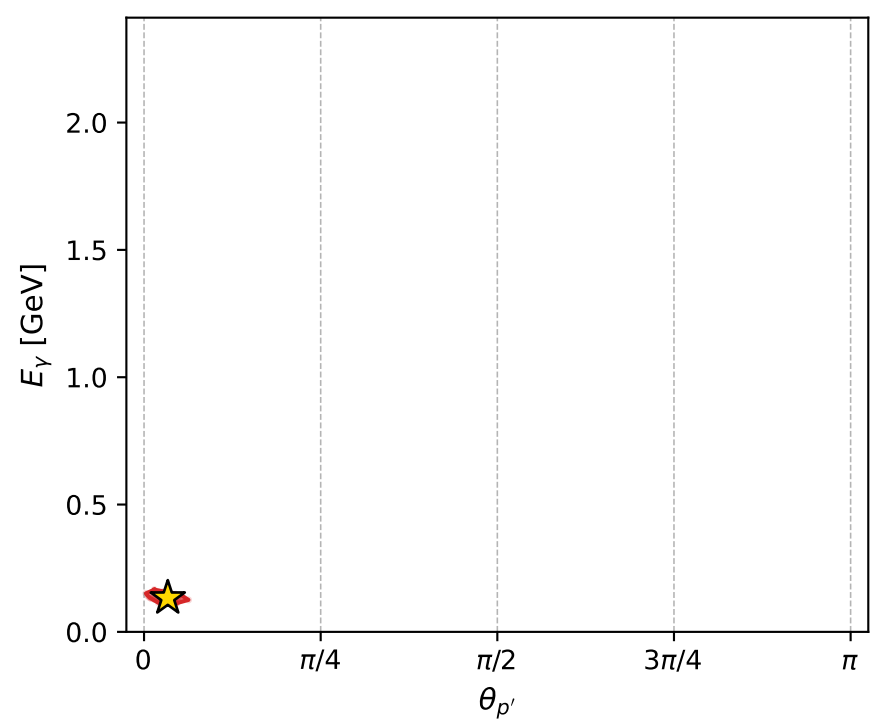
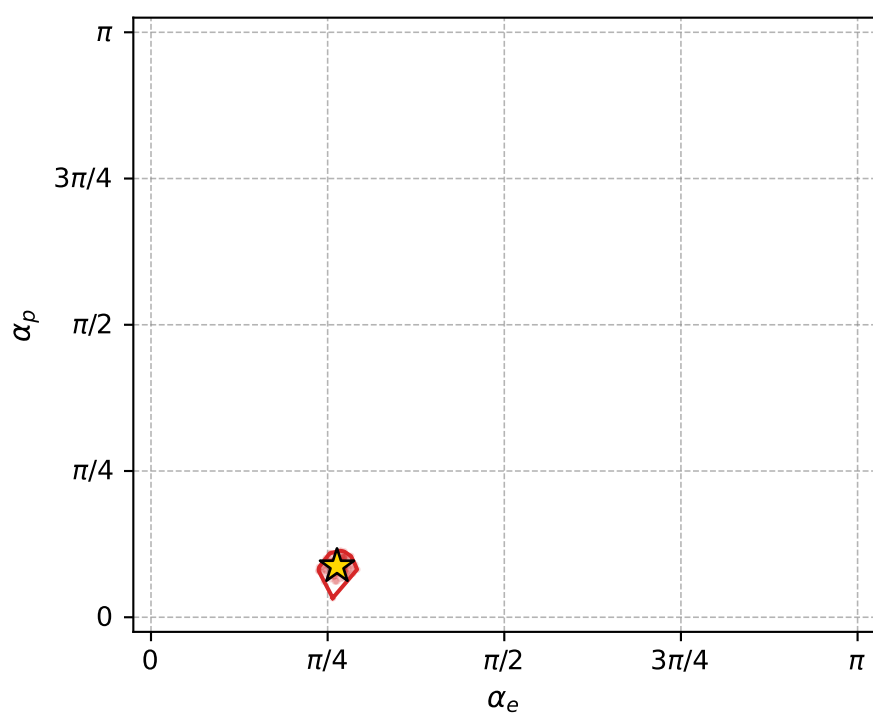
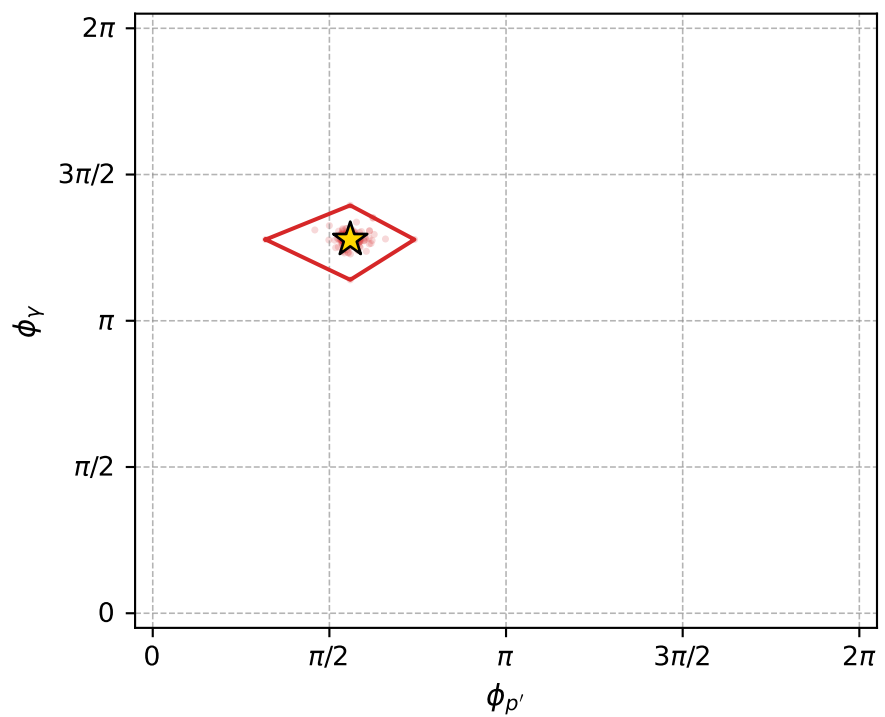
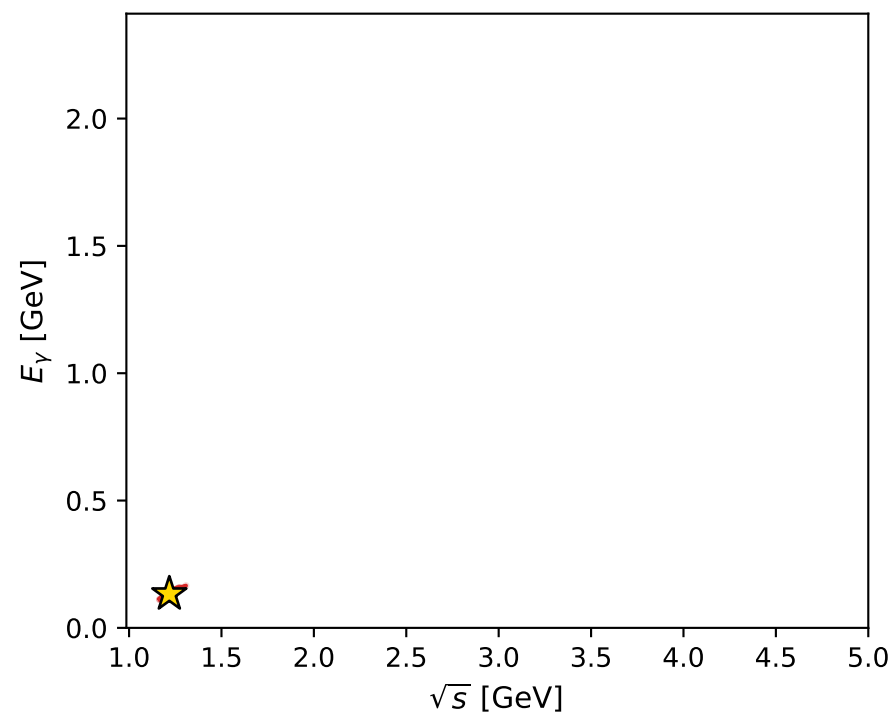
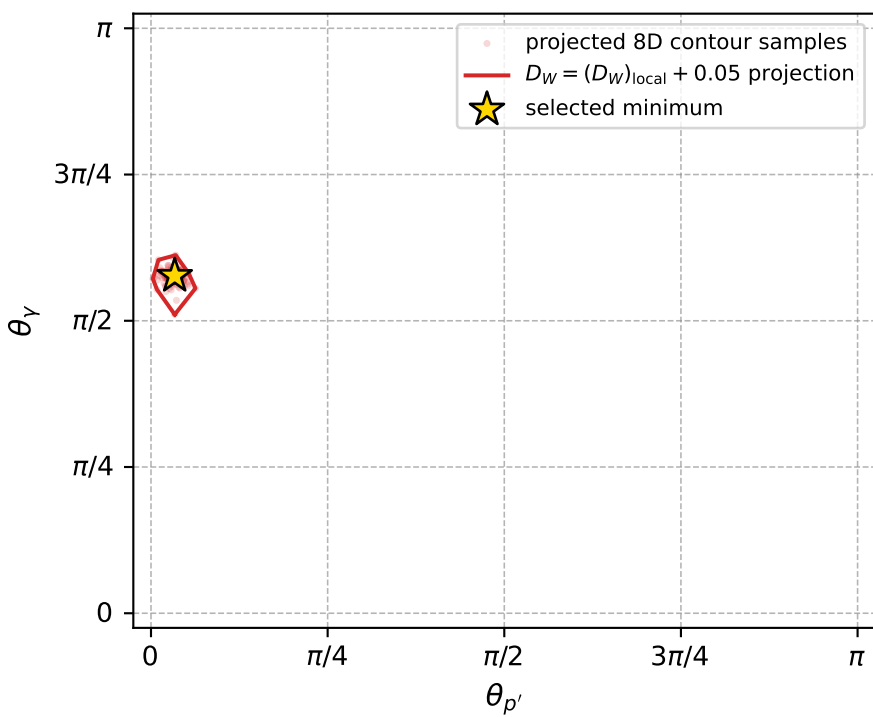
Transverse plane



Leading final-state amplitudes (N=6, retained=98.2%)



electron: polarization cluster P3, local minimum 821; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011115598$
 $D_W \text{ contour} = 0.061115598$
 above global minimum = 0.011099165
 8D contour samples = 254

selected phase-space point:

$\text{sqrt_s} = 1.217889$
 $\text{theta_p_out} = 0.1055549$
 $\text{theta_gamma_out} = 1.81197$
 $\text{qOut} = 0.1330198$
 $\text{phi_p_out} = 1.756364$
 $\text{phi_gamma_out} = 4.014282$
 $\text{alpha_e} = 0.8270048$
 $\text{alpha_p} = 0.2738888$

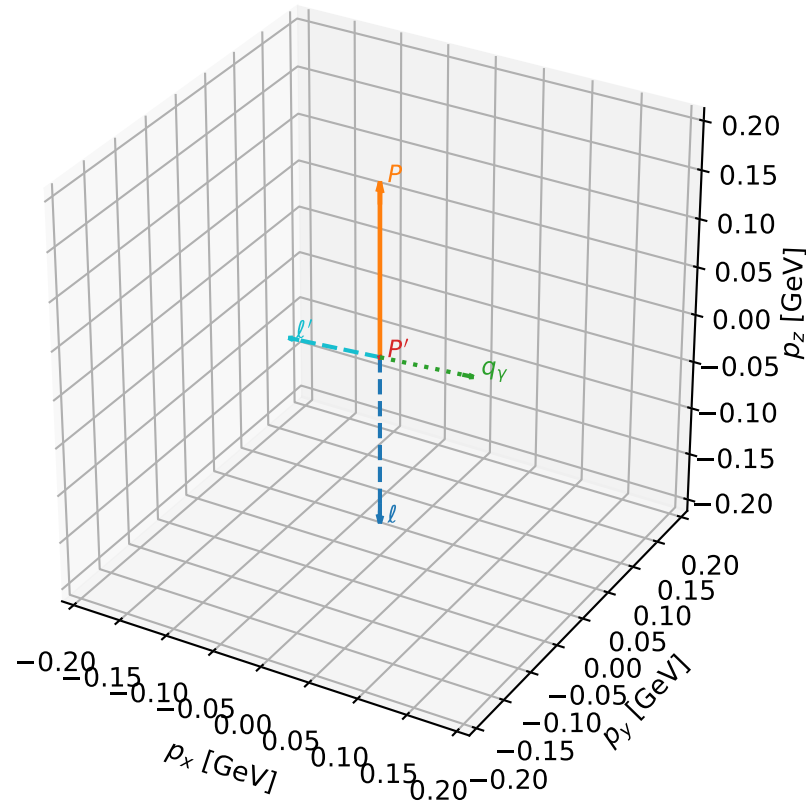
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_824 D_W=0.0112039
 region: polarization_cluster_03_local_minimum_824
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80643458$ rad
 incoming lepton: $0.692076|+\rangle + 0.721824|-\rangle$
 proton mixing angle: $\alpha_p=2.9445798$ rad
 incoming proton: $-0.980656|+\rangle + 0.195741|-\rangle$

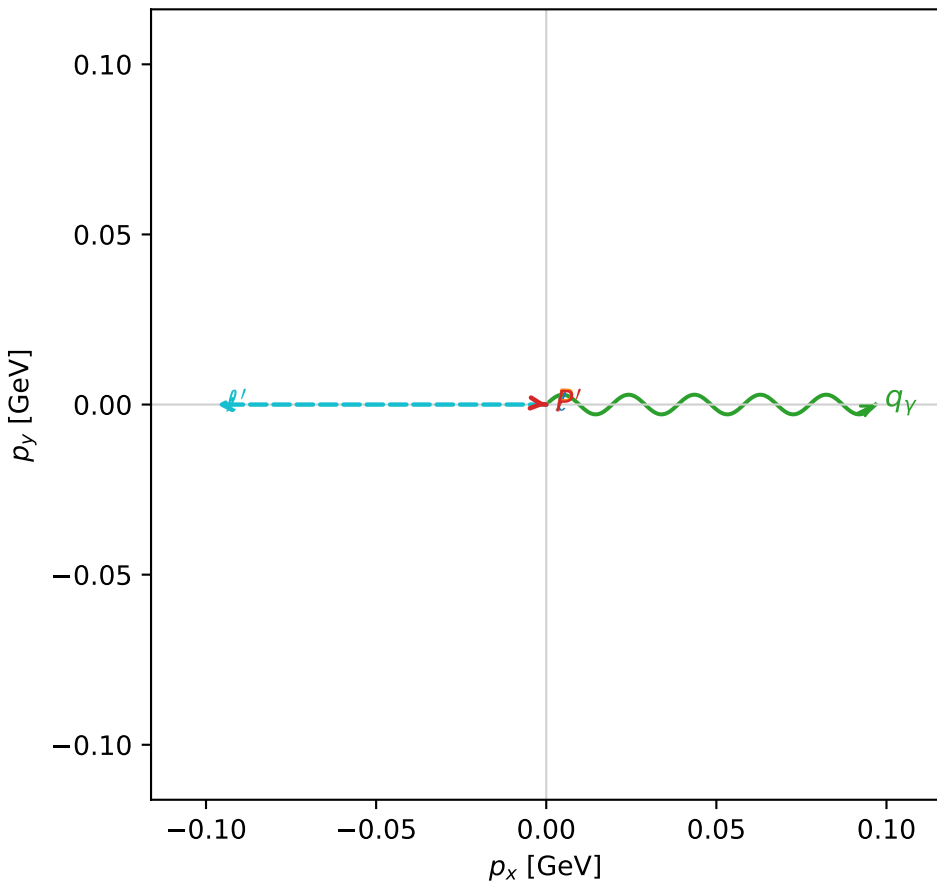
$s=1.2818$, $\sqrt{s}=1.13217$, $|\vec{P}|=0.177516$, $|\vec{P}'|=1.82863e-05$
 $\theta_{P'}=2.68037$, $\theta_V=1.49368$, $\phi_{P'}=6.18232$, $\phi_V=6.54819e-05$
 $E_V=0.097079$, $\theta_{\ell'}=1.64774$, $\phi_{\ell'}=3.14165$
 $Q^2=0.0318191$, $x_B=0.148731$, $t=-0.0312405$, $W^2=1.06196$, $y=0.532242$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1775$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1775$
 P $E= 0.9546$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1775$
 ℓ' $E= 0.0971$ $p_x= -0.0968$ $p_y= -0.0000$ $p_z= -0.0075$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_V $E= 0.0971$ $p_x= 0.0968$ $p_y= 0.0000$ $p_z= 0.0075$

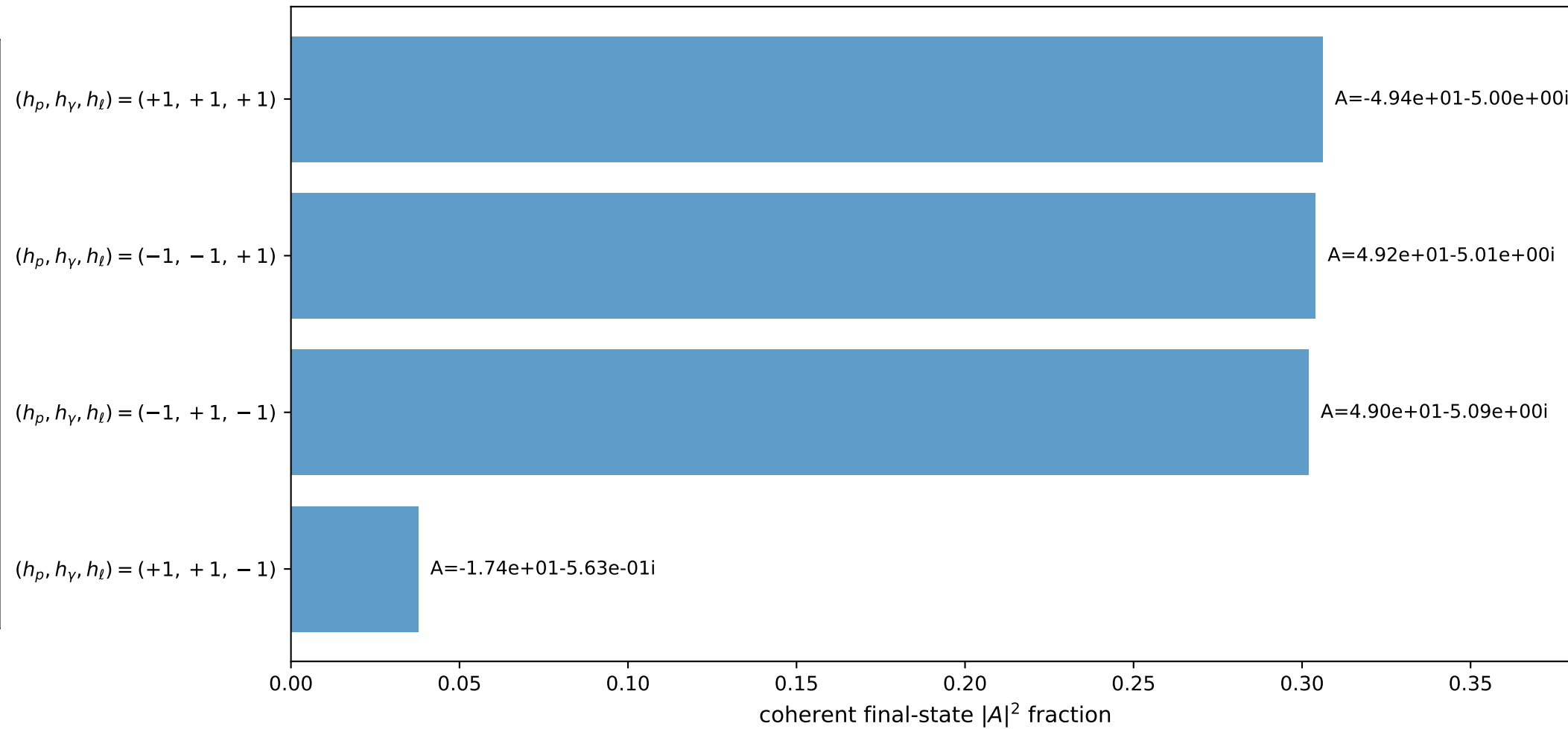
3D momenta



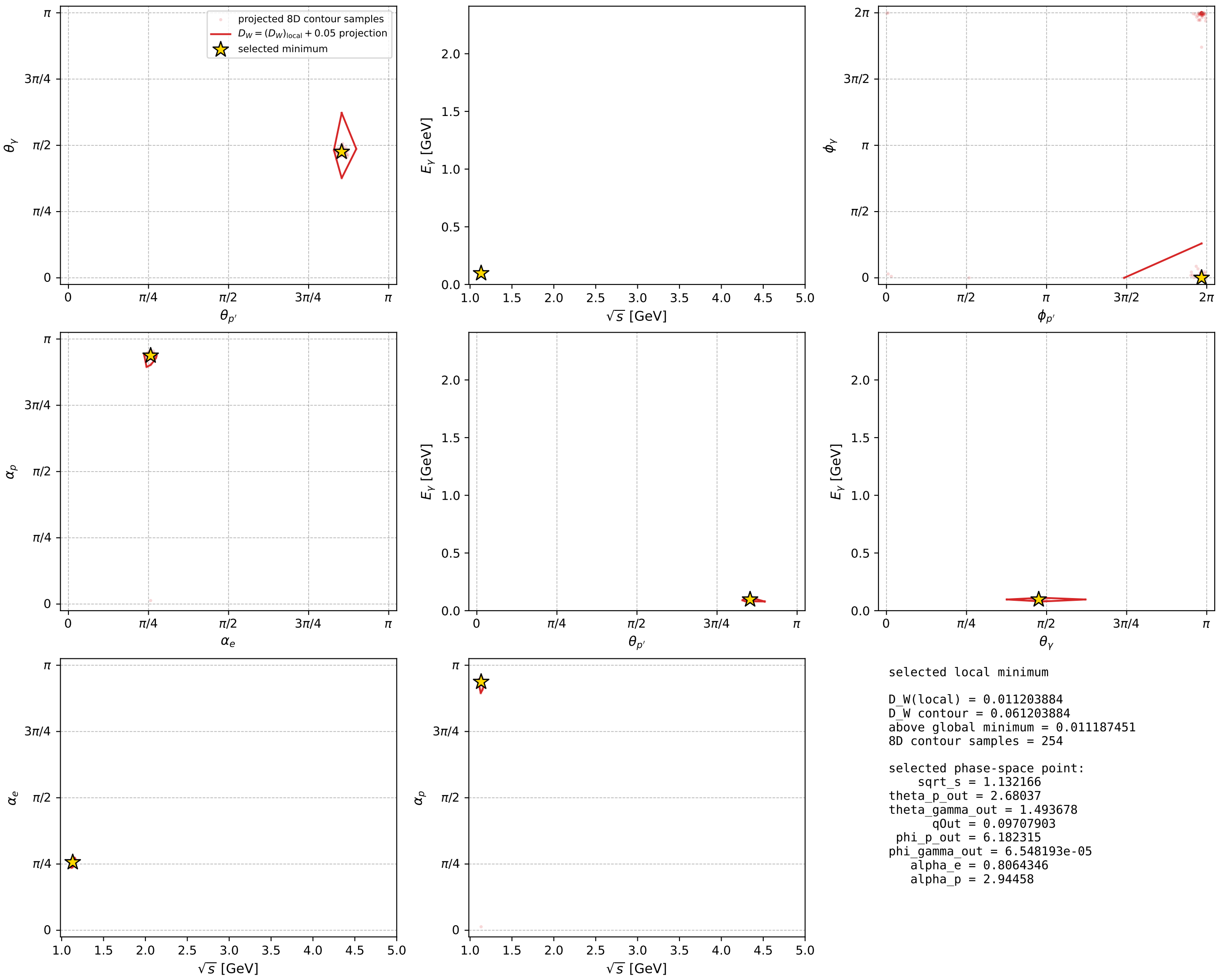
Transverse plane



Leading final-state amplitudes (N=4, retained=95.0%)



electron: polarization cluster P3, local minimum 824; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

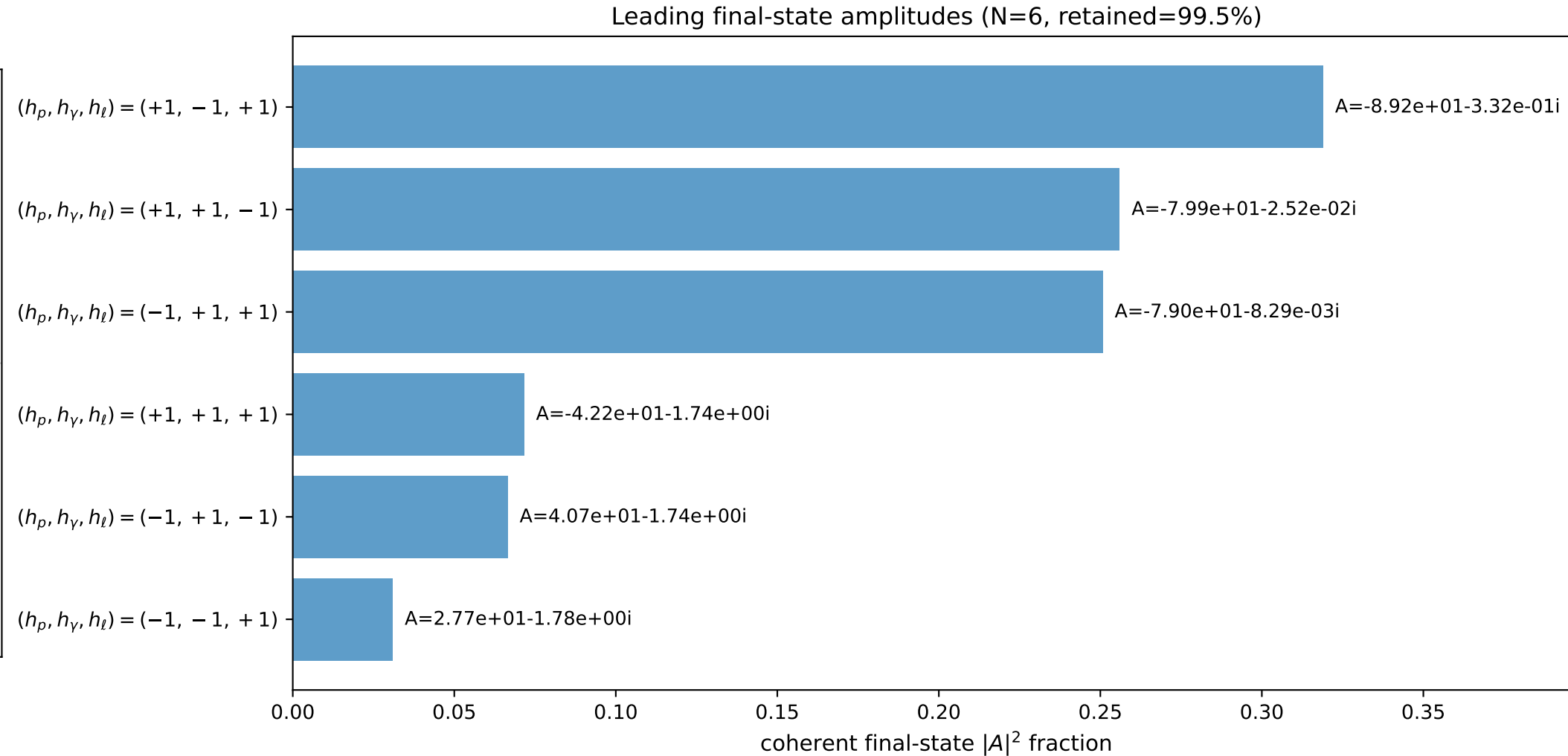
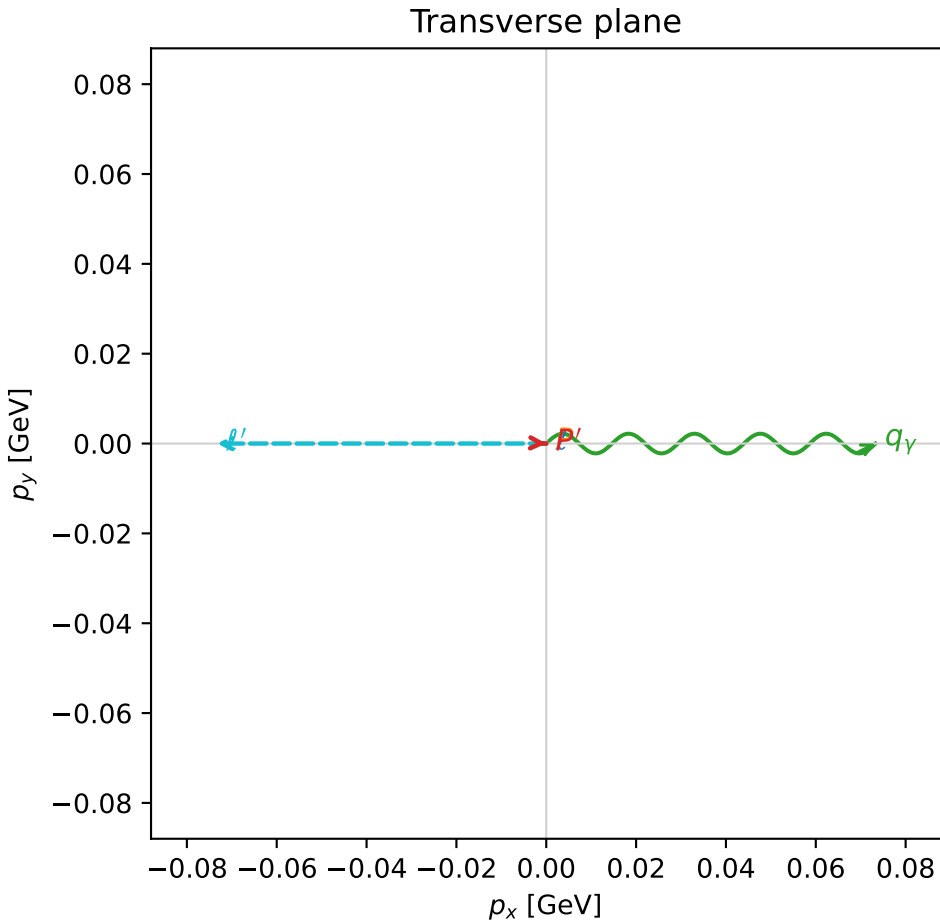
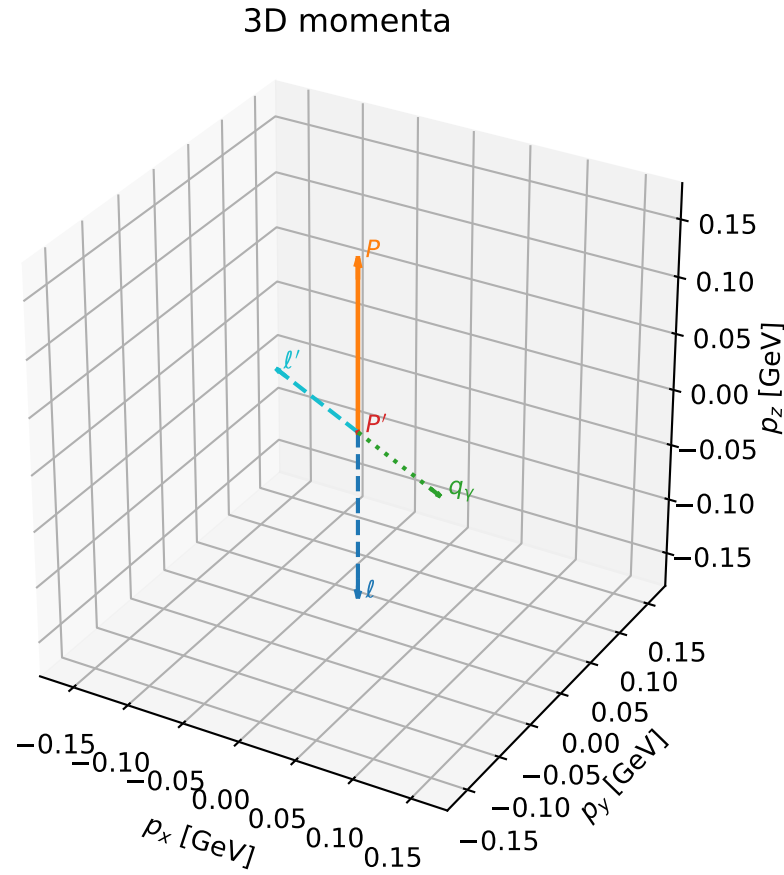


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

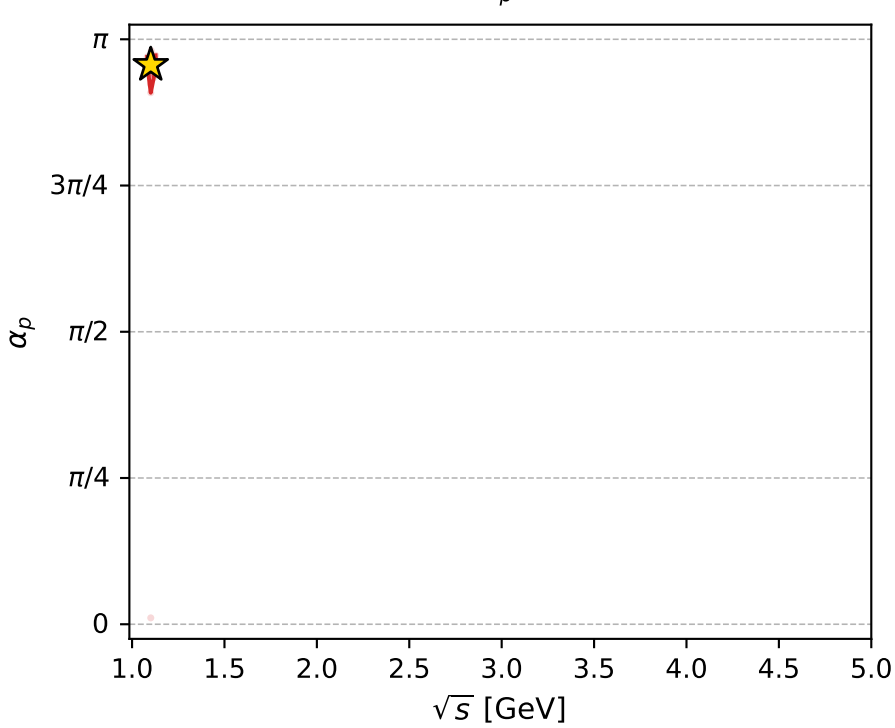
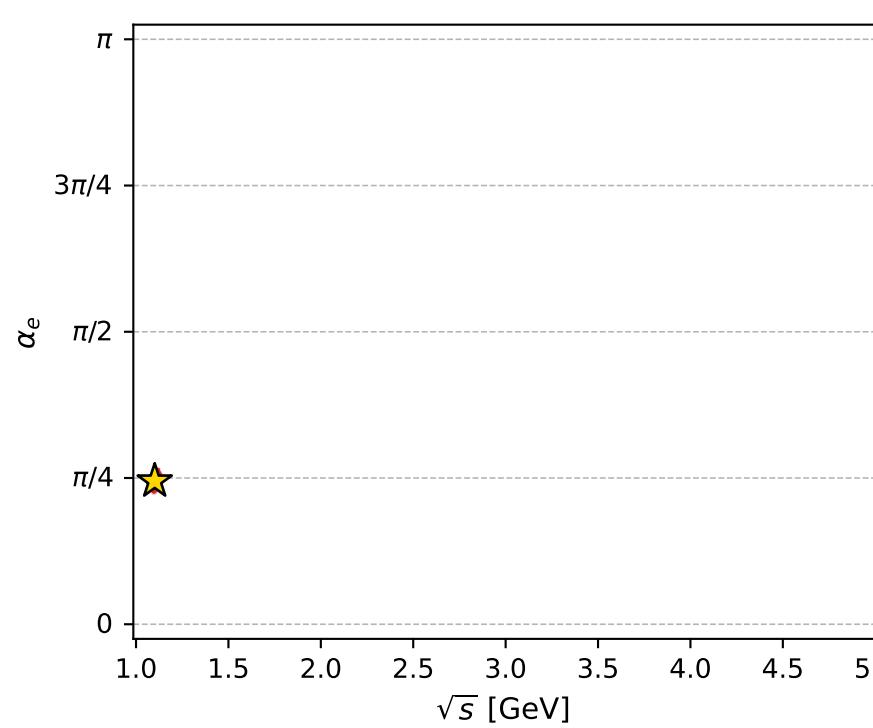
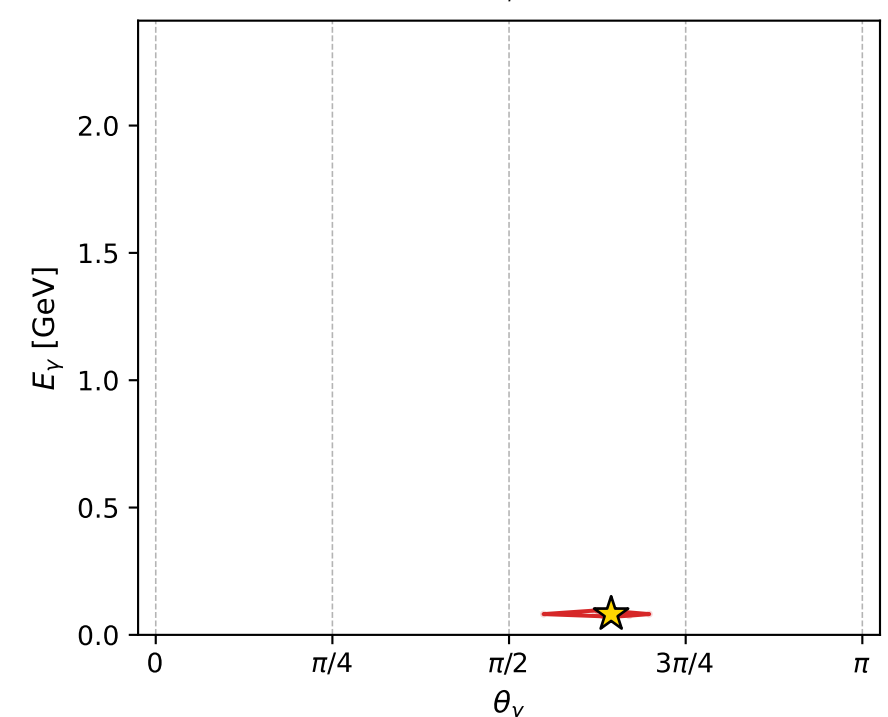
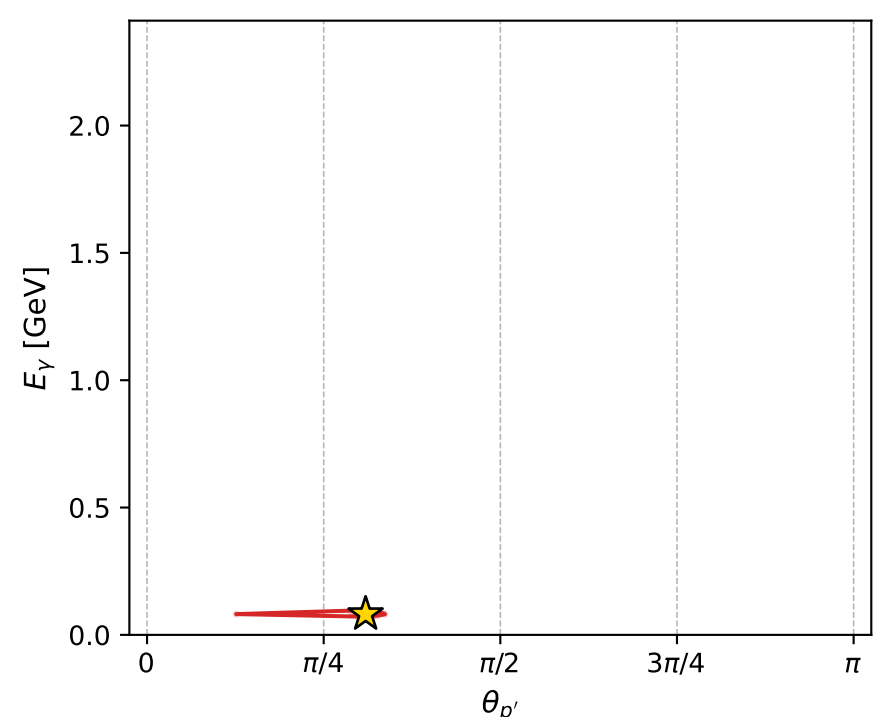
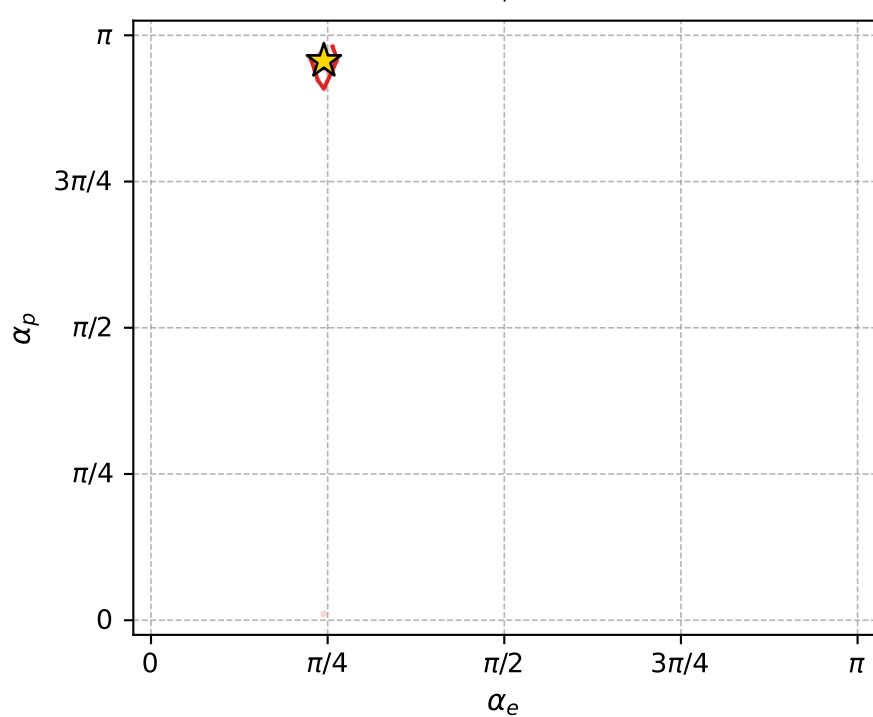
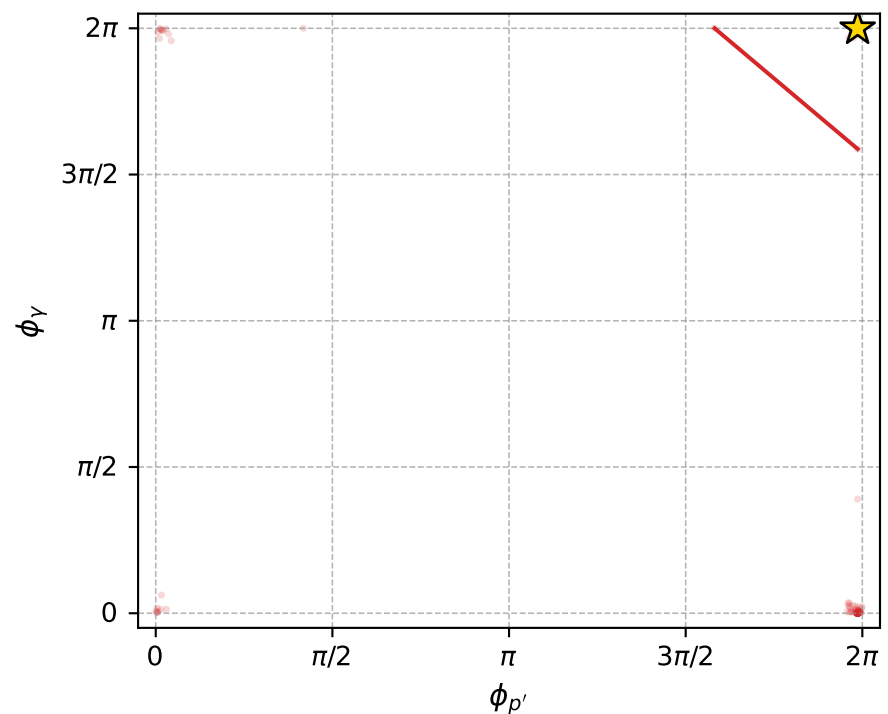
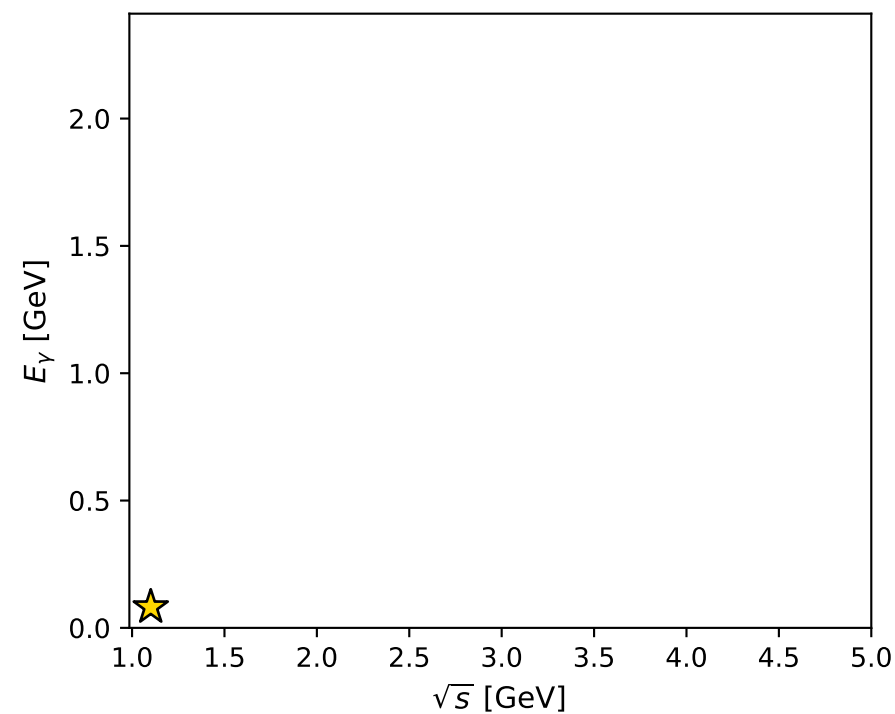
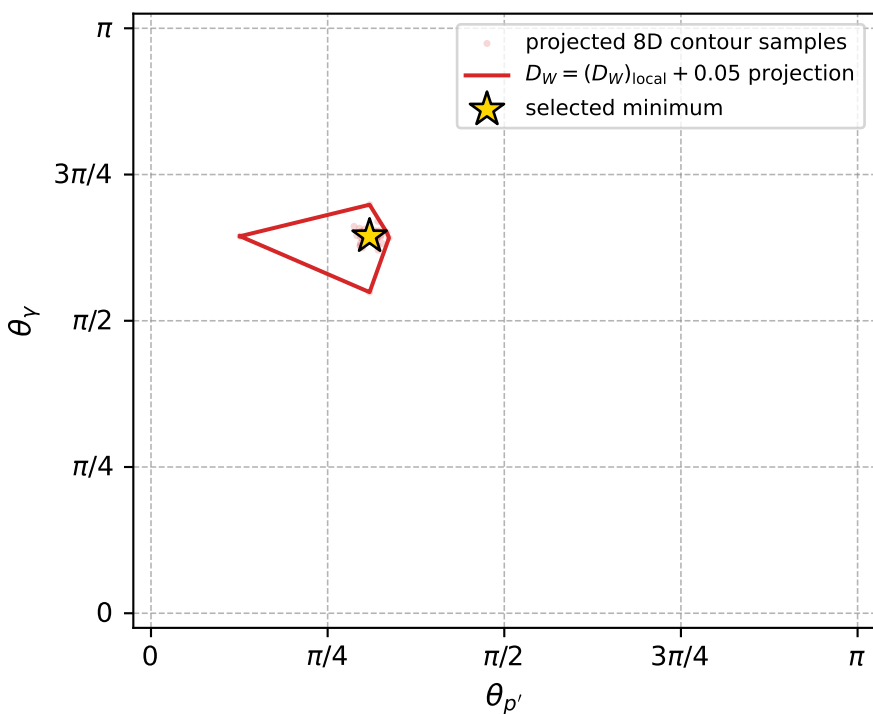
dw_mixing_angles_polarization_03_local_minimum_829 D_W=0.0112941
 region: polarization_cluster_03_local_minimum_829
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76880334$ rad
 incoming lepton: $0.718743|+\rangle + 0.695276|-\rangle$
 proton mixing angle: $\alpha_p=3.0039946$ rad
 incoming proton: $-0.990548|+\rangle + 0.137164|-\rangle$

$s=1.21254$, $\sqrt{s}=1.10115$, $|\vec{P}|=0.151067$, $|\vec{P}'|=2.11747\text{e-}06$
 $\theta_{P'}=0.97188$, $\theta_V=2.02479$, $\phi_{P'}=6.2416$, $\phi_V=6.28318$
 $E_V=0.0815761$, $\theta_{\ell'}=1.11682$, $\phi_{\ell'}=3.14159$
 $Q^2=0.0354561$, $x_B=0.188103$, $t=-0.0226748$, $W^2=1.03288$, $y=0.566558$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1511$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1511$
 P $E= 0.9501$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1511$
 ℓ' $E= 0.0816$ $p_x= -0.0733$ $p_y= 0.0000$ $p_z= 0.0358$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_V $E= 0.0816$ $p_x= 0.0733$ $p_y= -0.0000$ $p_z= -0.0358$



electron: polarization cluster P3, local minimum 829; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011294071$
 $D_W \text{ contour} = 0.061294071$
 above global minimum = 0.011277639
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.101155$
 $\text{theta_p_out} = 0.9718796$
 $\text{theta_gamma_out} = 2.02479$
 $\text{qOut} = 0.08157614$
 $\text{phi_p_out} = 6.241596$
 $\text{phi_gamma_out} = 6.283184$
 $\text{alpha_e} = 0.7688033$
 $\text{alpha_p} = 3.003995$

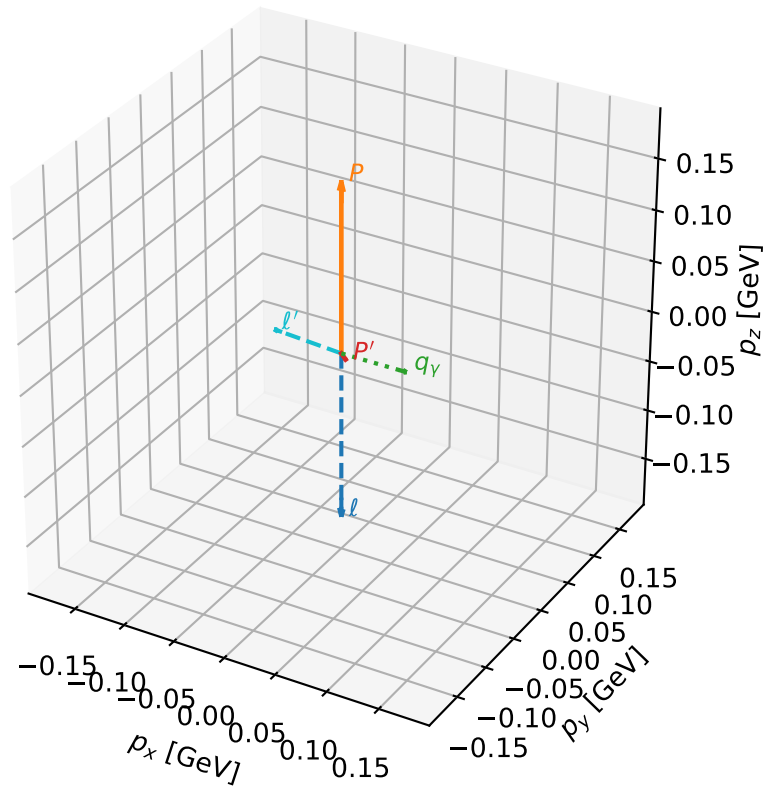
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_839 D_W=0.0116253
 region: polarization_cluster_03_local_minimum_839
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81111552$ rad
 incoming lepton: $0.68869|+\rangle +0.725056|-\rangle$
 proton mixing angle: $\alpha_p=2.9294071$ rad
 incoming proton: $-0.977573|+\rangle +0.210597|-\rangle$

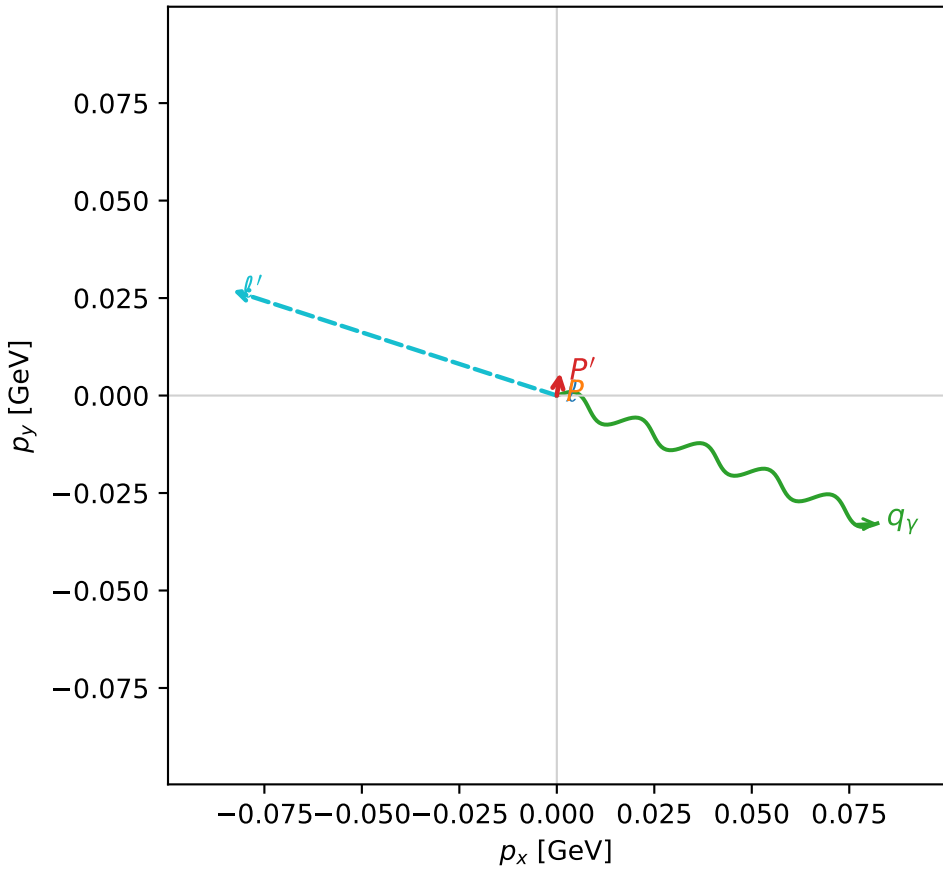
$s=1.24899$, $\sqrt{s}=1.11758$, $|\vec{P}|=0.165155$, $|\vec{P}'|=0.0105191$
 $\theta_{P'}=2.54466$, $\theta_Y=1.32836$, $\phi_{P'}=1.42117$, $\phi_Y=5.9038$
 $E_Y=0.0911783$, $\theta_{l'}=1.72064$, $\phi_{l'}=2.82816$
 $Q^2=0.024825$, $x_B=0.126333$, $t=-0.0300539$, $W^2=1.05152$, $y=0.532319$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1652$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1652$
 P $E= 0.9524$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1652$
 l' $E= 0.0883$ $p_x= -0.0831$ $p_y= 0.0269$ $p_z= -0.0132$
 P' $E= 0.9381$ $p_x= 0.0009$ $p_y= 0.0058$ $p_z= -0.0087$
 q_Y $E= 0.0912$ $p_x= 0.0822$ $p_y= -0.0328$ $p_z= 0.0219$

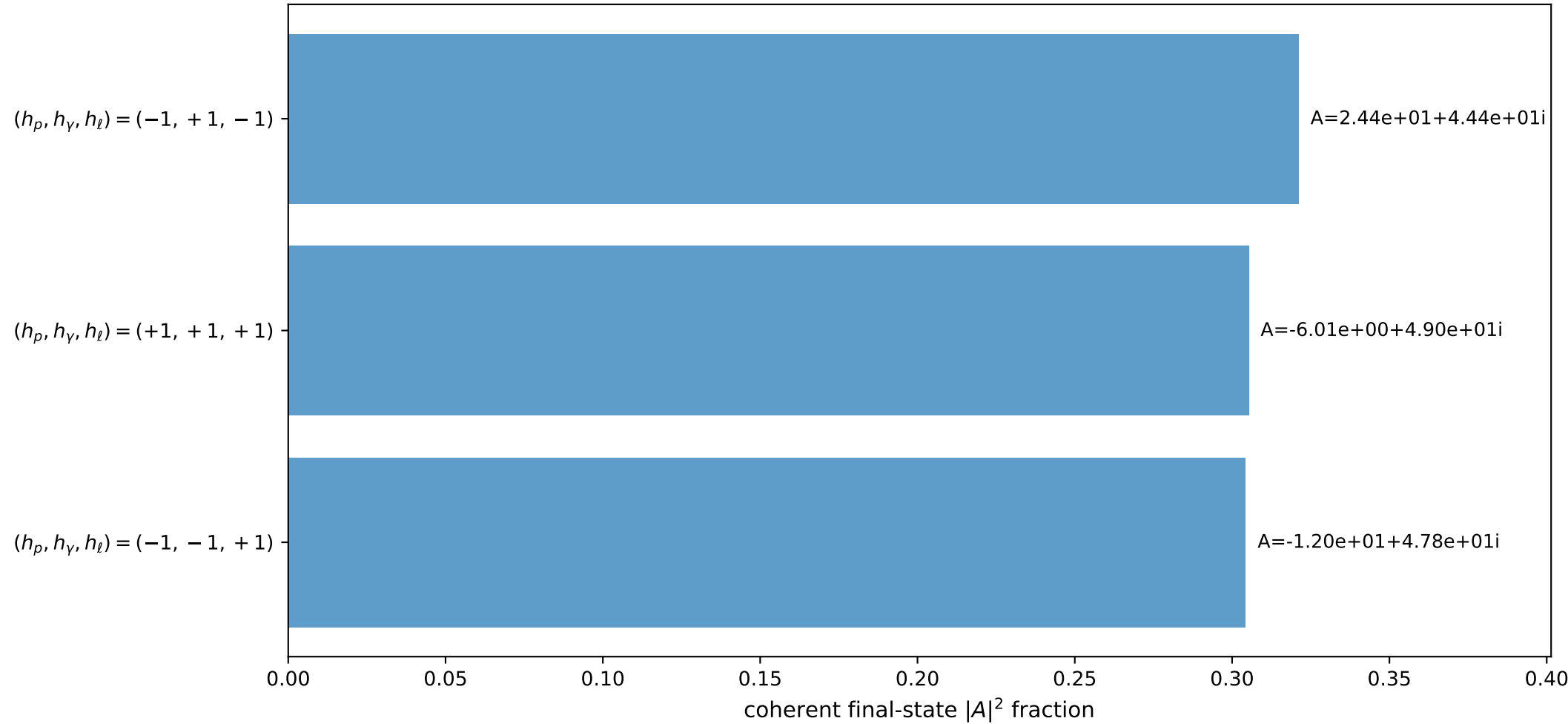
3D momenta



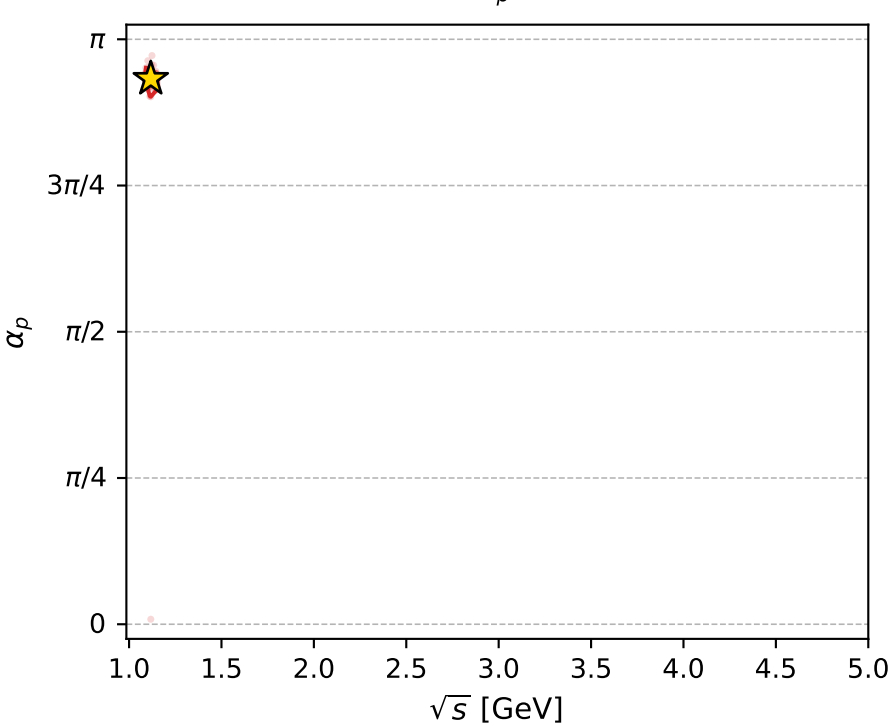
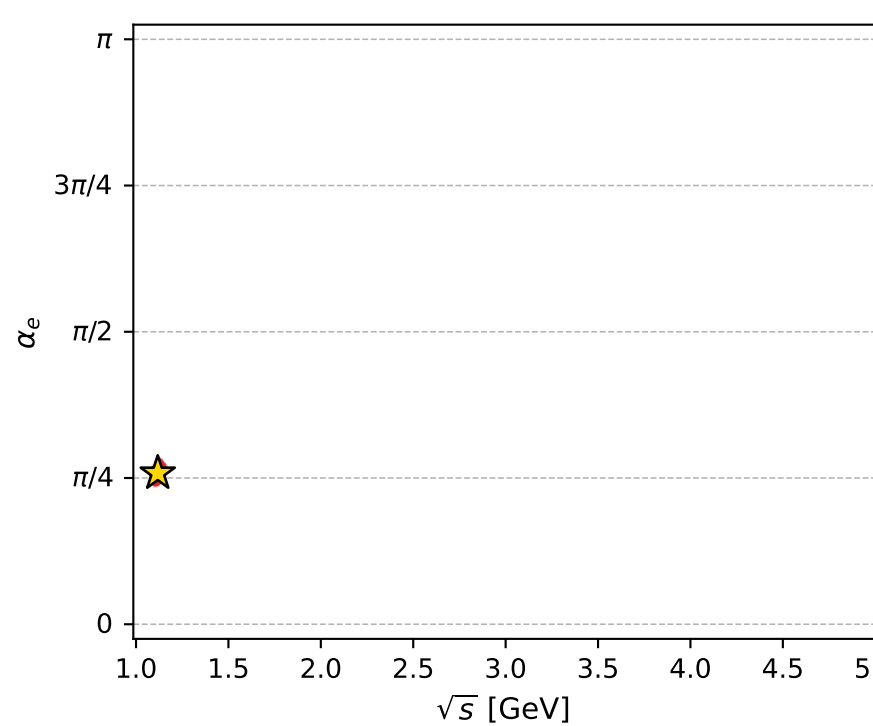
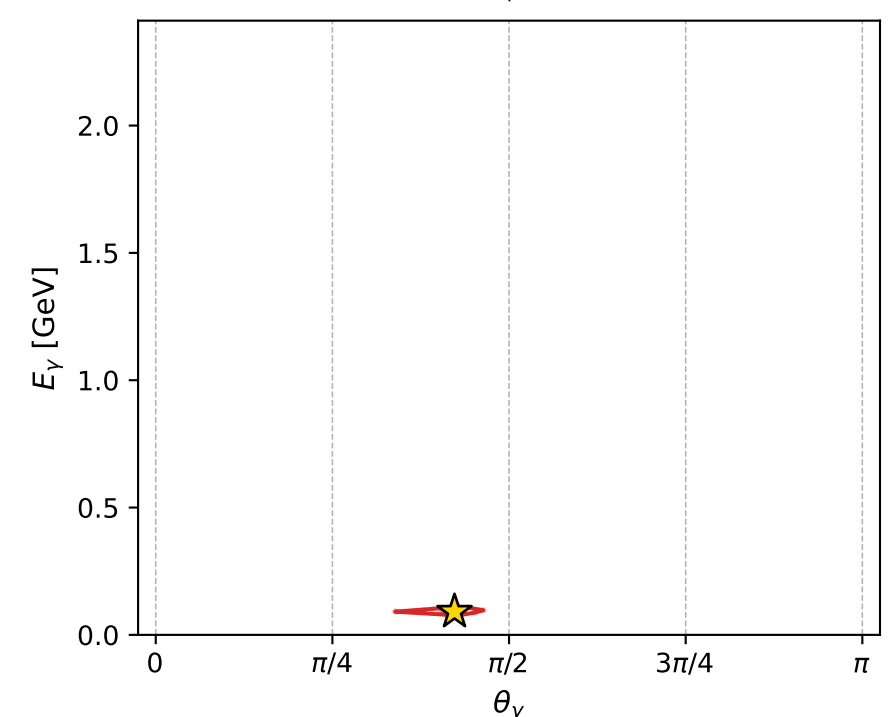
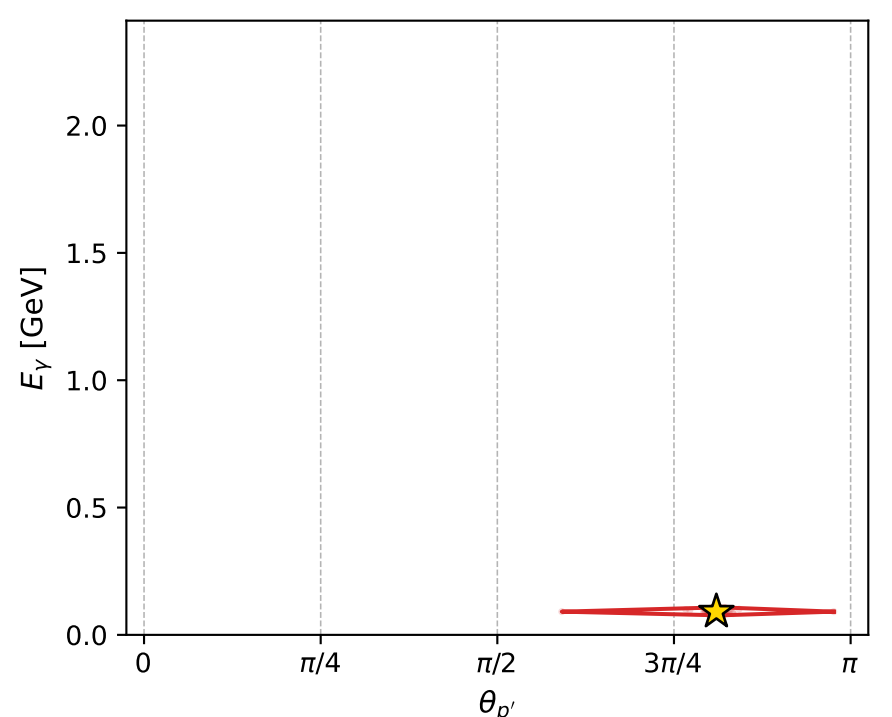
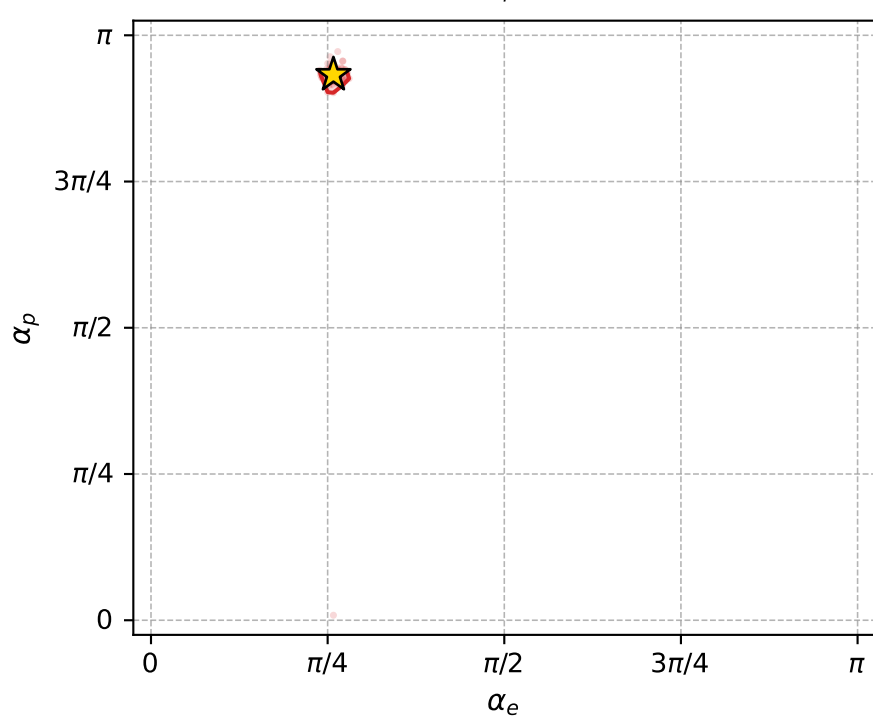
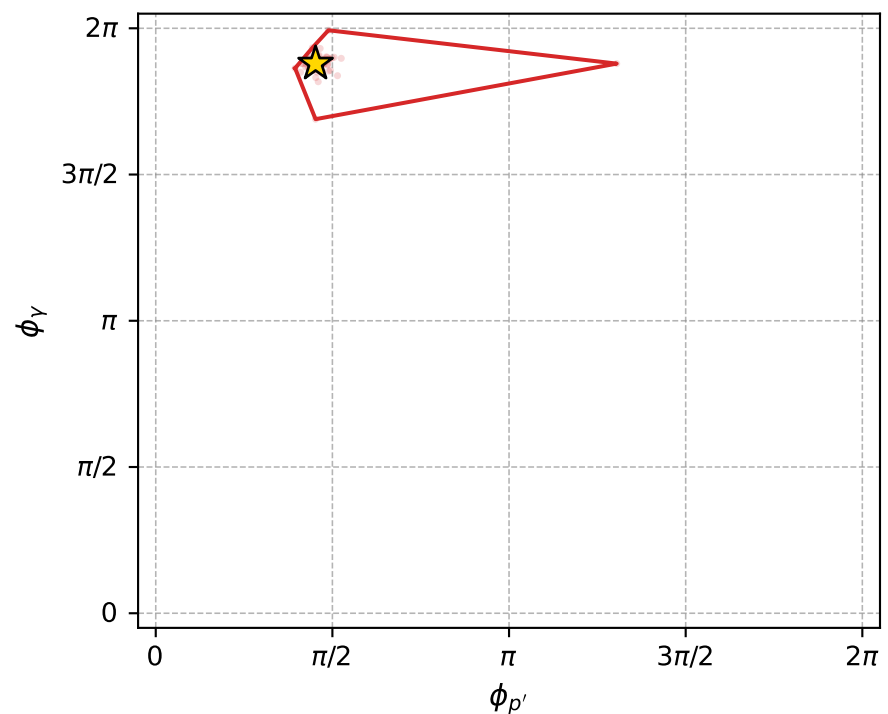
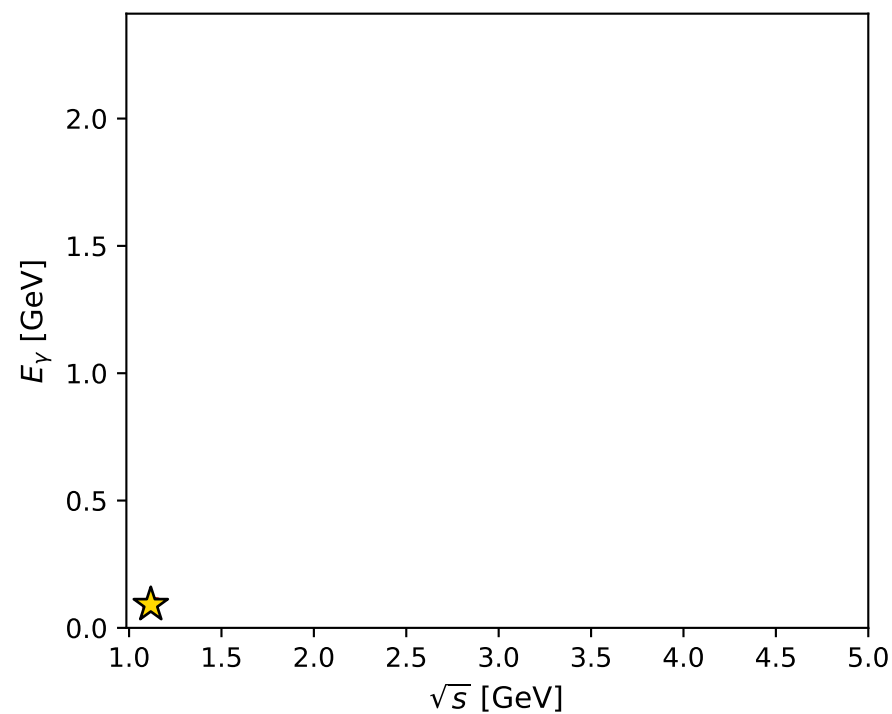
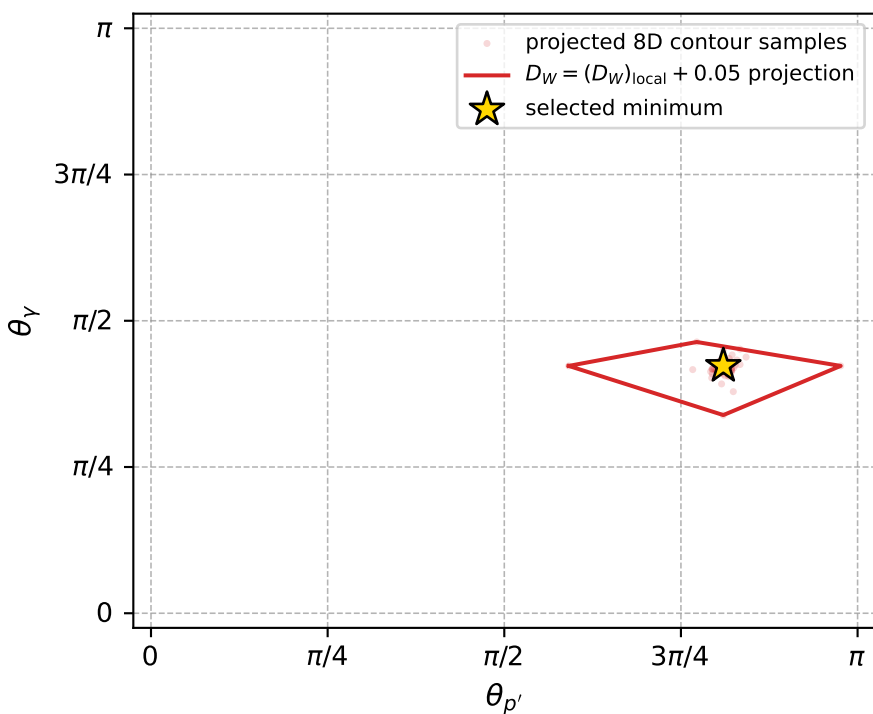
Transverse plane



Leading final-state amplitudes (N=3, retained=93.1%)



electron: polarization cluster P3, local minimum 839; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.011625306$
 $D_W \text{ contour} = 0.061625306$
 above global minimum = 0.011608874
 8D contour samples = 256

selected phase-space point:
 $\text{sqrt_s} = 1.117584$
 $\text{theta_p_out} = 2.544661$
 $\text{theta_gamma_out} = 1.328364$
 $\text{q0out} = 0.09117831$
 $\text{phi_p_out} = 1.421169$
 $\text{phi_gamma_out} = 5.903796$
 $\text{alpha_e} = 0.8111155$
 $\text{alpha_p} = 2.929407$

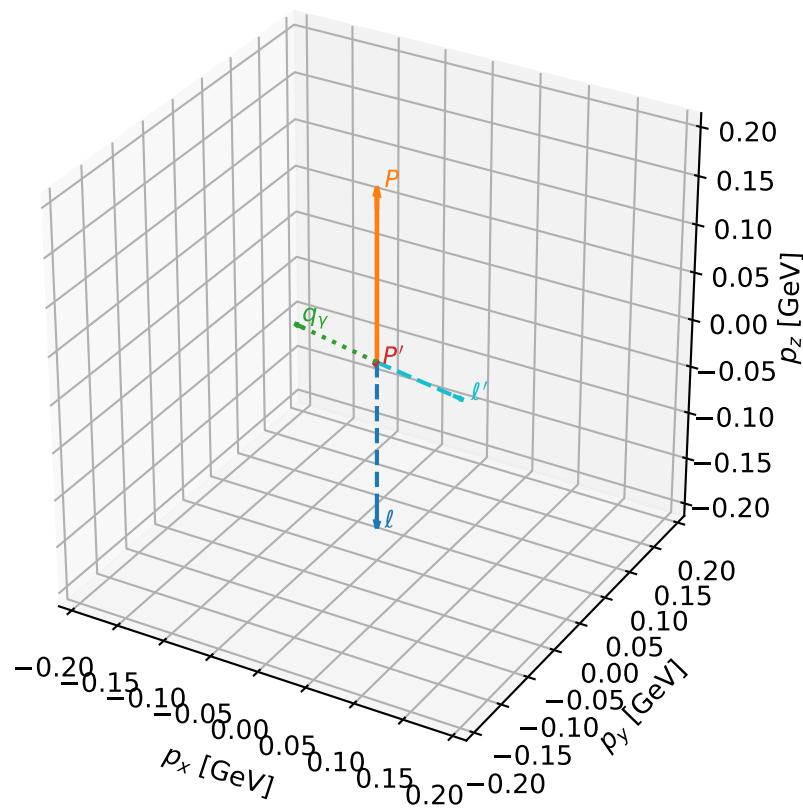
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_844 D_W=0.0118013
region: polarization_cluster_03_local_minimum_844
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.80012594$ rad
incoming lepton: $0.696616|+\rangle + 0.717444|-\rangle$
proton mixing angle: $\alpha_p=0.21446066$ rad
incoming proton: $0.977091|+\rangle + 0.21282|-\rangle$

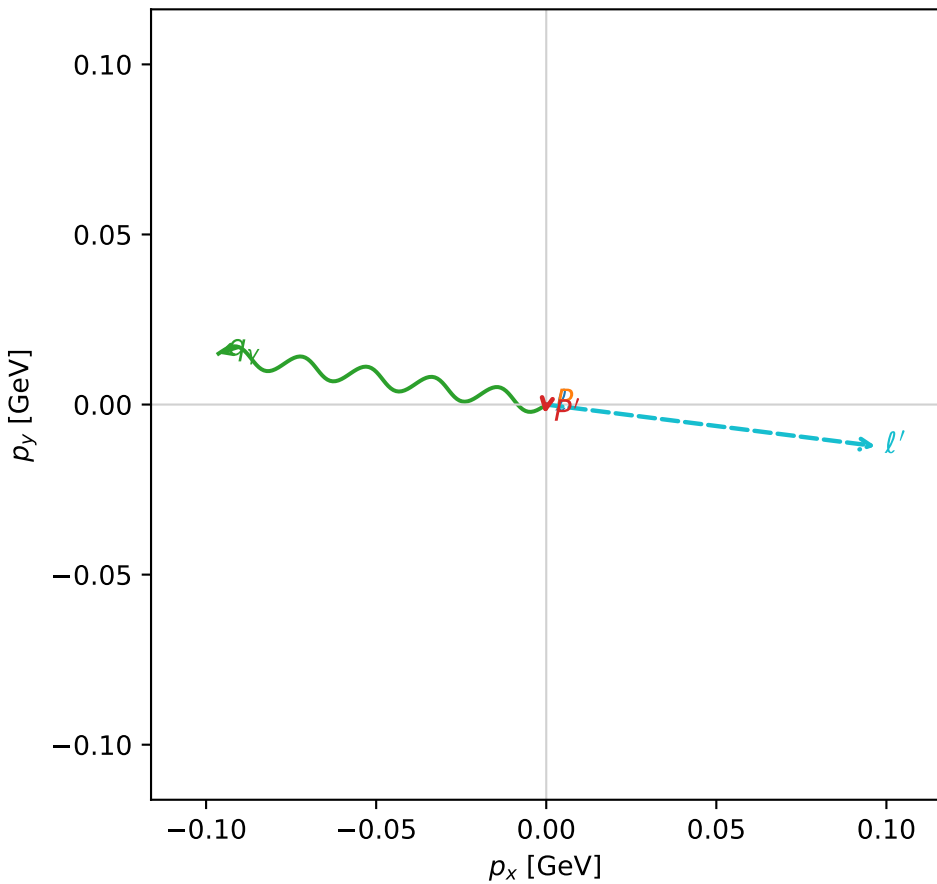
$s=1.28458$, $\sqrt{s}=1.13339$, $|\vec{P}|=0.17855$, $|\vec{P}'|=0.00287312$
 $\theta_{P'}=1.79311$, $\theta_V=1.51147$, $\phi_{P'}=4.53962$, $\phi_V=2.98743$
 $E_V=0.0976655$, $\theta_{\ell'}=1.62359$, $\phi_{\ell'}=6.15774$
 $Q^2=0.0330551$, $x_B=0.152839$, $t=-0.0318312$, $W^2=1.06306$, $y=0.534357$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1786$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1786$
 P $E= 0.9548$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1786$
 ℓ' $E= 0.0977$ $p_x= 0.0968$ $p_y= -0.0122$ $p_z= -0.0052$
 P' $E= 0.9380$ $p_x= -0.0005$ $p_y= -0.0028$ $p_z= -0.0006$
 q_V $E= 0.0977$ $p_x= -0.0963$ $p_y= 0.0150$ $p_z= 0.0058$

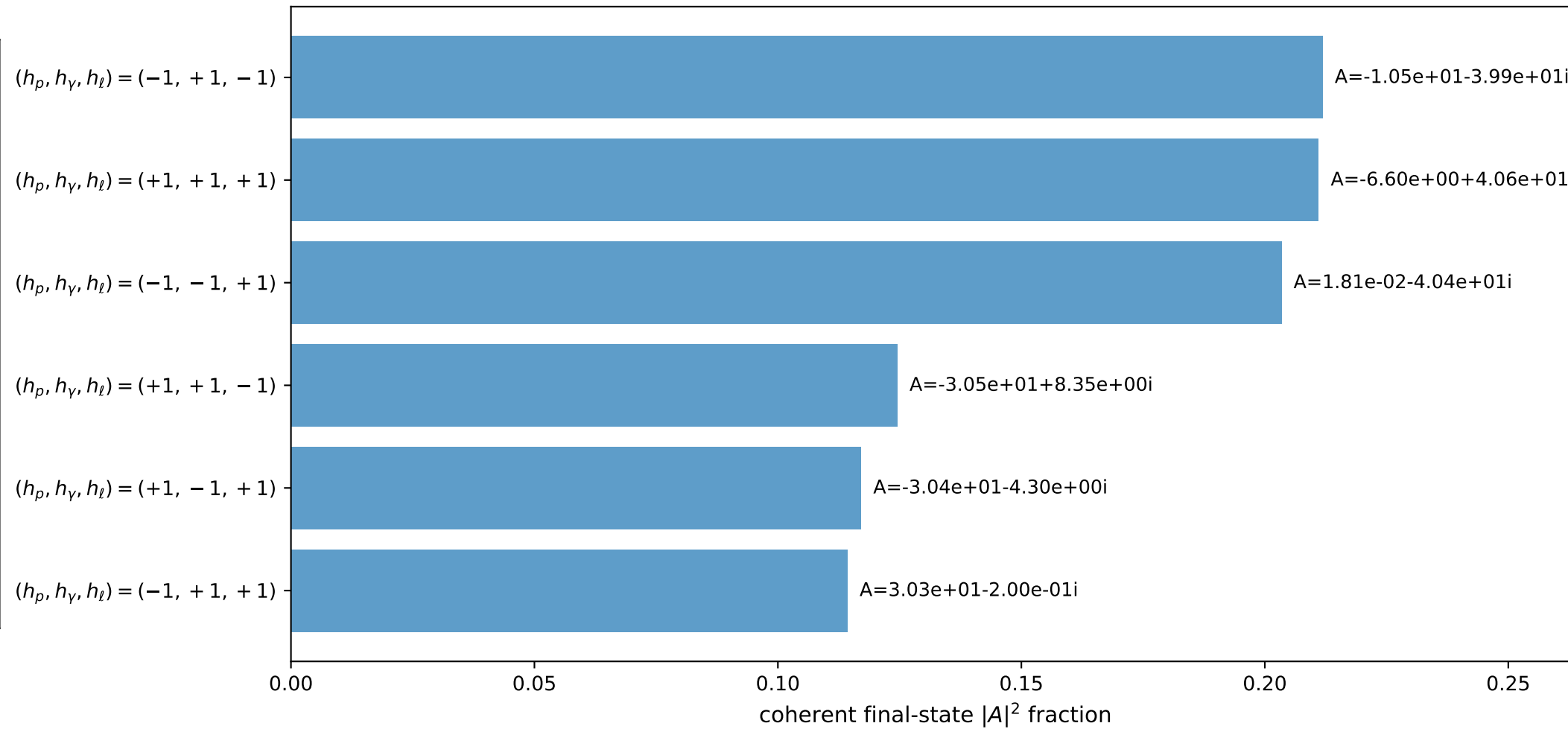
3D momenta



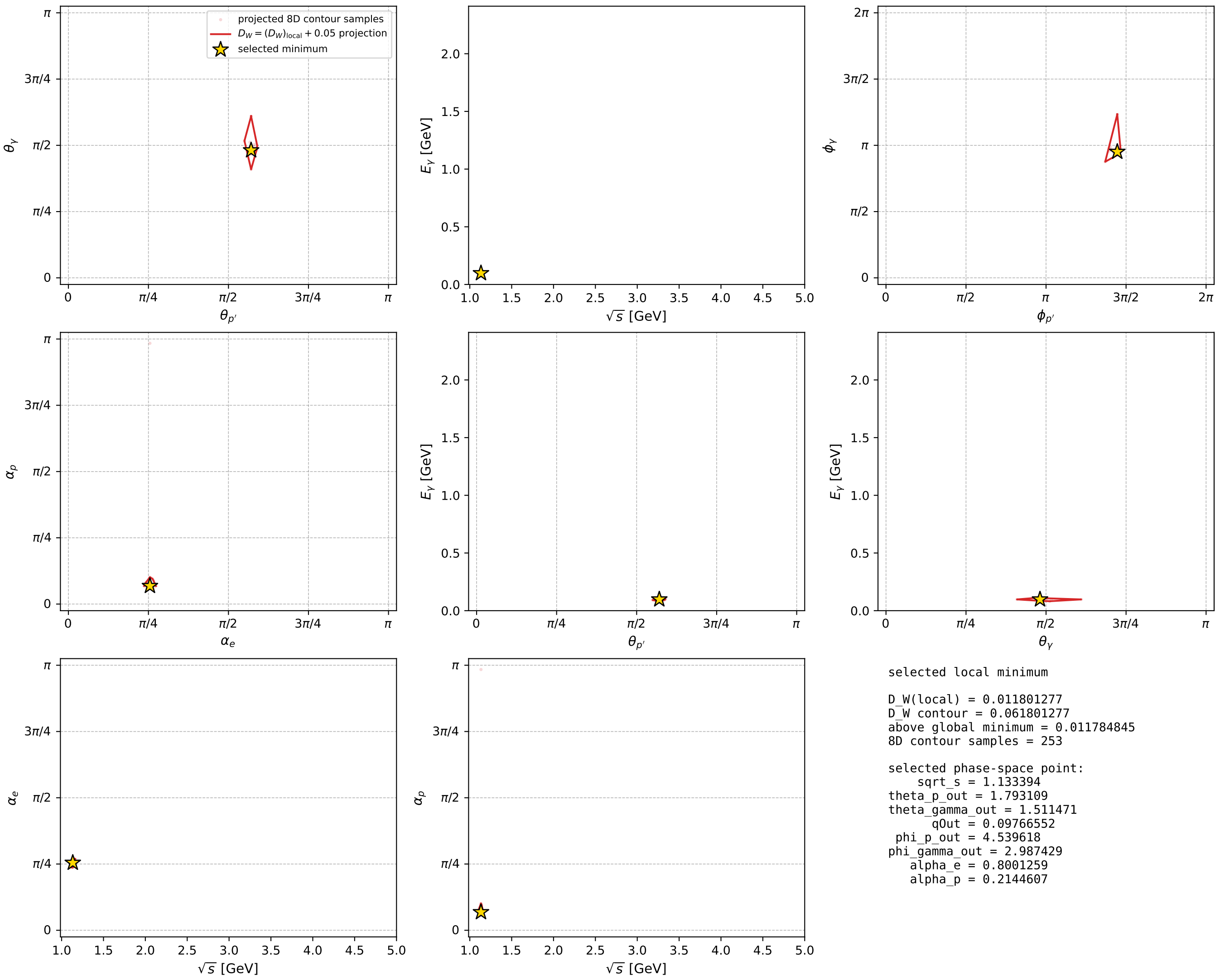
Transverse plane



Leading final-state amplitudes (N=6, retained=98.2%)



electron: polarization cluster P3, local minimum 844; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



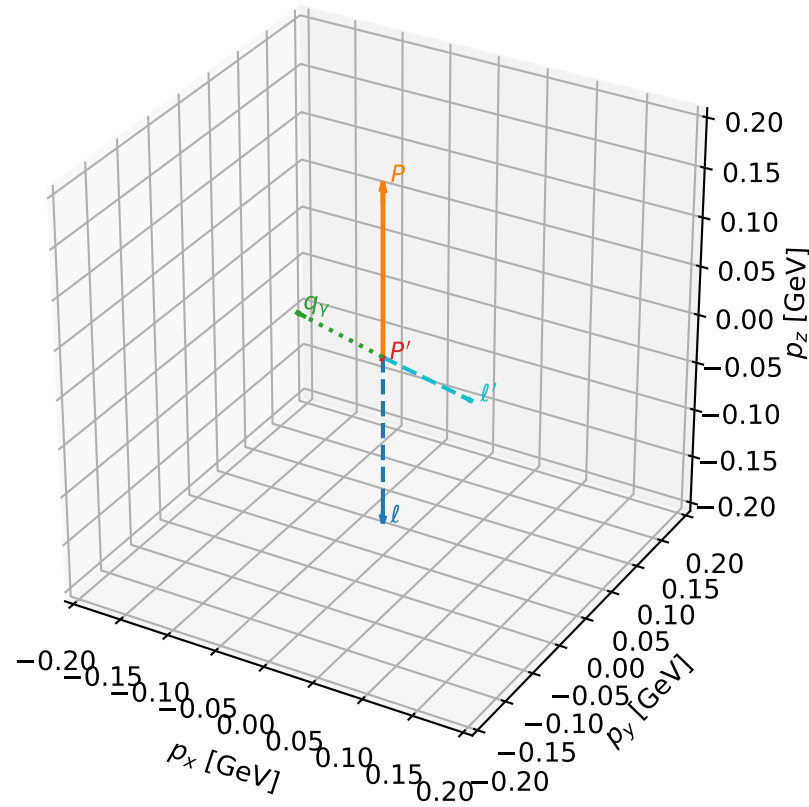
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_857 D_W=0.0121139
 region: polarization_cluster_03_local_minimum_857
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80882753$ rad
 incoming lepton: $0.690347|+\rangle +0.723478|-\rangle$
 proton mixing angle: $\alpha_p=0.20307519$ rad
 incoming proton: $0.979451|+\rangle +0.201682|-\rangle$

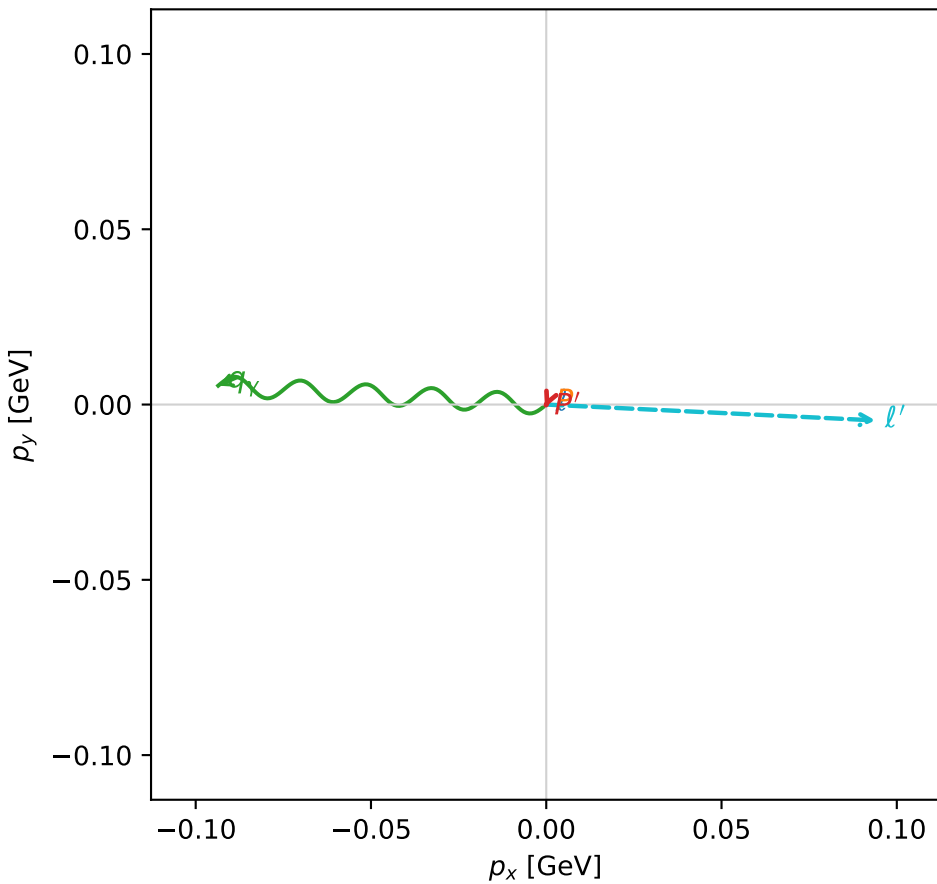
$s=1.27391$, $\sqrt{s}=1.12868$, $|\vec{P}|=0.174569$, $|\vec{P}'|=0.00200081$
 $\theta_{P'}=2.65343$, $\theta_V=1.38724$, $\phi_{P'}=4.29545$, $\phi_V=3.0841$
 $E_V=0.0953244$, $\theta_{\ell'}=1.73549$, $\phi_{\ell'}=6.23503$
 $Q^2=0.0278316$, $x_B=0.134673$, $t=-0.0308358$, $W^2=1.05867$, $y=0.524434$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1746$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1746$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1746$
 ℓ' $E= 0.0953$ $p_x= 0.0939$ $p_y= -0.0045$ $p_z= -0.0156$
 P' $E= 0.9380$ $p_x= -0.0004$ $p_y= -0.0009$ $p_z= -0.0018$
 q_V $E= 0.0953$ $p_x= -0.0936$ $p_y= 0.0054$ $p_z= 0.0174$

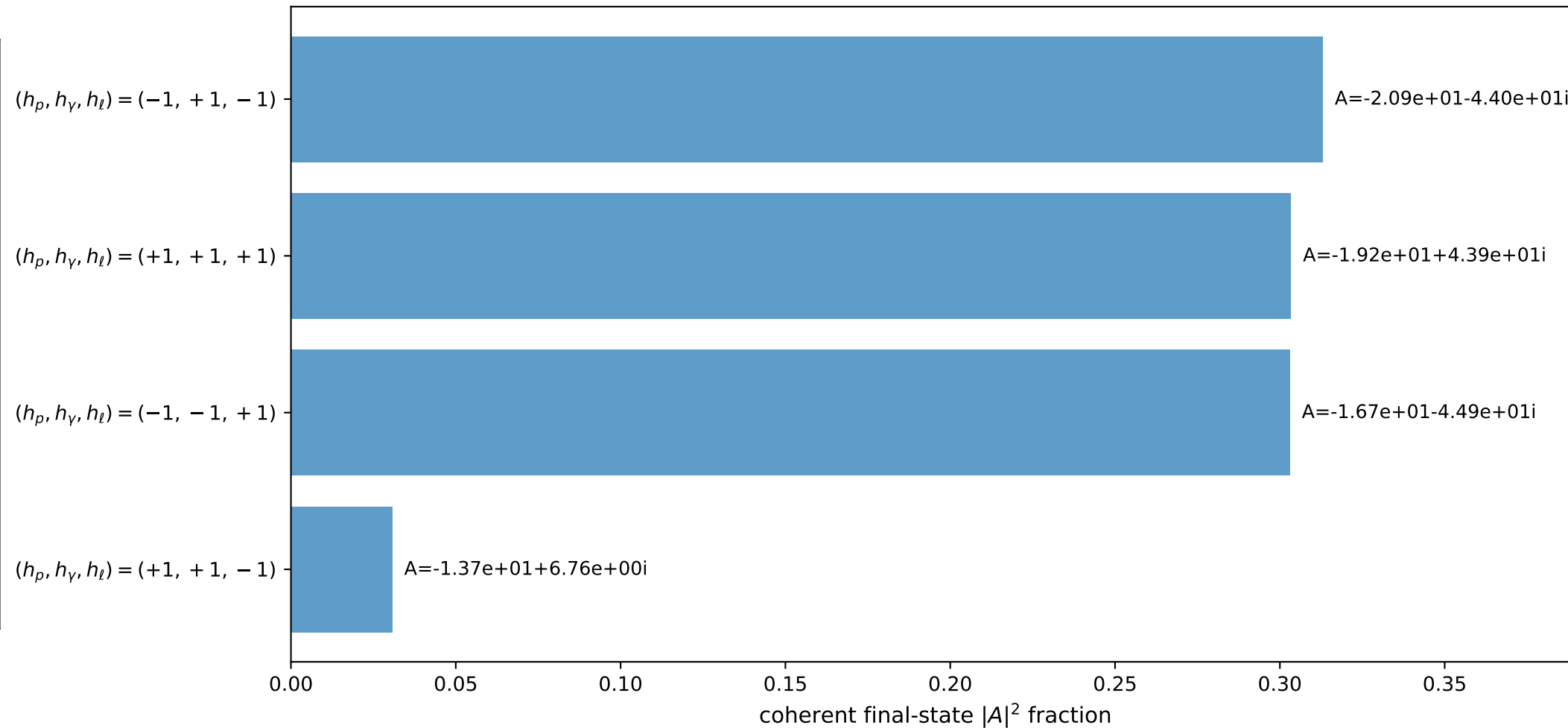
3D momenta



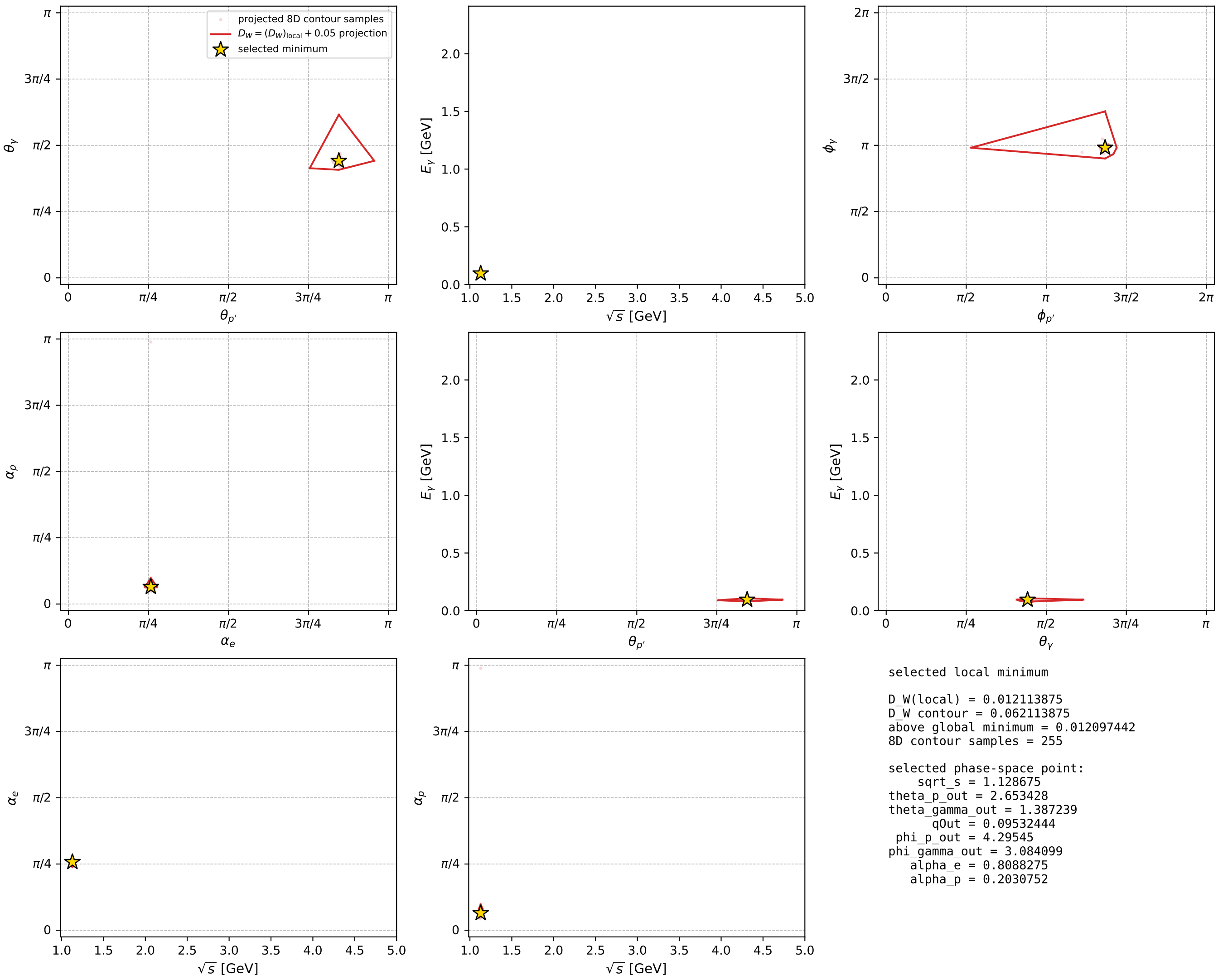
Transverse plane



Leading final-state amplitudes (N=4, retained=95.0%)



electron: polarization cluster P3, local minimum 857; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

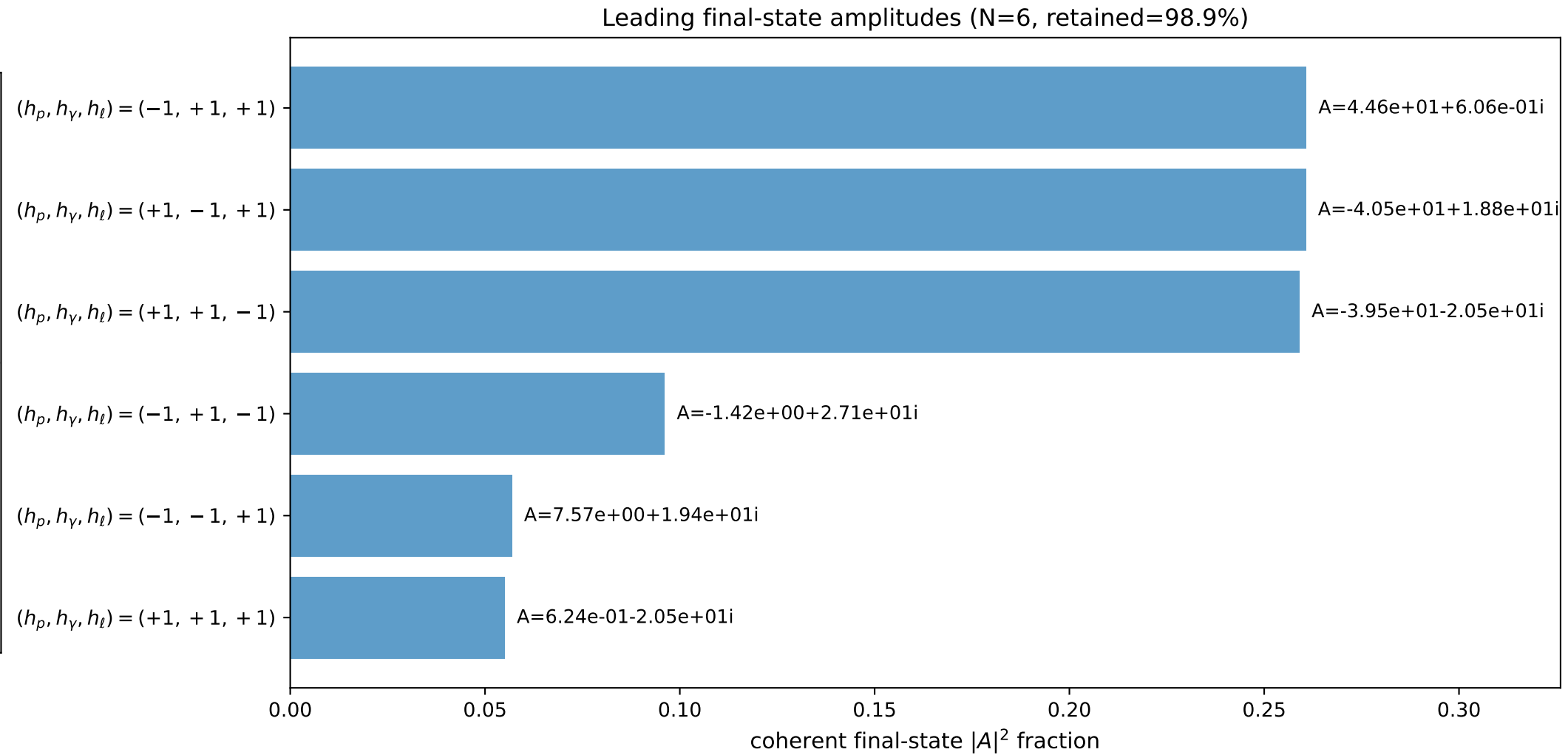
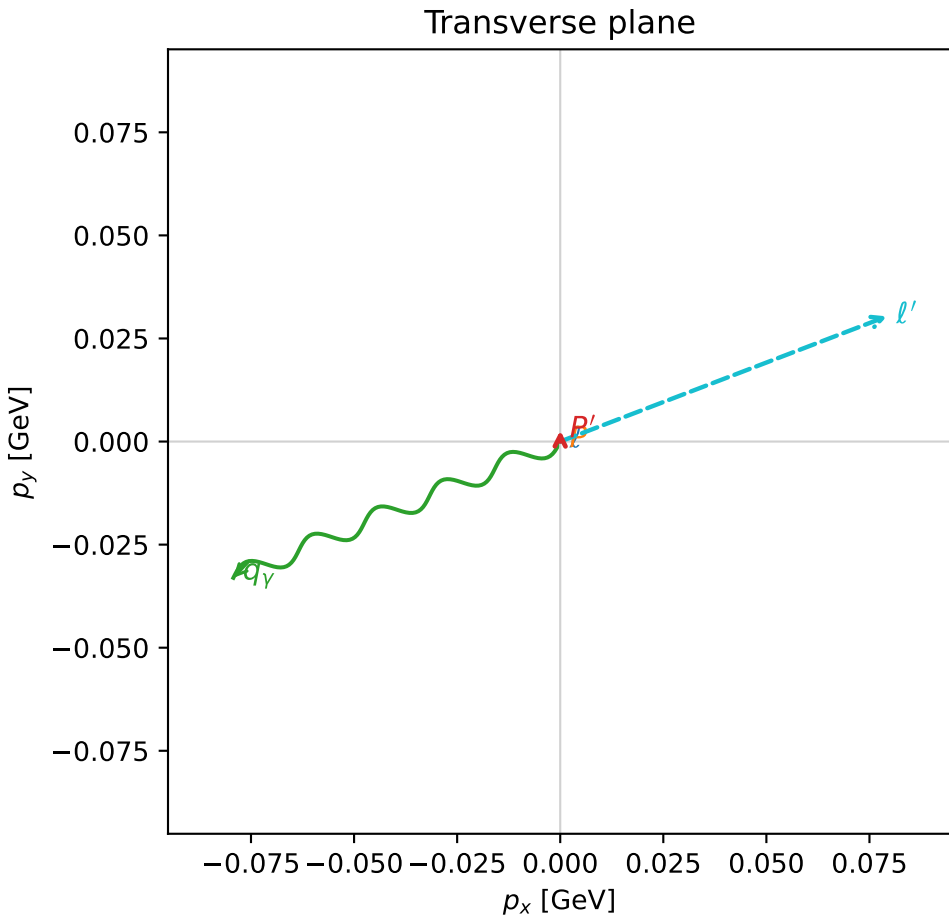
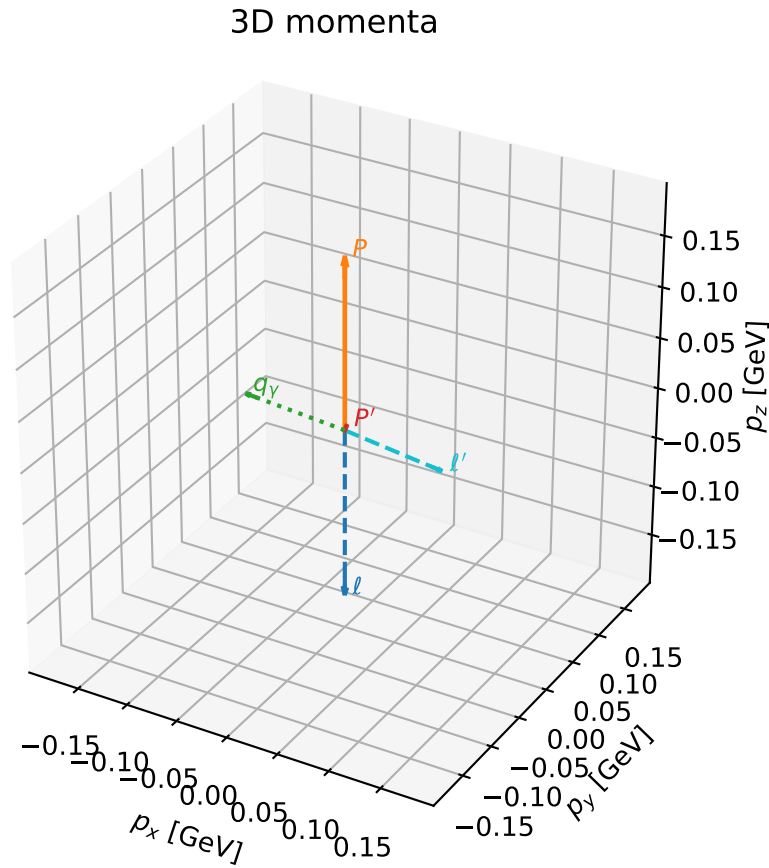


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

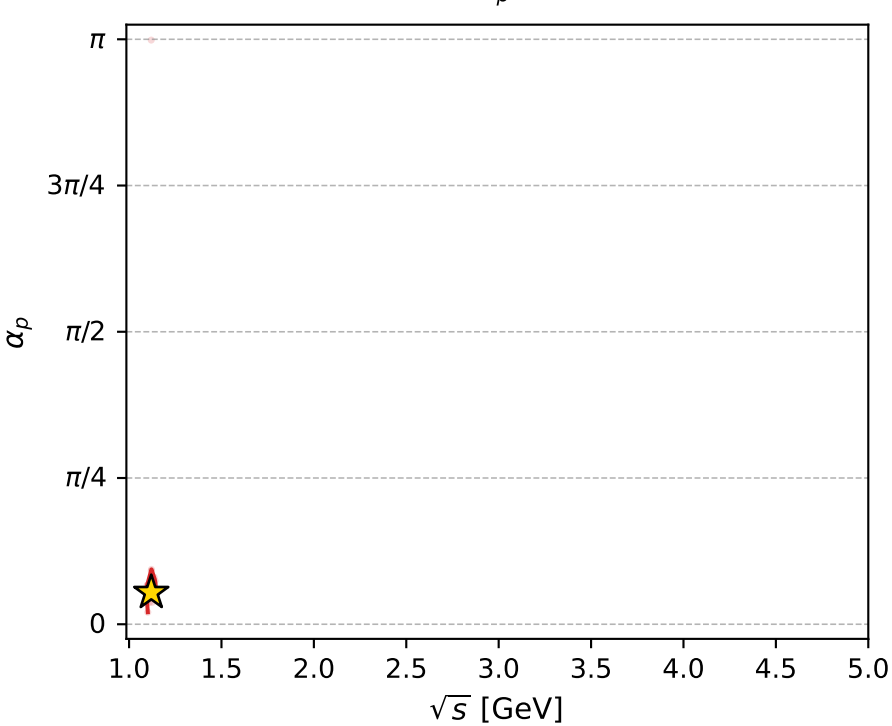
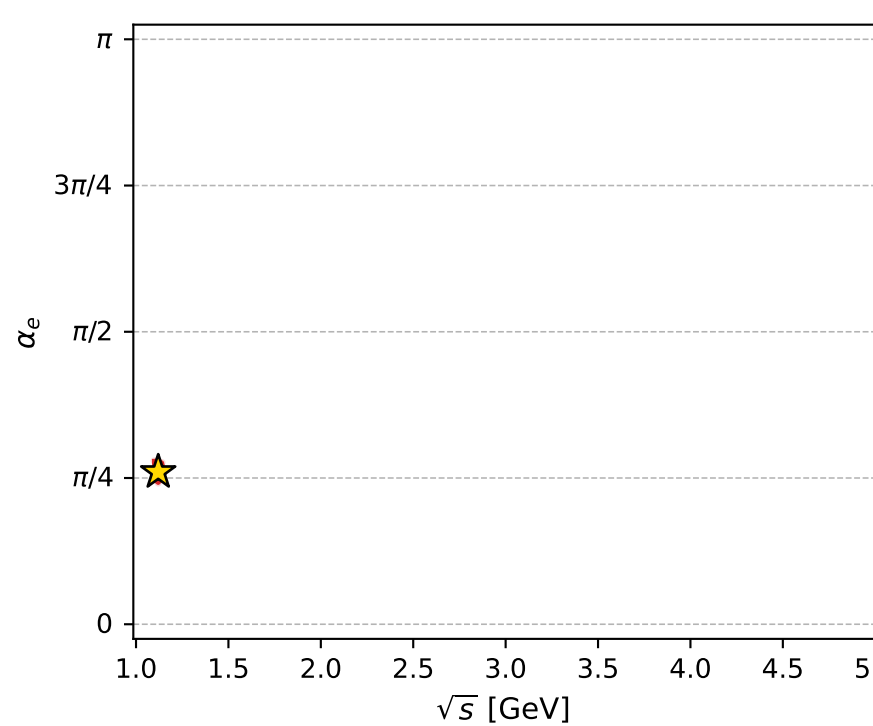
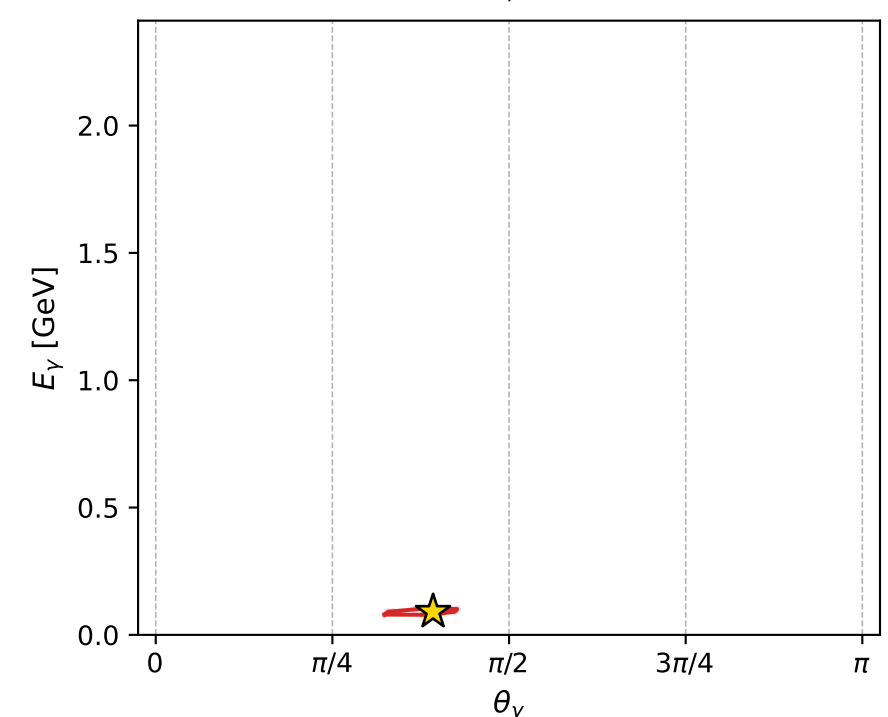
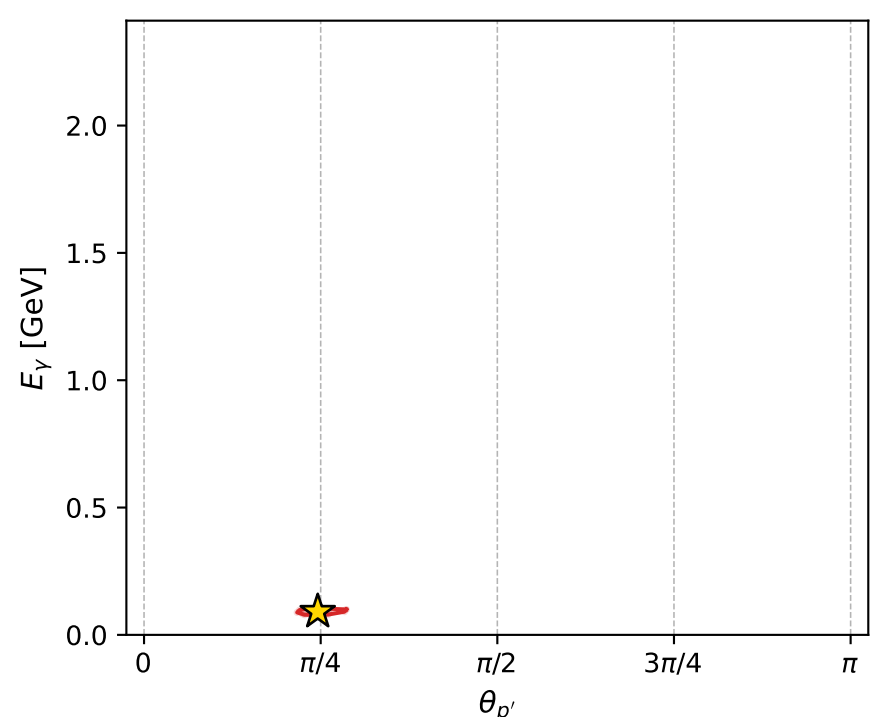
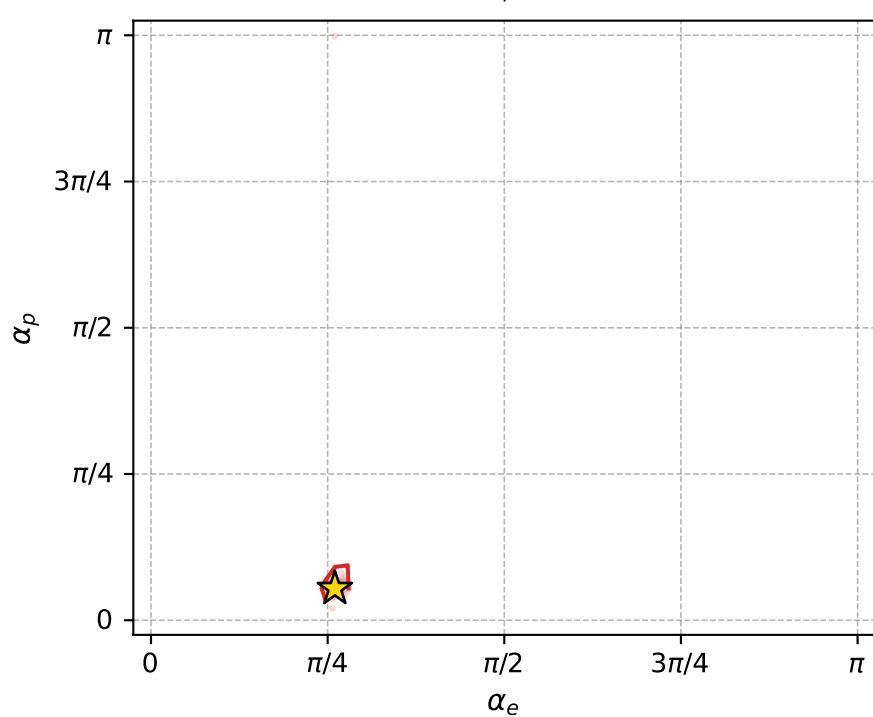
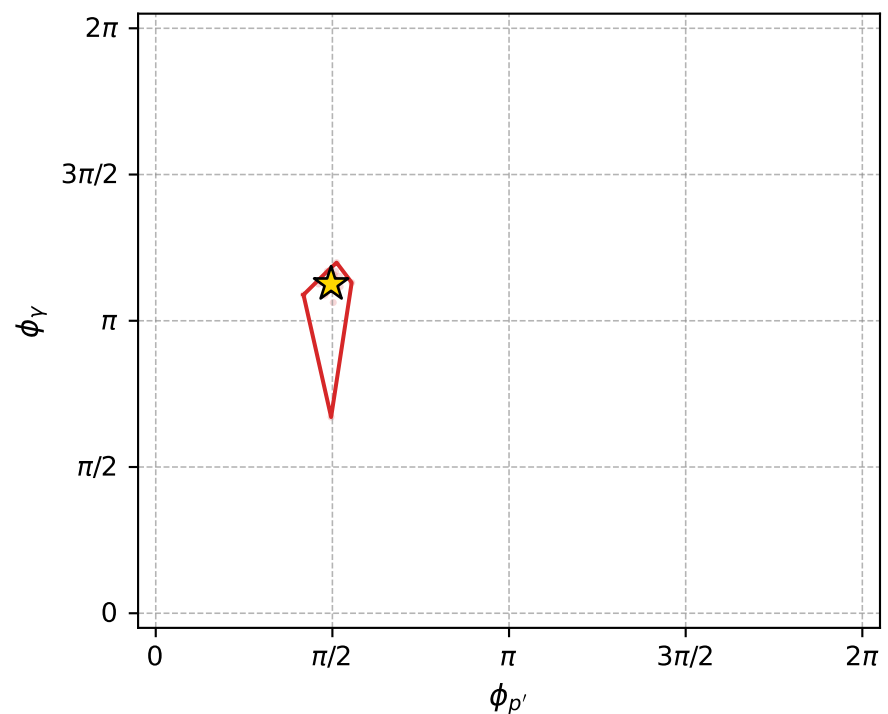
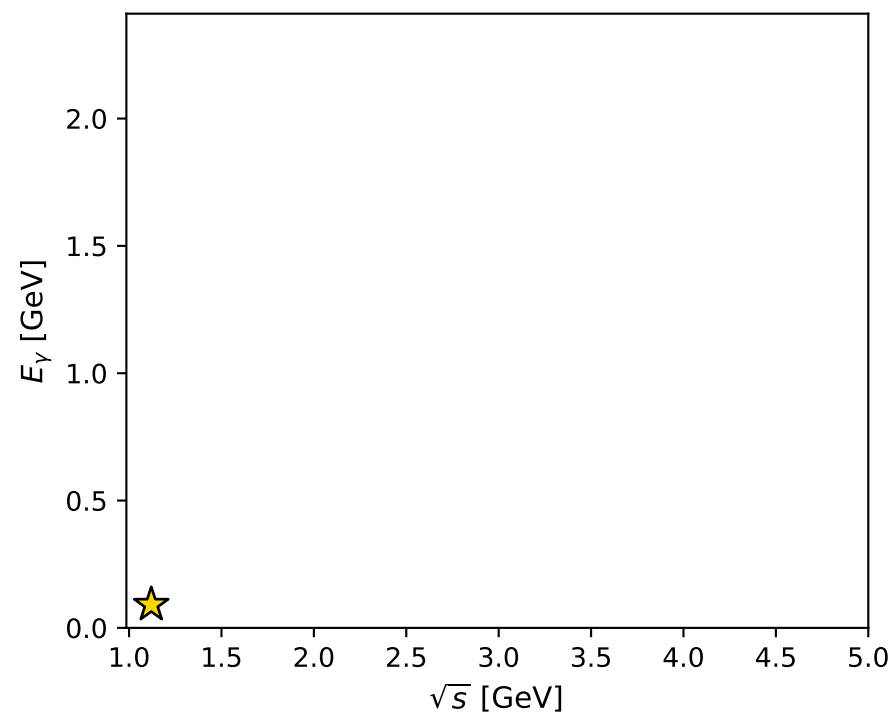
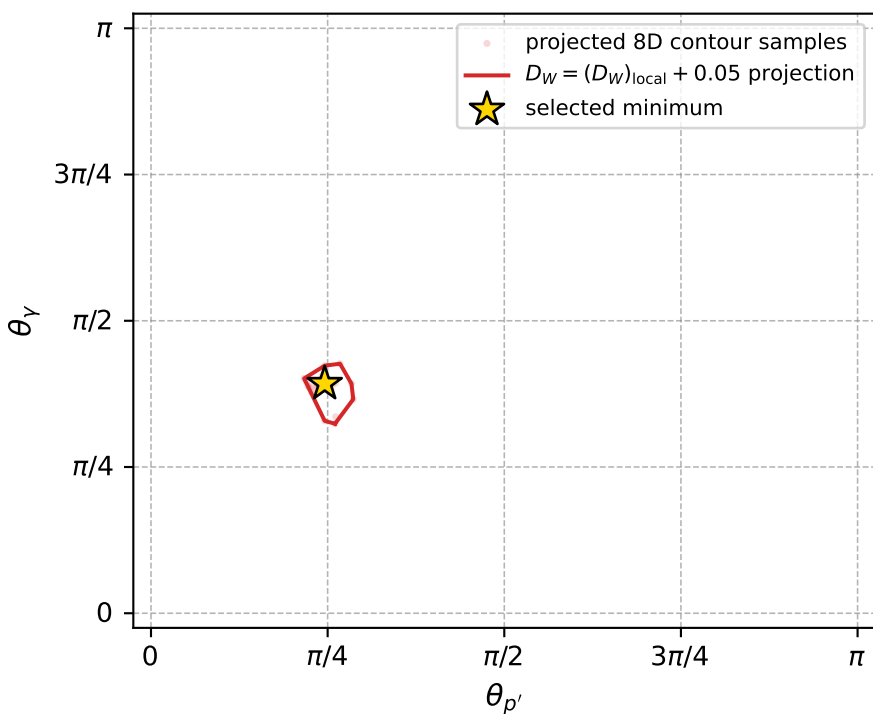
dw_mixing_angles_polarization_03_local_minimum_863 D_W=0.0122335
region: polarization_cluster_03_local_minimum_863
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81760093$ rad
incoming lepton: $0.683973|+\rangle +0.729507|-\rangle$
proton mixing angle: $\alpha_p=0.17041661$ rad
incoming proton: $0.985514|+\rangle +0.169593|-\rangle$

$s=1.25456$, $\sqrt{s}=1.12007$, $|\vec{P}|=0.167271$, $|\vec{P}'|=0.00384843$
 $\theta_{p'}=0.772292$, $\theta_Y=1.23294$, $\phi_{p'}=1.55938$, $\phi_Y=3.53659$
 $E_Y=0.0910334$, $\theta_{\ell'}=1.94096$, $\phi_{\ell'}=0.365928$
 $Q^2=0.0194357$, $x_B=0.102169$, $t=-0.0268535$, $W^2=1.05064$, $y=0.507675$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1673$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1673$
 P $E= 0.9528$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1673$
 ℓ' $E= 0.0910$ $p_x= 0.0792$ $p_y= 0.0304$ $p_z= -0.0329$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0027$ $p_z= 0.0028$
 q_Y $E= 0.0910$ $p_x= -0.0793$ $p_y= -0.0330$ $p_z= 0.0302$



electron: polarization cluster P3, local minimum 863; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012233537$
 $D_W \text{ contour} = 0.062233537$
 above global minimum = 0.012217104
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt}_s = 1.12007$
 $\text{theta}_p_{\text{out}} = 0.7722917$
 $\text{theta}_\gamma_{\text{out}} = 1.232944$
 $q_{\text{out}} = 0.09103339$
 $\text{phi}_p_{\text{out}} = 1.559378$
 $\text{phi}_\gamma_{\text{out}} = 3.536591$
 $\alpha_e = 0.8176009$
 $\alpha_p = 0.1704166$

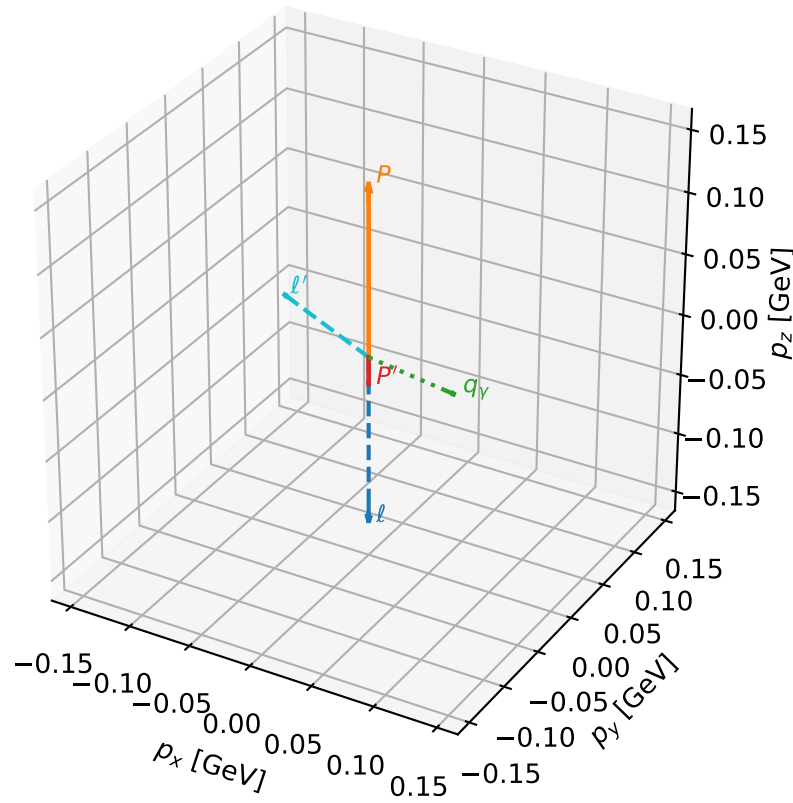
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_870 D_W=0.0123808
 region: polarization_cluster_03_local_minimum_870
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76538427$ rad
 incoming lepton: $0.721116|+\rangle + 0.692814|-\rangle$
 proton mixing angle: $\alpha_p=2.99225$ rad
 incoming proton: $-0.988869|+\rangle + 0.148788|-\rangle$

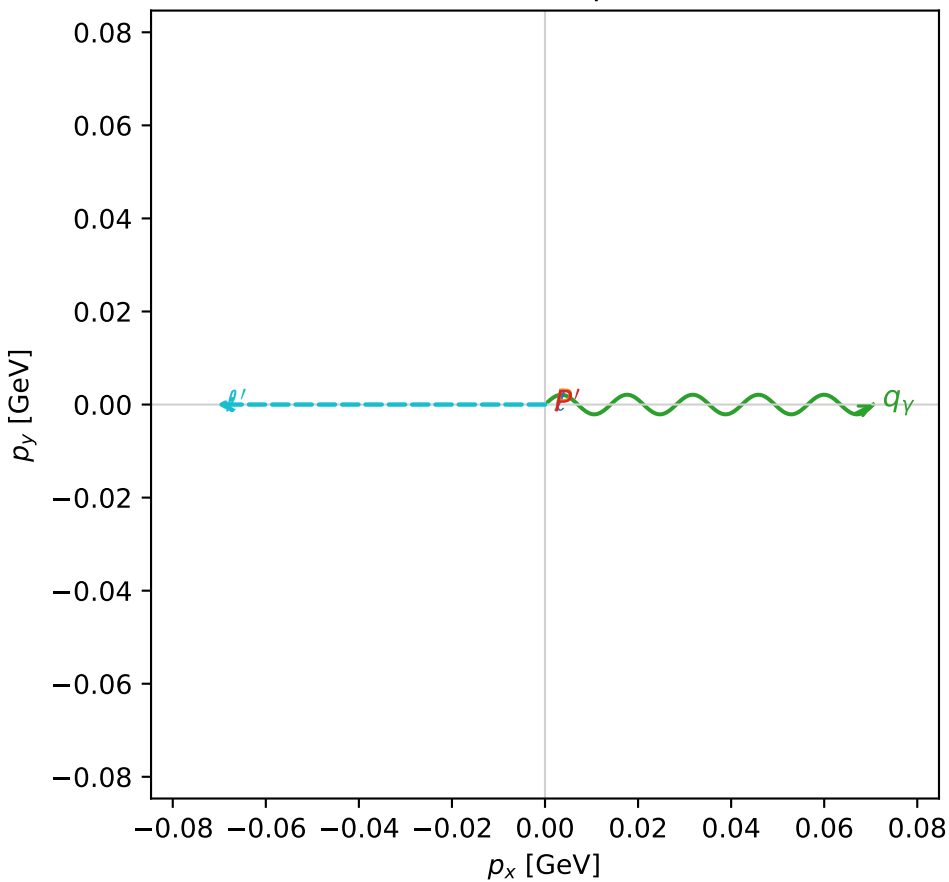
$s=1.18176$, $\sqrt{s}=1.08709$, $|\vec{P}|=0.138864$, $|\vec{P}'|=0.0220325$
 $\theta_{P'}=3.14159$, $\theta_V=1.71361$, $\phi_{P'}=5.23833$, $\phi_V=1.72592e-05$
 $E_V=0.0712811$, $\theta_{l'}=1.14291$, $\phi_{l'}=3.14161$
 $Q^2=0.0304738$, $x_B=0.186058$, $t=-0.0257884$, $W^2=1.01316$, $y=0.542491$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1389$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1389$
 P $E= 0.9482$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1389$
 l' $E= 0.0775$ $p_x= -0.0706$ $p_y= -0.0000$ $p_z= 0.0322$
 P' $E= 0.9383$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= -0.0220$
 q_V $E= 0.0713$ $p_x= 0.0706$ $p_y= 0.0000$ $p_z= -0.0101$

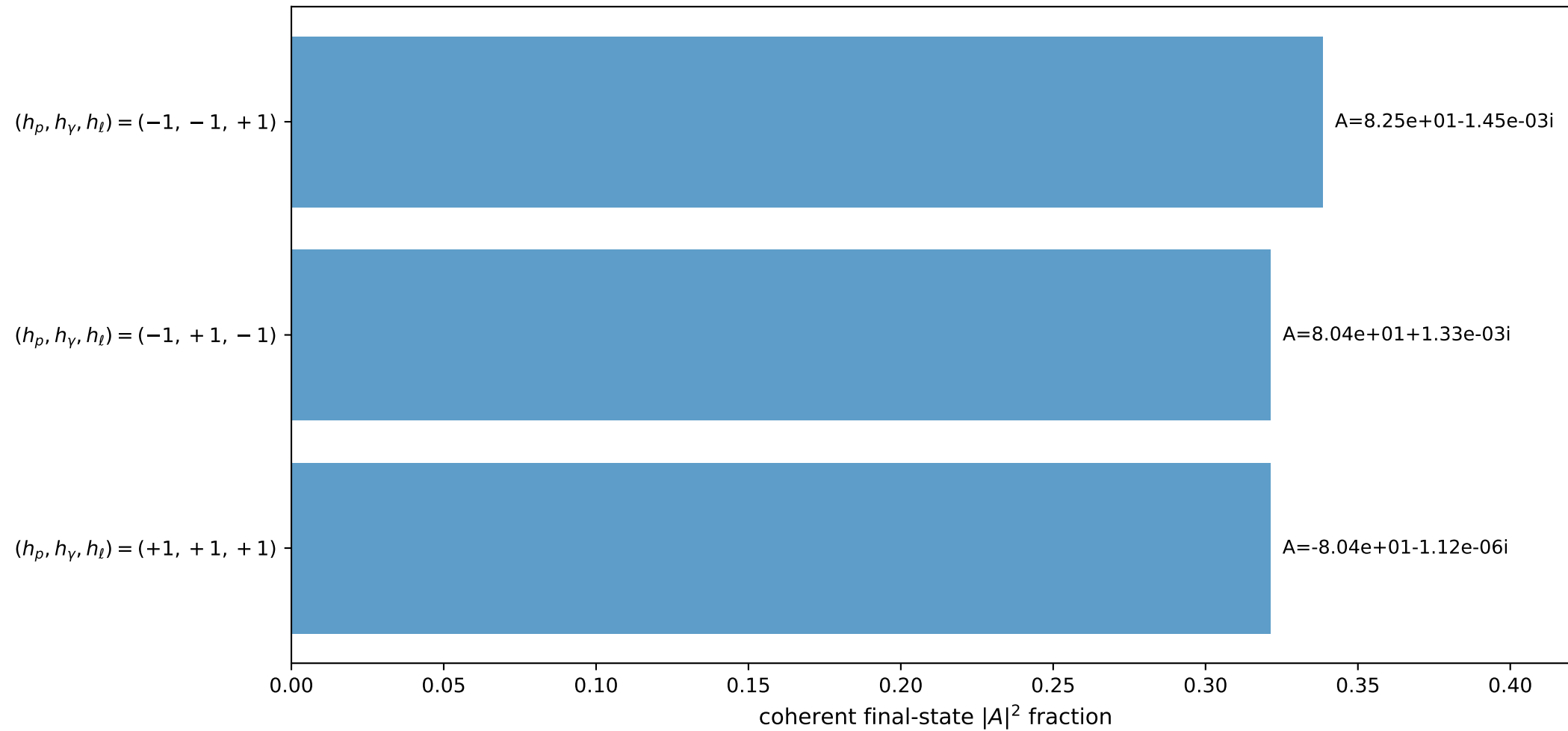
3D momenta



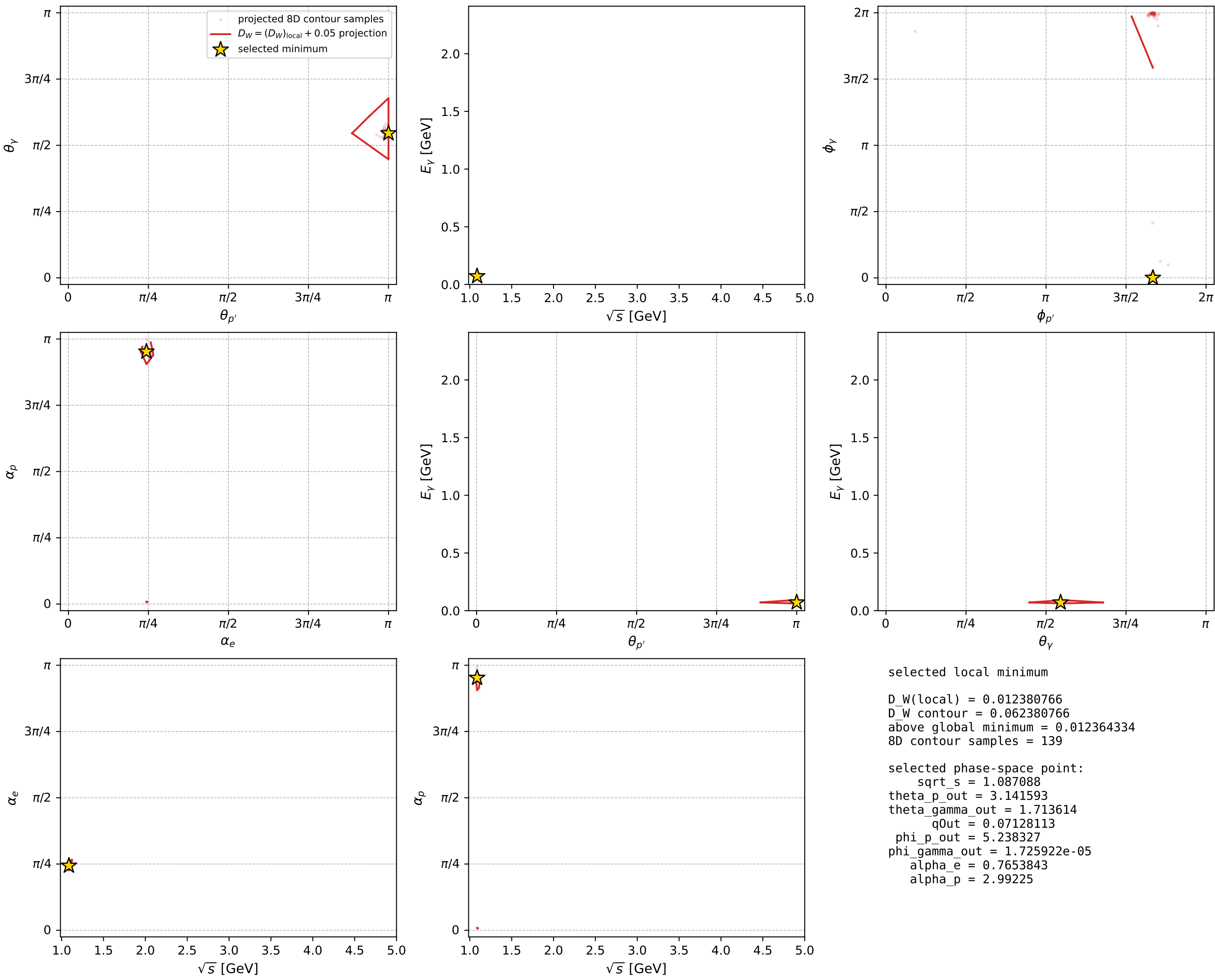
Transverse plane



Leading final-state amplitudes (N=3, retained=98.1%)



electron: polarization cluster P3, local minimum 870; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



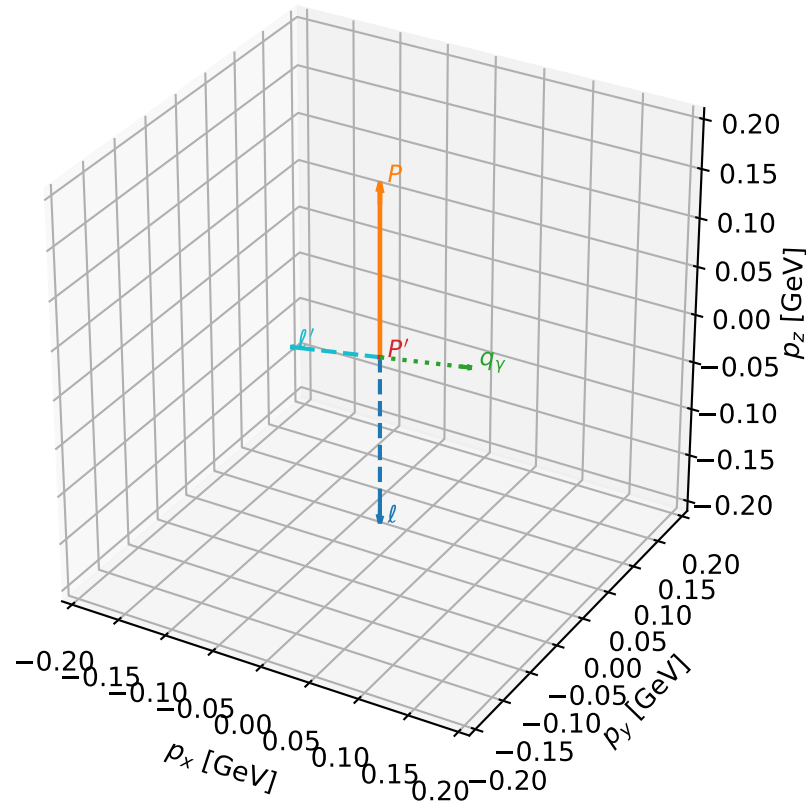
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_877 D_W=0.0127721
 region: polarization_cluster_03_local_minimum_877
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81544177$ rad
 incoming lepton: $0.685547|+\rangle + 0.728029|-\rangle$
 proton mixing angle: $\alpha_p=2.9540219$ rad
 incoming proton: $-0.98246|+\rangle + 0.186473|-\rangle$

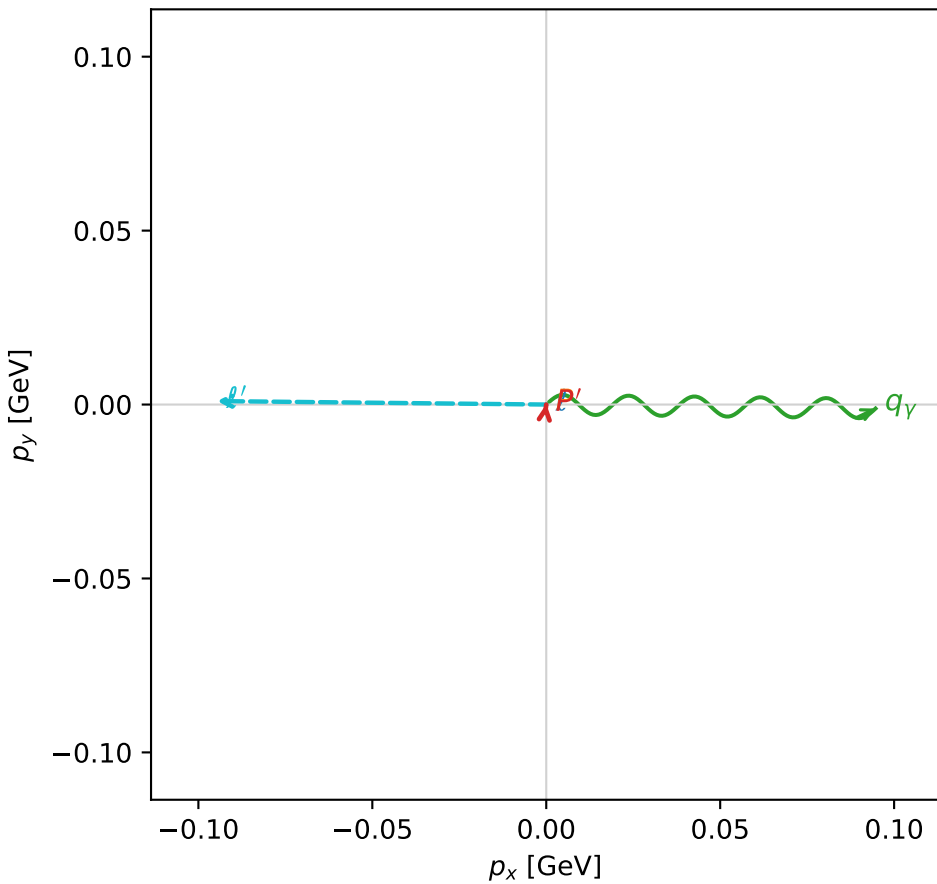
$s=1.27779$, $\sqrt{s}=1.1304$, $|\vec{P}|=0.176022$, $|\vec{P}'|=0.000167655$
 $\theta_{P'}=2.01784$, $\theta_V=1.39219$, $\phi_{P'}=1.44262$, $\phi_V=6.27108$
 $E_V=0.0961947$, $\theta_{\ell'}=1.74863$, $\phi_{\ell'}=3.13108$
 $Q^2=0.0278755$, $x_B=0.133801$, $t=-0.0307411$, $W^2=1.0603$, $y=0.523524$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1760$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1760$
 P $E= 0.9544$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1760$
 ℓ' $E= 0.0962$ $p_x= -0.0947$ $p_y= 0.0010$ $p_z= -0.0170$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= -0.0001$
 q_V $E= 0.0962$ $p_x= 0.0947$ $p_y= -0.0011$ $p_z= 0.0171$

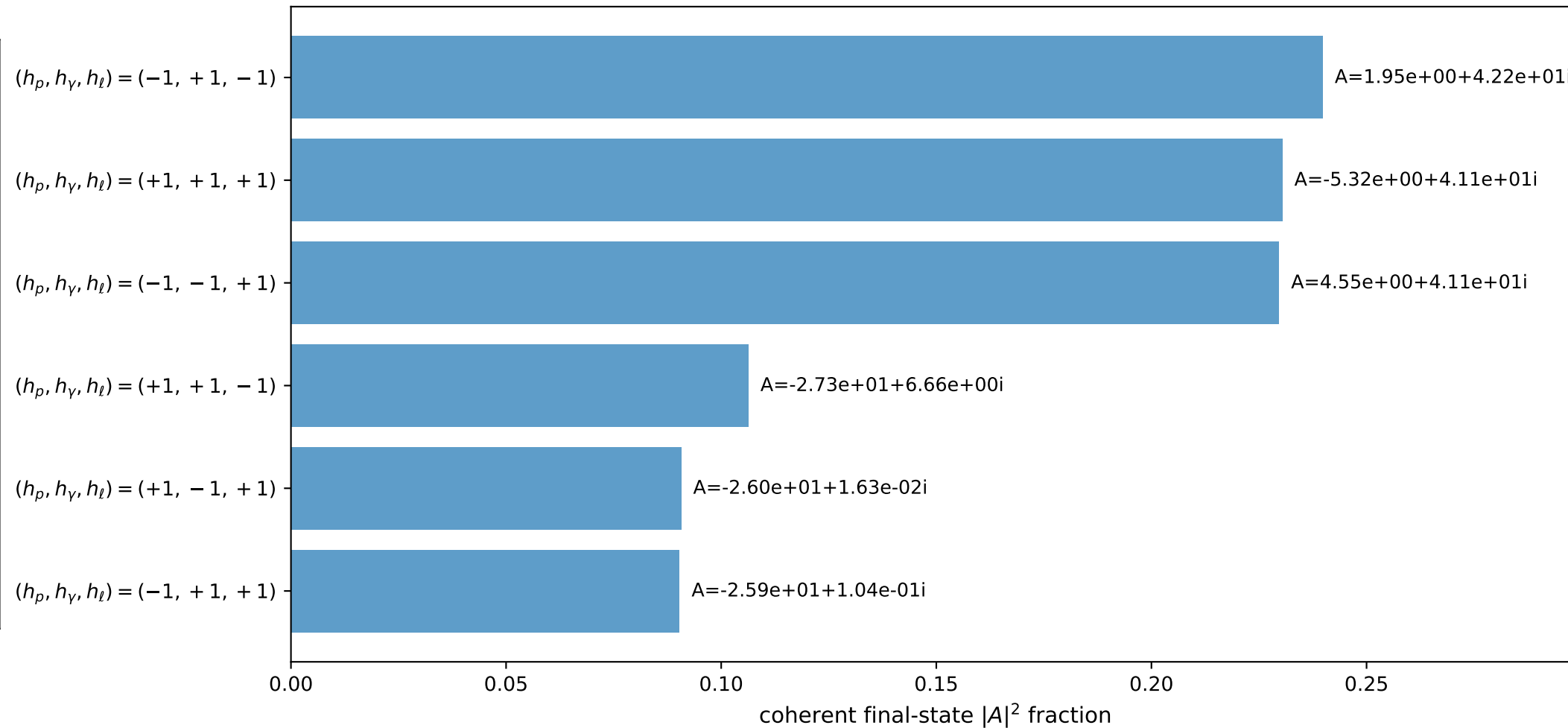
3D momenta



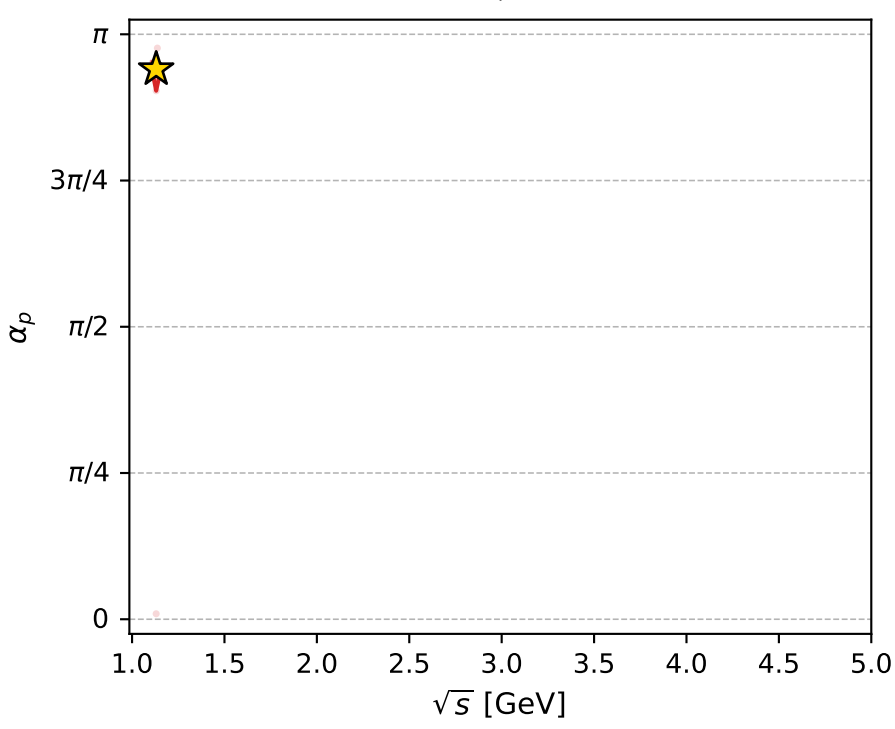
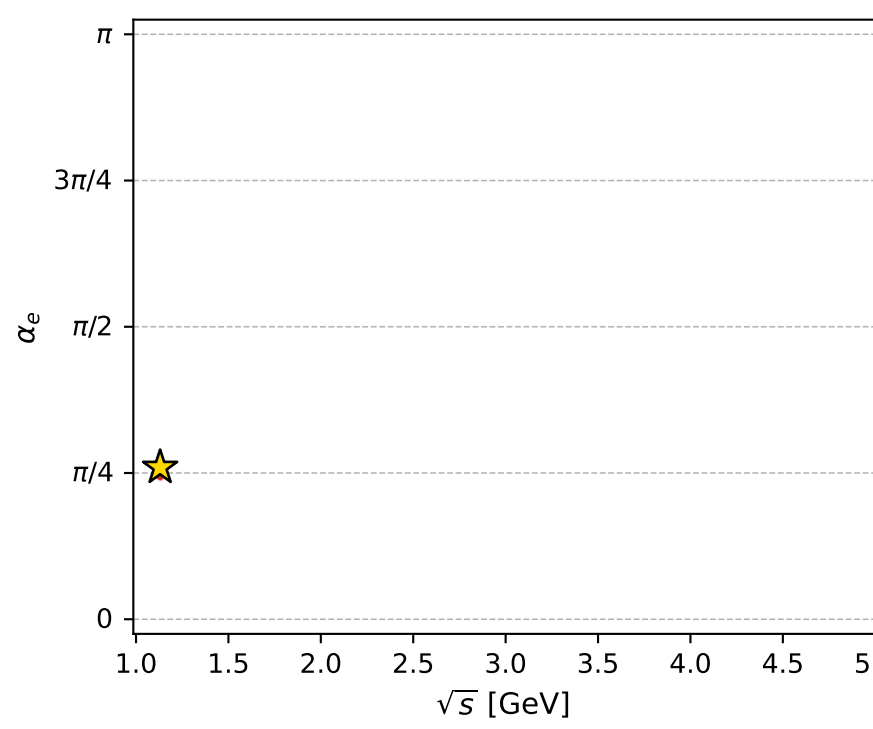
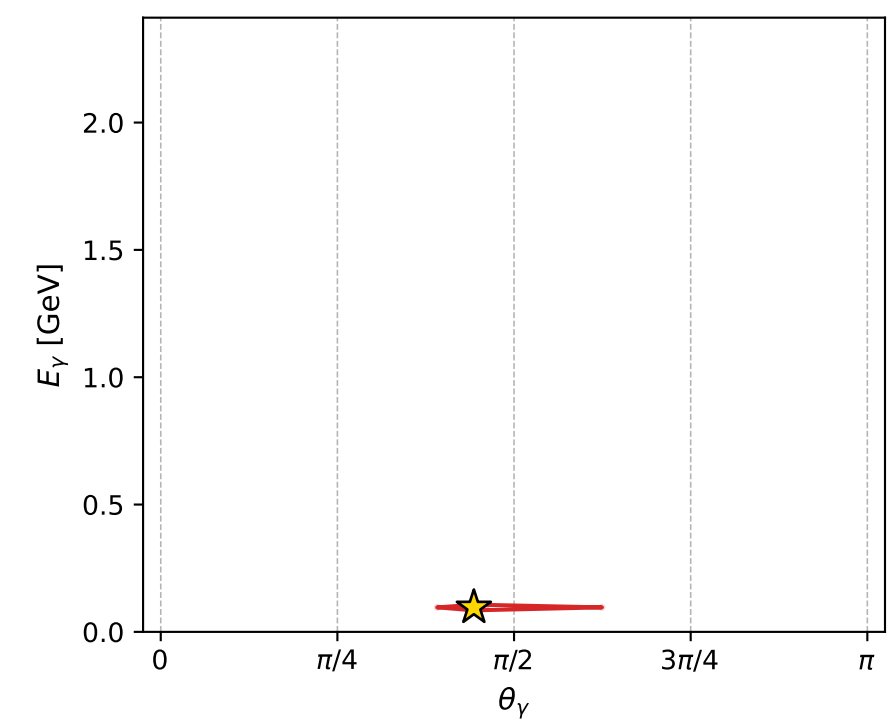
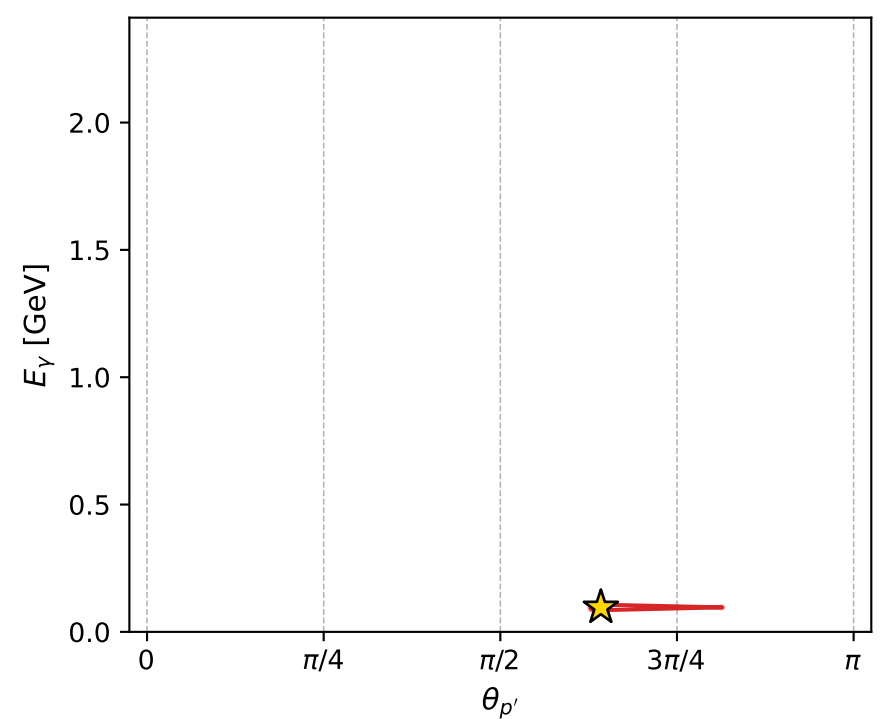
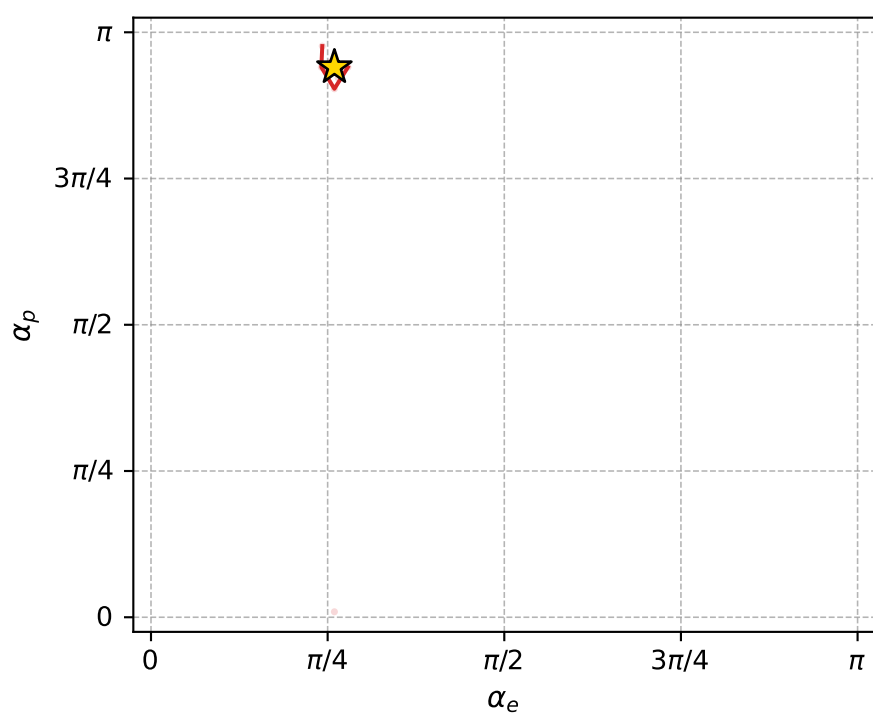
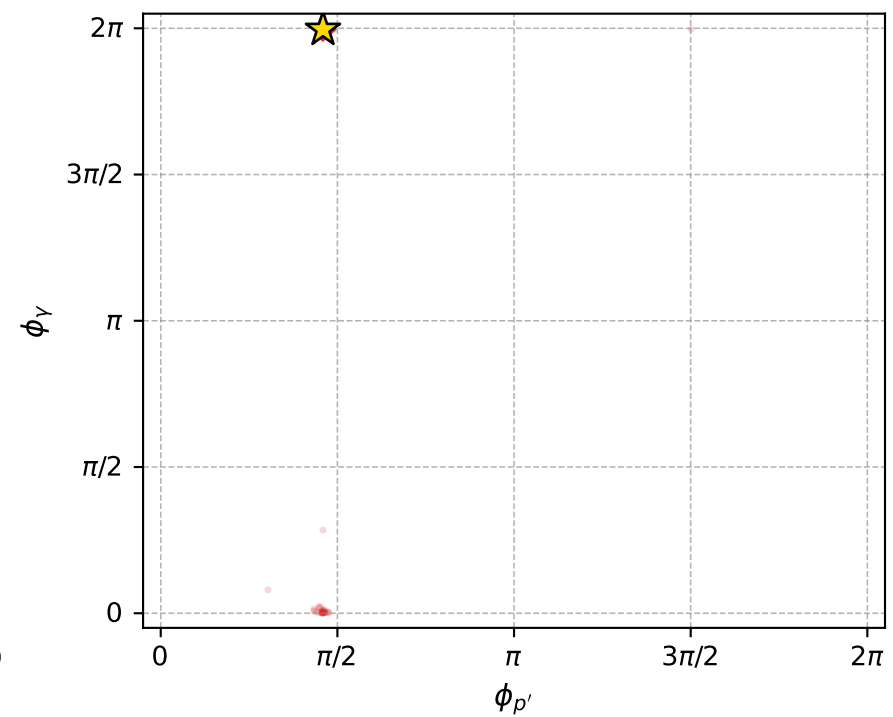
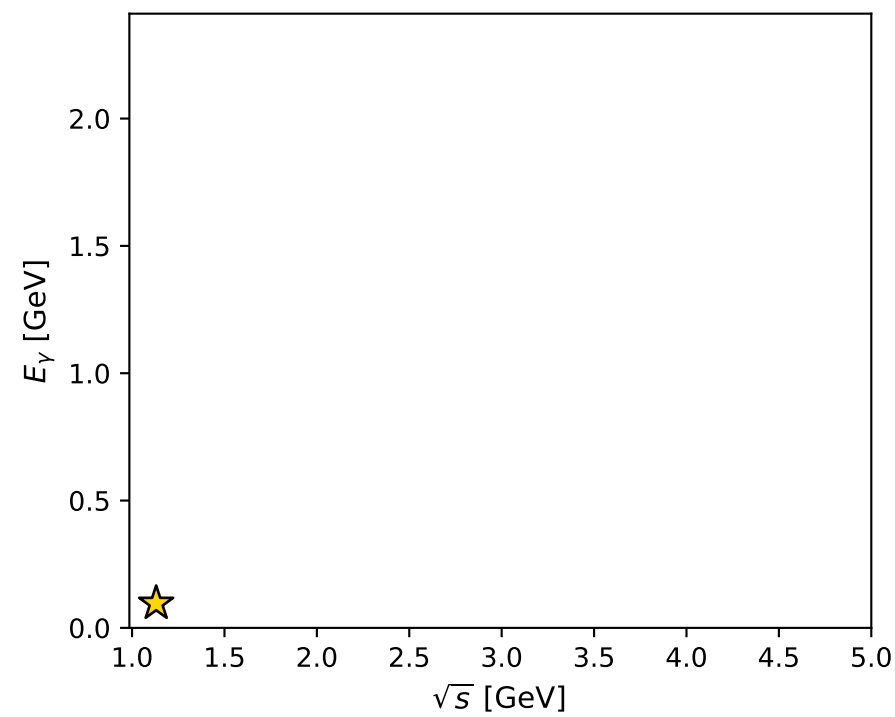
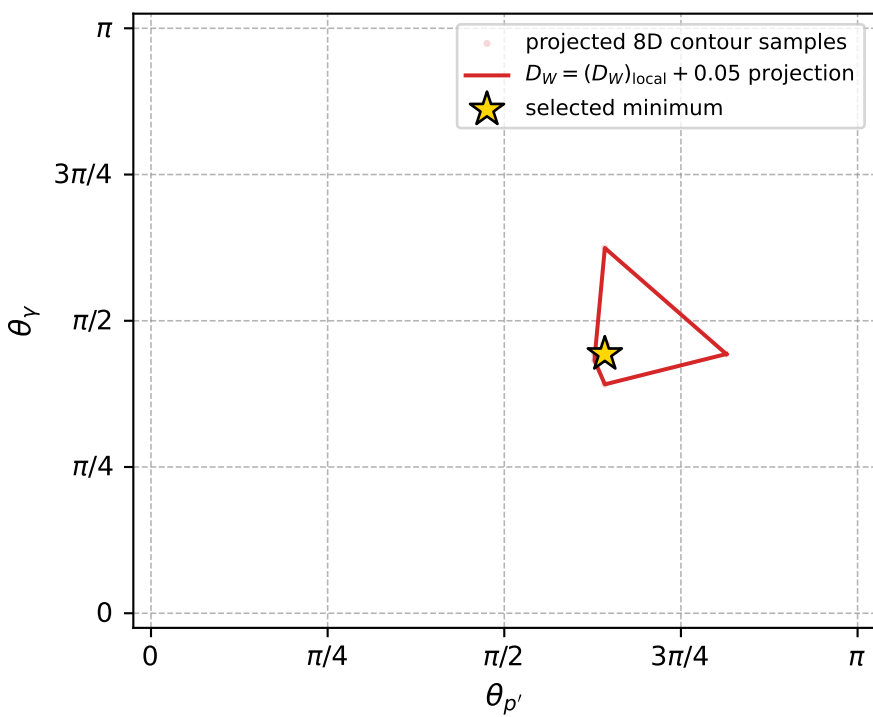
Transverse plane



Leading final-state amplitudes (N=6, retained=98.7%)



electron: polarization cluster P3, local minimum 877; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.012772115$
 $D_W \text{ contour} = 0.062772115$
 above global minimum = 0.012755682
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.130395$
 $\text{theta_p_out} = 2.017837$
 $\text{theta_gamma_out} = 1.392189$
 $\text{qOut} = 0.09619468$
 $\text{phi_p_out} = 1.442623$
 $\text{phi_gamma_out} = 6.271082$
 $\text{alpha_e} = 0.8154418$
 $\text{alpha_p} = 2.954022$

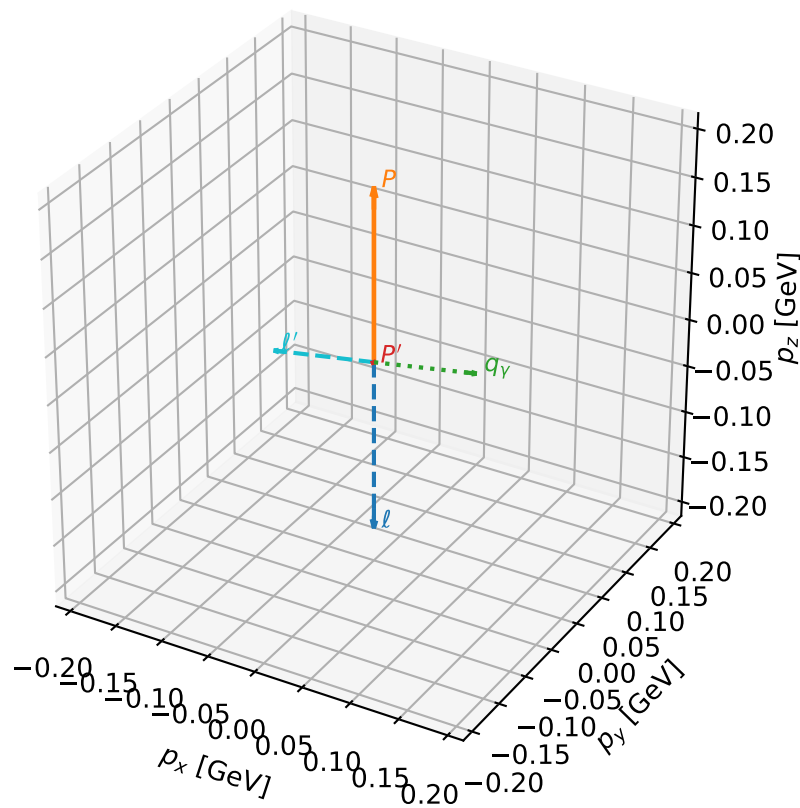
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_878 D_W=0.0128023
region: polarization_cluster_03_local_minimum_878
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.80667345$ rad
incoming lepton: $0.691904|+\rangle +0.72199|-\rangle$
proton mixing angle: $\alpha_p=2.9274399$ rad
incoming proton: $-0.977157|+\rangle +0.21252|-\rangle$

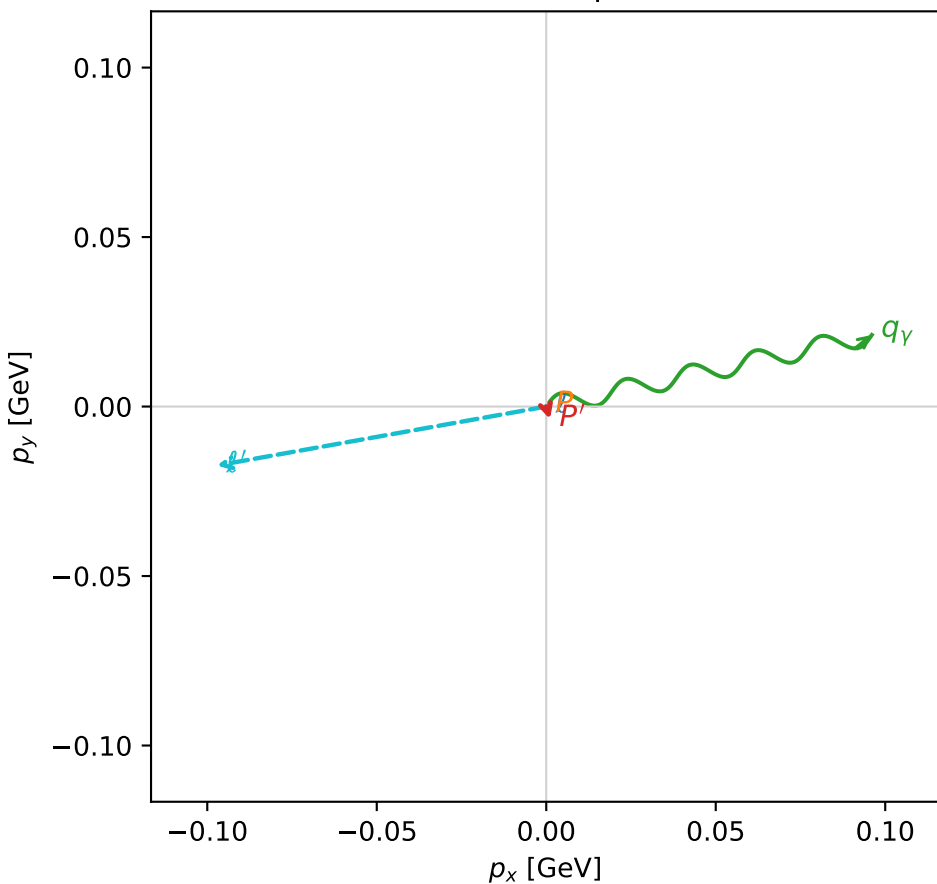
$s=1.28919$, $\sqrt{s}=1.13542$, $|\vec{P}|=0.180259$, $|\vec{P}'|=0.00426662$
 $\theta_{P'}=1.14662$, $\theta_V=1.51915$, $\phi_{P'}=5.00243$, $\phi_V=0.216422$
 $E_V=0.0984722$, $\theta_{\ell'}=1.63998$, $\phi_{\ell'}=3.31872$
 $Q^2=0.033204$, $x_B=0.152406$, $t=-0.0315842$, $W^2=1.06451$, $y=0.532235$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1803$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1803$
 P $E= 0.9552$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1803$
 ℓ' $E= 0.0989$ $p_x= -0.0972$ $p_y= -0.0174$ $p_z= -0.0068$
 P' $E= 0.9380$ $p_x= 0.0011$ $p_y= -0.0037$ $p_z= 0.0018$
 q_V $E= 0.0985$ $p_x= 0.0960$ $p_y= 0.0211$ $p_z= 0.0051$

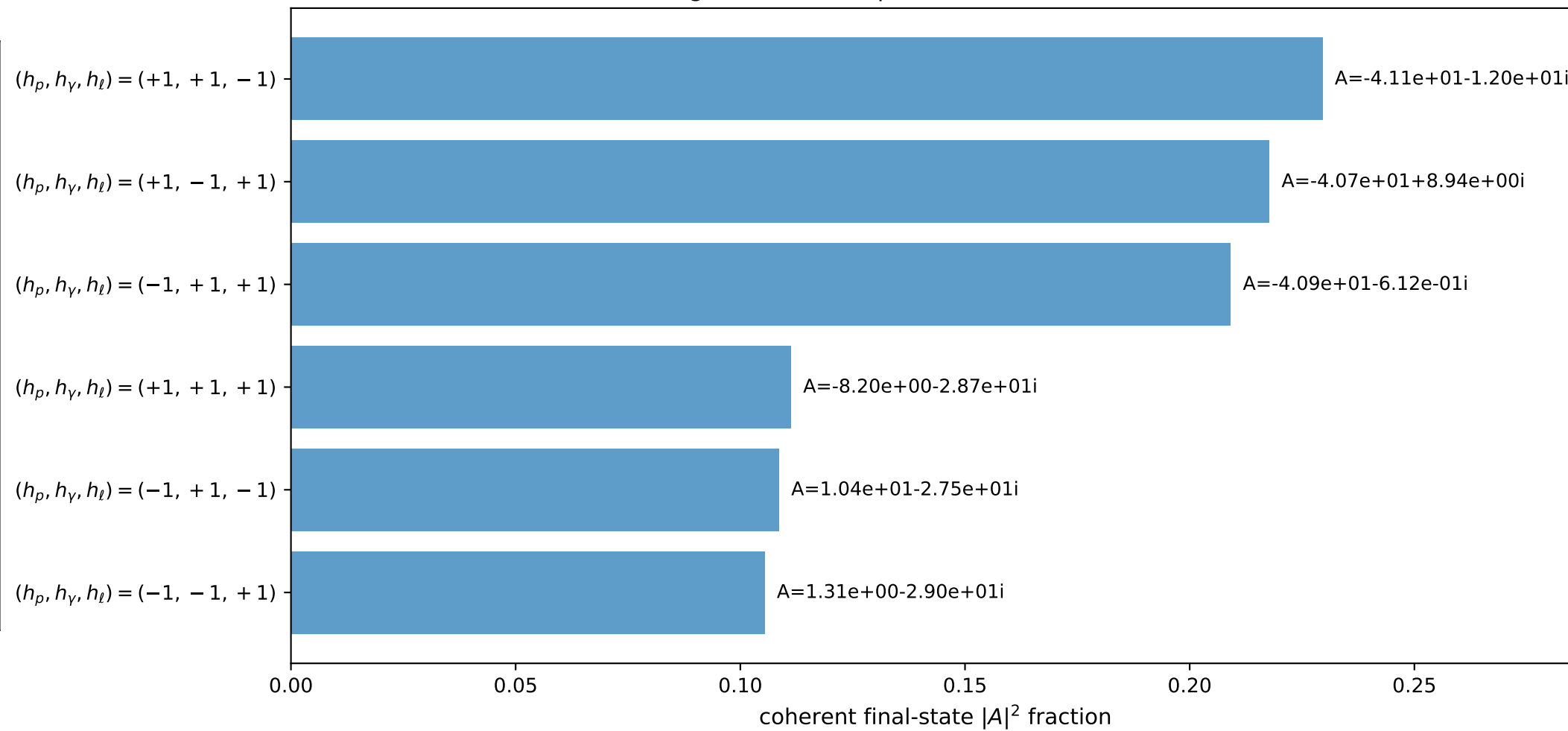
3D momenta



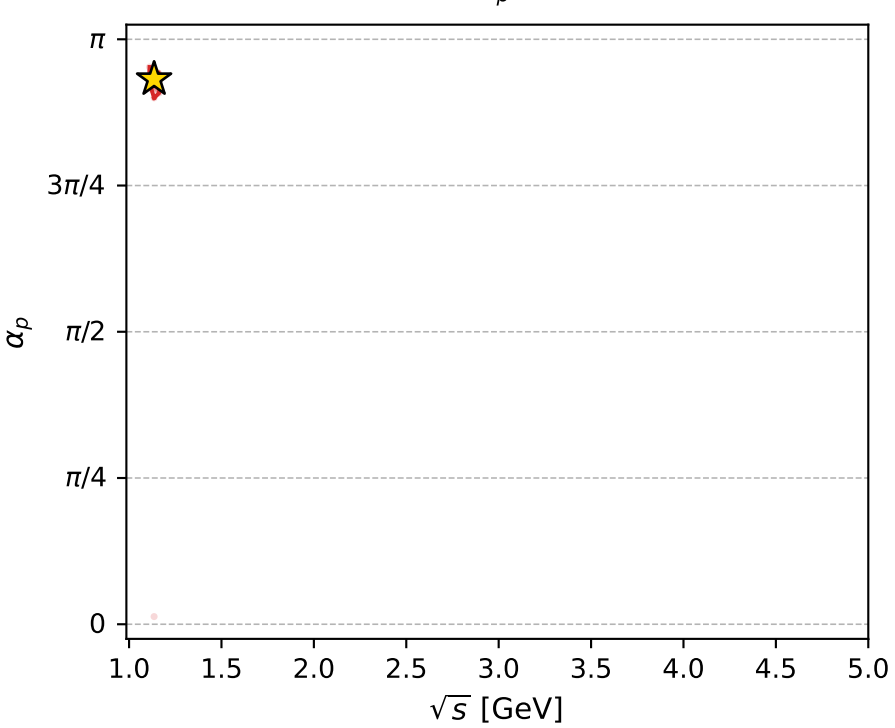
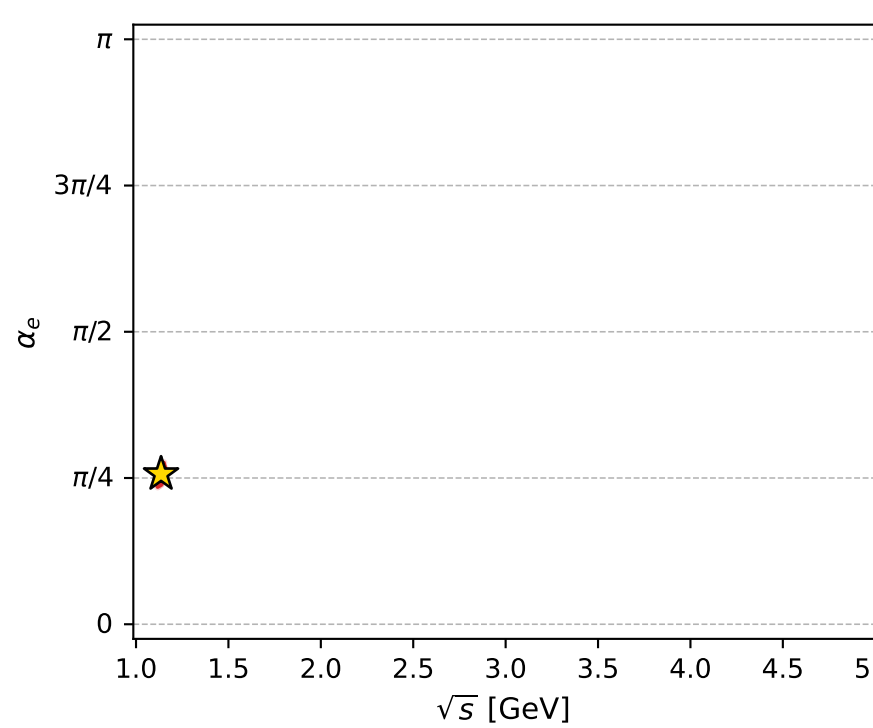
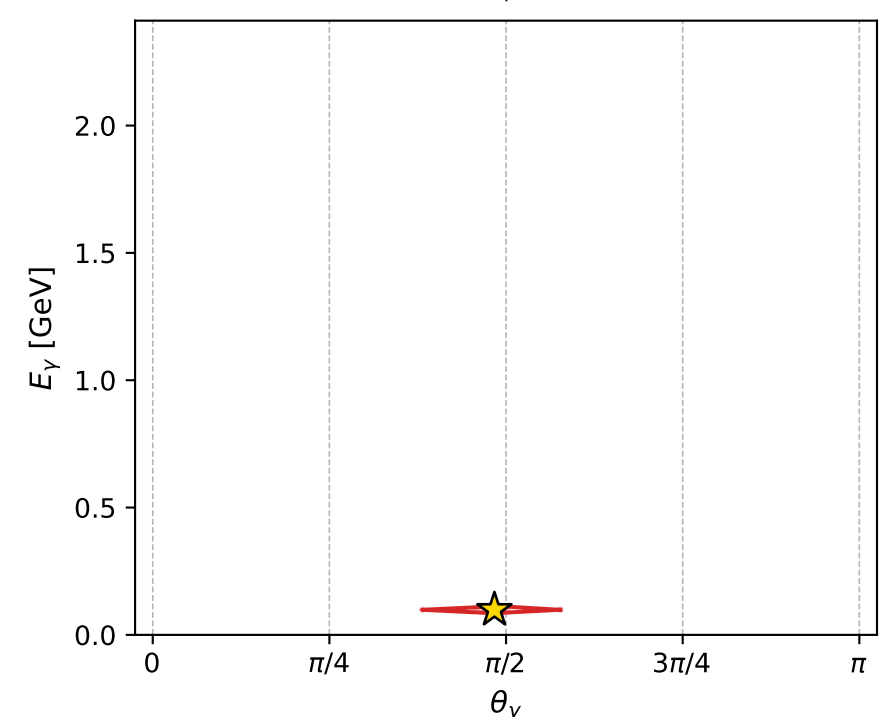
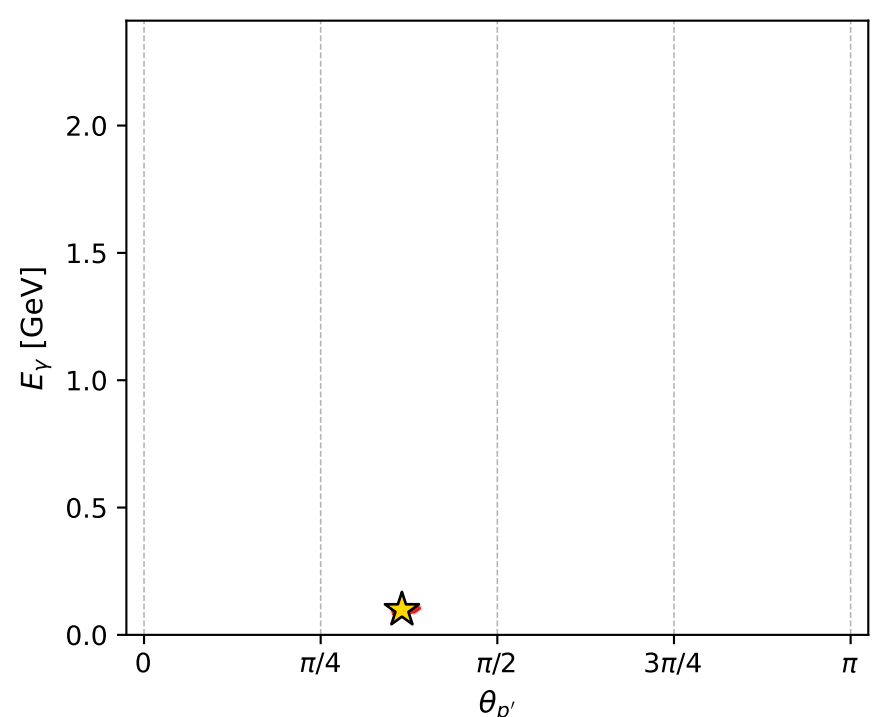
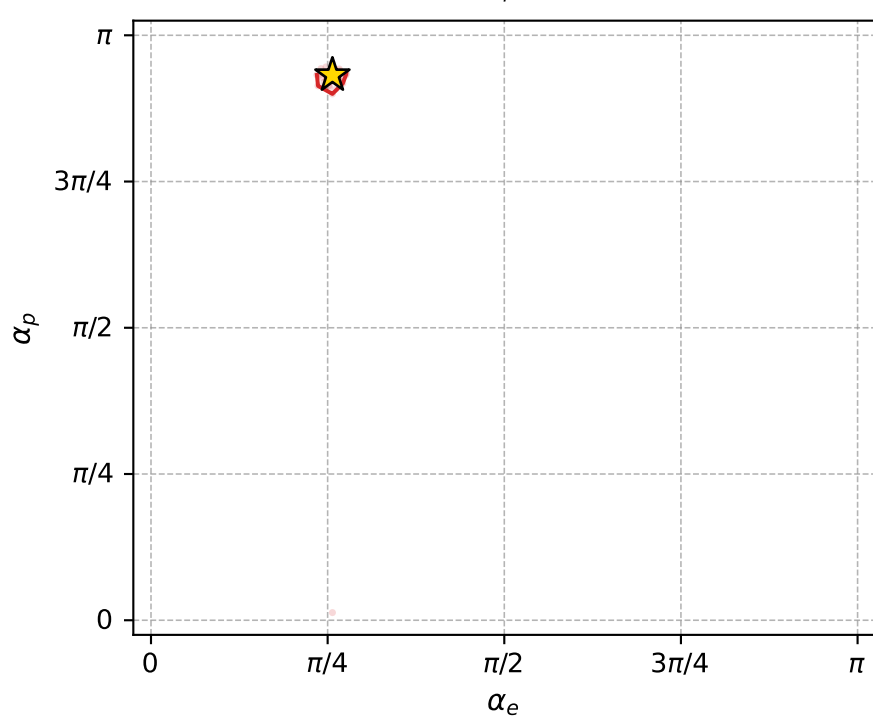
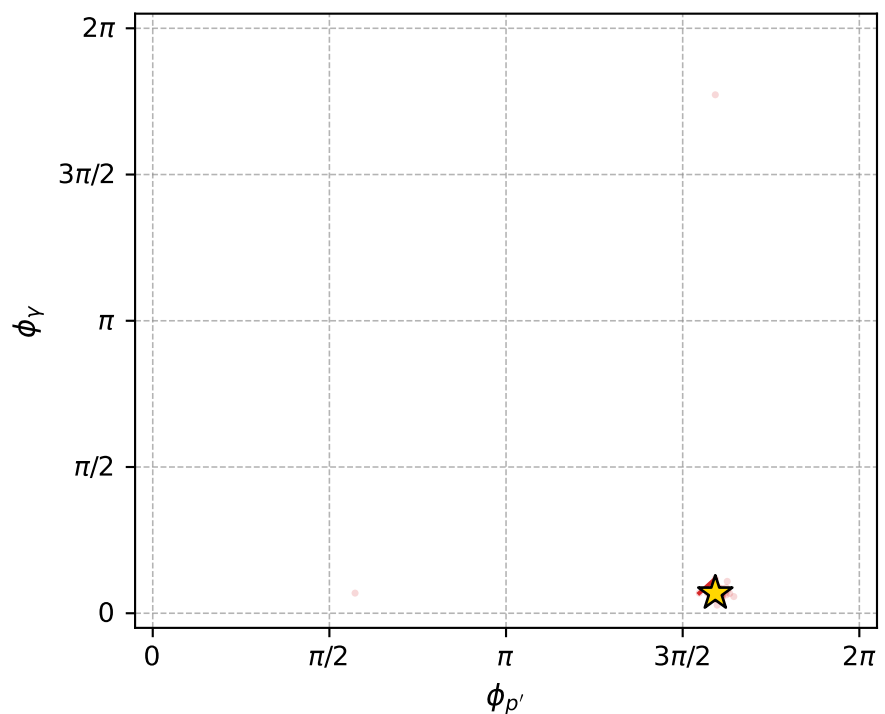
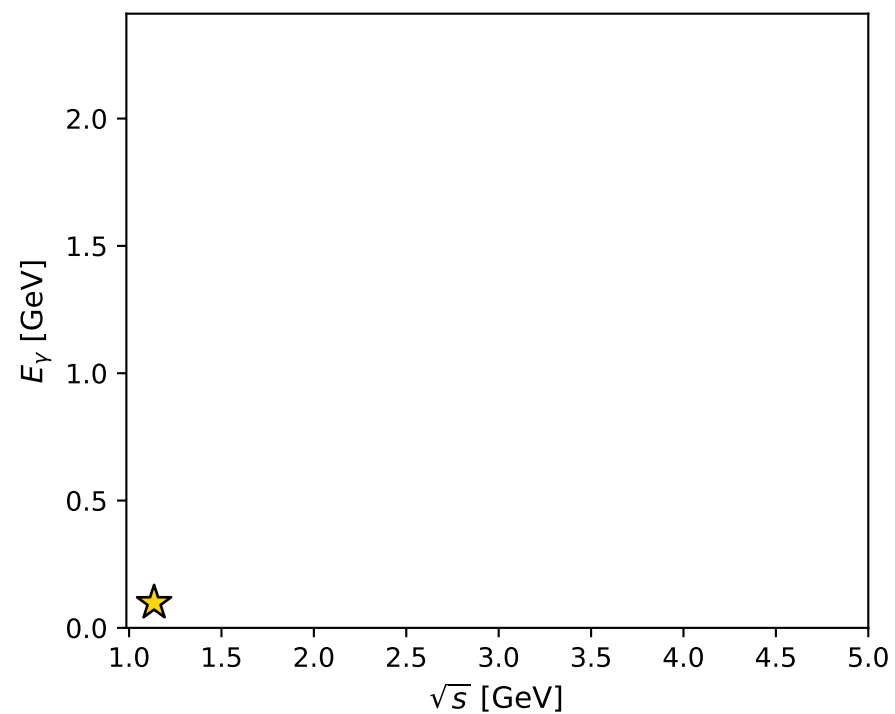
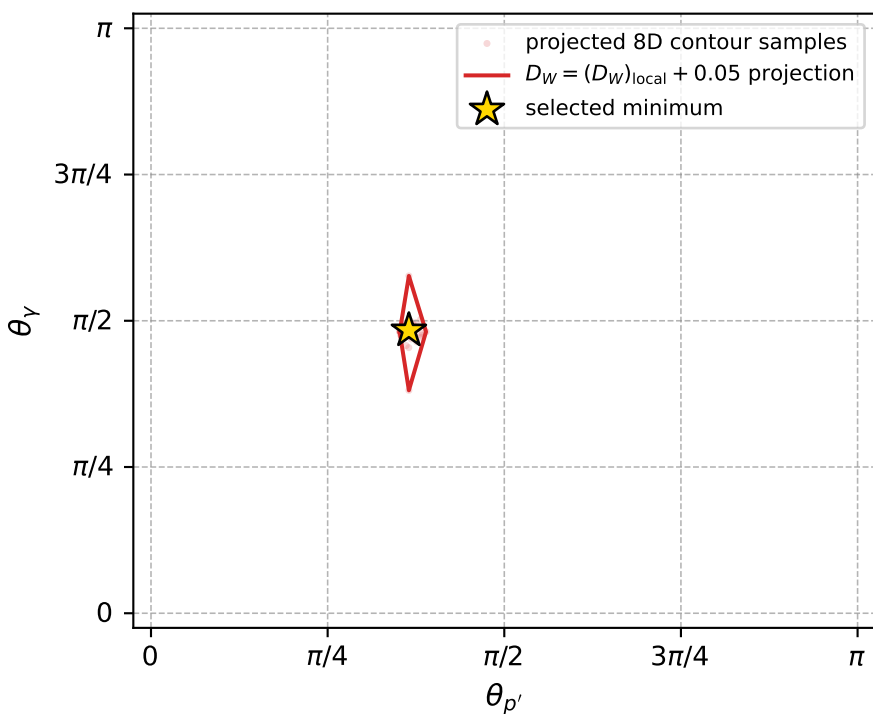
Transverse plane



Leading final-state amplitudes (N=6, retained=98.2%)



electron: polarization cluster P3, local minimum 878; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.01280226$
 $D_W \text{ contour} = 0.06280226$
 above global minimum = 0.012785827
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.135423$
 $\text{theta_p_out} = 1.146625$
 $\text{theta_gamma_out} = 1.519153$
 $\text{q0out} = 0.09847225$
 $\text{phi_p_out} = 5.002428$
 $\text{phi_gamma_out} = 0.2164223$
 $\text{alpha_e} = 0.8066735$
 $\text{alpha_p} = 2.92744$

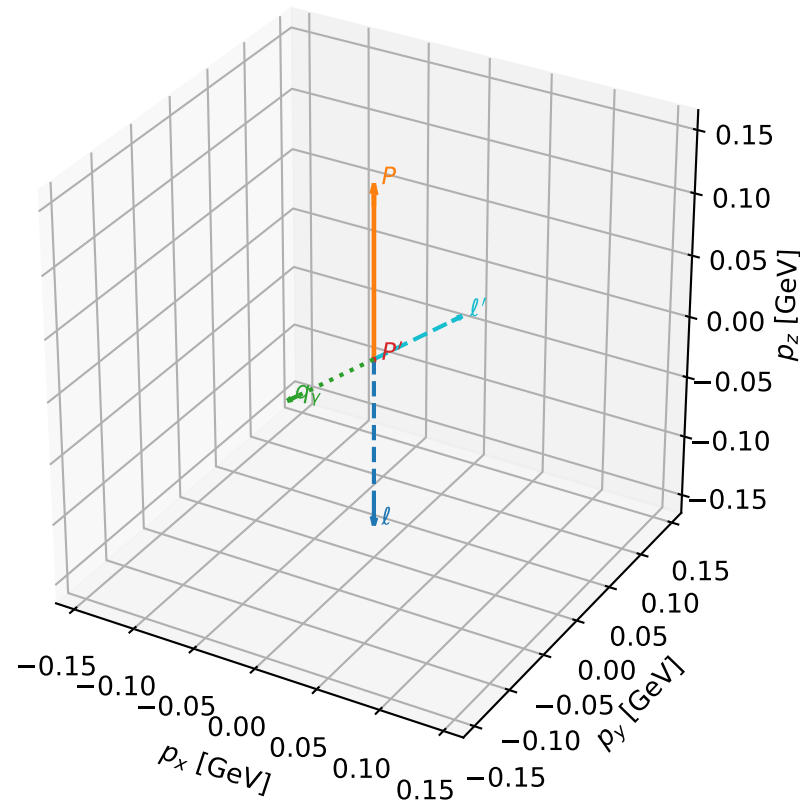
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_883 D_W=0.0129257
region: polarization_cluster_03_local_minimum_883
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.75380905$ rad
incoming lepton: $0.729087|+\rangle +0.684421|-\rangle$
proton mixing angle: $\alpha_p=0.10852539$ rad
incoming proton: $0.994117|+\rangle +0.108312|-\rangle$

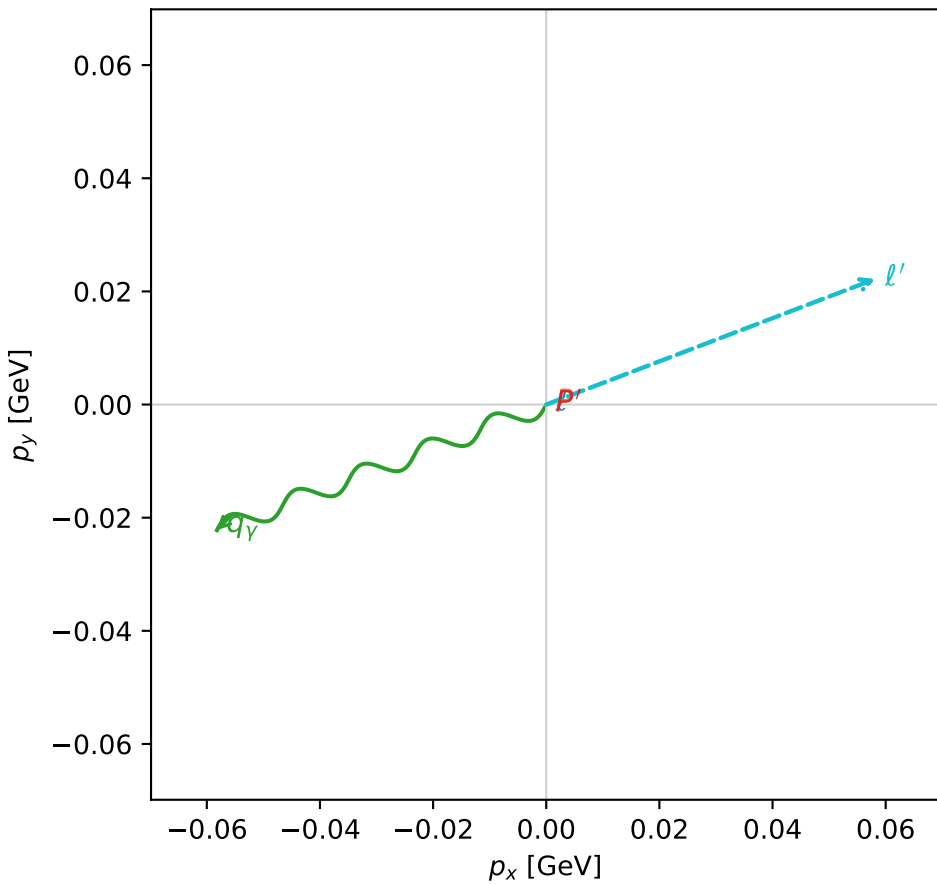
$s=1.17992$, $\sqrt{s}=1.08624$, $|\vec{P}|=0.138125$, $|\vec{P}'|=0.000534216$
 $\theta_{p'}=3.14159$, $\theta_V=2.13998$, $\phi_{p'}=4.09489$, $\phi_V=3.5062$
 $E_V=0.0739751$, $\theta_{\ell'}=0.995556$, $\phi_{\ell'}=0.364608$
 $Q^2=0.0316769$, $x_B=0.185884$, $t=-0.0191241$, $W^2=1.01858$, $y=0.567895$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1381$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1381$
 P $E= 0.9481$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1381$
 ℓ' $E= 0.0743$ $p_x= 0.0582$ $p_y= 0.0222$ $p_z= 0.0404$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0005$
 q_Y $E= 0.0740$ $p_x= -0.0582$ $p_y= -0.0222$ $p_z= -0.0399$

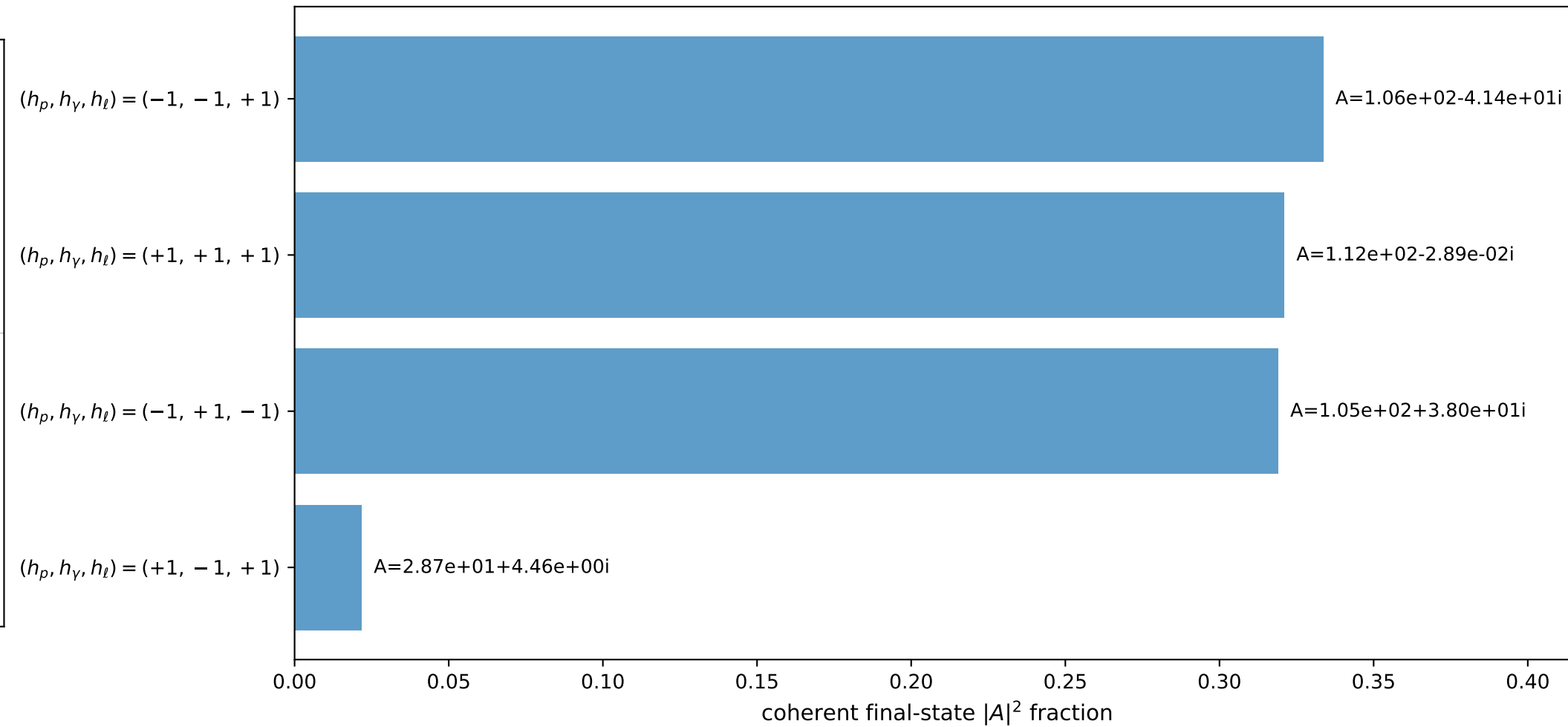
3D momenta



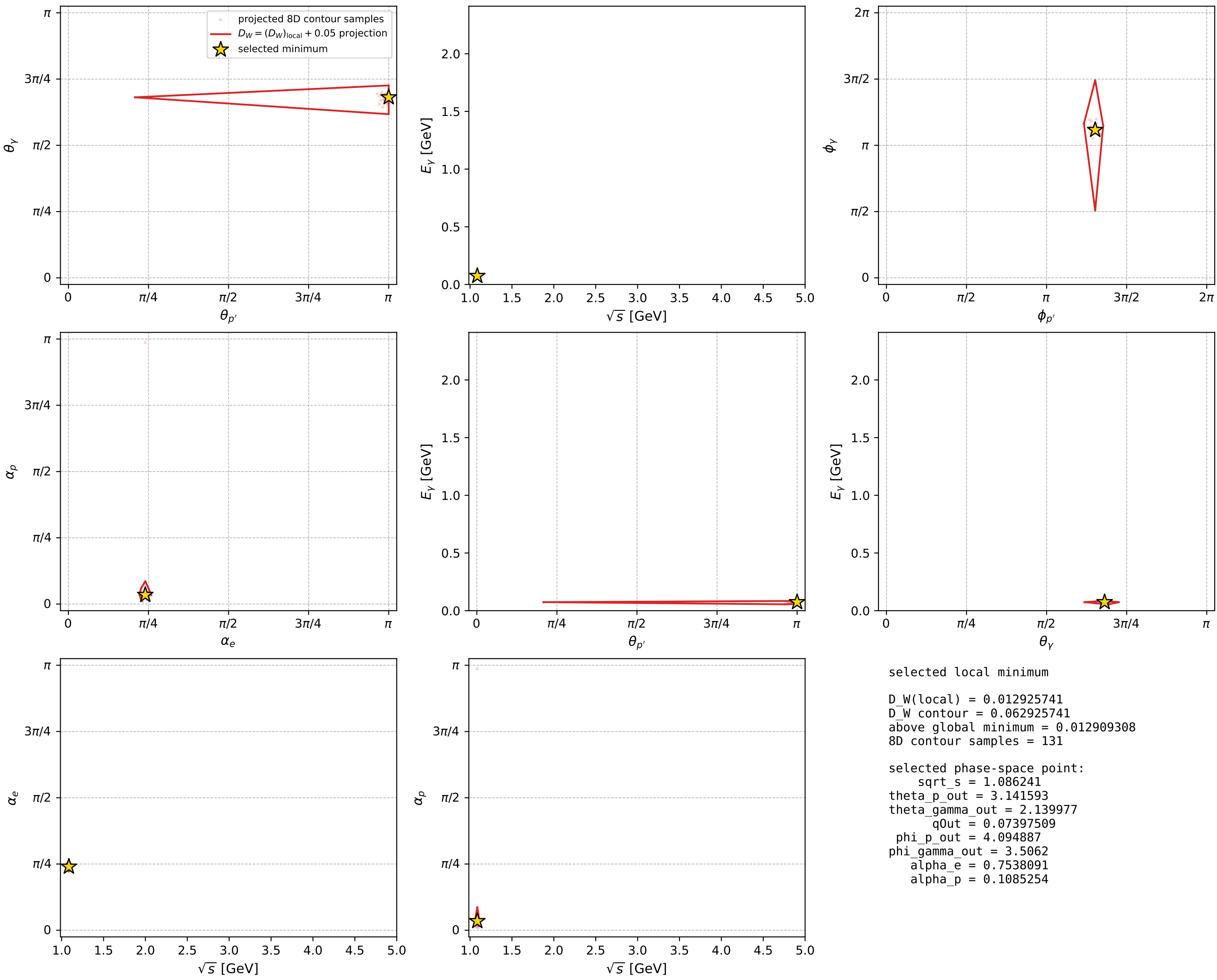
Transverse plane



Leading final-state amplitudes (N=4, retained=99.6%)



electron: polarization cluster P3, local minimum 883; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



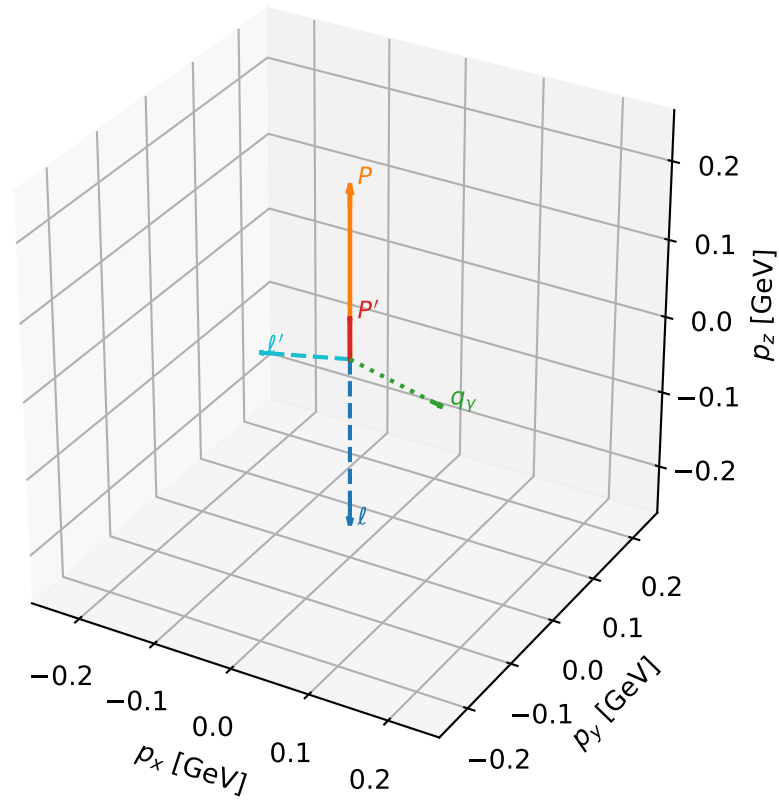
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_885 D_W=0.0130708
 region: polarization_cluster_03_local_minimum_885
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81698053$ rad
 incoming lepton: $0.684426|+\rangle + 0.729083|-\rangle$
 proton mixing angle: $\alpha_p=2.9786241$ rad
 incoming proton: $-0.98675|+\rangle + 0.162248|-\rangle$

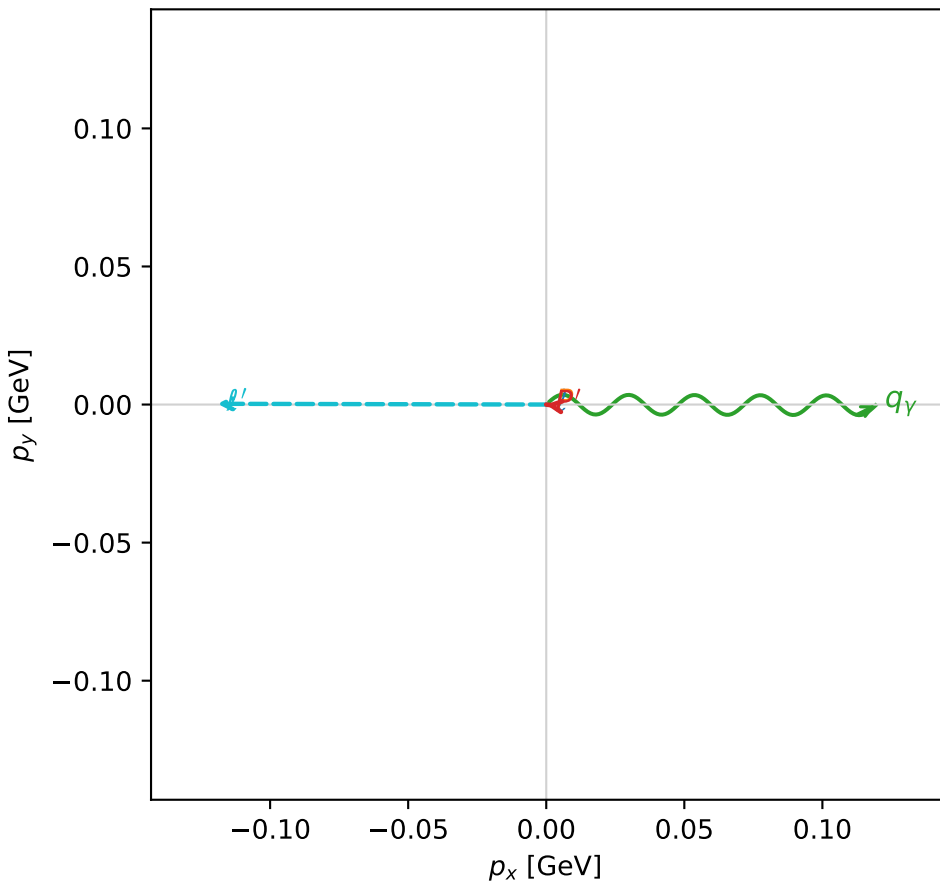
$s=1.40095$, $\sqrt{s}=1.18362$, $|\vec{P}|=0.220131$, $|\vec{P}'|=0.0526706$
 $\theta_{P'}=0.00278182$, $\theta_Y=1.79256$, $\phi_{P'}=3.00775$, $\phi_Y=6.28075$
 $E_Y=0.122259$, $\theta_{\ell'}=1.78392$, $\phi_{\ell'}=3.13932$
 $Q^2=0.0423086$, $x_B=0.153908$, $t=-0.0274667$, $W^2=1.11243$, $y=0.527529$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2201$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2201$
 P $E= 0.9635$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2201$
 ℓ' $E= 0.1219$ $p_x= -0.1191$ $p_y= 0.0003$ $p_z= -0.0258$
 P' $E= 0.9395$ $p_x= -0.0001$ $p_y= 0.0000$ $p_z= 0.0527$
 q_Y $E= 0.1223$ $p_x= 0.1193$ $p_y= -0.0003$ $p_z= -0.0269$

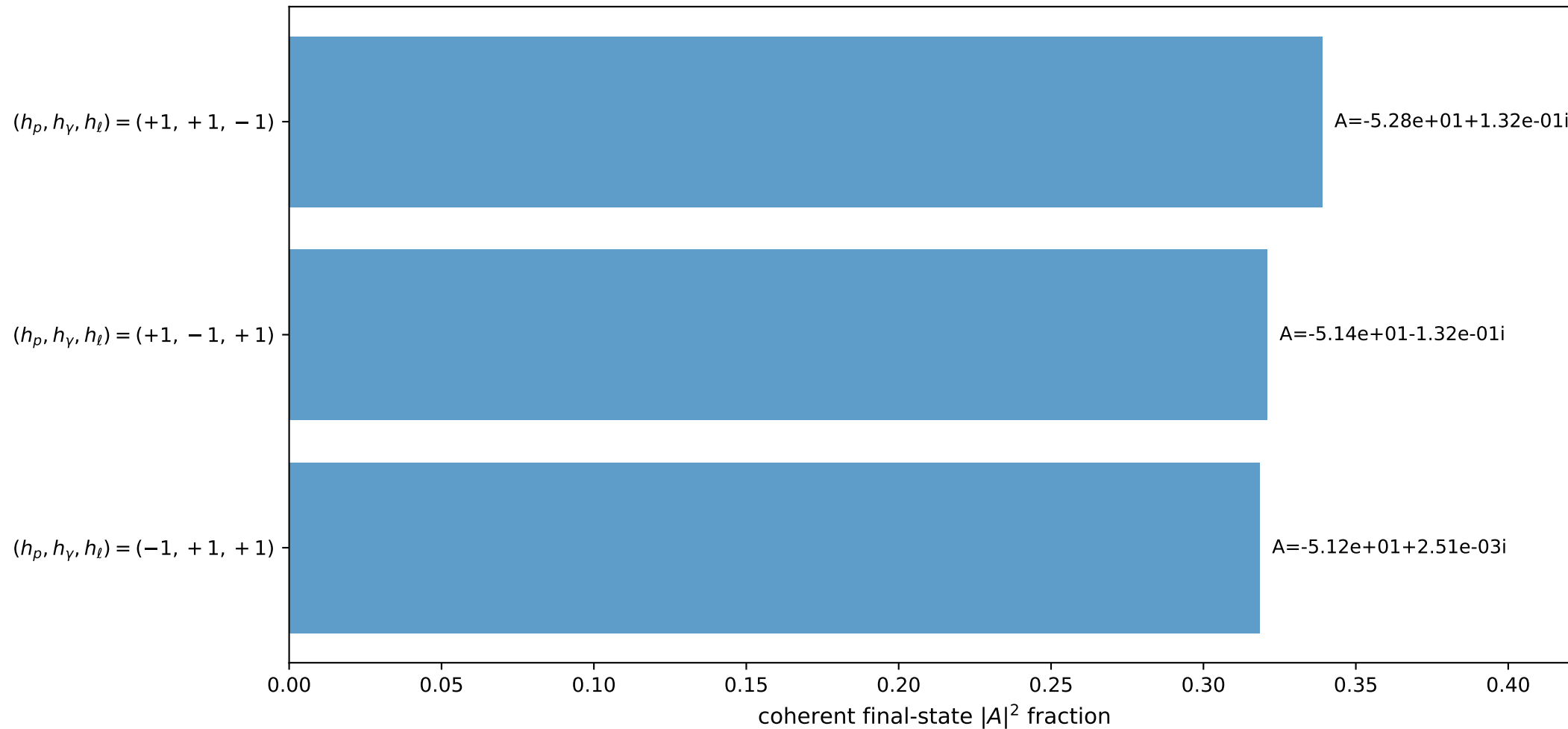
3D momenta



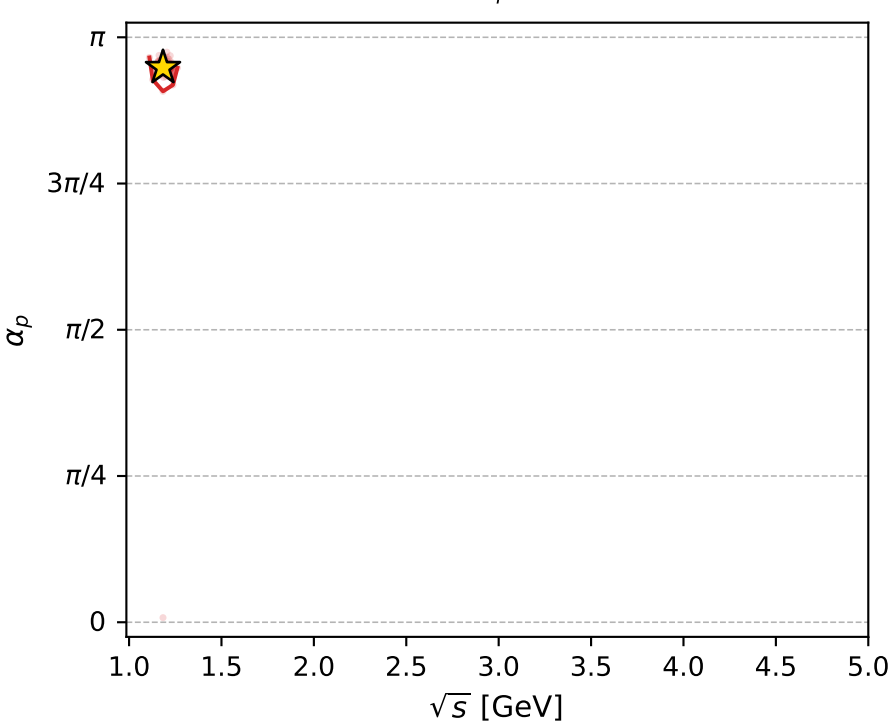
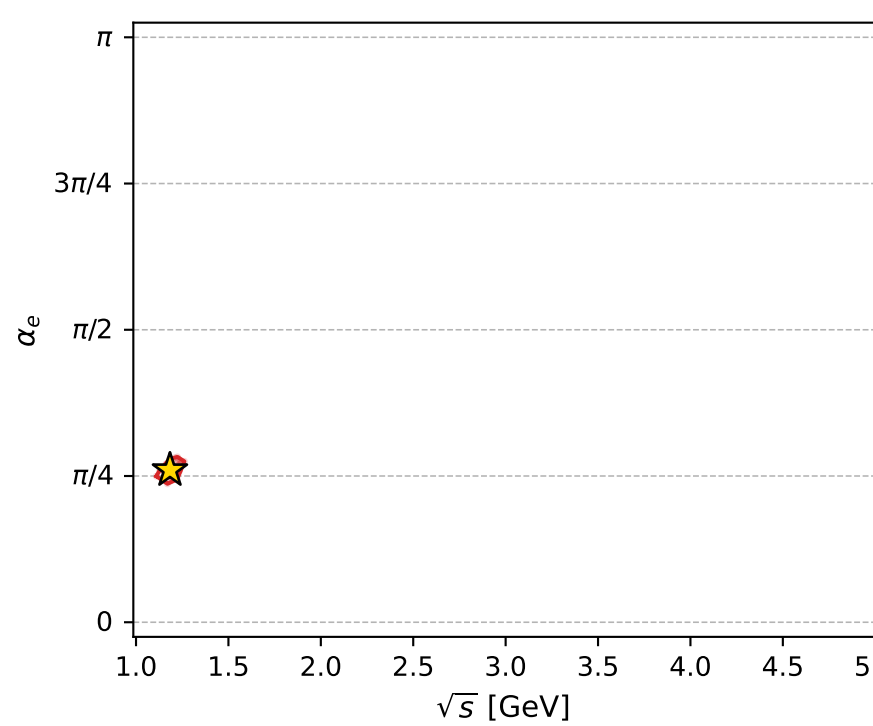
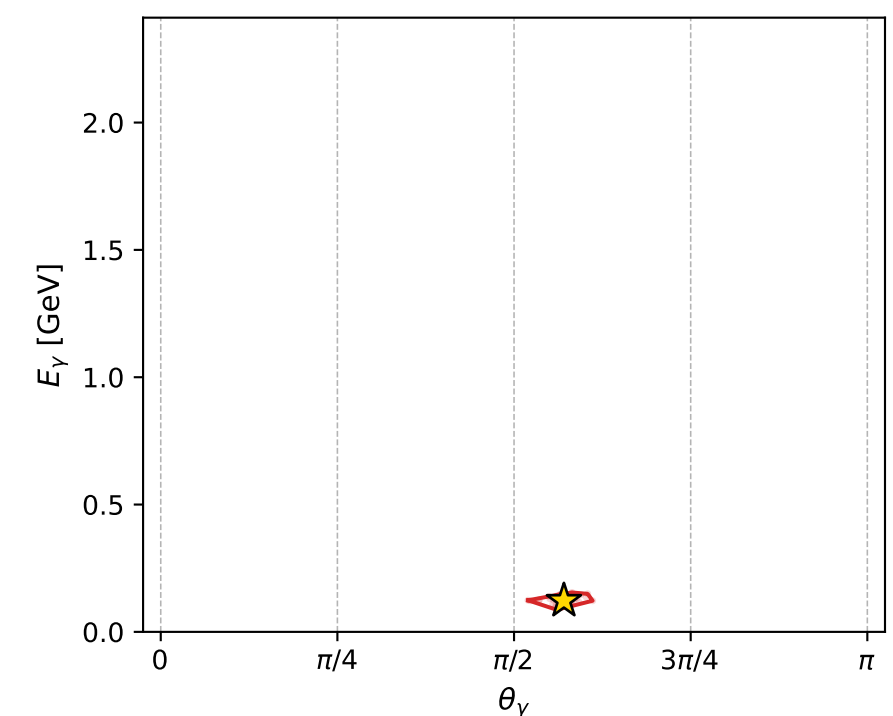
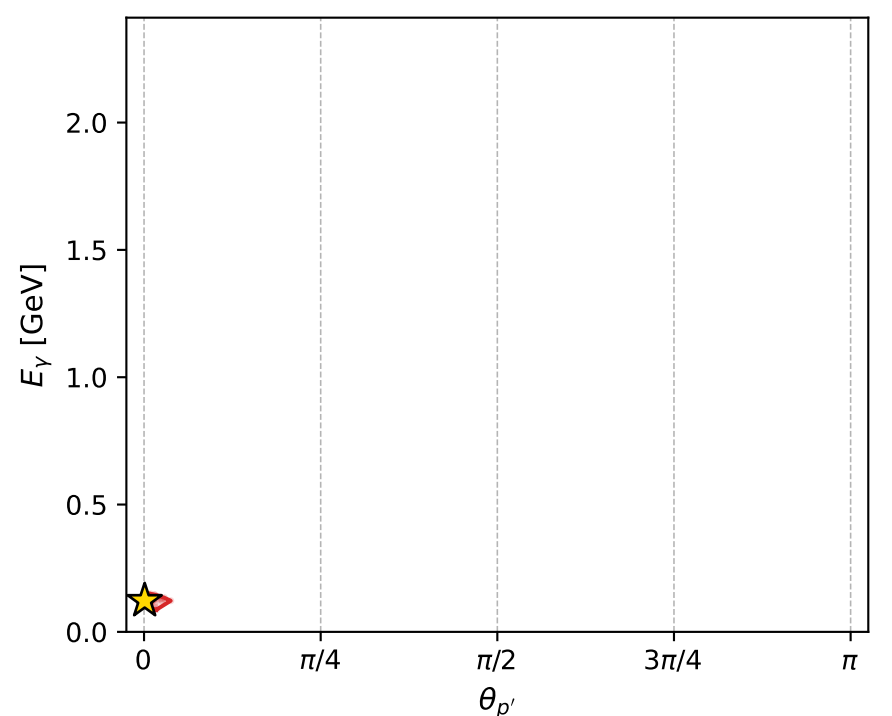
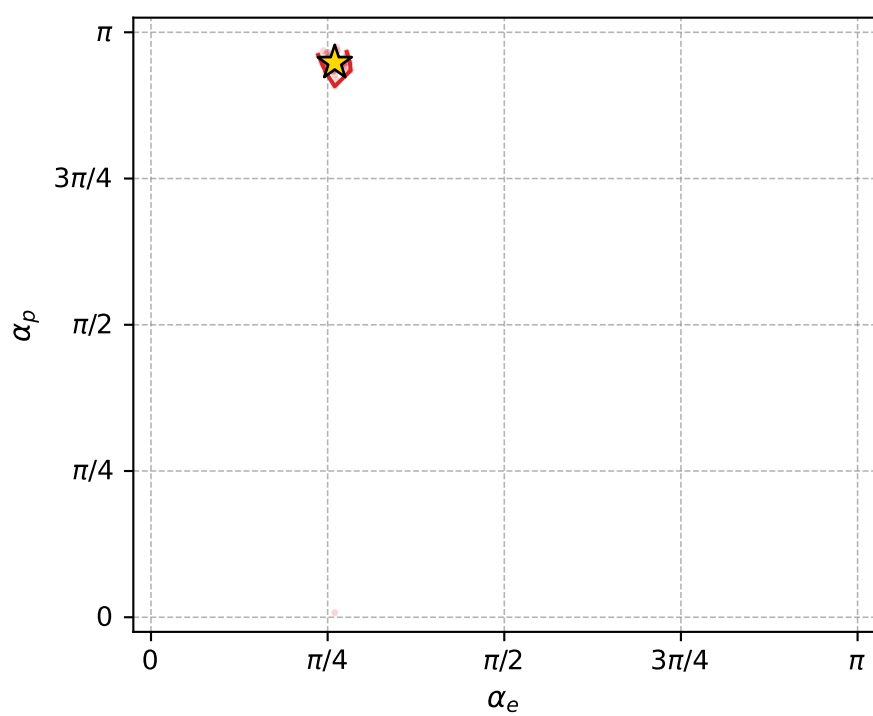
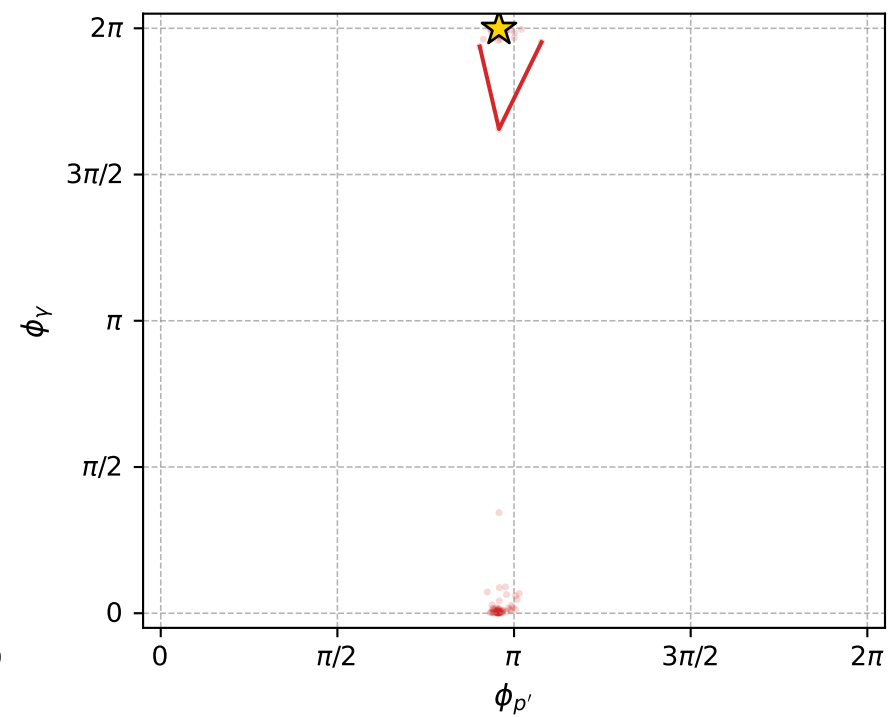
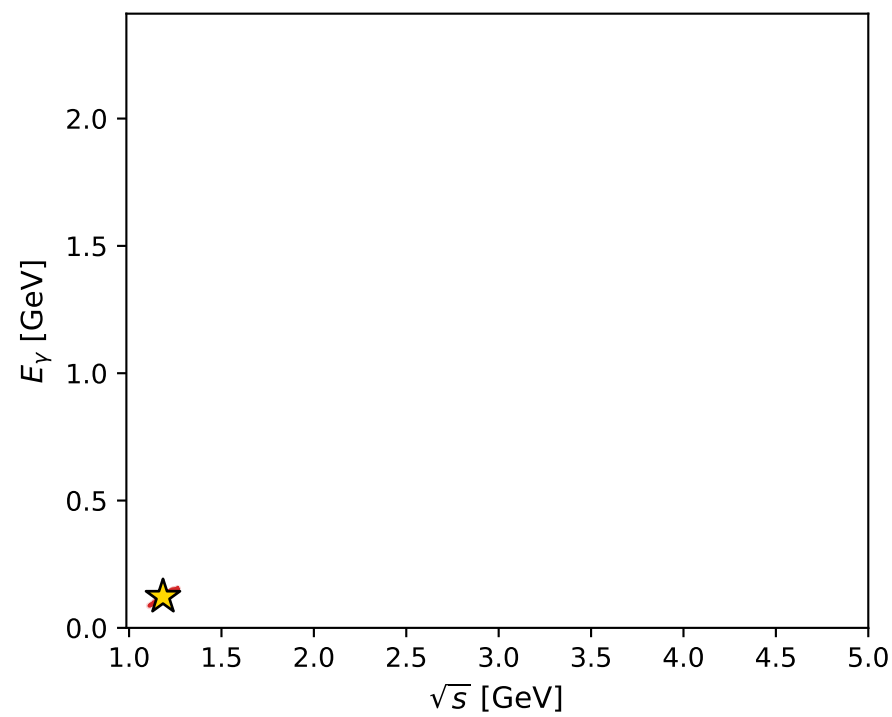
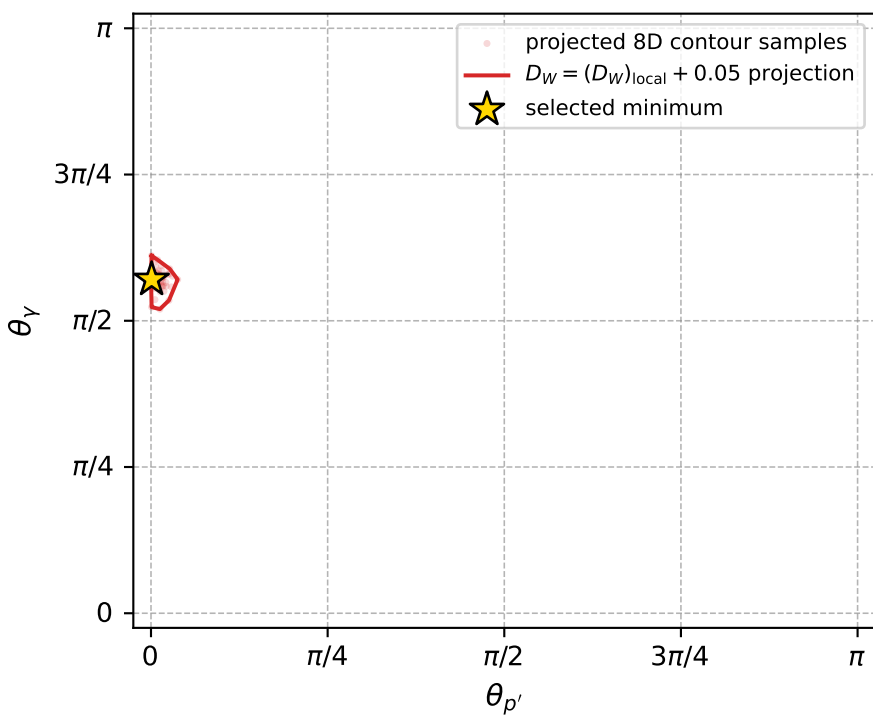
Transverse plane



Leading final-state amplitudes (N=3, retained=97.9%)



electron: polarization cluster P3, local minimum 885; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013070793$
 $D_W \text{ contour} = 0.063070793$
 above global minimum = 0.01305436
 8D contour samples = 165

selected phase-space point:
 $\text{sqrt_s} = 1.183615$
 $\text{theta_p_out} = 0.002781817$
 $\text{theta_gamma_out} = 1.792565$
 $\text{qOut} = 0.1222593$
 $\text{phi_p_out} = 3.007747$
 $\text{phi_gamma_out} = 6.280752$
 $\text{alpha_e} = 0.8169805$
 $\text{alpha_p} = 2.978624$

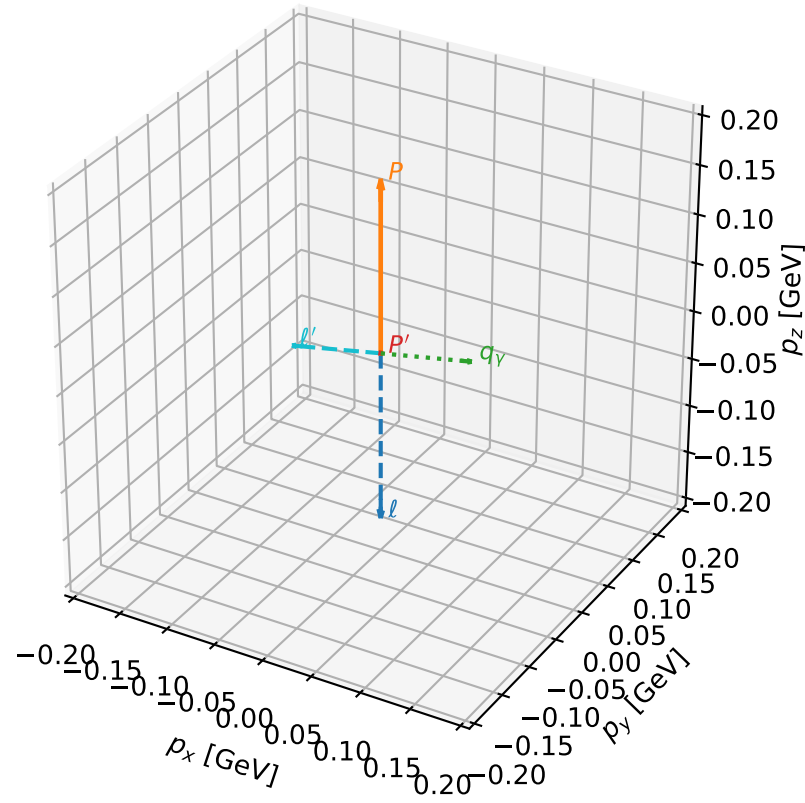
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_888 D_W=0.0132357
region: polarization_cluster_03_local_minimum_888
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81471988$ rad
incoming lepton: $0.686072|+\rangle +0.727533|-\rangle$
proton mixing angle: $\alpha_p=2.9588101$ rad
incoming proton: $-0.983342|+\rangle +0.181767|-\rangle$

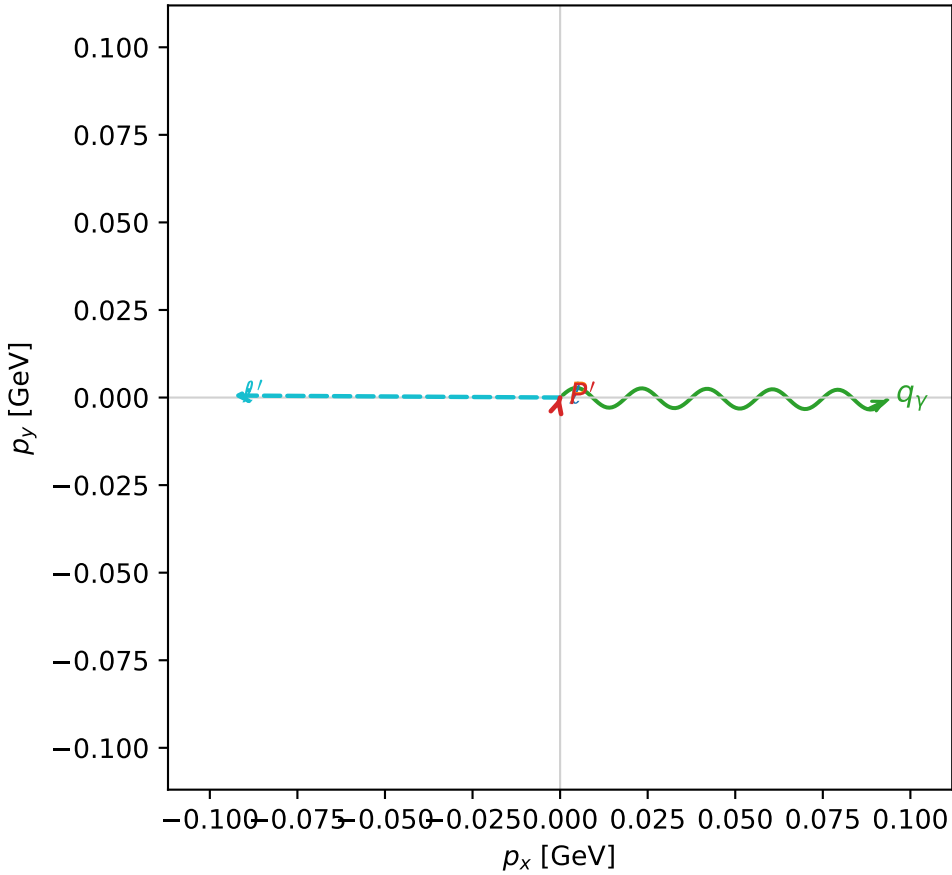
$s=1.27313$, $\sqrt{s}=1.12833$, $|\vec{P}|=0.174277$, $|\vec{P}'|=0.000416078$
 $\theta_{p'}=0.200961$, $\theta_Y=1.37346$, $\phi_{p'}=1.30137$, $\phi_Y=6.27608$
 $E_Y=0.0951137$, $\theta_{\ell'}=1.77228$, $\phi_{\ell'}=3.13535$
 $Q^2=0.0265461$, $x_B=0.129518$, $t=-0.029973$, $W^2=1.05826$, $y=0.521148$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1743$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1743$
 P $E= 0.9541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1743$
 ℓ' $E= 0.0952$ $p_x= -0.0933$ $p_y= 0.0006$ $p_z= -0.0191$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0001$ $p_z= 0.0004$
 q_Y $E= 0.0951$ $p_x= 0.0933$ $p_y= -0.0007$ $p_z= 0.0186$

3D momenta



Transverse plane

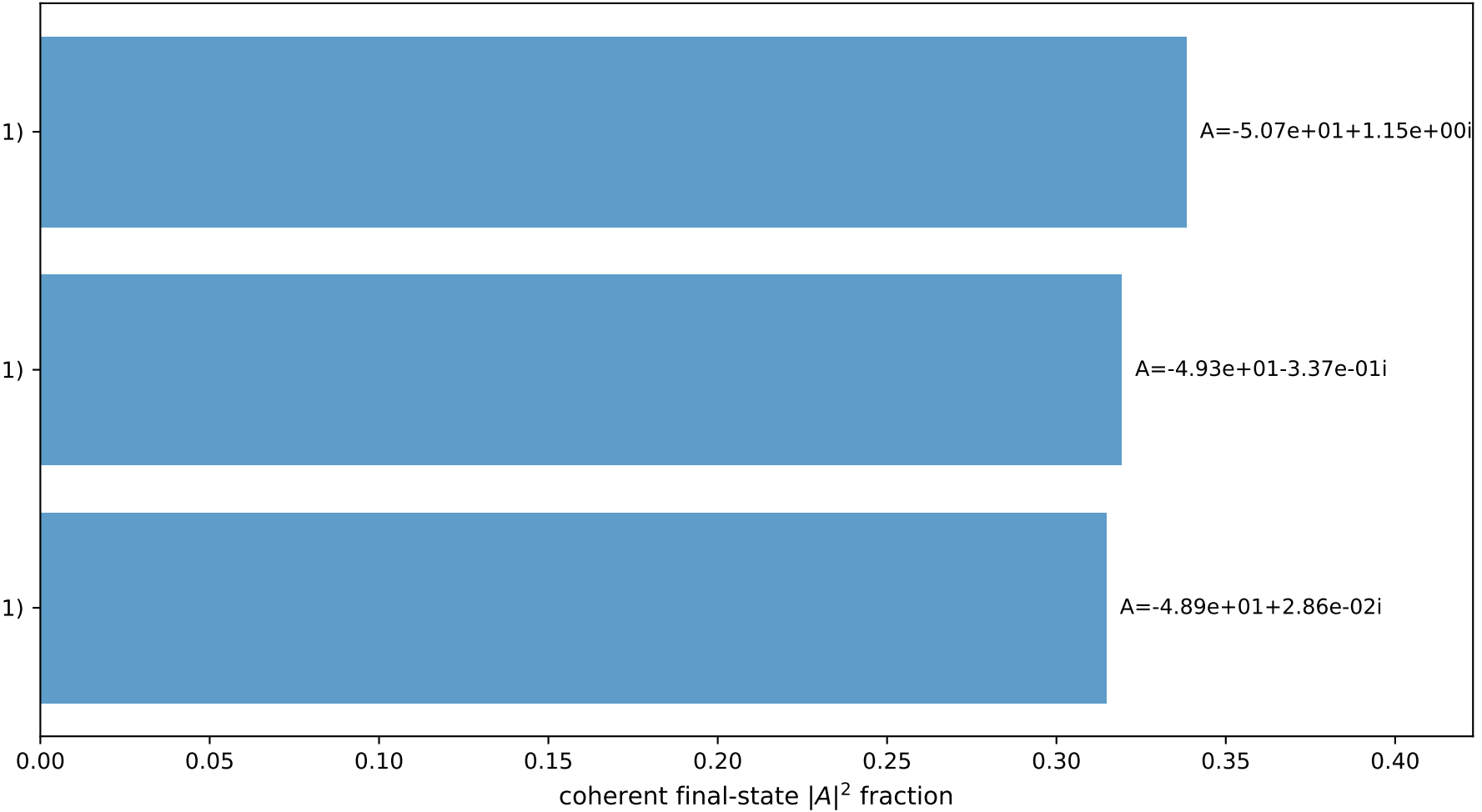


$(h_p, h_Y, h_{\ell}) = (+1, +1, -1)$

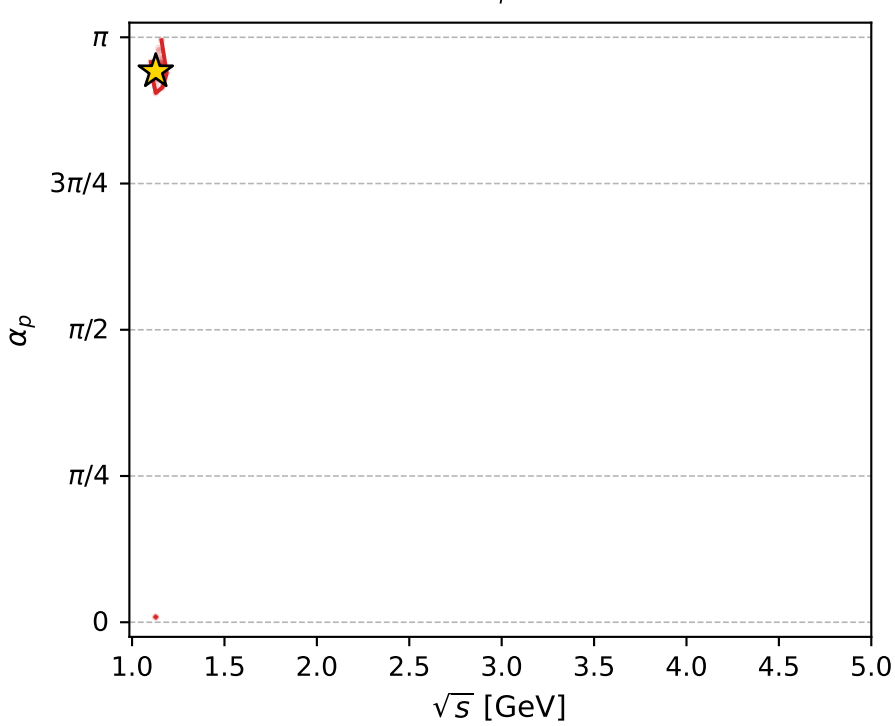
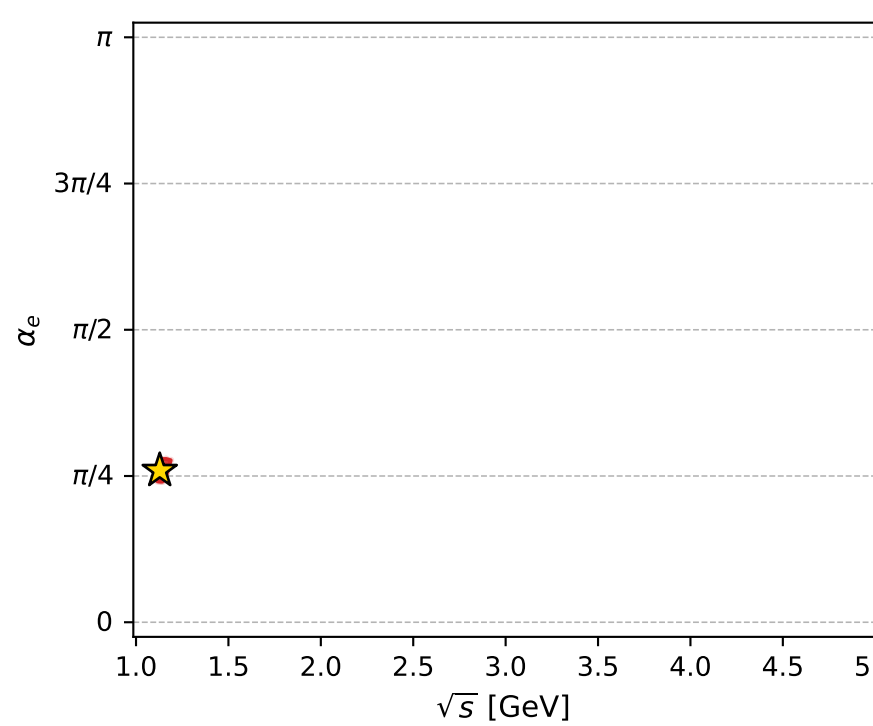
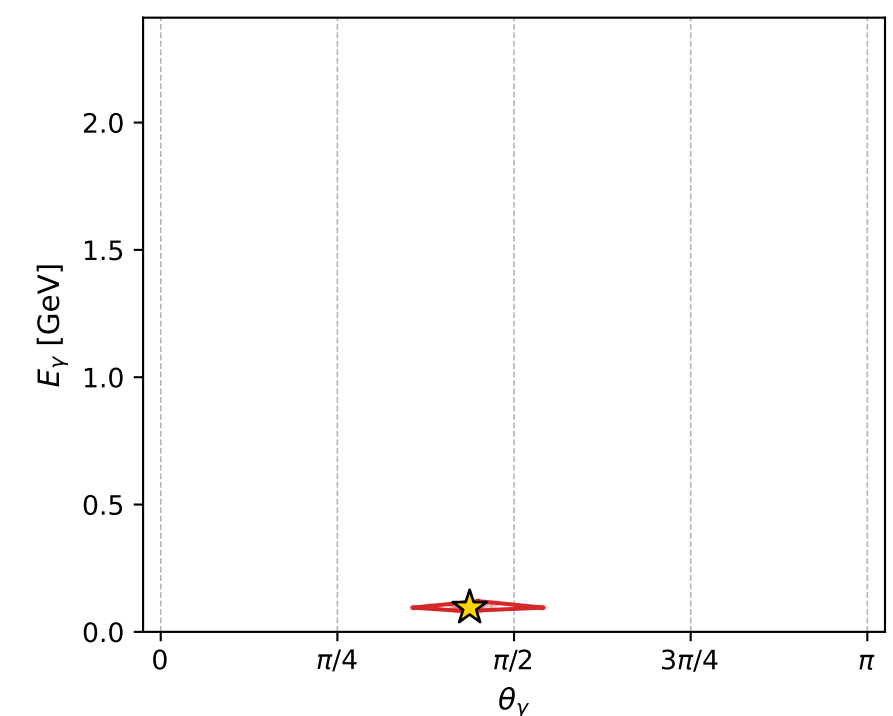
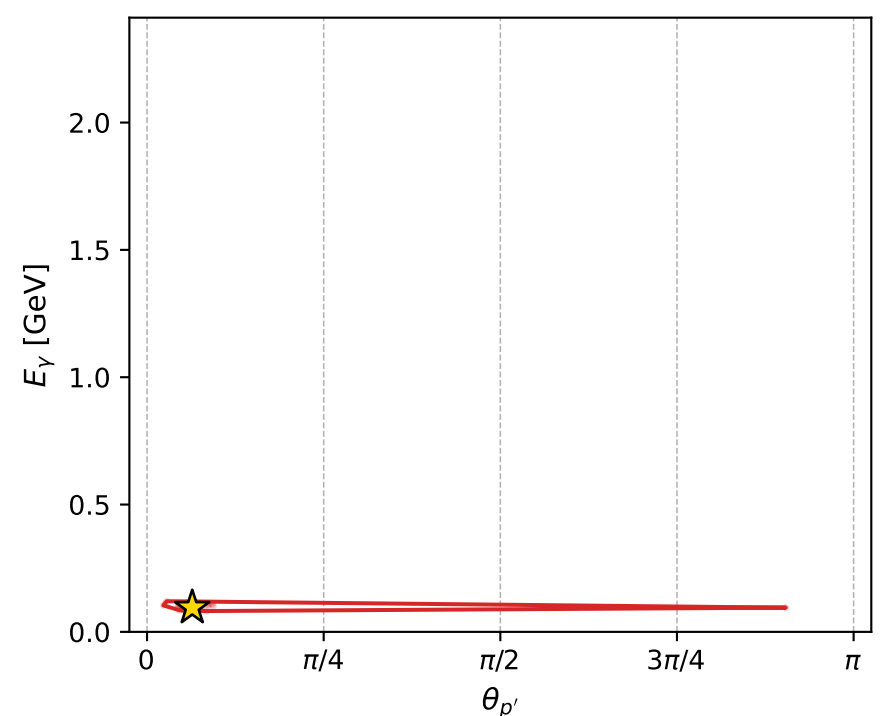
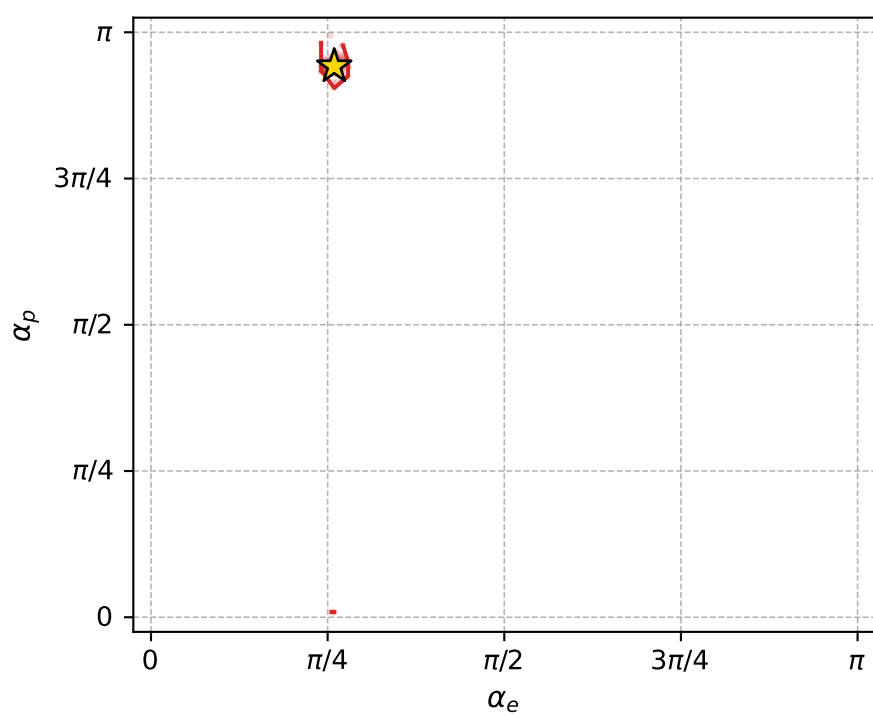
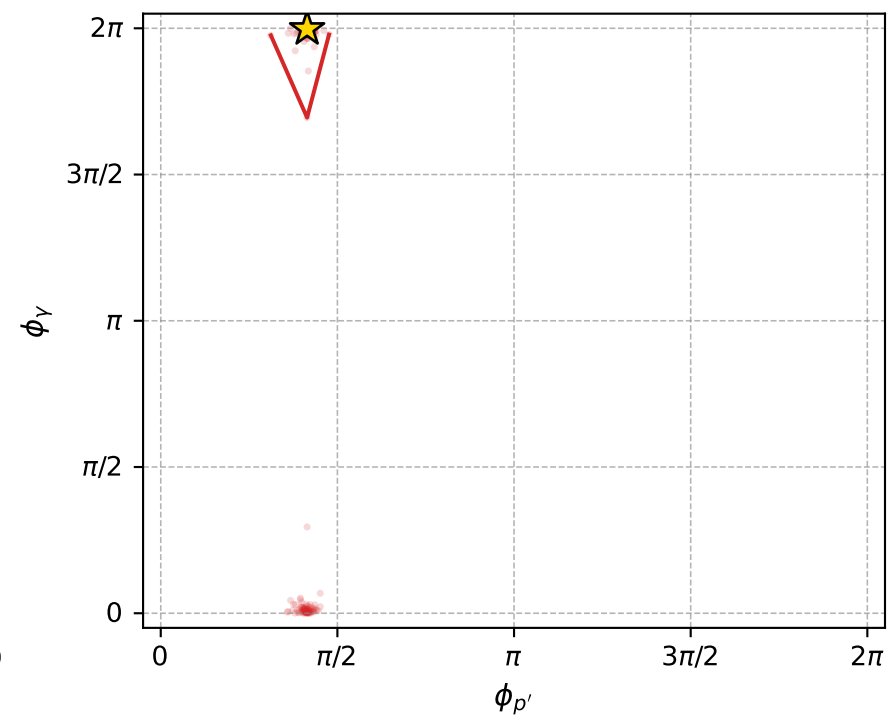
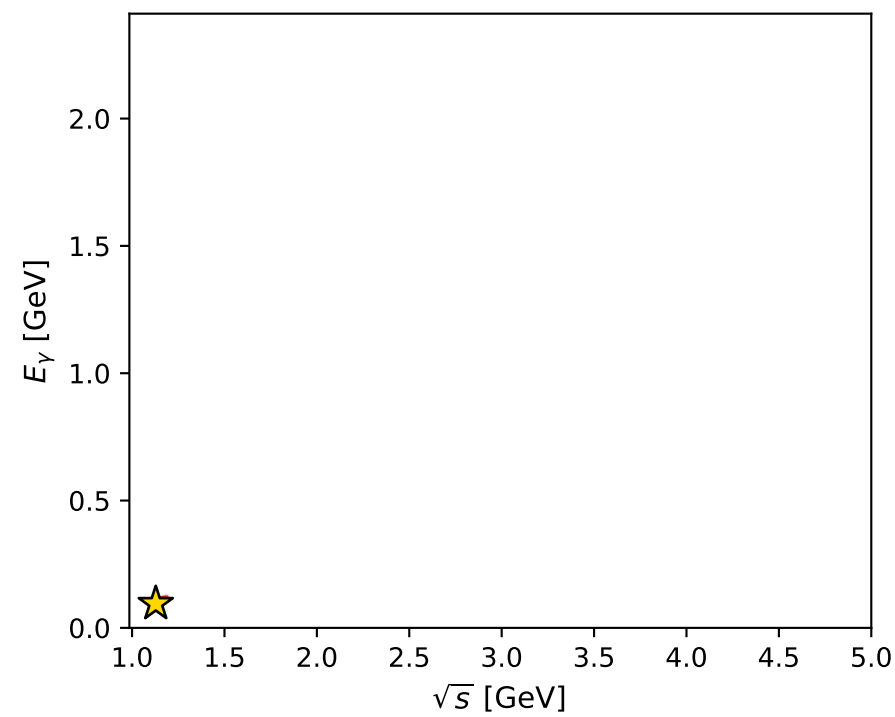
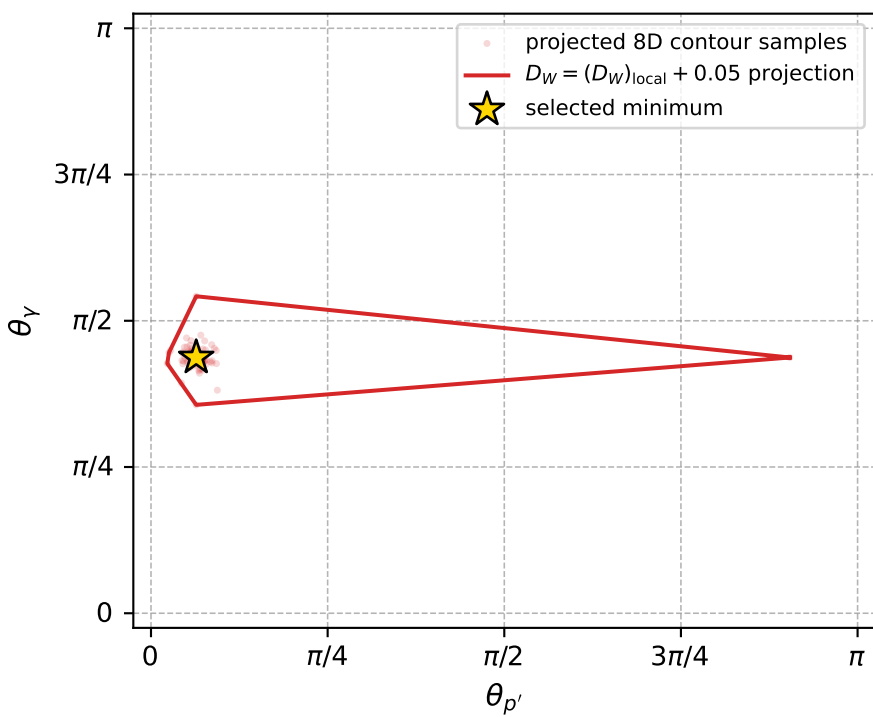
$(h_p, h_Y, h_{\ell}) = (+1, -1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, +1)$

Leading final-state amplitudes (N=3, retained=97.2%)



electron: polarization cluster P3, local minimum 888; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.01323573$
 $D_W \text{ contour} = 0.06323573$
 above global minimum = 0.013219297
 8D contour samples = 253

selected phase-space point:
 $\text{sqrt_s} = 1.128331$
 $\text{theta_p_out} = 0.2009611$
 $\text{theta_gamma_out} = 1.373462$
 $\text{qOut} = 0.0951137$
 $\text{phi_p_out} = 1.301366$
 $\text{phi_gamma_out} = 6.276082$
 $\text{alpha_e} = 0.8147199$
 $\text{alpha_p} = 2.95881$

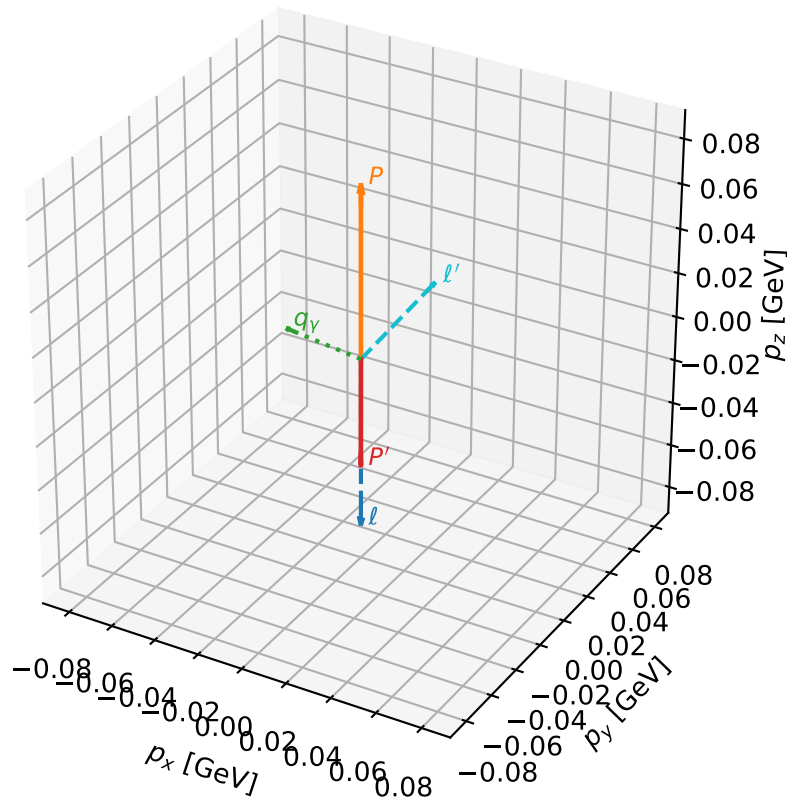
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_893 D_W=0.013397
region: polarization_cluster_03_local_minimum_893
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.73146653$ rad
incoming lepton: $0.744196|+\rangle + 0.667962|-\rangle$
proton mixing angle: $\alpha_p=0.19324878$ rad
incoming proton: $0.981385|+\rangle + 0.192048|-\rangle$

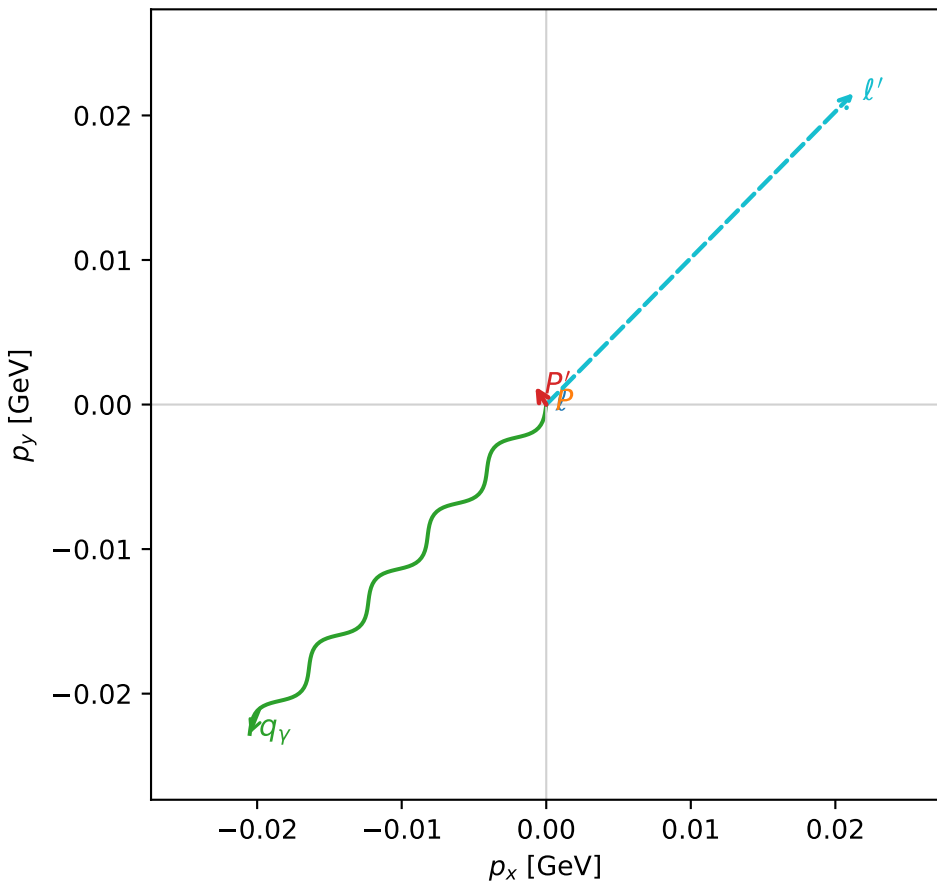
$s=1.03712$, $\sqrt{s}=1.01839$, $|\vec{P}|=0.0772153$, $|\vec{P}'|=0.049674$
 $\theta_{P'}=3.11242$, $\theta_V=1.00567$, $\phi_{P'}=2.11696$, $\phi_V=3.97936$
 $E_V=0.0363025$, $\theta_{\ell'}=0.786413$, $\phi_{\ell'}=0.791905$
 $Q^2=0.0112707$, $x_B=0.138417$, $t=-0.0160942$, $W^2=0.949999$, $y=0.517734$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0772$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0772$
 P $E= 0.9412$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0772$
 ℓ' $E= 0.0428$ $p_x= 0.0213$ $p_y= 0.0215$ $p_z= 0.0302$
 P' $E= 0.9393$ $p_x= -0.0008$ $p_y= 0.0012$ $p_z= -0.0497$
 q_Y $E= 0.0363$ $p_x= -0.0205$ $p_y= -0.0228$ $p_z= 0.0194$

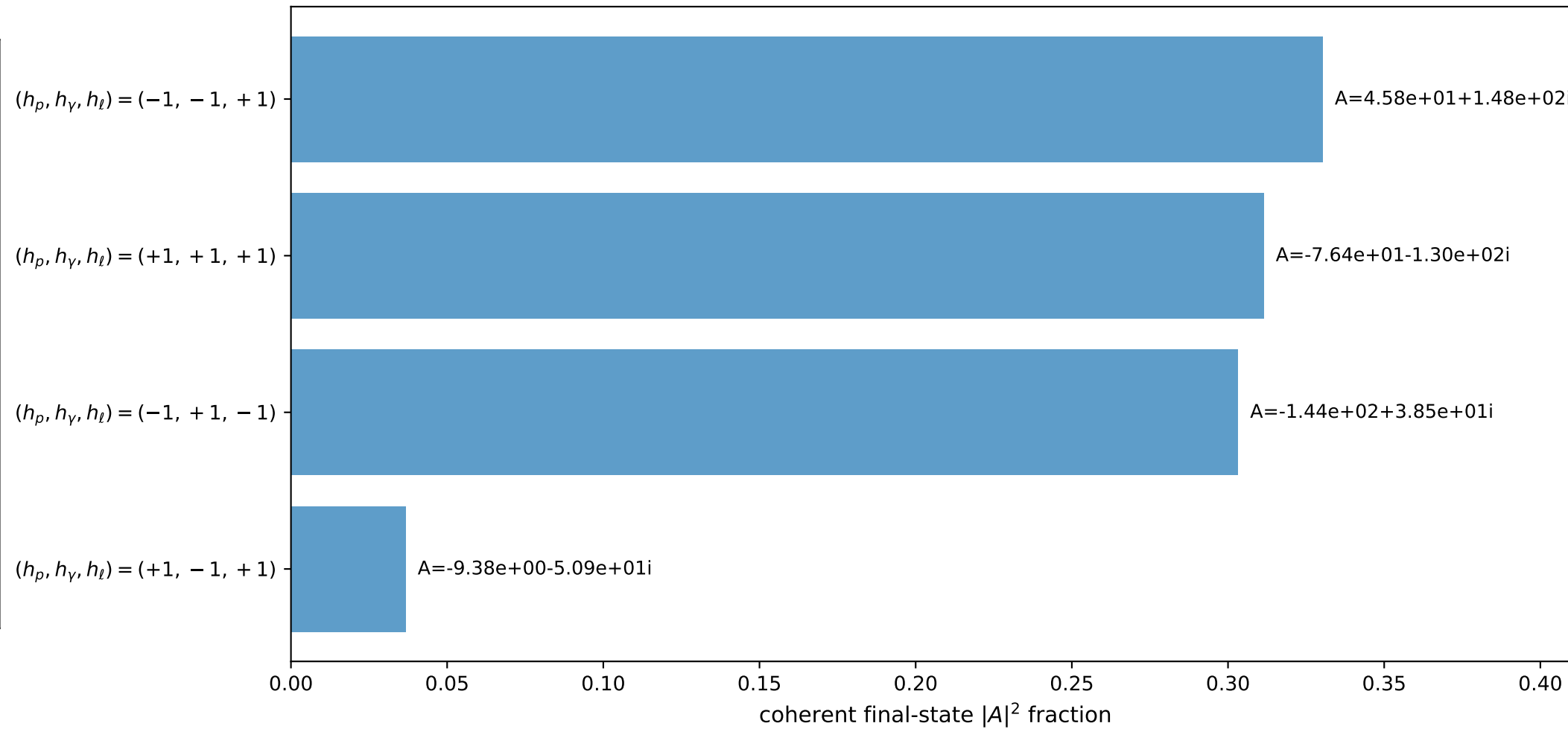
3D momenta



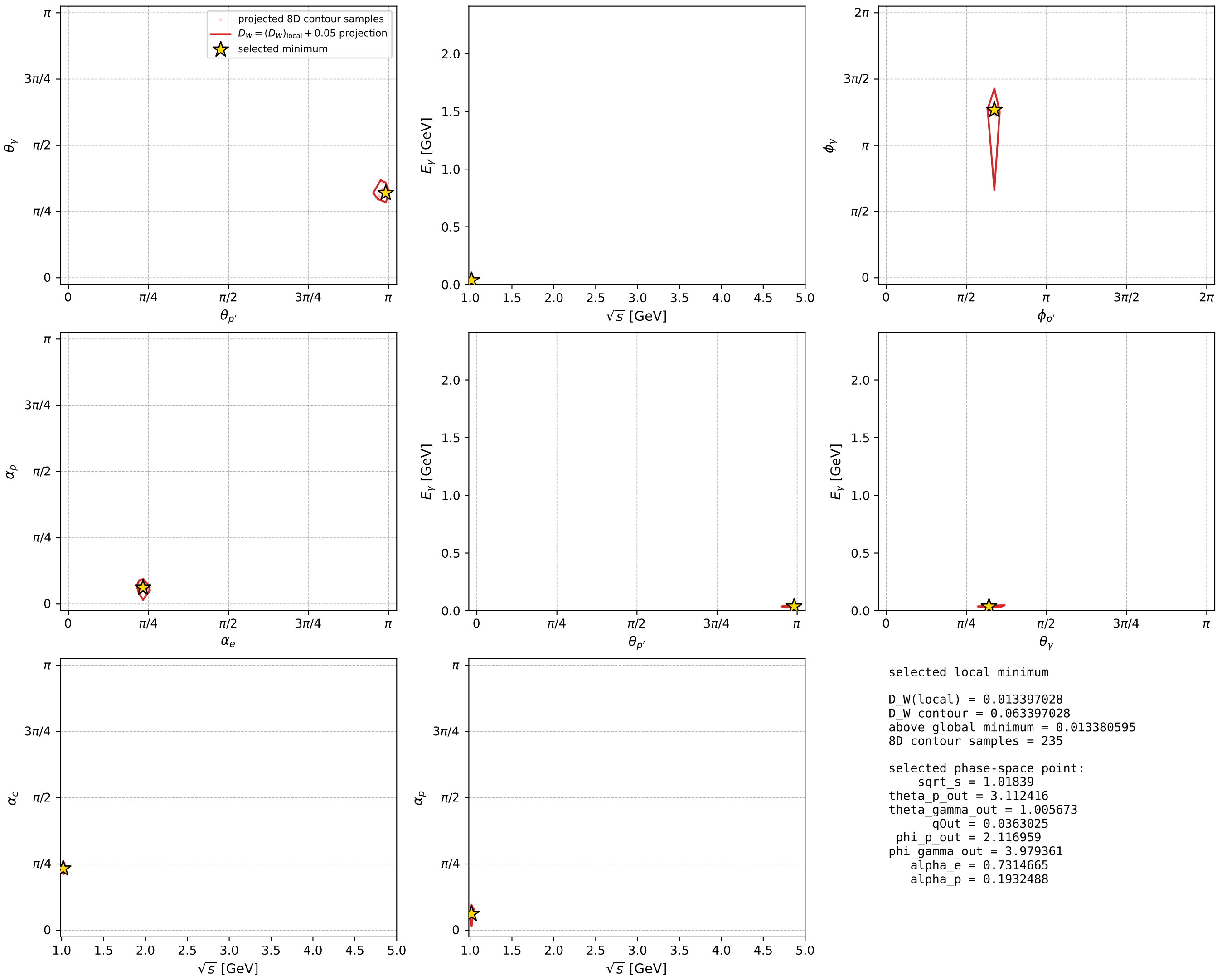
Transverse plane



Leading final-state amplitudes (N=4, retained=98.2%)



electron: polarization cluster P3, local minimum 893; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



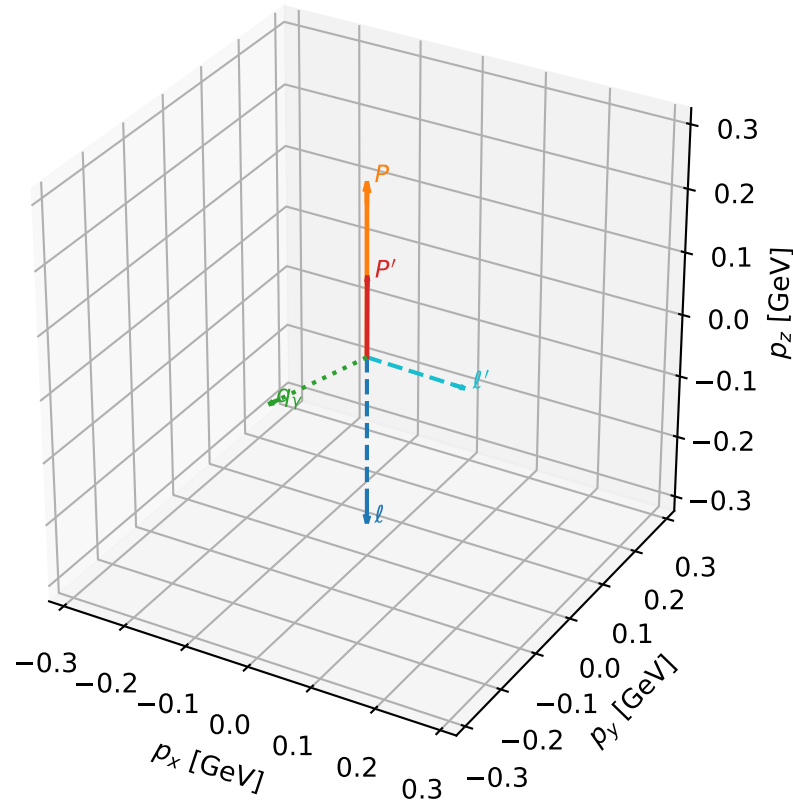
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_895 D_W=0.0135443
 region: polarization_cluster_03_local_minimum_895
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80868274$ rad
 incoming lepton: $0.690452|+\rangle + 0.723378|-\rangle$
 proton mixing angle: $\alpha_p=0.12832272$ rad
 incoming proton: $0.991778|+\rangle + 0.127971|-\rangle$

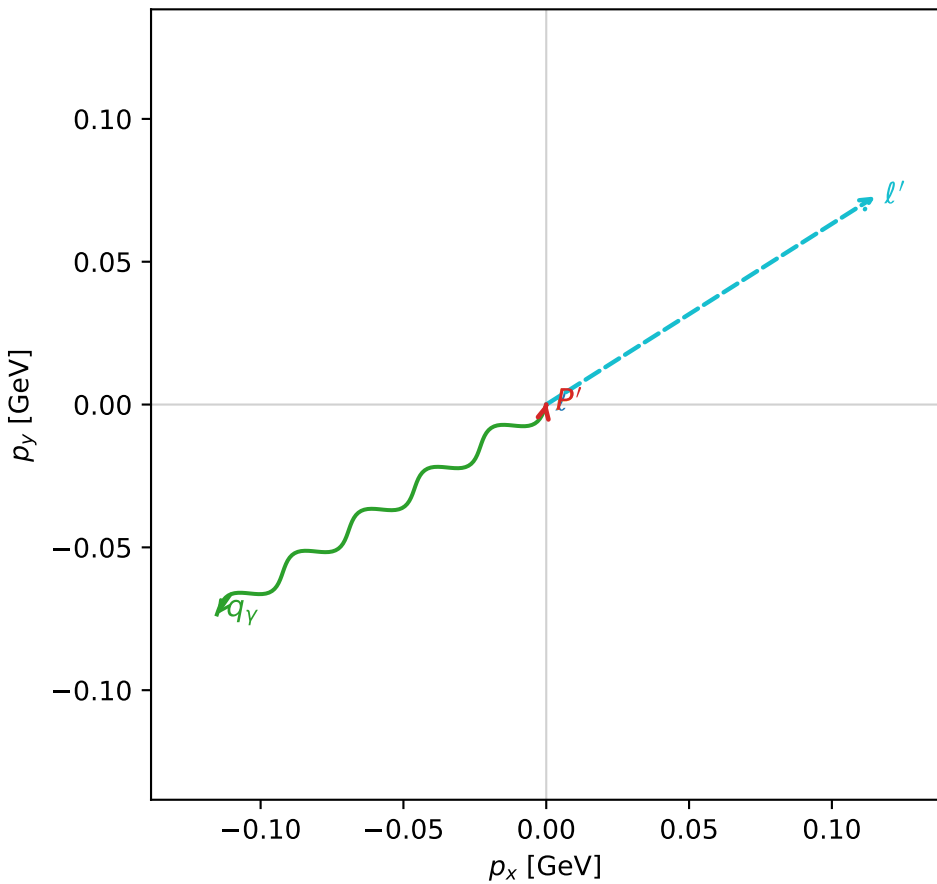
$s=1.55577$, $\sqrt{s}=1.24731$, $|\vec{P}|=0.270955$, $|\vec{P}'|=0.125499$
 $\theta_{P'}=0.00511446$, $\theta_Y=2.05708$, $\phi_{P'}=1.35178$, $\phi_Y=3.70899$
 $E_Y=0.154652$, $\theta_{\ell'}=1.94314$, $\phi_{\ell'}=0.564066$
 $Q^2=0.0504375$, $x_B=0.139555$, $t=-0.0202588$, $W^2=1.19082$, $y=0.534694$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2710$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2710$
 P $E= 0.9764$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2710$
 ℓ' $E= 0.1463$ $p_x= 0.1152$ $p_y= 0.0729$ $p_z= -0.0532$
 P' $E= 0.9464$ $p_x= 0.0001$ $p_y= 0.0006$ $p_z= 0.1255$
 q_Y $E= 0.1547$ $p_x= -0.1153$ $p_y= -0.0735$ $p_z= -0.0723$

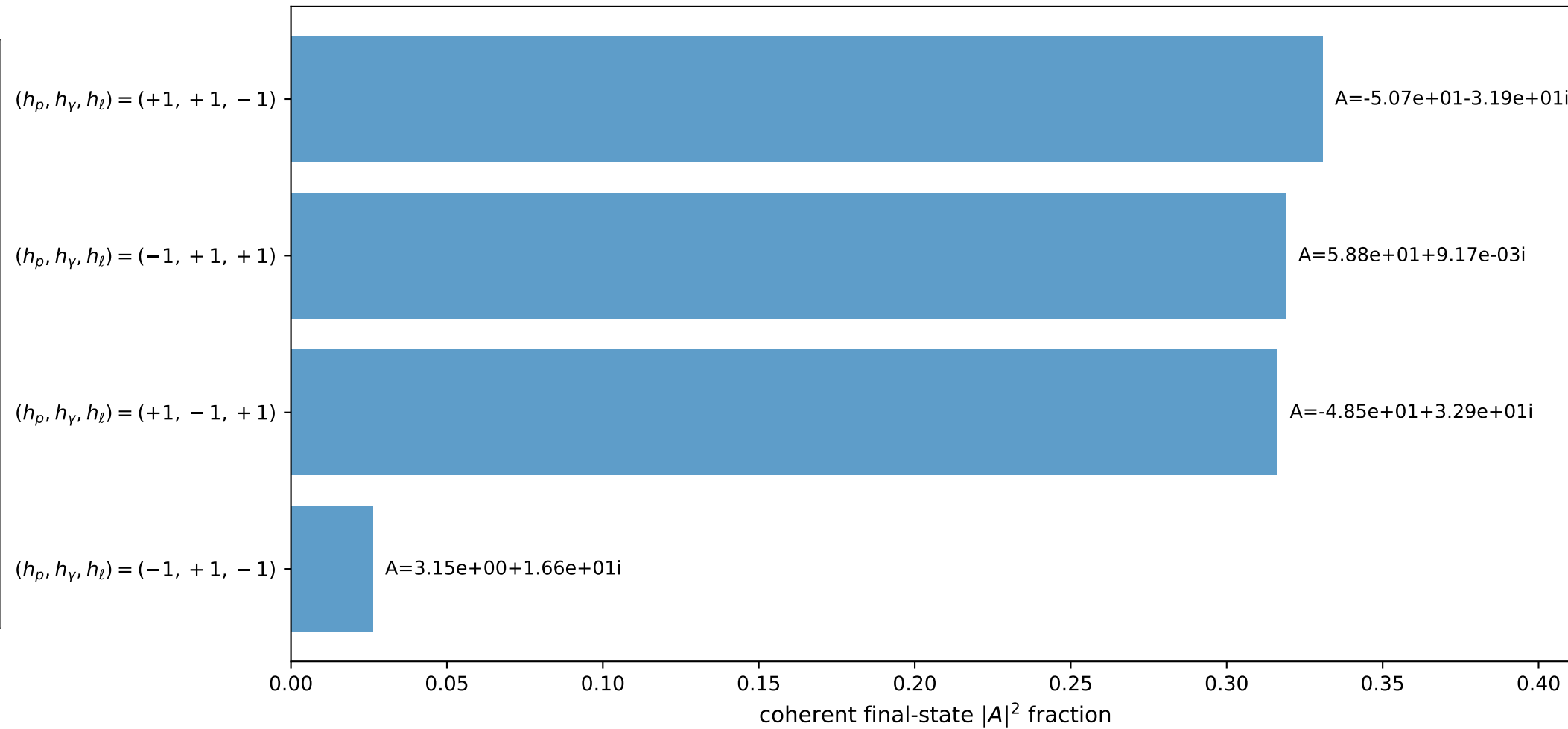
3D momenta



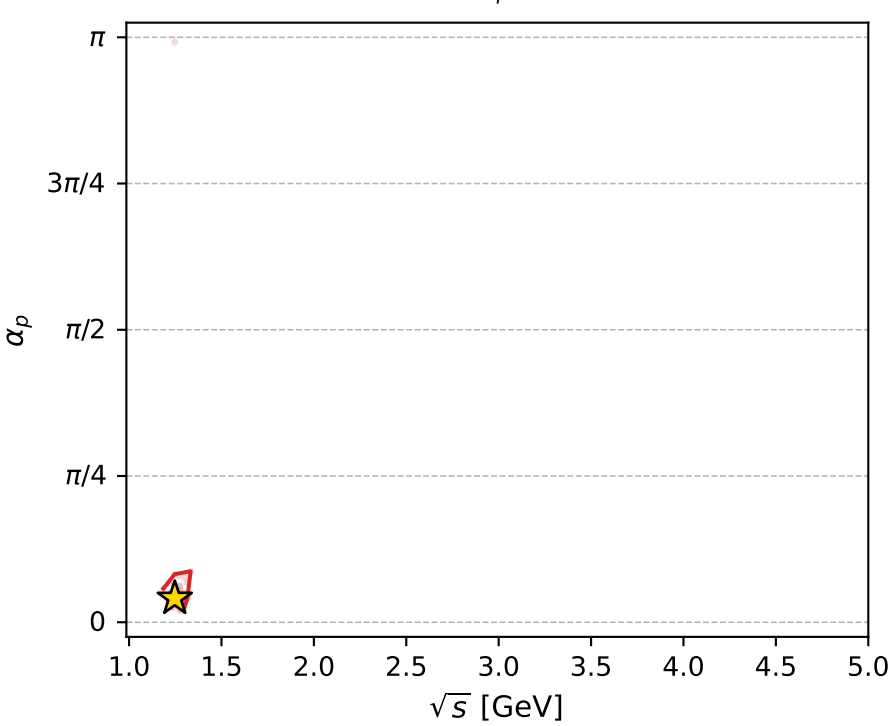
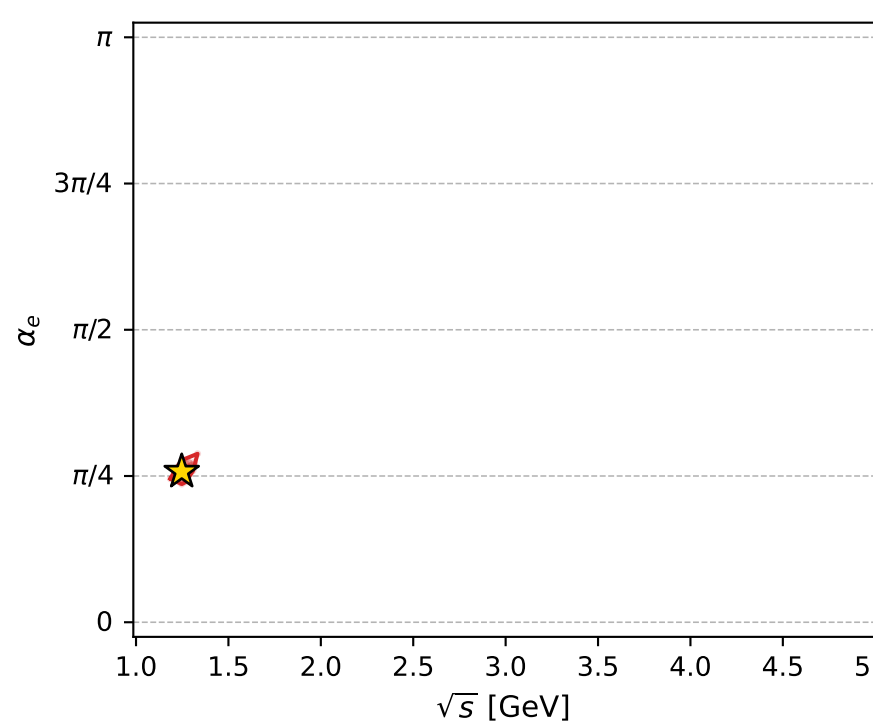
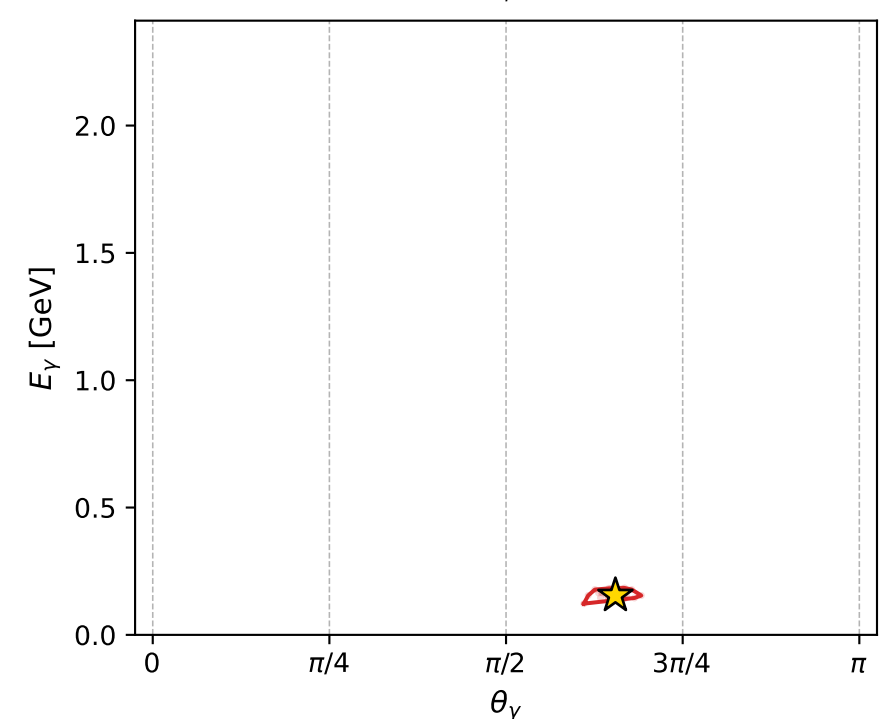
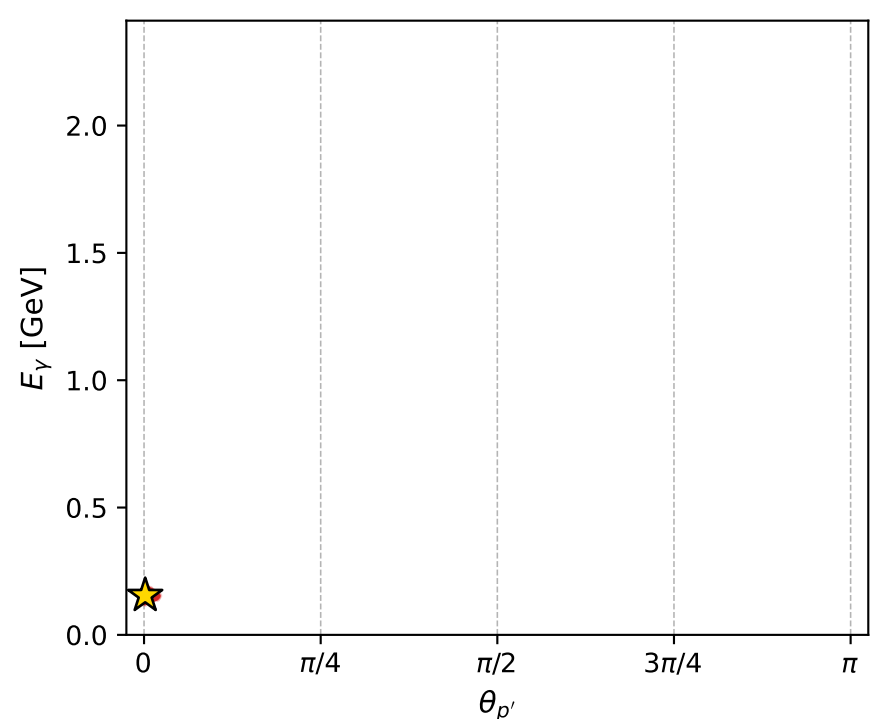
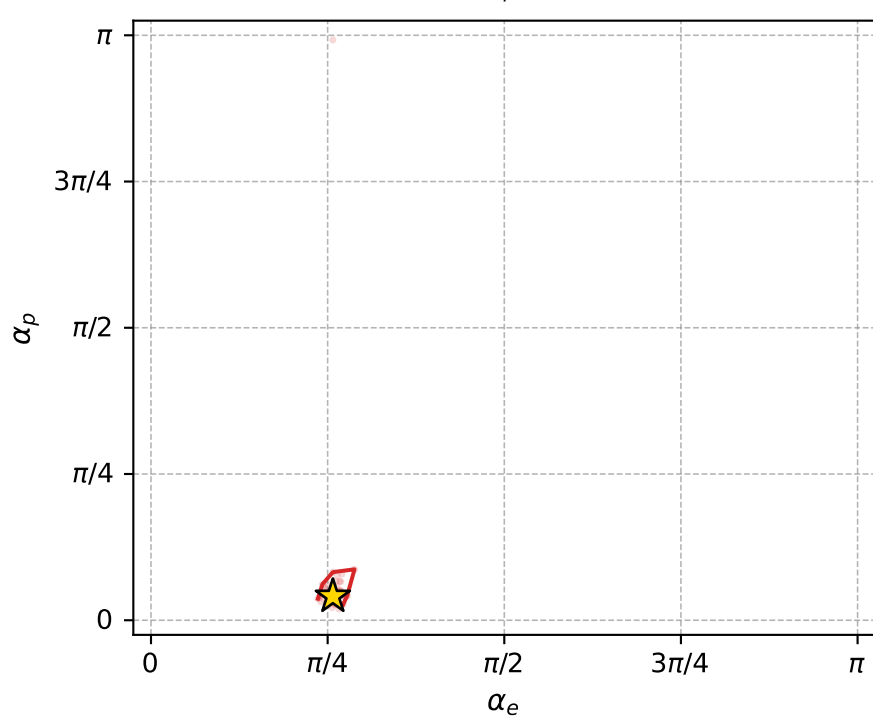
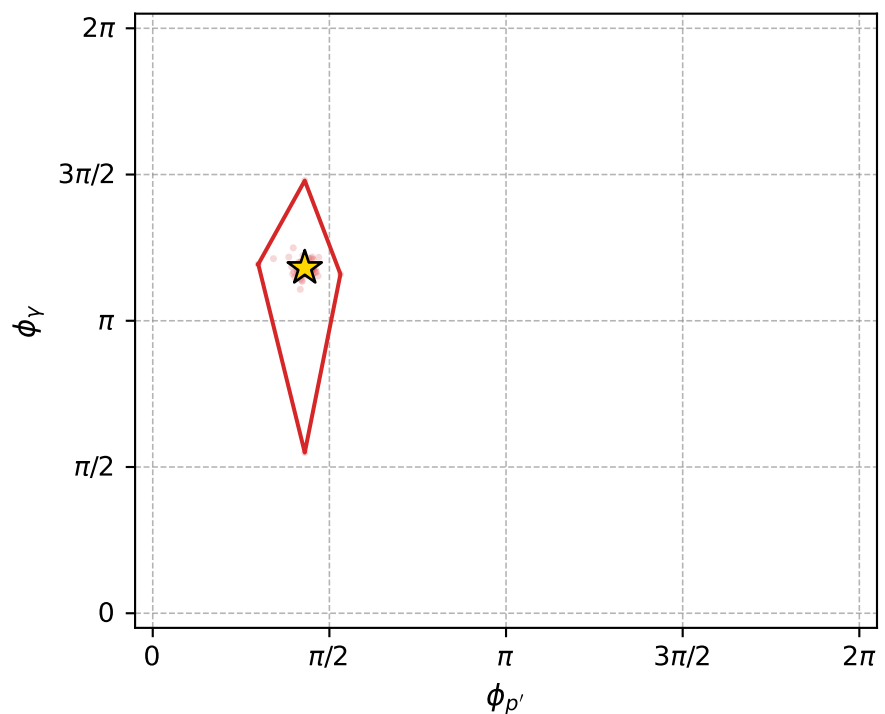
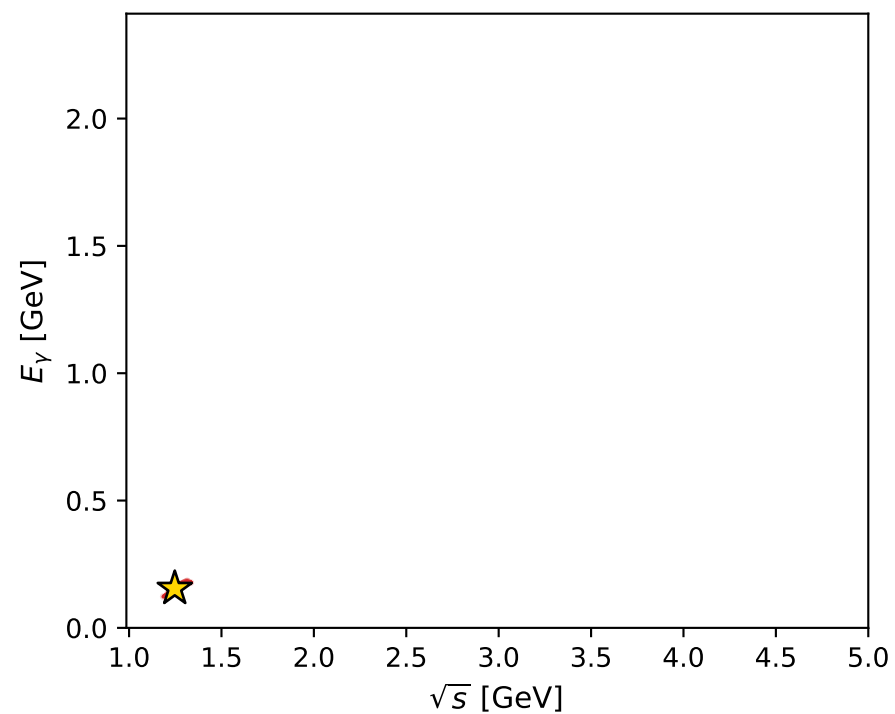
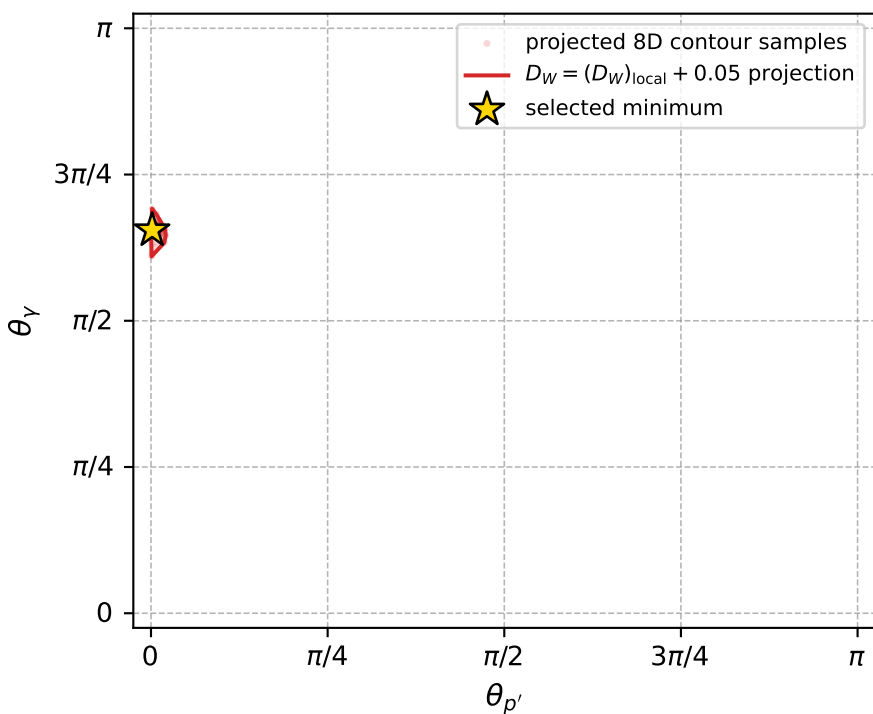
Transverse plane



Leading final-state amplitudes (N=4, retained=99.3%)



electron: polarization cluster P3, local minimum 895; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013544267$
 $D_W \text{ contour} = 0.063544267$
 above global minimum = 0.013527834
 8D contour samples = 200

selected phase-space point:

$\text{sqrt_s} = 1.247307$
 $\text{theta_p_out} = 0.005114462$
 $\text{theta_gamma_out} = 2.057079$
 $\text{qOut} = 0.1546522$
 $\text{phi_p_out} = 1.351775$
 $\text{phi_gamma_out} = 3.708986$
 $\text{alpha_e} = 0.8086827$
 $\text{alpha_p} = 0.1283227$

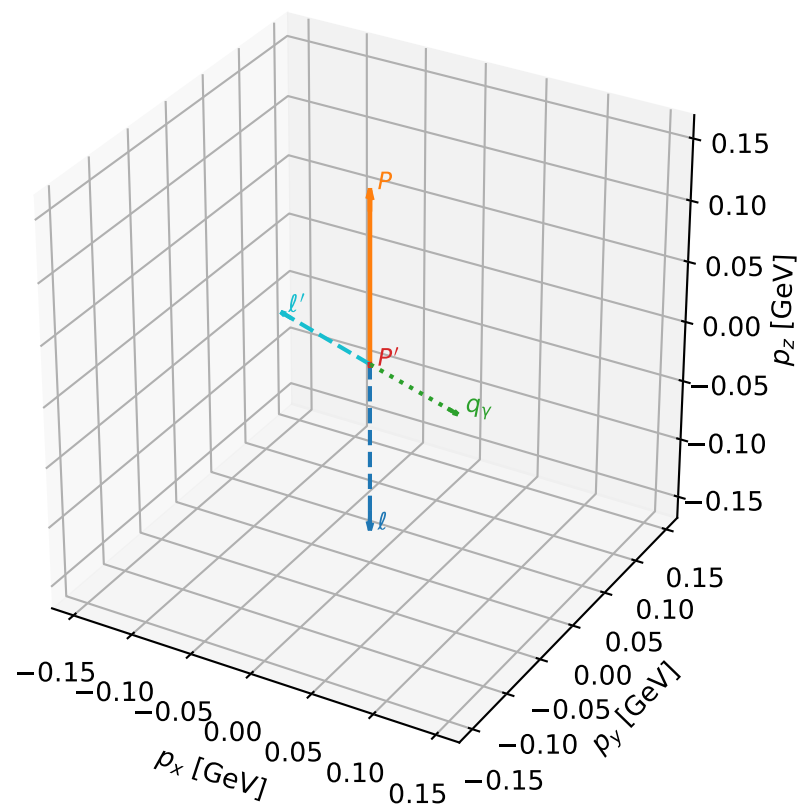
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_901 D_W=0.0136235
 region: polarization_cluster_03_local_minimum_901
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76804845$ rad
 incoming lepton: $0.719268|+\rangle + 0.694733|-\rangle$
 proton mixing angle: $\alpha_p=3.0166874$ rad
 incoming proton: $-0.992209|+\rangle + 0.124581|-\rangle$

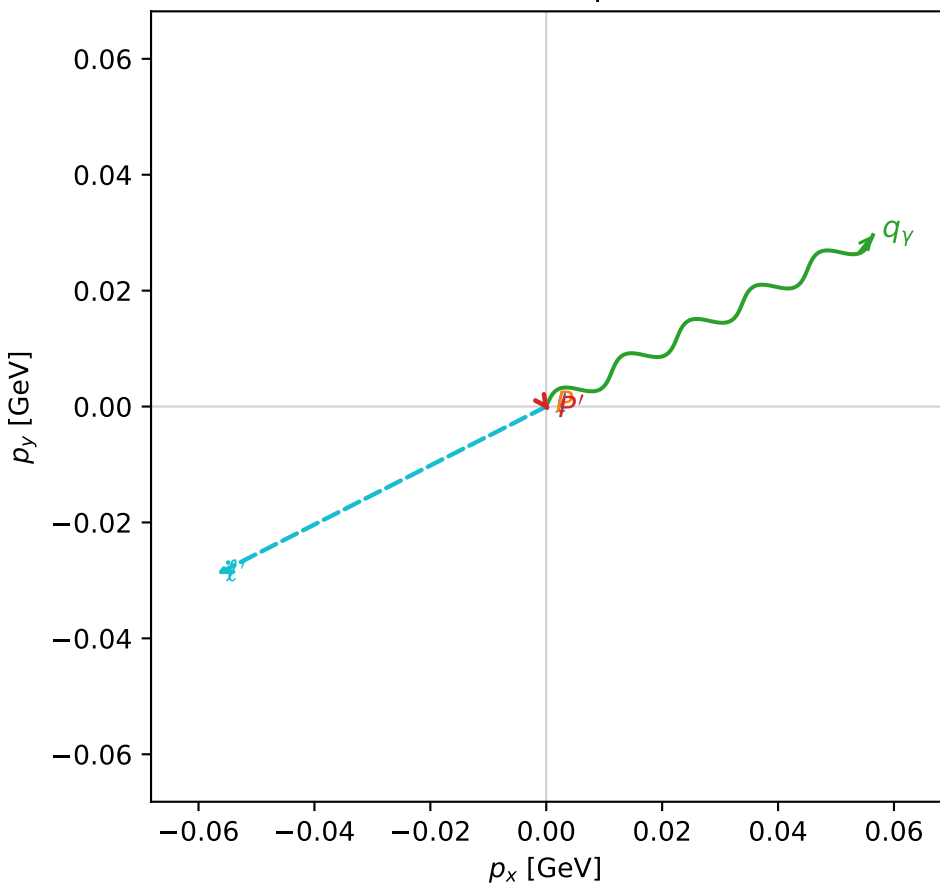
$s=1.18644$, $\sqrt{s}=1.08924$, $|\vec{P}|=0.140739$, $|\vec{P}'|=0.00111465$
 $\theta_{P'}=2.34702$, $\theta_Y=2.1358$, $\phi_{P'}=5.33854$, $\phi_Y=0.483143$
 $E_Y=0.0753584$, $\theta_{\ell'}=0.997938$, $\phi_{\ell'}=3.61239$
 $Q^2=0.0329351$, $x_B=0.189033$, $t=-0.0199183$, $W^2=1.02114$, $y=0.568268$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1407$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1407$
 P $E= 0.9485$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1407$
 ℓ' $E= 0.0759$ $p_x= -0.0568$ $p_y= -0.0289$ $p_z= 0.0411$
 P' $E= 0.9380$ $p_x= 0.0005$ $p_y= -0.0006$ $p_z= -0.0008$
 q_Y $E= 0.0754$ $p_x= 0.0564$ $p_y= 0.0296$ $p_z= -0.0403$

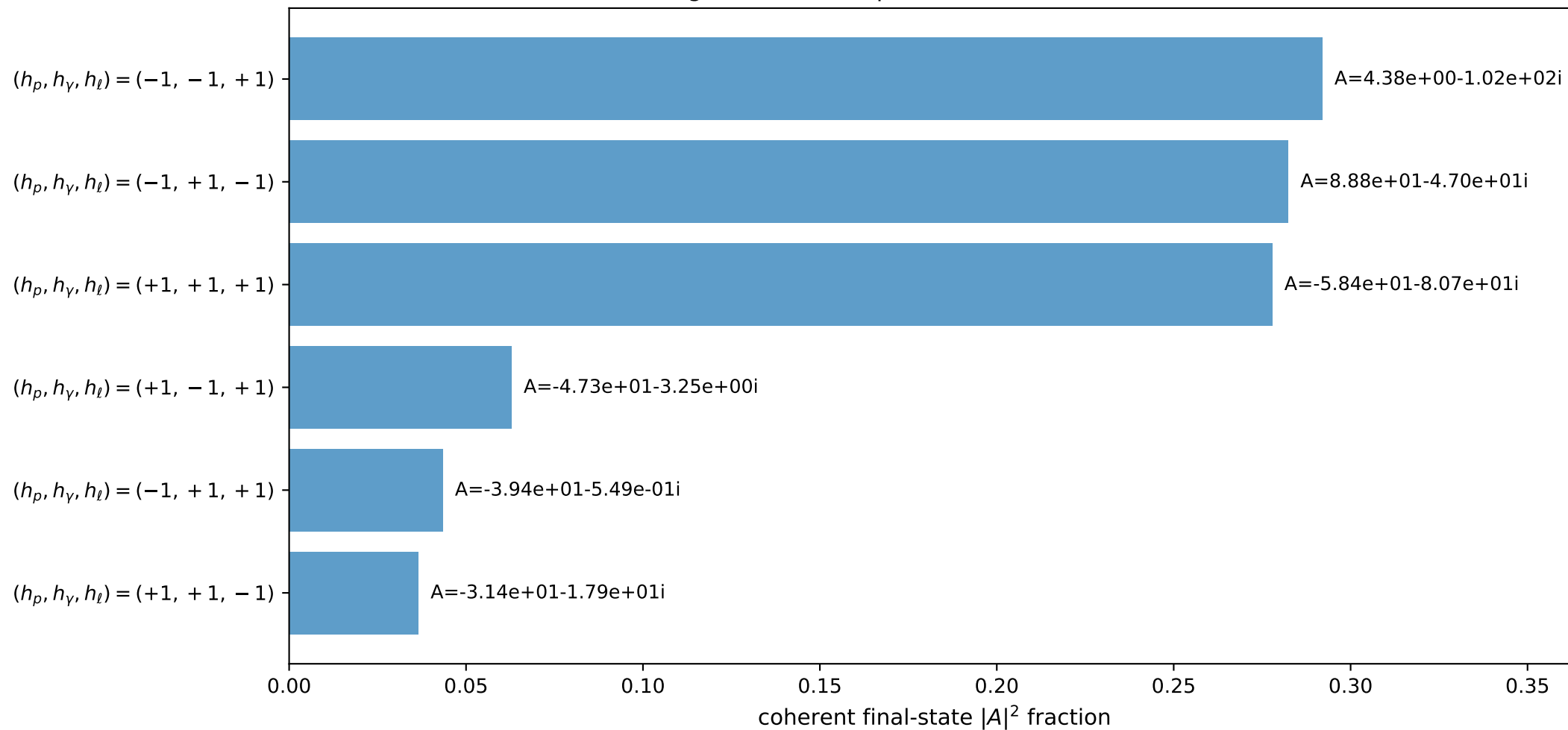
3D momenta



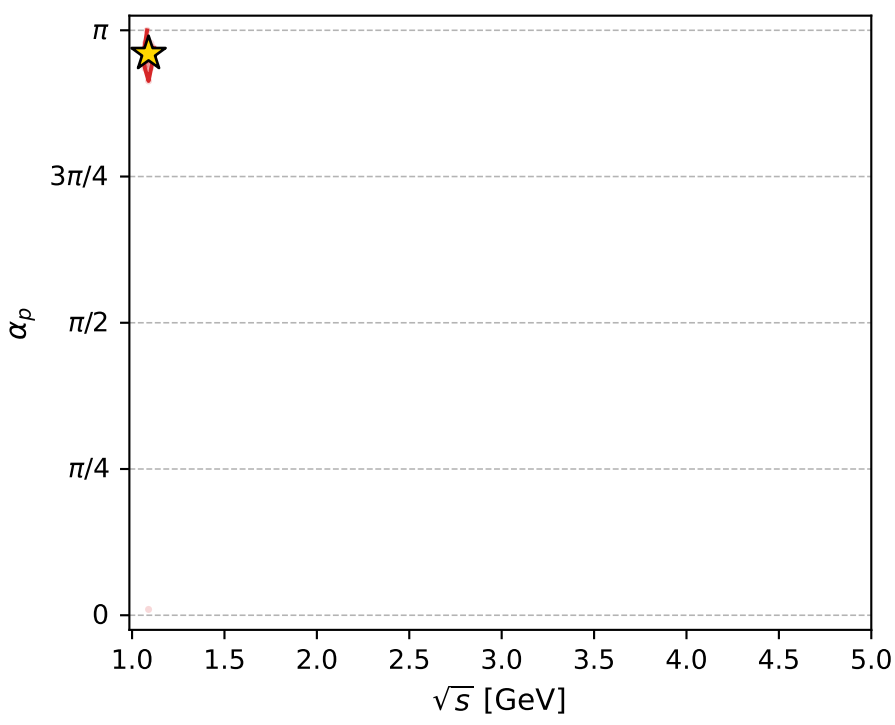
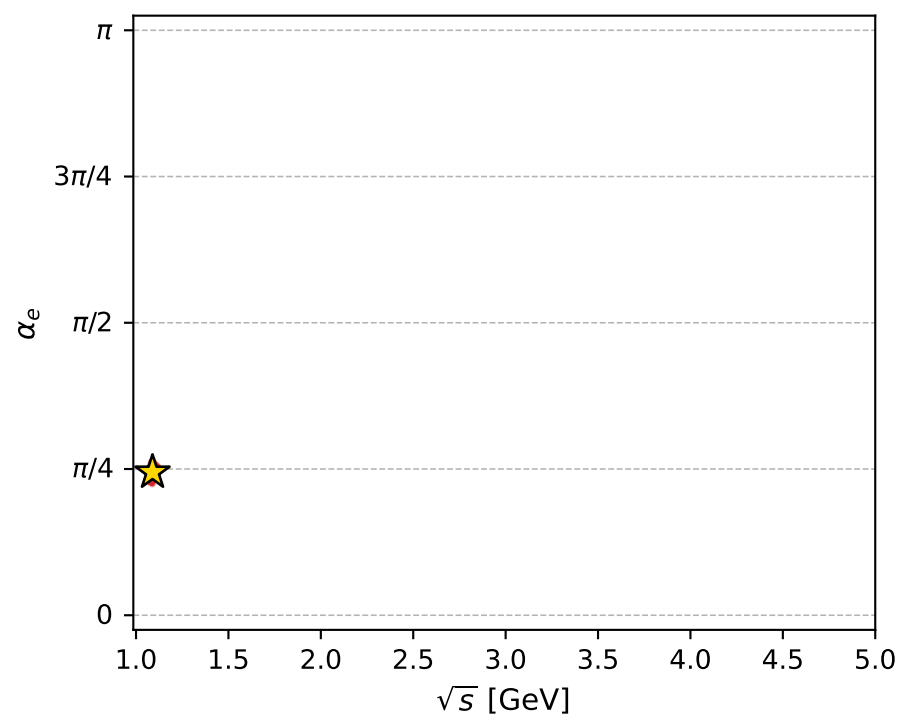
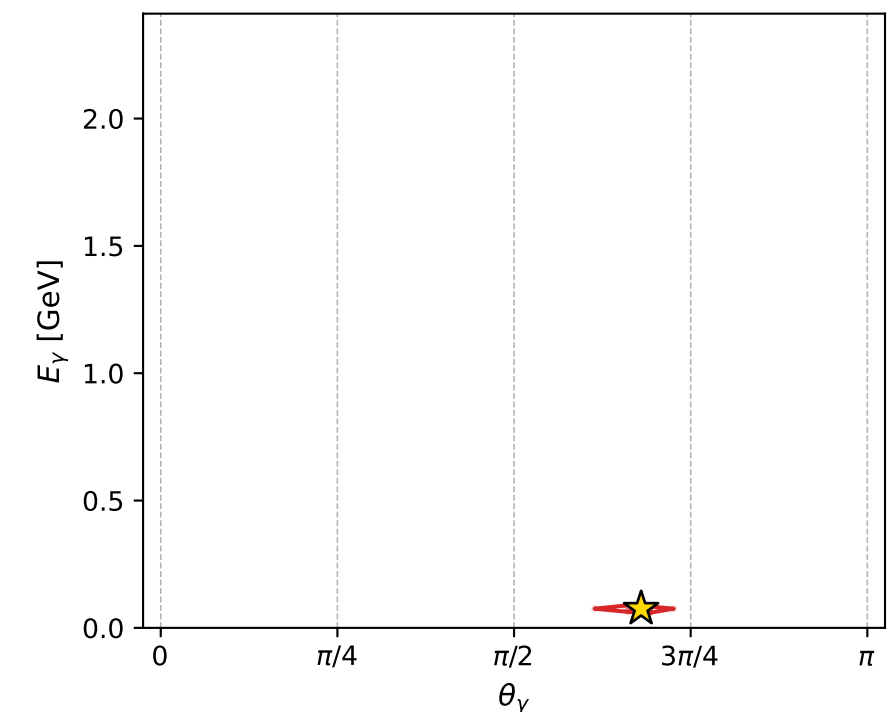
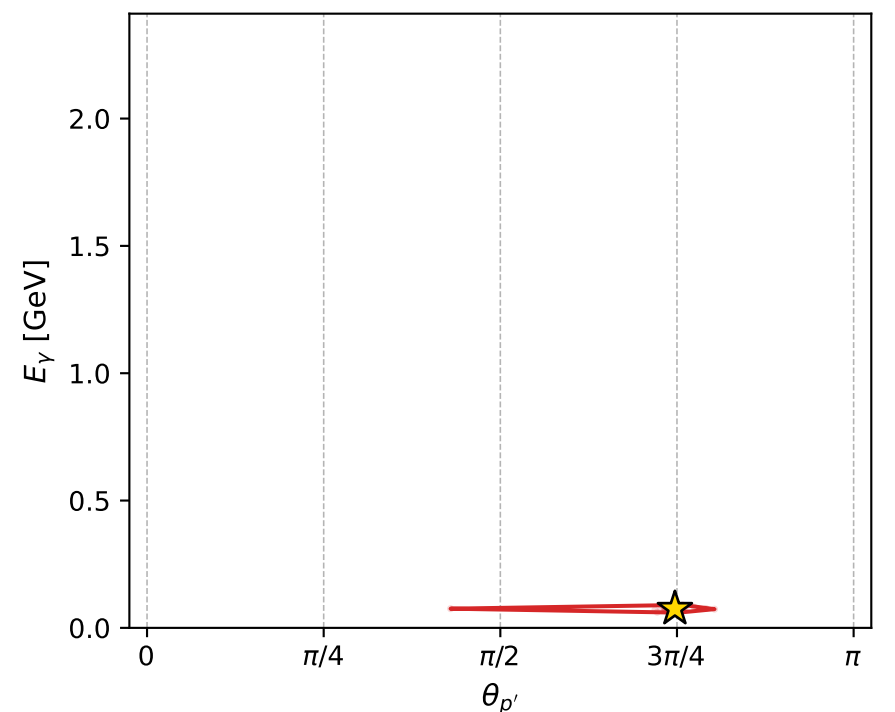
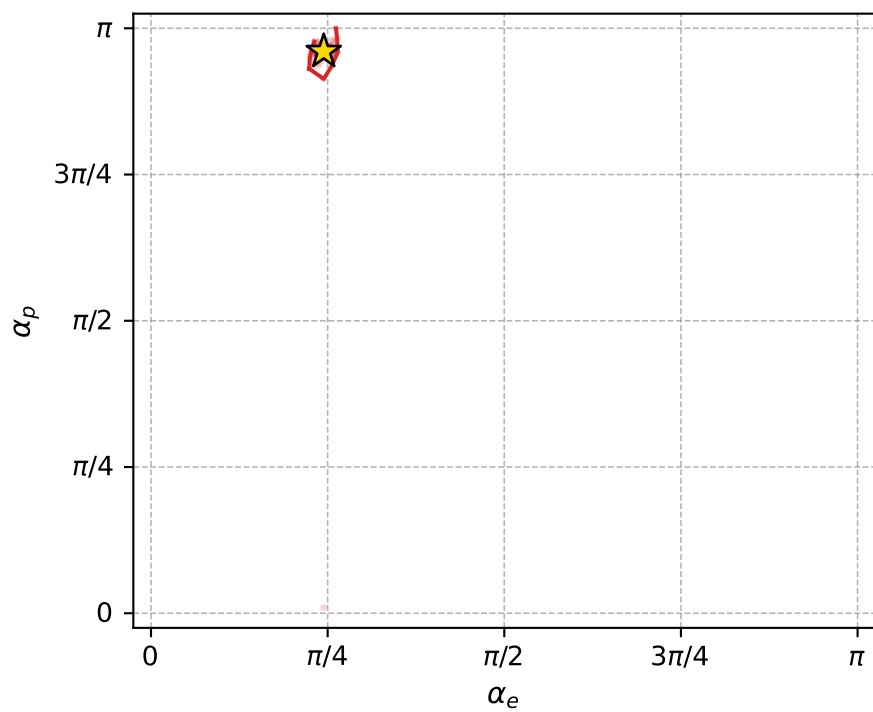
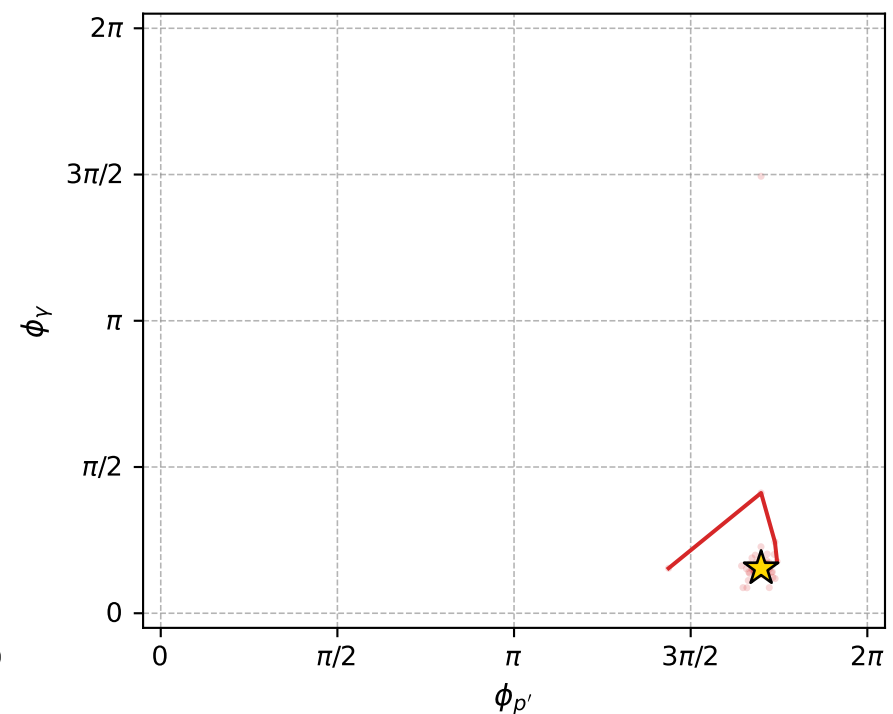
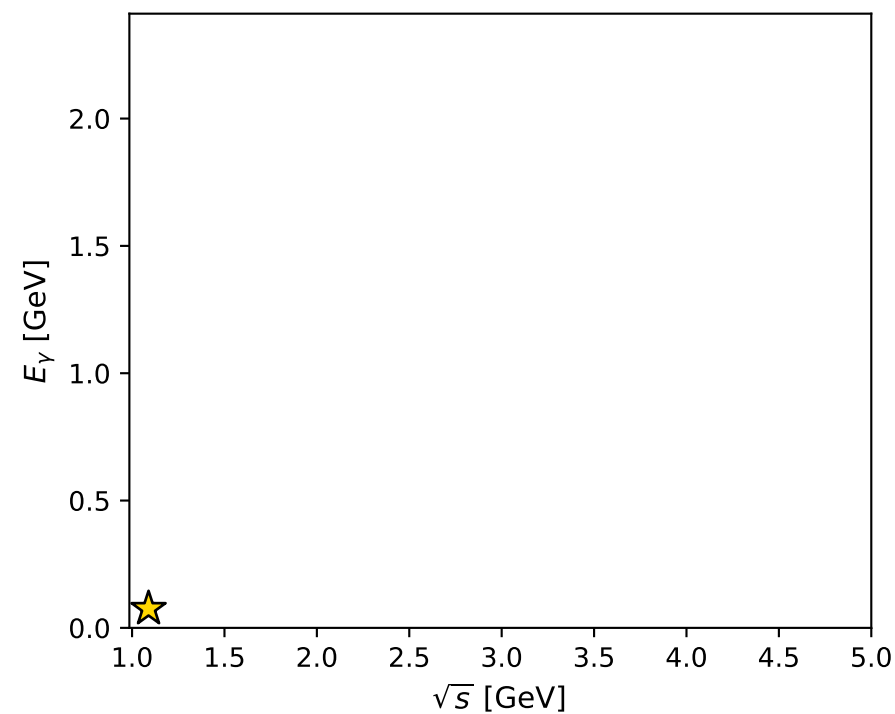
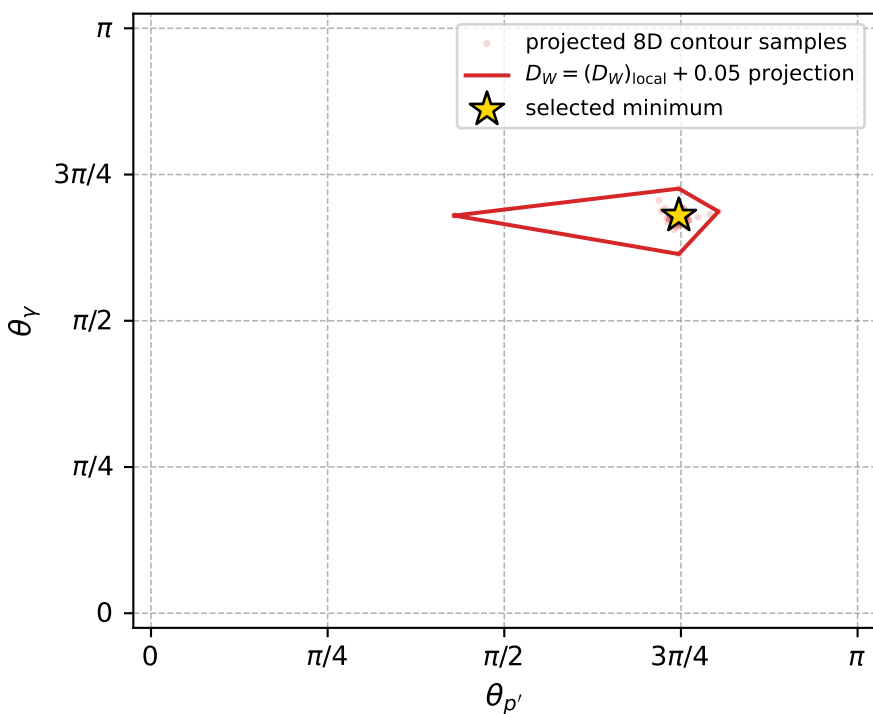
Transverse plane



Leading final-state amplitudes (N=6, retained=99.5%)



electron: polarization cluster P3, local minimum 901; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.013623497$
 $D_W \text{ contour} = 0.063623497$
 above global minimum = 0.013607064
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.089239$
 $\text{theta_p_out} = 2.347018$
 $\text{theta_gamma_out} = 2.135796$
 $\text{q0out} = 0.07535835$
 $\text{phi_p_out} = 5.338541$
 $\text{phi_gamma_out} = 0.4831427$
 $\text{alpha_e} = 0.7680484$
 $\text{alpha_p} = 3.016687$

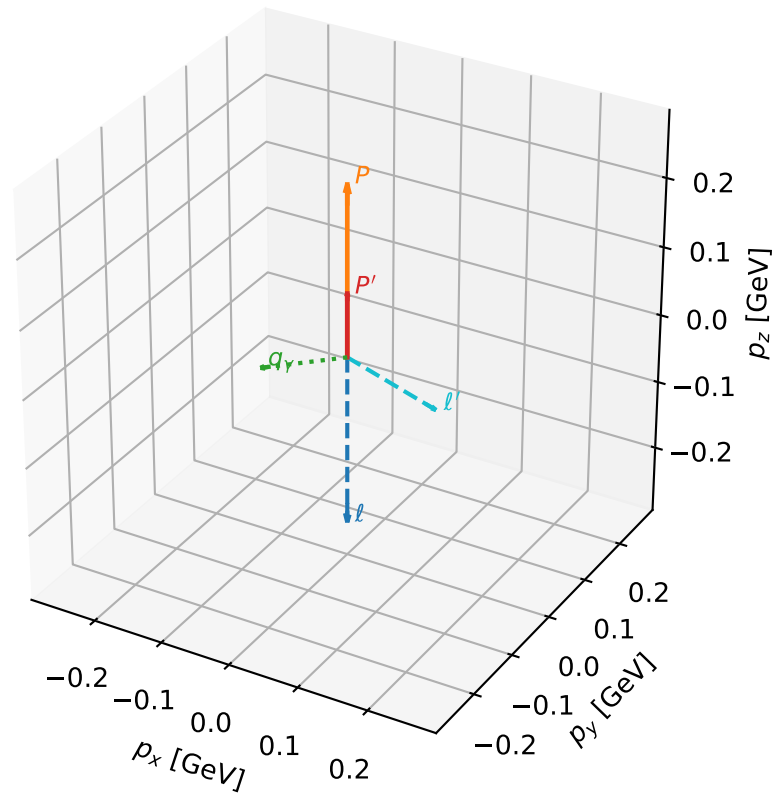
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_910 D_W=0.0140542
region: polarization_cluster_03_local_minimum_910
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81021801$ rad
incoming lepton: $0.689341|+\rangle +0.724437|-\rangle$
proton mixing angle: $\alpha_p=0.13099092$ rad
incoming proton: $0.991433|+\rangle +0.130617|-\rangle$

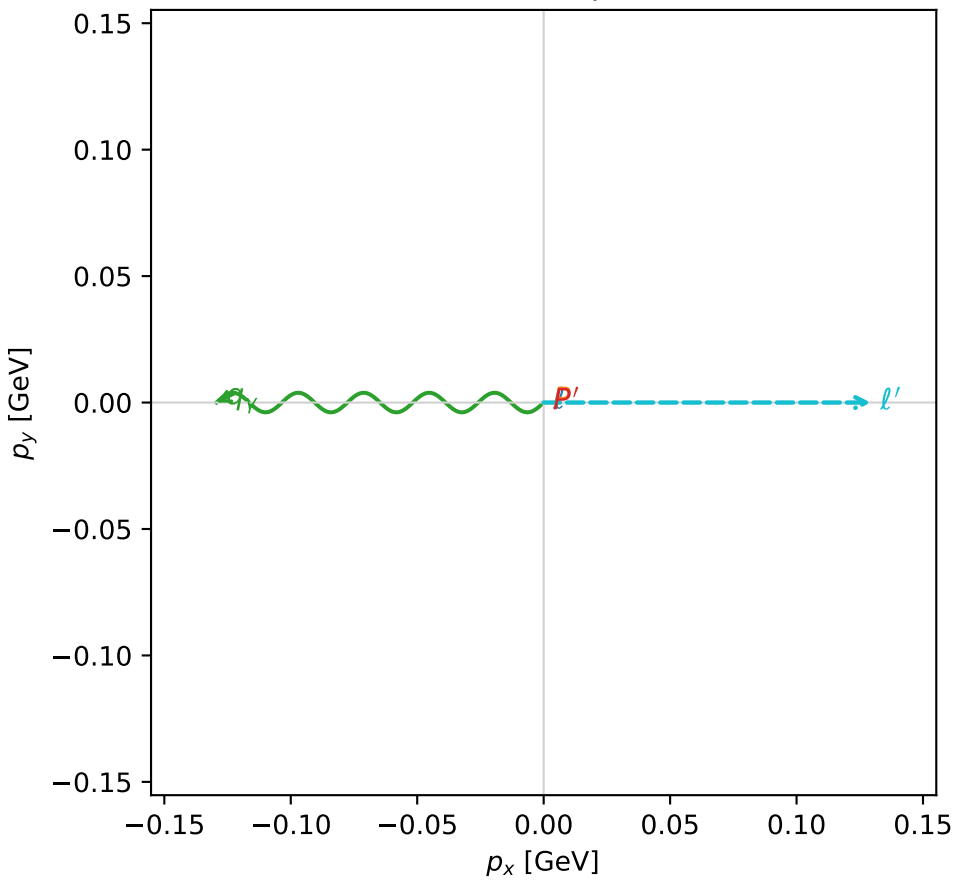
$s=1.48284$, $\sqrt{s}=1.21772$, $|\vec{P}|=0.247591$, $|\vec{P}'|=0.0926797$
 $\theta_{p'}=0$, $\theta_\gamma=1.96038$, $\phi_{p'}=3.1836$, $\phi_\gamma=3.14158$
 $E_\gamma=0.139858$, $\theta_{\ell'}=1.86755$, $\phi_{\ell'}=6.28318$
 $Q^2=0.047404$, $x_B=0.147721$, $t=-0.0232381$, $W^2=1.15334$, $y=0.53218$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2476$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2476$
 P $E= 0.9701$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2476$
 ℓ' $E= 0.1353$ $p_x= 0.1294$ $p_y= -0.0000$ $p_z= -0.0396$
 P' $E= 0.9426$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0927$
 q_γ $E= 0.1399$ $p_x= -0.1294$ $p_y= 0.0000$ $p_z= -0.0531$

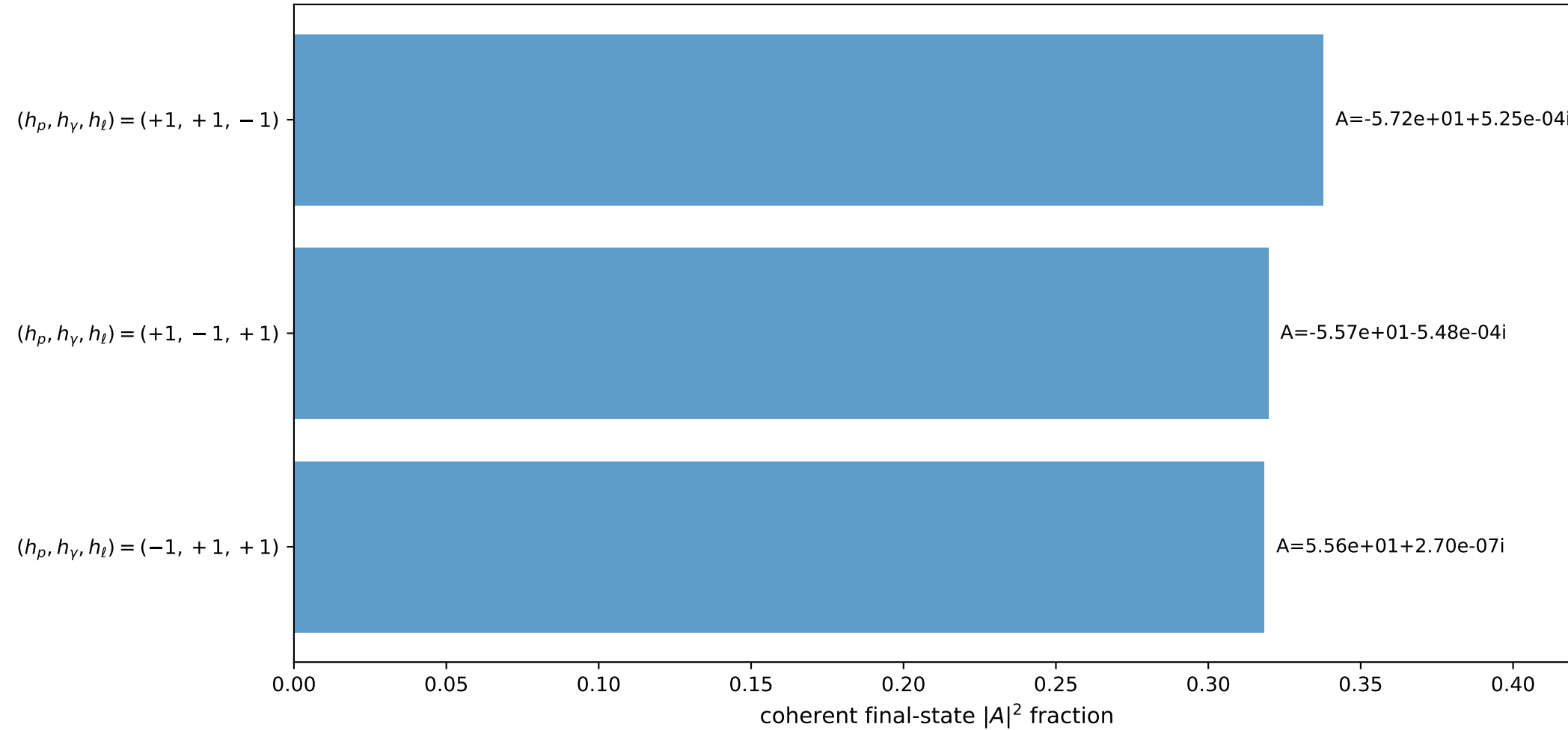
3D momenta



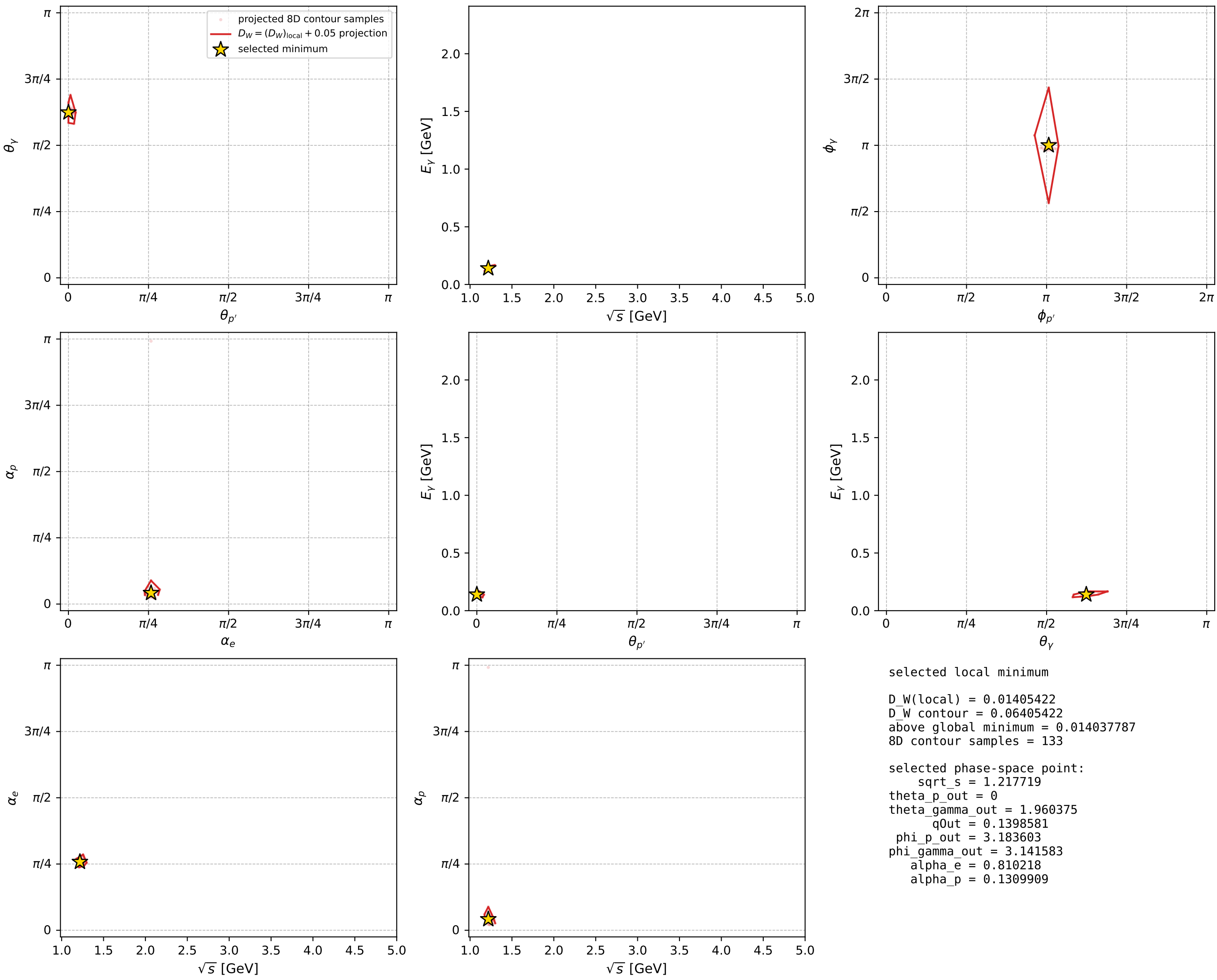
Transverse plane



Leading final-state amplitudes (N=3, retained=97.6%)



electron: polarization cluster P3, local minimum 910; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

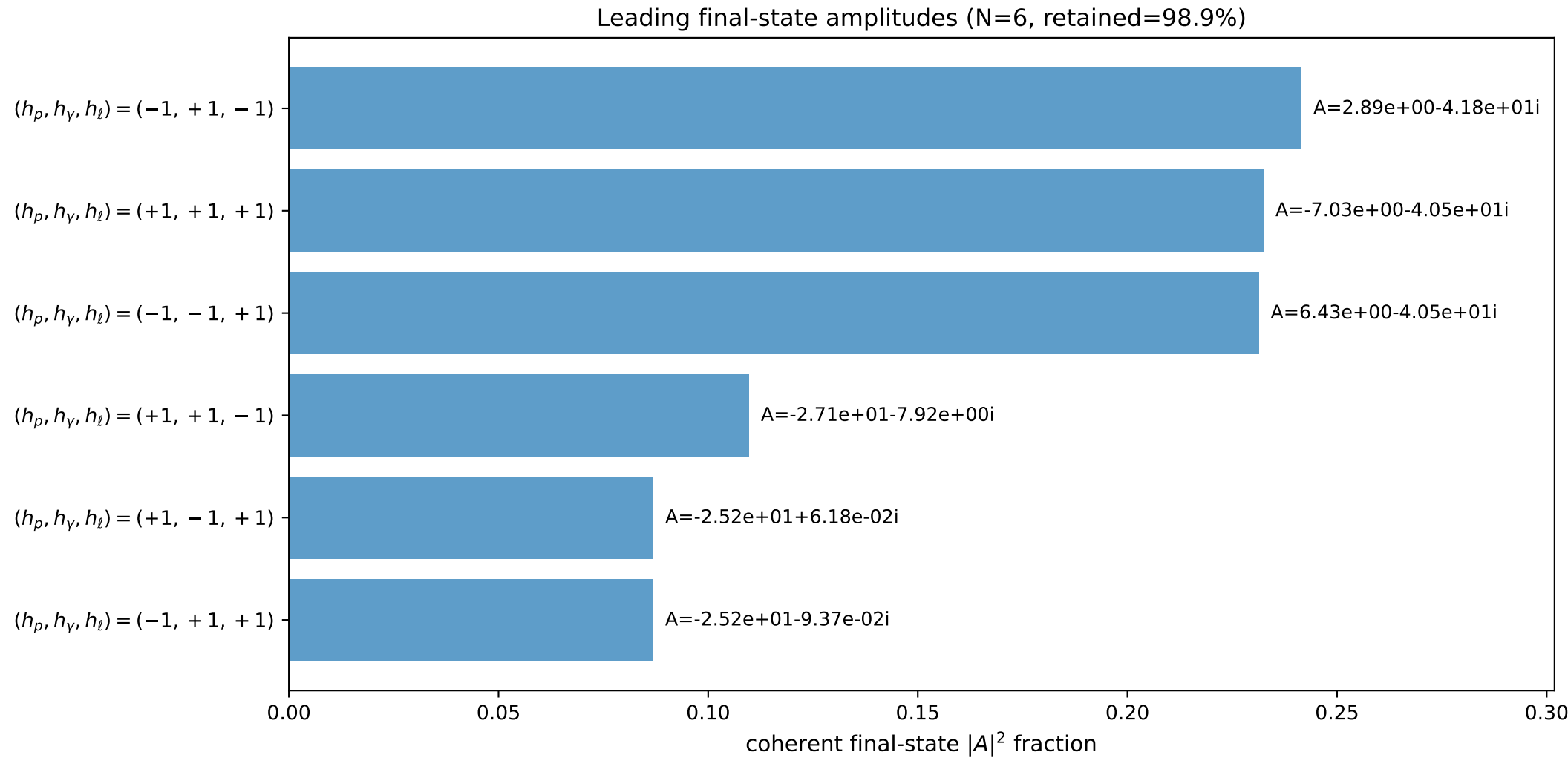
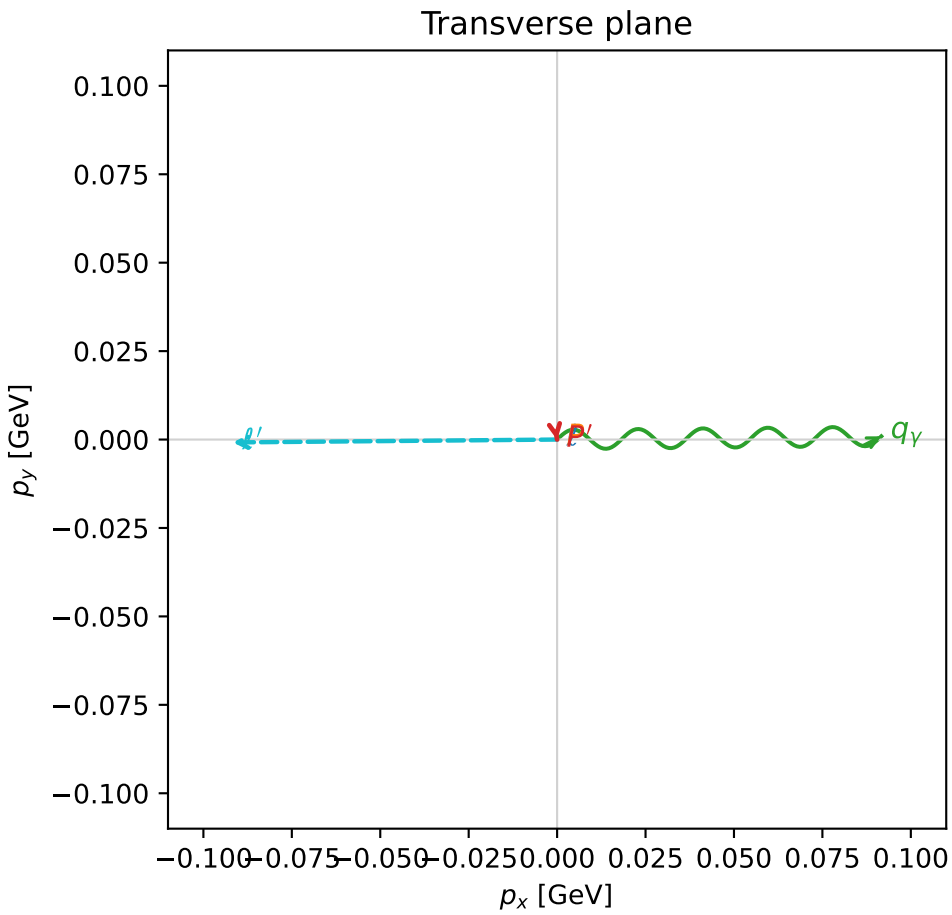
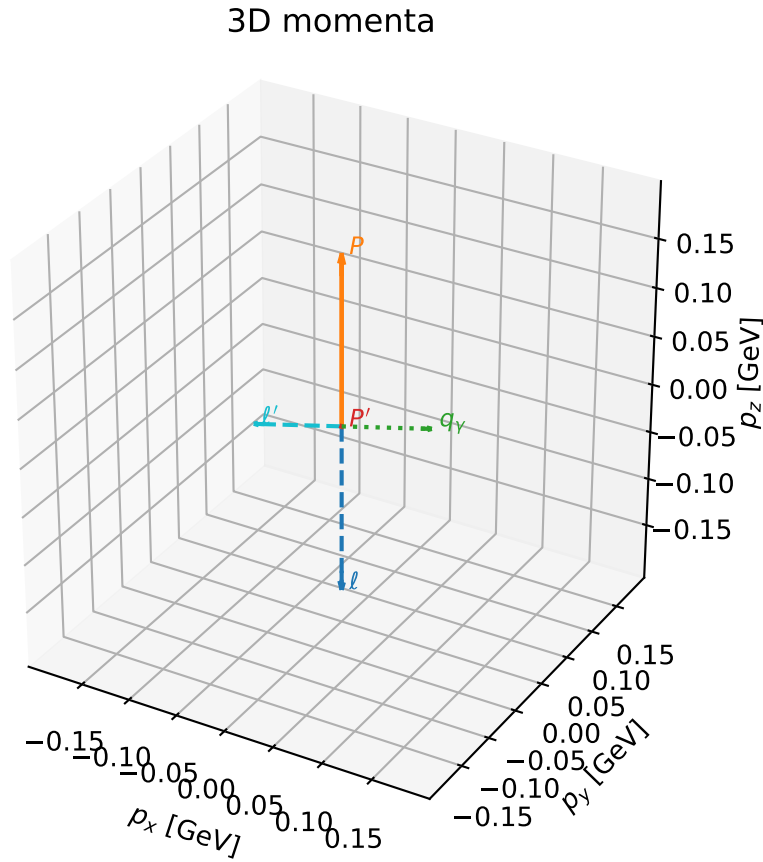


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

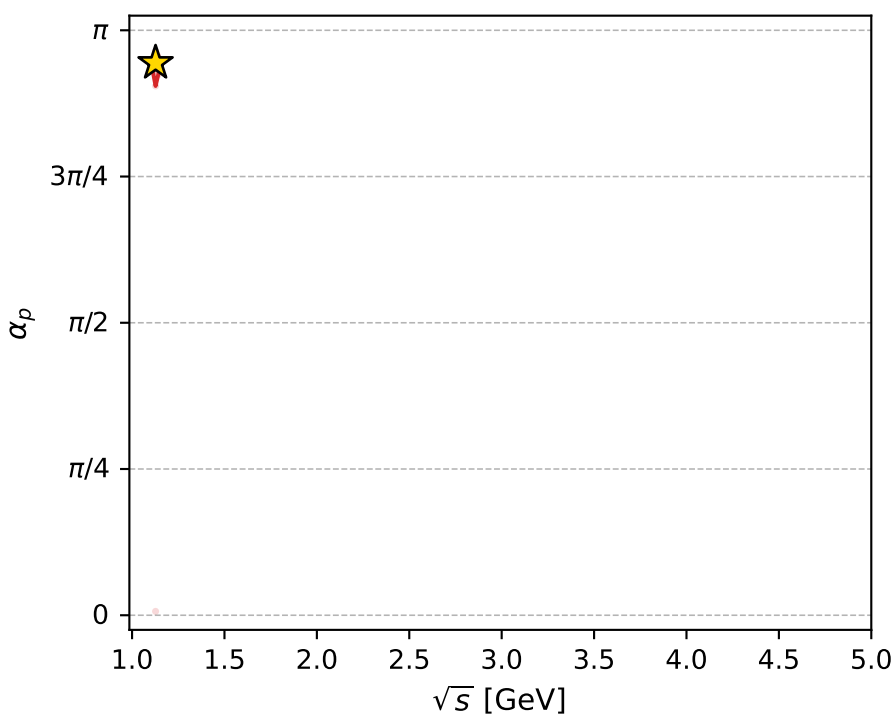
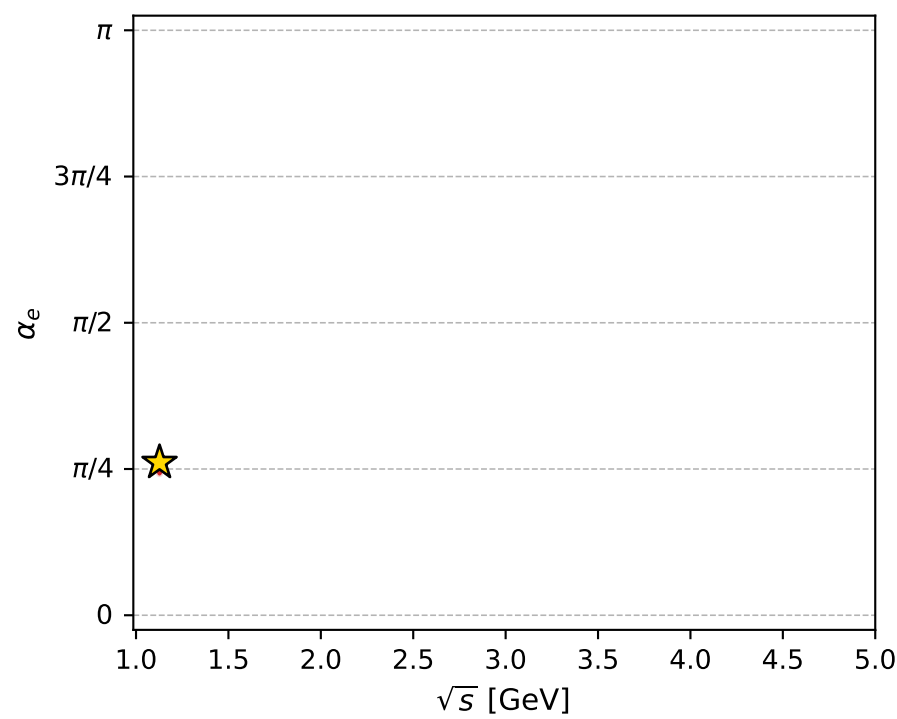
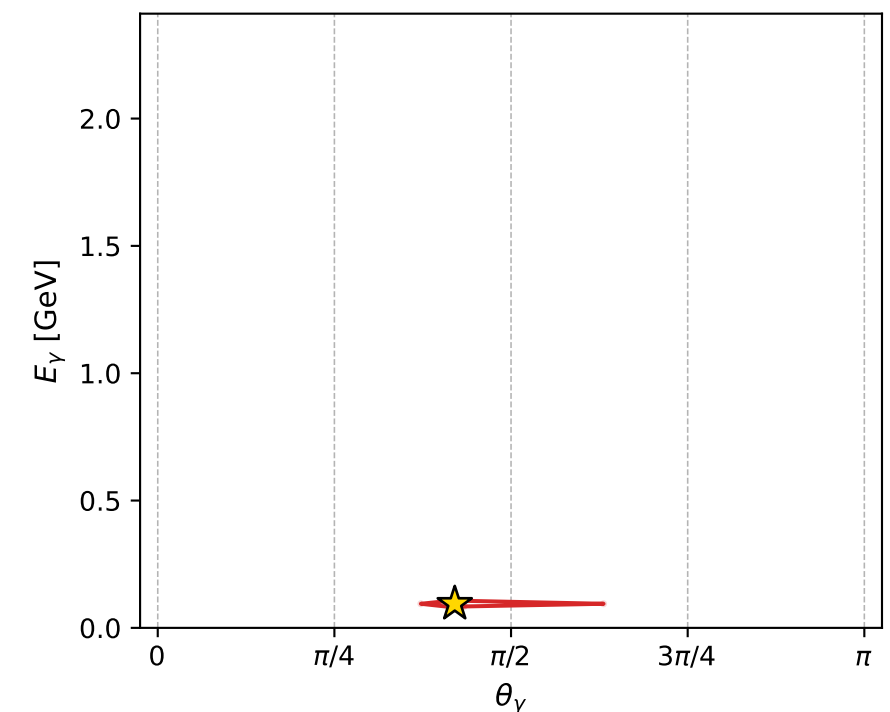
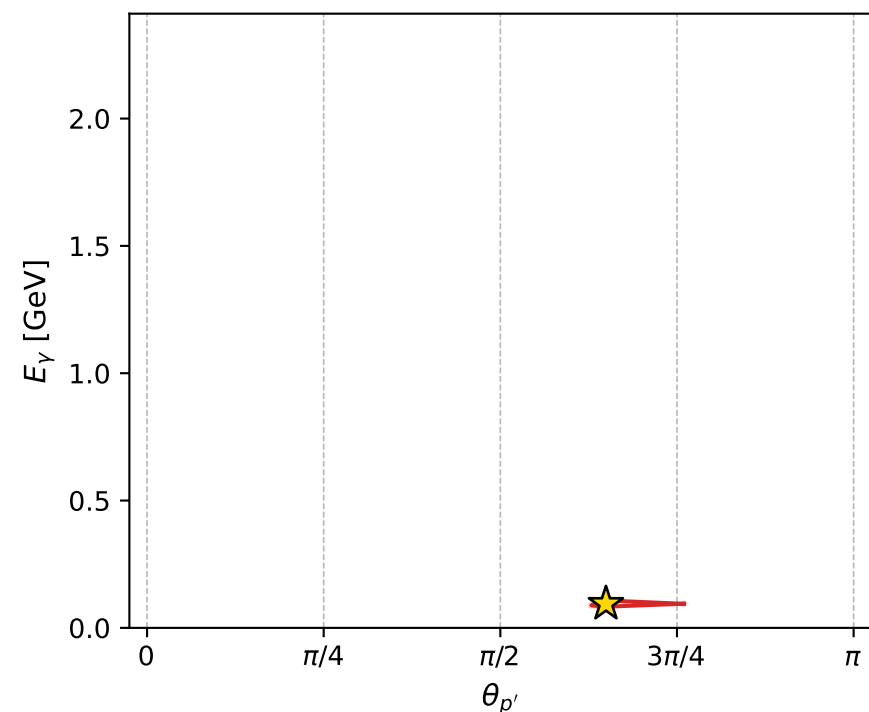
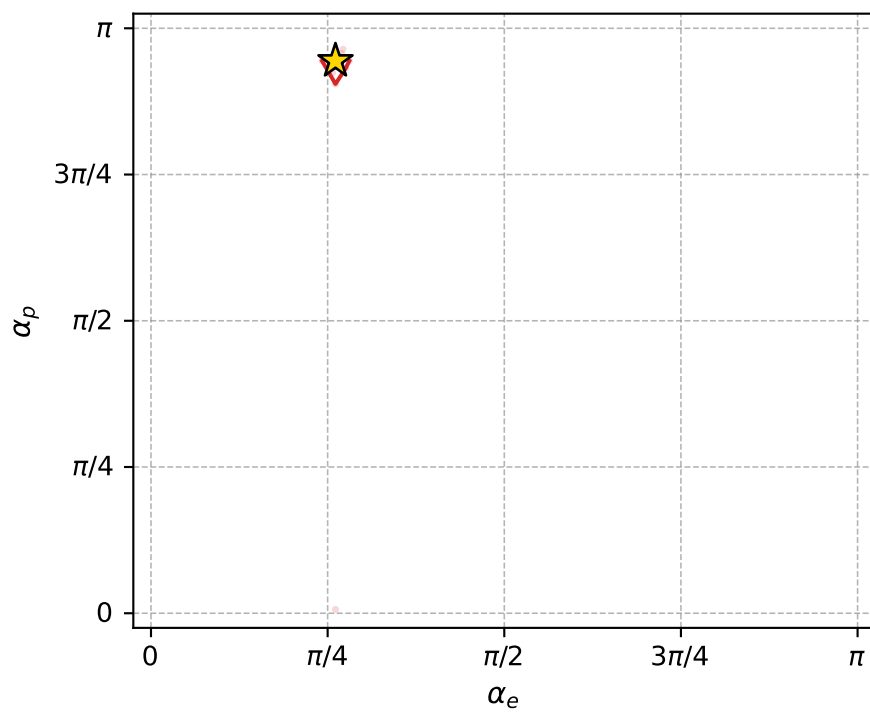
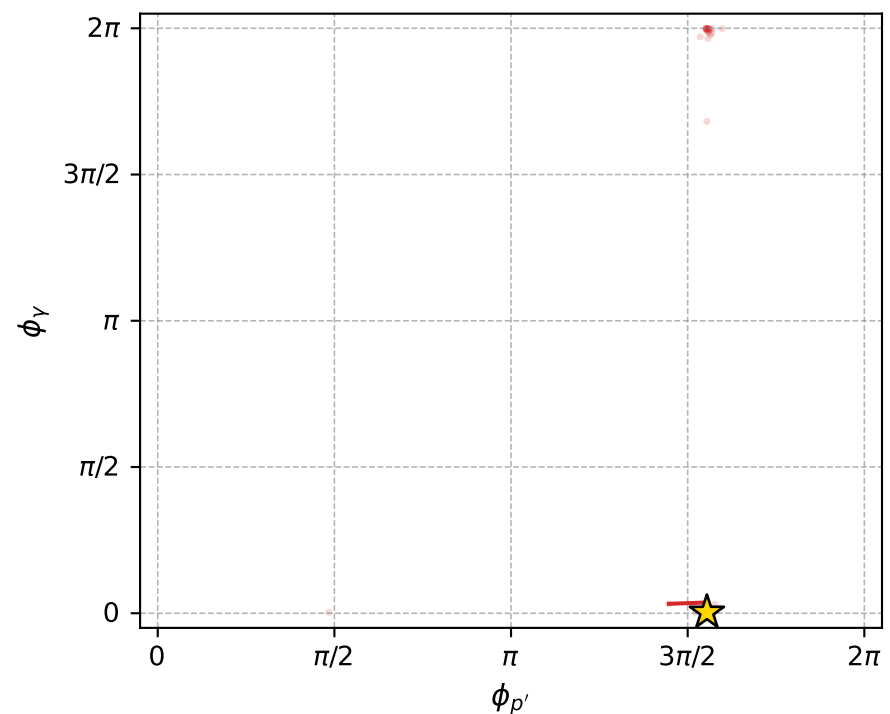
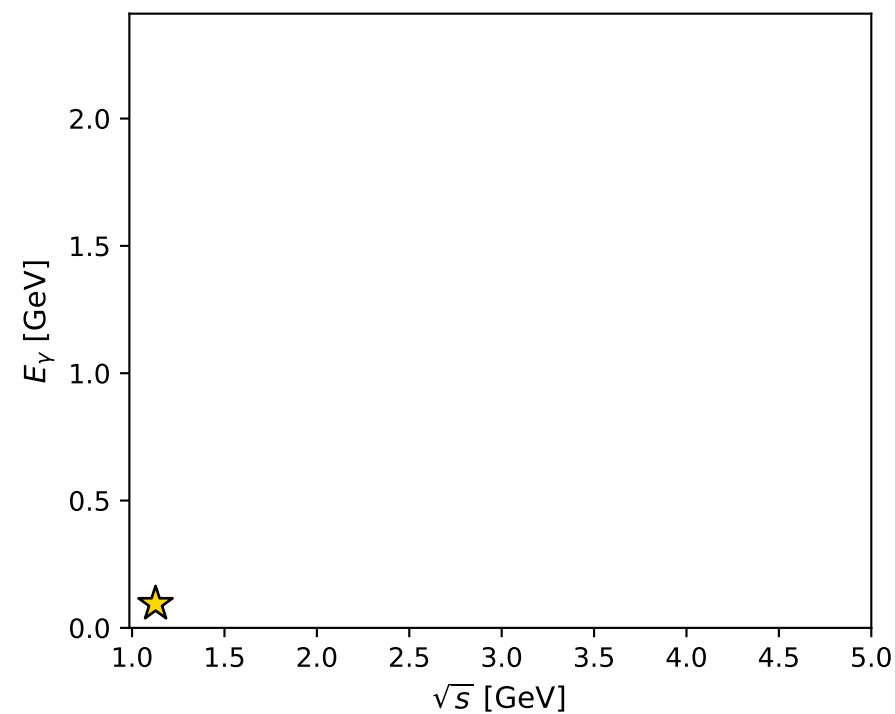
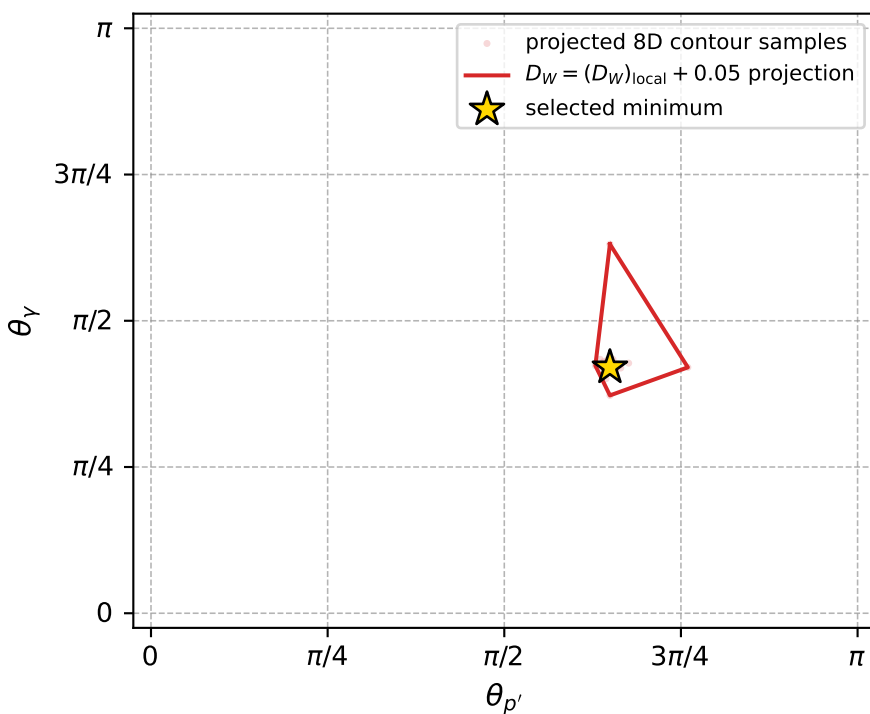
dw_mixing_angles_polarization_03_local_minimum_912 D_W=0.0141489
 region: polarization_cluster_03_local_minimum_912
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82019366$ rad
 incoming lepton: $0.68208|+\rangle +0.731278|-\rangle$
 proton mixing angle: $\alpha_p=2.9665183$ rad
 incoming proton: $-0.984714|+\rangle +0.174181|-\rangle$

$s=1.27063$, $\sqrt{s}=1.12722$, $|\vec{P}|=0.17334$, $|\vec{P}'|=0.000109657$
 $\theta_{p'}=2.04045$, $\theta_Y=1.32052$, $\phi_{p'}=4.88333$, $\phi_Y=0.00995711$
 $E_Y=0.0946095$, $\theta_{l'}=1.82052$, $\phi_{l'}=3.1505$
 $Q^2=0.0246943$, $x_B=0.122139$, $t=-0.0298119$, $W^2=1.05733$, $y=0.517368$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1733$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1733$
 P $E= 0.9539$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1733$
 l' $E= 0.0946$ $p_x= -0.0917$ $p_y= -0.0008$ $p_z= -0.0234$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= -0.0001$ $p_z= -0.0000$
 q_Y $E= 0.0946$ $p_x= 0.0917$ $p_y= 0.0009$ $p_z= 0.0234$



electron: polarization cluster P3, local minimum 912; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.014148859$
 $D_W \text{ contour} = 0.064148859$
 above global minimum = 0.014132426
 8D contour samples = 255

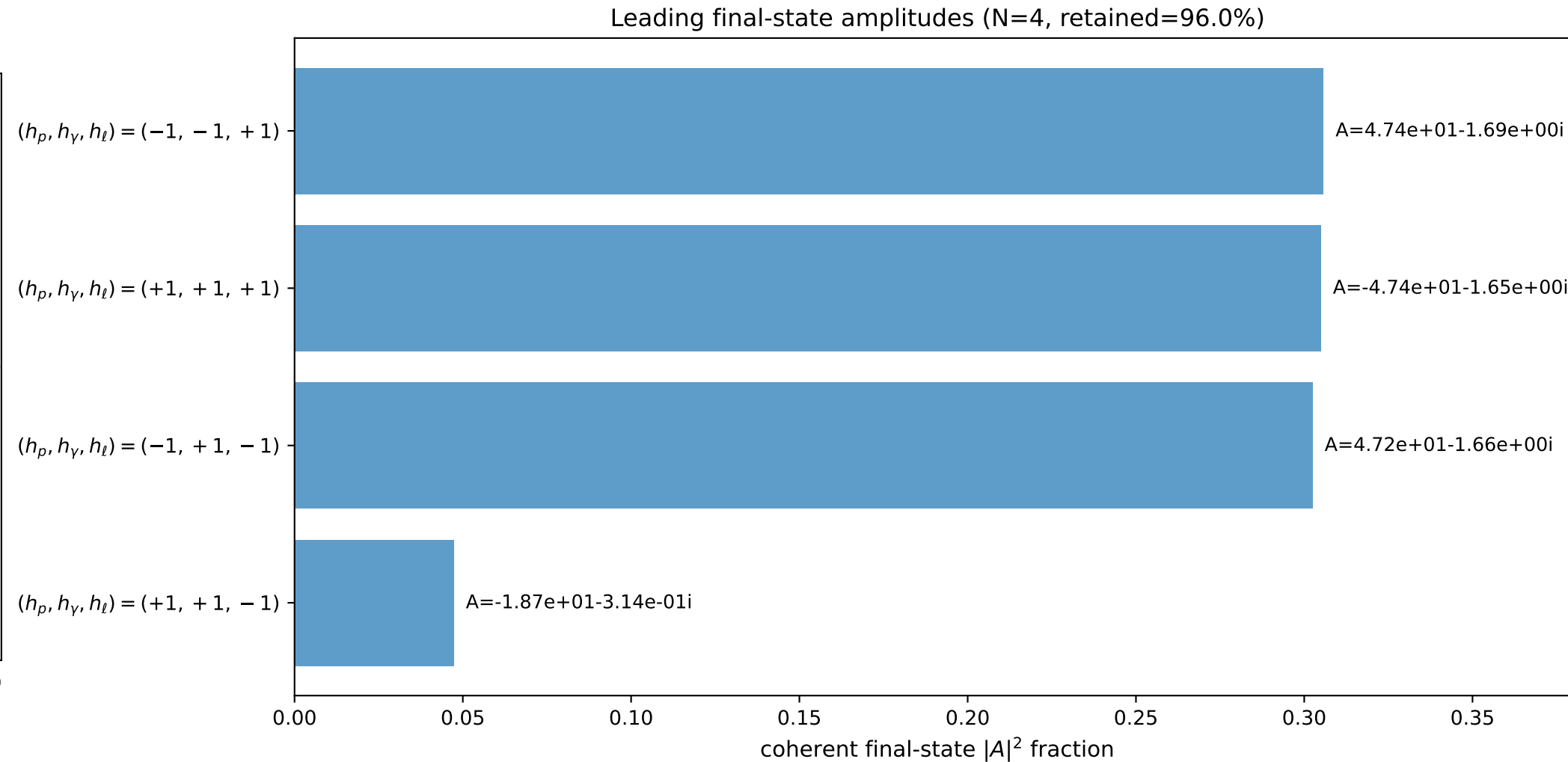
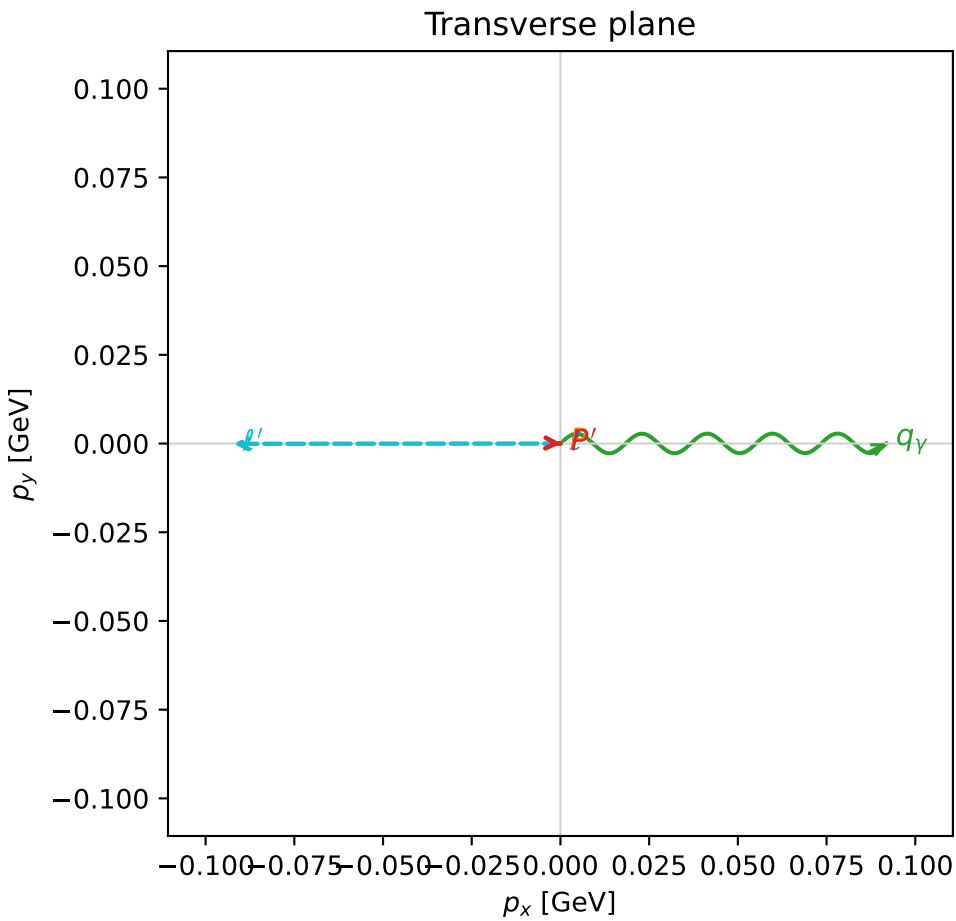
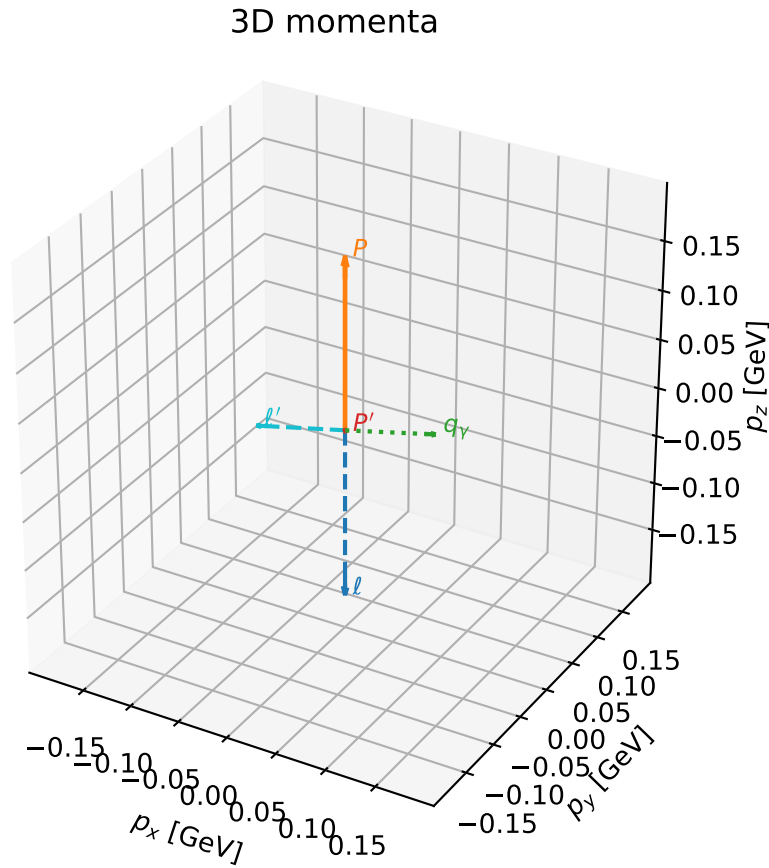
selected phase-space point:
 $\text{sqrt_s} = 1.127223$
 $\text{theta_p_out} = 2.04045$
 $\text{theta_gamma_out} = 1.320524$
 $\text{qOut} = 0.09460945$
 $\text{phi_p_out} = 4.883326$
 $\text{phi_gamma_out} = 0.009957107$
 $\text{alpha_e} = 0.8201937$
 $\text{alpha_p} = 2.966518$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

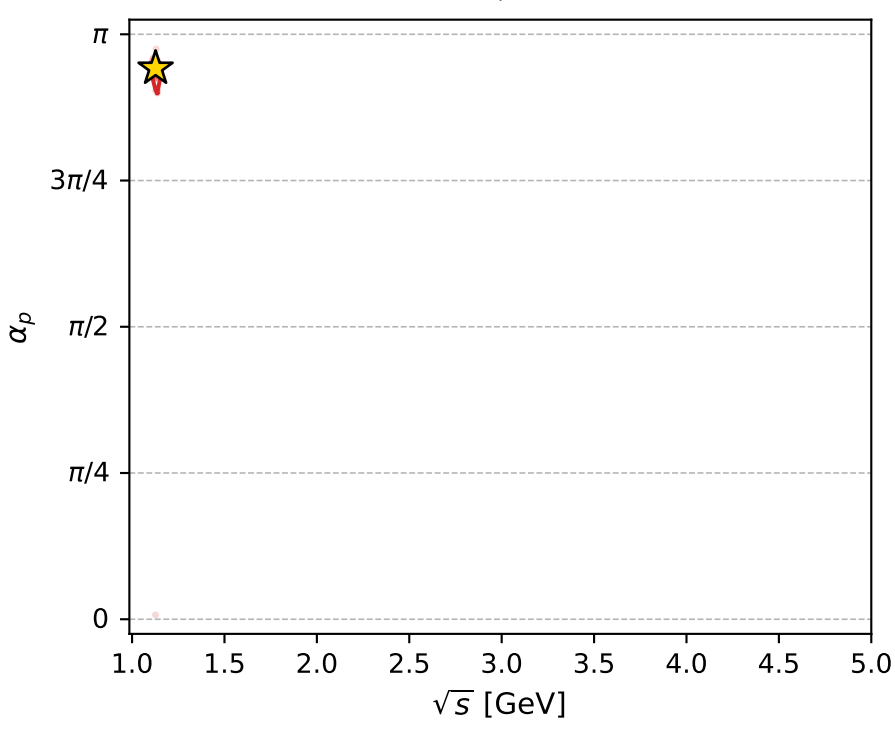
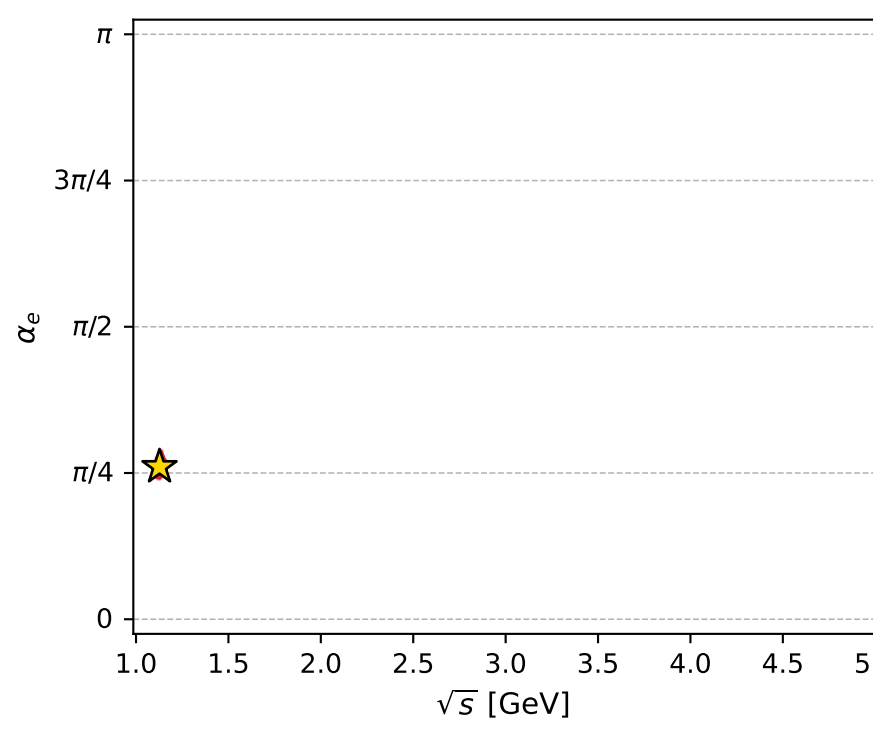
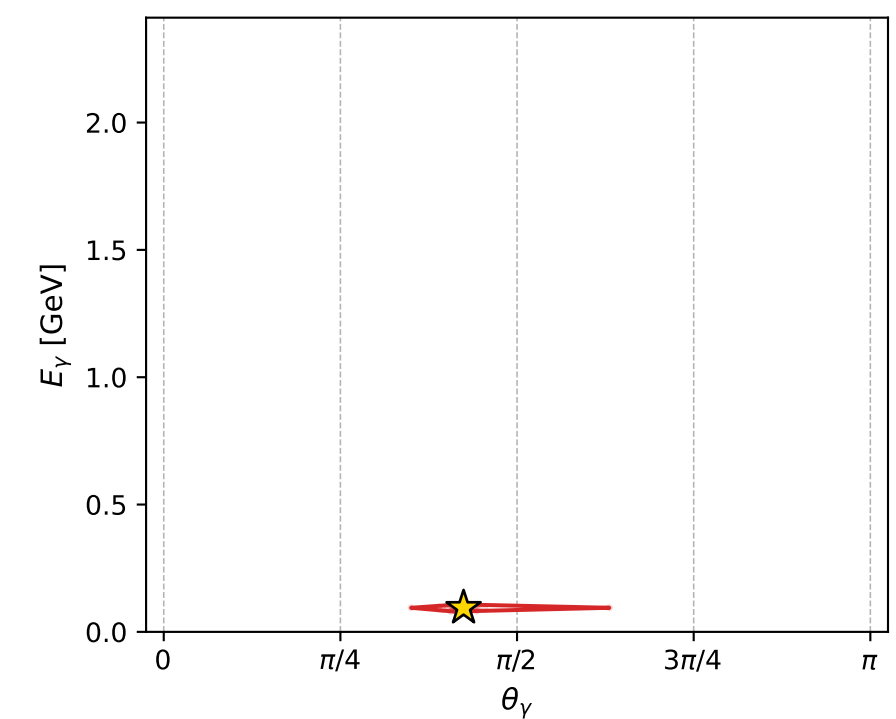
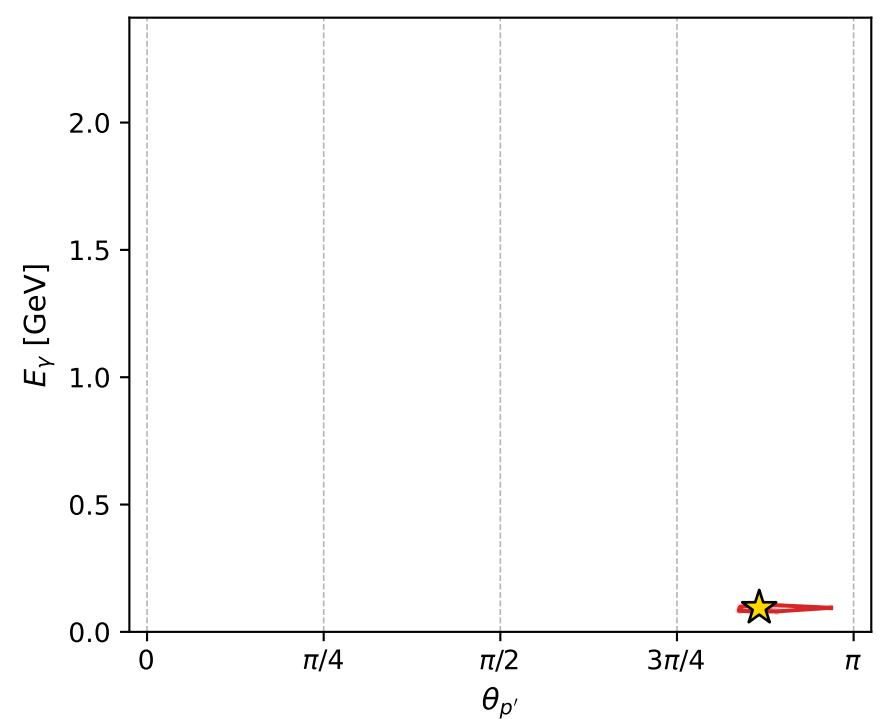
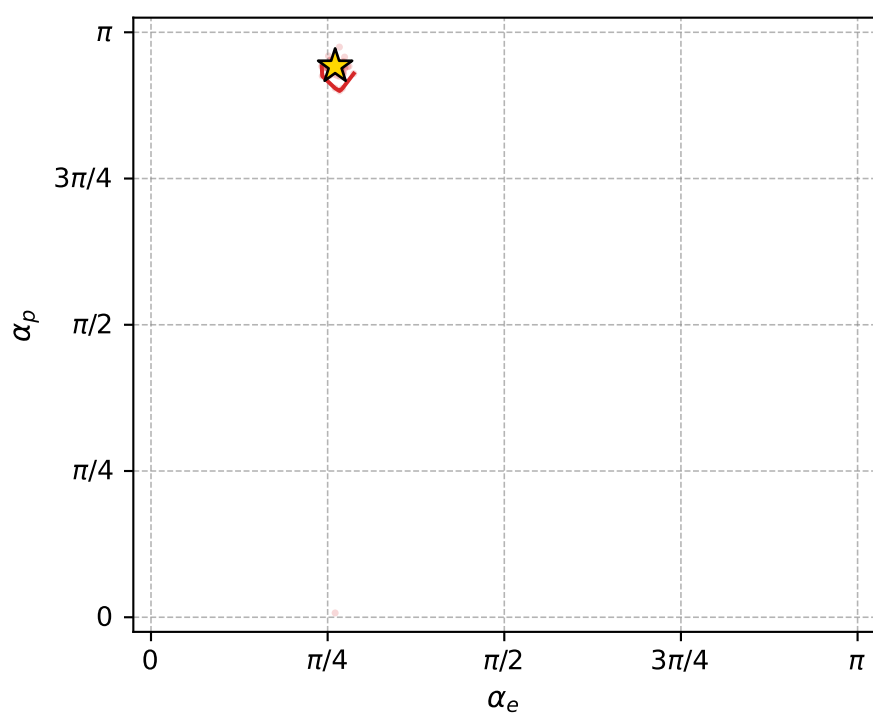
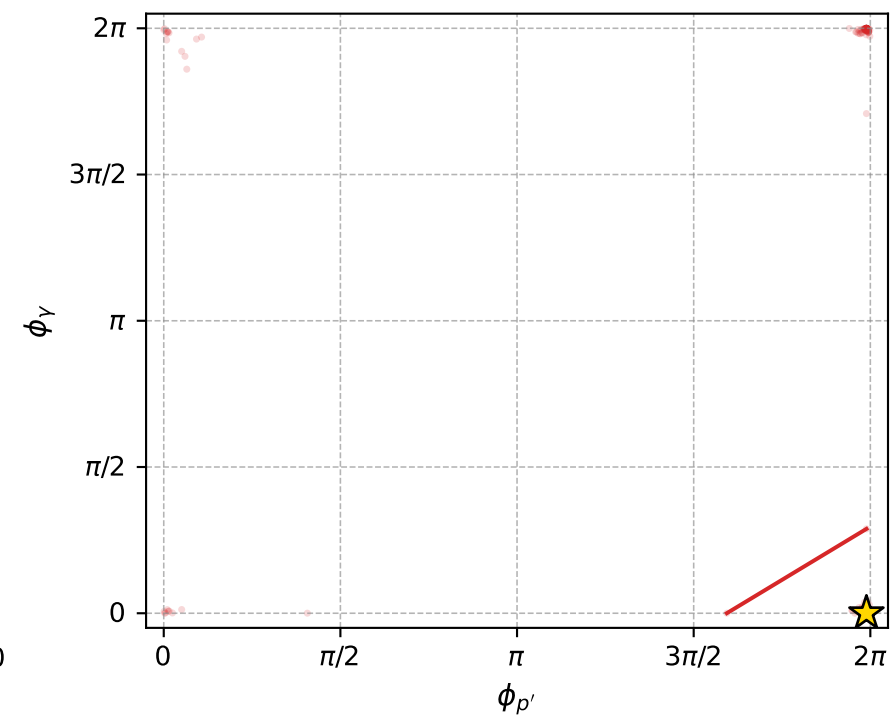
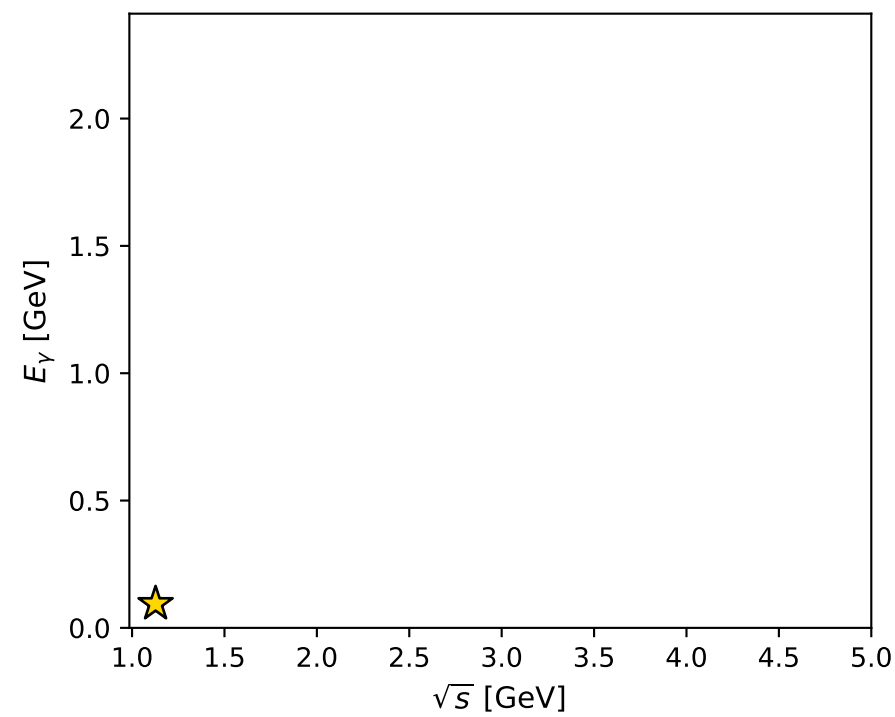
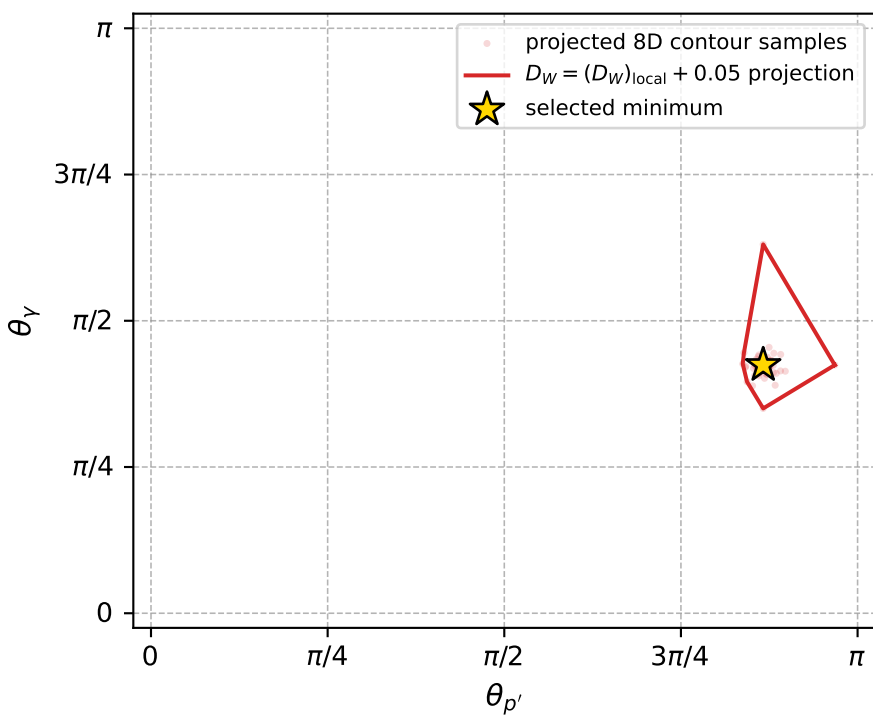
dw_mixing_angles_polarization_03_local_minimum_914 D_W=0.0141746
 region: polarization_cluster_03_local_minimum_914
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81869347$ rad
 incoming lepton: $0.683176|+\rangle +0.730254|-\rangle$
 proton mixing angle: $\alpha_p=2.9590454$ rad
 incoming proton: $-0.983384|+\rangle +0.181535|-\rangle$

$s=1.27069$, $\sqrt{s}=1.12725$, $|\vec{P}|=0.173362$, $|\vec{P}'|=0.00064534$
 $\theta_{p'}=2.72186$, $\theta_Y=1.33304$, $\phi_{p'}=6.24848$, $\phi_Y=0.000835405$
 $E_Y=0.0945639$, $\theta_{\ell'}=1.80185$, $\phi_{\ell'}=3.14233$
 $Q^2=0.0253108$, $x_B=0.124874$, $t=-0.0300066$, $W^2=1.05722$, $y=0.518597$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1734$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1734$
 P $E=0.9539$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1734$
 ℓ' $E=0.0947$ $p_x=-0.0922$ $p_y=-0.0001$ $p_z=-0.0217$
 P' $E=0.9380$ $p_x=0.0003$ $p_y=-0.0000$ $p_z=-0.0006$
 q_Y $E=0.0946$ $p_x=0.0919$ $p_y=0.0001$ $p_z=0.0223$



electron: polarization cluster P3, local minimum 914; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.014174557$
 $D_W \text{ contour} = 0.064174557$
 above global minimum = 0.014158124
 8D contour samples = 255

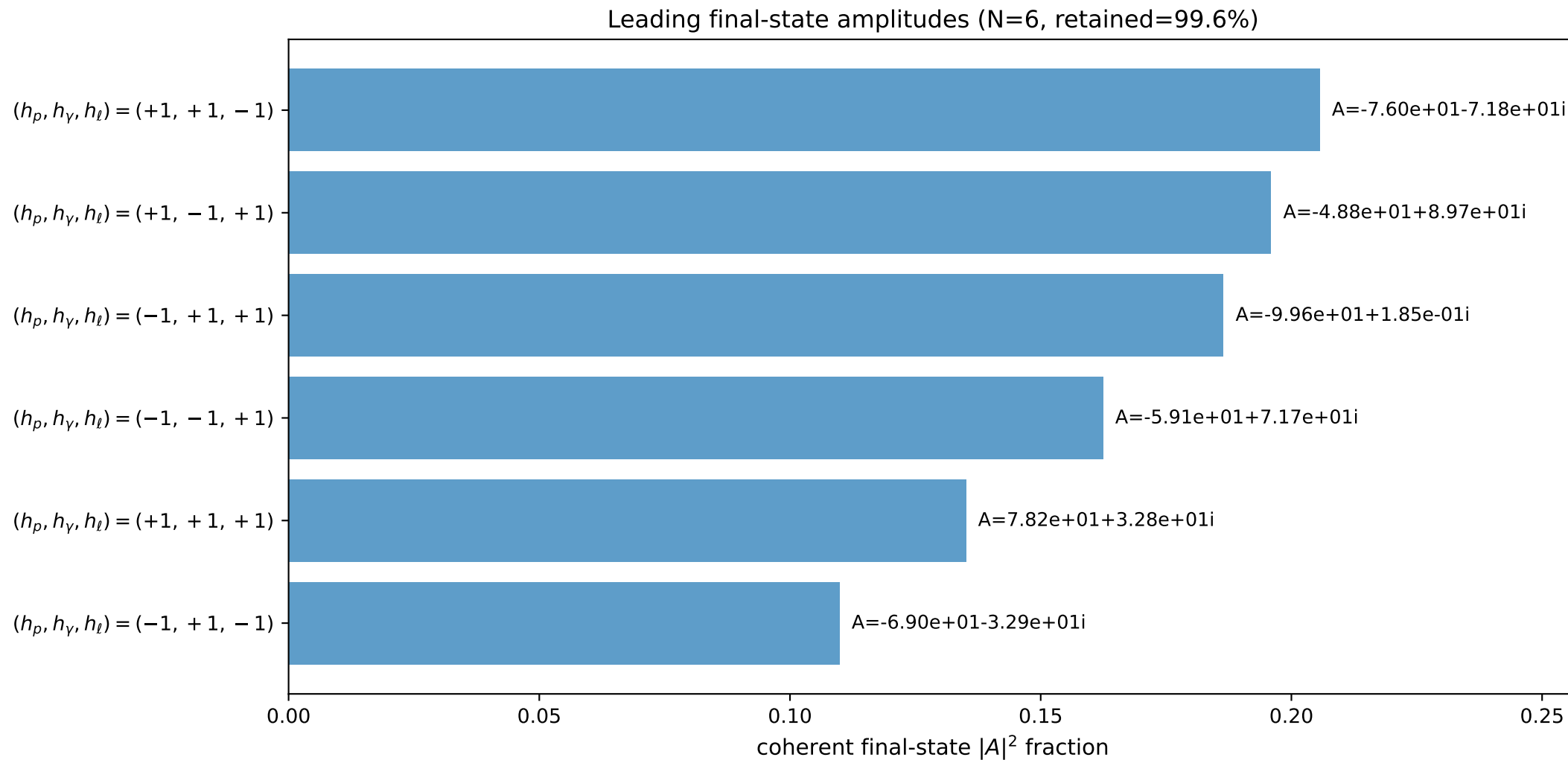
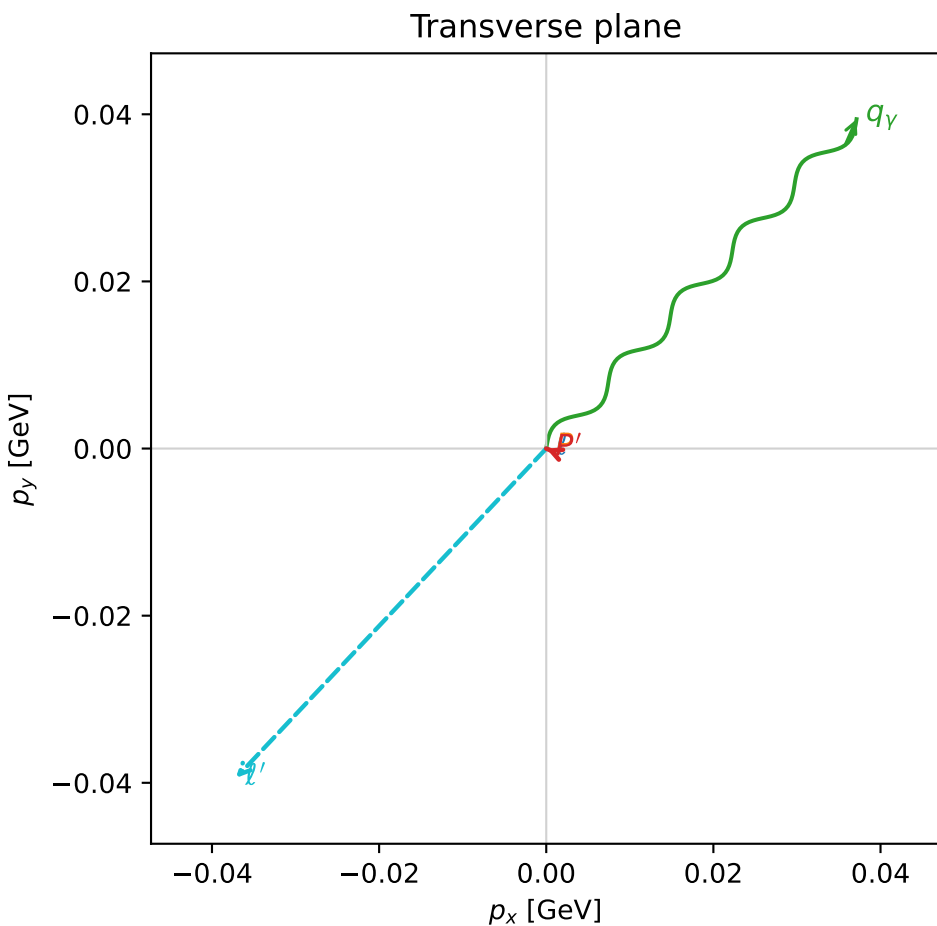
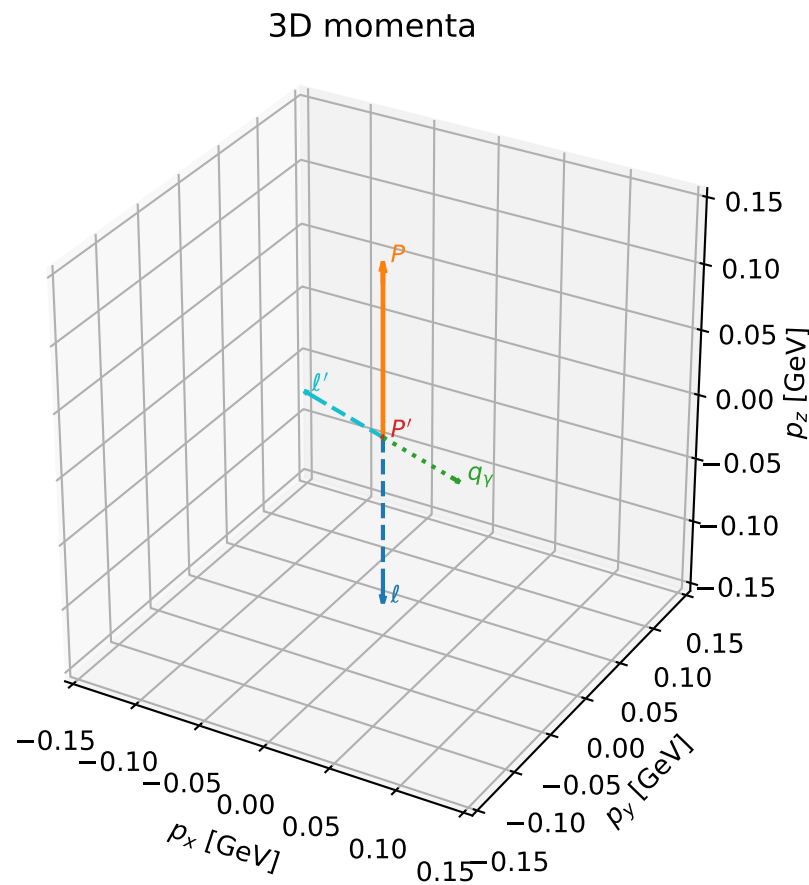
selected phase-space point:
 $\text{sqrt_s} = 1.127248$
 $\text{theta_p_out} = 2.721862$
 $\text{theta_gamma_out} = 1.333036$
 $\text{qOut} = 0.09456394$
 $\text{phi_p_out} = 6.248478$
 $\text{phi_gamma_out} = 0.0008354055$
 $\text{alpha_e} = 0.8186935$
 $\text{alpha_p} = 2.959045$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

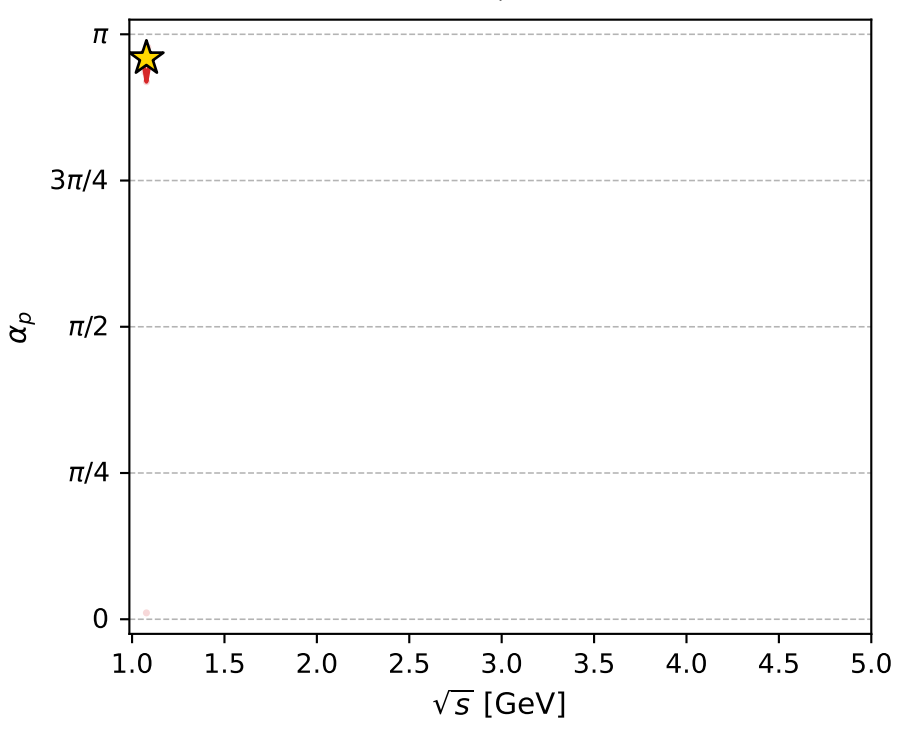
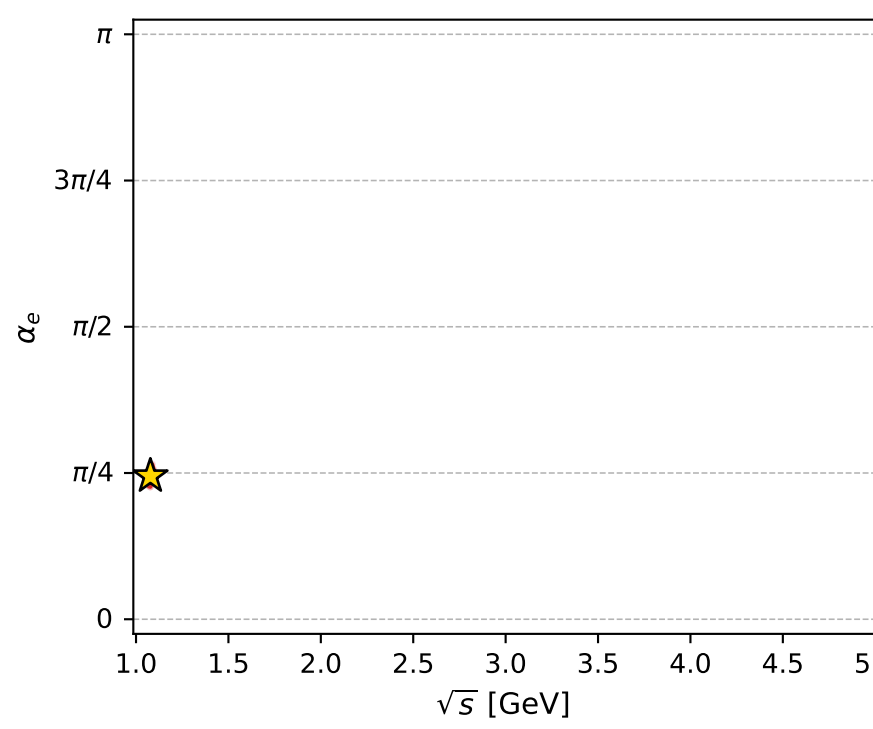
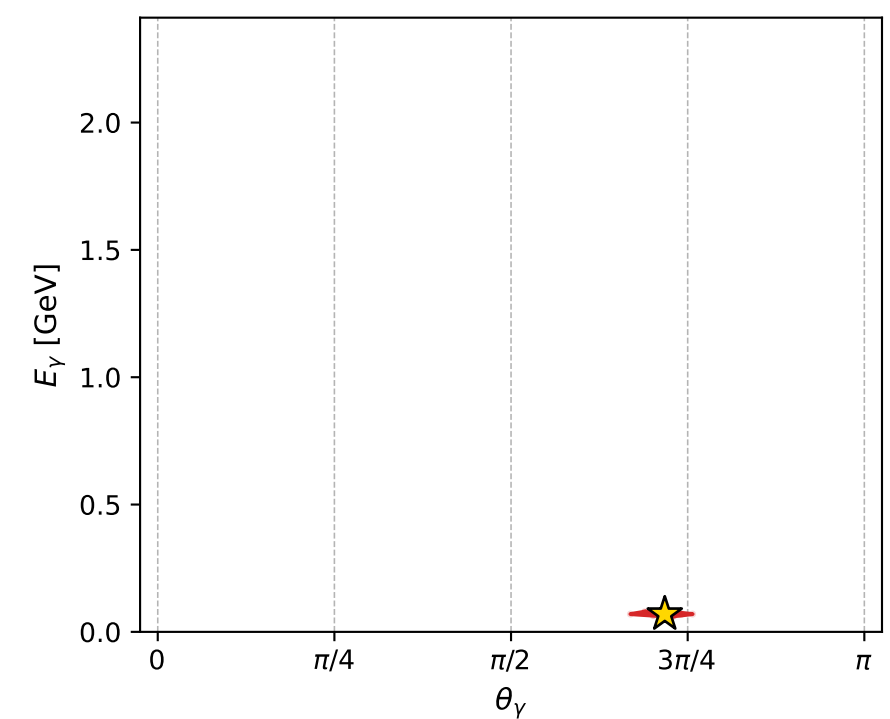
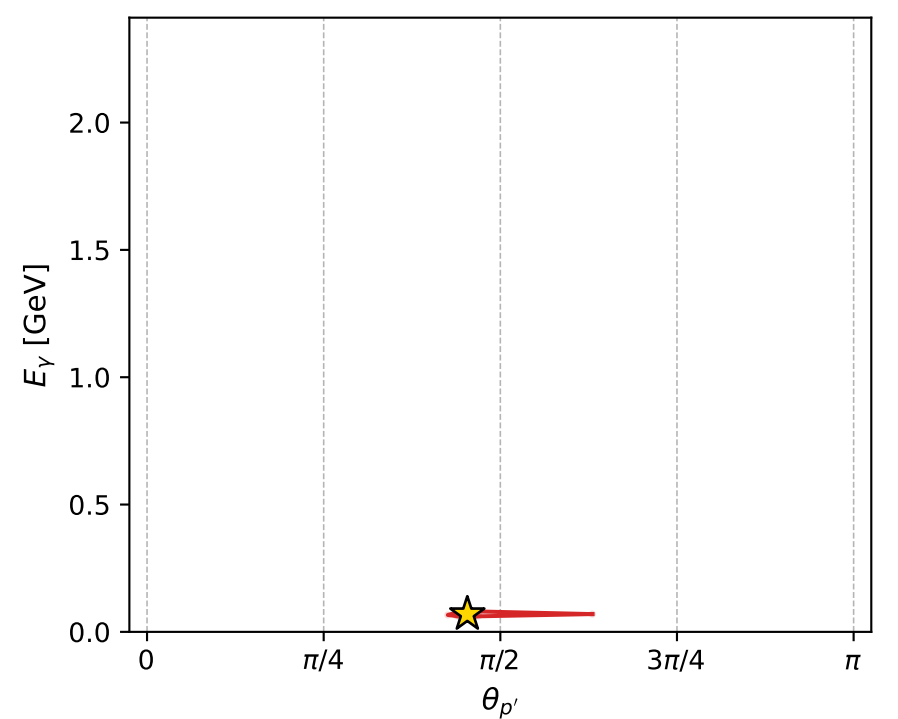
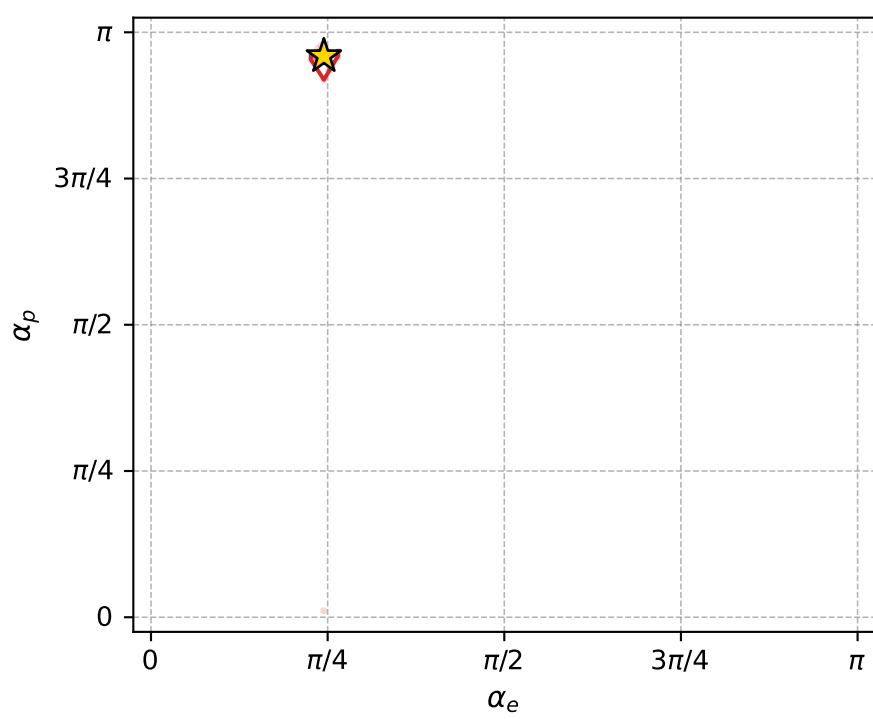
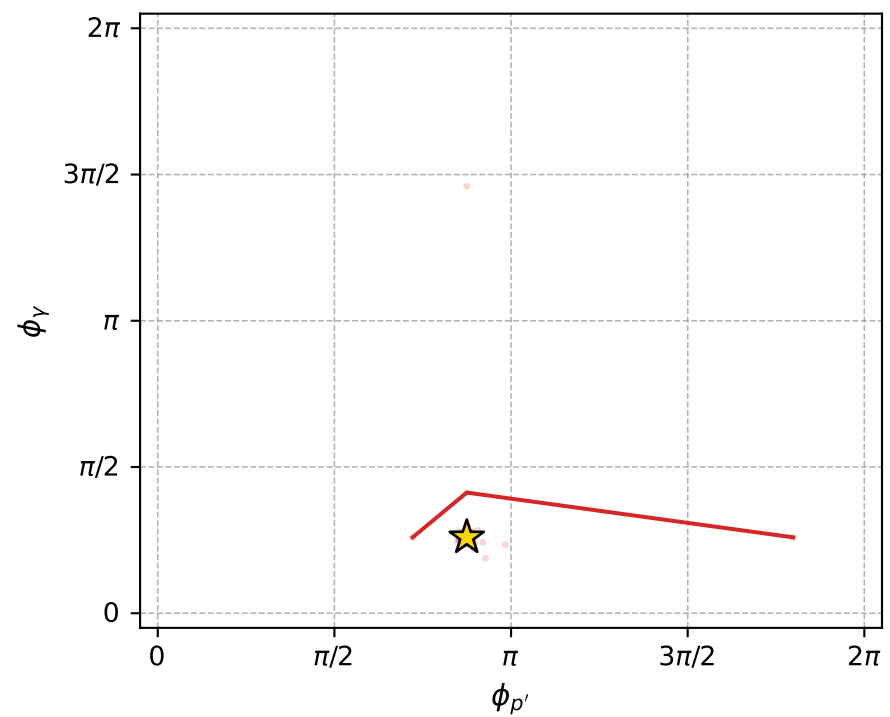
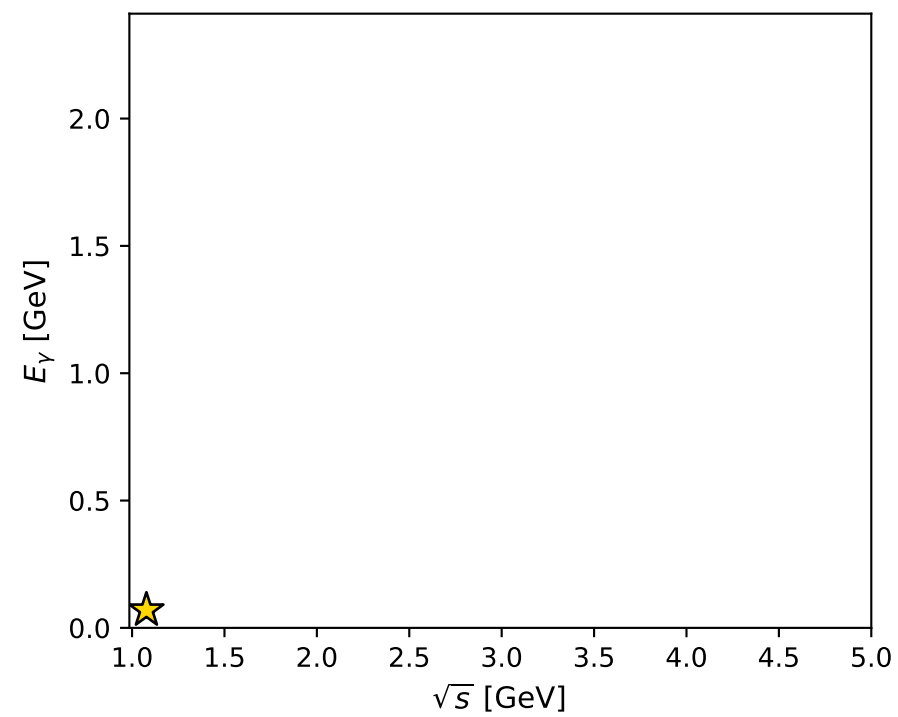
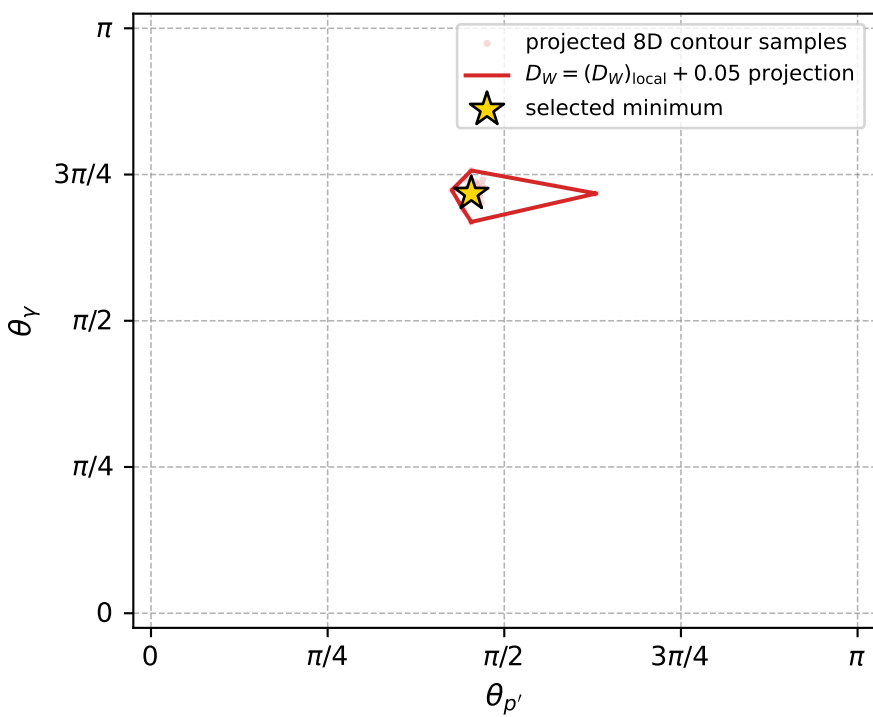
dw_mixing_angles_polarization_03_local_minimum_943 D_W=0.0155396
 region: polarization_cluster_03_local_minimum_943
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.7686046$ rad
 incoming lepton: $0.718881|+\rangle + 0.695133|-\rangle$
 proton mixing angle: $\alpha_p=3.0133567$ rad
 incoming proton: $-0.991789|+\rangle + 0.127885|-\rangle$

$s=1.16137$, $\sqrt{s}=1.07767$, $|\vec{P}|=0.130618$, $|\vec{P}'|=1.69213e-07$
 $\theta_{P'}=1.42401$, $\theta_V=2.2543$, $\phi_{P'}=2.7478$, $\phi_V=0.815177$
 $E_V=0.0698341$, $\theta_{\ell'}=0.887291$, $\phi_{\ell'}=3.95677$
 $Q^2=0.0297642$, $x_B=0.185132$, $t=-0.0169792$, $W^2=1.01085$, $y=0.571072$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1306$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1306$
 P $E= 0.9471$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1306$
 ℓ' $E= 0.0698$ $p_x= -0.0371$ $p_y= -0.0394$ $p_z= 0.0441$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0000$
 q_Y $E= 0.0698$ $p_x= 0.0371$ $p_y= 0.0394$ $p_z= -0.0441$



electron: polarization cluster P3, local minimum 943; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.015539621$
 $D_W \text{ contour} = 0.065539621$
 above global minimum = 0.015523188
 8D contour samples = 255

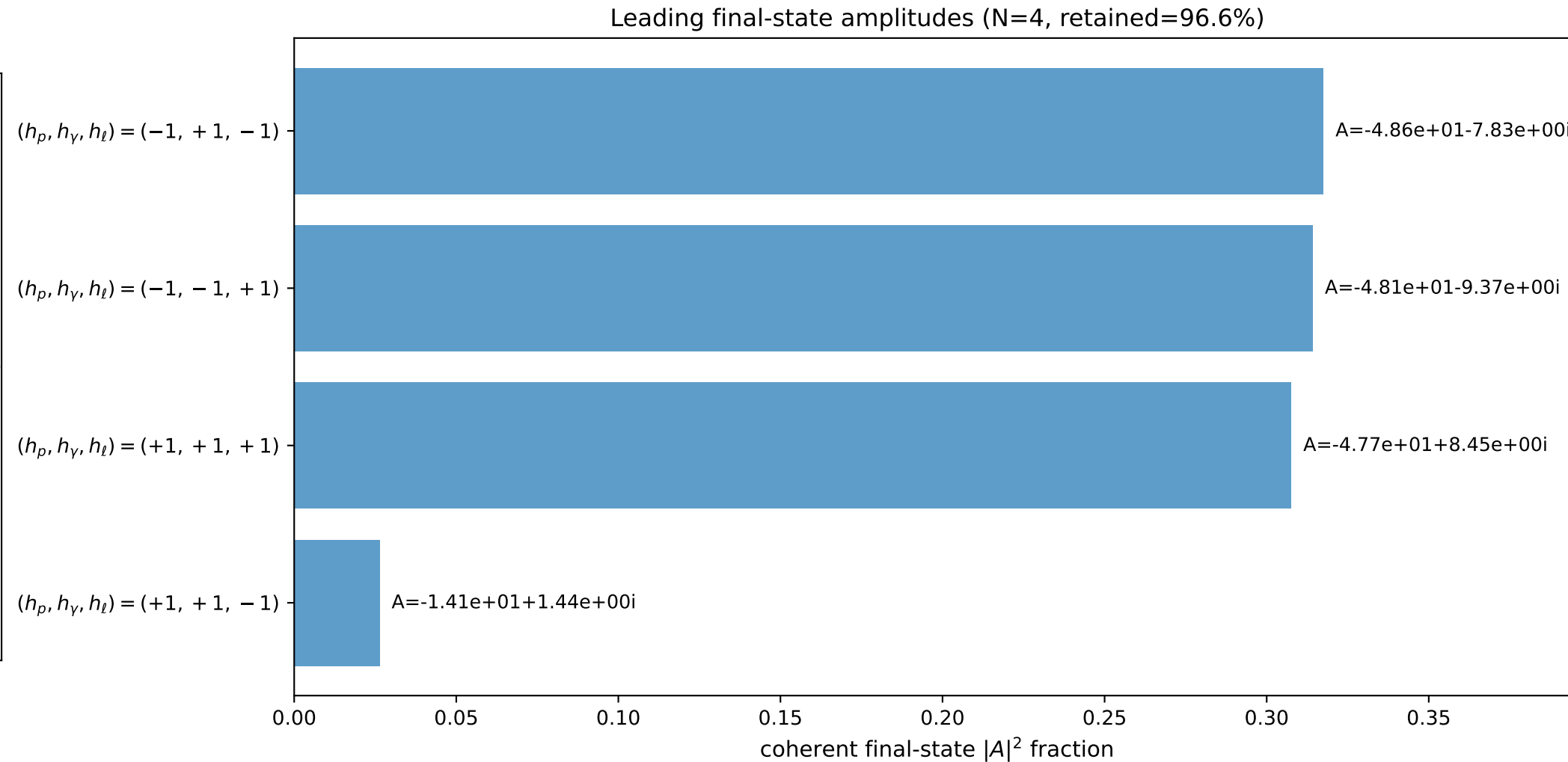
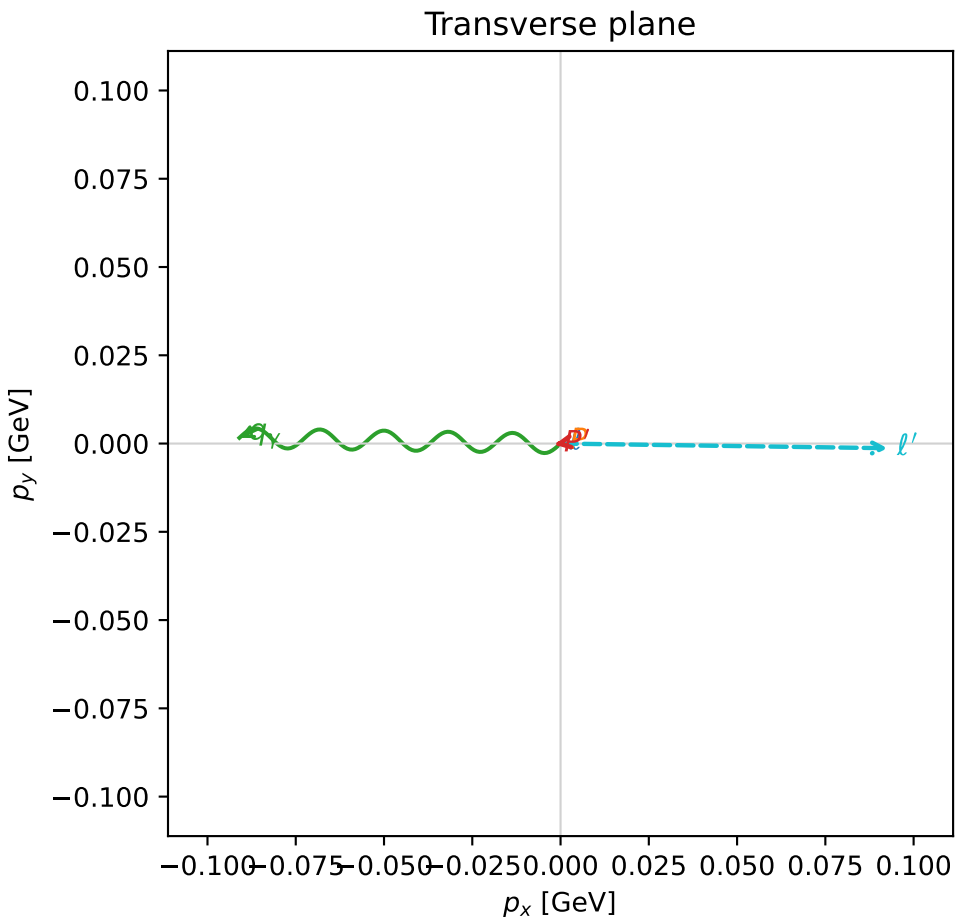
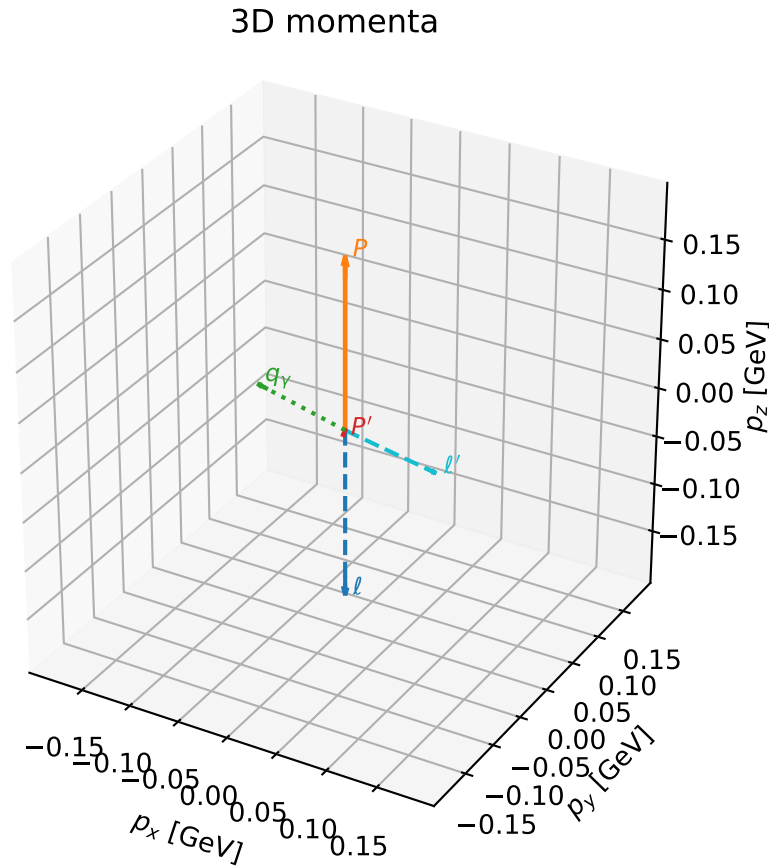
selected phase-space point:
 $\text{sqrt}_s = 1.07767$
 $\text{theta}_p_{\text{out}} = 1.424014$
 $\text{theta}_\gamma_{\text{out}} = 2.254301$
 $q_{\text{out}} = 0.06983412$
 $\text{phi}_p_{\text{out}} = 2.747796$
 $\text{phi}_\gamma_{\text{out}} = 0.8151774$
 $\alpha_e = 0.7686046$
 $\alpha_p = 3.013357$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

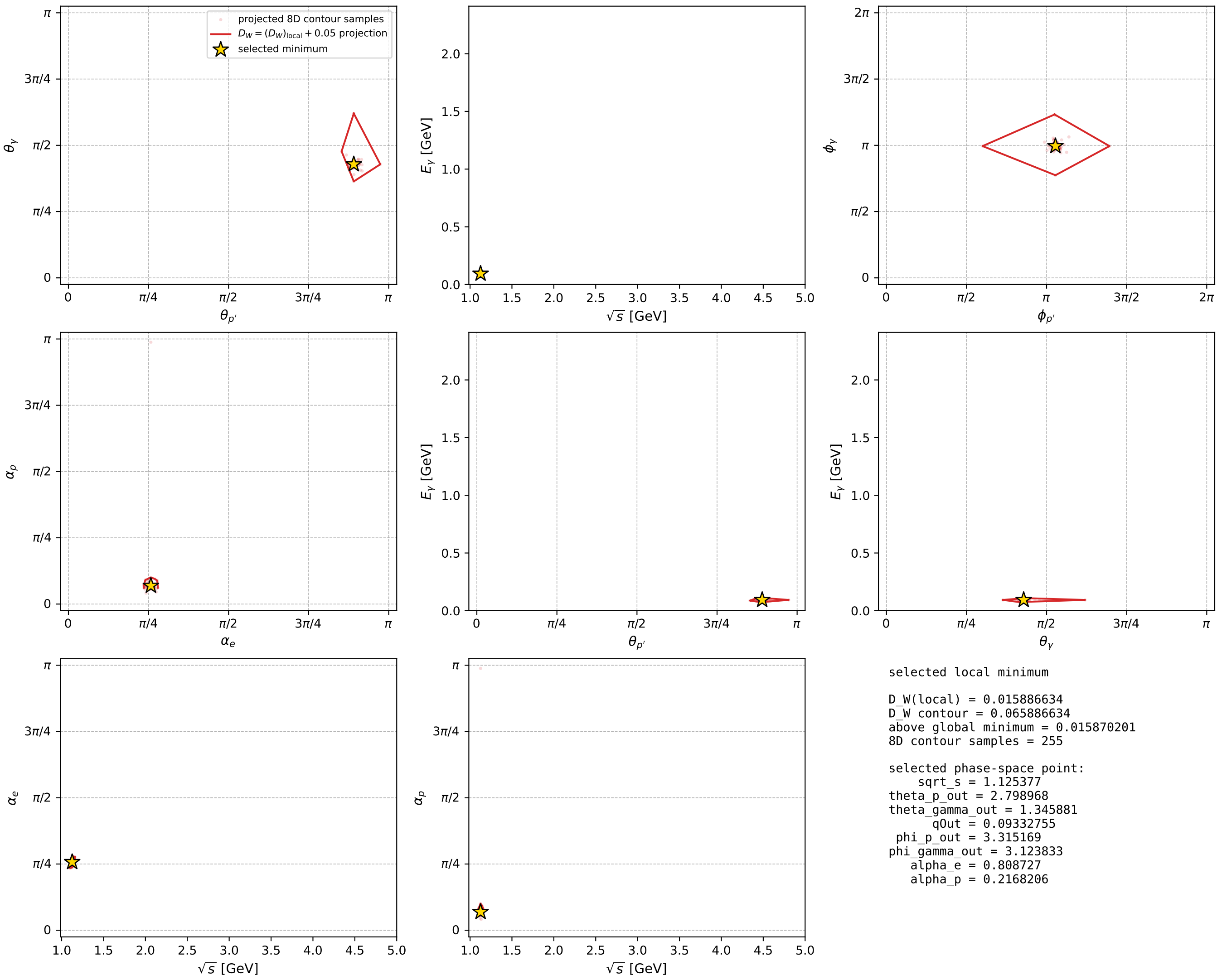
dw_mixing_angles_polarization_03_local_minimum_947 D_W=0.0158866
region: polarization_cluster_03_local_minimum_947
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.80872702$ rad
incoming lepton: $0.69042|+\rangle +0.723409|-\rangle$
proton mixing angle: $\alpha_p=0.21682056$ rad
incoming proton: $0.976586|+\rangle +0.215126|-\rangle$

$s=1.26647$, $\sqrt{s}=1.12538$, $|\vec{P}|=0.171777$, $|\vec{P}'|=0.00511121$
 $\theta_{p'}=2.79897$, $\theta_Y=1.34588$, $\phi_{p'}=3.31517$, $\phi_Y=3.12383$
 $E_Y=0.0933276$, $\theta_{\ell'}=1.74178$, $\phi_{\ell'}=6.26895$
 $Q^2=0.0268092$, $x_B=0.132858$, $t=-0.0309446$, $W^2=1.05482$, $y=0.521915$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1718$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1718$
 P $E= 0.9536$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1718$
 ℓ' $E= 0.0940$ $p_x= 0.0927$ $p_y= -0.0013$ $p_z= -0.0160$
 P' $E= 0.9380$ $p_x= -0.0017$ $p_y= -0.0003$ $p_z= -0.0048$
 q_Y $E= 0.0933$ $p_x= -0.0910$ $p_y= 0.0016$ $p_z= 0.0208$



electron: polarization cluster P3, local minimum 947; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



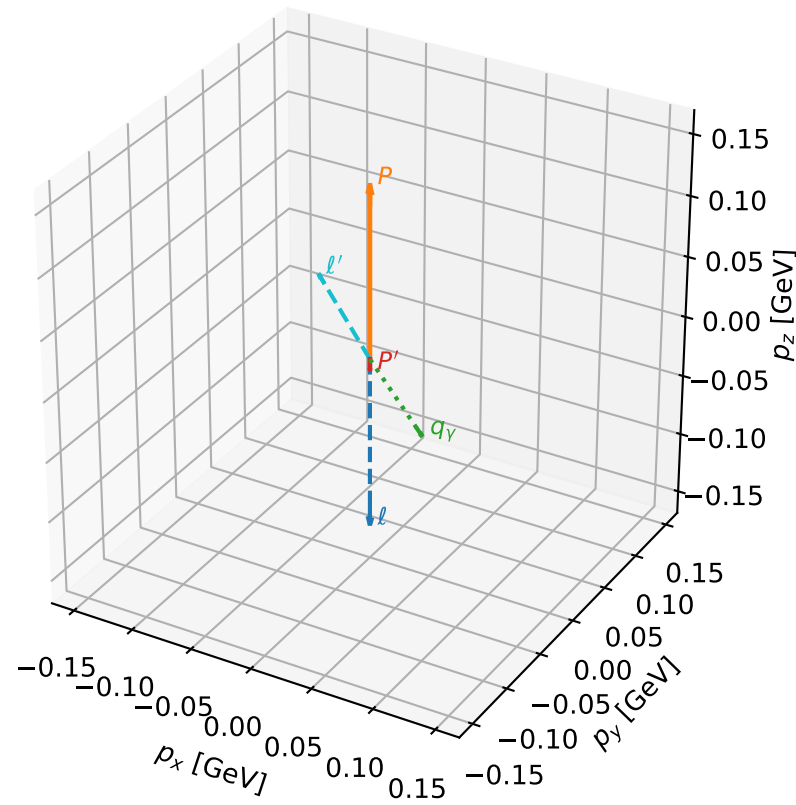
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_949 D_W=0.0159413
 region: polarization_cluster_03_local_minimum_949
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.77247231$ rad
 incoming lepton: $0.716187|+\rangle + 0.697908|-\rangle$
 proton mixing angle: $\alpha_p=3.0388823$ rad
 incoming proton: $-0.99473|+\rangle + 0.10253|-\rangle$

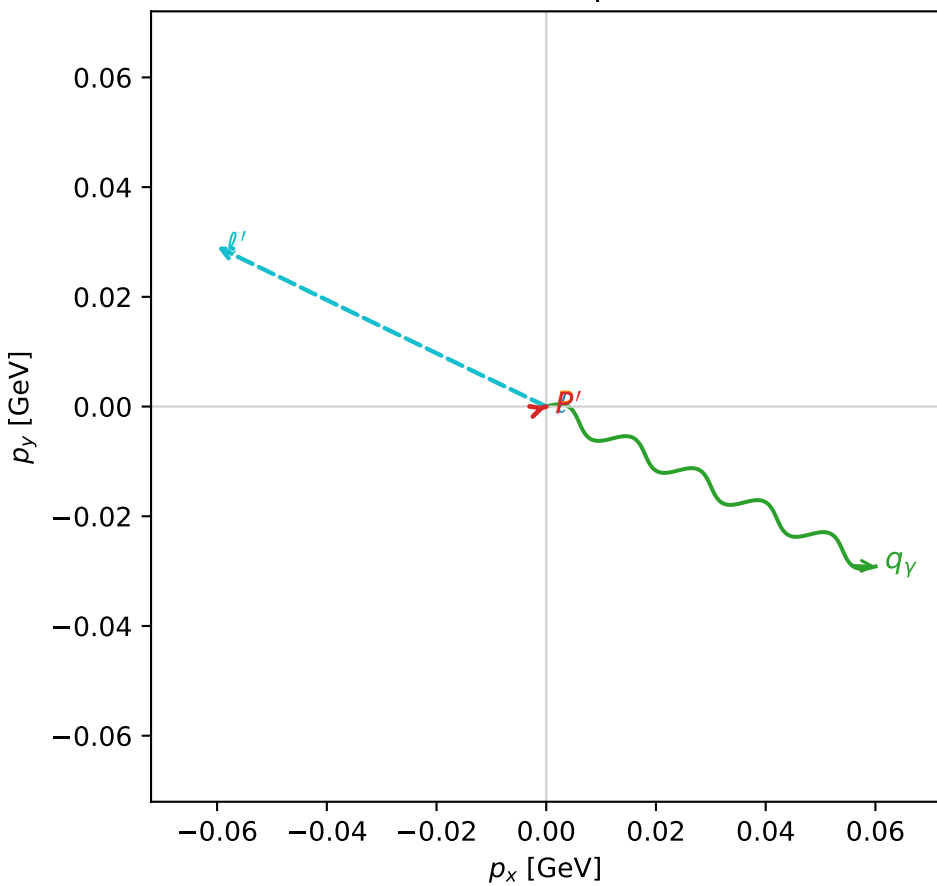
$s=1.18739$, $\sqrt{s}=1.08967$, $|\vec{P}|=0.141117$, $|\vec{P}'|=0.00815238$
 $\theta_{P'}=3.14159$, $\theta_V=2.01454$, $\phi_{P'}=0.313817$, $\phi_V=5.83115$
 $E_V=0.0738938$, $\theta_l=1.0322$, $\phi_{l'}=2.68956$
 $Q^2=0.0331965$, $x_B=0.193781$, $t=-0.0221708$, $W^2=1.01796$, $y=0.557021$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1411$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1411$
 P $E= 0.9486$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1411$
 l' $E= 0.0777$ $p_x= -0.0600$ $p_y= 0.0292$ $p_z= 0.0399$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0082$
 q_V $E= 0.0739$ $p_x= 0.0600$ $p_y= -0.0292$ $p_z= -0.0317$

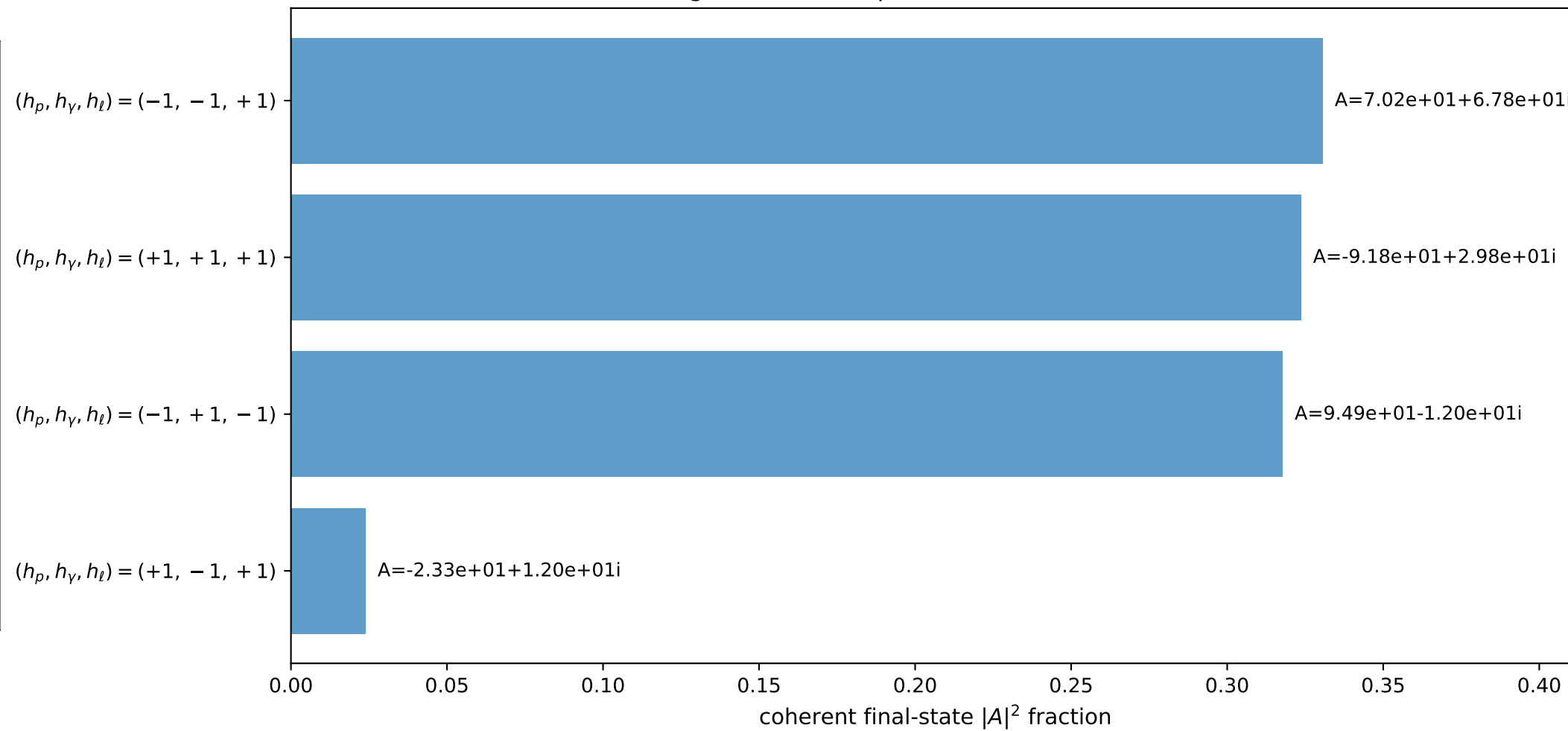
3D momenta



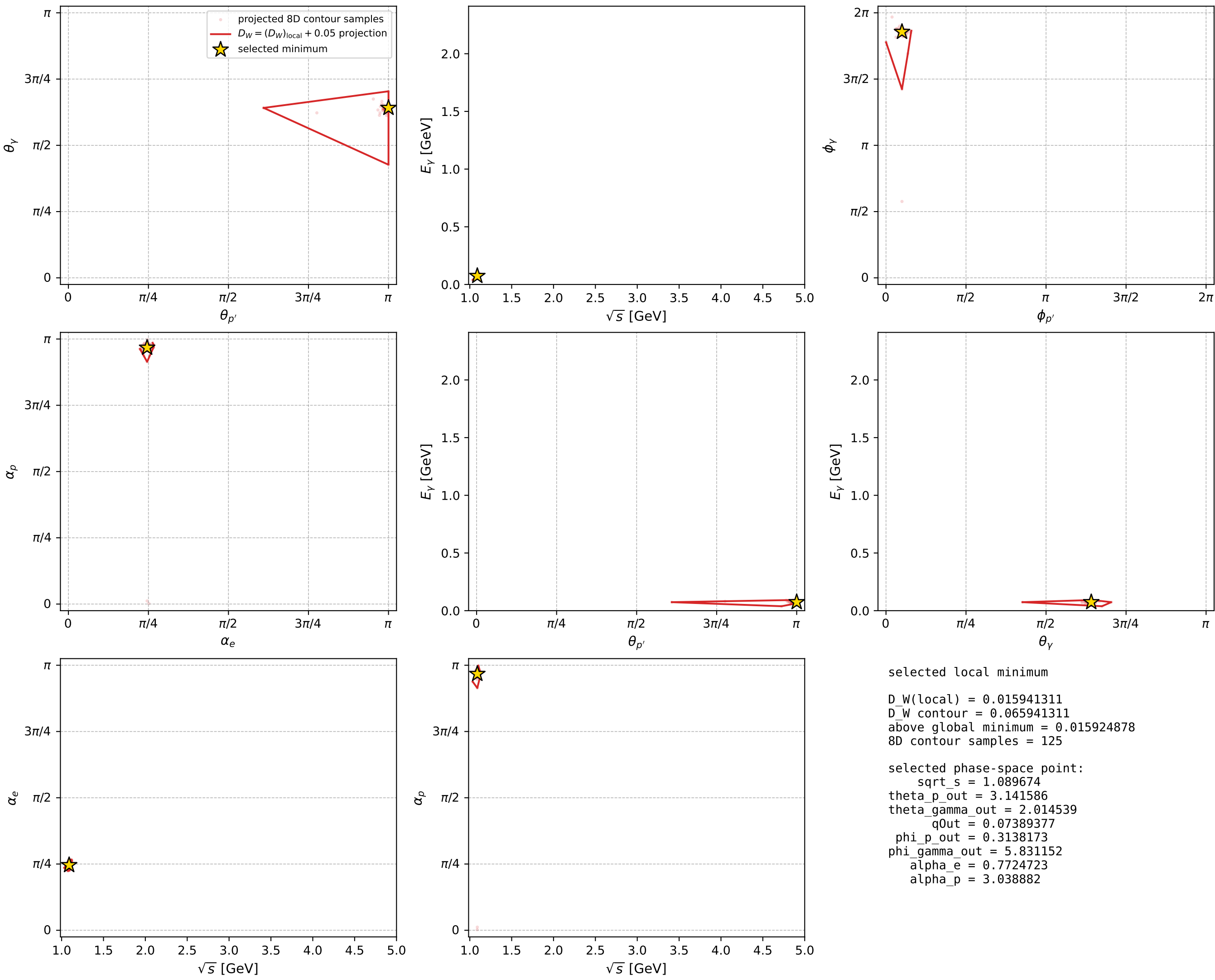
Transverse plane



Leading final-state amplitudes (N=4, retained=99.6%)



electron: polarization cluster P3, local minimum 949; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



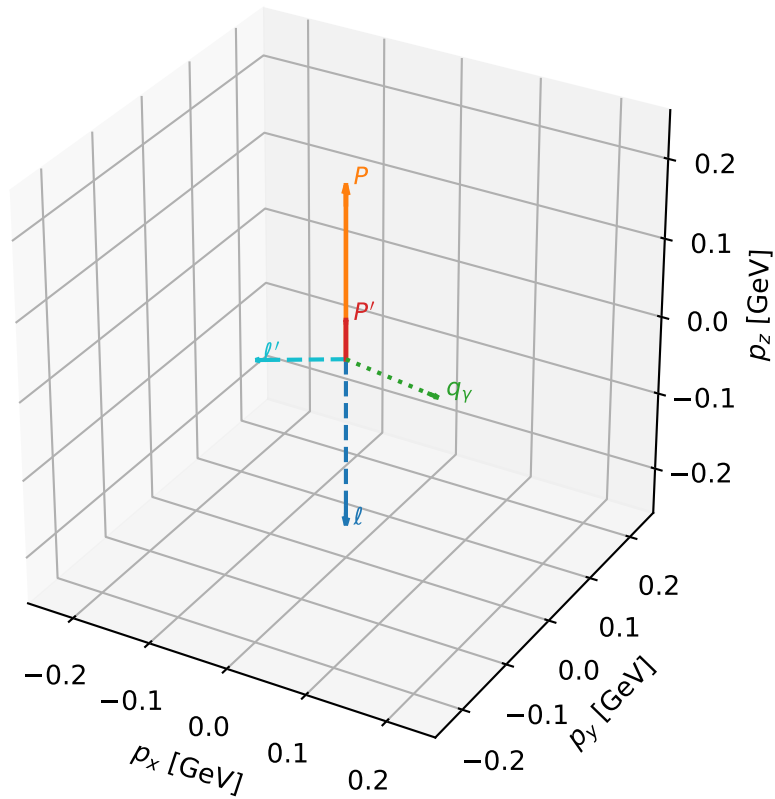
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_950 D_W=0.01601
region: polarization_cluster_03_local_minimum_950
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81877057$ rad
incoming lepton: $0.68312|+\rangle +0.730307|-\rangle$
proton mixing angle: $\alpha_p=2.9968052$ rad
incoming proton: $-0.989537|+\rangle +0.144282|-\rangle$

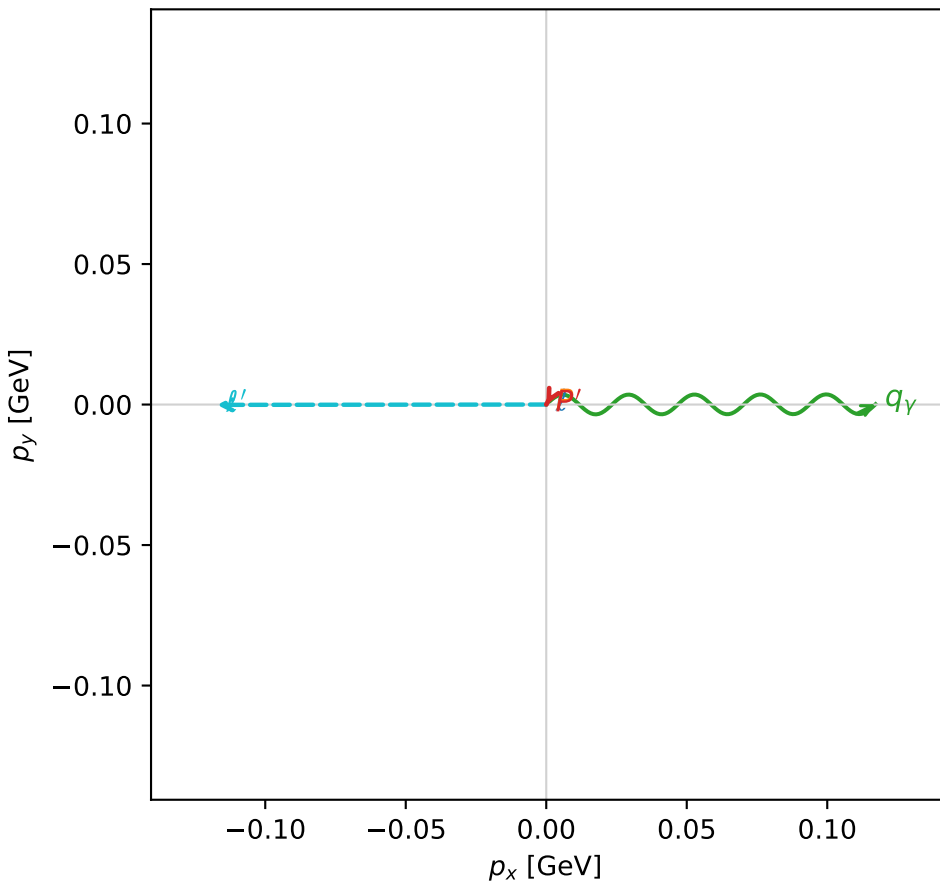
$s=1.39179$, $\sqrt{s}=1.17974$, $|\vec{P}|=0.216973$, $|\vec{P}'|=0.0497768$
 $\theta_{P'}=0.000110249$, $\theta_V=1.69747$, $\phi_{P'}=4.21159$, $\phi_V=0.00102048$
 $E_V=0.118149$, $\theta_{\ell'}=1.85983$, $\phi_{\ell'}=3.14257$
 $Q^2=0.0379363$, $x_B=0.145137$, $t=-0.0274049$, $W^2=1.10329$, $y=0.510567$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2170$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2170$
 P $E= 0.9628$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2170$
 ℓ' $E= 0.1223$ $p_x= -0.1172$ $p_y= -0.0001$ $p_z= -0.0349$
 P' $E= 0.9393$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0498$
 q_V $E= 0.1181$ $p_x= 0.1172$ $p_y= 0.0001$ $p_z= -0.0149$

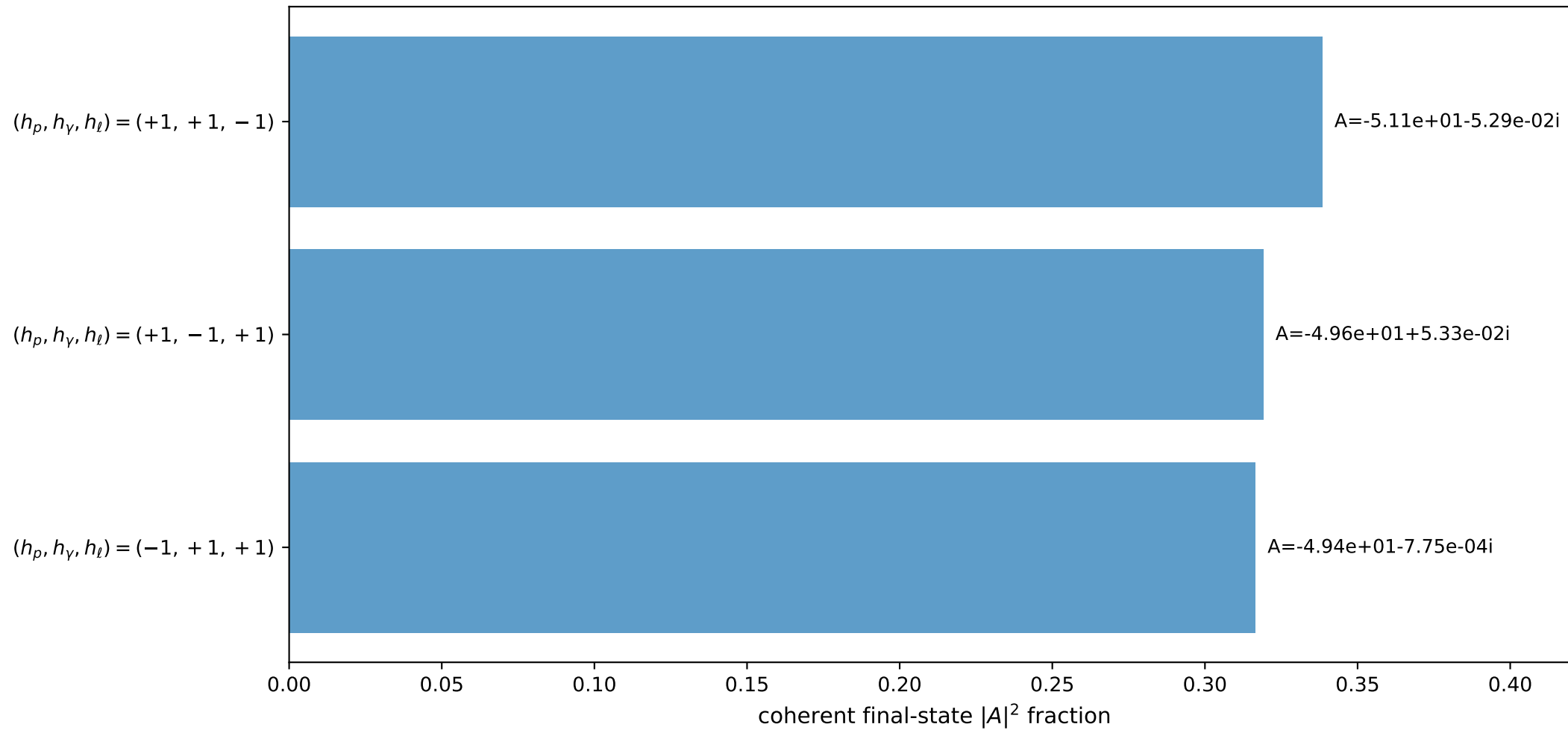
3D momenta



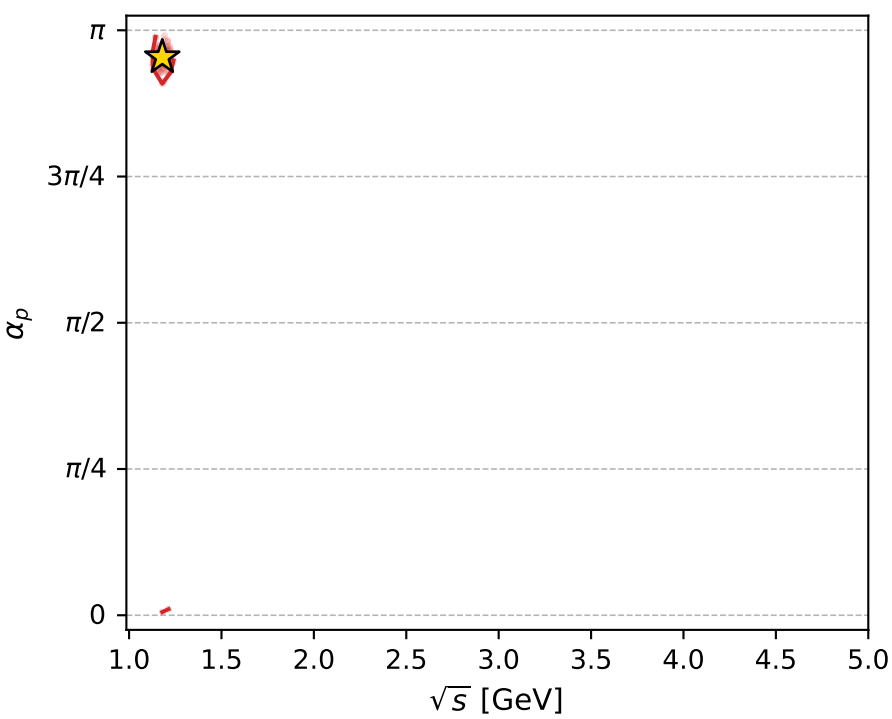
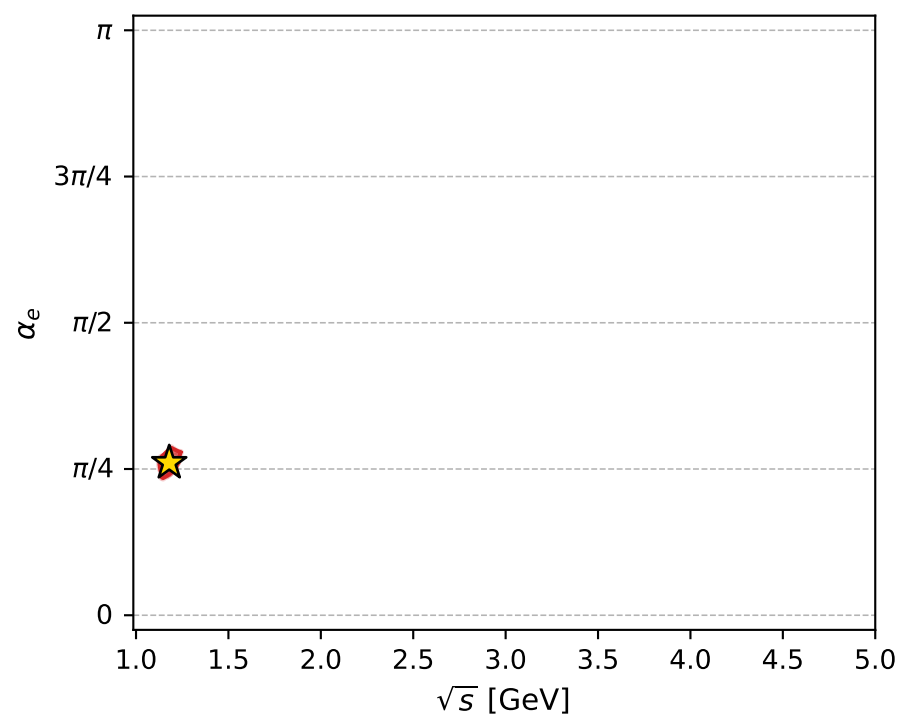
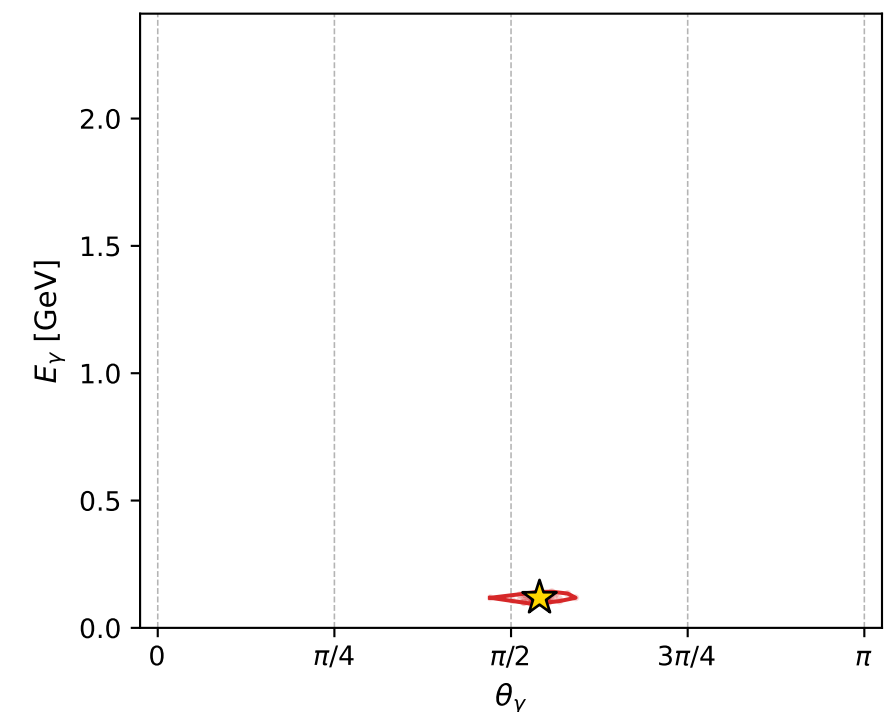
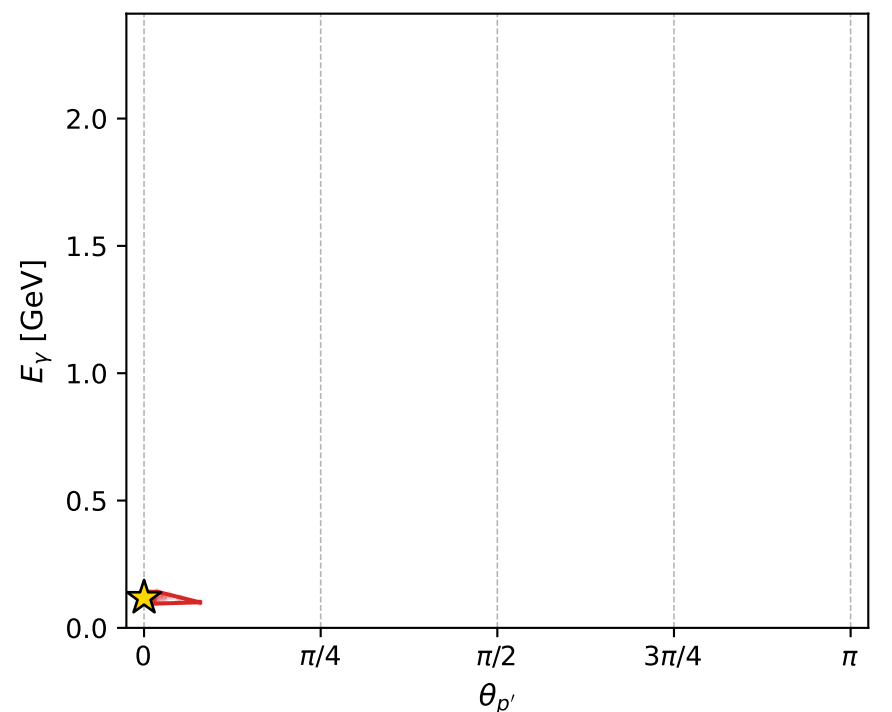
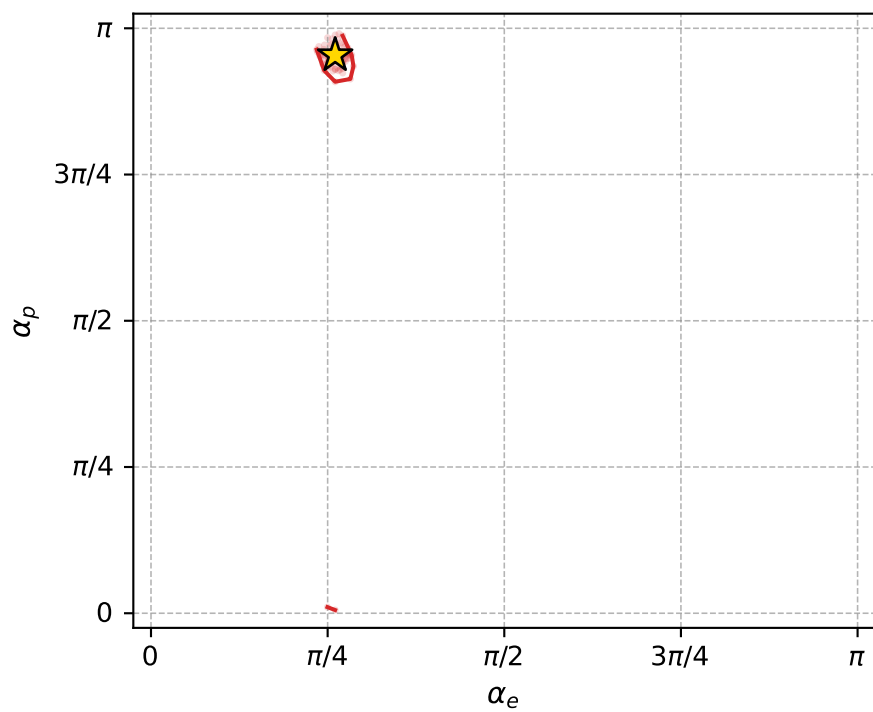
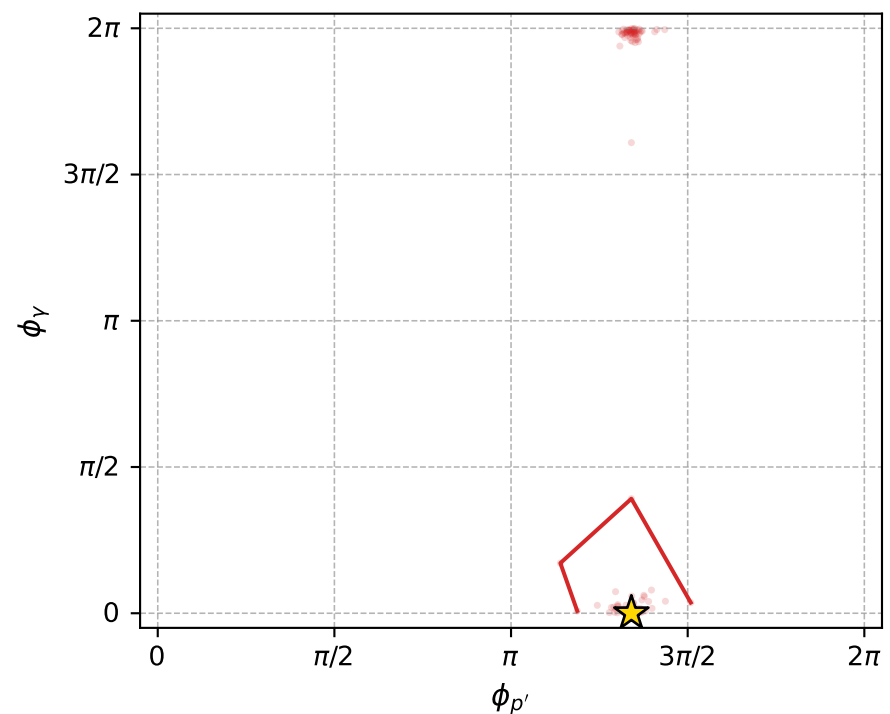
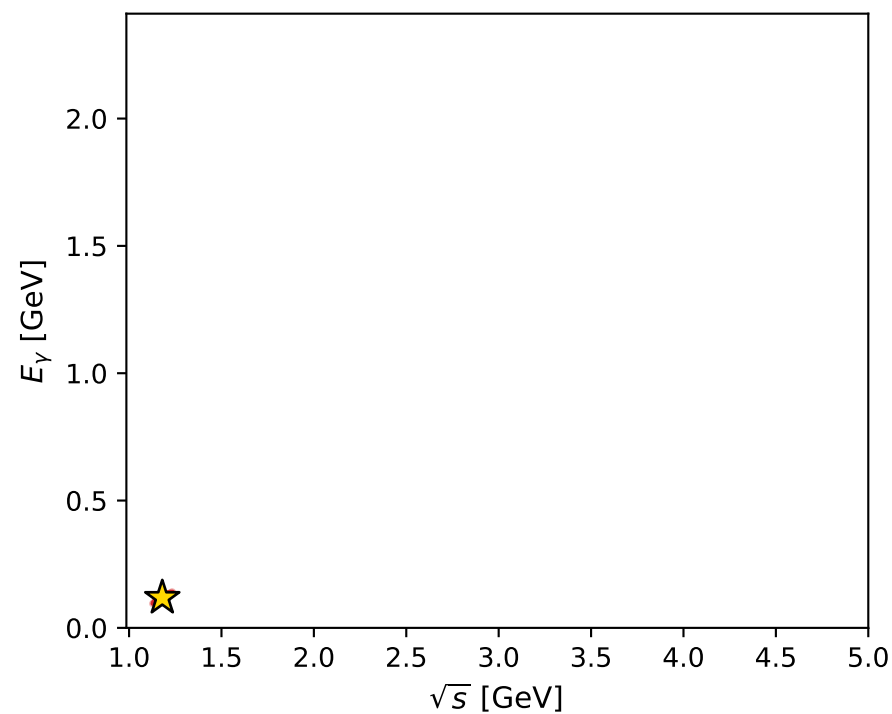
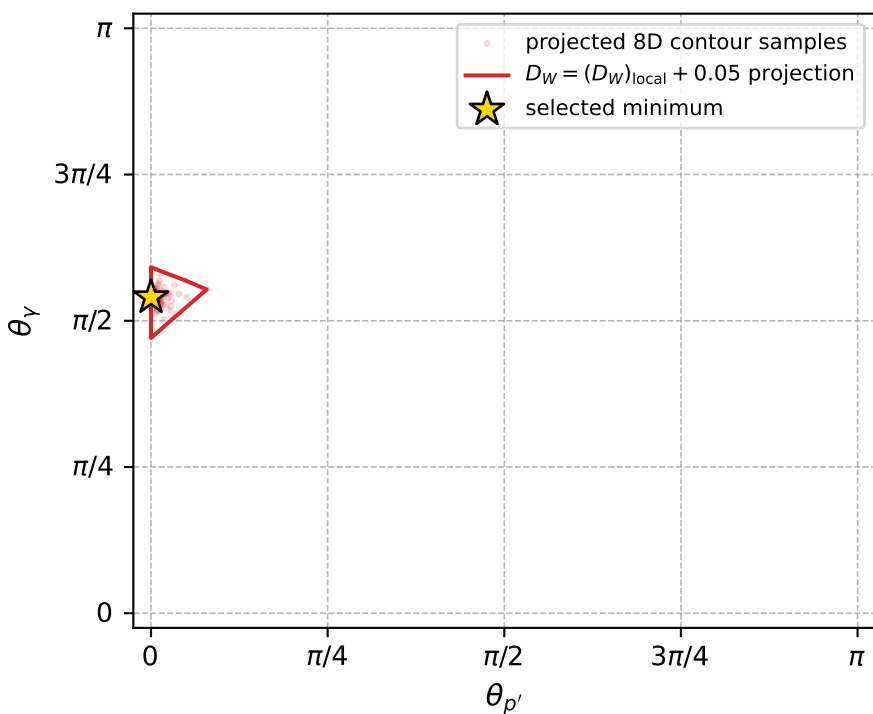
Transverse plane



Leading final-state amplitudes (N=3, retained=97.4%)



electron: polarization cluster P3, local minimum 950; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.016010042$
 $D_W \text{ contour} = 0.066010042$
 above global minimum = 0.015993609
 8D contour samples = 139

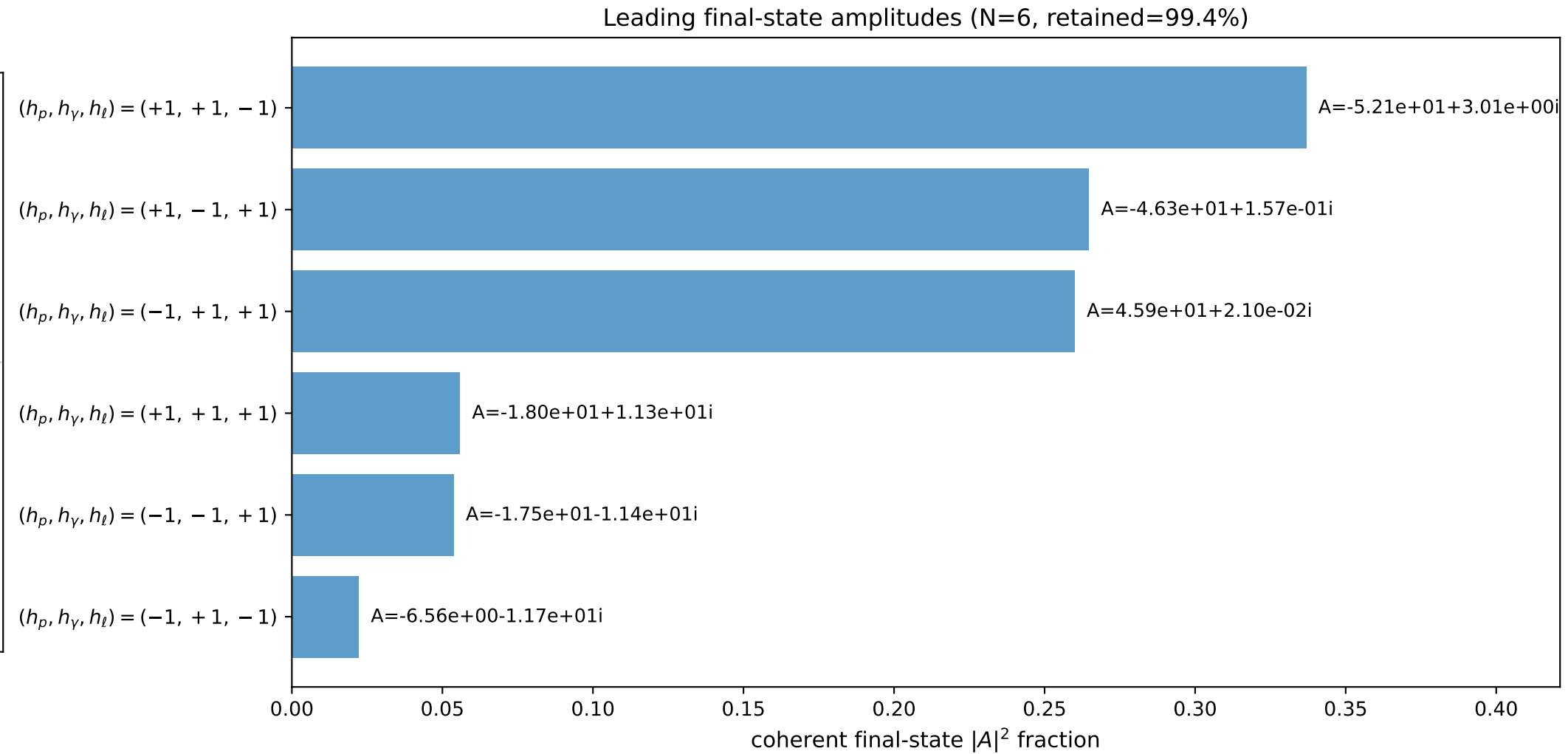
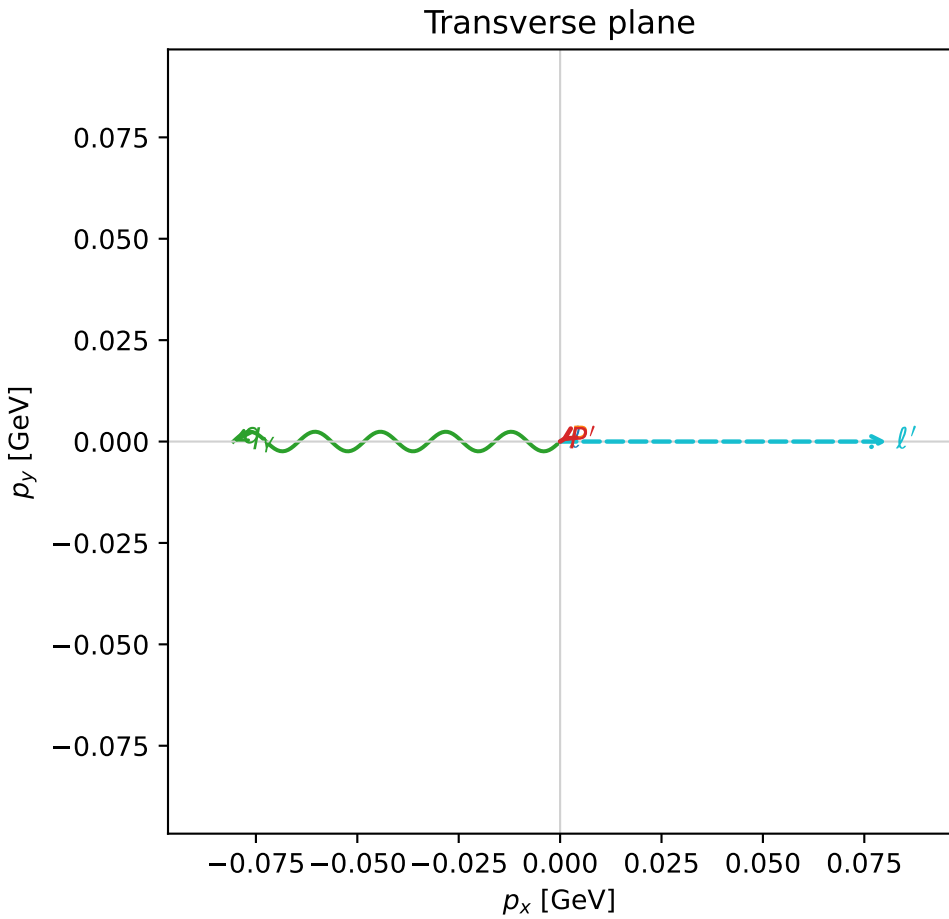
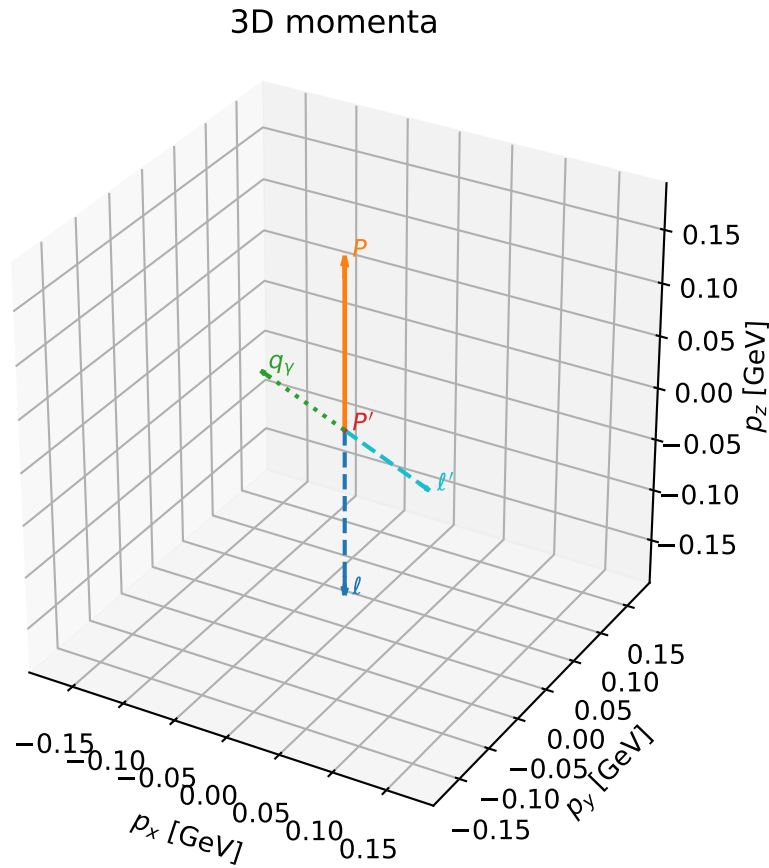
selected phase-space point:
 $\text{sqrt_s} = 1.179741$
 $\text{theta_p_out} = 0.000110249$
 $\text{theta_gamma_out} = 1.697474$
 $\text{qOut} = 0.1181491$
 $\text{phi_p_out} = 4.21159$
 $\text{phi_gamma_out} = 0.00102048$
 $\text{alpha_e} = 0.8187706$
 $\text{alpha_p} = 2.996805$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

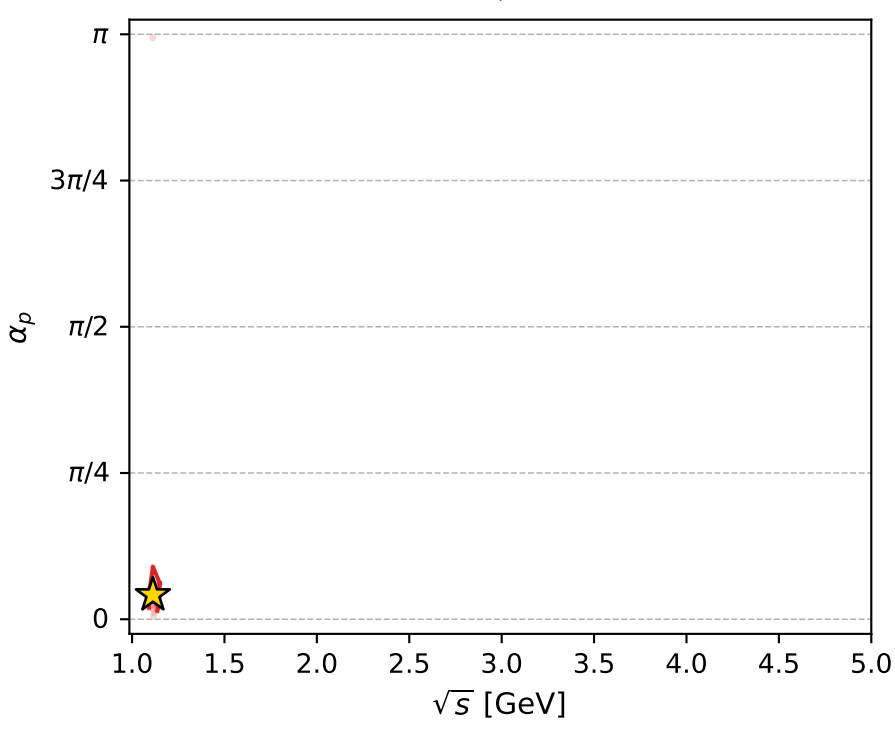
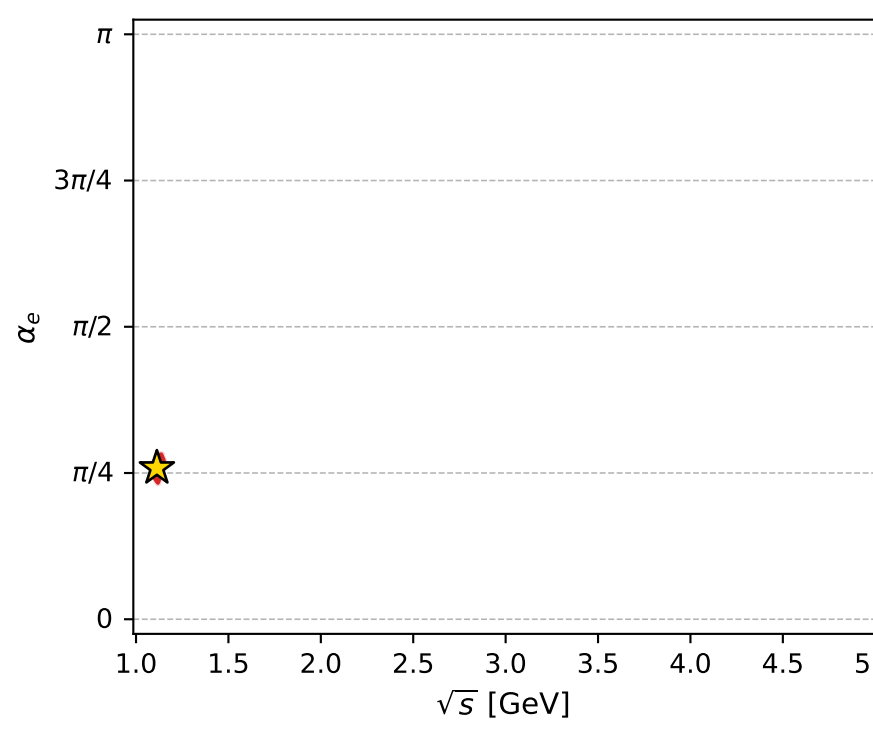
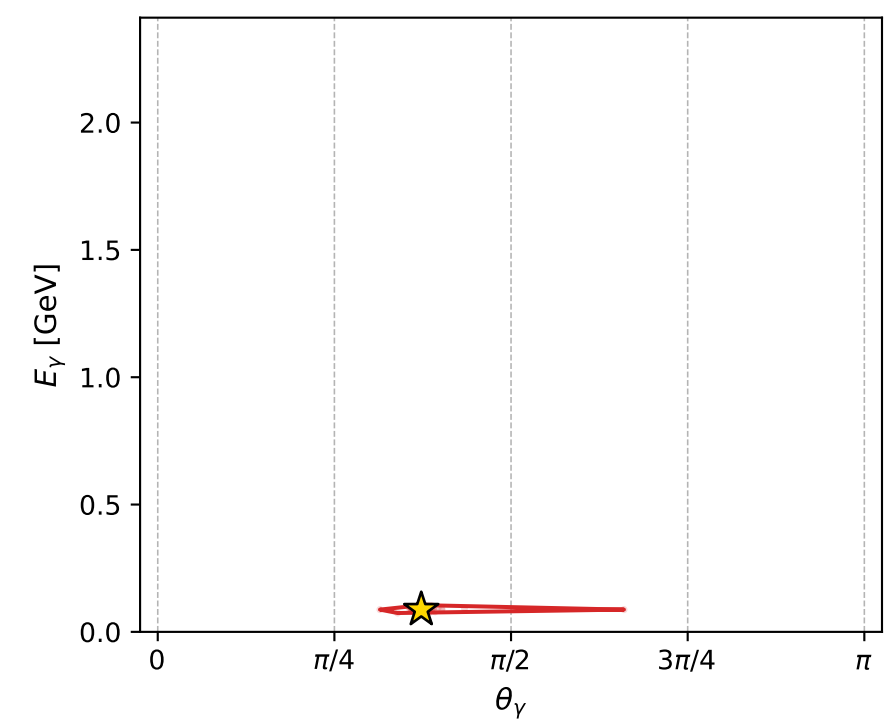
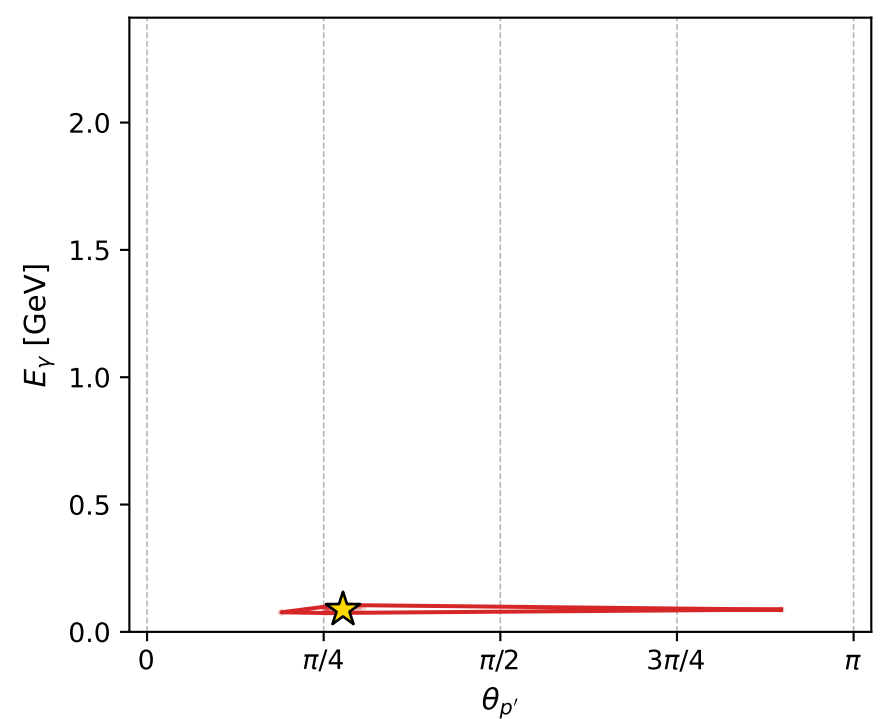
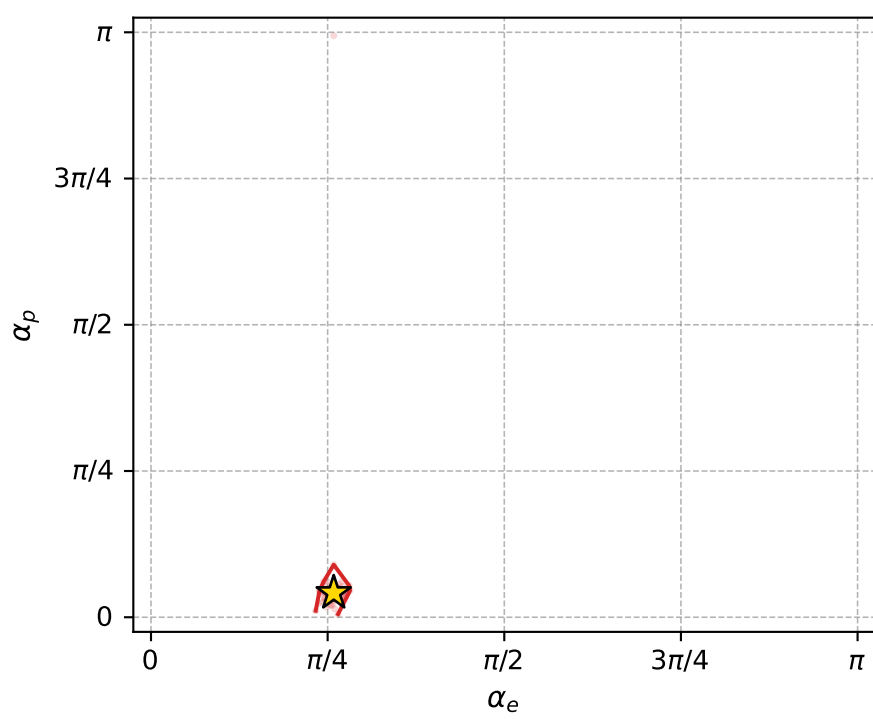
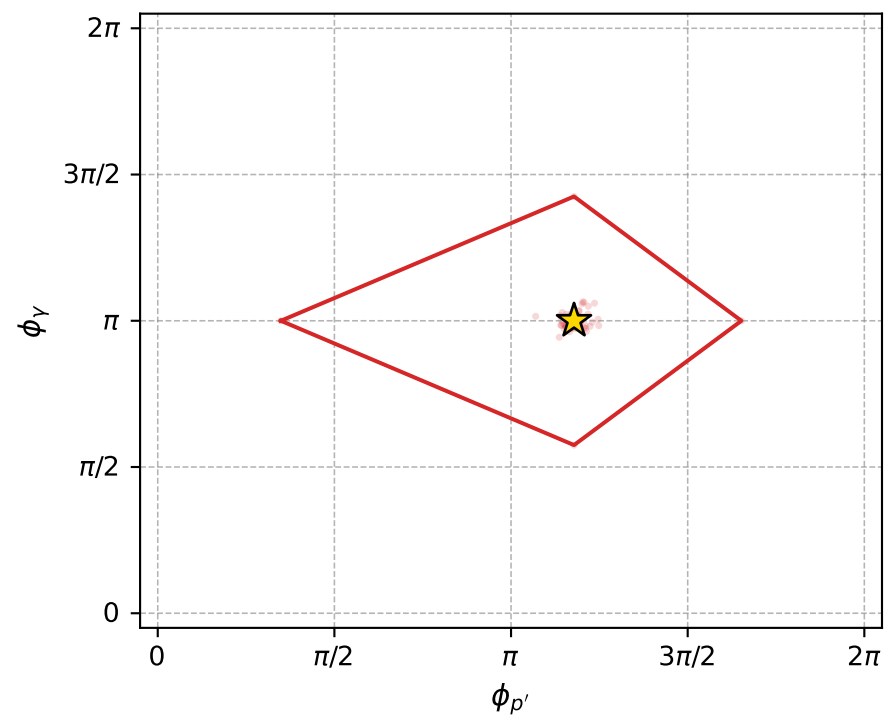
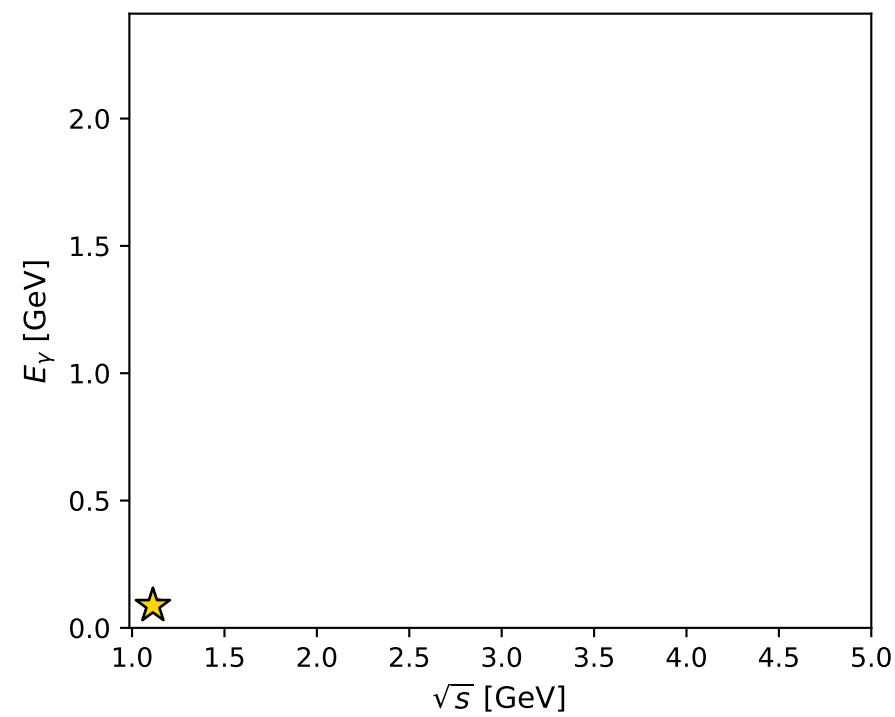
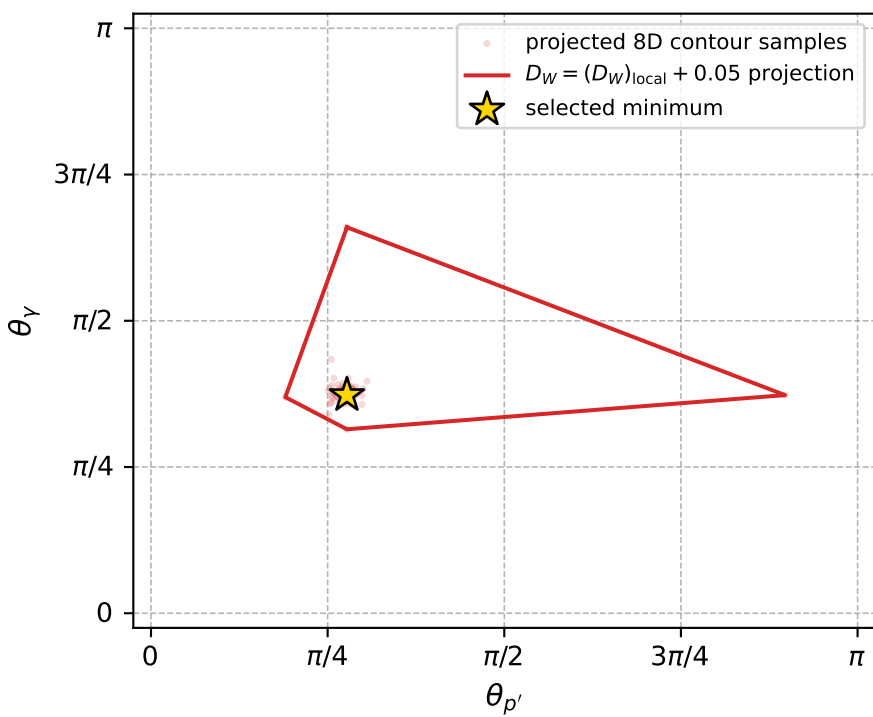
dw_mixing_angles_polarization_03_local_minimum_958 D_W=0.0164786
 region: polarization_cluster_03_local_minimum_958
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81246866$ rad
 incoming lepton: $0.687708|+\rangle +0.725987|-\rangle$
 proton mixing angle: $\alpha_p=0.13086151$ rad
 incoming proton: $0.99145|+\rangle +0.130488|-\rangle$

$s=1.23851$, $\sqrt{s}=1.11288$, $|\vec{P}|=0.161141$, $|\vec{P}'|=6.2775e-07$
 $\theta_{p'}=0.871254$, $\theta_\gamma=1.17158$, $\phi_{p'}=3.70173$, $\phi_\gamma=3.14146$
 $E_\gamma=0.0874401$, $\theta_{\ell'}=1.97002$, $\phi_{\ell'}=6.28305$
 $Q^2=0.0172269$, $x_B=0.0950373$, $t=-0.0257774$, $W^2=1.04388$, $y=0.50539$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1611$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1611$
 P $E= 0.9517$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1611$
 ℓ' $E= 0.0874$ $p_x= 0.0806$ $p_y= -0.0000$ $p_z= -0.0340$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0000$
 q_γ $E= 0.0874$ $p_x= -0.0806$ $p_y= 0.0000$ $p_z= 0.0340$



electron: polarization cluster P3, local minimum 958; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.016478649$
 $D_W \text{ contour} = 0.066478649$
 above global minimum = 0.016462216
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.112882$
 $\text{theta_p_out} = 0.8712538$
 $\text{theta_gamma_out} = 1.17158$
 $\text{qOut} = 0.08744014$
 $\text{phi_p_out} = 3.701728$
 $\text{phi_gamma_out} = 3.141459$
 $\text{alpha_e} = 0.8124687$
 $\text{alpha_p} = 0.1308615$

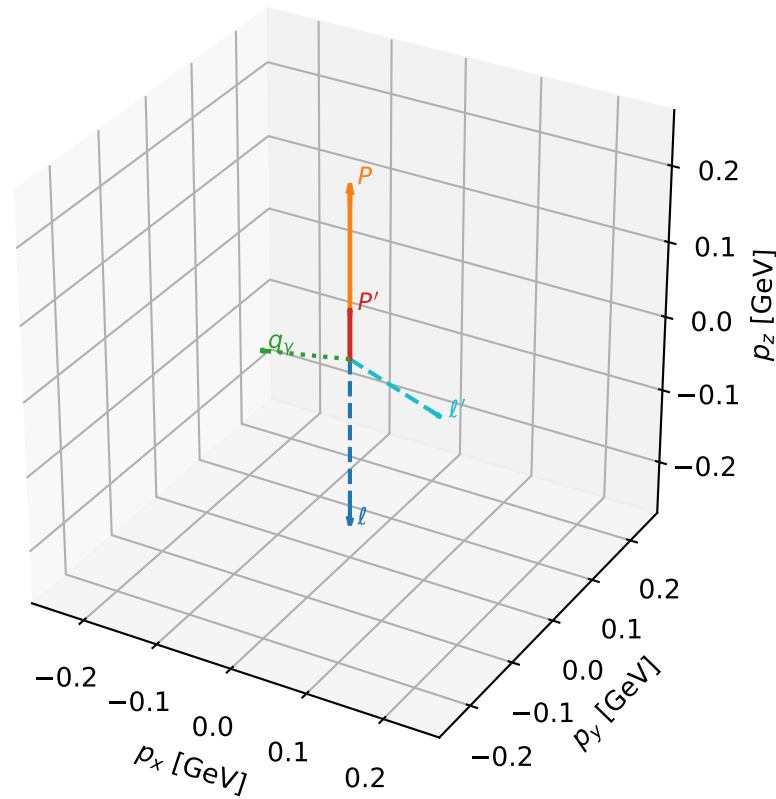
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_963 D_W=0.0166347
 region: polarization_cluster_03_local_minimum_963
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81066558$ rad
 incoming lepton: $0.689016|+\rangle + 0.724746|-\rangle$
 proton mixing angle: $\alpha_p=0.12552112$ rad
 incoming proton: $0.992133|+\rangle + 0.125192|-\rangle$

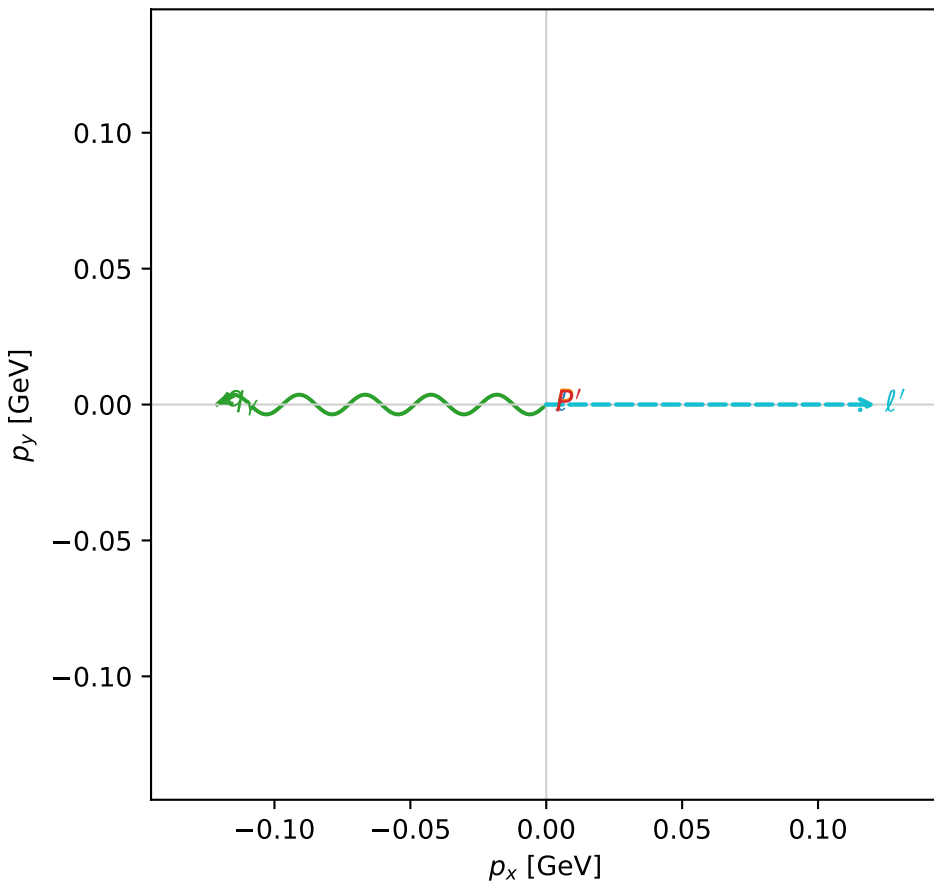
$s=1.42062$, $\sqrt{s}=1.1919$, $|\vec{P}|=0.226856$, $|\vec{P}'|=0.065164$
 $\theta_{P'}=0$, $\theta_V=1.75843$, $\phi_{P'}=3.0231$, $\phi_V=3.14157$
 $E_V=0.123339$, $\theta_{\ell'}=1.9056$, $\phi_{\ell'}=6.28316$
 $Q^2=0.0390834$, $x_B=0.142628$, $t=-0.0255301$, $W^2=1.11478$, $y=0.50672$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2269$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2269$
 P $E= 0.9650$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2269$
 ℓ' $E= 0.1283$ $p_x= 0.1212$ $p_y= -0.0000$ $p_z= -0.0422$
 P' $E= 0.9403$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0652$
 q_Y $E= 0.1233$ $p_x= -0.1212$ $p_y= 0.0000$ $p_z= -0.0230$

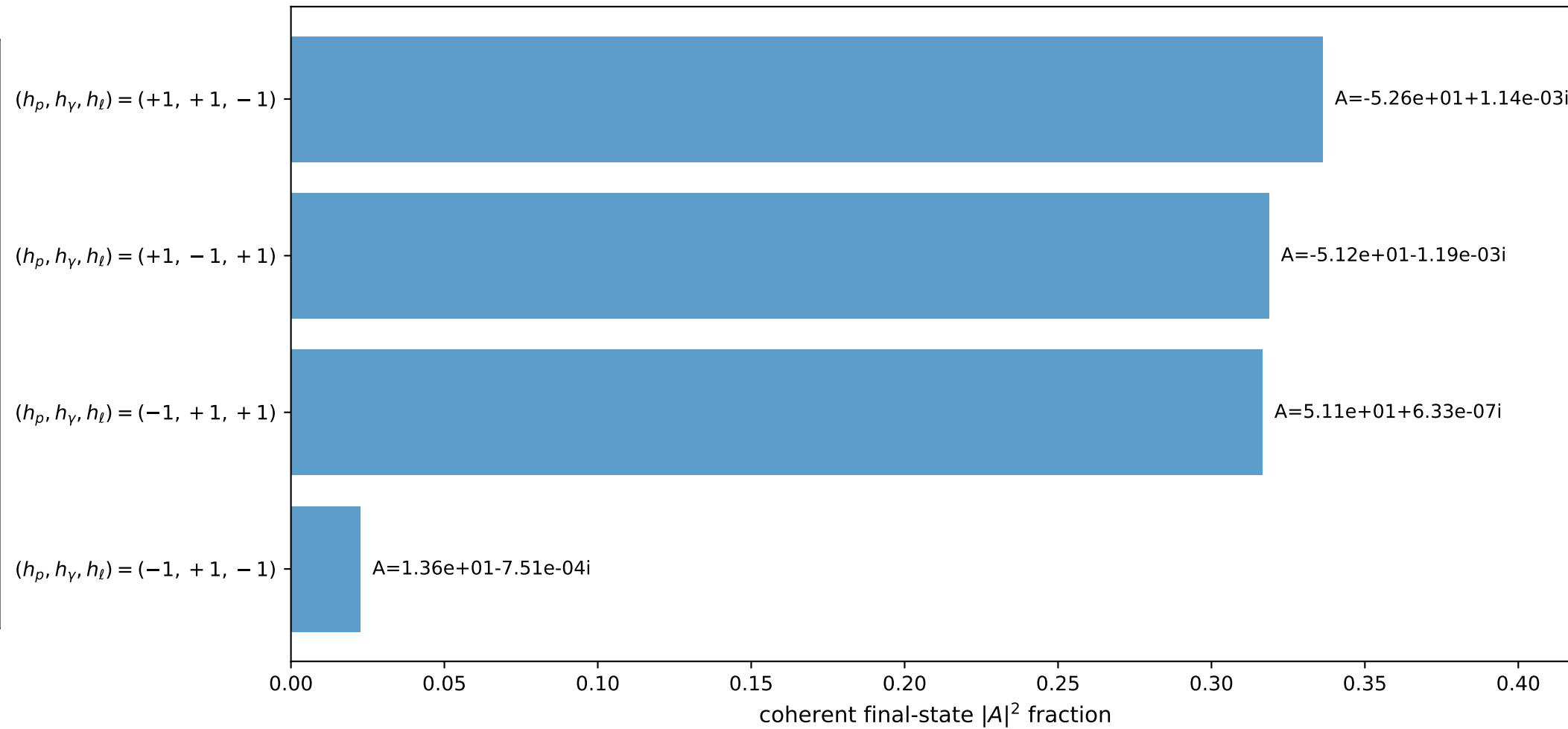
3D momenta



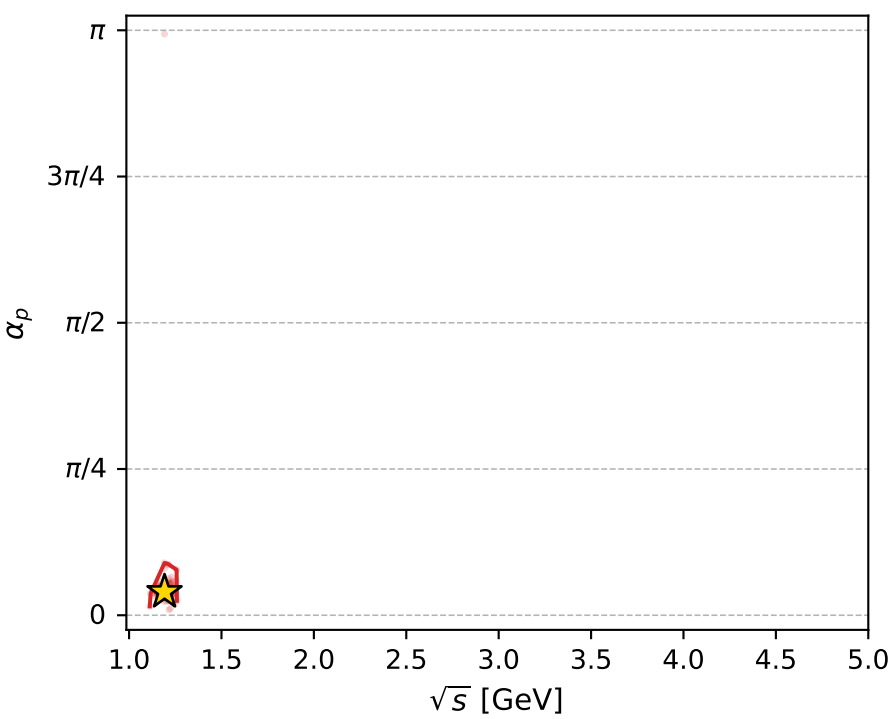
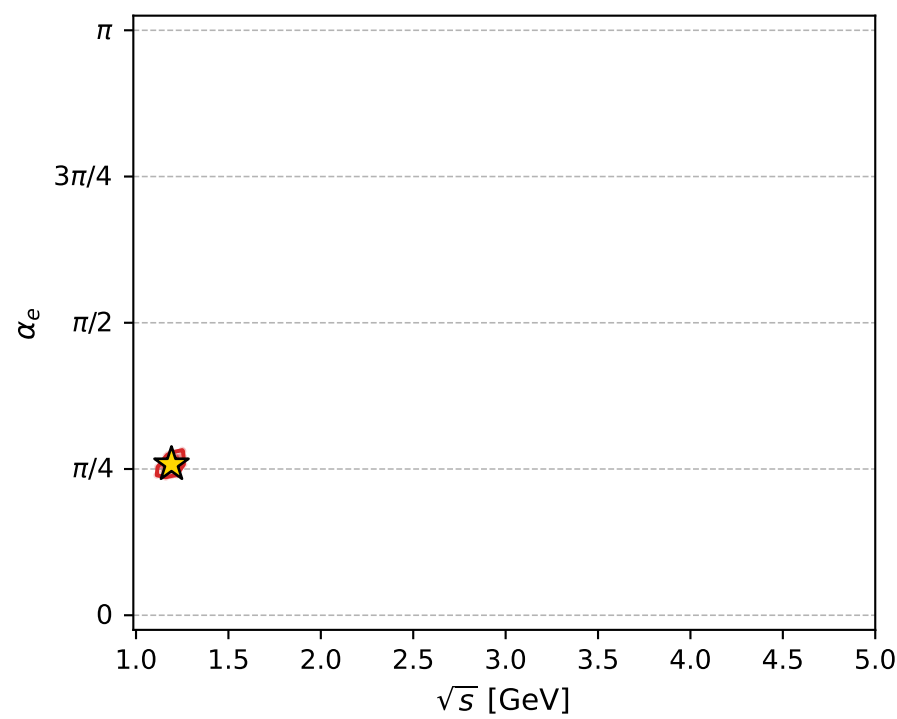
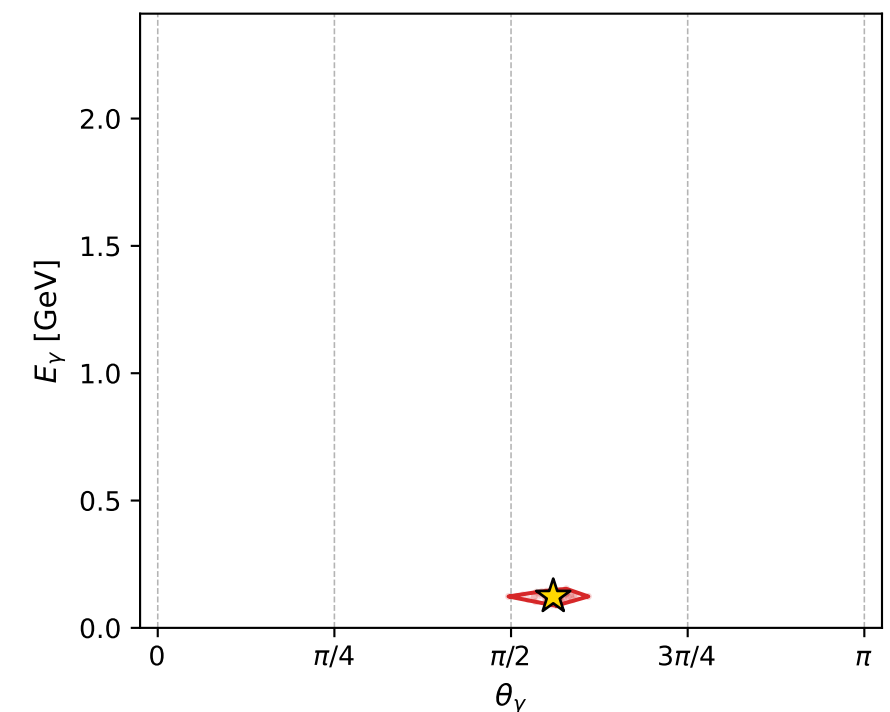
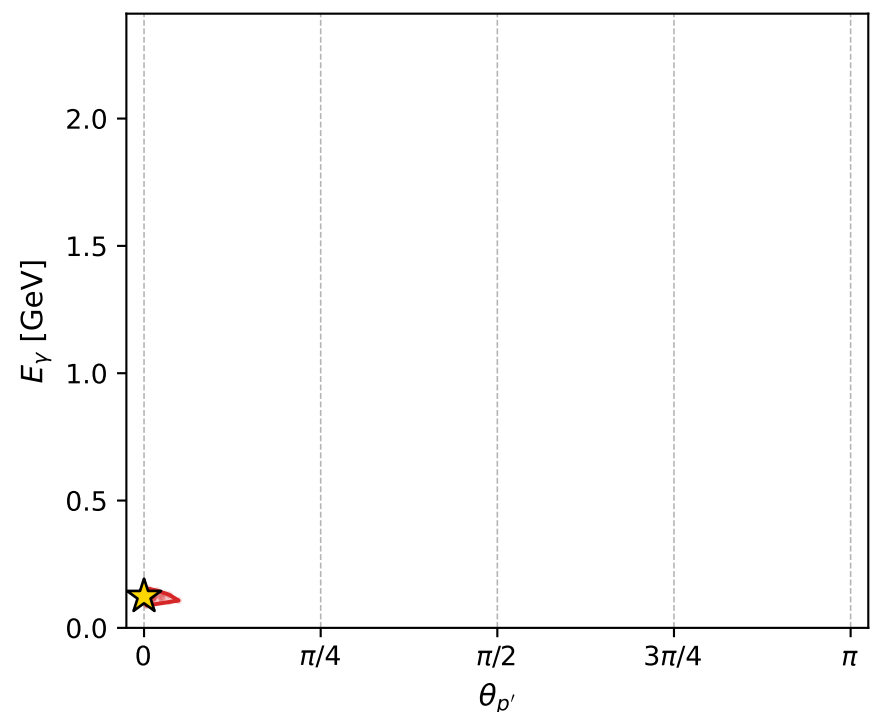
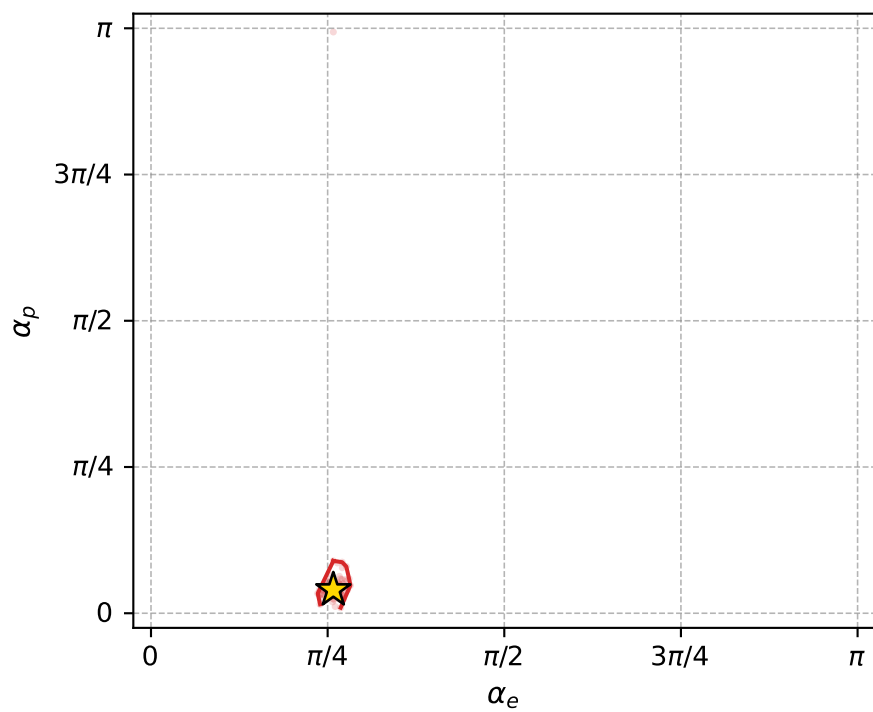
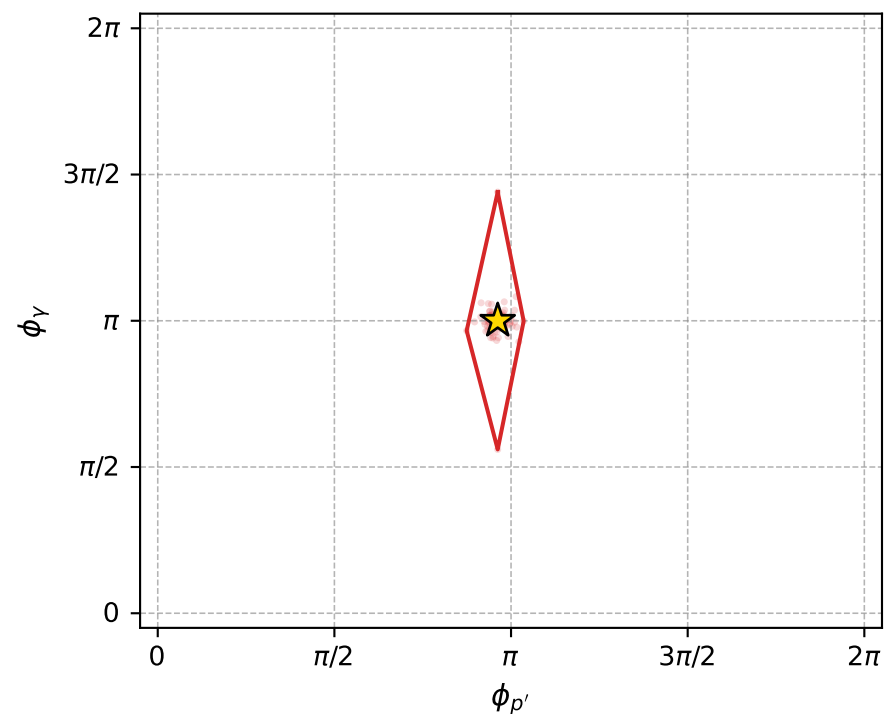
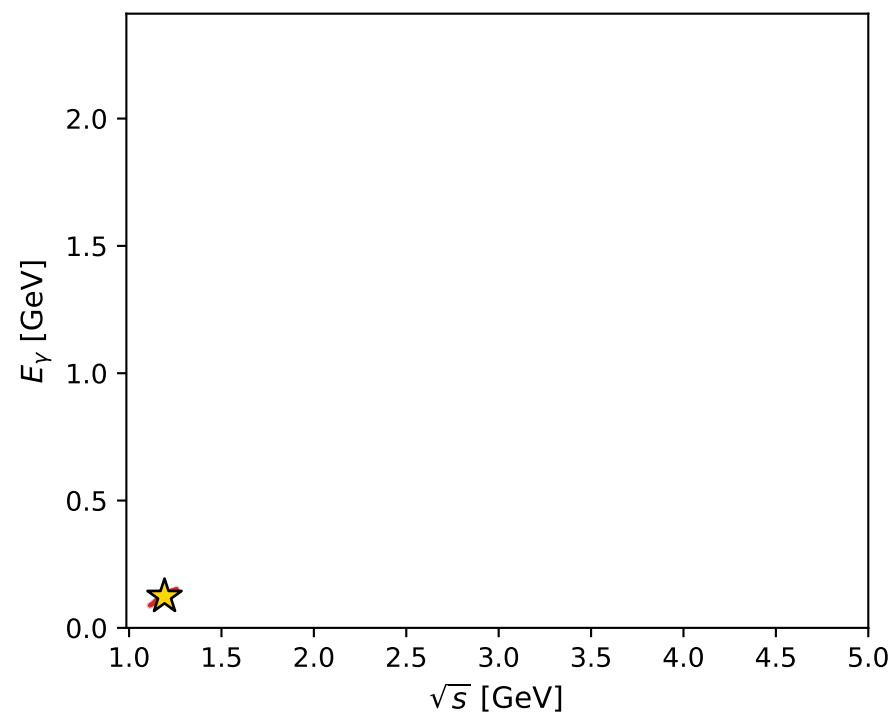
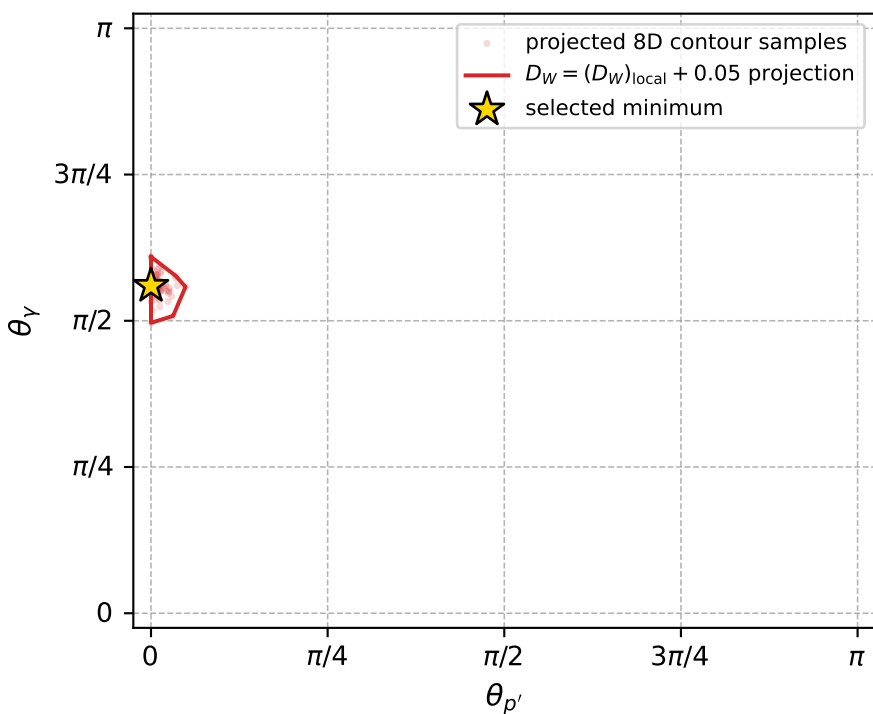
Transverse plane



Leading final-state amplitudes (N=4, retained=99.4%)



electron: polarization cluster P3, local minimum 963; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.016634662$
 $D_W \text{ contour} = 0.066634662$
 above global minimum = 0.016618229
 8D contour samples = 135

selected phase-space point:

$\text{sqrt}_s = 1.191899$
 $\theta_{p_out} = 0$
 $\theta_{\gamma_out} = 1.75843$
 $q_{out} = 0.1233392$
 $\phi_{p_out} = 3.023101$
 $\phi_{\gamma_out} = 3.14157$
 $\alpha_e = 0.8106656$
 $\alpha_p = 0.1255211$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_965 D_W=0.0166832
 region: polarization_cluster_03_local_minimum_965
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81364855$ rad
 incoming lepton: $0.686851|+\rangle \rightarrow +0.726798|-\rangle$
 proton mixing angle: $\alpha_p=0.14201446$ rad
 incoming proton: $0.989933|+\rangle \rightarrow +0.141538|-\rangle$

$s=1.7773$, $\sqrt{s}=1.33315$, $|\vec{P}|=0.33659$, $|\vec{P}'|=0.212996$

$\theta_{P'}=0$, $\theta_V=2.26574$, $\phi_{P'}=5.23278$, $\phi_V=3.9936$

$E_V=0.196857$, $\theta_{\ell'}=2.09262$, $\phi_{\ell'}=0.85201$

$Q^2=0.0588877$, $x_B=0.119863$, $t=-0.0140726$, $W^2=1.31225$, $y=0.54743$

four-momenta $[E, p_x, p_y, p_z]$ GeV:

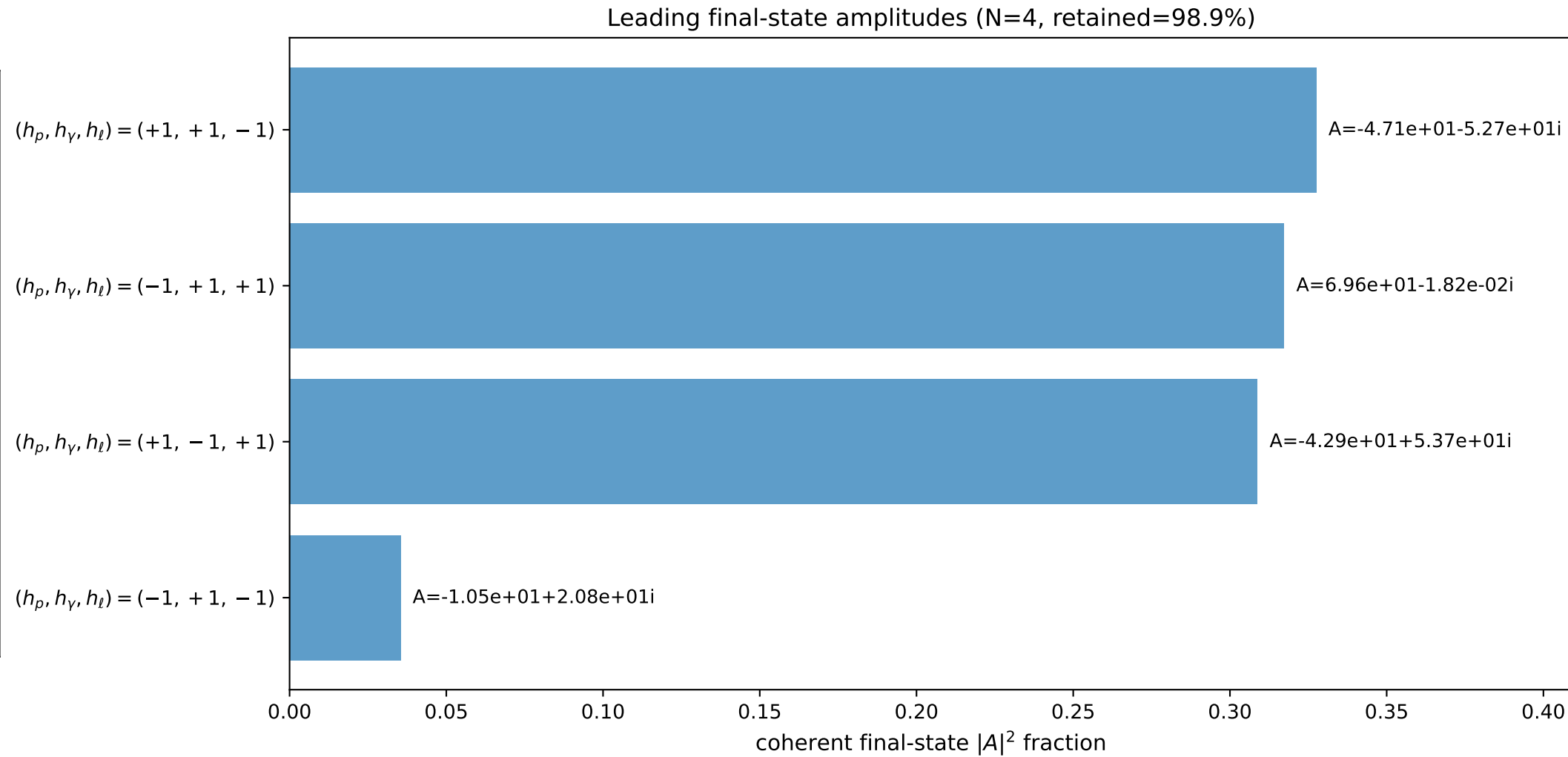
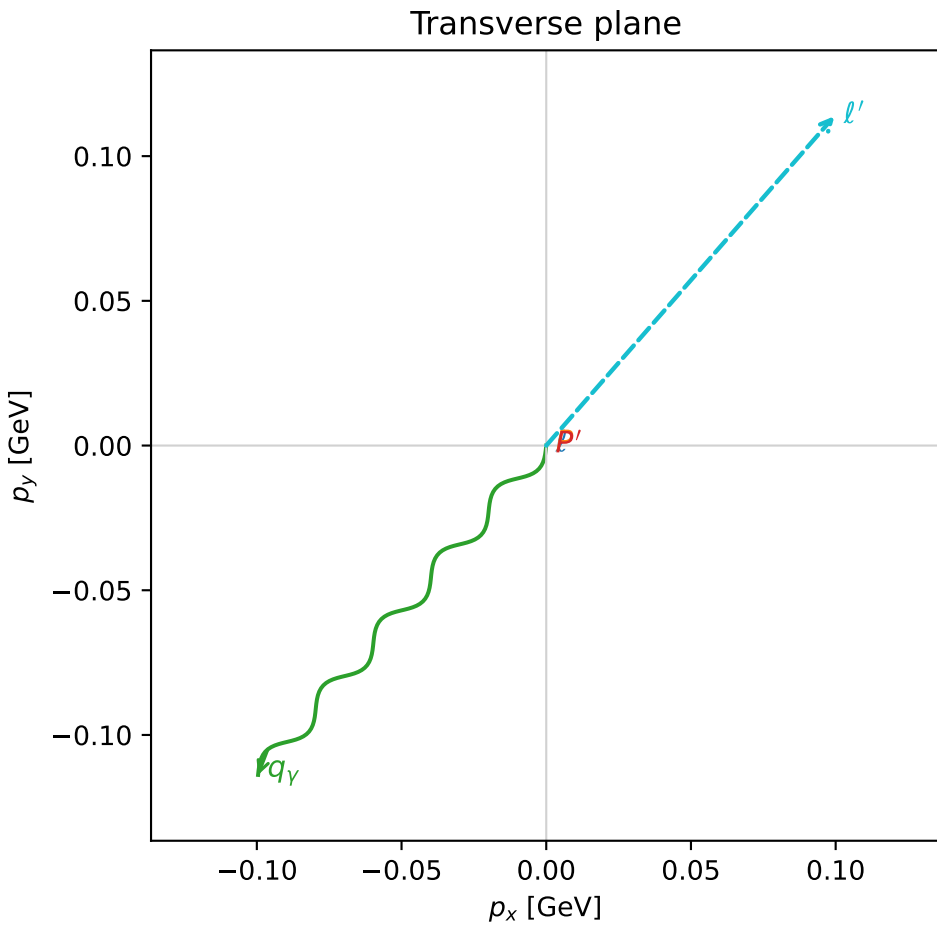
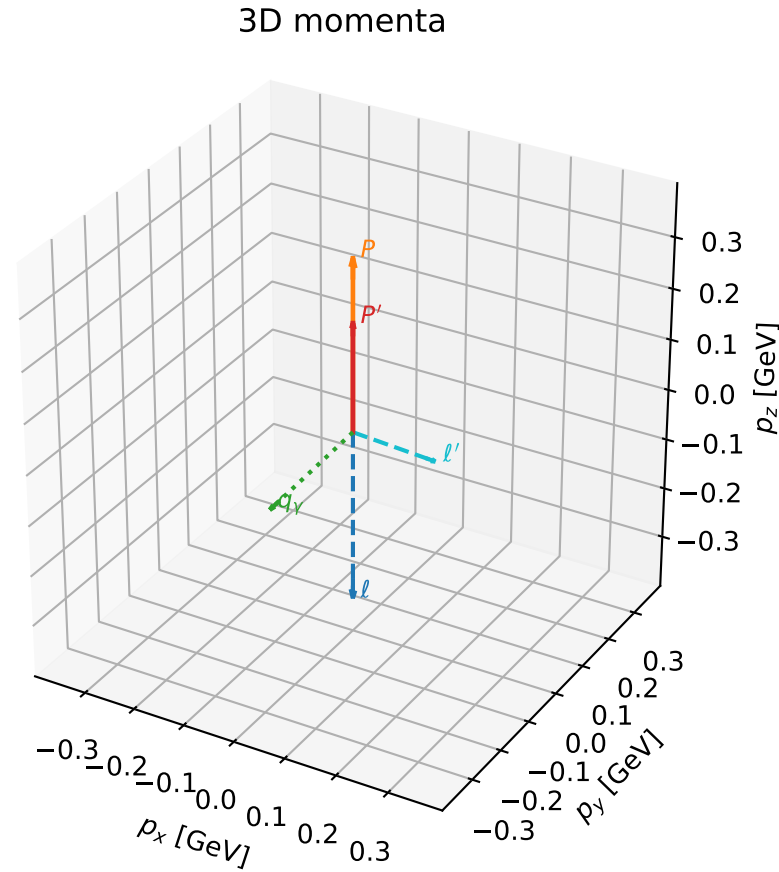
ℓ $E= 0.3366$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3366$

P $E= 0.9966$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3366$

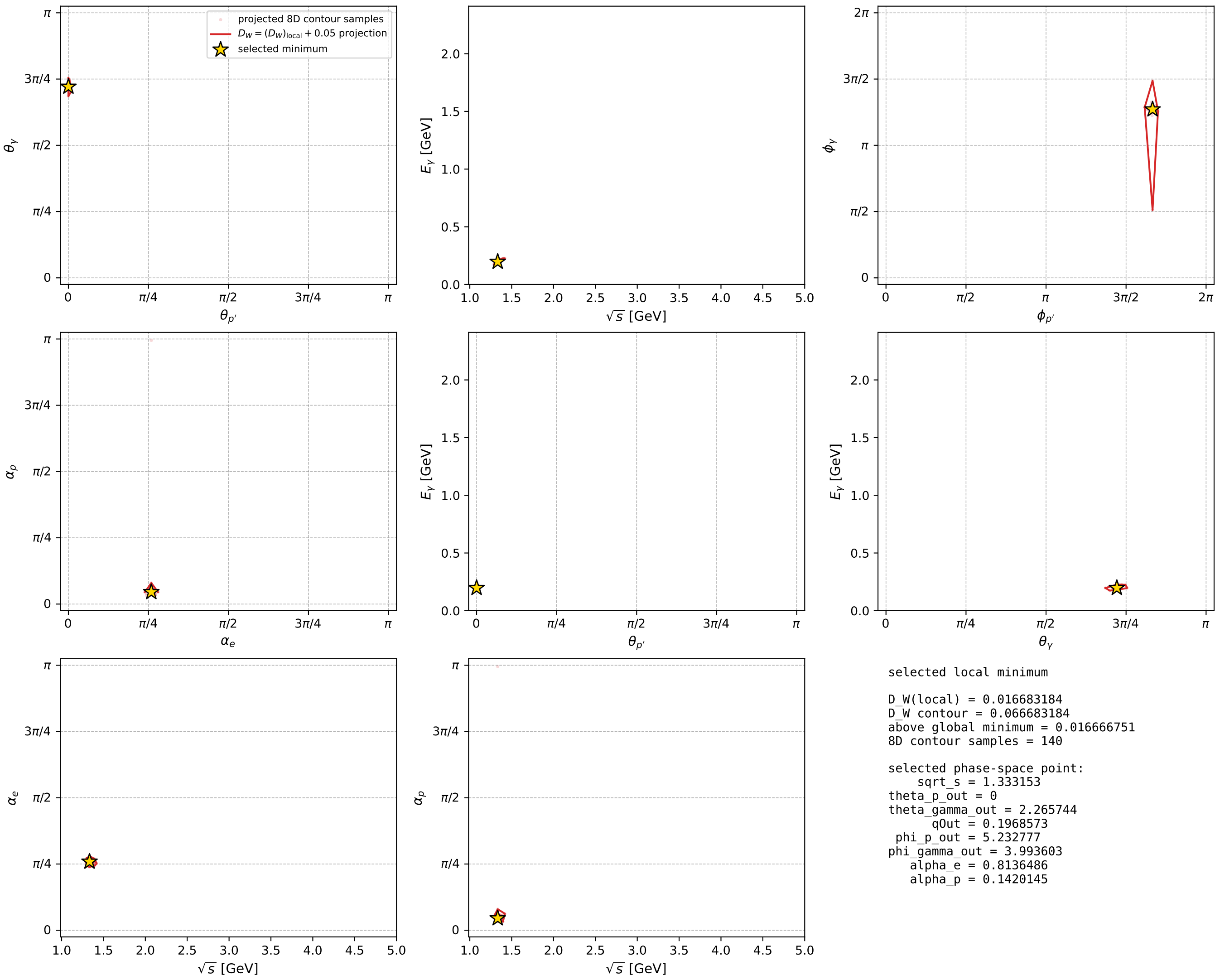
ℓ' $E= 0.1744$ $p_x= 0.0996$ $p_y= 0.1138$ $p_z= -0.0869$

P' $E= 0.9619$ $p_x= 0.0000$ $p_y= -0.0000$ $p_z= 0.2130$

q_V $E= 0.1969$ $p_x= -0.0996$ $p_y= -0.1138$ $p_z= -0.1261$



electron: polarization cluster P3, local minimum 965; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

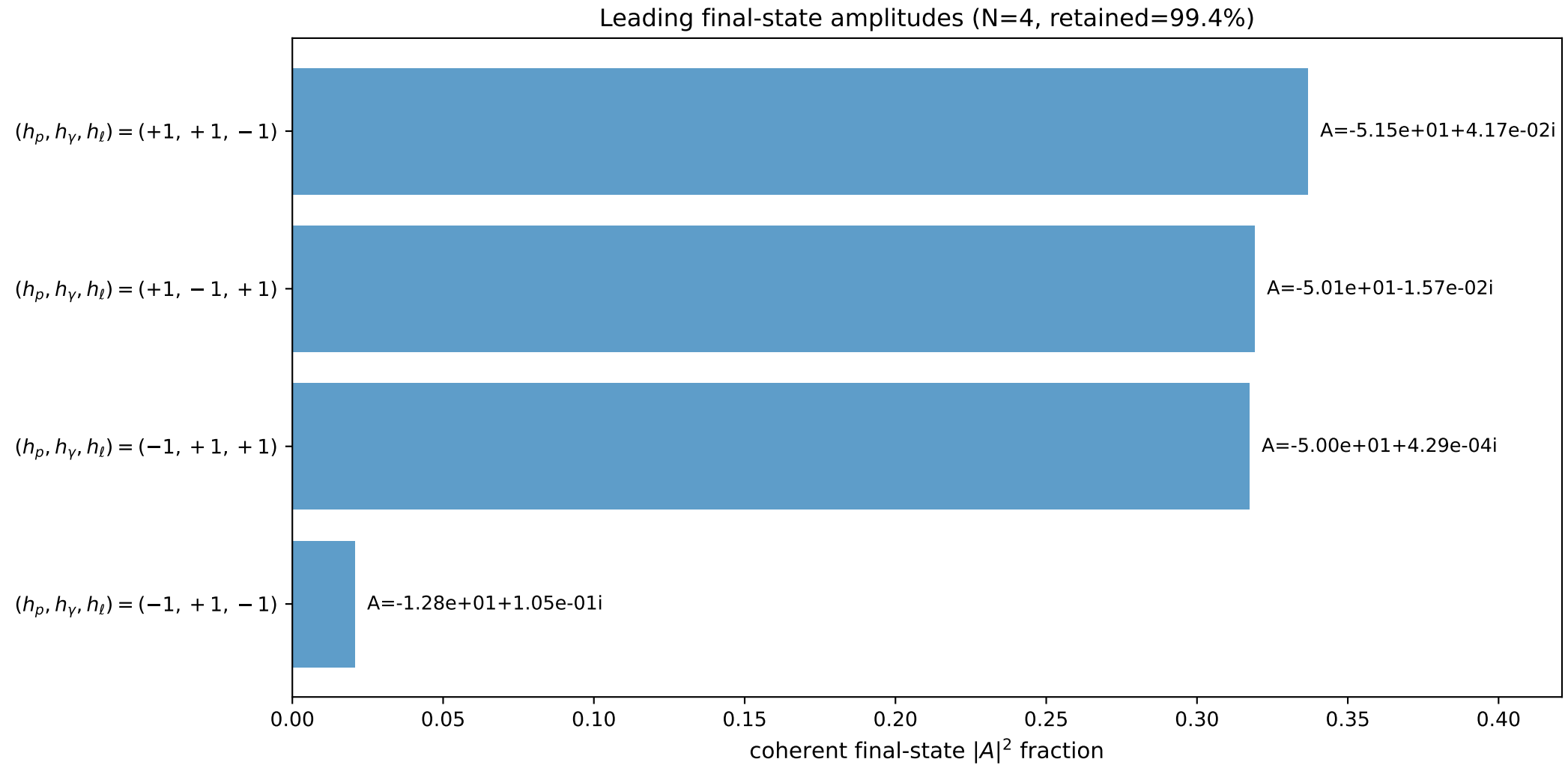
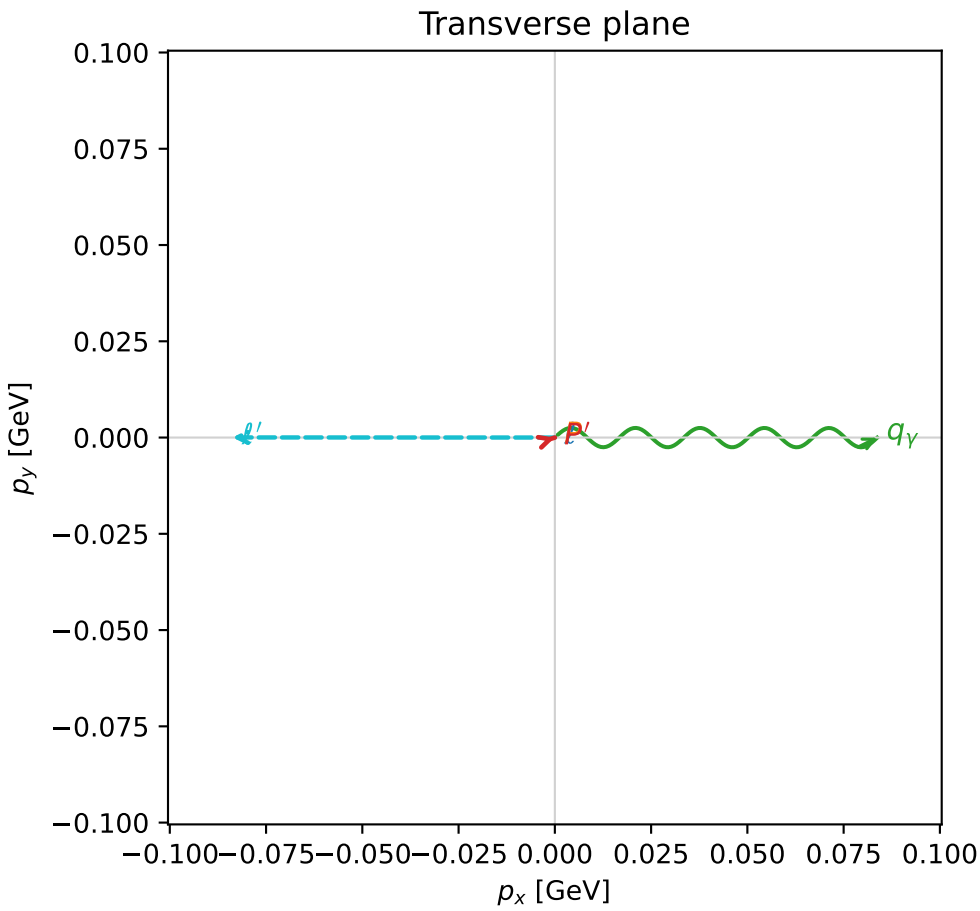
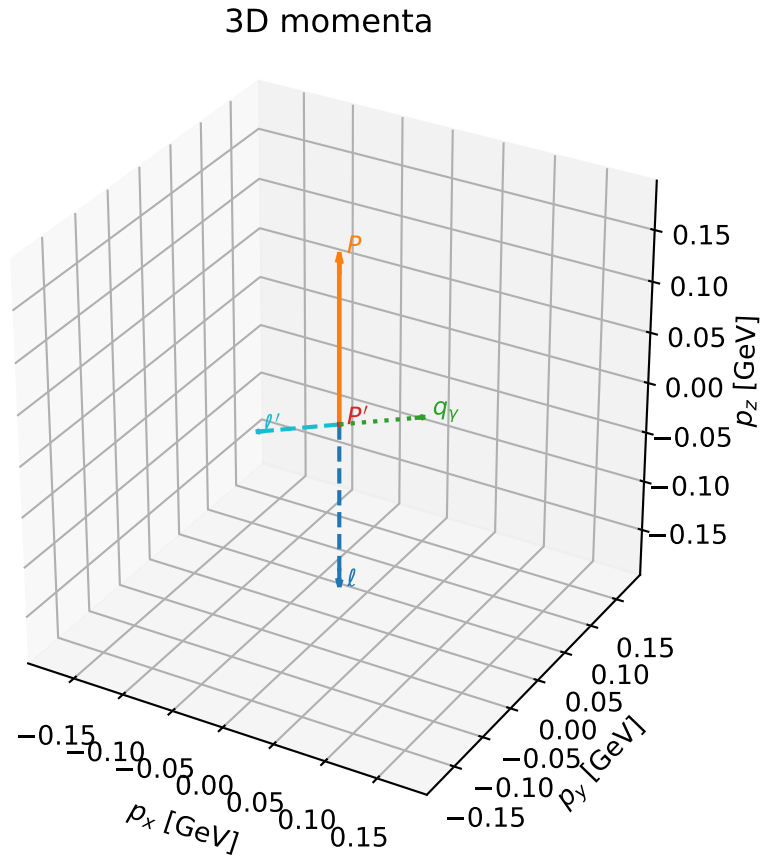


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

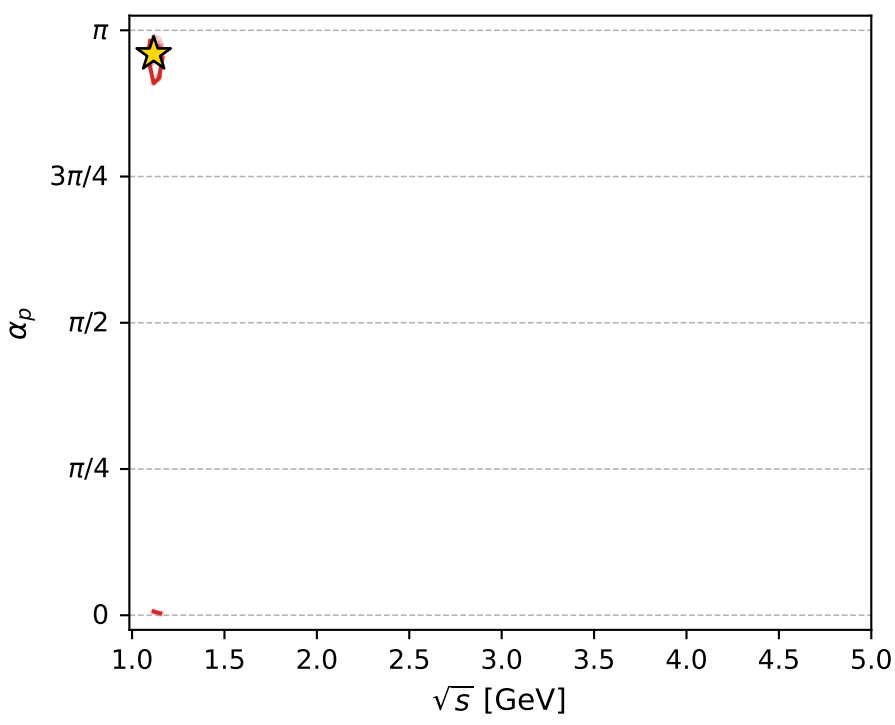
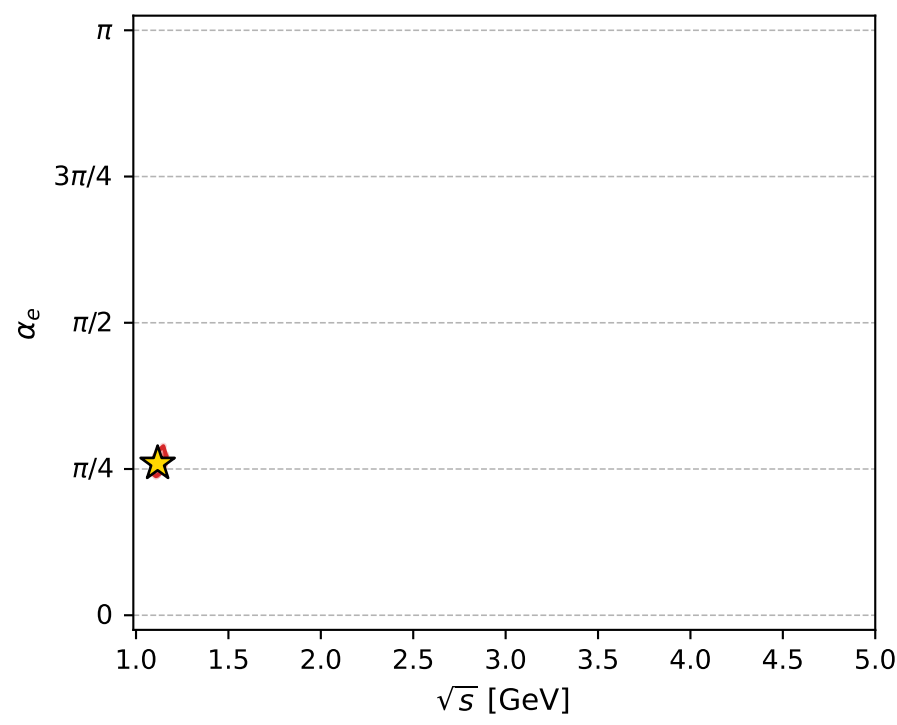
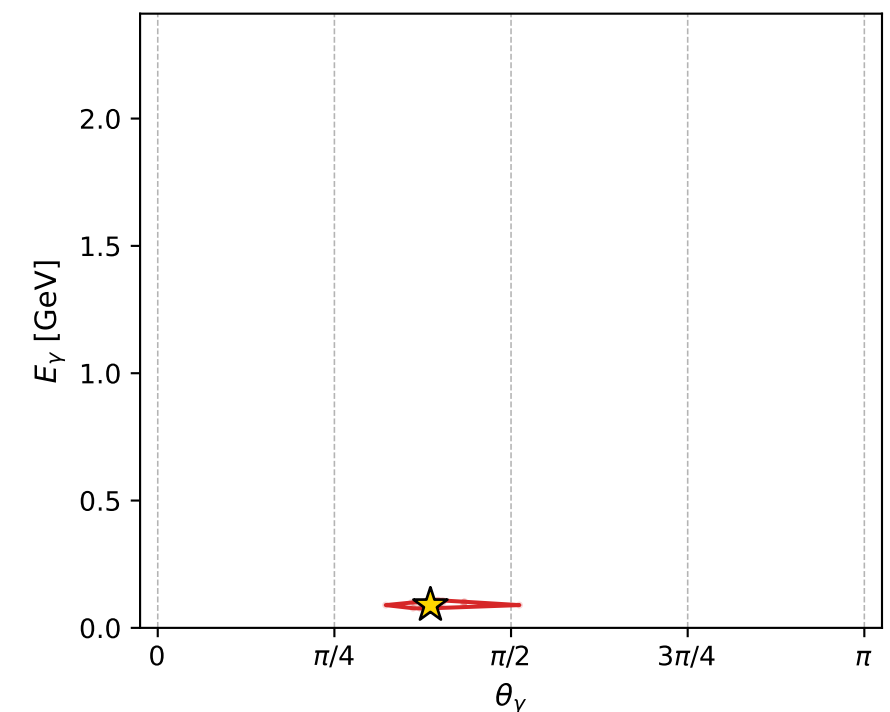
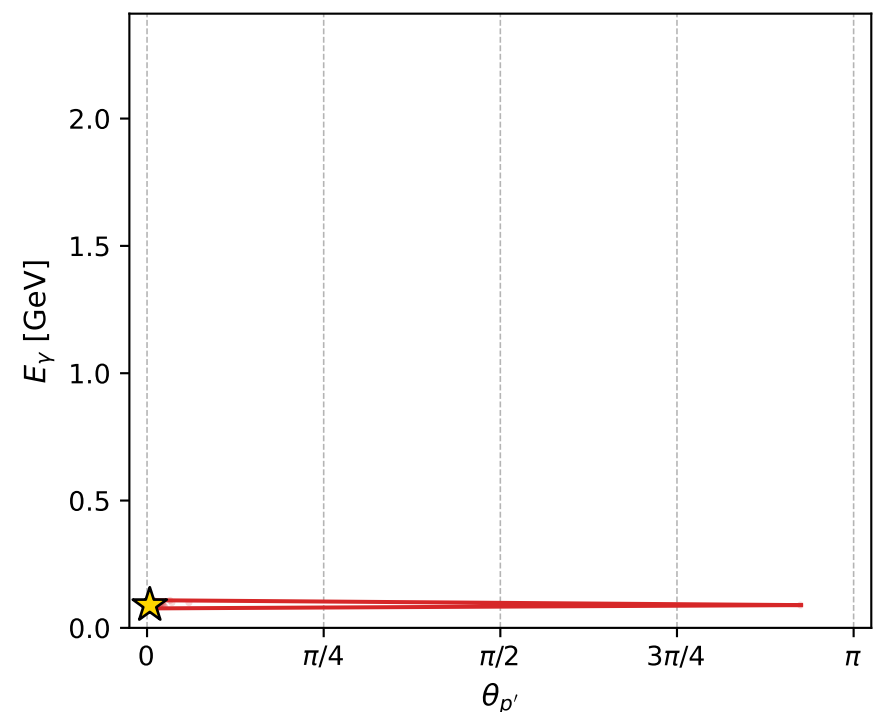
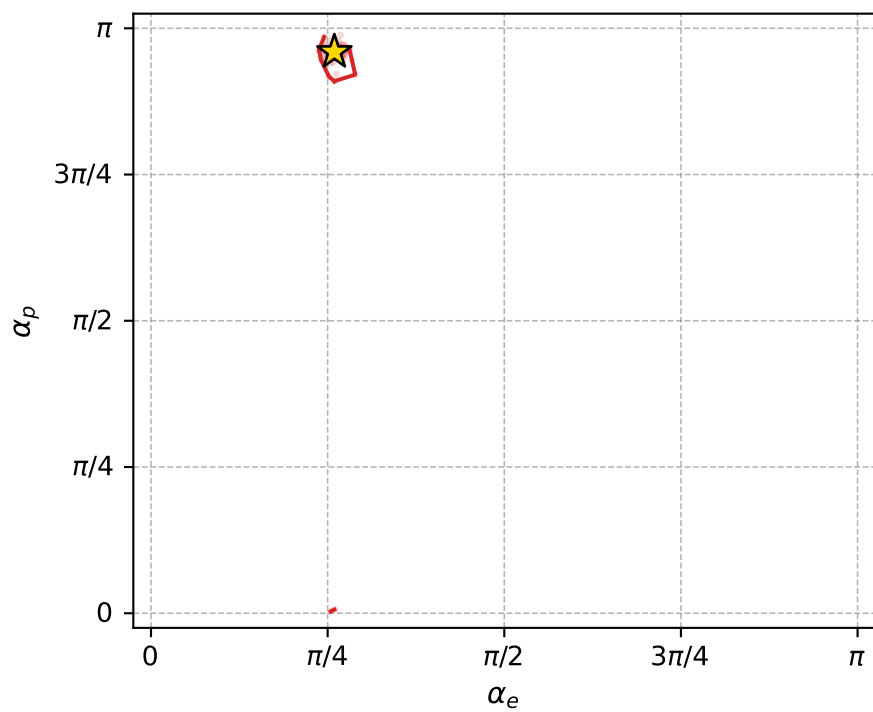
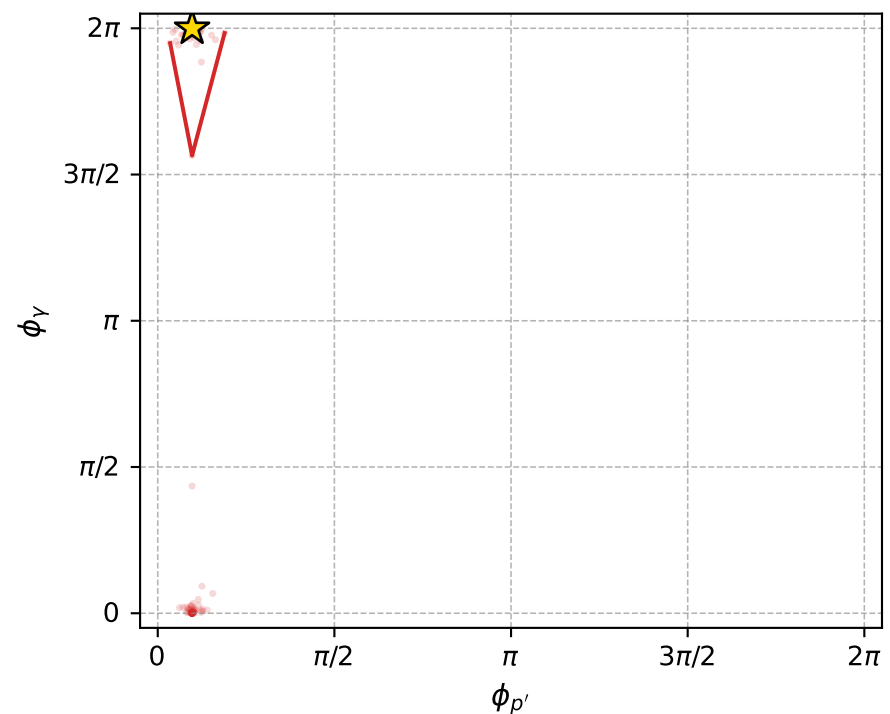
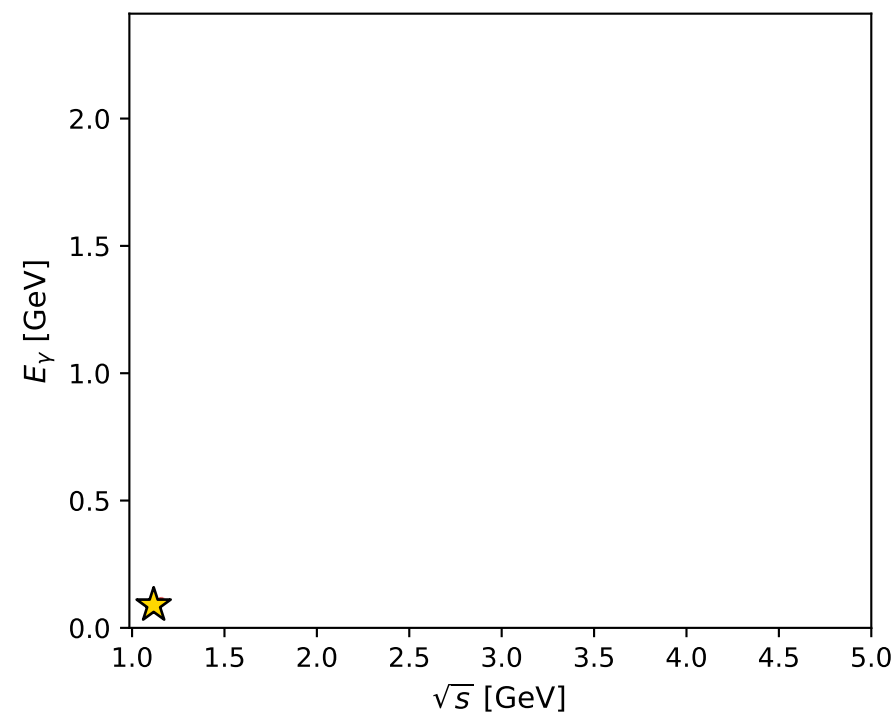
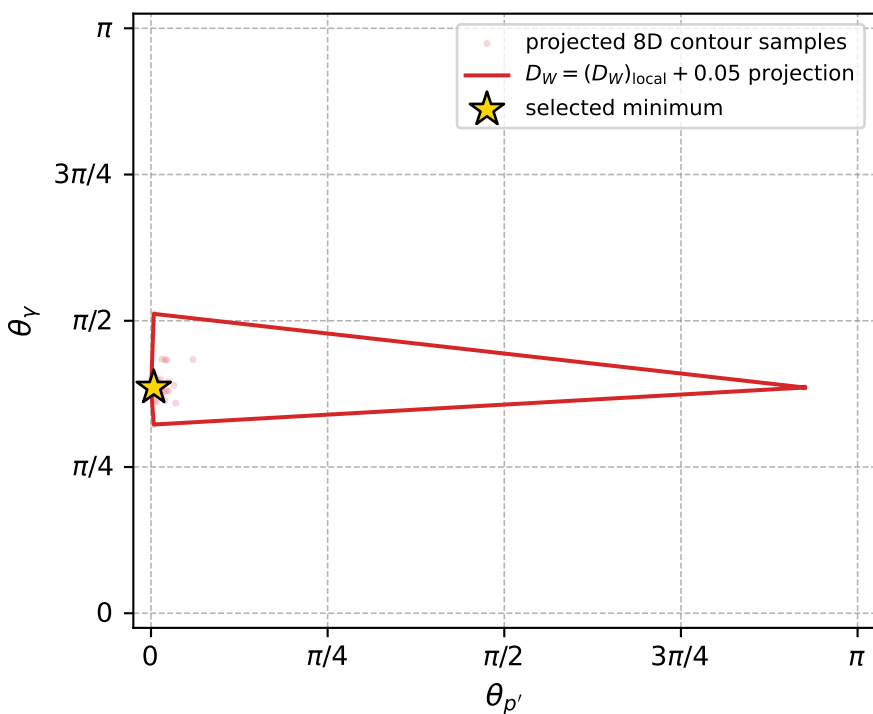
dw_mixing_angles_polarization_03_local_minimum_974 D_W=0.0171906
 region: polarization_cluster_03_local_minimum_974
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81534323$ rad
 incoming lepton: $0.685619|+\rangle +0.727961|-\rangle$
 proton mixing angle: $\alpha_p=3.0147067$ rad
 incoming proton: $-0.991961|+\rangle +0.126546|-\rangle$

$s=1.24742$, $\sqrt{s}=1.11688$, $|\vec{P}|=0.164554$, $|\vec{P}'|=0.000193532$
 $\theta_{P'}=0.012212$, $\theta_Y=1.21234$, $\phi_{P'}=0.305286$, $\phi_Y=6.28284$
 $E_Y=0.0894036$, $\theta_{\ell'}=1.93127$, $\phi_{\ell'}=3.14126$
 $Q^2=0.0190603$, $x_B=0.102053$, $t=-0.026809$, $W^2=1.04755$, $y=0.508112$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1646$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1646$
 P $E= 0.9523$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1646$
 ℓ' $E= 0.0895$ $p_x= -0.0837$ $p_y= 0.0000$ $p_z= -0.0316$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0002$
 q_Y $E= 0.0894$ $p_x= 0.0837$ $p_y= -0.0000$ $p_z= 0.0314$



electron: polarization cluster P3, local minimum 974; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017190571$
 $D_W \text{ contour} = 0.067190571$
 above global minimum = 0.017174138
 8D contour samples = 223

selected phase-space point:
 $\text{sqrt_s} = 1.116879$
 $\text{theta_p_out} = 0.01221198$
 $\text{theta_gamma_out} = 1.21234$
 $\text{qOut} = 0.08940359$
 $\text{phi_p_out} = 0.3052856$
 $\text{phi_gamma_out} = 6.282841$
 $\text{alpha_e} = 0.8153432$
 $\text{alpha_p} = 3.014707$

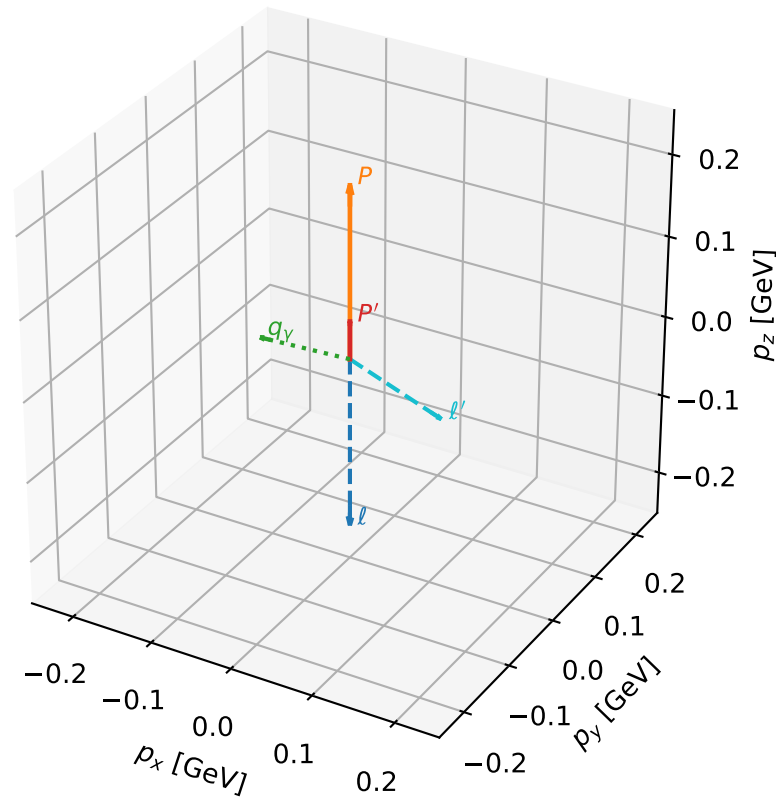
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_980 D_W=0.0173567
 region: polarization_cluster_03_local_minimum_980
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81192192$ rad
 incoming lepton: $0.688105|+\rangle +0.725611|-\rangle$
 proton mixing angle: $\alpha_p=0.12894437$ rad
 incoming proton: $0.991698|+\rangle +0.128587|-\rangle$

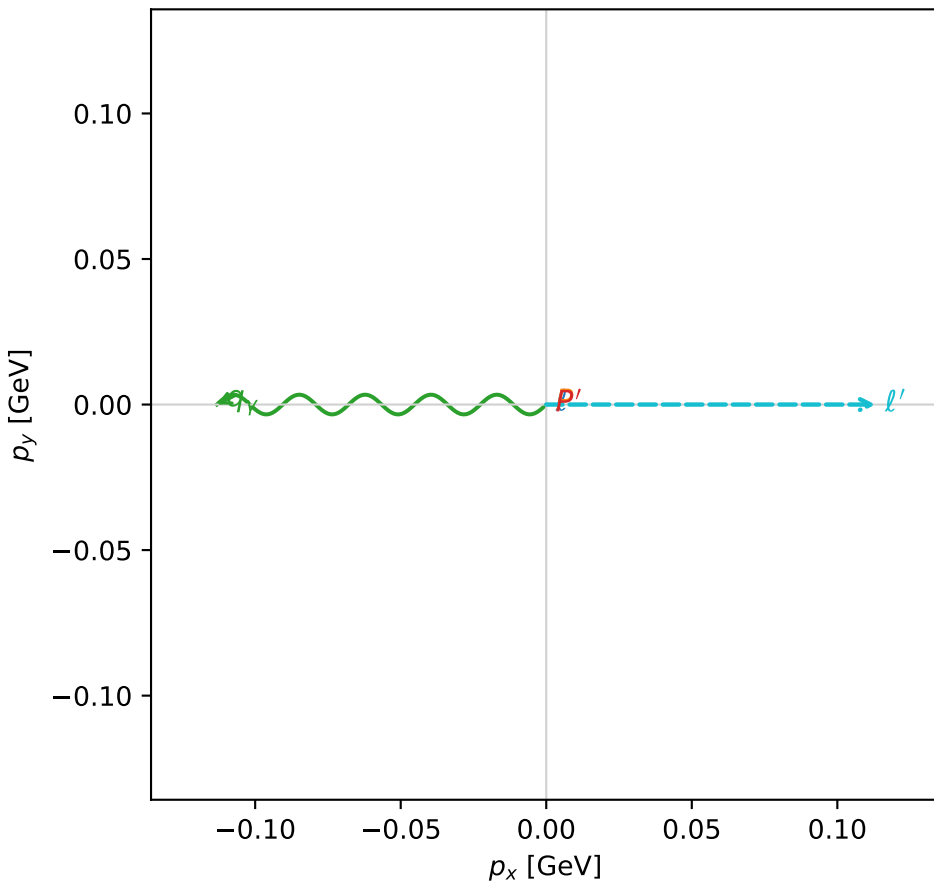
$s=1.37572$, $\sqrt{s}=1.17291$, $|\vec{P}|=0.211387$, $|\vec{P}'|=0.0466059$
 $\theta_{P'}=0$, $\theta_V=1.61638$, $\phi_{P'}=1.30886$, $\phi_V=3.1416$
 $E_V=0.11326$, $\theta_{\ell'}=1.92193$, $\phi_{\ell'}=5.28613e-06$
 $Q^2=0.0334199$, $x_B=0.135502$, $t=-0.0266526$, $W^2=1.09306$, $y=0.497376$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2114$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2114$
 P $E= 0.9615$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2114$
 ℓ' $E= 0.1205$ $p_x= 0.1131$ $p_y= 0.0000$ $p_z= -0.0414$
 P' $E= 0.9392$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0466$
 q_Y $E= 0.1133$ $p_x= -0.1131$ $p_y= -0.0000$ $p_z= -0.0052$

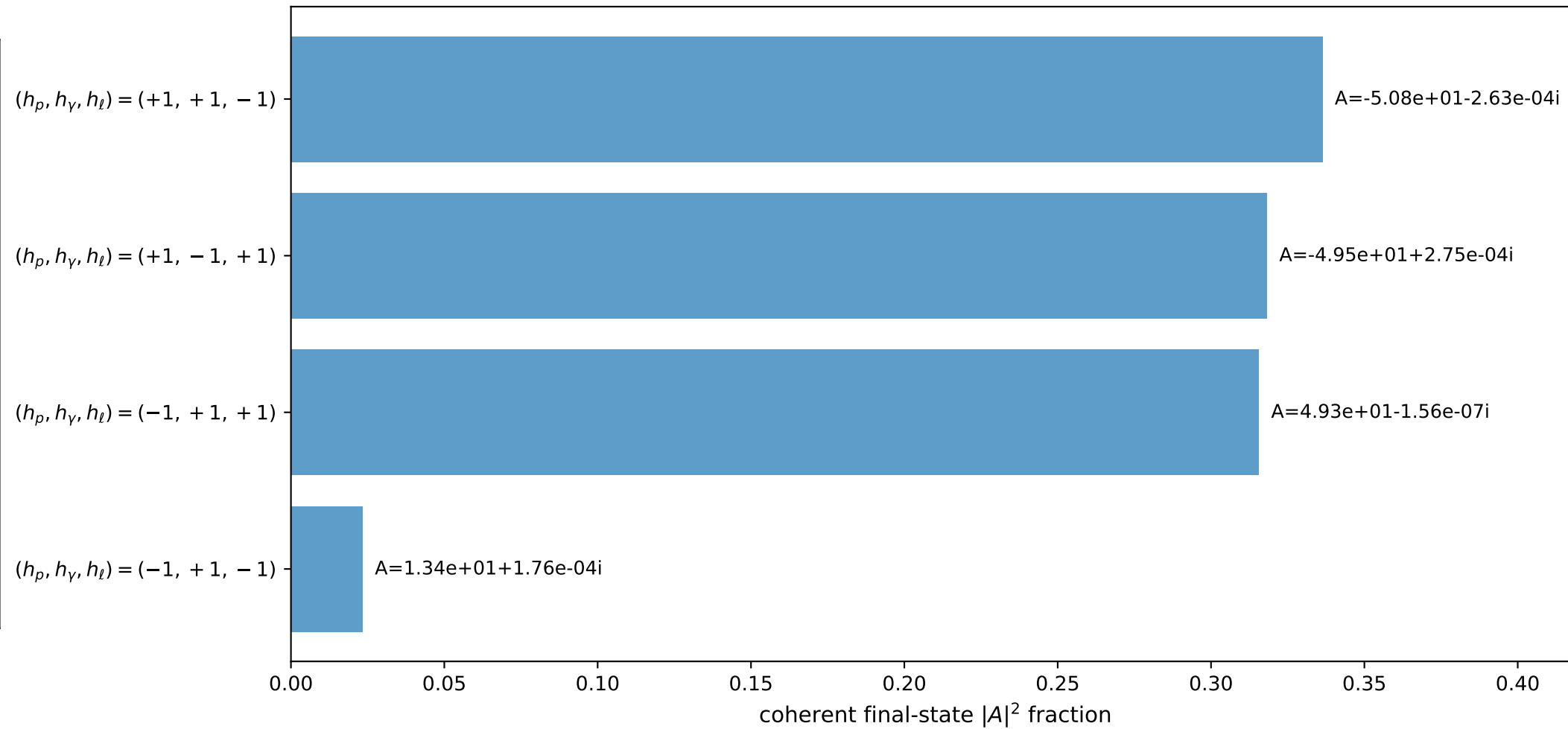
3D momenta



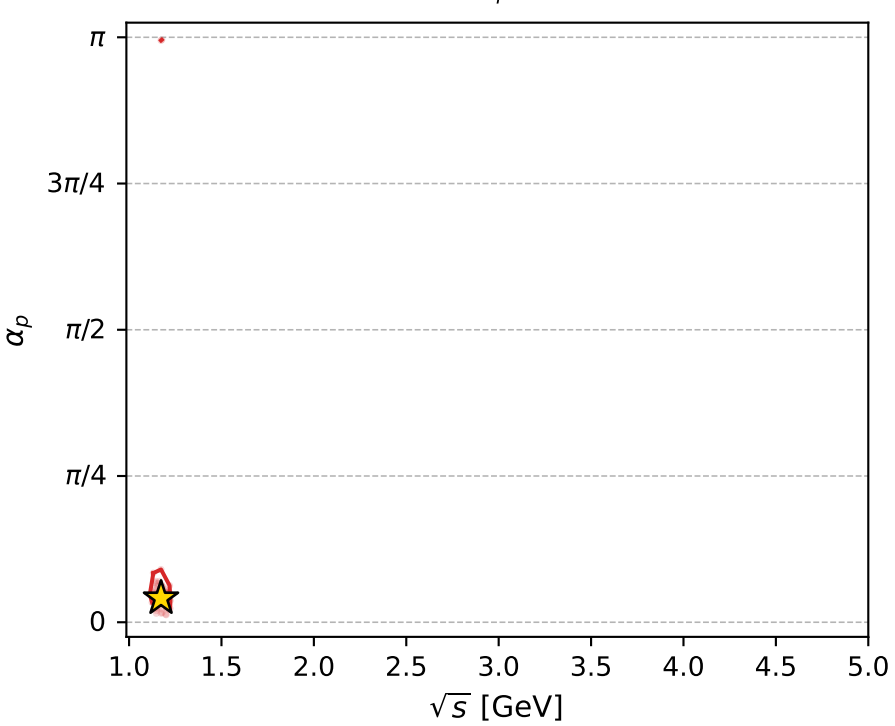
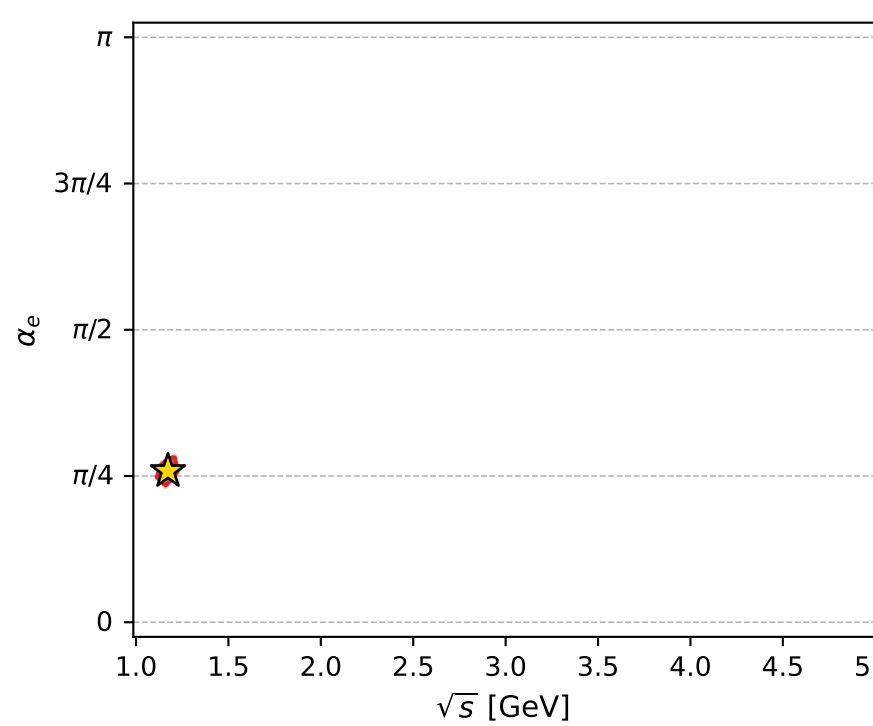
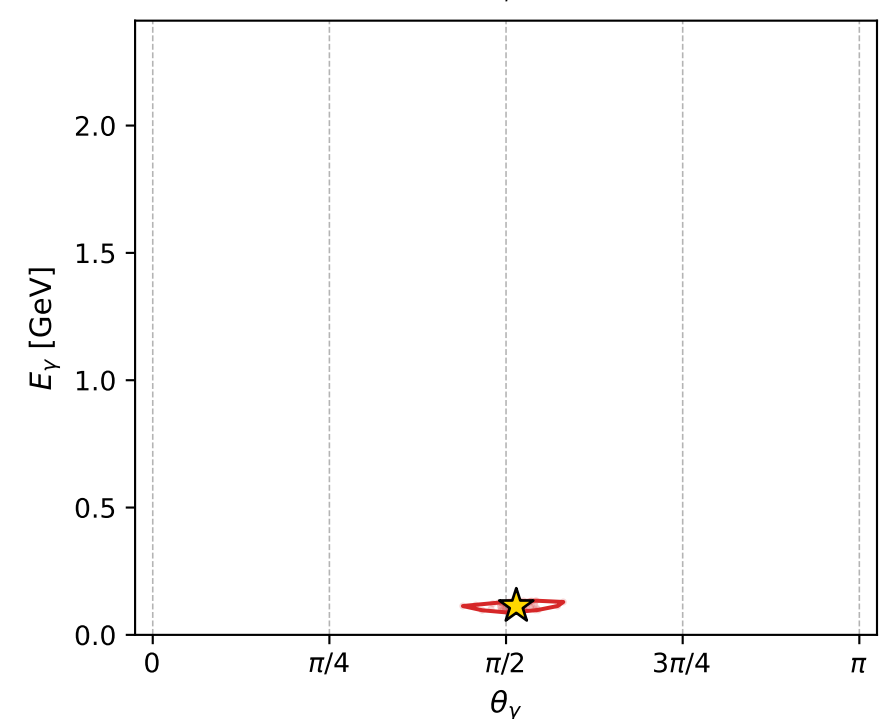
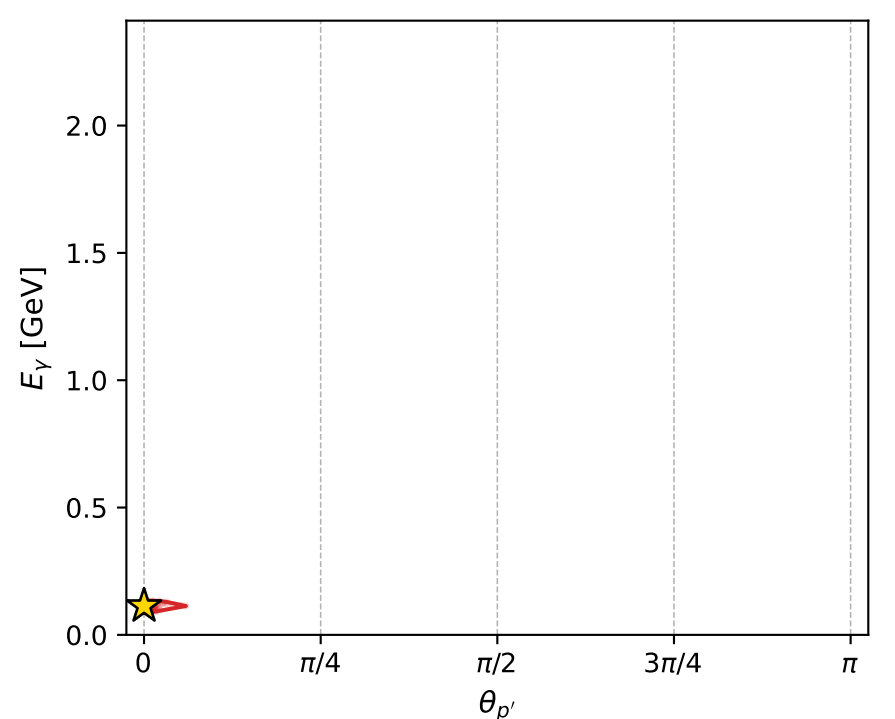
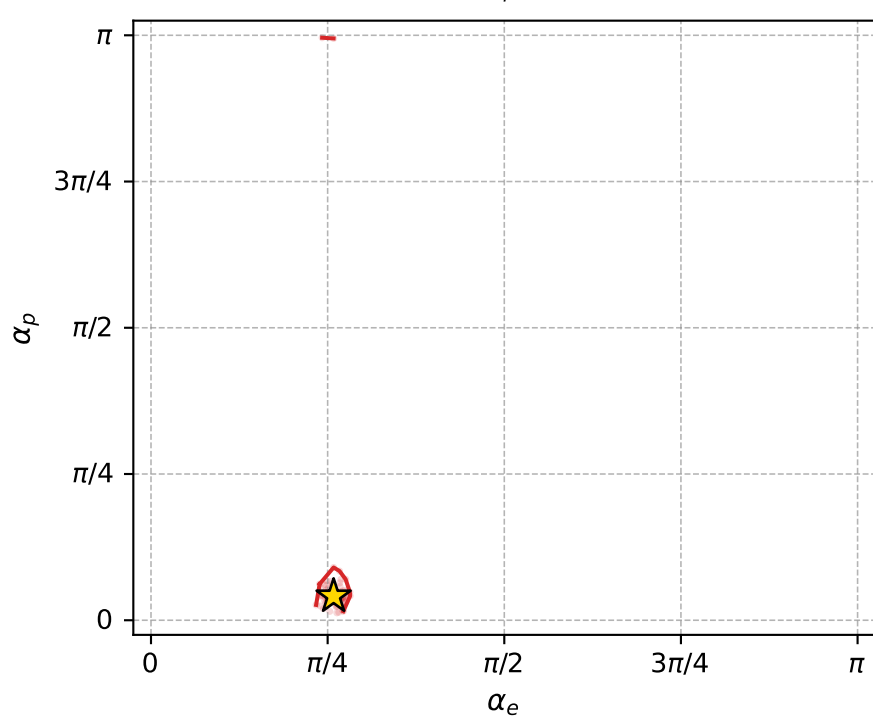
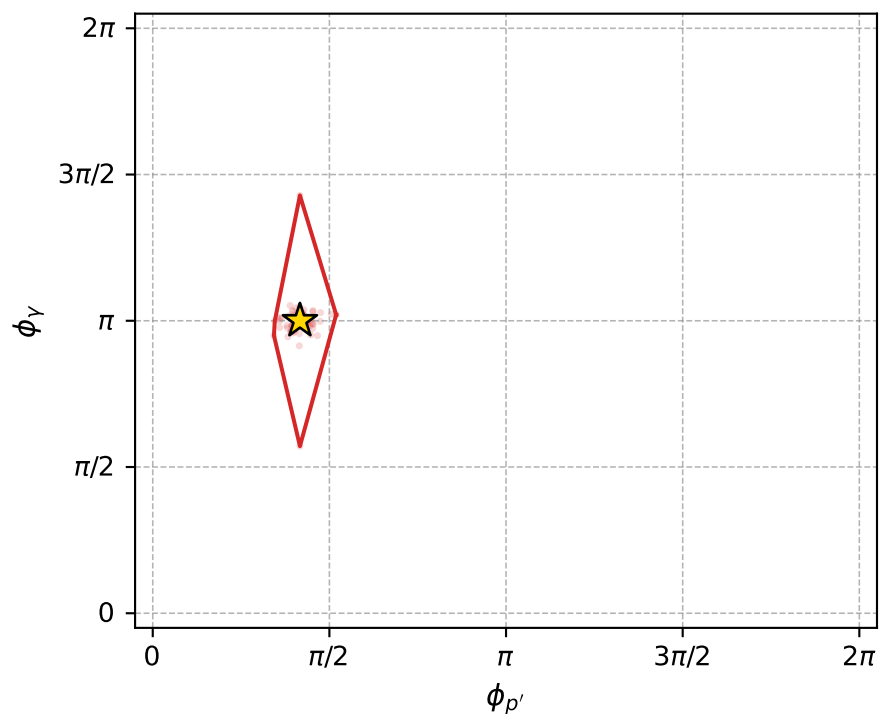
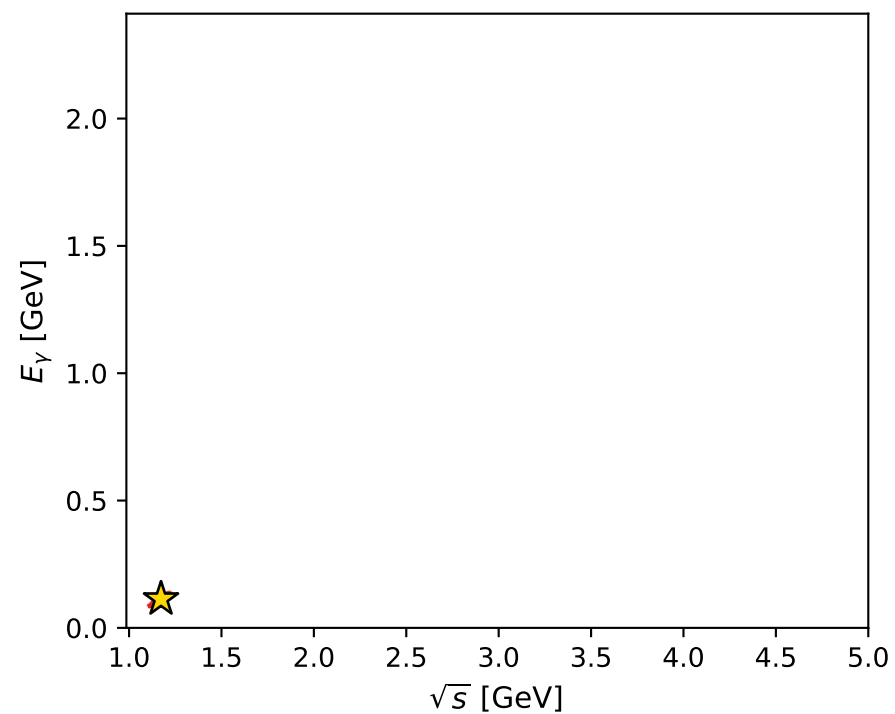
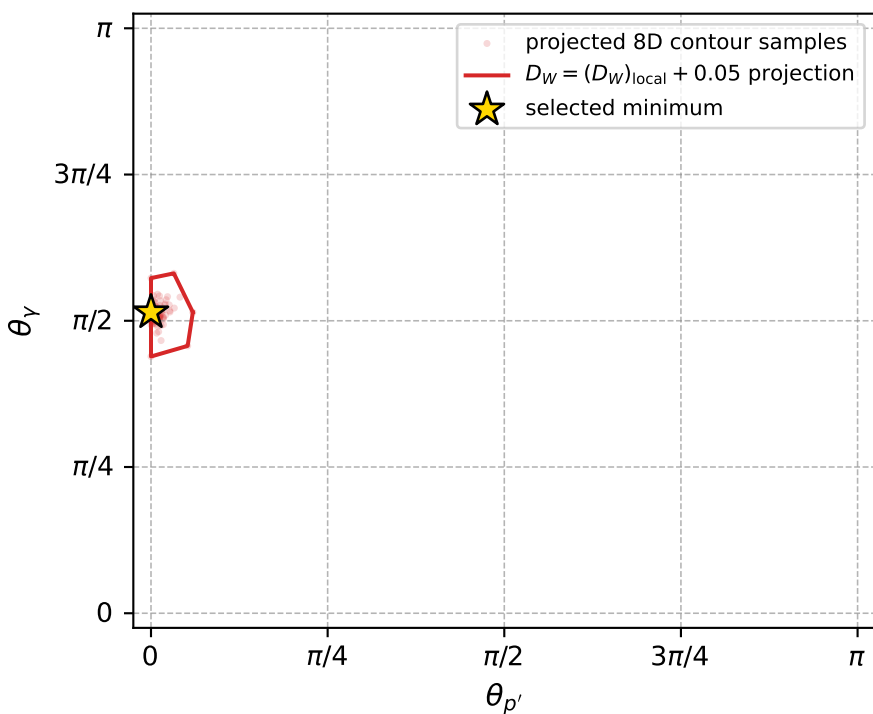
Transverse plane



Leading final-state amplitudes (N=4, retained=99.4%)



electron: polarization cluster P3, local minimum 980; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017356686$
 $D_W \text{ contour} = 0.067356686$
 above global minimum = 0.017340253
 8D contour samples = 139

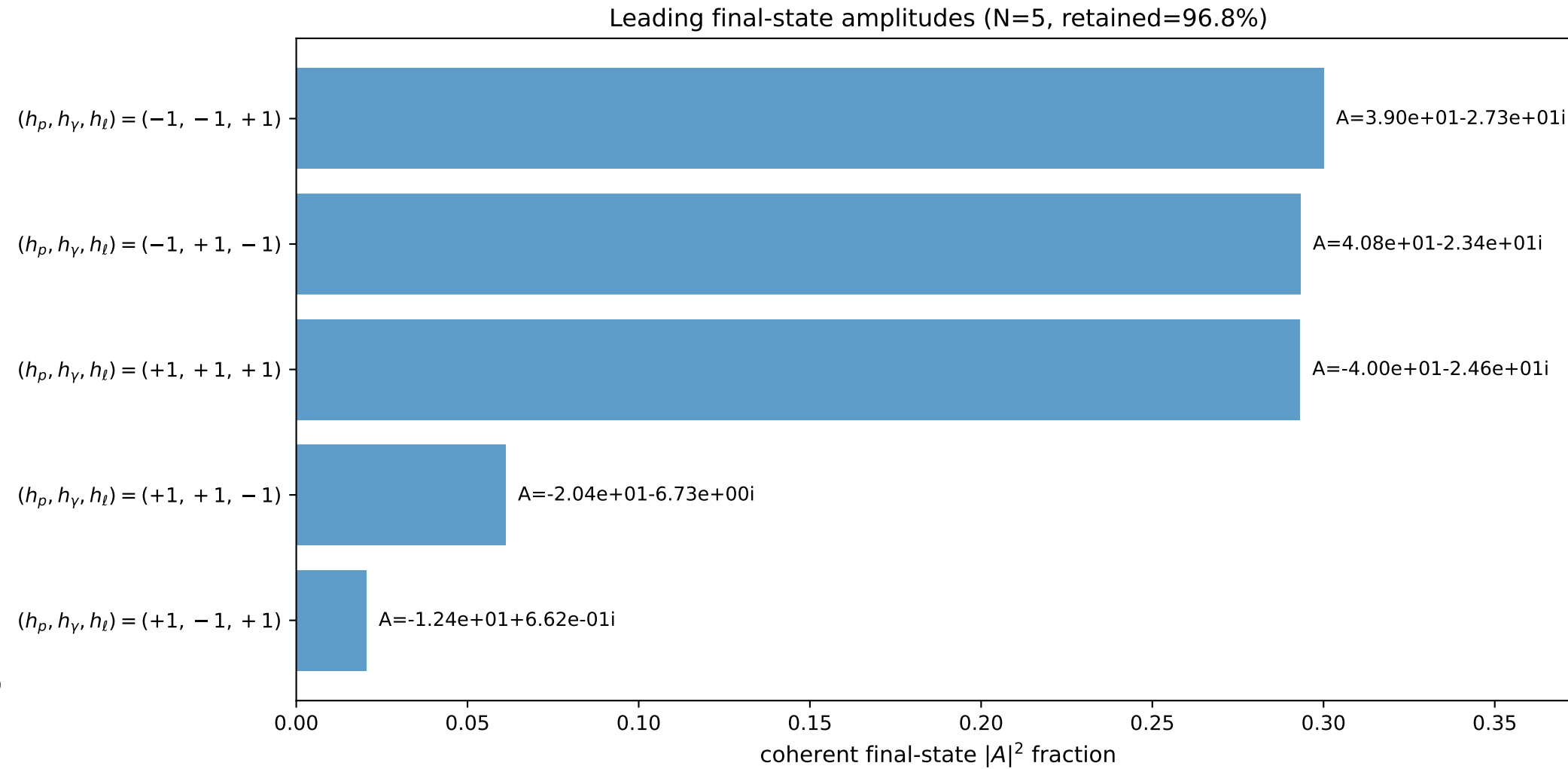
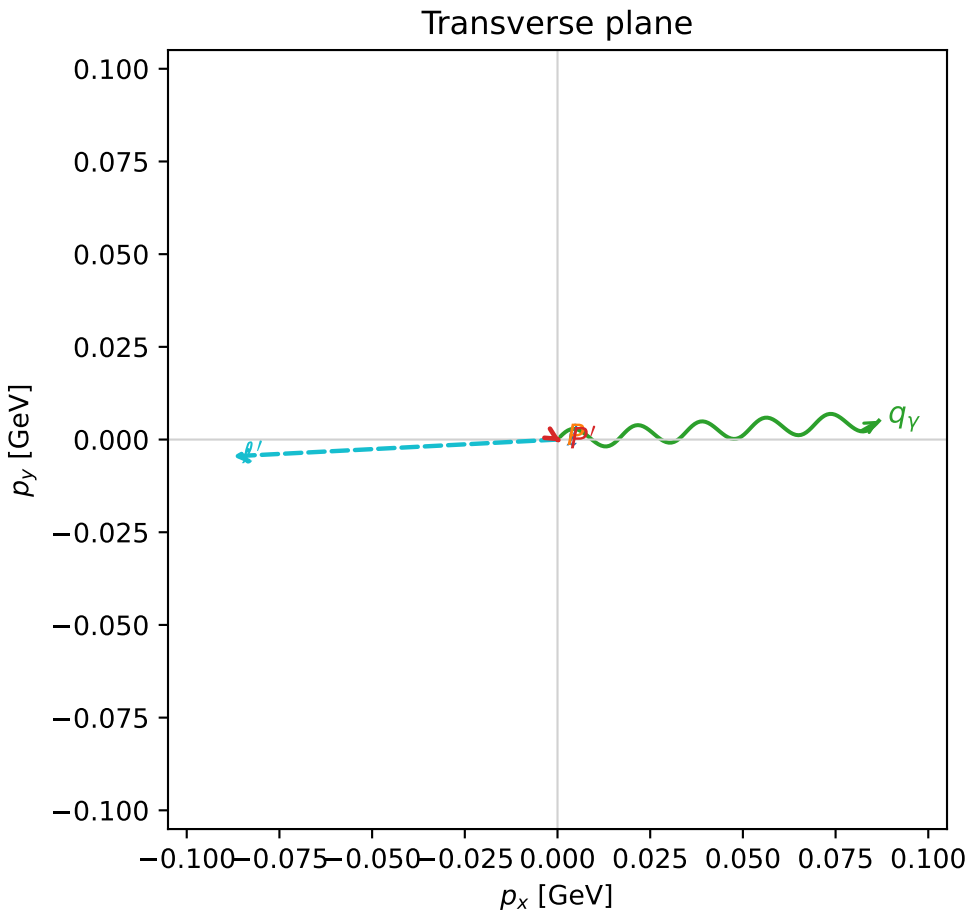
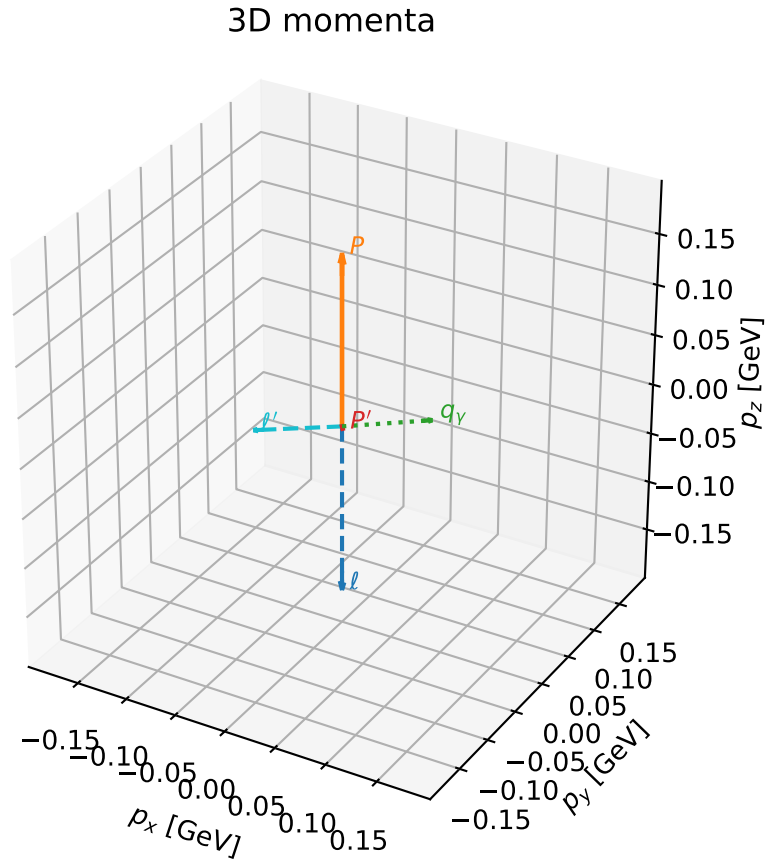
selected phase-space point:
 $\text{sqrt_s} = 1.172912$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.616376$
 $\text{qOut} = 0.1132596$
 $\text{phi_p_out} = 1.308864$
 $\text{phi_gamma_out} = 3.141598$
 $\text{alpha_e} = 0.8119219$
 $\text{alpha_p} = 0.1289444$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

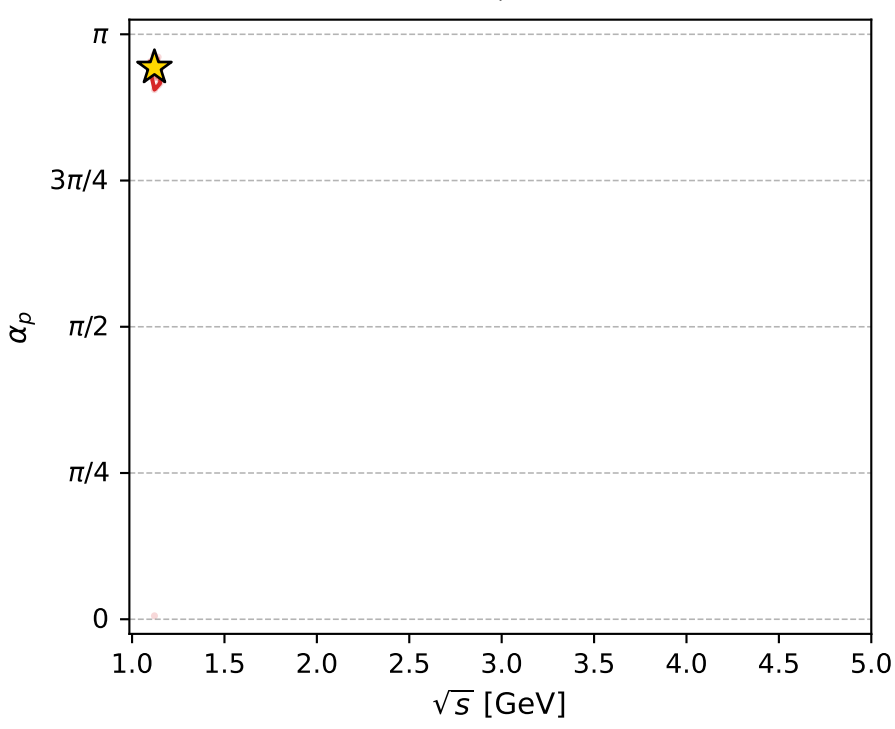
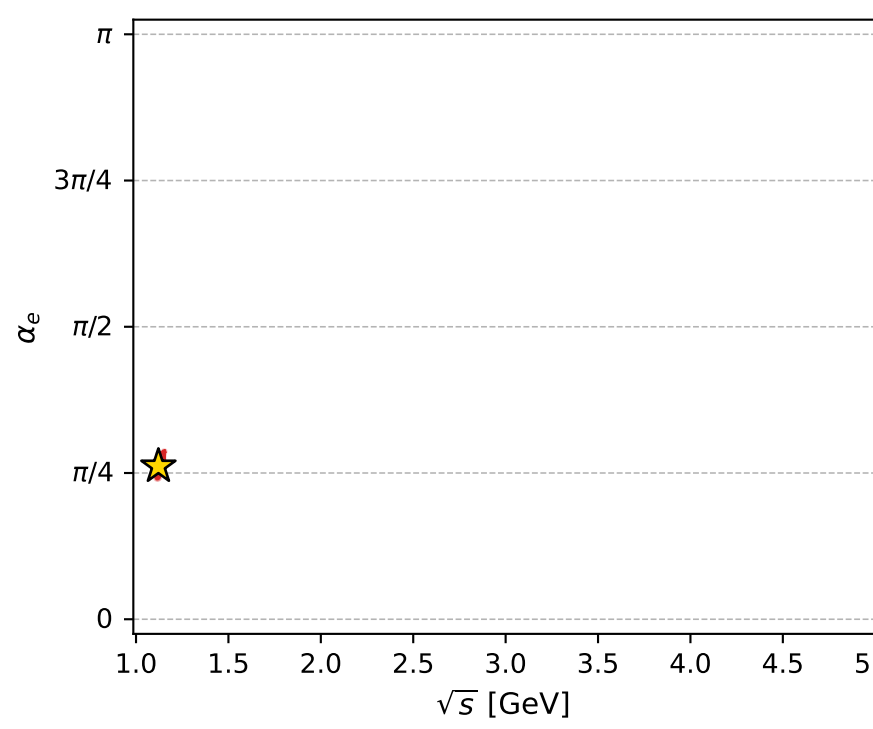
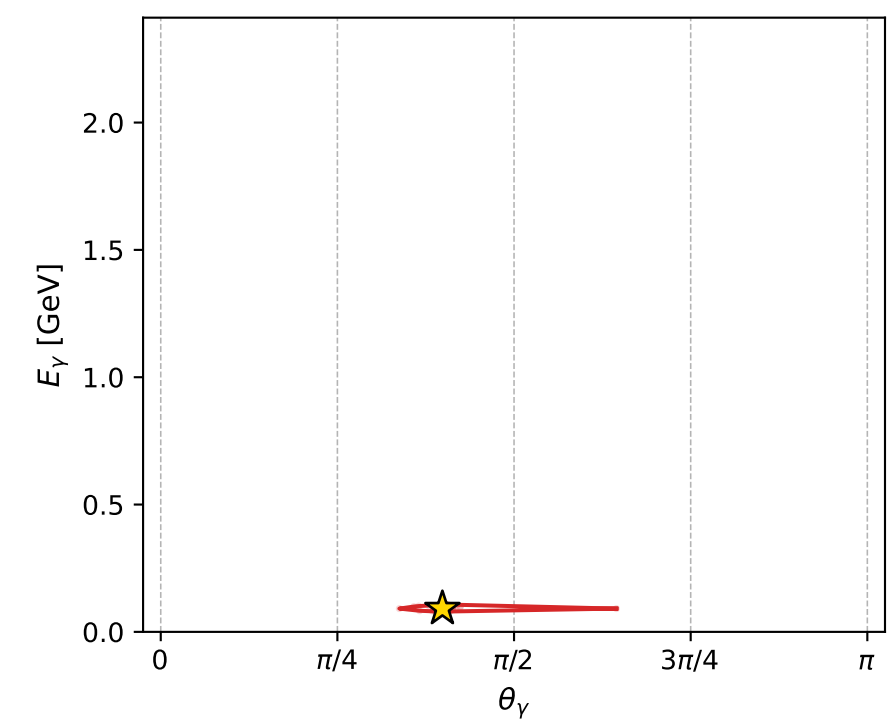
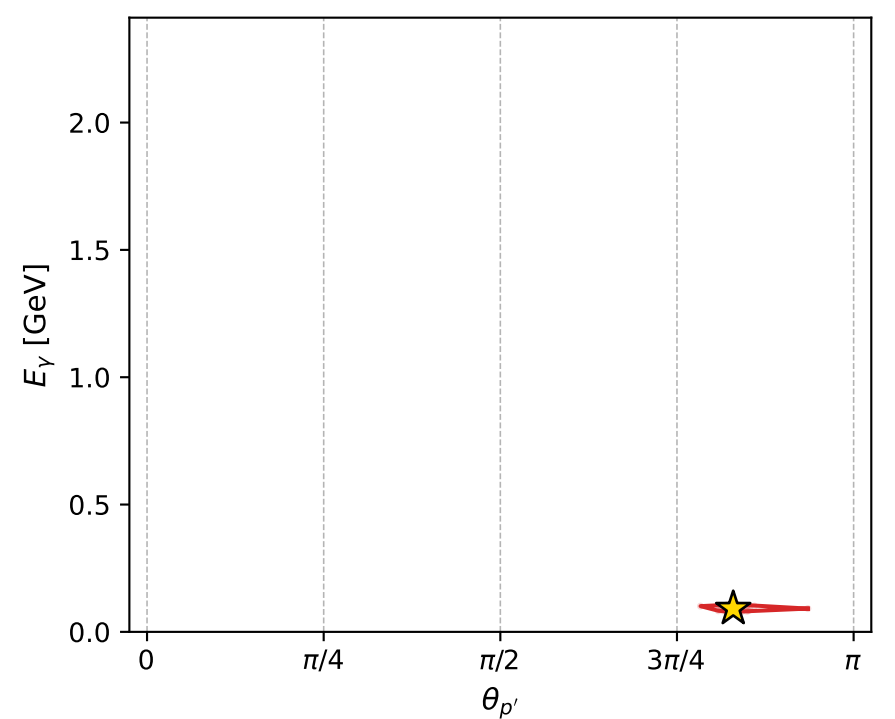
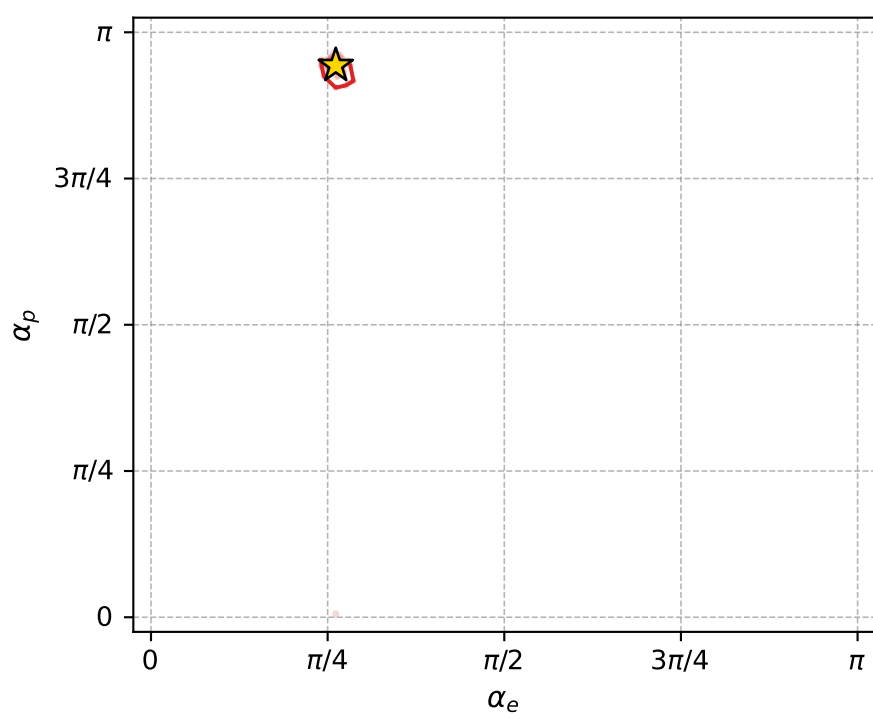
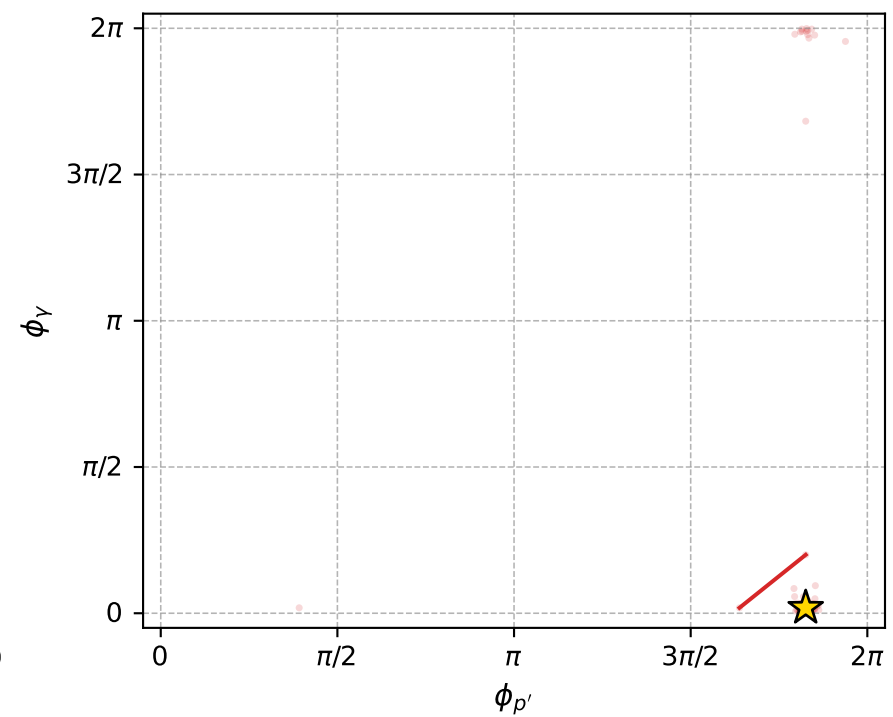
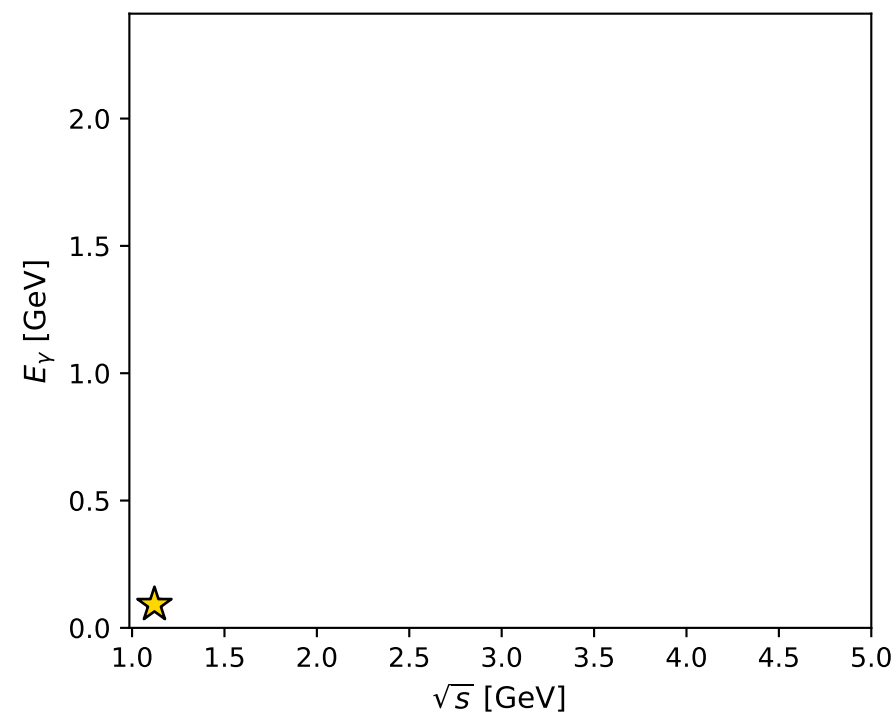
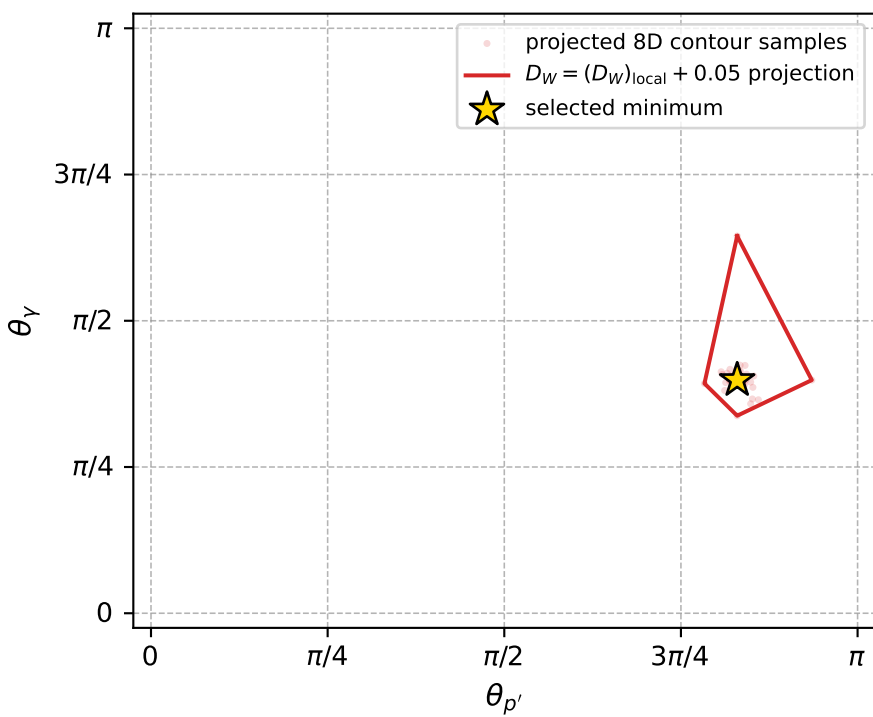
dw_mixing_angles_polarization_03_local_minimum_985 D_W=0.0176283
 region: polarization_cluster_03_local_minimum_985
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82151355$ rad
 incoming lepton: $0.681114|+\rangle + 0.732178|-\rangle$
 proton mixing angle: $\alpha_p=2.9630656$ rad
 incoming proton: $-0.984106|+\rangle + 0.17758|-\rangle$

$s=1.25689$, $\sqrt{s}=1.12111$, $|\vec{P}|=0.168157$, $|\vec{P}'|=0.00201124$
 $\theta_{p'}=2.60625$, $\theta_V=1.25227$, $\phi_{p'}=5.73523$, $\phi_V=0.0583372$
 $E_V=0.0914137$, $\theta_{l'}=1.86852$, $\phi_{l'}=3.19326$
 $Q^2=0.0217919$, $x_B=0.112773$, $t=-0.0286391$, $W^2=1.05129$, $y=0.512501$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1682$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1682$
 P $E= 0.9530$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1682$
 l' $E= 0.0917$ $p_x= -0.0875$ $p_y= -0.0045$ $p_z= -0.0269$
 P' $E= 0.9380$ $p_x= 0.0009$ $p_y= -0.0005$ $p_z= -0.0017$
 q_V $E= 0.0914$ $p_x= 0.0867$ $p_y= 0.0051$ $p_z= 0.0286$



electron: polarization cluster P3, local minimum 985; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017628293$
 $D_W \text{ contour} = 0.067628293$
 above global minimum = 0.01761186
 8D contour samples = 255

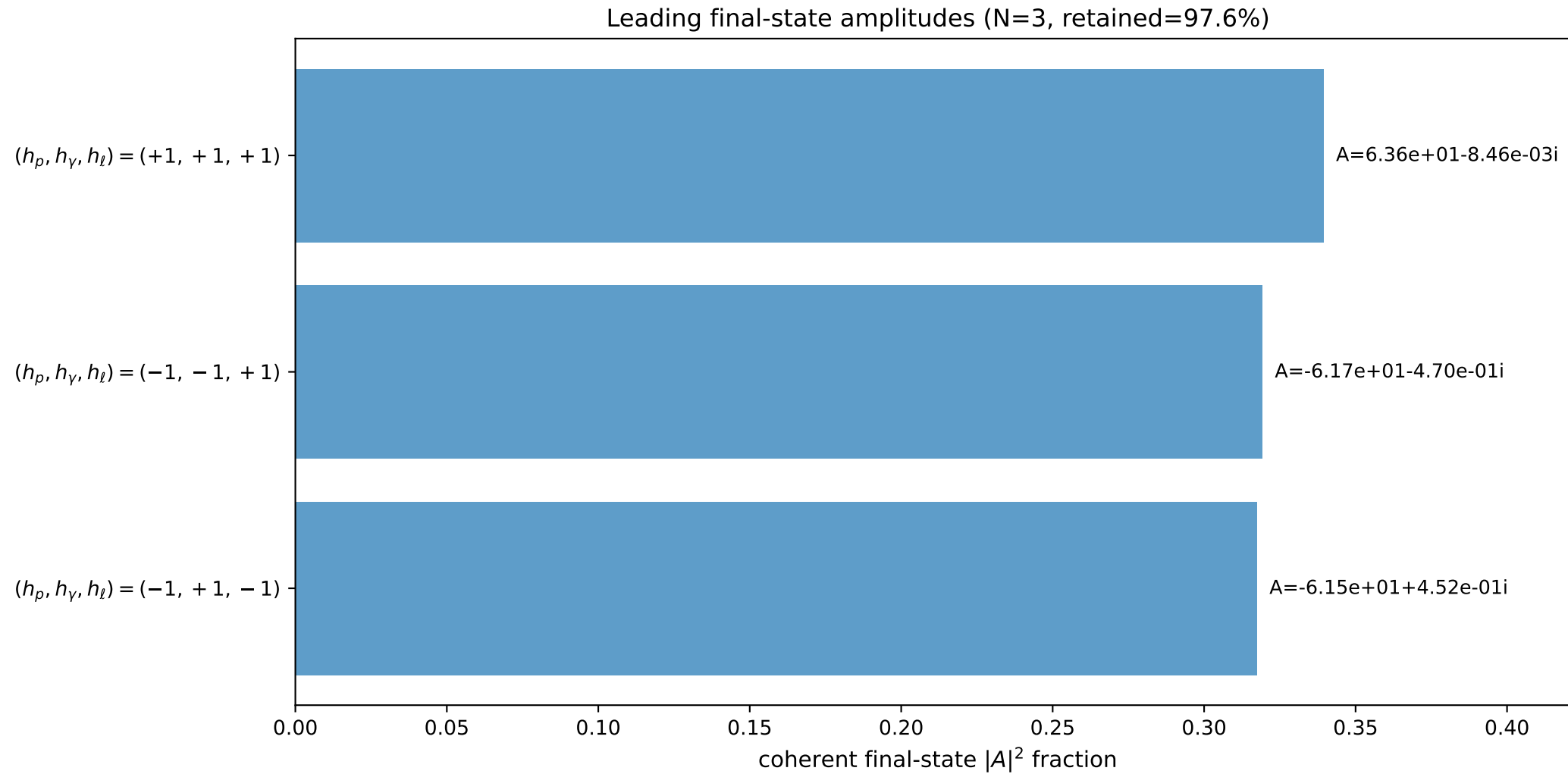
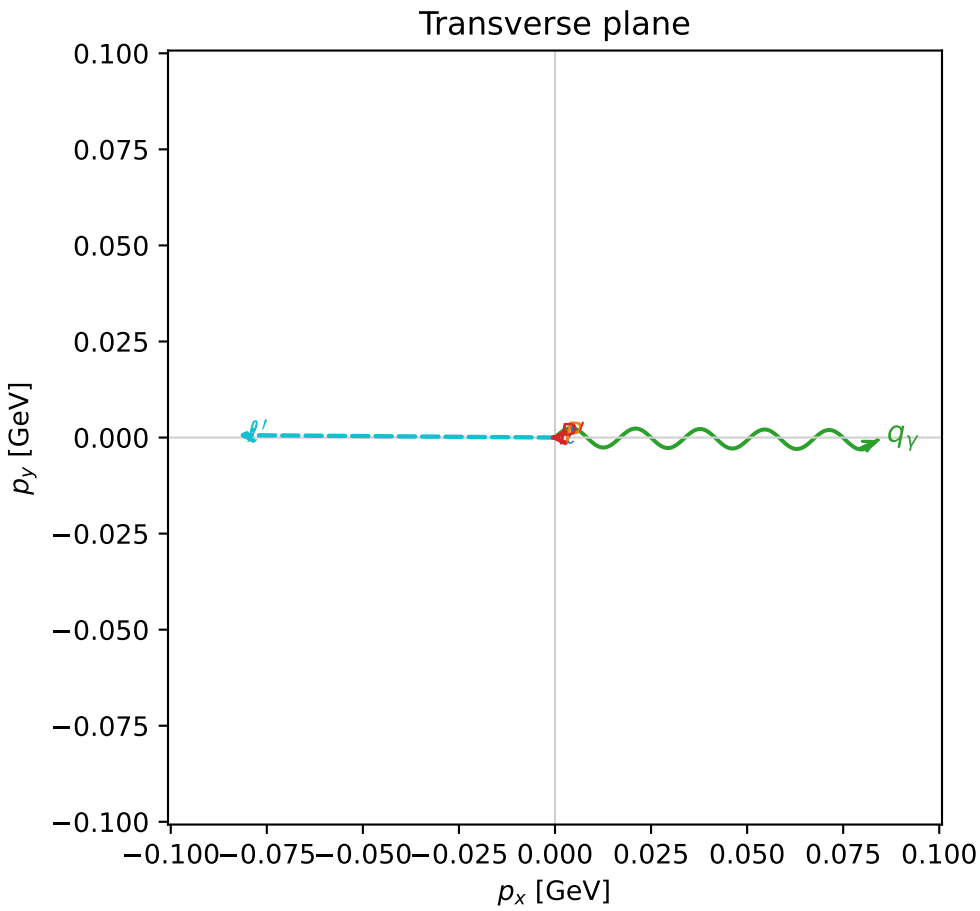
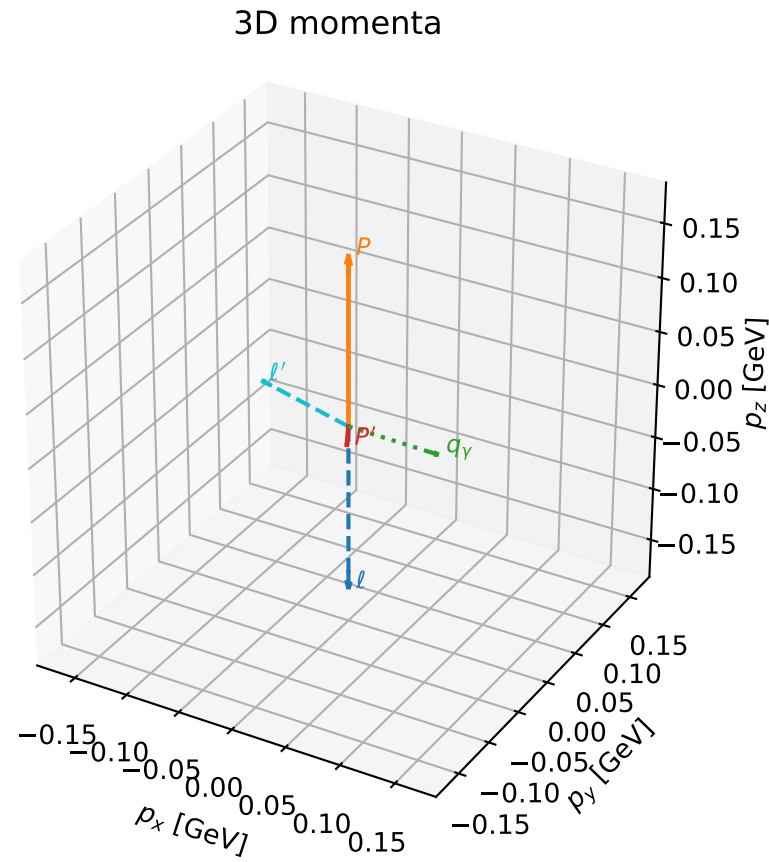
selected phase-space point:
 $\text{sqrt_s} = 1.121112$
 $\text{theta_p_out} = 2.606253$
 $\text{theta_gamma_out} = 1.252265$
 $\text{qOut} = 0.09141373$
 $\text{phi_p_out} = 5.735232$
 $\text{phi_gamma_out} = 0.05833715$
 $\text{alpha_e} = 0.8215136$
 $\text{alpha_p} = 2.963066$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

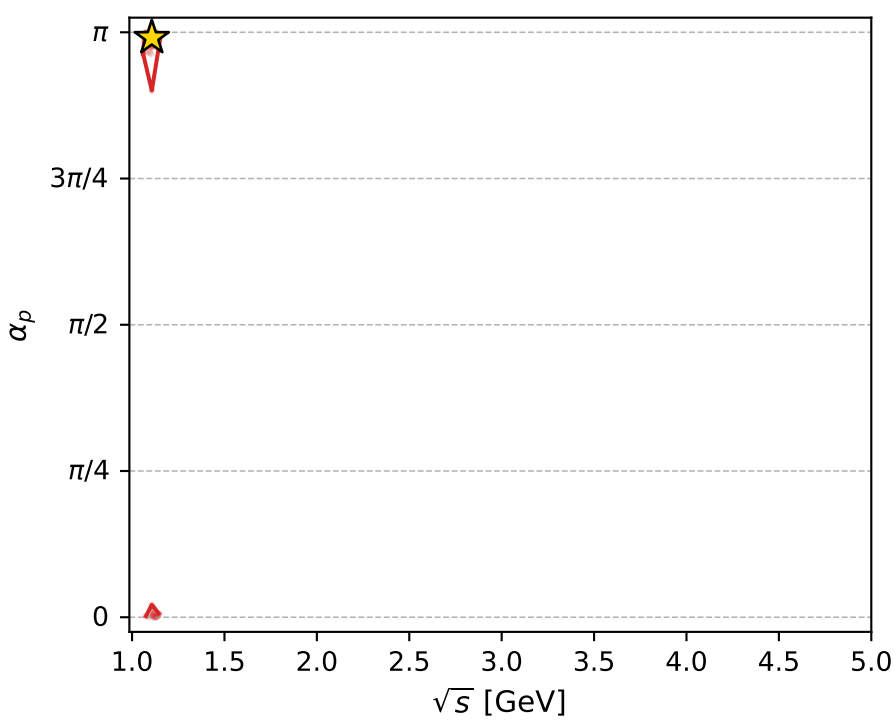
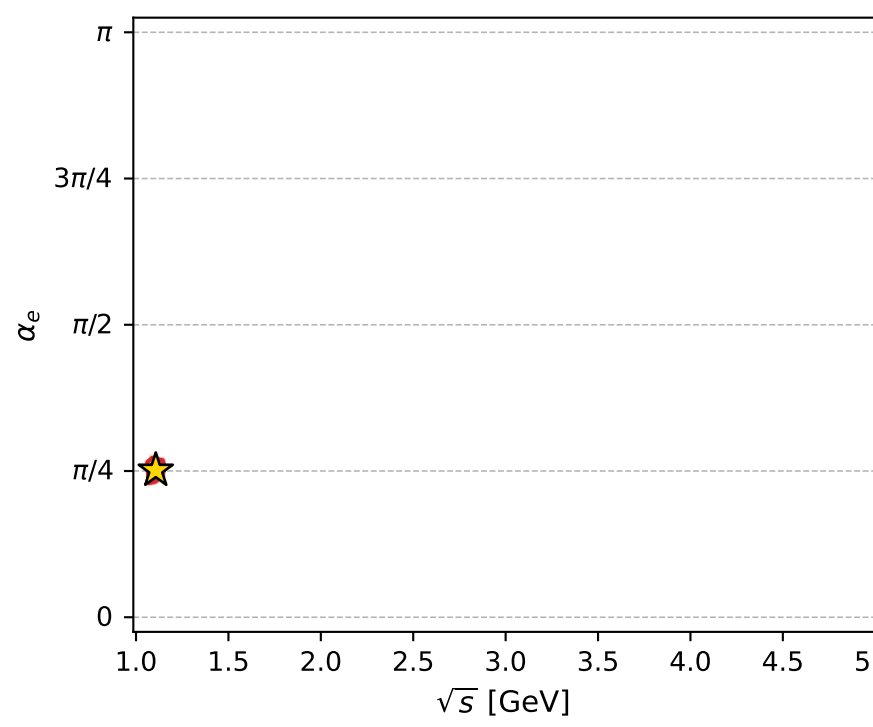
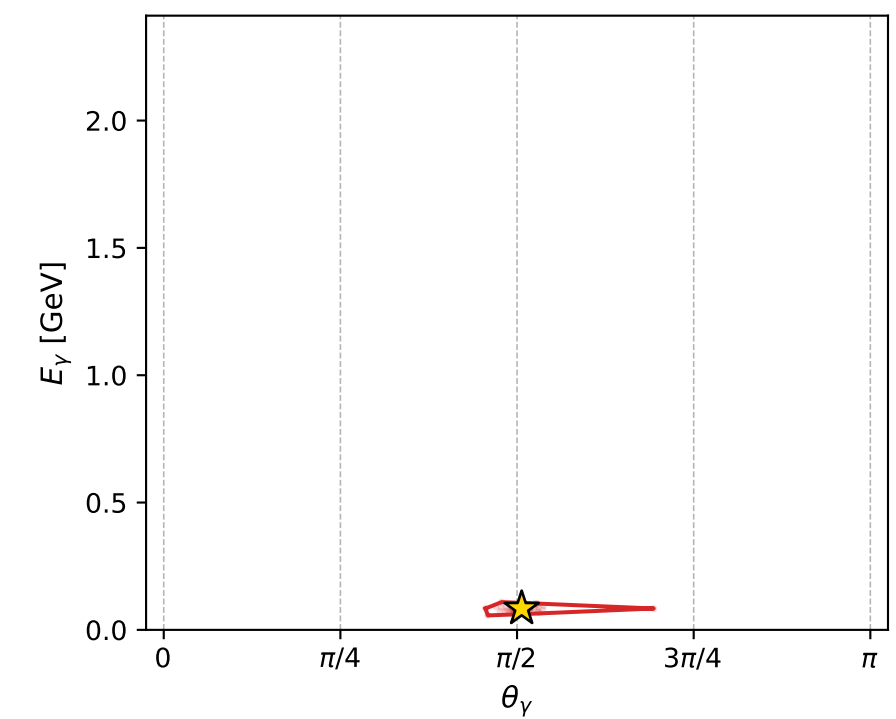
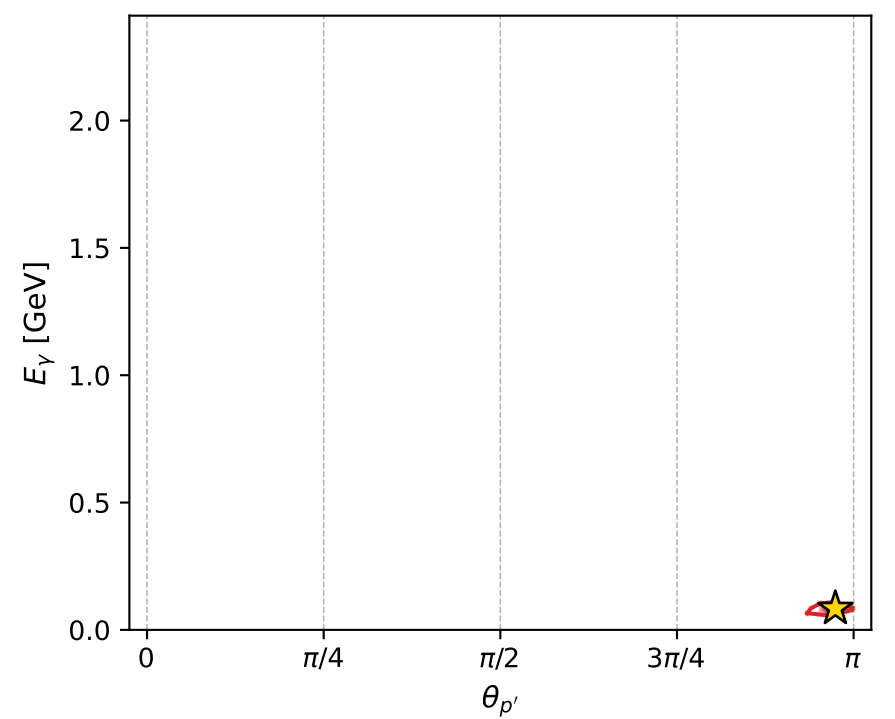
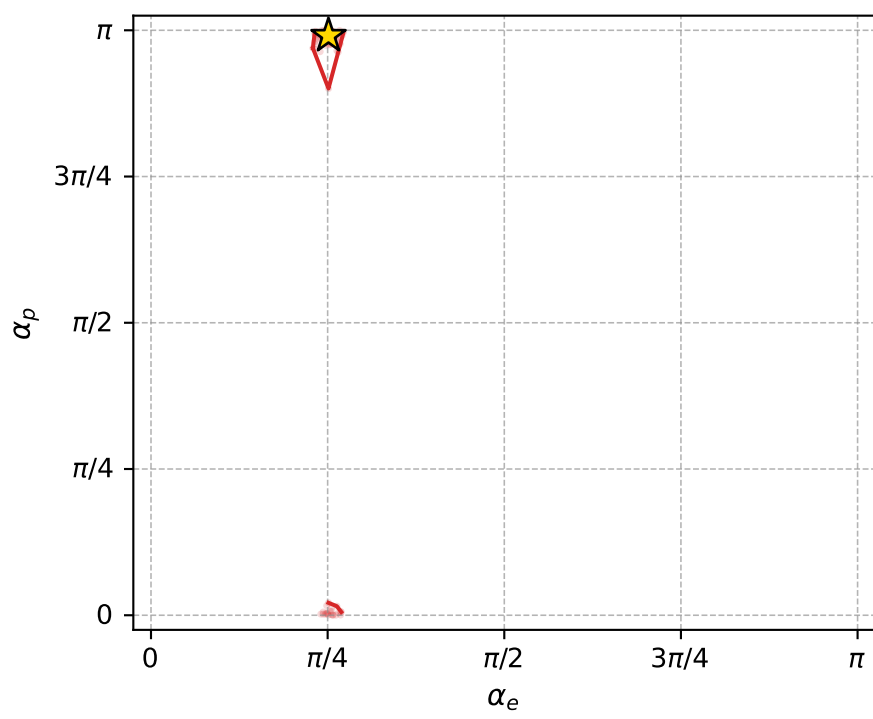
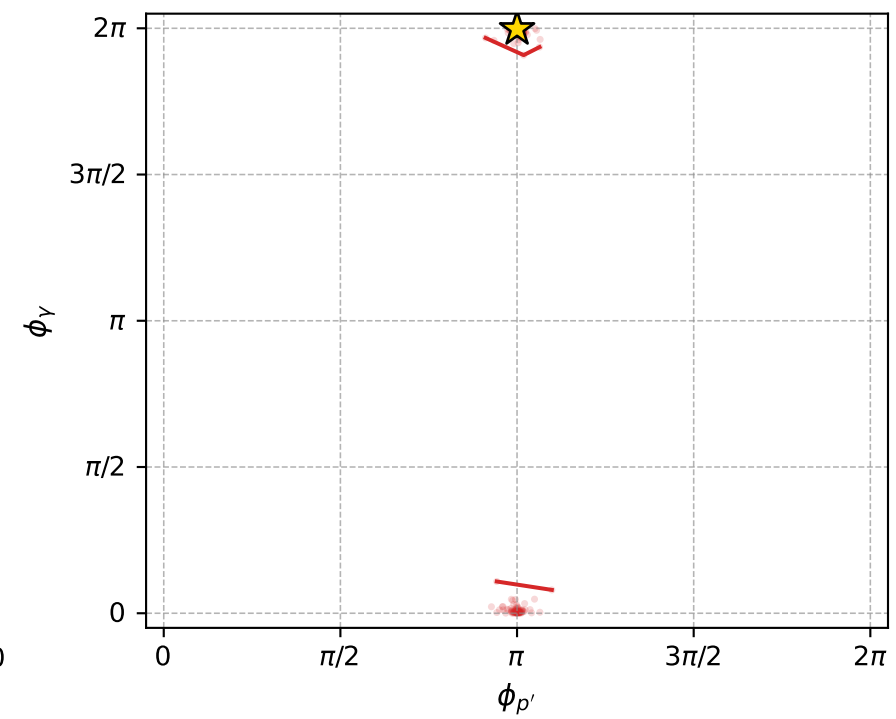
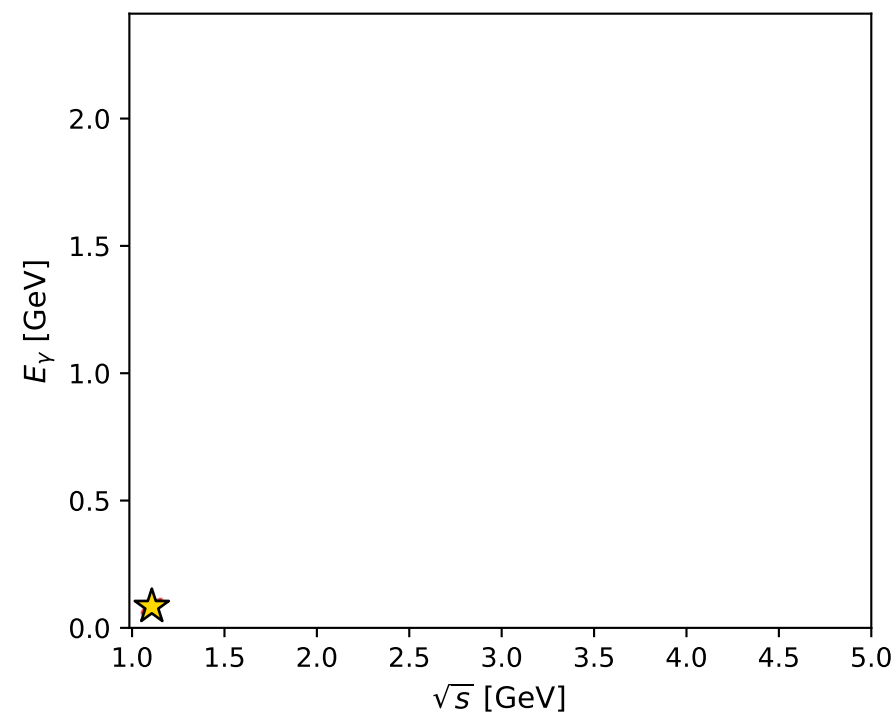
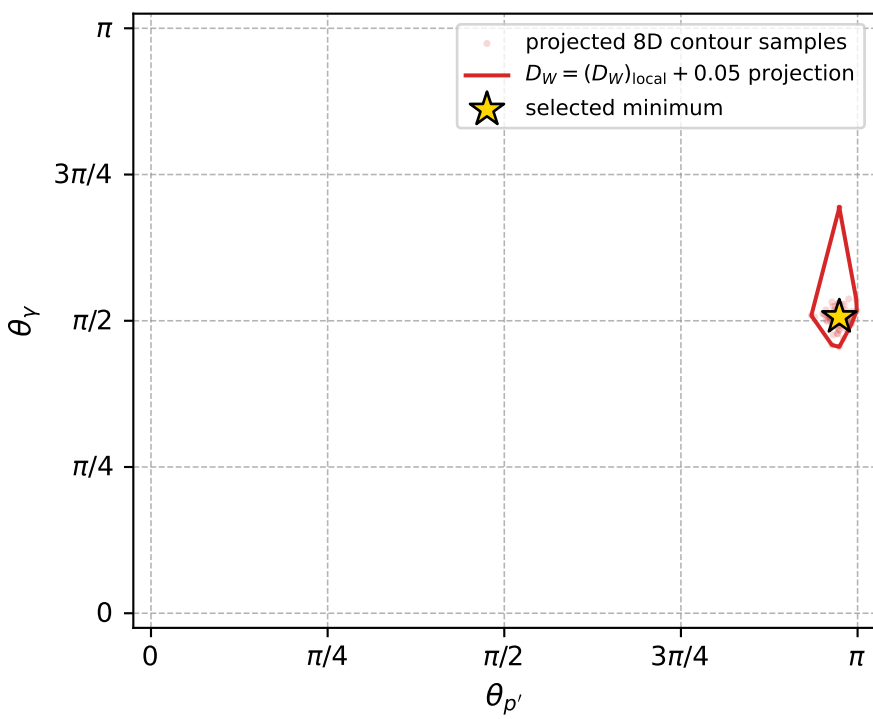
dw_mixing_angles_polarization_03_local_minimum_991 D_W=0.0179842
 region: polarization_cluster_03_local_minimum_991
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78916052$ rad
 incoming lepton: $0.704441|+\rangle + 0.709762|-\rangle$
 proton mixing angle: $\alpha_p=3.112534$ rad
 incoming proton: $-0.999578|+\rangle + 0.0290546|-\rangle$

$s=1.22514$, $\sqrt{s}=1.10686$, $|\vec{P}|=0.155981$, $|\vec{P}'|=0.0177058$
 $\theta_{p'}=3.06042$, $\theta_Y=1.59143$, $\phi_{p'}=3.14178$, $\phi_Y=6.27569$
 $E_Y=0.0839506$, $\theta_{l'}=1.34006$, $\phi_{l'}=3.13397$
 $Q^2=0.0324823$, $x_B=0.170796$, $t=-0.0299871$, $W^2=1.03754$, $y=0.550773$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1560$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1560$
 P $E= 0.9509$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1560$
 l' $E= 0.0847$ $p_x= -0.0825$ $p_y= 0.0006$ $p_z= 0.0194$
 P' $E= 0.9382$ $p_x= -0.0014$ $p_y= -0.0000$ $p_z= -0.0176$
 q_Y $E= 0.0840$ $p_x= 0.0839$ $p_y= -0.0006$ $p_z= -0.0017$



electron: polarization cluster P3, local minimum 991; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.017984173$
 $D_W \text{ contour} = 0.067984173$
 above global minimum = 0.01796774
 8D contour samples = 250

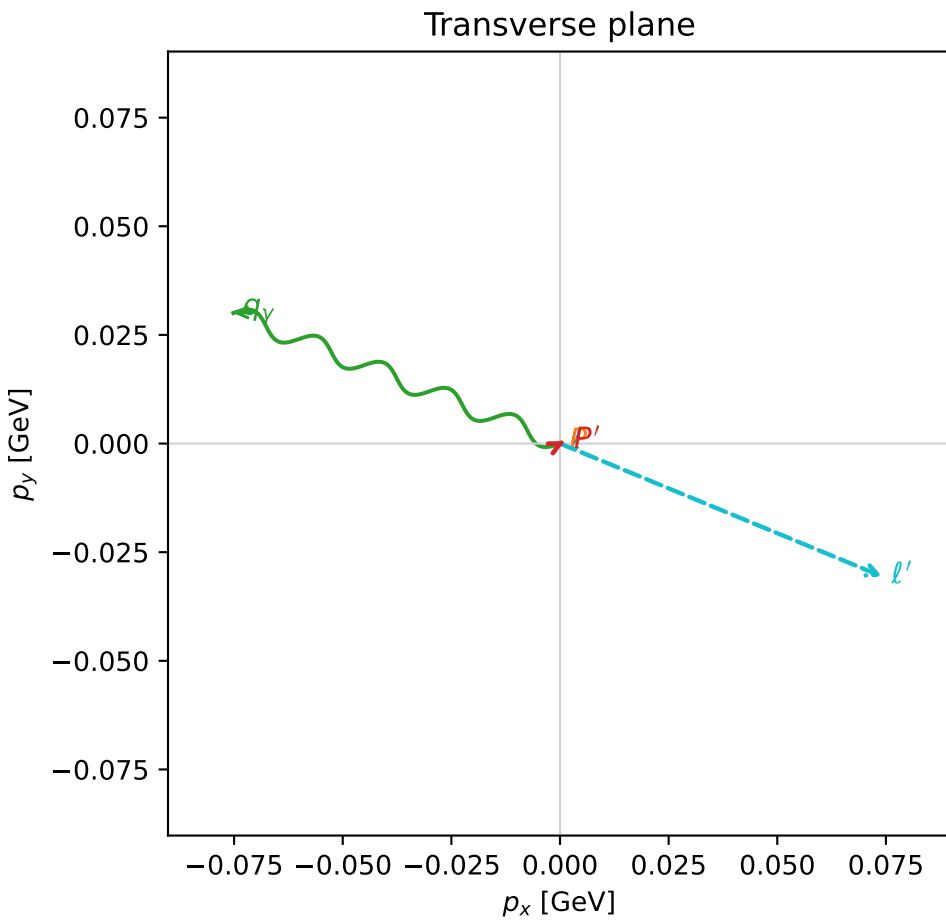
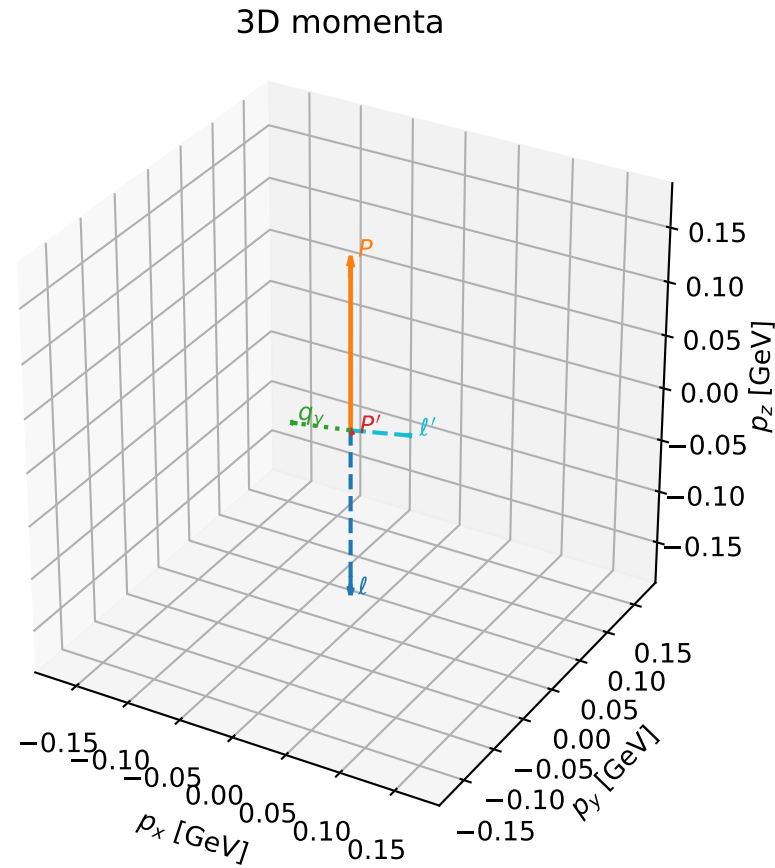
selected phase-space point:
 $\text{sqrt_s} = 1.106862$
 $\text{theta_p_out} = 3.060416$
 $\text{theta_gamma_out} = 1.591432$
 $\text{qOut} = 0.0839506$
 $\text{phi_p_out} = 3.141778$
 $\text{phi_gamma_out} = 6.275693$
 $\text{alpha_e} = 0.7891605$
 $\text{alpha_p} = 3.112534$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_994 D_W=0.0180986
region: polarization_cluster_03_local_minimum_994
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.77750627$ rad
incoming lepton: $0.712665|+> +0.701504|->$
proton mixing angle: $\alpha_p=0.028423775$ rad
incoming proton: $0.999596|+> +0.0284199|->$

$s=1.23267$, $\sqrt{s}=1.11026$, $|\vec{P}|=0.158894$, $|\vec{P}'|=0.00281842$
 $\theta_{p'}=2.70197$, $\theta_Y=1.91653$, $\phi_{p'}=0.467859$, $\phi_Y=2.76154$
 $E_Y=0.0860442$, $\theta_{\ell'}=1.19413$, $\phi_{\ell'}=5.89191$
 $Q^2=0.0374729$, $x_B=0.188427$, $t=-0.0258873$, $W^2=1.04124$, $y=0.563652$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1589$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1589$
 P $E= 0.9514$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1589$
 ℓ' $E= 0.0862$ $p_x= 0.0741$ $p_y= -0.0306$ $p_z= 0.0317$
 P' $E= 0.9380$ $p_x= 0.0011$ $p_y= 0.0005$ $p_z= -0.0026$
 q_Y $E= 0.0860$ $p_x= -0.0752$ $p_y= 0.0300$ $p_z= -0.0292$

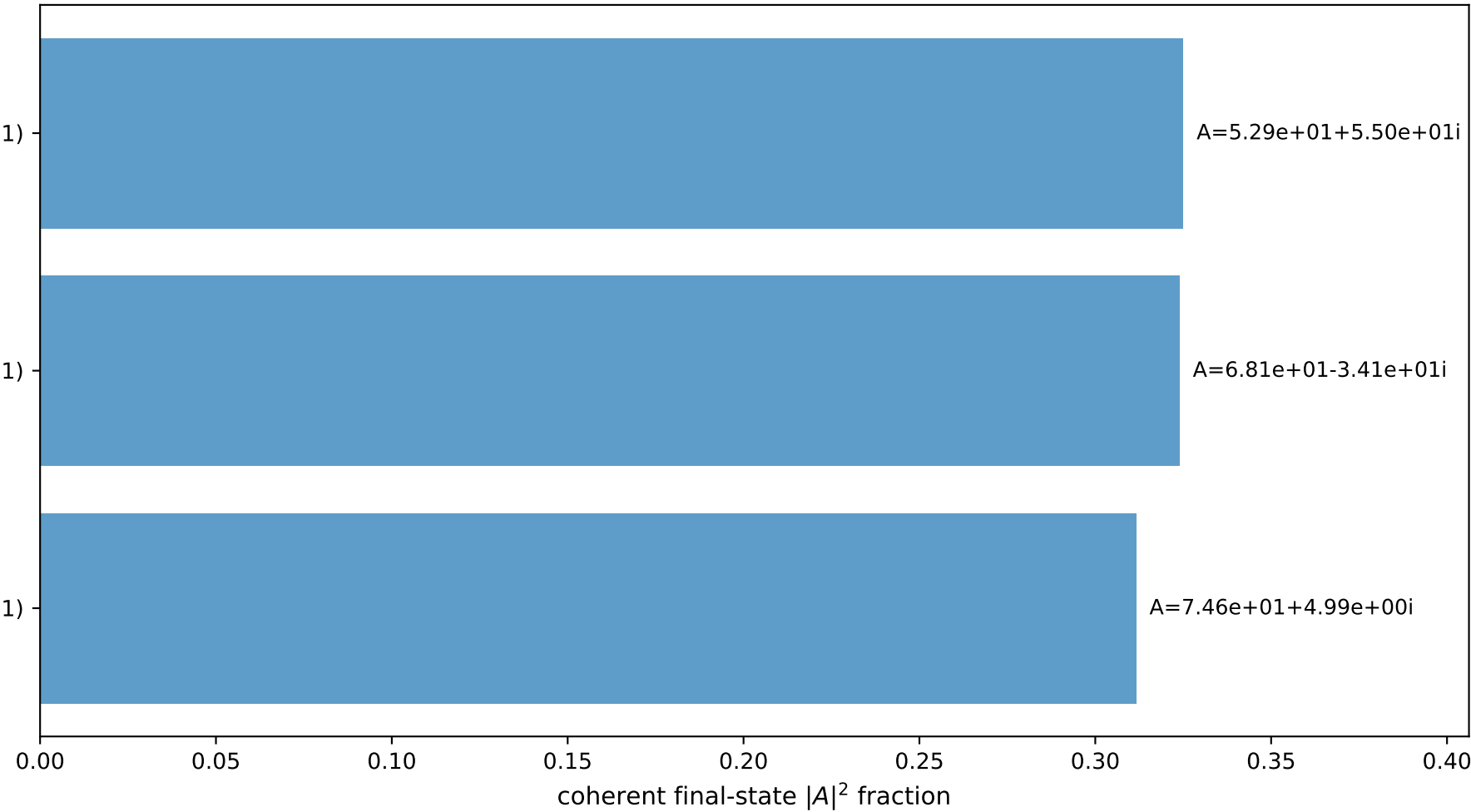


$(h_p, h_Y, h_{\ell}) = (-1, -1, +1)$

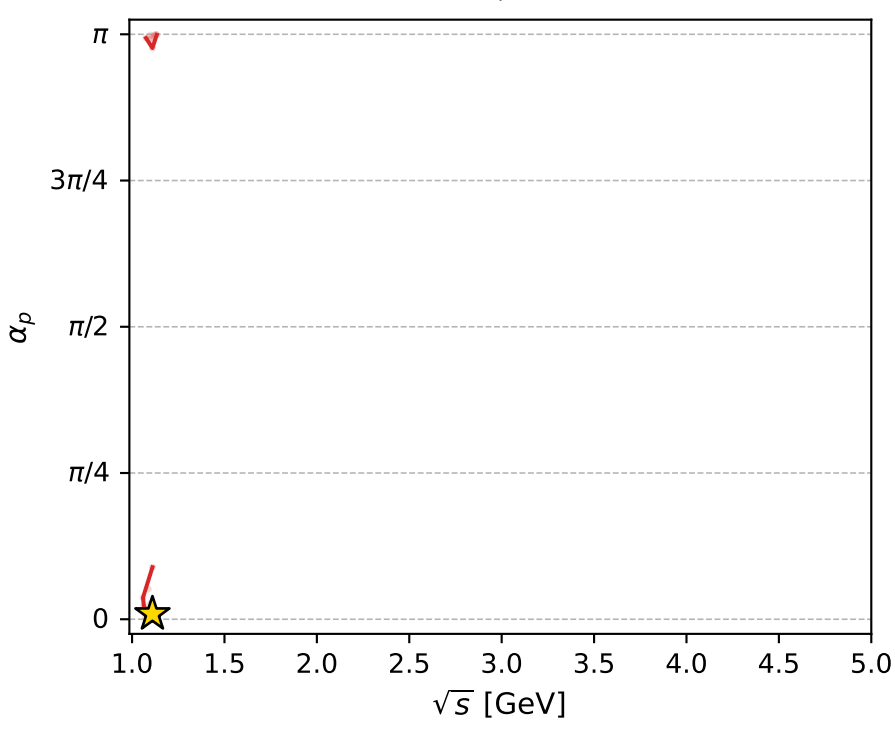
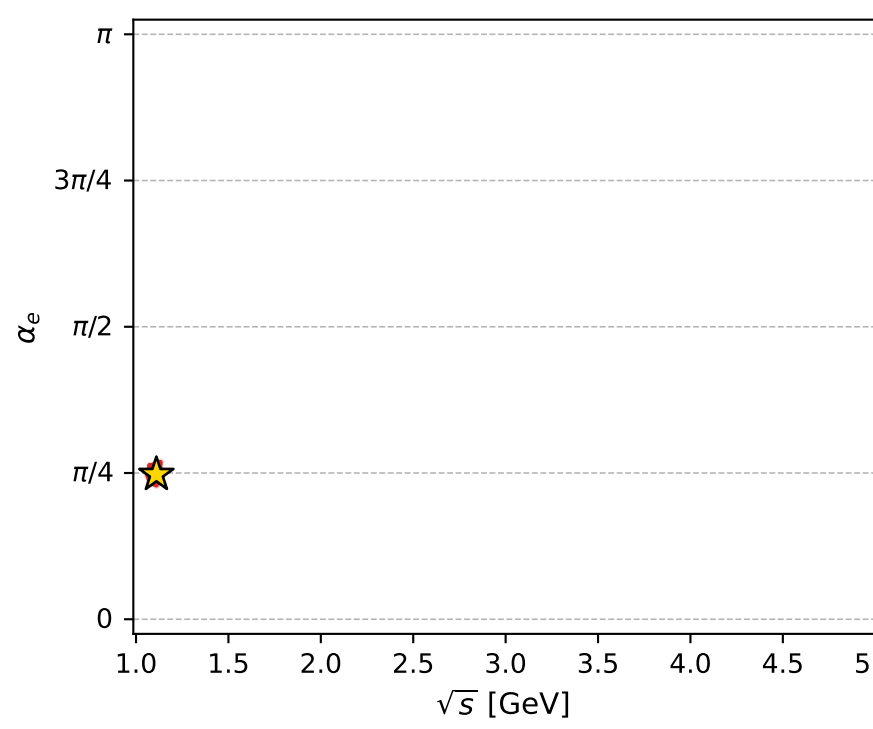
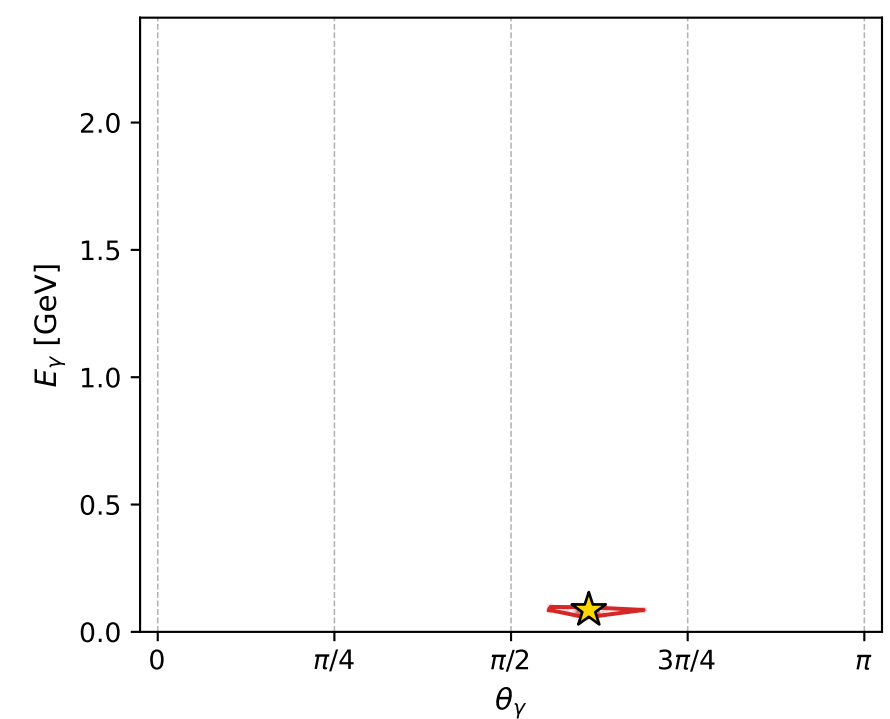
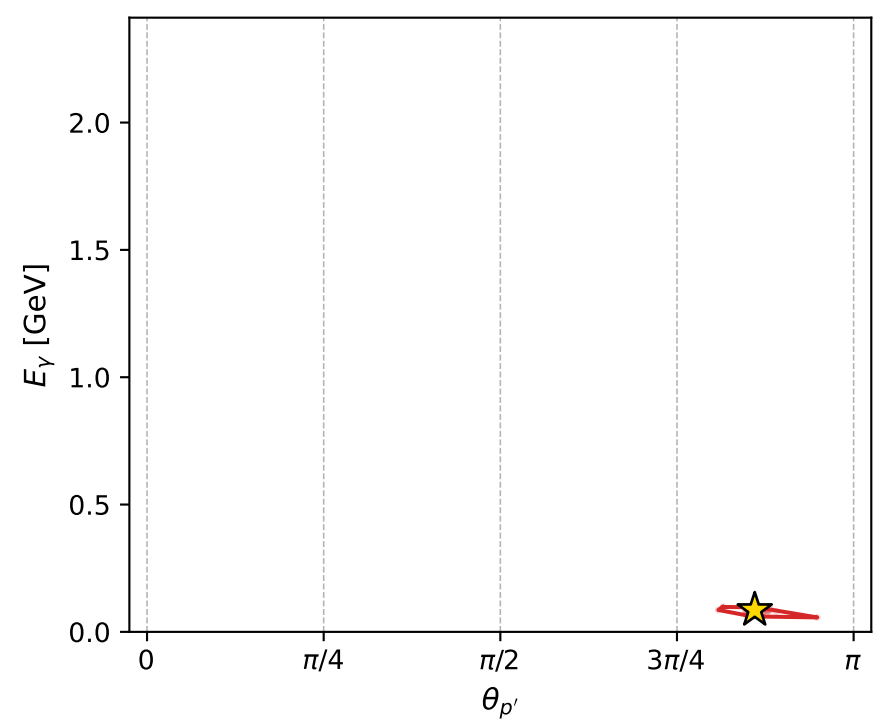
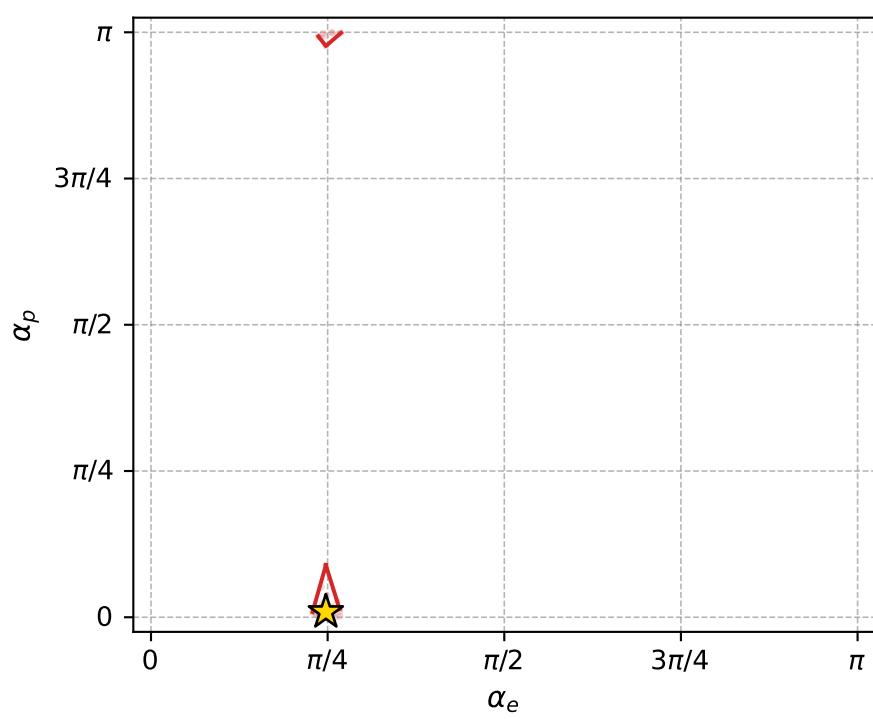
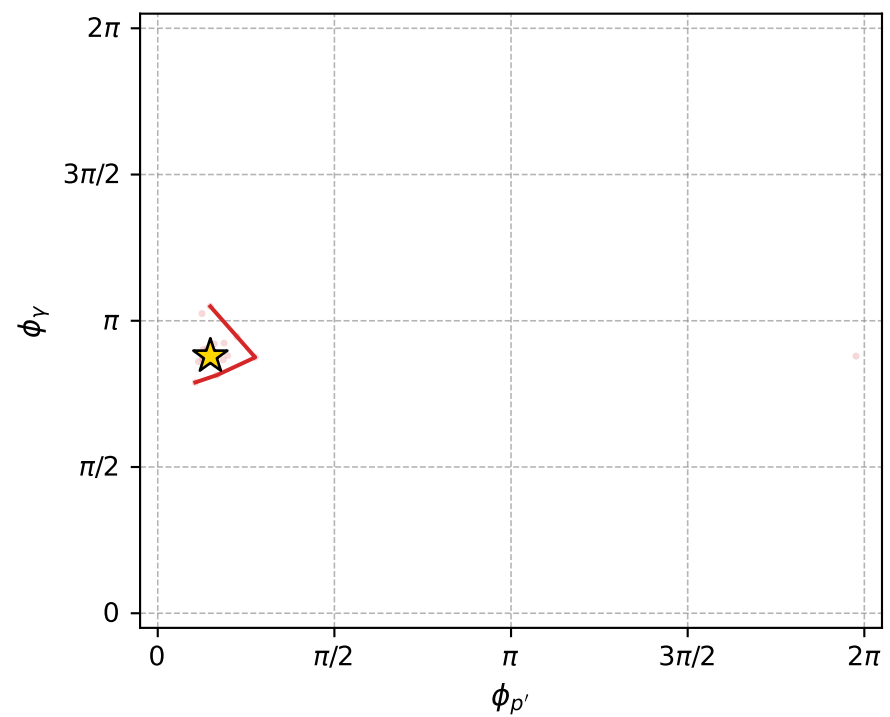
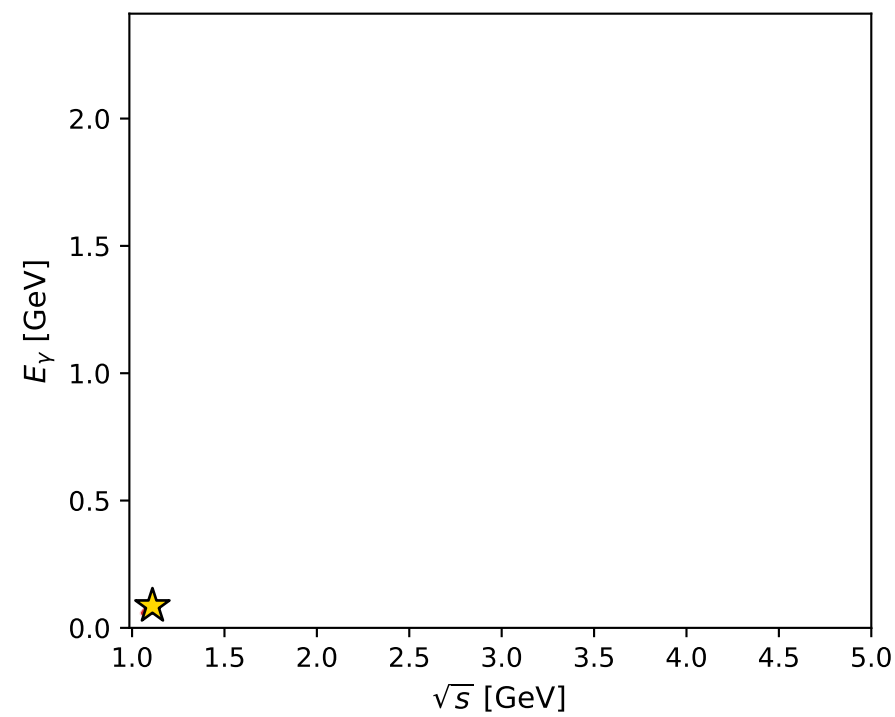
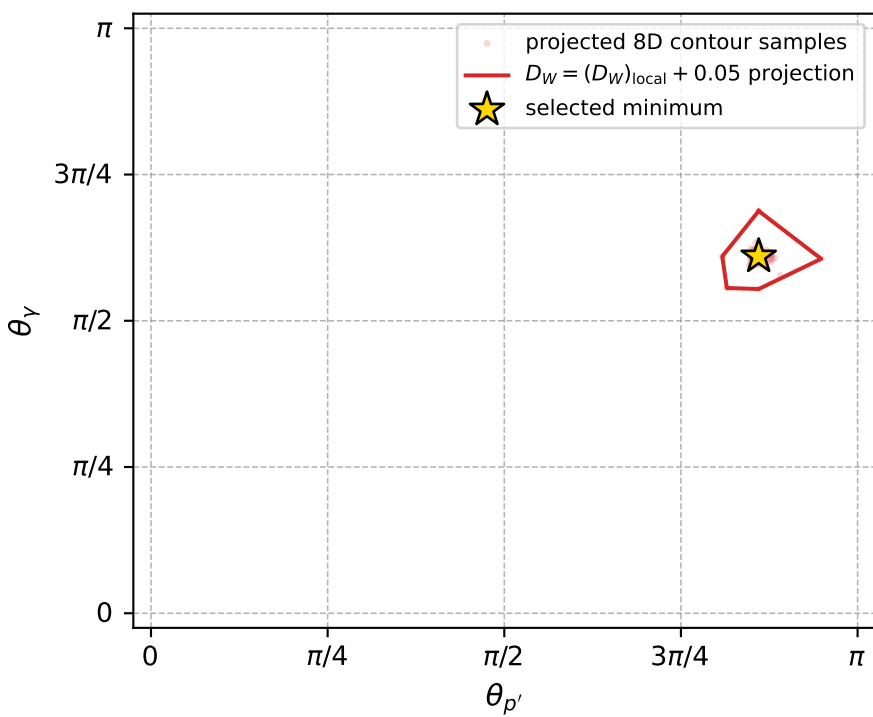
$(h_p, h_Y, h_{\ell}) = (+1, +1, +1)$

$(h_p, h_Y, h_{\ell}) = (-1, +1, -1)$

Leading final-state amplitudes (N=3, retained=96.0%)



electron: polarization cluster P3, local minimum 994; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.018098572$

$D_W \text{ contour} = 0.068098572$

above global minimum = 0.018082139

8D contour samples = 253

selected phase-space point:

$\text{sqrt_s} = 1.110258$

$\text{theta_p_out} = 2.701965$

$\text{theta_gamma_out} = 1.916527$

$\text{qOut} = 0.08604421$

$\text{phi_p_out} = 0.4678587$

$\text{phi_gamma_out} = 2.761535$

$\alpha_e = 0.7775063$

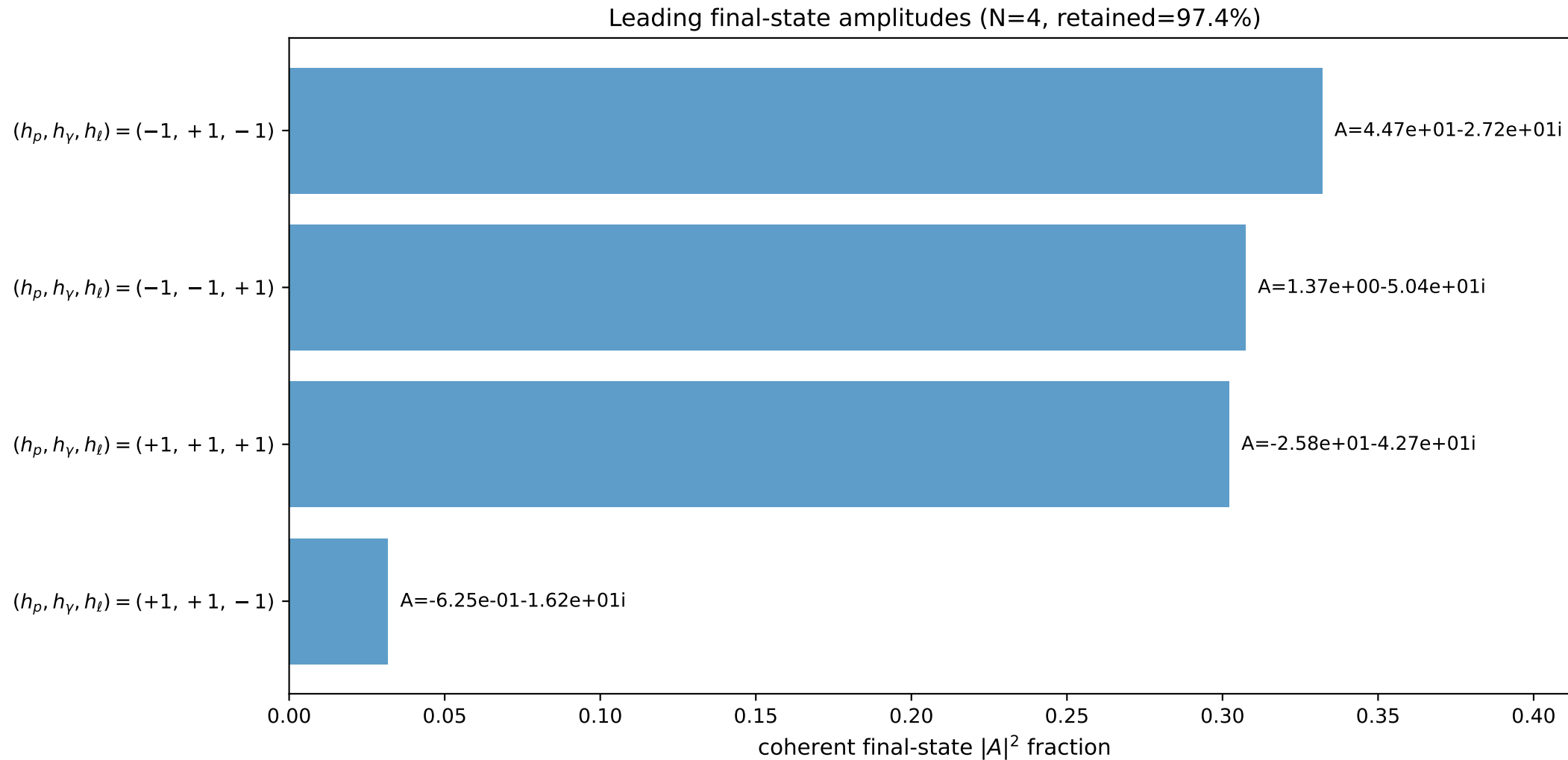
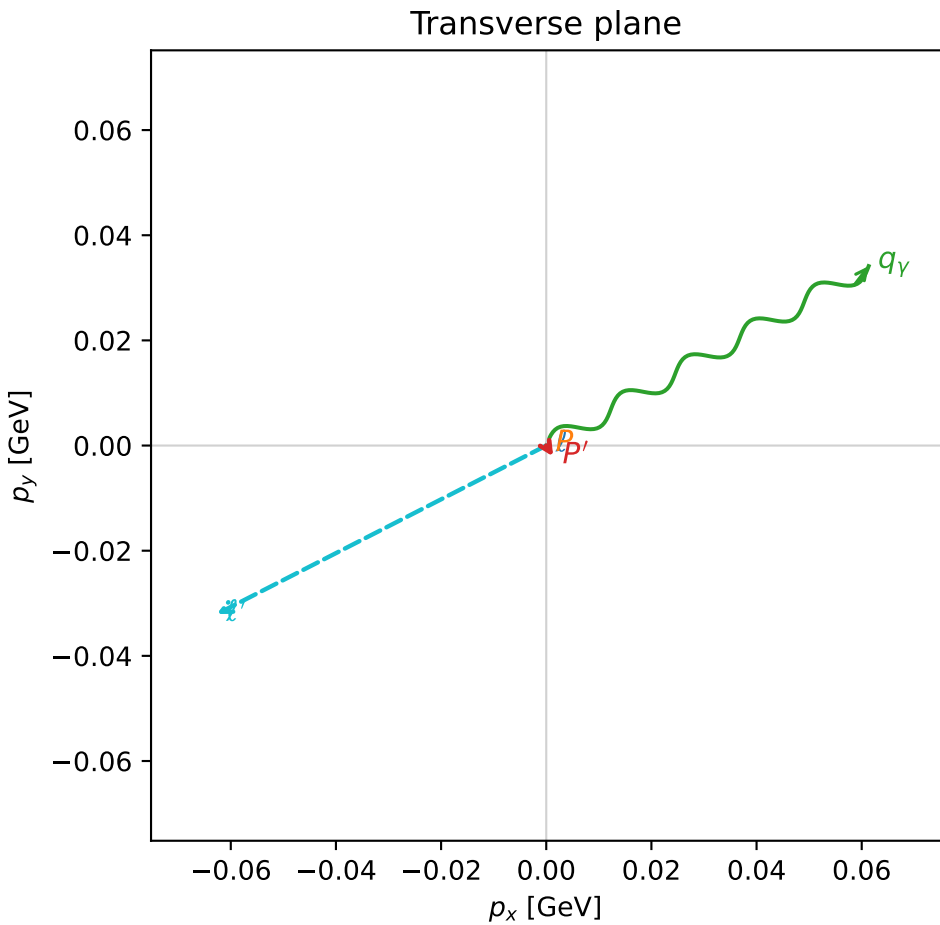
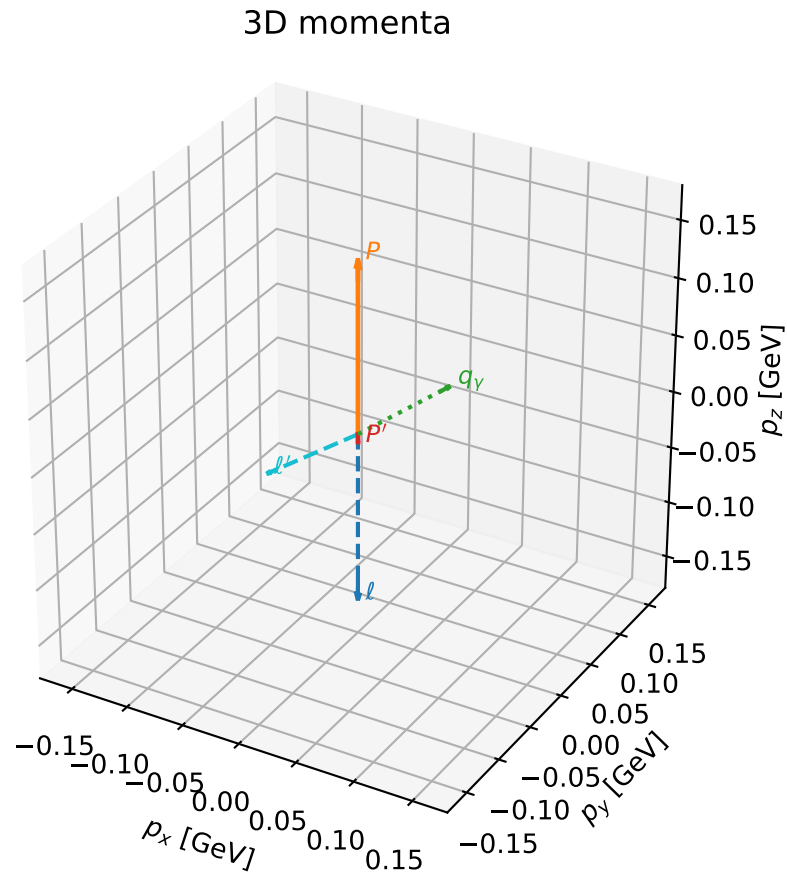
$\alpha_p = 0.02842378$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

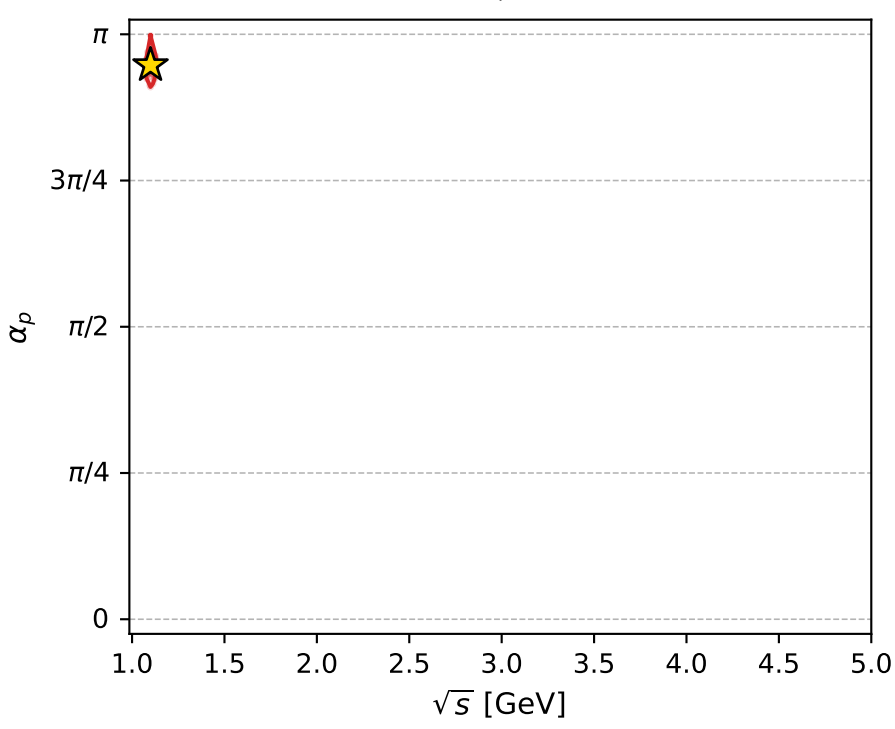
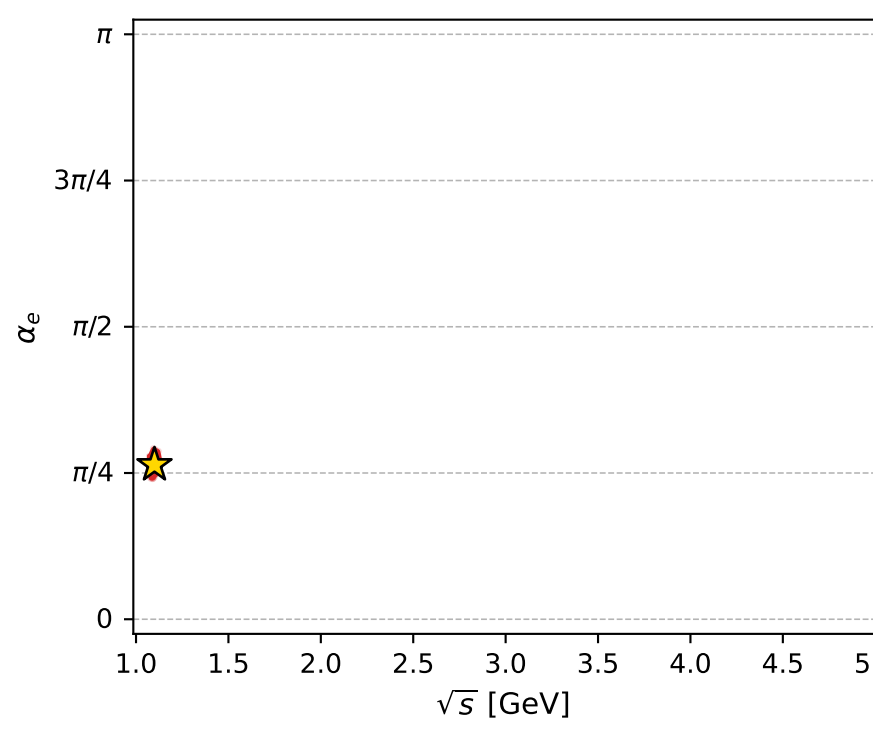
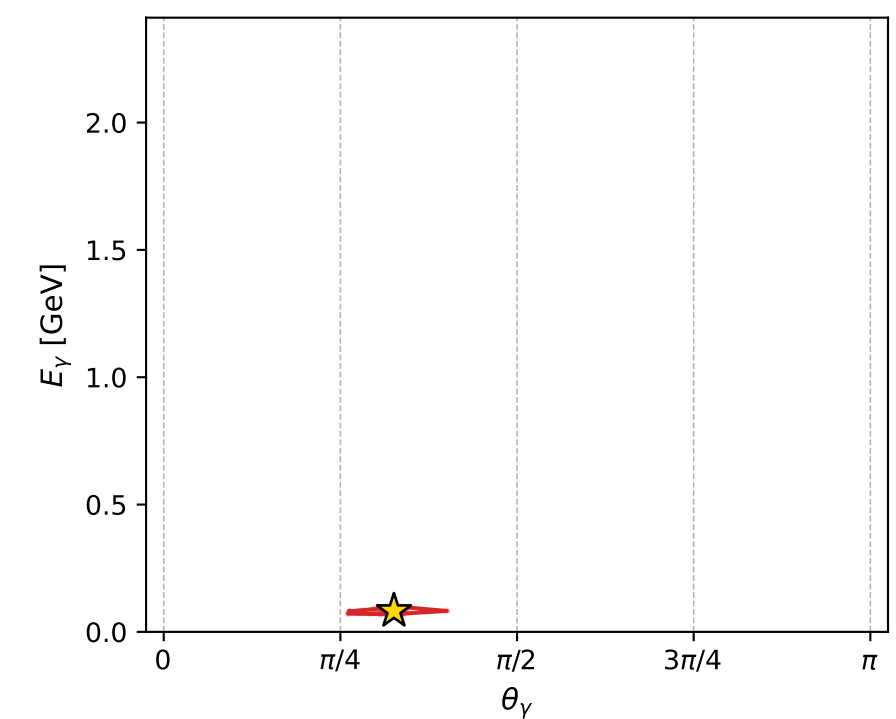
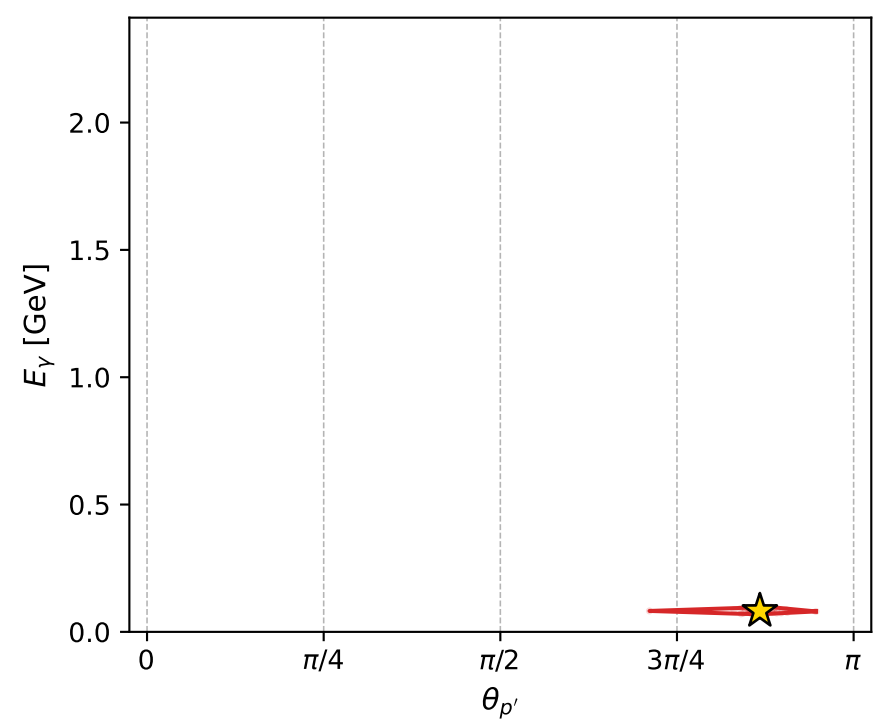
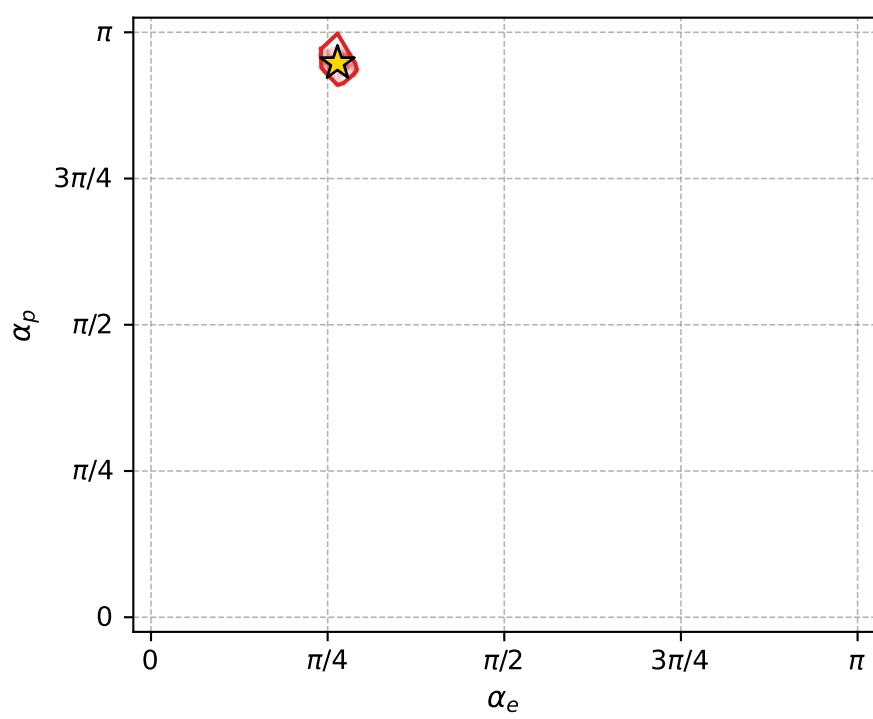
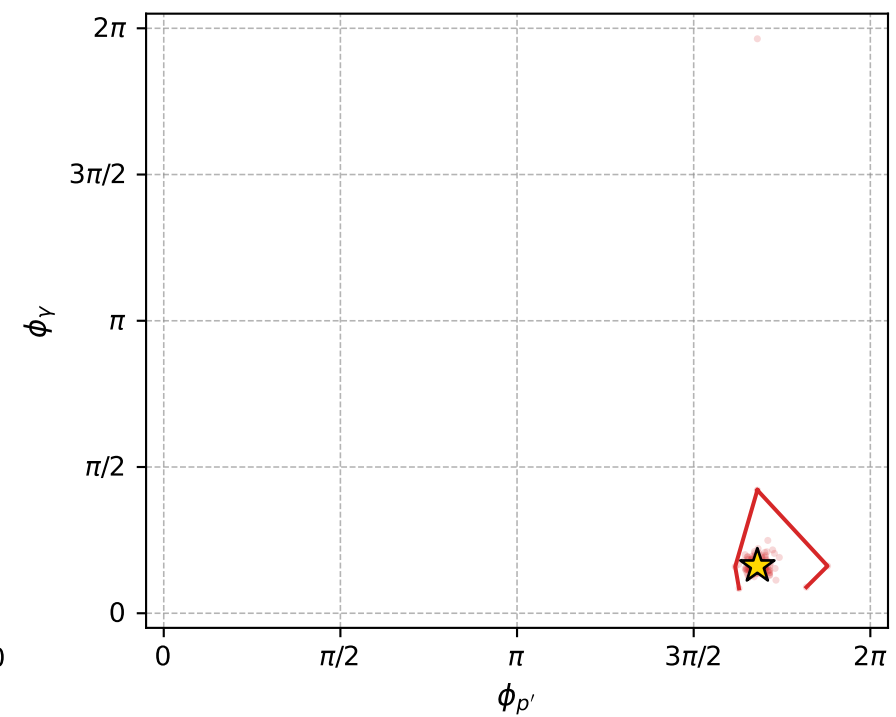
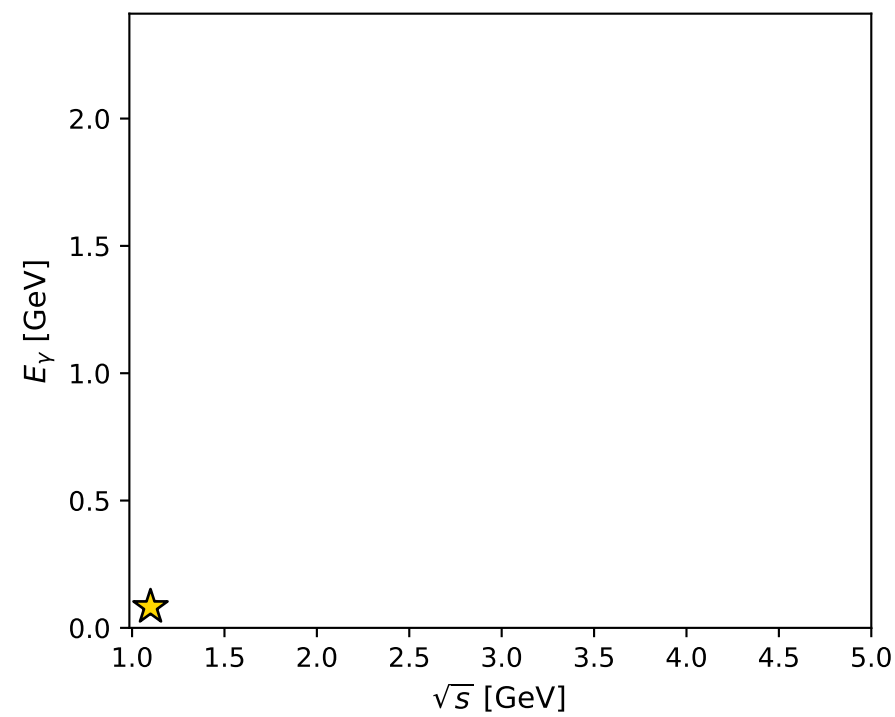
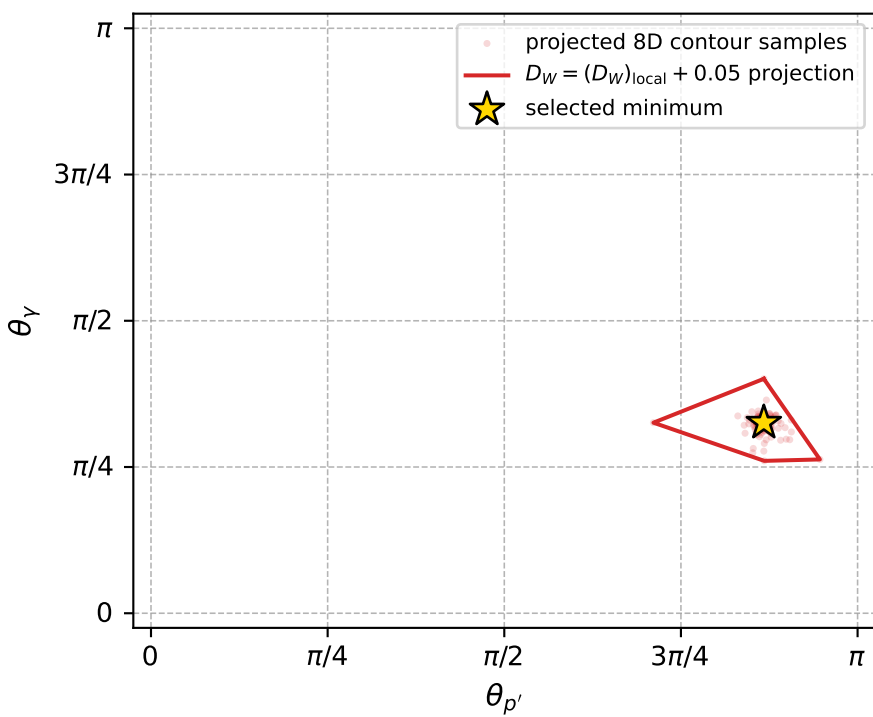
dw_mixing_angles_polarization_03_local_minimum_995 D_W=0.0181031
 region: polarization_cluster_03_local_minimum_995
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82915711$ rad
 incoming lepton: $0.675498|+\rangle + 0.737362|-\rangle$
 proton mixing angle: $\alpha_p=2.9757373$ rad
 incoming proton: $-0.986277|+\rangle + 0.165096|-\rangle$

$s=1.20955$, $\sqrt{s}=1.0998$, $|\vec{P}|=0.149895$, $|\vec{P}'|=0.00607097$
 $\theta_{P'}=2.72473$, $\theta_V=1.02347$, $\phi_{P'}=5.27853$, $\phi_V=0.507773$
 $E_V=0.0821791$, $\theta_{\ell'}=2.05732$, $\phi_{\ell'}=3.61448$
 $Q^2=0.0127053$, $x_B=0.0759294$, $t=-0.0240282$, $W^2=1.03447$, $y=0.507513$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1499$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1499$
 P $E= 0.9499$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1499$
 ℓ' $E= 0.0796$ $p_x= -0.0626$ $p_y= -0.0320$ $p_z= -0.0372$
 P' $E= 0.9380$ $p_x= 0.0013$ $p_y= -0.0021$ $p_z= -0.0056$
 q_V $E= 0.0822$ $p_x= 0.0613$ $p_y= 0.0341$ $p_z= 0.0428$



electron: polarization cluster P3, local minimum 995; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.018103131$
 $D_W \text{ contour} = 0.068103131$
 above global minimum = 0.018086698
 8D contour samples = 254

selected phase-space point:
 $\text{sqrt_s} = 1.099797$
 $\text{theta_p_out} = 2.724734$
 $\text{theta_gamma_out} = 1.023465$
 $\text{qOut} = 0.08217906$
 $\text{phi_p_out} = 5.27853$
 $\text{phi_gamma_out} = 0.5077727$
 $\text{alpha_e} = 0.8291571$
 $\text{alpha_p} = 2.975737$

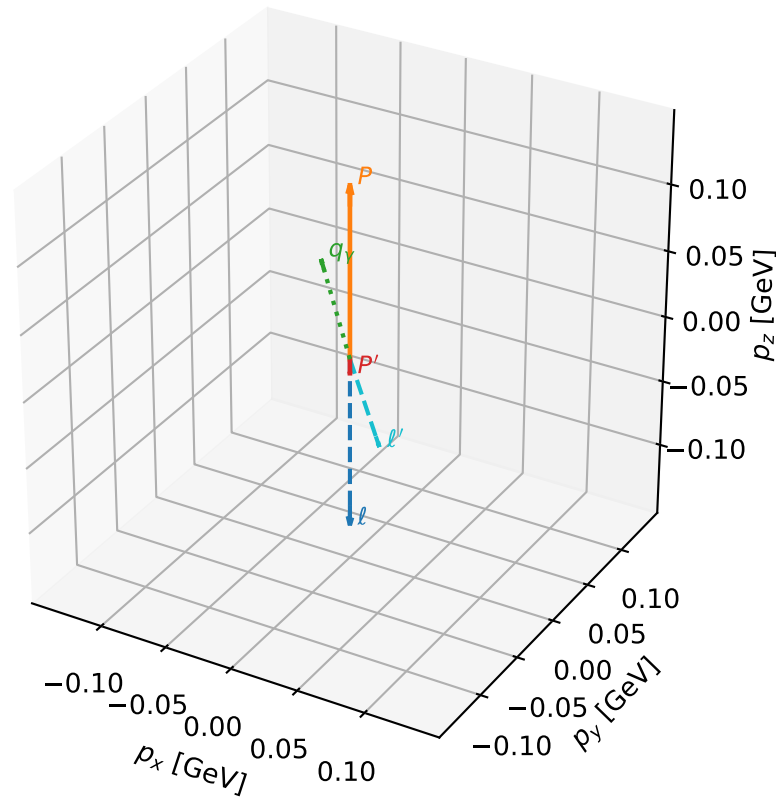
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_996 D_W=0.0181065
 region: polarization_cluster_03_local_minimum_996
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81338728$ rad
 incoming lepton: $0.687041|+\rangle +0.726619|-\rangle$
 proton mixing angle: $\alpha_p=0.13384661$ rad
 incoming proton: $0.991056|+\rangle +0.133447|-\rangle$

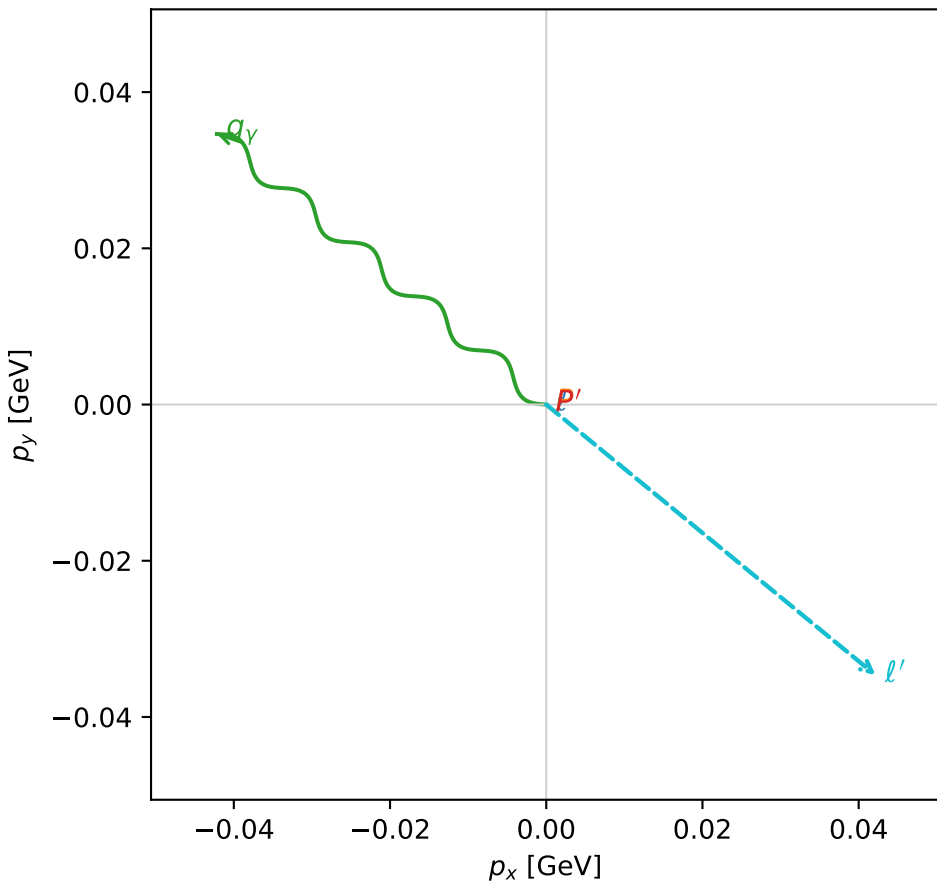
$s=1.15751$, $\sqrt{s}=1.07588$, $|\vec{P}|=0.129043$, $|\vec{P}'|=0.0106414$
 $\theta_{P'}=3.14159$, $\theta_V=0.857808$, $\phi_{P'}=2.75263$, $\phi_V=2.45387$
 $E_V=0.0721408$, $\theta_\ell=2.16091$, $\phi_{\ell'}=5.59546$
 $Q^2=0.00751809$, $x_B=0.0522571$, $t=-0.0194347$, $W^2=1.01619$, $y=0.518122$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1290$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1290$
 P $E= 0.9468$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1290$
 ℓ' $E= 0.0657$ $p_x= 0.0422$ $p_y= -0.0346$ $p_z= -0.0365$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= -0.0106$
 q_V $E= 0.0721$ $p_x= -0.0422$ $p_y= 0.0346$ $p_z= 0.0472$

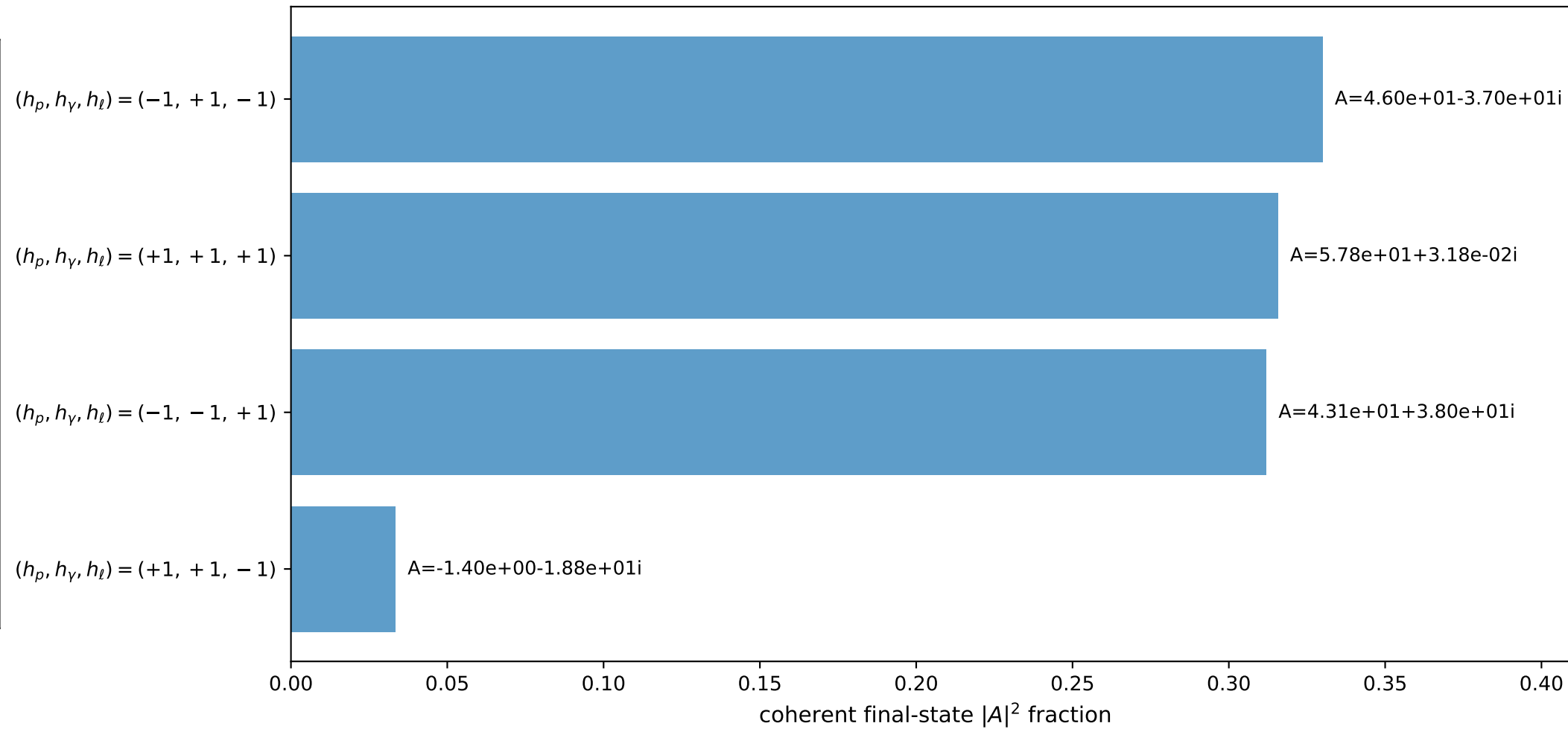
3D momenta



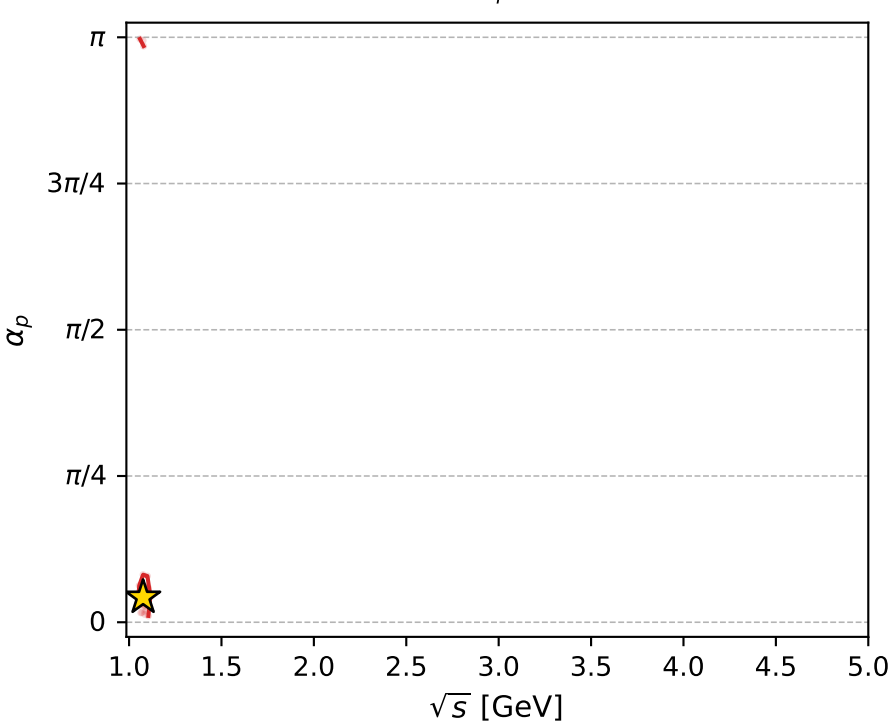
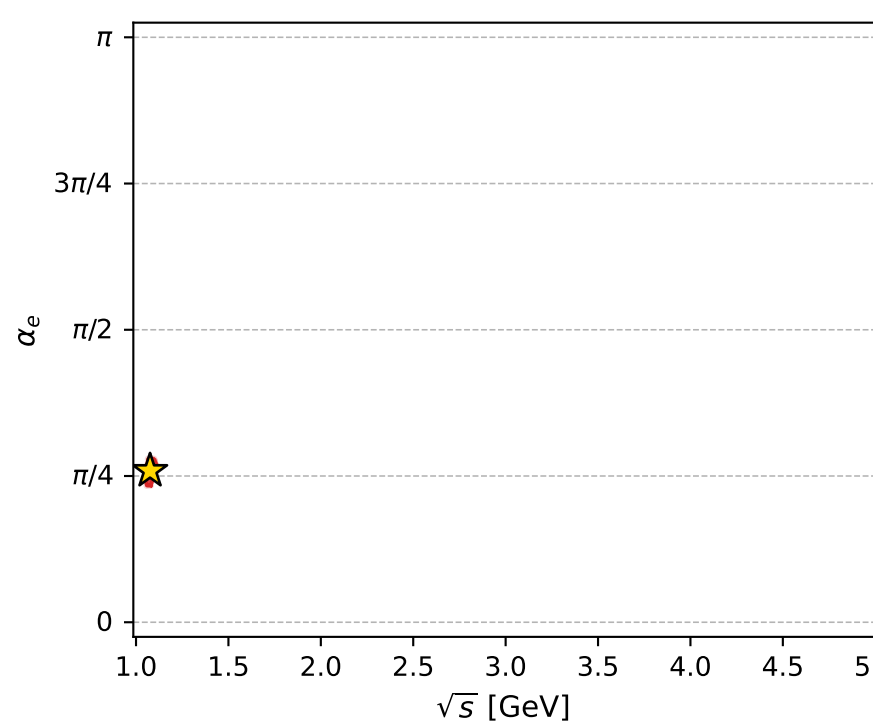
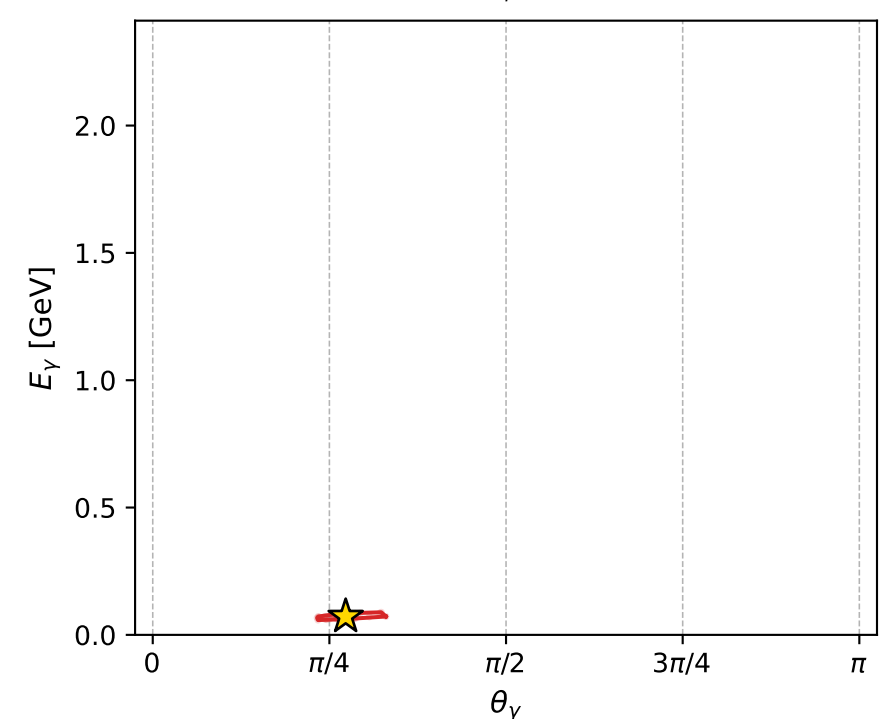
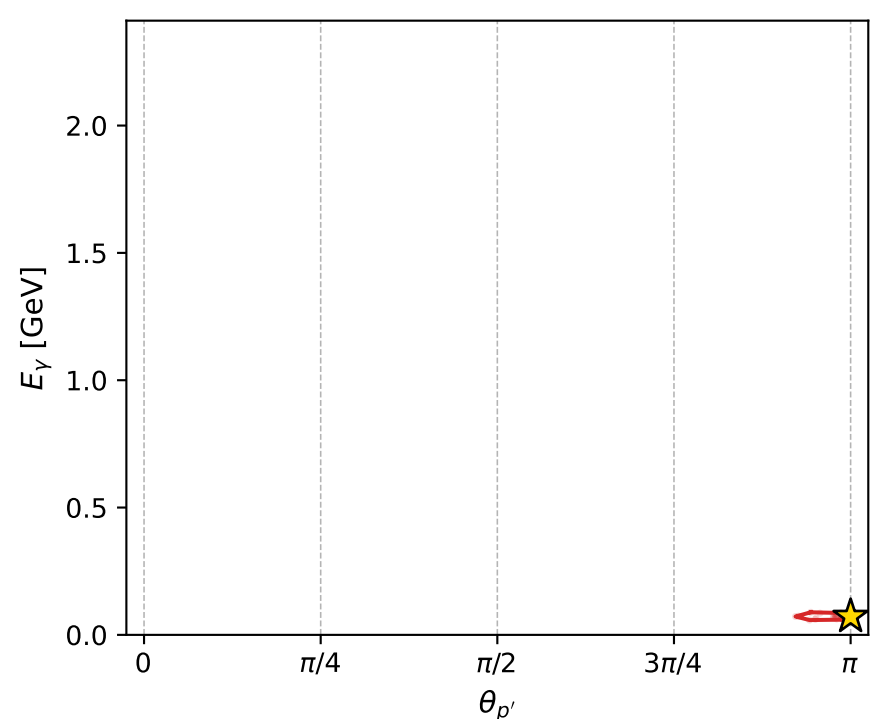
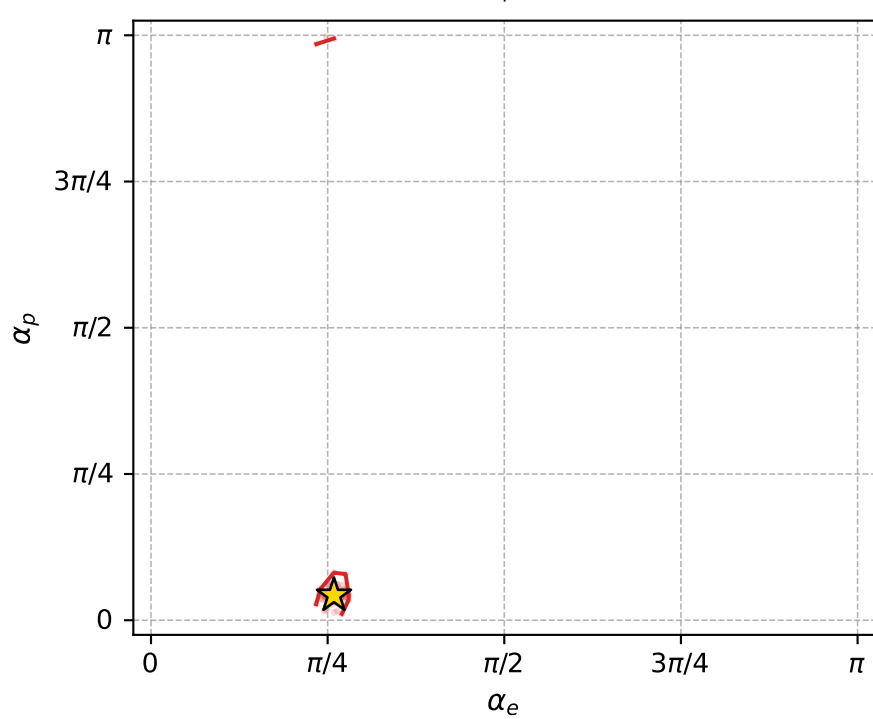
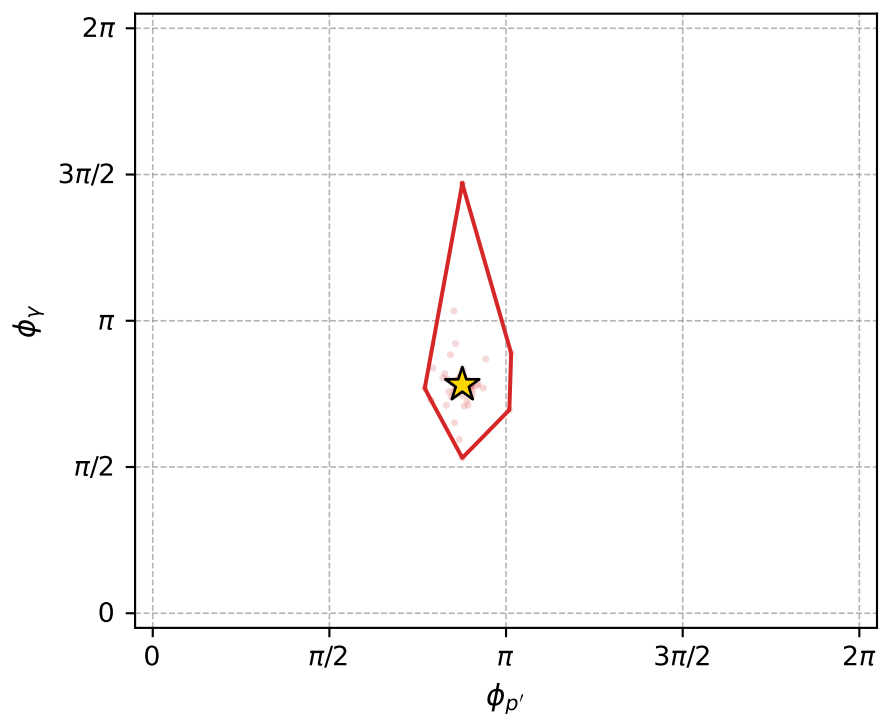
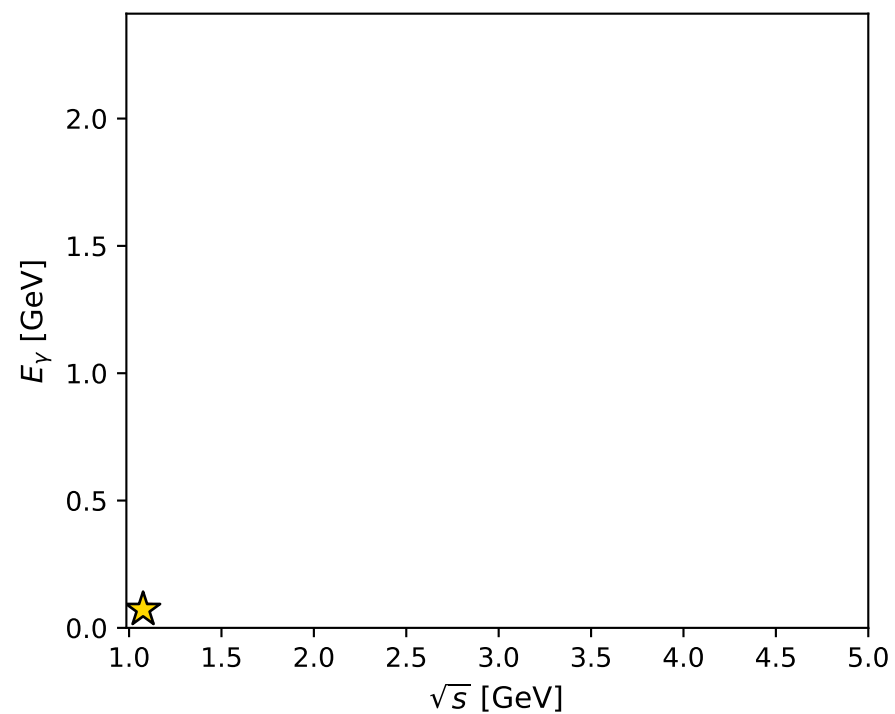
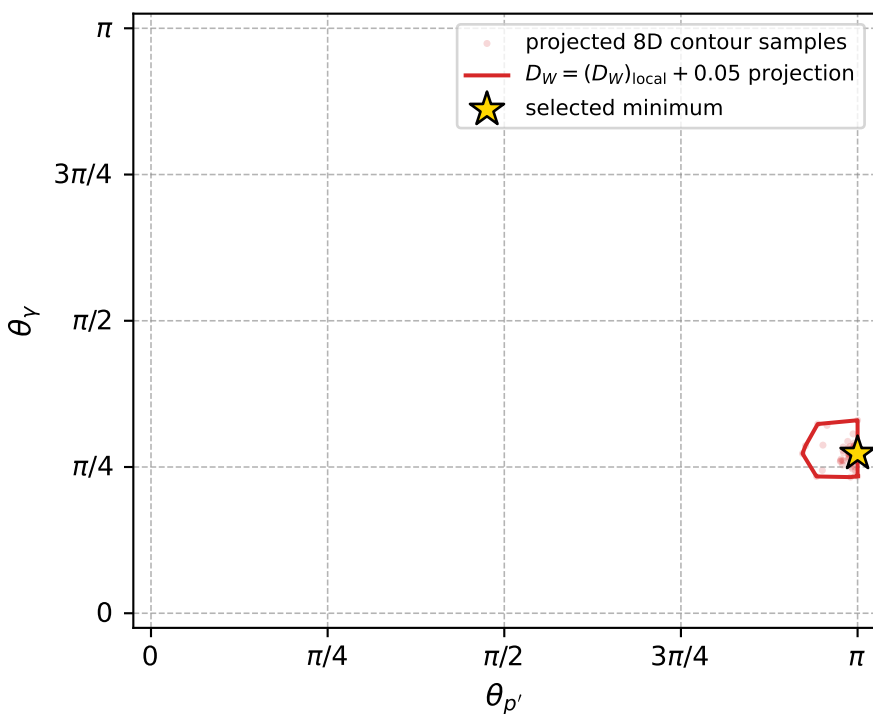
Transverse plane



Leading final-state amplitudes (N=4, retained=99.1%)



electron: polarization cluster P3, local minimum 996; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.018106538$
 $D_W \text{ contour} = 0.068106538$
 above global minimum = 0.018090105
 8D contour samples = 145

selected phase-space point:
 $\text{sqrt_s} = 1.075879$
 $\text{theta_p_out} = 3.141593$
 $\text{theta_gamma_out} = 0.8578075$
 $\text{qOut} = 0.07214084$
 $\text{phi_p_out} = 2.752631$
 $\text{phi_gamma_out} = 2.453868$
 $\text{alpha_e} = 0.8133873$
 $\text{alpha_p} = 0.1338466$

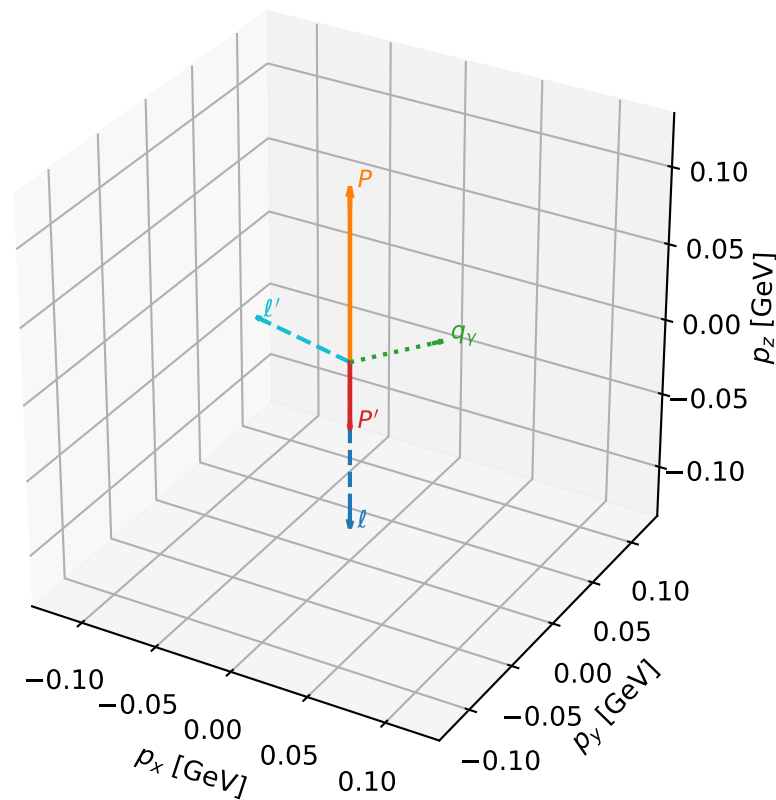
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1006 D_W=0.0187463
 region: polarization_cluster_03_local_minimum_1006
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.76617778$ rad
 incoming lepton: $0.720566|+\rangle + 0.693386|-\rangle$
 proton mixing angle: $\alpha_p=3.0194887$ rad
 incoming proton: $-0.992555|+\rangle + 0.121801|-\rangle$

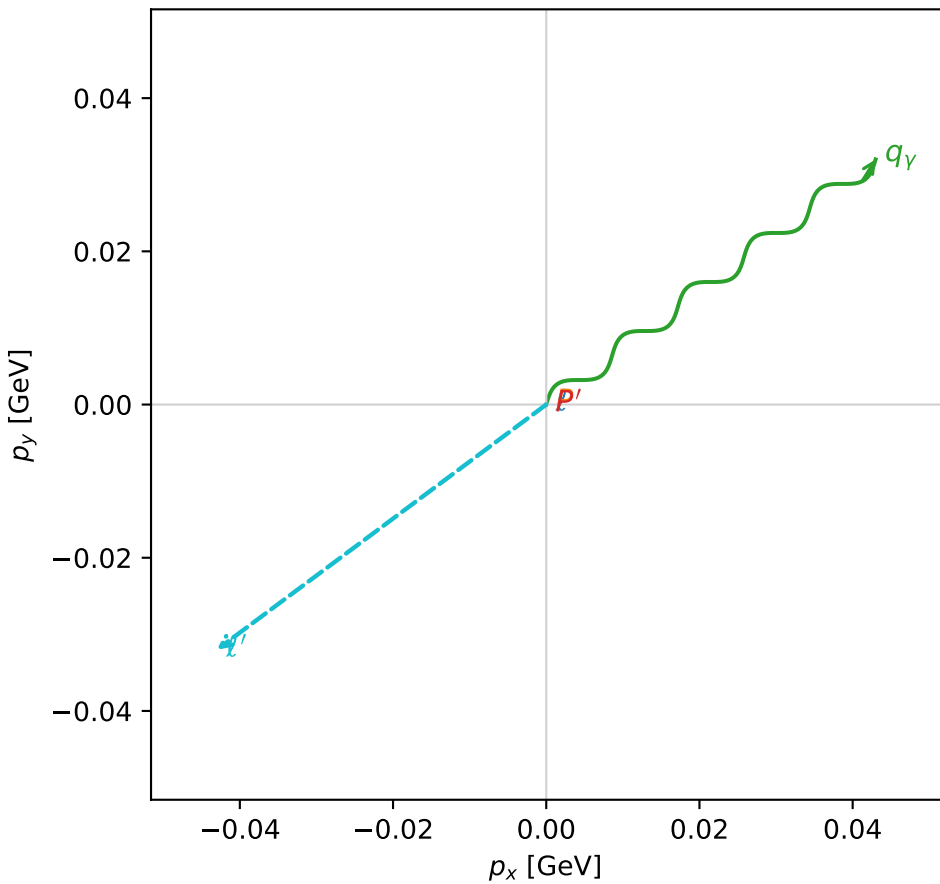
$s=1.11667$, $\sqrt{s}=1.05672$, $|\vec{P}|=0.112054$, $|\vec{P}'|=0.0437259$
 $\theta_{P'}=3.14159$, $\theta_V=1.37894$, $\phi_{P'}=0.202233$, $\phi_V=0.639847$
 $E_V=0.0545972$, $\theta_{\ell'}=1.01464$, $\phi_{\ell'}=3.78144$
 $Q^2=0.0216088$, $x_B=0.172795$, $t=-0.0242353$, $W^2=0.98329$, $y=0.528056$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1121$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1121$
 P $E= 0.9447$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1121$
 ℓ' $E= 0.0631$ $p_x= -0.0430$ $p_y= -0.0320$ $p_z= 0.0333$
 P' $E= 0.9390$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0437$
 q_Y $E= 0.0546$ $p_x= 0.0430$ $p_y= 0.0320$ $p_z= 0.0104$

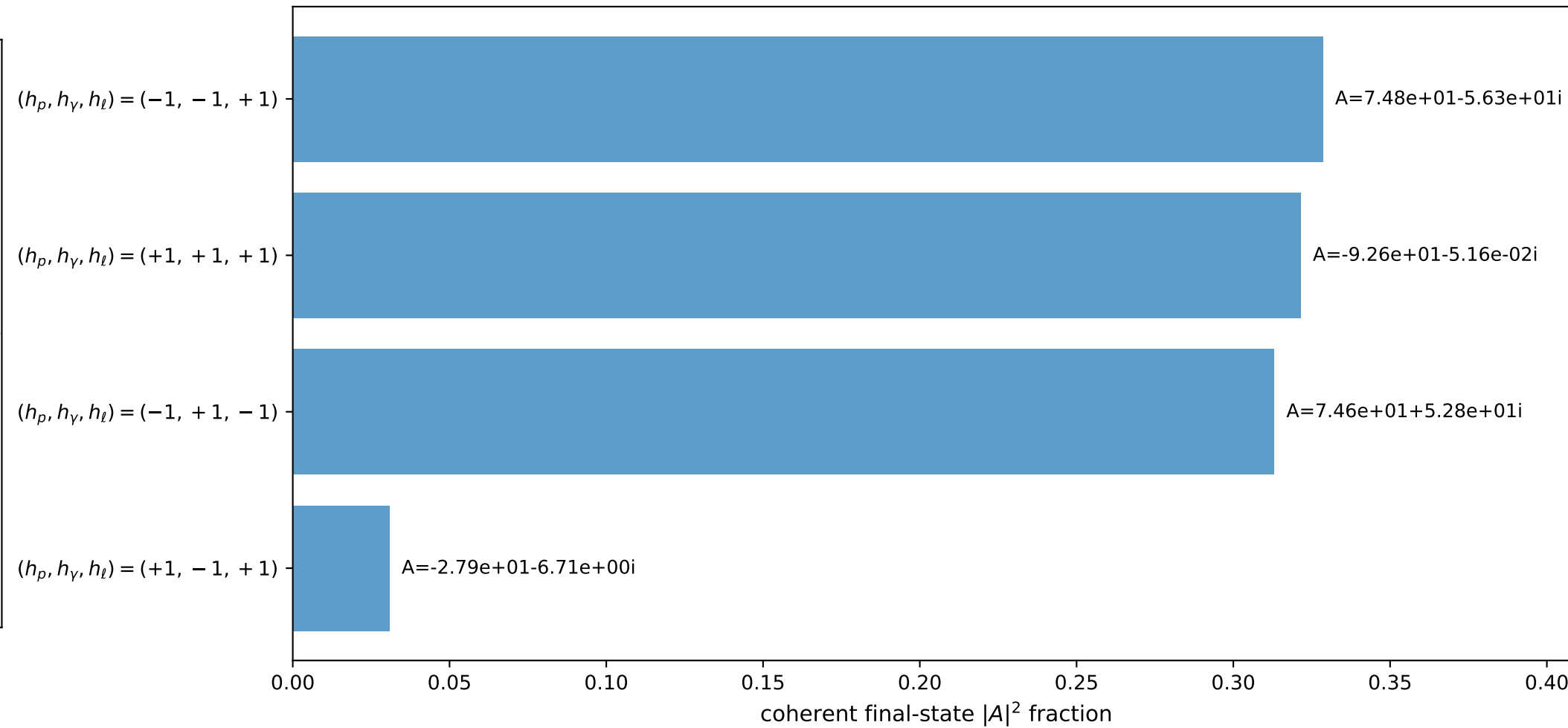
3D momenta



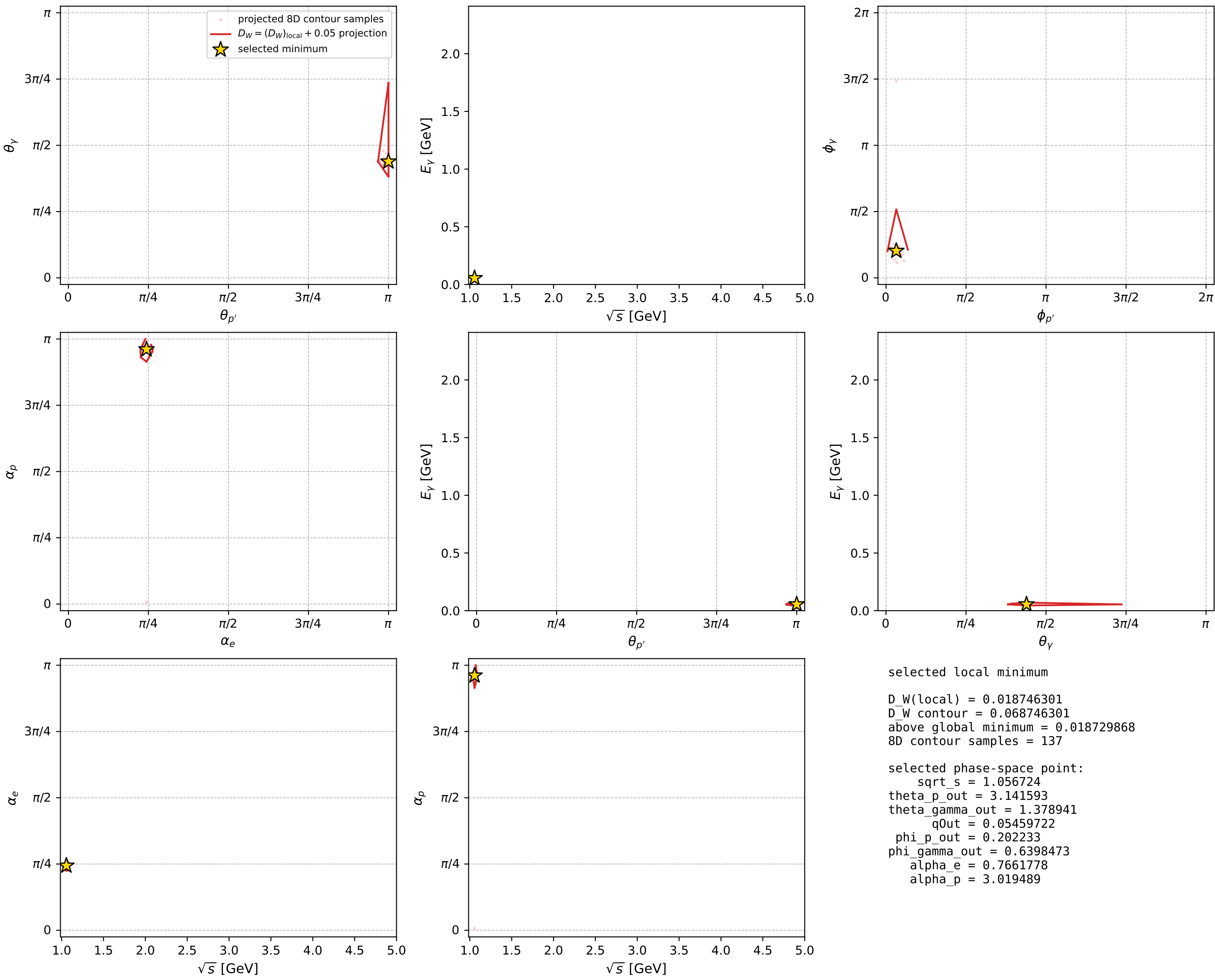
Transverse plane



Leading final-state amplitudes (N=4, retained=99.4%)



electron: polarization cluster P3, local minimum 1006; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



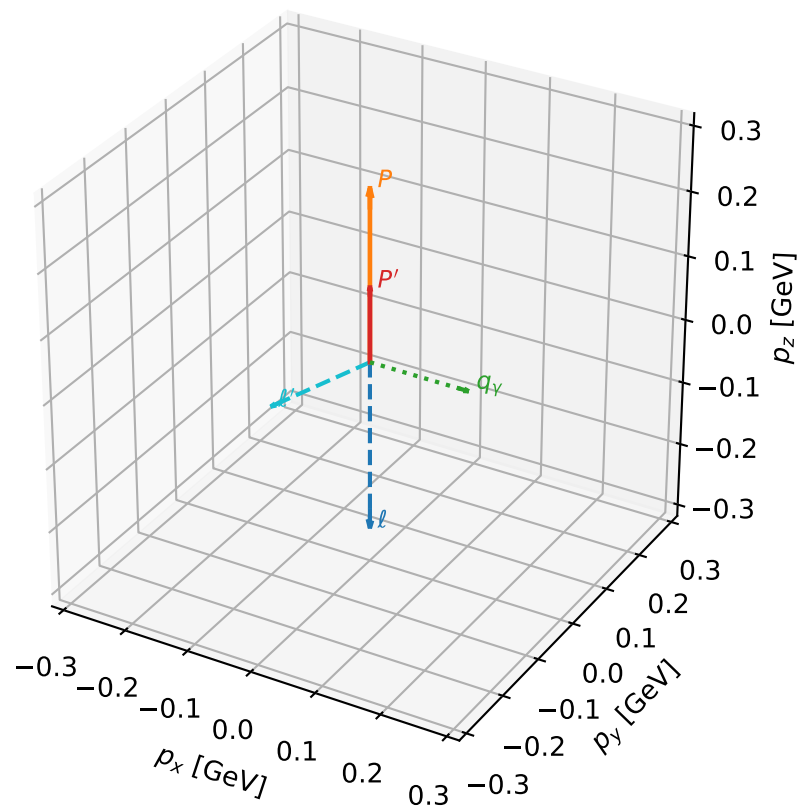
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1011 D_W=0.0189042
 region: polarization_cluster_03_local_minimum_1011
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81557799$ rad
 incoming lepton: $0.685448|+\rangle + 0.728122|-\rangle$
 proton mixing angle: $\alpha_p=3.0134082$ rad
 incoming proton: $-0.991796|+\rangle + 0.127834|-\rangle$

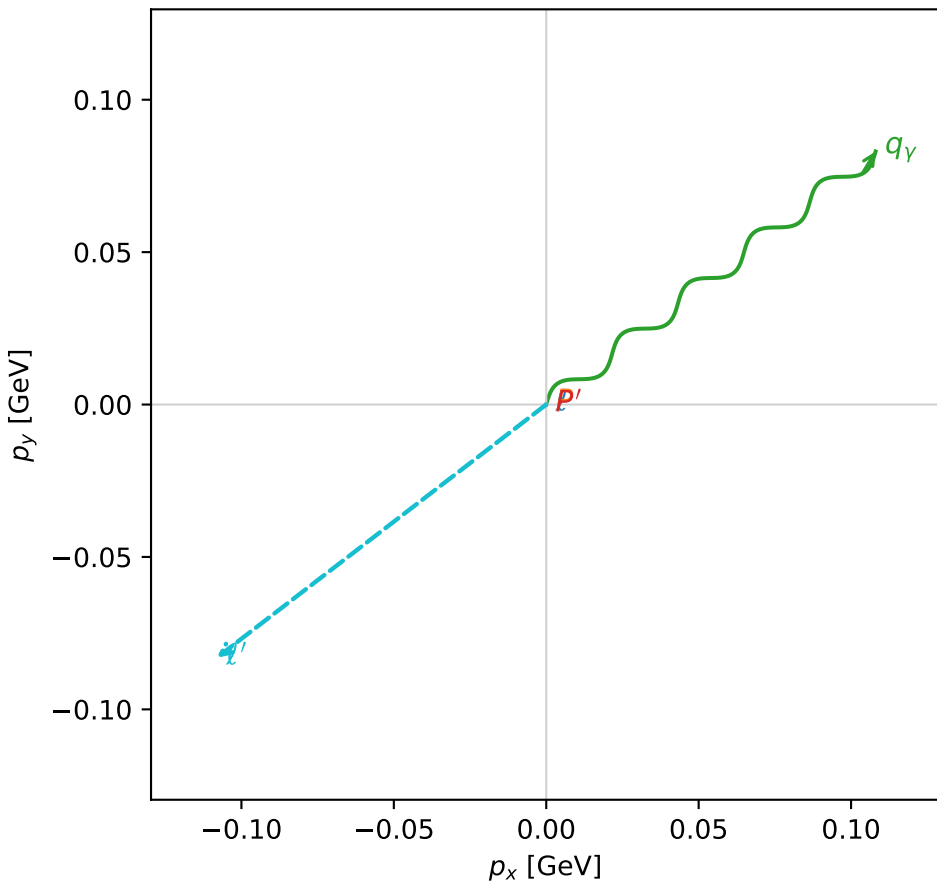
$s=1.54057$, $\sqrt{s}=1.2412$, $|\vec{P}|=0.266165$, $|\vec{P}'|=0.115719$
 $\theta_{P'}=0$, $\theta_V=1.96217$, $\phi_{P'}=2.30016$, $\phi_V=0.655199$
 $E_V=0.147406$, $\theta_{\ell'}=1.98244$, $\phi_{\ell'}=3.79679$
 $Q^2=0.0474789$, $x_B=0.140005$, $t=-0.0217387$, $W^2=1.17149$, $y=0.513257$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2662$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2662$
 P $E= 0.9750$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2662$
 ℓ' $E= 0.1487$ $p_x= -0.1080$ $p_y= -0.0830$ $p_z= -0.0595$
 P' $E= 0.9451$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.1157$
 q_V $E= 0.1474$ $p_x= 0.1080$ $p_y= 0.0830$ $p_z= -0.0562$

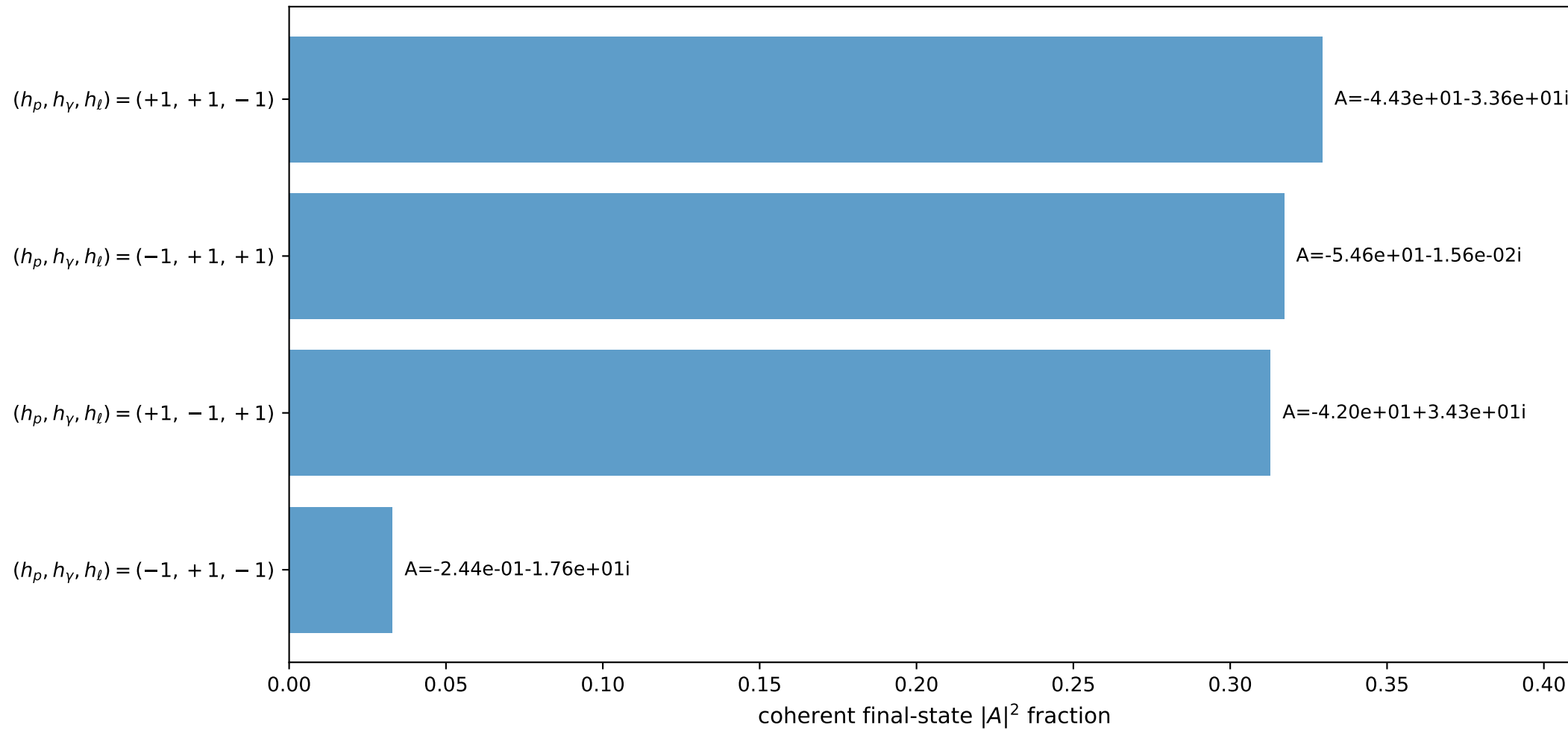
3D momenta



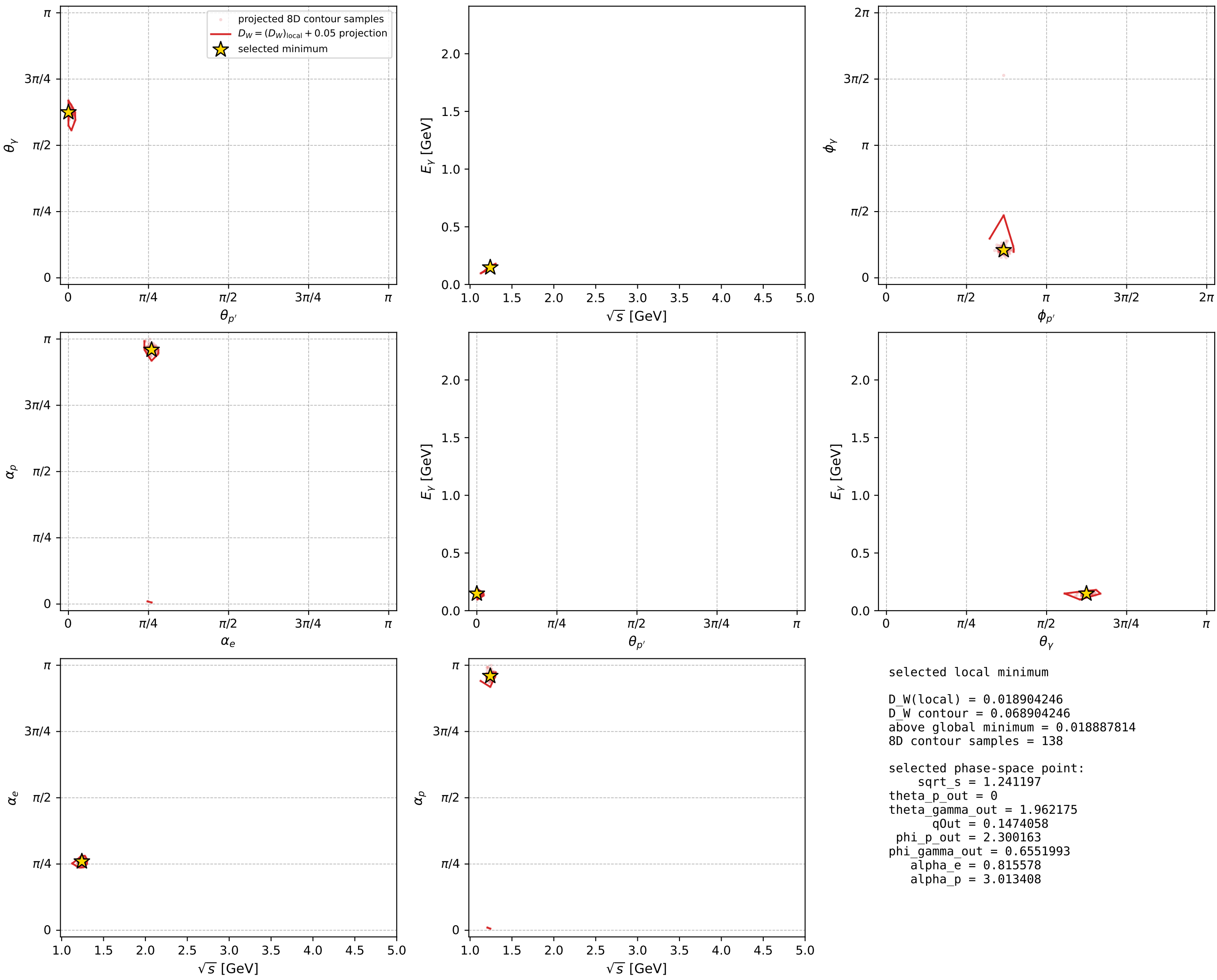
Transverse plane



Leading final-state amplitudes (N=4, retained=99.2%)



electron: polarization cluster P3, local minimum 1011; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

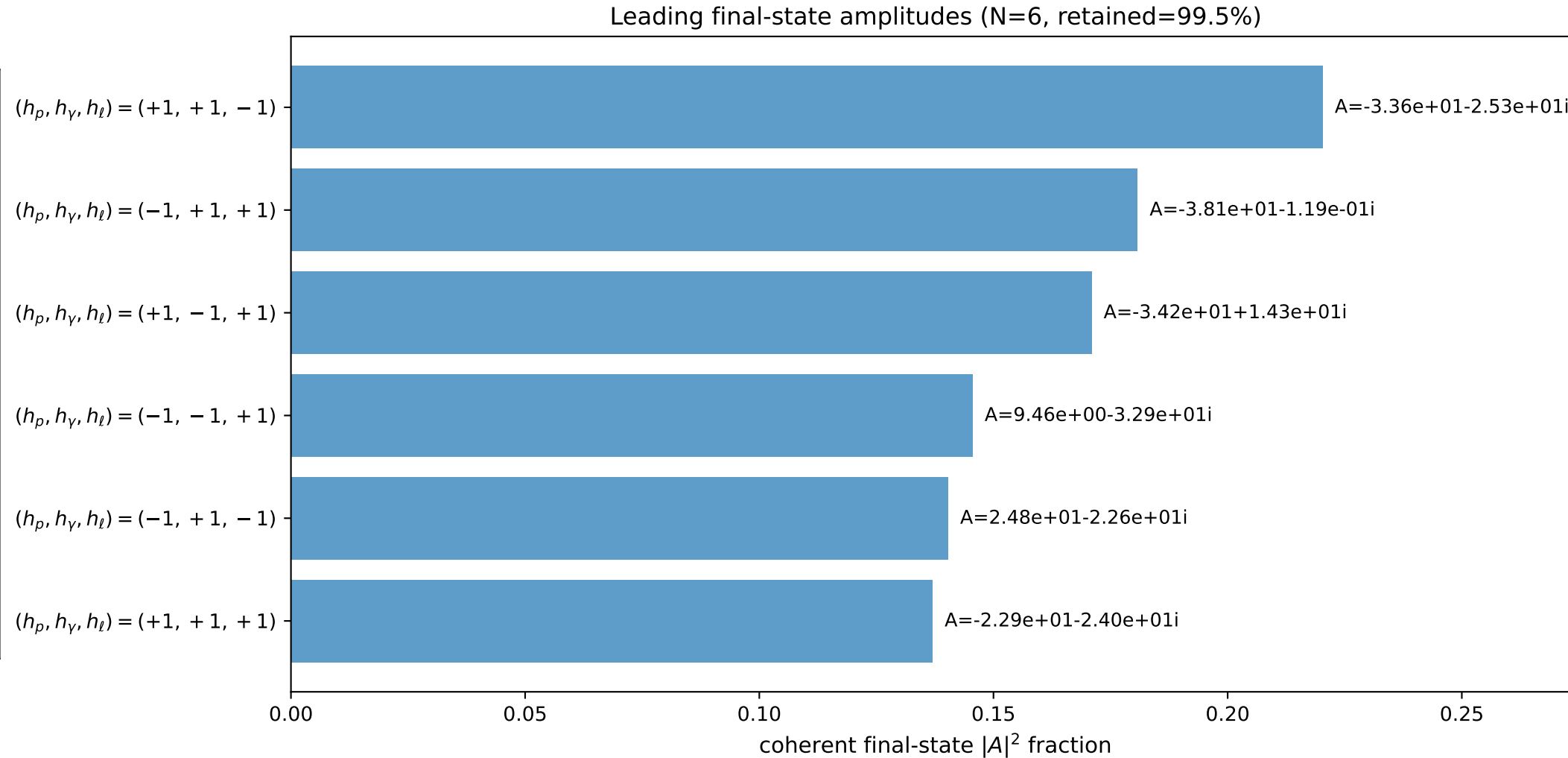
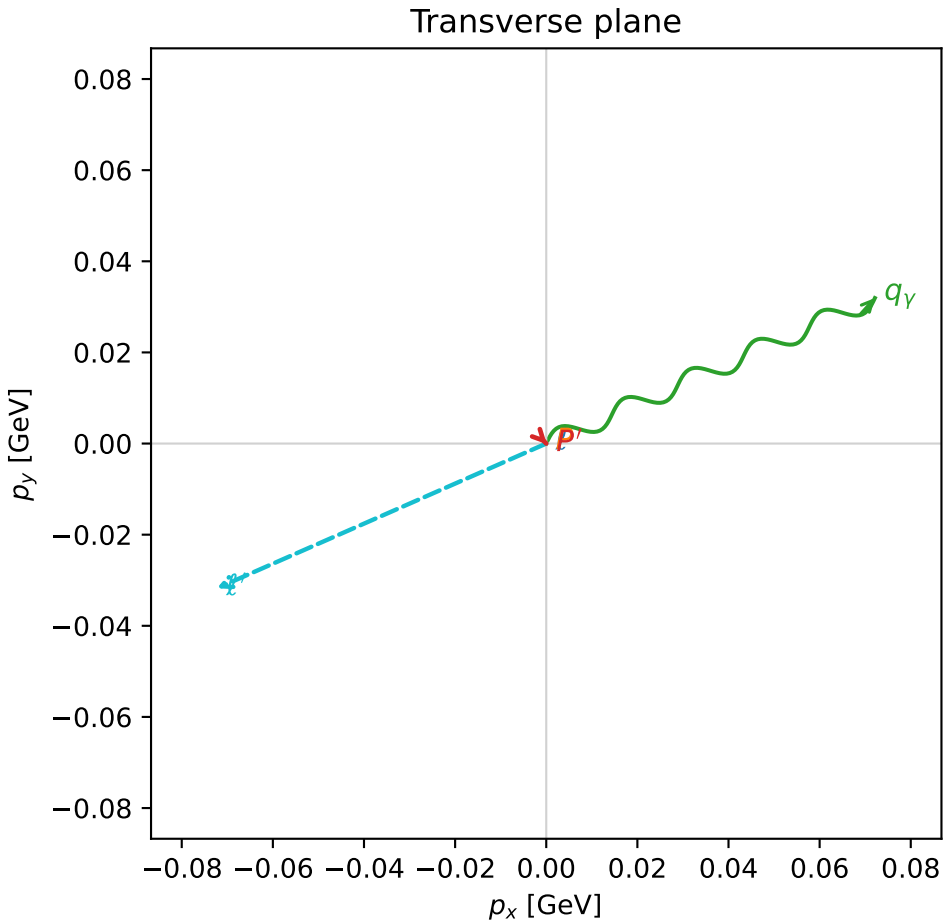
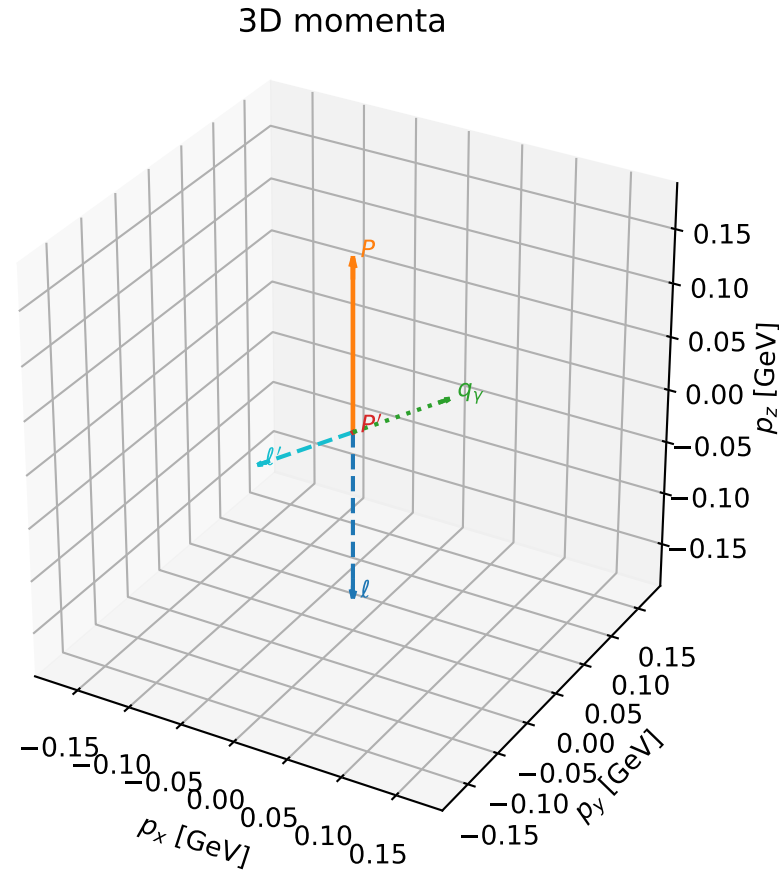


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

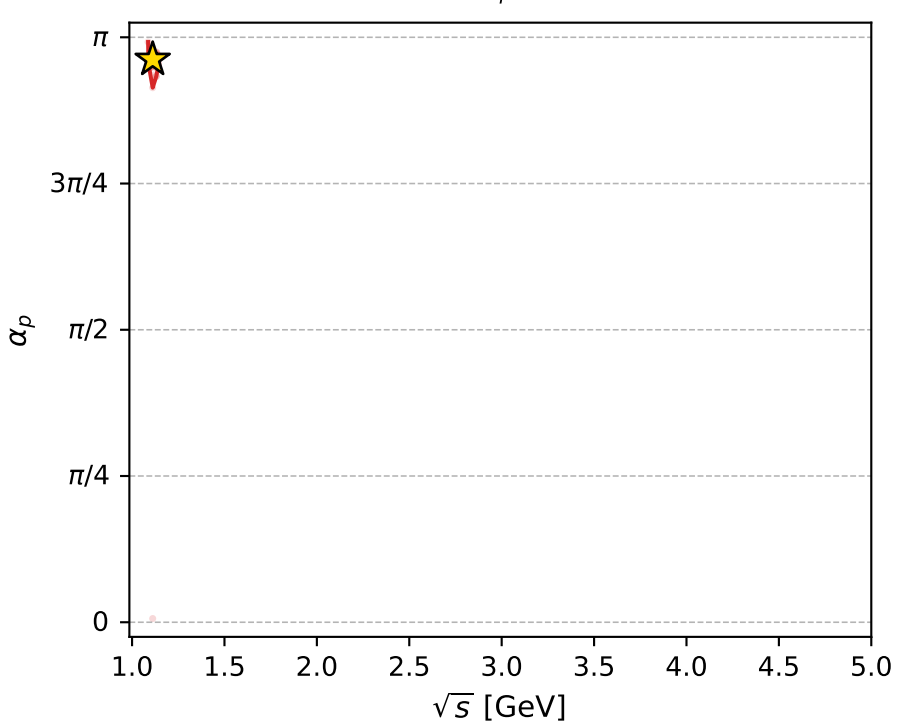
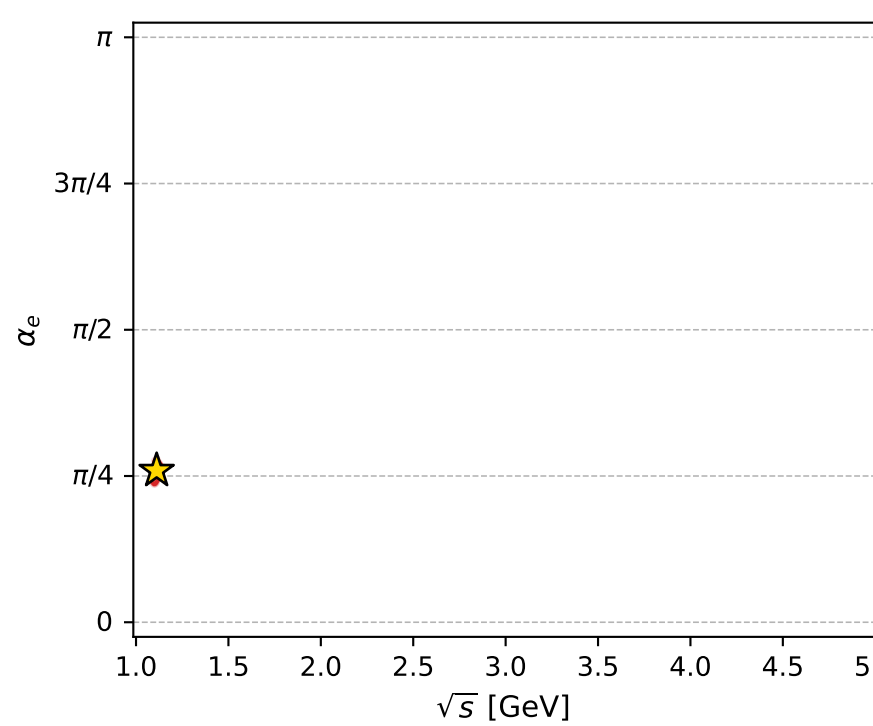
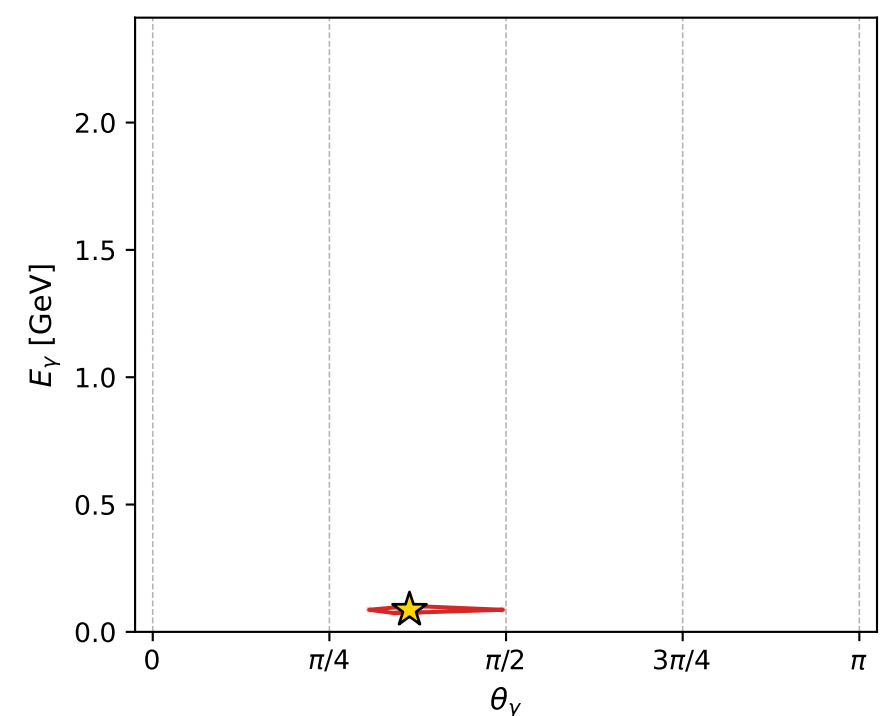
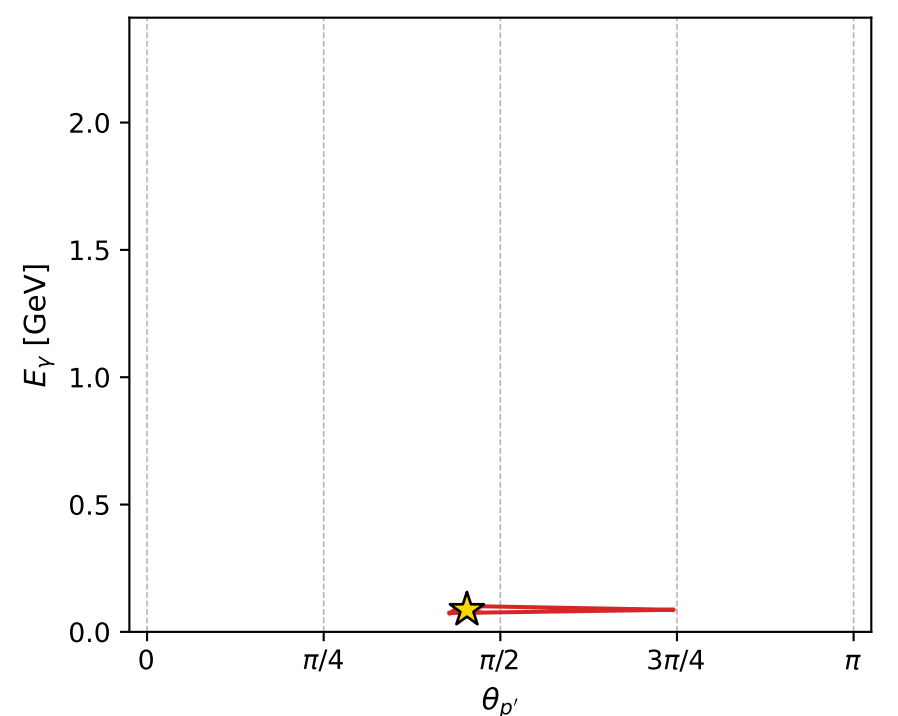
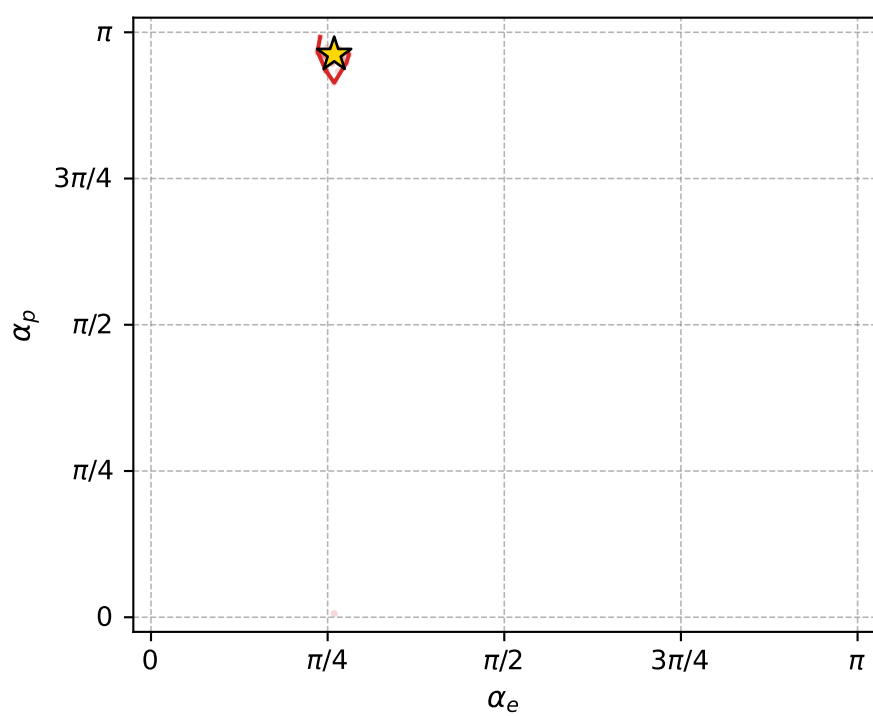
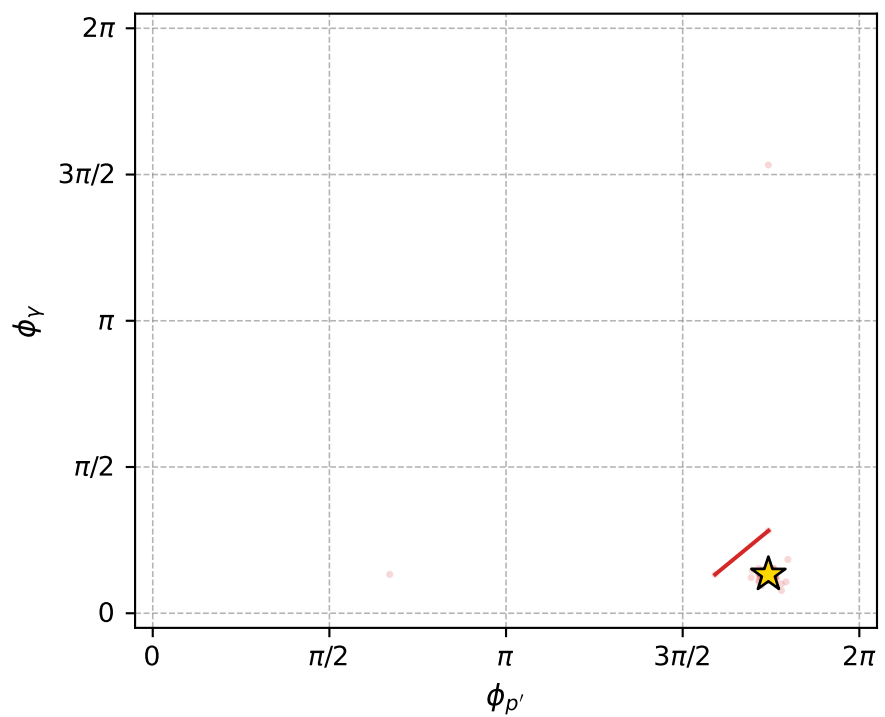
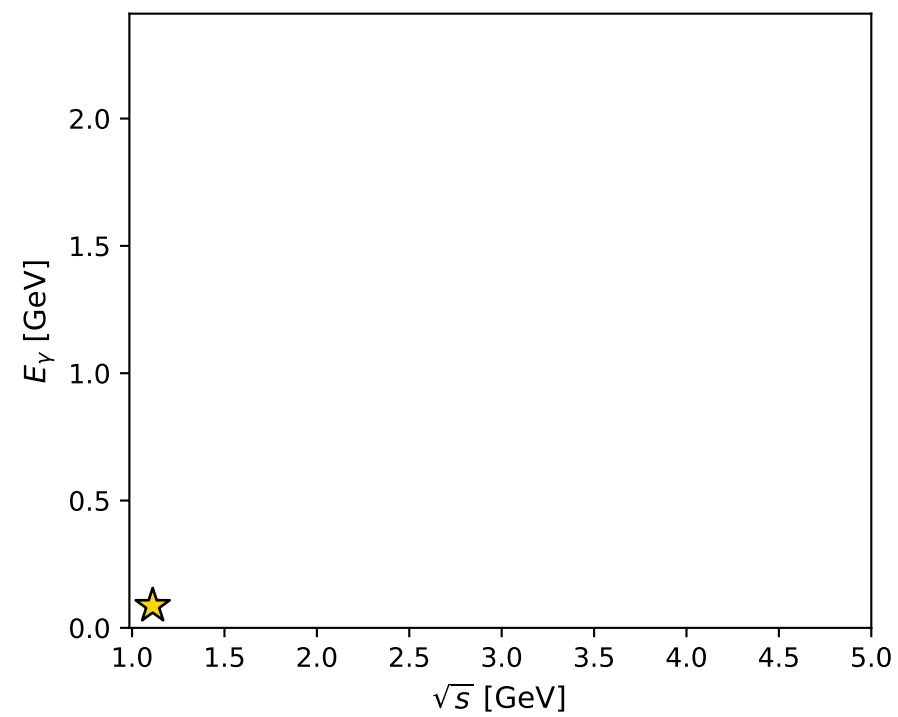
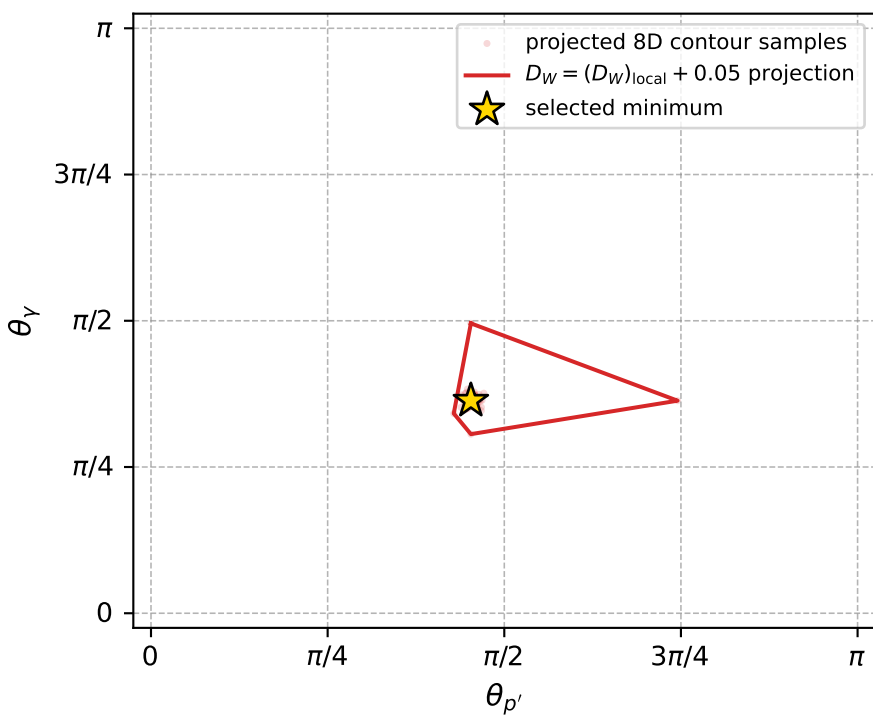
dw_mixing_angles_polarization_03_local_minimum_1016 D_W=0.0192511
region: polarization_cluster_03_local_minimum_1016
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81452042$ rad
incoming lepton: $0.686217|+\rangle + 0.727397|-\rangle$
proton mixing angle: $\alpha_p=3.0225306$ rad
incoming proton: $-0.99292|+\rangle + 0.118781|-\rangle$

$s=1.23561$, $\sqrt{s}=1.11158$, $|\vec{P}|=0.160028$, $|\vec{P}'|=0.000224796$
 $\theta_{P'}=1.42227$, $\theta_V=1.14157$, $\phi_{P'}=5.47454$, $\phi_V=0.416766$
 $E_V=0.0867486$, $\theta_l=2.00001$, $\phi_{l'}=3.55571$
 $Q^2=0.0162256$, $x_B=0.0906701$, $t=-0.0254145$, $W^2=1.04257$, $y=0.503$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1600$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1600$
 P $E= 0.9516$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1600$
 l' $E= 0.0868$ $p_x= -0.0723$ $p_y= -0.0318$ $p_z= -0.0361$
 P' $E= 0.9380$ $p_x= 0.0002$ $p_y= -0.0002$ $p_z= 0.0000$
 q_V $E= 0.0867$ $p_x= 0.0721$ $p_y= 0.0319$ $p_z= 0.0361$



electron: polarization cluster P3, local minimum 1016; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.019251139$
 $D_W \text{ contour} = 0.069251139$
 above global minimum = 0.019234706
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.111581$
 $\text{theta_p_out} = 1.422273$
 $\text{theta_gamma_out} = 1.141569$
 $\text{qOut} = 0.08674855$
 $\text{phi_p_out} = 5.474535$
 $\text{phi_gamma_out} = 0.4167662$
 $\text{alpha_e} = 0.8145204$
 $\text{alpha_p} = 3.022531$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

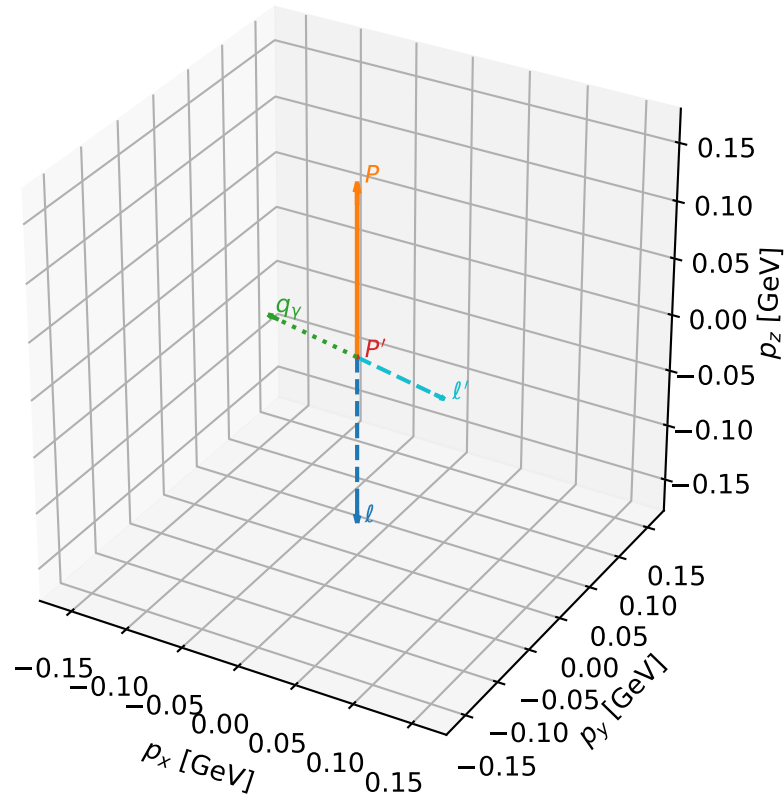
dw_mixing_angles_polarization_03_local_minimum_1029 D_W=0.0202389
 region: polarization_cluster_03_local_minimum_1029
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81675721$ rad
 incoming lepton: $0.684589|+\rangle +0.72893|-\rangle$
 proton mixing angle: $\alpha_p=0.13851662$ rad
 incoming proton: $0.990422|+\rangle +0.138074|-\rangle$

$s=1.20842$, $\sqrt{s}=1.09928$, $|\vec{P}|=0.149451$, $|\vec{P}'|=4.16257e-07$
 $\theta_{P'}=1.79203$, $\theta_Y=1.00733$, $\phi_{P'}=3.48142$, $\phi_Y=3.81294$
 $E_Y=0.0806405$, $\theta_{\ell'}=2.13426$, $\phi_{\ell'}=0.67134$
 $Q^2=0.0112295$, $x_B=0.0691$, $t=-0.0221955$, $W^2=1.03113$, $y=0.494588$

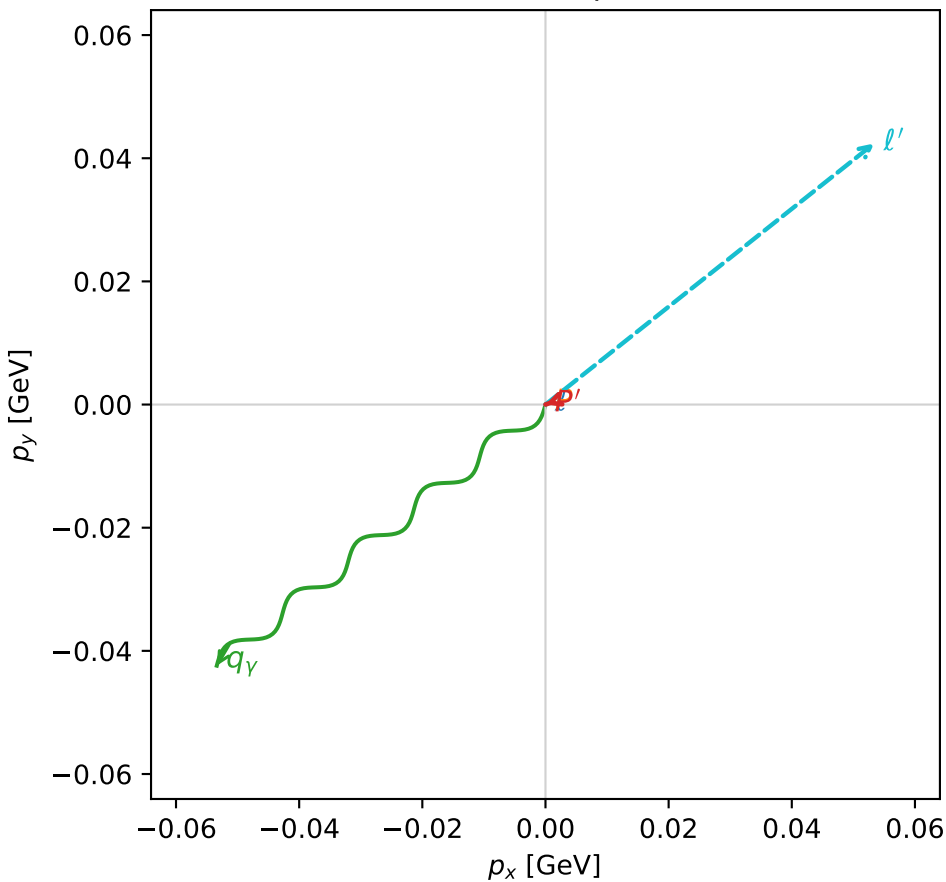
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 0.1495$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1495$
 P $E= 0.9498$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1495$
 ℓ' $E= 0.0806$ $p_x= 0.0534$ $p_y= 0.0424$ $p_z= -0.0431$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.0806$ $p_x= -0.0534$ $p_y= -0.0424$ $p_z= 0.0431$

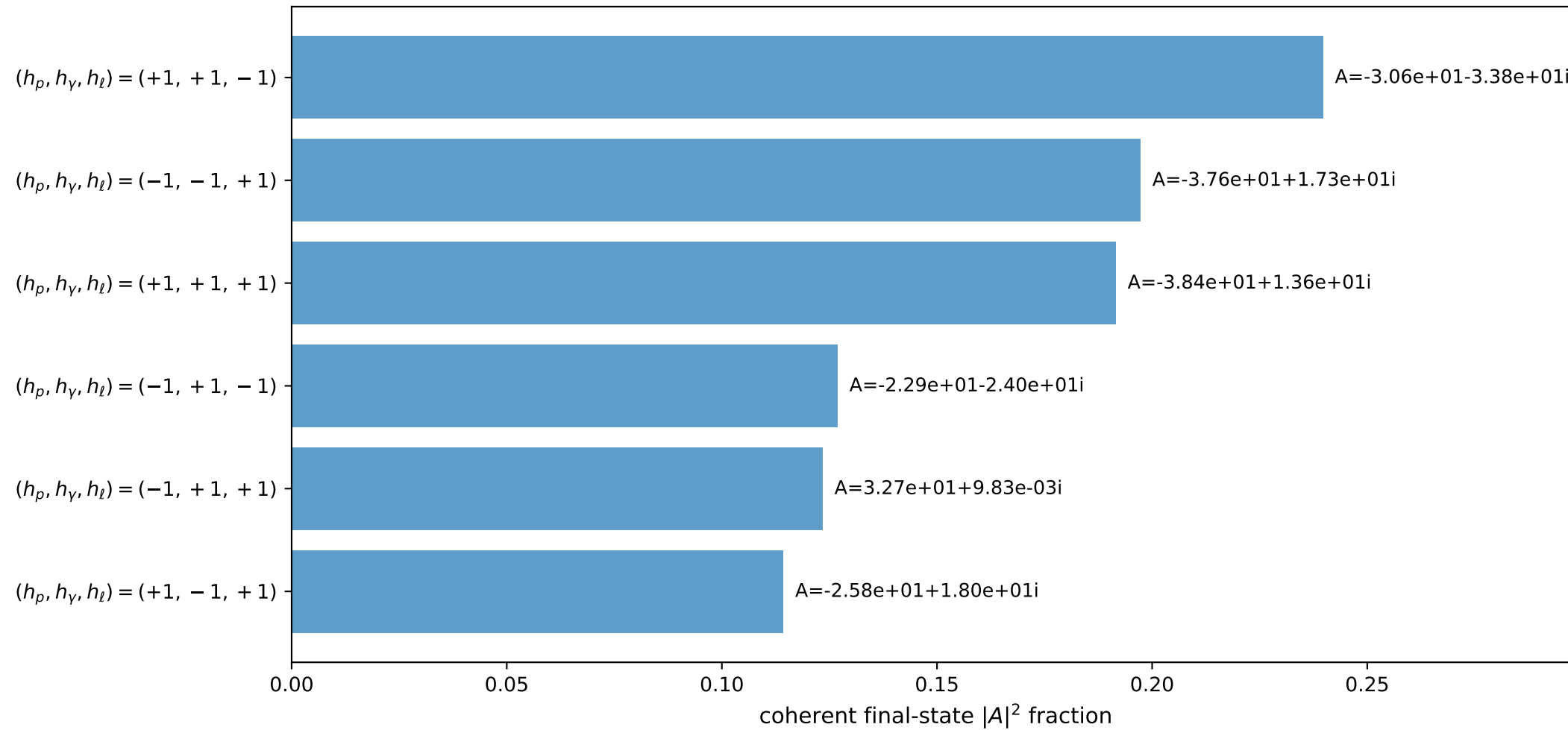
3D momenta



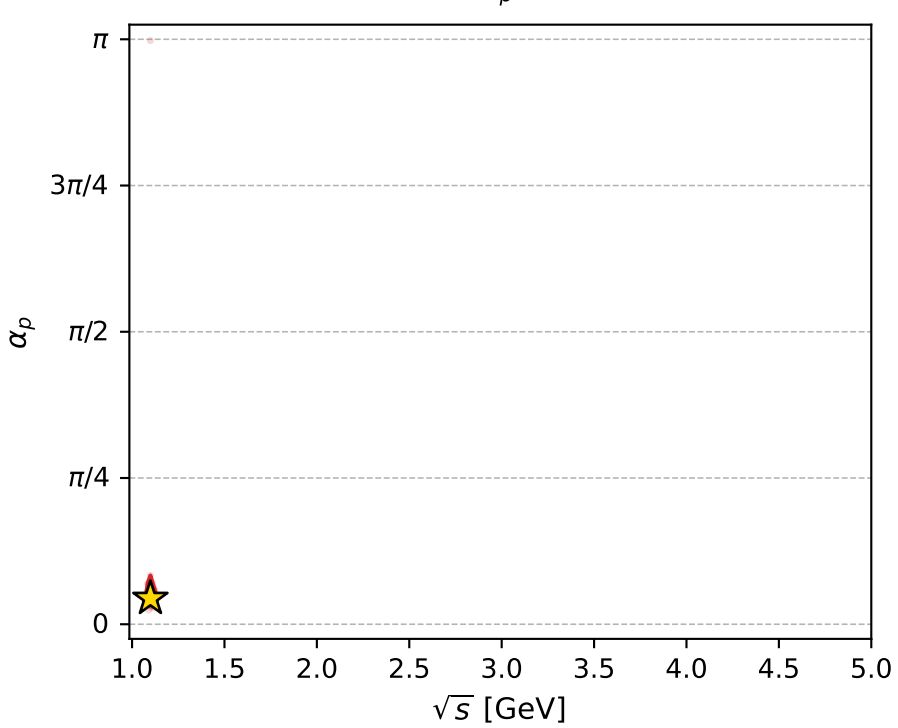
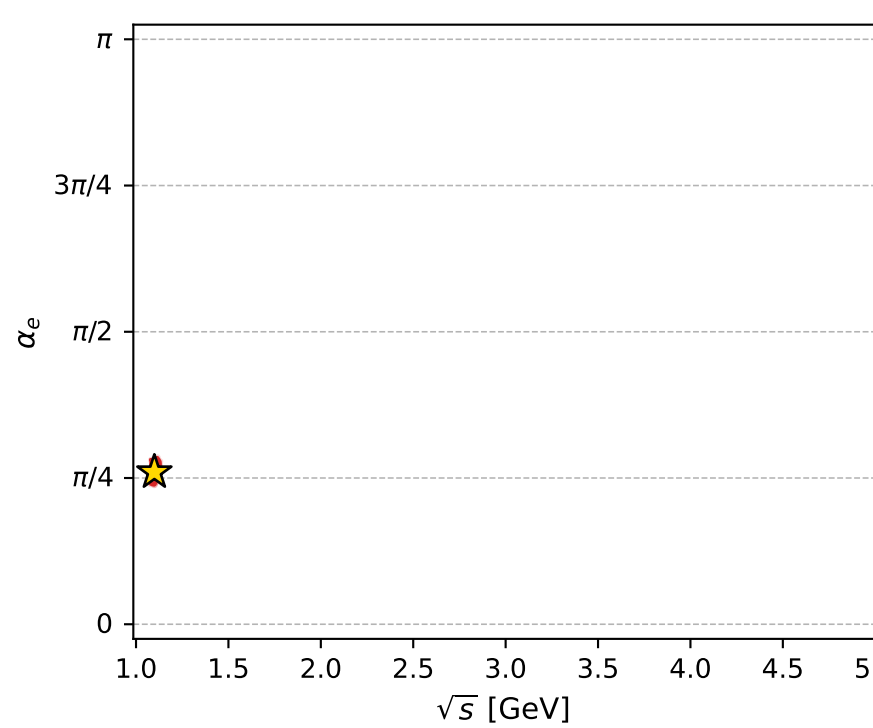
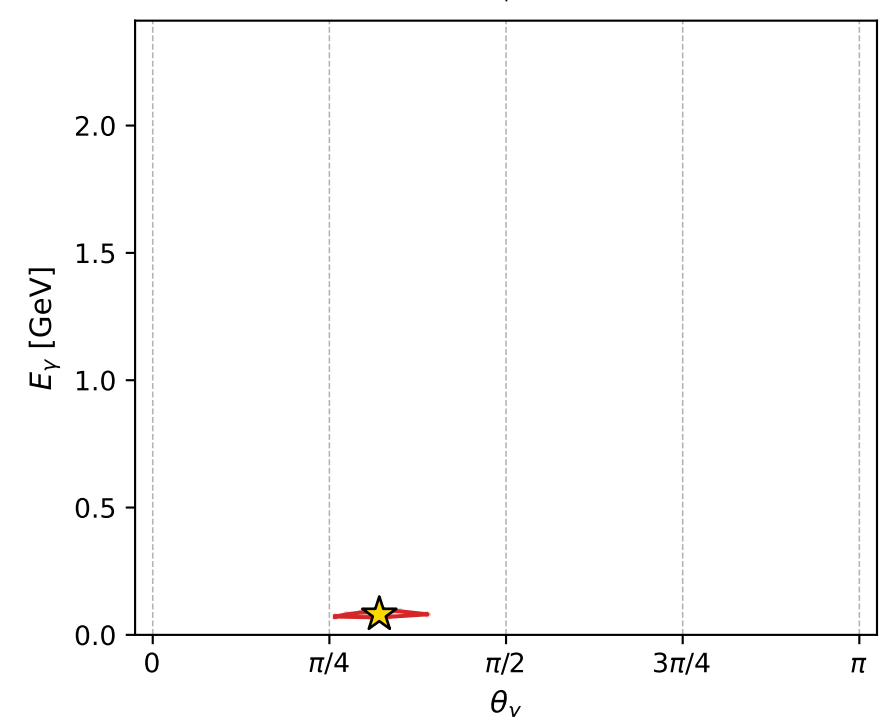
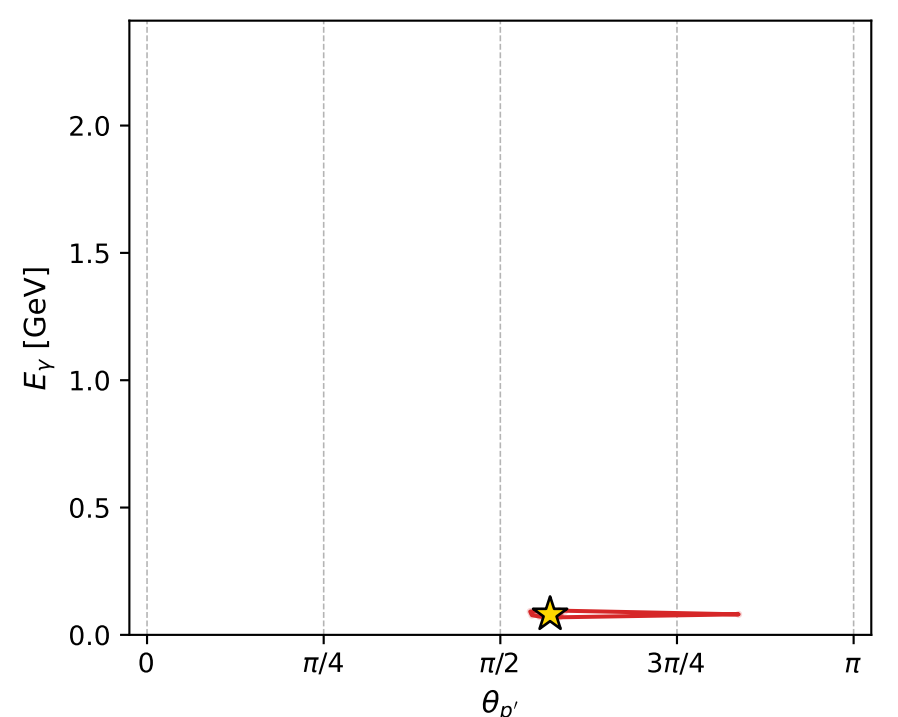
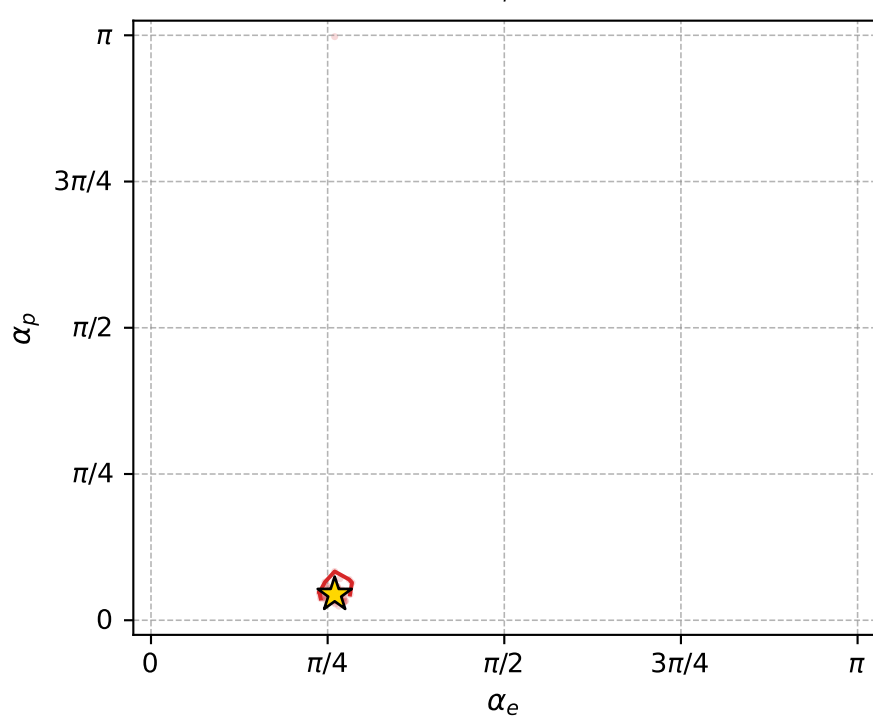
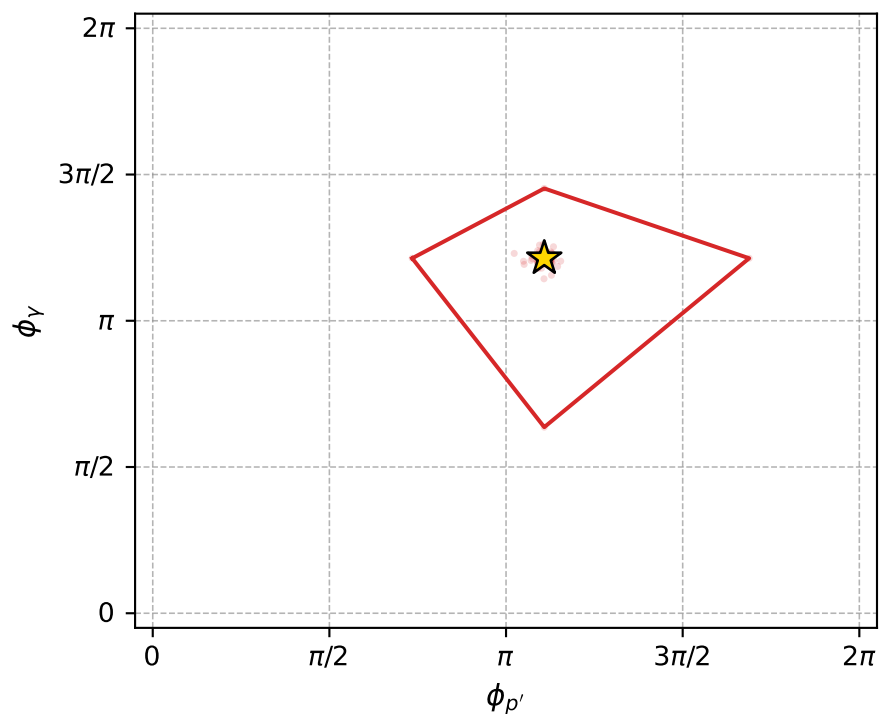
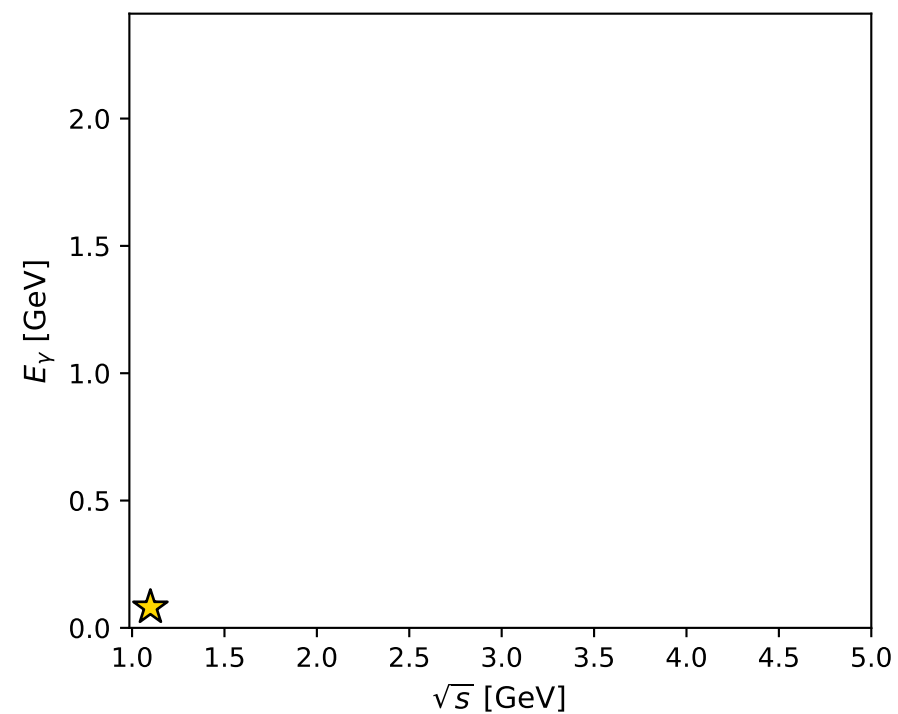
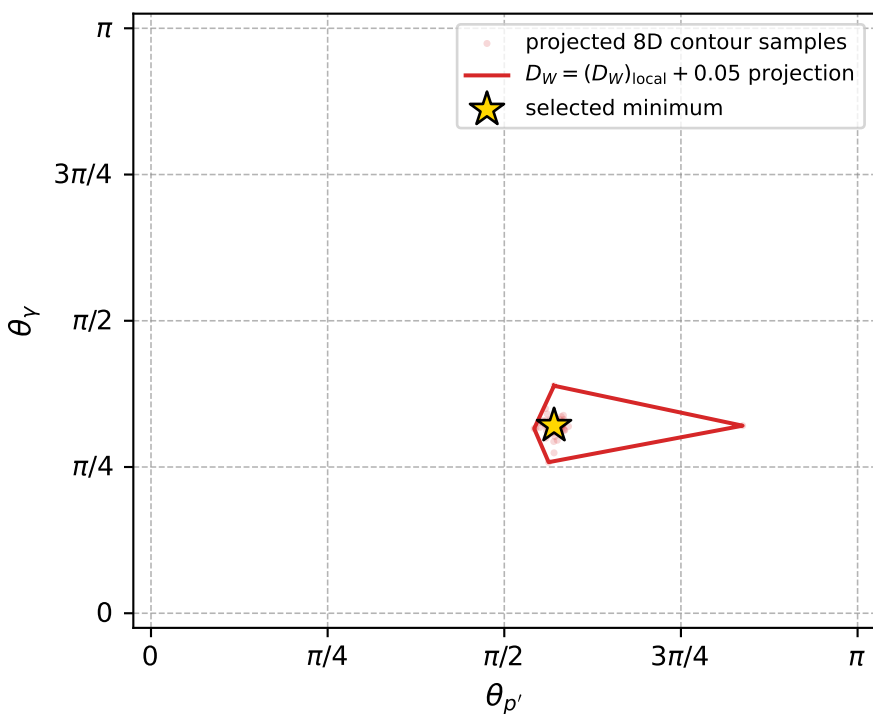
Transverse plane



Leading final-state amplitudes (N=6, retained=99.3%)



electron: polarization cluster P3, local minimum 1029; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.020238944$
 $D_W \text{ contour} = 0.070238944$
 above global minimum = 0.020222511
 8D contour samples = 255

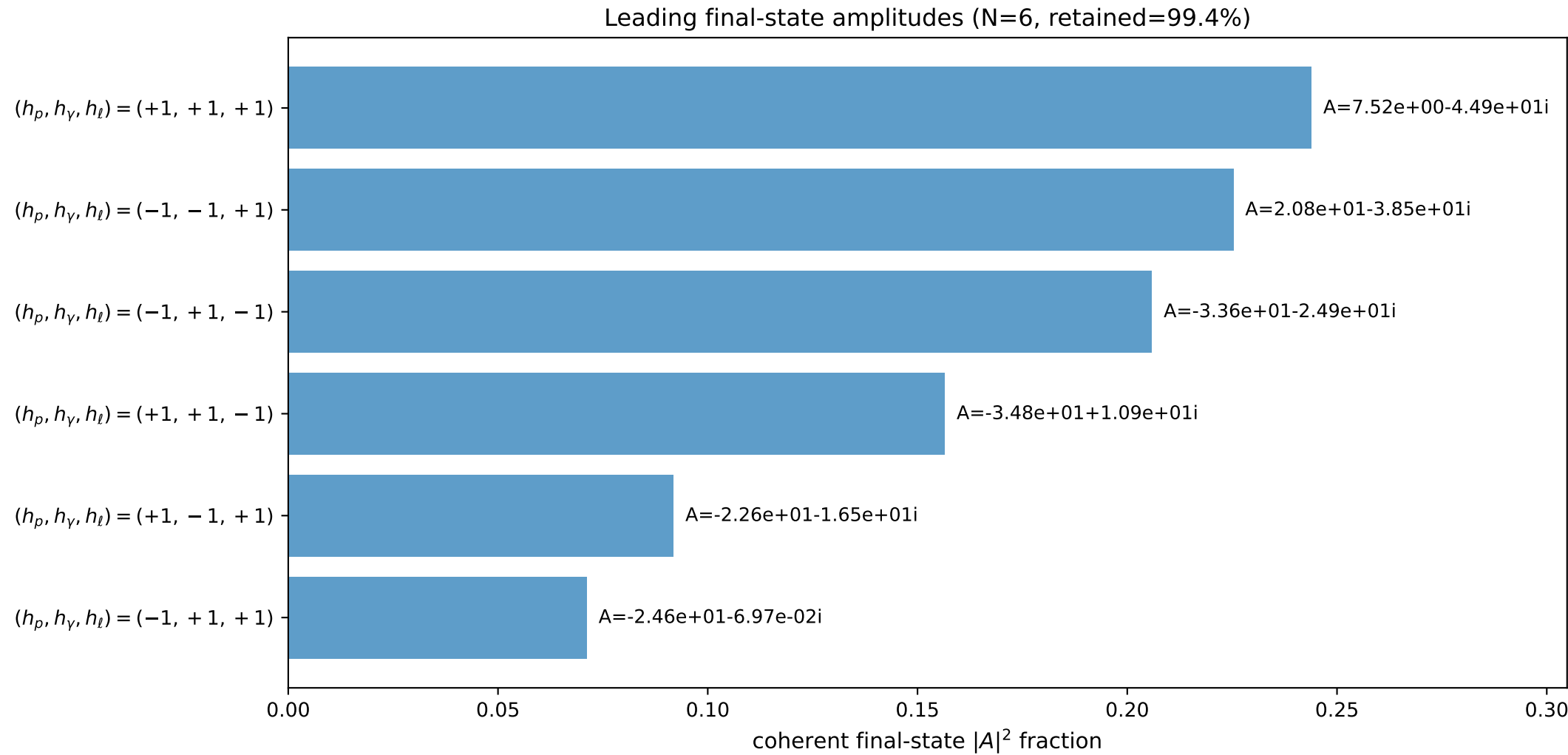
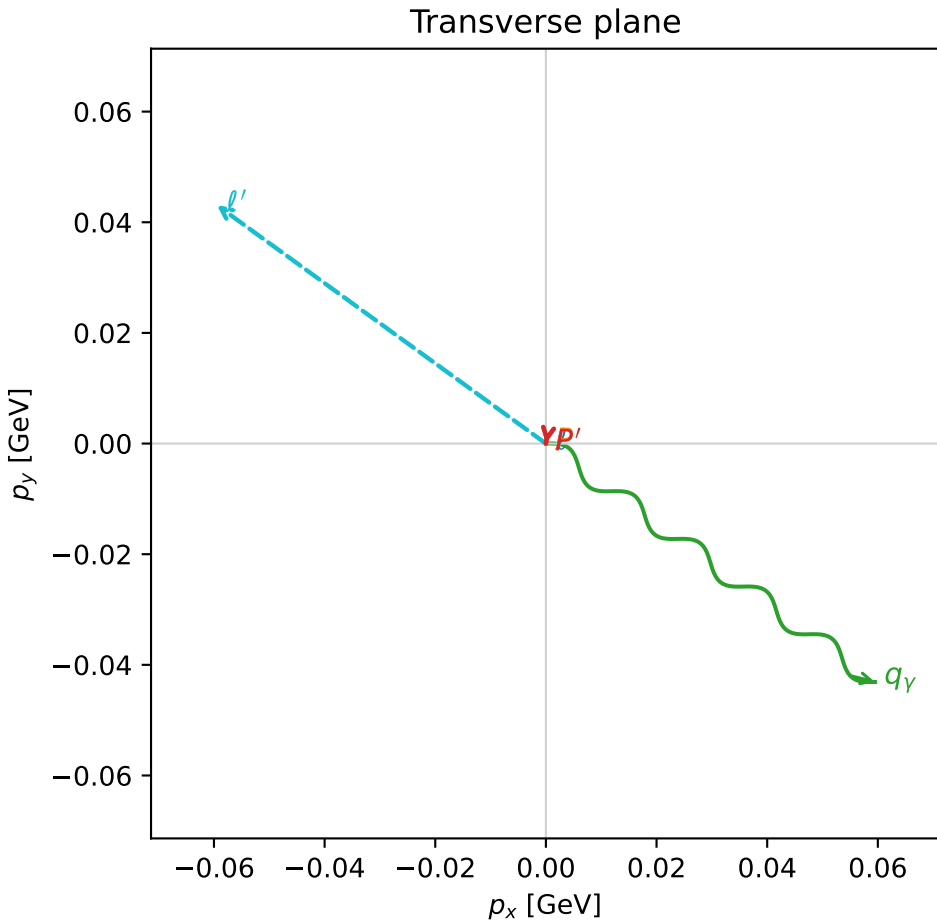
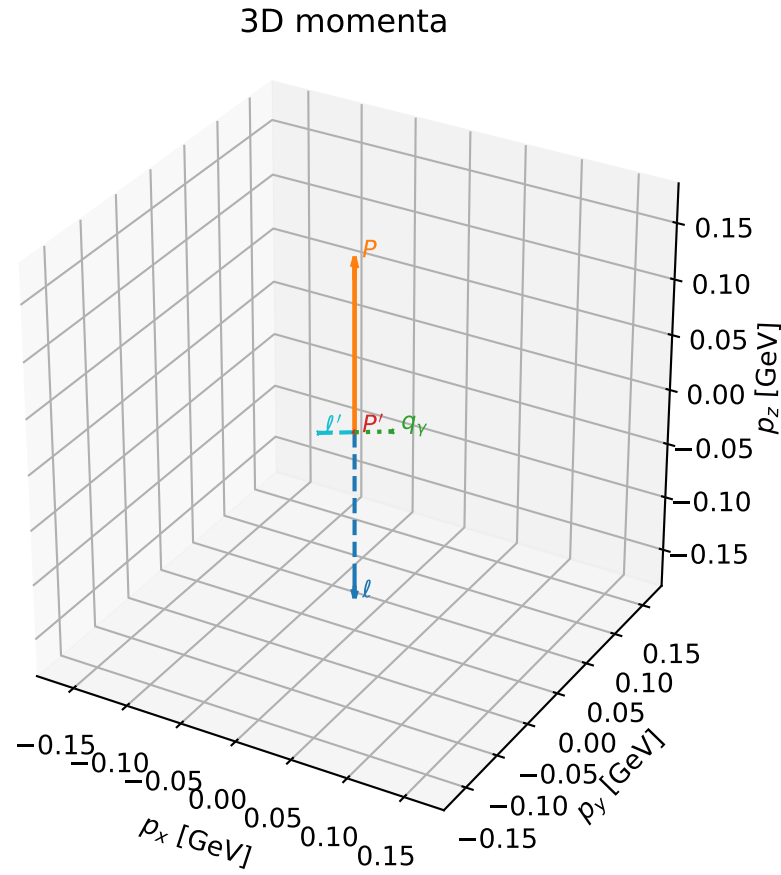
selected phase-space point:
 $\text{sqrt_s} = 1.099283$
 $\text{theta_p_out} = 1.792027$
 $\text{theta_gamma_out} = 1.007327$
 $\text{qOut} = 0.08064047$
 $\text{phi_p_out} = 3.48142$
 $\text{phi_gamma_out} = 3.812935$
 $\text{alpha_e} = 0.8167572$
 $\text{alpha_p} = 0.1385166$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

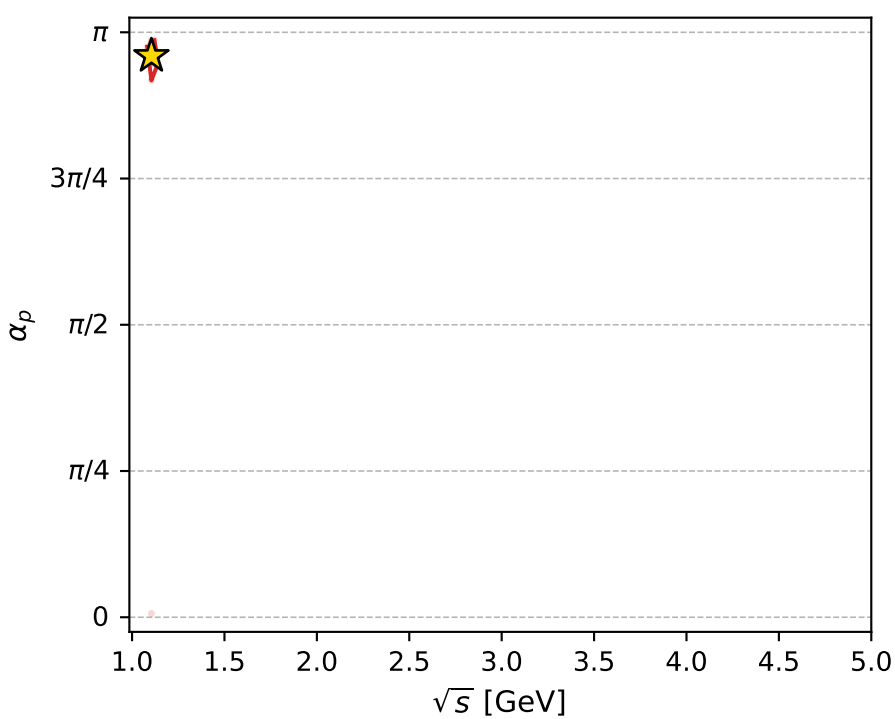
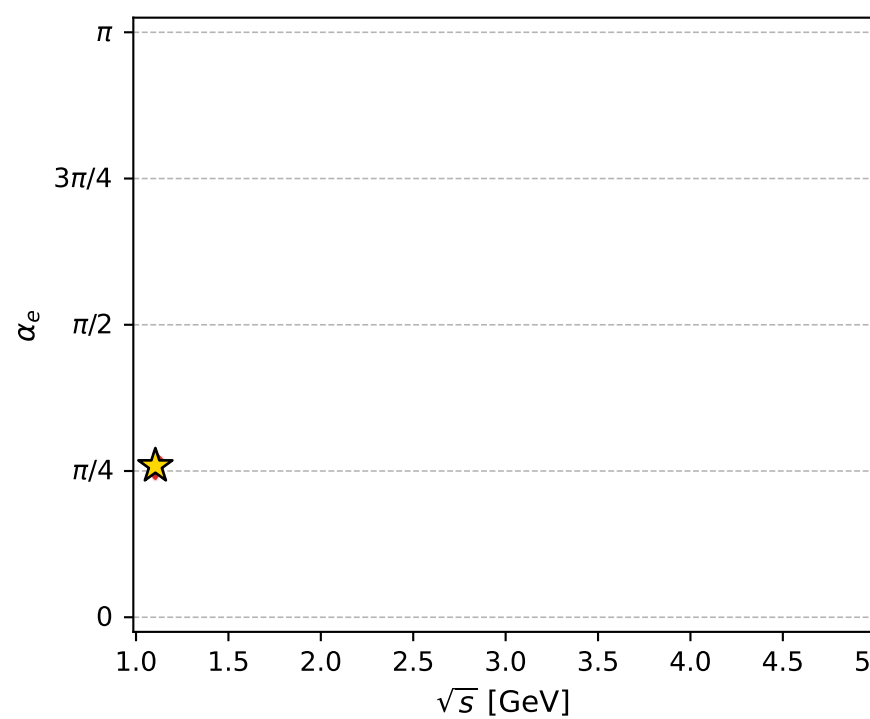
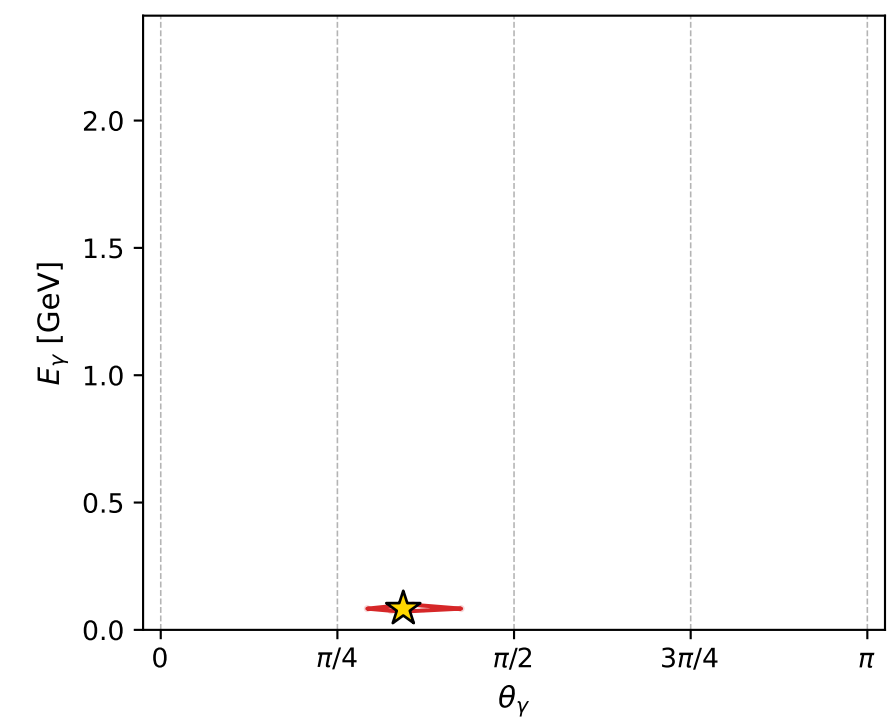
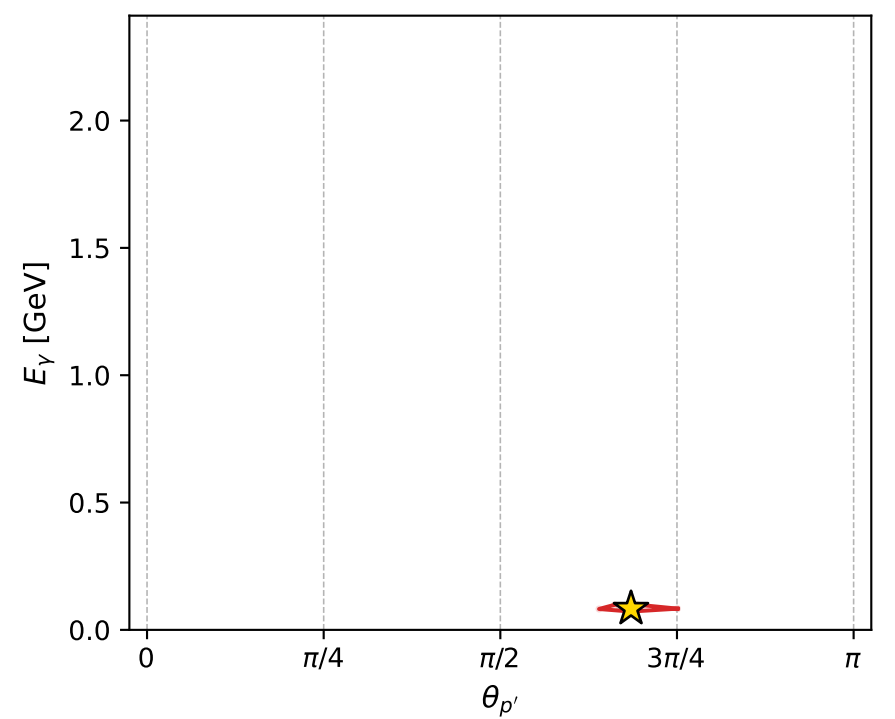
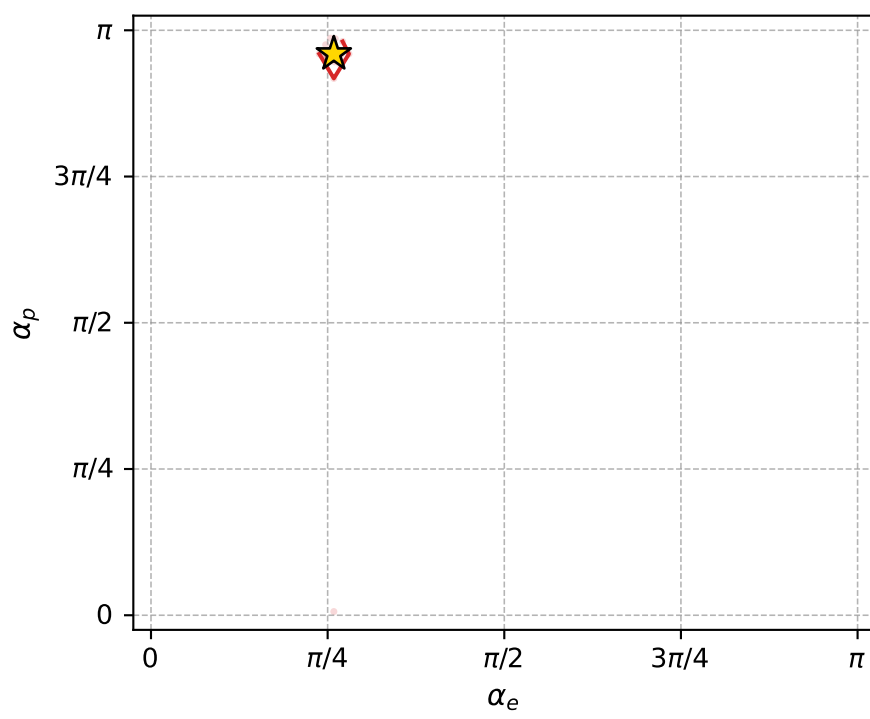
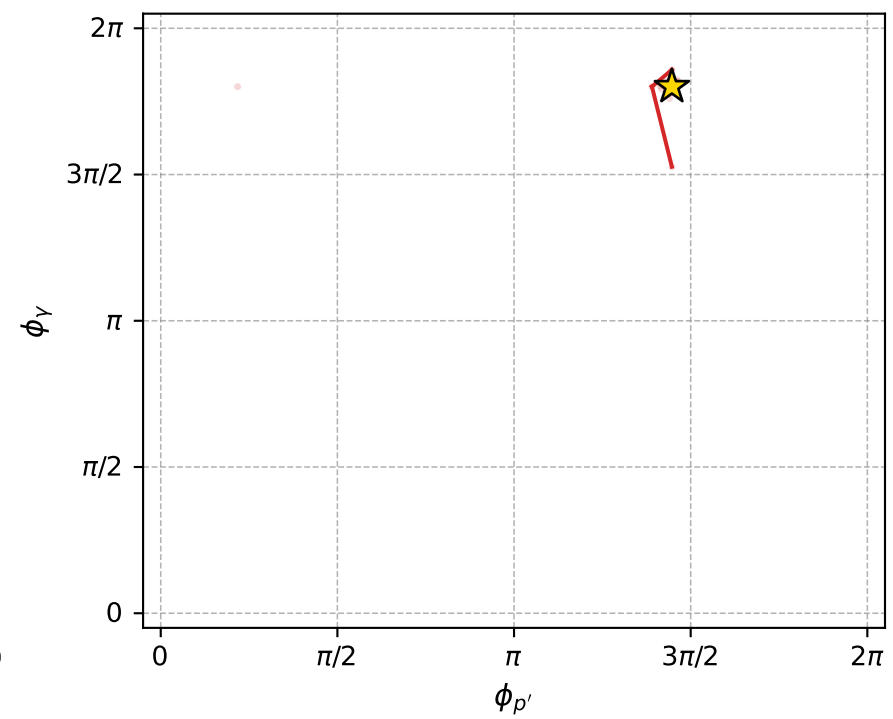
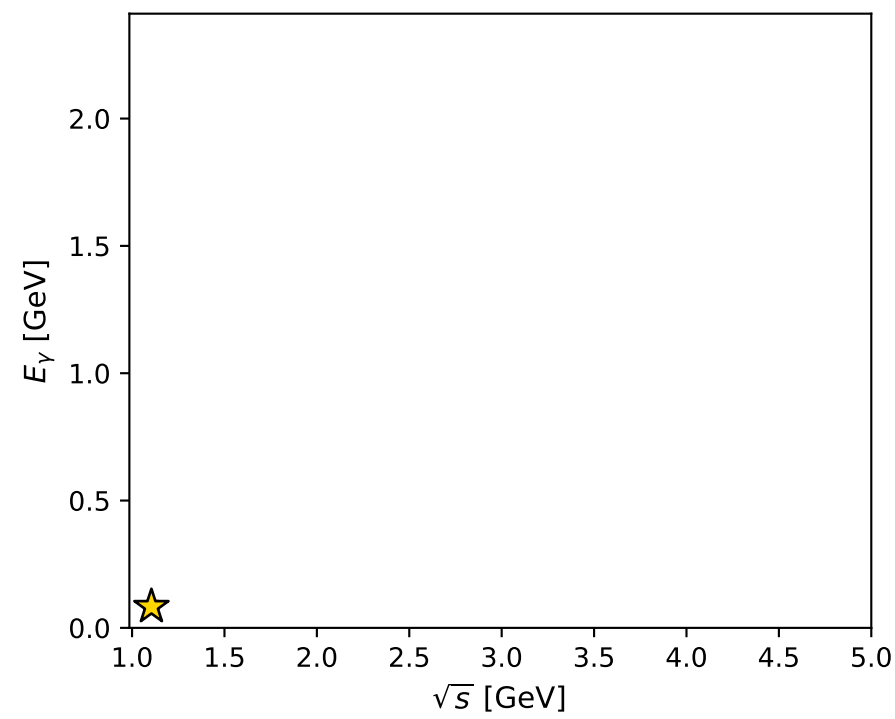
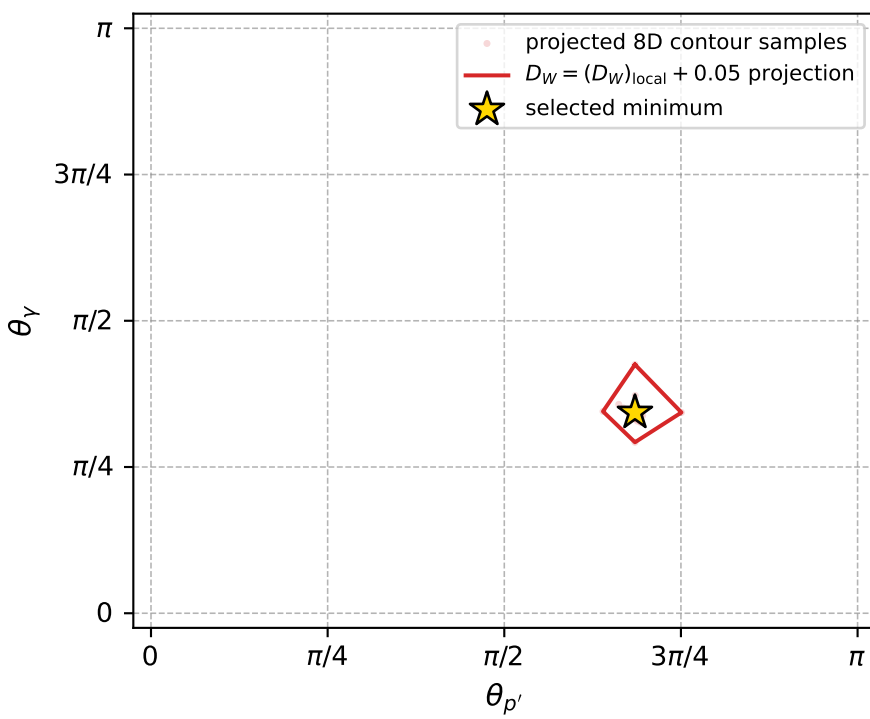
dw_mixing_angles_polarization_03_local_minimum_1033 D_W=0.0205327
region: polarization_cluster_03_local_minimum_1033
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.81295132$ rad
incoming lepton: $0.687358|+\rangle + 0.726319|-\rangle$
proton mixing angle: $\alpha_p=3.0146077$ rad
incoming proton: $-0.991948|+\rangle + 0.126644|-\rangle$

$s=1.22038$, $\sqrt{s}=1.10471$, $|\vec{P}|=0.154128$, $|\vec{P}'|=2.80085e-06$
 $\theta_{P'}=2.15181$, $\theta_Y=1.07832$, $\phi_{P'}=4.54516$, $\phi_Y=5.65621$
 $E_Y=0.0833527$, $\theta_{\ell'}=2.06325$, $\phi_{\ell'}=2.51459$
 $Q^2=0.0135462$, $x_B=0.0797228$, $t=-0.0235977$, $W^2=1.03621$, $y=0.498969$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1541$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1541$
 P $E= 0.9506$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1541$
 ℓ' $E= 0.0834$ $p_x= -0.0595$ $p_y= 0.0431$ $p_z= -0.0394$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0000$
 q_Y $E= 0.0834$ $p_x= 0.0595$ $p_y= -0.0431$ $p_z= 0.0394$



electron: polarization cluster P3, local minimum 1033; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.020532653$
 $D_W \text{ contour} = 0.070532653$
 above global minimum = 0.020516221
 8D contour samples = 255

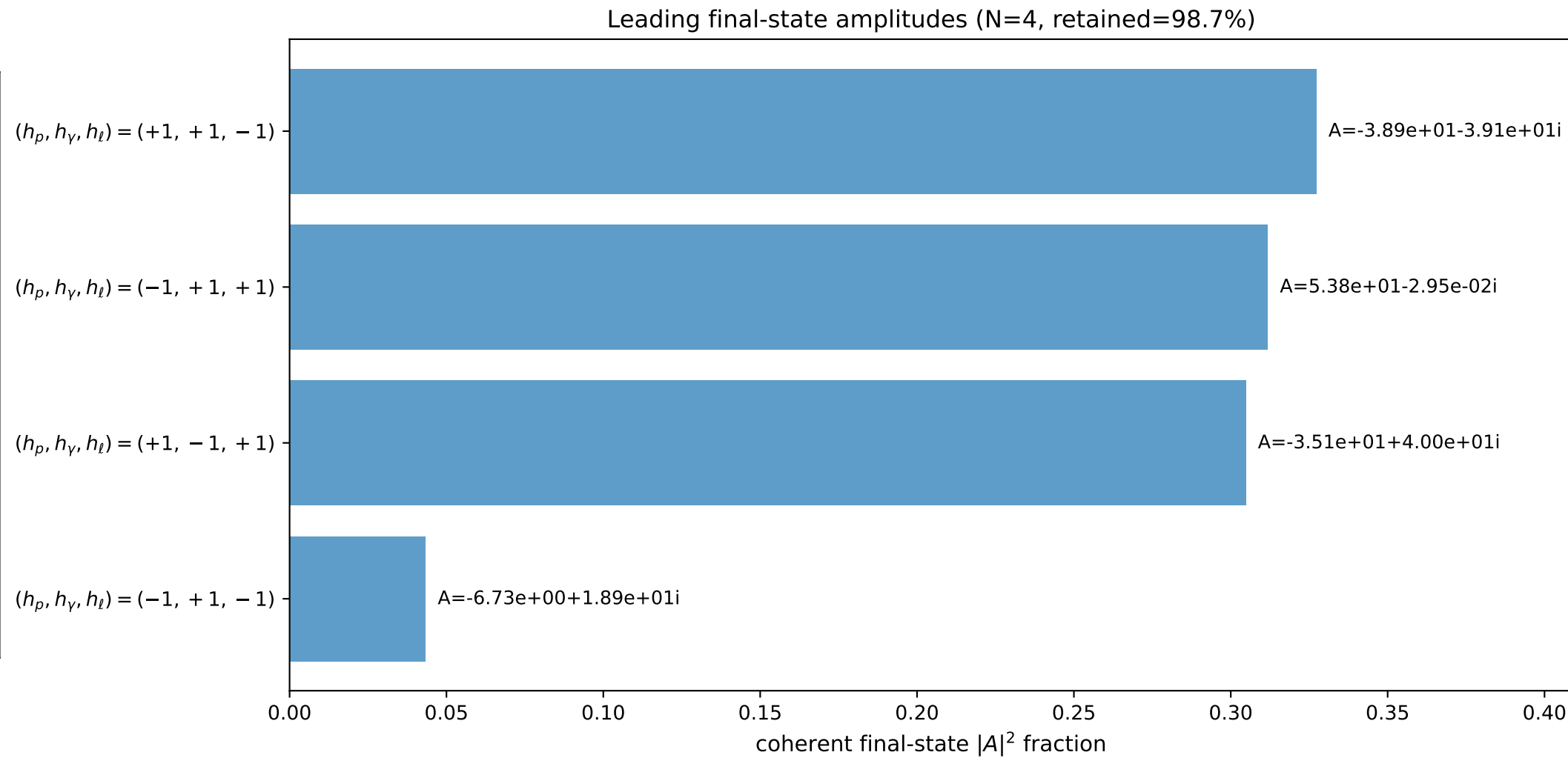
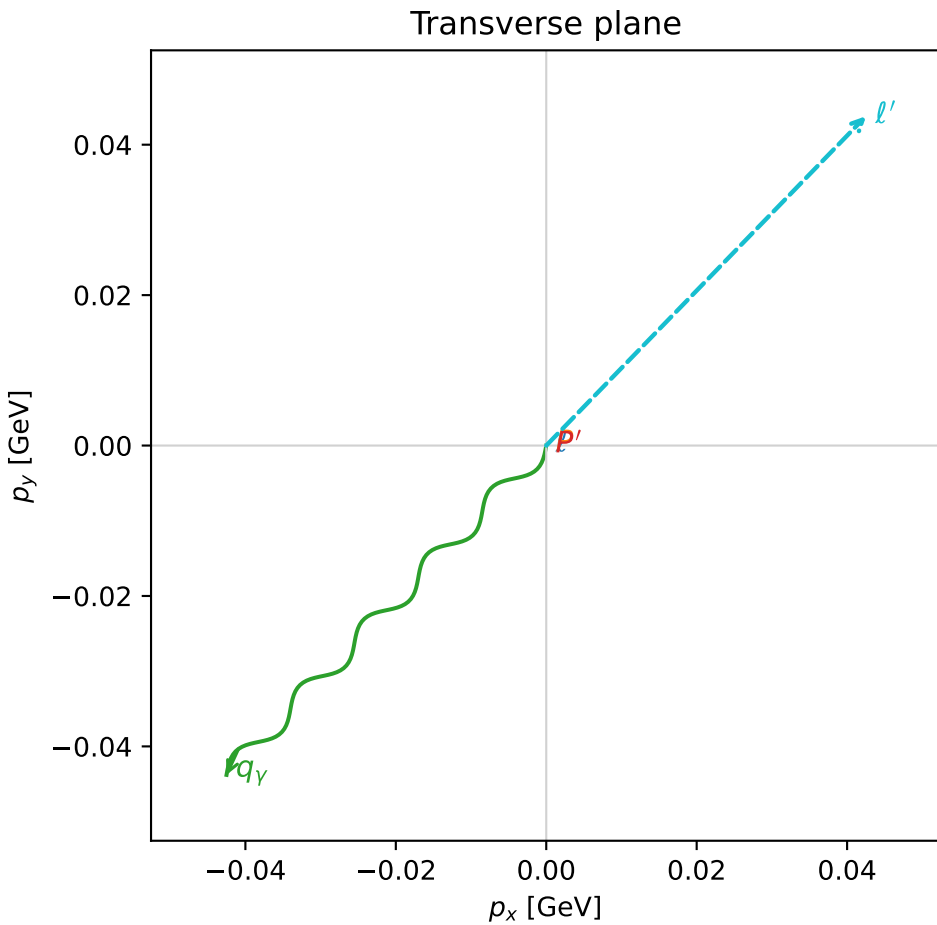
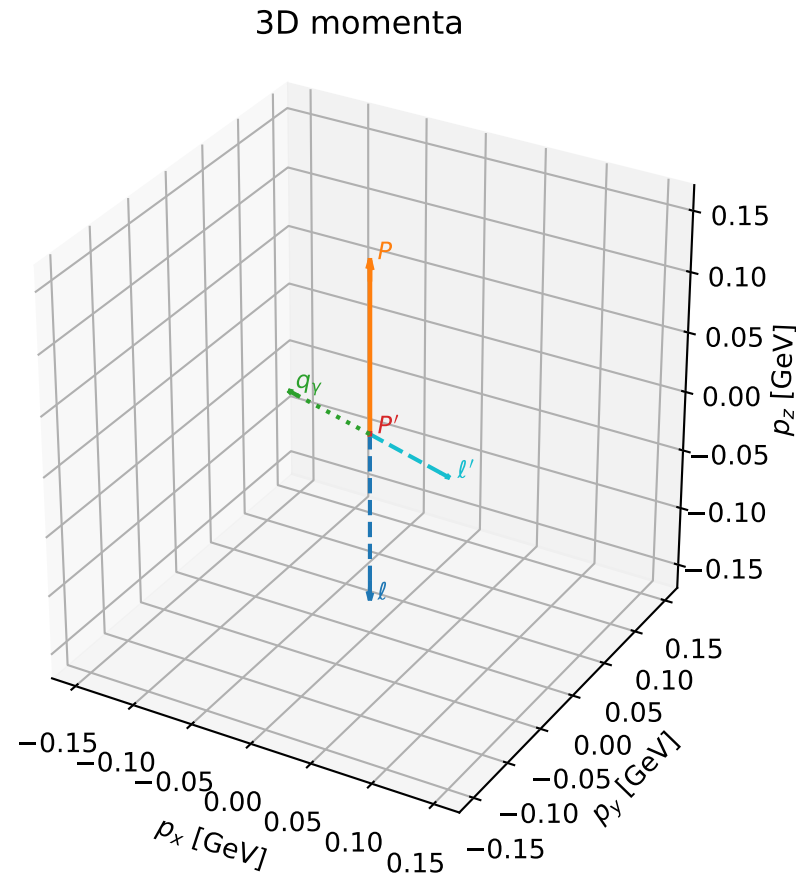
selected phase-space point:
 $\text{sqrt_s} = 1.104707$
 $\text{theta_p_out} = 2.151808$
 $\text{theta_gamma_out} = 1.078317$
 $\text{qOut} = 0.08335271$
 $\text{phi_p_out} = 4.545155$
 $\text{phi_gamma_out} = 5.656209$
 $\text{alpha_e} = 0.8129513$
 $\text{alpha_p} = 3.014608$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

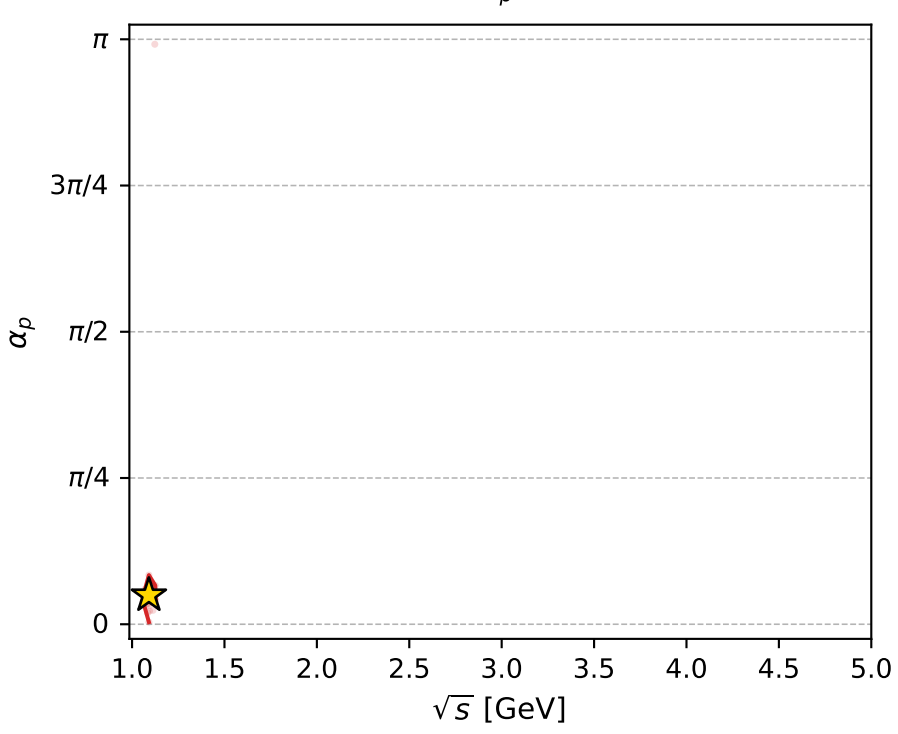
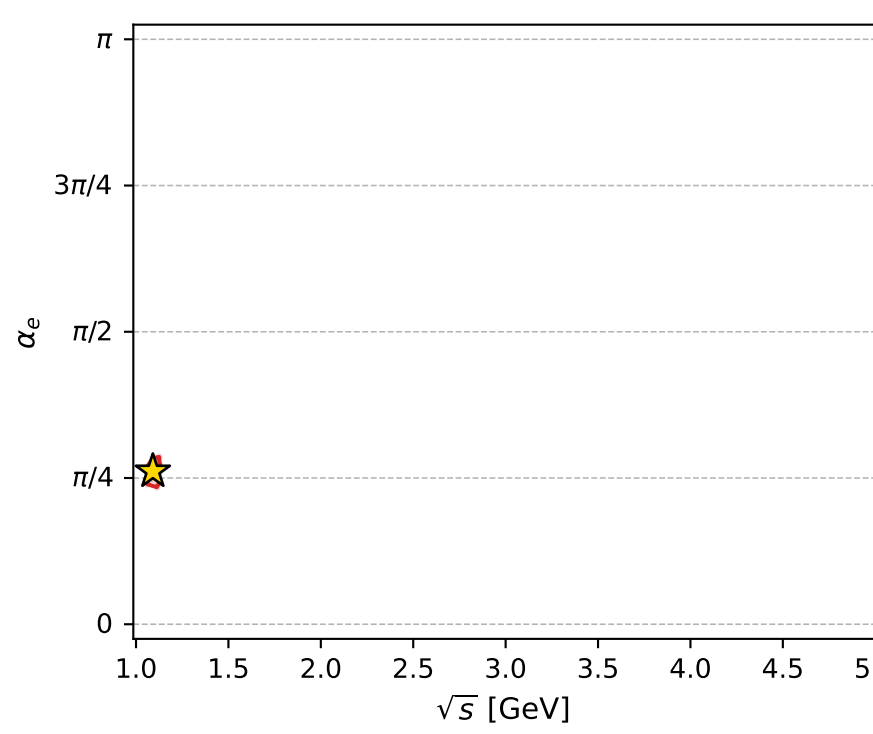
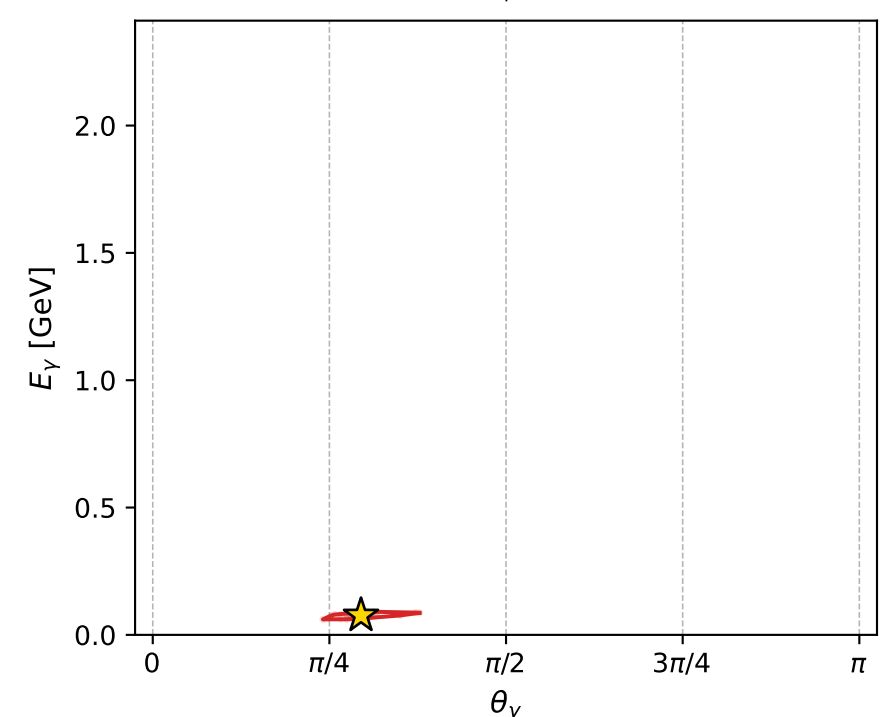
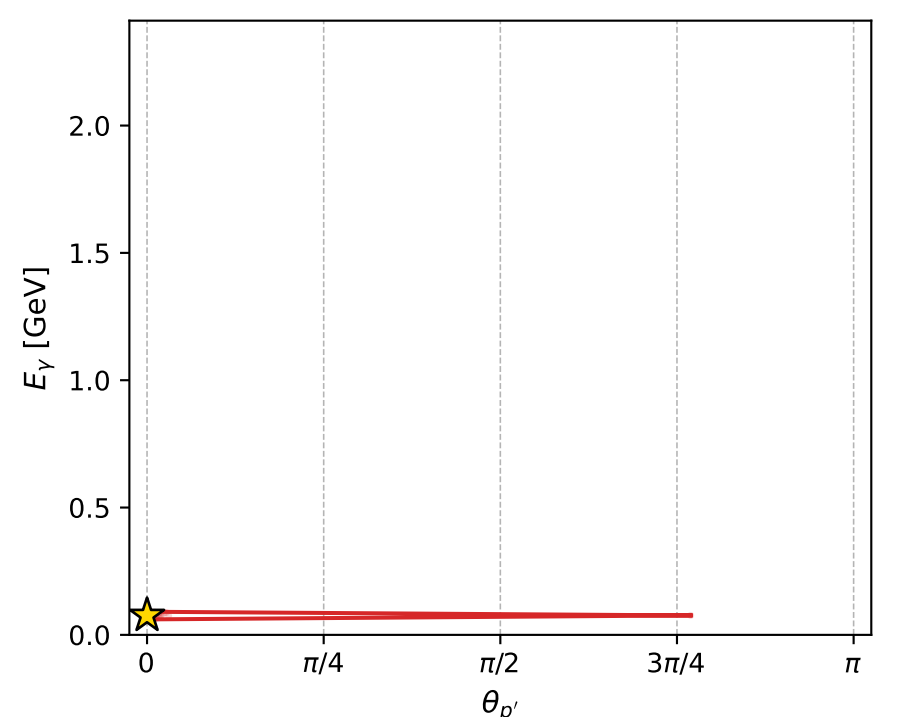
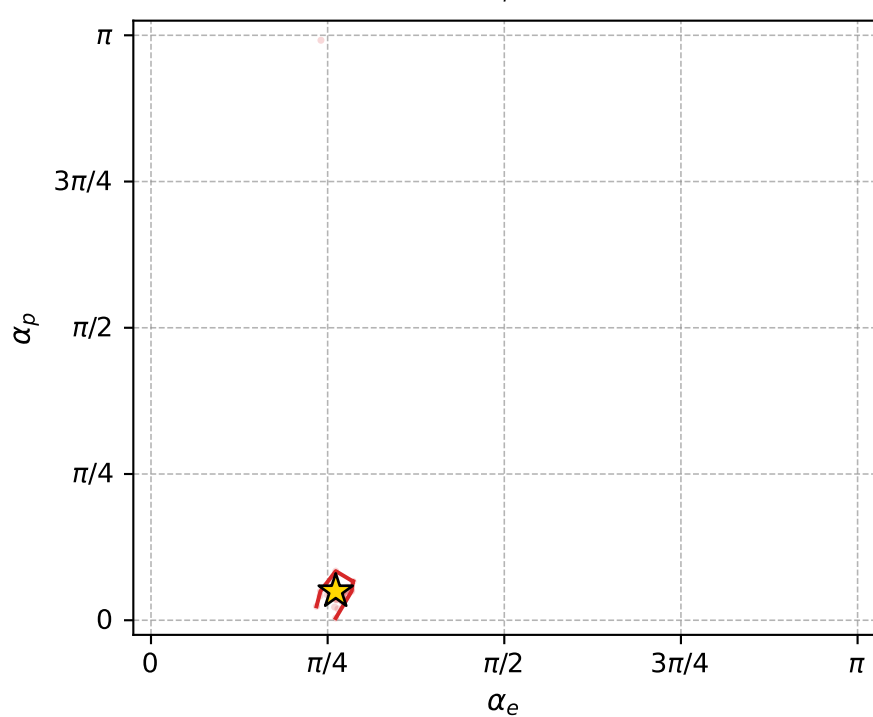
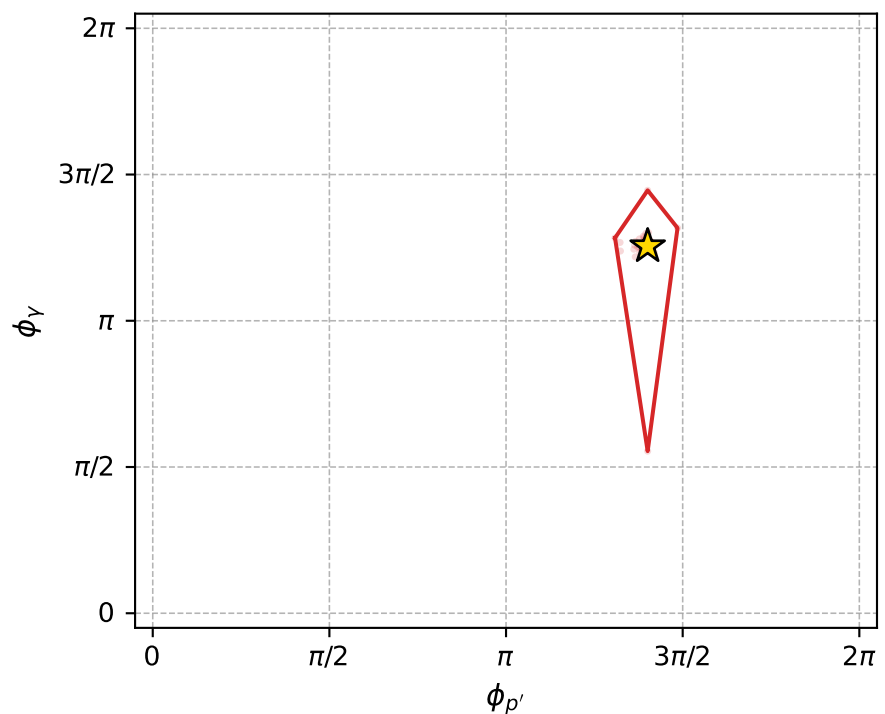
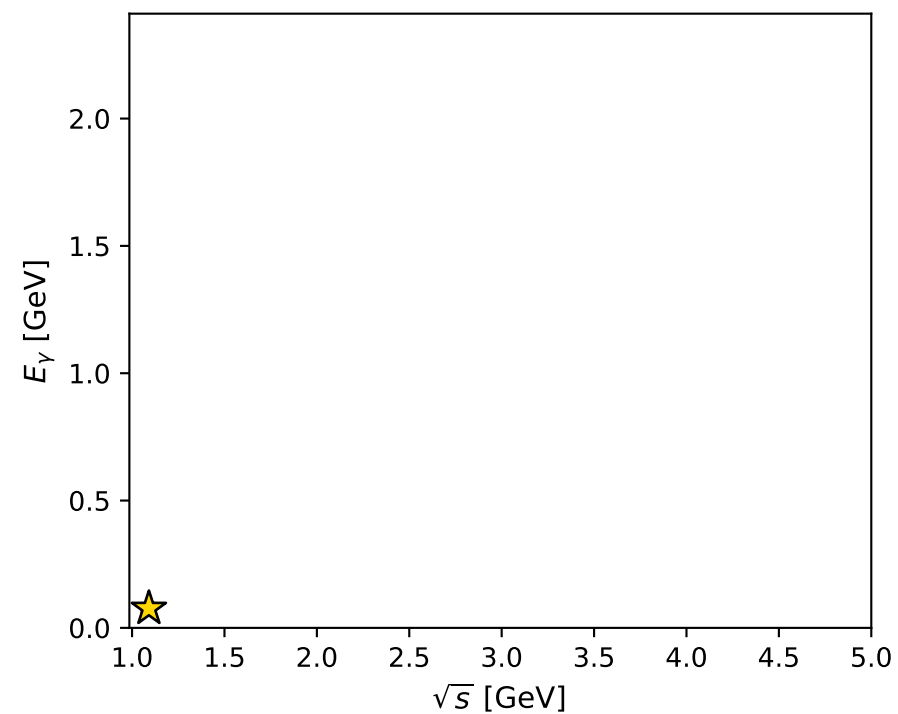
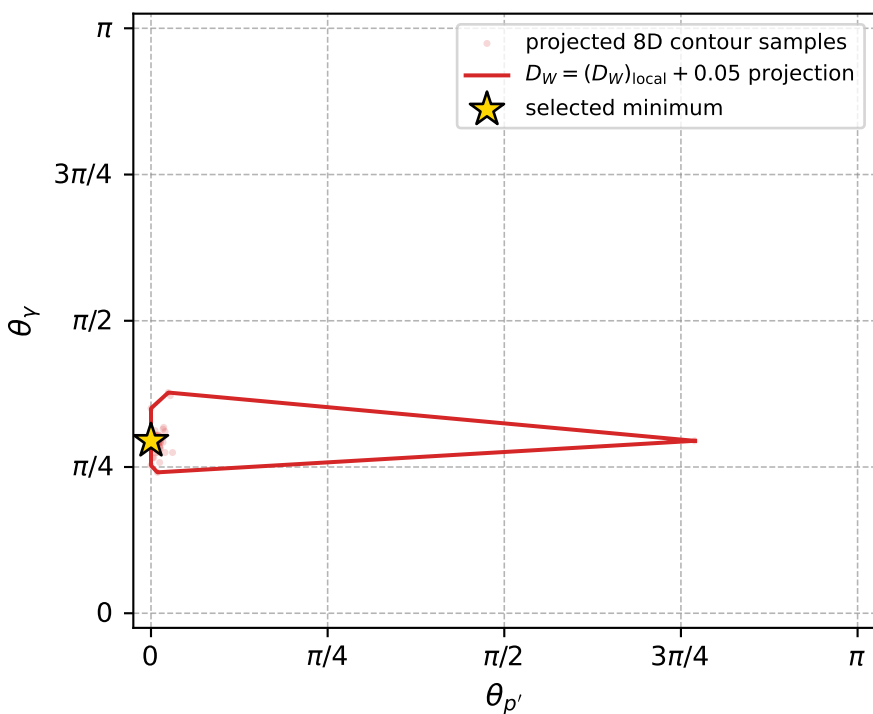
dw_mixing_angles_polarization_03_local_minimum_1046 D_W=0.0218584
region: polarization_cluster_03_local_minimum_1046
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.82137185$ rad
incoming lepton: $0.681218|+\rangle + 0.732081|-\rangle$
proton mixing angle: $\alpha_p=0.15572249$ rad
incoming proton: $0.9879|+\rangle + 0.155094|-\rangle$

$s=1.19066$, $\sqrt{s}=1.09117$, $|\vec{P}|=0.142422$, $|\vec{P}'|=0.000750587$
 $\theta_{P'}=0$, $\theta_V=0.925991$, $\phi_{P'}=4.40095$, $\phi_V=3.9418$
 $E_V=0.0763592$, $\theta_{\ell'}=2.22341$, $\phi_{\ell'}=0.800211$
 $Q^2=0.008593$, $x_B=0.056617$, $t=-0.0199553$, $W^2=1.02303$, $y=0.488308$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1424$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1424$
 P $E= 0.9488$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1424$
 ℓ' $E= 0.0768$ $p_x= 0.0425$ $p_y= 0.0438$ $p_z= -0.0466$
 P' $E= 0.9380$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= 0.0008$
 q_V $E= 0.0764$ $p_x= -0.0425$ $p_y= -0.0438$ $p_z= 0.0459$



electron: polarization cluster P3, local minimum 1046; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.021858403$
 $D_W \text{ contour} = 0.071858403$
 above global minimum = 0.02184197
 8D contour samples = 129

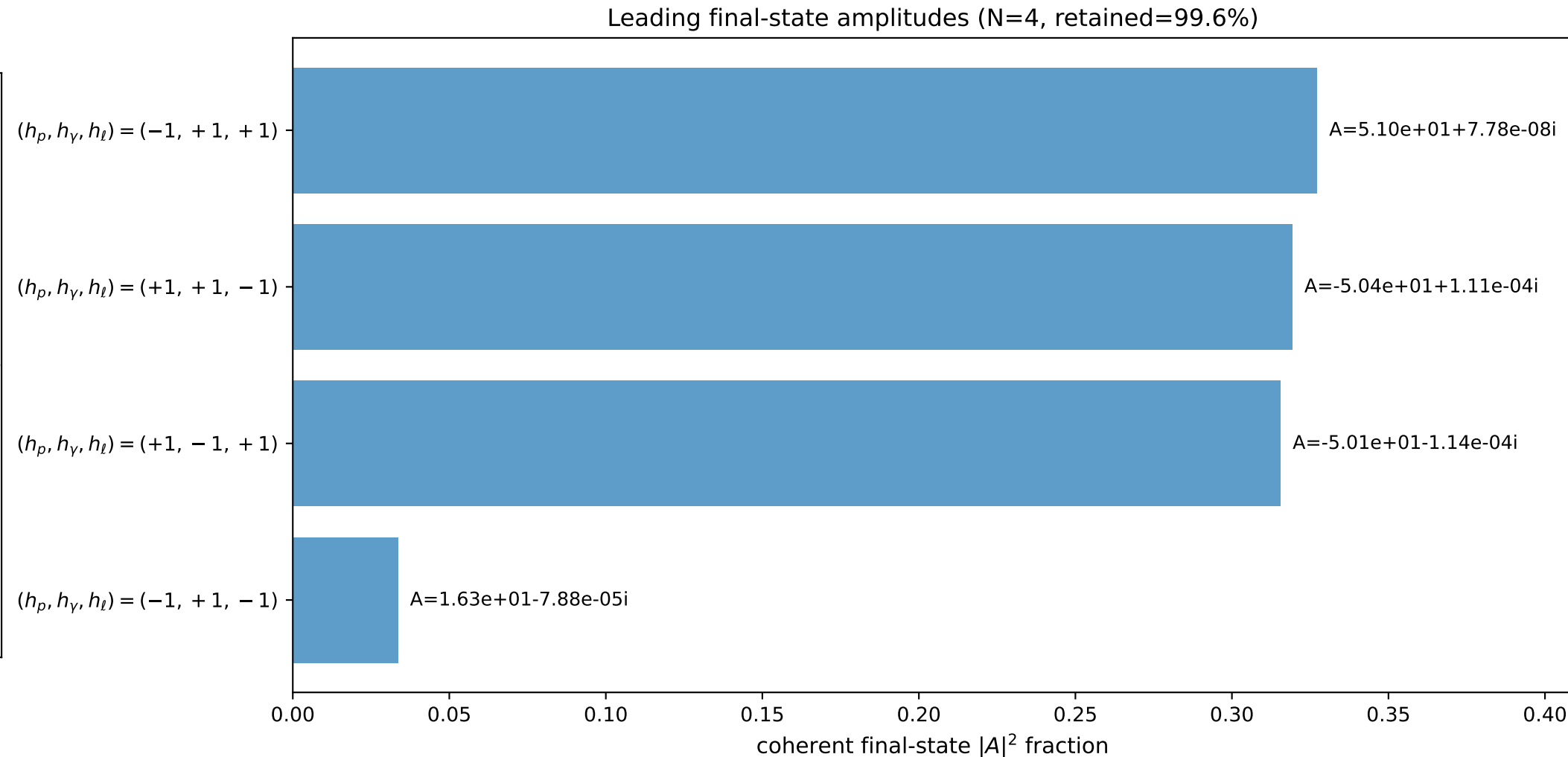
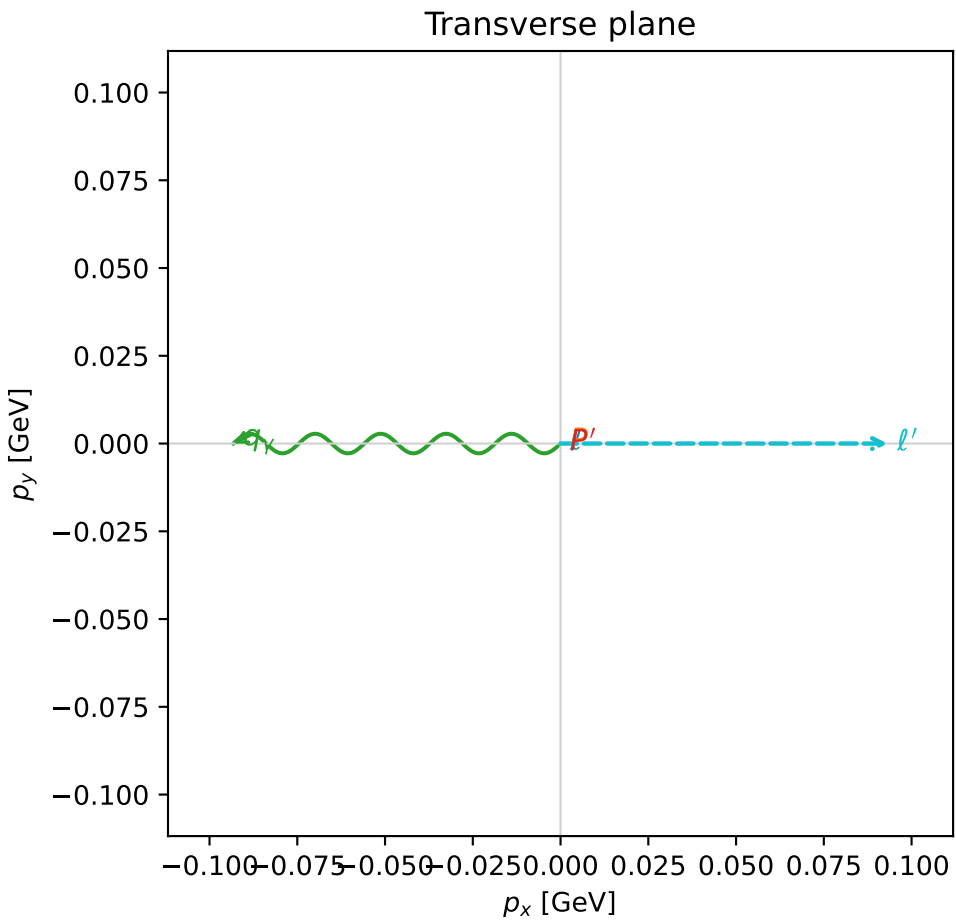
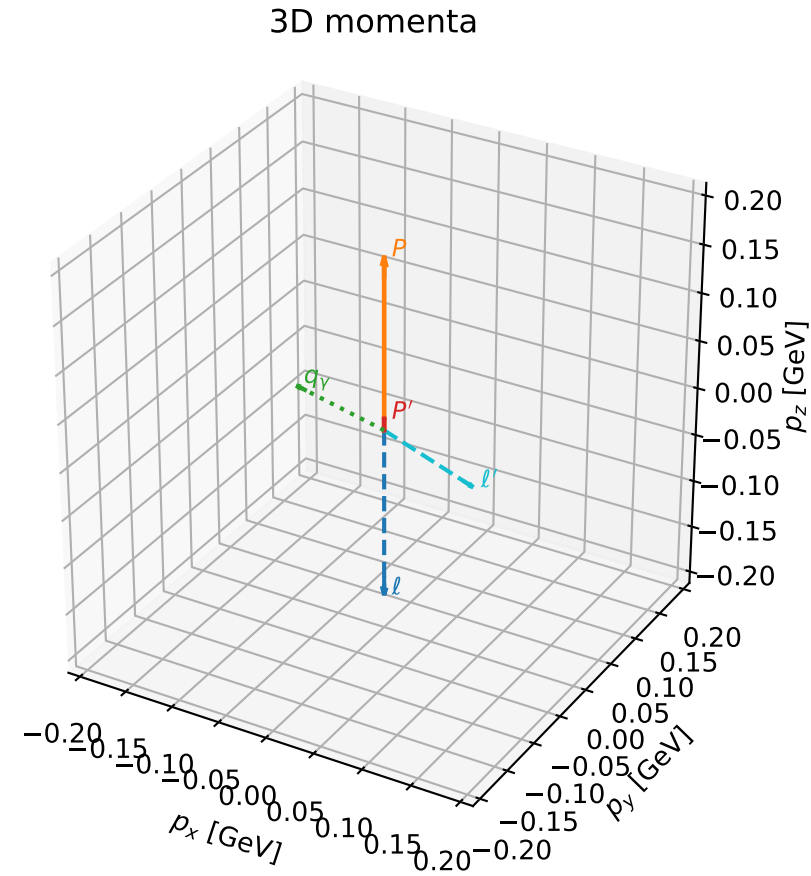
selected phase-space point:
 $\text{sqrt_s} = 1.091174$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 0.9259914$
 $\text{q0ut} = 0.07635922$
 $\text{phi_p_out} = 4.400952$
 $\text{phi_gamma_out} = 3.941804$
 $\text{alpha_e} = 0.8213718$
 $\text{alpha_p} = 0.1557225$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

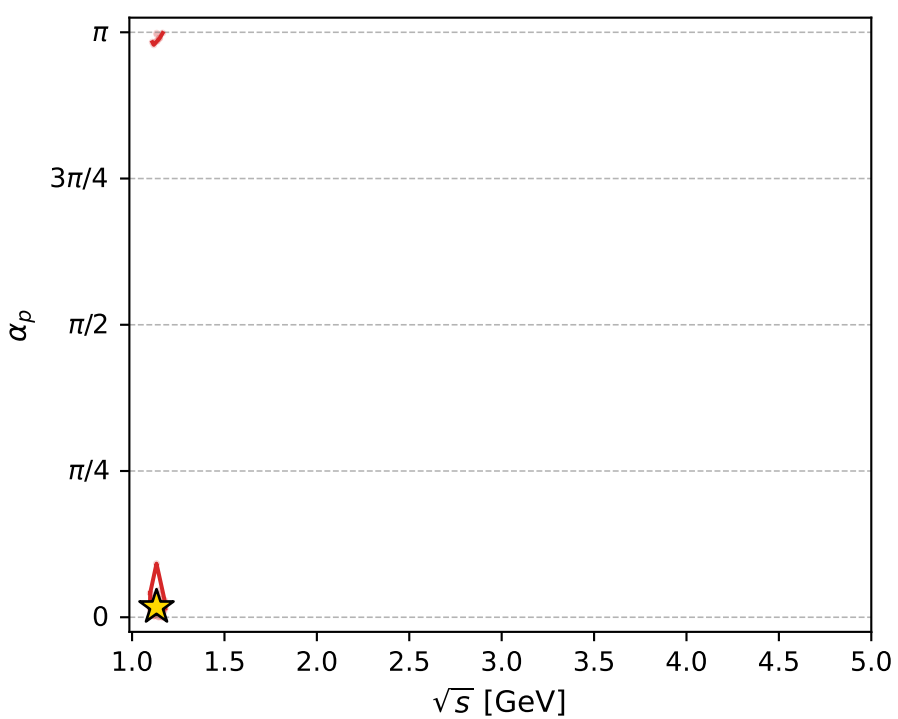
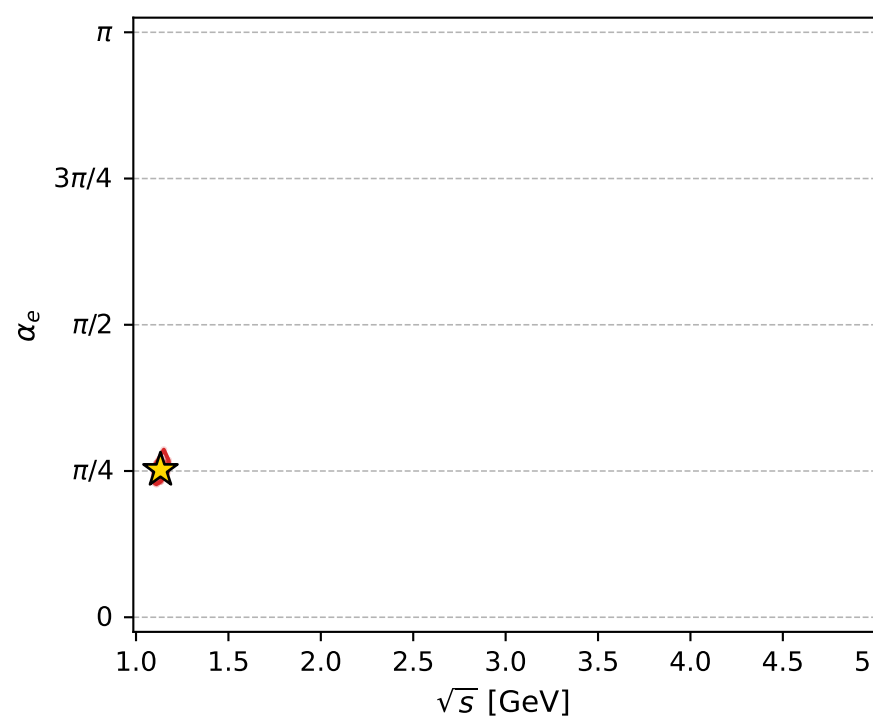
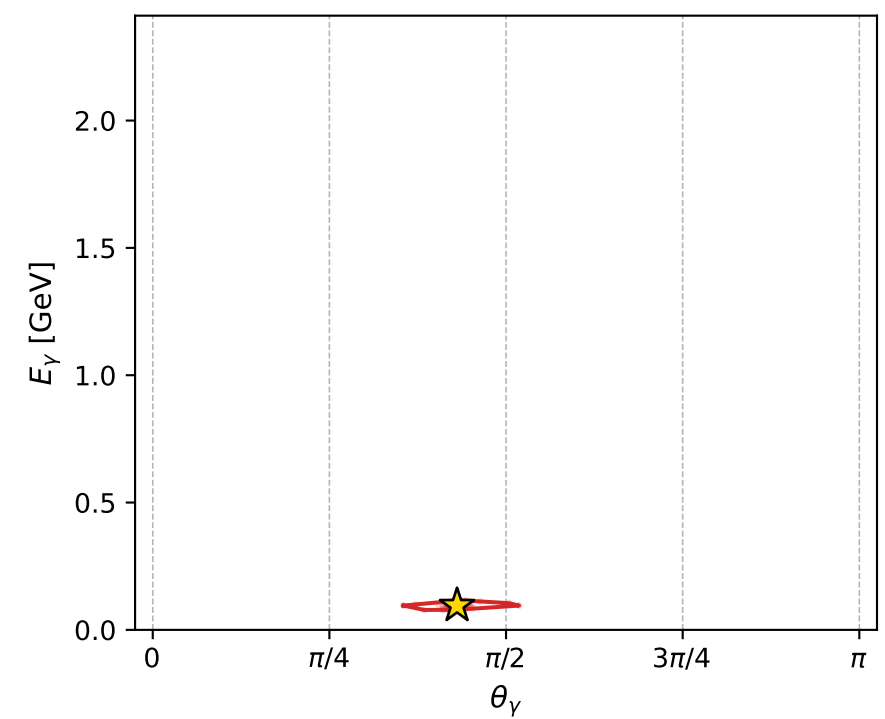
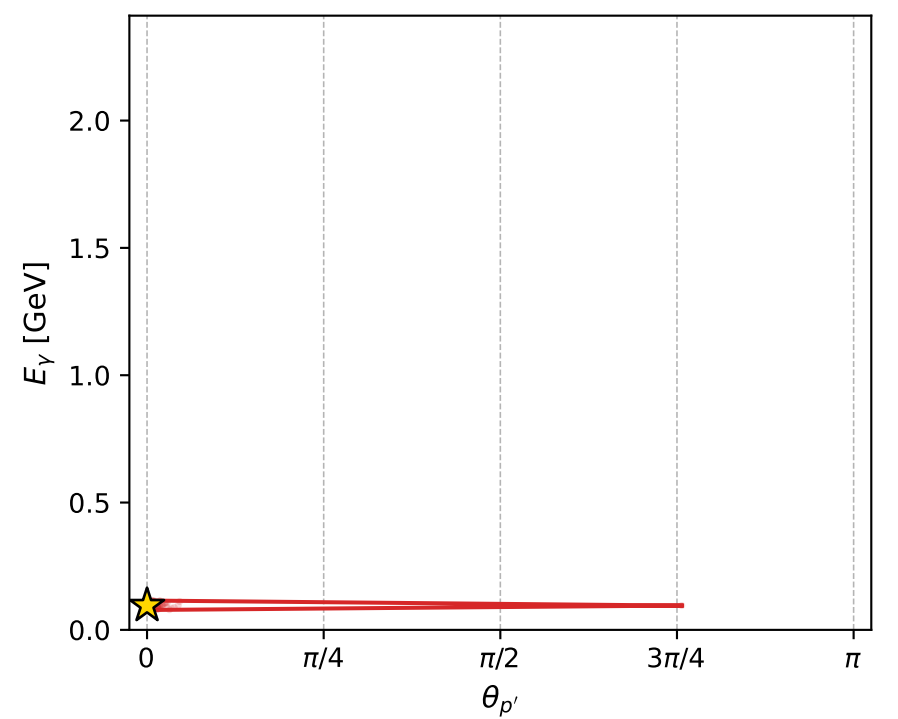
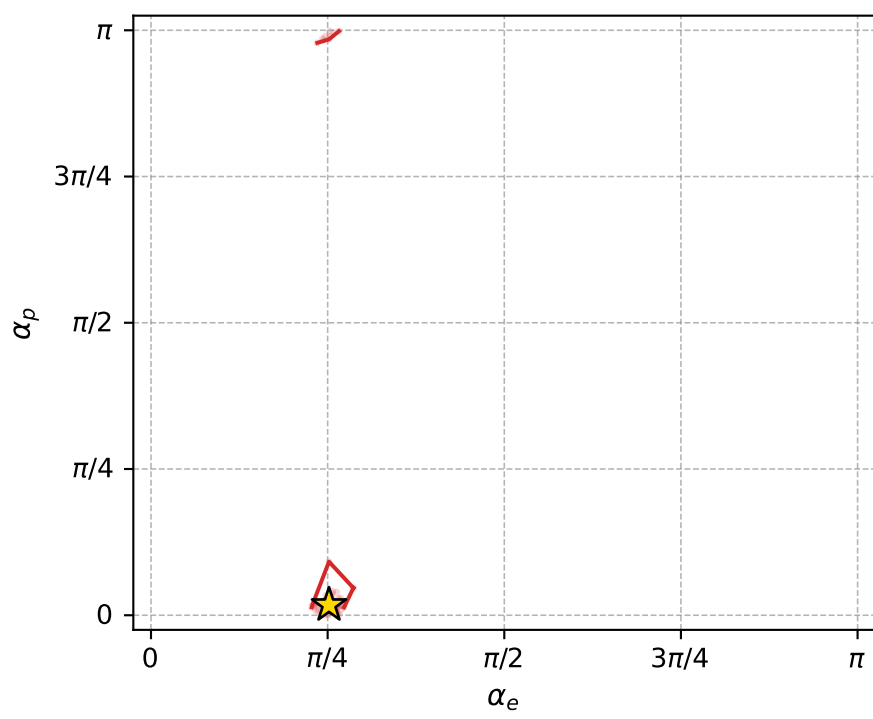
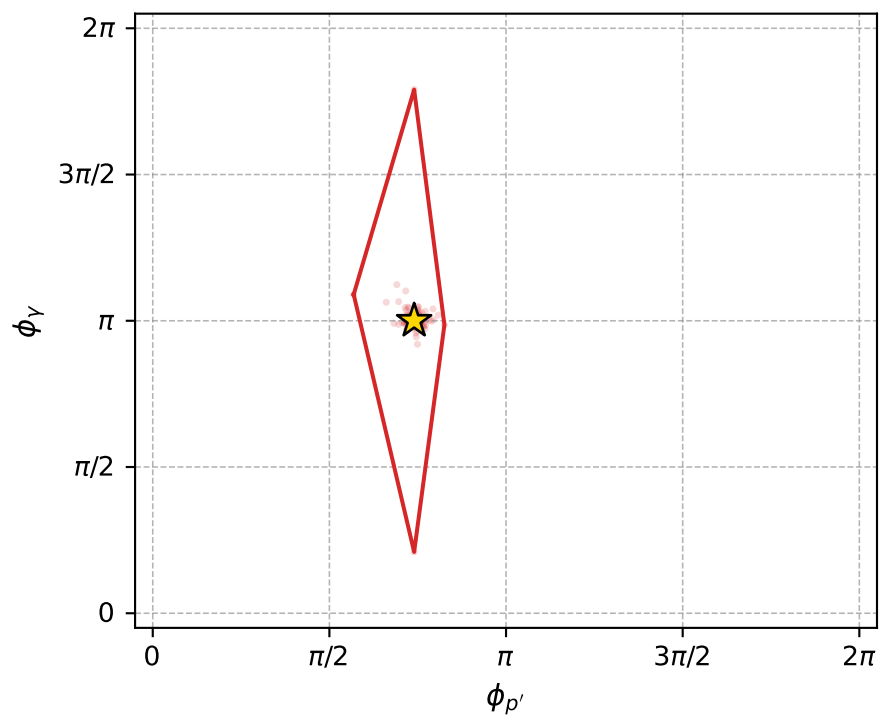
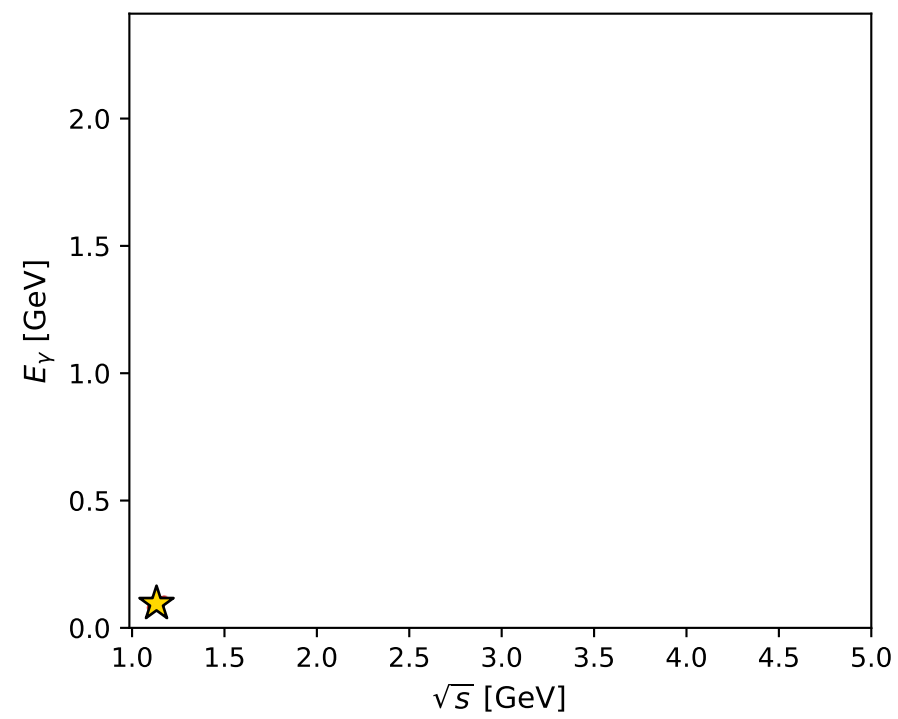
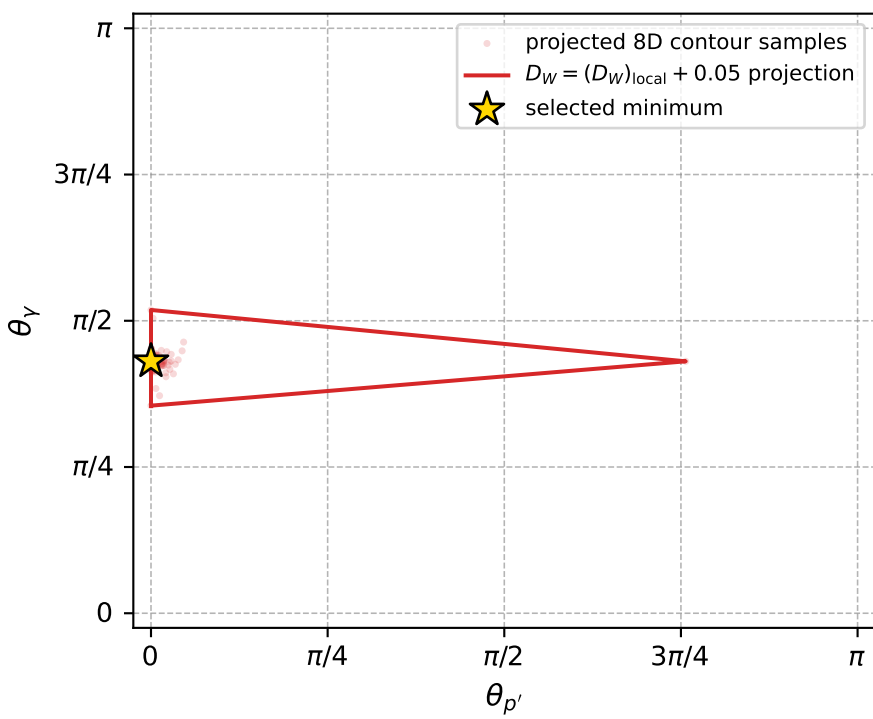
dw_mixing_angles_polarization_03_local_minimum_1047 D_W=0.0220028
 region: polarization_cluster_03_local_minimum_1047
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.79139452$ rad
 incoming lepton: $0.702854|+\rangle + 0.711334|-\rangle$
 proton mixing angle: $\alpha_p=0.055021403$ rad
 incoming proton: $0.998487|+\rangle + 0.0549936|-\rangle$

$s=1.28205$, $\sqrt{s}=1.13228$, $|\vec{P}|=0.17761$, $|\vec{P}'|=0.0120377$
 $\theta_{p'}=0$, $\theta_Y=1.35299$, $\phi_{p'}=2.3244$, $\phi_Y=3.14159$
 $E_Y=0.0954479$, $\theta_{\ell'}=1.9079$, $\phi_{\ell'}=6.28318$
 $Q^2=0.0234759$, $x_B=0.116186$, $t=-0.0271389$, $W^2=1.05842$, $y=0.502363$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1776$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1776$
 P $E= 0.9547$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1776$
 ℓ' $E= 0.0988$ $p_x= 0.0932$ $p_y= -0.0000$ $p_z= -0.0327$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0120$
 q_Y $E= 0.0954$ $p_x= -0.0932$ $p_y= 0.0000$ $p_z= 0.0206$



electron: polarization cluster P3, local minimum 1047; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.022002835$
 $D_W \text{ contour} = 0.072002835$
 above global minimum = 0.021986403
 8D contour samples = 142

selected phase-space point:
 $\text{sqrt_s} = 1.132278$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.352991$
 $\text{qOut} = 0.09544794$
 $\text{phi_p_out} = 2.324403$
 $\text{phi_gamma_out} = 3.14159$
 $\text{alpha_e} = 0.7913945$
 $\text{alpha_p} = 0.0550214$

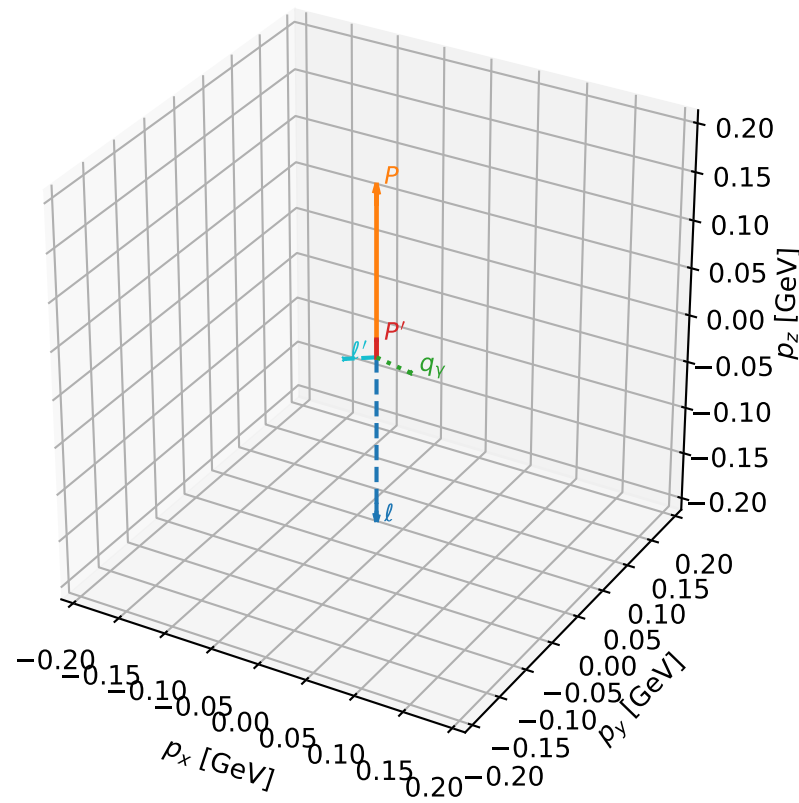
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1049 D_W=0.0224892
 region: polarization_cluster_03_local_minimum_1049
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.81978139$ rad
 incoming lepton: $0.682381|+\rangle +0.730997|-\rangle$
 proton mixing angle: $\alpha_p=3.0018672$ rad
 incoming proton: $-0.990254|+\rangle +0.139271|-\rangle$

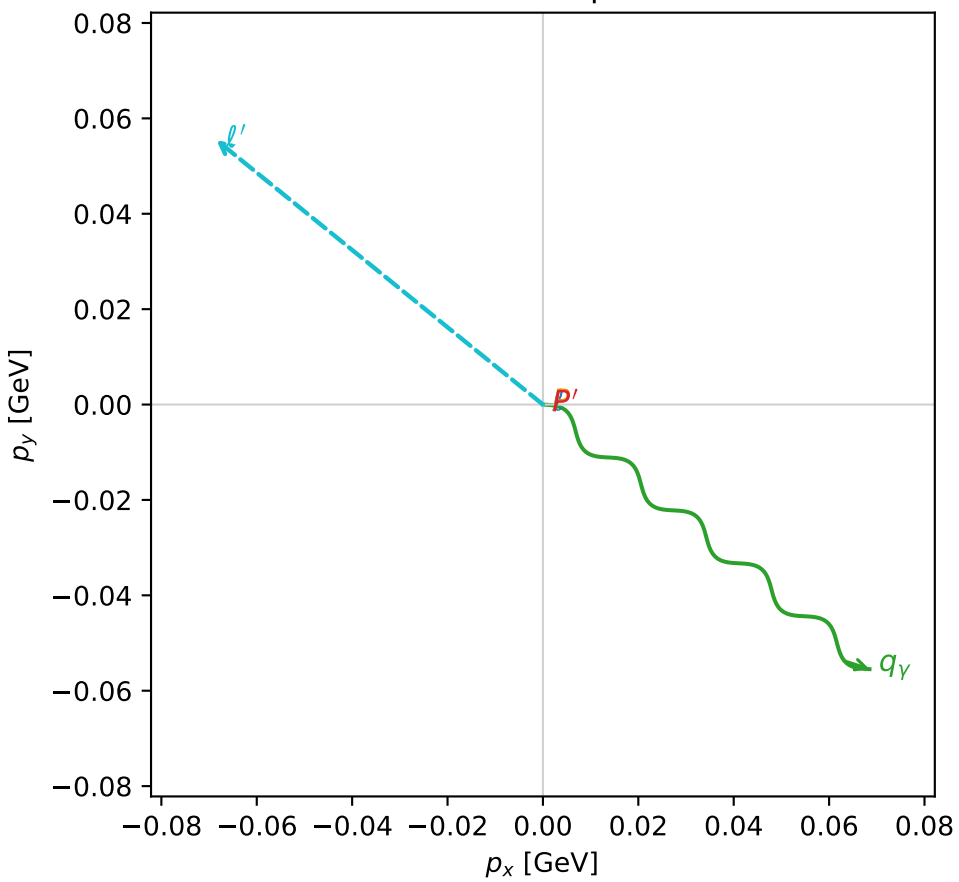
$s=1.28268$, $\sqrt{s}=1.13255$, $|\vec{P}|=0.177843$, $|\vec{P}'|=0.0184549$
 $\theta_{p'}=0$, $\theta_Y=1.23381$, $\phi_{p'}=3.66221$, $\phi_Y=5.60223$
 $E_Y=0.0933779$, $\theta_{\ell'}=2.08111$, $\phi_{\ell'}=2.46064$
 $Q^2=0.0183759$, $x_B=0.0954857$, $t=-0.0251312$, $W^2=1.05392$, $y=0.477733$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1778$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1778$
 P $E=0.9547$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1778$
 ℓ' $E=0.1010$ $p_x=-0.0685$ $p_y=0.0555$ $p_z=-0.0493$
 P' $E=0.9382$ $p_x=-0.0000$ $p_y=-0.0000$ $p_z=0.0185$
 q_Y $E=0.0934$ $p_x=0.0685$ $p_y=-0.0555$ $p_z=0.0309$

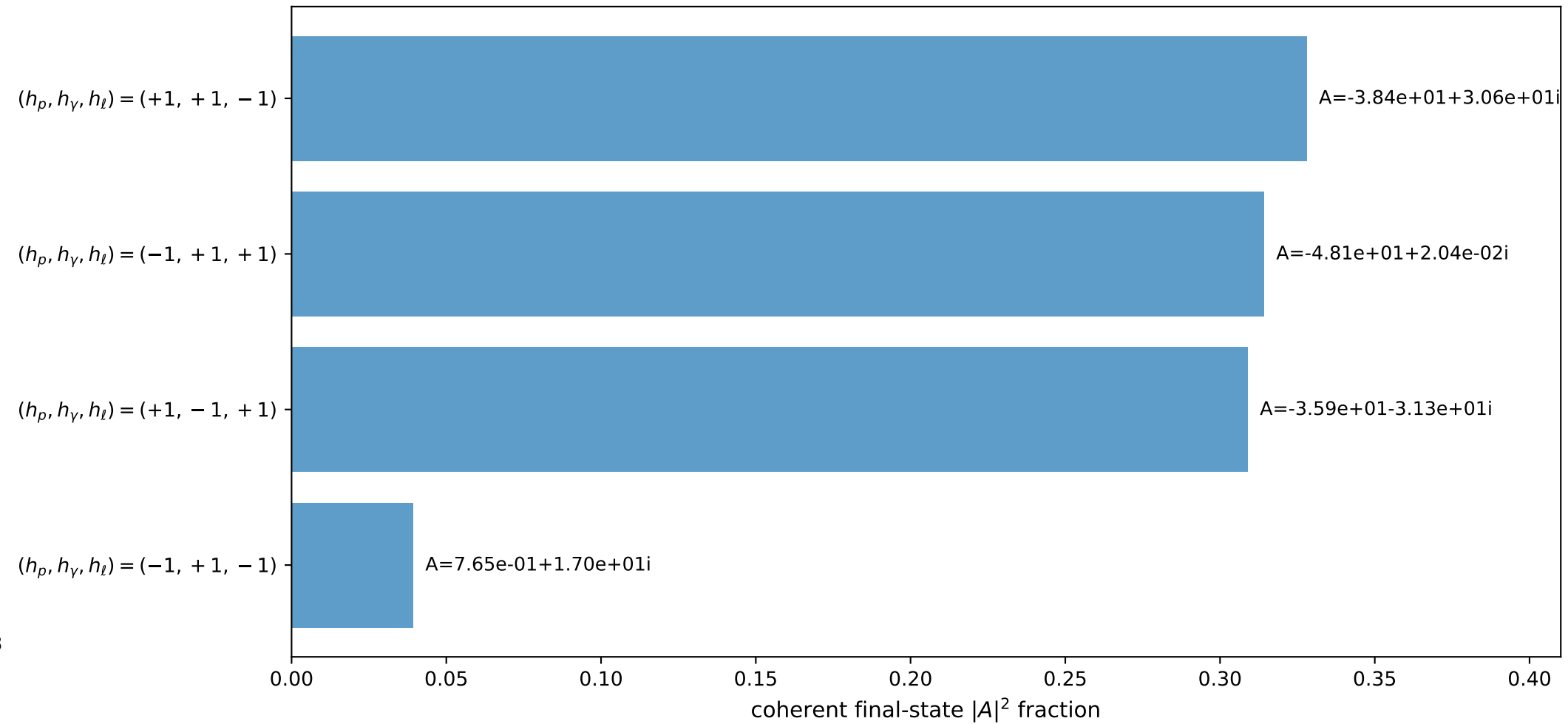
3D momenta



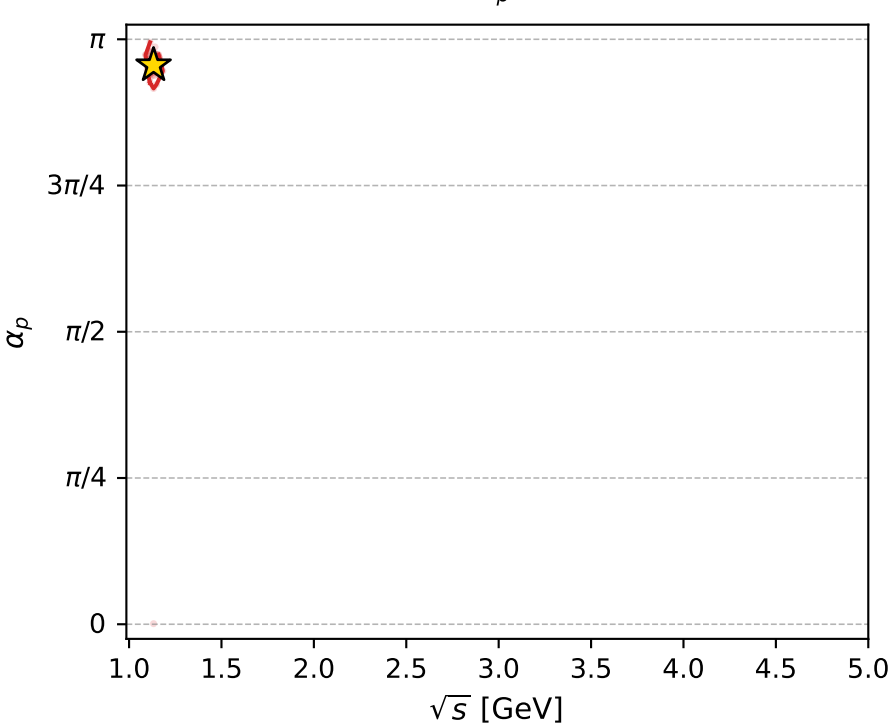
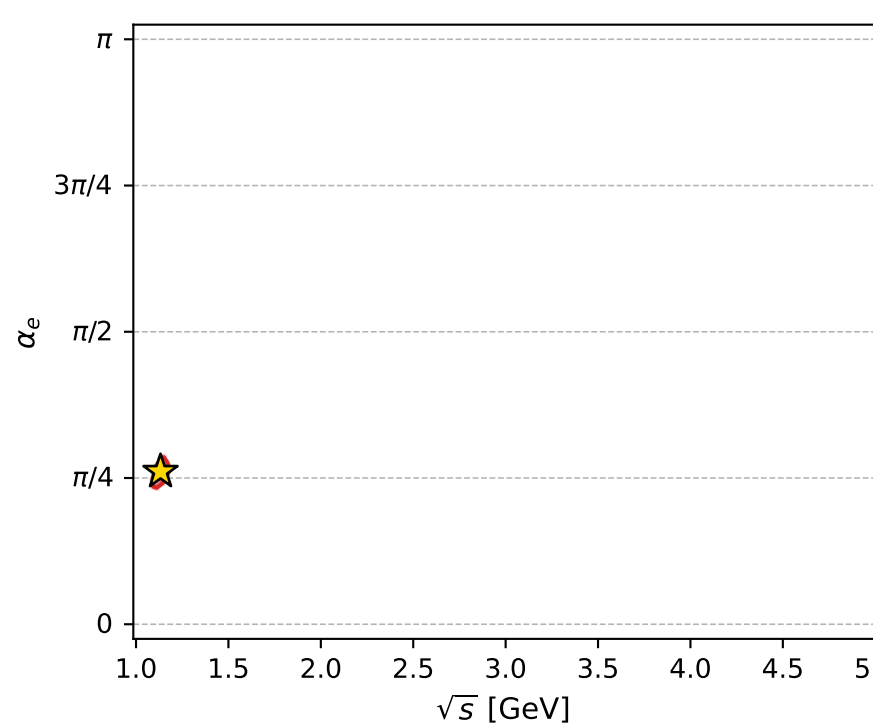
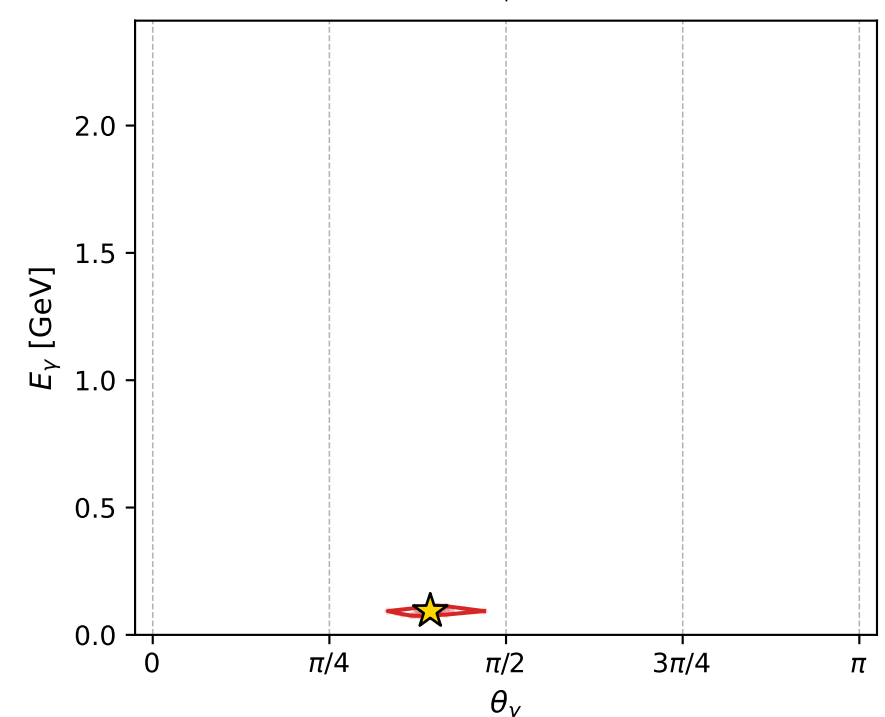
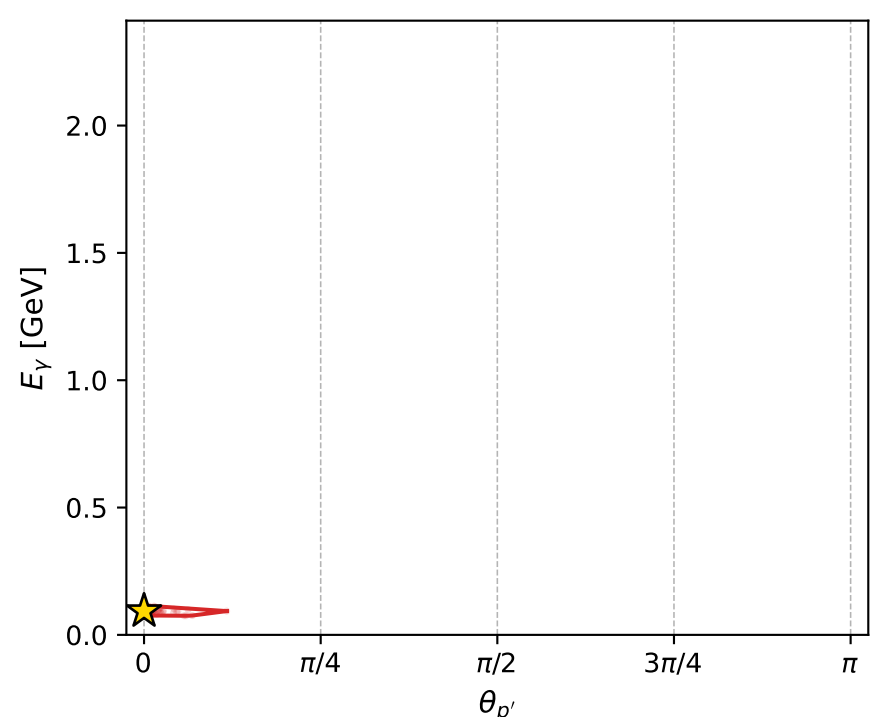
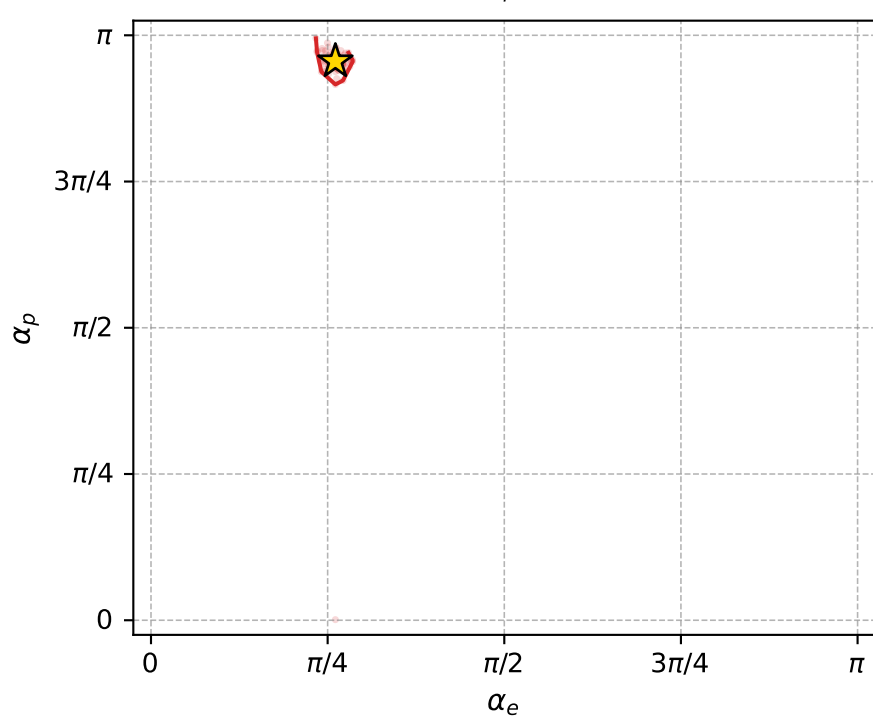
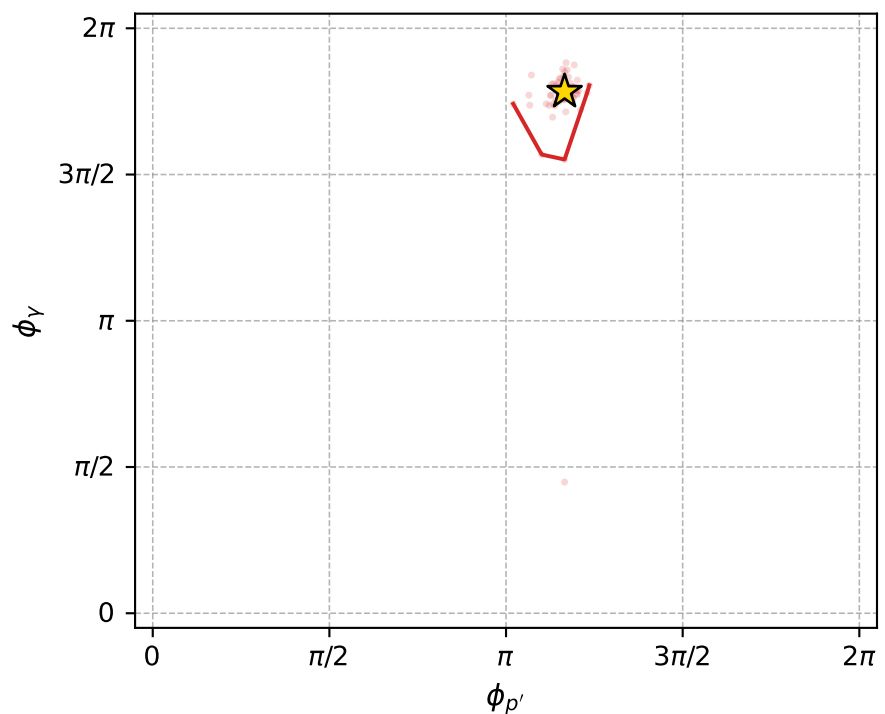
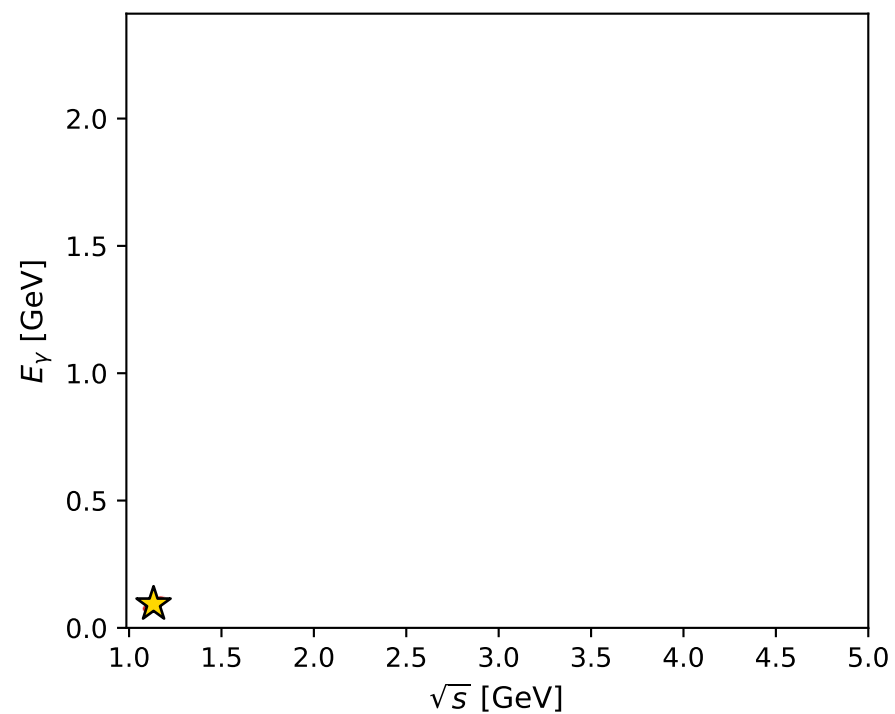
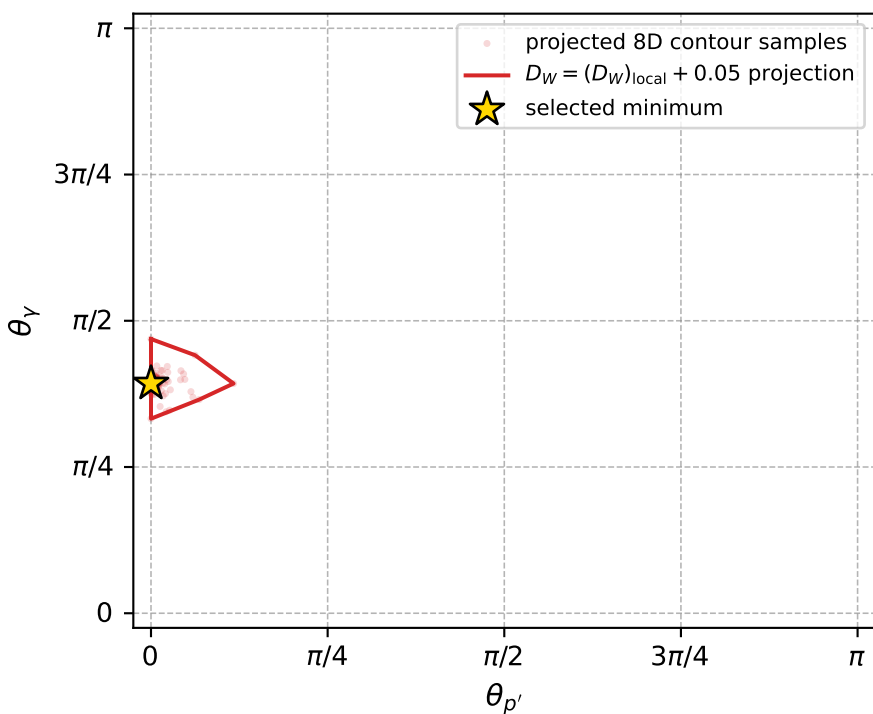
Transverse plane



Leading final-state amplitudes (N=4, retained=99.1%)



electron: polarization cluster P3, local minimum 1049; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.022489179$
 $D_W \text{ contour} = 0.072489179$
 above global minimum = 0.022472746
 8D contour samples = 133

selected phase-space point:
 $\text{sqrt_s} = 1.132554$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.23381$
 $\text{qOut} = 0.09337789$
 $\text{phi_p_out} = 3.66221$
 $\text{phi_gamma_out} = 5.602228$
 $\text{alpha_e} = 0.8197814$
 $\text{alpha_p} = 3.001867$

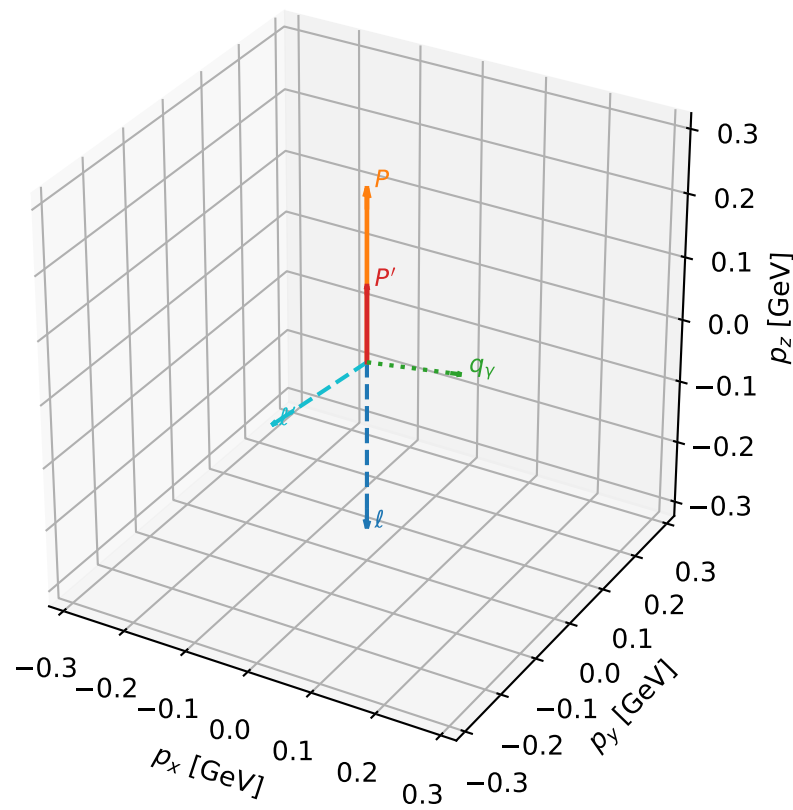
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1054 D_W=0.0230421
 region: polarization_cluster_03_local_minimum_1054
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82445874$ rad
 incoming lepton: $0.678954|+\rangle + 0.73418|-\rangle$
 proton mixing angle: $\alpha_p=2.9854691$ rad
 incoming proton: $-0.987837|+\rangle + 0.15549|-\rangle$

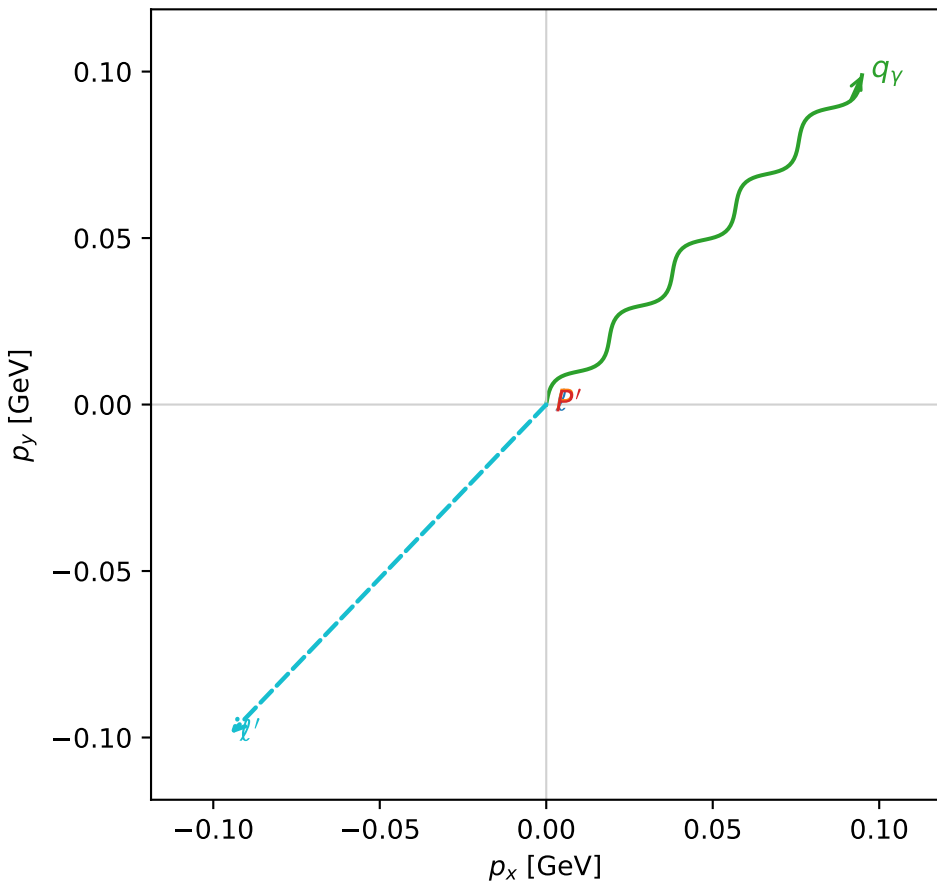
$s=1.55424$, $\sqrt{s}=1.24669$, $|\vec{P}|=0.270475$, $|\vec{P}'|=0.120041$
 $\theta_{P'}=0$, $\theta_V=1.867$, $\phi_{P'}=0.289594$, $\phi_V=0.806473$
 $E_V=0.143262$, $\theta_{\ell'}=2.08953$, $\phi_{\ell'}=3.94807$
 $Q^2=0.0430356$, $x_B=0.132815$, $t=-0.0216959$, $W^2=1.16084$, $y=0.480469$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2705$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2705$
 P $E= 0.9762$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2705$
 ℓ' $E= 0.1578$ $p_x= -0.0948$ $p_y= -0.0989$ $p_z= -0.0782$
 P' $E= 0.9456$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1200$
 q_V $E= 0.1433$ $p_x= 0.0948$ $p_y= 0.0989$ $p_z= -0.0418$

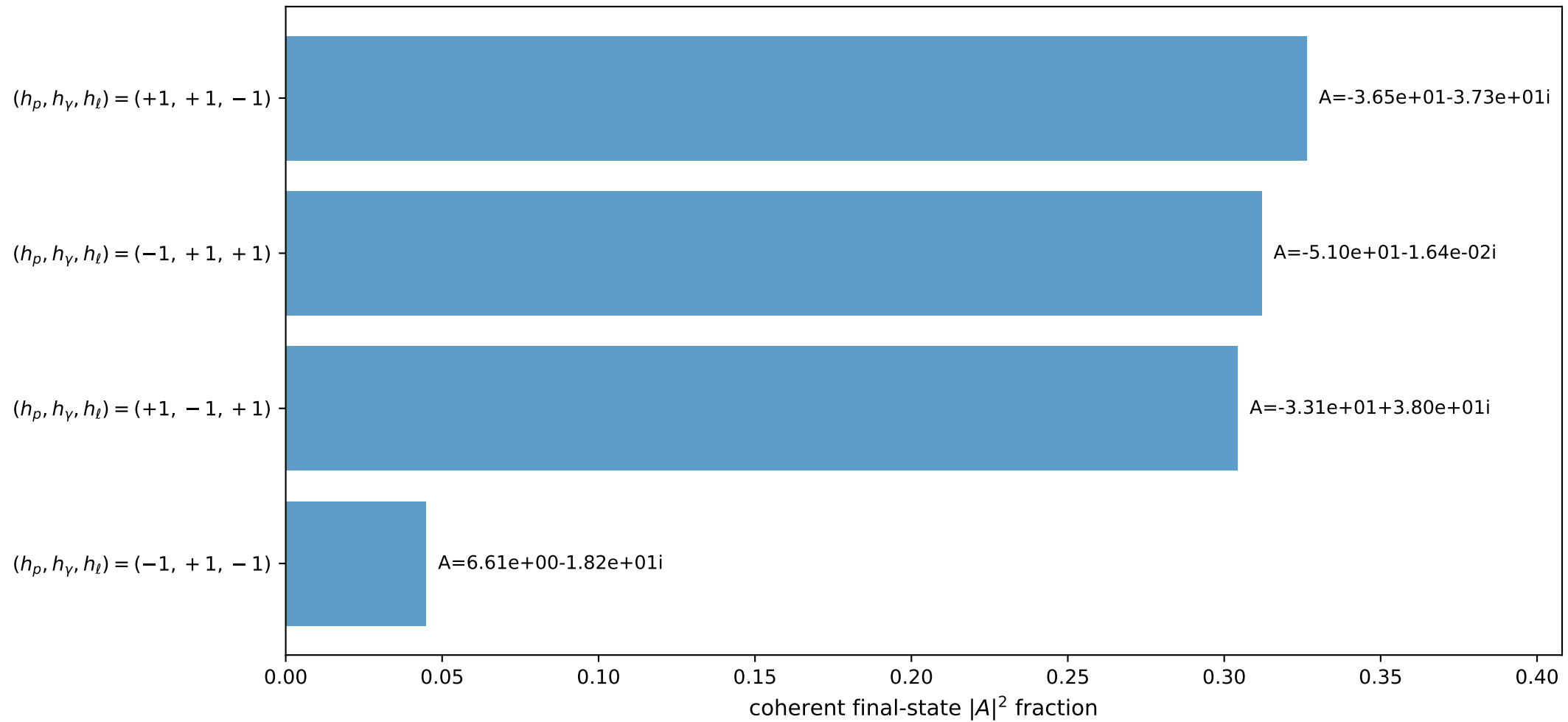
3D momenta



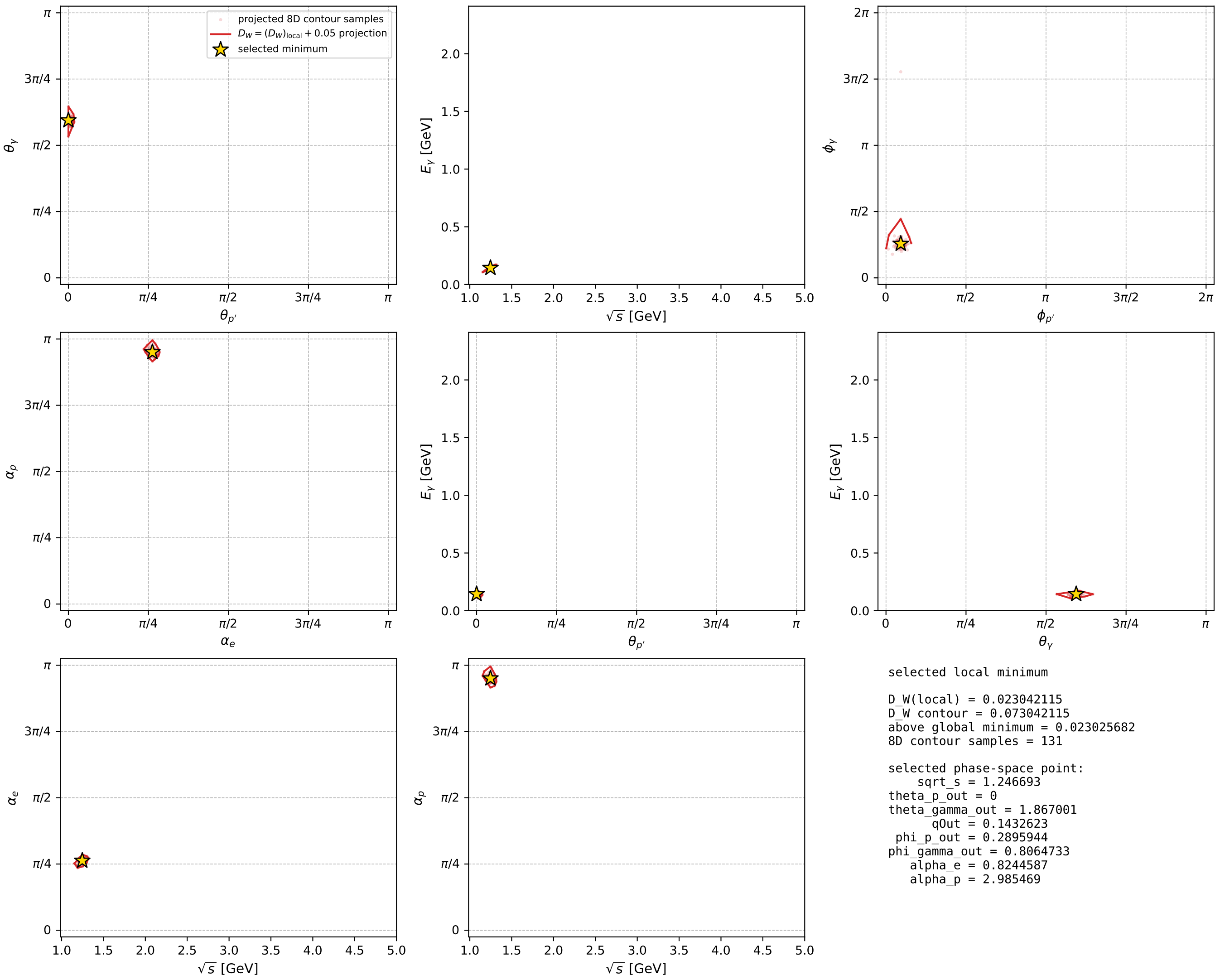
Transverse plane



Leading final-state amplitudes (N=4, retained=98.8%)



electron: polarization cluster P3, local minimum 1054; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



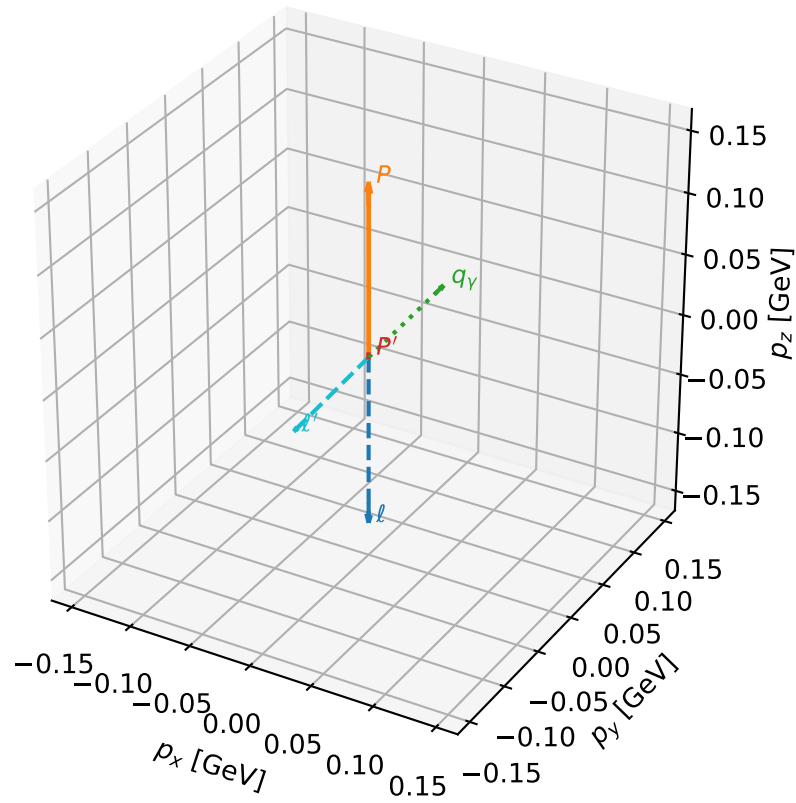
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1060 D_W=0.0237816
 region: polarization_cluster_03_local_minimum_1060
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.82699937$ rad
 incoming lepton: $0.677087|+\rangle + 0.735903|-\rangle$
 proton mixing angle: $\alpha_p=2.973027$ rad
 incoming proton: $-0.985826|+\rangle + 0.167768|-\rangle$

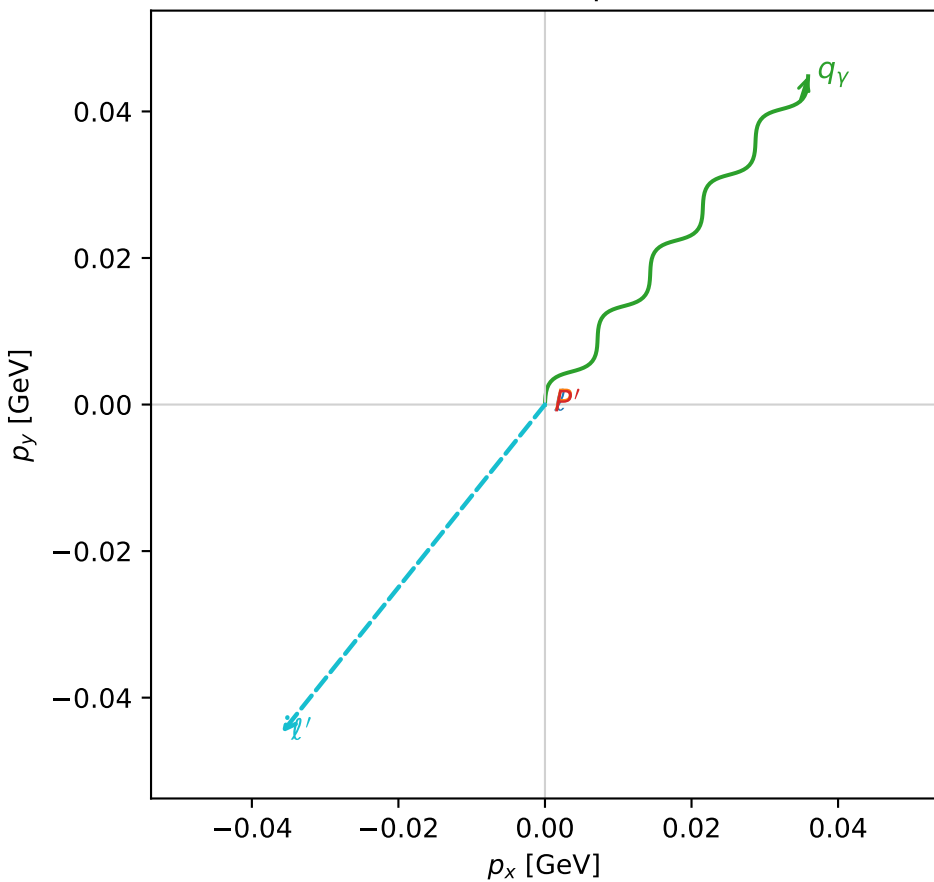
$s=1.18344$, $\sqrt{s}=1.08786$, $|\vec{P}|=0.139539$, $|\vec{P}'|=0.00211512$
 $\theta_{P'}=0$, $\theta_Y=0.884025$, $\phi_{P'}=0.803886$, $\phi_Y=0.895065$
 $E_Y=0.0742496$, $\theta_{\ell'}=2.2792$, $\phi_{\ell'}=4.03666$
 $Q^2=0.00737217$, $x_B=0.0503337$, $t=-0.0187788$, $W^2=1.01894$, $y=0.48243$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1395$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1395$
 P $E= 0.9483$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1395$
 ℓ' $E= 0.0756$ $p_x= -0.0359$ $p_y= -0.0448$ $p_z= -0.0492$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0021$
 q_Y $E= 0.0742$ $p_x= 0.0359$ $p_y= 0.0448$ $p_z= 0.0471$

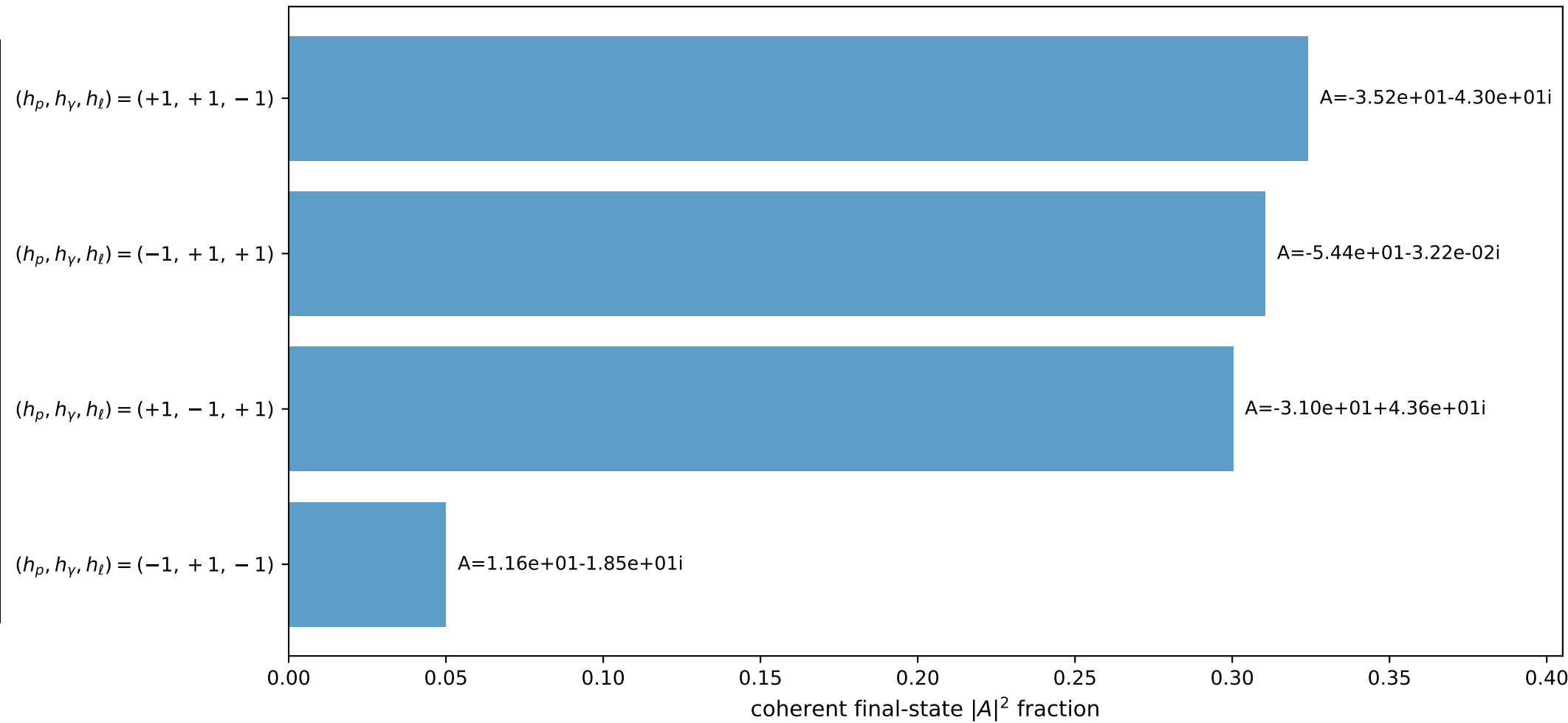
3D momenta



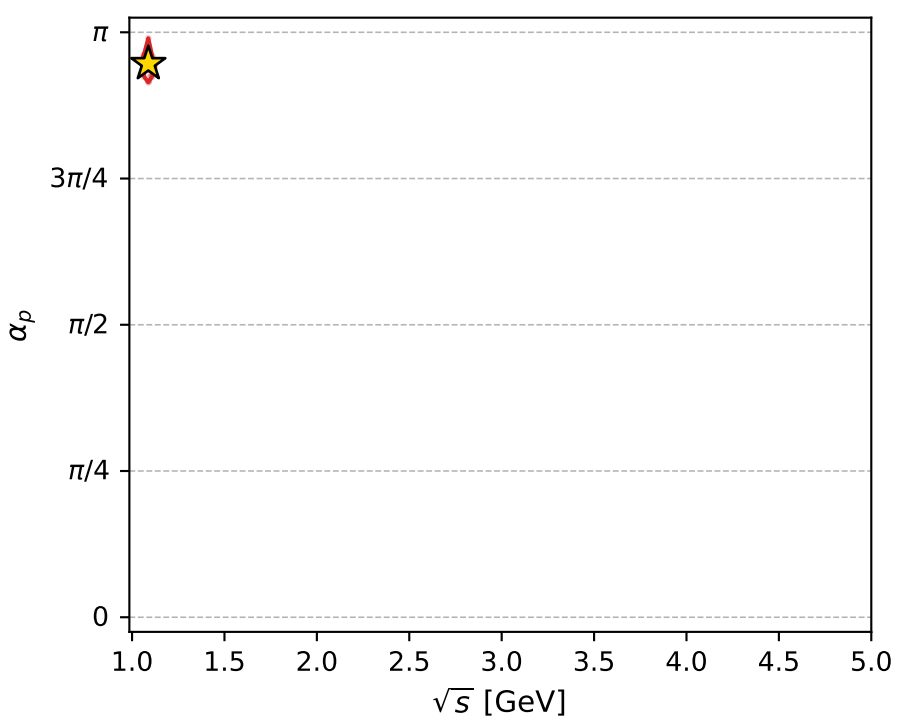
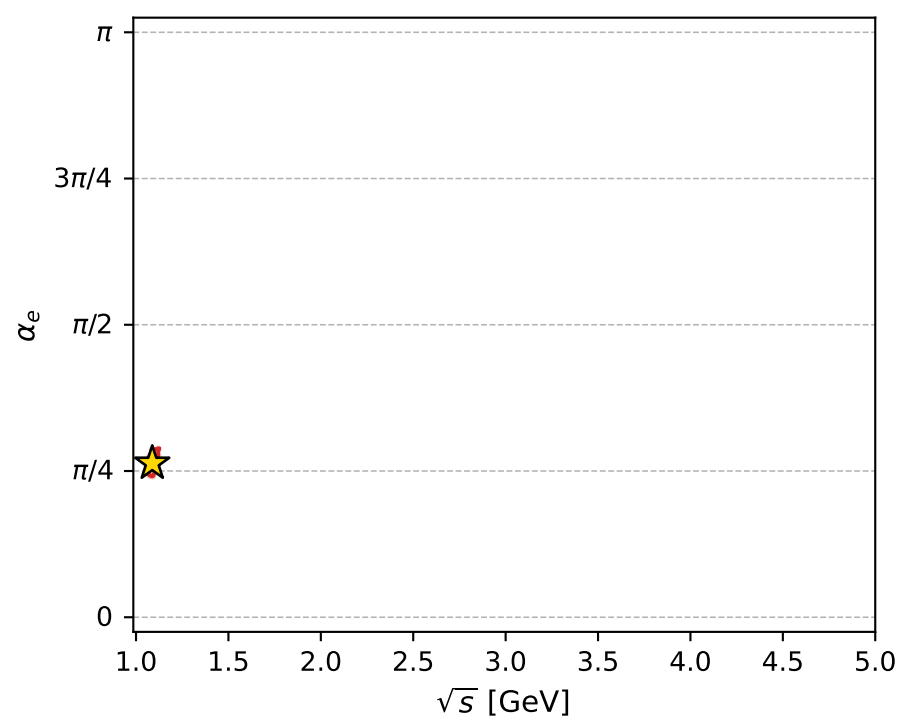
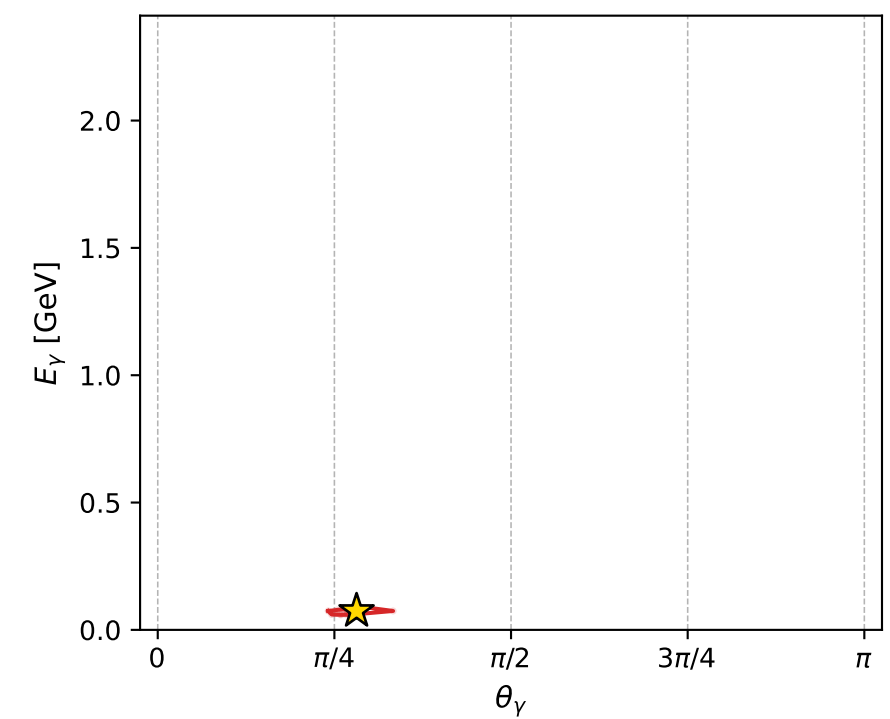
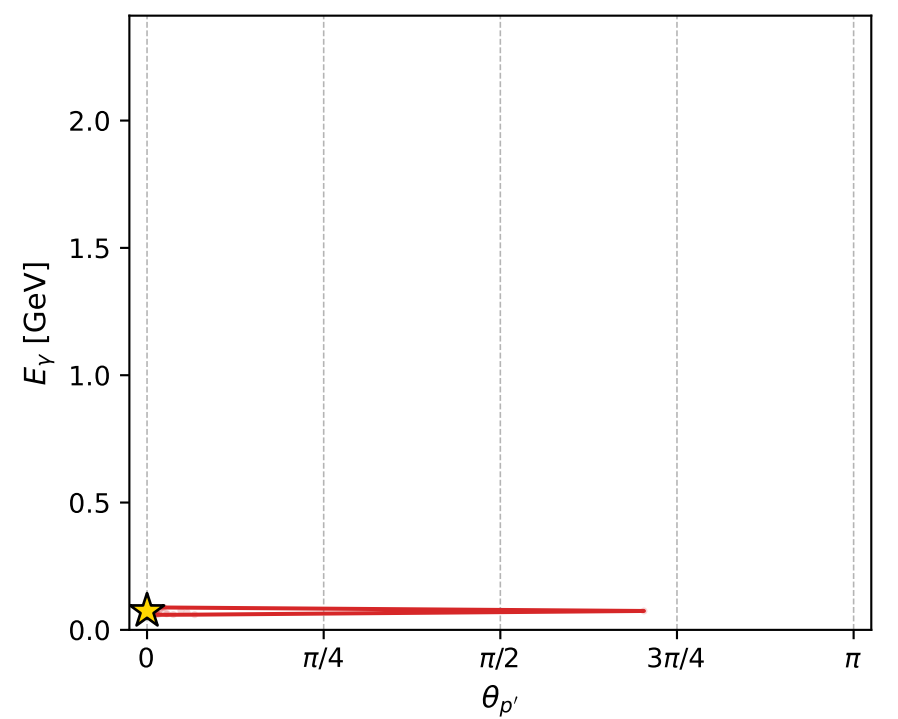
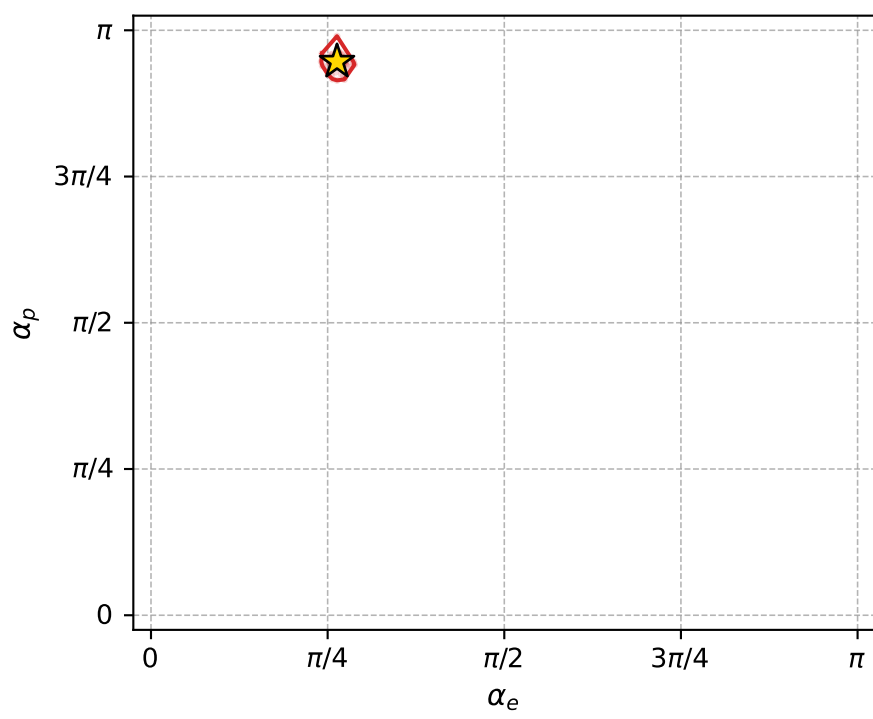
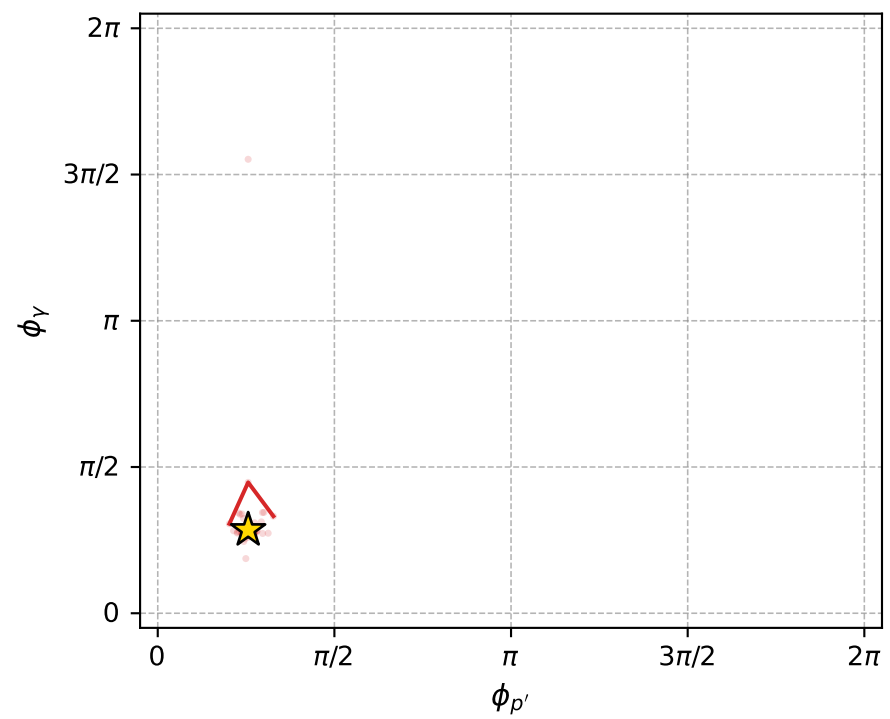
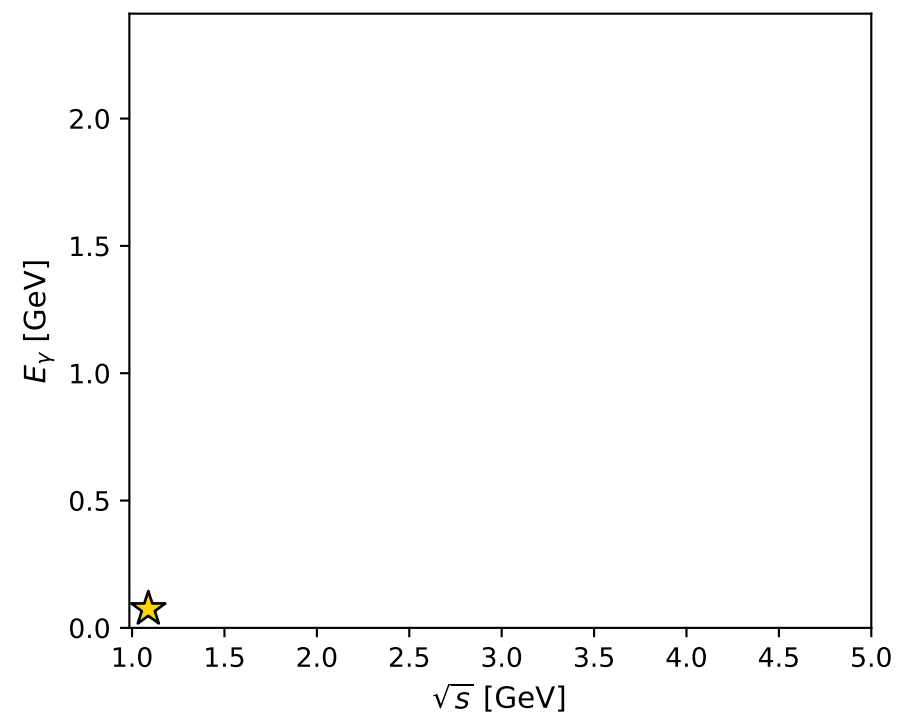
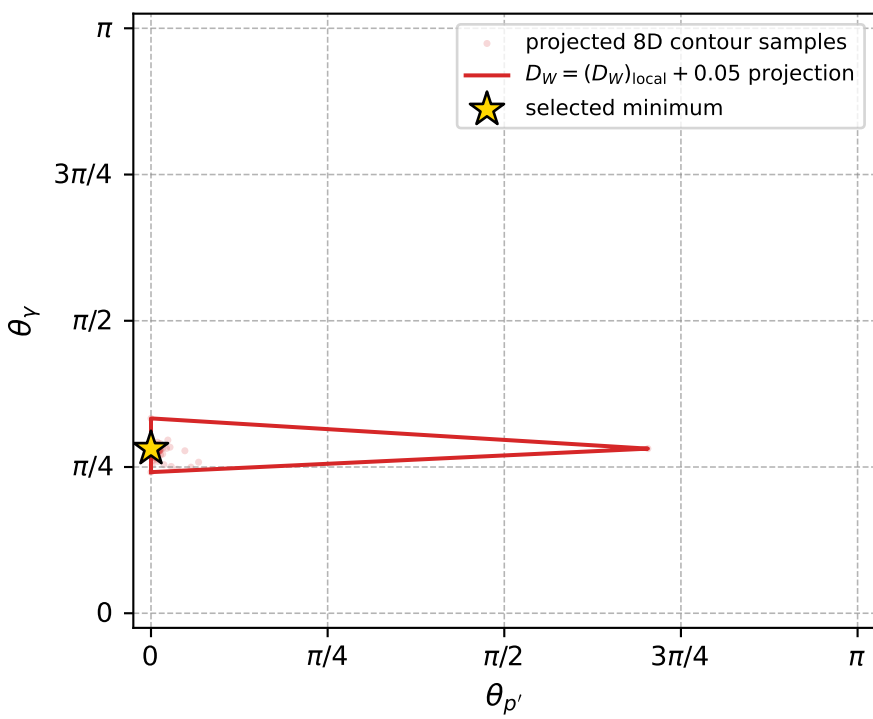
Transverse plane



Leading final-state amplitudes (N=4, retained=98.5%)



electron: polarization cluster P3, local minimum 1060; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.023781625$
 $D_W \text{ contour} = 0.073781625$
 above global minimum = 0.023765192
 8D contour samples = 136

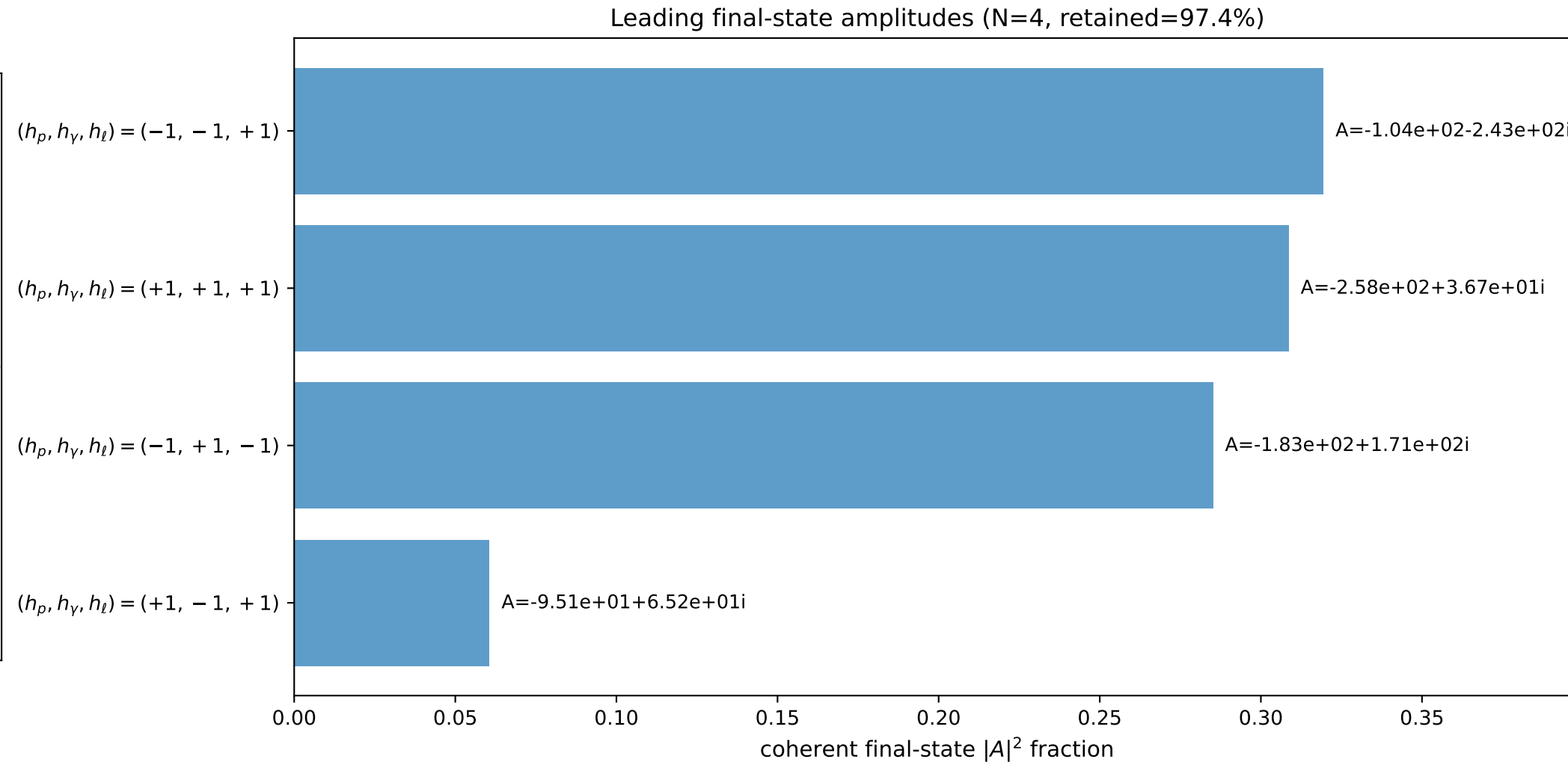
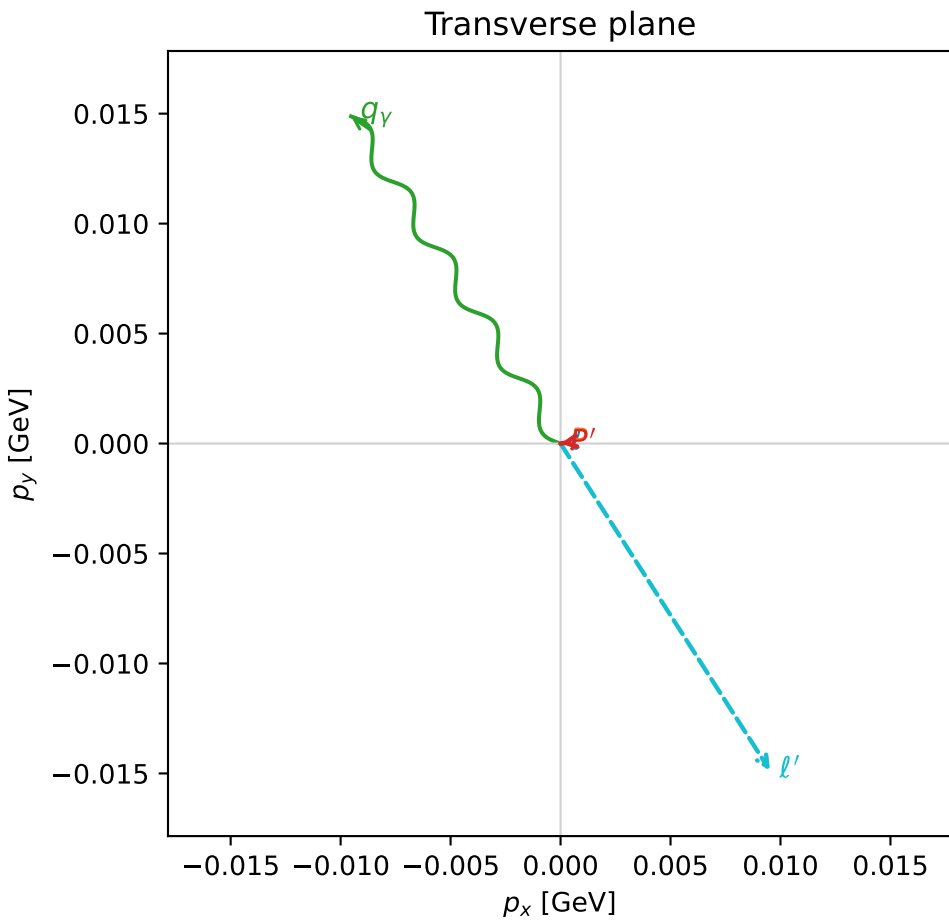
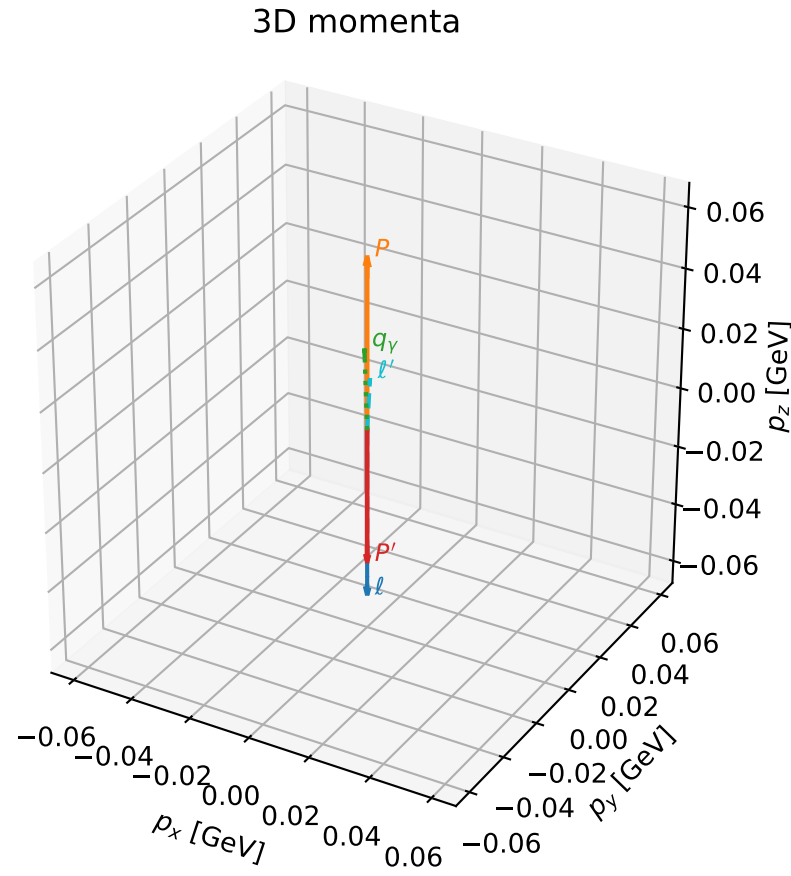
selected phase-space point:
 $\text{sqrt_s} = 1.087862$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 0.8840254$
 $\text{qOut} = 0.07424964$
 $\text{phi_p_out} = 0.8038856$
 $\text{phi_gamma_out} = 0.8950645$
 $\text{alpha_e} = 0.8269994$
 $\text{alpha_p} = 2.973027$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

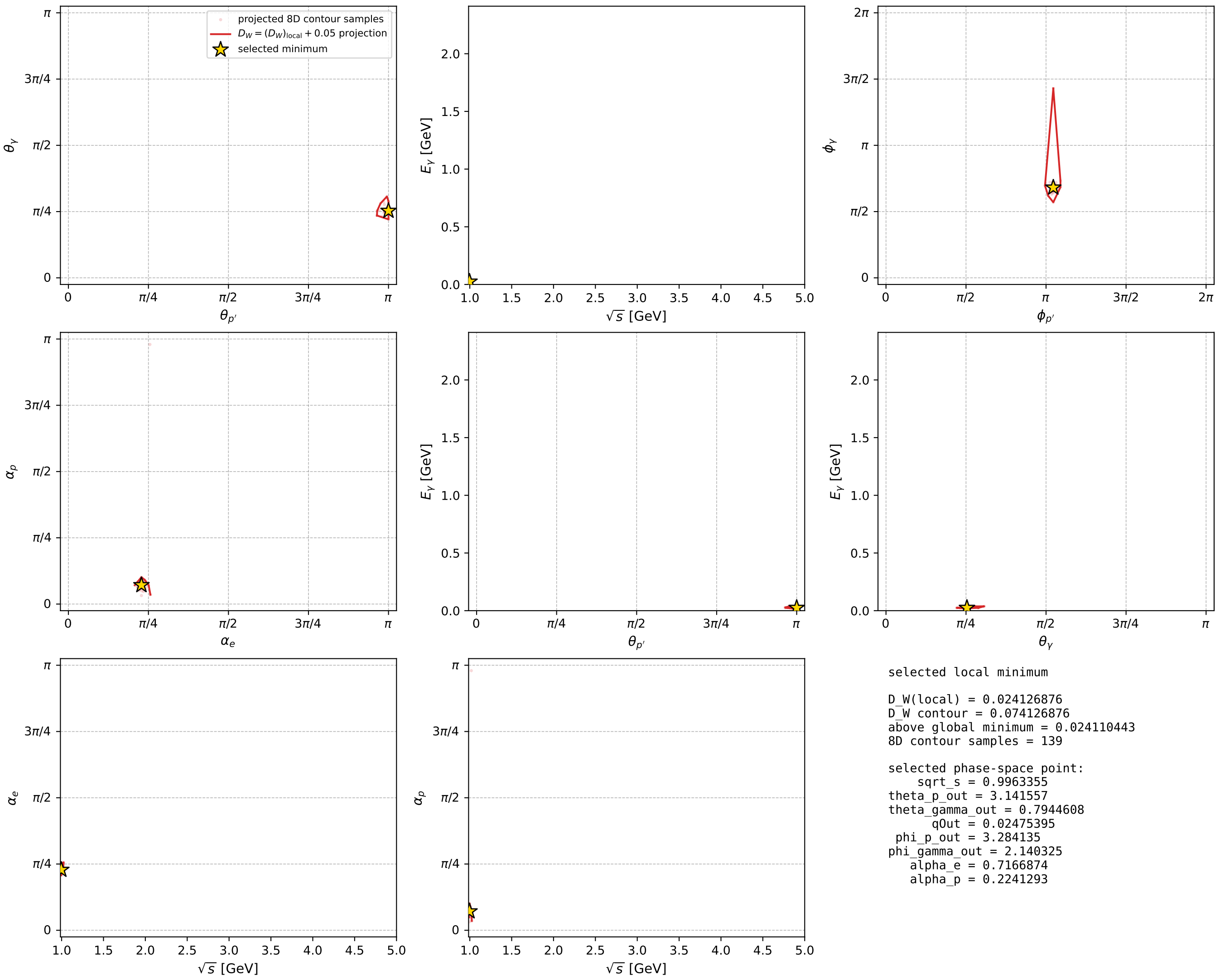
dw_mixing_angles_polarization_03_local_minimum_1062 D_W=0.0241269
 region: polarization_cluster_03_local_minimum_1062
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.71668745$ rad
 incoming lepton: $0.753986|+\rangle +0.656891|-\rangle$
 proton mixing angle: $\alpha_p=0.22412928$ rad
 incoming proton: $0.974988|+\rangle +0.222258|-\rangle$

$s=0.992684$, $\sqrt{s}=0.996335$, $|\vec{P}|=0.0566255$, $|\vec{P}'|=0.0446447$
 $\theta_{p'}=3.14156$, $\theta_Y=0.794461$, $\phi_{p'}=3.28414$, $\phi_Y=2.14032$
 $E_Y=0.0247539$, $\theta_{\ell'}=0.574235$, $\phi_{\ell'}=5.282$
 $Q^2=0.00677431$, $x_B=0.123587$, $t=-0.0102552$, $W^2=0.927884$, $y=0.485767$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.0566$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0566$
 P $E= 0.9397$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0566$
 ℓ' $E= 0.0325$ $p_x= 0.0095$ $p_y= -0.0149$ $p_z= 0.0273$
 P' $E= 0.9391$ $p_x= -0.0000$ $p_y= -0.0000$ $p_z= -0.0446$
 q_Y $E= 0.0248$ $p_x= -0.0095$ $p_y= 0.0149$ $p_z= 0.0173$



electron: polarization cluster P3, local minimum 1062; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



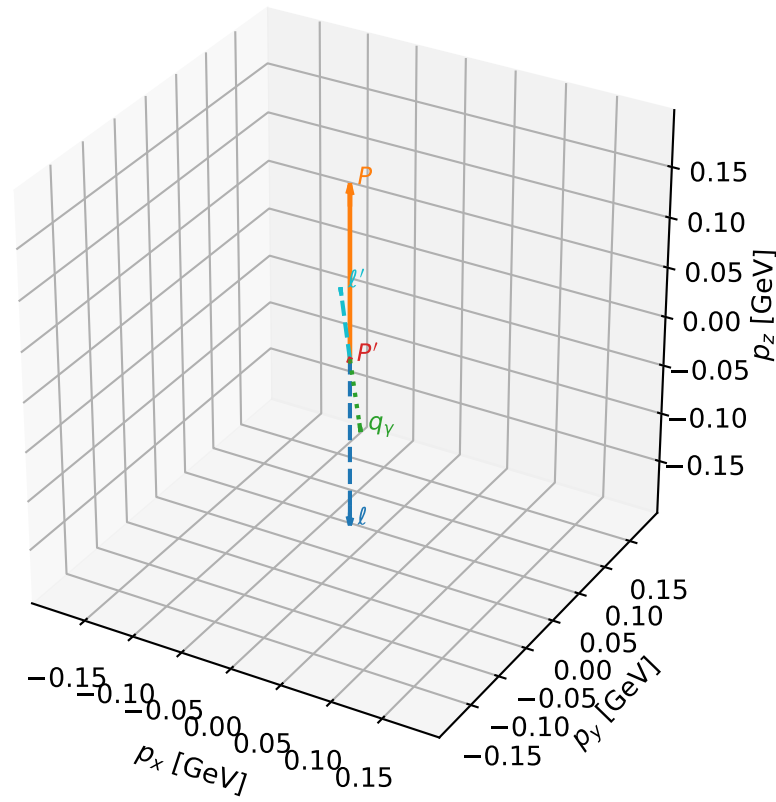
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1065 D_W=0.0242949
 region: polarization_cluster_03_local_minimum_1065
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78815883$ rad
 incoming lepton: $0.705152|+\rangle + 0.709056|-\rangle$
 proton mixing angle: $\alpha_p=3.1268559$ rad
 incoming proton: $-0.999891|+\rangle + 0.0147362|-\rangle$

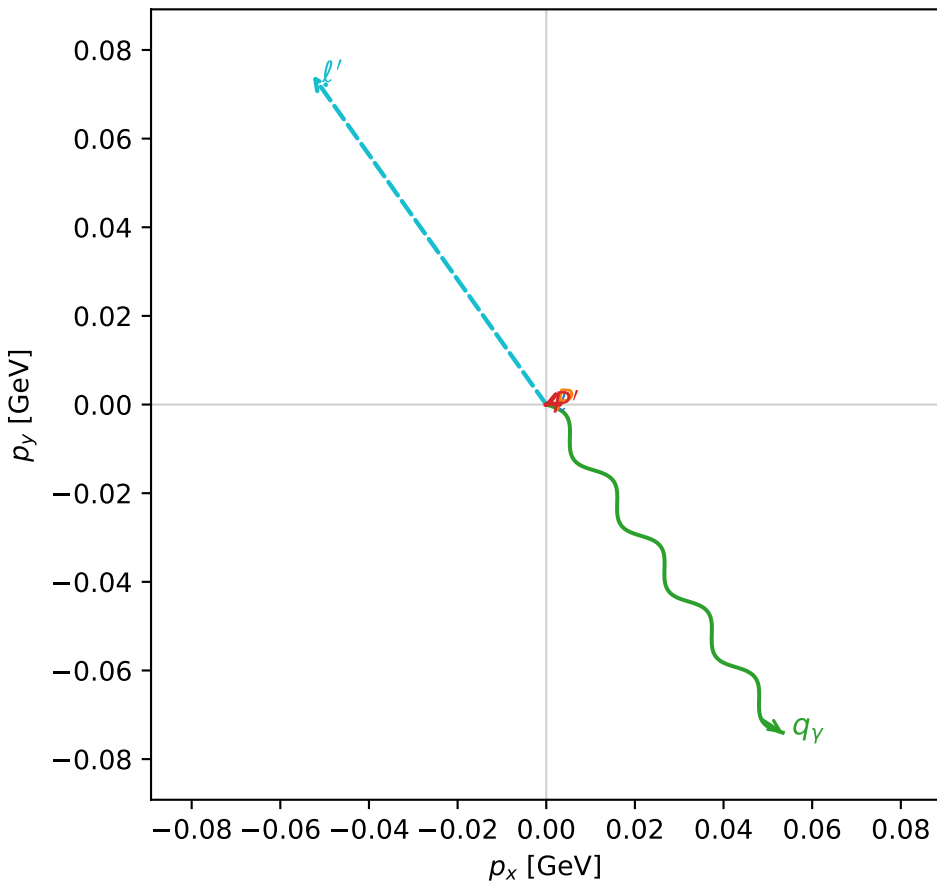
$s=1.26553$, $\sqrt{s}=1.12496$, $|\vec{P}|=0.171422$, $|\vec{P}'|=0.00195172$
 $\theta_{P'}=2.77687$, $\theta_V=1.7831$, $\phi_{P'}=3.55242$, $\phi_V=5.33691$
 $E_V=0.0933475$, $\theta_l=1.33913$, $\phi_{l'}=2.18785$
 $Q^2=0.0394615$, $x_B=0.183938$, $t=-0.0297732$, $W^2=1.05492$, $y=0.556246$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1714$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1714$
 P $E= 0.9535$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1714$
 l' $E= 0.0936$ $p_x= -0.0527$ $p_y= 0.0743$ $p_z= 0.0215$
 P' $E= 0.9380$ $p_x= -0.0006$ $p_y= -0.0003$ $p_z= -0.0018$
 q_V $E= 0.0933$ $p_x= 0.0534$ $p_y= -0.0740$ $p_z= -0.0197$

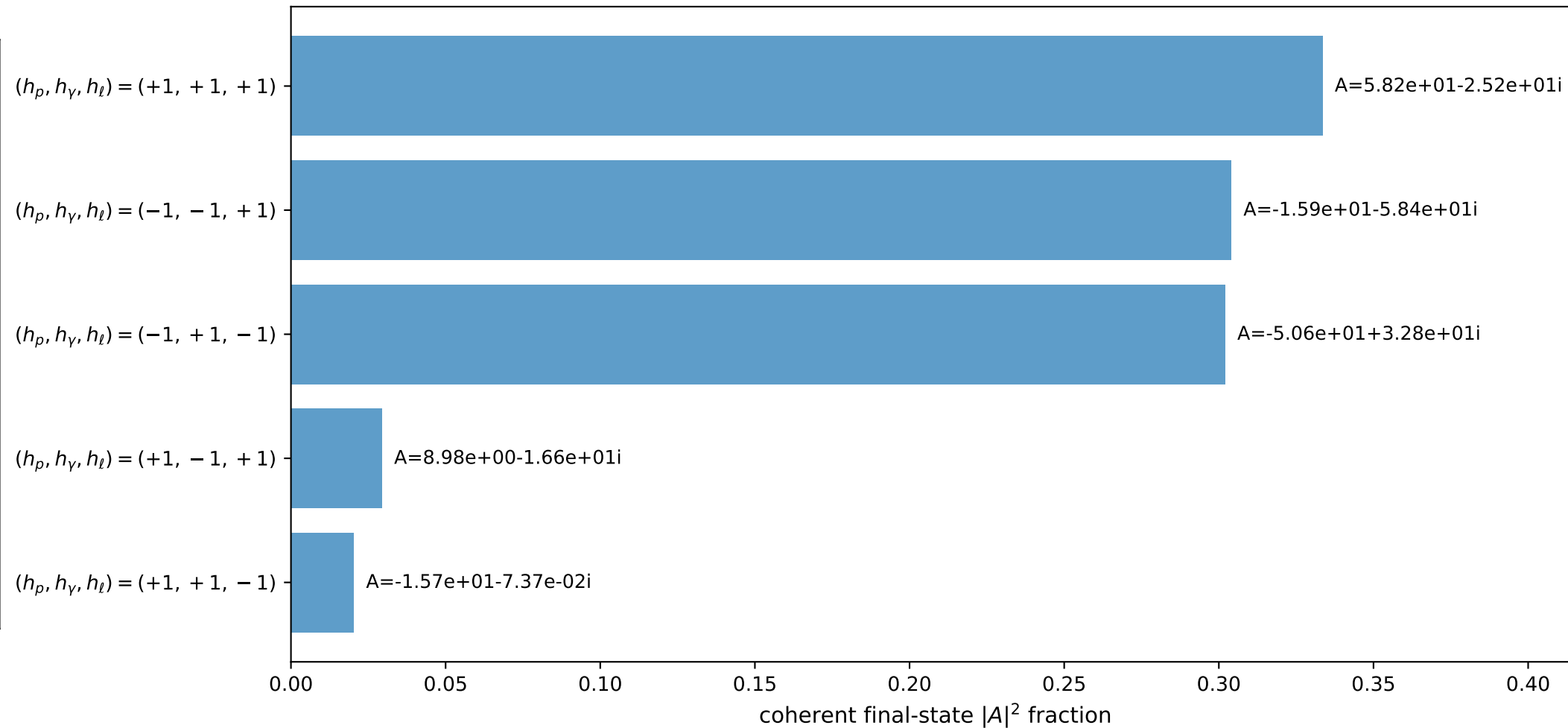
3D momenta



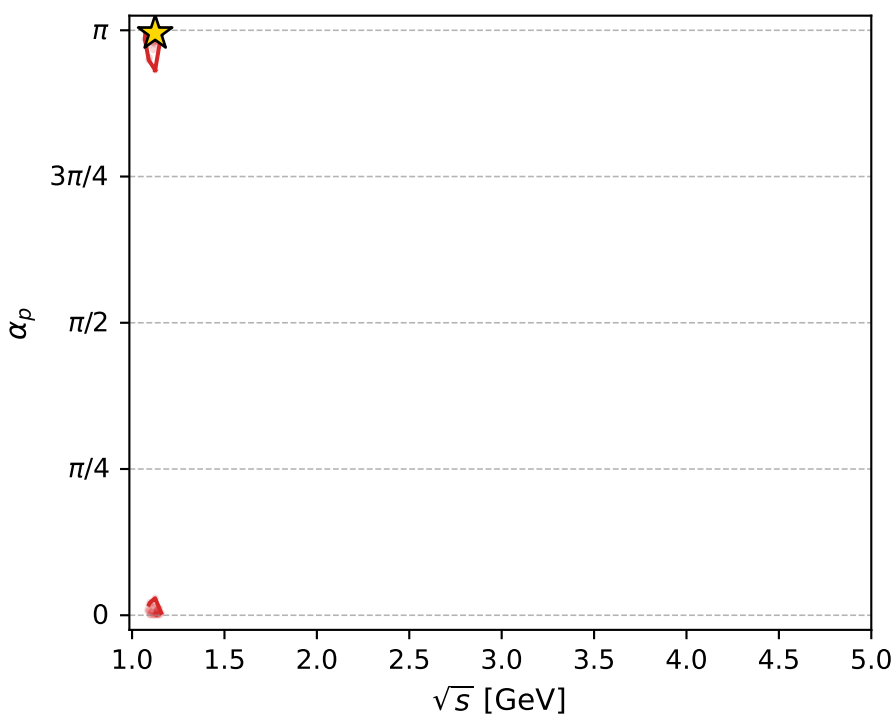
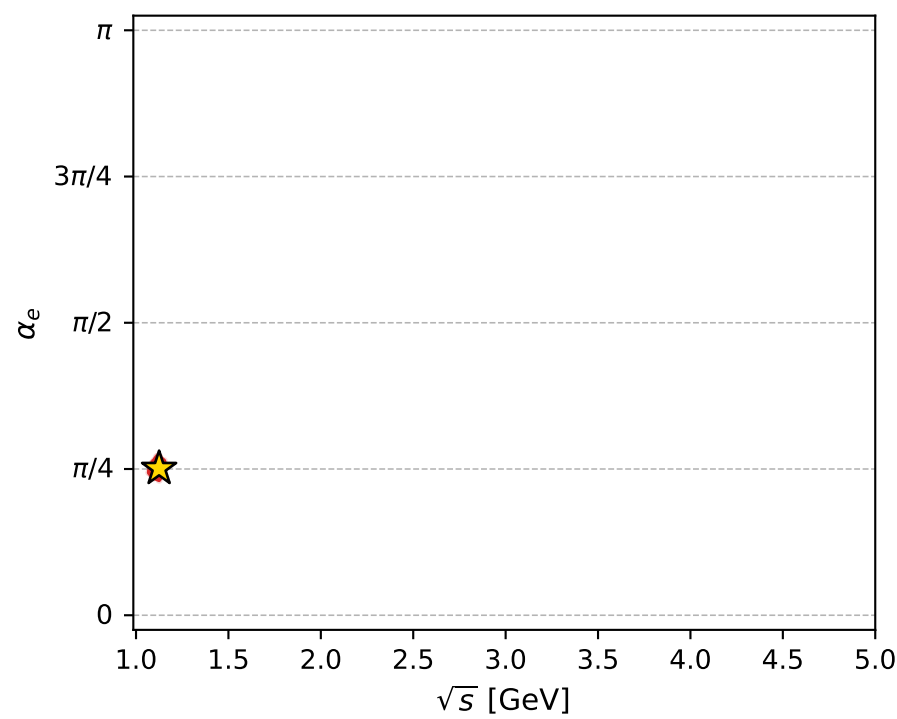
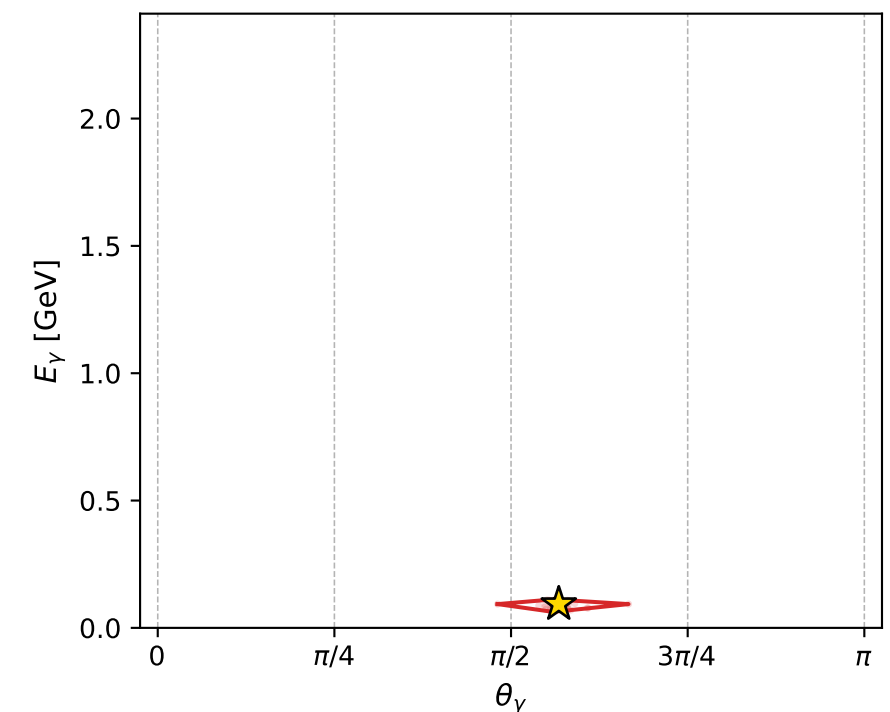
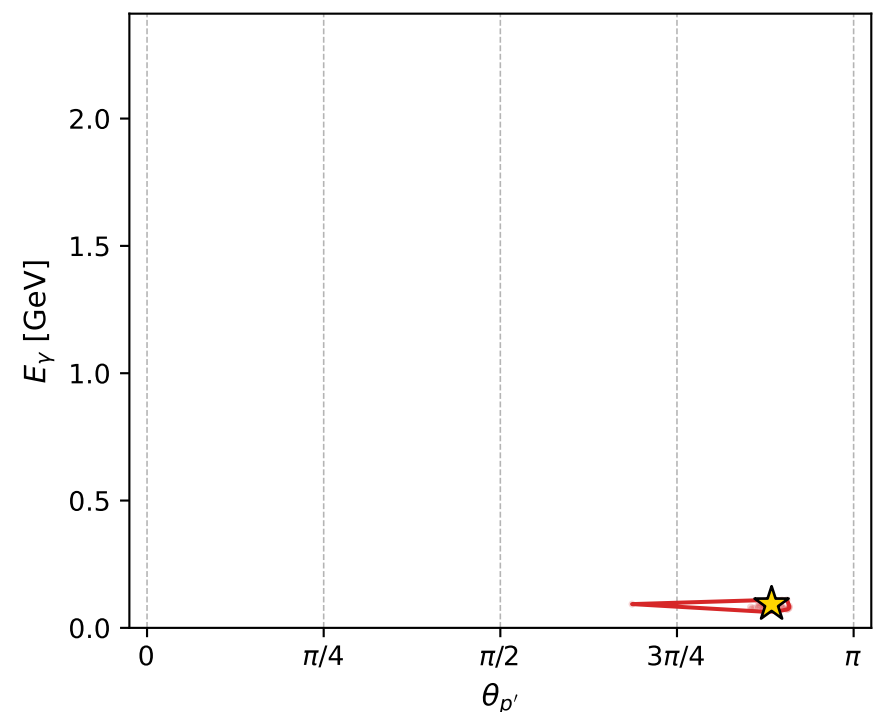
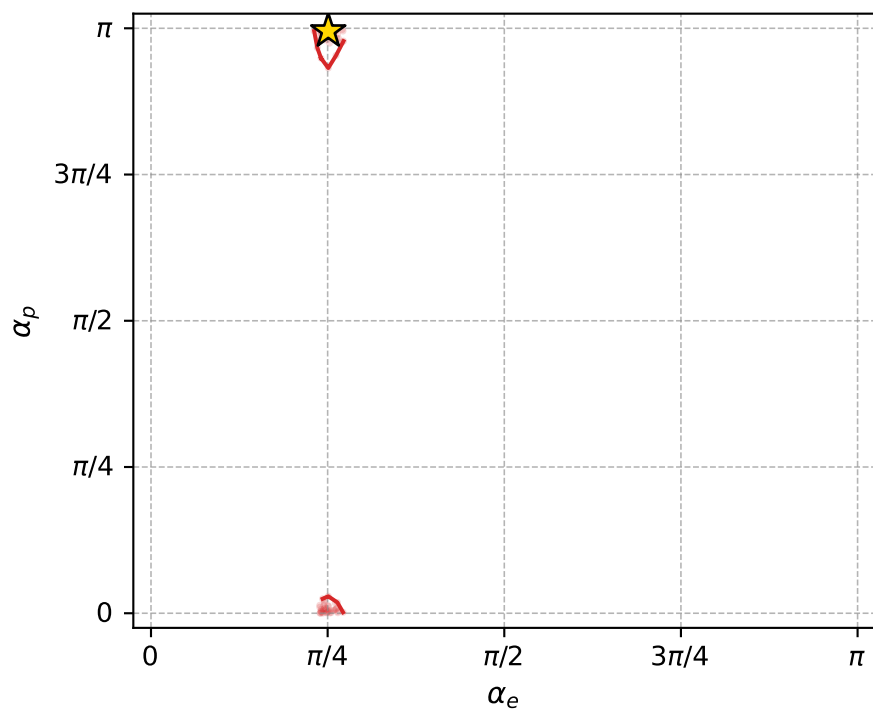
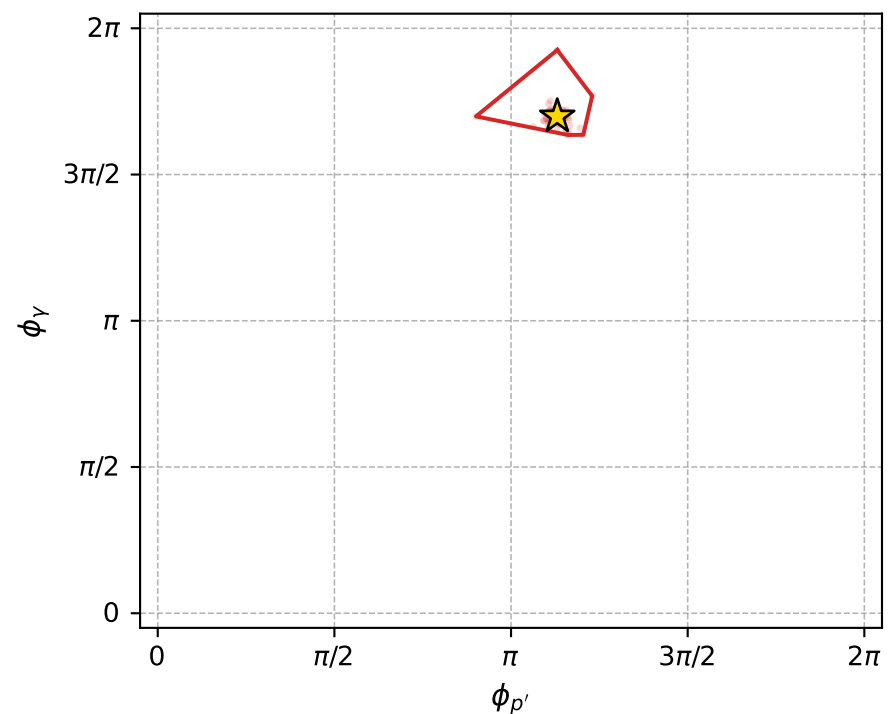
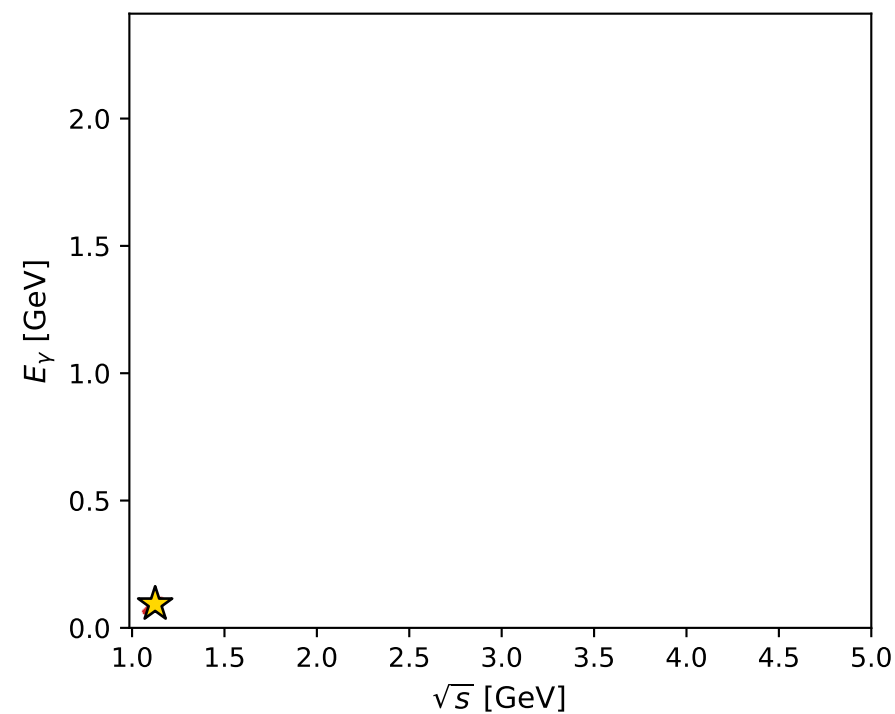
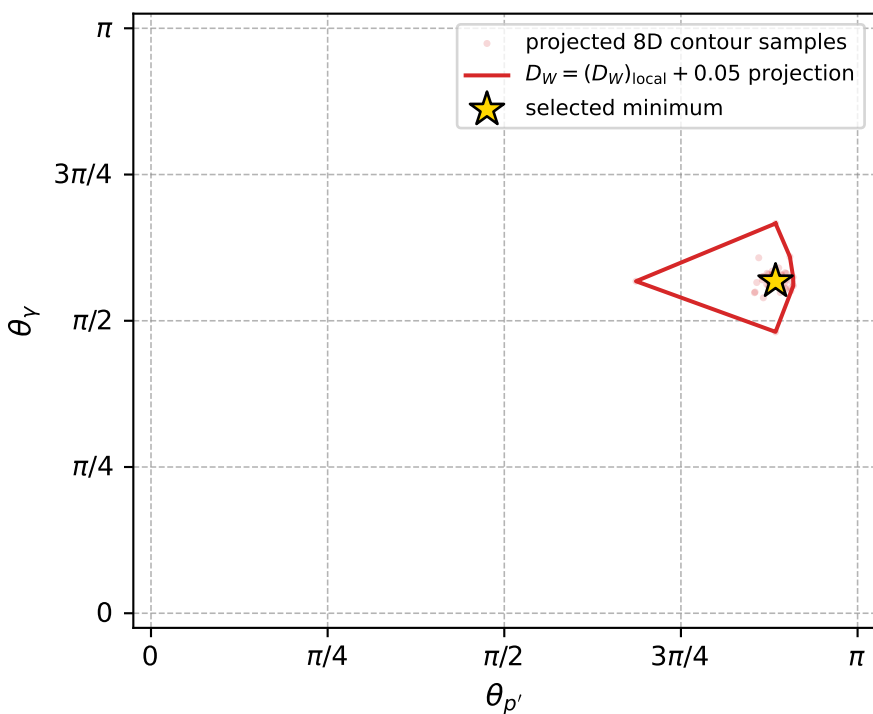
Transverse plane



Leading final-state amplitudes (N=5, retained=99.0%)



electron: polarization cluster P3, local minimum 1065; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.02429485$
 $D_W \text{ contour} = 0.07429485$
 above global minimum = 0.024278417
 8D contour samples = 253

selected phase-space point:
 $\text{sqrt_s} = 1.124958$
 $\text{theta_p_out} = 2.776866$
 $\text{theta_gamma_out} = 1.783098$
 $\text{qOut} = 0.09334747$
 $\text{phi_p_out} = 3.552421$
 $\text{phi_gamma_out} = 5.336909$
 $\text{alpha_e} = 0.7881588$
 $\text{alpha_p} = 3.126856$

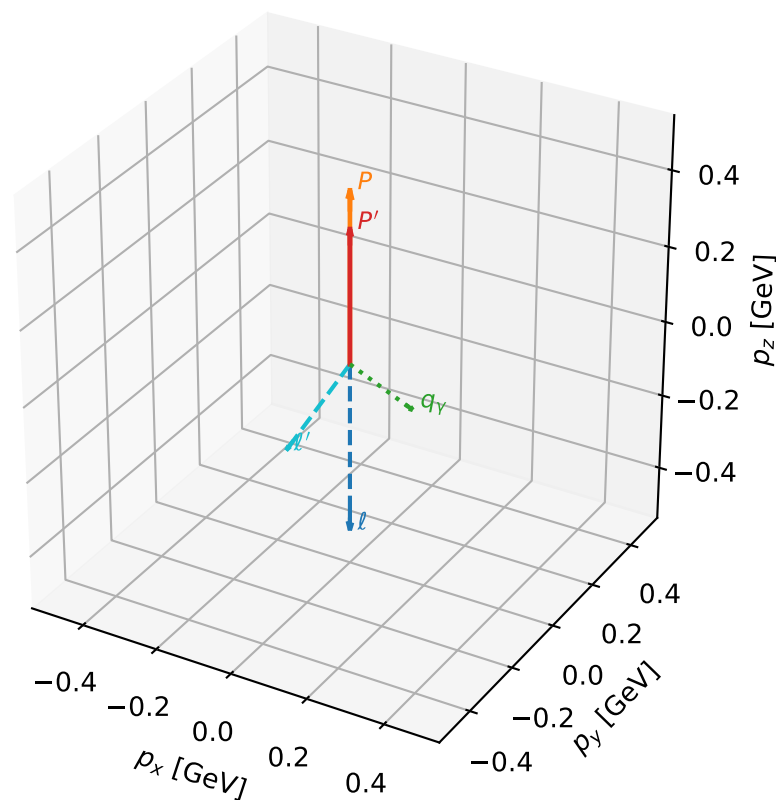
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1077 D_W=0.0250144
 region: polarization_cluster_03_local_minimum_1077
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.83795502$ rad
 incoming lepton: $0.668984|+\rangle + 0.743277|-\rangle$
 proton mixing angle: $\alpha_p=2.9359354$ rad
 incoming proton: $-0.978927|+\rangle + 0.204211|-\rangle$

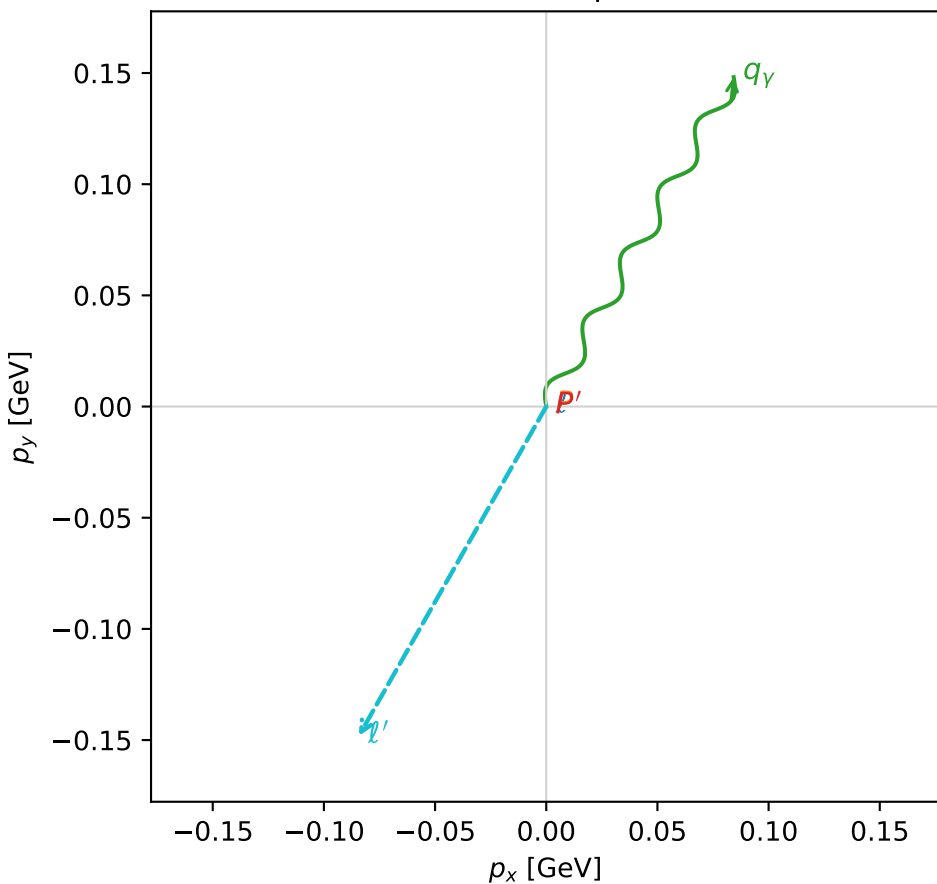
$s=2.22873$, $\sqrt{s}=1.49289$, $|\vec{P}|=0.451769$, $|\vec{P}'|=0.352953$
 $\theta_{P'}=0$, $\theta_V=2.36442$, $\phi_{P'}=1.44646$, $\phi_V=1.05342$
 $E_V=0.243023$, $\theta_{\ell'}=2.38269$, $\phi_{\ell'}=4.19501$
 $Q^2=0.0614057$, $x_B=0.0915378$, $t=-0.00825019$, $W^2=1.48926$, $y=0.497317$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.4518$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.4518$
 P $E= 1.0411$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.4518$
 ℓ' $E= 0.2477$ $p_x= -0.0843$ $p_y= -0.1481$ $p_z= -0.1797$
 P' $E= 1.0022$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3530$
 q_Y $E= 0.2430$ $p_x= 0.0843$ $p_y= 0.1481$ $p_z= -0.1733$

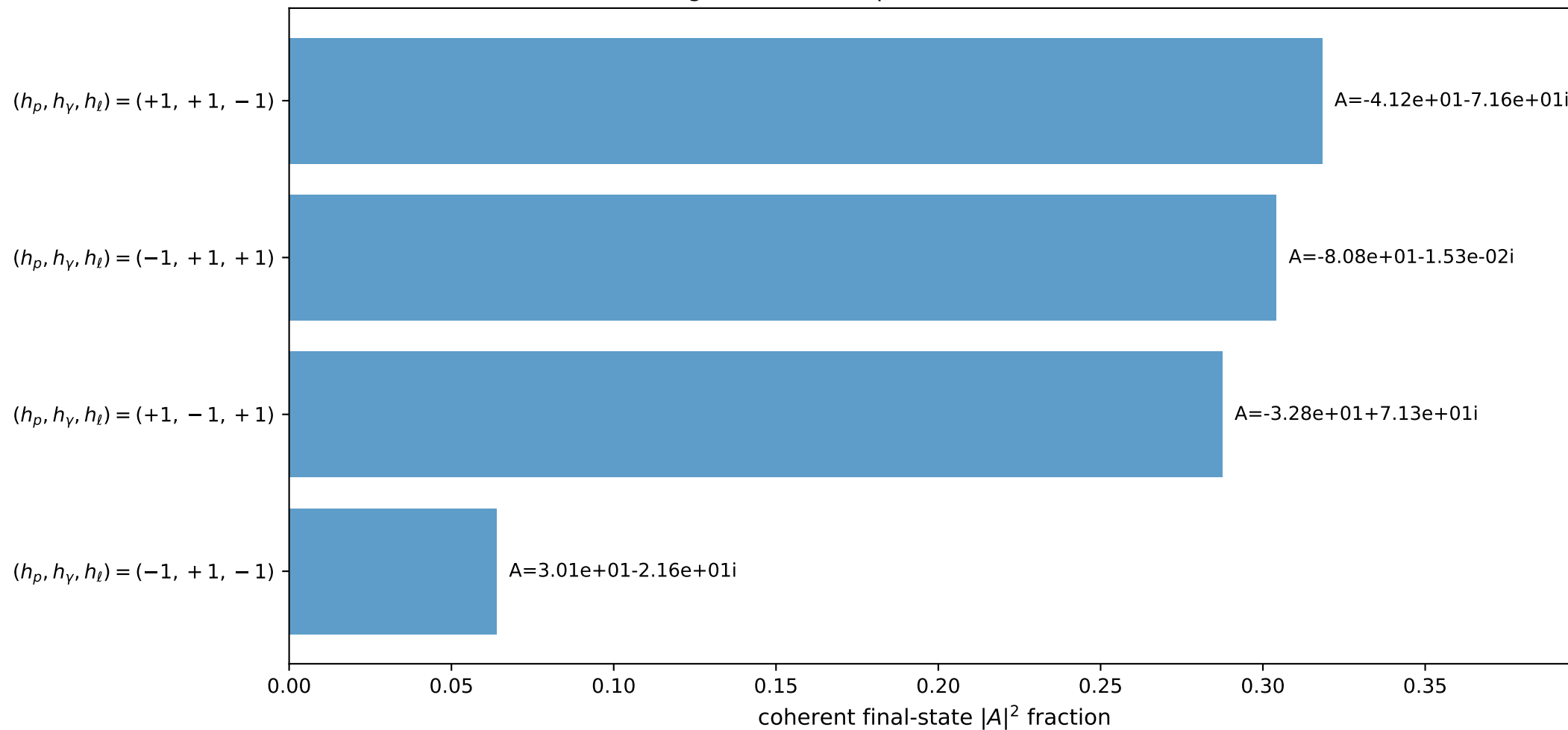
3D momenta



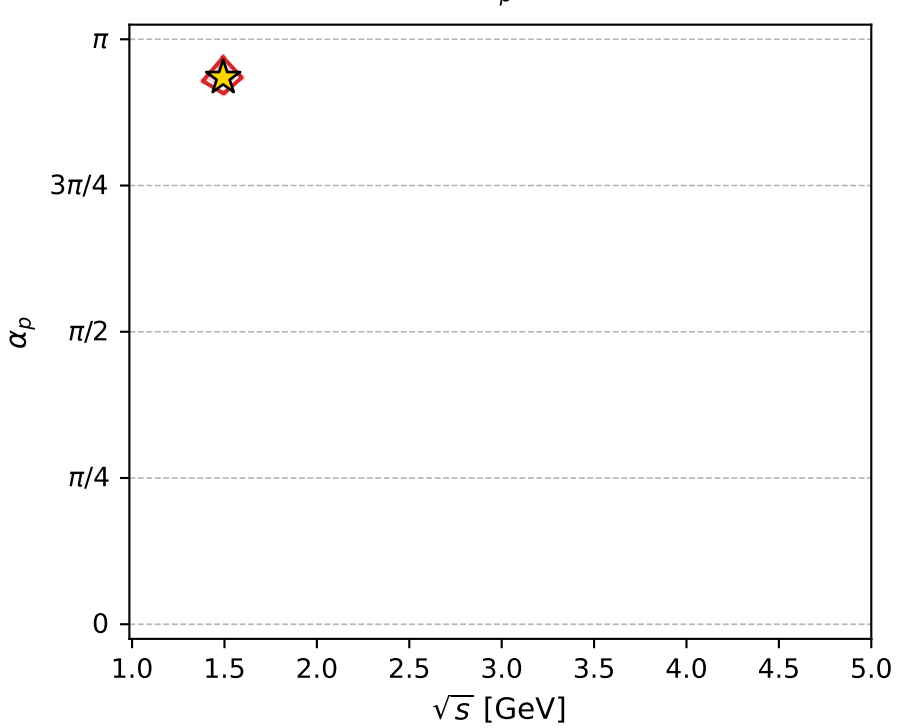
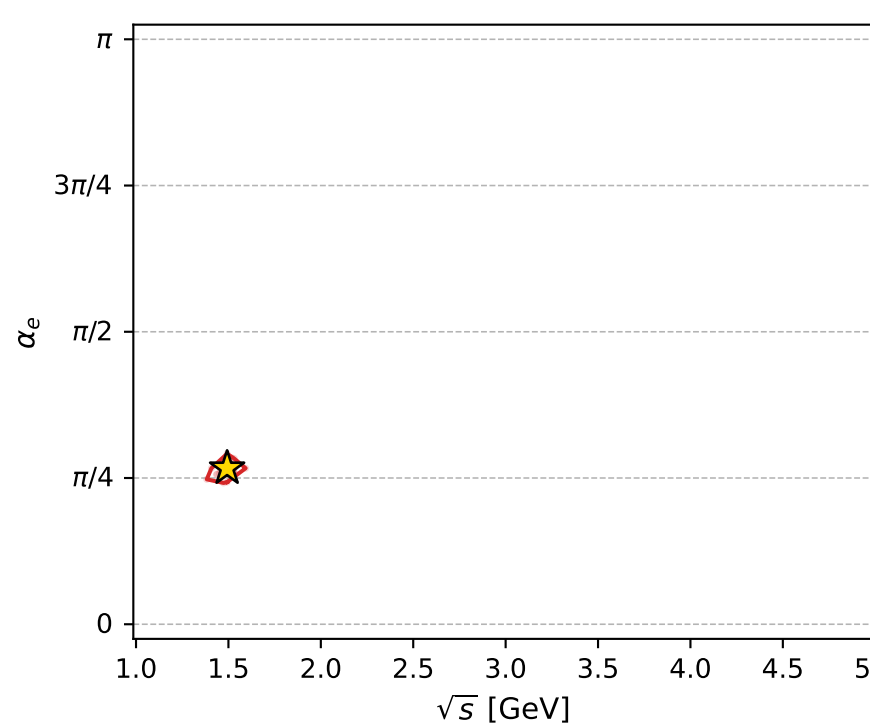
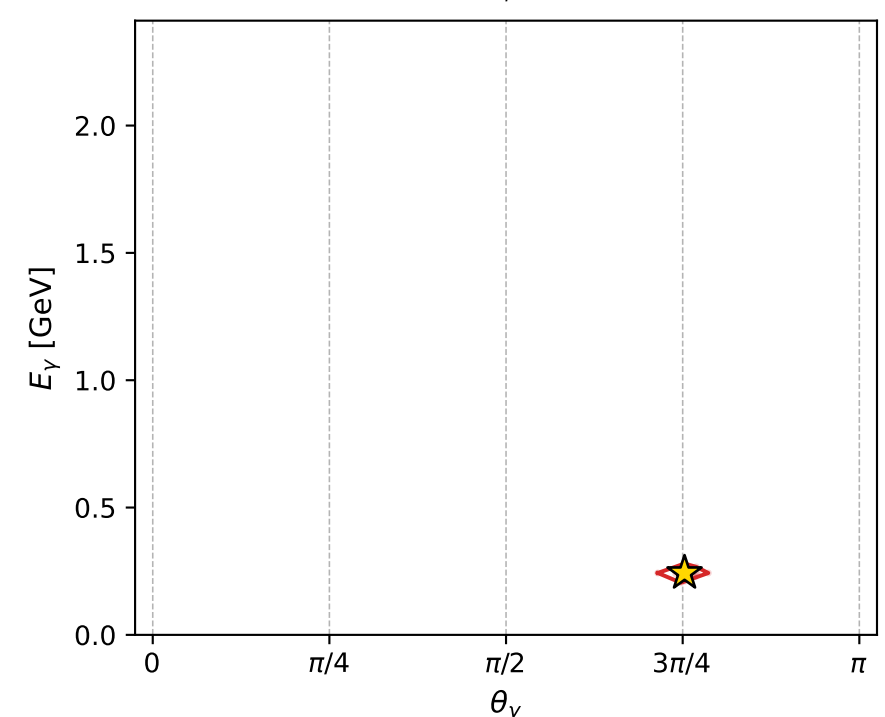
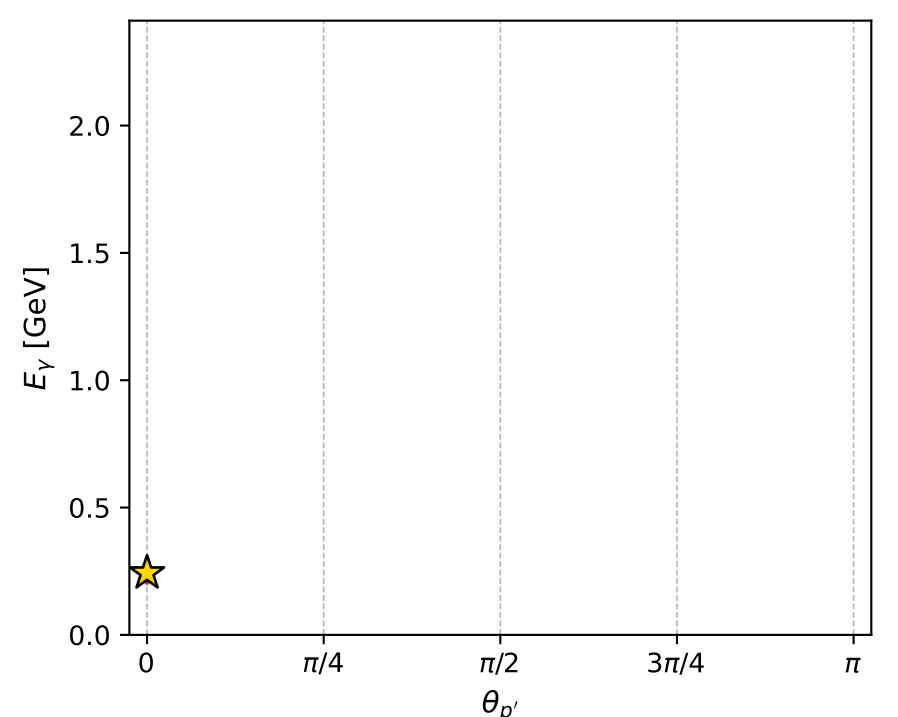
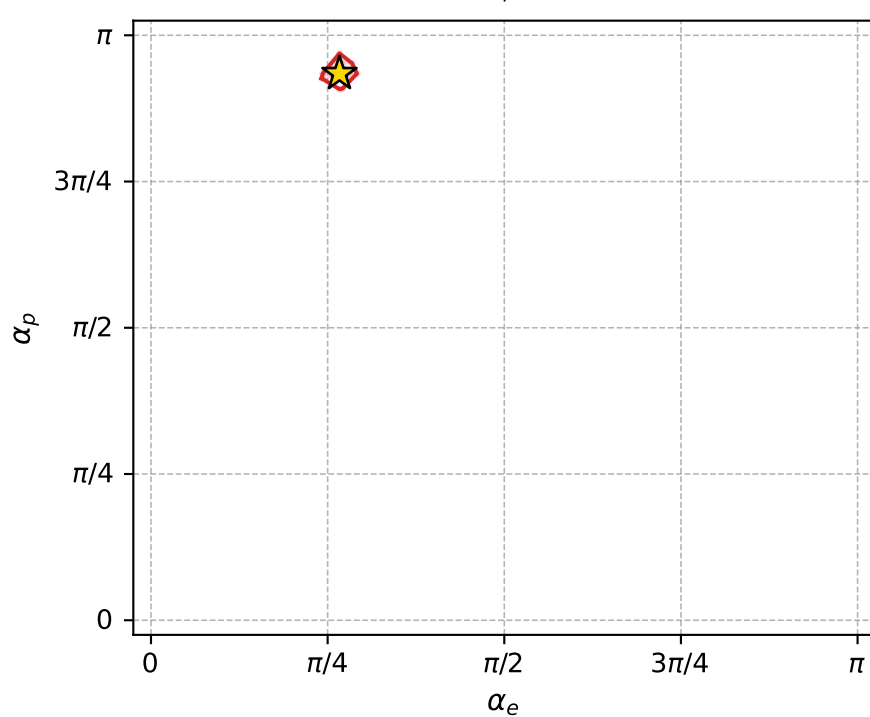
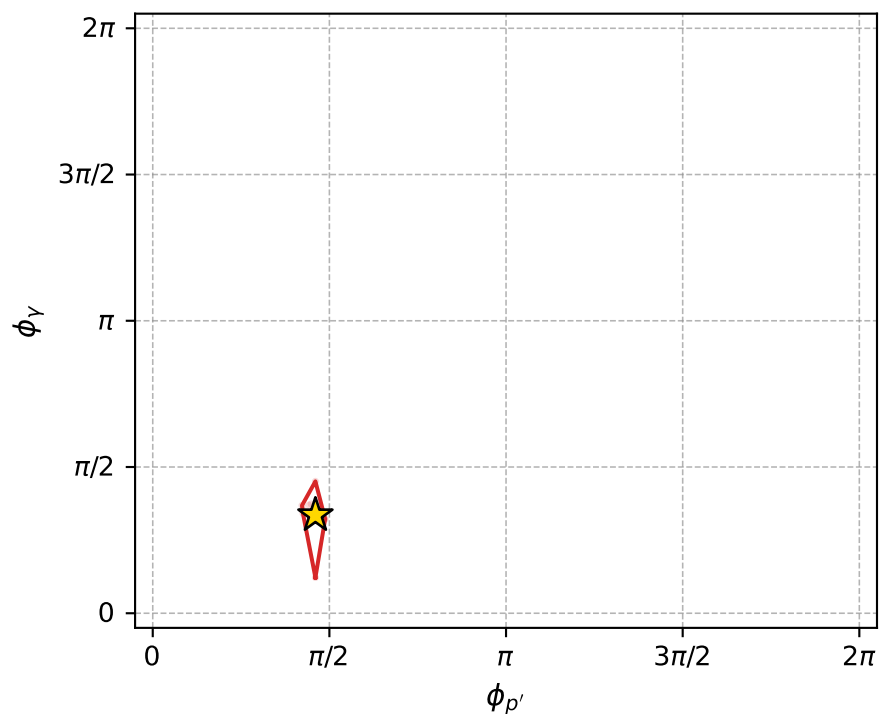
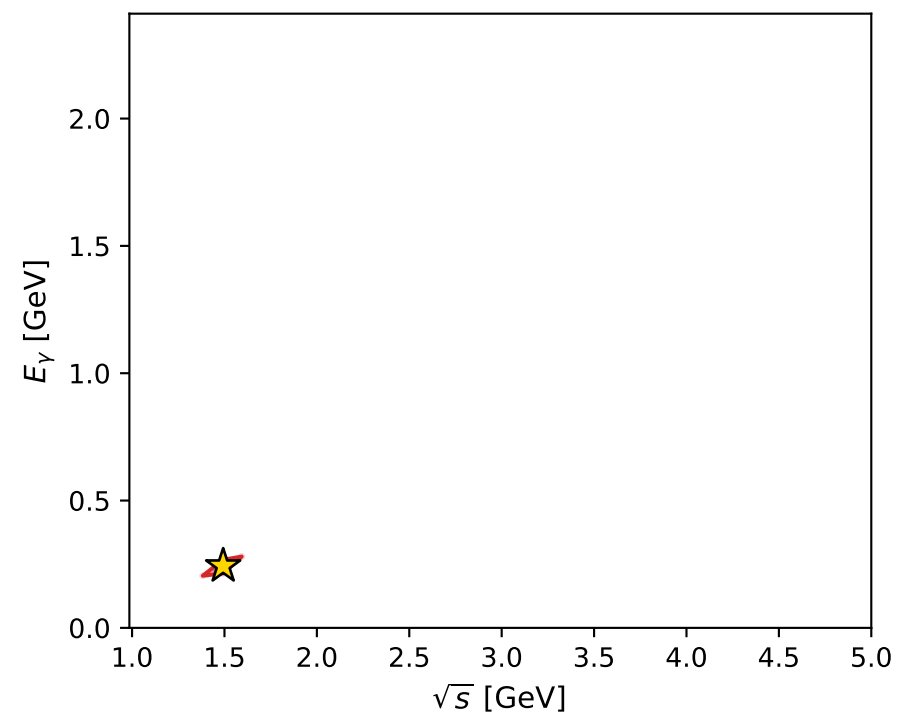
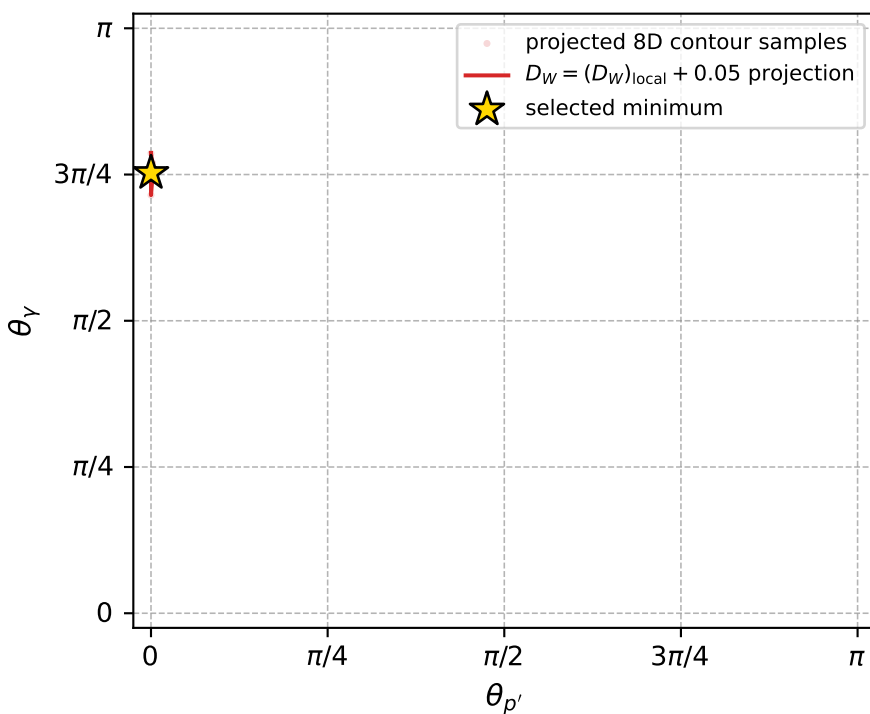
Transverse plane



Leading final-state amplitudes (N=4, retained=97.4%)



electron: polarization cluster P3, local minimum 1077; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.025014444$
 $D_W \text{ contour} = 0.075014444$
 above global minimum = 0.024998011
 8D contour samples = 128

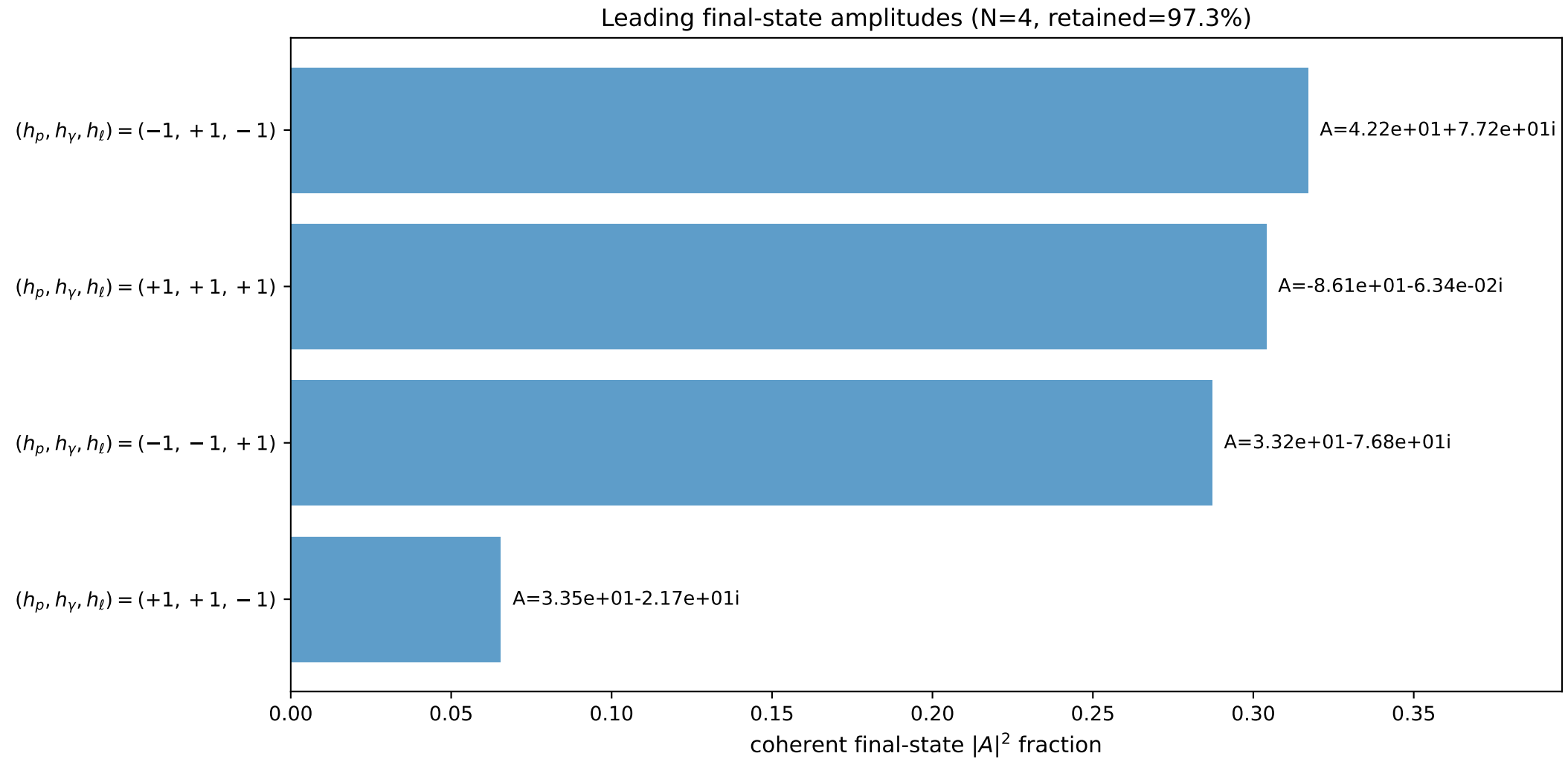
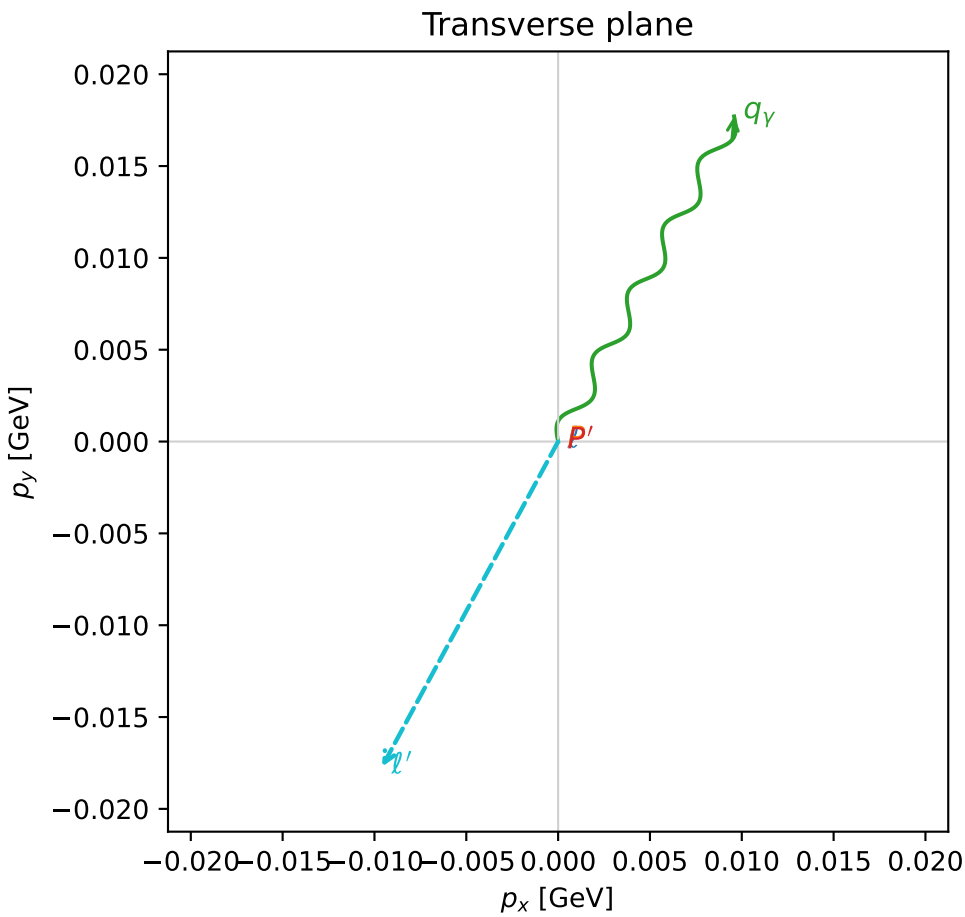
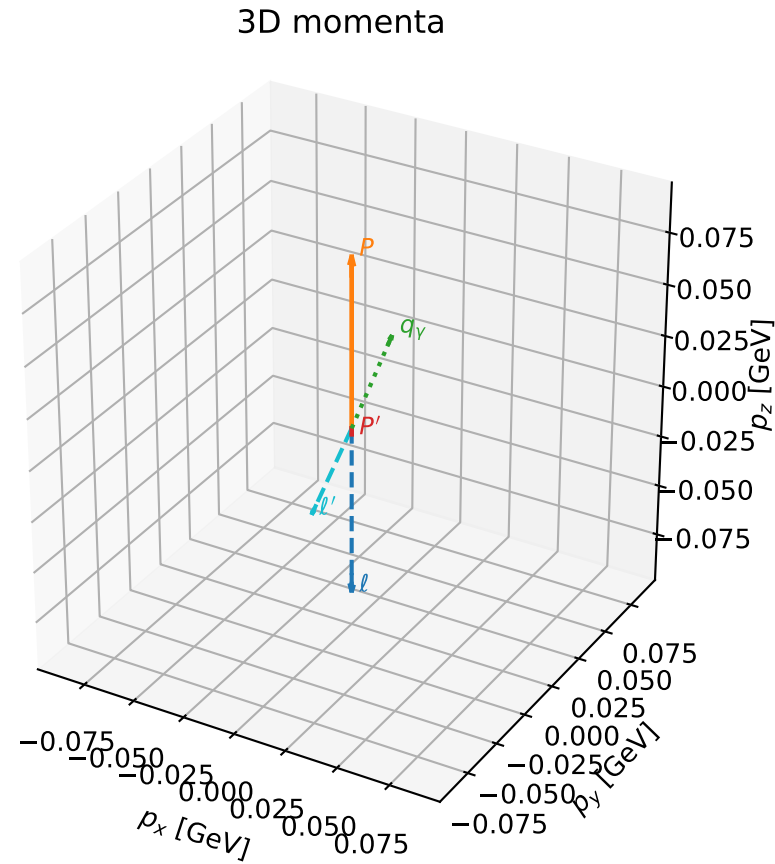
selected phase-space point:
 $\text{sqrt_s} = 1.492894$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 2.364423$
 $\text{qOut} = 0.2430232$
 $\text{phi_p_out} = 1.446458$
 $\text{phi_gamma_out} = 1.053417$
 $\text{alpha_e} = 0.837955$
 $\text{alpha_p} = 2.935935$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

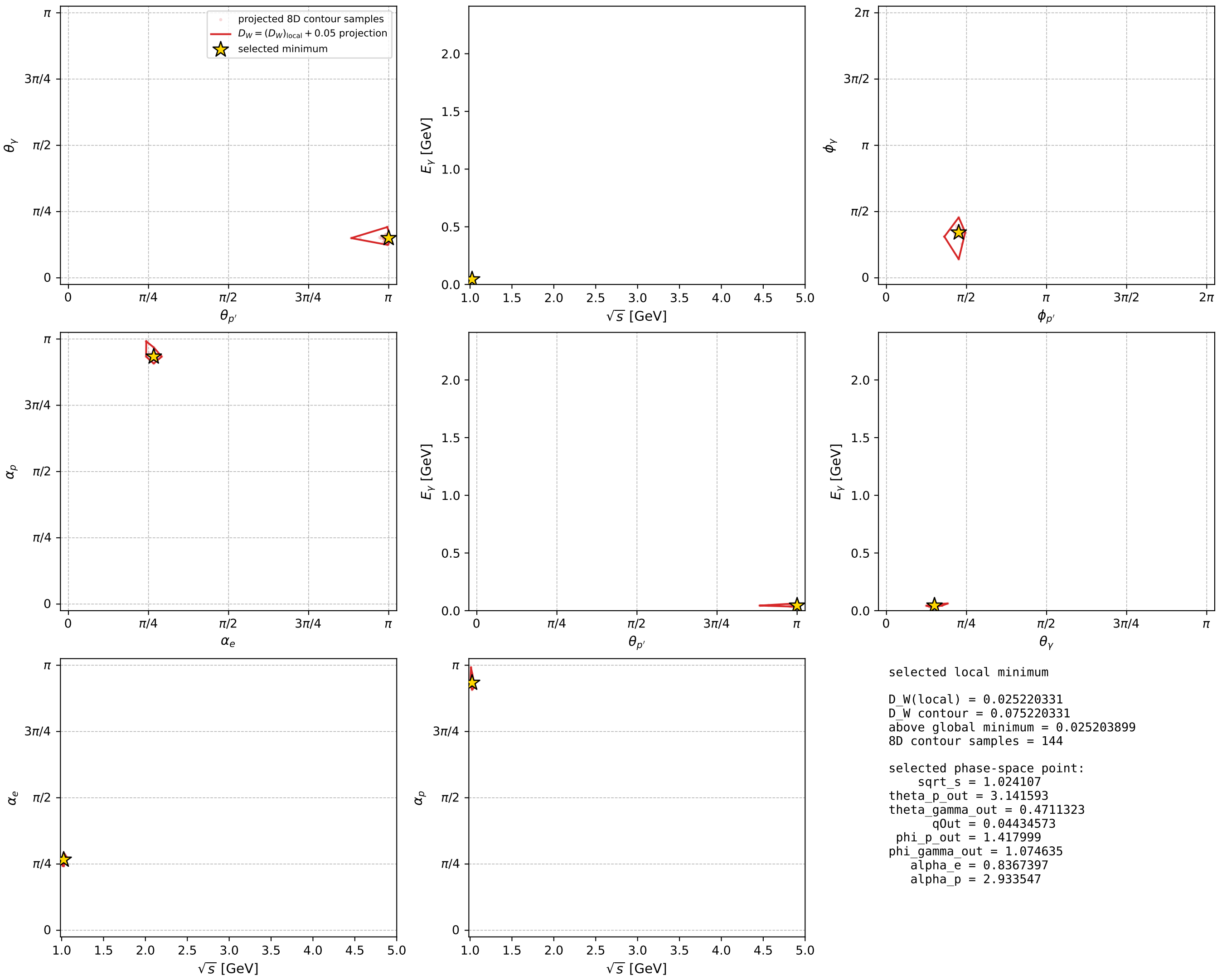
dw_mixing_angles_polarization_03_local_minimum_1081 D_W=0.0252203
 region: polarization_cluster_03_local_minimum_1081
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.83673966$ rad
 incoming lepton: $0.669887|+\rangle +0.742463|-\rangle$
 proton mixing angle: $\alpha_p=2.9335468$ rad
 incoming proton: $-0.978436|+\rangle +0.206548|-\rangle$

$s=1.04879$, $\sqrt{s}=1.02411$, $|\vec{P}|=0.0824855$, $|\vec{P}'|=0.00293306$
 $\theta_{P'}=3.14159$, $\theta_V=0.471132$, $\phi_{P'}=1.418$, $\phi_V=1.07463$
 $E_V=0.0443457$, $\theta_{l'}=2.63857$, $\phi_{l'}=4.21623$
 $Q^2=0.000853359$, $x_B=0.0101255$, $t=-0.00728326$, $W^2=0.963269$, $y=0.498833$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.0825$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0825$
 P $E= 0.9416$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0825$
 l' $E= 0.0418$ $p_x= -0.0096$ $p_y= -0.0177$ $p_z= -0.0366$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.0029$
 q_V $E= 0.0443$ $p_x= 0.0096$ $p_y= 0.0177$ $p_z= 0.0395$



electron: polarization cluster P3, local minimum 1081; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

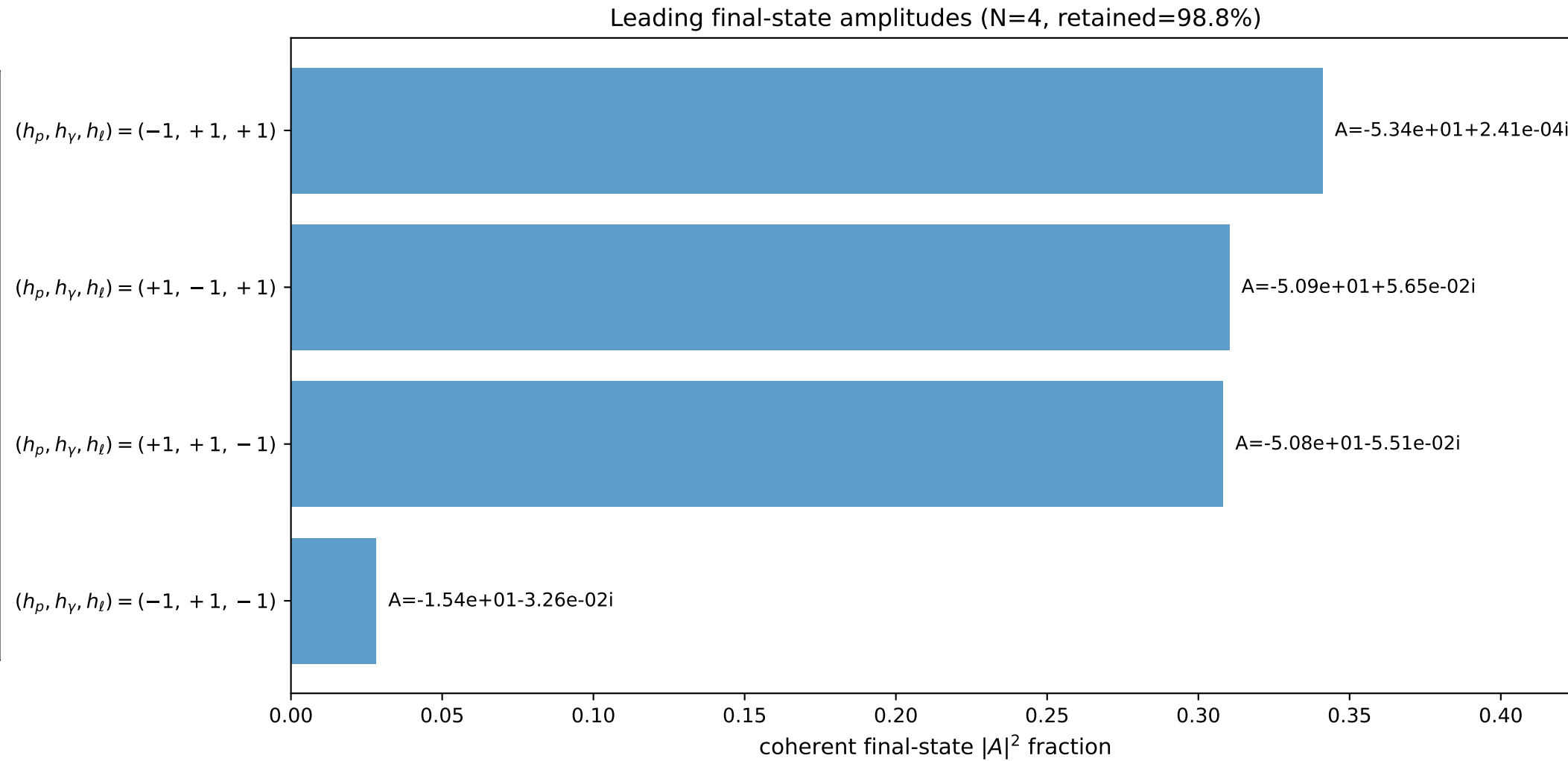
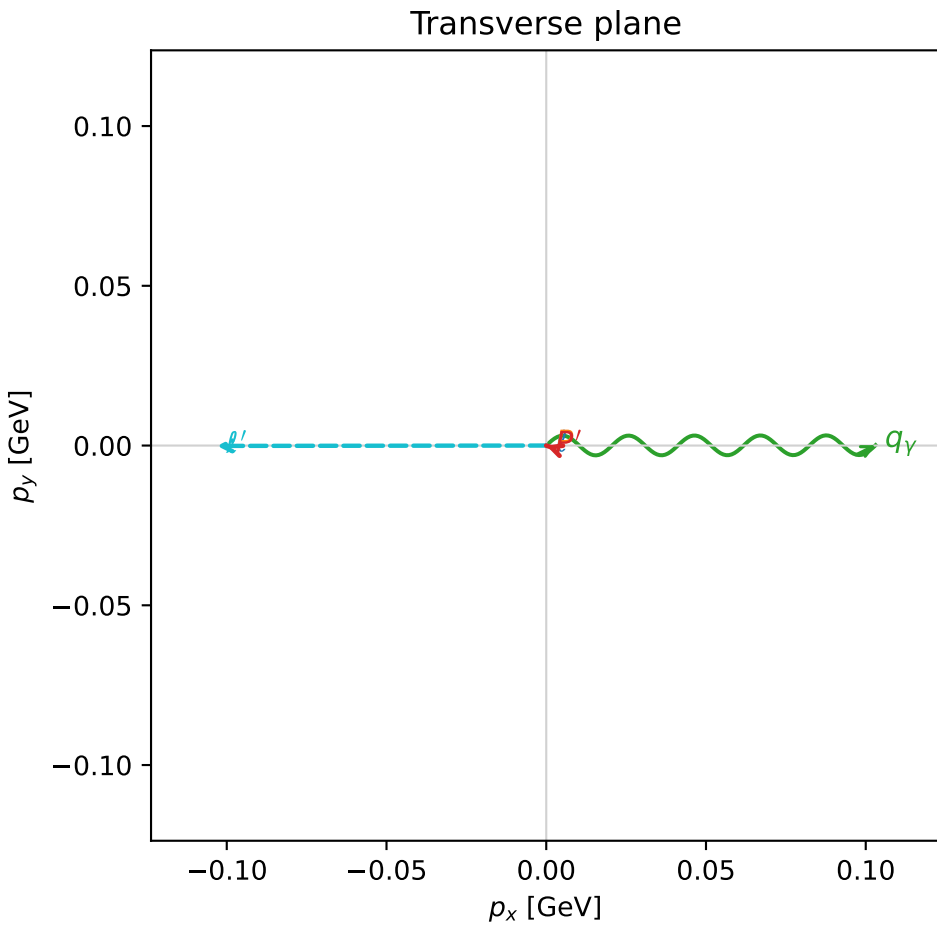
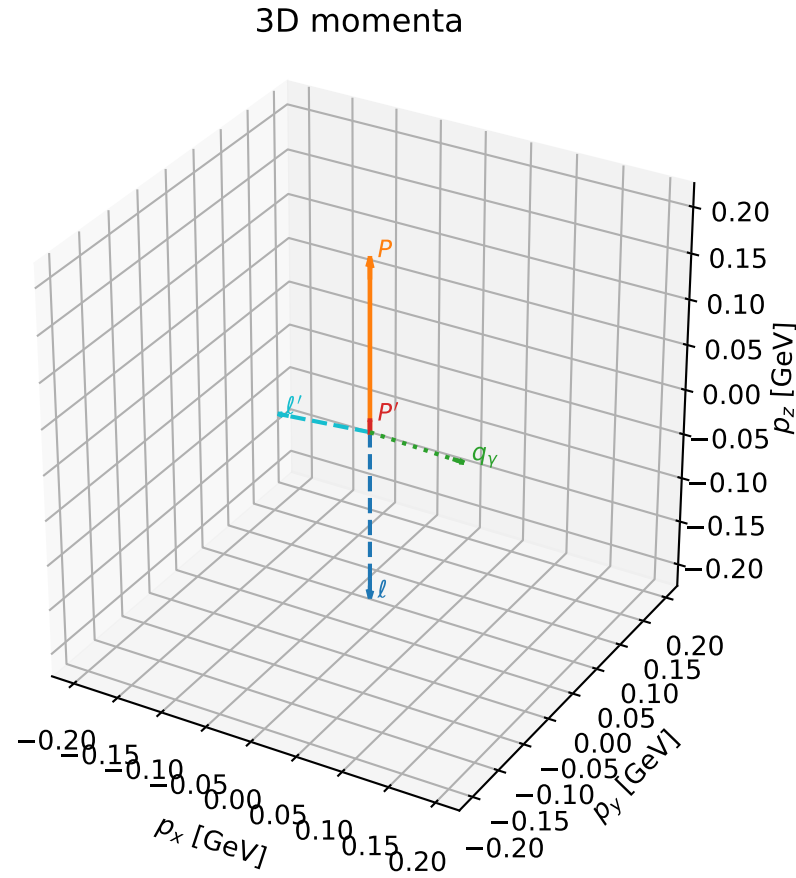


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

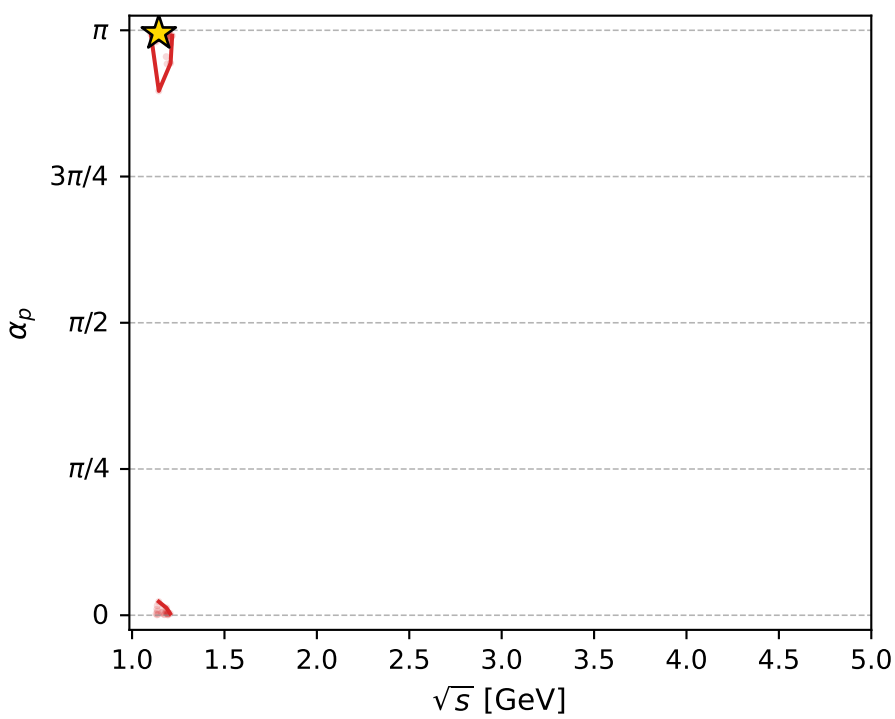
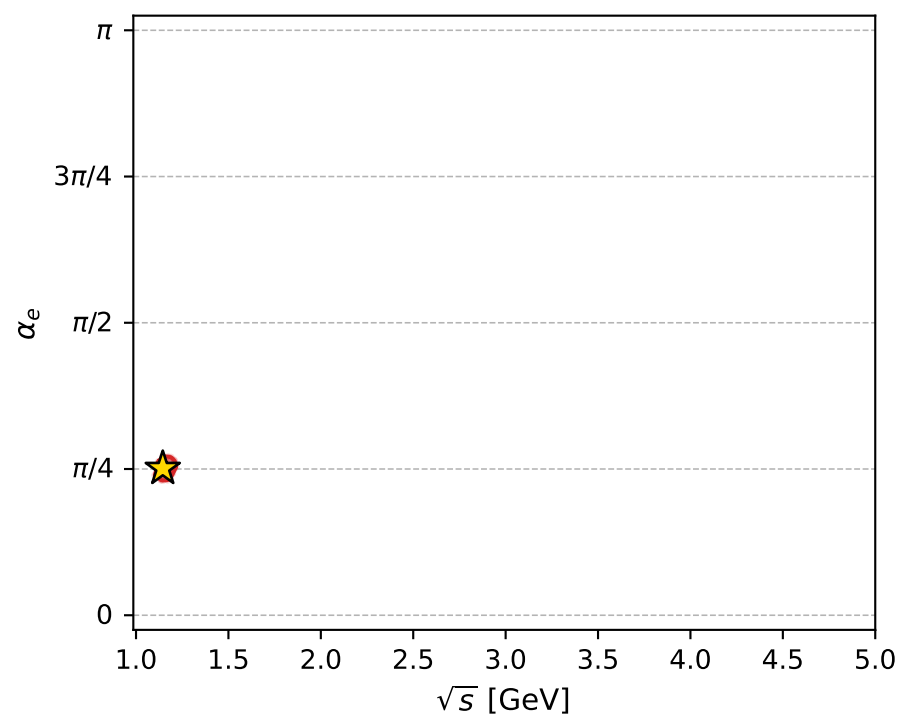
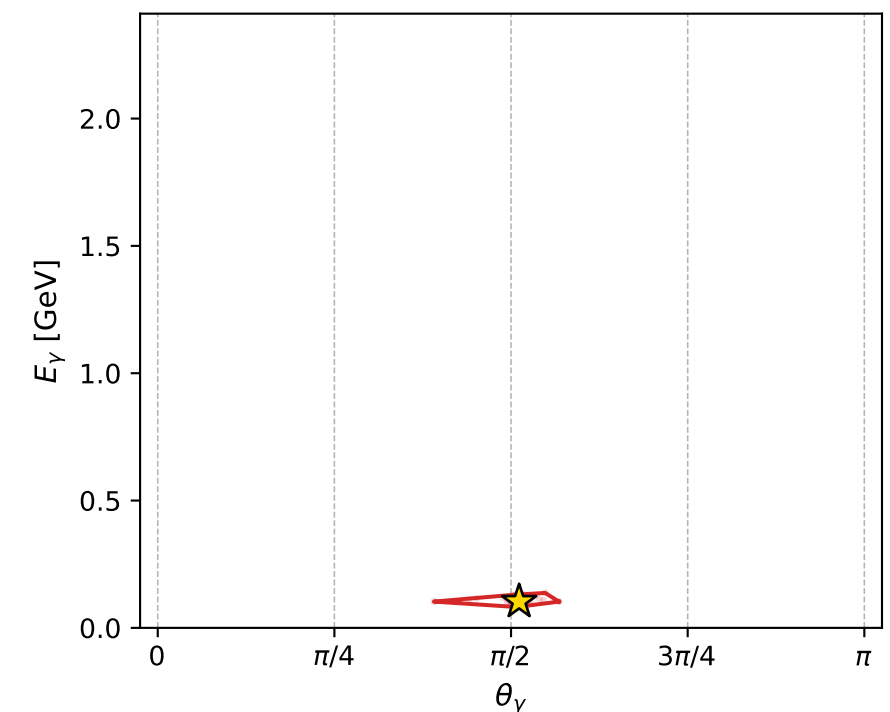
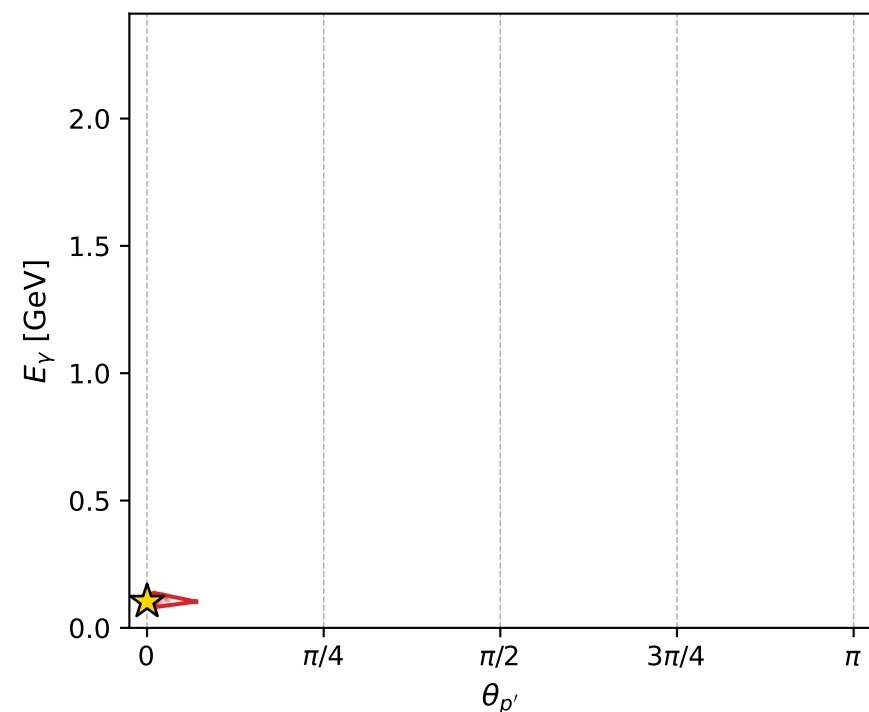
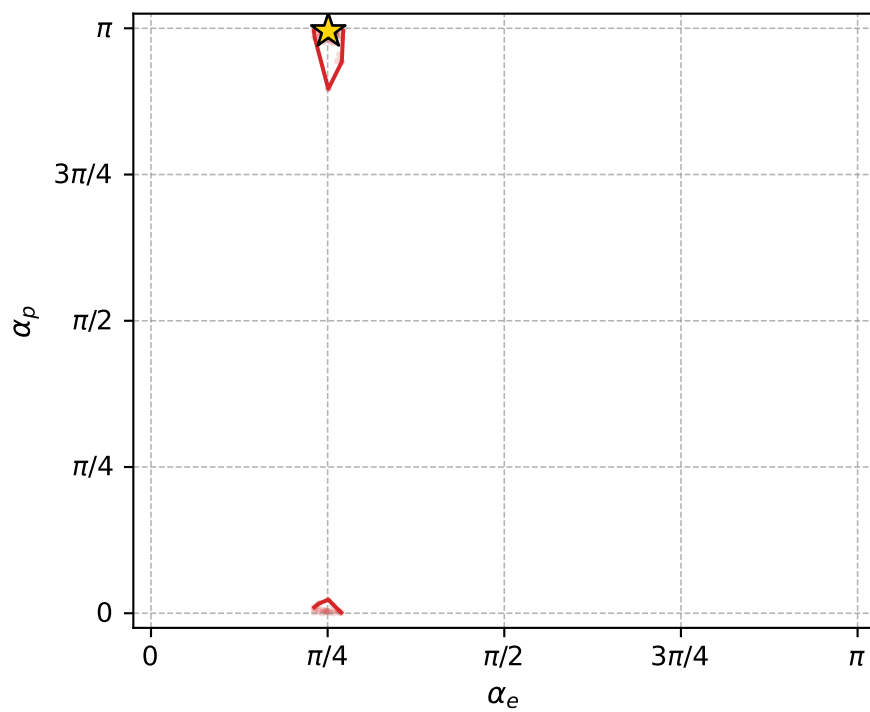
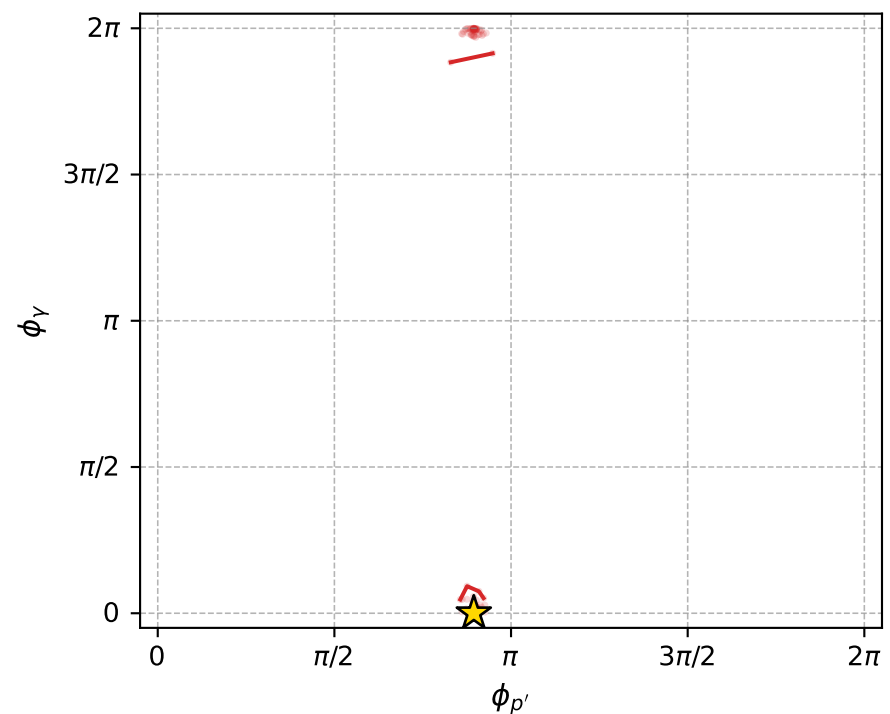
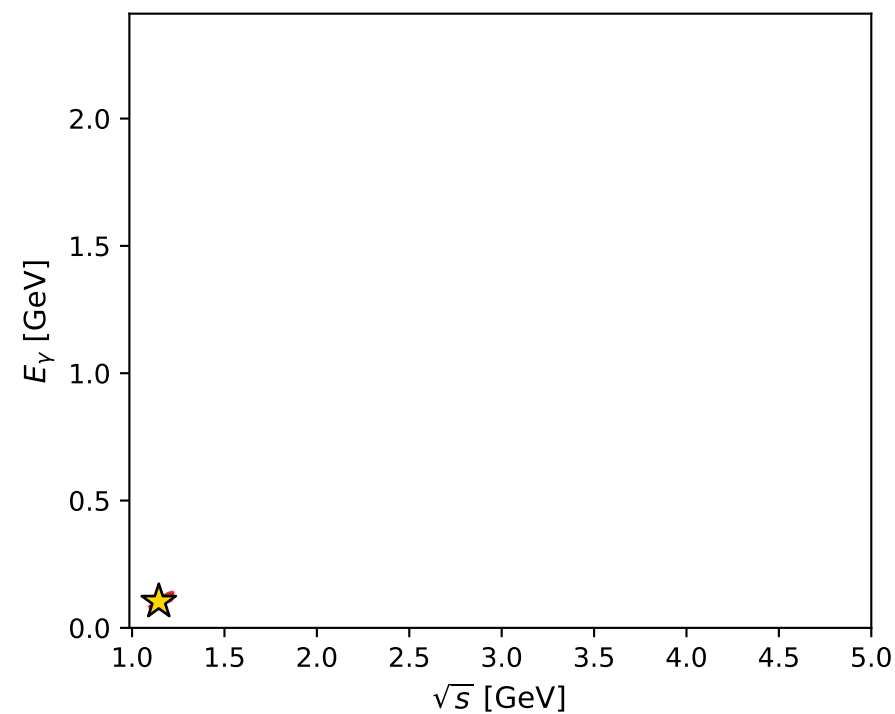
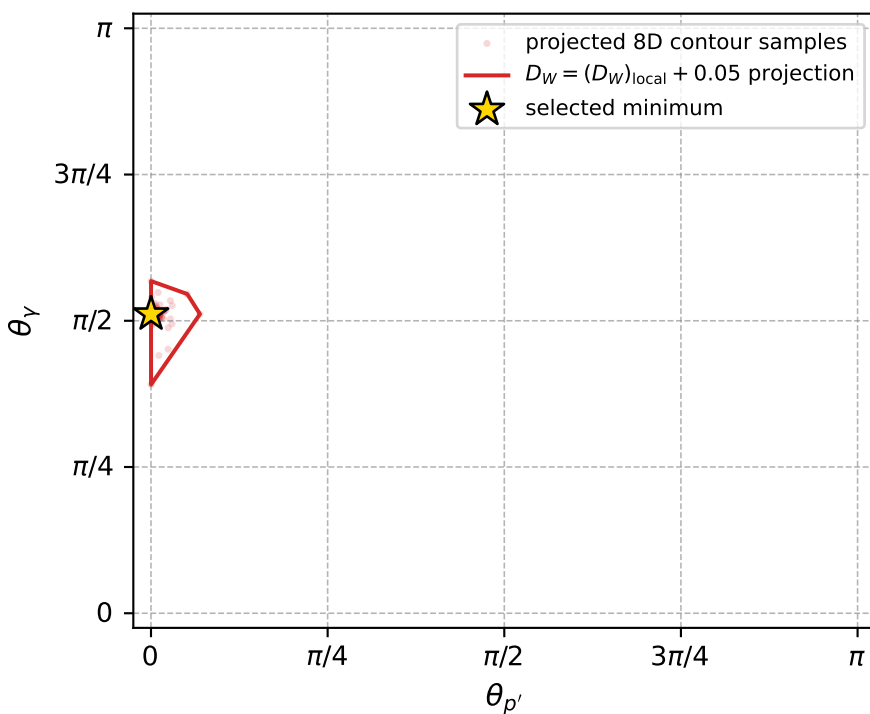
dw_mixing_angles_polarization_03_local_minimum_1082 D_W=0.0252263
 region: polarization_cluster_03_local_minimum_1082
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78828858$ rad
 incoming lepton: $0.70506|+\rangle +0.709148|-\rangle$
 proton mixing angle: $\alpha_p=3.1270994$ rad
 incoming proton: $-0.999895|+\rangle +0.0144928|-\rangle$

$s=1.31038$, $\sqrt{s}=1.14472$, $|\vec{P}|=0.188053$, $|\vec{P}'|=0.0128044$
 $\theta_{P'}=0.000255757$, $\theta_V=1.60678$, $\phi_{P'}=2.8105$, $\phi_V=0.00110035$
 $E_V=0.10315$, $\theta_{\ell'}=1.65878$, $\phi_{\ell'}=3.1427$
 $Q^2=0.0354995$, $x_B=0.154937$, $t=-0.0303669$, $W^2=1.07347$, $y=0.532178$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1881$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1881$
 P $E= 0.9567$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1881$
 ℓ' $E= 0.1035$ $p_x= -0.1031$ $p_y= -0.0001$ $p_z= -0.0091$
 P' $E= 0.9381$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.0128$
 q_V $E= 0.1031$ $p_x= 0.1031$ $p_y= 0.0001$ $p_z= -0.0037$



electron: polarization cluster P3, local minimum 1082; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.025226282$
 $D_W \text{ contour} = 0.075226282$
 above global minimum = 0.025209849
 8D contour samples = 137

selected phase-space point:
 $\text{sqrt_s} = 1.144718$
 $\text{theta_p_out} = 0.0002557574$
 $\text{theta_gamma_out} = 1.606782$
 $\text{qOut} = 0.1031497$
 $\text{phi_p_out} = 2.810502$
 $\text{phi_gamma_out} = 0.001100355$
 $\text{alpha_e} = 0.7882886$
 $\text{alpha_p} = 3.127099$

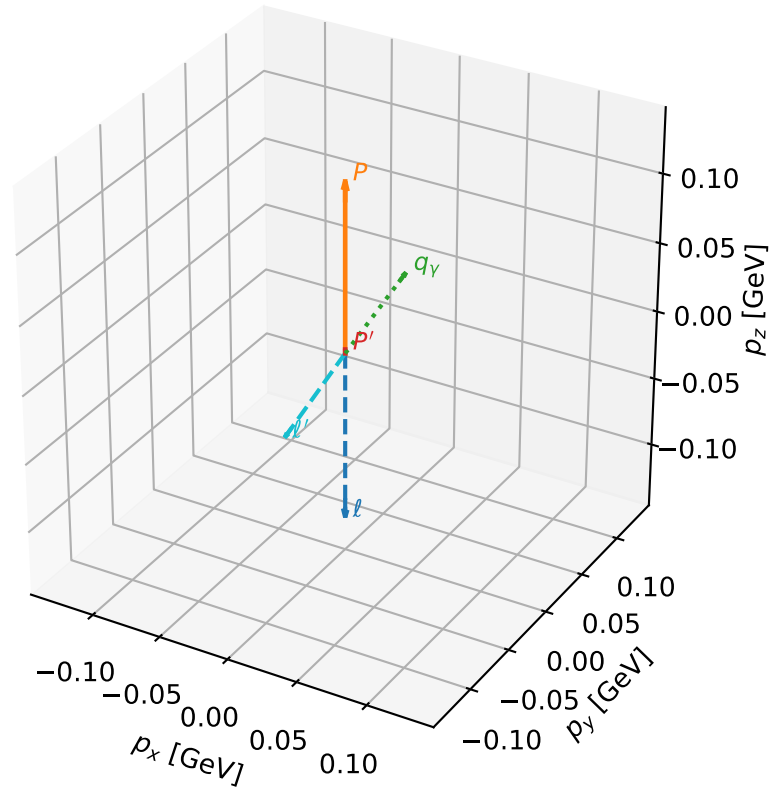
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1094 D_W=0.0264789
 region: polarization_cluster_03_local_minimum_1094
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.8349927$ rad
 incoming lepton: $0.671183|+\rangle + 0.741292|-\rangle$
 proton mixing angle: $\alpha_p=2.9469782$ rad
 incoming proton: $-0.981122|+\rangle + 0.193388|-\rangle$

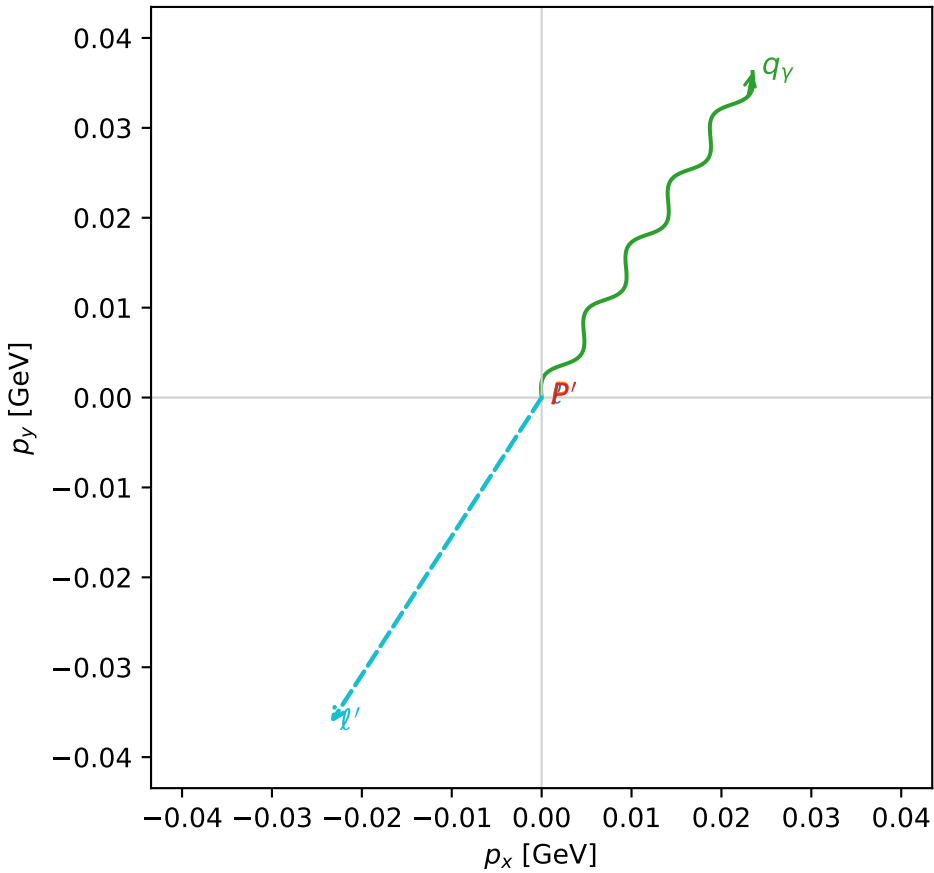
$s=1.14191$, $\sqrt{s}=1.0686$, $|\vec{P}|=0.122621$, $|\vec{P}'|=0.00277062$
 $\theta_{P'}=0$, $\theta_V=0.736191$, $\phi_{P'}=0.644641$, $\phi_V=0.995749$
 $E_V=0.064259$, $\theta_{l'}=2.43345$, $\phi_{l'}=4.13734$
 $Q^2=0.00391163$, $x_B=0.0314951$, $t=-0.0143005$, $W^2=1.00013$, $y=0.473915$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 0.1226$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1226$
 P $E= 0.9460$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1226$
 l' $E= 0.0663$ $p_x= -0.0235$ $p_y= -0.0362$ $p_z= -0.0504$
 P' $E= 0.9380$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.0028$
 q_V $E= 0.0643$ $p_x= 0.0235$ $p_y= 0.0362$ $p_z= 0.0476$

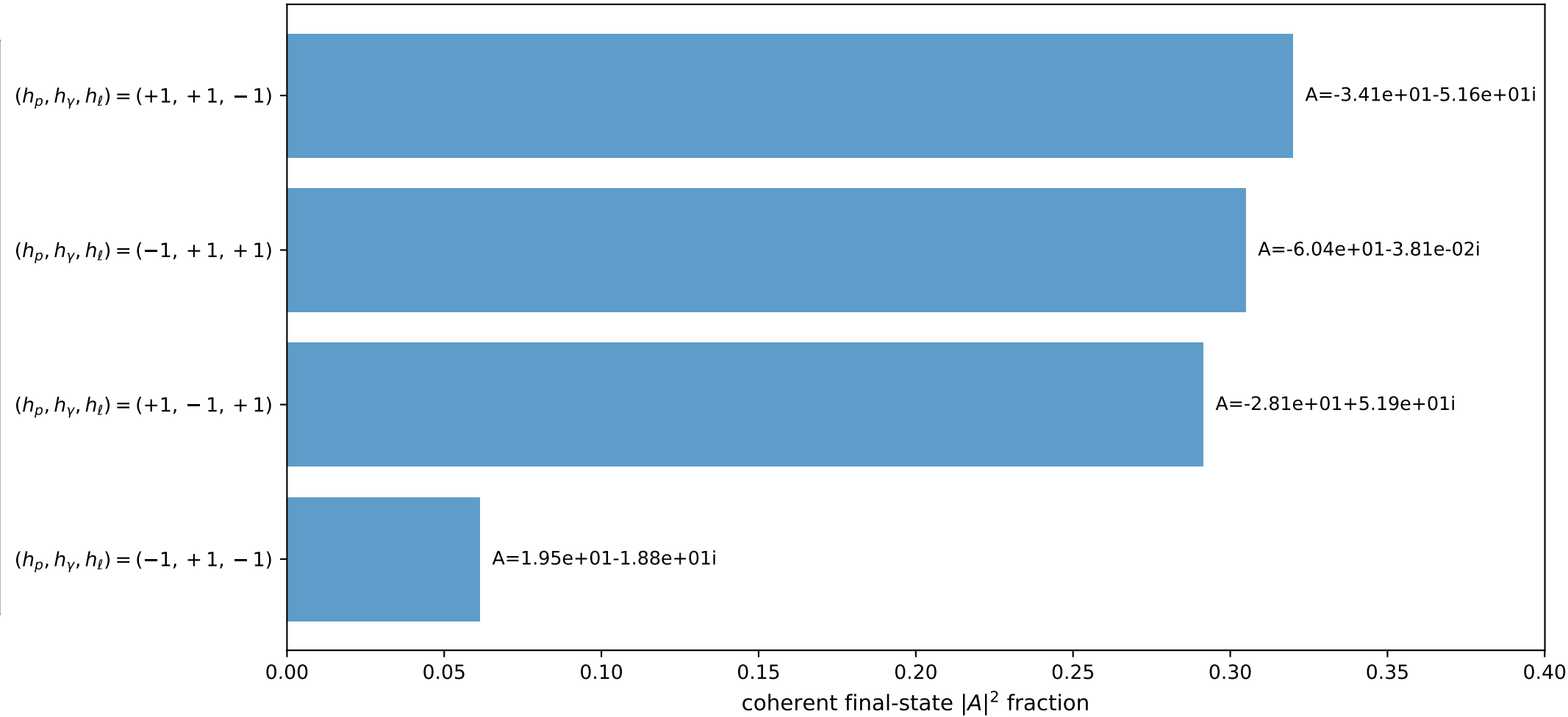
3D momenta



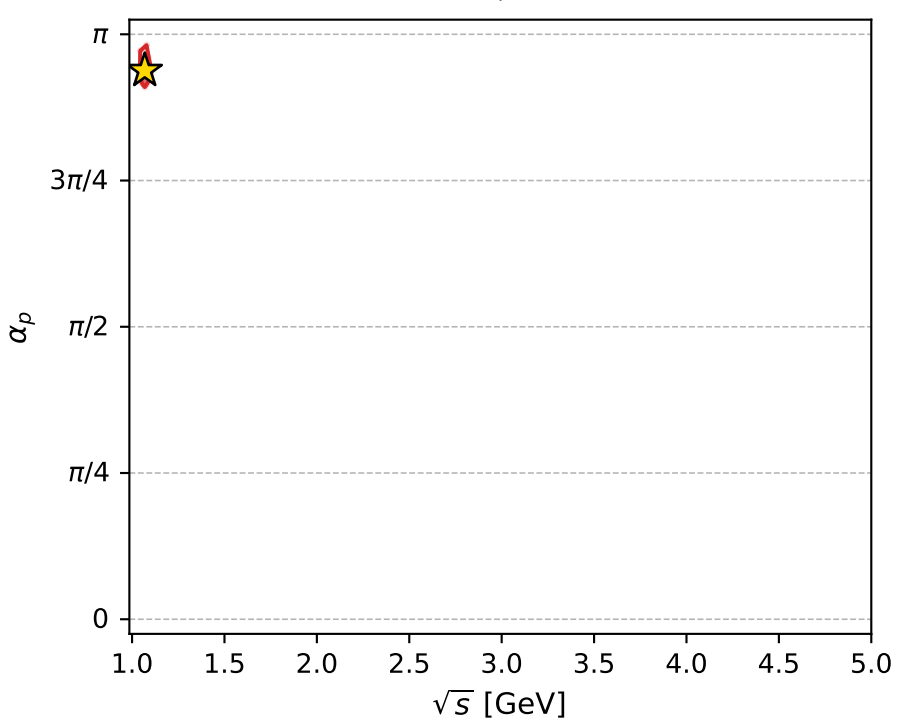
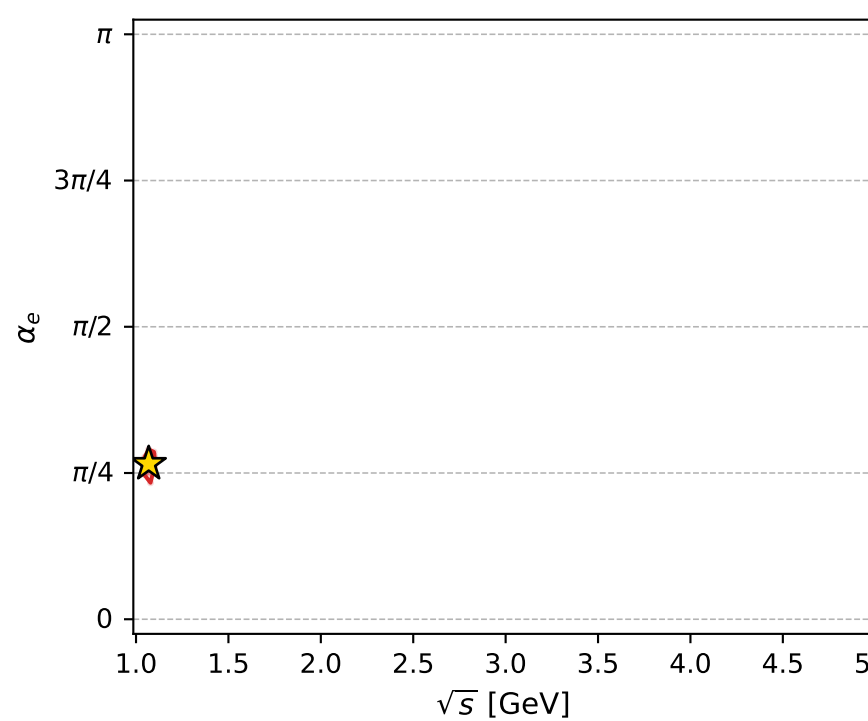
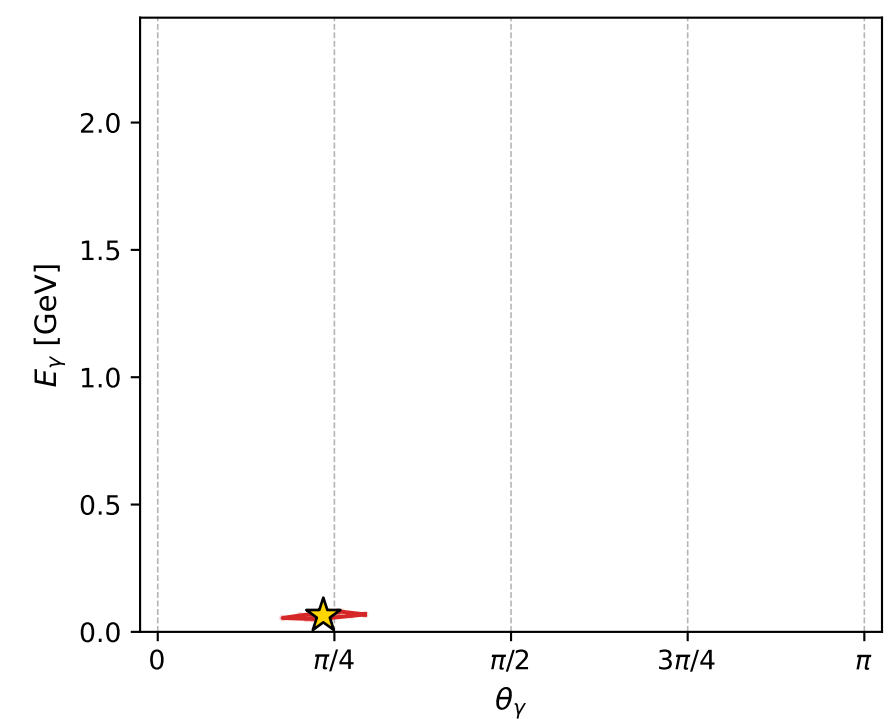
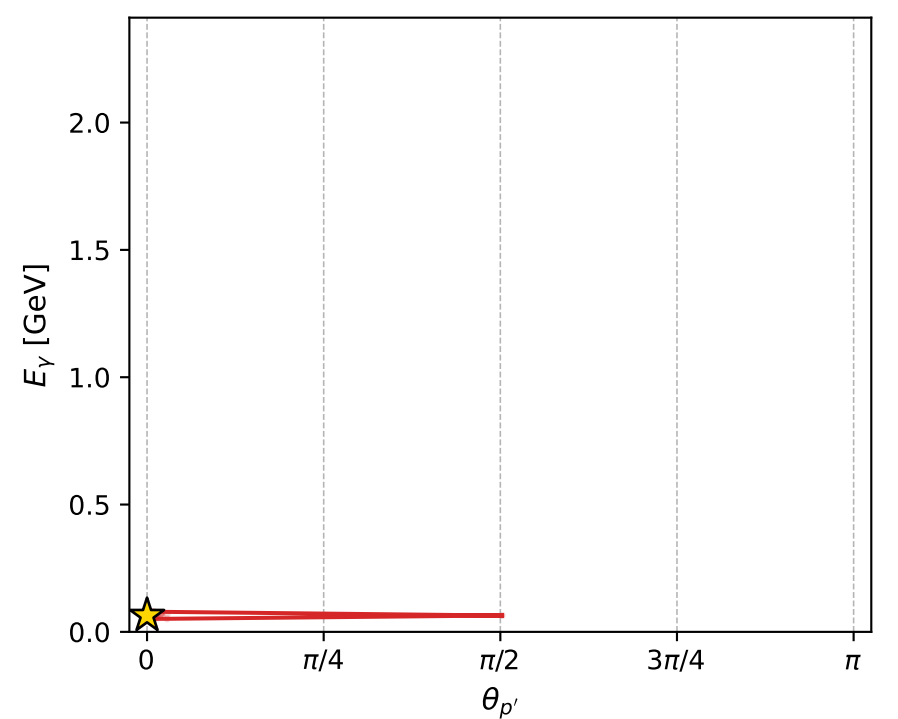
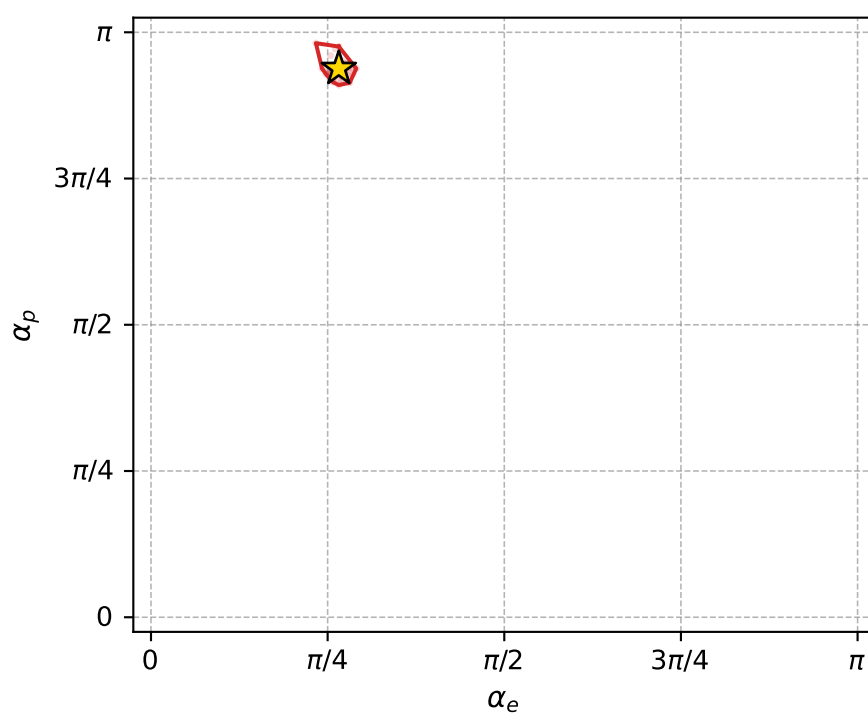
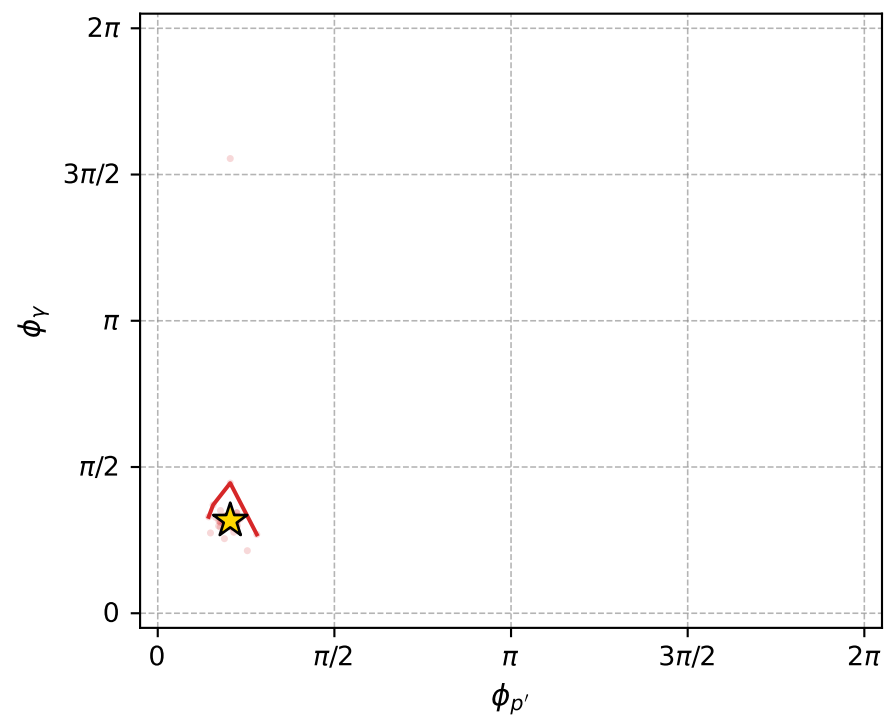
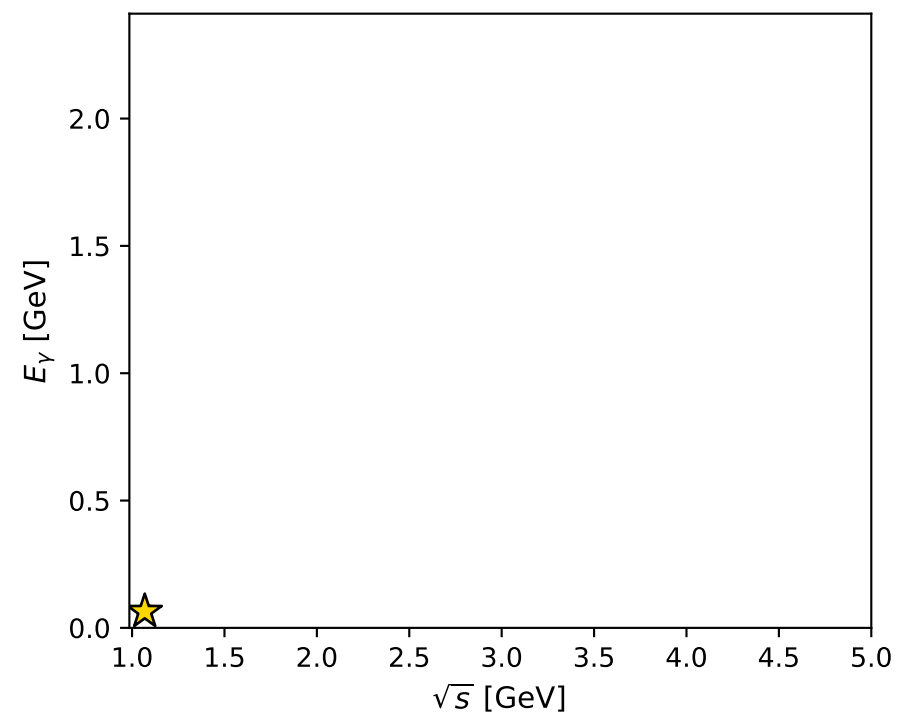
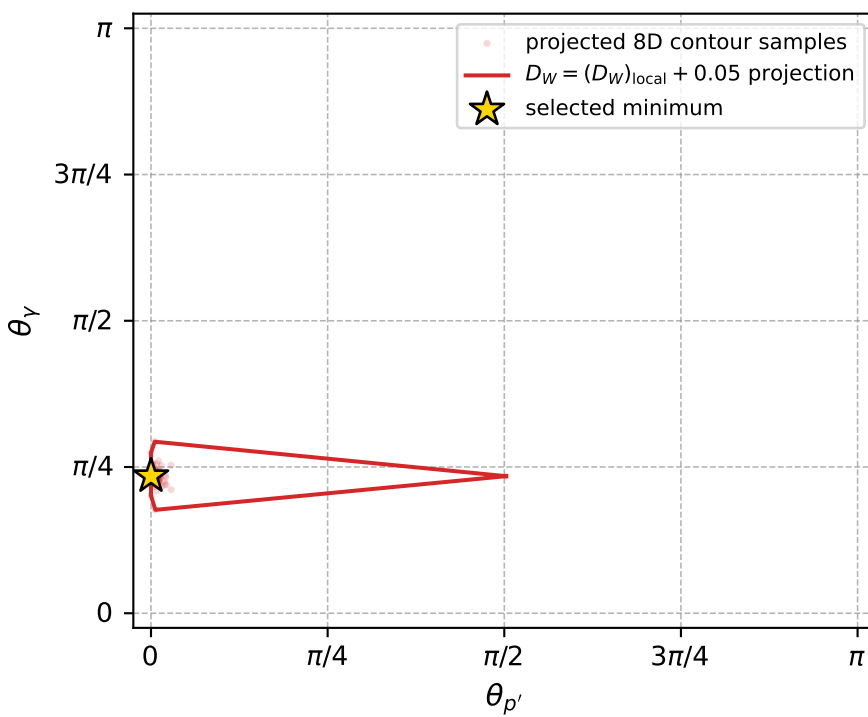
Transverse plane



Leading final-state amplitudes (N=4, retained=97.8%)



electron: polarization cluster P3, local minimum 1094; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.026478862$
 $D_W \text{ contour} = 0.076478862$
 above global minimum = 0.02646243
 8D contour samples = 133

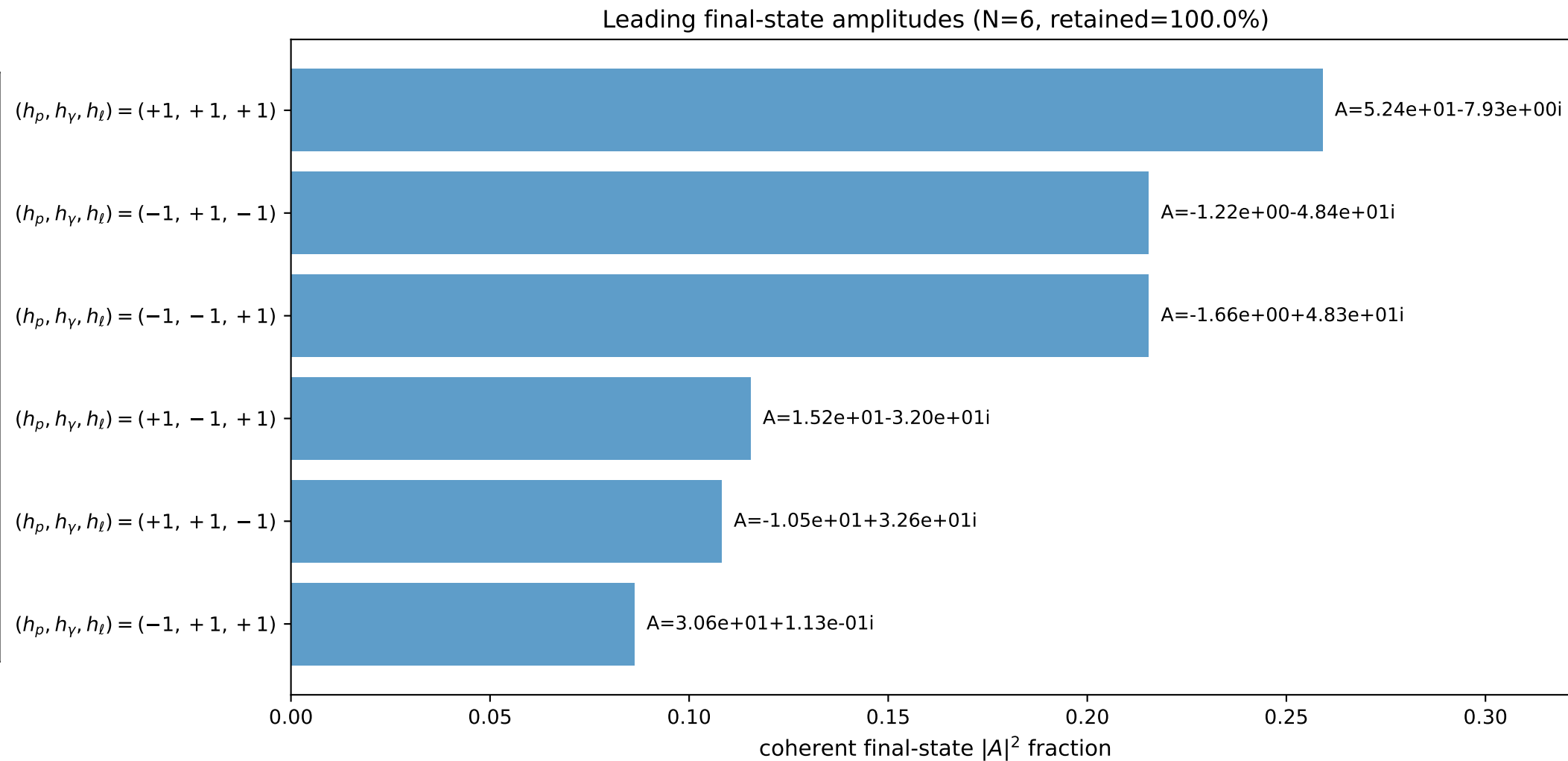
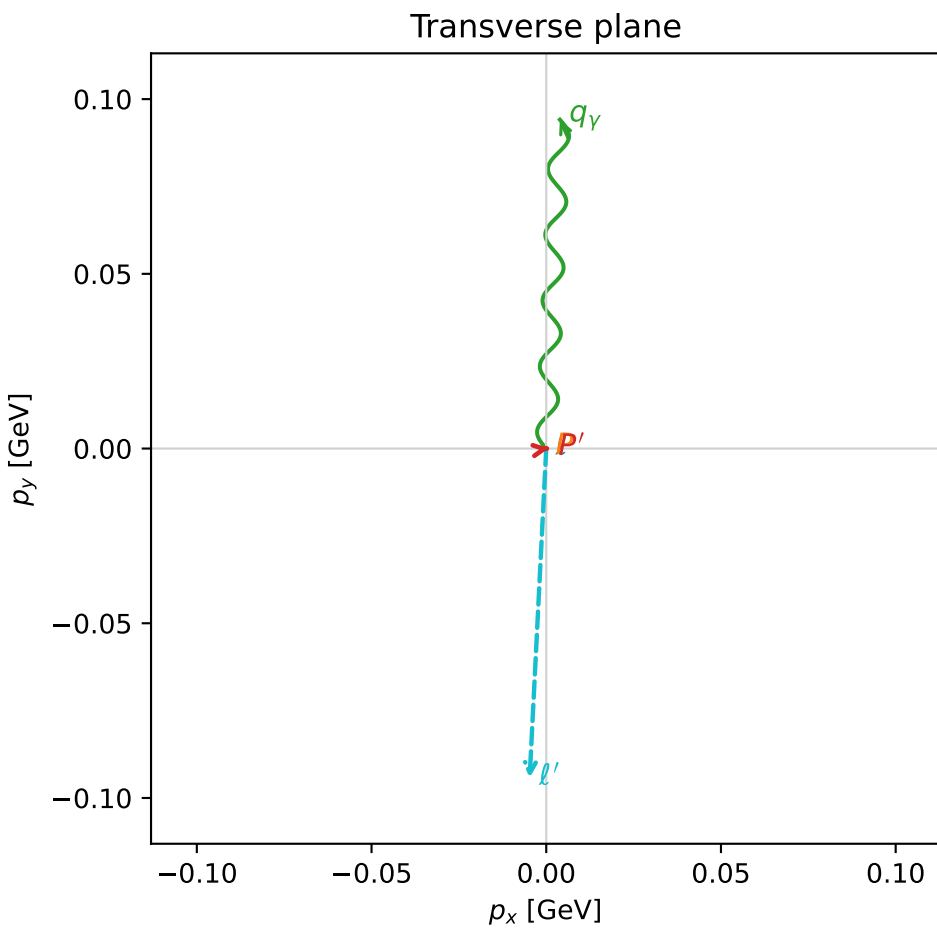
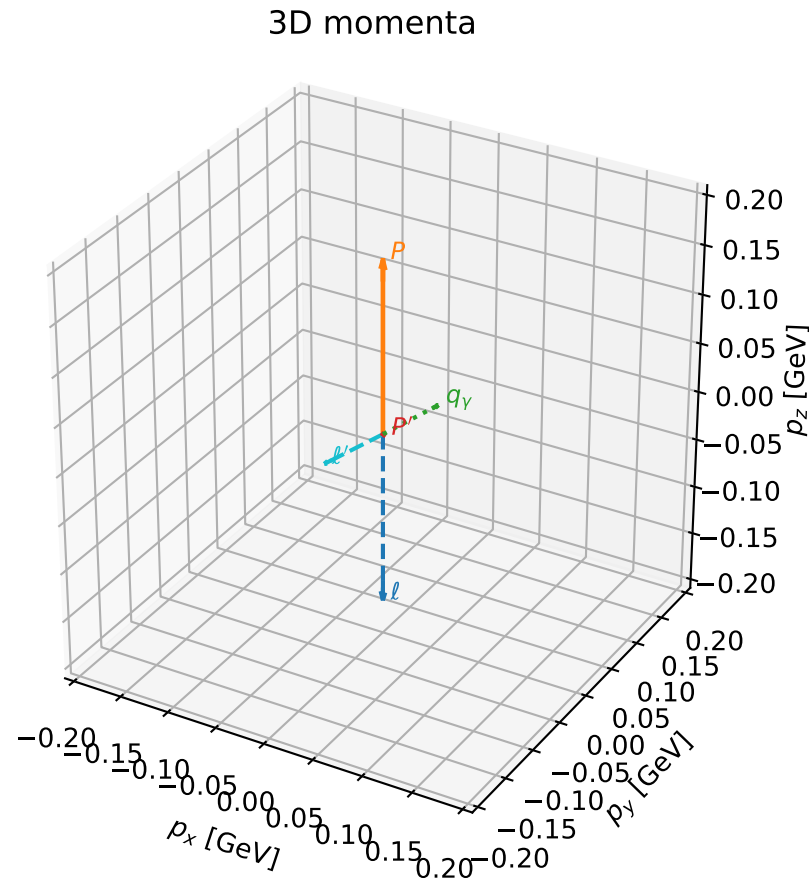
selected phase-space point:
 $\text{sqrt_s} = 1.068603$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 0.7361911$
 $\text{qOut} = 0.06425897$
 $\text{phi_p_out} = 0.6446411$
 $\text{phi_gamma_out} = 0.9957492$
 $\text{alpha_e} = 0.8349927$
 $\text{alpha_p} = 2.946978$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

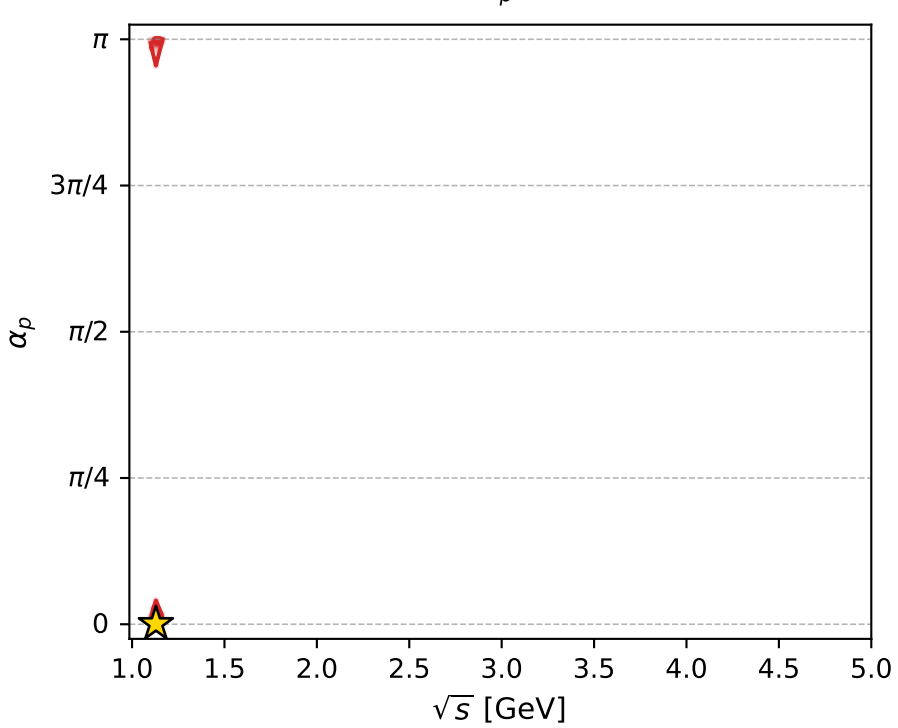
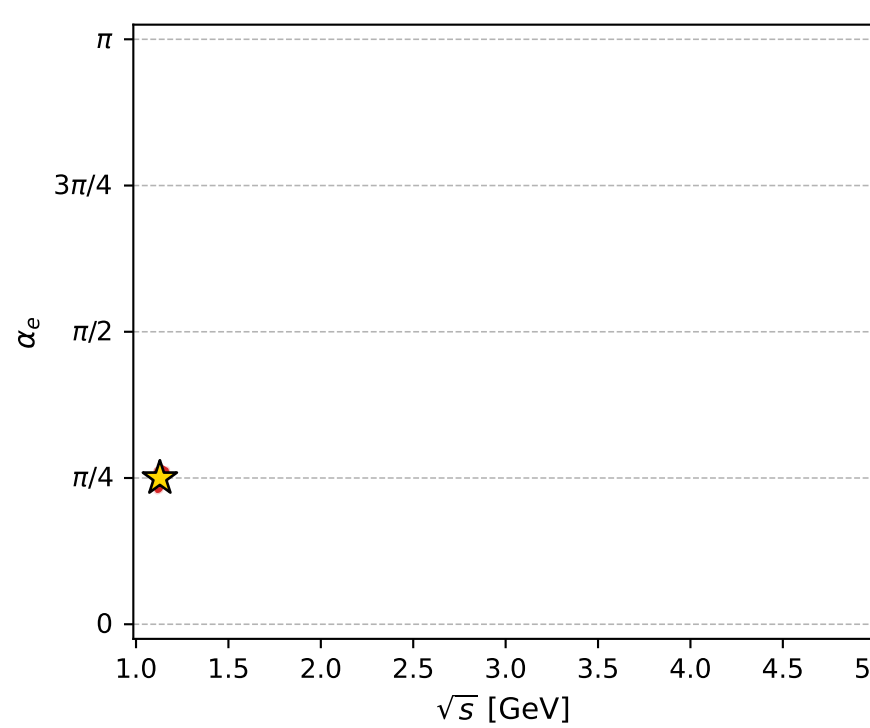
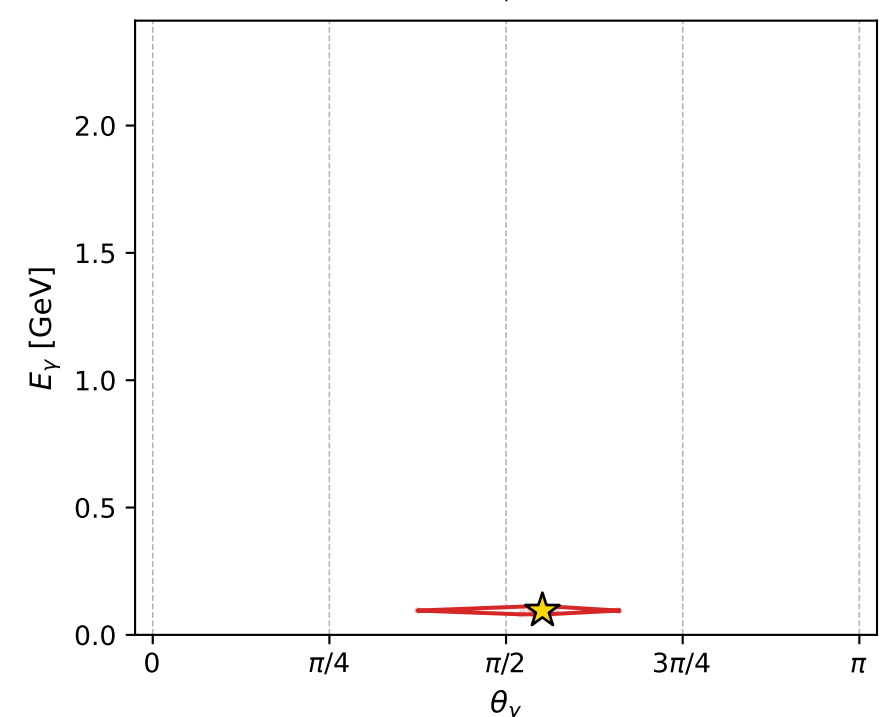
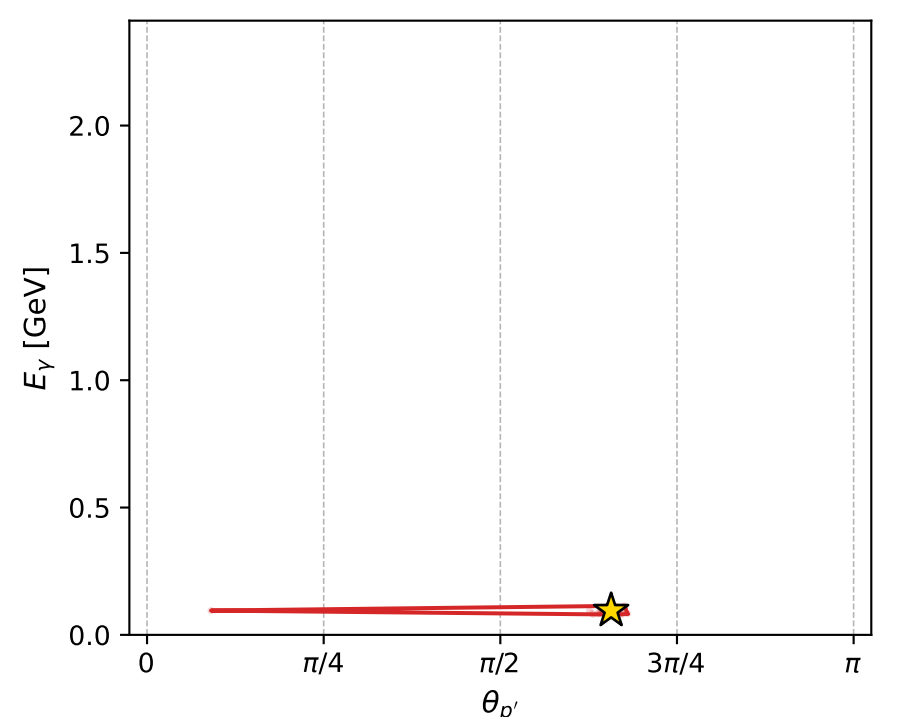
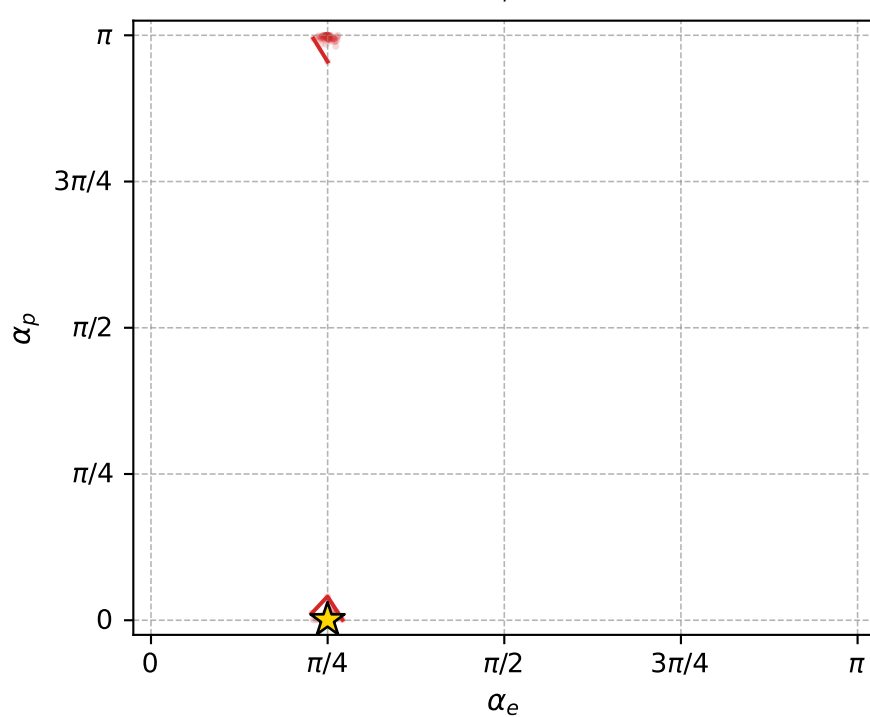
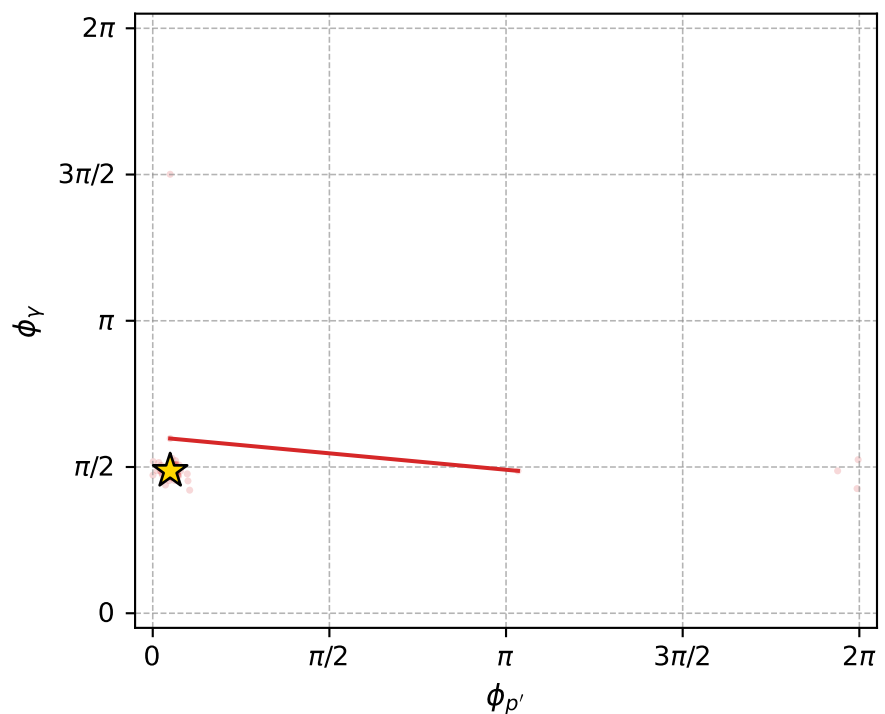
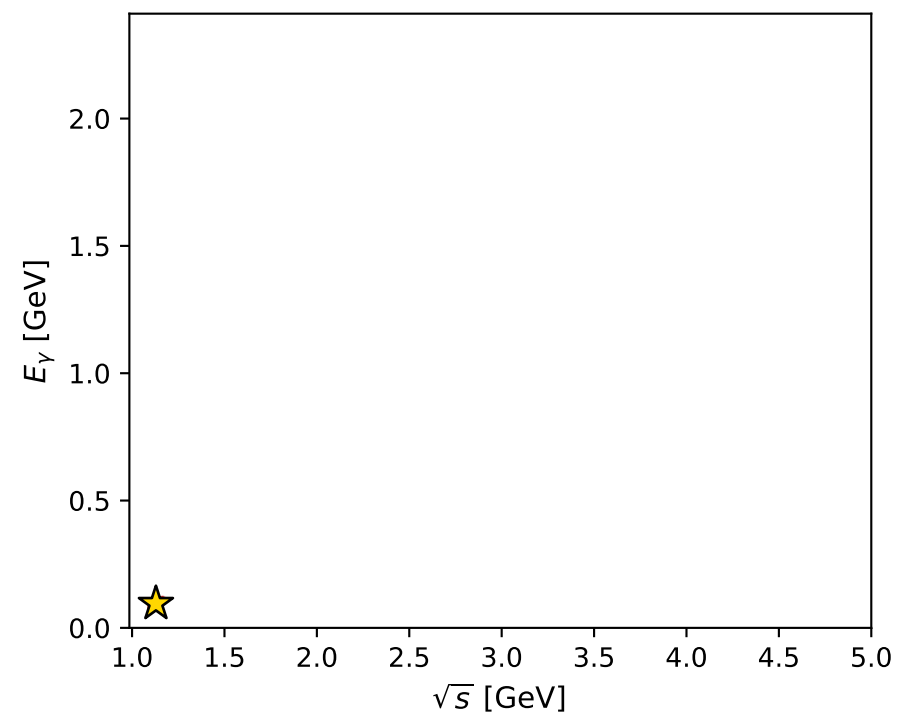
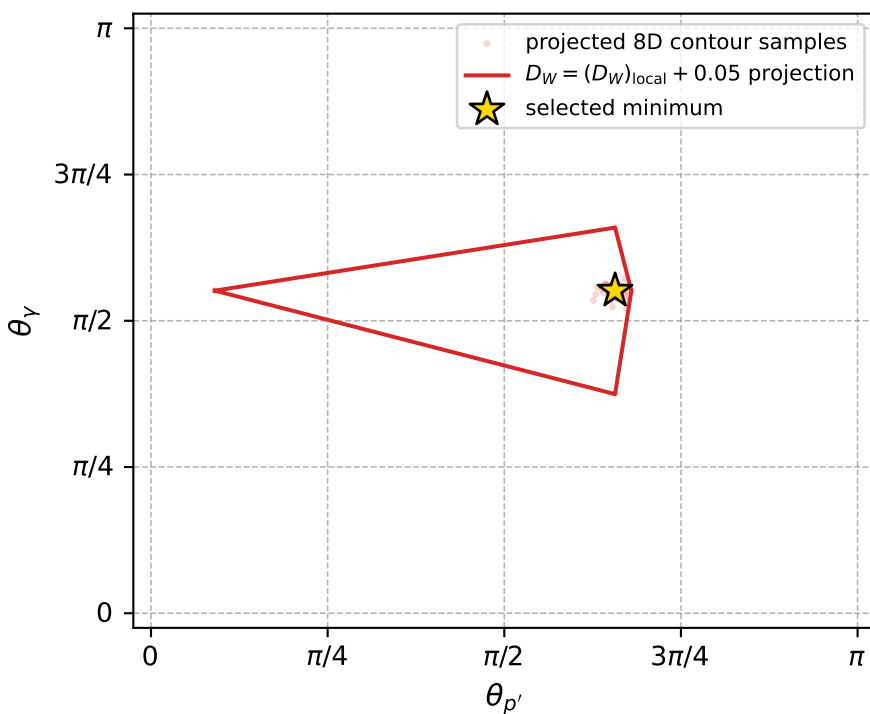
dw_mixing_angles_polarization_03_local_minimum_1096 D_W=0.0266862
 region: polarization_cluster_03_local_minimum_1096
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.78488527$ rad
 incoming lepton: $0.707469|+\rangle +0.706744|-\rangle$
 proton mixing angle: $\alpha_p=0.0023840874$ rad
 incoming proton: $0.999997|+\rangle +0.00238409|-\rangle$

$s=1.27495$, $\sqrt{s}=1.12914$, $|\vec{P}|=0.174958$, $|\vec{P}'|=0.000975106$
 $\theta_{P'}=2.06337$, $\theta_V=1.73249$, $\phi_{P'}=0.154759$, $\phi_V=1.52936$
 $E_V=0.0954447$, $\theta_{\ell'}=1.40464$, $\phi_{\ell'}=4.66202$
 $Q^2=0.039021$, $x_B=0.178971$, $t=-0.0305107$, $W^2=1.05885$, $y=0.55183$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E=0.1750$ $p_x=0.0000$ $p_y=0.0000$ $p_z=-0.1750$
 P $E=0.9542$ $p_x=0.0000$ $p_y=0.0000$ $p_z=0.1750$
 ℓ' $E=0.0957$ $p_x=-0.0048$ $p_y=-0.0943$ $p_z=0.0158$
 P' $E=0.9380$ $p_x=0.0008$ $p_y=0.0001$ $p_z=-0.0005$
 q_V $E=0.0954$ $p_x=0.0039$ $p_y=0.0941$ $p_z=-0.0154$



electron: polarization cluster P3, local minimum 1096; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.026686207$
 $D_W \text{ contour} = 0.076686207$
 above global minimum = 0.026669774
 8D contour samples = 255

selected phase-space point:
 $\text{sqrt_s} = 1.129135$
 $\text{theta_p_out} = 2.063366$
 $\text{theta_gamma_out} = 1.732486$
 $\text{qOut} = 0.09544469$
 $\text{phi_p_out} = 0.1547588$
 $\text{phi_gamma_out} = 1.52936$
 $\text{alpha_e} = 0.7848853$
 $\text{alpha_p} = 0.002384087$

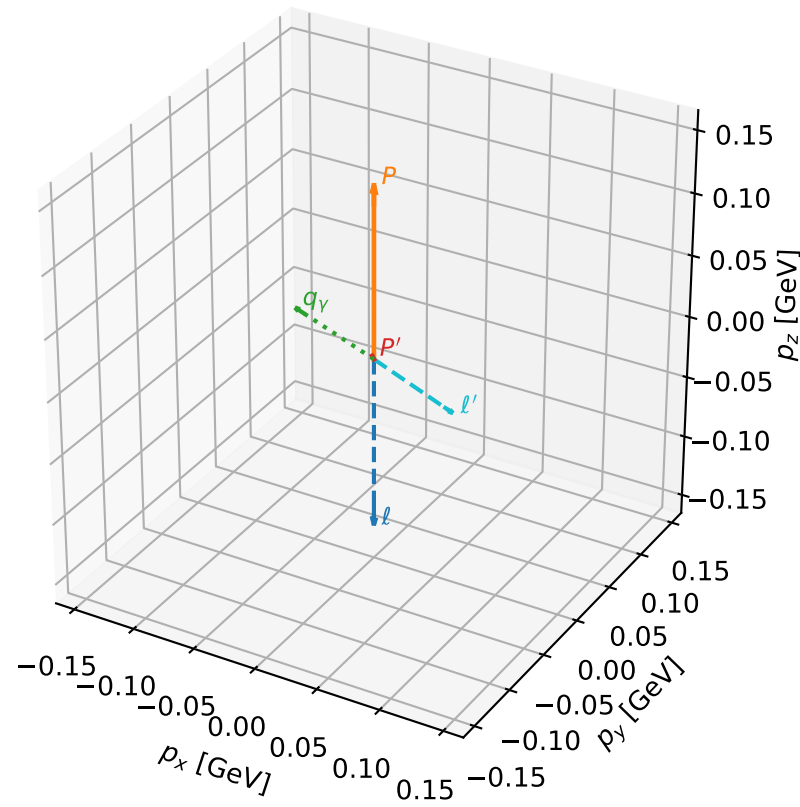
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_1101 D_W=0.0271493
 region: polarization_cluster_03_local_minimum_1101
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.84846861$ rad
 incoming lepton: $0.661133|+\rangle + 0.750269|-\rangle$
 proton mixing angle: $\alpha_p=0.20504957$ rad
 incoming proton: $0.979051|+\rangle + 0.203616|-\rangle$

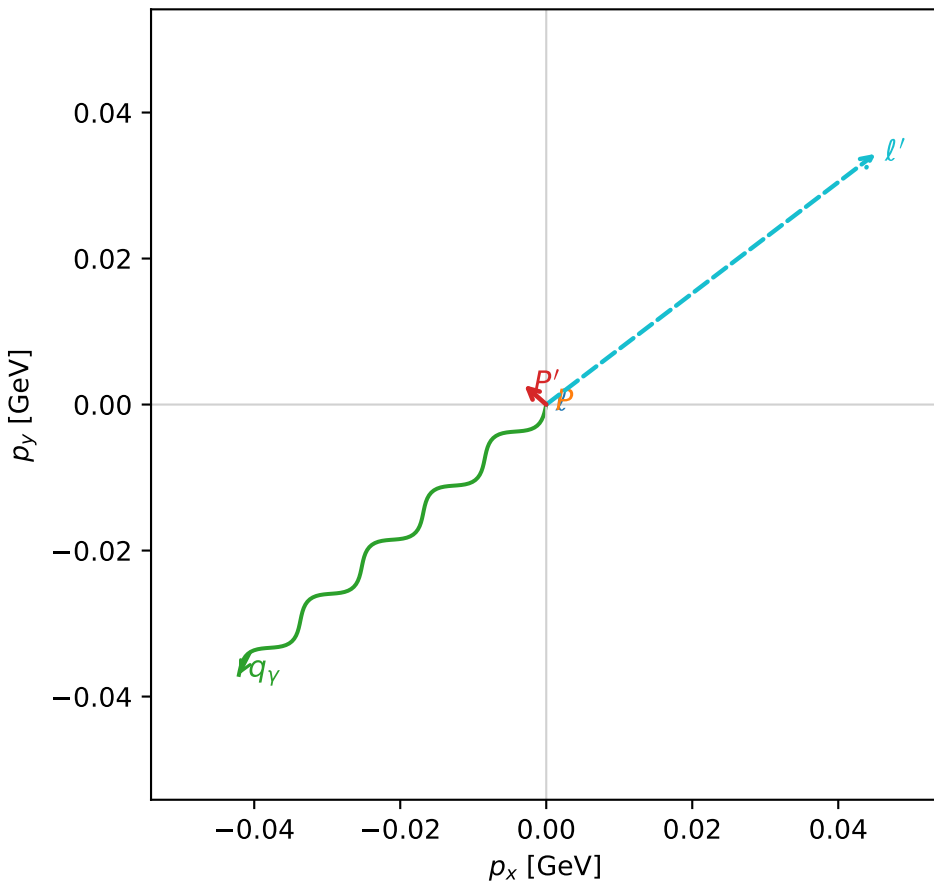
$s=1.18018$, $\sqrt{s}=1.08636$, $|\vec{P}|=0.13823$, $|\vec{P}'|=0.00406651$
 $\theta_{P'}=1.43234$, $\theta_V=0.863658$, $\phi_{P'}=2.41508$, $\phi_V=3.86293$
 $E_V=0.0737516$, $\theta_{\ell'}=2.27812$, $\phi_{\ell'}=0.650785$
 $Q^2=0.00722238$, $x_B=0.0496477$, $t=-0.0188664$, $W^2=1.01809$, $y=0.484366$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1382$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1382$
 P $E= 0.9481$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1382$
 ℓ' $E= 0.0746$ $p_x= 0.0451$ $p_y= 0.0344$ $p_z= -0.0485$
 P' $E= 0.9380$ $p_x= -0.0030$ $p_y= 0.0027$ $p_z= 0.0006$
 q_V $E= 0.0738$ $p_x= -0.0421$ $p_y= -0.0370$ $p_z= 0.0479$

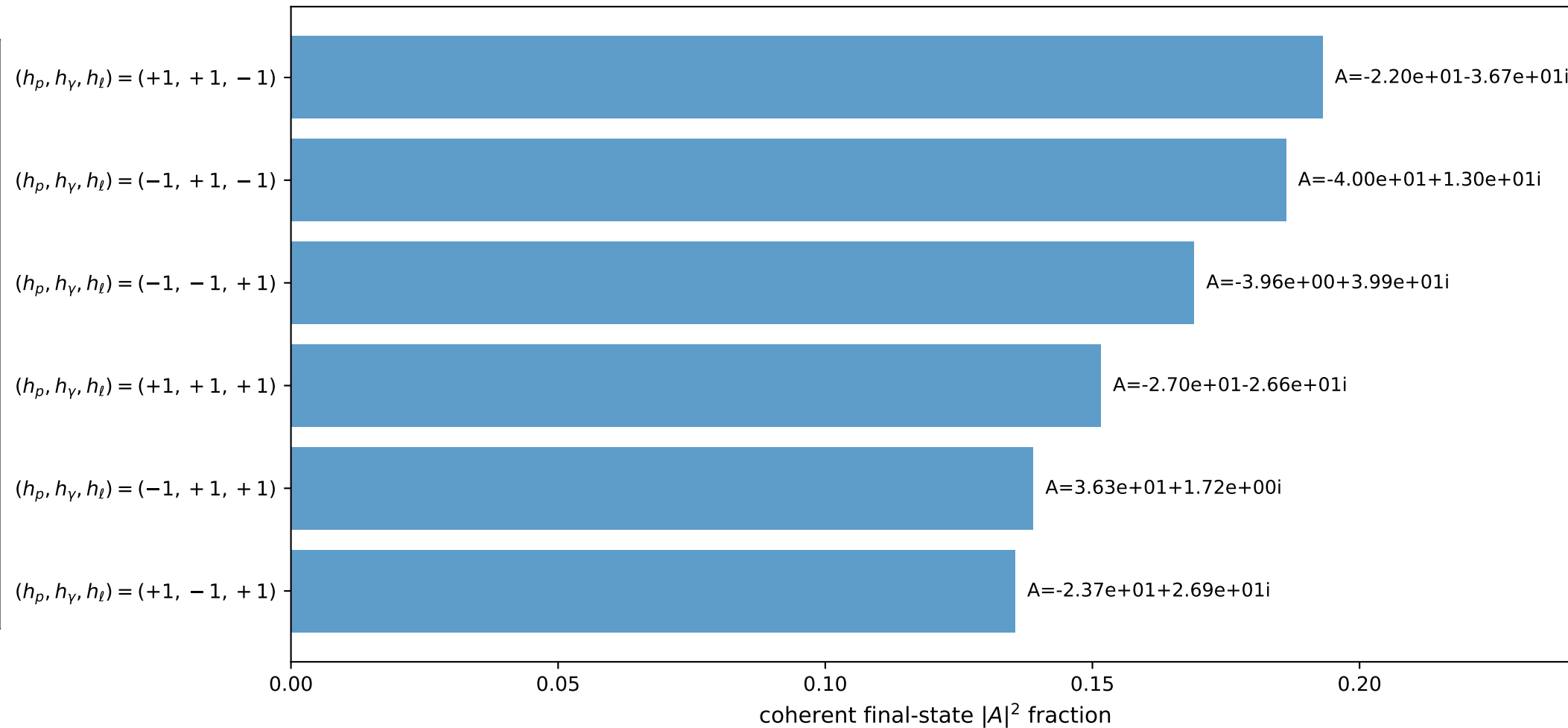
3D momenta



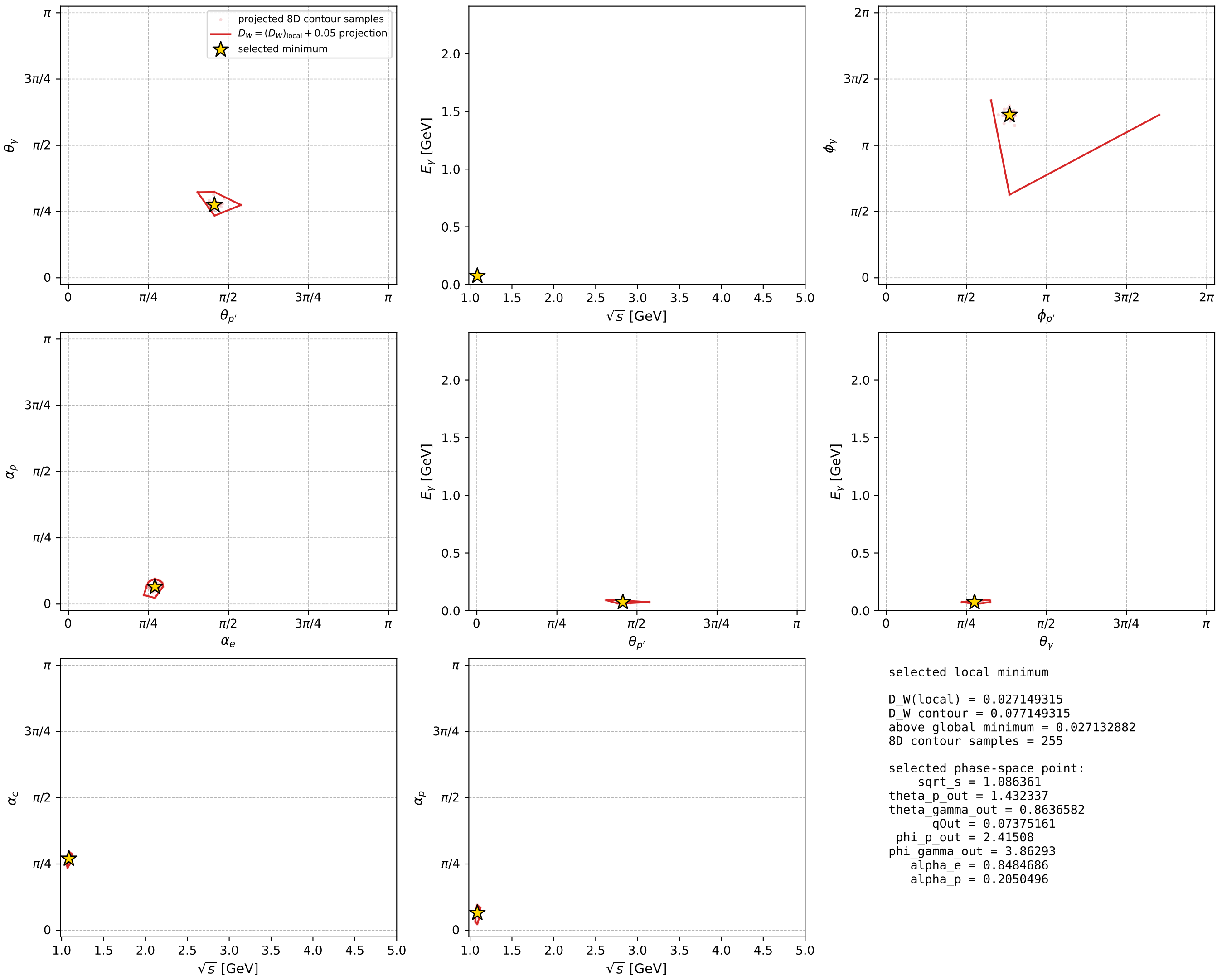
Transverse plane



Leading final-state amplitudes (N=6, retained=97.4%)



electron: polarization cluster P3, local minimum 1101; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

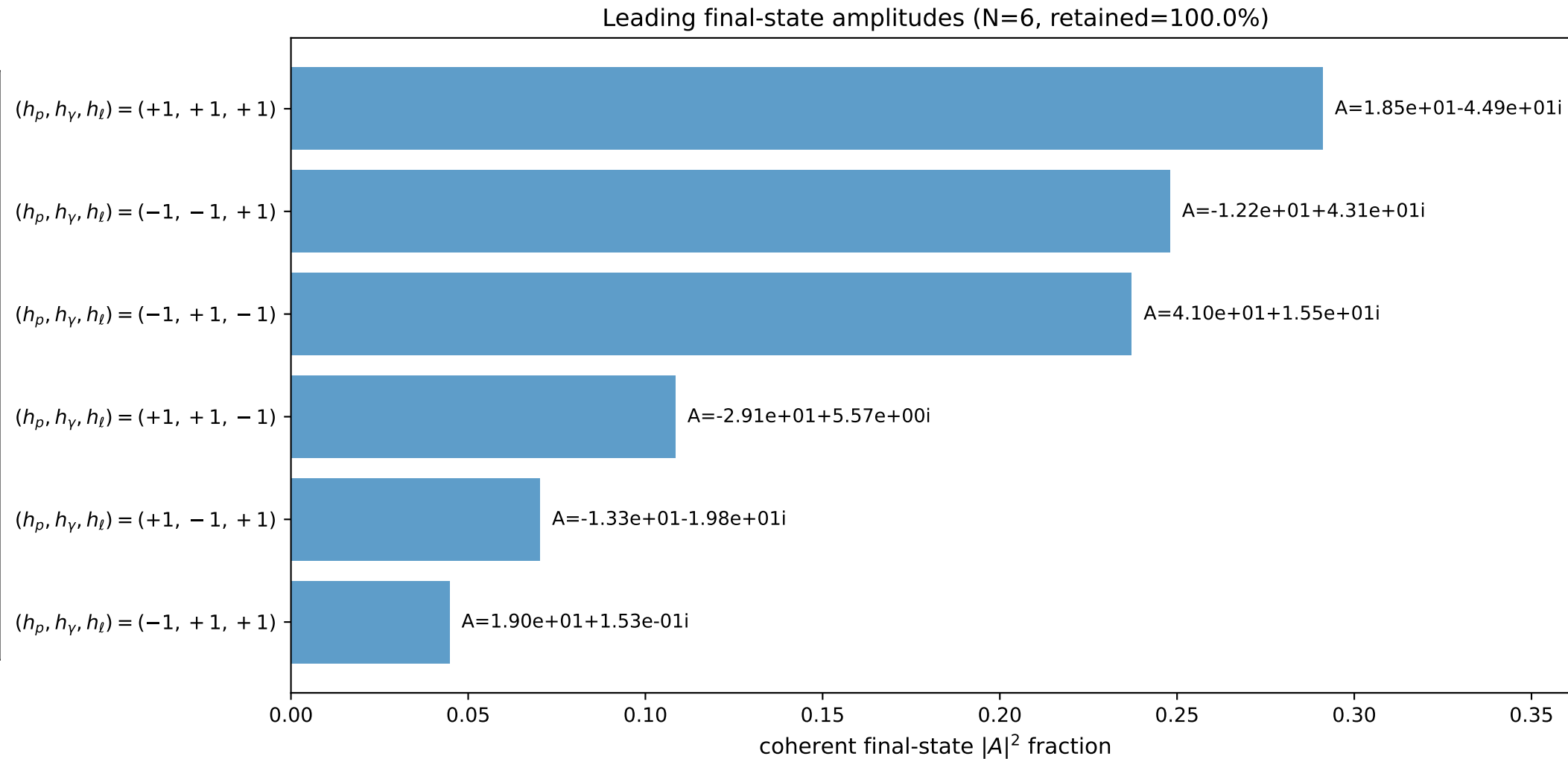
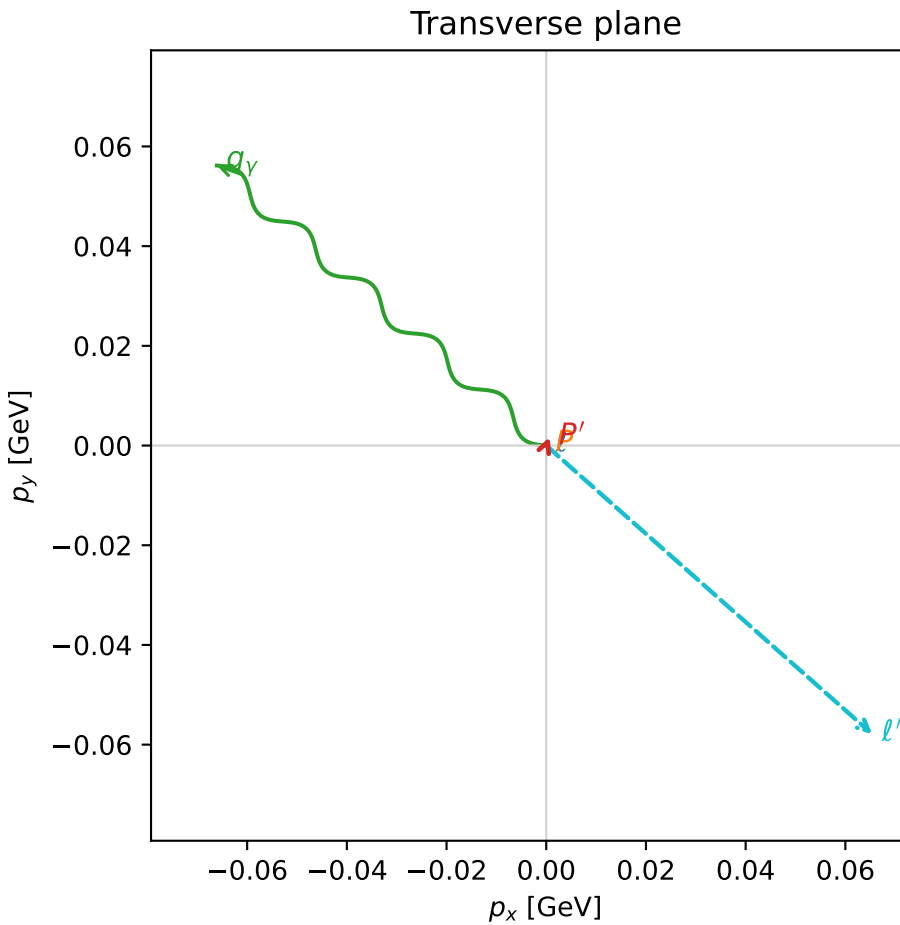
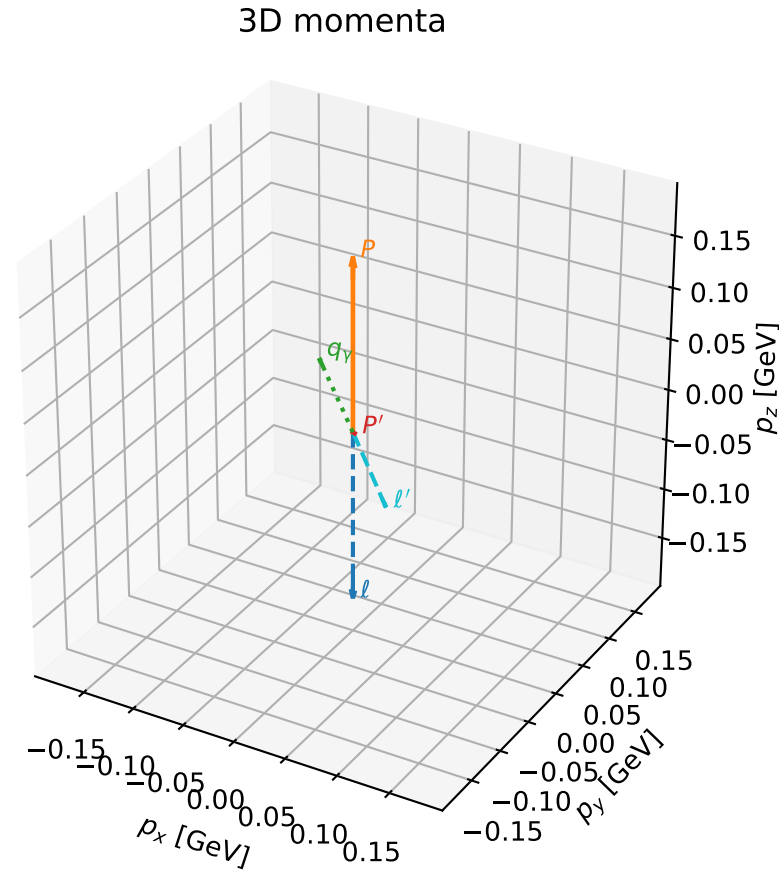


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

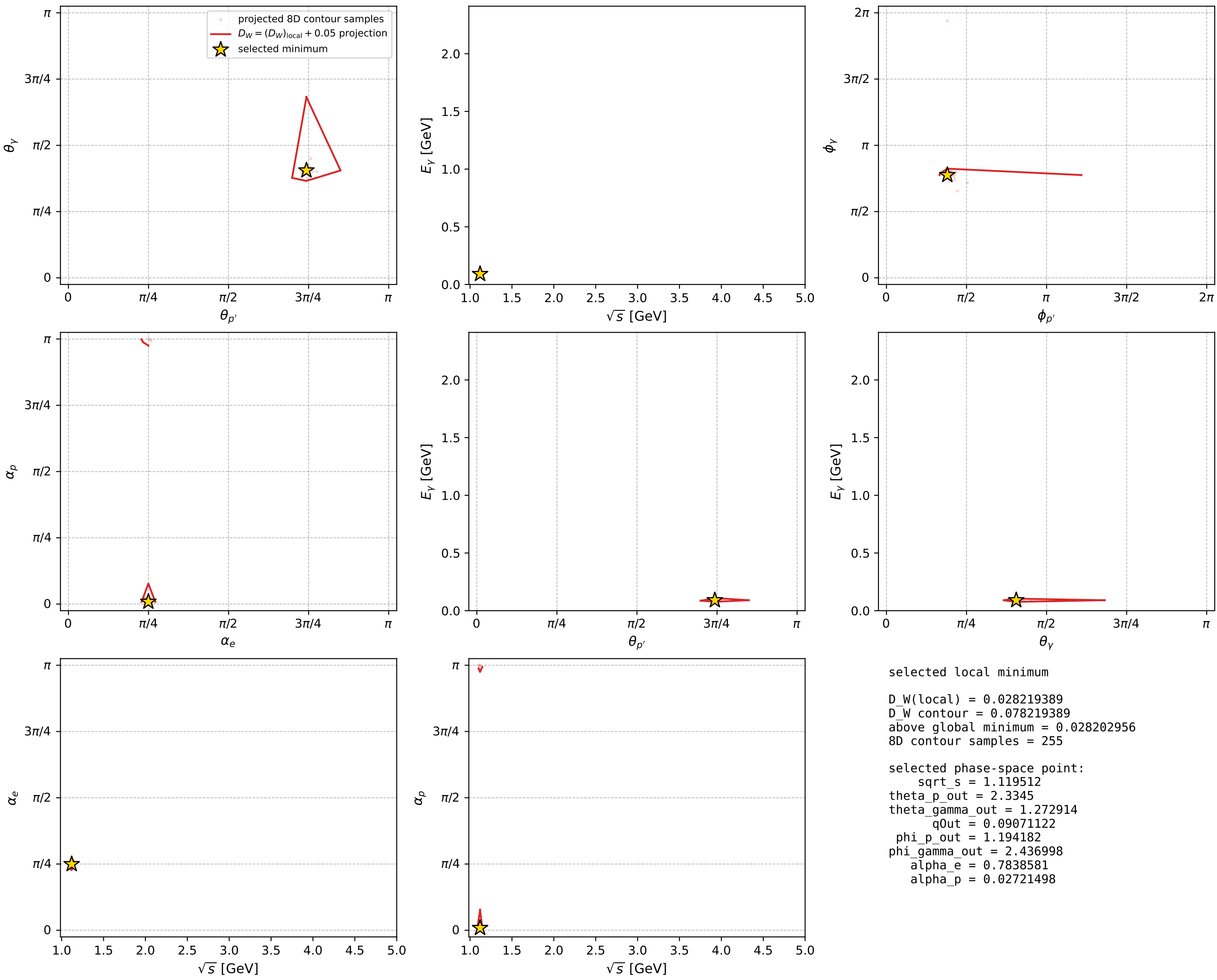
dw_mixing_angles_polarization_03_local_minimum_1109 D_W=0.0282194
region: polarization_cluster_03_local_minimum_1109
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: $\alpha_e=0.78385812$ rad
incoming lepton: $0.708195|+\rangle + 0.706017|-\rangle$
proton mixing angle: $\alpha_p=0.027214982$ rad
incoming proton: $0.99963|+\rangle + 0.0272116|-\rangle$

$s=1.25331$, $\sqrt{s}=1.11951$, $|\vec{P}|=0.166797$, $|\vec{P}'|=0.00253809$
 $\theta_{P'}=2.3345$, $\theta_V=1.27291$, $\phi_{P'}=1.19418$, $\phi_V=2.437$
 $E_V=0.0907112$, $\theta_{\ell'}=1.84823$, $\phi_{\ell'}=5.55871$
 $Q^2=0.0219932$, $x_B=0.114453$, $t=-0.0281967$, $W^2=1.05001$, $y=0.514533$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1668$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1668$
 P $E= 0.9527$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1668$
 ℓ' $E= 0.0908$ $p_x= 0.0654$ $p_y= -0.0579$ $p_z= -0.0249$
 P' $E= 0.9380$ $p_x= 0.0007$ $p_y= 0.0017$ $p_z= -0.0018$
 q_V $E= 0.0907$ $p_x= -0.0661$ $p_y= 0.0562$ $p_z= 0.0266$



electron: polarization cluster P3, local minimum 1109; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

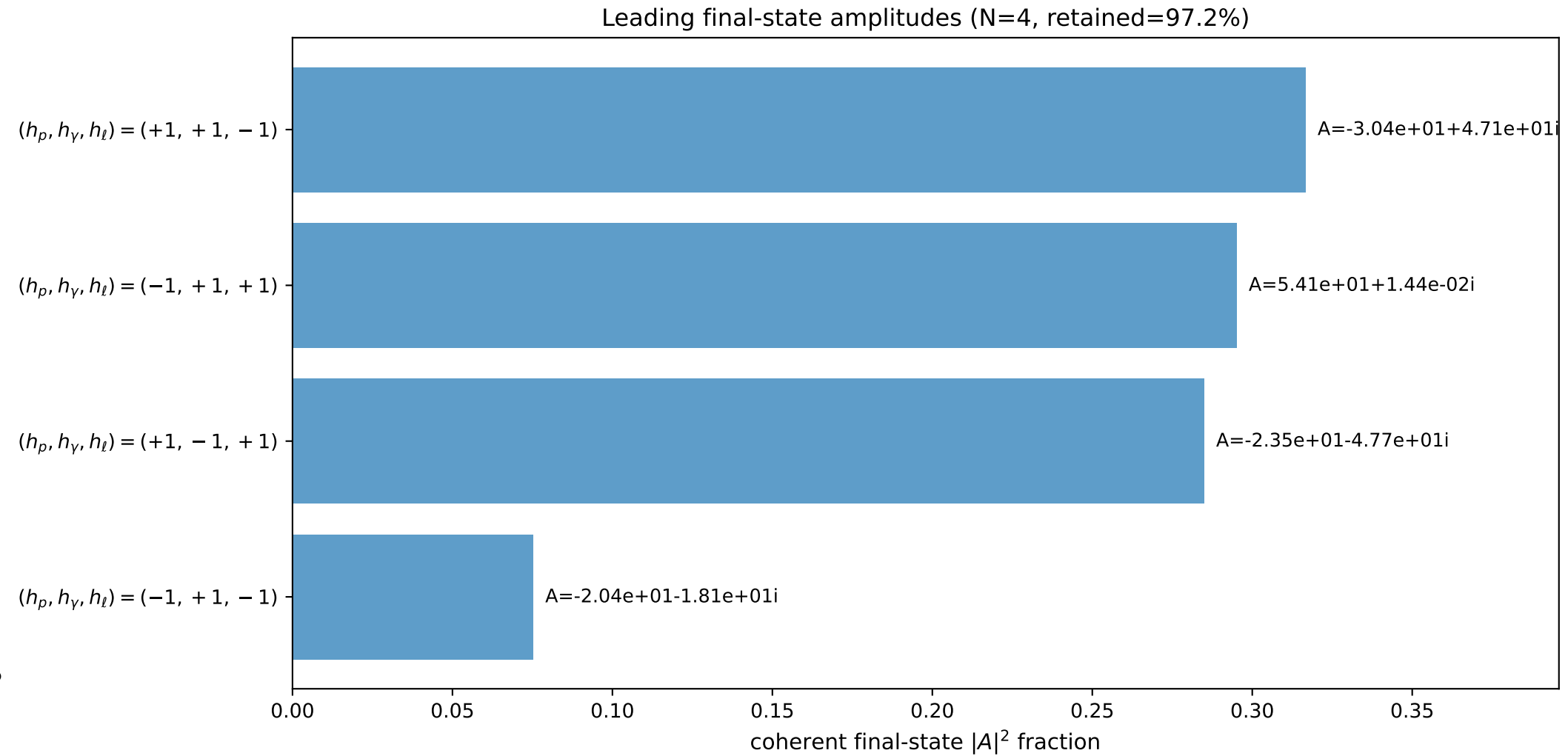
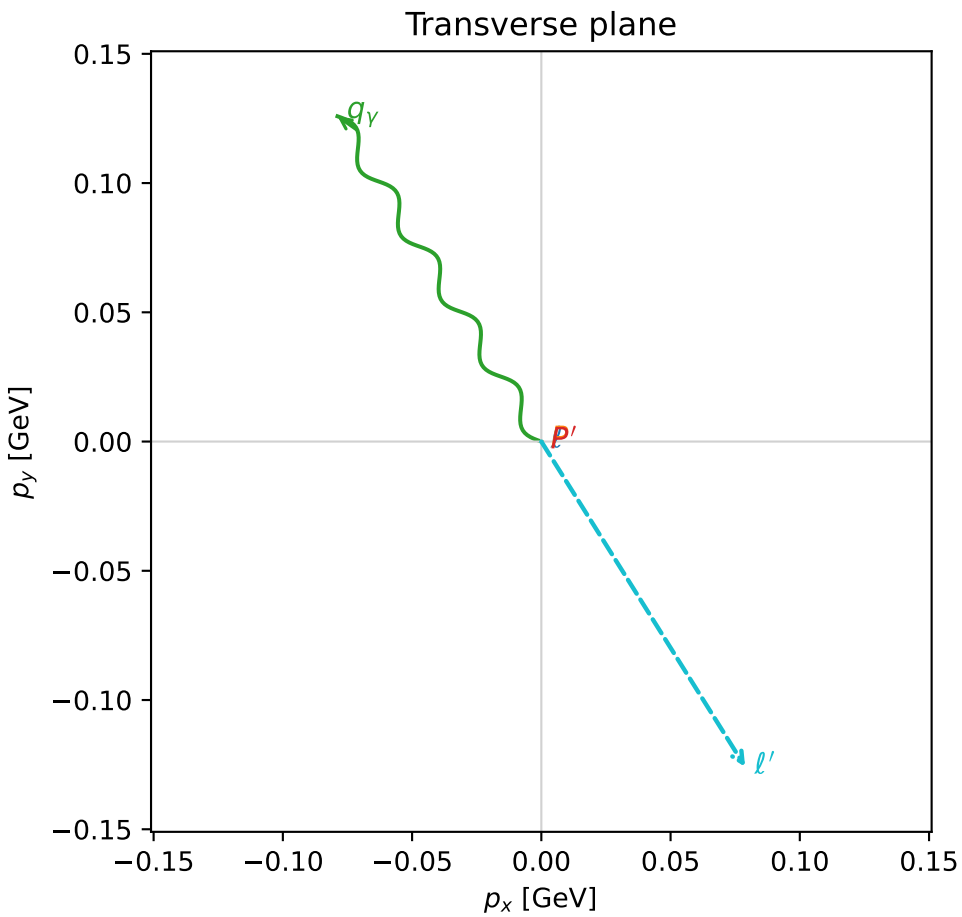
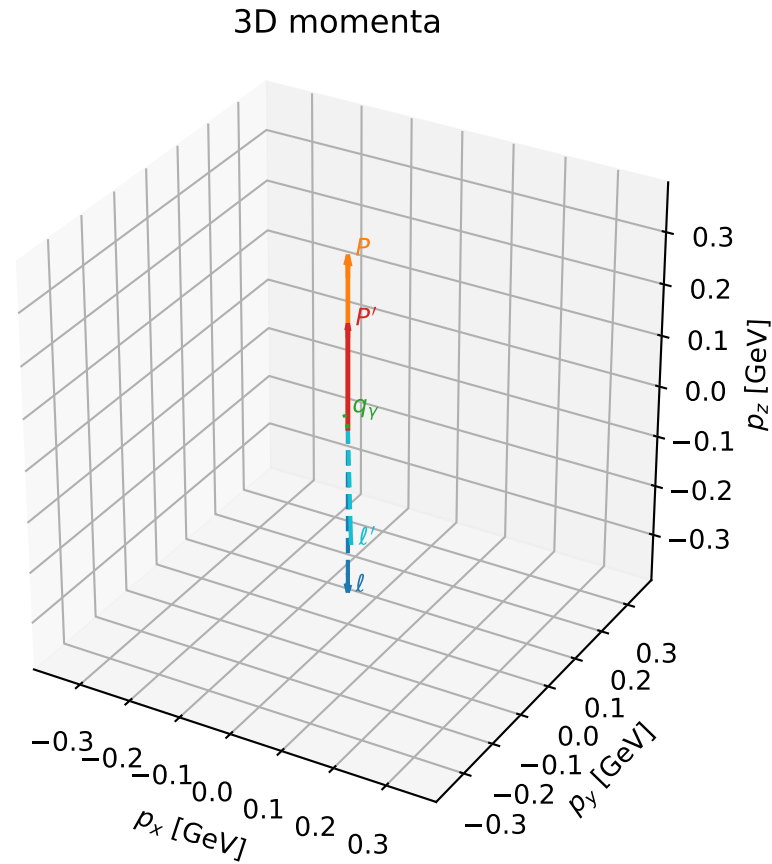


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

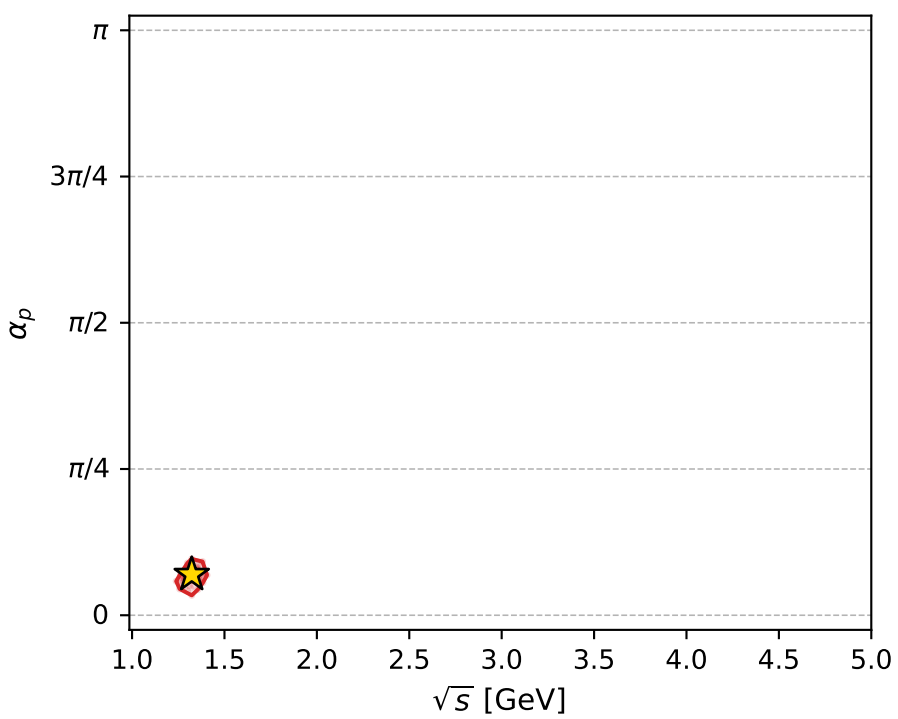
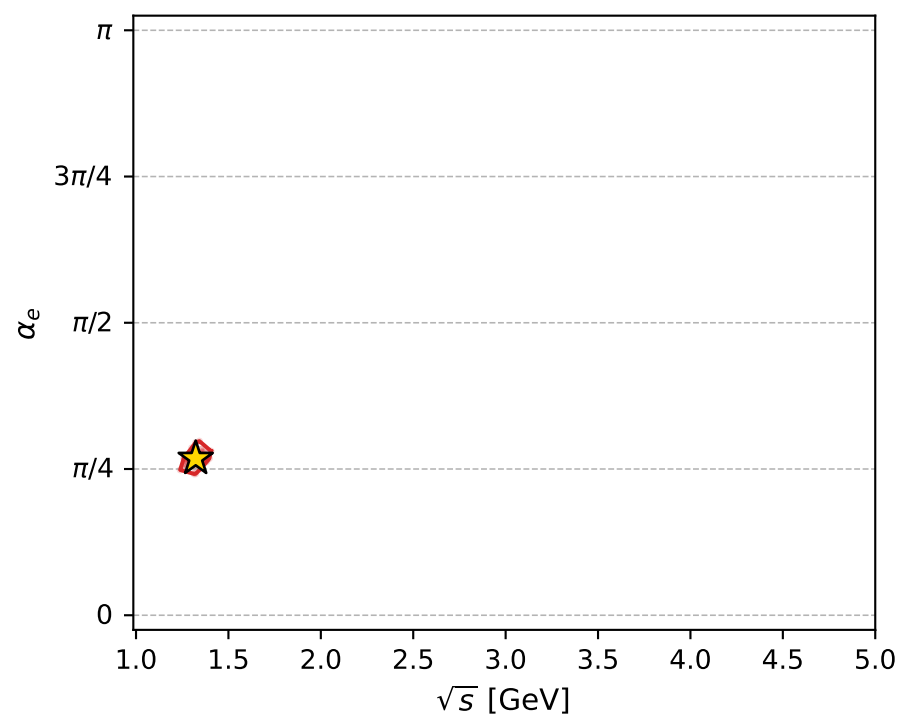
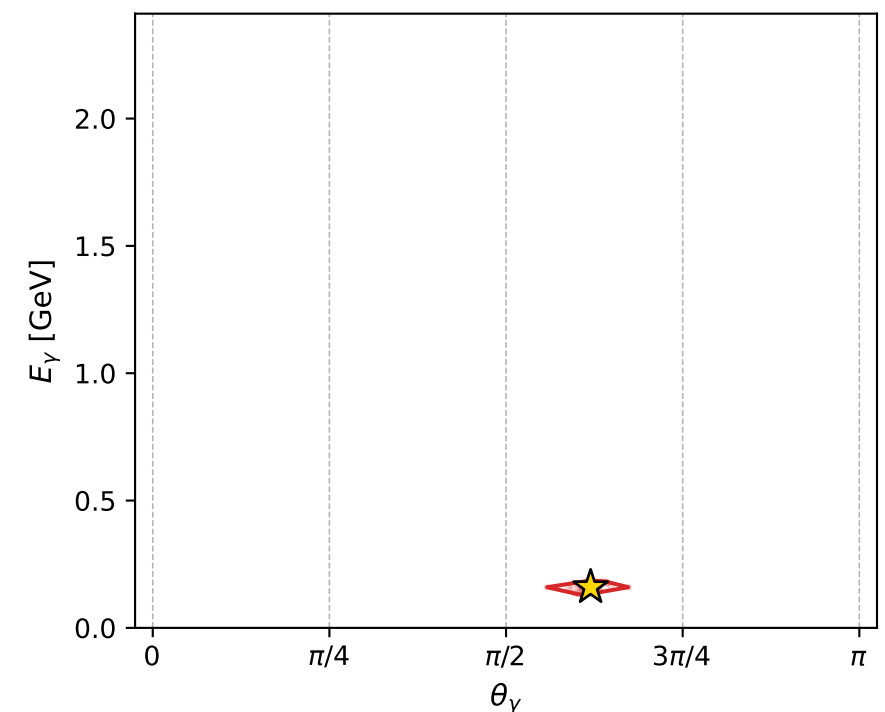
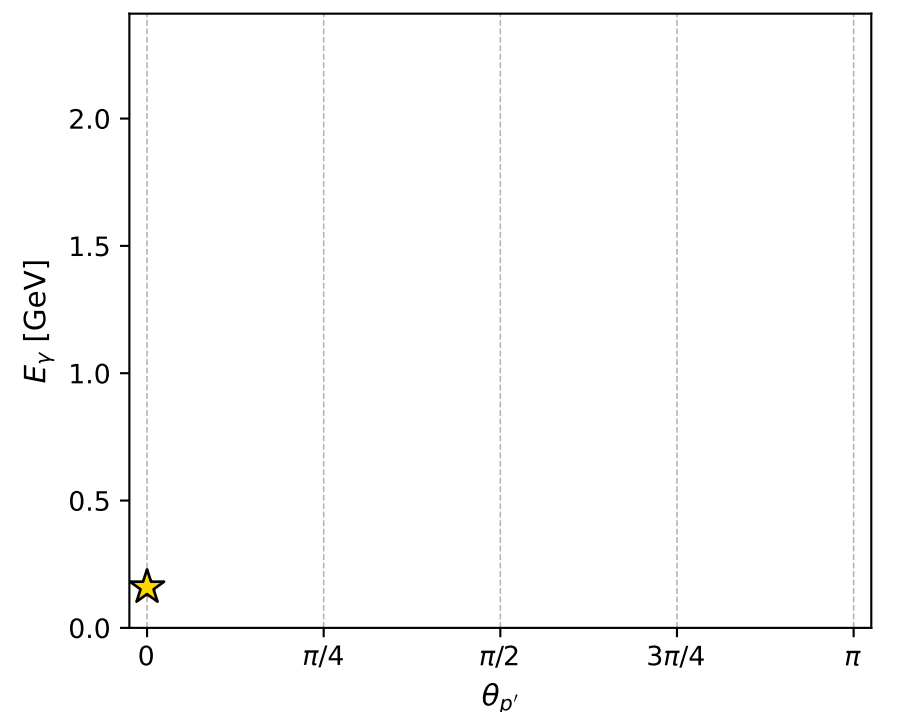
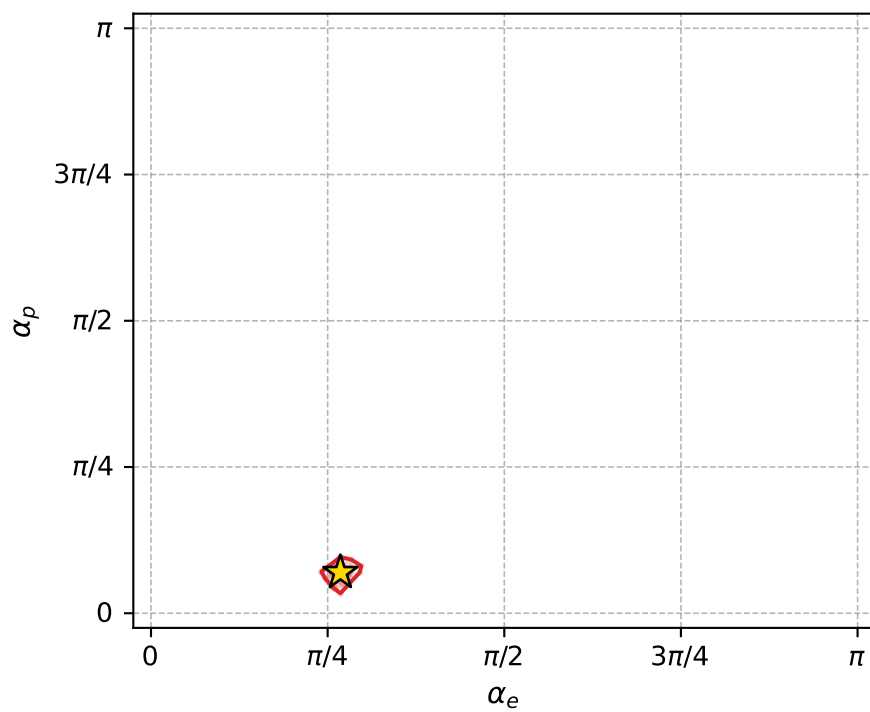
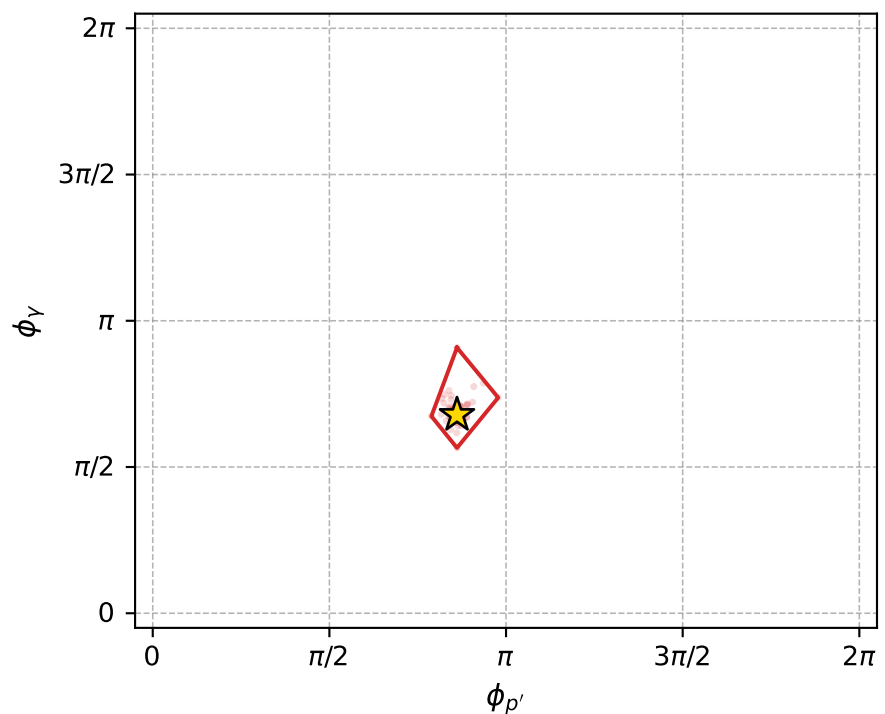
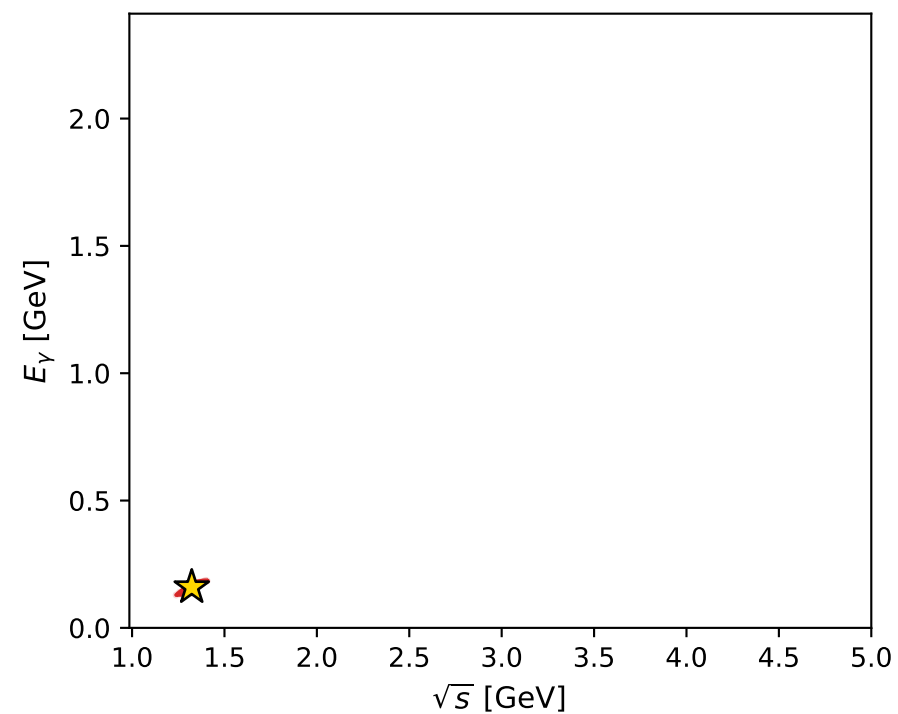
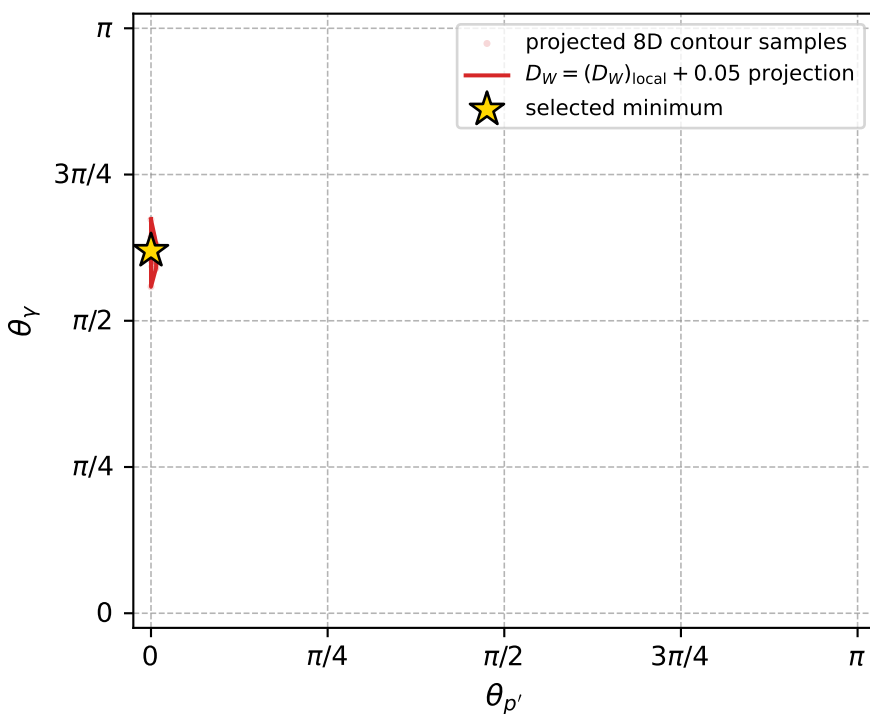
dw_mixing_angles_polarization_03_local_minimum_1126 D_W=0.0335594
 region: polarization_cluster_03_local_minimum_1126
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.84205121$ rad
 incoming lepton: $0.665934|+\rangle +0.746011|-\rangle$
 proton mixing angle: $\alpha_p=0.21844966$ rad
 incoming proton: $0.976235|+\rangle +0.216716|-\rangle$

$s=1.7502$, $\sqrt{s}=1.32295$, $|\vec{P}|=0.328945$, $|\vec{P}'|=0.19909$
 $\theta_{P'}=0$, $\theta_V=1.94676$, $\phi_{P'}=2.70549$, $\phi_V=2.13058$
 $E_V=0.159646$, $\theta_{\ell'}=2.32844$, $\phi_{\ell'}=5.27217$
 $Q^2=0.0420637$, $x_B=0.113205$, $t=-0.0156297$, $W^2=1.20935$, $y=0.426916$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3289$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3289$
 P $E= 0.9940$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3289$
 ℓ' $E= 0.2044$ $p_x= 0.0789$ $p_y= -0.1258$ $p_z= -0.1405$
 P' $E= 0.9589$ $p_x= -0.0000$ $p_y= 0.0000$ $p_z= 0.1991$
 q_V $E= 0.1596$ $p_x= -0.0789$ $p_y= 0.1258$ $p_z= -0.0586$



electron: polarization cluster P3, local minimum 1126; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.033559438$
 $D_W \text{ contour} = 0.083559438$
 above global minimum = 0.033543005
 8D contour samples = 140

selected phase-space point:
 $\text{sqrt_s} = 1.322952$
 $\text{theta_p_out} = 0$
 $\text{theta_gamma_out} = 1.946755$
 $\text{q0ut} = 0.1596457$
 $\text{phi_p_out} = 2.705486$
 $\text{phi_gamma_out} = 2.130581$
 $\text{alpha_e} = 0.8420512$
 $\text{alpha_p} = 0.2184497$